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Editors' Note

The *Handbook of Hispanic Linguistics* is intended to present the state of the art of research in all aspects of the Spanish language. It includes chapters on all main areas of language structure (phonology, morphology, syntax, and semantics), as well as chapters on sociolinguistic and psycholinguistic research related to Spanish. Research on both first- and second-language acquisition is also represented.

All chapters have undergone a review process. Our gratitude goes to several of our authors for acting as reviewers of chapters by other contributors. In addition, we are indebted to various other scholars for helping with this process, among them Inés Antón-Méndez, Xabier Artiagoitia, Kurt Blaylock, Giuli Dusias, Cecile McKee, Miguel Rodríguez-Mondoñedo, Ana Teresa Pérez-Leroux, Carme Picallo, Pilar Prieto, Andrés Saab, Victoria Vázquez, Erik Willis, and Mary Zampini.

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José Ignacio Hualde
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1 Geographical and Social Varieties of Spanish: An Overview

JOHN M. LIPSKI

1 Introduction

According to Spain's government-sponsored Cervantes Institute,[1] there are more than 400 million native or near-native speakers of Spanish in the world, distributed across every continent except Antarctica.[2] Spanish is the official language in twenty-one countries plus Puerto Rico; is the *de facto* first language for most of Gibraltar (Fierro Cubiella 1997; Kramer 1986); still maintains a small foothold in the Philippines, where it once enjoyed official status (Lipski 1987a); and is known and used on a regular basis by many people in Haiti (Ortiz López forthcoming), Aruba and Curaçao (Vaquero de Ramírez 1986), and Belize (Hagerty 1979). Moreover, in the country that harbors one of the world's largest native Spanish-speaking populations (effectively tied for second place with Colombia, Argentina, and Spain, and surpassed only by Mexico), the Spanish language has no official status at all. That country is the United States, which has at least 40 million native Spanish speakers, that is, some 10% of the world's Spanish-speaking population (Lipski 2008c).

All languages change across time and space, and Spanish is no exception. Although the Spanish language was relatively homogeneous in Spain circa 1500 – the time when Spanish first expanded beyond the boundaries of the Iberian Peninsula – it has diversified considerably as it spread over five continents during more than five hundred years. Many factors are responsible for the evolution of Spanish, including the natural drift of languages over time, contact with other languages, internal population migrations, language propagation through missionary activities, the rise of cities, and the consequent rural–urban sociolinguistic divisions, educational systems, community literacy, mass communication media, and official language policies. It is therefore not surprising that although the Spanish language retains

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a fundamental cohesiveness throughout the world, social and geographical variation is considerable. To explore all varieties of Spanish would require several volumes; the following sections offer an overview of regional and social variation in Spanish by means of a number of representative cases, selected to give a sense of the full range of possibilities.

2 Dialect divisions in Spain

Spain contains a complex array of regional and social dialects, but the most striking division – immediately noticeable by Spaniards and visitors alike – separates north and south. In the popular imagination, this translates to Castile–Andalusia, but to the extent that dialects exhibit geographical boundaries, the north–south distinction only approximately follows the borders between these historically distinct regions, while also encompassing other areas. The primary features used to impressionistically identify regional origins in Peninsular Spanish are phonetic: "southern" traits include aspiration or elision of syllable- and word-final/s/(e.g., *vamos pues* [ˈba.moh.ˈpu̯e] 'let's go, then'), loss of word-final/ɾ/(e.g., *por favor* [po.fa.ˈβo] 'please'), and the pronunciation of preconsonantal/l/as [ɾ] (e.g., *soldado* [soɾ.ˈða.o] 'soldier'). Traits widely regarded as "northern" include the apico-alveolar pronunciation [s°] of/s/, the strongly uvular pronunciation [χ] of the posterior fricative/x/(e.g., *caja* [ˈka.χa] 'box'), and the phonological distinction/θ/-/s/(e.g., *casa* [ˈka.s°a] 'house' -*caza* [ˈka.θa] 'hunting'). In reality, the regional distribution of these traits does not conform to a simple north–south distinction, since the traits intersect with one another and with additional regionalized features in fashions that cannot be reduced to a single geographical matrix. Most traditional dialect classification schemes for Peninsular Spanish cluster around historically recognized kingdoms and contemporary autonomous regions, albeit with considerable overlap of defining traits along border areas (e.g., Zamora Vicente 1967 and the studies in Alvar 1996). In contemporary Spain, at least the following geographically delimited varieties of Spanish can be objectively identified by linguists, as shown in (1):

(1) Geographically delimited varieties of Spanish:
- northern Castile, including Salamanca, Valladolid, Burgos, and neighboring provinces;
- northern Extremadura and León, including the province of Cáceres, parts of León, western Salamanca province, and Zamora;
- Galicia, referring to the Spanish spoken both monolingually and in contact with Galician;
- Asturias, especially inland areas such as Oviedo;
- the interior Cantabrian region, to the south of Santander;
- the Basque Country, including Spanish as spoken monolingually and in contact with Basque;
- Catalonia, including Spanish spoken in contact with Catalan;

- southeastern Spain, including much of Valencia, Alicante, Murcia, Albacete, and southeastern La Mancha;
- eastern Andalusia, including Granada, Almería, and surrounding areas;
- western Andalusia, including Seville, Huelva, Cádiz, and the Extremadura province of Badajoz – the Spanish of Gibraltar is also included;
- south-central and southwest Spain, including areas to the south of Madrid such as Toledo and Ciudad Real.

Features specific to this expanded list of regional varieties as well as socially-stratified variables within given areas will be presented in subsequent sections.

3 Dialect divisions in Latin America

There is no consensus on the classification of Latin American Spanish dialects due to the vast territorial expanse in question, the scarcity of accurate data on the speech of many regions, and the high degree of variability due to multiple language contact environments, internal migrations, and significant rural–urban linguistic polarization. In the popular imagination (e.g., as mentioned in casual conversations), Latin American Spanish dialects are defined by national boundaries, thus Mexican Spanish, Argentine Spanish, Peruvian Spanish, etc. Objectively, such a scheme cannot be seriously maintained, except for a few small and linguistically rather homogeneous nations. Rather, Latin American Spanish is roughly divided into geographical dialect zones based on patterns of settlement and colonial administration, contact with indigenous and immigrant languages, and relative proportions of rural and urban speech communities. For pedagogical purposes, the following classification, which combines phonetic, morphological, socio-historical, and language-contact data, provides a reasonable approximation to actually observable dialect variation in Latin America. This classification, shown in (2), is based on Lipski (1994), where the other dialect classifications are also discussed:

(2) Latin American Spanish dialect classifications:
- Mexico (except for coastal areas) and southwestern United States;
- Caribbean region: Cuba, Puerto Rico, Dominican Republic, Panama, Caribbean coast of Colombia and Venezuela, Caribbean coast of Mexico, and also Mexico's Pacific coast;
- Guatemala, parts of the Yucatan, and Costa Rica;
- El Salvador, Honduras, and Nicaragua;
- Colombia (interior) and neighboring highland areas of Venezuela;
- Pacific coast of Colombia, Ecuador, and Peru;
- Andean regions of Ecuador, Peru, Bolivia, northwest Argentina, and northeast Chile;
- Chile;
- Paraguay, northeastern Argentina, and eastern Bolivia;
- Argentina (except for extreme northwest and northeast) and Uruguay.

4 Major variation patterns: phonetics and phonology

Overviews of the pronunciation of Spanish in Spain are found in Alvar (1996) and for Latin America in Canfield (1981) and Lipski (1994). Among the most rapidly identifiable features separating regional and social varieties of Spanish are differences in pronunciation, both the realization of particular sounds and combinations of sounds, and the presence or absence of certain phonological oppositions. The following sections outline some of the more salient phonetic and phonological dimensions of Spanish dialect differentiation.

4.1 *Presence–absence of oppositions:* /s/-/θ/,/ɟ/-/ʎ/

In general, all regional and social varieties of Spanish share the same inventory of vowel and consonant phonemes, with two exceptions: the voiceless interdental fricative /θ/ and the palatal lateral /ʎ/ have geographically delimited distribution, and are absent in the remaining varieties of Spanish. The phoneme /θ/ occurs as an independent phoneme opposed to /s/ (e.g., *casa* [ˈka.sa] 'house' -*caza* [ˈka.θa] 'hunting') only in Peninsular Spain. The opposition /s/-/θ/ characterizes all Peninsular varieties of Spanish except for western and central Andalusia. In western Andalusia, the neutralization of /s/-/θ/ in favor of /s/ is known as *seseo*, and it typifies the speech of these provinces. Many speakers in rural areas and smaller towns throughout Andalusia neutralize the opposition in favor of [θ] (e.g., *mi casa* [mi ˈka.θa] 'my house'). This neutralization is known as *ceceo*, and is usually stigmatized by the speakers themselves and in neighboring urban areas; *ceceo* imitations figure prominently in the verbal repertoires of many Spanish comedians as well as in dialect literature. The opposition /s/-/θ/ is not found in the Canary Islands (where *seseo* is the norm), nor in any part of Latin America. In the residual Spanish still found in the Philippines, the opposition /s/-/θ/ occurs sporadically, given the varying Peninsular origins of the ancestors of Philippine Spanish speakers (Lipski 1987a). In Equatorial Guinea, the only officially Spanish-speaking nation in Africa, the opposition /s/-/θ/ is also variable since the Peninsular sources for Guinean Spanish came both from Castile (where the distinction is made) and from Valencia (where *seseo* used to prevail). Most Guineans, except for those who have lived extensively in Spain, are not consistent with respect to the /s/-/θ/ distinction (Lipski 1985a).

The palatal lateral phoneme /ʎ/ (written as *ll*) was once opposed to /ɟ/ (written as *y*) in all varieties of Spanish (e.g., *se calló* [se ka.ˈʎo] 'he/she became silent' -*se cayó* [se ka.ˈɟo] 'he/she fell down'). The opposition, with few minimal pairs to its credit, began to erode in favor of non-lateral pronunciations beginning in the sixteenth century, and today only a few Spanish-speaking regions maintain the distinction. The neutralization of /ʎ/-/ɟ/ in favor of the latter phoneme is known as *yeísmo*. In Peninsular Spain, /ʎ/ occurs as an independent phoneme in a few northern areas, but is rapidly disappearing today among younger generations. In the Canary Islands, /ʎ/ was retained robustly by all speakers until the final decades of the twentieth century, but is now rapidly fading. The phoneme/ʎ/is not present in the Spanish of Equatorial Guinea and is heard only occasionally in Philippine Spanish. In Latin America, the phoneme/ʎ/is maintained in all regional and social dialects of Paraguay and Bolivia, and in neighboring areas of northeastern and northwestern Argentina. In highland Peru, pockets of/ʎ/still remain, as they do in the central highlands of Ecuador. In Quito and other northern highland areas of Ecuador, the lateral pronunciation of/ʎ/gives way to a groove fricative pronunciation [ʒ], but the opposition/ʎ/-/ɟ/is still maintained (e.g., *halla* [ˈa.ʒa] 'he/she finds' – *haya* [ˈa.ɟa] 'that he/she may have') (Haboud and de la Vega 2008).

4.2 *Realization of coda consonants: /s/,/n/,/l/,/ɾ/*

In Spanish the greatest variation in the pronunciation of consonants occurs in post-nuclear position, often referred to as "coda" or "syllable-final." The post-nuclear or coda position is universally regarded as the weakest in terms of neutralization of oppositions, replacement by weaker versions of the consonant, such as approximants (sounds with very slight constrictions, weaker than fricatives) or vocoids (near-vowel sounds such as semivowels), depletion of all supralaryngeal features (meaning those features involving the action of the tongue, lips, pharynx, and velum), and total effacement (Hualde 1989a, 2005). Coda position is also the environment in which the greatest sociolinguistic differentiation of Spanish dialects typically occurs. The consonants most affected by coda-weakening processes are/s/,/ɾ/,/l/, and/n/.

By far the most common modification of Spanish coda consonants involves/s/, including aspiration to [h], deletion, and other instances of weakening. In Spain, syllable- and word-final/s/is aspirated or elided massively in the south, from Extremadura through Andalusia (including Gibraltar) (Lipski 1987b), Murcia, and parts of Alicante, but even in central and some northern regions (e.g., Cantabria), coda/s/is frequently aspirated. In the Canary Islands, weakening of coda/s/ occurs at rates comparable to Andalusia (Lipski 1985b). In Latin America, reduction of coda/s/reaches its highest rates in the Caribbean (Cuba, Puerto Rico, the Dominican Republic, Panama, Venezuela, coastal Colombia), as well as on the Mexican coast centering on Veracruz and Campeche. On nearly all Mexico's Pacific coast, final/s/is also reduced nearly as frequently as in the Caribbean (Moreno de Alba 1994). In Central America,/s/-reduction is massive in Nicaragua, and occurs at a lesser rate in El Salvador and Honduras. In South America, the entire Pacific coast from Colombia through Chile is a zone of heavy/s/-reduction. In Argentina and Uruguay,/s/-reduction is somewhat tempered in the large cities, but reaches high levels in provincial areas, as it does throughout Paraguay and eastern Bolivia. It is more economical to mention those Spanish-speaking areas where coda/s/ strongly resists effacement: most of northern Spain, most of Mexico, Guatemala, Costa Rica, and the highlands of Colombia, Peru, Ecuador, and Bolivia (Lipski 1984, 1986).

Found in some/s/-aspirating dialects is the aspiration of word-INITIAL postvocalic/s/, as in *la semana* [la.he.ˈma.na] 'the week.' Aspiration of word-initial/s/is most frequently found in the vernacular speech of El Salvador and much of Honduras (Lipski 1999a), and also in the traditional Spanish of northern New Mexico (Brown 2004). Rates of aspiration of word-initial/s/are considerably lower than those for word-final/s/-reduction, but there are no Spanish dialects in which word-initial/s/is aspirated while word-final/s/remains intact. Unlike aspiration of syllable- and word-final/s/, which is often just a regional trait with no negative connotations, aspiration of word-initial/s/is frequent only in colloquial speech in the regions where it occurs, and is predominantly found among less educated speakers.

In much of central Spain where reduction of coda/s/reaches only moderate levels, the phonetic result before a following consonant is a weak [ɾ] as in *los niños* [loɾ.ˈni.ɲos] 'the children.' This variant is not consistently found anywhere in Latin America.

Coda liquids/l/and/ɾ/are particularly susceptible to weakening processes in Spanish, and most weakening phenomena affect both consonants to some extent. In phrase-final position, the most common result is complete elision. Loss of phrase-final/l/and/ɾ/is common in southern Spain; it is also frequent in most regional and social dialects of the Canary Islands. In Latin America, deletion of word-final/ɾ/is common in eastern Cuba, Panama, the Caribbean coast of Colombia, much of Venezuela, along the Pacific coast of Colombia, Ecuador, and Peru, and in Afro-Bolivian Spanish (Lipski 2008a). In all of these regions, deletion of/ɾ/is associated with colloquial speech, but does not necessarily carry a heavy stigma, as indicated. Deletion of final/l/is less frequent in careful speech. In southern Spain (including the Canary Islands), the opposition of preconsonantal/l/and/ɾ/is tenuous, with neutralization in favor of [ɾ] constituting an Andalusian stereotype (e.g., *el niño* [eɾ.ˈni.ɲo] 'the child'). In some parts of the Canary Islands and occasionally in Murcia, coda/ɾ/is realized as [l] as in *puerta* [ˈpu̯el.ta] 'door.' The change of coda/ɾ/ to [l] is more common in the Caribbean, particularly in Puerto Rico and the Dominican Republic, in central Cuba and eastern Venezuela. Lateralization of/ɾ/, although occurring frequently in the aforementioned dialects, is often criticized, and forms the basis for jokes and popular cultural stereotypes. Found in western Cuba, the Caribbean coast of Colombia (and in the Afro-Colombian creole language Palenquero: Schwegler 1998: 265; Schwegler and Morton 2003), and parts of Andalusia is loss of word-internal preconsonantal coda liquids combined with gemination of the following consonant; when the following consonant is a voiced obstruent/b/,/d/, or/g/the resulting geminate is always a stop, not a fricative or approximant as normally occurs intervocalically. Examples include *algo* [ˈag.go] 'something,' *puerta* [ˈpu̯et.ta] 'door,' and *caldo* [ˈkad.do] 'soup.' Gemination is frequently depicted in dialect literature, always in portrayals of uneducated speakers, and is usually avoided in careful speech. Another regional variant is "vocalization" of coda liquids to semivocalic [i̯]; this occurs primarily in the Cibao region in the north of the Dominican Republic, and was once found occasionally in Cuba, Puerto Rico, and southeastern Spain (e.g., *por favor* [poi̯.fa.ˈβoi̯] 'please,' *capital* [ka.pi.ˈtai̯] 'capital'). This pronunciation is stigmatized and found in many literary stereotypes, particularly in the Dominican Republic.

Word-final nasal consonants are also subject to regional and social variation in Spanish. The most common alternative to the etymological [n] is a velar nasal [ŋ], which often disappears, leaving behind a nasalized vowel. Velarization of phrase-final/n/is the rule in Galicia and parts of Asturias, Extremadura, Andalusia, the Canary Islands, all Caribbean and Central American dialects, along the Pacific coast of Colombia, Peru, and Ecuador, and sporadically in the Andean highlands. Word-final prevocalic nasals are typically also velarized in these dialects, although with generally lower rates than for phrase-final/n/(Lipski 1986): *un otro* [uŋ.ˈo.tɾo] 'another.' Velarization is almost never explicitly acknowledged by naive (e.g., untrained in linguistics) speakers of any dialect, and many velarizing speakers are unable to accurately identify this sound in their own speech and that of other members of their speech community even when this pronunciation is brought to their attention.

In the Spanish of the Yucatan, Mexico, phrase-final/n/is often realized as [m], as in *Yucatán* [Øu.ka.ˈtam], *Colón* [ko.ˈlom] 'Columbus,' and *pan* [pam] 'bread' (Michnowicz 2008). This pronunciation has traditionally been associated with Maya-dominant bilinguals, but as Yucatan cities, particularly Mérida, grow in economic importance through tourism and light industry, many non-Maya-speaking residents have come to regard the labialization of word-final/n/with pride, as a marker of local identity. The change of final/n/to [m] also occasionally occurs in western Colombia (Montes 1979).

4.3 Realization of rhotics/ɾ/and/r/

Spanish has two rhotic ("r"-like) phonemes, the single tap/ɾ/and the trill/r/. All monolingual varieties of Spanish maintain this opposition in some form (e.g., in *caro* [ˈka.ɾo] 'expensive' vs. *carro* [ˈka.ro] 'cart,' *cero* [ˈse.ɾo] 'zero' vs. *cerro* [ˈse.ro] 'mountain'). Most Sephardic (Judeo) Spanish has lost the opposition/ɾ/-/r/, usually in favor of the tap. In the Afro-Bolivian Spanish dialect, this distinction is often neutralized in favor of the tap (Lipski 2008a), while in the Spanish of Equatorial Guinea, the tap–trill distinction is also tenuous, but is frequently neutralized in favor of the trill (Lipski 1985a) (e.g., *tres* [tres] 'three,' *pero* [ˈpe.ro] 'but'). The tap phoneme shows little regional or social variation except in coda position, where it is subject to the range of elision and neutralizations described in the preceding section. The combination/tɾ/fuses into an alveolar quasi-affricate, almost [ʧ], in the Andean highlands, Chile, Paraguay, northeastern Argentina, Costa Rica, Guatemala, and sometimes in New Mexico and parts of central Mexico. In these dialects, *otro* 'other' and *ocho* 'eight' are pronounced nearly identically.

The "trill"/r/, on the other hand, is subject to considerable regional and some social variation (Hammond 1999, 2000 offers a survey). The most common alternative to the alveolar trill is a voiced prepalatal fricative [ʒ], found throughout the Andean region (highland Ecuador, Peru, and Bolivia), as well as in much of northern Argentina, parts of Paraguay, and occasionally in Chile. In Central America, fricative/r/is common in Guatemala, and is often heard in Costa Rica, although an alveolar [ɹ] or retroflex [ɻ] approximant, quite similar to English *r*, is more commonly heard in Costa Rican Spanish. In much of highland Bolivia, bilingual (Aymara-speaking) individuals often realize /r/ as [z], effectively creating minimal pairs based only on voicing, such as *caso* [ˈka.so] 'case' vs. *carro* [ˈka.zo] 'cart' (Mendoza 2008: 221). In much of the Caribbean region, particularly the Dominican Republic, Cuba, and parts of Puerto Rico, a partially devoiced (often described as "pre-aspirated") trill is found: *carrera* [ka.hré.ɾa] 'race.' A velar fricative [x] or uvular trill [ʀ] is a frequent variant of /r/ in Puerto Rico, especially in rural and interior areas, although it is generally stigmatized (López Morales 1983); for some speakers, *jamón* 'ham' and *Ramón* 'Raymond' are virtually homophonous.

4.4 Unstressed vowel raising and vowel reduction

The raising of final atonic /o/ to [u] and /e/ to [i] is confined to a few regions of Spain and Latin America, and typifies rural speech. Typical examples include *nochi* < *noche* 'night,' *lechi* < *leche* 'milk,' *vieju* < *viejo* 'old,' *buenu* < *bueno* 'good.' Oliver Rajan (2007) and Holmquist (2001, 2005) document this trait for the speech of rural highland Puerto Rico. In Spain this pronunciation predominates in Galicia, but is occasionally found in other northern regions, possibly reflecting the raising of unstressed mid-vowels in Galician and Asturian-Leonese. Vowel-raising generally carries negative prestige, and is avoided by individuals seeking upward or outward mobility.

The reduction of atonic vowels (shortening, devoicing, and in the extreme case, elision), is characteristic of only a few Spanish dialects, all found in Latin America, and all the result of previous or contemporary contact with Native American languages. This behavior is found in some parts of central Mexico, and in the Andean highlands of Ecuador, Peru, and Bolivia. Phonological analyses are found in Lipski (1990) and Delforge (2008). The most common instances occur in contact with /s/, as in *pres(i)dente* 'president,' *(e)studiant(e)s* 'students.'

4.5 Vowel harmony

Vowel harmony is not common among the Romance languages, although metaphony (the raising of tonic vowels conditioned by final atonic vowels) frequently occurred in the development of Spanish. In a few Spanish dialects, all in Spain, harmony systems have emerged. The most robust patterns are found in the northern Cantabrian region, historically influenced by Asturian and Leonese dialects to the west. In the Montes de Pas dialect (Penny 1969a, 1969b, 1978; McCarthy 1984), all vowels in a word agree in tenseness or laxness (also known as [-ATR] or "minus advanced tongue root"), with laxness harmony being triggered by the masculine singular count suffix -[ʊ], producing alternations like those in (3):

(3)	[pʊ.ˈʎʊ.kʊ]	'young chicken'	[pu.ˈʎu.kus]	'young chickens'
	[ˈmɨ.ɟʊ]	'mine' (m. sg. count)	[ˈmi.ɟu]	'mine (m. non-count)

The tense–lax distinction is found for all vowels except /e/, which is transparent to laxing harmony. Pasiego also exhibits vowel harmony for the feature [high], in which all atonic vowels in a word must agree in height with the tonic vowel; the low vowel /a/ does not participate in height harmony, neither as a trigger (when in tonic position) nor as a target (when atonic) (4):

(4)	[ku.ˈmi.ða]	'food'	[bin.di.ˈθir]	'to bless'
	[ko.ˈlor]	'color'	[xe.ˈle.ʧo]	'fern'

To the east of Cantabria lies Asturias, whose regional dialects are known collectively as Bable. Metaphony is found in this region, whereby word-internal vowels raise under the influence of a word-final high vowel: /a/ becomes [e], /e/ becomes [i], and /o/ becomes [u]. Depending upon the particular dialect, metaphony can affect all vowels in the phonological word (including clitics), all vowels in the final foot (the tonic vowel plus post-tonic vowels), or only the tonic vowel. Hualde (1989b) and Walker (2004) provide theoretical accounts of these different harmony mechanisms. Some Asturian examples, included in (5), are:

(5)	[ˈbleŋ.ku]	'white (m. sg.)'	[ˈbla.ŋka]	'white (f. sg.)'
	[ˈpi.lu]	'hair (count sg.)'	[ˈpe.lo]	'hair (mass)'
	[kal.ˈdi.ɾu]	'pot'	[kal.ˈde.ɾos]	'pots'

Another form of vowel harmony is found in southeastern Spain, in the eastern Andalusian dialect cluster. In all of Andalusia, coda consonants are weak and frequently elided, particularly in word-final position. In most varieties of Spanish, vowels are laxed in closed syllables (with coda consonants), and in western Andalusian and Latin American dialects in which word-final coda consonants such as /s/ and /r/ are elided, the vowel in the now open syllable reverts to the tense vowels found in other open syllables. Eastern Andalusian is unique in that the lax vowel remains after word-final consonants have been elided. This is particularly noticeable with non-low vowels, and results in minimal pairs, as in (6):

(6)	/tjene/	[ˈtje.ne]	'have (3 sg.)'
	/tjenes/	[ˈtjɛ.nɛ]	'have (2 sg.)'
	/pero/	[ˈpe.ro]	'dog'
	/peros/	[ˈpɛ.rɔ]	'dogs'

For many speakers, laxing of the word-final vowel triggers vowel harmony, at least up to the stressed vowel and sometimes extending to pretonic vowels and even preposed clitics. Theoretical and phonetic accounts include Zubizarreta (1979), Sanders (1994), Corbin (2006), among many others.

5 Intonational differences: selected regional traits

Intonational patterns vary widely across Spanish regional and social dialects, and while most of the variation can be regarded strictly as subphonemic, meaning that they do not create oppositions based on different meanings, they provide unmistakable identification of these dialects. It is often the case that intonational sequences, referred to impressionistically as *el tono* 'tone' or *el cantado* 'the song,' provide the quickest and most reliable identification of a speaker's regional and social origins, even in the presence of background noise that masks individual vowels and consonants. Most work on Spanish intonation has been conducted within the framework of Autosegmental-Metrical Phonology, which describes prenuclear pitch accents, nuclear (phrase-final) pitch accents, and boundary tones as combinations of High and Low tones. Overviews can be found in Ladd (1996) and Gussenhoven (2004); for Spanish Beckman et al. (2002), Hualde (2002), and Sosa (1999). Within this framework, pitch accents – which fall on some but not all tonic syllables – are marked with an asterisk * for the tone most closely aligned with the tonic syllable. Leading or trailing tones may also be included if they form an integral part of the pitch accent configuration.

Most research on Spanish intonational patterns – including regional variants – has concentrated on pitch accent configurations that affect meaning (e.g., broad vs. narrow focus, and declarative vs. interrogative utterances). Less attention has been directed on intonational patterns that serve to identify regional and social dialects, although native speakers of Spanish can frequently identify familiar dialects more effectively based on intonation than on segmental or lexical traits. As the study of Spanish intonational patterns becomes increasingly nuanced, a more complete picture of the role of intonation in dialect differentiation will emerge. Two brief examples will illustrate the possibilities.

5.1 Pitch and tone: the Spanish of Equatorial Guinea and San Basilio de Palenque

Equatorial Guinea is the only Spanish-speaking country in sub-Saharan Africa. For most Guineans, Spanish is a second language, spoken in conjunction with one or more African languages. With the exception of Annobonese creole Portuguese (*fa d'ambú*) and Pidgin English (*pichi*), all Guinean languages have lexically specified High and Low tones on all vowels. As a consequence, Guineans tend to interpret Spanish pitch accents as phonologically High tones, and retain the high pitches even in connected speech. In a fashion similar to lexical tone languages, High pitch is always aligned with the tonic syllables, as in the sentence *Compramos con dinero en el mercado* 'We buy with money in the market' pronounced by a native speaker of Ndowé, a Bantu language spoken along the coast of Río Muni, on the African continent between Cameroon and Gabon (see Figure 1.1).

An even more drastic variant involving sequences of early-aligned high peaks on prenuclear accents comes from the Afro-Colombian village of San Basilio de Palenque, where the vernacular Spanish takes on many of the same suprasegmental traits as the local Afro-Hispanic creole language, Palenquero (Hualde and Schwegler 2008), itself formed several centuries ago in contact with Kikongo and

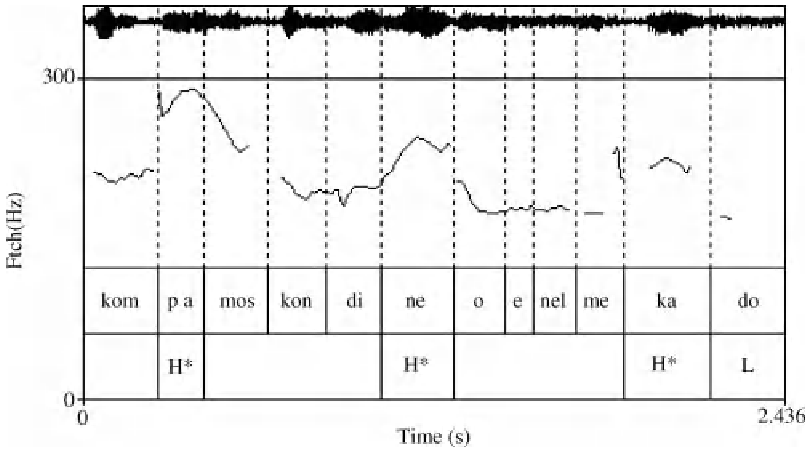

Figure 1.1 High pitch aligned on tonic syllables in *Compramos con dinero en el mercado* 'We buy with money in the market' (Ndowé, Bantu language).

other West and Central African lexical tone languages. In the following sentence, multiple high peaks with no downdrift and minimal tonal valleys between pitch accents make the utterance (pronounced in a normal nonemphatic conversation) sound excited or upset. The sentence is (allowing for local phonetic traits) *Yo me acordé que yo cargaba un treinta y ocho largo* 'I remembered that I was carrying a long-barrel .38 [revolver]' (see Figure 1.2).

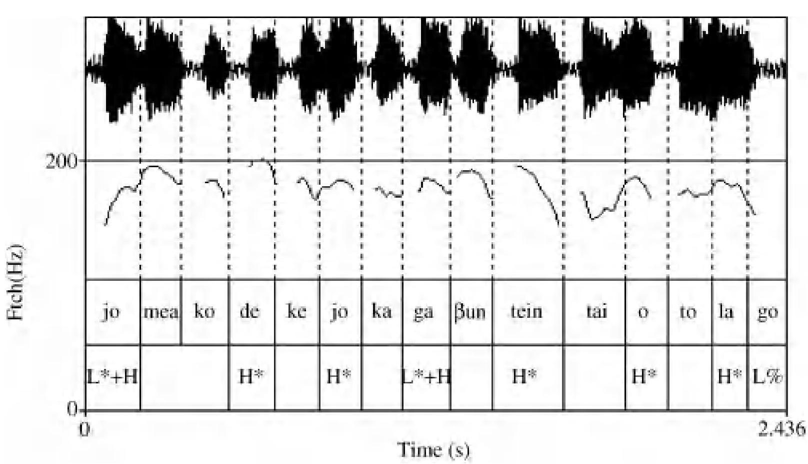

Figure 1.2 High peaks with no downdrift in *Yo me acordé que yo cargaba un treinta y ocho largo* 'I remembered that I was carrying a long-barrel .38 [revolver]' (Palenquero, Afro-Hispanic creole).

6 Regional and social morphosyntactic differentiation

Although sharing substantially the same basic grammatical patterns, Spanish varieties around the world diverge in terms of word order, the behavior of object clitics, and choice of verb tense, and mood, in addition to combinations directly attributable to contact with other languages. The following sections present a selection of morphosyntactic variables that differentiate Spanish dialects.

6.1 Object clitic agreement and doubling

Spanish of all regions permits a direct object noun phrase to be replaced by a clitic, regardless of the animacy of the DO; thus (7):

(7) Veo a Juan/el libro. 'I see John/the book.'
Lo veo. 'I see him/it.'

When the DO is a personal pronoun (i.e., [+animate]), both the clitic and the full pronoun may appear; indeed, for most dialects, if a personal pronominal DO occurs, a clitic must accompany it (8):

(8) Lo_i/$^{*}Ø_i$ veo a $él_i$ 'I see him.'

In a subset of Spanish dialects (particularly in the Southern Cone), clitic doubling of ([+definite]) DO NOUNS is also possible, and often even preferred (9):

(9) Lo_i/$Ø_i$ veo a $Juan_i$ 'I see John.'

In the Andean region, and sometimes extending into the Río de la Plata region, clitic doubling is not only found when the direct object is a pronoun or animate noun, but also occurs with inanimate [+definite] DOs. In monolingual and sociolinguistically unmarked varieties, the clitic agrees in gender and number with the direct object noun (10):

(10) No lo_i encontró a *su* $hijo_i$ 'She did not find her son.' (Peru)
La_i ves *una* $señora_i$ 'You see a woman.' (Peru)

Among Spanish-recessive bilinguals (speaking Quechua or Aymara) in the Andean region, invariant *lo* is often used to double all direct objects, irrespective of gender or number. The use of non-agreeing *lo* is widely regarded as an indicator of imperfect acquisition of Spanish, and is never found among monolingual Spanish speakers or balanced bilinguals (Godenzzi 1986, 1991a, 1991b; Mendoza 2008: 227). Some examples are shown in (11):

(11)	Cerráme*lo*$_i$ *la puerta*$_i$	'Close the door for me.' (Bolivia)
	¿Me *lo*$_i$ va a firmar *la libreta*$_i$?	'Are you going to sign the register for me?' (Salta, Argentina)
	Se *lo*$_i$ llevó una *caja*$_i$	'He carried off a box.' (Peru)

6.2 Regional word order patterns: interrogation; object–verb order

Spanish typically places the subject after the verb in interrogative sentences, both in absolute interrogatives (requiring a yes–no answer) and in phrases with interrogative words (12):

(12)	*¿Sabe usted a qué hora llega el avión*?	'Do you know what time the plane arrives?'
	¿Qué quiere Juan que le digamos?	'What does Juan want us to tell him?'

There is a cluster of dialects, including the Caribbean (Cuba, Puerto Rico, Dominican Republic, coastal Colombia, much of Venezuela, parts of Panama) in which subject–verb inversion does not occur in absolute interrogatives, nor in sentences with interrogative words, provided that the subject is a pronoun (13):

(13)	*¿Usted me puede ayudar*?	'Can you help me?'
	¿Dónde tú compraste esa bicicleta?	'Where did you buy that bicycle?'

In these same dialects, overt subject pronouns such as *tú* and *yo* are more frequent than in varieties of Spanish that do not present noninverted questions. Noninverted questions with interrogative words appear to be extending their domain of application, for example, in Dominican and Cuban Spanish, to occasionally encompass full nouns in subject position (e.g., Suñer 1994), as in *¿qué tu mamá quiere?* 'What does your mother want?' and *¿Dónde Juan compró eso?* 'Where did Juan buy that?' In the dialects of Spanish that exhibit noninverted questions, they are sociolinguistically unmarked and are used by nearly all speakers. In fact, inverted questions such as *¿Qué quieres tú?* 'What do you want?' may take on connotations of impatience or aggressiveness when used in these dialects.

Outside the Caribbean region, noninverted questions are occasionally found in the Spanish of the Canary Islands, in South American communities bordering on Brazil (since vernacular Brazilian Portuguese also exhibits noninverted questions), and in the traditional Afro-Bolivian dialect (Lipski 2008a). Along the Brazilian border (e.g., in northern Uruguay, northeastern Argentina, northern Bolivia, and eastern Paraguay), bilingual contact with Portuguese also results in occasional "*in situ*" questions, in which the interrogative word has not been moved to the front of the sentence (14):

(14)	*¿Naciste dónde?*	'Where were you born?'
	¿Esto cuesta cuánto?	'How much does this cost?'

Another regionalized shift in word-order patterns is found in the Andean region, principally the highlands of Ecuador, Peru, and Bolivia, where Spanish is in contact with Quechua and/or Aymara. In the latter two languages, the direct object normally comes before the verb; they are "O-V" languages, in contrast to the "V-O" pattern that typifies Spanish. Spanish-recessive bilingual speakers frequently place the direct object before the verb in configurations not usually found in the Spanish of other regions (15):

(15)	*Mi santo de mí lo han celebrado.*	'They celebrated my saint's day.'
	Dos hijitos tengo.	'I have two children.'
	Estico primer hijo es.	'This is my first child.'

Predicate nouns and adjectives as well as prepositional phrases are also placed preverbally, as in the following examples from highland Ecuador (16):

(16)	*Sembradita tengo la manzanilla.*	'I have camomile planted.'
	A cortar alfalfa mi mamá está yendo.	'My mother is going to cut alfalfa.'

Placing predicates in preverbal position, while occasionally possible in emphatic or topicalized sentences, is not the norm for the monolingual Spanish of any region, and in Andean dialects, O-V constructions are stigmatized and regarded as a demonstration of limited proficiency in Spanish.

6.3 Regionalized verb tense/mood usage

The choice of verb tenses and moods is relatively uniform across the Spanish-speaking world. There are only a small number of cases where regional or social variation can be consistently observed. Among the more noteworthy cases of variation in verb usage are the following.

First, a fundamental dichotomy separates Spain from most of Latin America as regards the interpretation of the preterite–present perfect distinction (e.g., as in *el jefe no llegó/no ha llegado hoy* 'the boss didn't/hasn't come today'). In most of Spain, the first sentence implies that the boss did not come and is not expected come, while the second sentence leaves open the possibility for a later arrival. In Spain, the present perfect can be used even when the moment of speaking is not included, as in *lo ha hecho ayer* '(he/she) did it yesterday.' In Latin America, the simple preterite (e.g., *llegó* 'arrived') does not necessarily exclude the present moment, so that *el jefe no llegó* could be construed as 'the boss hasn't arrived [yet]' (Alarcos 1947; Moreno de Alba 1988: 176–180; more recently Howe and Schwenter 2008; Schwenter and Torres-Cacoullos 2008).

Second, in Southern Cone dialects (Argentina, Uruguay, Paraguay, Chile, and Bolivia), and generally in Peru, Ecuador, and parts of Colombia, it is usual for subjunctive verbs in subordinate clauses to appear in the present tense even when the verb of the main clause requires past-tense reference (17):

(17)	Me pidió que le *haga* [*hiciera*] un favor.	'He asked me to do him a favor.'
	¿Sería posible que me *ayudes* [*ayudaras*] con mi tarea?	'Could you help me with my homework?'

This usage, while unremarkable in the areas mentioned above, is not acceptable in other Spanish-speaking countries, where a past subjunctive verb form is required.

Third, in much of the Andean region, the Spanish pluperfect indicative (*había* + PAST PARTICIPLE) is used to express information known only indirectly by the speaker, or deduced from indirect observation. Thus, a speaker who saw someone arrive can say *Llegaste anoche* 'You arrived last night,' while someone who has not witnessed the arrival, but encounters the interlocutor the following day and thereby deduces the arrival, can say *Habías llegado anoche*, literally 'You had arrived last night.' Similarly, an individual who reveals a previously unknown talent or ability might elicit a comment such as *Habías sabido montar a caballo* 'So you learned how to ride a horse.' Although this non-canonical use of the Spanish past perfect indicative is not a direct translation from Quechua or Aymara, the semantic nuances encoded by the pluperfect in Andean dialects corresponds to Quechua and Aymara evidentiality markers, which signal first-hand versus reported or deduced information (Laprade 1981; Mendoza 1991: 155–157, 196–203; Speranza 2006). Within the Andean region, these constructions are used by most speakers, irrespective of social class or bilingual language background.

Fourth, in highland Ecuador and occasionally in other Andean Spanish dialects, the Spanish future tense is used in imperative constructions: *comprarás el libro* 'buy the book,' *comerás todito* 'eat it all up.' As in true imperative constructions, clitic pronouns may follow the verb: *escribirás-me* 'write to me,' *dará-me lo que prometió* 'give me what you promised.' Although other varieties of Spanish occasionally employ future-tense verbs as imperatives (e.g., in the Ten Commandments), it has been suggested that the use of the future tense in Quichua[3] as a softening device for imperatives has contributed to the higher frequency of such constructions in highland Ecuadoran Spanish (Haboud 1998: 213–245; Haboud and de la Vega 2008: 177–178; Hurley 1995).

6.4 Contact-induced morphosyntax: northern Uruguay (Portuguese); Andean highlands (Quichua)

In much of the Spanish-speaking world, Spanish is in daily contact with other languages, in bilingual environments that are highly conducive to language mixing and transfer. Two representative cases are described below. The first involves Spanish in contact with the cognate language Portuguese at various points near the Brazilian border. The second entails contact between Spanish and Quichua in the Andean region of South America.

In Spanish-speaking South American countries that border on Brazil (that is, all except Uruguay, Ecuador, and Chile), at least some Portuguese is spoken near the

Brazilian border, with usage varying widely depending on the type of border and ease of crossing, the nature of the communities on either side of the border, commercial and familial relations with Brazil, and the availability of Brazilian mass media and schools. In northern Uruguay stable hybrid varieties have emerged since the end of the nineteenth century, known to linguists as Fronterizo or *dialectos portugueses del Uruguay* (DPU) and by the speakers themselves as *portuñol* (Elizaincín 1992; Elizaincín et al. 1987). This variety is grammatically closer to vernacular southern Brazilian Portuguese than to Spanish, but contains numerous Spanish lexical and functional items as well.[4] Other South American border regions also exhibit hybrid language behavior, although more frequently used with neighboring Brazilians than among fellow citizens, but in some regions there are native Portuguese-speaking enclaves within nominally Spanish-speaking border regions. This normally occurs in twin-city contexts with either an open and "dry" border (where one crosses the border simply by crossing a street), or an open border marked by a creek or small river, with no border controls to limit traffic between the two countries. Among the dry border areas where Spanish–Portuguese hybrid language is used by non-Brazilians are Rivera, Uruguay (bordering on Santana do Livramento), Bernardo de Irigoyen in Misiones Province, Argentina (bordering on Dionísio Cerqueira and Barracão), Pedro Juan Caballero, Paraguay (bordering on Ponta Porã), Capitán Bado, Paraguay (bordering on Coronel Sapuquaia), and Leticia, Colombia (bordering on Tabatinga), as well as Santa Elena do Uairén (Bolívar state) in Venezuela, only a few kilometers from an open land border with Brazil and the town of Pacaraima. Open borders represented by creeks or narrow rivers are found in Artigas, Uruguay (bordering on Quaraí), Bella Vista Norte, Paraguay (bordering on Bela Vista), Cobija, Bolivia (bordering on Brasiléia), and San Antonio, Misiones Province Argentina (bordering on Santo Antônio).

When attempting to speak Portuguese, whether to Brazilians or to other bilingual members of their own communities, Spanish-speakers in these border regions frequently mix the two languages freely and at times unconsciously, creating what is known locally as "portuñol"; only in northern Uruguay have these patterns coalesced into a stable language used freely among fellow Uruguayans. In the remaining border areas "portuñol" is more heterogeneous, being used principally to Brazilians or among descendants of Brazilians living outside the borders of their country. Some examples of spontaneous "portuñol" mixed language are given below; Spanish words are in regular type, Portuguese words are in italics, cognate Spanish–Portuguese words are in bold face, and hybrid forms not identical to either Spanish or Portuguese are in small capitals (Lipski 2009a, 2009b) (18):

(18) a. **cuando** yo *ia* en la **otra** escuela nosotros TENÍA **que ir** arriba por **un** barranco
'When I went to the other school, we had to climb up an embankment' {Bernardo de Irigoyen, Argentina}

b. nosotro TENÍA **que** *segurar* las **casa** sino *ía* **í para** abajo
'we had to secure the houses or else they would fall down' {Bernardo de Irigoyen, Argentina}

c. **porque** *não tem*, **como** le puedo *falar*, **vitrina**
'because there isn't how can I explain it, a show window' {Guayaramerín, Bolivia}

d. *mas* algunoh *brasileiro entendem* lo **que** hablamoh nosotro loh **boliviano**
'but some Brazilians understand the way we Bolivians speak' {Guayaramerín, Bolivia}

e. *eleh* tiene **que** *se* **adaptar a** las REGLA, verdá tiene **que** tener *tudo* DOCUMENTASÓN **ser** REGULARIZADU
'they [Brazilians] have to conform to the rules, right? they have to have all the documents in order' {Pedro Juan Caballero, Paraguay}

f. *quando fica velho* **a partir di cuarenta cinco cincuenta** año él *ja no pode mais*
'when one gets old, past forty-five or fifty' {Pedro Juan Caballero, Paraguay}

In nearly all Spanish-speaking communities on the Brazilian border, the vernacular Portuguese practice of marking plural/s/only on the first element of plural noun phrases (usually an article or other determiner) frequently carries over to Spanish; this can be observed in several of the preceding examples.

Quechua-Spanish contacts have resulted in numerous modifications in Andean Spanish, some of which have already been described (e.g., clitic doubling, atonic vowel reduction, object–verb word order). In the area of morphosyntax, most instances of Quechua influence are confined to Quechua-dominant bilinguals with little formal training in Spanish (e.g., Cerrón Palomino 1976; Rivarola 1990). Although increased access to Spanish-language education across the Andean region is reducing the number of individuals with limited abilities in Spanish, Quechua-dominant bilinguals are still numerous in parts of highland Ecuador and Peru, and to a lesser extent in Bolivia. When speaking Spanish, even to monolingual interlocutors, Quechua-dominant speakers often introduce Quechua case-markers and emphatic or topicalizing particles. The most common instance is the focus or affirmative particle -*ca*, which in Quechua can attach to nouns, pronouns, and verbs. The particle -*ca* is often used in Spanish by Quichua-dominant speakers, as in the following examples recorded in Imbabura province in northern Ecuador (19):

(19)	nosotros-*ca* ya no trabajamos	'we don't work any more'
	in Angla-*ca* sí hay capella	'in Angla there is a chapel'
	aura-*ca* sí tinimos	'now we have [it]'
	Otavalo-*ca* toda la veda vevemos	'we have lived in Otavalo all our lives'

The particle -*ca* has made its way into the monolingual Spanish dialect spoken in the Afro-Ecuadoran communities of the highland Chota/Mira and Salinas Valleys, in the provinces of Imbabura and Carchi (Chalá Cruz 2006; Lipski 2008b). These communities derive from Jesuit haciendas that had transferred to private ownership by the end of the eigthteenth century. Afro-Choteños are monolingual speakers of Spanish, but their ethnic dialect does include evidence of prior contact with Quichua. Some examples, shown in (20), include:

(20)	yo *ca* como nunca me ha salido	'since I have never seen [those apparitions]'
	yo *ca* no voy a ir	'I'm not going'
	él *ca* queriendo pegar a mí	'he wants to hit me'
	ahora *ca* ya no se ve eso	'that isn't seen any more'

The adverb *ta(n)* (< Spanish *también* 'also'), analyzed as a negative emphatic or indefinite marker by Muysken (1982: 110), or the homophonous affirmative suffix are also frequently found in Quichua-influenced Spanish, as in the following examples from Imbabura province, Ecuador (21):

(21) a. *maicito-ta trigo-ta cebada-ta tudito no madoramus, perdemos tudito*
'the corn, wheat, and barley didn't ripen, we lost everything'
b. *come trigo-ta, cebada-ta, todo come ese rata*
'it eats wheat, barley, that rat eats everything'
c. *borriguito-ca ya acaba, puerco-tan ya no hay*
'there are no more sheep, there are no more pigs'
d. *una arroba di papa-tan cargado, así sabíamos sober*
'carrying a sack of potatoes, that's how we would climb [to the village]'
e. *esi tiempo-tan todo istá cambiando-ta*
'in these times everything is changing'

Perhaps the most frequent stereotype of Quechua-induced morphosyntax in Andean Spanish is the use of the gerund instead of a finite conjugated verb form. It is commonly asserted that most such uses of the gerund represent transfer of the Quechua subordinating suffix *-shpa*, used when the subjects of the main and subordinate clauses are identical, and the suffix *-kpi* for dissimilar subjects (e.g., Haboud 1998: 207–210; Haboud and de la Vega 2008: 175–177). In spoken Spanish, however, the gerund occurs relatively infrequently, so it is not likely that Quechua-dominant speakers are actively translating *-shpa* combinations with gerunds in Spanish. More plausibly, gerund-based constructions are so frequently heard in bilingual Andean speech communities that Quechua-dominant speakers who acquire Spanish informally simply learn this predominant pattern without any implicit awareness of morphosyntactic equivalence (e.g., Muysken 1982). Some representative examples of the Andean use of the Spanish gerund from Imbabura province, Ecuador in (22) are:

(22) a. tractrur-ta *tenendo* platita-ca, tractor *ponindo*
'as for tractors, if [we] have money, [we] use a tractor'
b. todo *llamando* padrecito vene
'when [we] call for anything, the priest comes'

c. cosecha *acabando* toditu *acabando* ay vuelta sembramos cebada por ahi
'when the harvest is all over, we plant barley over there'

d. allá en la casa chiquitica *tenendo*
'there in that little house [I] have [*cuyes* = guinea pigs]'

Andean Spanish, again centering on Ecuador and southern Colombia, also uses the gerund in benefactive constructions, typically using *dar*: *dame comprando un periódico* 'buy me a newspaper,' *me dió abriendo la botella* 'he/she opened the bottle for me.' This type of construction may be a translation from Quichua, but is used widely by monolingual Spanish speakers as well. With the exception of *dar* + GERUND constructions, none of the contact-induced phenomena just described are found in the speech of monolingual Spanish speakers or Spanish-dominant bilinguals, and all carry connotations of marginality associated – often unfairly – with lack of educational opportunities in Spanish.

7 Lexical variation

Lexical differences among Spanish dialects are so numerous and all-encompassing as to elude easy classification. Linguists on both sides of the Atlantic often speak of "Americanisms" vs. "Peninsularisms," but to divide the lexicon in this fashion is a considerable oversimplification. There are, however, some common threads that lend substance to a rough Old World–New World lexical split. In addition to numerous borrowings from Native American languages, most of which have not entered the Peninsular Spanish lexicon, Latin American varieties contain several items of nautical provenance, introduced into the speech of future colonists during the long ocean crossings, and which have lost their nautical connotations in the Americas. These include *botar* (from 'bail water' to 'throw away'), *amarrar* (from 'belay an anchor line' to 'tie up [anything]'), *timón* (from 'rudder' to 'steering wheel' [in some countries]), and *arribar* (from 'make port' to simply 'arrive'). Several lexical items that are current throughout Latin America are considered archaic or have disappeared entirely from Spain: *platicar* 'to chat,' *cobija* 'blanket,' *pollera* 'rustic skirt,' *cabildo* 'municipal authority.' Words describing recent technological innovations – even when borrowed from English – often take different forms on either side of the Atlantic as well as among Latin American nations, for example, *ordenador* [Spain] – *computador/computadora* [Latin America], *teléfono móvil* [Spain and parts of Latin America] – *teléfono celular* [much of Latin America] 'cellular telephone.' 'Automobile' may be rendered as *coche* [Spain], *carro* [much of Latin America], *máquina* [Cuba], *auto* [Southern Cone], while words for 'large bus' include *autobús* [known everywhere], *autocar* [Spain], *guagua* [Caribbean and Canary Islands], *camión* [Mexico], *chiva* [Colombia], *movilidad* [Peru], *flota* [Bolivia], *colectivo* [Argentina], and *ómnibus* [Uruguay]. 'Fair-*complexioned, blonde' can be güero* [Mexico], *canche* [Guatemala], *chele* [Honduras, El Salvador, Nicaragua], *macho* [Costa Rica], *fulo* [Panama], *mono* [Colombia], *catire* [Venezuela], and *gringo* [much of South America]. Slang and taboo items further complicate the lexical profile of the Spanish language, as does dialect mixing resulting from demographic displacements, so that dialect classification via lexical criteria is a frustrating enterprise. In addition to a few core lexical items particular to each region, the main points of lexical stability involve morphological endings, especially diminutive suffixes, and choice of second-*person pronouns and the accompanying verb morphology.*

7.1 Variation in diminutive suffixes

All varieties of Spanish share the productive diminutive suffixes *-ito*/*-ita*, used to express a wide range of meanings from size to endearment to scorn; the suffixes *-illo*/*-illa*, while still relatively productive, are largely restricted to specialized or nondiminutive meanings (e.g., *ojillos* 'beady eyes,' *amiguillo* 'questionable friend,' *mundillo* 'closed clique,' *abogadillo* 'disreputable lawyer'). In addition, there are several regional diminutive suffixes that continue to be productive. In Spain *-iño*/*-iña* is frequent in Galicia, *-ino*/*-ina* can still be heard in Extremadura, *-ín* is productive in Asturias, *-ete* in Catalunya, Valencia and parts of Aragon, and *-ico*/*-ica*, once common in old Spanish and still used in Judeo-Spanish, is found in Aragon, Navarra, and Murcia, and in Latin America in Cuba, the Dominican Republic, Venezuela, Ecuador, Colombia, Costa Rica, and occasionally elsewhere (Lipski 1999b). In Spain, *-ico* can in principle be attached to any noun or adjective; in Latin America, *-ico* is restricted to words whose final consonant is /t/ (or occasionally the group /tɾ/): *momentico* 'just a moment,' *chiquitico* 'very little,' *maestrica* 'dear teacher.' The diminutive suffixes *-ingo*/*-inga* are productive in eastern Bolivia, and occasionally are heard elsewhere.[5] In Murcia and parts of Granada, *-iquio* / *-iquia* occasionally are found, while *-icho* / *-icha* appear from time to time in Aragon and La Mancha and surrounding areas.

7.2 Second-person subject pronouns and accompanying verb forms

Most varieties of Spanish exhibit a choice of second-person singular subject pronouns that roughly express the familiar–formal dichotomy, while only in Peninsular Spain is this distinction maintained in the plural, via the *vosotros–ustedes* choice. Pragmatic factors governing the choice of familiar vs. formal pronouns are complex and vary considerably across geographical regions, social classes, and age- and gender-defined cohorts. The availability of specific pronouns and the accompanying verbal morphology is largely defined by region, in a few instances intersected by social and ethnic variables. *Usted* and *ustedes* are found in all varieties of Spanish; all variation involves second-person pronouns expressing familiarity.

In Peninsular Spain, the second-person familiar subject pronouns are *tú* and *vosotros*. *Vosotros* was traditionally absent in western Andalusia and the Canary Islands, but is increasingly frequent in all of Andalusia. In Equatorial Guinea and the Philippines, both *ustedes* and *vosotros* are used, sometimes indiscriminately, as a reflection of the varied Peninsular origins of colonial settlers and administrators (Lipski 1987a, 2008d). In Latin America, *ustedes* is the only second-person plural pronoun, while among singular pronouns *vos* is the main alternative to the familiar *tú*. Páez Urdaneta (1981) offers an overview of *voseo* usage. *Vos*, originally a plural pronoun in Latin and old Spanish, always has singular non-formal reference in Latin American Spanish, and always combines with the object clitic *te*, rather than *os* – the clitic corresponding to *vosotros*. Every Latin American nation except Puerto Rico and the Dominican Republic has at least some speech communities where *vos* is used. Attitudes toward the use of *vos* vary widely. In Southern Cone nations and in parts of Central America (particularly Nicaragua following the 1979 Sandinista revolution: Lipski 1997), *vos* is generally accepted as the national standard and is freely used in advertising and in public discourse where familiar pronouns are appropriate. In countries where *vos* is confined to smaller regions, speakers may exhibit ambivalence, being proud of using a distinctive regional trait but often reluctant to use *vos* outside of their own speech community.

There are a few Latin American dialect zones in which *usted* predominates even when familiar reference is intended. This includes Costa Rica and parts of the Colombian interior. In the latter region *su merced* 'your mercy' sometimes shifts from a deferential form of address to an expression of extreme familiarity (Ruíz Morales 1987). Similar uses of *su merced* have been reported for the Dominican Republic (Pérez Guerra 1988).

8 Summary

The preceding sections have provided a glimpse into the range of variation that characterizes the Spanish language throughout the world. The descriptions are, of necessity, snapshots representative of the time of writing (mid-2010), and must be set against the backdrop of a rapidly changing world. The turn of the twenty-first century has witnessed rapid shifts in social communication patterns as well as increased demographic mobility, and the consequences for dialect variation are only beginning to be fully appreciated. One example of the linguistic "new world order" is increased exposure to national prestige norms in mass communication and telecommunications media. In much of Latin America, cable television is now readily available in peripheral regions of several countries where previously no national channels could be received. Mobile telephones and Internet access are now functional in numerous places where landline infrastructure would probably never have been possible, and the increased use of text messaging, blogs, chat rooms, and e-mail has provided easy communication links between isolated speech communities and compatriots far from home. On the one hand, greater exposure to national norms often results in the attenuation of regional and local dialect traits, but, on the other hand, the availability of chat rooms and blogs appears to be reinforcing the use of minority languages and dialects, such as Sephardic (Judeo) Spanish, Aragonese, Chabacano (Philippine Creole Spanish), and intertwined Spanish–English code-switching, to name only a few cases. Increased ethnic awareness and pride in previously stigmatized languages and dialects is also raising the profile and self-esteem of minority dialects, such as the local Spanish vernacular of San Basilio de Palenque, Colombia, (Morton 2005; Schwegler and Morton 2003), the once-stigmatized "liquid gliding" of /ɾ/ and /l/ in the Dominican Cibao region (Pérez Guerra 1991), the upsurge of Spanish–Quichua hybrids known as *media lengua* or *chaupi shimi* 'half-language' in northern Ecuador (Gómez Rendón 2008), and the rising popularity of *portuñol/fronterizo* speech in northern Uruguay (e.g., literary and cultural production such as Behares and Díaz 1998, Behares et al. 2006). Massive migrations from rural areas to major cities continue to occur throughout Latin America, with the result that cities like Lima, Guayaquil, Bogotá, Santa Cruz de la Sierra (Bolivia), Mérida and Tijuana (Mexico), and Caracas have become multi-dialectal mosaics in which rapid sociolinguistic evolution is all but inevitable. Increased communication – by electronic means or demographic proximity – thus embodies the potential both for rapid dialect leveling and for greater awareness and maintenance of dialectal features as identity markers. The only impossible outcome is the creation or retention of rigid geographical and social dialect boundaries.

NOTES

1 http://www.cervantes.es. Other estimates, such as those by Ethnologue (http://www.ethnologue.com) and UNESCO (http://www.unesco.org), also give totals around the 400 million figure.
2 Since Argentina unilaterally claims a large swath of Antarctica, and maintains small but permanent bases (as does Chile), one could arguably stretch the boundaries of the Spanish-speaking world past the Antarctic Circle.
3 In most of the Spanish- and English-speaking world, this language is referred to as *Quechua*, despite the fact that the language possesses only three vowel phonemes, /i/, /a/, and /u/. In Ecuador, the official name is *Quichua*; examples drawn from Ecuador will employ this variant.
4 A somewhat similar Portuguese–Spanish variety, albeit with different historical antecedents, is spoken in the Portuguese town of Barrancos, along the border with southwestern Spain. Navas Sánchez-Élez (1992) and Clements (2009: ch. 8) provide details.
5 For example, *fotingo* < *Ford-ingo* is a now dated expression used in Cuba to refer to a decrepit old automobile.

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2 The Spanish-based Creoles

J. CLANCY CLEMENTS

1 Introduction

As with any definition, the definition of a creole language, and its use to classify language varieties formed in contact situations, yields mixed results. In order to define what a creole language (or creole) is, we also need a definition of a pidgin language (or pidgin). A pidgin consists of speech forms that arise among speakers of two or more languages who need to communicate but do not share a common language. Over time, and depending on the nature of the interaction among the speakers of the different languages who are co-creating the pidgin, it may remain in an incipient phase in which speakers communicate using some rough conventions (e.g., a small set of common vocabulary), or it may progress further into a stable phase in which there is a conventionalized lexicon, as well as clear tense and aspect markers. At this stage, a pidgin is typically used for only certain functions, such as an in-group or trade language, or possibly as a lingua franca. If the contact situation supports it, a pidgin can also expand its functions, developing or adopting new vocabulary, increasing its range of expressions, and even undergoing standardization in order to be used as the language on radio, TV and/or in the print media. At this point, one speaks of an expanded phase of a pidgin (cf. Mühlhäusler 1986).

One key difference generally acknowledged between a pidgin and a creole is that the former does not have native speakers while the latter does. At any phase of a pidgin, it can acquire native speakers. That is, depending on the nature of the contact situation, a creole may develop from an incipient pidgin, from a stable pidgin, or from an expanded pidgin.

In this chapter, we discuss the creoles whose lexicon is currently largely from Spanish: Palenquero (PAL) in Colombia; Papiamentu (PAP) in Curaçao; and Zamboangueño (ZAM) in the Philippines (see Figures 2.1 and 2.2). There is evidence that these were originally derived from Portuguese-based creoles, a question we will address briefly.

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Figure 2.1 Northern South America and the southern Caribbean.

2 The lack of Spanish-based creoles

Based on Smith's (1994) extensive list of pidgins, creoles, and mixed languages extinct and currently spoken today, there are roughly 16 English-based creoles, 12 Portuguese-based creoles, 10 French-based creoles, but only three Spanish-based creoles. McWhorter (2000) attributes the dearth of Spanish-based creoles to various factors. First, he argues that PAL and PAP (the Atlantic creoles) were, in their origin, Portuguese-based creoles. Although this has not been entirely proven, what is indisputable is that there is a Portuguese component in both creoles and it seems to have been there since their respective formations (e.g., both creoles have the 3SG pronoun *ele* [< Ptg. *ele* 'he']). What is less clear is the precise relationship between PAL and PAP, or between these two creoles and the Portuguese-based creoles on the west coast of Africa, which are assumed to have constituted at least part of the input in the formation of PAL and PAP (for PAL, see Schwegler 1998: 229–237; Schwegler and Green 2007: 273; for PAP see Quint 2000; Jacobs 2009).

Second, McWhorter (2000) observes that if one assumes that Spain had little presence in Sub-Saharan Africa, no Spanish-lexified creoles could have developed from a diachronic perspective. McWhorter favors what he calls the "Afro-Genesis

Figure 2.2 The Philippines, with relevant cities and islands.

Hypothesis," which states that the Spanish-based creoles originated in Africa as Portuguese-based creole(s), and argues against the Limited Access Hypothesis (LAH), according to which PAL and PAP formed in South America under conditions in which there was inadequate access to Spanish, the target language. Last, following the LAH a creole would be expected to form under conditions in which enslaved or indentured workers greatly outnumber their overseers and consequently would not have access to the language of their overseers. McWhorter examines demographic figures from certain areas in colonial Colombia, Ecuador, Peru, Venezuela, and Mexico, arguing for each that the African population was indeed large enough to have fostered the formation of a creole language, but that in no case did it happen. While some scholars, such as Schwegler and Green (2007: 273), concede that in the hard-to-reach jungles of western Colombia (in the Chocó Department), thousands of black slaves toiled in mines under conditions that should have resulted in a creole, McWhorter's hypothesis has been challenged by others, such as by Lipski (2005: 281–287), who points out that:

> the fact that even the most uneducated and geographically isolated *chocoano* speaks grammatically standard Spanish (although including features typical of rural illiterate speakers worldwide) reveals that earlier barriers to access to full Spanish were completely penetrated, which does not exclude the possibility that prior to acquiring standard Spanish Chocó residents spoke some kind of Spanish-derived creole. (281)

With respect to the African-descendent communities in the Chota Valley area in Ecuador, Lipski (2005: 281–282) notes that since at least the mid-nineteenth century its inhabitants have always been free to travel throughout Ecuador, and that today the Pan-American highway passes through the region. Díaz-Campos and Clements (2008) question the Afro-Genesis Hypothesis for Venezuela based on historical considerations, and the argument for Peru is easily challenged based on Bowser's (1974: 94–95) statistics and accompanying discussion.[1] For Veracruz, Mexico, McWhorter's claim may still hold, but the sociohistorical details need to be studied to confirm or disconfirm it. In sum, then, McWhorter's Afro-Genesis Hypothesis has had limited reach because in each case the sociohistorical facts are complex and either were not conducive to creole formation or were conducive to the formation of a creole that subsequently could have disappeared (see Ortiz López 1998 for Cuba and Baxter and Lopes 2010 for the Tongas speech of São Tomé. See also Lipski 2009 for data on Afro-Bolivian Spanish, which seems to be historically linked to PAL).

So far, it has been pointed out that creoles are perhaps best understood through knowledge of their sociohistorical background, which typically would account for the particularities in each creole. As for how to account for the similarities among creoles, linguists such as Siegel (2008) and Clements (2009a) have appealed to processes of second language learning. We will come to describing the respective sociohistorical backgrounds of each creole, but we would first like to discuss some of the key processes involved in the formation of these creoles from the perspective of language change as an evolutionary process (Croft 2000; Mufwene 2001). In this way, we can understand what creoles have in common and why.

3 Language evolution and pidgin/creole formation and development

If we think of a speech community as a community of individuals who perceive themselves as speakers of the same language, then communicative intercourse in such a community can be seen as comparable with a community of interbreeding individuals (i.e., a species). With all its drawbacks, the analogy between a species defined as a group of interbreeding individuals, on the one hand, and a speech community defined as a group of intercommunicating individuals, on the other hand, allows us to ask questions about which lexical and structural features (the "genes" of a language) are selected and propagated in communicative intercourse.

This is especially relevant to contact situations involving some degree of social upheaval, as has often been the case in the formation of the colonial-language creoles, such as PAL, PAP, and ZAM (Croft 2000: 38; Mufwene 2001: 152–153).

In language acquisition and language use, frequency plays a crucial role in how we select lexical material and structure language. However, it also has to be acknowledged that frequency would not be as important as it is if the human mind did not function as it does. Among myriad other complex things, the human mind functions as a highly sophisticated pattern recognizer. If in dealing with linguistic and other input we assume that our minds work to create processing short cuts, these can be regarded in language learning as pattern generalizations over linguistic elements, extracted out of the input received by speakers in discourse. If the nature of the input changes, so too may the frequency of use of a given item and, in turn, the corresponding patterns. The essential point is that language is a dynamic system that is used to represent knowledge for the purpose of communication and that in such a system a varying amount of structure exists at any given time.

Among many other things, the formation of pidgins and creoles involves language acquisition, language processing, and language production, as well as innovation and propagation of linguistic forms. In the process of creole formation, we assume the Principle of Uniformity, common in the natural sciences, which for our purposes may be informally stated as: the same laws governing innovation and propagation, and language processing/production, apply in creole genesis as in any other type of language change. This principle would also apply to social and cognitive factors in language change and language use (cf. McColl Millar 2007: 284, 360–361). For language acquisition, the principle would mean that in the formation of a pidgin or creole, the co-creators would process input as second-language learners process input: they would acquire content words such as nouns, verbs, and adjectives before function words such as prepositions or verb auxiliaries, and they would acquire content words first because they denote entities in the world and such words tend to be more perceptually salient. That is, grammatical function words and bound inflections tend to be harder to perceive than content words because the former usually consist of at most a few sounds and are unstressed, even in deliberate and slow speech. In informal and rapid speech, speakers tend to shorten them further (Bates and Goodman 1999: 52; Zobl 1982).

Once a language variety has developed, among its lexical items there are some that are more resistant to change than others. This is due, in part, to the fact that they are deeply entrenched in the minds of the speakers through frequency of use, in particular, lower numbers, pronouns, and special words such as some wh-words, some spatial deictic particles, and the negative particle (Pagel et al. 2007). We return to this below.

If we assume that in creole-formation speakers are negotiating a system of communication in which lexical meaning is most important and grammatical meaning is deduced through the situation and the context (i.e., pragmatically, as suggested by Mühlhäusler 1986), they will acquire word forms that are most frequently used in discourse and most easily perceived or perceptually salient (Clements 2009a: 1–27). Frequency is defined as the number of times in a given corpus that a certain item or form appears. Following the Principle of Uniformity, we assume that the most frequently used word forms in spoken corpora are also the most frequently used forms in discourse in a contact situation and the ones most easily selected in forming a pidgin or creole.

For our purposes, the notion of perceptual salience is based on the ubiquity of CV structure and defined in relative terms, using the assumption that stressed syllables and free-standing morphemes are more easily perceived in the speech chain than unstressed syllables and clitics/affixes, respectively. This is stated in (1):

(1) Definition of Perceptual Salience:
  a. CV is more perceptually salient than VC, V
  b. stressed syllables are more perceptually salient than unstressed syllables
  c. free-standing morphemes are more perceptually salient than clitics/affixes

Based on the foregoing, we can say that the extent and nature of restructuring, and thus the relative importance of perceptual salience and frequency in shaping a newly-emerging language variety depends on the individual makeup of each of the contact situations. At the same time, we must acknowledge that the adult agents of creole formation already know one or more languages and that in creating a creole they introduce into the new language elements from their own language(s). This is variably called interference or transfer in the literature. As we will see, besides the languages spoken in a contact situation and the activities of language processing and production, the sociohistorical circumstances under which each of the creoles formed played an important part in their restructuring.

4 Restructured spanish and the sociohistorical background of PAL, PAP, and ZAM

Lipski (2005: 295–299) lists a series of traits typically used to argue for a prior creole source for PAL and PAP. We will include ZAM in the discussion because the traits, as it turns out, also are pertinent in this case. The relevant ones for our purposes are: (1) loss of case marking in the pronoun system (e.g., the loss of the clitic object pronouns); (2) no agreement in the noun phrase (gender or number) or verb phrase (number); (3) no definite articles; (4) loss of prepositions; (5) presence of the multi-functional preposition *na*; (6) presence of tense-aspect particles, such as *ya* for marking anteriority, *ta* for marking non-past, *ba* for past imperfectivity; *lo* (< Ptg. *logo* 'soon, after, right away') for marking future, etc.; (7) 3PL subject pronoun as plural marker (e.g., PAP *buki* 'book,' *buki-nan* [literally 'book' – 'they'] 'books'); (8) presence of the invariant 3SG pronoun *ele* with variants; (9) loss of copula (both PAL and PAP have copulas); (10) use of *tener* (or *tem*) instead of *haber* for existential expressions; (11) 2SG *vos* > *bo(s)* (< *vos*) instead of *tú*; and (12) use of subject pronouns as possessive pronouns

The presence of these features in the creoles in question means that there was significant restructuring of the lexifier language, whether it was Portuguese and/or Spanish. And such a degree of this restructuring implies that, in the respective environments in which the creole varieties developed, the circumstances were such that, for the creators of the creoles, the acquisition of grammar of the lexifier languages was more difficult than creating a new way to express grammatical relations and frame discourse temporally and/or aspectually. To understand why PAL, PAP, and ZAM formed as they did, we need to examine, even if briefly, the sociohistorical background of each of the creoles.

4.1 Palenquero

African slaves who had arrived in Cartagena, Colombia, and had subsequently escaped from plantations, founded Palenque de San Basilio and were instrumental in forming PAL. PAL is considered an exogenous creole in that its speakers were taken away from their homeland to another place where they were instrumental in its creation. Based on what we know about the actual make up of PAL, some of these slaves already spoke a Portuguese pidgin or creole. Historical documentation on the origin of the escaped slaves (also called maroons) who established Palenque points to an area in Africa where Bantu languages were/are spoken (Parkvall 2000: 137). Citing Holm (1989: 310), Parkvall notes that slaves fled the Colombian plantations around 1600. Patiño Roselli (2002) reports that many *palenques* (the term *palenque* translates roughly as "fortified settlement") were settled by maroons in the seventeenth century and that El Palenque de San Basilio (or Palenque) appears in colonial documents as one of the settlements of the area that survived a brutal campaign of repression launched by the Spanish in 1694. Schwegler and Green (2007) state that by 1700 Palenque was well established.

Moñino (p.c. July 1, 2010) notes that 4% of the PAL lexicon (roughly 200 words) is from Kikongo (a Bantu language), another 4% derives from Portuguese, and 90% is from Spanish. In his more recent research involving DNA testing, Moñino (p.c. July 1, 2010) shows preliminarily that, for the male inhabitants in the traditionally separated upper and lower neighborhoods of Palenque, the DNA evidence for males of the upper neighborhood locates their origin in a specific area of inland Congo, while the DNA evidence for the lower neighborhood males suggests that they come from many parts of West Africa. Together with the Kikongo lexical items and a number of other linguistic and cultural features in Palenquero (see Schwegler 1996), this evidence points convincingly to the influence of Kikongo in the creation of PAL.

The available documentation (e.g., Schwegler and Green 2007: 273) indicates that the Palenqueros not only maintained their physical isolation, but also their linguistic isolation in that, apart from maintaining Spanish from early on, they spoke their own language, which they passed onto their young. That is, the Palenqueros were bilingual in PAL and Spanish, and until recently this state of affairs had been maintained. Schwegler and Green also report that younger speakers are now choosing not to learn PAL, but still maintain a good passive knowledge of the language. Interestingly, both old and young Spanish–PAL speakers exhibit the same grammar in PAL. That is, there is no gradual shift toward the superstrate. Today, PAL is spoken actively by the older population in El Palenque de San Basilio and will most likely become extinct within a generation or two.

4.2 *Papiamentu*

PAP is currently spoken in the islands of Aruba, Bonaire, and Curaçao (ABC), located approximately 60 miles north of Venezuela. When the Dutch took the islands from Spain in the 1630s, the Dutch West India Company founded Willemstad and made it the capital of Curaçao. Between 1650 and 1713, Curaçao functioned as an important port – along with the settlements of Goree, Rufisque, and Cacheu on the Senegambian coast – for the Atlantic slave trade (Jacobs 2009), handling an increasing number of slaves, up to 20,000 in 1685. From 1700 to 1715, the number of slaves imported from Africa to Curaçao was about 3,500 to 4,000 a year. Slave importation continued, albeit illegally, up until 1778, when the last slave transport is said to have arrived in Curaçao. The great majority of the slaves were from the Congo, Angola, Togo, Benin, and Nigeria (Maurer 1998: 141). Two of the West Atlantic languages represented were Wolof and Edo, and the Bantu languages represented were Kikongo and Kimbundu, among others (Parkvall 2000: 145ff).

In the history of Curaçao, the presence of the Sephardic Jews also had an impact on the linguistic and demographic makeup of the island. Citing a wealth of sources, Jacobs (2009) notes that Sephardim in Cape Verde, along with those in Brazil and Amsterdam, worked side by side with the Dutch in the slave trade between west Africa and South America during the second half of the seventeenth century, and both the Dutch and the Sephardim were instrumental in the transfer of Upper Guinea Portuguese Creole from the African west coast and Cape Verde to Curaçao. By 1700 Papiamentu had already formed (cf. Parkvall 2000: 136 and sources therein).

Although all would agree that PAP is an exogenous creole, not everyone is in agreement on the precise origins of PAP. Maurer (1998) argues that PAP formed between 1650 and 1700 through the contact among three groups. In the African group, there were African slaves who spoke mostly the Bantu languages of the Congo and Angola, and Kwa languages of Ghana, Togo, and Benin. From the Europeans there were two groups: those from west-central Europe, most notably Dutch, but also German speakers, and the Sephardic Jews. Quint (2000) and Jacobs (2009) go further, arguing that Upper Guinea Creole Portuguese (which includes Cape Verdean Creole) is to a large extent the basis for PAP and they provide linguistic and historical evidence for the genetic link between the two varieties, through the agency of the Dutch and the Sephardic Jews, a theory that expands on Goodman (1987), Kouwenberg and Muysken (1994), among others. Today Curaçao is self-governing and there is discussion of Curaçao becoming a separate country within the Kingdom of the Netherlands, as is Aruba, but it has not yet taken place.

4.3 Zamboangueño

The term *Chabacano*, *Chavacano*, or *Chabakano* – originally meaning 'uncouth' (Grant 2009) – is used to refer to the Spanish-based creoles spoken in the Philippines, mainly Ermitaño in Ermita in central Manila (now extinct), Caviteño in Cavite north of Manila (with a small group of native speakers), and ZAM spoken in and around Zamboanga City on the southern island of Mindanao. Due to space considerations, I will concentrate on ZAM, the language of the most thriving creole-speaking community of the three, but the reader is encouraged to consult Sippola (2010). It is generally agreed that ZAM formed after 1719, when the Spaniards returned to a fort in Zamboanga, Fort Pilar, which they had abandoned about 50 years earlier. Lipski (1992) reviews the various hypotheses about the formation of ZAM and the reader is encouraged to consult it for more detailed information.

Both Lipski (1992) and Fernández Rodríguez (2006a) posit an origin of ZAM largely independent of the Spanish-based creoles in Manila. Both propose a strong influence early on of Philippine languages. In his proposal of the different stages of development of ZAM, Lipski posits an influence from Caviteño Creole Spanish in the second half of the eighteenth century, due to migrations of Cavite civilians to Zamboanga at that time. This would account for the Portuguese material found in ZAM. The strong Visayan element, according to Lipski, may have come into ZAM starting at the turn of the twentieth century. Recently, Fernández Rodríguez (p.c. July 10, 2010) has found evidence that suggests that ZAM may have formed away from Zamboanga, on the island of Panay, near Iloilo City. Unlike PAL and PAP, ZAM is an endogenous creole in that it was not formed in a plantation culture with slaves brought in from afar, but rather it seems to have formed gradually its speakers were/are Austronesian language speakers for most part. Based on the 2000 Philippine census, ZAM is claimed as a mother tongue by 358,729 people, more than any other Spanish-based creole in the world (Rubino 2008: 279).

4.4 Summary of the socio-historical background

Given the nature of the sociohistorical development in each of the areas where a creole has developed, it seems that in all three cases there is evidence suggesting that McWhorter (2000) may be at least partially correct in his claim that the three Spanish-based creoles under discussion are Portuguese-based in that we can be reasonably confident that all creoles under discussion display vestiges of Portuguese origin, possibly from a Portuguese pidgin or creole or from Portuguese itself, as Bickerton (2002) has argued.

Based on the sociohistorical background in each case, we summarize this section with the following observations. PAL formed within a community of African maroons some/most of whom most likely spoke a Kikongo-influenced Portuguese-based creole (possibly akin to Sãotomense). The speakers of PAL have been bilingual in PAL and Spanish for centuries, because of which we would expect certain elements from Spanish, forms and structures to possibly have found their way into PAL. PAP emerged from input from various communities, the numerically dominant ones overall being the African slaves and the Dutch and the Sephardim who constituted the link to the Upper Guinea Portuguese creole. For a long time, PAP has been influenced by Dutch and, more recently, by Spanish. ZAM also formed from heterogeneous input, including material from numerous Philippine languages, Spanish, Caviteño, and most recently English. While the Spanish influence waned significantly in the twentieth century, English took its place. Moreover, the ever-present Philippine languages wield considerable influence, because many ZAM speakers are bi- or trilingual in ZAM, English and/or one or more Philippine languages. Given this situation, we would also expect to find in ZAM elements (forms and/or structures) from Philippine languages and from English.

In the next section we compare key elements and structures of each of the three creoles and discuss to what extent we can account for their presence based on general mechanisms of language acquisition and language change, as well as the sociohistorical situation of each of the creoles.

5 A comparison of some linguistic features of PAL, PAP, and ZAM

In comparing key elements of the three creoles, we suggest that many of the similarities found in PAL, PAP, and ZAM can be accounted for straightforwardly by invoking the nature of acquisition whereby content morphemes are acquired before functional morphemes and more perceptually salient and/or frequently used forms are favored in acquisition over less perceptually salient and/or frequently used forms. From the list of traits mentioned above, we think that the following phenomena can be accounted for, namely: (1) no agreement in the noun phrase (gender or number) or verb phrase (number); (2) no definite articles; (3) loss of prepositions; (4) presence of the multifunctional preposition *na*; (5) loss of case marking in the pronoun system (i.e., loss of the clitic object pronouns); and (6) the selection of *ya*, *ta*, *ba*, and *lo* as tense-aspect markers. Now, let us turn to the discussion of these key elements in PAL, PAP, and ZAM. We will deal with the noun phrase and the verb phrase, in that order.

5.1 The noun phrase

Based on the foregoing, for nouns and noun phrases (NPs) we would expect the Portuguese- or Spanish-bound pluralization morpheme *-s* not to survive as the default marker because it is bound and only consists of a single sound. Moreover, we would expect any type of number or gender agreement, such as we find in Spanish *las casas bonitas* (the-FEM-PL house-FEM-PL pretty- FEM-PL 'the pretty houses'), not to show up in the creoles. If pluralization were to develop later, we would expect it to be marked with a free form containing at least a CV structure for reasons of perceptual salience.

These expectations are largely borne out by the data. In none of the creoles is the plural *-s* is found as the default plural marker. However, all the creoles possess a plural marker containing a CV structure. The PAL plural marker is *ma* (< Kikongo *ma* 'plural noun-class marker' and/or Ptg./Sp. *mais/más*), shown in (2). In PAP, *nan* (< presumably *nan* 'they' in PAP) marks the plural, shown in (3), as well as *e* (< Sp. *el/él* [el] 'the-MASC, he-MASC'). In ZAM the default plural marker is *maga*, *mana*, or *maŋa* (< Tagalog *maŋá* or *mga* 'plural'), as in (4), while *-s* (< Sp. *-s* 'plural') is only found in some frequently used nouns, and according to Forman (1972: 112–113) is disappearing.

(2) Ma pelo asé ndrumí mucho.
PL dog HAB sleep a.lot
'The dogs sleep a lot.'

(Schwegler and Green 2007: 291)

(3) aña -nan
year PL
'years'

(Kramer 2004: 181)

(4) maga ermáno
PL brother
'brothers'

(Forman 1972: 112)

With regard to (in)definiteness marking in the NP, PAL has retained no definite articles from Portuguese or Spanish. Instead, it marks definiteness with a bare noun and indefiniteness with invariable *un* (< Sp. *un* 'a-MASC.SG' (5)). In contrast, PAP has both definite and indefinite invariable articles (6), as does ZAM (7). Thus, the expectation that there be no definite articles is true for PAL, but not for PAP or ZAM.

(5) un ma kusa aí
INDEF PL thing there
'some things there'

(Schwegler 1998: 261)

(6) e/un kas
the/a house
'the/a house'

(Maurer 1998: 155)

(7) Yo un dalága || el Čabakáno antigwa
I INDEF young.girl || the Chabacano ancient
'I' am a young girl || 'the ancient Chabacano'
(Lipski and Santoro 2007: 389)

The subject and object pronominal systems for the three creoles are shown in Table 2.1. In PAL, apart from in the 1SG, the subject–object distinction is not formally maintained, but it maintains the informal–formal address distinction, though only in the singular. The pronouns are derived either from Portuguese (*ele [< ele]*), Portuguese and/or Spanish (*bo [< vos]*), Spanish (*uté [< usted]*, *utere [< ustedes]*, *suto [<nosotros]*, *hende* (< *gente* 'people'), Kikongo (*enú* (< *éenu* '2EMPH'), *ané* (see Schwegler 1998: 260) or Portuguese, Spanish and Kikongo for *mi* (< Sp. *mi*, Ptg. *mim*, Kik. *-áami* '1SG').

In PAP, there is no formal distinction anywhere between subject and object pronouns, nor between formal and informal address forms. These traits make it the simplest system of the three creoles. It is noteworthy that, with the exception of 3PL, PAP has the same pronominal system as Upper Guinea Creole Portuguese, which has *(a)mi*, *(a)bo*, *el*, *(a)nos*, *(a)bos*, *(a)elis* (Jacobs 2010).

ZAM makes a clear formal distinction is all persons between subject and object pronouns except in the 1PL and 2PL where the distinction is made lexically. The marker *ko(n)* is from Spanish, but we know that several relevant Philippine languages use the morpheme {ka + G/N/NC/CV} to mark oblique case for proper nouns (Tagalog and Ilonggo *kay*, Cebuano *kang*, Waray-Waray and Tausug *kan*, Magindanaon *kaní*). The phonological and rough semantic correspondence between these forms, Portuguese *com*, and Spanish *con*, most likely played a role in ZAM marking objects with *kon* (cf. Fernández Rodríguez 2006b). Also noteworthy is the distinction in ZAM between inclusive and exclusive 1PL forms, also found in many of the Philippine languages, including Tagalog and Visayan, both of which historically had an impact on ZAM. ZAM also distinguishes formal vs. informal address forms, also found in the Philippine languages.

Both PAL and PAP largely use the (object) pronominal forms as possessive determiners. The exceptions are in the 2SG/PL where PAL has *sí* (without the formal–informal distinction) and PAP has *su* as the 3SG/PL form. By contrast, as Table 2.2 shows, ZAM has both a long and a short set of possessive determiners marked with prefix *di-*, as well as a long and short set without *di-* in the 1SG/PL, the 2SG/PL neutral, and the 3PL, and it maintains the inclusive–exclusive distinction in this set.

With regard to possessive determiner order, in PAL the possessive determiners follow the noun (*amigo mi* [lit. friend my] 'my friend,' as is the default position for possessive determiners in Kikongo (Bentley 1887: 582). In PAP, they precede the noun (*mi kas* 'my house'). In ZAM, the long forms (i.e., *dimío* [longer] vs. *dimí* 'my' [shorter]) can precede as well as follow the noun they determine and are also used as possessive as pronouns. The short forms always precede the noun they determine.

Table 2.1 Pronominal systems of PAL, PAP, and ZAM PAL (Schwegler 1998; Schwegler and Green 2007), PAP (Kouwenberg and Ramos-Michel 2007), and ZAM (Lipski and Santoro 2007).

	PAL	*PAP*	*ZAM*
1sg			
subject	yo, i-	mi (EMPH: ami)	yo (-yo)
object	mi	mi (EMPH: ami)	ko(n)migo
2sg fam			
subject	bo	bo, -bu (EMPH: abo)	ébos (-bos)
object	bo, -o	bo, -bu (EMPH: abo)	-kombós
2sg neutral			
subject	—	—	tú
object	—	—	kontígo
2sg form			
subject	uté	bo, -bu (EMPH: abo)	usté
object	uté	bo, -bu (EMPH: abo)	kon-usté
3sg			
subject	ele	e (EMPH: ele), su	éle, lé
object	ele, -e, -lo, -o	e (EMPH: ele), su	konéle
1pl			
subject	suto, (ma) hende	nos	kamé (exclusive) kitá (inclusive)
object	suto, (ma) hende	nos	kanámon (exclusive) kanáton (inclusive)
2pl fam			
subject	utere	boso	—
object	utere	boso	—
2pl neutral			
subject	—	—	kamó
object	—	—	kanínyo, kon-bosótros
2pl form			
subject	enú (archaic)	boso	ustédes
object	enú (archaic)	boso	kon-ustédes
3pl			
subject	ané	nan	silá
object	ané, -lo, -o	nan	kaníla

Table 2.2 Possessive determiners in ZAM (Forman 1972; Lipski and Santoro 2007).

1SG	dimío, mío, dimí, mí
1PL	diámon (excl), diáton (incl), ámon (excl), áton (incl)
2SG FAM	dibós
2SG NEUTRAL	ditúyo, túyo, ditú, tú
2PL NEUTRAL	ínyo
2SG FORMAL	diusté
2PL FORMAL	diustédes
3SG	disúyo, diíla
3PL	disú, íla

There are some striking similarities in the creoles. All three share the 3SG pronoun *ele* (< Ptg. *ele* 'he, him [after prepositions]') which is of Portuguese origin and all share reflexes of *vos* (< Ptg. 2PL 'you', Sp. 2PL 'you'). Finally, it is interesting to note that, with the possible exception of PAL *-lo*, there are no reflexes of Spanish or Portuguese object clitic pronouns. This is predicted given that they are quasi-bound and unstressed morphemes. PAL *-lo* is not surprising given the long history of Spanish-PAL bilingualism.

As alluded to above, the adjectives in all three creoles are invariable. As in Spanish and Portuguese, however, adjectives in PAL, PAP, and ZAM generally follow the noun, in PAL especially if it is plural (Schwegler and Green 2007: 294; Lipski and Santoro 2007: 390–391; Kramer 2004: 182–183).

5.2 *The verb phrase*

Based on our discussion above, for verbs and the verb phrase we would expect the Portuguese/Spanish conjugation system not to survive. The forms we would expect to find their way into the creoles are the most frequently used paradigm verb forms in discourse, which are the 3SG for the finite forms and the infinitival form for the non-finite forms (cf. Clements 2009a, 2009b for the relative distribution of finite and nonfinite forms in spoken language). Given that in discourse, present-tense forms are overall more frequently used than simple past-tense forms, one would expect the latter not to be retained. If they were retained, one would expect the 3SG form to survive. One last prediction regarding verb forms involves the well-known verb classes in both Portuguese and Spanish, *-a*, *-e*, and *-i* morphological verb classes and their respective suffixes. Given that inflectional morphology is not predicted to be retained in naturalistic L2 acquisition in a heterogeneous contact situation, and therefore not predicted to form part of a pidgin or creole formed in such a situation, we would not expect verb classes and their corresponding morphology to be retained. If, however, any of the verb classes were retained, frequency considerations would allow us the prediction that the *-a* class and corresponding forms would be retained.

Table 2.3 Copulas in PAL, PAP, and ZAM (Schwegler and Green 2007; Kouwenberg and Ramos-Michel 2007; Lipski and Santoro 2007).

	PAL	*PAP*	*ZAM*
Present tense	tá é	tá	– (equative)
Present or past	fwé, sendá	–	takí, talyí, talyá (locative, existential)
Past tense	ta-ba, era, fwé-ba, sendá-ba	tabata	– (equative)

Having reviewed what we expect to find in the verb forms, we discuss three topics: copulas, verb forms, and tense-mood-aspect particles. Table 2.3 contains the PAL, PAP, and ZAM copulas. While PAL and PAP have an expressed copula form in general, ZAM has an unexpressed equative copula, as shown in (8). This is also the case in many of the Philippine languages, including Tagalog (Schachter 1987: 942).

(8) Soltéro el anák disúyo.
bachelor DEF son 3SG POSS
'His son is a bachelor.'

(Forman 1972: 161)

PAL maintains somewhat the distinction found in Portuguese and Spanish between *ser* and *estar*: the forms *é* and *fwé* are used to refer to more permanent states, while the copula *tá* is used to refer to more temporary states. However, both copulas can be used with locatives, which is not the case in the lexifier languages, unless reference to an event is being made. In PAP, no reflex of *ser* is found. Rather, there is only one copula form.

As is predicted, all the forms for which we know the origin are from the 3sg form of the verb: *é* (< Ptg. *é* 's/he, it is,' Sp. *e(s)* 's/he, it is'), *fwé* (< Spanish *fue* 's/he, it was'), *éra* (< Ptg. *era* 's/he, it was,' Sp. *era* 's/he, it was'), *tá* or *ta* (< Ptg. *esta* 's/he, it is,' Sp. *está* 's/he, it is'). Interestingly, a possible origin for the past marker *ba* is the *-a* verb class imperfective suffix *-ba* in Spanish and/or its Portuguese counterpart *-va*. Of note is that it has CV structure and is more perceptually salient than *-a*, the corresponding imperfective suffix from the *-e* and *-i* verb classes (e.g., *bebí**a*** '1/3SG IMPF' from *beber* 'drink,' *subí**a*** '1/3SG IMPF' from *subir* 'go up'), not to mention more frequently used given the strong preponderance of *-a*-class verbs in both Spanish and Portuguese.

Table 2.4 The tense-mood-aspect particles in PAL, PAP, and ZAM.

	PAL	*PAP*	*ZAM*
Present	Ø/ta	Ø/ta	ta
Present progressive	ta, ta___ -ndo	(no specific form)	(no specific form)
Present habitual	asé sabé	(no specific form)	(no specific form)
Preterit	a (completive)	a	ya- (perfective) -ya (completive)
Imperfect	asé-ba (habitual) sabé-ba (habitual) ta-ba (progressive)	tabata	ta
Past perfect		(no one form)	—
Future	tan tan-ba (from past)	lo lo-ba (from past)	ay

With regard to the base verb form selected for the verbal system in each of the creoles, it is either the 3SG form or the infinitival form, or both forms, in the case of ZAM. Illustrative examples are given in (9):

(9) PAL: ablá
PAP: kanta
ZAM: kantá

For the most part, the tense-mood-aspect markers, shown in Table 2.4, developed from either auxiliary verbs/copulas, main verbs, or adverbs: *ta* < Ptg., Sp. *esta*, *está*, *asé* < Sp. *hacer* 'do,' *sabé* < Ptg., Sp. *saber*, *ya* < Sp. *ya* 'already,' *a* < Sp. *ha* '3SG PRS PERF AUX,' *tan* (< possibly early PAL *ta anda* 'be going'), *lo* < Portuguese *logo* 'right away, afterwards,' *ay* (< Spanish *hay* 'there is/are'). In these cases, an element with a certain function in the lexifier language or an earlier stage of the creole has developed into a TMA marker, which is not uncommon in the languages of the world. For instance, the English future auxiliary *will* developed from Old English main verb *willan*, *wyllan* 'to wish, to desire, to want.' The exception to this is the marker *ba*, which arguably is a reflex of the imperfect suffix *-va* (Portuguese) and/or *-ba* (Spanish) of the *-a* verb class, possibly influenced, as well by, the imperfect copula form *estava* (Portuguese) and/or *estaba* (Spanish).

The survival of inflectional morphology in creole languages is not common and would not be predicted assuming, as we have, that content morphemes survive rather than functional morphemes in creole formation. However, the frequency and perceptual salience argument would predict that if an imperfect affix were to survive, of the two candidates, *-ba*/*-va* (e.g., *esta**<u>ba</u>***) and *-a* (e.g., *subí**<u>a</u>*** '1/3SG IMPF go up') *-ba* would be preferred because with its CV structure it is perceptually more salient, and it is more frequently used in discourse, given the preponderance of *-a* class verbs in Portuguese and Spanish relative to the number of *-e* and *-i* class verbs. Given the early influences on PAP and PAL from Portuguese and (Sephardic) Spanish, it is possible that the perfective particle *a* in these languages could have had the form [ha] (< Ptg. and Sp. 3SG AUX *ha*) at the time it became part of the grammar. If this were not the case, it would not be predicted to be selected because it would be perceptually of little salience as a single low vowel.

Before concluding, two brief observations are in order. First, although this is somewhat simplified, the presentational verbs, the equivalent of English 'there is/are' are *(a)ten* (< Ptg. *tem* '3SG PRS have') in PAL, *tin* (< Ptg. *tinha* '1/3SG IMPF have' in PAP, and *tyéne* (< Sp. *tiene* '3SG PRS have') in ZAM. In this last case, a case could be made that *tyéne* relexified the Portuguese form *tem* because in the Asian Portuguese creoles the reflexes of Portuguese *tem*; *têm* 's/he, it has; they have' occur much more commonly as a presentational verb, and it is not that common in Spanish. Second, both PAL and PAP exhibit SVO constituent order, whereas ZAM displays VSO order, a clear case of influence from the Philippine languages which all exhibit VSO order.

We want to return to the list of 12 phenomena mentioned at the outset in Section 4 above. Based on our discussion so far, we have provided an account for the first six by appealing to the acquisition of content before grammatical morphemes and to the notions of frequency and perceptual salience. What about the remaining six phenomena?: (7) 3PL subject pronoun as plural marker (e.g., PAP *buki* 'book,' *buki-nan* 'books'); (8) presence of the invariant 3SG pronoun *ele* with variants; (9) loss of copula (both PAL and PAP have copulas); (10) use of *tener* (or *tem*) instead of *haber* for existential expressions; (11) 2SG *vos* > *bo(s)* (< *vos*) instead of *tú*; and (12) use of subject pronouns as possessive pronouns. For (7), this is a trait found in various Atlantic creoles, possibly due to the influence of West African languages. Traits (8), (10), and (11) stem from Portuguese. Trait (9) does not apply to PAL and PAP, and it applies to ZAM due to influence from the Philippine languages. Finally, the use of subject pronouns as possessives (trait (12)) can be accounted for by the Principle of Economy, though this same principle is inoperative in ZAM. The reason might be found in the nature of the contact situations: PAL and PAP speakers formed their respective communities on a continent other than the one of their African heritage. That is, PAL and PAP are exogenous creoles. By contrast, the ZAM speakers formed their community in the midst of various indigenous, and from a typological standpoint, relatively homogeneous, Austronesian languages spoken in the area where ZAM formed and has developed. That is, ZAM is an endogenous creole. Thus, one could argue that the scope of transfer of native Austronesian language structure into ZAM has been more of a possibility given the multilingualism of the area.

6 Concluding remarks

In this brief overview of the history and structure of the three Spanish-based creoles, Palenquero, Papiamentu, and Zamboanqueño, we have attempted to account for the structure of the creoles using an evolutionary approach to language change and language acquisition. We argued that the creation of these creoles initially involved the Principle of Uniformity in second language acquisition, but that in environments of considerable social upheaval the acquisition of grammar of the targeted lexifier languages was more difficult than creating a new way to express grammatical relations and frame discourse temporally and/or aspectually. We appealed to the notions of frequency and perceptual salience to account for the selection of certain verb forms as base forms, the loss of inflectional morphemes and prepositions, and the development of a multifunctional preposition with CV structure. We assumed input from a Portuguese source – pidgin, creole, or otherwise – to account for the Portuguese elements in our creoles. And we assumed the Principle of Economy to account for PAL and PAP using essentially the same pronominal system for subject, object, and possessive pronominal expression. The question emerged why ZAM did not follow the same Principle of Economy. It was suggested that it might have to do with the distinction that PAL and PAP are exogenous creoles, having developed in a plantation culture among a slave population uprooted from their ancestral home, whereas ZAM is an endogenous creole, one created (apparently) over time by speakers in the midst of various indigenous, and from a typological standpoint, relatively homogeneous, Philippine languages.

Because of the synoptic nature of this chapter, many issues involving the genesis and development of PAL, PAP, and ZAM have not been touched upon. There is a wealth of historical information being discovered (e.g., Jacobs 2009) that continues to shed new light on the Africa–Palenque and Africa–Curaçao connections. Moreover, DNA testing adds another dimension to the question of who the creators of these creoles were and who has influenced them over time. In short, there is still a lot of work to be done on the history, linguistic connections, and genetic makeup of these communities that will keep us busy for decades to come.

NOTE

1 If we take Bowser's (1974: 94–95) statistics regarding the size of the plantations (whom McWhorter 2000: 12 also cites) seriously, one could conceivably question McWhorter's claim in this case. Bowser states that the plantation sizes were relatively modest, rarely exceeding 40 slaves, which would make the Limited Access Hypothesis less applicable.

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3 Spanish Among the Ibero-Romance Languages

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1 Introduction

In this chapter, "Ibero-Romance" is purely geographically defined and will be understood as referring to the Romance languages of Spain and Portugal and their former empires. From the linguistic point of view, this is an arbitrary subdivision of the Romance-speaking world, since some of the Romance varieties of the Peninsula have features in common with varieties spoken north of the Pyrenees. Thus, Catalan exhibits many lexical parallels with the Romance of southeastern France (Colón 1976: 39–52) and Castilian shares some phonetic changes with Gascon, notably the weakening of Latin initial/f/and the merger of Latin/b/and/w/. On such grounds it has been strenuously argued that Catalan in particular is not an 'Ibero-Romance' language at all (Rohlfs 1979: 255–260; for a more balanced view see Badia i Margarit 1975 [1964]: 31–51), a conclusion which resonates with Catalans' resentment of the historic encroachment and dominance of Castilian. It is also important to state that the terms "language," "dialect," and "variety" will be used almost interchangeably. The only objectively definable distinction that can be drawn is between, on the one hand, languages, etc., which have *Ausbau* status (Kloss 1967), that is to say, have been "elaborated" for administrative, legal, and literary purposes and have typically been established as official languages of particular states, and on the other hand, languages, etc., which do not have such a status.

This chapter is divided into two sections which give a contrastive account of the history and principal features of Castilian vis-à-vis other Ibero-Romance languages and evaluate, respectively, the role played by linguistic contact among the Ibero-Romance languages in the formation and evolution of Castilian and the extent of ongoing contact between Castilian and other Ibero-Romance languages in the present day.

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2 The historical dimension

2.1 "Origins"

All the present-day Ibero-Romance languages are usually represented as having originated in varieties of Latin spoken in those territories of the northern part of the Iberian Peninsula which were not subject to sustained Moorish occupation following the invasion of 711 CE. These varieties were then carried southwards in the process of the expansion of the northern Christian states known as the Reconquest by northerners who resettled the conquered areas. The origins of Castilian lie in the eighth-century Romance of a frontier region of the Asturian kingdom occupying the north of the modern province of Burgos, between the Ebro valley and the Cantabrian mountains, bounded to the northeast by Basque-speaking territories. It is likely that the city of Burgos in due course provided a context for linguistic leveling and became a focus of linguistic prestige (Bustos Tovar 2005: 275–276), since it was the spearhead for further southerly expansion and the center, first of an independent county in 932 CE, and then of a kingdom in 1065 CE. The proximity of the hinterland of Castile to the Basque Country has led to a number of substratist hypotheses being adduced to explain some of the features which distinguish Castilian from its Romance neighbors (Jungemann 1955). At the same time, the frontier situation of Castile and its relative isolation from Asturias and Leon has encouraged an excessively nationalist and anthropomorphized approach to be taken to the development of Castilian (Lapesa 1981: 184–186). It has been characterized as exhibiting independent features (mostly phonetic, since sound-change was the main focus of interest in comparative and historical linguistics in the early twentieth century) which drove a "wedge" (Menéndez Pidal 1976 [1926]: 69) between the linguistic continuum which stretched transversally from west to east. Thus, for example, the 'f > h' change (e.g., FARĪNA(M) > *harina*/aˈɾina/'flour') is not shared with León or Aragon; conversely, while areas of León and Aragon (and Catalonia) palatalize initial/l/ (e.g., LŬPU(M) > León./ˈʦobu/, Cat. *llop* 'wolf'), Castilian does not (*lobo*). However, it is probably not an exaggeration to say that there is not a single phonetic feature in the early development of Castilian that cannot be accounted for in terms of natural phonetic development and is not attested elsewhere in the Romance-speaking world.

The hypothesis of a "dual" origin for Castilian based on an admixture of the language of Burgos and that of Toledo was promoted by Criado de Val (1960). This suggests that Castilian was significantly affected by contact with what is generally known as "Mozarabic," the Ibero-Romance varieties which continued to be used in Al-Andalus (the Moorish Peninsular territories) either in advance of the Reconquest when southerners took refuge in the Christian north, or in the process of the Reconquest itself as northern settlers came into contact with Mozarabic-speaking inhabitants of the reconquered territories. However, it has proved difficult to find concrete linguistic evidence of such contact.

2.2 *The problem of early texts*

Our vision of the early Ibero-Romance languages, or more accurately, the vernacular Romance of the Iberian Peninsula, is derived, on the one hand, from reconstruction on the basis of modern forms and, on the other hand, from the evidence of written documents. The two are not always in agreement, and the delicate philological process of reconciliation has often envisaged the possibility of "influence" or "hesitation" to reflect what we might more neutrally call variation. For example, the *Poema de Mio Cid* shows alternation in the perfect stem of *ser* (< Lat. FUI-) between *o~ue* (*commo si fuesse en montaña* 'as if he were on the mountainside', l.61, but *commo si fosse delant* 'as if I were present', l.2137), and even in a number of cases where *ue* is written, the assonance pattern suggests/o/. This, and other, alternations have been variously interpreted: *o* may represent a diphthong/wo/, which was a feature of early Leonese and Aragonese, suggesting dialect mixture or even a non-Castilian provenance for the text. By contrast, Lapesa's (1985 [1980]) vigorous defense of the Castilian provenance of the *Poema de Mio Cid* hypothesized that/(w)o/was an archaism which was not entirely consistently updated by the scribe and was even deliberately cultivated to elevate the tone: that is to say, that a phenomenon which had sometimes been interpreted as a regional variant was actually a stylistic variant. Literary texts prior to the Alfonsine period generally present such apparent dialect mixture. One strand of philological inquiry has been to try and discern the "true" Castilian which underlies the supposedly scribally corrupted forms: see, for example, Alarcos Llorach's (1948) discussion of the sources of the *Libro de Alexandre* (thirteenth century), which concludes that non-Castilian features in two of the (significantly discrepant) surviving manuscripts were introduced by copyists.

The language of the prose documents emanating from the royal scriptoria of Alfonso X (reigned 1252–1284) is often taken to mark the first emergence of the notion of a standard language in Castile. Yet it is, in fact, probably more accurate to say that this activity is most significant for its substantial elaboration of a Romance vernacular as an *Ausbau* language, since translation necessitated the introduction of new technical and abstract vocabulary and the extension of the syntactic possibilities of Romance. In the matter of relations between the Romance of Castile and that of other Ibero-Romance areas, it is once again interesting to assess to what extent the language of the Alfonsine corpus can be identified with the Romance of any one Castilian area, or whether it represents some kind of koinéization or dialect mixture. Fernández Ordóñez (2005: 403–409) comes to the conclusion that while the majority of the large number of translators, emenders, and scribes were based in Seville or Toledo, the "norm" of Alfonsine prose is not straightforwardly identifiable with the Romance vernacular of either of these places, and that the texts show much dialectal variation, both from work to work and within the same work, perhaps precisely because of the collaborative nature of the translation process in which a number of different people would intervene to produce the finished "Castilian" text.

2.3 "*Secondary" dialectalization*

It is usually supposed that as the northern states expanded their territory southwards in the Reconquest, "secondary" dialectalization of the three Ibero-Romance varieties associated with the ultimately most powerful states (Castilian in Castile-León, Portuguese in Portugal, and Catalan in the Crown of Aragon) occurred, other northern varieties eventually being territorially or politically eclipsed. Although we know of some particular associations between non-Castilian-speaking areas of the north and the southern territories reconquered by Castile (the settlement of people from León and Navarre in Seville and people from León in Jerez) (see García Martín 2008: 57), any specific linguistic impact on *Ausbau* Castilian by these Ibero-Romance varieties has proven difficult to identify. Such features as the neutralization of /s/ and /θ/, the aspiration and loss of syllable-final /s/, the weakening of /s/ to [h], and the neutralization of syllable-final /ɾ/ and /l/ can all be seen as natural, more "advanced" developments within Castilian.

Yet there are areas on the frontiers of the former northern kingdoms which exhibit in their secondary developments the results of contact between the languages of the north and therefore cannot easily be categorized as direct developments any one of them: Zamora Vicente (1967) labeled these 'hablas de tránsito' (transitional dialects), but they are perhaps more appropriately thought of as the product of a certain amount of hybridization. One such is the traditional speech of Murcia, which was repopulated from the northeast and for a brief time (1296–1304) was politically a part of Aragon. Murcian exhibits the Catalan-type palatalization of initial /l/ in some words (*llengua* corresponding to Cast. *lengua* 'tongue'), though other features, such as the aspiration of syllable-final /s/, are shared with much of Andalusia. There are also features which are individual to Murcian: in some areas /ʎ/ and /ʝ/ are neutralized as [ʎ] (rather than the more usual [ʝ] of Castilian *yeísmo*), a phenomenon known as *ultralleísmo*; there is devoicing of /ʤ/ to /ʧ/ (*minchar* 'to eat' is cognate with Cat. *menjar*), and the form of the diminutive is *-icho* (probably a development of the stereotypical Aragonese *-ico*).

2.4 *Standardization*

In the medieval period, preference for different Romance varieties is almost entirely a function of politics, though the use of conventional literary languages such as Galician and Provençal should also be noted. It must be insisted that this was not a "battle of languages," as it is sometimes figuratively represented, but the ability of speakers of a particular variety to imbue their variety with prestige. The consequence was the elaboration of such preferred varieties for official, legal, and technical purposes, usually with the concomitant creation of a literary tradition whose prestige may linger. We can also see that varieties which are eventually unpreferred become unprestigious. Leonese as a chancery language was rapidly eclipsed by the ascendancy of Castile and the consequent promotion of, and preference for, Castilian. The only sense that can thereafter be given to the notion of "Leonese" is a cluster of varieties spoken in an area defined politically by the former kingdom of León and, possibly, by the varieties of other areas which share linguistic features with Leonese and do not form a continuum with Castilian or Portuguese (Mirandês and the Fala of Extremadura, see Section 3.2.1). By the sixteenth century, stereotypical features of Leonese provided the basis of the stage style *sayagués*, used for humorous purposes in Castilian drama to suggest rusticity.

Further west, the political separation of Portugal from Galicia in the late twelfth century and the union of Galicia with Castile-León spelled the similar eclipse of Galician within Castile-León, though the strong existence of a prestigious literary Galician tradition of secular lyrical poetry lasted as long as the genre for which it was the vehicle. The decision by Afonso III (reigned 1248–1279) to use Romance (Portuguese) instead of Latin as the language of the Portuguese chancery was contemporaneous with similar actions by Fernando III (reigned 1217–1252) and continued by Alfonso X in Castile, and by Jaume I (reigned 1213–1276), who established Catalan as the chancery language of the Crown of Aragon, a confederation which comprised Aragon, Catalonia, Valencia, and the Balearic Islands in present-day Spain. Also in the east, the prose works of the Mallorcan polymath Ramón Llull (1232?–1315/16) contributed to the elaboration of Catalan in much the same way as the texts of the Alfonsine corpus to that of Castilian. Catalan was later relatively quickly eclipsed by the adoption of Castilian after the union of the Castilian and Aragonese crowns in the late fifteenth century and the progressive Castilianization of Aragon, the Catalan élite adopting Castilian for both official and literary purposes even though the use of Catalan in an official context was not formally prohibited until 1714 (Galmés de Fuentes 2001).

Association of these emerging standards with a geographical location, as we have already seen, is not always straightforward. Castile had no capital as such, and the tradition that the geographical locus of the standard was Toledo was probably later wishful thinking (González Ollé 1987); in Portugal, while the establishment of the capital in Lisbon coincided with the definitive switch from Latin to Portuguese as a chancery language, Coimbra seems also to have continued to be a center of linguistic prestige. By contrast, the linguistic uniformity of medieval Catalan texts has often been remarked upon (Russell-Gebbett 1965: 23) and must have been centered on eastern Catalan, and on Barcelona in particular.

The Renaissance is characterized by the interest of humanist scholars in describing the vernaculars and the consequent production of the first, partial codifications together with "secondary responses" to language (Bloomfield 1944). The role of printing in standardizing and increasing awareness of written practice is also of great importance (Eisenstein 1979: 117–118). In Spain, Antonio de Nebrija's 1492 *Gramática de la lengua castellana* was the first of a series of grammars and treatises on Castilian; his *Diccionario latino-español* of 1492 (Nebrija 1979 [1492]) and his *Vocabulario español-latino* of c. 1495 (Nebrija 1951 [1495] and 1973 [1516]) were the beginnings of a lexicographical tradition stretching via Covarrubias's *Tesoro de la lengua castellana o española* of 1611 to the dictionaries of the Real Academia Española. Orthography was treated as part of the *Gramática*, but was dealt with more fully in the *Reglas de orthographia en la lengua castellana* (1517): these works established explicitly the phonemic basis, adhered to ever since, of Spanish spelling. Here, then, were already for Castilian the three cornerstones of language standardization. The Portuguese parallels date from somewhat later: the *Gramática da Linguagem Portuguesa* by Fernão de Oliveira (1536) and the *Ortografia e origem da língua portuguesa* by Duarte Nunes de Leão (1576), with the first dictionary, by Morais Silva, not appearing until 1789 (Baxter 1992: 12).

In the modern, "Academic" period, when there is a concern for the more exhaustive codification of languages and the limitation of variation, the lack of parallel between Spanish and Portuguese has become more marked. One of the consequences of the arrival of the Bourbon king, Philip V, in Spain was the foundation in 1713 of the Real Academia Española (RAE), ostensibly in imitation of the Académie Française, but actually taking a more proactive role in standardization, while at the same time being much more permissive in attitude. In the course of the eighteenth century, the RAE published the *Diccionario de autoridades* (1726–1739), an Orthography (1741), and a Grammar (1771) (Sarmiento 1984). These have been continuously updated ever since, and today the written norm for the whole of the Spanish-speaking world is provided by the Asociación de Academias de la Lengua Española headed by the Real Academia Española (see Section 2.5). The last spelling revisions were made in 2003; the most recent printed dictionary is the twenty-second edition of 2001 (now regularly updated on the Internet), and the most recent grammar is the 2009 *Nueva gramática de la lengua española: Morfología/ Sintaxis I, Sintaxis II* (Madrid: Espasa Libros). There is also the more overtly normative *Diccionario panhispánico de dudas*. Although today prescriptive standards are also laid down by leading newspapers and the broadcasting media (Stewart 1999: 21–28), these often derive from the RAE publications. By contrast, the Portuguese Academia das Ciências, founded in 1779, has concerned itself primarily with spelling reform and produced only the beginnings of a dictionary; responsibility for language planning matters in Portugal has now passed to a new Comissão Nacional da Língua Portuguesa (Baxter 1992: 12–14). The Academia Brasileira de Letras, founded in 1897, in addition to occupying itself with spelling reform, also maintains a dictionary. Thus, in the Portuguese-speaking world the notion of a standard grammar is to be found only in respected but unofficial codifications, most notably Cunha and Lindley Cintra (1984/1985).

In the nineteenth and twentieth centuries, Catalan and Galician underwent revivals and elaboration into *Ausbau* languages, which are today co-official with Castilian in Catalonia, Valencia, the Balearic Isles (Catalan), and Galicia (Galician). Standardization has been achieved relatively rapidly, in a way which mirrors that of Castilian, by the establishment of regulatory language planning bodies and the publishing of standard orthographies, dictionaries, and grammars. The initial formulation of the Catalan standard by Pompeu Fabra (1868–1948), while adopting an orthography which was pandialectal, including distinctions (such as *a/e*, *o/u* in stressed syllables) that are not found in Barcelona, tended towards eastern Catalan phonological, morphological, and syntactic variants with a rather eclectic vocabulary motivated by a policy of *depuració*, or difference from Castilian – by the nineteenth century, many Castilian words had infiltrated spoken Catalan (Saldanya et al. 2006; Bibiloni 1999). The Galician standard, motivated by difference not only from Castilian but also from Portuguese, has been more eclectic and cannot be identified with any one geographical area (Carballo Calero 1976: 75–81). In the wake of the Catalan and Galician revivals, other Ibero-Romance varieties, for which there were still native-speakers in the late twentieth century, have also been partially codified and elaborated, most notably Asturianu and Aragonés.

2.5 *The sense of speech community*

A surprisingly little-studied point of comparison among the *Ausbau* Ibero-Romance languages is the extent to which there is "unity in diversity" and a sense of speech community. In this respect, Castilian and Portuguese, by far the largest in terms of diasporic expansion, differ considerably. The Spanish-speaking world was politically fragmented into a number of independent republics as a result of the majority of the Empire's emancipation from Spain in the early nineteenth century, and it might have been expected that this would be accompanied by corresponding linguistic fragmentation. However, the 1847 Grammar of the Venezuelan statesman and philosopher Andrés Bello (1781–1865) can be credited with having begun a movement in Latin American prescriptive linguistic writing which significantly inhibited this process. Bello saw a parallel between the break-up of the Spanish Empire and that of the Roman Empire, and the danger that was presented to linguistic unity as militating against Americanism. His solution, adopted by later influential linguists, most notably the Colombians Rufino José Cuervo (1844–1911), Miguel Antonio Caro (1845–1909), and Marco Fidel Suárez (1855–1927), and the Mexican Rafael Ángel de la Peña (1837–1906), was to recommend following the Peninsular norm in an overarching *norma culta*. The Latin American Academies, which were formed beginning with the Colombian Academy in 1871 founded by Caro, collaborated with the Spanish Real Academia from the outset, and since 1951 there have been four-yearly meetings of the Asociación de Academias de la Lengua Española. Such collaboration on linguistic corpus planning has intensified, and the latest official Orthography, Dictionary, and Grammar have been undertaken in the name of all the academies. At the same time, it is openly acknowledged that Spanish has a pluricentric norm, and these documents strive to reflect educated usage across the Spanish-speaking world.

By contrast, there appears to have been no such inclination in nineteenth-century Brazil to bring the language into conformity with the European Portuguese norm. There was no perceived issue of linguistic fragmentation within what had remained a single political entity; it is perhaps also relevant that there was no strong academic tradition in Brazil (the first full university was founded only post-independence, while Spanish-speaking America boasted a steady stream of foundations beginning with the Universidad de San Marcos in Lima founded in 1551). In the twentieth century, the Modernist movement rejected tradition and insisted on Brazilian originality, with literary authors (e.g., Mário de Andrade) writing in a way which more closely approximated spoken Brazilian Portuguese. Since this time, there has been a general favoring of the distinction of Brazilian Portuguese from European Portuguese. The result is that even in spelling, there is no official agreement, despite a long history of attempts at reform and unification. The latest formal agreement between the Portuguese and Brazilian Academias is the Novo Acordo of 1990, though in practice Portugal follows the 1945 agreement, while Brazil follows the *Vocabulário Ortográfico da Língua Portuguesa* approved by the Academia Brasileira in 1943. This and the striking differences in syntax, especially with regard to pronoun position, mean that books in Portuguese are generally published in separate Brazilian and European editions.

The priority for other Peninsular Ibero-Romance speech communities has been the corpus and status planning which will secure their survival through official promotion. This has led to the aspiration to create monocentric standards based on eclectic choices. This has been possible for Galician, which is co-official with Castilian in only one autonomous region. Catalan, however, has proved a rather different proposition, since varieties of what may broadly be called "Catalan" range over three autonomous regions, the so-called *Països Catalans*. The co-official language of the Balearic Isles is designated *catalá*, though its regulating body is the Universitat de les Illes Balears rather than the Barcelona Institut d'Estudis Catalans. In Valencia, the co-official language is designated *valenciá* and is regulated by the Acadèmia Valenciana de la Llengua; although this body recognizes the unity of Valencian and Catalan, this question has generated a bitter politically-inspired polemic (Sempere 1995).

3 Inter-dialectal contact

3.1 The influence of other Ibero-Romance languages on Castilian

While the diversification of Castilian in the reconquered territories of the Iberian Peninsula and in Spain's overseas empire has received considerable scholarly attention, the development of Castilian in other Romance-speaking areas of Spain has only relatively recently become a focus of serious investigation, largely thanks to sociolinguistic interest in contact phenomena. The general conclusions to be drawn to date are that contact influence is strongest in the Castilian of bilingual speakers for whom the other language is dominant (hence, the speech of uneducated rural speakers is typically more prone to influence than that of educated urban speakers). Thus, *seseo* in the Castilian of Catalan-speaking areas is found only in "low" rural sociolects among speakers for whom Catalan is dominant (Blas Arroyo 2005b: 1071). In Galicia, the phenomenon of *geada* (rendering of Castilian /g/as [x]) is found only in uneducated speech (Echenique Elizondo and Sánchez Méndez 2005: 506). Yet there is also evidence of influence on the speech of educated monolingual speakers (Ramallo 2007), and also of lingering residual influence in areas where the other language is no longer actively used. It has been noted, for example, that generally in the Castilian of Catalan-speaking areas the semantic ranges of *venir/ir*, *traer/llevar* are different from the standard. *Venir*, like Cat. *venir*, is used to indicate motion from the speaker's location to the interlocutor's location (so *Mañana vendré a visitarte* 'I will come and visit you tomorrow' rather than standard *Mañana iré a visitarte*, lit. 'I will go and visit you tomorrow'). The ranges of Cast. *llevar* and *traer* are both covered by Cat. *portar*, denoting transport both to the speaker's location (Cast. *traer*) and from the speaker's location to the interlocutor's location (Cast. *llevar*) with the consequence that *llevar* is used for standard *traer* and vice versa (so *Te llevo un regalo* for standard *Te traigo un regalo* 'I have brought you a present,' and *Voy a traerte un libro a tu oficina* 'I'm going to bring a book to your office' for standard *Voy a llevarte un libro a tu oficina*, lit. 'I'm going to take a book to your office'); such usage is accepted to the point of being a marker of Catalan identity (Vann 1998: 264). Similarly, even among educated Castilian speakers in Galicia, there is a general absence of compound verb forms, with the preterite being used for the perfect (e.g., *Lo hice ahora mismo* rather than standard *Lo he hecho ahora mismo* 'I've just done it'), and the *-ra* verb form having a wide and distinctive range of functions which include that of the standard pluperfect (so *Me dijo que llegara el día anterior* for standard *Me dijo que había llegado/llegó el día anterior* 'He told me he had arrived the previous day'; see Rojo 2005: 1095).

The effect of contact does not always tend away from the standard. In the Castilian of Catalan-speaking areas, it has been noted (Echenique Elizondo and Sánchez Méndez 2005: 300) that Catalan speakers who make the etymological distinction between /ʎ/ and /ʝ/ transfer this to Castilian, thus apparently bucking the very generalized trend in Castilian towards *yeísmo*. Catalan speakers of Castilian also preserve intervocalic /d/ to a greater extent (Blas Arroyo 2005b: 1080), and the availability of a wider range of syllable-final consonants, especially voiceless plosives, in Catalan also appears to facilitate the maintenance of "learned" consonantal groups in Castilian (e.g., *perfecto* [perˈfekto] rather than such weakening of /k/ as [perˈfeθto] or [peɾˈfeto] which is common in Castile and elsewhere), and the final consonants of foreign loanwords (e.g., *carnet*, *carnets* [kaɾˈnet], [kaɾˈnets], which preserve a more transparent relationship between spelling and phonology rather than standard [kaɾˈne], [kaɾˈnes]) (Blas Arroyo 2005b: 1080).

Consciousness of contact influence among speakers may also result in hypercorrection in a desire to approximate the Castilian standard. Thus, Abuín Soto (1971: 185) and Rojo (2005: 1095–1096) report that because of awareness of use of the preterite for the perfect in the Castilian of Galicia (see above), the perfect is sometimes used by speakers in contexts where standard Castilian uses only the preterite (e.g., *Colón ha descubierto América* 'Columbus discovered America'). Also, the compound pluperfect may be used as the equivalent of the *-ra* form in a general past function where the preterite would be the expected standard form in Castilian (e.g., *cuando había ido a Madrid* for standard *cuando fue a Madrid* 'when he went to Madrid').

It can often be difficult to identify whether a change is contact-induced or is in line with more widely occurring tendencies in Castilian: this is a general procedural problem when there is a high degree of structural congruence between the donor and host language, as in the case of the Ibero-Romance languages. Velarization of final /n/ in Galicia is plausibly attributed to the influence of Galician, but it is also a widespread phenomenon in many other Castilian varieties (Penny 2000: 151). Commentators on the Castilian of both Catalan-speaking areas (Echenique Elizondo and Sánchez Méndez 2005: 301) and of Galicia (Cotarelo Vallador 1927: 97) have noted avoidance or anomalous use of the relative adjective *cuyo* 'whose,' which has no equivalent in either Catalan or Galician, and its substitution by *del cual*, etc.; but *cuyo* is also infrequent in spoken Castilian (Blas Arroyo 2005b: 1080). Contact influence often consists of a shift in markedness: the greater frequency of the *haber de* + infinitive paraphrase in the Castilian of Catalan-speaking areas and of the *tener* + past participle paraphrase in the Castilian of Galicia is likely to have been encouraged by contact, even though these constructions do exist in standard Castilian. *Haber de* + infinitive has survived strongly as a future in some areas of Latin America (Kany 1951: 153), and *tener* with a past participle which agrees with the direct object, most often indicating a resultant state, is strongly attested throughout the history of Castilian and is part of standard usage (Harre 1991: 94–128). Similarly, in the Castilian of Catalan-speaking areas, *agradar* 'to appeal to,' which in Castilian is less frequent and belongs to a higher register than the cognate Cat. *agradar* 'to please,' which is comparable in usage to Cast. *gustar*, is used with a greater frequency than in the standard language. Features which are widespread in Castilian, but regarded as substandard, are sometimes erroneously attributed to contact with other Ibero-Romance languages. Blas Arroyo (1999: 43–68) observes that pluralization of the existential verb *haber* with a plural predicate (*habían muchos coches en la calle* for standard *había muchos coches en la calle* 'there were many cars in the street') is a stereotype of Catalan/Valencian speakers of Castilian; but in fact such "agreement" is very common elsewhere and is also regarded as substandard in Catalan (Pountain 2006: 100).

With the above provisos, the following are some of the more systematic stereotypical features of the Castilian of other Ibero-Romance-speaking areas which have been widely identified in the literature as due to, or encouraged by, previous or ongoing contact and appear to be relatively strongly embedded. It must be stressed, however, that they are not generally prestige forms, and normative pressure prevents their diffusion more widely within the Spanish-speaking world. Significant for a general theory of language change is that such contact-induced change appears to affect all linguistic levels, and that while many of these features pertain to the speech of bilinguals in Catalan-speaking areas and in Galicia, there is also plenty of evidence of "legacy" contact in areas which today are largely non-bilingual, such as Asturias and Aragon.

3.1.1 Phonology Castilian has the least complex vowel system of the standard Ibero-Romance languages (a five-term /i e a o u/ system in both tonic and atonic position). Historically, this is the outcome of the extensive diphthongization, in both open and closed syllables, of Latin tonic short (open) /ĕ/ (/ɛ/) and /ŏ/ (/ɔ/) to /je/ and /we/, a feature shared with neighboring Romance varieties in León and Aragon, but not with Catalan, Galician, and Portuguese, where such diphthongization does not take place and which have at least a seven-term system tonically (/i e ɛ a ɔ o u/). The Castilian of these regions exhibits a greater difference than in the standard between stressed and unstressed vowels, the former tending to be lower and the latter higher (Badia i Margarit 1975 [1964]: 154), though the raising of final vowels is widespread within the Spanish-speaking world (Penny 2000: 133–134). There is, however, no evidence of the creation of new systematic phonemic contrasts. In Catalan-speaking areas in which/a/and/e/in atonic syllables are respectively raised and centralized and the distinction hence neutralized, there is a tendency for the same process to carry over to Castilian (thus *sentencias* [sənˈtenθjəs] 'sentences'). The raising of atonic/a/to/e/is a legacy of Bable (Asturianu) characterizing the Spanish of Asturias, giving verb forms such as /ˈmiɾen/for *miran* and feminine plurals such as/ˈbakes/for *vacas*.

Consonantal phenomena are most noticeable in the Castilian of Catalan-speaking areas. Here, syllable-final/l/is velarized as [ɫ], and final/d/, which is fricative in Castilian and is generally undergoing further weakening to the point of total disappearance, undergoes apparent fortition to [t] in parallel with the historical development and present-day sound pattern of Catalan (thus, *ciudad* [θjuˈδat] corresponding to Cat. *ciutat* [sjuˈtat] 'city, town'). Final/s/is also frequently voiced before a following vowel (*los árboles* [lozˈaɾβoles] 'the trees').

The Castilian of Aragon exhibits a tendency to convert proparoxytones into paroxytones, in common with all surviving Aragonese Romance varieties (Zamora Vicente 1967: 221). Thus, *árboles* is rendered as [aɾˈβoles] and the first person plural imperfect inflection *-ábamos* as [aˈβamos].

3.1.2 Morphology and syntax Historically, all the Ibero-Romance languages preserved three demonstratives derived from modified reflexes of Latin ECCE + ISTE 'this (of mine),' ECCE + IPSE 'that (of yours),' and ECCE + ILLE 'that (of someone else).' In some Catalan-speaking areas, this has been reduced to a binary demonstrative system with the loss of the former middle term *aqueix* (leaving *aquest* 'this' vs. *aquell* 'that'); in parallel with this, Castilian speakers in these areas often use *este* for *ese*, thus tending towards a similar elimination of the middle demonstrative (*este* 'this' vs. *aquel* 'that').

The verbal morphology of common irregular verbs is often the product of contact. In the Castilian of Galicia, the present subjunctive forms *dea* for standard *dé* 'give' and *estea* for standard *esté* 'be' are encountered, and in Asturias, different morphological forms of the second and third person singular of the present tense of *ser* 'be' are common: *yeres*, *yes* for standard *eres*, *es*. As regards verb form usage, Catalan and Castilian differ significantly in strategies for expressing the future; in Catalan, there is no 'go' future corresponding to Castilian *ir a* + infinitive – or, at least, the imitation of such a 'go' future is normatively resisted (see Section 3.2.2 below). The Castilian of Catalan speakers shows more extensive use of the inflected future (e.g., *haré* 'I will do') to express future time-reference than is usual in the spoken Castilian of other areas, where this form has intentional and epistemic modal, rather than simple future temporal, value (the epistemic modal value of the

Castilian future observable in *serán las ocho* 'it must be eight o'clock' is not paralleled in Catalan). Another feature of verbal syntax in the Castilian of Catalan speakers is the use of the future indicative for the present subjunctive in future-referring temporal clauses, in parallel with Catalan (e.g., *Cuando vendrás, iremos al cine* 'When you come, we will go to the cinema,' rather than standard *Cuando vengas*, ...).

Some differences in clitic pronoun position are probably due to contact, although the history of Castilian has shown ongoing variation in this area. In Castilian and Catalan, the Tobler-Mussafia law of clitic pronoun placement, which constrained clitics from appearing in absolute initial position, has been progressively abandoned, and in the modern language, clitics are generally proclitic, being only enclitic with positive imperative, infinitive, and gerund verb forms (e.g., Cast. *¡hazlo!* 'do it!' but *¡no lo hagas!* 'don't do it!,' *hacerlo* 'to do it,' *haciéndolo* 'doing it'); in auxiliary + infinitive or gerund structures, the pronoun may be enclitic to the infinitive/gerund or proclitic to the auxiliary (e.g., Cast. *quiero saberlo ~ lo quiero saber* 'I want to know (it)'). In the Castilian of Asturias, resistance to initial placement of clitic pronouns can still be observed, and clitics tend to be postposed to an initial finite verb in positive declarative sentences. The Castilian rule that *se* stands first in any sequence of clitics is sometimes broken in the Castilian of Galicia, with such orders as *Te se cayó el libro* 'you dropped the book' for standard *Se te cayó el libro* (Abuín Soto 1971: 185). In the Castilian of Aragon, there is a particular kind of *leísmo*: the use of *le(s)* rather than *lo(s)* and *la(s)* in the specific context of a preceding *se*. A particular pronominal function observable in the Castilian of Galicia and paralleled in Galician is the extended use of the "dative of interest" in which an atonic indirect object pronoun appears to signal the involvement of an interlocutor rather than representing an argument of the verb (*Le somos una nación, ¿sabe usted?* 'We're a nation (I'm telling you)': example from Seco 1989: 164).

In the Castilian of Aragon, subject forms of personal pronouns, especially the second person singular, are sometimes used as prepositional objects. Thus, *para tú* 'for you,' which is paralleled in Aragonese Romance varieties (Zamora Vicente 1967: 253), is used for standard *para ti*.

Variation in affective diminutive suffixes is very obviously attributable to contact, and a strong marker of regional variety. The suffix *-iño* is used in the Castilian of Galicia, *-ín* in Asturias, and *-ico* in Aragon.

In the Castilian of Barcelona, *que* is often used to introduce a polar (yes–no) question. This is also a well-known feature of central Catalan (Blas Arroyo 2005a: 564), for example, *Oye, ¿que te has hecho daño?* for standard *Oye, ¿te has hecho daño?* 'Hey, have you hurt yourself?' (Grau Sempere 2005; Hualde 1992: 2, who regards the construction as "stereotypical of a Catalan background"). In the Castilian of Valencia, Blas Arroyo (2005b: 1078–1079) attributes the suppression of *desde hace* in 'ago' adverbials (e.g., *tengo el carnet de conducir siete años* for standard *tengo el carnet de conducir desde hace siete años* 'I've had my driving license for seven years') to the parallel construction in colloquial Catalan.

3.2 The influence of Castilian on other Ibero-Romance languages

3.2.1 Portuguese The influence of Castilian on Portuguese has tended to be played down by scholars. Piel 1989 [1976] identifies a number of lexical borrowings from the period of the union of the two crowns (1580–1640), when contact was most intense, some of which can be grouped into categories which reflect particular aspects of Spanish culture (military terms: *cabecilha*, *caudilho*, *guerrilha* 'ringleader, warlord, guerrilla (insurgent)'; dress: *boina*, *mantilha* 'beret, mantilla (veil)'; society: *tertúlia*, *chiste* 'meeting, joke'); and others of which appear to be more general (*faina*, *trecho*, *moreno* 'work, period, brunette/tanned or dark-skinned'), but also notes that the extent of this influence is, as in Catalan (see Section 3.2.2), probably masked by the ease with which borrowings can be adapted to Portuguese phonological patterns and by semantic calquing.

A somewhat different phenomenon of considerable current interest is Portunhol, or code-switching between Portuguese and Spanish in Brazil. There has been an increased interest in Spanish in Brazil since the 1980s because of Brazil's membership, alongside several Spanish-speaking countries, in the Regional Trade Agreement Mercosur/Mercosul and the usefulness of Spanish in trade with the United States and even the EU; it is now compulsory for Brazilian secondary schools to offer Spanish as a foreign language. Portunhol is perhaps best characterized as a fashion cultivated for humorous or licentious purposes; unlike code-switching between Spanish and English in the United States, it does not obey the free morpheme constraint, according to which a switch may not occur between a bound morpheme and a lexical form unless the latter has been phonologically integrated into the language of the bound morpheme (Sankoff and Poplack 1981: 5), so that a form such as *terê* 'I will have' shows the Portuguese verb-stem *ter-* with the Spanish inflection *-ê (-é)*; in this respect Portunhol is more of a language hybrid (Lipski 2006).

Indeed, the close proximity of Castilian- and Portuguese-speaking areas in parts of the Iberian Peninsula and South America has led to a number of interesting cases of what is arguably hybridization. It is important to distinguish the languages that are discussed below from those Romance languages in the northern Peninsular dialect continuum which impressionistically lie "between" Castilian and Portuguese. Such are Mirandês, spoken in Miranda do Douro and parts of Vimioso, which has co-official status with Portuguese in this area, and the Fala of the Valle de Jálama (Cáceres), spoken in Valverde del Fresno, Eljas, and San Martín de Trevejo. The latter in all probability represents an enclave of Galician/Leonese colonization established in the twelfth–thirteenth centuries, separated geographically from the Galician speech community and subject to increasing contact with Castilian.

In the Portuguese town of Barrancos (Beja), alongside Portuguese and Castilian, a language known as *barranquenho* is spoken, in which a Portuguese base has systematically adopted a number of Spanish features in marked contrast to neighboring varieties: aspiration or loss of syllable-final/s/, loss of final/r/, neutralization of the opposition between/b/and/v/, use of the *-ito* diminutive of Spanish as opposed to the *-inho* diminutive of Portuguese, and use of the present subjunctive in future-referring temporal clauses rather than the Portuguese future subjunctive. Barrancos was a disputed territory from the thirteenth century, and the present border was only definitively fixed in 1926; it was populated by Castilian-speaking settlers from Badajoz, who were still in the majority in the sixteenth–seventeenth centuries, and it has continuing cultural affinities with Spain. There is thus a strong sense of Barranquenho identity and the individuality of Barranquenho as a language: while Portuguese is the official language, Spanish and Barranquenho are used in everyday conversation, and it is likely that a strong motivation for the use of Barranquenho is cryptolectic.

In South America, the Fronteiriço of the Uruguayan-Brazilian border has attracted considerable attention (Hensey 1972, 1982; Elizaincín 1976, 1992). As with Barranquenho, Fronteiriço occurs in an originally disputed territory between Spain and Portugal which was resolved in 1825 with the formation of the state of Uruguay; but there were large numbers of Portuguese-speaking settlers, and Portuguese continued to be used, especially in the northern frontier towns. Fronteiriço does not have the same systematicity as Barranquenho, and variation between Portuguese and Spanish forms suggests that it involves a significant element of code-switching. But, as in Portunhol, the free morpheme constraint is not observed, yielding verb forms like *chegó* 'he arrived' (Portuguese stem *cheg-* with Spanish preterite inflection *-ó*), and there are some apparently totally original developments: in one text collected by Elizaincín (1992: 151–152), there is a verb-form *cheó* 'he filled,' which is neither the *llenó* of Spanish nor the *encheu* of Portuguese, but apparently a formation from the basic Portuguese adjectival stem *chei(o)*, perhaps paralleling the Spanish formation of the verb from the corresponding adjective, and the Spanish verb inflection.

3.2.2 Castilian and Catalan Upper-class Catalans have a history of active bilingualism since at least the sixteenth century, and until the nineteenth century this tended to be an asymmetric bilingualism with a leaning towards Castilian. In the second half of the nineteenth century and the first third of the twentieth century, Catalan was cultivated by the industrial bourgeoisie and ever since has enjoyed a higher status. The imposition of Castilian during the Franco dictatorship and the influx of large numbers of monolingual Castilian-speaking immigrants from other areas of Spain, especially Andalusia, again strengthened the presence of Castilian. Today, all Catalan speakers are active bilinguals. In the 1980s the term *català light* came to be used to denote Catalan with many Castilian borrowings and features, and more specifically the everyday spoken Catalan used in the media (Kailuweit 2002). Such constant contact with Castilian has naturally led to borrowing to which Catalan corpus planners have always been sensitive. The everyday use of Catalan (as distinct from the official written language) is riddled with Castilianisms on all levels. The *Servei lingüístic* of the Universitat Oberta de Catalunya (2007) has recently published a list of some 480 Castilianisms which we may take to be sufficiently common to have aroused disapproving comment. The list includes not only easily identifiable simple lexical substitutions such as *despedir* for *acomiadar* 'to bid farewell; to dismiss' and *aliviar* for *alleugerir* or *alleujar* 'to alleviate' (both *despedir* and *aliviar* appear in the Alcover and Moll *Diccionari català-valencià-balear*, however, which demonstrates their antiquity as Castilian borrowings), but also Castilian words which have been adapted to Catalan patterns such as *desperdici* (Cast. *desperdicio*) for *deixalla* 'waste'. There are also cases of semantic calques, where a Catalan word has a different range of meaning in parallel with its Castilian cognate: while Cast. *propio* corresponds to Cat. *propi* in the sense of 'own (belonging to oneself),' the meaning of 'same' (Cat. *mateix*) is castigated. The influence of Castilian also extends to syntax: *anar a* + infinitive is used to form a 'go'-future (see 3.1) in parallel to Cast. *ir a* + infinitive; *de* + infinitive is used to express a conditional protasis.

The influence of Castilian is also apparent in particular sociolects. In Barcelona, *bleda*, which is associated with upper-class female speakers, is characterized by use of the clear Castilian alveolar-palatal articulation of *l* ([l]) instead of the dark Catalan velar articulation ([ɫ]). *Xava*, which is associated with working-class speakers, is marked by the pronunciation of Catalan [ə] as [a] (whether [ə] corresponds to written *e* or *a*). Both these phenomena have spread within Barcelona and can be seen as examples of linguistic leveling since in both these cases the outcome tends towards the simpler system of Castilian.

4 Conclusion

The history of Spanish is to be understood first of all as the elaboration of an *Ausbau* language based on the Romance vernacular of Castile, though not necessarily identifiable with any one locality. In this, Castilian and other standardized Ibero-Romance languages have much in common. However, Spanish contrasts significantly with other Ibero-Romance languages in the aspiration to maintain a single speech community, which has encouraged the modern acceptance of a pluricentric norm. At the same time, the ongoing contact between Spanish and other Ibero-Romance languages offers much scope for further research and may be expected to be of importance to an understanding of contact-induced language change, bilingualism, and linguistic hybridization.

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4 Spanish in Contact with Amerindian Languages

ANNA MARÍA ESCOBAR

1 Introduction

Contact between Spanish and Amerindian languages began with the arrival of Columbus on the island of *Hispaniola* (now the Dominican Republic and Haiti) in 1492. The subsequent conquest and expansion of the Spanish crown in the sixteenth century was rapid, taking less than a century (Boyd-Bowman 1968). The sociohistorical characteristics of the contact, however, have varied through time and geographical space. The contact between Spaniards and indigenous populations led to armed confrontations, as well as to the spread of diseases (smallpox and chickenpox, in particular) for which the indigenous peoples did not have biological defenses, provoking forced migrations and deaths (Cook 1998). The conflicts and the impact of diseases continued as the Spaniards expanded their dominance into Meso-America, the northern continent, and South America. Researchers calculate that between 25% and over 50% of the Amerindian population perished due to wars and/or exposure to new diseases, recuperating its numbers only in the eighteenth century (Sánchez Albornoz 1973: 22, 108).

The first language contact was between Spanish and Arahuac languages in the Caribbean, lasting more than two decades before the conquest of the Aztec empire (*Antillean period*, between 1492 and 1519). Various Spanish clergy wrote the first Amerindian grammars during the sixteenth and early seventeenth centuries. These grammars (and dictionaries) described the languages of the most powerful indigenous groups of the time, in Mesoamerica (Nahuatl, Phurhépecha, Hñähñu or Otomi, Mayan), in the Colombian region (Chibcha or Muisca), in the Andean region (Quechua, Aymara), in the Chilean region (Mapudungun), and in the Paraguayan region (Guarani). These languages were used as *lingua franca* to catechize (by the missionaries), and to politically control (by the Spanish crown) the indigenous populations.

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The complexity of the contact situation in Latin America has been described in sociohistorical terms using three main criteria (Granda 1995): (1) the type of Spanish settlement; (2) its geographical location; and (3) the size and complexity of the Amerindian community. The type of Spanish settlement is determined by the intensity of contact between the colonial settlement and the Spanish crown. Granda distinguishes central settlements (capitals of Viceroyalty, e.g., Mexico City, Lima), from peripheral settlements, which had less contact with the metropolis and the capitals (e.g., US Southwest, Central America, Chile, Paraguay, and the Buenos Aires region until the mid-eighteenth century). Whether the settlement was on the coast or in the mountains was also relevant, since coastal settlements had more contact with the metropolis due to port access (e.g., Veracruz, Portobelo, Cartagena, Callao). Last, Granda stresses the social complexity and territorial extensions of these populations during colonial times – characteristics which also figured in the success of efforts by indigenous populations to deter Spanish advances (e.g., the Mayan in Southern Mexico and Central America, the Mapuche in Chile) and/or to maintain the use of their language and customs (e.g., Nahuatl, Mayan, Quechua, Guarani, Mapudungun).

The Bourbon reforms of the eighteenth century were intended by the Spanish crown to more strongly enforce Hispanization and assimilation of the indigenous, among other sociopolitical reforms. However, the reforms only exacerbated the social situation, leading to the well-known uprisings, characteristic of this period, which, in turn, led to independence movements. The formation of the new Spanish-speaking countries in the early nineteenth century led to hopes for a new phase in the contact between Spanish and the Amerindian languages. However, the non-fulfillment of this unspoken promise resulted in populist movements at the turn of the twentieth century in most of Latin America, which brought attention to the Amerindian populations. Towards the end of the twentieth century and the beginning of the twenty-first century, the demands from leaders of the Amerindian populations for *multicultural*, *multiethnic*, and *multilingual policies* began to be taken into consideration by the national governments, with the understanding that there is a need for a new type of *citizenship*, which is not discriminatory to its constituents (López 2008). The consequent emerging visibility of the Amerindian populations in the political arena has gone hand in hand with the emergence of strong and structured indigenous political organizations and leaders, such as in Chiapas (Mexico), Ecuador, Bolivia, and Guatemala, to name a few (cf. López 2009). The presence of Amerindian languages and their speakers in the sociopolitical life of the countries is giving rise in the twenty-first century to a wider tolerance (if not acceptance) of Spanish-Amerindian language bilingual communities; a renewed desire among indigenous youth to relearn the language of their parents or ancestors; and at the national level, a gradual move in the educational arena, from localized indigenous Bilingual and Intercultural Education to a national Intercultural Education (López 2010).

2 Amerindian languages

In 1492 there were around 2,000 languages in the Americas (Willey 1971). Today only 550 to 700 languages survive (Campbell 1997). The exact number of speakers of Amerindian languages is not clear, although calculations are available in census data, and on the *Ethnologue* website. Much is related to the fact that being an individual of indigenous origin is today a matter of self-identification. Indigenous peoples can be not counted by means of political acts that make the indigenous *invisible*, or by the indigenous individual's own personal alienation (cf. López 2009). The *Sociolinguistics Atlas of Indigenous Populations in Latin America* calculates that of 534 communities, 103 or 20% are monolingual in Spanish (Sichra, in progress, cited in López 2009: 420).

While the Americas hold 8% of the population of the world, 89% of the languages spoken in Central America and South America have fewer than 100,000 speakers; making the languages especially vulnerable (UNESCO interactive map). *Ethnologue* (2009) calculates that 82% of the languages of the world fall in this category, and 50% have fewer than 10,000 speakers. Some researchers estimate that if the situation does not change, 50% of the languages of the world will disappear by the end of the twenty-first century (Romaine 2007). This worrisome prospect is enhanced by the expansion of modern technologies, globalization efforts, and the diffusion of the few languages which participate in these processes, which together represent only 4% of the languages of the world.

Speakers of Amerindian languages in Latin America number between 40 and 50 million, and represent 10% of the total population of the region, but 40% of the rural population (López 2009). The Latin American countries with the largest indigenous populations are three in the Andean region (Peru, Bolivia, and Ecuador) and Guatemala. However, some countries have greater linguistic diversity with respect to Amerindian languages, such as Mexico, Peru, Colombia, Venezuela, and Bolivia, to name those which exhibit the most variation (Table 4.1).[1]

The most important Spanish contact varieties in Latin America, by virtue of the numbers of speakers involved and their urban presence in the twentieth and twenty-first centuries, are Spanish in contact with Quechua, with Guarani, with Mayan, with Aymara, with Nahuatl, and with Mapudungun.

Quechua has the greatest number of speakers in the Americas (between 10 and 12 million speakers) in a territory which spreads from southern Colombia to northwest Argentina, and includes parts of Ecuador, Peru, Bolivia, and northeast Chile. Speakers of Mayan languages constitute the second largest group, with more than 6 million speakers. Mayan languages are spoken in southern Mexico (Chiapas and the Yucatan Peninsula) and Guatemala. Guarani, spoken mainly in Paraguay but also in bordering regions in Brazil, Bolivia, and Argentina, is third, with somewhat less than 5 million speakers. Aymara, spoken in the Andean region in southern Peru, Bolivia, northwest Argentina and northeast Chile, is next, with approximately 2.5 million speakers. Nahuatl, spoken in central Mexico, has approximately 1.5 to over 2 million speakers, while Mapudungun, spoken in southern Chile and

Table 4.1 Amerindian languages spoken in Latin American countries (based on Ethnologue 2009)[2].

Country	*Amerindian languages*	*Language families*
Argentina	17	Araucanian, Aymaran, Chon, Lule-Vilela, Mataco-Guaicuru, Puelche, Quechuan, Tupi
Bolivia	37	Arawakan, Aymaran, Canichana, Cayubaba, Chapacura-Wanham, Itonama, Leco, Macro-Ge, Mataco-Guaicuru, Movima, Panoan, Quechuan, Tacanan, Tsimané, Tupi, Uru-Chipaya, Yuracare, Zamucoan
Chile	7	Alacalufan, Araucanian, Aymaran, Kunza, Quechuan, Rapanui, Yámana
Colombia	80	Andoque, Arawakan, Barbacoan, Camsá, Carabayo, Carib, Chibchan, Choco, Guahiban, Maku, Páez, Puinave, Quechuan, Salivan, Ticuna, Tinigua, Totonacan, Tucanoan, Witotoan
Costa Rica	6	Chibchan
Ecuador	13	Barbacoan, Chibchan, Choco, Jivaroan, Quechuan, Totonacan, Tucanoan, Waorani, Zaparoan
El Salvador	4	Lenca, Mayan, Misumalpan, Uto-Aztecan
Guatemala	24	Arawakan, Mayan
Honduras	7	Arawakan, Chibchan, Lenca, Mayan, Misumalpan, Tol
Mexico	291	Algic, Hokan, Huavean, Mayan, Mixe-Zoque, Oto-Manguean, Seri, Tarascan, Tequistlatecan, Totonacan, Uto-Aztecan
Nicaragua	7	Arawakan, Chibchan, Monimbo, Misumalpan, Oto-Manguean
Panama	7	Chibchan, Choco
Paraguay	15	Mascoian, Mataco-Guaicuru, Tupi, Zamucoan
Peru	92	Arauan, Arawakan, Aymaran, Cahuapanan, Candoshi-Shapra, Harakmbet, Hibito-Cholon, Jivaroan, Muniche, Panoan, Peba-Yaguan, Quechuan, Tacanan, Taushiro, Totonacan, Tupi, Urarina, Witotoan, Zaparoan
Venezuela	43	Arawakan, Arutani-Sape, Carib, Guahiban, Pumé, Salivan, Tupi, Warao, Yanomam, Yuwana

Figure 4.1 Major Amerindian languages in contact with Spanish in the twentieth century.

southwestern Argentina, has a population between 500,000 and 1 million (see Figure 4.1).

Lexical borrowings from Amerindian languages entered Spanish mainly during the colonial period (cf. Boyd-Bowman 1971). These cultural borrowings, mostly nouns, include lexical items referring to the flora and fauna of the region, as well as to local social organization, food, agriculture, clothing, folklore, and religion (see Table 4.2).

Following, we examine the main grammatical features of Spanish in contact with Quechua, Mayan, Guarani, Nahuatl, and Mapudungun.

3 Contact features: grammatical

Grammatical features of Amerindian languages have influenced varieties of Spanish spoken in regions where indigenous languages are also spoken. In evaluating contact-induced grammatical features in Spanish, it is important to distinguish features found only in second-language speakers of Spanish who have the Amerindian language as their first language. Some of these second-language features are found throughout Latin America – among them, omission of

Table 4.2 Lexical borrowings from Amerindian languages into Spanish.

Arahuac or Taino	*ají* ‘hot pepper,’ *barbacoa* ‘barbecue,’ *cacique* ‘indigenous leader,’ *hamaca* ‘hammock,’ *canoa* ‘canoe,’ *maíz* ‘corn,’ *tabaco* ‘tobacco,’ *huracán* ‘hurricane’
Nahuatl	*cuate* ‘friend,’ *chamaco* ‘child,’ *tomate* ‘tomato,’ *chocolate* ‘chocolate,’ *chicle* ‘chewing gum,’ *aguacate* ‘avocado,’ *tiza* ‘chalk,’ *coyote* ‘coyote,’ *chichis* ‘breasts’
Mayan	*xix* ‘bread crumbs,’ *mulix* ‘curly,’ *chem* ‘eyesleep,’ *xux* ‘beehive,’ *xoy* ‘stye,’ *xic* ‘armpit,’ *huech* ‘armadillo,’ *makkum* ‘stew,’ *pech* ‘tick,’ *tucha* ‘leg calf’
Quechua	*papa* ‘potato,’ *palta* ‘avocado,’ *puna* ‘high altitudes,’ *chacra* ‘ranch,’ *choclo* ‘corn,’ *soroche* ‘high altitude sickness,’ *cancha* ‘large outdoor space,’ *alpaca* ‘alpaca’
Guarani	*jaguar* ‘jaguar,’ *tapioca* ‘tapioca,’ *petunia* ‘petunia,’ *tucán* ‘tucan,’ *mangangá* ‘an insect,’ *maraca* ‘maraca (a musical instrument)’
Mapudungun	*poncho* ‘poncho,’ *guata* ‘stomach, belly,’ *pichín* ‘urine,’ *piñén* ‘body dirt,’ *laucha* ‘mouse,’ *peuco* (a type of hawk), *chuico* ‘big vase’

grammatical expressions, such as articles (e.g., _ *fines de semana es igual* ‘(the) weekends are the same’) and prepositions (*estos muchachos* __ *nuestras familias* ‘these guys (of) our families’).

Agreement markers are usually left out, such as those for gender in the determiner phrase (*el oreja* ‘the-MASC ear-FEM’) and number (*niños sucio*_ ‘dirty-(PL) children’), as well as subject-verb person number agreement (*es necesario que ellos habla su quechua pues* (‘it is necessary that they speak (SING.) their Quechua, then’). Regularization patterns of verbal morphological paradigms are also common (e.g., verb conjugations: *ponieron* $<$ *pusieron* ‘they put’, *hacerán* $<$ *harán* ‘they will do’), as commonly found in acquisitional studies.

In addition to second-language features, Spanish contact varieties present other characteristics attributable to language contact, and are presented in the following sections.

3.1 Contact with Quechua[3]

Spanish in contact with Quechua can be found principally in Peru, Bolivia, and Ecuador (ordered by Quechua-speaking population size). Smaller populations are found in southwest Colombia, northern Argentina, and northeastern Chile. Although in the Peruvian and Bolivian Andes, Spanish is also in contact with Aymara, another Quechuaymaran language, the Aymara-speaking population is smaller. The majority of Quechua speakers are Quechua–Spanish bilinguals.

The degree and type of contact between Spanish and Quechua varies in each of these countries, leading to some differences in the contact features found in different parts of the Andean region. For example, regarding Quechua grammatical borrowings, the use of the focus marker *-ga* (-ka) in Spanish is commonly found in Ecuador but not in Peru. More widely found in the Andean region are the diminutive *-cha* (e.g., *señora-cha* 'miss'), the nominal plural *-kuna* (e.g., *oveja-kuna* 'sheep-PL'), and the vocative *-y* (e.g., *vida-y* 'my life (for affection)').

Spanish grammatical features common to the Andean region are characteristic of the dialect known as Andean Spanish. This dialect is spoken by monolingual speakers and by native Quechua–Spanish bilinguals. The most important features follow.

Phonological characteristics include the assibilation of /r/ (e.g., *risa* [řísa] 'laugh,' *salir* [salíř] 'to exit'), as well as the use of consonantal variants which suggest the favoring of strengthening tendencies in this contact situation, as in the case with /s/ in coda position (e.g., *Cu*[s]*co*, *casa*[s] 'houses'), the strengthening of /x/ before /e, i/ (e.g., [x]*inete* 'rider,' *mu*[x]*er* 'woman'), and voiced stops as stops in intervocalic position, (e.g., *ma*[d]*era* 'wood'). The opposite tendency is found with unstressed vowels, which tend to be weakened or omitted (e.g., *todos* [tóds] 'everybody', *pues* [ps] 'well (discourse marker)').

Suprasegmental features have not been studied in depth across the region, except for an Andean intonational contour pattern, which has been studied only for Peru (O'Rourke 2005). There is also a tendency to stress the penultimate syllable, following a Quechua pattern (e.g., *vámonos* > *vamonos* 'let's go,' *plátano* > *platano* 'banana').

A phenomenon which has often been cited, but not studied in detail, is the omission of the third-person object clitic in answering questions (1a) or when the direct object is dislocated to the left of the utterance (1b).

(1) a. *¿Sabes que el señor Quispe se murió? – No ___ he sabido.*
'Did you know that Mr. Quispe died? – No, I did not know.'
b. *Al maestro ___ saludó en la plaza.*
'He/she greeted the teacher in the plaza.'

An extension of this phenomenon is *loismo* in direct object function for third person, without number- or gender-agreement markers (2) (Pozzi-Escot 1975), which is also found in other varieties of Spanish in contact.

(2) *Fui a ver la carretera. Ya lo habían arreglado.*
'I went to see the highway. They had already had it fixed.'

Although reported for both second-language and native Spanish, the specifics of *loismo* in Andean Spanish have only been studied descriptively; more research is needed.

A related phenomenon, also found in Porteño Spanish, is the use of the third-person direct object clitic with its nominal referent present in the utterance (*lo visité* *a mi papá* 'I visited my father'). The third-person possessive marker accompanying the noun is usually present with its genitive phrase (3a). It can also be found, although less frequently, with the first-person possessive marker (3b).

(3) a. *su hermano de mi prima*
Lit. 'her/his brother of my female cousin'
b. *mi chacra de mí*
'my ranch' Lit. 'my ranch of me'

A non-subject pronoun is commonly found with intransitive verbs of movement and verbs of change in all grammatical persons (4) (example from Muntendam 2006).

(4) *Anoche mismo <u>me</u> estuve soñando mi vestido blanco.*
'Only last night, I was dreaming (<u>myself</u>) of my white dress.'

Other uses include expressions with discursive function; among them, the frequently found diminutive *-ito* in nouns and adjectives, but also in gerunds (*corriendito*, 'running' said of a child), numerals (*unito* 'only one'), adverbs (*lejitos* 'pretty far away'), and pronouns (*ellita* 'she, the little one'). The diminutive is frequently used to express modesty and deferential courtesy, suggesting parallels with some Quechua suffixes (limitative suffix *-lla*, affective suffix *-yku*, and in combinations with the inchoative *-ri* and progressive *–chka*) (Escobar 2001). Courtesy is also expressed in Andean Spanish in the frequent use of titles with first names (*Doctor Gustavo* 'Doctor Gustavo,' *Señor Pedro* 'Mr. Pedro,' *Licenciado Juan* 'Licensed Juan').

Spatial deictic adverbials tend to appear with the redundant preposition *en* 'in' (e.g., *en ahí* 'in there,' *en acá* 'in here'), suggesting a calque from Quechua locative *-pi*. Another spatial expression, a demonstrative, is commonly found in combination with a possessive preceding a noun (*esos mis hijos* 'those my children').

Innovative semantic and pragmatic functions (less overt features) are found mainly in verbal and adverbial expressions, and are derived from grammaticalization processes triggered by the contact situation. Innovative functions include the use of *estar* + GERUND with present function (5) and with telic verbs (6).

(5) *¿En qué parte de Estados Unidos <u>está viviendo</u> Susana?*
'Where in the United States is Susana living?'

(6) *El sábado <u>estoy saliendo</u> de vacaciones.*
'On Saturday I am leaving on vacation.'

A better-known case is the contrast of the present perfect and the pluperfect in the Andean region. The pluperfect functions as a reportative when it appears with a finite or gerundive expression of the verb *decir* 'to say' within the topic unit (*dice* 'he says,' *que dice* 'that he says,' *diciendo* 'saying'). In these varieties, the present perfect contrasts with this verbal form in the narrative by expressing the evidential function of 'experienced' information (7), which resembles evidential distinctions found in Quechua.

(7) *Juan había vivido en Lima*
'(I was told that) Juan lived in Lima.' (Lit. 'Juan had lived in Lima.')

The contrast of the pluperfect (for reported information) and the present perfect (to mark personal experience) has been documented for Bolivia and Peru. For Ecuadorean Andean Spanish, however, the present perfect has a reportative function instead.

Some adverbial expressions can also express evidential function, as is the case with *pues* 'well' to mark experienced information (Zavala 2001); and can appear with *siempre* 'always,' *sí* 'yes,' and *así* 'this way' to reinforce its meaning (example taken from Manley 2007: 204–206) (8).

(8) *Porque para el trabajo, hay veces con la gente que no sabe hablar quechua, no puedes conversar así sí pues.*
'Because for work, there are times with the people who don't know how to speak Quechua, you can't converse this way yes well.'

Constituent order in Spanish is somewhat flexible, but Andean Spanish is described as favoring adverbials (9) and objects (10) in preverbal position, with adverbials and prepositional phrases tending to appear at the beginning of utterances. These patterns – characterized as marking focus – are attributed to Quechua's OV word order.

(9) *De las 5 de la mañana hasta las 11 ha hecho trabajar.*
'from 5 in the morning to 11, he has made [us] work.'

(10) *tu chiquito oveja véndeme* (Luján et al. 1986)
'your little sheep sell [it] to me'

Social characteristics of the contact situation in the region are contributing to the diffusion of contact features throughout the Andean countries (cf. Escobar 2011).

3.2 *Contact with Mayan languages*[4]

Mayan languages are spoken mainly in southern Mexico and Guatemala. Contact with Mayan languages in Mexico occurs mainly in the states of Yucatan, Quintana Roo, Campeche, and Chiapas. According to the Mexican census, the states with the largest percentages of Amerindian populations are: Yucatan (59%), Oaxaca (48%), Quintana Roo (39%), Chiapas (28%), Campeche (27%), and Hidalgo (24%). Of the 22 languages spoken in Guatemala, 20 are Mayan, and 41% of its 14 million inhabitants are of indigenous origin.

Language contact in the Yucatan Peninsula has been studied in more detail than in other Mayan regions (Suárez 1945; Lope Blanch 1987, 1993). Atypically for Amerindian languages, Yucatec Maya has some prestige within the peninsula. The contact features presented here refer mainly to Yucatec Spanish.

The only grammatical borrowing mentioned in the literature is the case of the diminutive marker *chan* before the noun (e.g., *chan niño* 'small child-MASC'), although not found as frequently as the Spanish suffix *-ito* (Suárez 1945).

Mayan words allow a wider variety of word-final consonants than Spanish, and these consonants are preserved in lexical borrowings (Lope Blanch 1987). Some examples are found in (11).

(11)	/tʃ/	*pech*	'tick'
	/ʃ/	*xux*	'beehive'
	/p/	*tup*	'younger son'
	/t/	*xet*	'leporine lip'
	/k/	*xik*	'armpit'
	/m/	*makkum*	'stew'

A characteristic intonation for Yucatec Spanish has also been noted (Lope Blanch 1987: 39). The shifting of stress to word-final position in nouns used as vocatives (e.g., *hijo* > *hijó*, *niña* > *niñá*) (Lope Blanch 1987: 94), and vowel lengthening have likewise been described. All three features are said to be influenced by Mayan, but no empirical studies have been done.

A well-known phonological feature of Yucatec Spanish is the use of bilabial [m] for/n/in word-final position (*Yucata*[m] 'Yucatan', *jamó*[m] 'ham'); attributed to the fact that in Yucatec Mayan, nasals in coda position are pronounced as bilabials (Alvar 1969: 60). A recent sociolinguistic study suggests that Yucatec Spanish speakers are using this feature more frequently to express regional identity (Michenowicz 2007). A similar social function in Yucatec Spanish (Michonowicz 2009) has been attributed to the maintenance of voiced stops as stops in intervocalic position (e.g., *ca*[b]*allo* 'horse').

The tendency to change the word-initial fricative/f/to the stop/p/in Yucatec Spanish is attributed to the lack of/f/in the Mayan consonant system (e.g., *feliz* > [pelís]) (Suárez 1945), but more studies are needed. Further study is also needed for the described tendency to maintain/s/in word-final position, which competes, especially in coastal areas, with the Caribbean variant (i.e., weakening of the consonant).

The non-resyllabification of word-final/s/and a following word-initial vowel has been described as characteristic of Yucatec Spanish (*ibas allá* [í-bas-a-yá] 'you went there'; example from Alvar 1969: 40). Lope Blanch considers this phenomenon to be related to glottalization between words (1987: 106, 123), whereby a glottal stop is inserted between a word-final vowel and a word-initial consonant (*tenía quince* 'I was fifteen'), between a word-final vowel and a word-initial stressed vowel (*habla el* *hombre* 'the man speaks'), or between a word-final/s/and a word-initial stressed vowel (*los hombres* 'the men').

The morphosyntactic features most characteristic of Yucatec Spanish include an extensive use of the diminutive – not only with nouns and adjectives, but also with gerunds, adverbs, and pronouns – similar to what occurs in the Andean region.

Other features include the frequent use of a reduplicated superlative suffix *-ísimo* (*riquisisísimo* 'very very tasty'), and the common use of a redundant possessive marker before the possessed noun accompanied by the genitive construction (*su tapa de la olla* lit. 'his lid of the pot').

The third-person direct object clitic is also found redundantly with the post-nominal direct object expression without markers for gender and number (12), and the indefinite article is found before the possessive marker accompanying a noun (*un mi sombrero* lit. 'a my hat').

(12) *Lo llamé a Juan.* Lit. 'I called him John.'
Lo metió el libro en el cajón. Lit. 'He placed it the book in the drawer.'

The Maya monolingual population is the highest in Guatemala, at 43.6%; while in Mexico, it is only 9.8% (Lopez 2010: 4).

3.3 Contact with Guarani[5]

Language contact between Spanish and Guarani represents a special type of contact situation. While Spanish is the official language of the country and Guarani is the *national language*, bilingual speakers make up around 89% of the national population. Paraguayan Spanish presents very particular features, and makes use of many Guarani lexical and grammatical borrowings. Bilinguals use a hybrid variety known as *Jopara* (cf. Zajícová 2009), which is not a stable variety, but rather varies in the extent to which it incorporates Guaraní-influenced features. The presence of contact features is dependent on where the interaction takes place (urban or more rural), and on the sociodemographic characteristics of the speakers (whether young urban or rural migrants in the city) (Zajícová 2009). Jopara is mainly an oral variety, although it can also be found in written form.

Guarani, an agglutinative Tupi language, has SVO order when the subject is specified, and SOV order when it is not. Guarani grammatical expressions are widely found in Paraguayan Spanish and Jopara. Most grammatical expressions are free morphemes in Guarani. These include evidential markers such as the reportative *ndaje* 'it is said' (13) and the affirmative marker *voi* 'well' (13).

(13) *La niña ndaje no comía más casi dos días voi y por eso estaba un poco desnutrida, pero el tua ánga* ('stepfather') *igual le pegaba.*
'It is said that the girl would not eat for more than two days. Well, for that reason she was somewhat malnourished, but the stepfather would still hit her.'

Other evidential markers include those for doubt (*nungá*), and for inference (*ko/nikó/nió, katú*), as well as the epistemic marker for counterfactual information (*gua'ú*). Nonetheless, not all Guarani discourse markers are borrowed. Among those evidential markers that are not borrowed are the suffixes for reportative (*-je*), and for inference (*-po/-nipo/-pipo*). However, other grammatical suffixes can be borrowed. These include the weak imperative *na* (14), the strong imperative *ke/oke* (15), the vocative *nde* (16), the interrogative *pa* (17), and the intensifier *nió* (18) (examples from Granda 1982).

(14) *Abrocháme <u>na</u> acá el vestido.*
'Button up IMP here the dress.'

(15) *Andáte ahora mismo <u>oke</u>.*
'Go right now IMP.'

(16) *¿De dónde salí, <u>nde</u> tipo?*
'Where does he come from, VOC the guy?'

(17) *¿Entendíte, <u>pa</u>?*
'Did you understand INTERROG?'

(18) *Tenemo <u>nió</u> teléfono ahora.*
'We have INTENS a telephone now'

Borrowed bound expressions include the emphatic/augmentative *-ite* (*plata en efectivo<u>ite</u>* 'money in real bills-EMP'), the diminutive with pragmatic function *-í* (*virum<u>í</u>* 'real money-DIM (lit. little money)'), and the nominal plural marker *-kuera* (*sus amigo<u>kuéra</u>*'his friend-PL'), among others.

Phonological features found in Spanish and attributed to contact with Guarani include a tendency to insert a glottal stop to separate hiatus (*alcohol* > [alkoʔól] 'alcohol', *caí* > [kaʔí] 'I fell') (Granda 1982: 152), which resembles a similar phenomenon in Yucatec Spanish.

Different from other varieties of Spanish, a simple rhotic appears in word-initial position, while/b/is pronounced as a [v] (e.g., *burro* > [v]*urro, caballo* > *ca*[v]*allo*). The phoneme corresponding to orthographic/*y*/is produced as a prepalatal affricate or fricative in any position (e.g., *mayo* [máʤo] or [máʒo] 'May', *yo* [ʤo] or [ʒo] 'I').

At the morphosyntactic level, the indefinite article appears with the possessive marker before the noun (*un mi amigo* lit. '<u>a my</u> friend'), and the possessive marker can appear accompanying a noun when the genitive prepositional phrase is also present (*su casa de Juan* lit. 'his house of John'), as seen in other contact varieties.

Some innovative adverbial functions are also found. Frequent constructions are *todo ya*, literally 'all already,' used for emphasis (19), and cases of double negative (20).

(19) *Ya trabajé todo ya.* Lit. 'I already worked all already.'
Tu hijo creció todo ya. Lit. 'Your son grew all already.'

(20) *Nada no te dije.* Lit. 'Nothing I did not say.'
Nadie no vino. Lit. 'Nobody did not come.'

Both patterns are regarded as calques on Guarani constructions.

3.4 *Contact with Nahuatl*[6]

Spanish in contact with Nahuatl, a southern Uto-Aztecan language, is spoken mainly in communities found in central Mexico in the states of San Luis de Potosí, Hidalgo, Puebla, Veracruz, Tlaxcala, Guerrero, Morelos, and in the Federal District. Nahuatl is an agglutinative language, where derivation and compounding are productive, and where word order is fairly free.

The only grammatical borrowing mentioned in the literature is the suffix *-eco* for origin (*yucateco* 'from the Yucatan' < Nah. *-écatl*) (Lope Blanch 1991: 169). Nonetheless, some syllable patterns found in Nahuatl borrowings are attributed to Nahuatl syllable structure. The sequence/tl/(also found in Greek borrowings in Spanish) is pronounced in the same syllable (*a-tlas* 'atlas,' *a-tle-ta* 'athlete,' *ix-tle* 'type of plant') (Lope Blanch 1991: 101), unlike other Spanish varieties. The palatal affricate/ʃ/is also found in borrowings from Nahuatl (*mixiote* > *mi*[ʃ]*iote* 'type of meat dish') (Lope Blanch 1991: 97).

Regarding morphosyntactic features, the diminutive is frequently found in nouns, adjectives, and adverbs (e.g., *apenitas* 'just a little'). The nominal plural marker tends to appear in agreement with the possessor and not with the possessed in genitive constructions, following the Nahuatl pattern (21) (Flores Farfán 1992: 64).

(21) *Fueron a sus casa.* 'They went to their house. (each to their own house)'

Similar to other varieties of Spanish in contact, the possessive marker appears before the possessed noun when the genitive prepositional phrase is present (22) (Flores Farfán 1992; Hess 2000).

(22) *Ayer se quemó su casa de María.* Lit. 'Her house of María burned yesterday.'

Hess (2000: 115) finds that a possessive (instead of a definite article) appears in nouns that are marked for prototypical possession by their possessors (23).

(23) *El doctor le curó su pierna.* 'The doctor took care of his (broken) leg.'

With nominal direct objects, *lo* is used preceding the verb, without gender or number marking (24). Flores Farfán explains that it functions as a marker of the transitivity of the verb, which in Nahuatl is obligatory (1992: 58–59)

(24) *¿no lo vieron mi llave?* 'Haven't you seen (it-FEM.SG) my key?'

The use of a verbal periphrasis with *saber* 'to know' to express non-past habitual is also reported as due to contact (25) (Flores Farfán 1992: 51). Analytic verbal expressions have also been reported for other contact varieties.

(25) *Saben vender cerveza toda la noche.* 'They sell beer the whole night.'

In the literature, the Nahuatl language is also known as Aztec or Mexicana.

3.5 Contact with Mapudungun[7]

Contact between Spanish and Mapudungun began in the sixteenth century. The Mapuche people maintained a stronghold but were pushed south by the Spaniards, and are now found mainly in the south-central part of Chile and the southwestern part of Argentina. Although Mapudungun is not spoken by many in Argentina, a stable ethnolect is the norm in the ethnic community (Acuña and Menegotto 1997). On the Chilean side, where the community is more numerous, Mapudungun is still spoken by a small monolingual community and a large bilingual community.

Guntermann *et al.* (2009) find that 62% of the population 10 years and older does not have competence in Mapudungun, although 11% have high competency in the language. It is believed that the high urbanization of the Mapuche population (63%) in Santiago, Temuco, Concepción, and Osorno is contributing to the low vitality of the language and has fostered quicker integration into the Spanish-speaking society and culture (Wittig 2009). Nonetheless, many individuals are making efforts to relearn the language.

Mapudungun, an Araucan-Chon language, is an agglutinative SVO language. Transitive verbs have a complex system of relational markers, similar to other Amerindian languages (Salas 2010). As the largest indigenous community in Chile, Mapudungun was recognized at the end of the nineteenth century as influential in what is known as the Chilean intonation (Lenz 1893; Alonso 1953: 289). However, this feature has yet to be analyzed empirically.

Other features mentioned in Lenz' work include the alveolar pronunciation of/t/and/d/preceeding/r/(e.g., *entre* 'between,' *madre* 'mother'), which is characteristic of Chilean Spanish, and the assibilated pronunciation of/r/in word-initial position (*ropa* > [ř]*opa* 'clothes'), after/l/or/n/(*alrededor* > *al*[ř]*ededor* 'around'), and in combination with/t/and/d/(*trapo* > *t*[ř]*apo* 'cloth'). Nonetheless, Alonso (1953) argued against a Mapudungun influence in the development of these features of Chilean Spanish. No empirical studies, however, have been done in support of either position. Acuña and Mengotto (1997) suggest that there is a need to review the features presented by Lenz, since they are also found in the Argentinean region where an indigenous Mapuche population resides.

Other phonological phenomena which are more widely accepted as characteristic of the language contact situation are the alternation of/b/and/f/(e.g., *bruja* 'witch' > *fruja*, *cebolla* 'onion' > *cefolla*) and the insertion of an epenthetic vowel in /kl/ clusters (a sequence not found in Mapudungun) (e.g., *clavo* 'nail' > [kal]*avo*, *Claudia* > [kal]*audia*).

Some argue that the omission of the /s/ in word-final position is a contact feature, as sibilants cannot appear in this position in Mapudungun (Acuña and Menegotti 1997) (e.g., *má_ o meno_* 'more or less,' *eso_ animalito_ son de Dio_* 'those little animals belong to God'). Number marking, when present, tends to appear in the modifier preceding the noun (e.g., *los brazo_* 'the-PL arms'), including quantifiers which do not require a number marker in other varieties of Spanish (e.g., *cuatros año_* 'four years').

Morphosyntactic features include the omission of the direct object clitic in answering questions (26).

(26) *¿Viste a Antonio? – No, no __ vi.* 'Did you see Antonio? – No, I did not see (him).'

Similar to other contact varieties, the direct object clitic *lo* appears with a post-verbal direct object nominal (27), and does not have gender or number marking.

(27) *Lo echan la manzana* (*las manzanas*). 'Lit.: They throw it the apples.'

The use of *loismo* with direct-object function is also found in this contact variety (28).

(28) *El nene agarra la foto y lo hace pedazo.* 'The young boy takes the photo (FEM) and breaks it (MASC) into pieces.'

The *lo* is not only used for third person, but in the Argentinean variety is also used instead of *nos,* the first person plural pronoun for direct object (Acuña and Menegotto 1997).

4 Spanish in contact with Amerindian languages

Similarities and differences found in Spanish varieties in contact with Amerindian languages are due to typological characteristics of Spanish and the Amerindian language, to cross-linguistic tendencies in language change, as well as to socio-historical characteristics of the contact situation. Several linguistic phenomena seem to be *repeated* in the various language contact situations. These include the type of borrowings, especially of grammatical markers, and phonological and morphosyntactic characteristics (see Table 4.3). The empty cells in Table 4.3 indicate a lack of available data for that feature.

Feature similarity has been used in the literature as an argument to suggest that these might not be contact-induced features. Another argument has been

Table 4.3 Linguistic characteristics in varieties of Spanish in contact with Amerindian languages.

	Quechua	*Maya*	*Guarani*	*Nahuatl*	*Mapudungun*
Borrowings					
Lexical	√	√	√	√	√
Grammatical	(√)	(√)	√	(√)	
Phonological					
stops > stops/V_V	√	√			
/s/in word final	√	√			
Morphosyntactic					
Diminutive	√	√	√	√	
Evidentials	√		√		
Possessive	√	√	√	√	
Loísmo	√ (Ecua. *leísmo*)	√	(*leísmo*)	√	√
Double DO	√	√	?	√	√
Double Det.	√	√	√		

(√) = few.

the occurrence of these features in monolingual varieties of Spanish. My objective here is to stress that we cannot disregard these features as potential contact-induced phenomena without further detailed linguistic analyses.

The diminutive is described as characteristic of Latin American Spanish. In Spanish in contact with Quechua and Maya, *-ito* is the common expression. In Guarani, however, it is frequently found with the Guarani suffix *-í*. It has been argued that the phonological similarity, and the fact that Guarani–Spanish contact favors grammatical borrowing, might explain this use. Nonetheless, Suárez (1945) has suggested that for Yucatec Spanish, the Mayan expression *chan* with diminutive function is also used sometimes before nouns, with equivalent functions to those in Mayan. In the Andean region, the Quechua diminutive suffix *-cha* has also been reported in some areas. In analyzing the functions that the diminutive has in these contact varieties of Spanish, it is clear that in addition to size, the diminutive is used to express courtesy and modesty, functions found in the Amerindian languages as well, and consistent with universal tendencies of diminutive expressions.

These pragmatic functions suggest that the higher frequency of use of the diminutive in these Spanish varieties is a way of serving the discursive functions necessary in communication in such communities, especially in oral discourse. In addition, discourse markers in Guarani, Quechua, and Mayan are highly complex and relevant in face-to-face communication. Discourse constraints seem to also influence and account for other discourse-related functions, such as evidential meanings, and possession.

Another common feature is the use of *lo* without number and gender markers. Palacios (2005) compared its use in data from Spanish in contact with several languages: with Maya in Guatemala, with Quechua in Ecuador and Peru, and with Guarani in Paraguay. She found that while the contact varieties in Peru and Guatemala use *lo* without distinguishing number, gender, or animacy, contact varieties in Ecuador and Paraguay use *le* with direct object function without distinguishing gender, animacy, or case, distinguishing only number. Palacios proposes two pronominal systems that would represent different stages of the grammaticalization of this expression, where the use of *le* for direct object in Ecuador and Paraguay would represent a later stage than the use of *lo* in the contact varieties of Peru and Guatemala.

In Spanish in contact with Nahuatl, *lo* appears with transitive verbs accompanied by a nominal direct object. Flores Farfán (1992) explains that in these contexts *lo* expresses the transitivity of the verb, calquing a pattern from Nahuatl. In Spanish in contact with Quechua in Peru, *lo* appears not only with transitive verbs, but also with movement intransitive verbs. Some have suggested that it marks direction, calquing a pattern from Quechua. In a study on Spanish in contact with Mapudungun in Argentina, Acuña and Menegotto (1997) also describe a *loismo* phenomenon in this variety. Similar to the Peruvian case, *lo* can appear with both transitive and intransitive verbs. In their data, *lo* can also be used for first person plural in direct- object function (29), as well as with intransitive pronominal verbs (30) (found in other Latin American varieties as well).

(29) *Chau, me voy a la escuela, mañana <u>los</u> (nos) veremo(s).* 'Bye, I'm going to school, tomorrow we will see <u>each other</u>.'

(30) *Nosotros <u>lo(s)</u> (nos) vinimo(s) en junio.* Lit. 'We came <u>ourselves</u> in June.'

In the indirect object pronoun paradigm of this contact variety, *le* is extended from third person to first person plural as well.

From these examples from five contact varieties, it is clear that the use of the pronominal forms *lo* and *le* requires further study in contact varieties of Spanish. Moreover, criteria such as verb type, presence/absence of object nominal expression, and position of nominal expression, in addition to the criteria mentioned above, are some of the factors that need to be considered in the analysis. In these contact varieties, *loísmo* and *leísmo* must be studied alongside structures with a redundant direct object pronoun accompanying a postverbal nominal direct object.

While more research is needed to address the specific syntactic, semantic, and pragmatic features underlying contact uses, it is clear that the analysis of Spanish in contact with Amerindian languages is still in its infancy. Other features that require more study in order to uncover the true similarities and differences of use in the different contact varieties are the use of the possessive marker with the genitive construction and the use of two determiners before the noun.

5 Sociolinguistic characteristics

Historically, Mexico, Guatemala, Bolivia, Peru, and Ecuador are considered countries with important indigenous populations (see Table 4.4), although the national percentages vary.

Social and political changes in the region since the end of the twentieth century have led national governments to be more supportive of Amerindian languages and populations, although the type of support varies. In the last decades, bilingual intercultural educational programs have been created (in Bolivia, Ecuador, Mexico, Guatemala, and Colombia) in response to the demands from the indigenous peoples themselves. However, these initiatives are more of a political response than an academic or social justice concern (López 2009). Receiving literacy education in the native language is a human right (Article 14, United Nations, 2007), but it is also an important means to empower indigenous communities (López 2009). However, high levels of illiteracy remain a social reality in Latin America (Table 4.4).

At the onset of the twenty-first century, Mexico had an illiteracy rate in the indigenous population of 31% (mainly female); which is five times the national percentage (6%). Its indigenous population constitutes 7% of the national population. The urbanization rate of this population is high at 36%, although not as high as in Bolivia (53%), or Peru (44%) (see Table 4.4). Migration occurs mainly from the

Table 4.4 Sociolinguistic comparison of countries with high indigenous populations in Latin America.

	Mexico	*Guatemala*	*Bolivia*	*Peru*	*Ecuador*
National population[8] (in millions)	103 [2005]	14 [2010]	8.2 [2001]	28 [2007]	14 [2010]
Indigenous population[9]	7% [2005]	41% [2002]	62% [2001]	32% [2001]	7% [2001]
Indigenous population under 15[10]	39%	46%	39%	n.a.	41%
National urbanization[11]	77%	49%	62%	76%	66%
Urban indigenous population[12]	36%	32%	53%	44%	18%
Indigenous population poverty[13]	3.3/9	2.8/9	2.2/9	1.8/9	2.1/9
National illiteracy[14]	6%	31%	11%	11%	9%
Indigenous population illiteracy	31%	48%	13%	n.a.	28%

southern states, which are the poorest and have larger indigenous populations, to central and northern states, and especially to Mexico City.

National illiteracy in Guatemala is 31%, but 48% in the indigenous population and 75% among rural indigenous women. The illiteracy percentages also vary according to indigenous community (e.g., the Chorti population is 50% illiterate; the Kaqchikel population 20% illiterate). According to the 2002 census, 41% of the Guatemalan population is indigenous, with 45% monolingual in the Amerindian language. Few bilingual and bicultural programs exist today in Guatemala.

In the Andes, Bolivia's indigenous population was around 62% in 2001 (15 years and older), and 13% of this population was illiterate. Peru has a national illiteracy of 11%, but it rises to 34% for rural women, and is higher in regions where speakers of an Amerindian language are in the majority (e.g., Ancash, Cusco, Huancavelica, Apurímac, and Ayacucho). Ecuador's indigenous population represents 7% of the national population (2001 Census), but is 33% according to the official Confederation of Indigenous Nationalities of Ecuador (CONAIE). The national illiteracy rate is 9%, but 28% in the indigenous population, with higher percentages (up to 40%) among rural indigenous women.

In an effort to approach the *illiteracy problem* in Latin America, some countries are trying to replace Hispanization programs in the indigenous populations with intercultural education programs (López 2009: 434). Literacy, however, is helpful to indigenous communities, only if the education programs are also integrated into their social and political life (López 2009: 438–439).

6 Final remarks

The end of the twentieth century witnessed intense population movements in Latin America, especially from rural to urban areas. The earlier strict correlation between linguistic varieties and regions is no longer applicable today. Migration has led to increased urbanization, changing the profile of these countries from mainly rural at the turn of the twentieth century to highly urban at the turn of the twenty-first century (see Table 4.4).

Urbanization has led to the emergence of newly-formed urban social networks, bringing languages and cultures of the rural areas to the cities. While the Amerindian languages continue a diglossic relationship with Spanish in the cities at the macro-level, new contact phenomena are emerging within these urban indigenous communities at the social-network level (cf. Altamirano and Hirabayashi 1999). Different degrees of intensity of dialect and language contact are emerging in Latin American cities, changing the sociolinguistic profile of the countries in the process (Escobar and Wölck 2009). Studies report increases in lexical and grammatical borrowings from Spanish into Amerindian languages and vice versa, as well as a rise in code-switching phenomena and continua of styles, from varieties closer to the Amerindian language to those closer to Spanish (cf. Zavala and Bariola 2007; Zajícova 2009, Shappeck 2010). These new contact situations are cases of *covert* *language contact* (Escobar 2007), whose study can help us distinguish more clearly semantic and pragmatic innovations, which are less transparent than phonological and lexical.

Who is *indigenous* is defined internationally by self-identification. However, this varies according to the indigenous community the individual belongs to and to the status of the Amerindian languages within particular national borders. Only if social and literacy programs in Latin American countries take into consideration the language and culture of the indigenous population will they help recognize and fortify the indigenous language and culture, as well as the citizenship of its population and its contribution to the national society. Considering that the Latin American indigenous population is younger than the nonindigenous population (ECLEC 2006), national intercultural education programs can help maintain the Amerindian languages in this young population, as well as help raise a new indigenous citizen. These new situations suggest further increment of bilingual populations and the formation of new Spanish contact varieties and new urban varieties of the Amerindian languages. The multiethnic, multilingual and multicultural characteristics of the social history of Latin America is reflected in the diversity of its Spanish varieties, which merit more study, paying special attention to the diversity of its speakers.

NOTES

1 Brazil has 193 languages.
2 Cuba, Dominican Republic, Puerto Rico, and Uruguay have no Amerindian languages.
3 Information based on Toscano Mateus (1953); Pozzi-Escot (1975); Mendoza and Minaya (1975); A. Escobar (1978); Luján et al. (1984); Mendoza (1991); Klee (1996); Haboud (1998); A. M. Escobar (2000); de los Heros (2001); Granda (2001); Cerrón (2003); Sánchez (2003); Klee and Caravedo (2006).
4 Information based on Suárez (1945); Lope Blanch (1987, 1991, 1993); Martín (1985).
5 Information based on Granda (1980, 1982, 1988, 1991, 1999); Quant and Irigoyen (1980); Dietrich (1995); Zajicová (2009).
6 Information based on Lope Blanch (1987, 1991, 1993); Farfán (1992).
7 Information based on Oroz (1966); Acuña and Menegotto (1997); Fernández (2010); Salas (2010).
8 Information based on country's official census data.
9 Information based on ECLAC (2006).
10 Information based on ECLAC (2006: 179), with census data from 1999 to 2001.
11 Information based on UNICEF for 2008 (http://www.unicef.org), except for Bolivia and Peru, which were taken from their national census data (2001 and 2007, respectively).
12 Information based on ECLAC (2006) for all countries, except Peru, which is taken from López (2010: 4).
13 13 Information based on ECLAC (2006: 150), with census data from 1999 to 2001. The poverty scale goes from 1 to 9 (highest poverty level).

14 Illiteracy rates are based on studies for the respective countries published in López and Hanemann (2009): Mexico (Schmelkes et al. 2009); Guatemala (Verdugo and Raymundo 2009); Bolivia (Carrarini et al. 2009); Peru (Zúñiga 2009); and Ecuador (Yánez 2009).

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5 The Phonemes of Spanish

REBEKA CAMPOS-ASTORKIZA

1 Introduction

This chapter describes the phonemes of Spanish, focusing on their distribution. Special attention is paid to some of the major dialectal variants found across the Spanish-speaking world. The Spanish vocoids (i.e., vowels and glides) are discussed first in Section 2, followed by consonantal phonemes in Section 3, which are grouped together according to their manner of articulation. Section 4 discusses the status of some quasi-phonemic contrasts in Spanish, and section 5 concludes with some suggestions for future research based on recent findings. The approach taken in this chapter is to present traditional descriptions of the Spanish sound inventory together with some of the most recent developments in the field of Spanish laboratory phonology, which shed light on long-standing issues regarding the Spanish phonemes and their realization.

2 Vocoids

2.1 Vowels

Vowels are produced with a relatively open vocal tract so that the air flows unimpeded since no obstacle is present. This is the main articulatory difference between vowels and consonants. Also, vowels are characterized by default vocal fold vibration. Different vowel qualities result from differences in vocal tract configuration during speech production, namely from different shapes and position of the tongue and lips. These give rise to the articulatory terms used in the classification of vowels: tongue height, tongue backness, and lip rounding. It should be noted that these articulatory descriptions have acoustic correlates, which are described as differences in vowels' resonance frequencies, dependent upon vocal tract shape and size (for an acoustic description of the Spanish vowels, see Quilis 1993). Spanish has a five vowel inventory (Table 5.1) and makes contrastive use only of the height and backness features, given that rounding is predictable from backness.

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Table 5.1 Spanish vowel inventory.

	Front	*Central*	*Back*
High	i		u
Mid	e		o
Low		a	
	unrounded		rounded

Navarro Tomás (1977) claims that the Spanish mid vowels/e, o/have open variants (i.e., lower variants), when they are in contact with [r], before [x], in a syllable closed by any consonant, except/e/, which does not open if the syllable is closed by [m, n, s, d, θ], and in rising diphthongs with a palatal glide. However, instrumental studies based on acoustic data have failed to provide evidence supporting this claim and conclude that there are no systematic distinctions in degree of openness for the mid vowels (e.g., Monroy 1980; Martínez Celdrán 1984a: 288–301; Morrison 2004). An interesting study of the Spanish mid-vowel allophony is presented in Martínez Celdrán and Fernández Planas (2007), where the authors analyze acoustic and articulatory data to test the hypothesis that mid vowels are open or close depending on the phonetic context. In line with previous findings, results from their acoustic analysis indicate that there is no systematic openness. On the other hand, the articulatory data show that there is a significant difference in openness according to the contexts mentioned by Navarro Tomás. Based on these results, the authors conclude that the allophonic distinction between open and close mid vowels exists in Spanish, although it is not manifested in the acoustic analysis, possibly due to the nonlinear relationship between articulation and acoustics (Martínez Celdrán and Fernández Planas 2007: 188).

Nasalization is not contrastive among Spanish vowels, although some descriptions note that vowels can be partially nasalized in contact with nasal consonants (Navarro Tomás 1977: 39; Hualde 2005: 123; Piñeros 2006: 161). More precisely, according to Navarro Tomás (1977: 39), vowels between two nasal consonants and vowels in word-initial position followed by a nasal consonant are subject to the highest degree of nasalization. Interestingly, in some Caribbean and Andalusian dialects, we may find word-final nasal deletion with nasalization of the preceding vowel (e.g., *pan* [ˈpãŋ] ~ [ˈpã] 'bread,' *tapón* [taˈpõŋ] ~ [taˈpõ] 'cork') (Terrell 1975; Vaquero 1996; Hualde 2005: 123). In these cases, allophonic vowel nasalization could be seen as developing a contrastive role due to loss of the conditioning environment (i.e., the nasal consonant).

Although, the Spanish vocalic system has been described as being fairly stable across dialects (Quilis and Esgueva 1983; Morrison and Escudero 2007), there are situations where the Spanish vowels display some degree of variation in their realization. Vowel variation has been found in situations of language contact. For instance, Guion (2003) found a reduced vowel system for bilingual Spanish–Quechua speakers in Ecuador, and Willis (2005) encountered vowel variation in Southwest US Spanish compared with Mexican Spanish. O'Rourke (2010) found vowel differences between Lima and Cuzco speakers, the latter being in a context of more intense contact with Quechua. Furthermore, the author found that Cuzco speakers show different vowel patterns depending on their knowledge of Quechua, so that native Spanish–Quechua bilingual speakers displayed a smaller vowel space than Spanish monolinguals. In noncontact situations, unstressed mid vowels seem to be most prone to variation. Raising of mid vowels in word-final position has been documented for western Puerto Rican Spanish (e.g., Navarro Tomás 1974 [1948]; Holmquist 1998, 2001; Oliver 2007; see Chapter 1 for mid-vowel raising in Peninsular varieties). Willis' (2008) study of the Dominican Spanish vocalic system shows that mid vowels present much variation and tend to overlap with neighboring high vowels.

A process affecting mid vowels that has received much attention in the literature is unstressed vowel reduction (i.e., shortening, devoicing and perceptual deletion of vowels in unstressed positions). This process is characteristic of Mexican and Andean Spanish, and it has been impressionistically documented in studies such as Lope Blanch (1963), Canellada and Zamora (1960), and Perissinotto (1975) for Mexican Spanish; Hundley (1983) and Lipski (1990) for Andean Spanish. These studies agree that the process is gradient and variable, that it targets mainly mid vowels, especially /e/, that it occurs most frequently in vowels preceding a word-final coda /s/, and that it is favored by faster speech rates. In an effort to acoustically define the process, Delforge (2008) presents quantitative, acoustic data regarding vowel reduction in conversational speech from Cuzco, Peru. This author finds that the most common realization of reduced vowels is completely devoiced and the least common is apparent elision. Delforge notes that reduction does not result in vowel centralization but rather in vowel devoicing, suggesting that reduction does not have any major effect on vowel quality. The data show that while word-medially /e/ and the high vowels have the highest rates of devoicing, in word-final syllables closed by /s/ and open syllables before a pause, all vowels show high rates. In addition, devoicing is highly favored by a contiguous /s/ but reduced vowels also occur preceding or following other voiceless consonants (including assibilated /ř/). Furthermore, word position affects the process so that devoicing is more common word-finally than in any other position. It should be noted that in Delforge's data, vowel devoicing does not seem to be dependent upon speech rate.

(1) Examples of unstressed vowel reduction in Andean Spanish (Delforge 2008)

Cusqueña	[kusˈkeɲa]	'a beer brand'
viajes	[ˈbjaxe̥s]	'trips'
casi todo	[ˈkasi̥ˈtoðo]	'almost all'
alpakas	[alˈpakḁs]	'alpacas'
estos	[ˈesto̥s]	'these'

Based on her results, Delforge (2008) analyzes unstressed vowel reduction in Andean Spanish within the framework of Articulatory Phonology (Browman and Goldstein 1989) and argues that the process stems from different degrees of gestural overlap between consonants and vowels in Andean Spanish. On the other hand, Lipski (1990), working within an autosegmental framework, represents unstressed vowel deletion as the result of loss of [-consonantal] from the vowel featural representation due to phonetic shortening and devoicing. Both Delforge (2008) and

Lipski (1990) argue that /e/ is the vowel that most frequently devoices because of its status as [+coronal], a feature that it shares with /s/, the most common trigger of devoicing. They argue that this articulatory similarity between /e/ and /s/ favors the interaction between these two sounds, resulting in higher devoicing rates.

2.2 Glides

Spanish has a front or palatal high glide and a back or labiovelar high glide. Glides occur in tautosyllabic combinations of vocoids and differ from vowels in that they are nonsyllabic. They may occur in rising diphthongs (i.e., a glide [j, w] followed by a nuclear vowel as in *miel* [ˈmjel] 'honey' and *cuatro* [ˈkwa.tɾo] 'four') or in falling diphthongs (i.e., a nuclear vowel followed by a glide [i̯, u̯], as in *veinte* [ˈbei̯n.te] 'twenty' and *caucho* [ˈkau̯.tʃo] 'rubber'). In the former case, glides are referred to as semiconsonants and in the latter, as semivowels. Note that the phonetic symbols used to represent Spanish glides tend to reflect this dichotomy, although not all studies follow this symbol usage (see e.g., Hualde 2005). Occasionally, mid glides [e̯, o̯] may occur in colloquial speech as a result of reducing a hiatus to a diphthong, for example *cohete* [ˈko̯e.te] 'firework' and *línea* [ˈli.ne̯a] 'line' (Navarro Tomás 1977: 160). In some areas, these sequences may be further reduced giving rise to forms such as [ˈkwe.te] and [ˈli.nja]. This process of vowel raising and diphthongization has been reported in Latin American and Peninsular Spanish (see, among others, Moreno de Alba 1994; Jenkins 1999; Hualde and Prieto 2002; Face and Alvord 2004; Alba 2006; Garrido 2007; Hernández 2009). The phonemic status of glides in Spanish is discussed in Section 4.1 below.

3 Consonant phonemes

Consonantal sounds are produced with some degree of constriction in the vocal tract. Three articulatory parameters are used to classify consonants: place of articulation, manner of articulation, and voicing. Table 5.2 includes the Spanish consonant inventory organized according to these three parameters. As I discuss below, some of these sounds do not occur in all dialects.

3.1 Obstruents

3.1.1 Stops Oral stop or plosive consonants are characterized by a complete interruption of the airflow. A complete closure of the vocal tract is followed by a release, which may be accompanied by a burst as the airflow rushes out. Spanish includes a series of voiceless stops /p, t, k/ and another of voiced ones /b, d, g/. The difference among their members lies in their place of articulation, bilabial, dental, and velar, respectively. Spanish voiced plosives are usually voiced throughout the closure (unlike English ones that tend to lack full voicing in certain contexts). In some contexts, voicing might be partial during the closure but it

Table 5.2 Spanish consonant inventory.

	bilabial		*labiodental*		*interdental*		*dental*		*alveolar*		*alveopalatal*		*palatal*		*velar*	
	vlss	*vd*	*vlss*	*vd*	*vlss*	*vd*	*vlss*	*vd*	*vlss*	*vd*	*vlss*	*vd*	*vlss*	*vd*	*vlss*	*vd*
Stop	p	b					t	d						ɟ	k	g
Fricative			f		θ				s						x	
Affricate											tʃ					
Nasal		m								ŋ				ɲ		
Lateral										l				ʎ		
Rhotic-tap										ɾ						
Rhotic-trill										r						

vlss = voiceless.
vd = voiced.

always starts before the release. Spanish voiceless plosives lack aspiration; that is, the release of the closure occurs at the same time or slightly before the beginning of voicing for the following vowel (for more on the VOT of Spanish plosives, see Abramson and Lisker 1973; Williams 1977; Casteñada Vicente 1986).

After continuants (i.e., vocoids and continuant consonants), voiced plosives are not produced with a full closure but rather with some approximation of the articulators (see voiced obstruents in Chapter 6 for more details). These continuant allophones, represented as [β, ð, ɣ], are characterized by uninterrupted airflow, with a variable degree of constriction depending on context, style, speech rate, and dialect. Some authors characterize them as phonetically approximants (e.g., Martínez Celdrán 1984b, 1991, 2004).

Voiceless stops are realized as voiced in certain dialects, mainly in Cuban Spanish (Ruiz Hernández and Mirayes 1984; Quilis 1993: 222–224) and Canary Islands Spanish (Trujillo 1980, Oftedal 1985, Marrero 1988). Voicing of voiceless stops has also been documented in several Peninsular varieties (e.g., Torreblanca 1976 for Toledo, Machuca Ayuso 1997 for Barcelona, Lewis 2001 for Bilbao, and Martínez Celdrán 2009 for Murcia). Voicing may occur in word initial or internal position, and it might be subject to stylistic restrictions (cf. Martínez Celdrán 2008). Spectrographic evidence from several studies shows that this weakening process is gradient and that the resulting realizations may range from partial voicing to lenited pronunciations with more or less constriction (Torreblanca 1976; Oftedal 1985; Machuca Ayuso 1997; Lewis 2001; Martínez Celdrán 2009). The examples in (2) illustrate this point. It should be noted that some dialects, namely Colombian (Lewis 2001) and Argentine Spanish (Colantoni and Marinescu 2010) do not participate in this weakening process.

(2) Voicing of voiceless stops (from Quilis 1993)

Orthography	*Standard*	*Voicing dialects*	
campana	[kamˈpana]	[kamˈbana]	‘bell’
pizarra	[piˈsara]	[biˈsara]	‘blackboard’
tacón	[taˈkon]	[taˈɣon]	‘heel’
cuatro	[ˈkwatro]	[ˈkwaðro] ~ [ˈkwað̥ɾo]	‘four’

The voicing contrast among stops is usually neutralized in coda position so that stops in this position are differentiated only by their place of articulation. The realization of plosives in this context may range from a voiceless stop in emphatic pronunciations to a voiced approximant or noncontinuant in a more neutral or conversational style, with other possible intermediate realizations (Navarro Tomás 1977; Hualde 2005: 146). The precise production of these sounds includes different degrees of constriction and voicing depending on style and phonetic environment. For example, stress is reportedly a relevant factor, and neutralization tends to be more likely when the syllable is not stressed (Navarro Tomás 1977: 77). The examples in (3) show the standard orthography and some of the possible productions for words containing a coda stop. It should be noted that the possible pronunciations for the orthographic pairs *p-b*, *t-d* and *k-g* in coda are exactly the same, illustrating the loss of contrast in that context.

(3) Neutralization of voicing among stops in coda position

Orthography	*Variable productions*	
apto	[ˈapto ~ ˈab̥to ~ ˈabto ~ ˈaβto]	‘apt’
absoluto	[apsoˈluto ~ ab̥soˈluto ~ absoˈluto ~ aβsoˈluto]	‘absolute’
atmósfera	[atˈmosfeɾa ~ ad̥ˈmosfeɾa ~ adˈmosfeɾa ~ aðˈmosfeɾa]	‘atmosphere’
admitir	[atmiˈtiɾ ~ ad̥miˈtiɾ ~ admiˈtiɾ ~ aðmiˈtiɾ]	‘admit’
doctor	[dokˈtoɾ ~ dog̊ˈtoɾ ~ dogˈtoɾ ~ doɣˈtoɾ]	‘doctor’
dogma	[ˈdokma ~ ˈdog̥ma ~ ˈdogma ~ ˈdoɣma]	‘dogma’

This voicing neutralization is reflected in the Spanish lexicon, which does not exploit the voicing contrast among plosives in coda position; that is, we do not find lexical items that only differ in the voicing of their coda stops (Quilis 1993: 204). Note that this is relevant only for word-internal codas since word-finally, only/d/ occurs in Spanish words (there are some exceptions that include mainly borrowings, e.g., *déficit* ‘deficit,’ *club* ‘club,’ etc.).

Quilis (1993: 205) proposes three archiphonemes/B, D, G/for stops in coda position. These archiphonemes do not contrast in voicing and have a variable phonetic realization depending on the factors mentioned above. Quilis’ archiphonemic analysis bears some resemblance with an underspecification approach to neutralization, under which coda stops would be underspecified for the voice feature, resulting in a variable realization of voicing.

The neutralization facts are somewhat different in some Castilian dialects. In these varieties,/g/in coda position tends to be pronounced as a voiceless velar fricative [x] (e.g., *ignorante* [ixnoˈɾante] ‘ignorant,’ *pragmático* [pɾaxˈmatiko] ‘pragmatic’). This results in the preservation of the contrast between/k/ and/g/in coda position, for example in items such as *doctor* [doɣˈtoɾ] vs. *dogma* [ˈdoxma] (Hualde 2005: 148). In the same dialectal region, coda/d/is sometimes pronounced as a voiceless (inter)dental fricative [θ] (e.g., *red* [ˈreθ] ‘net,’ *verdad* [beɾˈðaθ] ‘truth’), giving rise to the neutralization between coda/d/and/θ/ (González 2002, 2006).

Vocalization of coda stops also results in neutralization of the voicing contrast. This type of vocalization is found in some dialects, where voiced and voiceless stops in syllable-final position are realized as glides. Quilis (1993: 220) points out that vocalization is more common among labial and velar obstruents, given that the Spanish glides share more properties with them than with dental obstruents. Data from vocalization in Chilean Spanish shows that/p, b/are vocalized into [w],/t, d/into [j], and/k, g/into either [w] or [j], depending on the dialect (see examples in (4)) (Lenz 1940; Oroz 1966 cited in Martínez-Gil 1997). Vocalization can be seen as another weakening process of stops in coda position.

(4) Vocalization of coda stops in Chilean Spanish

Orthography	*Standard*	*Dialectal Chilean*	
apto	/ˈapto/	[ˈawto]	'apt'
objeto	/obˈxeto/	[owˈxeto]	'object'
étnico	/ˈetniko/	[ˈejniko]	'ethnic'
admirar	/admiˈɾaɾ/	[ajmiˈɾaɾ]	'to admire'
doctor	/dokˈtoɾ/	[dojˈtoɾ] ~ [dowˈtoɾ]	'doctor'
dogma	/ˈdogma/	[ˈdojma] ~ [ˈdowma]	'dogma'

An extreme case of coda stop neutralization is found in Caribbean dialects, where coda plosives tend to be subject to neutralization not only in voicing but also in their place of articulation. In these cases, any syllable-final plosive may be realized as a velar consonant or a glottal stop. This can be seen in the colloquial Caribbean pronunciation of *admitir* [aɣmiˈtiɾ], *submarino* [sukmaˈɾino], and *étnico* [ˈeʔniko] 'admit, submarine, ethnic,' which in a more standard pronunciation would be [aðmiˈtiɾ], [suβmaˈɾino], and [ˈeðniko] (Guitart 1976: 23; Zamora and Guitart 1982: 109). Morgan (2010: 197) notes that this type of place neutralization among coda plosives is more extended than previously reported and can be found in dialects other than the Caribbean.

Coda stops are often subject to deletion in Peninsular Spanish, even in the speech of educated speakers, and we can find pronunciations such as *obsesión* [oseˈsjon] vs. [oβseˈsjon] 'obsession,' *taxi* [ˈtasi] vs. [ˈtaɣsi] 'taxi'. Deletion of coda plosives is also present in Latin American Spanish but to a lesser extent, being more restricted to rural areas (Hualde 2005: 147).

3.1.2 Affricates and fricatives Fricative consonants are characterized by a continuant flow of the air stream through a narrowing in the vocal tract. This articulatory configuration results in frication noise, which is the main acoustic characteristic of this group of sounds. Spanish fricatives include/f, θ, s, x/. The phoneme/f/is found only in onset position, and its realization is most frequently as a labiodental fricative (e.g., *farola* [faˈɾola] 'lamppost,' *sofá* [soˈfa] 'sofa'). There are, however, reports that in Caribbean dialects this phoneme might be realized as a voiceless bilabial fricative [ɸ], especially before the diphthong [we] (e.g., *afuera* [a.ˈɸwe.ɾa] 'outside') (Jiménez Sabater 1975; Vaquero 1996). The voiceless alveolar fricative/s/has two main types of articulation depending on the dialect. Castilian Spanish has an apico-alveolar production, and Andalusia, Canary Islands, and Latin American Spanish has a predorso- or lamino-alveolar realization (see Quilis 1993: 248–251 for other less common realizations).

Word- and syllable-final/s/is subject to weakening in many Spanish dialects, resulting in aspiration or loss of this segment (see studies cited below and, among many others, Terrell 1979, 1986; Alba 1982; Lipski 1984, 1985; Amastae 1989; Carvalho 2006; File-Muriel 2009; see also Chapter 6 for more on/s/weakening and its interaction with other segmental phenomena). The process of/s/weakening can be found in Southern Peninsular varieties and in Canary Islands Spanish.

This process is also widespread in Latin America, with the exception of the highlands of central Mexico and Guatemala, central Costa Rica, and the Andean region (Lipski 1994; Hualde 2005: 161). The degree and frequency of /s/ weakening and its phonetic result vary across dialects (Terrell 1977, 1979; Bybee 2000; Torreira 2006), with Caribbean varieties reportedly showing the highest rates of aspiration and deletion (Bybee 2000; Hualde 2005: 161). The process of /s/ weakening is also subject to much variation due to other social and linguistic factors. Style and socioeconomic status have been identified as conditioning the degree of weakening, with higher rates in casual styles and among less educated speakers (Lipski 1985; Alba 2004). Weakening is more frequent in some phonological contexts than in others (Bybee 2000; Hualde 2005: 161–163): preconsonantal position, whether word-internally or across words, is the environment that most favors weakening (e.g., *este* [ˈeh.te] ~ [ˈe.te] 'this,' *las camas* [lah ˈka.mah] ~ [la ˈka.mah] 'the beds'), followed by pre-pausal position (e.g., *vamos* [ˈba.moh] 'let's go'). The environment with the least frequency of /s/ weakening is before a vowel (e.g., *las olas* [lah ˈo.lah] 'the waves'), although there are some dialects, for example in New Mexico Spanish, where pre-vowel /s/ is subject to high rates of aspiration even at the beginning of a word (e.g., *la semana* [la he.ˈma.na] 'the week') (Brown and Torres Cacoullos 2002; Brown 2005).

The interdental fricative /θ/ is found only in some dialects, mainly those spoken in central and northern Spain, which show a phonemic distinction between /s/ and /θ/ (e.g., *casa* [ˈkasa] 'house' vs. *caza* [ˈkaθa] 'hunting'). Most dialects, including those found in Andalusia, Canary Islands, and most of Latin America, lack the interdental fricative phoneme /θ/, which in these dialects corresponds to /s/ (e.g., *casa* [ˈkasa] 'house' vs. *caza* [ˈkasa] 'hunting'). This is called *seseo*. A smaller number of dialects do not show a contrast between these two phonemes, but the sound they have is a dental fricative, very similar to /θ/. This is called *ceceo* and can be found in Eastern Andalusia and some parts of Central America.

The voiceless velar fricative /x/ is subject to much dialectal variation, the main realizations being [x, χ, h, ç]. In Castilian Spanish, /x/ has a more retracted realization, being frequently described as uvular [χ] especially when followed by a back vowel (e.g., *junta* [ˈχunta] 'meeting', *ajo* [ˈaχo] 'garlic') (Navarro Tomás 1977: 142; Hualde 2005: 154). In Andalusia, the Canary Islands, the Caribbean, and Central America, /x/ is pronounced as a laryngeal /h/ (e.g., *caja* [ˈkaha] 'box' and *gente* [ˈhente] 'people'). In Chilean Spanish, /x/ preceding a front vowel /i/ or /e/ has an anterior articulation, being produced as the palatal fricative [ç] (e.g., compare *gente* [ˈçente] 'people' and *gira* [ˈçiɾa] 'it spins' with *jarrón* [xaˈron] 'vase') (Lipski 1994: 201; Hualde 2005: 155). The velar realization [x] is found in the rest of the dialects (i.e., Mexico and most of South America).

Affricates are characterized by two phases in their articulation. They start with a complete closure, quickly followed by a slight opening of the vocal tract, through which air rushes out resulting in friction. Spanish has one affricate phoneme /tʃ/, a voiceless alveopalatal that occurs only in onset position (e.g., *pecho* [ˈpe.tʃo] 'breast,' *chico* [ˈtʃi.ko] 'boy'). In some dialects, the voiceless affricate is realized as a voiceless prepalatal fricative [ʃ] (e.g., *cacho* [ˈkaʃo] 'piece,' *chino* [ˈʃino] 'Chinese'). This can be seen as an instance of articulatory weakening, by which the affricate loses its closure phase but retains its frication release. These fricative variants can be found in northern Mexico, the Dominican Republic, Puerto Rico, Cuba, Chile, and Andalusia.

Orthographic *ll*, *y*, and *hi* when followed by a non-high vowel, represent a voiced palatal obstruent /ɟ/, whose pronunciation is subject to much variation. Its degree of constriction may vary depending on the environment, style, speech rate, and dialect, ranging from a stop to an approximant (see Aguilar 1997). The voiced palatal continuant [ʝ] usually occurs after a vowel or a continuant consonant (e.g., in *maya* [ˈmaʝa] 'Mayan,' *la llave* [la ˈʝaβe] 'the key,' *la hierba* [la ˈʝerβa] 'the grass'). We tend to find allophonic realizations of this palatal sound as a voiced affricate or plosive [ɟ] when it occurs after a lateral or nasal consonant or after a pause. However, affricate productions are often found in intervocalic position in Mexican and Caribbean varieties (Jiménez Sabater 1975: 108–110; Lope Blanch 1989, 1996; Martín Butragueño (in press)). Furthermore, in certain dialectal areas, most prominently in New Mexico and northern Mexico, the palatal obstruent has a very weak pronunciation and may be deleted between /i/ or /e/ and another vowel (e.g., *anillo* [aˈnio] 'ring,' *cabello* [kaˈβeo] 'hair') (Canfield 1981: 80; Lipski 1990; Alvar 1996). The phonemic status of [ʝ] ~ [ɟ] is discussed in Section 4.2 below.

Argentinean Spanish has developed a different pronunciation for [ʝ] ~ [ɟ]. In this variety, we find a voiced or voiceless prepalatal fricative [ʒ ~ ʃ] in contexts where the palatal obstruent is used in other dialects (e.g., *maya* [ˈmaʒa] 'Mayan,' *llave* [ˈʒaβe] 'key'), but note that orthographic *hi-* is not subject to this development, and it is pronounced like in other varieties (i.e., *hierba* [ˈʝerβa] 'grass' vs. *yerba* [ˈʒerβa] 'mate leaf'). This type of pronunciation is referred to as *žeismo* or *rehilamiento*. The voiceless variant seems to be a newer development, and devoicing is more common among younger generations, women, and members of the middle class (Wolf and Jiménez 1979; Guitarte 1955; Lipski 1994: 170).

3.2 Sonorants

3.2.1 Nasals Nasal consonants are produced with an oral closure and a lowered velum, which allows the air to flow through the nasal cavity. There are three nasal phonemes in Spanish /m, n, ɲ/, which contrast in their place of articulation (i.e., bilabial, alveolar, and palatal, respectively). As for their distribution, the three nasals can be found word-medially, in onset position, giving rise to minimal triplets such as *cama* /ˈkama/ 'bed' vs. *cana* /ˈkana/ 'gray hair' vs. *caña* /ˈkaɲa/ 'cane.' Word-initially, the palatal nasal has a very limited occurrence, mainly in borrowings from indigenous languages (e.g., *ñame* [ˈɲame] 'yam') and some words of Leonese origin since Leonese palatalizes initial /n/ (e.g., *ñublado* [ɲuˈβlaðo] 'cloudy') (Hualde 2005: 173–174). In coda position, there is neutralization and nasals assimilate to the place of articulation of a following consonant (see nasal assimilation in Chapter 6). Word-finally, both /n/ and /m/ may occur, although /m/ is limited to a handful of borrowings (e.g., *álbum* 'album'), and in these cases, it tends to be pronounced as [n]; however, note that a word-final nasal is realized as bilabial [m] is some varieties of Colombian and Yucatán Spanish (Hualde 2005: 176).

In some Latin American and Peninsular varieties, a word-final nasal before a pause is realized as velar [ŋ]. Velarization of final nasals is frequently accompanied by nasalization of the preceding vowel (e.g., *pan* [ˈpãŋ] 'bread,' *corazón* [koɾaˈsõŋ] 'heart'). This phenomenon is widely found across the Spanish-speaking world, including the Caribbean, the Pacific coast of South America, Canary Islands, and the Spanish regions of Asturias, Galicia, León, Extremadura, and Andalusia (see, among others, Alonso et al. 1950; Malmberg 1965: 3; Zamora Vicente 1967: 416; Jiménez Sabater 1975: 116–119; López Morales 1980; Quilis and Graell 1992; Quilis 1993: 239–242).

3.2.2 Liquids: laterals and rhotics The term liquid includes lateral and rhotic sounds, which tend to pattern together. For example, these sounds share similar distributional properties in Spanish: only a lateral or a rhotic may appear after another consonant in a complex onset (see Chapter 7). Each type of liquid will be discussed in turn below.

Laterals are produced by creating a closure at some point in the middle part of the vocal tract and leaving the sides of the tongue open (both or just one side, depending on the speaker), allowing the airflow to escape through that opening. Spanish has primarily one lateral phoneme, voiced alveolar /l/, which can occur in any syllabic position (e.g., *lago* [ˈla.ɣo] 'lake', *ala* [ˈa.la] 'wing,' *claro* [ˈkla.ɾo] 'clear,' *mal* [ˈmal] 'bad'). In coda positions, /l/ assimilates in place of articulation to a following consonant articulated with the front of the tongue (i.e., not to labials or velars; see lateral assimilation in Chapter 6).

Some dialects have a second lateral phoneme, namely the palatal lateral /ʎ/. It can be found mainly in the speech of older speakers in north and central Spain, northern Argentina, Paraguay, and the Andean Region (Canfield 1981: 6–7; Hualde 2005: 180). This sound, which only occurs in onset position, corresponds to orthographic *ll* and contrasts with /ʝ/, which corresponds to orthographic *y*. Most dialects lack this distinction since they only have the palatal obstruent phoneme. This situation is called *yeísmo*, where /ʎ/ has merged with the obstruent /ʝ/, and lexical distinctions based on the /ʎ/ vs. /ʝ/ contrast have been lost; see examples in (5). In dialects with *lleísmo*, on the other hand, a contrast between /ʎ/ and /ʝ/ exists and we find lexical contrasts based on this opposition. The examples in (5) illustrate these two situations.

(5)	*Orthography*	*'Yeísta' dialects*	*'Lleísta' dialects*	
	haya	[ˈaʝa]	[ˈaʝa]	'had'
	halla	[ˈaʝa]	[ˈaʎa]	'finds'
	cayó	[kaˈʝo]	[kaˈʝo]	'fell'
	calló	[kaˈʝo]	[kaˈʎo]	'became quiet'

Spanish also has two rhotics, a tap /ɾ/ (*vibrante simple*) and a trill /r/ (*vibrante múltiple*). Both of them are alveolar, the tip of the tongue and the alveolar ridge being the active and passive articulators, respectively. But each rhotic involves a different production mechanism: the tap consists of a single alveolar closure, while the trill is produced with several, quick closures. However, the trill is not just a series of taps since the articulatory configuration responsible for it differs from that for the tap (but cf. Harris 1983: 67–68). Trills require more "muscular tension" (Navarro Tomás 1977: 122–123) and greater articulatory precision (Recasens 1991). In fact, Spanish trills involve the Bernoulli effect in their production; that is, the rapid closures are the result of the aerodynamic conditions created in the vocal tract, while taps are produced by a single, voluntary movement of the tongue tip (Martínez Celdrán and Fernández Planas 2007: 149–151).

The two Spanish rhotics are contrastive only in intervocalic position within a word. In this context, we can find minimal pairs such as *perro* /ˈpe.ro/ 'dog' vs. *pero* /ˈpe.ɾo/ 'but,' *carro* /ˈka.ro/ 'cart' vs. *caro* /ˈka.ɾo/ 'expensive.' Word-initially and after a heterosyllabic consonant, only /r/ occurs (e.g., *rosa* [ˈrosa] 'rose,' *sonrisa* [son.ˈri.sa] 'smile'). In complex onsets, only /ɾ/ is possible (e.g., *broma* [ˈbɾoma] 'joke,' *centro* [ˈsentɾo] 'center'). In coda position, there is some variation. When there is a following vowel and resyllabification takes place, we find only /ɾ/. On the other hand, if there is no resyllabification because of a following consonant or pause, then either the tap or the trill may occur, although the most common realization tends to be the tap. The trill is usually restricted to emphatic productions, although some dialects tend to prefer trill realizations of coda rhotics in general (and also of rhotics in onset clusters; Alonso 1945; Hualde 2005: 182).

Rhotics are not always produced with a complete closure. We can find continuant variants that range from fricative to approximant realizations depending on the constriction degree (Blecua 2001; Martínez Celdrán and Fernández Planas 2007: 157). These pronunciations might occur in any position and are more likely to occur in conversational styles. Furthermore, rhotics are subject to an array of non-normative pronunciations depending on the dialect. Assibilation of rhotics is present in a number of geographical areas, including the Andean region, parts of Mexico and Central America, Paraguay, and northern Argentina (e.g., Argüello 1978; Lipski 1994; Moreno de Alba 1994; Bradley 2004; Serrano 2006; Lastra and Martín Butragueño 2006). These continuant variants are generally voiced, although there are also voiceless instances, and show frication in their high frequencies, which gives them a strident-like quality (Quilis and Carril 1971). Within the Spanish dialectological tradition, the symbol [ř] is used to represent assibilated rhotics. They may occur in different contexts, depending on style and dialect (e.g., *verde* [ˈbeřðe] 'green,' *carro* [ˈkařo] 'car'; see Bradley 1999, 2004 for an articulatory-based analysis of Ecuadorian assibilation). Assibilation also affects the consonant cluster /tr/, which is produced as a devoiced affricate [tř̥]. This variant is found in Chile, Costa Rica, Ecuador, and La Rioja region of Spain (Quilis 1993: 352–354).

Velar or dorsal realizations of rhotics are found in several Caribbean varieties, especially in Puerto Rican Spanish. These dorsal variants are articulated with the tongue postdorso and range from a velar fricative [x] to a uvular rhotic [ʀ]. They may be either voiced or voiceless (e.g., *rico* [ˈxiko] ~ [ˈʀiko] 'rich,' *corro* [ˈkoxo] ~ [ˈkoʀo] 'I run') (Vaquero and Quilis 1984; Quilis 1993: 350–351). In Caribbean dialects, we can also find pre-aspirated realizations of trills. These variants are produced with some pharyngeal friction followed by an alveolar tap or trill, and they are subject to some degree of devoicing (Quilis 1993: 351–352; Hualde 2005: 187). The symbol [hr] is used to represent these pre-aspirated sounds (e.g., *tierra* [ˈtjehra] 'earth,' *Ramón* [hraˈmon] 'Raymond'). Willis (2006, 2007) presents spectrographic data from Dominican Spanish showing that these pre-aspirated rhotic realizations are better characterized as containing pre-breathy voice, and he argues for using the IPA symbols [ɦɾ] or [ɦr] depending on the number of taps that follow the pre-breathy portion.

3.2.3 Neutralization of liquids Neutralization of liquids in coda position is found in several dialects, in which the difference between a lateral and a rhotic is lost, and they lack lexical distinctions based on this contrast (e.g., *mar* [ˈmaɾ] 'sea' vs. *mal* [ˈmal] 'bad'). The resulting sound varies depending on the region and even within the same speaker (Hualde 2005: 188). In some dialects, coda /l/ is pronounced as a rhotic. This is called rhotacism and can be found in the Canary Islands (Marrero 1988), Andalusia (Quilis-Sanz 1998), and to a lesser extent in Puerto Rico and in some regions of the Dominican Republic (Quilis 1993: 325–326). Other varieties neutralize into a lateral (i.e., coda rhotics are realized as /l/). This phenomenon is called lambdacism and it has been attested in Cuba, Dominican Republic, and Puerto Rico. Lambdacism is reportedly most frequent word-finally and is highly disfavored before a nasal consonant (Quilis 1993: 356). In Puerto Rican Spanish, the result of lambdacism has been impressionistically described as an intermediate sound between a rhotic and a lateral (e.g., Navarro Tomás 1948: 76; Hualde 2005: 188). Recent experimental work characterizes this sound as an approximant and shows that it is different from [l] in its acoustic features, more precisely in its formant trajectories and duration, although the precise acoustic differences vary across speakers (Paz 2005; Simonet et al. 2008). Furthermore, Paz (2005) shows that Puerto Rican speakers can perceive the difference between lexical coda /l/ and /r/, even in cases where the difference has supposedly been neutralized, while speakers of other varieties, namely Argentine Spanish, have difficulty perceiving the difference. Based on these results, Simonet et al. (2008) conclude that liquid neutralization in Puerto Rican Spanish is incomplete and less common than previously reported.

Vocalization of coda rhotics and laterals is common among lower socioeconomic levels in the Dominican region of El Cibao (Jiménez Sabater 1975: 90–105; Alba 1990). In this variety, coda liquids are realized as a palatal glide [j]. Also, some dialects delete coda liquids with or without gemination of the following consonant (see total assimilation of liquids in Chapter 6).

(6)

Orthography	*Standard*	*Rhotacism*	*Lambdacism*	*Vocalization*	
algo	[ˈalɣo]	[ˈaɾɣo]	[ˈalɣo]	[ˈajɣo]	'something'
aquel	[aˈkel]	[aˈkeɾ]	[aˈkel]	[aˈkej]	'that'
porque	[ˈpoɾke]	[ˈpoɾke]	[ˈpolke]	[ˈpojke]	'because'
comer	[koˈmeɾ]	[koˈmeɾ]	[koˈmel]	[koˈmej]	'to eat'

4 Quasi-phonemic contrasts

4.1 Glides vs. high vowels

Many studies analyze glides as allophones of the high vowels rather than as independent phonemes (Navarro Tomás 1977; Alarcos 1965; Quilis & Fernández 1985; Hualde 2005). This is based in the observation that high vowels are realized as glides when they are unstressed and next to another, nonidentical vowel (e.g., *cuento* [ˈkwento] 'tale,' *pienso* [ˈpjenso] 'I think,' *veinte* [ˈbei̯nte] 'twenty'). Thus, glides and high vowels seem to be in complementary distribution. However, in some dialects, there seem to exist near-minimal pairs which are based in a contrast between glides and high vowels (Aguilar 1999; Hualde and Prieto 2002; Hualde 2004). In these words, we find unstressed high vowels next to other vowels, where a glide would normally be expected. The examples in (7) include some of these pairs, where the word in the left column contains an exceptional unstressed high vowel.

(7)

a. Exceptional high vowel			b. Expected glide		
dueto	[du.ˈe.to]	'duel'	*duelo*	[ˈdwe.lo]	'duet'
huida	[u.ˈi.ða]	'looks after'	*cuida*	[ˈkuj.ða]	'escape'
pié	[pi.ˈe]	'I chirped'	*pie*	[ˈpje]	'foot'

Some authors argue that words like those in (7a) should be lexically marked as exceptional, without positing an independent phonemic category for glides (e.g., Hualde 1997; Harris and Kaisse 1999). However, Hualde (2004) presents evidence that these exceptional words are not random since there are some factors that explain what positions show exceptional syllabification. He argues that there is contrast between glides and high vowels, but it is limited to certain positions where exceptional syllabification applies; see also Martínez Celdrán (1989: 78–84, 93–96) for further discussion of glides as a different phonemic category.

Further evidence against the analysis of glides as allophones of high vowels comes from contrasts such as *cambia* [ˈkam.bja] 'it changes' vs. *varía* [ba.ˈɾi.a] 'it varies.' Spanish verbs in the present indicative form are always stressed in the penultimate syllable. Thus, some authors argue that the contrast between verbs such as *cambiar* and *variar* is the result of an underlying distinction between high vowels and glides (Harris 1969: 122–125, 1983, 1989). However, other analyses maintain that the difference between such verbs stems from the contrast between high vowels underlyingly specified as nuclear (e.g., in *varía*) and high vowels without any syllabic specification (e.g., in *cambia*). Under this view, there is no need to posit a phonemic distinction between glides and high vowels (Cressey 1978: 78–79; Roca 1997; see also Hualde 1997 for further discussion).

4.2 The phonemic status of [ʝ̞ ~ ɟ]

There are two main positions with respect to the phonemic status of the sound /ʝ̞/in Spanish. On the one hand, this sound can be seen as an allophone of the high vowel/i/. This view is based on the analysis of/ʝ̞/as the result of strengthening the glide [j] in syllable-initial position. On the other hand,/ʝ̞/can be treated as an independent consonantal phoneme. Evidence for this analysis comes from two sources, namely minimal pairs that, for some speakers, rely on the contrast between/j/and/ʝ̞/and words that lack [j] fortition. Some (near)-minimal pairs that show a contrast between [j] and [ʝ̞] are illustrated in (8).

(8) Minimal pairs showing the [j] vs. [ʝ̞] contrast

desierto [de.ˈsjeɾ.to]	'desert' vs.	*deshielo* [des.ˈʝ̞e.lo]	'thawing'
abierto [a.ˈbjeɾ.to]	'open' vs.	*abyecto* [ab.ˈʝ̞ek.to]	'abject'

Supporters of the allophonic analysis of/ʝ̞/argue that these minimal pairs can be explained by referring to the relation between syllabic division and morphology (e.g., Hualde 1997). The words in the right column contain a prefix, which requires a syllable boundary right after it so that/ʝ̞/occupies a syllable-initial position and thus, strengthening takes place.

Regarding the second argument for the phonemic status of/ʝ̞/(i.e., words that do not show/ʝ̞/fortition), Hualde (2004) reports that educated speakers tend to avoid word-initial strengthening of words that begin with orthographic *hie* (e.g., *hiena* 'hyena') (see also Navarro Tomás 1977). In these cases, educated speakers disfavor a noncontinuant pronunciation in contrast with words beginning with orthographic *ll*, *y*, for which the whole range of constriction degrees is possible, including stop realizations. Some examples from Castilian Spanish taken from Hualde (2004) illustrates this point in (9), where we can see that the range of pronunciations for the high vowel/i/, the palatal glide/j/, and the palatal obstruent/ʝ̞/only partially overlaps (the symbols have been modified; Hualde uses [i̯] for all palatal glides). Here, we can talk about a quasi-phonological contrast induced by the orthography (see Hualde 2004 for further discussion).

(9) Range of possible pronunciations in Castilian Spanish

hiato	'hiatus'	[i.ˈa] ~ [ˈja]
hiena	'hyenna'	[ˈje] ~ [ˈʝ̞e]
yema	'yolk'	[ˈje] ~ [ˈʝ̞e] ~ [ˈɟe]

5 Conclusion

This chapter has provided a general overview of the phonemes of Spanish and their main dialectal variants, including references to some of the major 'classics' in the field and to the most recent developments in the study of the Spanish sound system (see also Chapter 6 for further phonological processes). One of the main advances in recent years has been the use of laboratory or experimental approaches to the analysis of Spanish phonology and phonetics. Based on previous impressionistic, but nonetheless valuable, descriptions of the language, researchers have been able to give a more accurate and detailed picture of the Spanish sounds and their variation across the Spanish-speaking world using acoustic and articulatory data. Furthermore, new experimental methodologies allow us to integrate results from different linguistic subfields, for instance from sociolinguistics, language acquisition, and psycholinguistics, into the study of speech production and perception.

Some of the areas that have greatly benefited from laboratory approaches in recent years include the study of the Spanish sound inventory in contact situations such as Spanish in contact with indigenous languages in Latin America, with English in the United States, and with other Romance languages and Basque in the Iberian Peninsula. In this chapter, I mentioned some of the issues recently explored in relation to language contact, including the vocalic system and/s/aspiration. Inter- and intra-dialectal rhotic variation is another area that has attracted a great amount of attention in recent years, especially from an experimental perspective. Several studies (see those cited above on rhotics in section 2.2.2) have presented evidence indicating that rhotic production is subject to more variation than previously reported. In fact, in some dialects, innovative realizations such as assibilated rhotics seem to be in the process of replacing canonical productions. This reconfiguration of the system has consequences for the phonemic analysis of rhotics that need to be explored. Recent empirical data are also challenging traditional analyses of neutralization in Spanish by showing that these neutralizations are in fact incomplete, as we saw for liquid neutralization in Puerto Rican Spanish in Section 3.2.3. In light of these results regarding incomplete neutralization, the supposed voicing neutralization of Spanish stops in coda position is called into question. Instrumental techniques could help answer this question and bring some insight into an area of Spanish phonology where much variation has been reported but where the exact realizations of coda stops and their distribution are far from clear.

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6 Main Phonological Processes

FERNANDO MARTÍNEZ-GIL

1 Introduction

In his illuminating discussion of the vowel and consonant systems in human languages, Ladefoged (2005: 4) points out that "two factors, articulatory ease and auditory distinctiveness, are the principal constraints on how the sounds of languages develop." The type of phonological phenomenon that most typically reflects ease of articulation is segmental assimilation. This chapter deals primarily with assimilatory phenomena found in the segmental phonology of Spanish. It surveys four types of assimilatory processes that have engendered the most theoretical interest in the relatively recent literature. Some of them arise by partial assimilation, such as homorganic nasal and lateral assimilation, spirantization, and voicing assimilation; others involve complete assimilation (gemination), including gemination induced by/s/-deletion in Andalusian Spanish and complete assimilation of syllable-final liquids in Cuban Spanish.[1]

This chapter is organized as follows. Partial assimilation phenomena in Spanish are discussed in Sections 2–4: Section 2 deals with homorganic nasal and lateral assimilation; Section 3 examines the spirantization voiced obstruents; and Section 6.4, voicing assimilation of coda obstruents. Section 5 analyzes two types of complete consonant assimilation found in the Andalusian and Cuban varieties of Spanish. Finally, Section 6 presents some concluding remarks.

2 Nasal and lateral assimilation

Spanish nasals exhibit a three-way phonemic contrast in point of articulation (*PA*),/m/~/n/~/ɲ/, in syllable-initial position, both word-initially and word-medially (see Campos-Astorkiza, Chapter 5, above). In coda position before a pause (i.e., utterance-final position) or before a vowel in the following word, nasal PA contrasts are neutralized (cf. Harris 1984a): in most standard dialects, the nasal PA is coronal [n]: *pa*[n] 'bread,' *u*[n] *amigo* 'a friend' (cf. **pa*[m], **pa*[ɲ],

The Handbook of Hispanic Linguistics, First Edition. Edited by José Ignacio Hualde, Antxon Olarrea, and Erin O'Rourke.

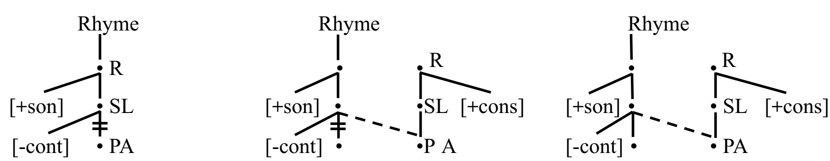
**u*[m] *amigo*, **u*[ɲ] *amigo*, etc.). In the so-called *velarizing* dialects, nasals surface as velar [ŋ]: *pa*[ŋ], *u*[ŋ] *amigo*. Elsewhere, coda nasals undergo homorganic assimilation to a following consonant, both within words and across word boundaries, adopting seven distinct points of articulation: bilabial, as in *a*[m]*bos* 'both,' *u*[m] *peso* 'a weight'; labiodental, as in *e*[ɱ]*fermo* 'sick-MASC.', *u*[ɱ] *fin* 'an end'; interdental (only in standard Castilian), as in *o*[n̪]*ce* 'eleven,' *u*[n̪] *zapato* 'a shoe' (*c*, *z* = /θ/) dental, as in *cua*[n̪]*do* 'when,' *u*[n̪] *día* 'a day'; alveolar, as in *ga*[n]*so* 'goose,' *u*[n] *río* 'a river'; palatal, as in *a*[ɲ]*cho* 'wide MASC.,' *u*[ɲ] *yeso* 'a plaster'; and velar, as in *te*[ŋ]*go* 'I have,' *u*[ŋ] *gato* 'a cat-MASC.' The lateral/l/also assimilates to the PA of a following coronal consonant: interdental (only in standard Castilian), as in *a*[l̪]*zar* 'to raise,' *e*[l̪] *circo* 'the circus'; dental, as in *sa*[l̪]*to* 'jump,' *e*[l̪] *día* 'the day'; alveolar, as in *bo*[l]*sa* 'bag,' *e*[l] *río* 'the river'; and palatal, as in *co*[ʎ]*cha* 'bedspread,' *e*[ʎ] *yeso* 'the plaster'. Unlike nasals, however, /l/does not assimilate to consonants with either labial or velar PA, as in *cu*[l]*pa* 'blame,' *e*[l] *barco* 'the ship,' *go*[l]*fo*, 'gulf,' *e*[l] *fin* 'the end,' *ta*[l]*co* 'talc,' *e*[l] *gato* 'the cat-MASC.'

The data on nasal assimilation has served as critical empirical evidence for several breakthroughs in the development of phonological theory. Thus, Hooper (1972, 1976) demonstrates that certain facts of Spanish nasal assimilation cannot be adequately captured without reference to syllabic boundaries, against the standard generative framework of Chomsky and Halle (1968), in which the syllable was granted no theoretically relevant status. In particular, Hooper notes that while nasals assimilate in PA to a following consonant both within and across word boundaries, as just described, they assimilate to glides only across word boundaries: *u*[ɲ#j]*elo* 'an ice,' *u*[ŋ#w]*evo* 'an egg,' but not within words: *n*[j]*eto* 'grandson,' *n*[w]*evo* 'new-MASC.' (cf. *[ɲj]*eto*, *[ŋw]*evo*). As Hooper argues, the difference is determined primarily by the presence vs. absence, respectively, of a syllabic boundary between the two segments.[2] Second, nasal assimilation has often been used as an illustration of the inadequacy that often emerged in the classic generative model between rule naturalness and rule simplicity: although a very natural rule (very frequent across languages), nasal assimilation is extremely complex to formulate in linear notation (the rule in Harris 1969: 16, for example, uses the alpha notation to require concord of four PA features in its context and its structural change). With the advent of multilinear phonology, a simple solution to this problem became available: all assimilation processes are uniformly captured with one single mechanism: autosegmental spreading. In conjunction with feature constituents, countenanced by the universal feature hierarchy, assimilation of a single feature now becomes formally just as simple as the assimilation of a natural class of features, such as those included under PA. And, third, assimilation of/n/followed by a bilabial consonant in colloquial Spanish commonly gives rise to co-articulation, resulting in a nasal with a simultaneous occlusion at the alveolar and the bilabial PA: *i*[n͡m]*móvil* 'motionless,' *co*[n͡m] *Pablo* 'with Pablo,' etc. (Harris 1969: 14–16; Navarro Tomás 1977: 89, 113). Nasals realized with a simultaneous velar and bilabial or labiodental PA have also been reported for velarizing varieties: *i*[ŋ͡m]*móvil*, *co*[ŋ͡m] *Pablo*, *e*[ŋ͡ɱ]*fermo* 'sick' (Harris 1969:

14–16). Co-articulated segments were particularly problematic for linear analyses because their characterization in terms of linearly-arranged feature matrices entailed contradictory feature specifications. In autosegmental representation, they are simply analyzed as nasals associated to two PA nodes: one is underlying; the other is spread from the following consonant.

In autosegmental phonology, there is one basic parameter for expressing phonological processes: either *delete* or *insert* association lines. Neutralization is carried out by means of feature delinking (= *delete*); assimilation, by feature spreading (= *insert*). Thus, formulated autosegmentally within the feature hierarchy, as in (1) (where *R* = Root; *L* = Laryngeal; *SL* = Supralaryngeal), coda nasal PA neutralization is achieved by delinking of the PA node, as in (1a), and subsequent assignment by default of the coronal anterior PA (alternatively, by default assignment of velar PA in velarizing dialects). Homorganic nasal assimilation is usually expressed as the spreading of the PA node from the following consonant with simultaneous delinking of the nasal's PA (1b), an operation that incorporates PA neutralization (1a); finally, a co-articulated nasal is created if spreading does not entail delinking of the target's PA, as in (1c) (see Goldsmith 1981; Harris1984a; Hualde 1898a, 1989b; Piñeros 2006; for a somewhat different analysis of co-articulation, see Núñez-Cedeño and Morales-Front 1999: 80; Kenstowicz 1994: 485–488).

(1) a. *PA neutralization* b. *PA delinking* c. *Co-articulation*



Since the early generative accounts (Harris 1969; Cressey 1978), there is general agreement that homorganic assimilation of nasals and laterals in Spanish is one and the same assimilatory phenomenon, and therefore the two should be formalized into a single statement. This outcome is achieved if nasals and laterals are characterized as the phonological class of noncontinuant sonorants, defined by the features [+ son, -cont]. The potential generation of velar or labial laterals, including co-articulated ones by the application of (1b)/(1c), is ruled out by appealing to markedness constraints on PA (Cressey 1978: 66–67; Hualde 1989b).

A typical account of homorganic nasal assimilation within Optimality Theory (*OT*) appeals to the domination of AGREE-PA (a markedness constraint requiring that a sequence of two consonants agree for PA features), and the positional faithfulness constraint ONSET IDENT-PA (which protects the input–output identity of PA features for onset consonants) over IDENT-PA, which simply demands input–output PA faithfulness (see Baković 2001). The ranking AGREE-PA, IDENT ONSET-PA > > IDENT-PA effectively enforces regressive PA assimilation (Lombardi 1999; Baković 2001; see Piñeros 2006 for a slightly different OT approach). Neutralization of coda nasals and laterals in prepausal and prevocalic contexts in standard varieties is achieved by ranking the PA markedness hierachy *Dorsal > > *Labial > > *Coronal over Ident-PA to thus enforce a coronal PA across words (Piñeros 2006).

To conclude this section, a remark must be made on nasal velarization. Trigo (1987) first suggested that nasals described as "velar" in traditional descriptions are actually debuccalized (i.e., placeless). Following Trigo, Baković (2001) proposes the syllabic markedness constraint NasCodaCond, which bars coda nasals from bearing PA features for velarizing dialects (see also Bermúdez-Otero forthcoming: 23); "velarization" as debuccalization results from the ranking NasCodaCond > > Ident-PA. However, this view of nasal velarization is at odds with the nasal co-articulation facts discussed earlier: if velarized nasals were, in fact, the phonetic manifestation of debuccalization, we would expect PA assimilation, not co-articulation.

3 Voiced obstruents

Underlying voiced obstruents in Spanish exhibit two sets of allophonic realizations in complementary distribution. As illustrated in (2), they surface as stops in utterance-initial position (i.e., word-initially after a pause), and after homorganic sonorants (nasals and /l/) (2a–b); elsewhere (i.e., after continuants, including vocoids, heterorganic laterals, rhotics, and fricatives), they are realized as spirants (2c–f), a phenomenon commonly referred to as *spirantization*.

(2)	a.	[b]arco 'ship' [b]ámonos 'let's go!' [d]iente 'tooth' [g]ato 'cat'	b.	a[mb]os 'both' cua[n̪d]o 'when' sue[l̪d]o 'salary' fa[ŋg]o 'mud'	c.	cue[β]a 'cave' co[ð]o 'elbow' la[ð]o 'side' la[ɣ]o 'lake'
	d.	ár[β]ol 'tree' per[ð]er 'to lose' car[ɣ]a 'load'	e.	des[β]iar 'to deflect' des[ð]e 'from, since' mus[ɣ]o 'moss'	f.	ol[β]idar 'to forget' al[ɣ]a 'seaweed'

Voiced obstruents are also spirantized in coda position, since, given general conditions on Spanish codas, the preceding segment will necessarily be a vowel (for clarity, relevant syllable boundaries are indicated by periods): *a*[β.ð]*omen* 'abdomen,' *a*[β].*negado* 'self-sacrificing,' *a*[ð.β]*erso* 'adverse,' *a*[ð].*mirar* 'to admire,' *i*[ɣ].*norar* 'to ignore,' *esti*[ɣ].*ma* 'stigma,' etc.

Since Lozano (1979), most derivational analysis of spirantization assume the *archiphonemic* hypothesis, according to which voiced obstruents lack any value for the feature [cont] underlyingly (Mascaró 1984, 1991; Harris 1984b, 1985; Hualde 1989b, 1991; Branstine 1991; Palmada 1997, among others). Under this assumption, the set of non-affricate underlying obstruents in Spanish would be have the specifications for voicing and continuancy shown in (3), where unspecification for the feature [cont] is indicated by uppercase symbols (/θ/occurs only in Castilian Spanish).

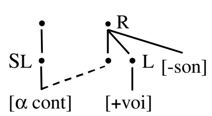
(3) The archiphonemic analysis of Spanish obstruents:

	p	t	k	f	s	(θ)	x	B	D	G
[voi]	–	–	–	–	–	–	–	+	+	+
[cont]	–	–	–	+	+	+	+			

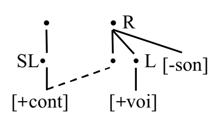
There is a theory-internal argument for an archiphonemic analysis: while the feature [cont] serves to establish an underlying distinction between voiceless stops and fricatives (/p, t, k/vs./f, s-θ, x/), for voiced obstruents it can be entirely predicted from the preceding environment, and therefore it ought to be regarded as non-distinctive. Following a basic tenet of generative phonology, predictable information is not encoded in underlying forms, but rather supplied by phonological rules. Accordingly, it has been generally assumed that voiced obstruents lack any specification for continuancy in underlying representations.

The surface distribution of Spanish voiced obstruents has been among the most intensely debated topics of Spanish phonology in the last decades. Two fundamental theoretical issues are at the heart of the debate: (a) the nature of the underlying segments; and (b) the formal mechanism(s) that account for their surface distribution. While early generative treatments posited underlying stops plus a spirantization rule that changed them into their corresponding spirants in the appropriate environments (Harris 1969; Cressey 1978), the most influential autosegmental analyses of the stop-spirant distribution of Spanish voiced obstruents fall roughly into one of three major categories, illustrated in (4) (adapted here to the feature hierarchy model of representations): (a) in the *continuant-spreading* hypothesis, [-cont] is spread from a preceding segment (4a); [-cont] is supplied by a special rule in utterance-initial position (Mascaró 1984); (b) in the *spirantization* hypothesis, [+cont] is spread from a preceding segment (4b); in all remaining contexts (utterance-initial and after non-continuants), [-cont] is supplied by default (Harris 1984b, 1985; see also Goldsmith 1981); and (c) in the *fortition* hypothesis, [-cont] is spread from a preceding homorganic non-continuant sonorant (nasal or lateral) created by PA assimilation in a prior stage of the derivation; [+cont] is assigned by default elsewhere, except utterance-initially, where a special rule assigns voiced obstruents the value [-cont] (Hualde 1989b, 1991; see also Branstine 1991).[3]

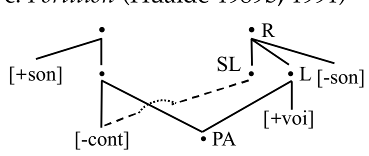
(4) a. *Continuant spreading* (Mascaró 1984)



b. *Spirantization* (Harris 1984b, 1985)



c. *Fortition* (Hualde 1989b, 1991)



Because the approaches in (4) provide clearly different answers to the fundamental issue regarding the phonological principles that determine the stop-spirant surface distribution in Spanish, it would be useful to evaluate their merits and shortcomings.

In the continuant-spreading approach (4a), the surface specification of voiced obstruents is essentially governed by the continuancy value of the preceding segment; the (admittedly *ad hoc*) rule needed to account for the stop allophones in utterance-initial position suggested in Mascaró (1984: 293), would be unnecessary, under the natural assumption that in the absence of an immediately preceding segment, voiced obstruents are assigned [-cont] by default. Since homorganicity is not recognized as a relevant factor in this analysis, a problem evidently arises for [cont]-spreading in post-lateral position; namely, in order to derive a stop (cf. (2b)), /l/must be specified as [-cont] before/D/, but in order to obtain spirants before/B, G/(cf. (2f)), it must be [+cont], a stance in fact adopted in Mascaró (1984). Two main objections have been raised against this proposal: (a) its apparent circularity (Hualde 1989b: 28); and (b) from a theoretical point of view, a dual feature specification is highly problematic, if not prohibited outright (Hualde 1991: 102). Mascaró's proposal, however, seems to be solidly justified on articulatory grounds. As this author argues, a lateral is produced by raising the tongue tip to block the airflow in the mid-section of the mouth, a gesture that characterizes stops, while simultaneously allowing the airflow to escape (on both sides of the tongue), as in continuants (Mascaró 1984, 1991). Lateral constriction typically takes place at the coronal PA, but it is not found at either the labial or (exclusively) velar PA. Thus, from a phonetic point of view, it is hardly surprising that laterals behave as stops at the coronal PA, but as continuants elsewhere (see Holt 2002 for an attempt to incorporate this dual behavior into the feature hierarchy). In any event, it seems apparent that it is the [-cont] property of/l/ at the coronal PA what crucially determines the stop realization of a following/D/, not homorganicity, as claimed by the fortition hypothesis (4c). Finally, in the continuant spreading account, the fact that voiced obstruents surface as stops after nasals follows from the progressive assimilation to the nasal [-cont] value; homorganicity of nasals to a following voiced obstruent plays no role in this analysis.

There are reasons to presume that the crucial condition in nasal-voiced obstruent sequences is the nasal's [-cont] specification, not its homorganicity. Mascaró (1991: 171) points out that when a speaker makes an effort to pronounce a non-coronal nasal in loan words, a following voiced obstruent invariably surfaces as a stop, not a spirant, even though the cluster is not homorganic:

(5)	álbu[m d]e fotos	'photo album'	(cf. *álbu[m ð]e fotos)
	Vietna[m d]el sur	'South Vietnam'	(cf. *Vietna[m ð]el sur),
	parki[ŋ b]arato	'cheap parking'	(cf. *parki[ŋ β]arato)

There are other cases in which heterorganic obstruent-voiced obstruent sequences are realized as either surface stops (more deliberate speech) or as spirants (more informal registers). Surface forms in which such sequences disagree for continuancy are typically disallowed, as illustrated in (6). Obviously, such data runs counter to the predictions of the fortition hypothesis.

(6) fú[tb]ol or fú[ðβ]ol 'football' (cf. *fú[tβ]ol, *fú[ðb]ol, etc.)
Ma[kd]onalds or Ma[ɣð]onalds 'McDonalds' (cf. *Ma[kð]onalds, *Ma[ɣd]onalds, etc.)
ané[kd]ota or ané[ɣð]ota 'anecdote' (cf. *ané[kð]ota, *ané[ɣd]ota, etc.)

The spirantization hypothesis (4b) enjoys an obvious advantage over the other two approaches in that the default assignment of [-cont] accounts for the occurrence of stop allophones in utterance-initial position; no additional rule is needed. However, this hypothesis suffers from three seemingly intractable problems that render it untenable. First, recall that the feature [-cont] is crucial to classify nasals and laterals as a natural class, therefore allowing a unitary account of nasal and lateral homorganic assimilation. This characterization, however, is at odds with the assumption made in the spirantization approach that/l/is [+cont], which is needed to account for the spirants in [lβ] and [lɣ] sequences (cf. (2f) above), a clearly contradictory state of affairs. The second issue is the problematic appeal to geminate inalterability in order to block [+cont]-spreading in homorganic [l̪d] sequences, on the grounds that they share PA, and thus form a subsegmental geminate (Harris 1984; Kenstowicz 1994: 488) since it is unclear how the shared PA of the [l̪d] sequence would prevent the spreading of [+cont] from/l/onto the following/D/. Geminate inalterability effects typically emerge when the structural description of a rule refers to timing slots in the prosodic skeleton (Hayes 1996), which is evidently not the case in the spirantization rule (4b). And third, since in the spirantization approach the stop allophones are generated by default, their (systematic) occurrence after nasals and laterals is regarded as merely accidental, thus failing to draw any connection with a preceding non-continuant segment. To the extent that only the stop allophones occur after [-cont] segments, a linguistically relevant generalization seems to be missed.

The fortition approach, on the other hand, crucially appeals to the homorganicity of nasals and laterals in the structural description of the rule that derives the stop allophones (4c). As suggested earlier, the occurrence in certain registers of heterorganic nasal-voiced stop clusters (e.g., *albu*[m d]*e fotos*) presents an obvious problem for this analysis because it wrongly predicts that voiced stops should be realized as spirants when preceded by a heterorganic nasals. Another shortcoming of this approach is that because [+ cont] is postulated as the default value for voiced obstruents, a special rule is needed in order to account for stops in utterance-initial position (see Hualde 1991), arguably a "brute-force" solution that otherwise has no apparent motivation in Spanish phonology.

Especially problematic for all three approaches in (4) is the existence of certain American Spanish dialects in which, in addition to the contexts in which they occur in standard varieties, voiced obstruents surface as stops when preceded by non-nuclear segments (a glide and/or a consonant), as illustrated in (7) (see Flórez 1965; Resnik 1975; Montes Giraldo 1982, 1995; Canfield 1981; Zamora Munné and Guitart 1982; Amastae 1986, 1989, 1995; Moreno de Alba 1988, among others). The data in (7) is problematic for all three analyses in (4) since it is not clear how they can be adequately modified in order to accommodate such a distribution.

(7)

a. After glides:			b. After consonants:		
standard	*dialectal*		*standard*	*dialectal*	
va[jβ]én	va[jb]én	'oscillation'	ár[β]ol	ár[b]ol	'tree'
cu[jð]ar	cu[jd]ar	'to look after'	ol[β]idar	ol[b]idar	'to forget'
tra[jɣ]o	tra[jg]o	'I bring'	res[β]alar	res[b]alar	'to slide'
de[wð]a	de[wd]a	'debt'	ar[ð]e	ar[d]e	'it burns'
ra[wð]o	ra[wd]o	'swift, rapid'	des[ð]e	des[d]e	'from, since'
a[wð]az	a[wd]az	'audacious'	car[ɣ]a	car[g]a	'load'
ca[wð]al	ca[wd]al	'flow'	al[ɣ]o	al[g]o	'something'
a[wɣ]urio	a[wg]urio	'omen, presage'	ras[ɣ]ar	ras[g]ar	'to tear'

If such difficulties were not serious enough, there is a compelling case for positing voiced stops at the underlying level, against both the archiphonemic hypothesis and the postulation of underlying spirants/approximants. The argument is related to phonotactic constraints on Spanish complex (i.e., biconsonantal) onsets (see Colina, Chapter 7, below). It is well known that underlying voiceless stops/p, t, k/in Spanish may cluster with a following liquid in syllable onsets, but their fricative counterparts, with the exception of/f/, may not.[4]

The exclusion of sequences such as */sr/, */sl/, */θr/, */θl/, */xl/, and */xl/as onset clusters is a systematic property of Spanish syllable structure phonotactics, as these clusters fail to satisfy a minimal sonority distance (Harris 1983, 1989; Martínez-Gil 1996, 1997, 2001; Colina 2006, 2009). Instructively, voiced obstruents are spared from this minimal sonority distance restriction: they freely combine with a following liquid in complex onsets, even when they surface as spirants: *a*.[β]*ril* 'April,' *ha*.[β]*lo* 'I speak,' *la*.[ð]*rón* 'thief;' *a*.[ɣ]*rio* 'bitter;' *si*.[ɣ]*lo* 'century;' etc. (/dl/ clusters are excluded for independent reasons; see Harris 1983). Thus, at some relevant (non-surface) phonological level, voiced obstruents and voiceless stops must share a common phonological class property that defines them as a natural class, and distinguishes them from voiceless fricatives with regards to complex onset phonotactics.

It is apparent that the common property is the feature [-cont]. Because phonotactic conditions on core syllabification rules in Spanish must have direct access to underlying representations (Harris 1983, 1989; Hualde 1989d), it follows that voiced obstruents must be stops at this level. Under this assumption, the phonotactic generalization on biconsonantal onsets is quite simple: "Spanish complex onsets may consist of an *underlying* oral stop or/f/followed by a liquid" (or, alternatively in terms of sonority: "Spanish onsets consist of at most of two consonants that differ maximally in sonority," where the sonority "cut" among obstruents is the plus or minus specification for the feature [cont]; see Martínez-Gil 2001). Such a generalization cannot be readily captured under either an archiphonemic approach or one that posits underlying voiced spirants /β, ð, ɣ/ (as in Danesi 1982; see also Baković 1997), where the asymmetry between voiced spirants and voiceless fricatives in their ability to cluster with a following liquid can only be regarded as an accident, and thus the viability of either hypothesis is seriously undermined. While there is articulatory and acoustic evidence that the spirant allophones are actually voiced approximants, not voiced fricatives (Martínez Celdrán 1984, 1991, 2004), it is equally clear that phonetic facts do not necessarily correlate with phonological organization. The complex onset phonotactics just discussed suggest that Spanish voiced obstruents behave *phonologically* as stops rather than approximants.

Two early OT analyses of the stop-spirant distribution in Spanish are Baković (1997), in which voiced obstruents are analyzed as approximants underlyingly, and Carreira (1998), who assumes a feature-hierarchy configuration in which the feature [cont] is articulator-dependent. Both crucially depend, in some form or another, on a homorganicity condition, and thus they face difficulties analogous to the fortition approach (4b) in accounting for data such as (5)–(7) above. Yet a viable OT alternative enjoys two primary advantages over a derivational account. First, because constraints in OT operate exclusively on surface forms, there is no requirement that voiced obstruents bear a particular underlying value for the feature [cont] (as required in OT in any event by the principle known as *Richness of the Base*). Namely, the spirant-stop distribution can be obtained, whether archiphonemes, stops, or spirants are assumed underlyingly, if IDENT-[cont], which demands input-output identity for the value of the feature [cont]), is dominated by two markedness constraints which require the following: (a) that /d/ be [-cont] at the coronal PA; and (b) that voiced obstruents agree for the feature [cont] with a preceding segment. And second, the domination of these two markedness constraints over the phonotactic constraint requiring that the first member of a bisegmental onset be a stop or /f/ ensures that voiced obstruents will surface as spirants when followed by a tautosyllabic liquid. Finally, voiced obstruents are disfavored in utterance-initial position, and thus surface as stops as the elsewhere (default) case, by a high enough ranking of the well-known markedness constraint against voiced fricatives over voiced stops (see González 2003 for a relatively similar account).

4 Voicing assimilation

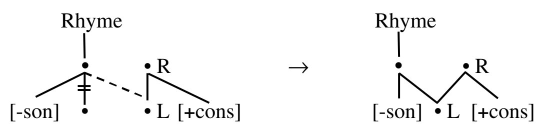
Coda obstruents in Spanish assimilate in voicing to a following consonant (see Alonso 1945; Harris 1969; Hooper 1972, 1976; Navarro Tomás 1977; Hualde 1989b, and references therein). The basic data, given in (8), illustrate the process of voicing assimilation (*VA*) for obstruent-consonant clusters that differ in voicing underlyingly, namely, the voiceless continuants /f, θ, s/ followed by a voiced consonant

(8a); the voiceless stops /p, t, k/ followed by a voiced consonant (8b); and voiced obstruents followed by a voiceless consonant (8c):[5]

(8)	a.	*afgano*	[avˈɣano]	'Afghan'	b.	*hipnosis*	[iβˈnosis]	'hypnosis'
		juzgar	[xuðˈɣar]	'to judge'		*ritmo*	[ˈr̄iðmo]	'rhythm'
		hallazgo	[aˈɟaðɣo]	'finding'		*fútbol*	[ˈfuðβol]	'football'
		desde	[ˈdezðe]	'from, since'		*étnico*	[ˈeðniko]	'ethnic'
		isla	[ˈizla]	'island'		*técnica*	[ˈteɣnika]	'technique'
	c.	*absurdo*	[aɸˈsurðo]	'absurd'				
		objeto	[oɸˈxeto]	'object'				
		adquirir	[aθkiˈrir]	'to acquire'				
		adjunto	[aθˈxunto]	'attached'				
		Agfa	[ˈaxfa]	(name)				

Standard autosegmental analyses of VA (Hualde 1989b; Martínez-Gil 1991) propose the operation in (9), which spreads the laryngeal features of a consonant onto a preceding coda obstruent, simultaneously delinking the obstruent's laryngeal specification.

(9)



As shown in Martínez-Gil (2002), the data in (8b–c) is particularly interesting from a derivational perspective because the interaction of *VA* and spirantization (*Spir*) gives rise to a rule-ordering paradox. Namely, in order to derive voiced spirants from underlying voiceless stops (8b), VA must precede SPIR, but precisely the opposite order would be required in order to derive voiceless spirants from underlying voiced obstruents (8c). The paradox, illustrated in (10) with the alternative derivations of the representative items *ritmo* (8b) and *adquirir* (8c), presents a serious challenge for a rule-based account since no particular ordering of the two rules yields the desired results:

(10)	a.	*U.R.s*:	/r̄it.mo/	/ad.ki.rir/	b.	*U.R.s*:	/r̄it.mo/	/ad.ki.rir/
		(1) *VA*:	r̄id.mo	at.ki.rir		(1) *Spir*:	—	að.ki.rir
		(2) *Spir*:	r̄ið.mo	—		(2) *VA*:	r̄id.mo	aθ.ki.rir
		Output:	r̄ið.mo	*at.ki.rir		*Output*:	*r̄id.mo	aθ.ki.rir

In Martínez-Gil (2002), it is shown that no direct formal connection is to be drawn between the ordering paradox in (10) and the assumption in (10) that voiced obstruents are underlying stops plus a (feature-changing) SPIR rule (as in Harris 1969), and that the data in (8) also presents insurmountable difficulties for the archiphonemic hypothesis. It is further argued that the problems posed by the interaction of SPIR and VA are merely artifacts of serial derivations, and that a simple and straightforward account of the data in (8b–c) can be reached in a constraint-based approach. In the OT account proposed therein, the interaction of Spir and VA is captured by postulating the domination of two markedness constraints that enforce spirantization and voicing assimilation over conflicting identity constraints that demand input–output faithfulness for the features [cont] and [voice], respectively: Spir bars voiced stops after continuant segments; regressive VA is achieved by the interaction of the positional faithfulness constraint *Ident-Onset*-[voice] (= the [voice] value of an onset output must be identical to that of its input) (Beckman 1997) and *Agree*-[voice] (a consonant sequence must agree in voicing); regressive voicing assimilation ensues if *Ident-Onset*-[voice] outranks *Agree*-[voice] (Lombardi 1999). However, the data in (8c) still present a problem for this analysis since it does not discriminate between the two candidates *a*[t]*quirir* and *a*[θ]*quirir*, both of which satisfy Spir and VA but violate Ident-[voice]. In Martínez-Gil (2002), the decision is left to the Coda-Cond(ition), which favors continuant obstruents over stops in coda position, a constraint motivated by the fact that *ceteris paribus* coda continuants in Spanish are conspicuously favored over stops. A higher ranking of the Coda-Cond over faithfulness to input [voice] correctly selects *a*[θ]*quirir* over **a*[t]*quirir*.

5 Complete assimilation

One of the best studied examples of complete assimilation in Spanish occurs in a number of dialects that aspirate and optionally delete syllable-final/s/. In Andalusian Spanish, deletion of coda/s/gives rise to two main patterns of consonantal gemination, dubbed here *Pattern A* and *Pattern B* and illustrated in (11) (see Hualde 1989a, 1989b; Gerfen 2001, 2002; Morris 2002; Campos-Astorkiza 2003, 2007, and references therein). In Pattern A, assimilation is complete and the outcome is a plain geminate. In Pattern B, on the other hand, when the aspirated/s/is followed by a voiceless obstruent, the sequence surfaces as a preaspirated voiceless geminate, as in (11a); if it is a sonorant, the result is a partially-voiced geminate (a geminate whose implosive portion is voiceless), as in (11b), an indication that the laryngeal features of the aspirated /s/ are preserved.

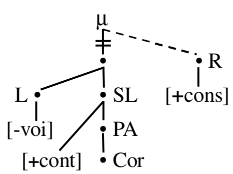
(11)		*Pattern A*	*Pattern B*	
a.	caspa	ca[pp]a	ca[^{h}pp]a	'dandruff '
	esfera	e[ff]era	e[^{h}ffẽra]	'sphere'
	más cerca	má[θθ]erca	má[^{h}θθ]erca	'closer'
	costa	co[tta]	co[^{h}tt]a	'coast'
	más [x]ente	má[xx]ente	má[^{h}xx]ente	'more people'
	mosca	mo[kk]a	mo[^{h}kk]a	'fly (insect)'
b.	isla	i[ll]a	i[l̥l]a	'island'
	mismo	mi[mm]o	mi[m̥m]o	'same, self '
	asno	a[nn]o	a[n̥n]o	'donkey'

Interestingly, when the consonant that follows the deleted /s/ is a voiced obstruent (which, it will be recalled, is spirantized in this context), the outcome geminate is a voiceless fricative, not a voiced one, as one might expect, as illustrated in (12) (for the sequence [-sβ-], the result is variably [ff] or [ɸɸ]).

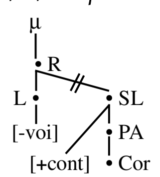
(12)	resbalar	re[ff]alar/re[ɸɸ]alar	'to slip, slide'	(cf. *re[ββ]alar)
	desde	de[θθ]e	'since, from'	(cf. *de[ðð]de)
	musgo	mu[xx]o	'moss'	(cf. **mu*[ɣɣ]*go*)

A standard autosegmental treatment of the two patterns of gemination in (12) is proposed in Hualde (1989b), adapted in (13) following the syllabic representations of moraic phonology:

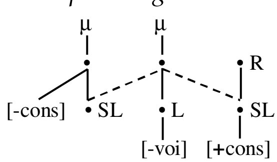
(13) a. *Gemination*:



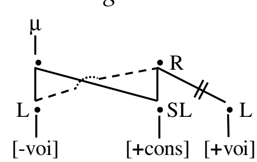
b. /s/-*Aspiration*:



c. *Pre-aspirated gemination*:



d. *Devoicing*:



Gemination in Pattern A is straightforward: it involves the spreading-and-delinking operation shown in (13a), whereby the mora affiliated to a coda /s/ is linked to the Root node of a following consonant; /s/-deletion is factored into the rule by the simultaneous dissociation of this segment from its original moraic unit. An alternative way to interpret this type of gemination is that it arises as compensatory lengthening (Campos Astorkiza 2003, 2007), whereby the deletion of coda /s/ leaves its mora available for reassociation to the following consonant.[6] As for Pattern B, Hualde (1989b) analyzes /s/-aspiration as debuccalization, formulated as delinking of all supralaryngeal features (13b). The resulting [h] provides the input for the creation of preaspirated geminates, generated when the supralaryngeal features are simultaneously spread from both the preceding vowel and the following consonant onto the aspirated segment, as in (13c). When the following consonant is a sonorant (cf. (11b) above), the phonetic manifestation of preaspiration is a delay in voice onset time within the initial portion of the geminate sonorant. Finally, the voiceless geminate fricatives in (12) are derived when voiced obstruents that have undergone (13c) are targeted by an additional operation that spreads the laryngeal features from the aspirated segment onto the Root node of the consonant to its right, delinking this segment's [+ voice] specification, as in (13d).

An alternative, and arguably simpler, characterization of the preaspirated geminates in (11) can be obtained under the assumption that in addition to [voice] the laryngeal features of fricatives include [+ spread glottis] (Vaux 1998): preaspirated geminates would arise when the floating [+ spread glottis] specification, set free by/s/-deletion, is reassociated to the laryngeal features of the following consonant. This is essentially the idea adopted in the OT accounts of Andalusian/s/ -aspiration and deletion in Morris (2002), Lloret (2008), and Lloret and Jiménez (2009), further supported by the fact that even when coda/s/is deleted, the [+ spread glottis] feature manifests itself by exerting a lowering (laxing) effect on the preceding vowel. Finally, the devoicing mechanism (14d), designed to account for the emergence of geminate voiceless fricatives in (12), can be dispensed with on the grounds that it overlooks the well-established fact that of all obstruents, geminate voiced fricatives such as [ββ], [ðð], [ɣɣ] are the most marked among the obstruents (Thurgood 1993; Kirchner 2001; Podesva 2002; Kawahara 2007). In general, voiced geminate fricatives are highly disfavored across languages due to the difficulty of maintaining a long turbulent airflow in the vocal tract needed for producing a geminate fricative while simultaneously increasing subglottal air pressure necessary to initiate vocal fold vibration (Ohala 1983; Kirchner 2001; Hayes and Steriade 2004). In sum, the failure of gemination to create voiced geminate fricatives in/s/-voiced obstruent sequences in (12) can be attributed to geminate markedness considerations rather than a devoicing effect induced by /s/-aspiration.

There are two basic OT approaches to gemination in Andalusian Spanish. The first one would derive plain geminates by domination of some sort of a CODA-COND disfavoring coda/s/, combined with a mora faithfulness constraint (MAX-μ) over input–output faithfulness (cf. Gerfen 2002: 199). As a slight variation of this analysis, under factorial typology, preaspirated geminates would emerge as optimal if IDENT-[+ spread glottis] dominates the CODA-COND (cf. Morris 2002). Alternatively, we might appeal to some version of *AGREE*, requiring that a sequence of two consonants agree for all features, constrained by a higher ranking of *IDENT-Onset* (demanding input–output identity of all features for onset consonants, thus enforcing total regressive assimilation), over feature faithfulness constraints. The derivation of preaspirated geminates would ensue if IDENT-[+ spread glottis] is ranked over *AGREE*.

Although perhaps the best studied, the Andalusian Spanish data in (11)–(12) is not the only complete assimilation process in contemporary Spanish. Another well-known instance of total assimilation affects coda liquids/l, r/, both within and across word boundaries, in the Cuban variety of Spanish spoken in Havana. Two distinct patterns of liquid assimilation have been reported by Guitart (1976, 1980, 2004). The first one, illustrated in (14), involves complete assimilation (*St. Sp.* = Standard Spanish; *Hav. Sp.* = Havana Spanish).[7]

(14)	*St. Sp.*	*Hav. Sp.*			*St. Sp.*	*Hav. Sp.*	
a.	cuerpo	cue[pp]o	‘body’	b.	cur[β]a	cu[bb]a	‘curve’
	culpa	cu[pp]a	‘blame’		sal[β]a	sa[bb]a	‘(s)he saves’
	carta	ca[tt]a	‘letter’		ar[ð]e	a[dd]e	‘it burns’
	saltar	sa[tt]ar	‘to jump’		fal[d]a	fa[dd]a	‘skirt’
	corcho	co[tt ʃ]o	‘cork’		ver lluvia	ve[ɟɟ]uvia	‘to see rain’
	colchón	co[tt ʃ]ón	‘mattress’		el yeso	e[ɟɟ]eso	‘the plaster’
	barco	ba[kk]o	‘boat’		pur[ɣ]a	pu[gg]a	‘purge’
	palco	pa[kk]o	‘stage box’		al[ɣ]o	a[gg]o	‘something’
		St. Sp.	*Hav. Sp.*			*St. Sp.*	*Hav. Sp.*
c.	marfil	ma[ff]il	‘ivory’	d.	hermano	he[mm]ano	‘brother’
	golfo	go[ff]o	‘gulf’		alma	a[mm]a	‘soul’
	ser rojo	se[rr̄]ojo	‘to be red’		verla	ve[ll]a	‘to see her’
	farsa	fa[ss]a	‘false’		carne	ca[nn]e	‘meat, flesh’
	bolsa	bo[ss]a	‘bag’		vulnerar	vu[nn]erar	‘to harm’
	surgir	su[hh]ir	‘to arise’		dar ñame	da[ɲɲ]ame	‘to give yam’
	el jamón	e[hh]amón	‘the ham’		el ñame	e[ɲɲ]ame	‘the yam’

According to Guitart (2004), this pattern was highly stigmatized in pre-revolutionary Havana and occurred mainly in the speech of individuals with a low socioeconomic/educational level. Since the Cuban revolution, it has gained prestige, especially among middle-aged and younger speakers, and at present is no longer distinctly bound to socioeconomic or educational status.

The second pattern, described in Guitart (1976, 1980, 2004; see also Harris 1984, 1985, and Kenstowicz 1994: 423–424), was typical of individuals with a higher education or socioeconomic status in pre-revolutionary times, and still survives in the speech of older individuals currently living in both Cuba and the United States. The additional complexities of the second pattern are as follows: (a) the liquids /r, l/ are turned into the retroflex when located in a syllable rhyme (*mar* → *ma*[ɖ] ‘sea’, *abril* → *abri*[ɖ] ‘April’); b) retroflexion spreads rightwards onto a following coronal consonant if this consonant is itself followed by a vocoid (*arde* → *a*[ɖɖ]*e* ‘it burns,’ *caldo* → *ca*[ɖɖ]*o* ‘broth’; *carta* → *ca*[ɖʈ]*a* ‘letter’; *alto* → *a*[ɖʈ]*o* ‘high-MASC.’; *carne* → *ca*[ɖɳ]*e* ‘flesh, meat’; *farsa* → *fa*[ɖʂ]*a* ‘farce’; *bolsa* → *bo*[ɖʂ]*a* ‘bag’; *el yeso* → *e*[ɟ̣ɟ̣]*eso* ‘the plaster;’ *corcho* → *co*[ɖʈ ʃ]*o* ‘cork;’ *colchón* → *co*[ɖʈ ʃ]*ón* ‘mattress;’ etc.). However, if the following coronal consonant is followed by /r/ (the only tautosyllabic consonant allowed after coronal obstruents in Spanish), assimilation in retroflection does not occur (*ser droga* → *se*[dd]*roga* ‘to be drug(s),’ *el dragón* → *e*[dd]*ragón* ‘the dragon’). Since liquids assimilate in PA and nasality to a following *heterorganic* consonant, geminates result whenever the following consonant is voiced (obstruent or sonorant), in a manner analogous to (14b–c). Retroflexion aside, the second pattern is further different from the first one in that coda liquids do not assimilate in voicing to a following consonant (cf. *cuerpo* → *cue*[bp]*o* ‘body,’ *culpa* → *cu*[bp]*a* ‘blame,’ *barco* → b*a*[gk]*o* ‘boat,’ *palco* → *pa*[gk]*o* ‘stage box,’ etc., in addition to the other relevant examples mentioned earlier to illustrate progressive retroflection). Another peculiarity of the second pattern is that PA assimilation does not take place at all when the following consonant is alveolar. Because /l, r/ are also alveolar, the condition that both consonants differ in all PA features is not met, as can be seen in the examples *carne* → *ca*[ɖɳ]*e* 'flesh, meat,' *farsa* → *fa*[ɖʂ̬]*a* 'farce,' etc., given earlier.

An account of gemination in (14), whether formulated in an autosegmental or an OT framework, would essentially follow the formal mechanisms outlined earlier for Pattern I in Andalusian Spanish, now limiting the target of gemination to the class of liquids. As for the second pattern, the reader is referred to Harris (1985) for specific details on this author's autosegmental analysis. It is not clear at this point how the complexities of the second pattern should be adequately handled in an OT approach, and thus the topic is left for future research.

An interesting question arises, however, in regard to the outcome of liquid-voiced obstruent sequences in both gemination patterns as to why the resulting geminate is not affected by spirantization. As we saw earlier, such sequences in Andalusian Spanish are realized as geminate voiceless fricatives instead of their highly marked voiced counterparts. In both patterns of gemination in Cuban Spanish, the potential generation of geminate voiced spirants is systematically averted by the mapping liquid-voiced obstruent sequences into voiced stop geminates. In Harris (1984, 1985) and Kenstowicz (1994: 423–424), this result is attained by appealing to geminate inalterability: once the geminate is derived by turning, for example, *arde* 'it burns' into *a*[dd]*e*/*a*[ɖɖ]*e*, the reasoning goes that the spirantization rule can no longer apply because only the first portion of the geminate satisfies its structural description. As remarked earlier, this approach is problematic since the spirantization rule does not include any reference to prosodic timing, an essential prerequisite for appealing to geminate inalterability. Here again, the failure of spirantization to affect the resulting geminate can be readily attributed to the highly marked status of voiced spirant geminates. From an OT perspective, this proposal is quite straightforward: assuming an undominated ranking of the markedness constraint banning voiced spirant geminates, in combination with a high enough ranking of IDENT-[voice], a potential surface form like **a*[ðð]*e* will be less optimal than its competitors *a*[dd]*e*/*a*[ɖɖ]*e*.

6 Conclusion

This chapter has offered a critical overview of a number of processes found in the segmental phonology of Spanish that involve partial and complete assimilation which have undoubtedly attracted some of the most intense theoretical interest and debate in recent decades, within both the serial (including autosegmental) and the constraint-based OT frameworks. The main purpose of this work has been to present a comprehensive view of the relevant data and discuss the most influential accounts available in the relatively recent literature in an attempt to reveal their insights, highlight their contribution to our understanding of a particular phenomenon, expose potential shortcomings, and in some instances, sketch some general guidelines for a constraint-based account, in line with much current theoretical research in phonology. It should be apparent from the discussion in this chapter that although past contributions have significantly improved our understanding of the segmental phonology of Spanish, many issues have not been adequately resolved, and still await future research.

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NOTES

1 Space limitations preclude the discussion of a more inclusive set of phonological phenomena in Spanish, for which the reader may consult standard treatises such as Zamora and Guitart (1982), D'Introno et al. (1995), Hammond 2001, or Hualde (2005).

2 In many (perhaps most) dialects, a syllable-initial glide undergoes consonantization, resulting in what is sometimes described as a 'hardened glide': *u*[ɲ#ɟ]*elo*, *u*[ŋ#gʷ]*evo* 'an ice, an egg.'

3 Amastae (1986) presents an alternative autosegmental treatment in which spirantization, which targets voiced obstruents in coda position, is fed by the application of a previous rule that renders them ambisyllabic. A problem for this approach, however, is that there is otherwise no independent motivation for ambisyllabicity in Spanish. Some phonologists have questioned whether lenition phenomena actually involve feature assimilation, as entailed by the approaches in (4), and propose an analysis in terms of movements along a scale of phonological strength, ultimately motivated by ease of articulation (Foley 1977; see Gnanadesikan 1997; Kirchner 2001, 2004 for OT reformulations of this idea).

4 The apparently exceptional status of /f/ is discussed in Martínez-Gil (2001).

5 Sequences of obstruents that agree in voicing underlyingly are omitted in (8) because VA here would be vacuous. The data in (8) reflects the pronunciation of many educated northern Peninsular Spanish speakers, including my own, in a moderately informal register, although some of the specific details of such data may not be shared by other speakers of Castilian Spanish. A pattern of VA analogous to (8) can be found in the variety of standard Mexican Spanish described in Harris (1969: 40–45).

6 Support for an interpretation in terms of compensatory lengthening is provided by an alternative to consonant gemination found in some /s/-deleting dialects (e.g., Cuban

Spanish), in which deletion results in lengthening of a preceding vowel, resulting in surface vowel quantity distinctions: *p*[e]*cado* 'sin' vs. *p*[e:]*cado* 'fish,' *p*[a]tillas 'sideburns' vs. *p*[a:]*tillas* 'pills,' *b*[u:]*que* 'boat' vs. *b*[u:]*que* 'I look for-SUBJ.,' *cas*[a:] 'house' vs. *cas*[a:] 'houses,' *v*[e:] '(s)he sees' vs. *v*[e:] 'you see' (see Resnick and Hammond 1975; Hammond 1978, 1986; Núñez-Cedeño 1988; Núñez-Cedeño and Morales-Front 1999). A similar process of compensatory vowel lengthening triggered by /s/-deletion is found in Andalusian Spanish (Alonso et al. 1950: 219; Lloret 2008).

7 A more restricted type of gemination targets /r/ before /l, n/ in Chilean Spanish (Oroz 1966: 138): /-rl-/ → [-ll-], as in *perla* → *pe*[ll]*a* 'pearl,' *burla* → *bu*[ll]*a* 'mockery,' and /-rn-/ → [-nn-], as in *carne* → *ca*[nn]*e* 'meat, flesh,' *horno* → *ho*[nn]*o* 'furnace.'

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7 Syllable Structure

SONIA COLINA

1 Introduction: basic concepts

1.1 Definition and significance

A syllable can be defined as a sound or sounds grouped around a peak of sonority or prominence. Despite its relatively late introduction as a unit of phonological organization (Hooper 1972; Kahn 1976), the syllable has played a critical role in phonological theory. A quick review of the phonological processes of Spanish (see Chapter 6) reveals that most require reference to the syllable and/or syllabic constituents. Furthermore, without reference to syllabic structure, one would fail to see that apparently unrelated segmental phenomena, such as nasal assimilation (e.g.,/kanbio/[ˈkam.bio] 'change'), nasal velarization (e.g.,/tren/[treŋ]'train'),/s/ aspiration and deletion (e.g.,/dos/[doh] [do] 'two'),/s/voicing assimilation (e.g.,/ isla/[ˈiz.la] 'island'), and obstruent deletion (e.g.,/atraktibo/[a.tɾa.ˈti.βo]), are all triggered by the inability of the coda to license specific featural specifications. The syllable is also relevant with respect to nonsegmental aspects of the phonology. For instance, stress is built upon syllables, and, consequently, many stress generalizations in Spanish, such as the *three-syllable window* and the *branching condition* (Harris 1983 and many others after him), cannot be formulated without reference to the syllable. In addition, phonotactic restrictions hold over syllables and syllabic components: for example, the ill-formedness of hypothetical **muersto* can only be explained as the result of an excessive number of segments in the rhyme (three being the maximum), given that/ue/and/rs/sequences are by themselves licit strings (Harris 1983: 9–10).

Syllabic structure figures prominently in the phonological competence of the native speaker: native speakers of all languages have intuitions about syllable structure and syllabification in their languages. In Spanish, these intuitions are generally rather strong and clear. This is not surprising as syllabification in Spanish is quite unambiguous, with a few exceptions to be addressed later in this chapter.

Given its wide scope, the aim of this chapter is to present an overview of the state of the art in Spanish syllabification rather than a detailed comparison of competing

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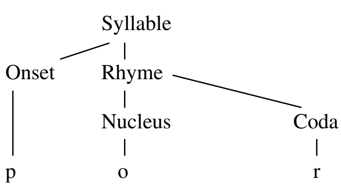
analyses (see Colina 2009 for an explicit comparison of serial and derivational accounts of syllabification). The content is organized as follows: after reviewing syllabic constituents and basic syllabification mechanisms in Section 1, sonority is discussed in Section 2; the presentation then covers specific syllabic components, onsets (Sections 3 and 4), nuclei (Section 5), and codas (Section 6); syllabification across words is considered in Sections 4 and 5; finally, Section 7 discusses the role of morphology in syllabification, in particular with respect to the coda position, morphological boundaries, and certain types of epenthesis.

1.2 Formal representation of the syllable: syllabic constituents

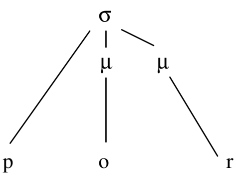
Syllables are comprised minimally of a nucleus, and maximally of an onset, nucleus, and coda. The nucleus and coda together are referred to as the rhyme. The nucleus is the peak or most sonorous part of the syllable; the less sonorous segments that precede the nucleus constitute the onset, and those following the nucleus are the coda (1). Under a moraic model of representation (for Spanish, see Dunlap 1991; Hualde 1994), the rhyme is the moraic component of the syllable, that is, the part that counts for stress computation; the nucleus is always dominated by a mora, whereas the coda contains elements that can be moraic (with their own mora) (2a) or nonmoraic, depending on the language (2b) (Hualde 1999a: 180).[1]

The onset comprises the nonmoraic segments preceding the nucleus. X-bar theory has also been used to represent syllabic structure (Hualde 1991). Under this model of syllabic representation, a nucleus projects three constituents: N, N′ and N″. N″ corresponds to the syllabic level. Onset segments are attached directly to N″, whereas nuclear and coda segments are attached to N and N′ respectively. N′ constitutes the rhyme (3).

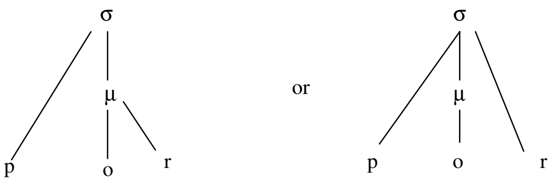
(1)



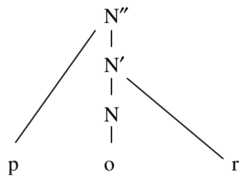
(2a)



(2b)



(3)



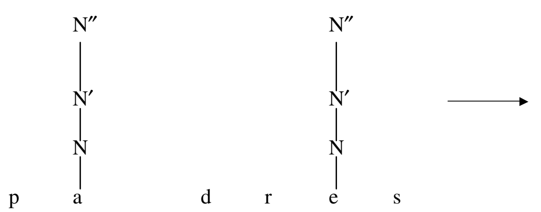
Serial and nonserial phonological approaches have been used to explain Spanish syllabification. Throughout this chapter, reference will be made to work framed in both. In a serial model of syllabification, syllabic structure is built by means of the ordered application of rules (Hualde 1991; Harris 1991), as shown in (4):

(4) Serial model of syllabification
1. Identify Nucleus: [-consonantal] segments project a syllable (N, N′ N″).
2. Complex Nucleus: attach a prevocalic glide to the left of the nucleus under the N′ node.
3. Onset Rule: attach a consonant to the left of the nucleus under the N″ node.
4. Complex Onset Rule: adjoin a second consonant to the left of the nucleus under the N″ node if the result is a well-formed consonantal cluster.
5. Coda Rule: adjoin a segment to the right of the nucleus under N′.
6. Complex Coda Rule: adjoin a consonant to the right of the nucleus under N′
7. Adjoin/s/: adjoin/s/under N′.

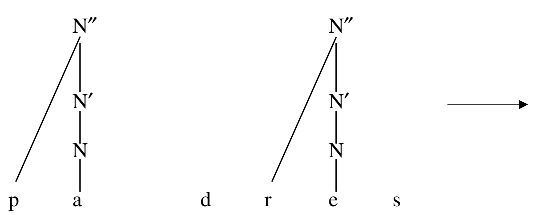
An illustrative derivation of the word/padres/is given in (5).

(5)

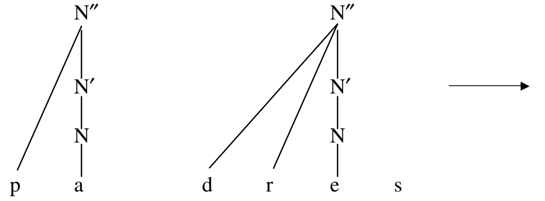
Rule 1



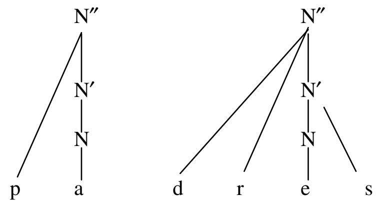
Rule 2 (N/A)
Rule 3



Rule 4



Rule 5



In a nonderivational model, such as Optimality Theory (OT) (McCarthy and Prince 1993; McCarthy 2002), phonological outputs, including their syllabification, result from the interaction of universal constraints ranked in a language-specific fashion. Constraints can be violated under domination of more highly ranked constraints. However, constraints can only be minimally violated, that is, violation only occurs when it is required by the need to satisfy a conflicting, more highly ranked constraint. *Gen* generates all possible analyses (candidates) for a given input, and the function *Eval* selects the optimal candidate, the one that best satisfies the language-specific ranking.

Constraints are of two types: faithfulness constraints require preservation of the input; and markedness constraints favor unmarked structure. Syllabification is seen as the result of a conflict between constraints requiring preservation of input segments (e.g., DEP-IO 'no epenthesis,' MAX-IO 'no deletion,' IDENTITY (x) (IDENT for short) 'the specification for feature x in the input must match that of the output'), and constraints banning marked syllable structure (e.g., ONSET 'syllables must have onsets,' *CODA 'syllables cannot have codas,' *COMPLEX 'No complex syllabic components'). For example, given the Spanish input /ala/, the ranking ONSET over DEP-IO guarantees the output ['a.la] with no initial epenthesis or deletion, despite the onsetless first syllable. ONSET also selects syllabification of the intervocalic consonant with the second vowel, rather than *['al.a].

2 Sonority and syllabic structure

Sonority, a complex notion that loosely described corresponds to perceptual prominence and/or degree of stricture, plays an important role in how segments are organized into syllables. As mentioned above, the syllable is a sound or sounds grouped around a sonority peak. The relative sonority of phonological segments is encoded in a universal hierarchy (6), where phonological classes are given a ranking by means of a relative sonority index:

(6)

Stops <	Fricatives <	Nasals <	Liquids <	Glides <	Vowels
1	2	3	4	5	6

Syllables exhibit a contour based on the universal sonority scale according to which the syllable rises in sonority towards the nucleus, the sonority peak, and then decreases towards the coda (*Sonority Sequencing Principle* Clements 1990). The rise tends to be maximal from the onset to the nucleus, and minimal in the transition from the nucleus to the coda.

While the relative sonority of one phonological class versus another is universal (e.g., fricatives are always more sonorous than stops), the number of sonority-based classes in the sonority scale varies cross-linguistically; for instance, some languages group stops and fricatives together in the sonority class of obstruents, while for others, the stops must be differentiated from fricatives in terms of sonority. For Spanish, Harris (1989a, 1989b) proposes a sonority hierarchy with three sonority classes: obstruents, nasals, and liquids. However, on the basis of the phonotactics of onset clusters (see Section 3), Martínez-Gil (2001) argues for a subset of distinctions that separates obstruents into stops and fricatives: Stops < Fricatives < Nasals < Liquids < Glides < Vowels.

As the reader can infer from the previous discussion, serial analyses of syllabification, for which syllable structure is the result of the application of a series of ordered rules, such as (5), incorporate sonority into syllabification by imposing sonority conditions on specific segments referred to in syllabification rules. However, sonority scales cannot be formalized into a rule-based theory of the syllable, appearing in the form of separate, external mechanisms and conditions, which are needed in addition to rules. Colina (2009) shows that a nonderivational model of phonology, such as OT, is able to formalize the generalizations that relate syllabic positions to sonority classes by means of the harmonic alignment of two prominence scales, namely, the universal sonority scale and the syllable-position prominence scale (Nucleus > Coda > Onset) (7). The syllable-position prominence scale combines with the sonority scale to produce a constraint hierarchy (McCarthy 2002: 21): the most prominent syllable position (Nucleus) is aligned with the most sonorous segment (Vowel) and vice versa, so that the most highly ranked constraint is the one that penalizes parsing the most sonorous segment in the least prominent position (8). A constraint hierarchy ranks constraints so that the most highly ranked constraint is the least likely to be violated. Thus, if the highest ranked constraint penalizes the association of a vowel with onset, all other segments will make better onsets than a vowel (8):

(7) Sonority as alignment of syllabic positions with sonority scales

Nucleus >….. > Onset
| |
Vowel >…..> Obstruent

(8) *NUC/obstruent ≫ *NUC/nasal ≫ *NUC/lateral ≫ *NUC/glide ≫ *NUC/vowel
*ONSET/vowel≫*ONSET/glide≫*ONSET/lateral≫*ONSET/nasal≫*ONSET/obstruent
*CODA/obstruent≫*CODA/nasal ≫ *CODA/liquid ≫ *CODA/glide ≫ *CODA/vowel

Sonority also plays a crucial role in the well-formedness of onset clusters: generally, in Spanish, two onset consonants must exhibit the maximal sonority distance possible among consonants (see Section 3).

3 Onsets and onset clusters

Across languages, the least marked syllable type consists of a consonantal onset and a vocalic nucleus, CV (e.g., *ma.no* 'hand'). Closed syllables are universally marked with respect to open syllables. Accordingly, an intervocalic consonant is universally syllabified with the second vowel, as this configuration results in two open syllables. Syllabifying the intervocalic consonant with the first vowel, *VC.V (e.g., **man.o*), would produce a closed syllable; in addition, the second syllable would be onsetless and therefore more marked than in V.CV. Serial models of syllabification capture these facts by means of a Nucleus Rule that projects a nucleus from each vowel and an Onset Rule that attaches a consonant to the left of the nucleus (Hualde 1991). In order to obtain the correct syllabification, namely V.CV (vs. *VC.V), the Onset Rule must apply after the Nucleus Rule (also after the Complex Nucleus in Spanish) and before the Coda Rule that attaches a consonant or glide to the right of the nucleus. In a nonserial model of syllabification, the combined effect of a universal constraint that requires that syllables have onsets (ONSET) and another that bans codas (*CODA) results in the syllabification of an intervocalic consonant with the second vowel. As argued by Colina (2009), one of the advantages of the optimality theoretic account is that it captures the universal preference for onsets by means of a universal constraint; reference to a universal rule (i.e., the Onset Rule) in a serial account in which rules are language specific remains problematic.

Onsets are not required in Spanish: the minimal syllable consists of a vocalic nucleus (e.g., *u.na* 'one'). In other words, onsetless syllables are not repaired through epenthesis or deletion. However, as shown in Section 4, a word-initial onsetless syllable can acquire an onset through *resyllabification* of a word-final consonant or through *syllable merger*, a process through which a word-final vowel followed by a vowel-initial word becomes a diphthong (Section 5). Stops, fricatives, nasals, and liquids are possible onsets in Spanish, but not glides or vowels. In other words, all [+ consonantal] segments can generally be parsed in the onset. Prevocalic glides that are not preceded by a consonant surface as [+ consonantal] in most dialects (9c–d):

(9)	a. perd-er	'to lose'	perd-[ˈi̯e]ron	'they lost'
	b. com-er	'to eat'	com-[ˈi̯e]ron	'they ate'
	c. cre-er	'to believe'	cre-[ˈʝe]ron	'they believed'
	d. o-ir	'to smell'	o-[ˈʝe]ron	'they smelled'

The process in (9c–d) has been interpreted as a rule of *onset strengthening* or *glide consonantization*, which turns a [−consonantal] segment into [+consonantal] for reasons of sonority, as a glide would be too sonorous for the onset position in most Spanish dialects (Hualde 1989a, 1991, 1997; Harris and Kaisse 1999; Colina 2009).[2] In an OT model, the consonantization facts are explained as the violation of an Ident constraint imposed by the satisfaction of the highly ranked Onset constraint; Ident is violated because the [−consonantal] specification of the input does not match that of the output [+consonantal] (Colina 2009).[3]

A series of two consonants in word-medial position is preferably syllabified as an onset cluster, as long as the resulting sequence constitutes a well-formed cluster. Possible onset clusters are those that can appear word initially; that is, a stop or/f/ followed by a liquid. The cluster */dl/is not possible in any dialect, while/tl/is acceptable in some varieties of Spanish, such as Mexican Spanish, and most varieties of Latin American Spanish. The ill-formedness of *dl and *tl is related to dissimilarity requirements, as the two consonants have the same point of articulation, and, in the case of *dl, the consonants also agree in voicing. The preference for syllabifying as many consonants as possible in the onset (vs. coda + onset) is known as *onset maximization*.

As mentioned above, the main principles governing the well-formedness of onset clusters are related to sonority: the first member of the cluster must be a member of the set of the least sonorous onsets, and the second member is drawn from the class of most sonorous onsets. In other words, a complex onset in Spanish consists of at most two consonants that differ maximally in sonority. This principle, often known as the Maximal Sonority Distance (Colina 2009), Minimal Sonority Distance (Martínez-Gil 1997), or the Complex Onset Condition (Martínez-Gil 2001), is in accordance with Clements' Sonority Cycle (1990), which states that the rise in sonority is maximal from the onset to the nucleus and minimal from the nucleus to the coda. The only fricative that is allowed as the first member of the onset cluster is/f/. Martínez-Gil (2001) argues that/f/ shares one characteristic with stops: namely, the lack of specification for [+continuant]. Stops are [−continuant], and/f/, although [+continuant], does not need to be underlyingly specified as such, as no language uses the feature [continuant] contrastively in labiodentals. He concludes that the underlying presence of [+continuant] in fricatives other than/f/contributes to sonority in Spanish, thus making these segments (except for/f/) too sonorous to be the first consonant in a cluster. In a serial account, a Complex Onset Rule attaches a second consonant to the left of an onset consonant; in order to obtain only possible clusters, this rule necessitates a condition that specifies that the rule will apply as long as the resulting cluster is well formed (Hualde 1991). The Complex Onset Rule is ordered before the Coda Rule (4).

In OT, onset maximization is the result of the domination of a constraint that bans codas (*CODA) over one that prohibits complex onsets (*COMPLEX N); this serves to eliminate syllabic markedness (closed syllables) at the expense of the onset, as long as it does not violate sonority requirements for the onset. Sonority requirements on the onset are captured by the MSD (Maximal Sonority Distance) constraint (Martínez-Gil 1997; Colina 2009). Note that the fact that spirantization turns the cluster-initial obstruent into an approximant constitutes a violation of MSD, as in the output, the cluster is made up of an approximant + liquid; as a result, the constraints responsible for spirantization must be more highly ranked than MSD. Chilean Spanish exhibits a different behavior, as it avoids MSD violations in the output by syllabifying the underlying obstruent in the coda; this underlying obstruent surfaces as a coda glide (vocalization) to avoid a coda obstruent (Martínez-Gil 1997) (see Chapter 5). Unlike other languages such as English, Spanish does not allow /s/ + obstruent onset clusters, which are avoided through epenthesis in word-initial position and word-medially, by a heterosyllabic parsing (/s/ in the coda and the obstruent in the onset) (e.g., *ins.cri.bir* 'to inscribe,' *e̲s.cri.bir* 'to write'). This is an active process of epenthesis as demonstrated by loans (e.g., *stop* [es.ˈto], *slogan* [ez.ˈlo.ɣan]).

4 C#V and C#CV sequences across words

Section 3 focused on the word-internal environment; however, some of the processes mentioned there can be found at work across word boundaries as well. The principle according to which an intervocalic consonant syllabifies with the second vowel also applies across words: when a word-final coda is followed by a vowel-initial word, the word-final consonant is syllabified as the onset to the word-initial vowel, *la.s o.tras* 'the others.' The consequence is that the first syllable of the second word is no longer onsetless. This phenomenon is traditionally known as *resyllabification*. Some derivational accounts of resyllabification include the proposal of a rule of resyllabification (Harris 1983) (10) and a second postlexical application of the Onset Rule (Hualde 1989a, 1991).

(10) Resyllabification

$$\begin{array}{c}[+\text{cons}]\\ | \\ \text{R}\end{array} \longrightarrow \begin{array}{c}[+\text{cons}]\\ | \\ \text{O}\end{array} \;/\; \_\_\_\_\_\_\; \#\, \text{V}$$

In contrast with a word-final consonant followed by a vowel-initial word, a word-final consonant followed by a consonant-initial word fails to resyllabify, even if the resulting cluster would be well formed. One can infer that, since the second word already has an onset, resyllabification is unnecessary; yet a rule-based model cannot incorporate this insight into its account of resyllabification. The rule-based account states that, unlike the Onset Rule which applies lexically and postlexically, the Complex Onset Rule only has a lexical application (Hualde 1991); or, alternatively, that the rule only applies when the consonant is followed by a vowel (9) (Harris 1983). However, this does not explain why the Complex Onset Rule needs to be restricted to the lexical environment; in other words, what is the crucial difference between the Onset Rule and the Complex Onset Rule with regard to the across-the-word context? Why is a closed syllable preferred to a complex onset in the across-the-word environment, but not word-medially? An OT analysis (Colina 1995, 1997, 2009) brings forth an important insight: at stake is the interaction between conflicting tendencies; the preference for onsets and the dislike for codas, on the one hand, and the need to align syllabic and morphological constituents, on the other hand. Misalignment of the syllable and the word is sacrificed in resyllabification to obtain an onset for an otherwise onsetless syllable; but, when the word is consonant-initial and the syllable is not onsetless, there is no need to misalign the word and the syllable.

5 Nuclei and complex nuclei: diphthongs, vowels, and glides

Only vowels are possible nuclei in Spanish. Sequences of nonconsonantal segments are well formed in Spanish, yet their syllabification is possibly the one area of Spanish syllable structure that entails the most difficulties for the analyst. It is also the area in which native speaker intuitions are less clear.

Spanish exhibits a clear preference for syllabifying a sequence of two vocoids in the same syllable, as a *diphthong*, consisting of a glide ([−vocalic] vocoid), and a vowel ([+vocalic] vocoid). (The term *vocoid* will be used herein to refer to the category of nonconsonantal segments that includes vowels and glides). The glide can be the first or the second segment in the diphthong, resulting in a rising diphthong (e.g., *m*[ˈje]*do* ‘fear’), or a falling diphthong (e.g., *p*[ˈej]*ne* ‘comb’), respectively.

There is general agreement in the phonological literature that postconsonantal prevocalic glides are part of the nucleus and postvocalic glides are in the coda. Some strong arguments in favor of the nuclear status of prevocalic glides are (1) some mid vowels alternate with diphthongs in stressed position (e.g., *v*[e]ˈ*nimos* ‘we’re coming,’ *v*[ˈje]*nen* ‘they come’). Since diphthongization is dependent on stress and stress relies on syllabification (Harris 1985), the source element [e] must be in the nucleus when diphthongization occurs. For the resulting glide to be parsed in the onset, an additional rule would be required to move it to the onset; yet no independent motivation for this rule that moves a glide to an already filled onset is known (Harris and Kaisse 1999: 128). (2) A rhyme in Spanish can contain at most three segments; a syllable can have up to five segments, if two of those are in the onset (Harris 1983) (e.g., *clien.te* ‘client’). Since the sequence [ˈmwe] itself is well formed, as seen in *muerte* ‘death,’ the ill-formedness of **muersto* can only be explained by assuming that the postconsonantal prevocalic glide in **muersto* is in the rhyme (thus in the nucleus, given its prevocalic position) (Harris 1983) (3) Spanish has one well-known lexical restriction on identical vocoids in the rhyme, *ii, *uu *ji, *ij, *wu, *uw (Harris 1983). Therefore, the two high front segments in **escritori-ito* 'desk' (dim), being ill-formed, must both be in the rhyme (nucleus), rather than in the onset and rhyme, respectively. A postvocalic high vocoid, as in *tramoy-ista* 'trickster,' is acceptable because it occupies the onset, as demonstrated by its consonantization (Harris and Kaisse 1999: 128). (4) In hypochoristic formation, segments parsed in the onset cannot become nuclear in the hypochoristic (e.g., *Petronio*, *Petro *Petr*). Prevocalic glides, however, are copied as a full vowel in nuclear position (e.g., [da.ˈnjel] [ˈda.ni]) (Colina 1996; Hualde 1999a).

The syllabic affiliation of postvocalic glides is not as problematic as that of prevocalic glides: whether they are part of the nucleus or the coda, postvocalic glides are in the rhyme and have similar status regarding other phonological processes such as stress computation. Yet there is at least one strong argument in favor of the coda affiliation of postvocalic glides. Only sonorants or [s] are allowed after a coda glide (e.g., *vein.te* 'twenty,' *béis.bol* 'baseball'). Since coda obstruents other than [s] are acceptable in many varieties of Spanish (e.g., *ob.so.le.to* 'obsolete,' *dig.no* 'worthy'), the only way to explain their ill-formedness after a glide is to assume that the glide is the first segment of the coda and that the restrictions on the consonant following it are in fact restrictions on coda clusters.

In most dialects, sonority is the criterion that determines which vocoid becomes the nonvocalic part of a diphthong. The vowel category in the sonority scale presented in (6) must be subdivided in high vowels, mid vowels, and low vowels, from lowest to highest sonority. Generally, the most sonorous vowel is the nuclear (nonglide) vocoid. Similarly, the least sonorous vocoid, namely a high vocoid, makes the best glide, unless it bears stress. A stressed high vocoid followed or preceded by any vocoid is syllabified in two syllables (*hiatus*) (e.g., *día* [ˈdi.a] 'day'). If no high vocoid is available, the sequence will be normally pronounced in hiatus in careful speech (e.g., *teatro* [te.ˈa.tro] 'theater'), but it can become a diphthong in fast and/or casual speech, in which case the next most sonorous vocoid (mid) will be selected for gliding (e.g., [ˈte̯a.tro]). A sequence of two identical vocoids is normally reduced to one (e.g., *alcohol* [al.ˈkol] 'alcohol'), unless in very careful speech in which they are pronounced in hiatus (e.g., [al.ˈko.ol]). A low vocoid is rarely (if ever) nonmoraic. Any hiatus can become a diphthong postlexically, across words, in casual or fast speech. Serial analyses account for diphthongs through a gliding or denuclearization rule that delinks the mora attached to an unstressed high vowel in contact with another nonhigh vowel (Hualde 1994; Harris and Kaisse 1999). The Complex Nucleus Rule or the Coda Rule will then apply to the resulting nonmoraic vocoid.

Diphthongs are the result of avoiding an onsetless syllable by creating a complex nucleus or an additional coda, depending on the sonority of the vocoids. An OT account captures this through the ranking: ONSET ≫ *COMPLEX NUCLEUS ≫ *CODA. The generalization regarding selection of the target for gliding can be obtained in a straightforward manner through a hierarchy of constraints, shown in (11), that requires the association of moras with vocoids on the basis of sonority. The constraint requiring association of the most sonorous vocoid with a mora is the most highly ranked: low/μ ≫ mid/μ ≫ hi/μ.[4]

(11) Glide formation: high glides regardless of position
low/μ, ONSET,≫ mid/μ ≫ hi/μ, *COMPLEX NUCLEUS ≫ *CODA

One dialect that does not resort to sonority to select the target for gliding is Chicano Spanish. The main criterion is position within the syllable: the first vocoid is the target for gliding, generally across words (cf. Martínez-Gil 2000 for data and serial and OT analyses; Colina 2009 for an OT analysis).

As mentioned above, diphthongization and gliding (also known as *syllable merger*) apply across words; the motivation for this process becomes clear in an OT account. In the across-the-word environment, a diphthong is the result of avoiding an onsetless syllable by creating a complex onset or an additional coda *and* misaligning the word and syllable, as shown by the ranking in (12). This explanation allows the OT analyst to formalize the connection between resyllabification and syllable merger; however, a serial model, for which rules are language-specific, would have to resort to principles that would have to be considered universal (i.e., Onset Maximization for resyllabification; a glide is [+high] as a default, etc.).

(12) Glide formation across words: no low glides
low/μ, Onset ≫ mid/μ≫ hi/μ, *COMPLEX NUCLEUS, *CODA, ALIGN-LST [5]

Although a sequence of two vocoids in which one of the two is high and unstressed normally surfaces as a diphthong, a few exceptions exist, giving rise to a few near-minimal contrasts, for example, *dien.te* 'tooth' vs. *cli.en.te* 'customer'; *dia.go.nal* 'diagonal' vs. *di.a.blo* 'devil'. These data acquire relevance within the wider context of the phonetic status of glides: does Spanish have a contrast between glides and vowels? Are glides underlying and therefore part of the phonemic inventory of Spanish? If not, how can one explain the exceptional forms? For the purposes of the present chapter, this issue bears on whether glides and vowels can be obtained from general principles of syllabification, or if, on the other hand, syllabification information can be contrastive.

The position espoused here is that these contrasts are exceptional (i.e., lexically determined) and that, therefore, Spanish does not have an underlying contrast between glides and vowels. This is in agreement with work by Hualde (1997, 1999b) and Roca (1991, 1997, 2006) and contra Harris (1969, 1989a) and Hualde (1991, 1994). This view is based on the fact that many exceptional hiatuses can be explained through: (a) analogy, when a high vocoid is stressed in a morphologically related word (e.g., *v*[i.ˈa.]*ble* 'viable,' cf. *vía* 'way'; see Colina 1999 for an OT analysis); (b) the presence of intervening morphological boundary, such as a prefix or suffix (*bi-* ,*-oso*, *-al*, *-ario*) (e.g., *man*[u.ˈa]*l* 'manual'). Those hiatuses that cannot be accounted for through analogy and/or morphological factors constitute a small list that lends itself well to exceptional treatment. Hualde (1999b) has shown that exceptional hiatuses are not randomly distributed in the lexicon and that some tendencies can found: (a) exceptional hiatuses are almost always word-intial (e.g., *ci.á.ti.ca* 'sciatica'); (b) stress falls on second vowel of the hiatus or on the following syllable, but not any further to the right (e.g., *di.á. blo* 'devil,' *di.a.fráɡ.ma* 'diaphragm,' but *dia. pa.són* 'tuning fork'); (c) the tautomorphemic sequences *ie*, *ue* are rare as hiatus; (d) exceptional hiatuses are never of falling sonority; and (e) there is no hiatus involving /uV/ (e.g., *cua.tro* **cu.atro* 'four,' after a velar). If there were a true contrast between vowels and glides, one would expect a more random, general distribution of hiatuses, affecting both rising and falling sequences.

Finally, it must be noted that some traditional arguments in favor of the glide/vowel contrast are framework-specific, as they are crucially dependent on rule ordering: for instance, it is argued that for */sá.u.ri.o/[ˈsaw.ɾjo] 'saurus' not to violate the three-syllable window, according to which stress in Spanish cannot be placed further to the left than the antepenultimate syllable, one of the two glides needs to be underlying (Roca 2006). This argument, however, does not apply in a parallel model in which only the output is evaluated, as [ˈsaw.ɾjo] is stressed on the penultimate syllable and does not violate the three-syllable window (Hualde 1997). Similarly, some researchers (Harris 1969, 1989b) have argued that, since the present indicative is always stressed on the penultimate, the difference between [am.ˈpli.o] 'I widen' and [ˈkam.bjo]'I change' can only be explained through a contrast in the underlying representation/anplio//kanbjo/. This does not present a problem for analyses that assume that stress is a lexical property of words (Hualde 1997) or those that evaluate outputs, such as OT, since [am.ˈpli.o] and [ˈkam.bjo] carry stress on the penultimate syllable.

Exceptional hiatuses have been formalized in serial analyses as marked exceptions to the gliding rule, in other words, as exceptions to the rules of syllabification (Hualde 1997). In agreement with Roca (1997) and Harris and Kaisse (1999: 133), I contend that what is exceptional is not whether a rule is prevented from applying, nor that a high vocoid is marked as nonsyllabic, but that high vocoids are marked as syllabic. In other words, only the exceptional high vocoids are marked (as syllabic). The lack of contrast between vowels and glides is a reflection of their being specified only as [−consonantal]; syllabification and sonority constraints select a glide or a vowel, moraic or nonmoraic in the output, unless a mora (μ) (alternatively, N(ucleus), [+syllabic], or syllable head, as in Harris and Kaisse 1999) is exceptionally present in the input (exceptional hiatus). This account is preferable to an exception to the gliding rule because it takes advantage of the fact that it is precisely nonpredictable (exceptional information) that appears in the underlying representation, rather than having to mark exceptionality in rule application.

6 Coda consonants and coda restrictions

Word-internally most Spanish dialects allow the following in the coda: nasals with the same point of articulation as the following consonant (e.g., *cambio* [ˈkam.bio] 'change');/l/or/r/(neutralized in some Caribbean dialects) (e.g., *malta* 'malt,' *marca* 'mark');/s/(e.g., *pista* 'clue'); and/θ/(e.g., *pi*[θ]*ca* 'pinch') in dialects with /θ/in their inventories. Coda oral stops appear in some dialects in formal registers and under the influence of spelling; they tend to be neutralized with regard to the features [voice] and [continuant], hence there is no contrast possible between the coda obstruents in *concepción* 'conception' and *obsesión* 'obsession'; *étnico* 'ethnic' and *administrar* 'to administer'; *técnica* 'technique' and *dogmático* 'dogmatic' (Alonso 1945; Hualde 1989b). In word-final position, only coronal consonants are possible: the sonorants /-l/, /n/, and /-r/ (*papel*, *camión*, *amor*) and the obstruents /-s/, /θ/ (*mes*, *pez*), and /d/ (virtud).[6] The phoneme /d/ can be realized as [ð], [θ], [t] or be deleted. Other consonants are rare and may appear in unassimilated borrowings such as *club*, *frac*, *bulldog*, *album*, *chef* (Hualde 1999a). Although word-medial stops and fricatives are acceptable in some dialects, as shown above, word-final obstruents are unusual in all dialects (with the exception of /s/).

As mentioned before, closed syllables are marked, as languages exhibit a universal tendency to avoid codas. This universal dislike for codas surfaces also in languages that permit them, like Spanish, in the form of significant restrictions on what segments can appear in this position. Codas, like other syllabic components, are subject to the sonority contour: since sonority decreases from the nucleus, with the decrease being minimal in the transition from the nucleus to the coda, generally, glides make the best codas, followed by liquids, nasals, and then obstruents (fricatives and stops in that order). According to this hierarchy, obstruents are the worst codas and the most likely to be deleted, as is in fact the case in many varieties of Spanish.

In addition to the sonority contour, coda restrictions affect individual segments and their features: the coda is known not to license certain features and/or feature combinations, such as place of articulation and voice. This inability to license segmental features can result in featural deletion or in parsing of the illegal feature through a contiguous segment that is not in the coda. Examples of this in Spanish are: (1) nasal and lateral assimilation, where it can be argued that assimilation results from a doubly linked PA node, which is thus licensed through the onset, managing to escape coda restrictions without being deleted; and (2) /s/ voice assimilation, where voicing is a consequence of the association of the laryngeal node of the coda /s/ to that of a following voiced onset segment.

Piñeros (2006: 147) shows that in nasal place assimilation, as well as in the processes of neutralization, velarization, and absorption found in some dialects, the constraints AGREE(place), PLACE HIERARCHY and ALIGN-C(nasal) "conspire to undermine place features of nasal consonants in the syllable coda." ALIGN-C(nasal) is the alignment equivalent of *CODA: it requires all consonants, and in this case, the feature nasal, to be aligned with the left edge of the syllable; thus in order to satisfy ALIGN-C, all consonants should be in the onset and none in the coda. Piñeros' optimality theoretic analysis reveals one reason why some features are not licensed in the coda: deleting or not parsing some features partially satisfies the ALIGN-C constraint (*CODA in our analysis); a partially parsed coda is a better coda than a fully specified one, which would have more features in the coda position that are not aligned with the onset. Fewer features in the coda mean neutralization of contrast in the coda, where contrast preservation is not as important as in prosodically prominent positions, such as the onset.

Coda licensing restrictions also interact with context-free segment markedness: for instance, voiced obstruents are marked due to the difficulty of combining vocal cord vibration and oral cavity constriction; thus, in dialects where they are not altogether deleted, coda obstruents become less marked through devoicing or by parsing the [+voice] feature through a following voiced onset segment; continuancy is also neutralized, and obstruents often surface as fricatives or approximants because they make better codas than stops. Some dialects, such as Chilean, exhibit obstruent vocalization in the coda, a process that can be understood as replacing a highly marked coda obstruent with the least marked coda, a glide (Martínez-Gil 1997; Piñeros 2001). In an OT account, this is the result of changing the [+consonantal] feature of the input into [−consonantal] in the output (a violation of IDENT(consonantal)).

Existing serial analyses of the coda processes mentioned above (i.e., nasal assimilation,/s/aspiration, voicing assimilation, obstruent devoicing, and spirantization) consist of separate rules for each process, each including the coda as part of its structural description. This approach misses the generalization that it is the marked status of the coda (as a syllabic constituent) that triggers the apparently unrelated processes. In other words, derivational analyses fail to capture the connection between the above processes and the markedness of codas, resulting in a tendency to eliminate codas or replace them with less marked ones (by deleting features, modifying them, or parsing them through another position, etc.). These connections are brought to the foreground in an OT model equipped with a *CODA constraint (or ALIGN-C), Faithfulness constraints (IDENT, MAX), a sonority hierarchy, and segmental markedness constraints that ban marked featural combinations (e.g., *OBSTRUENT VOICE) (Piñeros 2001; Colina 2009, among others). (For more on segmental processes in the coda see Chapter 6.)

Spanish allows some coda clusters: glide + consonant (e.g., [ˈbejn].*te* 'twenty,' [awk.ˈsi].*lio* 'help'); sonorant +/s/(e.g., *pers.pec.tive* 'perspective,' *ins.taurar* 'institute,' *sols.ticio* 'solstice'); obstruent +/s/(e.g., *abs.tener* 'abstain' *ads.cribir*, 'ascribe'). Nasal +/s/and obstruent +/s/clusters are often reduced to only/s/ in casual and informal speech; this is expected in the case of the sequence obstruent +/s/, as the/s/is more sonorous and therefore a better coda than the obstruent. However, sonority cannot be considered the criterion governing reduction in sonorant +/s/clusters, as nasals are more sonorous than/s/and should be retained. A unified explanation can be found in the proposed cross-linguistic status of/s/as an adjunct (e.g., cf. for Italian, Davis 1990; Ancient Greek, Steriade 1988). Evidence for the adjunct status of/s/lies in the behavior of s +obstruent word-initial clusters, which undergo epenthesis (e.g., *inscribir* vs. *escribir* 'inscribe' vs. 'write'), while other word-initial clusters tend to be repaired through deletion (e.g., *psicología*/sikoloxia/'psychology'). If/s/is an adjunct with special status, possibly on the basis of its phonetic salience, one can understand its preservation through e-epenthesis. The adjunct account also sheds light on coda-cluster restrictions in Spanish; if/s/is an adjunct with special status and can thus be considered exceptional, the relevant generalization is that Spanish does not allow coda clusters.

7 Syllable structure and morphology

7.1 *Morphology and the coda position*

Morphology often interacts with syllabification. In some dialects, processes that affect the coda, such as nasal velarization and /s/ aspiration, can occur in onset prevocalic positions as the result of the resyllabification of a word- or prefix-final consonant (i.e., across a prefix or word boundary, but not a suffix): for example, *des+hecho* [de.ˈhe.tʃo] 'undone,' *las alas* [la.ˈha.lah] 'the wings'; *bien* [ˈbjeŋ] 'good,' *bienes* [ˈbje.nes] 'good-pl,' but *bien+estar* [bje.ŋes.ˈtar] 'well-being,' *in+humano* [i.ŋu.ˈma.no] 'inhumane' (Hualde 1989a, 1991; Colina 2009). This has been explained as a consequence of rule ordering. For instance, a form like [de.ˈhe.tʃo] reflects the order: aspiration, prefixation, and resyllabification. Optimality theoretic analyses see these facts as the attempt to reduce allomorphy (i.e., preserving identity and satisfying IDENT constraints) (Colina 1999, 2006; Wiltshire 2006) by having, for example, [deh] as the only allomorph of the prefix /des/, regardless of its context. Dialectal variation has also been reported (Kaisse 1997, 1998) with regard to prefixes (e.g., [de.ˈhe.tʃo] vs. [de.ˈse.tʃo]). Rule-based models argue for differences in the order of the rules across dialects, while constraint-based analyses ascribe variation to the prosodic nature of prefixes (which in some dialects are said to behave like prosodic words, requiring identity of their output form in various contexts) and to variation in the ranking of the constraints.

7.2 *Epenthesis and morphology*

Another area in which morphology becomes relevant for syllabification is in regard to some types of so-called epenthesis. Generally in Spanish, an illegal coda or coda cluster in word-final position undergoes deletion: for example, *club* [ˈklu] *[ˈklu.βe] 'club,' *chalet* [tʃa.ˈle] *[tʃa.ˈle.te] 'chalet.' However, in a significant number of *-e* final words, *-e* (unlike *-a* and *-o*) follows illegal consonants and consonant clusters (e.g., *nube* 'cloud,' *parte* 'part'), but it is less common after the set of consonants allowed in word-final position /d, l, n, r, s, θ/ (e.g., *prole* 'offspring,' *sede* 'seat'). This has been traditionally interpreted as evidence for word-final epenthesis (Harris 1986a, 1986b, 1991, 1999; Colina 1995, among others), the main argument being that final epenthesis captures an important generalization regarding the syllable structure of Spanish: namely, that /tʃ, x, p, t, k, b, g, f/ are not syllabifiable word-finally. On this basis, Harris (1999) assigns words ending in *-e* preceded by an unsyllabifiable consonant and those ending in a syllabifiable consonant to a special morphological class. While membership in the class is not predictable, the distribution of *-e* or no terminal vowel is predictable for the members of the class, with *-e* appearing after consonants or consonant clusters which would be unsyllabifiable without it. Consequently, Harris (1999) argues for a rule of "morphological" epenthesis that inserts *-e* after an unsyllabifiable word-final consonant in the location of a terminal element; this is considered "morphological" epenthesis because the rule needs to know the location of terminal vowel (Harris 1999). Others (Morin 1999; Colina 2003; Bonet 2006) contend that the presence word-final *-e* after illegal consonants or consonant clusters stems from a rule of epenthesis that was active at an earlier stage in the history of the language, but that no longer is, deletion being preferred instead. The true generalization is not about the rule of epenthesis, but about illegal coda consonants and clusters, which Modern Spanish, in contrast with earlier varieties, avoids through deletion.

Plural epenthesis has also been argued by some to be ruled by morphology (Colina 2003, 2006; Bonet 2006). Traditional accounts of plural formation in Spanish analyze [e] in the [es] allomorph as the result of phonological epenthesis, needed to repair the unsyllabifiable cluster that results from attaching plural /s/ to a final consonant (e.g., *tren, tre.nes* *[tɾens] 'train/s'). More recently, however, Colina (2006) has argued that plural [e] cannot purely be an effect of phonological epenthesis, as there are examples of native words where the supposedly "ill-formed" clusters surface intact in the singular (e.g., *seis* [ˈsejs], but *ley* *[ˈlejs] [ˈle.ʝes]). Colina's proposal is that plural *-e-* appears in the plural of morphological words only; in other words, a condition for the [es] allomorph is that the plural be based on a singular output that has morphological word status in the Spanish lexicon; consonant-final bases which are not part of the morphology of Spanish, such as xenonyms, can therefore form their plurals by selecting the [s] allomorph, as in *fan, fans*. The [es] plural allomorph reduces syllabic markedness by providing a terminal element for a morphological word. Also within a morphological approach, Bonet (2006) makes the case that plural [e] is part of the subcategorization frame of the relevant lexical entries, thus reinterpreting it as a terminal element.

7.3 Conclusion

The purpose of this chapter was to present an overview of the state of the art on Spanish syllable structure. It examines our current understanding of the role of sonority, syllabic constituents, syllabic restrictions on segmental content, syllabification across words, and the interaction between syllabification and morphology, also covering controversial issues, such as the status of glides and of plural 'epenthesis'. Reference is made to various theoretical models.

NOTES

1 What type Spanish belongs to is a controversial matter.
2 There is significant dialectal variation as to the degree of constriction of this consonantal segment: fricative, affricate, stop, approximant (Hualde 1997; Harris and Kaisse 1999).
3 Under the assumption that the input is a vocoid not specified as [+/−consonantal], the faithfulness constraint violated would be Dep(consonantal), which inserts the feature [consonantal] not present in the input.

4 Alternatively, the hierarchy could consist of negative constraints, e.g. *hi/μ. The constraint banning the association of the least sonorous vocoid with a mora is the most highly ranked: *hi/μ ≫ *mid/μ ≫ *low/μ.
5 mid/μ is unranked with respect to *COMPLEX NUCLEUS, *CODA, ALIGN-LST.
6 Exceptionally, /-x/, *reloj* 'clock,' and /-t/, *cénit* 'zenith' can appear in final codas (Hualde 1999a; Harris 1983 and Roca p.c. disagree).

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8 Stress and Rhythm

JOSÉ IGNACIO HUALDE

1 Definition and functions of stress

Stress may be defined as the greater prominence that a given syllable receives over the rest of the syllables in a domain. This domain is the prosodic word in the case of word-level stress. The prosodic word may include a morphological word plus unstressed clitics. In an example like *los profesores de matemáticas* 'the mathematics professors,' for instance, there are two prosodic words, in each of which one syllable receives prominence: [*los profes**o**res*] [*de matem**á**ticas*]. A defining property of stress is thus its culminative function (Trubetzkoy 1973: 181). It has been claimed that equally or more important is the obligatory character of stress: in languages with word-level stress (stress languages), every prosodic word must have stress on one syllable (Hyman 2006). In Sections 5 and 6, we will consider some apparent exceptions to the culminativity (only one syllable with primary stress in the domain) and obligatoriness (at least one syllable with primary stress in the domain) of stress in Spanish.

The most prominent stress in an utterance is known as the nuclear stress. In Spanish, like in other Romance languages, nuclear stress generally falls on the last word of the utterance: *quiero coMER* 'I want to eat,' *quiero comer PAN* 'I want to eat bread,' *quiero comer pan con QUEso* 'I want to eat bread with cheese' (we set the syllable with nuclear stress in capital letters). By this, we mean that speakers tend to perceive the last lexically stressed syllable as more prominent than the rest. This is unlike in English, where the placement of nuclear stress is more variable and subject to specific rules (cf. e.g., *I want to eat a PIZza* vs. *I want to EAT something*; see Ladd 2008). In this chapter, we will concentrate on word-level stress in Spanish. We will also consider the issue of secondary stresses or prominences below word-level stress (Section 7).

The basic parameters of Spanish word-level stress have been clear for a long time. In terms of Trubetzkoy's functions, stress has a distinctive function in Spanish since it serves to distinguish words: *c**a**nto* 'I sing' vs. *cant**ó*** 's/he sang'; ***sá**bana* 'bed sheet'

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vs. *sabana* 'savannah'; *Gales* 'Wales' vs. *galés* 'Welsh'; *plato* 'dish' vs. *plató* 'cinema stage.'[1] The fact that the position of stress is represented in the standard orthography of the language when it deviates from the general rules bears witness to this distinctive, phonemic value. Trubetzkoy's demarcative function is also served in Spanish by stress to a certain extent, although not as clearly as in languages with fixed initial or final stress, since the stressed syllable must be one of the last three syllables of the word (i.e., there is a "three-syllable window"), and it is most usually the penult.

The rest of this chapter is structured as follows. In Section 2, we consider the patterns of stress of different word types. Section 3 is a brief overview of the assumptions of Metrical theory, as they apply to the analysis of stress in Spanish. In Section 4, we discuss the topic of whether stress assignment in Spanish is sensitive to quantity distinctions. Section 5 deals with stress in compounds. Section 6 considers the distinction between lexically stressed and unstressed function words. Section 7 is concerned with the nature and patterns of secondary stress. Section 8 discusses the acoustic correlates of Spanish stress. Finally, Section 9 addresses the issue of the connection between stress and rhythm.

2 Patterns of word-level stress in Spanish

2.1 *The three-syllable window*

The most important generalization regarding the placement of word-level stress in Spanish has already been mentioned. It is the "three-syllable window" at the end of the word: all lexical words have stress on one of their last three syllables. We find oxytonic words like *alacrán* 'scorpion,' paroxytonic words like *cucuracha* 'cockroach,' and proparoxytonic words like *libélula* 'dragonfly,' but there are no words with stress farther back from the end, like **cúcaracha*. There are thus only three possible stress patterns, counting from the end of the word. This is a restriction that is found in several other languages, including Modern Greek, Cairene Arabic, and Dutch, to name a few. The domain for this constraint in Spanish includes inflectional suffixes. This constraint is what causes stress shift in the plural of *régimen*, which is *regímenes*, not **régimenes*, which would violate the restriction. Notice, however, that there is no general repair mechanism that would apply to the plural of other words with the same structure such as *ómicron*, *asíndeton*, and *Júpiter*, where there is general uncertainty among speakers regarding the plural (see Hualde 2005: 207). Enclitics are invisible for this constraint: *dándomelos* 'giving them to me,' with stress on the same syllable as *dando* 'giving.' On the other hand, adverbs in *-mente*, such as *rápidamente* 'quickly,' are not true exceptions since these are compounds with two stresses: *rápidamente*.

Establishing other generalizations requires distinguishing among lexical classes and some morphological analysis of words (but see Section 4). In particular, verbs must be separated from nouns, adjectives, and adverbs. Grammatical or function words must also be treated apart from lexical words.

2.2 *Stress patterns of nouns, adjectives, and adverbs*

Stress, as mentioned, is phonologically contrastive, unpredictable, or "free" in Spanish (with the restriction on freedom of placement imposed by the three-syllable window), as we can see by comparing examples like the toponyms *Gólgota*, *Bargota*, and *Bogotá*. Nevertheless, it is not by any means the case that all three patterns are equally frequent in all classes of words and independent of morphological and syllabic structure. Regarding nouns and adjectives first, the clearest generalizations emerge when we consider words in the singular, and, furthermore, when we separate other inflectional suffixes from the base (Hooper and Terrell 1976; see also Harris 1983: 100; Roca 1988, 1999, 2005; Oltra-Massuet and Arregi 2005).

The stress falls on the same syllabic nucleus in the plural as in the singular: *american(os)*, *universidad(es)*, *apóstol(es)* – leaving aside the case of words like *régimen*, already mentioned, and the single unexplained exception of *carácter/caracteres*. Notice that keeping the stress on the same vowel in the plural as in the singular means that we have variation between singular and plural counting from the end of the word in the case of words with singulars ending in a consonant (e.g., *león* with final stress and *leones* with penultimate stress; *árbol* with penultimate stress and *árboles* with antepenultimate stress). In words with a regular feminine and where the feminine adds a syllable to the masculine, we find the same consistency regarding the position of the stress if we leave aside inflection: *portugués*, *portugues(a)*. It is thus clear that by considering the position of stress with respect to the uninflected base without inflectional suffixes, we can achieve a greater generalization. The unstressed final vowel of nouns and adjectives is an inflectional suffix and is deleted in derivation: *alt-o*, *alt-a*, *altur-a*, *alt-itud*.

The generalization is that the vast majority of nouns and adjectives have stress on the syllable containing the last vowel of the base: *american(o)*, *calabaz(a)*, *rinoceront(e)*, *universidad*, *regulación*. That is, most words ending in a vowel in the singular are paroxytonic (i.e., with word-penultimate stress), and most words ending in a consonant are oxytonic (with word-final stress). Over 95% of nouns and adjectives follow this simple accentual rule. We thus see that the Spanish lexicon somehow fails to take full advantage of the possibilities for establishing contrasts among words in the position of the stress that are allowed by the system.

Much less commonly, the stress may fall one syllable before the last one of the base: *húngar(o)*, *epístol(a)*, *ánad(e)*, *apóstol*, *césped*. This pattern thus includes words ending in a vowel in the singular that are proparoxytonic (i.e., with antepenultimate stress) and words ending in a consonant that are paroxytonic. This pattern is not randomly distributed in the lexicon. A few specific endings are particularly common in proparoxytonic words. These include the so-called superlative suffix *-ísim(o/a)* (*buenísimo* 'very good,' *purísima* 'very pure,' *generalísimo* 'generalissimo'), the prestressing suffix *-ic(o/a)* (*mágico* 'magic,' *magnífico* 'magnificent,' *fonética* 'phonetics,' *fonológico* 'phonological'), and the also prestressing ending *-(c)ul(o/a)* (*ridículo* 'ridiculous,' *espectáculo* 'spectacle,' *libélula* 'dragonfly,' *esdrújula* 'proparoxytonic'). As for paroxytonic words ending in a consonant, paroxytonic stress is as common as oxytonic stress with the endings *-en* (*examen* 'exam.' *virgen* 'virgin,' *volumen* 'volume' vs. *amén* 'amen,' *retén* 'stop; reserve,' *sostén* 'bra') and *-il* (*útil* 'useful,' *frágil* 'fragile,' *difícil* 'difficult' vs. *fusil* 'rifle,' *sutil* 'subtle,' *juvenil* 'youthful') (Aske 1990). There are also some restrictions on proparoxytonic and paroxytonic stress that have to do with syllable structure. These are discussed in Section 4.

Finally, it would appear that there are two sets of exceptional items. First, some words ending in a vowel have final stress: *Panamá*, *colibrí*, *dominó*, *café*, *menú*. These words would not constitute exceptions, falling under the general rule, if we assume that the final vowel is part of the base (Harris 1983: 117), which, in general, is a reasonable assumption, cf. *maní* 'peanut,' *manisero* 'peanut seller,' not **manero* (although this is not so clear in the case of *Panamá*, *panameño*). The other set of exceptions are words ending in a consonant (i.e., without an inflectional suffix) and with antepenultimate stress, such as *régimen*, *Júpiter*, *Álvarez*, *déficit*, *análisis*.

The stress patterns of adverbs are essentially the same as we find for adjectives and nouns: final stress in consonant-final words and penultimate stress in vowel-final words (e.g., *ayer*, *después*, *mañana*, *luego*), although some very common vowel-final adverbs have final stress (e.g., *aquí*, *allá*, *así*). In this respect, it is interesting to note that in some cases the endings of adverbs are treated like the inflectional suffixes of nouns and adjectives in derivation: *lej-os* 'far,' *lej-ano* 'distant.' The stress patterns discussed in this section are summarized in Table 8.1 below. Adverbs ending in *-mente* are compounds, as already mentioned, and will be discussed in Section 5.

2.3 Stress patterns of verbs

Verbs follow different stress rules than other words (Oltra-Massuet 2005; Roca 2005, 2006, and references therein). Furthermore, the rules are different for different tenses. Regarding stress, we can distinguish two main groups of tenses: present tenses (including the present indicative, present subjunctive, and imperative), where stress is regulated counting from the end of the word (e.g., *explico*, *explicamos*) and all other tenses, where the stress falls on the syllable that contains a specific morpheme: the theme vowel in some tenses (e.g., *explic-**a**-ba*, *explic-**á**-ba-mos*, *explic-**a**-se*, *explic-**á**-se-mos*) and the tense marker in the future indicative and conditional (e.g., *explic-a-ré*, *explic-a-re-mos*, *explic-a-ría-mos*). Exceptionally, irregular or "strong" preterits have stress on the last syllable of the root in the first and third person singular forms (e.g. *anduve* 'I walked' vs. regular *anulé* 'I annulled').

Table 8.1 Stress patterns in Spanish.

Pattern type	*Examples*
Regular pattern: stress on last vowel of base	*varapal-o, animal, maní*
Marked pattern: one syllable earlier	*espátul-a, apóstol*
Exceptional pattern: two syllables earlier	*Júpiter*

The stress pattern of present tenses is essentially penultimate, except for the *vosotros* and *vos* forms, which have final stress. Present tense stress can thus be assimilated to that of nouns, adjectives, and adverbs. An important qualification is that, with the one exception of *estar* (*est**á*** vs. regular *h**a**bla*), all verbs have exactly the same stress pattern in the present tense. In particular, antepenultimate stress is excluded (which causes shift of the stress in verbs derived from proparoxytonic nouns and adjectives; e.g. *l**í**quido* 'liquid,' *liqu**i**do* 'I liquidate').

The columnar stress of other tenses, on the other hand, requires reference to the position of specific morphemes, rather than to distance from the end of the word. This morphological conditioning of the position of the stress is in some way reminiscent of what we find in morphological derivation, where the last derivational suffix in the word always determines the location of the stress, either on the suffix itself or on the preceding syllable (e.g. ***á**tomo*, *at**ó**mico*, *atomicid**a**d*). The difference is that in verbs it is not necessarily the last suffix that imposes the stress pattern, but some specific one. More details are given in Chapter 12, below.

3 Metrical structure and stress

With the development of Metrical Theory in the 1970s and 1980s (Liberman 1975; Liberman and Prince 1977; Hayes 1980, 1995; Halle and Vergnaud 1987; Halle and Idsardi 1995), work on Spanish stress increased dramatically (Harris 1983, 1991; Roca 1988, 1999; Lipski 1997, among many others). Some more recent work on Spanish stress has adopted the formalism of Halle and Idsardi (1995) (Harris 1995; Oltra-Massuet and Arregi 2005; Roca 2005). An OT analysis can be found in Roca (2006). Work on Spanish stress has also been pursued from other very different perspectives, such as ruleless analogical modeling, see Eddington (2000) and Aske (1990). Here we will briefly discuss the fundamental assumptions and mechanisms of Metrical Theory and their application to the analysis of the stress system of the Spanish language, without entering into details of specific analyses, given the fact that a number of substantially different formulations of the Spanish stress rules within Metrical Theory have been offered by different authors and even by the same author in different works.

An original claim of Metrical Theory is that stress is related to the rhythm of languages (Liberman 1975; Liberman and Prince 1977). The term "metrical" derives in fact from traditional poetic terminology. We noticed before that, in addition to word-level stress, we can have higher degrees of stress when we consider words in phrases. It is also possible sometimes to distinguish several levels of prominence within words (secondary stresses in addition to primary stress). Consider, for instance, the example in (1):

(1)

							x	
			x				x	
	x	.	x	.	x	.	x	.
	G**u**	mer	sin	do,	G**u**	mer	sin	do

In a possible rendition of this example, we may perceive a rhythmic structure with stresses on alternating syllables, higher stress on the penultimate syllable of each of the two words and a still higher level of stress on the penultimate syllable of the phrase. Under the assumption that stress is relative prominence within a domain, syllables are first grouped in binary feet with initial prominence (trochees), feet are grouped at the word level, where the last foot receives greatest prominence, and words are grouped into prosodic phrases, with again highest prominence on the last word. Degrees of prominence are represented in (1) by placing an x on the prominent syllable at each of the three levels that are recognized here. In (2) the metrical constituency that we are assuming is added to the representation:

(2)	(						x	)	Phrase-level prominence
	(		x	)	(		x	)	Word-level prominence
	(x	.)	(x	.)	(x	.)	(x	.)	Foot-level prominence
	Gu	mer	sin	do	Gu	mer	sin	do	Syllables

An attractive aspect of Metrical Theory is the possibility it offers of capturing the typology of stress systems in the languages of the world as resulting from the interaction of a small number of parameters (Halle and Vergnaud 1987; Hayes 1995, among others). By assigning either iambic (right-headed) (. x) or trochaic (left-headed) (x .) feet, starting from either the beginning or the end of the domain, we would have the following possible patterns of foot-level stress for a domain with an uneven number of syllables in (3), under the assumption that extra syllables remain unmetrified:

(3)	trochees from	(x	.)	(x	.)		iambs from	(.	x)	(.	x)	
	beginning	**ta**	ta	**ta**	ta	ta	beginning	ta	**ta**	ta	**ta**	ta
	trochees from		(x	.)	(x	.)	iambs from		(.	x)	(.	x)
	end	ta	**ta**	ta	**ta**	ta	end	ta	ta	**ta**	ta	**ta**

Another parameter would determine whether primary word-level prominence is assigned to the first or to the last foot. For instance, if the last foot has primary stress, the pattern with trochees assigned starting from the end of the word would produce *taˌtataˈtata*, perhaps the pattern of a possible rendition of Spanish *riˌnoceˈronte*.

Yet another parameter in Metrical Theory is extrametricality or invisibility of an element to the parsing of syllables into feet. In Latin, the stress predictably fell on either the penultimate or the antepenultimate syllable. Unlike in Spanish, it could never be placed on the final syllable (except, of course, for monosyllabic words). Regarding the position of the stress with respect to the end of the word, we thus find that in Latin, like in Spanish, the stress cannot go beyond the antepenult, but in addition, the final syllable is invisible or extrametrical.

A final parameter in the metrical typology of stress systems that we need to consider is quantity-sensitivity. In some languages, there is a contrast between heavy and light syllables that is relevant for stress assignment. Latin was one of these languages. In Latin, in words of three or more syllables, stress fell on the penultimate syllable if this syllable was heavy, where a heavy syllable was a syllable that contained either a long vowel, like in *catēna*, *consulōrum*, or a syllable-final consonant, as in *perfecta*. If the penultimate was light, stress fell on the antepenultimate syllable, as in *tabula*, *consulēs*. In a quantity-sensitive language, a heavy syllable cannot occupy the weak position in a metrical foot. The definition of Spanish with respect to these parameters has been the object of much debate within metrical analyses. Different opinions have been voiced regarding whether Spanish stress is quantity sensitive, whether the most adequate analysis of Spanish stress involves iambic or trochaic feet, and in the use of extrametricality (see Oltra-Massuet and Arregi 2005; Roca 2005 for recent discussion of these issues with reference to earlier work).

Whereas the idea of positing a connection between stress and rhythm is an interesting one, further work has tended to show that rules of word-level stress assignment cannot be easily conflated with the rhythmic patterns of languages. Thus, for Spanish, Roca (1986) pointed out that, to the extent that one may be able to perceive secondary ("foot-level") stresses, they are assigned at the phrase-level, mostly on alternating syllables in the phrase (but see Section 7), not separately on individual words within phrases, so that word-level stress cannot be built upon these secondary stresses (as was done in Harris 1983). More generally, van der Hulst (1997) has argued that, typologically, the rules of primary-stress tend to target edges of words and result in very different generalizations from the ones that are postulated for alternating, rhythmic stress. This has tended to weaken the metrical hypothesis, or, at least, to weaken the connection between linguistic stress and rhythm present in earlier metrical work. Nevertheless, the intense scrutiny that the stress patterns of Spanish received in metrical stress theory has resulted in the discovery of a number of facts and generalizations that must be taken into account in the description and phonological analysis of the language.

Even if we abandon the basic rhythmic assumptions of Metrical Theory, it is still possible and desirable to capture typological generalizations regarding word stress assignment with a small number of parameters (see e.g., van der Hulst 1999). This is a desideratum of any theory of stress since we do find a very small number of stress patterns and constraints that tend to reappear in the languages of the world. For instance, both word-initial stress and word-penultimate stress are extremely common patterns in languages from diverse families and geographical areas.

4 Quantity-sensitivity and related restrictions in Spanish stress patterns

The first orthographic rules of the Spanish Academy assumed that Spanish had preserved the quantity-conditioned contrasts in stress placement of Latin (Real

Academia Española 1726), a view disputed by other authors, such as Larramendi (1729). As mentioned before, in Latin, stress could not be antepenultimate if the penultimate syllable had either a phonologically long vowel or a syllable-final consonant. Spanish has lost the contrast between phonologically long and short vowels, so that only the difference between open and closed syllables might be relevant. We find in fact that antepenultimate stress is rare when the penultimate syllable ends in a consonant. That is, with antepenultimate stress we find words like *sábado*, but generally not words like **dómingo*. This is perhaps not too surprising when we consider the origin in Latin of most of the Spanish vocabulary and the absence of diachronic processes that might have produced such a pattern. Exceptions (already pointed out by Larramendi 1729) are the toponym of non-Latin origin *Frómista* (in Spain) and borrowings from other languages. Borrowings from English, foreign surnames, and toponyms, such as *W**a**shington*, ***A**nderson*, are an open class of exceptions, although sometimes common nouns are adapted, *b**á**squetbol* ~*bastquetb**o**l* 'basketball.' A related fact is the absence of antepenultimate stress when the penultimate has a diphthong, again with the exception of some toponyms like *Pátzcuaro* (in Mexico).

A disputed question is how much importance to give to these facts in the analysis of Spanish stress. Is Spanish stress fundamentally quantity-sensitive – even if it lacks a contrast between short and long vowels – given the fact that antepenultimate stress in words with a "heavy" penultimate syllable is only marginal? (Harris 1983, 1995). Or, should we instead consider that the rarity of the pattern is simply a matter of historical residue, a lexical gap, without synchronic relevance for the analysis? (Roca 1990). An opinion that has been voiced is that evidence can be gathered by asking speakers for acceptability judgments on nonce words with the relevant structure or by asking people to pronounce nonce words (Bárkányi 2002; Alvord 2003). An alternative view, from the formal side, is that since rules can have exceptions, these experiments do not inform us about the most adequate formalism (Oltra-Massuett and Arregi 2005).[2]

Another interesting restriction in the distribution of antepenultimate stress is that there is no antepenultimate stress in words ending in a diphthong (Harris 1983: 107–108). There is, for instance, *Meli**á***, *Mar**í**a*, and *Em**i**lia*, but not ****É**milia*. It should be noted that, in Latin, vocalic sequences of rising sonority were always syllabified as hiatus, so that a word like *Italia* had antepenultimate stress in Latin (*i.t**a**.li.a*). Perhaps the only exceptions to this generalization are the words *ventr**í**locuo* and *grand**í**locuo*. If the word ends in a falling diphthong (i.e., there is a word-final glide), the restriction is even stronger: stress can only be final (Harris 1983: 118), as in *samur**á**i*, *conv**o**y*, except for some borrowings like *h**o**ckey* and *y**ó**quey* 'jockey.' A question is whether these generalizations based on syllable structure (which are never exceptionless) should be given a different status from other lexical patterns, such as the fact that words ending in *-(c)ul(o/a)* tend to be proparoxytonic.

5 Stress in compounds

Compounds can be divided into two groups regarding their stress. On the one hand, in some compounds, each member maintains its stress; in others, on the other hand, only the last member of the compound preserves stress (Hualde 2006/7). These compounds can be described according to the following groupings: (a) compounds without stress deletion, which include most N + N compounds (*pez esp**a**da* 'swordfish,' *v**e**rde ol**i**va* 'olive green'), and adverbs in *-mente* (*r**á**pidam**e**nte* 'quickly,' *senc**i**llam**e**nte* 'simply,' *igu**a**lm**e**nte* 'likewise'); and (b) compounds with stress deletion, which include V + N compound nouns (*lavapl**a**tos* 'dishwasher'), some N + N compounds (*telar**a**ña* 'spider web'), numbers (*cinco m**i**l* 'five thousand'), compound names (*Pedro Ju**a**n*, *Victoria Eug**e**nia*), some N + Adj and Adj + N compound nouns (*yerbabu**e**na* 'mint,' *mediod**í**a* 'noon'), and N + Adj compound adjectives (*barbilamp**i**ño* 'beardless').

Preservation of the stress of each member can be taken as an indication of phrasal structure, as opposed to the fusion of the members of the compound in a single prosodic word (Hualde 2006/7). Some compounds in group (b) minimally contrast with segmentally identical phrases. This is the case with V + N compounds, the contrast being established by deletion of the stress of the verb in the compound (e.g., *lavapl**a**tos* 'dishwasher' vs. *l**a**va pl**a**tos* 's/he washes dishes'). Notice also the minimal pair *mediod**í**a* 'noon' vs. *m**e**dio d**í**a* 'half a day.' In the case of N + N compounds, those that undergo stress deletion pluralize at the end of the compound, whereas those with stress preservation have plural inflection after the first noun (e.g., *telar**a**ñas* 'spider webs,' *bocac**a**lles* 'street entrances' vs. *cami**o**nes cist**e**rna* 'tanker trucks,' *p**e**rros l**o**bo* 'wolf dogs'), which is another indication of greater internal cohesion in the former group. Regarding the phrasal status of adverbs in *-mente*, note that this element can have scope over two coordinated adjectives, as in *lenta y cuidadosamente* 'slowly and carefully.'

6 Unstressed words

Among function words, there is a contrast between unstressed and stressed words. Unstressed words are words that do not bear stress in the context of the phrase. Leaving aside enclitic pronouns, they are all proclitic elements. Somewhat surprisingly, the distribution between stressed and unstressed words in different classes of function words is somewhat idiosyncratic in Spanish (Navarro Tomás 1977 [1918]: 189–194; Quilis 1993: 390–395; Real Academia Española 1973: 69–74; Hualde 2006/7).

Stressed determiners in the noun phrase include indefinite articles (e.g., ***u**na pal**a**bra* 'a word'), demonstratives (***e**sta pal**a**bra* 'this word'), and quantifiers (*p**o**cas pal**a**bras* 'few words'). On the other hand, definite articles (*las pal**a**bras* 'the words') and possessives (*mi pal**a**bra* 'my word,' *nuestra pal**a**bra* 'our word') are unstressed (but possessives are stressed in areas of Leonese substrate). When modifying a numeral, stressed determiners become unstressed, with some optionality in this case (*unos dos* 'about two,' *estos dos* 'these two,' *otros dos* 'other two'). Among adjective modifiers, *medio* and *casi* are (often) unstressed (*medio dormido* 'half asleep,' *casi nuevo* 'almost new'), but *muy* and *poco* are stressed (*muy nuevo* 'very new,' *poco sincero* 'little sincere').

Prepositions are all unstressed (*para los estudiantes* 'for the students,' *hacia la sierra* 'towards the mountains,' *contra todos* 'against everyone,' *sobre la mesa* 'on top of the table,' *para con tus amigos* 'in relation to your friends'), with the single exception of *según* 'according to' (*según tu hermano* 'according to your friend'). Single-word conjunctions are also unstressed (e.g., *pero para tus amigos no* 'but not for your friends'). This creates a contrast, which is orthographically reflected, between question words/relative pronouns and conjunctions: *no sé cómo lo hicieron* 'I don't know how they did it' vs. *como lo hicieron mal, no duró* 'since they did it wrong, it did not last.'

The quantifiers *más* 'more' and *menos* 'less' are also conjunctions with the meanings of 'in addition to' and 'except for,' respectively. As conjunctions, they are stressless. This time, the contrast is not reflected orthographically: *necesito menos problemas* 'I need fewer problems' vs. *menos problemas, necesito de todo* 'except for problems, I need everything.' The same contrast arises with the time adverbs *mientras* 'in the mean time' and *luego* 'later,' which are unstressed when used as conjunctions: *mientras hablábamos* 'as we were talking' vs. *mientras, hablábamos* 'in the meantime, we were talking'; *luego llegaron a tiempo* 'therefore, they arrived on time' vs. *luego llegaron ellos* 'later, they arrived.' Phrasal conjunctions or locutions, on the other hand, may be either stressed or unstressed. For instance, *aun cuando* 'even though' and *puesto que* 'since' are normally unstressed, but the near synonymous expressions *a pesar de que* 'even though' and *dado que* 'since' are stressed. The adverb *apenas* 'barely' is stressed also when used as a temporal conjunction (e.g. *apenas salí de mi casa empezó a llover* 'I had hardly left my house, when it started to rain'). This would seem to be an exception to the statement that single-word conjunctions are unstressed, and would be an argument for analyzing it as two words, despite its orthographic representation, reflecting its etymology (*a penas* 'with pain').

Subject pronouns are stressed (e.g., *yo dije* 'I said'). Proclitic object pronouns are always unstressed (e.g., *te lo dije* 'I told you'). Enclitic object pronouns are also normally unstressed (e.g., *para decírtelo* 'to tell you,' but they may receive "secondary" stress; see Section 7).

In citation form or under contrastive focus, where they would be final in the prosodic phrase, unstressed words receive stress on the penultimate syllable or on their only syllable. In these contexts, the contrast between stressed and unstressed words is obliterated (e.g., *contra es una preposición* '*contra* is a preposition,' *luchó CONTRA todos* 's/he fought AGAINST everybody'). In a possible analysis, unstressed words have an "underlyingly" stressed syllable, but this stress is deleted by their incorporation in the same prosodic word as a following content word (Hualde 2009). In this analysis, syntactic phrases with a single stress like *para con tus amigos* 'in relation to your friends' are single prosodic words which undergo the same process of stress deletion that is observed in compounds like *lavaplatos* 'dish washer.'

7 Secondary stress

Starting with Navarro Tomás (1972 [1918]: 195–196), several authors have made the claim that, in addition to the syllable with primary stress, it is possible to perceive a weaker degree of prominence, or secondary stress, on several other syllables in the prosodic word or phrase (Stockwell et al. 1956; Harris 1983: 85–86; Roca 1986). For Navarro Tomás, secondary stress is assigned to the initial syllable of the prosodic word, unless it immediately precedes the syllable with primary stress, as in *ˌconversaˈción* 'conversation,' *ˌpor la maˈñana* 'in the morning', and also on alternating syllables, counting from the syllable with primary stress, as long as it does not fall on a syllable adjacent to the initial, as in *ˌcontraˌproduˈcente* 'counterproductive,' *ˌcontra ˌlo traˈtado* 'against the agreement'. The other authors mentioned above, although differing in the details of the description, also see secondary prominence on the initial syllable and/or on alternating syllables from the one bearing primary stress. Native speakers' intuitions are less than clear on this matter of secondary stress, which would not be lexically contrastive. Experimental work has failed to find evidence for secondary stress (Prieto and van Santen 1993; Díaz-Campos 2000).

Our position is that secondary stress in Spanish is not an inherent property of certain syllables (unlike primary stress) or an obligatory phenomenon. Rather, it is a rhetorical device. For rhetorical purposes, syllables other than the one bearing lexical stress can be given prominence (see Wallis 1951; Bolinger 1962; Quilis 1993; Hualde 2006/7, 2010). This phenomenon is frequent in certain types of public speech, such as lectures, and on radio and television. Although the phenomenon still requires more research, several patterns can probably be distinguished. In one rhetorical pattern, there is secondary stress two syllables before the primary, generally in prosodic words that are not final in the turn or intonational phrase: *para los traˌbajaˈdores* 'for the workers.' In another rhetorical stress pattern, mostly used before a pause and considered more emphatic, secondary stress is placed, instead, on the initial syllable: *ˌsolidariˈdad* 'solidarity.' Finally, alternating stresses, counting from the syllable with primary stress, are also found, but the frequency of this pattern appears to be very low: *ˌpara ˌlos traˌbajaˈdores*. Against earlier assumptions, observation of public speech (in addition to the experimental work reported in Hualde 2010) shows that secondary stress may be placed on the initial syllable of the word, even if the lexical stress is on the second, as in the last word in this example: *los ˌpasaˈjeros con desˈtino a ˌSalaˈmanca, ˌBarceˈlona, ˌSeˈvilla …* 'passengers traveling to Salamanca, Barcelona, Seville …'. Syllables with secondary stress bear a prominent pitch accent, whereas lexical or primary stress is cued by duration in words with secondary stress.

A different phenomenon is stress on enclitics. As mentioned in Section 1, enclitics do not cause stress shift, unlike derivational suffixes, and are invisible with respect to the three-syllable window restriction, unlike both derivational and inflectional suffixes. On the other hand, they may receive stress when phrase final. The imperative *v**á**monos* 'let's go,' for instance, may be realized with two pitch-accents, and perceived prominence on two syllables, *v**á**mon**o**s*, or even with a pitch accent only on the final syllable, *vámon**o**s*. Enclitic stress is particularly frequent in Argentinian Spanish (Moyna 1999), but it is also found in Peninsular Spanish and some other varieties; see Real Academia Española (1973: 71): *acércaté*, *acérquenmé*, *acerquémonós*, and from literary texts: *ayudanós* 'help us,' *dexamé* 'let me.' Enclitic stress always falls on the last syllable of the word and appears to be impossible immediately after the lexically stressed syllable of the verb. That is, we may find, for instance, *d**í**mel**o*** 'say it to me,' but not *?*d**i**m**e*** 'tell me,' or, using Argentinean *voseo* forms, *cant**á**mel**o*** 'sing it to me,' but not *?*cant**a**l**o*** 'sing it.' Enclitic stress is mostly found in imperatives, but it is also possible with other verbal forms that take enclitics (e.g., *para dec**í**rtel**o*** 'to say it to you') (Colantoni et al. 2010).

8 Acoustic correlates of stress

We have defined stress as relative prominence. Stress is thus primarily a perceptual notion. A number of different acoustic correlates contribute to the perception of prominence on a given syllable, although not all of them are always present in the acoustic signal of a given token (Ortega-Llebaria and Prieto 2011). In fact, there can be tokens where the lexically stressed syllable has no identifiable acoustic correlates since, after all, decoding of language is a function of both the information present in the signal and the previous knowledge of the language that the listener has (i.e., word recognition).

A very important fact that contributes to the perception of prominence is that stressed syllables function as anchoring points for intonational pitch accents. In Spanish, most words in the discourse bear a pitch accent, and these pitch accents are aligned with respect to the lexically-stressed syllables, which act as designated anchoring points for these intonational elements. We find deviations from this rule in contexts where secondary stresses are employed (see Section 7). Notice that the observation that stressed syllables tend to bear pitch accents does not mean that stressed syllables will have higher pitch than other syllables. Since the most common pitch accents in Spanish declaratives are rising, with the tonal rise continuing onto the following syllable, very often the posttonic will have higher pitch than the stressed syllable (see Chapter 9, below). In fact, it appears to be the rise in pitch extending over the syllable from a low point at the beginning, rather than the position of the peak, that serves to characterize the stressed syllable as prominent. In addition, in phrase-final position in declaratives, pitch rises tend to be very small or may be completely absent, so that pitch is not a very reliable cue of stress in this position in the phrase. Stress syllables also tend to show greater duration. This durational cue becomes more important when tonal cues are unreliable or not present, such as the phrase-final position in declaratives, as just mentioned, and in parentheticals, which tend to be realized in a flat low tone

(Ortega-Llebaria and Prieto 2007). Notice, however, that unstressed phrase-final syllables are often longer than preceding stressed ones (because of phrase-final lengthening), so that the greater duration of the stressed syllable is relative to the pretonic syllables. A rise in pitch is normally accompanied by a rise in intensity, so that this feature also serves as a cue for stress.

When pronounced in isolation, in citation form, words are accompanied by a pitch accent entirely realized within the stressed syllable, and this syllable will show higher pitch, greater duration, and greater intensity. In citation form, thus, all three cues – pitch, duration, and intensity – serve to identify the lexically stressed syllable within the word.

In some languages, there are strong segmental cues of lexical stress. In particular, the vowel inventory may be severely restricted in unstressed syllables. In Spanish, we do not find these restrictions, and all five vowel phonemes may occur in both stressed and unstressed syllables. Nevertheless, all vowels are somewhat centralized when unstressed, with respect to the more peripheral position in the vowel space that they occupy when stressed (Martínez Celdrán and Fernández Planas 2007: 188–191). Notice, in addition, that the diphthongs *ie*, *ue* are essentially restricted to stressed syllables (with few exceptions), which causes numerous morphophonological alternations of the type represented by *p**ue**de/pod**e**mos*, *qu**ie**re/quer**e**mos*.

9 Spanish rhythm

A traditional observation is that languages impressionistically differ in their rhythm, an aspect of language to which infants appear to be attuned (Nazzi et al. 1998). In some languages, rhythm appears to be regulated by alternating stresses, so that intervals between stressed syllables would tend to be equalized, whereas in other languages there is more durational constancy among all syllables. This has led to a typology of rhythm types where languages could be classified as either "stress-timed" or "syllable-timed" (to which "mora-timed" is sometimes added). In more recent work, languages have been placed along a scale which goes from perfect stress-timing at one ideal extreme to perfect syllable-timing at the other (see Kohler 2009 for the history of the development of these ideas). Stress-timing would be found in languages with great durational and intensity differences between prominent and nonprominent syllables, where unstressed syllables are greatly reduced and compressed so that stress beats would give the impression of occurring at regular intervals, thus marking the rhythm of the language. The impression of syllable-timing, on the other hand, would be most strongly conveyed by a language without stress and where all syllables had very similar structures (e.g., CV.CV.CV).

Metrical theory, which, as we saw in Section 3, posits a connection between prominence at several levels of linguistic structure and rhythm, was initially proposed and developed on the basis of a language like English, which has been taken as a paradigmatic example of stress-timing. Although Spanish has word-level stress, this lexical stress plays less of a role in regulating the rhythm of the language than in English. This is primarily so because Spanish does not have the extreme reduction of unstressed vowels that we find in English. The very notion of stress foot is hard to justify for Spanish since we can have long stretches of unstressed syllables (Toledo 1988; Nolan and Asu 2009). For instance, in *los estudiantes de nuestras universidades* 'the students of our universities,' which is not an anomalous example, we find eight unstressed syllables between two stressed ones.

Several attempts have been made to find acoustic cues that would correlate with subjective impressions of rhythm, generally focusing on the relative duration of different sequences. Thus, Ramus et al. (1999), in an influential paper, consider the proportion of the duration of the text taken up by vowels (%V) and the standard deviation of consonantal intervals (ΔC) as correlates whose combination defines rhythmic classes. In another influential paper, Grabe and Low (2002) consider the variability in vowel duration and in the duration of intervocalic intervals, calculating the difference between adjacent elements (Pairwise Variability Indexes or PVI). In these studies, Spanish appears as clearly different from English and other Germanic languages. This is also the case in Nolan and Asu (2009), where PVI is also used and two Spanish dialects, Peninsular and Mexican, are taken into account. (On the other hand, some authors have questioned the status of Spanish as a syllable-timed language; see Pointon 1980; Borzone de Manrique and Signorini 1983; Pamies Bertrán 1999.)

It should be noted that these different acoustic measurements conflate phonological facts of syllable structure with phonetic effects having to do with the realization of stress, both of which may contribute separately to the overall rhythmic impression of the language (Dauer 1983, 1987; Cummins 2002). Variability in the duration of consonantal intervals will be conditioned by the possibilities offered by the language regarding the complexity of onset and coda clusters; that is, in a language that allows up to three or four consonants both in the onset and in the coda of the syllable, we will find more variability than in a language with only CV syllables. Regarding variability of vocalic portions, this will be greater in a language with a phonological contrast between short and long vowels and with numerous diphthongs in all positions than in a language without long vowels or diphthongs; languages with these contrasts limited to certain positions, such as stressed syllables, will occupy an intermediate point in the scale.

It is at least likely that we may obtain a more accurate impression of the rhythm a given language, defined in these terms, from a phonological description of its syllable structure (and the relative frequency of each possible structure) rather than from acoustic measurements taken from short texts, which may not represent well the variability present in the language, in addition to the fact that there may be great variability in production across speakers and styles (Arvaniti 2009; Arvaniti and Ross 2010). On the other hand, besides the influence of syllable structure on perceived rhythm, we may also find that two languages differ in the amount to which unstressed syllables are reduced with respect to stressed ones. That is, there can be differences in duration (as well as intensity and segmental quality) between adjacent syllables that are due to the phonetic implementation of stress and are independent of syllable structure. A language where unstressed syllables are greatly reduced with respect to stressed ones (such as English) will sound more stress-timed than a language with less reduction in unstressed syllables (such as Spanish) and much more so than a language without stress. Another element that may contribute to the perception of rhythm is the distribution and shape of pitch events. If we take two hypothetical languages with the same syllable structure and similar amounts of reduction of unstressed syllables, but in one of them the penultimate syllable of every word carries a pitch accent and in the other pitch accents are less regularly distributed, chances are that the general rhythmic impression of these two languages would be quite different.

It is also important to keep in mind that not all speech is rhythmic, or equally rhythmic. In any language, we may produce speech that is more or less rhythmic, but the way to make speech more rhythmic appears to differ among languages. In English, greater rhythmicity is obtained by the temporal spacing of lexically stressed syllables (Cummins 2002). In Spanish, instead, rhythm is created by assigning pitch accents to lexically unstressed syllables in a regular fashion. This seems to be an important difference between the two languages. As we saw in Section 7, in Spanish, lexically-unstressed syllables can be recruited as beat-bearers in purposefully rhythmic speech, as in *los p**a**sajeros con dest**i**no a S**a**lam**a**nca* . . . 'passengers traveling to Salamanca . . .' It has been claimed that rhythmic regularity in language facilitates processing (Kohler 2009); the addition of "secondary" or nonlexical stresses in public speech in Spanish would thus serve this communicative function by creating greater rhythmic regularity.

We may also have displacement of the beat with respect to lexical stress in songs, as in *t**e**ngo una muñ**e**ca vestid**a** de az**u**l* 'I have a doll dressed in blue,' where the prominence is displaced from the lexically stressed penultimate syllable of *vestida* to the last one in order to accommodate the tune (Janda and Morgan 1988; Morgan and Janda 1989), something that is generally impossible in a language like English.

10 Conclusions

The stress patterns of Spanish have been the object of intense scrutiny, with a proliferation of competing analyses, especially within Metrical Theory. Here we have avoided issues of formalism, focusing instead on descriptive generalizations. A foundational premise of Metrical Theory is that stress contributes to establish the rhythmic structure of the language by creating alternating patterns of prominence. In Spanish, word-stress does not appear to be built on lower prominences, at the foot level, unlike in English. Nevertheless, rhythmicity in speech may be created by anchoring pitch-accents on lexically unstressed syllables.

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NOTES

1 In general, we will use stress marks only when they are part of the standard spelling of words. When necessary for clarity, we will set the vowel of the stressed syllable in bold. When discussing patterns of secondary stress, primary (or lexical) and secondary stress will be indicated in addition with IPA diacritics placed immediately before the stressed syllable.
2 For quantity sensitivity in Old Spanish, see Martínez-Gil (2010).

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9 Intonation in Spanish

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1 Introduction

Intonation is a suprasegmental characteristic of speech that pertains to features of sound above the level of vowels and consonants. The changes in prosody or melodic contour are assigned a certain grammatical and/or pragmatic meaning according to the context in which it is used, such as an emphatic statement or a question expressing surprise. Languages may vary according what contours are used in each instance. In Spanish, the goal of intonation studies has been to determine which contours and intonation features are common across Spanish dialects and which ones serve to distinguish these dialects (e.g., in broad terms, Peninsular vs. Latin American, or by region, Caribbean Spanish vs. Andean Spanish). In addition, the ways in which Spanish intonation may have developed and continue to develop in these different dialects has been of interest, in particular in terms of language contact and language acquisition. The goal of this chapter, then, is to first describe intonation itself in terms of its properties of sound, and to introduce how this may be used to express differences in meaning. Then, in Section 2, the intonation features that have been examined for Spanish are summarized, including some of the main findings. In Section 3, the ways is which Spanish dialects have been found to differ will be described. Next, in Section 4, Spanish in contact with other languages and its affect on intonation is addressed. Finally, language acquisition and the development of Spanish intonation are discussed in Section 5.

1.1 Intonation: sound and meaning

Prior to describing Spanish intonation in particular, a brief description of intonation in terms of its physical properties and how it is measured may be useful. From an acoustic perspective, as speech is modulated, a sound wave is produced that is perceived as pitch, which may be higher or lower at different points in an utterance. The fundamental frequency or F0 is the basic repetition of this sound wave; one full wave pattern is measured repeatedly in time in cycles per second or Hertz (Hz).

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Shorter patterns repeat more often and produce more cycles in one second, which results in a higher fundamental frequency and a higher perceived pitch; longer patterns have fewer cycles in one second, a lower F0, and lower perceived pitch (see Quilis 1981 for a survey of additional definitions of intonation that vary according to the focus of the researcher).

Languages use differences in fundamental frequency in distinct ways. Those languages that use changes in pitch to distinguish between word-level meaning are considered tone languages. On the other hand, languages which distinguish between sentence-level meaning and other pragmatic meanings are intonation languages. For example, in Spanish the sequence of words *trabajas, en, la, biblioteca* 'you-work, in, the, library' can be uttered as a statement or a question using the same word order, but differ only in the intonation contour used. In many varieties, the statement *Trabajas en la biblioteca* would end in a lower pitch level, while the question *¿Trabajas en la biblioteca?* would have a different final contour than statements (e.g., a higher final pitch level, or a rise to a higher pitch during the stressed syllable followed by a sharp fall, as in Castilian Peninsular Spanish and Puerto Rican Spanish, respectively). Intonation, together with stress and rhythm, are the concern of studies in prosody, which strive to understand the (inter)connection of speech characteristics, their organization, and structure, above the level of the segment. For a background on intonation research in general, see Cruttenden (1997); Ladd (1996); Gussenhoven (2004); Prieto (2003); see Jun (2005) for recent instrumental work on intonation in several languages.

2 Spanish intonation structure

The structure of Spanish intonation can be discussed according to certain features of the contour with respect to the type of utterance in which it is found. Several neutral and non-neutral utterance types have been described in Navarro Tomás (1944) and Quilis (1981, 1993). In this section we describe four basic utterance types: declaratives, interrogatives, exclamatives, and imperatives, as well as utterances with a particular emphasis or interest, that is, utterances under focus and/or topicalization. For each of these utterance types, the contour features related to stressed syllables and those appearing at the end of the utterance are described, that is, the pitch accents and boundary tones. The description here is based on the findings for Castilian Peninsular Spanish (hereafter Castilian Spanish); descriptions of other dialects are included in Section 3.

Fundamental research on Spanish intonation can be found in Navarro Tomás (1918, 1944), based on readings of literary texts, and Quilis (1981, 1993), based on more naturalistic speech samples. An overview of Spanish intonation research also appears in Kvavik and Olsen (1974). However, much of the more recent work on Spanish intonation has employed a laboratory phonology approach in which a computerized instrumental analysis is made of high-quality speech recordings. From the waveform, a spectrogram and fundamental frequency contour are produced, which are then used to identify associations between tonal targets and specific structure within an utterance following the Autosegmental Metrical model (Pierrehumbert 1980). That is, the stressed syllable, word boundaries, the edge of phrases, etc., are locations where turning points within the contour may be observed, such as target lows (L) or highs (H). See Hualde (2002, 2003b) for a description of this model and its use with Spanish. Within this approach, a notation system has been developed for Spanish, based on the Tones and Break Indices (ToBI) system used for English and other languages (Beckman and Hirschberg 1994). The Spanish version, Sp_ToBI, and its revisions (Beckman et al. 2002; Sosa 2003a; Face and Prieto 2007; Estebas-Vilaplana and Prieto 2008) are employed with the possibility of being able to compare findings across studies of Spanish (and other languages) (e.g., see Prieto and Roseano 2010 for a comparison of ten Spanish dialects).

2.1 Declarative sentences

In declarative sentences in Spanish, the utterance often contains several pitch movements which appear associated with the stressed syllable of content words (e.g., nouns, verbs, adjectives, adverbs), while function words (determiners, prepositions, conjunctions, etc.) do not normally receive sentence-level stress except when a particular emphasis is made. The changes in fundamental frequency for a stressed syllable have specific configurations which may include high (H) and low (L) tonal targets; these configurations, or pitch accents, may be monotonal or bitonal (e.g., H^* or $L+H^*$, where the asterisk indicates a stronger association of the tone with the stressed syllable in the case of bitonal pitch accents). The final pitch accent of the utterance is considered to be in nuclear position in Spanish for neutral utterances while non-final pitch accents are described as being in prenuclear position. The description of pitch accents in Spanish can then be divided between those observed in prenuclear and nuclear position.

In prenuclear position in Spanish declaratives, there is generally a high target observed associated with the stressed syllable, often preceded by a low tonal target at or near the beginning of the stressed syllable (Prieto 1998). However, the timing of the high (or peak) is often after the end of the stressed syllable. This feature was described by Navarro Tomás as follows: "Frequently the unstressed syllable following the first accented syllable is somewhat higher than the previous one, with a difference of one or two semitones. After this slight elevation, the tone returns to a mid level in the following syllables" (1944: 67, my translation).[1] Laboratory research has observed this feature of non-final peak displacement in Peninsular Spanish (see de-la-Mota 1997; Face 2002a; Llisterri et al. 1995; Martínez-Celdrán et al. 2003; Prieto and Torreira 2007 for factors affecting displacement, such as stress pattern, syllable type, syntactic boundary, and number of intervening unstressed syllables, among others; and Prieto et al. 1995 for similar work with Mexican Spanish).

The representation of this type of $L+H$ 'rising' tone has been discussed in terms of which tone is more strongly associated with the stressed syllable, and should receive the asterisk notation (e.g., $L^* + H$, $L + H^*$, or $(L + H)^*$; see Hualde 2002 for summary). In recent proposals, Face and Prieto (2007) and Estebas-Vilaplana and Prieto (2008), a three-way distinction is given that employs $L + H^*$ for those prenuclear pitch accents whose peak is reached at (or before) the end of the stressed syllable, $L^* + H$ for those in which a low tone is realized during the stressed syllable while the rise to the peak does not begin until after the end of the stressed syllable, and $L + >H^*$ for cases when the rise to the peak does begin during the stressed syllable but is not reached until afterwards (using $>$ to indicate this delay).[2] In any case, the feature of post-tonic peak alignment for prenuclear (non-final) pitch accents is observed in several varieties of Spanish and may be considered a common characteristic for Spanish (with several notable exceptions discussed in the sections below).

In nuclear position, a common characteristic observed for pitch accents is the timing of the peak, which appears during the stressed syllable and may be preceded by a low target. The pitch accent is then assigned the configuration of $L + H^*$. Other features which may affect the realization of the nuclear (and prenuclear) pitch accents in Spanish declaratives are downstep, indicated with !, and final lowering. That is, subsequent peaks may be lower than previous peaks, with the final peak being markedly lower, which may be due to final lowering in addition to downstep (Prieto et al. 1996; Face 2002a). As noted in Face (2003c), downstep and final lowering may be more frequent in laboratory speech than semi-spontaneous speech, such that the presence or absence of these features may need to be explored (e.g., how final lowering may be used to signal 'predictable' information in the different types of speech; see Face 2010 for further discussion). In several cases, the tonal target in nuclear position is considered low, such that a L^* notation is given (Estebas-Vilaplana and Prieto 2010 for Castilian Spanish).

The boundary tones for declaratives in Spanish can be divided into two levels of structure, the highest level or intonational phrase, and below this level, the intermediate phrase. A broad focus declarative with no particular emphasis generally ends in a low pitch level; the end of this full intonational phrase is then marked with a low final boundary tone, or L% (for examples, see Section 2.5). The intermediate phrase may be marked by a high or low (H^- or L^-) either earlier in the utterance, or at the end of the utterance preceding the final boundary tone (e.g., L^- L%). Evidence for the presence of this level of phrasing is found by Nibert (2000) to disambiguate between sentences with two possible interpretations. Others have also argued for the presence of intermediate phrases, for example, to separate old and new information (Hualde 2002), while some have opted for more complex final boundary tones (e.g., HL% in non-neutral statements; see Estebas-Vilaplana and Prieto 2008, 2010).

2.2 *Interrogatives*

Interrogatives in Spanish can be divided between those that are requesting affirmation or negation, that is, yes–no questions, and those that are seeking specific information, that is, pronominal questions (*¿Quién?*, *¿Qué?* 'Who?,' 'What?,' etc.), also known as wh-questions. The type of prenuclear pitch accent within these interrogatives is similar to that of declaratives, i.e., post-tonic alignment of peaks. However, scaling differences have been observed in comparison to declaratives (Prieto 2003). The nuclear pitch accents may differ from declaratives according to question type and direction of the final contour. In many cases, the target(s) of the final pitch accent are considerably lower, especially when followed by a high final boundary tone. Some describe the nuclear pitch accent as L^*, while others have considered the peak to be downstepped ($L + !H^*$, with the exclamation indicating a lowered peak) (Prieto and Roseano 2010). The nuclear pitch accent may be relatively lower when followed by a final rise than when followed by a final fall. Typically, pronominal questions in Spanish end in a fall, while yes-no questions (also known as absolute interrogatives) often end in a rise. However, pragmatic differences, such as politeness being expressed in pronominal questions with a final rise, and other dialect differences have been observed, such as falls in Caribbean Spanish yes-no questions (see Section 3). Estebas-Vilaplana and Prieto (2010) find a final fall in neutral wh-questions, but a final rise in wh-questions expressed with additional speaker involvement in Castilian Spanish.

Face (2004) offers a detailed comparison between absolute interrogatives and broad focus declaratives as a first step in identifying the multiple cues available to the listener between utterance type. In his study of Madrid Spanish speakers, Face (2004) finds phonetic differences in initial peak scaling in absolute interrogatives, as well as the absence of medial pitch accents for the majority of speakers. Also, there is alignment of a low tone in the nuclear, utterance-final position (which may be a potentially contrasting pitch accent L^* or $L^* + H$ vs. $L + H^*$ in declaratives).

An inventory of boundary tones for Spanish is given in Estebas-Vilaplana and Prieto (2008, 2010), many of which are may appear in neutral and non-neutral questions: monotonal boundary tones H%, L%, and M% (for mid-level tones), bi- and tri-tonal boundary tones, LH%, HL%, HH%, and LHL% have been suggested as potential final boundary tones for Spanish. However, differences between what is considered a mid-level phonologically distinct tone compared to a H% or L% with phonetic undershoot needs to be established, as does the difference between H% and HH% (see 2008: 278 for examples). Also, the possibility of an initial high boundary tone has been explored. However, Gurlekian and Toledo (2009) do not find an initial peak height difference in absolute interrogatives, rejecting the notion of an initial high boundary tone, contrary to Navarro Tomás (1967: 226), but rather find that these interrogatives differ from declaratives in the phrasing used in each of the modalities (see example of a yes–no question in Section 2.5).

2.3 *Exclamatives and imperatives*

Exclamatives and imperatives are two types of non-neutral utterances that differ in intonation from the declaratives and interrogatives described above. First, exclamatives may be based on a statement or question, such as *¡No te creo!* 'I don't believe you!,' or *¡¿Qué quieres que haga?!* 'What do you want me to do?!' For both non-neutral utterance types, exclamatives and imperatives, differences in scaling are found, with a higher tone reached (Navarro Tomás 1967: 231, 234; Quilis 1981). In Castilian Spanish, differences in target height are observed for exclamatives, such as an upstepped higher peak in nuclear position (Estebas-Vilaplana and Prieto 2010). Pitch accent configuration may differ as well: tonic-aligned prenuclear peaks appear in exclamatives, while the nuclear pitch accent is falling ($H+L^*$) in imperatives (ibid). In addition, scaling of peaks was shown to be significantly higher in exclamatives and imperatives compared to declaratives (Prieto 2003).

Willis (2002) also investigates imperatives for Mexican Spanish and finds a differential recruitment of prosodic cues according to speaker, including increased duration and local pitch range, reduced deaccenting, and a "marked" pitch accent with an early-aligned peak. While these local features may be employed to distinguish between imperatives and declaratives, they are found with other utterances types as well. Therefore, Willis (2002) coincides with Kvavik (1987, 1988) in not finding a distinct intonation pattern, but suggests that the frequency of usage of certain cues may depend on the given modality.

2.4 Narrow focus and topicalization

Utterances with narrow focus are those which give particular emphasis to a portion of the utterance or proposition. Additionally, the narrow focus may be contrastive when compared with a prior proposition or option. The narrow focus may be placed on the last element in an utterance or earlier in the utterance. In Spanish, the pitch accent for elements under focus may differ from elements not under focus. In non-final position, the peak is aligned during the stressed syllable (or $L+H^*$), it may be higher, and it may be followed by a L- phrase accent, and lower or deaccented peaks (e.g., in Madrid Spanish, Face 2002a, 2002b); see example in Section 2.5 below. (However, see Section 3 for narrow focus in Dominican Spanish; see Face 2003b for the role of syntactic constituency in contrastive focus.) In cases of topicalization, the element of interest is moved to the left (or right) periphery (Contreras 1976: 81). The phrase may be then set off with a tonal marking, such as an intermediate phrase (also known as a phrase accent), which may be high or low. More examination of the interaction between intonation and syntax is needed for Spanish (for work in this area, see Domínguez 2004; Fant 1984; Silva-Corvalán 1983).

2.5 Illustration of Spanish intonation contours

In this section, we see examples of intonation contours as produced by a male Castilian Spanish speaker. The stressed syllables are underline in the caption and appear with primary stress marking (ˈ) in the figures. Figure 9.1 is a short, broad focus declarative, while Figure 9.2 is a broad focus declarative containing a subject. Figure 9.3 is a yes-no question, while Figure 9.4 is an example of a declarative with contrastive focus on the subject.

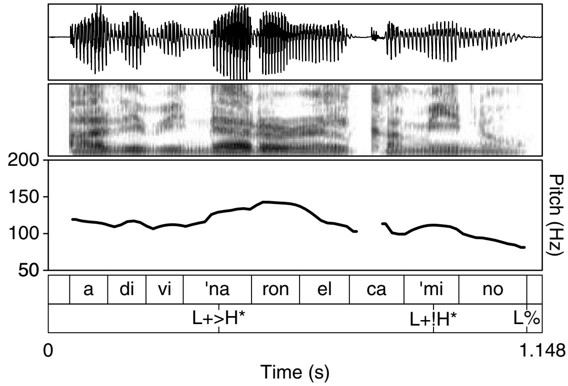

Figure 9.1 Broad focus declarative *Adivinaron el camino.* 'They figured out the way.'

For the declaratives in Figures 9.1 and 9.2, the prenuclear peaks are posttonic, giving a L + >H* pitch accent. The final peak in nuclear position is considerably lower in both cases: in Figure 9.1, the peak is downstepped, or L + !H*; in Figure 9.2, the target is low, or L*. Both end a low boundary tone or L%. Figure 9.3 includes an example of a yes–no question, which ends in a high boundary tone, or HH% following the notation in Prieto and Roseano (2010). Finally, in the example of constrative focus on the subject in Figure 9.4, the peak is aligned within the stressed syllable, giving a L + H* pitch accent; the following peaks are lower, including a downstepped prenuclear peak !H* and a low nuclear pitch accent L* in final position. Like the previous declarative, this declarative with subject contrastive focus ends in a L% boundary tone. The examples included are given to illustrate the

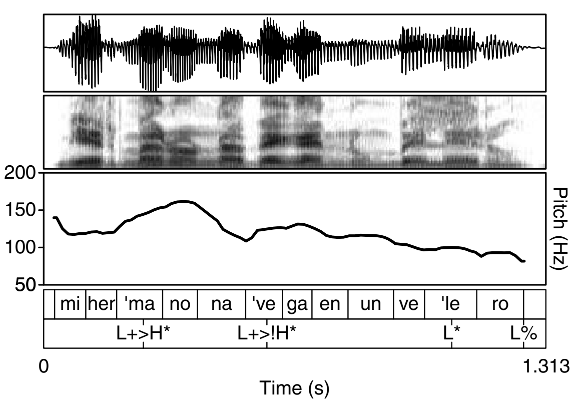

Figure 9.2 Broad focus declarative *Mi hermano navega en un velero.* 'My brother navigates in a sailboat.'

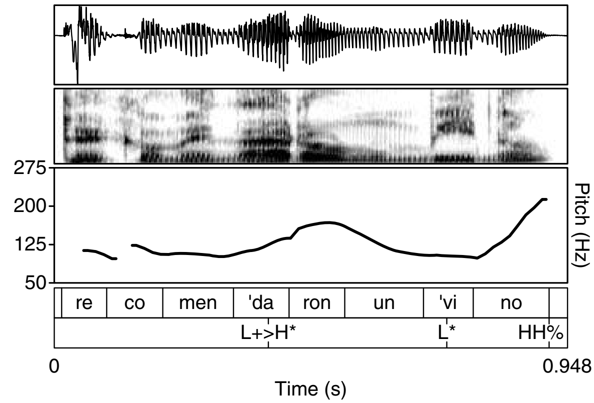

Figure 9.3 Yes–No question *¿Recomendaron un vino?* 'Did they recommend a wine?'

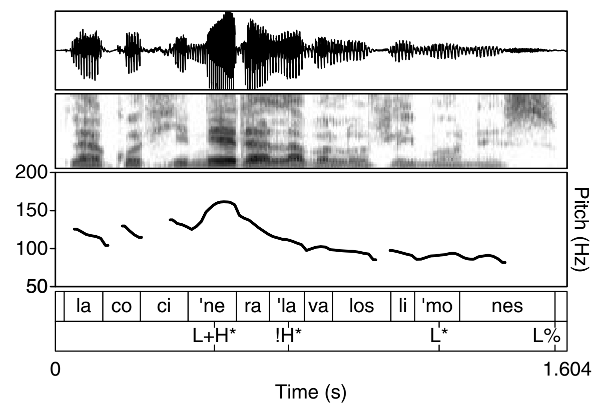

Figure 9.4 Declarative with contrastive focus on subject *LA COCINERA lava los limones.* 'THE COOK washes the lemons.'

structure of Spanish intonation as discussed in the previous sections (for more examples of Castilian Spanish, see Estebas-Vilaplana and Prieto 2010).

3 Dialect differences in intonation

Differences in Spanish intonation may develop through the diffusion of internal changes, through contact with other languages, and as the result of (second) language acquisition (the latter two sources are discussed in Sections 4 and 5 below). Intonation in varieties of Spanish has been noted in the literature by several researchers (Alcoba and Murillo 1999; Kvavik and Olsen 1974; Quilis 1993).

The work by Sosa (1999) marks the implementation of the Autosegmental Metrical model to compare several varieties of both Peninsular and Latin American Spanish. Prieto and Roseano (2010) offer a compilation of recent cross-dialect research in which utterances in neutral and more marked contexts are compared across several varieties of Spanish using Sp_ToBI notation as well as iconic representations of each of the pitch accents and terminal contours. In this way, an overview of dialects may shed light on which contours are common across Spanish and which may represent regionally distinct differences.

While a detailed description of dialect differences extends beyond the scope of the present work, a few general observations can be made. Dialect differences in Spanish intonation can often be observed in the choice of pitch accent in prenuclear and nuclear position, as well as in the boundary tone used in yes–no and pronominal questions. That is, while many varieties show post-tonic alignment of peaks in non-final position, a number have been observed with tonic-aligned peaks (see Section 4 on language contact below). Also, the configuration of the final nuclear pitch accent may be either a low (or falling) tone, or a high (or rising) tone (e.g., Castilian vs. Mexican Spanish) (see Prieto and Roseano 2010). In Puebla Mexican Spanish, the nuclear pitch accent distinguishes between declaratives and absolute interrogatives ($L + H^*$ and L^*, respectively), in addition to the final rise in the questions, rather than systematic differences in tonal levels earlier in the utterance (Willis 2005). Last, the final boundary tone in yes–no questions in Caribbean varieties is often a fall or circumflex, while the final boundary tone in pronominal questions may be high (e.g., in Mexican Spanish) (see Sosa 1999 for sample pitch tracks). However, Willis (2004) finds final configurations in yes–no questions for Dominican Spanish that differ from previous reports, including a low nuclear pitch accent followed by either a high plateau, a rise followed by a slight fall, or a continuous rise. In wh-questions, Sosa (2003b) finds that Mexican and Colombian Spanish speakers produce more final rising contours in read speech than do Puerto Rican and Venezuelan speakers, who tend to use a falling contour for wh-questions.

Other intonation features have also been examined related to interrogatives in different dialects. An example is Alvord (2009), in which absolute interrogatives in Miami Cuban Spanish are found to have a higher pitch range (compared to declaratives). From this, Alvord concludes that range rather than final contour (fall or rise) distinguishes between utterance type. Adding support to Face's (2005) findings regarding the role of scaling in the perception of Madrid interrogatives, Alvord (2009) notes that pitch scaling may need to be indicated in cases where it is potentially phonological, although this type of notation is not currently part of the AM model. Other recent work on interrogatives includes Manchego Peninsular Spanish yes/no questions and the timing of early and late final rises, distinguished as H^* H% and L^* H%, respectively (Henriksen 2010).

Dialect differences may also be found in the type of pitch accent employed in non-neutral utterances. For Dominican Spanish, Willis (2003a, 2003b) finds the usage of a late low target well into the stressed syllable followed by a rise after the end of the stressed syllable for prenuclear pitch accents in broad focus declaratives (described as a 'late Low late High'). On the other hand, in contrastive focus, the low aligned at the beginning of the stressed syllable with a posttonic peak is found (or 'early Low late High'), which is considered the broad focus prenuclear pitch accent in Peninsular Spanish. The findings on Dominican Spanish demonstrate how the same type of configuration may have a neutral meaning in one variety, but a marked meaning in another.

The following are several studies divided according to region which represent just a portion of recent research that has been conducted on intonation in dialects of Spanish. Some research on Peninsular Spanish intonation includes: *Castilian Spanish* (Cabrera Abreu and García Lecumberri 2003; Face 2002a, 2003a; Martínez-Celdrán and Fernández-Planas 2003; Toledo 2007, 2008), *Barcelona Spanish* (Rao 2009; Romera et al. 2008), *Basque Spanish* (Elordieta 2003; Elordieta and Calleja 2005), *Majorcan Spanish* (Simonet 2008, 2010). Studies of Latin American Spanish intonation include: *Latin American varieties* (McGory and Díaz-Campos 2002; Sosa 2003b, 2003c), *Argentinian Spanish* (Colantoni 2011; Colantoni and Gurlekian 2004), *Chilean Spanish* (Cid Uribe and Ortiz-Lira 2000; Ortiz-Lira 2003), *Mexican Spanish* (Avila Hernández 2003; Martín Butragueño 2003, 2004; Willis 2005), *Peruvian Spanish* (O'Rourke 2005), *Venezulean Spanish* (Mora 2003). Some work on Caribbean Spanish includes: *Dominican Spanish* (Willis 2003a, 2003b, 2004, 2007), *Puerto Rican Spanish* (Armstrong 2010). Varieties of Spanish as spoken in the United States is another area to be explored, for example, *Miami Cuban Spanish* (Alvord 2007, 2009, 2010). Spanish intonation dialectology is a promising area of growth that will contribute to other areas of study as well (e.g., sociolinguistics, language contact, and bilingualism). Coordinated efforts such as the work by Prieto and Roseano (2010) using a shared systematic approach will facilitate making generalizations and distinctions between dialects.

4 Intonation in Spanish in contact with other languages

In addition to dialect differences in Spanish intonation, which may be due to historical developments or other internal factors, Spanish in contact with other languages is another means by which variation in Spanish intonation may develop. Previously claims have been made that indigenous languages have influenced the development of regional intonation patterns in dialects of Latin American Spanish (Henríquez Ureña 1938; Malmberg 1948, among others; see Sosa 1999: 241–245 for summary). Recent instrumental work on Spanish and other languages now can address these issues from the standpoint of laboratory phonology. For example, Gussenhoven and Teeuw (2008) have examined the H tone in Yucatec Maya. This work may in turn advance the research on Mexican Spanish and address the prior limitations recognized by Lope Blanch (1987) of establishing "dependencies" or influence between the two languages since knowledge of both intonation systems is needed. In another example, O'Rourke (2004, 2005) finds support for the case of contact: regional differences in the tonic alignment of non-final peaks in Peruvian

Spanish were observed, with several Cuzco speakers showing more instances of tonic alignment than Lima speakers. There findings on Cuzco Spanish coincide with tonic alignment patterns observed in Cuzco Quechua (O'Rourke 2009). Earlier alignment of non-final peaks is also observed in Spanish in contact with Basque (Elordieta 2003), and also Spanish historically in contact with Italian (Colantoni and Gurlekian 2004); see Colantoni 2011 for further discussion on the potential outcomes of contact, including convergence, simplification, creation of 'mixed' systems, and the assignment of differential meaning.

In his analysis of Majorcan Catalan-Spanish bilinguals, Simonet (2008) finds evidence for borrowing of intonation features in that some groups of Spanish-dominant bilinguals (younger females) show more Catalan-like concave (HL*. . .L%) contours than other Spanish-dominant speakers, who show a more Spanish-like convex contour (LH*. . .L%) in Spanish read declaratives; borrowing was also observed for yes/no questions of Catalan-like falling contours into Spanish by Spanish-dominant bilinguals, suggesting convergence towards the behavior of Catalan-dominant bilinguals. Spanish may also influence the development of another prosodic system, for example Spanish in contact with Basque (Hualde 2003a, 2003c), or in a case of creolization, in the development of Palenquero (Hualde and Schwegler 2008).

5 Intonation and language acquisition

Cross-linguistic studies on Spanish intonation have been conducted for different purposes, from characterization of the differences between the two languages to determining the challenges faced by the L2 learner. For example, several studies have focused on the differences between English and Spanish (e.g., García Lecumberri 1995, 1996; Gutiérrez Díez 2008, Gutiérrez-Bravo 2002). Also, Estebas-Vilaplana (2008) compares English and Spanish final contours within the AM framework. With respect to the learner, Kvavik and Olsen (1974) call for intonation research on Spanish, recognizing the gap between segmental and suprasegmental fluency achieved by undergraduate students, the former meeting target-like goals while the latter lagging distantly behind as one of the last areas to learn (see Kvavik 1976 for further discussion). With newer descriptions of Spanish intonation that are now being produced, a re-emphasis on the pedagogical implications and potential innovations is an area that can be further developed.

From an acquisition standpoint, Henriksen et al. (2010) examine how Spanish intonation is developed by English speakers in a study abroad context. Lleó and Rakow (2006, 2010) and Lleó et al. (2004) also address child language acquisition issues for Spanish and German monolinguals and bilinguals. For example, young bilingual children show later prenuclear peak alignment in Spanish in the direction of the target language but not to the same extent as their monolingual peers, with some instances in the opposite direction (i.e., earlier aligned peaks similar to German H*). In another study, Prieto et al. (in press) examine the relationship between prosodic development and lexical and grammatical development in

Catalan- and Spanish-acquiring children. While they find that target-like intonation patterns are observed at the onset of speech (e.g., tune-to-text alignment), other features such as tonal scaling are developed later; also, after reaching the 25-word point, a jump in the number of nuclear configurations used is found. With respect to L2 acquisition of Spanish intonation, currently little is known, in particular with adult learners. By determining not only the potential challenges faced by the L2 or bilingual Spanish learner but also the actual stages of development of the Spanish target-like system, we may gain a greater understanding of the structure of each language (and the features that must be modulated in order to use intonation to express distinct sociopragmatic meanings in Spanish).

6 Summary and future directions

In this chapter, we have seen some basic characteristics of Spanish intonation and have examined how Spanish intonation may differ across dialects, in contact with other languages, and in language acquisition. We have seen that a significant amount of work has focused on declaratives and interrogatives. More recently, this work has expanded both in terms of the types of utterances analyzed, including more emphatic and non-neutral contexts, and the range of dialects analyzed (see Prieto and Roseano 2010). In declaratives and interrogatives in Spanish, posttonic prenuclear peaks are common, although lower tonal targets appear in nuclear position either as a target low L^* or downstepped peak. Less work has been conducted on exclamatives and imperatives, although higher tonal height is one characteristic that has been reported, while several other features may be recruited in the case of imperatives. In utterances with contrastive focus, earlier peak alignment within the stressed syllable, in addition to other strategies, such as the use of a high or low phrase accent, and higher peaks, have been found. Dialects can be distinguished according to choice of pitch accent (e.g., tonic-aligned prenuclear peaks in contact varieties, such as Argentinian, Basque, and Peruvian Spanish), as well as the choice of boundary tones, particularly in questions (e.g., Caribbean Spanish may fall in absolute interrogatives for some varieties, but show a plateau or rise in others, such as Dominican Spanish). Spanish in contact has shown an influence of other languages on the development of prosodic variation (e.g., a preference of Spanish-dominant bilinguals to use Catalan-like final contours). However, the acquisition of Spanish intonation is relatively less-explored, with the noted exception of child language acquisition research with German and Spanish monolinguals and bilinguals, and Catalan-speaking and Spanish-speaking children previously mentioned.

Briefly, some future directions in research are the following. As the features of basic contours are established, other more marked contours and contexts can be examined. Also, although the majority of work discussed has focused on production, perception of Spanish intonation is another area to be explored (Face 2005; Ortega-Llebaria and Prieto 2009; Ronquest and Díaz-Campos 2007). Studies focusing on pragmatics (Armstrong 2010; Escandell Vidal 1999; Labastia 2006) and sociolinguistic issues (Alvord 2009, 2010; Enbe and Tobin 2008; Moreno Fernández 1999; Simonet 2008, 2010) may also contribute to our understanding of Spanish intonation. In addition, the way in which Spanish intonation is similar or different to other Romance languages should continue to be examined (e.g., see Elordieta et al. 2005 on phrasing in Spanish and European Portuguese, Prieto et al. 2005 on pitch accents in Romance, Ferreira 2009 on Peninsular Spanish in comparison to Brazilian Portuguese). As research on Spanish intonation in itself is a rich field that offers vast opportunities due to the large population of native speakers, bilinguals, and language learners, we must strive to establish connections with other languages and disciplines, so that the findings on Spanish may contribute to our overall understanding of intonation, its use, and meaning.

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NOTES

1 "Es frecuente que la sílaba débil que sigue inmediatamente a la primera acentuada resulte algo más alta que ésta, con diferencia de uno o dos semitonos. Después de esta ligera elevación, el tono vuelve a acomodarse al nivel medio en las sílabas sucesivas" (Navarro Tomás 1944: 67).

2 While all three of these pitch accents may not appear in any given variety of Spanish (or Romance), and may therefore not be phonologically distinct, use of this common notation with the same conventional meaning allows for cross-comparisons. Challenges as to how to assign a starred tone are encountered however, in cases for example of dialects of Basque Spanish, in which a continuum of tonal targets (and potential associations) is observed between varieties in comparison with Madrid Spanish (Elordieta and Calleja 2005). See also Willis (2009) for a discussion of abstraction in Spanish phonology and intonation.

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10 Morphophonological Alternations

DAVID EDDINGTON

1 Introduction

Morphophonology is the study of how word formation interacts with phonology. The domain of phonology proper is concerned with identifying phonemes, the allophones of each phoneme, and the context in which the allophones appear. For example, [d] and [ð] are both allophones of/d/: [d] appears postpausally and postnasally while [ð] occurs elsewhere. This phonological alternation is generally considered exceptionless, which contrasts it with morphophological processes that have many exceptions. In morphophonology, not only may allophones of the same phoneme be involved in an alternation, but allophones of different phonemes may alternate. So, although [t] and [s] belong to different phonemes they alternate in the morpheme/perßert-/'pervert' when it is followed by different affixes:/perßert+ir/'to pervert,'/perßers+o/'perverted.'

Morphophonological alternations are quite common in both the derivational and inflectional morphology of Spanish. For instance, the alternation between [t] and [θ] (e.g.,/inyekt+ar/~/inyekθ+jon/'inject, injection') has received attention by researchers (Harris 1969; Núñez 1993) as have the [o] ~ [we] and [e] ~ [je] alternations (e.g.,/tost+ar/~/twest+an/, 'to toast, they toast,'/tjen+e/ ~/ten+emos/'it has, we have' (Bybee and Pardo 1981; Carreira 1991; García-Bellido 1986; Eddington 2006; Harris 1969, 1977, 1978, 1985, 1989; Halle et al. 1991; St. Clair 1971). On the other hand, alternations such as [f] ~ [Ø] as in/fum+ar/ ~/um+o/'to smoke, smoke,' and [ð] ~ [ß] as in/ßiß+ir/~/ßið+a/'to live, life' are rarely discussed in any systematic fashion.

Why some morphophonemic alternations attract more attention than others certainly has to do with how many words contain them, although no cutoff point has been established to separate the significant from the insignificant (Esau 1974: 10). For some, significant patterns are those that are most amenable to rule-based analysis (Chomsky 1965: 42; Chomsky and Halle 1968: 335; Hurford 1977: 575).

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Rather than grapple with the issue of significance, and given the purpose of the present volume, I have chosen to address some of the morphological alternations based on how much attention they have received in the literature. Therefore, I will discuss diphthongization, diminutive allomorphy, velar and coronal softening, and depalatalization.

2 Diphthongization

Diphthongization refers to the alternation of [o] with [we] as in *b[wé]no ~ b[o]ndád* 'good, goodness' and [e] with [je] as in *pim[jé]nta ~ pim[e]ntéro* 'pepper, pepper shaker.'[1]

Historically, these alternations arose from the Romance vowels/ɔ/and/ɛ/. When unstressed,/ɔ/became/o/and/ɛ/became [e]. However, in stressed position they diphthongized:/ɔ/became [we] and/ɛ/became [je] which gave rise to the contemporary morphophonemic alternation that is largely conditioned by stress. In verbal forms, analogy sometimes interrupted this regular evolution (Penny 2002: 182–184). Since the phonemes/ɔ/and/ɛ/no longer exist, it is difficult to predict which instances of unstressed [o, e] alternate with stressed [we, je] (e.g. *c[wé]nto ~ c[o]ntámos* 'I count, we count,' *v[jé]ne ~ v[e]nís* 's/he comes, you come') and in which cases there is no stress-conditioned alternation (e.g. *t[ó]sen ~ t[o]séis* 'they cough, you cough,' *of[e]ndér ~ of[é]nsa* 'to offend, offense'). The existence of stressless diphthongs (e.g. *p[we]blíto ~ p[wé]blo* 'small town, town,' *m[jé]do ~ m[je]dóso* 'fear, afraid') is an additional issue that must be dealt with.

In traditional generative analysis, all allomorphs are derived from a unique underlying representation. A critical part of an analysis in this tradition, then, is to distinguish which vowels participate in the diphthongization alternations and which do not. As a result, a number of ingenious diacritics and rule-systems have been proposed. These include diacritic marks on the vowels, empty vowel slots, and cyclic rule application (Carreira 1991; García-Bellido 1986; Harris 1969, 1977, 1978, 1985, 1989; Halle et al. 1991). While such methods allow everything, including apparent exceptions, to be accounted for, they have the disadvantage of being completely abstract; they lack any surface validity and thus escape potential falsifiability. As a result, it is impossible to prove one analysis is more correct than another on an empirical basis.

If linguistics is a branch of cognitive science (Chomsky 1968: 1), the goal of morphophonological analysis is to determine what generalizations have relevance for actual speakers as well (Pilleux 1980:115). As far as diphthongization in Spanish is concerned, Bybee and Pardo (1981; see also Eddington 1998) wondered how synchronically valid diphthongization is. They presented Spanish speakers with sentences containing nonce verbs such as *biérca*, *muéna*, *monár* that demonstrate diphthongization and elicited responses that entailed manipulating the diphthongization alternation. They found that many subjects provided responses that go against a rule of diphthongization (e.g. *biercó* rather than the expected *bercó*). Rule systems derive both [je] ~ [e] and [we] ~ [o] by means of a unitary process, yet the experimental data demonstrate that [je] ~ [e] is much more productive, suggesting that the two alternations are independent of each other in cognitive processing.

Another question that merits investigation is how speakers determine when simple vowels that alternate with diphthongs are realized as diphthongs and when they are not. Abstract mechanisms proposed by rule analyses are not helpful because they have no tangible counterparts in the speech signal. Are there surface-apparent clues to diphthongization? Eddington (1996, 1998) investigated this issue by examining ten derivational suffixes and the relationship they have to diphthongization in the stem of the words to which they are affixed. Only words that have morphemic relatives with and without a diphthong (e.g., *diente* ~ *dentista* 'tooth, dentist') were considered in the study.

Usage suggest that words with the productive diminutive, superlative, and augmentative suffixes, *-(c)ito*, *-zuelo*, *-(c)illo*, *-ísimo*, and *-azo*, generally appear with diphthongs (e.g., *fuerte* ~ *fuertezuelo*, 'strong, somewhat strong,' *vieja* ~ *viej(ec)illa* 'old woman, little old woman'). A dictionary search for common words ending in the less productive suffixes *-al*, *-(i)dad*, *-ero*, *-oso*, and *-ista* reveals a great deal of variety. For instance, all of the common words ending in *-al*, *-(i)dad*, *-ero*, contain simple vowels (e.g., *dental*, *novedad*, *herrero*) compared with 83% of words with *-oso* (*vergonzoso*) and 50% of those ending in *-ista* (*dentista*). This was determined experimentally (see Eddington 1996).

In order to test whether the relationship between suffix and diphthongization was psychologically significant, native Spanish speakers were asked to state their preferences for diphthongs or simple vowels in a series of neologisms. For example, was a person who worked scattering *estiércol* an *estiercolero* or an *estercolero*? The percentage of simple vowel responses given by the subjects reflects quite well the percentage of simple vowels that appear in common words in usage and the dictionary search. In other words, the presence of a diphthong or simple vowel in a neologism is dependent to some extent on how often simple vowels or diphthongs appear in common extant words in the language ending in a particular suffix.

Suffixes are not the only surface-apparent clue to diphthongization. Albright et al. (2001) studied diphthongization in Spanish verbs and show that the phonemic makeup of the verb stem provides clues to how likely a verb is to have a diphthongizing stem. They examined 1,689 verbs with [e] and [o] in the stem computationally (Albright and Hayes 1999). The algorithm they applied yielded probabilities that verbs with a certain phonological shape would be diphthongizing verbs. For example, based on diphthongizing verbs such as *cerrar* 'to close' and *enterrar* 'to bury' the model predicts that verbs ending in *-errar* have a high probability of being diphthongizing. On the other hand, no verbs ending in *-echar* (e.g., *echar* 'to throw,' *aprovechar* 'to take advantage of') are diphthongizing, which means that [e] ~ [jé] alternation is not likely to appear in similar verbs. The prediction of their model significantly correlated with responses given by subjects to nonce words. For example, given *lerrámos* the subjects preferred the diphthongized inflection *liérro* over *lérro* when the crucial syllable was stressed.

When the results of these studies are considered together, they suggest that diphthongization can be extended to some degree to nonce words and neologisms.

The experimental data support a model of processing in which speakers make use of the properties of the words they have experienced, such as the suffix the word ends in and the phonological makeup of the word, in order to handle the alternation between diphthongs and simple vowels.

3 Diminutive formation

Variation in diminutives ending in *-ito/a(s)* is of two types. First, there is allomorphy in the suffix itself which appears as either *-ito*, *-ecito*, or *-cito* (e.g., *normal+ito*, *rey+ecito*, *joven+cito* 'normal, king, young man') or the feminine or plural forms of these (e.g., *normal+ita*, *rey+ecitos*, *joven+citas* 'normal, kings, young women'). The diminutivized stem itself varies depending on whether the final vowel is retained or not. In *Lupe* > *Lup+ita* 'Lupe' and *minuto* > *minut+ito* 'minute' the final vowel is lost, while in *Andrés* > *Andres+ito* and *llave* > *llave+cita* it is retained. According to some researchers (Jaeggli 1980; Méndez-Dosuna and Pensado 1990) many cases of diminutivization do not entail loss of the final vowel when *-ito* is affixed (*minuto* > *minut+ito*). Instead, it is viewed as infixation of *-it-* (*minuto* > *minut+it+o*) in the same way *Carlos* > *Carl+it+os* and *Víctor* > *Vict+it+or* appear to involve an infix.

Spanish diminutivization has received attention in a number of different frameworks. Jaeggli (1980) treats it in a classical generative model, while Colina (2003), Elordieta and Carreira (1996), and Miranda (1999) approach it in optimality theory. Eddington (2002) accounts for it in an exemplar model. The key point that Prieto (1992) and Crowhurst (1992) make is that diminutivization is partially governed by prosody. Two of the more problematic cases, diminutives of singular stems that end in *-s* and cases of infixation as in *Víctor* > *Vict+it+or*, are dealt with respectively by Bermúdez-Otero (2007) and Méndez-Dosuna and Pensado (1990). Ambadiang (1996, 1997), Colina (2003), and Harris (1994) argue that diminutive formation is morphological, not phonological in nature.

The bulk of the work on Spanish diminutives centers on finding the phonological or morphological context that governs which suffix (or infix) is applied as well as the formal mechanisms for deriving the diminutives. In one sense, diminutivization is a very regular process that appears to be conditioned by phonology. This means that it is possible to describe the majority of the forms with a small number of generalizations. On the other hand, the more data one considers the more one encounters diminutives that escape simple phonological explanation. In part this is due to dialectal differences, but there is individual variation as well. Even when these are eliminated from consideration, a sizeable number of odd cases remain. This prompted Harris (2004) to abandon the quest for predictability based on phonology and conclude that "diminutivisation is riddled with arbitrary lexical choices" (2004: 186).

Is there a way to test the extent to which the diminutive allomorphs are predictable based on non-abstract properties? The 2,447 diminutives taken from Eddington's (2002) search of the Corpus del Espanol (Davies 2002) may serve as a test set to this end. All approaches to diminutives should account for these data, and it would be ideal if each of the extant analyses could be compared against this data set. Unfortunately, none of them deals with the full range of diminutive forms in the data set. A further difficulty is that many crucially depend on entities that cannot be tested because they are not surface-apparent (e.g., constraints, Colina 2003; Miranda 1999; epenthetic vs. terminal *-e* and resyllabification, Crowhurst 1992). For this reason, I focus on surface-apparent characteristics.

The generalizations in Table 10.1 (taken from Eddington 2002) represent an attempt to describe diminutivization in the simplest and most theory-neutral way possible based on three characteristics of the base form. The most predictive part of a stem as far as diminutive allomorphy is concerned is the word-final phone. However, the number of syllables in words ending in *-e* is a conditioning factor as well (Crowhurst 1992; Elordieta and Carreiras 1996; Prieto 1992): 67% of bisyllabic words ending in *-e* are of the type *traje* > *trajecito*. Those with three or more syllables tend to be of the *elefante* > *elefantito* type in that 88% are formed in this manner. Grammatical gender also plays a part in that, with few exceptions (e.g., *la manita/o*, *el mapita*), the final phone of a diminutive is *-a* if feminine and *-o* if masculine. This is indicated in Table 10.1 as *depends on gender* (d.o.g.). A respectable 94% of the diminutives may be predicted with these simple generalizations. So few words take *-ecito* (e.g., *tren+ecito*, *pan+ecito*) that no generalization is made about these forms.

The question that arises is how to account for the 6% that do not fit into these generalizations. Part of it is due to the corpus the data are derived from that includes diminutives from many different countries; diminutivization is known to

Table 10.1 Generalizations that predict diminutive allomorphs based on the final phone, gender, and number of syllables of the base word.

Final Phone	*Process*	*Example*	*Total No.*	*% Correct*
/o/	Add *-ito* to stem minus *-o*	*carro* > *carrito*	948	96
/a/	Add *-ita* to stem minus *-a*	*pena* > *penita*	1,044	96
/e/	Add *-cito/a* to bisyllabic stem minus *-e*, d.o.g.	*llave* > *llavecita*	90	86
/e/	Add *-ito/a* to stems with 3 + syllables minus *-e*, d.o.g.	*elefante* > *elefantito*	72	99
/r/	Add *-cito/a* to the stem, d.o.g.	*pastor* > *pastorcito*	58	83
/n/	Add *-cito/a* to the stem, d.o.g.	*colchon* > *colchoncito*	104	94
/l/	Add *-ito/a* to the stem, d.o.g.	*Isabel* > *Isabelita*	75	97
/s/	Add *-ito/a* to the stem, d.o.g.	*Andrés* > *Andresito*	28	50
Other phoneme	Add *-ito/a* to the stem, d.o.g.	*reloj* > *relojito*	28	39
			2,446	94

d.o.g. = depends on gender.

vary regionally. There is also variation even within the same speech community (Crowhurst 1992; Jaeggli 1980; Prieto 1992) and within individual speakers (Harris 1994). This may be partially responsible for the doublets that exist in the database (e.g., *carne* > *carnecita/carnita* 'meat,' *chofer* > *chofercito/choferito* 'chauffeur,' *tren* > *trencito/trenecito* 'train,' *piedra* > *piedrita/piedrecita* 'stone'). In these cases, one is predicted by the generalization while the other escapes it into the other 6%.

Abstraction representations such as differing prosodic templates (Crowhurst 1992; Prieto 1992) and constraint reranking (Colina 2003) have been proposed to account for these differences, as have different morphological parses (Bermúdez-Otero 2007). There are, however, some concrete characteristics that are related to the dialectal differences such as whether the stem contains the diphthongs [je] and [we] (Prieto 1992). In these cases, some varieties prefer the *-ecito/a* allomorph and others *-ito/a* in these cases (e.g., *pueblecito/pueblito*, 'small town,' *tienda* > *tiendecita/tiendita*, 'small store').

Other cases escape the broad generalizations in Table 10.1 because the internal morphological structure of the word may need to be taken into consideration. A case in point are *corona* and *llorona* 'crown, crying woman,' both of which are feminine and end in *-a*. The diminutive forms, however, are quite different (*lloroncita*, *coronita*). A Google search reveals that *lloronita* is not unknown, but is about ten times less frequent than *lloroncita*. Harris (1994) argues that the differing morphological structure of the words is responsible for the different diminutive forms: *coron+a* and *llor+on+a*, while Colina (2003) attributes the difference to each belonging to a distinct morphological class. Another possibility is that *lloroncita* is based on the masculine form *lloroncito* rather than on *llorona*.[2]

Morphology has also been attributed a part in the creation of the diminutive form of words that end in *-s*, but are not plurals (e.g., *lejos* 'far,' *Carlos* 'Charles,' *garrapatas* 'tick'). Only 50% of these in the database are formed by adding *-ito/a* to the word (e.g., *adiós* > *adiosito* 'goodbye,' *Jesús* > Jesusito 'Jesus'). Others appear to have an infixed morpheme (*lejos* > *lejitos*, *Carlos* > *Carlitos*). Bermúdez-Otero (2007) suggests that much of the variation may be accounted for by dividing such words into two morphological categories. Words with athematic stems such as *virus* are treated as having no internal morphological structure. As a result, *-ito/a* is added directly to the base (*virus* > *virusito*), while pseudoplurals such as *Carlos* and *Lucas* are parsed as *Carl+o+s* and undergo diminutivization as do true plurals such as *galletas* (*gallet+a+s* > *gallet+it+a+s*, 'cookies,' *Carl+o+s* > *Carl+it+o+s*).

Perhaps the most difficult diminutives to account for are the handful of forms whose base ends in *-or* and *-ar* and that have penultimate stress. These include *Víctor* > *Victítor* 'Victor,' *azúcar* > *azuquítar* 'sugar,' *Héctor* > *Hectítor* 'Hector,' and *ámbar* > *ambítar* 'amber.' They are similar to pseudoplurals in that, despite their monomorphemic status, they are inflected as if they had an internal morpheme boundary (e.g., *azúc+ar*). All approaches to diminutives require some sort of manipulation of the formal apparatus in order to account for these unusual forms.

In sum, diminutive forms are largely predictable based on their gender, number of syllables, and phonological form, although dialectal and individual variation is attested. Appeals to morphological structure and abstract mechanisms have been resorted to in order to describe both dialectal variation as well as the forms that do not appear to be predictable phonologically.

4 Velar and coronal softening

The lexicon of Spanish is replete with morphologically related forms that exemplify velar and coronal softening in derived environments. Velar softening denotes an alternation between the velar stops /g/ and /k/, and the fricatives /θ/ and /x/ (or /s/ and /x/ in dialects without /θ/), as shown in (1).

(1)	/g/~/θ/	*distin/g/uir*	'to distinguish'	*distin/θ/ión*	'distinction'
		grie/g/o	'Greek'	*gre/θ/iano*	'Grecian'
	/g/~/x/	*ma/g/o*	'magician'	*ma/x/ia*	'magic'
		conyu/g/al	'marital'	*cónyu/x/e*	'spouse'
	/k/~/θ/	*Costa Ri/k/a*	'Costa Rica'	*costarri/θ/ense*	'Costa Rican'
		católi/k/o	'Catholic'	*catoli/θ/ismo*	'Catholicism'

Coronal involves an alternation between the coronal stops /t/ and /d/ and the fricative /s/, and in those dialects that contain it, the fricative /θ/ as well, as demonstrated in (2).

(2)	/t/~/s, θ/	*inyec/t/ar*	'to inject'	*inyec/θ/ión*	'injection'
		Mar/t/e	'Mars'	*mar/θ/iano*	'Martian'
	/t/~/s/	*emi/t/ir*	'to emit'	*emi/s/or*	'emitter'
		perver/t/ir	'to pervert'	*perver/s/o*	'perverted'
	/d/~/s/	*alu/d/ir*	'to allude'	*alu/s/ión*	'allusion'
		exten/d/er	'to extend'	*exten/s/ivo*	'extensive'
	/d/~/s, θ/	*aba/d/*	'abbot'	*aba/θ/ial*	'abbatial'

The large number of lexical items that participate in these alternations may be the reason they have received attention from so many investigators (Harris 1969, 1983; Morin 1997, 2002; Núñez-Cedeño 1993; Pilleux 1980; Spencer 1988; Wieczorek 1990).

Most studies of coronal and velar softening take one of three approaches to the issue. The first has as its goal to demonstrate that the alternations are the result of rules that depend primarily on phonology, that transform underlying representations into surface forms (Harris 1969, 1983; Núñez-Cedeño 1993). Apparent exceptions to these mechanisms are explained by appeals to differing phonetic and morphological contexts, as well as to a series of morphological and phonological elements used as diacritics. The second approach is principally a reaction to the first (Morin 1997, 2002; Pilleux 1980; Spencer 1988; Wieczorek 1990). These researchers argue that there are so many exceptions to the postulated rules, and so much *ad hoc* use of formal mechanisms that are essentially diacritic marks, that velar and coronal softening must be a lexicalized rather than a productive processes. The third approach represents attempts to determine the degree to which the alternations are synchronically productive by experimental means (Morin 1997, 2002; Núñez-Cedeño 1993).

In Harris' (1969) approach to velar softening, he proposes that underlying /k/ becomes /θ/ before front vowels (e.g., *místi/k/o* ~ *misti/θ/ismo* 'mystic, mysticism'). In like manner /g/ becomes /x/ before front vowels (e.g., *larin/g/oscopio* ~ *larin/x/e* 'laryngoscope, larynx'). The first difficulty with proposing front vowels as the motivating context is the existence of so many cases of velar stops before front vowels (e.g., *Puerto Ri/k/o* ~ *puertorri/k/eño* 'Puerto Rico, Puerto Rican,' *bode/g/a* ~ *bode/g/ero* 'wine cellar, wine producer'). Coronal softening suffers from many exceptions as well. For example, in some words it fails to apply before front vowels (e.g., *compe/t/ir* ~ *compe/t/idor* 'to compete, competitor'). In other cases, it paradoxically applies before back vowels (e.g., *perver/t/ir* ~ *perver/s/o* 'to pervert, perverse'). One way that Harris maintains the apparent regularity of velar softening is by assuming a phonological diacritic such that words with /k, g/ before front vowels are underlyingly /k^w, g^w/. These labiovelars are then deleted once they have served their function of blocking velar softening.

Another mechanism that Harris uses to explain apparent counterexamples is morphological boundaries. Accordingly, softening applies to *indi/k/ar* ~ *índi/θ/e* 'indicate, index' but not to *arran/k/ar* ~ *arran/k/e* 'to start, start' because they have different morphological boundaries in the underlying representation (i.e., *arrank#e* vs. *indik+e*). In essence, # blocks softening. The lack of true morphological motivation for different boundaries has been pointed out by many (Pilleux 1980; Spencer 1988; Wieczorek 1990). Morin (1997) extended and fortified this criticism by demonstrating that softening cannot be tied to particular suffixes either. For example, *-e*, *-ía*, and *-ismo* appear with softening in some words (e.g., *api/k/al* ~ *ápi/θ/e* 'apical, apex,' *aboga/d/o* ~ *aboga/θ/ía* 'lawyer, practice of law,' *católi/k/o* ~ *catoli/θ/ismo* 'Catholic, Catholicism'). However, in other cases no softening is observed (e.g., *arran/k/ar* ~ *arran/k/e*, 'to start, start,' *aba/d/*~ *aba/d/ía* 'abbot, abbey,' *taba/k/o* ~ *taba/k/ismo,* 'tobacco, nicotine addiction').

I have discussed two of the diacritic mechanisms used by Harris in his attempt to describe velar and coronal softening as processes that are mostly exceptionless, at least at an abstract level. A full treatment of the entire apparatus necessary to do this is beyond the scope of the present chapter. Both Harris, and the lengthy but the lesser-known treatment by Núñez-Cedeño (1993), require a great deal of formal manipulation and use of abstract entities. Accounting for softening in a more contemporary framework such as optimality theory does not eliminate the need for diacritics (see Morin 2002).

Observable data show that many instances of softening alternations exist. These may be conditioned by front vowels or particular morphemes. However, any generalization of this nature must admit a good number of counterexamples. For this reason, many researchers (Morin, 1997; 2002; Pilleux 1980; Spencer 1988; Wieczorek 1990) have concluded that softening alternations are lexicalized rather than playing a synchronically active role in Spanish phonology.

Historical considerations also place doubt on the validity of softening alternations. Rule analyses assume directionality in the alternations. For instance, underlying/g/surfaces as/x/and not vice versa; underlying/k/is transformed into/θ/and not the other way around. However, Pilleux (1980) notes cases in which it must move in the opposite direction. Historically, *farin/x/e* 'farynx' and *Gre/θ/ia* 'Greece' are older words upon which the more modern words *farin/g/ólogo* and *Gre/k/olatino* are based, yet these assume a phonological derivation completely opposite of that assumed in velar softening rules (i.e.,/x/ > /g/,/θ/ > /k/). The most likely explanation for velar (but not coronal) softening may be orthography. In Spanish orthography <c> is pronounced/k/before <o, u, a> and as/θ/before <i, e>. In like manner <g> is/g/before <o, u, a> and/x/before <i, e>. Accordingly, adding *-ólogo* to *faring-* places <g> before <o> at which point the spelling norm dictates the pronunciation as/g/.

Implicit in many accounts of softening is that it is the result of historical evolution. However, Morin (2002) demonstrates that softening alternations are not due to diachronic derivational processes, but the result of massive borrowing of learned words from Latin. That is, in related words such as *par/t/e ~ par/θ/ial* 'part, partial' the latter did not derive from the stem of the former by adding the suffix *-ial*. Instead, the popular word *parte* has had a continuous history in Spanish, while *parcial* was a learned word borrowed in about the fifteenth century. Phonetic evolution of the borrowings eventually yielded the modern pronunciation and softening alternations. In short, "words with these apparent alternations were either integral borrowings from Latin, and/or reflect the spelling pronunciations of Spanish at the time they entered the language as learned words" (Morin 2002: 157). Since softening was not the result of a historical derivational process, one cannot assume that the derivational relationships between the softening alternations have survived into Contemporary Spanish.

Morin (2002) tested the synchronic validity of softening by asking Spanish speakers to add the suffixes *-ente*, *-ino*, *idad*, *-ico*, *-ense*, *-ismo*, *-ista*, *-ía*, and *-iano* to nonce words such as *semedo* and *semoca* to determine whether the stem final/t, k, d, g/ would be softened. None of the test questions containing stem-final/t/or/d/were softened by any of her 32 test subjects. That is, the coronal in *seme/d/o* was not softened into *seme/s/iano* or *seme/θ/iano*. Velar softening was somewhat more productive in that/k/was softened in 30% of the answers (e.g., *semoca* > *semo/θ/ino*) and/g/in 13% of the answers. It appears that these instances of velar softening are due to orthography. In the study, nonce words ending in/k/and/g/appeared next to suffixes beginning with orthographic high vowels. In the dialect of the subjects tested *ge, gi* are pronounced [xe, xi] and *ce, ci* are pronounced [θe, θi]. On the other hand, orthographic <t> and <d> do not vary their pronunciation based on the following vowel. This explains why coronal softening was not productive at all compared to velar softening, which appears to be dependent on orthographic convention.

Núñez-Cedeño (1993) carried out a similar study to determine the synchronic validity of coronal softening through experimental means. In spite of the methodological limitations and statistical difficulties in his study (see Eddington 2004), the results are worth mentioning. As in Morin, Spanish speakers were asked to add suffixes that could trigger softening to stems ending in/t/and/d/. Only 3.4% of his eight subjects' responses demonstrated coronal softening. In another study, he presented the subjects with nonce verbs such as *enfurtir* and had them rate a number of suffixed forms derived from the base verb in terms of their acceptability (e.g., *enfursión*, *enfurtión*, *enfurtición*, *enfusión*). The change/t, d/ > /s/that is predicted by coronal softening was rated as acceptable in only 25% of the cases. The most acceptable option was to leave the/t, d/unchanged. Both of these findings suggest only a modicum of productivity for coronal softening.

In short, softening alternations are not the result of historical phonetic evolutions that have survived into Contemporary Spanish, nor are they very active synchronically either. The evidence for the productivity of velar softening appears to be due to orthographic factors. Formal attempts to make softening appear synchronically valid postulate mechanisms to account for the alternations, but are obliged to explain away the large numbers of counterexamples by appealing to abstract entities that cannot be verified. Since softening alternations are not predictable based on surface apparent criteria, they must be lexicalized in the words that exemplify them.

5 Nasal and velar depalatalization

The alternation between/n/and/ɲ/and between/l/and/ʎ/are illustrated by the examples below shown in (3):

(3)	*donce/ʎ/a*	'damsel'	*donce/l/*	'young nobleman'
	aque/ʎ/a	'that'	*aque/l/*	'that'
	be/ʎ/o	'beautiful'	*be/l/dad*	'beauty'
	desde/ɲ/ar	'to disdain'	*desdé/n/*	'disdain'
	te/ɲ/ir	'to dye'	*ti/n/te*	'dye'
	re/ɲ/ir	'to quarrel'	*re/n/cilla*	'quarrel'

These alternations were first noted by Alonso (1945) and have been treated in a number of formal frameworks: generative phonology (Contreras 1977); cyclic phonology (Harris 1983); lexical phonology (Wong-Opasi 1987); distributed morphology (Harris 1999); and optimality theory (Baković 1998, 2001; Lloret and Mascaró 2007). The proposal is that the palatals are underlying, but are depalatalized when they fall into the coda. However, that the coda is the relevant context is not apparent from an inspection of the surface forms. In the case of/*des.de.ɲar*/~/*des.den*/'to disdain, disdain' the nasal palatal is apparently depalatalized in the coda, but the palatal would be expected in/*des.de.nes*/'disdains.' In order to account for this discrepancy a wide variety of formal mechanisms have been appealed to, many of which place the palatals in the coda at some stage of the derivation. While formal accounts of depalatalization may be adequate descriptions that relate a number of lexical items, the questions of how pervasive depalatalization is in the Spanish lexicon, as well as how productive it is must be addressed.

Lloret and Mascaró (2007) compiled an extensive listing of word pairs that contain these alternations in order to determine their productivity. They point out that many purported instances of depalatalization involve words that are very semantically distant (e.g. *tropel* ~ *atropellar* 'mob, run over'). Others, such as *útil* ~ *utillaje* 'useful, tool' have related lexical items that do not participate in the alternation (e.g., *utilidad* 'utility'). Still others, such as *catalán* ~ *Catalúña* 'Catalan, Catalonia' appear to have independent rather than shared bases. Once such cases are eliminated, the relevant data are reduced to three cases of/n/~/ɲ/(*desdén* ~*desdeñar* 'disdain, to disdain,' *don* ~ *doña* 'Mr., Mrs.,' *champán* ~ *champañería*, *champañera* 'champagne, champagne shop, champagne$_{\mathrm{Adj}}$') and seven of/l/ ~/ʎ/(*él* ~ *ella* 'he, she,' *aquel* ~ *aquello* 'that, that,' *doncel* ~ *doncella* ~ *doncellez* 'nobleman, maiden, maidenhood,' *clavel* ~ *clavellina* 'nation, pink,' *piel* ~ *pellejo* 'skin, hide,' *Sabadell* ~ *sabadellense* Sabadell, from Sabadell,' *beldad* ~ *bello* 'beauty, beautiful').

Lloret and Mascaró recognize that the limited lexical presence of these alternations casts doubt on the productivity of a process of depalatalization. A study by Pensado (1997) also suggests a lack of productivity. Her subjects performed a number of experiments to test whether depalatalization is recognized and can be applied by native Spanish speakers. In the first experiment, subjects saw drawings of animal-like creatures and a machine that made them. The nonce animal names contained/l/and/n/in the coda (e.g., *enapil*, *sirapén*), while the corresponding verbs had/ʎ/and/ɲ/in the onsets (*enapillar*, *sirapeñar*). For example, *Esto es un enapil* 'This is a *enapil*'; *Esto es una máquina de enapillar* 'This is a machine to *enapillar*.' The subjects' task was to form plurals of the nonce words by add *-es*, past participles of the words by adding *-ado*, and adjectives by adding *-oso*.

Past participles and adjectival formation resulted in some surprises. One group of test subjects saw the sentence containing the noun *enapil* before the sentence containing the verb *enapillar*. The other group was shown the verb before the noun. The order in which they were presented the sentences containing *sirapén* and *sirapeñar* varied as well. What is interesting is that the order of presentation influenced the outcome. For example, if the last nonce word seen by the subjects was *enapil*, subjects tended toward the adjective *enapiloso*, but when *enapillar* was the last nonce word seen, the trend was toward *enapilloso*. In other words, the choice was affected by order of presentation, and speakers had no qualms about putting either the palatals or the nonpalatals in the syllable onset.

Since past participles are based on verbal stems, *enapillado* and *sirapeñado* would be the expected responses to the nonce words. In spite of this, many subjects chose *enapilado* and *sirapenado*, completely disregarding the depalatalization allomorphy they were presented in the questionnaire. If depalatalization were a productive process, Pensado's subjects should have been able to apply it to the nonce words. However, the large degree of inconsistency in their answers, coupled with the fact that many answers were based on the phonological shape of the last nonce word presented to them suggest that the subjects were unaware of the depalatalization alternations and how to apply them. While Pensado's data are interesting, it should be noted that there are a number of flaws in the exposition and methodology of her study (see Eddington 2004; Lloret and Mascaró 2007) which future research needs to take into account.

Another critic of depalatalization is Harris (1999). Although he supported a process of depalatalization in his earlier writings (Harris 1983), it proved problematic when incorporated into a distributed morphology model (Harris 1999). His reason for abandoning depalatalization is not based on historical reasoning or data suggesting it is not active. Instead, he finds that depalatalization results in exceptions to his rules which predict final /e/ epenthesis incorrectly and derive *él* as /eʎe/ rather than /el/.

Only a handful of lexical items exemplify depalatalization. This, coupled with the experimental evidence that it is not applied to new lexical items, suggests that depalatalization is not a productive process. Nevertheless, Lloret and Mascaró (2007) argue that it must be an active process because it productively applies when words are adopted into Spanish. For example, Catalan *Co*/ʎ/ and *se*/ɲ/ become /kol/ and /sen/. They note that orthographic adaptation would render /ɲ/ as /ni/, but phonetic adaptation results in /n/. An innovative aspect of their treatment is that they expand the /ʎ/~/l/ and /ɲ/~/n/ alternations to include /m/~/n/ (e.g., *Abrahán ~ abrahámico*, *íten ~ itemización*). In borrowings from other languages (e.g., *ron* < *rum*) word-final /m/ is normally realized as /n/ in Spanish. Rather than depalatalization, they speak of centralization of these phones to an alveolar place of articulation when they appear in the coda.

Lloret and Mascaró (2007) make a good case for productive depalatalization word-finally in borrowed words. However, this fact does not in any way constitute evidence that depalatalization is a productive word-internal process as well. In a like manner, it has no bearing on whether words such as *bello ~ beldad* are derived from a common stem such as /beʎ-/ in the mental lexicon of Spanish speakers. This claim must be established independently. A similar error is made when one assumes that because the /l/~/ʎ/ and /n/~/ɲ/ alternations may be described by the same formal mechanism, the productivity of one is evidence for the productivity of the other. As already noted, Lloret and Mascaró (2007) include /m/~/n/ alternations in their account and speak of centralization. If Spanish speakers are found to pronounce word-final /m/ as [n] with a high degree of frequency, this has no bearing whatsoever on whether Spanish speakers derive *don* and *doña* from the same underlying representation. The quest for elegance and simplicity in formal descriptions requires incorporation of as many similar alternations as possible into the description. However, actual processing suggests that speakers do not always account for such alternations in the most concise way (Bybee and Pardo 1981; Hale 1973).

Analyses that consider the /l/~/ʎ/ and /n/~/ɲ/ alternations to be the result of the same process of depalatalization must surmount another obstacle. In varieties of Spanish that contain /ʎ/, a stronger case for a unitary process is possible since both alternations entail only a change in place of articulation. However, most Spanish varieties lack /ʎ/ (Lipski 2004), and in its place have a nonlateral palatal which means that /l/ alternates with /j/, /ʃ/, or /ʒ/. These phones not only differ from /l/ in place of articulation, but in manner (lateral vs. approximant or fricative), and in the case of /ʃ/, in terms of voicing as well. Therefore, in most varieties of Spanish, the notion of a unitary process that only affects place of articulation, and accounts for both *don* ~ *doña*, and *él* ~ *ella* is untenable.

Given the small number of lexical items that participate in depalatalization, it has attracted a disproportionate amount of attention. While the process appears to be productive word-finally, mostly in Catalan borrowings, experimental evidence suggests that speakers are not able to extend it to pairs of nonce words that exemplify it. Its productivity in this context is probably due to a phonotactic constraint against/m, ʎ, ɲ/in codas rather than to a morphophonological alternation. Additionally, the lack of/ʎ/in the majority of the Spanish-speaking world argues against a unitary process of/ʎ/and/ɲ/depalatalization. In my view, it should be placed alongside other nonsystematic alternations such as those in *amigo ~ amistad*, *resucitar ~ resurrección*, and *herejía ~ herético*.

6 Conclusions

In the present chapter I have discussed a number of morphophonological alternations in Spanish emphasizing the role of surface-apparent traits in the alternations. I argue that the somewhat productive status of diphthongization appears to be due analogy to existing words. I also show that diminutive allomorphy is largely predictable based on the base's final phone, gender, and number of syllables. Little evidence exists to support coronal softening as an active process, while the productivity of velar softening appears to be orthographic in nature. Depalatalization, on the other hand, only occurs in borrowings where palatals would appear in word-final position, but this phonotactic constraint is not evidence for a unique underlying/ʎ/for both *bello* and *beldad*, for instance.

NOTES

1 Accent marks indicate spoken stress and do not follow orthographic conventions.
2 Thanks to J. I. Hualde for this insight.

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11 Derivation and Compounding

SOLEDAD VARELA

1 Derivation: types, suffixation, prefixation

1.1 Types of derivation

In the word-formation process known as derivation, we can distinguish between: (a) affixal derivation in which there is addition of affixes to a lexeme (*in-constante* 'inconstant,' *transporta-ble* 'transportable'); and (b) nonaffixal derivation or back-formation based on the elimination of some morphological part of the lexeme (*retén* 'stop, checkpoint' from *retener* 'to retain'). Akin to this second procedure are the so-called thematic or postverbal forms in which a verb lexeme is converted into a noun just by adding an unstressed vowel to the root. This final vowel may coincide with the theme vowel of the verb, as in *guard-a-r* 'to guard' > *guard-a* 'guard,' although most times the derived noun shows up with a different vowel: *atrac-a-r* 'to attack' > *atrac-o* 'robbery'; *empuj-a-r* 'to push' > *empuj-e* 'pressure.' In some cases the same verbal root admits any of the three vowels: *costar* 'to cost' > *cost-e*, *cost-o*, *cost-a*(*s*) 'cost.'[1] In all Spanish dialects, word formation by means of affixation is prevalent.

1.2 Suffixation: categorial and semantic selection

Lexical derivation through suffixation is the most productive, general, and varied of Spanish word-formation procedures. Spanish has a large number of suffixes with variable meanings, and all the main lexical categories (V, N, A) accept this type of derivation. Moreover, resource-to-suffixation is common to all language varieties – technical and scientific, legal and administrative, as well as literary – and to all registers or levels of formality, and suffixed words appear pervasively both in speech and writing.

In the suffixation process, the "word marker" of the base noun is always suppressed. Word markers are morphological segments which are specifically

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associated with nominal derivation. They are formed mainly by one of the three vowels *a*, *o*, *e* (*esponj*(*a*) 'sponge' > *esponj-oso* 'spongy,' *cor*(*o*) 'choir' > *cor-al* 'choral,' *cond*(*e*) 'count' > *cond-ado* 'county'), marginally also by the vowels *i* and *u* (*tax*(*i*) > *tax-ista* 'taxi-driver,' *trib*(*u*) 'tribe' > *trib-al* 'tribal'), and by some complex endings formed by any vowel + *s*: (*mecen*(*as*) > *mecen-azgo* 'patronage,' *Carl*(*os*) 'Charles' > *carl-ista* 'Carlist,' *diabet*(*es*) > *diabét-ico* 'diabetic,' *tes*(*is*) 'thesis' > *tes-ina* 'minor thesis,' *vir*(*us*) > *vír-ico* 'viral') (Roca 2005).

In the case of adjectives with gender variation, the vowel ending (*-o* for the masculine and *-a* for the feminine) is an inflectional morpheme, a mark of agreement, and, as such, it is absent whenever there is further derivation of the adjective (*sólid-* 'solid' > *solid-ific-ar* 'solidify'). Only in the case of the adverbial suffix *-mente* does the adjective in the base seem to show the feminine gender vowel *-a* (*sólid-a-mente* 'solidly'), but this is a historical remnant of a noun phrase in which the adjective showed agreement with the Latin ablative noun *mente* 'mind,' the origin of the present day suffix *-mente*.

As in the case of lexemes, noun suffixes also are endowed with an inherent gender or may show gender motion. For instance, the suffixes *-ción* and *-dad* transmit the feminine gender, while other suffixes, like *-er-*, can adopt the masculine or the feminine form (*zapat-er*{*-o*/*-a*} 'shoemaker'). Suffixes also have their own lexical category. This fact has two important consequences: suffixes determine the lexical category of the complex word they form; and they impose constraints on the lexical category of their input bases. When a suffix changes the category of its base, it is common to refer to "heterogeneous derivation," as in the case of *demostra*$_V$-*ción*$_N$ 'demonstration' and *demostra*$_V$-*ble*$_A$ 'demonstrable,' where the corresponding suffixes convert the verb stem *demostra-* into a noun and an adjective, respectively. Some suffixes do not change the lexical category of the base, but may still modify some subcategorial features of it. For instance, the suffix *-er*{*o*/*-a*} above applies to a [-animate] N and gives a [+animate] N. Some suffixes, finally, respect the lexical category of the base yielding a so-called homogeneous derivation, as is the case of the suffix *-ito* which, when attached to a noun or an adjective, maintains the category of the base (*caballo* 'horse' > *caball-ito* 'little horse'; *pequeño* 'small' > *pequeñ-ito* > 'quite small').

Suffixes also have the property of selecting the base to which they attach in accordance with its lexical category. As we saw above, *-ción* is a noun suffix which selects verbs. On the other hand, the noun suffix *-dad* subcategorizes for adjective bases (*bon*$_A$-*dad*$_N$ 'goodness') and the adjective suffix *-oso* selects nouns (*bondad*$_N$-*oso*$_A$ 'kind'). Suffixes are also sensitive to subcategorial features of their bases. For instance, *-ble* selects transitive-agentive verbs. Thus, from *domesticar una fiera* 'to domesticate a wild beast,' we can form *la fiera es domesticable* 'the wild beast is tamable,' but **bostezable*, **nacible* or **llegable* are ill-formed because the verb base (*bostezar* 'to yawn,' *nacer* 'to be born,' and *llegar* 'to arrive,' respectively) does not meet the transitivity condition. Transitive verbs that denote a state and therefore do not have an agentive subject are also bad candidates for *-ble* derivation; for instance, from *tener fiebre* 'to have fever' we do not obtain **la fiebre es tenible* 'fever is have+*ble*.' Some transitive psych verbs of the *temer* class ('to fear') accept the suffix *–ble*, although they are experiencer verbs not agentive verbs. However, most of the resulting adjectives are lexicalized with an evaluative or appreciative meaning: *temible* 'dreadful, unpleasant,' *adorable* 'adorable, charming,' *detestable* 'detestable, awful.' Note that from *Juan teme los truenos* ('J. fears thunders' = 'J. is afraid of thunders') we cannot get **Los truenos son temibles* ('Thunders are fear-*ble*').

Selection of the lexical base by the suffix may also depend on some categorical distinction of a semantic nature. For instance, the suffix -(*i*)*dad*, which forms quality or state nouns out of adjectives, only selects qualitative adjectives (*peligroso* 'risky' > *peligros-idad* 'riskiness'), and rejects relational adjectives (*aéreo* 'aerial' > **aer-idad* 'aerial+*idad*'). In the cases where -*dad* adjoins to a relational adjective, the resulting N usually refers to a collectivity: *cristiano* > *cristian-dad* 'Christendom.' This restriction is easily recognized in such cases where the same adjective can have the two interpretations, one qualitative and another relational. For example, *familiar* denotes a quality in *trato familiar* 'familiar relationship,' with the meaning 'intimate, close,' but it denotes something related or pertaining specifically to the family in *planificación familiar* 'family planning.' Only in the first case is it possible to form the noun *familiaridad* (*del trato*) 'familiarity (of the relationship).' In some cases, morphology distinguishes these two interpretations with different suffixes applied to the same nominal base. Thus, in the following pairs the first adjective has a qualitative interpretation and the second a relational one: (*aspecto*) *caball-uno* 'horse-like aspect'/(*cría*) *caball-ar* 'horse breeding'; (*labios*) *carn-osos* 'fleshy lips'/(*industria*) *cárn-ica* 'meat industry'; (*brazos*) *muscul-osos* 'muscular arms'/(*tono*) *muscul-ar* 'muscle tone.'

A special case of derivation is the so-called evaluative suffixation, a very productive type of derivation mainly applying to nouns and adjectives. This semantic class comprises a fixed number of suffixes that transmit a diminutive, augmentative, or pejorative meaning. Variation among Spanish dialects is particularly high in this field, so that the preference for one form or another is regionally determined. The suffixes -*ito*, -*illo*, -*ín*, -*ino*, -*iño*, or -*ico* usually convey a diminutive interpretation when attached to a nominal base (*perr*{-*ito*, -*illo*, -*ín*} 'little dog'). On the other hand, -*azo* or -*ón* characteristically have an augmentative meaning (*jef-azo* 'big boss,' *notici-ón* 'big news'), and -*ucho* or -*ajo* may transmit a pejorative meaning (*papel-ucho* 'bad paper,' *hierb-ajo* 'bad grass'). However, evaluative meanings cannot be categorically associated with a specific suffix, because the evaluative or affective connotation that they may transmit depends on the lexical base to which they adjoin and also on the speaker's intention and the context in which they are produced. So, the suffix -*azo* can denote big size – thus with an augmentative meaning – in the noun *perrazo* 'big dog,' but transmits a connotation more akin to the diminutive or even the derogatory meaning in the nominalized adjective *un buenazo* 'a good person'; and, it adds a purely affective connotation in the noun *padrazo* 'indulgent father.'

The lexical base to which the evaluative morpheme is attached preserves its basic notional content both in nouns and in adjectives: *montañ-ita* = 'mountain + little,' *grand-ón* = 'big + very.' The evaluative suffix -*ito* may also attach to simple adverbs (*despac-ito* 'slowly+*ito*'), and, especially in Latin American Spanish, to a limited set of gerunds (*calland-ito* 'keeping quiet+*ito*'). Being evaluative, these suffixes pertain to the "homogeneous derivation," that is, they do not change the lexical category of the base. However, it is quite common for evaluative suffixes to become lexicalized, in which case the relation to a primitive evaluative meaning can be completely lost, as in the case of *bombilla* 'lamp-glass' or *mechón* 'lock (of hair).' Evaluative suffixes are inserted after all the other derivative suffixes and just before the inflectional ones: *tost-ad-it-o-s* 'toast-ed-*it*-masc-plur.' When the evaluative suffix applies to a noun, it often imposes the canonical vowel that characterizes feminine nouns (*-a*) vs. masculine nouns (*-o*) even if the "word marker" shown in the base is different: *man-o*$_{Nfem}$ > *man-it-a* 'little hand' (in Peninsular Spanish), *el jef-e*$_{\mathrm{Nmasc}}$ > *el jef-az-o* 'big boss.' Diminutive suffixes may be preceded by an interfix. The presence and the form of the interfix (*-c-* or *-ec-*) depend on the number of syllables of the base (*sol-ec-ito* 'little sun') as well as on its phonological make-up (*jardin-c-ito* 'little garden,' *rued-ec-ita* 'little wheel').

Spanish does not have comparative suffixation. The old Latin comparative suffixes are not active any longer, although they appear in some residual formations with a comparative meaning: *mejor* 'better,' *peor* 'worse,' *mayor* 'bigger,' *menor* 'smaller.' The only morpheme with a degree meaning that has remained in Spanish is superlative *-ísim*{*-o*/*-a*}. Applied to a qualitative adjective, this suffix denotes the quality in its higher or more intense degree: *buen-ísimo* 'very good.' Some adverbs also accept the superlative suffix: *cerqu-ísima* 'very near.' More restrictively, some nouns that denote kinship relations may manifest this suffix in order to convey the meaning of an intense influence of the person holding this relation to a political authority (*yern-ísimo* 'son-in-law+*ísimo*'), or just to convert the noun into a superlative adjective (*padr-ísimo* 'father+*ísimo*,' 'cool!').

1.3 Prefixation

Contrary to suffixes, prefixes do not belong to a precise lexical category and they respect the lexical category of the base to which they attach: *ventana*$_{\mathrm{N}}$ > *contra-ventana*$_{\mathrm{N}}$, 'counter-window' = 'shutter,' *contar*$_{\mathrm{V}}$ > *re-contar*$_{\mathrm{V}}$ 'to count again,' *fino*$_{\mathrm{A}}$ > *extra-fino*$_{\mathrm{A}}$ 'extra-fine.' However, some prefixes like *anti-*, *pro-* and others with a quantitative meaning like *mono-* or *tri-* seem to convert the noun to which they attach into an adjective. The relevant cases are the ones where a prefixed noun can modify another noun, like in *minas antipersona* 'anti-person mines' or *bandera tricolor* 'tricolor flag.' Such a modification is not possible if the noun is not prefixed (**minas persona*, **bandera color*), and therefore some authors have referred to them as "transcategorial prefixes." However, the prefixed noun does not generally agree in number with the noun it modifies, as would be expected if it were a real adjective: *banderas tricolor*(**es*). On the other hand, these constructions alternate with others where the prefixed noun has clearly been turned into an adjective by means of a specific suffix: *minas antiperson-al-es* 'anti-personnel mines.' Taking into account these facts, we suggest that the prefix does not have the ability to change the category of the base and that the prefixed N is in an appositive relation with the head-N of the NP. The function of the prefix preceding the noun is to emphasize the modifying value of the N to which it attaches. Note that the semantic capacity of some prefixes for transmitting an enhancing value to the complex word that they form is also attested in other cases. For instance, deverbal adjectives in *-do* or *-ble* may derive into an adverb with the suffix *-mente* only if their respective bases appear qualified by the prefix *in-* and are thus completely adjectivized: *opinada* > **opinadamente/in-opinada* > *inopinada-mente* 'unexpectedly,' *creí-ble* > **creíble-mente/in-creíble* > *increíble-mente* 'unbelievably.'

Whereas suffixes force the canceling of the final unstressed vowel of the base or word-marker (Section 3.2), prefixes tend to preserve their phonological identity: *autoevaluar* 'to auto-evaluate,' not **aut(o)evaluar* or **auto(e)valuar*. This may be so even in the case of confluence of two identical vowels (*pre-estreno* 'preview'), most regularly when vowel coalescence would induce confusion with another word (*semiinconsciente* 'almost unconscious' vs. *semiconsciente* 'almost conscious'). Sometimes the prosodic independence of prefixes contravenes even Spanish general syllabification patterns, as in *subregional* not **su-bregional* 'subregional' (cf. *co-bre* 'copper,' not **cob-re*). Also, contrary to suffixes, which frequently modify the stress position of the base word (*leˈal* 'loyal' > *leal-ˈtad* 'loyalty'), prefixes do not interfere in it (*leˈal* > *des-leˈal* 'disloyal').

Although the same prefix may apply to different lexical categories (*super-hombre* 'superman,' *super-valorar* 'to overvaluate,' *super-fácil* 'extremely easy'), prefixes tend to specialize for a specific lexical category in accordance with their semantics. Thus, since what we disclaim or highlight are, in general, properties or qualities, the negative prefix *in-* and the intensifier *re-* attach to qualitative adjectives (*in-culto* 'uncultivated,' *re-bonito* 'very pretty'). On the other hand, size and quantity affect objects that have a dimension and are countable; therefore, the prefixes *mini-* and *mono-* select only nouns (*mini-crisis* 'micro-crisis,' *mono-cultivo* 'single-crop') or denominal adjectives (*mono-silábico* 'mono-syllabic'). Finally, reversative *des-* attaches only to verbs that denote actions that may be undone (*des-enchufar* 'to unplug'), and iterative *re-* only to actions that can be repeated (*re-conducir* 'to re-conduct').

In many cases, Spanish prefixes correspond to actual prepositions of Spanish, or else to Latin or Greek prepositions that have not entered into Spanish as free or separable morphemes. Thus, some prefixes can be identified with a Spanish preposition because of their form as well as their function (*sobre-volar* 'to overfly,' *entre-sacar* 'to pick out') and others only by virtue of their function (*super-poner* 'to superpose,' *inter-poner* 'to interpose'). However, not all prefixes coincide with prepositions or are etymologically related to them, as is the case of *dia-* or *bi-* which are always realized as affixes that precede the lexical base, that is, as prefixes. On the other hand, prefixes that in some instances may be considered prepositional take more frequently an adverbial value (*sobre-alimentar* 'to overfeed,' *ultra-rápido* 'extra fast'). In my opinion, they should be classified as a type of affix that combines with a lexical base in a process of derivation, and not as a kind of free form that enters into a compounding process.

Prefixes should be distinguished from "learned stems" which very frequently appear preceding the lexical base. Some of them are very productive in general vocabulary (*filo-comunista* 'filo-communist'), and others appear mostly in technical and scientific vocabulary (*hemo-globina* 'hemoglobin'). Learned stems, contrary to prefixes, can also show up at the right edge of the lexical base (*biblió-filo* 'bibliophile'), can take affixes by themselves (*á-grafo* 'a-graph, illiterate,' *graf-ía* 'graphy'), and combine with another learned stem to form a compound (*filo-logía* 'philology'). All these facts make them different from real prefixes, and therefore we will consider them lexemes entering into composition (Section 3.1). Prefixes should be also differentiated from clipped words, like *demo*(*cracia*) 'democracy' or *euro*(*pa*/*peo*) 'Europa/European,' which appear in first position combined with another lexical base and constitute a compound: *demo-cristiano* 'Christian-Democrat,' *euro-parlamentario* 'European-deputy.' Sometimes, prefix and clipped word are formally identical. In *auto-gobierno* 'self-government' we have a prefix, but in the compound *auto-escuela* 'driving school' we have the shortened variant of the noun *auto(móvil)* 'automobile.'

Prefixes combine with suffixes in a formation known in the Romance linguistic tradition as "parasynthesis" An example is the verb *en-torp-ec-er* 'to benumb, to make torpid,' supposedly formed by the simultaneous adjoining of the prefix *en-* and the suffix *-ec-* to the adjective stem *torp-*. If not realized as a joint process, we would not obtain a complete word: neither **en-torpe* nor **torp-ecer* are possible forms. In the *Nueva gramática de la lengua española* (2009), this kind of structure, which is very productive in Spanish, is analyzed as containing a verbalizing "discontinuous morpheme": *en-* … *-ec-* or *a-* … *-iz-* (as in *a-terror-iz-a-r* 'to terrify'). In some parasynthetic formations without any specific suffix (*en-fri-a-r* 'to cool,' *a-barat-a-r* 'to make cheaper'), it is commonly assumed that the thematic vowel of the verb acts as a derivational suffix. This capacity of the thematic vowel is also attested in some N > V derivations without suffixation: *sal* 'salt' > *sal-a-r* 'to salt.'

2 Derivation argument structure, aspect, and affix ordering

2.1 *Derivation and argument structure*

Some prefixes, when adjoined to a lexical base, change their Argument Structure (AS). For instance, when the simple verb *callar* 'to keep quiet' takes the prefix *a-*, this introduces obligatorily an agentive subject in a causative construction: *Las voces de protesta callaron* 'Protest voices stopped' > *El alcalde acalló las voces de protesta* 'The mayor silenced the protest voices.' Similarly, the prefix *co*(*n*)- changes the AS of the base verbs *vivir* 'to live' or *operar* 'to operate,' forcing the presence of a comitative complement: *Juan convive* *(*con su novia*) 'J. lives-together with his girlfriend,' *Juan coopera* *(*con los bomberos*) 'J. cooperates with the firemen.'[2]

Regarding suffixes, a well-known case is that of deverbal nouns derived by means of suffixes like *-ción*, *-miento*, *-do*, and others which denote an event and take grammatical arguments. For morphologists who place word formation in the lexicon, their aspectual and distributional peculiarities result from the "inheritance" of the properties of the verb that lies at their base. For those who favor a syntactic account of word formation, the nominal's peculiarities are also a consequence of its internal structure, through the combination of the noun suffix with (some part of) the syntactic projection of the verb lexeme.

As in other languages, two main types of derived nominals are recognized in Spanish: (1) eventive nominals, with AS; and (2) result/object nominals lacking AS. The notions of event and result are usually represented under the same form of the noun (*donación* 'donation,' *estacionamiento* 'parking,' *estampado* 'printing'), but sometimes morphology differentiates these notions: *invención* 'invention'/*invento* 'thing invented' = 'invention,' *conservación* 'conservation'/*conserva*(*s*) 'preserved foods.' Some derived nominals also resort to different allomorphs of the base: one allomorph if the deverbal makes reference to the action and another one when it refers to its result, as in *apertura*$_{\mathrm{Nevent}}$ *de las negociaciones* 'opening of the negotiations' vs. *abertura*$_{\mathrm{Nobject/result}}$ (*de la pared*) 'hole of the wall.' Certain prefixes also permit us to distinguish the nominal's condition: they can be associated to the noun if it denotes an action or an event (*reapertura* 're-opening,' *coproducción* 'co-production'), but they are not compatible with the derived nominal if this makes reference to the action's object or the action's result (**reabertura* 're-hole,' **coproducto* 'co-product').

2.2 *Affixation and aspect*

The nature of the verbal action, that is, what we understand generally as aspectual properties, is determinant in the adjoining of some affixes to their bases as well as in the aspectual behavior of the resulting derived word. For instance, some nouns in *-dor* derive from atelic activity verbs (*fumar* 'to smoke' > *fuma-dor* 'smoker'), and transmit a habitual-continuous aspect. This is the reason why they can be modified by frequency adjectives, such as *constante* 'constant,' *continuo* 'continuous,' and the like. They differ in aspectual terms from V N compounds (Section 3.3), with which they are usually compared since both word-formation processes give rise to agent (and instrument) Ns. However, a NP like *los constantes fumadores* 'the constant smokers' is perfectly constructed, but **los constantes cuidacoches* 'the constant car-watchers' is not, because in this case the aspectual marks of the V do not percolate to the nominal construction. Other deverbal formations are also aspectually determined: (a) adjectives in *-nte*, which select atelic verbal bases, either states or activities (*puente colgante* 'suspension bridge,' *rodillo deslizante* 'sliding roller') (Cano in press); (b) adjectival participles in *-do*, which are aspectually perfective and correspond to telic verbs yielding stage-level adjectives (*los cansados viajeros* 'tired travelers') and secondarily individual-level adjectives (*trabajo muy descansado* 'very relaxing job') (Varela 2003); (c) adjectival gerunds in *-ndo* (*agua hirviendo* 'boiling water'), dynamic-durative from the aspectual point of view (Fábregas 2008).

Eventive nominals have the well-known particularity of being affected by the same aspectual restrictions as the verbs in their base. In *la llegada de Juan* {**durante* *una hora/*hasta el anochecer*} 'John's arrival {during one hour/until nightfall},' we have an example of a noun derived from a verb of accomplishment that, consequently, does not admit a temporal complement headed by *durante* or *hasta*. On the contrary, the nominal in *la búsqueda de aparcamiento* {*durante una hora/hasta el anochecer*} 'John's search for a parking place {during one hour/until nightfall}' derives from an activity predicate, and, as expected, readily accepts such prepositional complements.

Prefixes are also sensitive to aspectual distinctions. For instance, the prefix *re-* of repetition selects productively telic verbal bases which produce a result (*re-abrir la ventana* 'to re-open the window'). However, *re-* accepts neither stative verbal bases (**re-pertenecer* 're-belong') nor activity verbs that do not contemplate the event culmination (**re-nadar* 're-swim') (Martín García 1998).

2.3 Productivity and affix ordering

Some combinations of suffixes are very productive. This is the case of the adjective suffix *-ble* (represented by its allomorph *-bil*) followed by the noun suffix *-idad* (*renta-bil-idad* 'profitability') and of the sequence *-al* + *-iza* + *-ción* (*peaton-al-iza-ción* 'the act of making [a space] apt for pedestrians'). Some suffixes are grouped in pairs in such a way that, for many morphologists, there is not derivation on one of them but the substitution of one suffix by the other one: *-nte/-ncia* (*obedie-nte* 'obedient'/ *obedie-ncia* 'obedience') or *-ista/-ismo* (*marx-ista* 'Marxist'/*marx-ismo* 'Marxism'). The solidarity among the two suffixes is also shown by further derivations. For instance, if an *anti-…-ismo* is created to indicate the opposition to a specific doctrine or movement, the corresponding *anti-…-ista* will be formed (*anticonsumismo* 'anti-consumption+*ismo*'/*anticonsumista* 'anti-consumption+*ista*'). The relation among them is of a semantic nature, as can be seen by the fact that they create clear cases of "bracketing paradoxes" (Spencer 1991): an *anticonsumista* is not someone who is against *consumistas* but against *consumo*.

The possibilities for prefix combination are very limited and subject to very strict semantic restrictions. Prefixes with a prepositional value are internal to prefixes with an adverbial value (*re-ex-portar* 'to re-ex-port,' *des-en-cadenar* 'to unchain'). Two prefixes with a prepositional content can be also concatenated (*contra-en-dosar* 'to counter-endorse,' *co-a-sociarse* 'to co-associate'), as well as two prefixes of the adverbial type (*super-in-moral* 'super-in-moral,' *ex-vice-presidente* 'ex-vice-president'). What we will not encounter is the combination of prepositional prefix + adverbial prefix since the first one, having scope over the argument structure of the base and affecting its event dimension, must be next to the base. On the other hand, adverbial prefixes inherit the argument and event structure of the base and only affect its semantic content, hence their external position. Inside some syntactic constructions, prefixes corresponding to the same lexical field, either antithetic (*pro y antigubernamental* 'pro- and anti-governmental') or synonymous (*pre y proto-histórico* 'pre- and proto-historical'), can be coordinated with ellipsis of the common base they share.

3 Compounding: constituents, traditional classifications, and types

3.1 Constituents

Compounding in Spanish, as in the rest of Romance languages, is less productive than derivation and is much more restricted than in the Germanic languages. We may define compounding as a word-formation process by which two or more lexemes combine to constitute a complex word, which is a lexical unit from a semantic, phonological, and functional point of view. For instance, the noun lexeme *pel-* 'hair' followed by the linking element *-i-* and the adjective lexeme *roj-* 'red' combine to form the adjective compound *pelirrojo* 'red-haired.' By lexeme, we understand here not only uninflected forms of independent words, as the stems *pel-* and *roj-* of the example above, but also the so-called learned stems of Greco-Latin origin. In fact, all Romance languages have continued to produce "neoclassical compounds" by means of these components, combined either among themselves (*logopeda* 'speech therapist') or combined with an actual word of the language (*panamericanismo* 'Pan-Americanism').

Compounding by means of Greco-Latin stems is special in respect to its constitution. When two of these constituents combine together, the origin of the second one determines the kind of linking vowel appearing between them: if the second stem is of Greek origin, the linking vowel is usually *-o-* (*mare-ó-grafo* 'tide-graph'); if the second constituent is of Latin origin, the linking vowel is in general *-i-* (*gran-í-voro* 'grain-eater'). Another peculiarity of this kind of compounding is that with A compounds denoting nationalities, the last adjective appears in its native form and the rest of the adjectives in their learned or Latin form: *hispanofrancés* 'Hispanic-French'/*francoespañol* 'French-Spanish.'

The lexical categories that enter into compounding are the major ones: V, N, and A. There are also a few cases where Adv(erb) appears as a compound constituent, either modifying a verb (*malvender* 'to sell off cheap,' *malgastar* 'to waste') or a deverbal adjective (*bienhablado* 'well-spoken,' *malnacido* 'badly-born' = 'obnoxious'). The Prep(ositions) entering into compounding constitute a closed category, no longer productive, in which Prep can coordinate with other prepositions (*desde* < *de es de* 'from') or with a conjunction (*conque* 'whereupon'). As far as Prep N combination is concerned, our view is that these formations are better analyzed as a derivational construct. Besides the reasons adduced in Section 2.3, it is worth noticing that prepositions that do not alternate with prefixes, such as *desde* 'from' or *hacia* 'towards,' never combine with N to form a compound.

The most productive categories in the constitution of compounds are N, V, and A, but the categories of the resulting word are almost exclusively N or A. Combinations of V and N (*pararrayos* 'stop lightnings' = 'lightning conductor') or even the few compounds in which two Vs combine (*subeibaja* 'goes-up-*i*-goes-down' = 'seesaw') always produce nouns. Only a few combinations of N V – following the Latin model and no longer productive – yield verbs (*maniatar* 'to hand-cuff'), as well as the very few instances of Adv V already mentioned (*maltratar* 'to ill-treat').

3.2 Traditional classifications of compounding

Compounds are classified as endocentric and exocentric according to the position of the head (i.e., taking into account whether it is inside the complex unit or outside of it). Two notions of head are mixed with the endo/exocentricity issue. One refers to the category of the complex word as percolated from its head. Another one is the hyponym/hypernym semantic relation between the compound and its head. From the first point of view, we consider that all productive compound types are endocentric, although the possibility of lexicalization or metaphoric change of any existing word often makes semantic composition opaque so that semantic endocentricity is blurred. For instance, *malapata* 'bad-leg' = 'hard luck' is categorially a feminine noun, as is its head (*pata* 'leg'), but semantically we are unable to identify the compound with the noun on its head without the necessary encyclopedic information. Contrary to other languages where the head of the compound is defined according to its position (the right constituent in English, as in derivation), in Spanish, as well as in the rest of the Romance languages, it is not possible to identify the head on the basis of its position because it is located at the left in some compounds (*aguardiente* 'water-burning' = 'brandy') and at the right in other cases (*justiprecio* 'fair-price'). The head is thus identified with the constituent that imposes its category to the whole compound and that is a hypernym of the complex (Rainer and Varela 1992).

In the Spanish tradition, it is common to distinguish between subordinate and coordinative compounds. The first type is represented by compounds with a verbal or a deverbal head (*perniquebrar* 'to break the leg of someone,' *limpiabotas* 'boot-cleaner') that take an argument as their complement. Other compounds showing a head-modifier relation (*vanagloria* 'vainglory,' *malvivir* 'to live poorly') or a head-complement relation (*hojalata* 'tinplate,' *telaraña* 'spiderweb') are usually recognized as a subgroup of the subordinate type. As for coordinative compounds, the constituents belonging to the same lexical category can appear in juxtaposition (*aguanieve* 'water-snow' = 'sleet') or can show a linking vowel between them (*roj-i-blanco* 'red-*i*-white').

Another traditional classification is that between lexical and syntactic compounds. The first type includes compounds that manifest themselves as a phonological and morphological unit. They show one single primary stress, and their inflectional markers are placed at the end of the complex word ([*mujeres*] *peliˈrroj-a-s* '[women] hair-red$_{+\text{fem}+\text{plur}}$'). The second type manifests its constituents as orthographically separated. Compounds of this second type have more than one primary stress and the inflectional markers always appear on the head of the word (*ˈverdes boˈtella* 'green$_{+\text{plur}}$ bottle' = 'bottle-green', *ˈfaldas-pantaˈlon* 'skirt$_{+\text{plur}}$ pant' = 'split-skirts,' *ˈojos de ˈbuey* 'eye$_{+\text{plur}}$ of ox' = 'portholes'). Syntactic compounds are also distinguished from lexical compounds in the relations among their constituents and the functional elements that are allowed inside the compounding construct. So in N + N compounds, the second noun is in an appositive relation with the head noun (*paquete bomba* 'package bomb' = 'mail-bomb'), a syntactic relation not found inside the so-called lexical compounds. Moreover, some syntactic compounds introduce the complement constituent by means of the preposition *de* (*cuello de botella* 'neck of bottle' = 'bottleneck'), a constituent not present in lexical compounding.

The barrier between lexical and syntactic compounds is neither clear nor inflexible. For instance, it is not uncommon that "syntactic" compounds become "lexical" through the addition of a suffix: *barrio bajo* 'low neighborhood' = 'slum' > *barriobaj-ero* 'related to slums.' Sometimes, the complex underneath the derived compound is not even attested as a compound on its own: *quince años* 'fifteen years' > *quinceañ-ero* 'fifteen-year-old.' There is also evidence of the coexistence of the two patterns: *guardias civiles/guardiaciviles* 'Civil Guards,' showing a tendency to treat frequent compounds of the syntactic type as a single word.

3.3 Compound types

3.3.1 V N > N Traditionally described as containing a V constituent in initial position, this pattern constitutes the largest subgrouping of Spanish compounds, as well as the most productive and regular. It has a great vitality in all dialectal variants of Spanish as a means to form agentive nouns (*limpiabotas* 'boot-cleaner'), instrumental nouns (*abrelatas* 'can opener'), and only marginally, locative nouns (*guardamuebles* 'furniture repository'). This nominal construction is analyzed very differently depending on the approach to word-formation, that is, whether it is considered a lexical or a syntactic process. In the first case, there is also disagreement as to whether these compounds are exocentric or endocentric. Following a lexical approach, I consider that these are synthetic compounds formed by a nominalized verb followed by a complement N, so that the pattern should be described as $\mathbf{N_{dev}N}$ rather than V N. To my understanding, the head of the construct – thus an endocentric compound – is the deverbal noun in first position. Verbs that can nominalize in this construction are mostly of the agentive-transitive type, a fact difficult to explain if the source of the compound were a reduced sentence, as sustained by some authors (Kornfeld 2009).[3] Specifically, I consider that the internal nominal affix is responsible for the agentive meaning conveyed by the whole complex, an assumption that may account, among other things, for the fact that we get compounds like *crecepelo* ('grow-hair' = 'hair tonic') or *giradiscos* ('turn-discs' = 'record player'), while the underlying verbs (*crecer* 'to grow' and *girar* 'to turn,' respectively) are not agentive, as shown by the ungrammaticality of the corresponding VP constructions: **crecer el pelo* 'to grow the hair,' **girar el disco* 'to turn the disk.' Note that there are some instances of deverbal nouns – maybe former compounds lacking the object-N – in which the thematic vowel transmits the agentive value by itself, as in *el guarda* ('the guard'), *el caza* ('the hunter' = 'the fighter-plane'), *el limpia* ('the cleaner'), *el okupa* ('the squatter'). Moreover, there are other compounds in which the verb plus its thematic vowel give rise to an agentive N, as in the case of some $\mathrm{N_{dev}Adv}$ (*mandamás* 'domineering-more' = 'bossy') or $\mathrm{AdvN_{dev}}$ combinations (*malqueda* 'badly-doer' = 'unreliable'). An approach whereby the compound corresponds to an underlying sentential structure does not account either for the fact that the possible compounds are determined by the argument structure of the underlying verb, and, among other things, they do not accept light verbs: **el dapaseos* 'the takes-walks' vs. *Juan da paseos* 'J. takes walks.'[4]

We consider this compounding pattern as the mirror image of the English synthetic compounding which also produces agentive (*cuidacoches* ↔ *car watcher*) or instrumental nouns (*lavaplatos* ↔ *dish-washer*). In Old Spanish, this compound type followed the Latin order, showing exactly the same pattern as English, with the deverbal suffixed N at the right of the complex word (*cuentadante* 'count-giver' = 'accountant'). Nowadays, this pattern is experiencing some revitalization, presumably due to the influence of the English language (*dermoprotector* 'skin-protector,' *ruidofabricante* 'noise-constructer' = 'noisemaker' (cf. Varela and Felíu 2003).

Another piece of evidence with respect to the parallelism between this type of Spanish compound and the English synthetic compounds comes from language acquisition research. As described in Clark (1998), children acquiring English as L1 first produce root compounds with the sentence order (head + complement): *throw-button*. Afterwards, they go through a stage in which they add the appropriate affix to the verb, but still maintain the primitive order: *thrower-button*. It is not until approximately age five when they get both affix and order correctly: *button-thrower*. According to Lardiere (1998), native speakers of Spanish acquiring English as L2 also coin at first synthetic compounds with the VO order (*eater-flies*); that is, with the head at the left side of the complex word, in accordance with the productive pattern of Spanish, but with an overt derivational agentive affix as in English. These two observational facts are, to my understanding, evidence of the parallelism between the $N_{dev}N$ compounding pattern of Spanish and the N N_{dev} compounding pattern of English.

3.3.2 N N > N "Lexical" compounds of this type are not very numerous (coordinative: *carr-i-coche* 'wagon-*i*-car' = 'covered wagon,' subordinate: *bocacalle* 'entrance to a street'). However, in Peninsular Spanish we can currently see the coinage of new compounds following the English pattern, that is, N N compounds in which the first N – usually reduced to a two-syllable constituent – acts as a determiner of the second one: *bolsilibro* 'pocket-book,' *publicesta* 'publicity-basket.' N+N combinations of the "syntactic" type are much more productive though. Two subtypes of juxtaposed compounds can be identified in the latter case. In one subtype, the two Ns are coordinated, a fact usually marked by the use of the hyphen (*salón-comedor* 'living room-dining room'), and the semantics of the compound is obtained through the total sum of the two Ns. In the other subtype, the second N is in an appositive relation to the first N (*pantalón campana* 'pant bell' = 'bell-bottom pants,' *perro pastor* 'dog sheep' = 'sheep dog'). In this case, the N in apposition denotes only some properties of the entity expressed. Therefore, in the case of *pantalón campana*, *campana* ('bell') only conveys the meaning of "hollow, round (i.e., 'bell-shaped')," not the other properties that it could have. One formal property that makes it possible to differentiate between these two classes of nominal compounds derives from agreement. In the coordinated type, in case there is an animate entity, both Ns must show the same gender and the same number: *reinas-filósofas* 'queen$_{+\text{fem}+\text{plur}}$-philosopher$_{+\text{fem}+\text{plur}}$.' This requirement is not observed in the case of appositive compounds: *perra pastor* 'dog$_{+\text{fem}}$sheep,' not **perra pastora* 'dog$_{+\text{fem}}$sheep$_{+\text{fem}}$,' and *pantalones campana* 'pants bell,' not **pantalones campanas* 'pants bells.' One particular subclass of the appositive compound type is that in which the N in apposition emphasizes or intensifies some of the first constituent's properties. In this case, we get Ns like *estrella* 'star,' *clave* 'key,' *prodigio* 'prodigy,' *modelo* 'model' combined with all kind of nouns, animate or inanimate, which they modify, showing that they are entities that have a special or privileged position inside their class: {*palabra*, *problema*, *decisión*, *hombre*} *clave* '{word, problem, decision, man} key,' {*oferta*, *juez*, *pasajero*, *cliente*} *estrella* '{offer, judge, passenger, client} star.' Another particular subclass of the appositive compound type is the one integrated by color names. In this case, the second N specifies the name of the color in first position. The modifier N can denote a material substance (*rojo vino* 'red wine'), a flower or a fruit (*rojo amapola* 'red poppy,' *amarillo limón* 'yellow lemon'), a precious stone (*rojo rubí* 'red ruby'), or any other entity characteristically identified by a certain color, even in a figurative sense (*rojo pasión* 'red passion').

3.3.3 A A > A The combination of two qualitative adjectives, directly coordinated (*claroscuro* 'light-dark') or linked by the vowel -*i*- (*agr-i-dulce* 'sour-*i*-sweet') is very limited, except for color adjectives (*blanquiazul* 'white-blue,' *verdinegro* 'green-black'), a combination that is subject to very strict phonological conditionings. The first constituent must have two syllables counting the linking vowel -*i*- (*verd~~e~~-i-azul* 'green-*i*-blue,' *roj~~o~~-i-blanco* 'red-*i*-white'). One-syllable adjectives ending in a consonant are also excluded from the first position even when they form a two-syllable constituent with the linking vowel (**gris-i-negro* 'grey-*i*-black'). However, the combination of adjectives denoting nationalities is relatively productive. Due to pragmatic factors, these compounds are sometimes ambiguous between a coordinative interpretation (*relaciones francocanadienses* 'French-Canadian relations') or a subordinative one (*ciudadano francocanadiense* 'French-Canadian citizen'). In the case of relational adjectives, it is not rare for more than two As to be coordinated, usually separated by a hyphen (*político-económico-social* 'political-economic-social'). Adjective compounds show a high level of cohesion, as demonstrated by the occurrence of shortened forms (*verbonominal* 'verbal-nominal') or learned allomorphs (*francoespañol* 'French-Spanish') in initial position, and the manifestation of gender and number morphemes at the end of the construct in all cases (*asuntos político-económic-o-s* 'affaires political-economic$_{+\text{masc},+\text{plur}}$,' *relaciones franco-aleman-a-s* 'relations French-Germanic$_{+\text{fem},+\text{plur}}$').

3.3.4 N A/A N > N Some of these combinations show total amalgamation, either in the order A N (*buenaventura* 'good fortune') or in the order N A (*montepío* 'mountain pious' = 'charitable fund'). A characteristic of this type of compound is its high level of lexicalization that often prevents its semantics from being deduced from the meaning of its parts (*llave inglesa* 'key English' = 'monkey wrench,' *oro negro* 'black gold').

3.3.5 N-i-A > A It is a quite productive pattern, especially in literary Spanish. This compound is formed by a nominal stem, followed by the linking vowel *-i-*, and a simple A or an AdjPart (*cuellilargo* 'neck-*i*-long' = 'long-necked,' *pelirrizado* 'hair-*i*-curled' = 'curly-haired'). It has the particularity – not expected in sentential syntax – that the combination of N and A yields an A. In fact, it is the noun that is selected by the adjective, delimiting the property denoted by the adjective (Gil 2006). From a semantic point of view, the internal N is usually confined to elements of inalienable possession, mainly body parts. From a phonological point of view, it is in general restricted to a two-syllable constituent, counting the linking vowel *-i-*, and in some cases it presents reduced forms (*cabiz-bajo* 'head-low' = 'crestfallen,' not **cabeci-bajo*).

4 Internal structure of compounding, inflection and derivation, phrases, and recursivity

4.1 Internal structure

None of the category combinations found in compounds are unknown of sentence syntax, nor are the relations between the compound components. Thus, for some linguists compounds are not morpholexical objects, but syntactic objects showing a "reduced" syntax. The only difference that may be pointed out is that in Spanish compounding there are vestiges of the Latin order with the complement-head disposition (*terrateniente* 'landowner') instead of the syntactic normal order in which the complement follows the head. As said in relation to $N_{dev}N$ compounds, this structure with the determinant in the right position – also typical of technical and scientific terminology based on learned stems – has recently flourished in newly coined compounds, most probably due to the influence of the English compounding pattern, and manifests itself not only in complement-head compounds (*euroconversor* 'Euro-converter'), but also in adjunct-head compounds (*videovigilante* 'video-watchman').

Contrary to derivational suffixes which can absorb different arguments (*-dor* in *corre-dor* binds the subject-agent of the verb), the constituents of the compound fill argument positions (*lava* [___]$_N$ > *lavaplatos* 'wash-dishes' = 'dishwasher'). Inside compounds, the argument required by a predicate is not optional (*para**(*brisas*) 'stop-breezes' = 'windshield'), as is in the case of the derivative (*escritor* (*de novelas*) 'writer of novels'). In a way, compounds are a more compact type of lexical construction than derivatives. For instance, nonhead constituents in derivation can govern an argument outside the lexical construct: *conduc-tor de camiones* 'conductor of trucks' = 'truckdriver.' In the case of compounds, however, argument dependency is completely saturated inside the lexical complex, and no part of the compound can project outside the construct: *[[*cuarto*] *de estar*] *dormido* 'room-to-stay asleep,' *[[*aparca*] *coches*] *en batería* 'car-parker side by side.' Also, some derived words, namely, adjectival participles and event nominals, can govern purpose clauses because they imply an agent: (*un edificio*) *acondicionado para admitir cableado eléctrico* '(a building) prepared to admit electric cable'; *la edificación de un* *aula para que jueguen los niños* 'the construction of a classroom for the children to play.' Once inserted inside a compound, this option is unavailable. So, a compound with an adjectival participle as one of its constituents cannot govern a final clause: **el aireacondicionado para evitar el calor* 'the air-conditioning to avoid the heat.' The same holds true with respect to a compound containing a deverbal noun, be it agentive (**un limpiabotas para no morirse de hambre* 'a boot-cleaner for not dying of hunger') or eventive (**la dermoprotección para que el sol no penetre en la piel* 'the skin-protection so that the sun does not penetrate into the skin').

4.2 Inflection and derivation of compounds

Lexical compounds show inflection at the right edge of the construct, independently of the place of the head: *pequeñoburgues-es* 'little-bourgeois$_{+\text{plur}}$,' *aguafuerte-s* 'water-strong$_{+\text{plur}}$' = 'etchings.' The obvious dual nature of inflection (a morphological element with syntactic pertinence) accounts for the distinction that has been observed between internal inflection, present in some Spanish compounding (*el* [*lavaplato-s*] 'the$_{+\text{sing}}$ [washdishes$_{+\text{plur}}$]'), and external inflection, which has syntactic relevance and triggers agreement (*los* [*parasol*]-*es* 'the$_{+\text{plur}}$ [stop-sun]$_{+\text{plur}}$'). In order to consolidate the amalgamation of constituents inside the compound, all internal remains of inflectional markers that are not syntactically pertinent tend to be deleted: *Estado-~~s~~ Unid~~o~~ -~~s~~* + *ense* 'States United + *ense*' > *estadounidense* 'US citizen,' *paragu-~~a~~ -~~s~~* + *ero* 'stop-waters' + *ero* > *paragüero* 'umbrella stand.' In nonorthographical or syntactic compounds, the inflection that is pertinent for agreement is always marked on the head of the word (*ciudad-es dormitorio* 'town$_{+\text{plur}}$ dormitory' = 'dormitory towns'), and, in the case of N+A compounds, it is marked on both constituents (*llave-s inglesa-s* 'key$_{+\text{fem}+\text{plur}}$English$_{+\text{fem}+\text{plur}}$' = 'monkey wrenches').

As we have seen, the constituents of the compound are frequently linked by means of a vowel. This linking vowel is not to be confused with the real inflectional marker, as the lack of agreement between constituents shows: *Latin-o-américa* 'Latin-*o*-America' vs. *América latin-a* 'America Latin$_{+\text{fem}}$' or *sord-o-mud-o* 'deaf-*o*-mute$_{+\text{masc}}$' = 'deaf and dumb' next to *sord-o-mud-a* 'deaf-*o*-mute$_{+\text{fem}}$.' The whole compound can be further derived, as in [[[*medio*]$_\text{A}$[*ambient~~e~~*]$_\text{N}$]$_\text{N}$*al*]$_\text{A}$ 'environment+*al*,' but it can also include a derived word, as in [[*agua*]$_\text{N}$[[*mar*]$_\text{N}$*i-na*]$_\text{A}$]$_\text{N}$ 'aqua-marine.' When two Ns in composition are internally suffixed, it is not rare that the suffix belonging to the first N be deleted in order to get the compound more consistently amalgamated: *cant~~ante~~-autor* > *cantautor* 'singer-author' = 'song-writer.' Compound amalgamation favors also haplology of phonological material of the first constituent, not necessarily coincident with a morphological unit. Its reduction to a two-syllable trochee is especially frequent, as seen in newly coined compounds: *publi~~citario~~+reportaje* 'publicity-report' > *publirreportaje*.

4.3 Compounds and phrases

Inside the compound there are cyclic domains to which syntax does not have access. In morphology, the relevant domain is the morphological word (DiSciullo and Williams 1987). The existence of such a domain, nonvisible for proper syntax, explains several of the restrictions that affect the compound and do not affect phrases: (a) no anaphoric relations are allowed: although in Spanish syntax a bare noun can be the antecedent of a pronoun (*Hacen falta corchos$_i$para las botellas. ¿Quién los$_i$ compra?* 'There is need of corks for the bottles. Who buys them?'), it is impossible to refer to a bare N inside a compound by means of a pronoun (**Este* [*saca*[*corchos$_i$*]] *los$_i$ deja perfectos* 'This pull-out-corks leaves them perfect'); (b) the constituents of compounds in coordination are not subject to ellipsis (*Compré un lava**(*platos*) *y un secaplatos*. 'I bought a wash-dishes and a dry-dishes'); (c) compound constituents lack syntactic independence, as shown by the impossibility of attributing a modifier or an adjunct to only one of them (**lavablancosplatos* 'wash-whitedishes,' **lavabienplatos* 'washwell-dishes'); (d) ungrammaticality also arises if the modifier is coordinated with another modifier (**bajoyaltorrelieve* 'low-and-high relief'), or if the constituents are interrupted by a parenthesis (**agua—digamos—marina* 'aqua—let's say—marine').

4.4 Productivity and recursivity

Compounding in Spanish is much less productive than derivation and shows very limited recursivity. The more productive pattern among subordinate compounds is the $N_{dev}N$ type (*lavavajillas* 'dish-washer'), limited to the adjunction of another N_{dev} and thus yielding a compound of the same type (*abrillantalavaplatos* 'polish-wash-dishes' = 'dishwasher polish,' *limpiaparabrisas* 'clean-stop-breezes' = 'windshield wipers'). Next, but at considerable distance, is the pattern N*i*A (*pelirrojo* 'red-haired'), which lacks recursivity. Among coordinate compounds, the most productive one is the A A type, also the only pattern that shows larger recursivity (*eurolatinoamericano* 'European-Latin-American').

It has been noticed that nominal composition in Spanish, as in the rest of the Romance languages, is subject to some restrictions that do not affect Germanic compounding. The more relevant differences are the following: (a) lack of productivity; (b) absence of certain relations among the constituents of N+N compounds, such as part-whole (**cadena reloj* 'watch-chain') or container-content relationships (**caja plátano* 'banana box'; (c) lack of recursivity (**perro policía mascota* vs. *pet police dog*); (d) adjunction to the right of the head constituent (*<u>perro</u> policía*), as opposed to a language like English with left adjunction (*police <u>dog</u>*). These two last restrictions have been related to the presence of the word-marker in a language like Spanish (Piera 1995). Such a feature introduces a further level of structural complexity among the head constituent and its modifier – as shown in the segmentation proposed with the double bracket among constituents – so that adjunction to the left is blocked (*... [[per-]WM]) and can be done only to the right ([[per-]WM]...). Recursivity is also prevented by the presence of a double bracket between the first modifier and a potential second one (*[[per-]WM] [[polici-]WM]] [[mascot-]WM]]...).

Interestingly, the lack of productive N+N root compounds in Romance languages has been related to an independent parameter: "The Compounding Parameter" [TCP] (Snyder 2001). Spanish is claimed to be a [-TCP] language, as opposed to [+TCP] languages like English. This feature is also considered responsible for the existence of the following 'complex predicate' constructions: (a) verb-particle (*John handed his paper in*); (b) make-causative (*John made Mary smile*); and (c) transitive resultatives (*Helen wiped the table clean*). That is, in [+TCP] languages like English, all lexical categories (N, Prep, V, and A) can adjoin to another lexical category constituting a kind of compound formation in the syntax. Languages like Spanish do not have such an option, and this accounts for the unproductivity of root compounding as well as for the absence of parallels to the other three constructions.

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NOTES

1 Another type of non-affixal derivation is shortening or clipping, that is, the suppression of some phonetic material of an existing word, sometimes traceable to a morphological boundary (*tele* < *televisión*), but most often not (*cole* < *colegio* 'school') since the rule is that the clipped word should be a two-syllable trochee. Although clipping is mostly used in colloquial language and for hypocorisms, the shortened variant may sometimes displace the complete word in standard usage. For example, in Peninsular Spanish *moto* is normally used for *motocicleta* 'motorcycle.'

2 Next to these cases, there are *con-* prefixed verbs which have a comitative complement, but do not relate to an underived verbal base: *confesar con* 'to confess with' (**fesar* does not exist). At the same time, other verbs that carry the prefix *con-* are semantically opaque in relation to their respective bases and do not take a comitative complement: *llevar* 'to carry' > *conllevar* 'to bear, to live with,' *seguir* 'to follow' > *conseguir* 'to obtain' (I thank an anonymous reviewer for pointing out these data).

3 According to Kornfeld (2009), these compounds would be originally adjectives which are reanalyzed as nouns by virtue of nominal ellipsis (*e*): *una máquina traga monedas* > *una e traga monedas* > *una tragamonedas* ('a machine swallows coins' > 'a *e* swallows coins' = 'a slot machine'). However, this analysis does not work as seen when the second noun is masculine and it is the form *uno* of the pronoun that must appear substituting the elliptical head-N: *un e lava platos* > **uno lavaplatos* ('a *e* washes dishes' > a washes-dishes' = 'a dishwasher').

4 Thematic roles are also relevant: the type of psych verbs entering this construction are not true experiencer verbs but verbs whose external argument conveys a volitional component and can thus be considered as agentive. For example, in a potential compound as *adoraestatuas* 'adore-statues,' the verb *adorar* ('to pay reverence') has the agentive meaning compatible with a passive: *la estatua es adorada* 'the statue is worshiped' (I am grateful to an anonymous reviewer for pointing out these data).

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12 Morphological Structure of Verbal Forms

MANUEL PÉREZ SALDANYA

1 Introduction

Verbal inflexion or conjugation is the set of forms that a verb takes in order to express grammatical notions such as person, number, tense, etc. Verbal inflection is rather complex in Spanish if we compare it with the inflection of nouns and adjectives. The inflectional paradigm of a noun in Spanish includes only two forms, one for the singular and one for the plural (e.g., *casa* and *casas*). Adjectives have four forms, since, in addition to singular and plural, they may have different forms in the masculine and feminine gender (e.g., *alto*, *alta*, *altos*, *altas*). In comparison, a verb has 62 different forms in Spanish if we count only simple forms (*canto*, *cantas*, *canta*, …), and up to 118 forms if we add compound forms with *haber* as an auxiliary (*he cantado*, *has cantado*, *ha cantado*, …). Regarding simple forms, there are 8 verb tenses (one of which, the imperfect subjunctive, has double forms) with six personal forms, one mood, the imperative, which has only second person forms (singular and plural), and three nonpersonal forms, one of which, the participle, is inflected for gender and number. For each simple form there is a compound form, except for the imperative and the participle.

In this chapter, I first consider the grammatical categories existing in the Spanish verb (Section 1.1) and how the inflected forms are organized (Section 1.2). The remaining sections focus on the morphological properties of each grammatical category: person and number (Section 2); time, aspect, and mood (Section 3); and conjugation (Section 4). Finally, Section 5 is concerned with the main morphological irregularities found in verbs.

1.1 Grammatical categories of verbal forms

The great complexity of verbal inflexion is due to the fact that a number of grammatical categories are expressed in the verb – in particular, person, number,

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tense, aspect, and mood. Person and number signal agreement with the subject of the clause; tense, aspect, and mood refer to the event (the localization of the event in time and the way it is conceptualized by the speaker). To these categories with a syntactic function, we may also add the conjugation marker or theme vowel, which varies according to specific patterns of conjugation. Consider, for instance, the verbal form *cantábamos* 'we were singing; we used to sing' from *cantar* 'to sing.' As in all inflected verbal forms, we may distinguish two components: on the one hand, the root, *cant*-, is the lexical element, and, consequently, carries the meaning that is reflected in dictionaries; on the other hand, we have the ending or inflectional suffix, -*ábamos*, by means of which different grammatical categories are expressed.

In the specific example we are analyzing, the ending includes three different marks: -*á*-, -*ba*- and -*mos.* Starting from the end, the suffix -*mos* refers to the person and number of the subject (PN): it is used in sentences where the subject refers to the speaker (first person) together with other people (plural). The suffix -*ba*-, in turn, provides information on tense, aspect, and mood (TAM): it is the marker of the imperfect indicative, a form that indicates that the event of singing is situated in the past (past tense or preterit), and is presented as an ongoing process or as an event that was repeated habitually (imperfective aspect) and as an asserted fact (indicative mood).

Finally, the suffix -*á*- is a theme vowel (or conjugation marker) which appears next to the root in different forms of verbs that show the same inflectional patterns as *cantar*. In this sense, the term *theme* or *thematic base* is used to refer to the segment formed by the root and the theme vowel together. Based on the theme vowel and other inflectional properties, verbs are grouped in three conjugations or inflectional classes. To the first conjugation belong verbs whose infinitive ends in -*ar* (like *cantar*, *estar* 'sing, be'); to the second, those with infinitives in -*er* (*temer*, *hacer* 'fear, do/make'), and to the third, those with infinitives in -*ir* (*partir*, *venir* 'leave, come').

Besides belonging to a specific conjugation, verbs may be regular or irregular. They are regular if their inflected forms follow a predictable and recurrent pattern, as is the case with *cantar*, *temer*, and *partir*, which I will use as models in my analysis of regular inflection. On the other hand, verbs like *estar*, *hacer*, and *venir* are irregular because they present forms that depart from the general patterns. Almost all first conjugation verbs are regular. In addition, this is the most numerous conjugation, with about 90% of all verbs, according to Alcoba (1999: 4936), and the only one that nowadays is productive and incorporates new verbs. Nevertheless, some of the most frequent Spanish verbs belong to one of the other two conjugations.

1.2 Verb tenses and nonpersonal forms

The set of inflected forms of a lexeme constitute the inflectional paradigm of that unit. In the verb, paradigms are organized according to grammatical meanings and form. To begin with, we may distinguish personal or finite forms, which are inflected for person/number (PN) and tense/aspect/mood (TAM), from nonpersonal or nonfinite forms, which do not include either person or tense reference. Nonpersonal forms are the infinitive, the gerund, and the participle; the rest are personal. The participle differs from the other nonpersonal forms because it has gender and number. In their turn, the personal forms are classified according to mood (indicative, subjunctive, or imperative) and, within the indicative and subjunctive moods, according to tense, where the term 'tense' is used in a broad sense to also include aspect. The personal forms that have the same TAM reference shape a verb tense (or a *screeve*, in Carstairs-McCarthy 2000: 597 terminology).

From a formal point of view, we may also distinguish simple forms from compound forms, which are created with the auxiliary *haber* and the participle. A debatable issue is whether these forms are part of the inflectional paradigm of verbs, or, rather, are aspectual periphrases, similar to those with *tener* + participle, for instance.[1] In support of the latter point of view, we may mention the fact that the two verbs of the compound form maintain a certain syntactic independence, which allows for the insertion of a lexical element in certain cases (e.g., *Habíamos* ya *pensado en todo eso* 'we had already thought of all that'). On the other hand, the participle with *haber* never agrees with the direct object or the subject, unlike what happens in other periphrases with the participle (e.g., *Ya las tenemos* acabadas, *las cartas* 'we already have the letters finished (feminine plural participle),' vs. *Ya las hemos* acabado, *las cartas* 'we have already finished (invariable participle) the letters'). The invariable participle takes the least marked form: masculine singular (see note 2 for the concept of markedness).

In Table 12.1, I list and exemplify the range of personal and nonpersonal forms of the verb *cantar* (shown in the third person singular, except for the imperative, where the second person singular is given). Simple forms are on the left and compound forms on the right.

Almost all simple forms have a compound counterpart that adds the meaning of anteriority in time and perfect(ive) aspect. The participle, which already expresses these meanings, does not have a compound form, and neither does the imperative, which is a single-tense mood.

2 Person and number markers

2.1 *Person and number oppositions*

The category of *person* establishes a ternary opposition related to the participants in the discourse. The first person refers to the speaker, the second to the listener, and the third to a different entity, which may be human or not. The category of number is binary and is superimposed to that of person. In that way, we have three persons in the singular and three in the plural. The first person plural refers to the speaker together with some other person or persons, including the listener(s) or not. The second person plural includes the listener(s) and other persons other than the

Table 12.1 Tenses and nonpersonal forms.

Nonpersonal forms	
infinitive: *cantar*	perfect infinitive: *haber cantado*
gerund: *cantando*	perfect gerund: *habiendo cantado*
participle: *cantado*	—
Indicative	
present: *canta*	perfect: *ha cantado*
imperfect: *cantaba*	pluperfect: *había cantado*
preterit (or simple past): *cantó*	past anterior: *hubo cantado*
future: *cantará*	future perfect: *habrá cantado*
conditional: *cantaría*	conditional perfect: *habría cantado*
Subjunctive	
present: *cante*	perfect: *haya cantado*
imperfect: *cantara* or *cantase*	pluperfect: *hubiera* or *hubiese cantado*
future: *cantare*	future perfect: *hubiere cantado*
Imperative	
canta	—

speaker. Finally, the third person refers to a set of entities (people, animals, or things) excluding the participants in the speech act.

Verbs agree in person and number with the subject of the clause. If the subject is a noun phrase, the verb shows third person agreement: *el niño* canta *bien* (3sg)/*los niños* cantan *bien* (3pl) 'the boy sings well/the boys sing well.' A plural noun phrase may also agree in first or second person if it refers to a group that includes the participants in the speech act. In *Los mexicanos {cantamos/cantáis} muy bien* 'We/you Mexicans sing very well,' *cantamos* (1pl) is used if the speaker is Mexican, and *cantáis* (2pl) is used if the listeners are Mexican. Personal pronouns also have features of person and number, with which verbs agree. For instance, the pronoun *yo* 'I' triggers first person singular agreement whether or not it is explicit: *Yo canto muy bien* ~ *Canto muy bien* 'I sing very well.'

Something must be said in this respect regarding the forms of address *usted* and *ustedes*, and similarly *vos*. These pronouns refer to the listener(s) and are thus semantically second person singular (*usted*, *vos*) or plural (*ustedes*). However, *usted* and *ustedes* trigger third person grammatical agreement (*usted canta*, *ustedes cantan*), since etymologically they derive from the noun phrase *vuestra merced* 'your grace.' The second person singular form *vos*, used in several areas of Latin America (see Chapter 13, below), triggers etymologically second person plural agreement (*vos cantáis* ~ *cantás*) because it derives from the Latin second person plural pronoun vōs, although related object pronouns are singular (*vos te lavás* 'you wash yourself'). The examples in (1) illustrate person and number agreement with the present indicative of *cantar*.

(1) Verbal agreement with personal pronouns:
1sg: (yo) canto 'I sing'
2sg: (tú) cantas 'you-sg sing'
3sg: (él/ella, usted) canta 'he/she/you-sg sing'
1pl: (nosotros/nosotras) cantamos 'we sing'
2pl: (vosotros/vosotras) cantáis 'you-pl sing'
(vos) cantás 'you-sg sing' (from the same source as *cantáis*)
3pl: (ellos/ellas, ustedes) cantan 'they/you-pl sing'

2.2 *Regular and irregular marks*

Person and number are expressed, in most verbal paradigms, by the following forms in (2) (see Ambadiang 1993: 174–178 for different theoretical proposals regarding the analysis of these markers).

(2) Person and number markers:
1sg: Ø
2sg: -s
3sg: Ø
1pl: -mos
2pl: -is
3pl: -n

The plural is marked with respect to the singular and the second person singular is marked with respect to the other two persons, in accordance with cross-linguistic tendencies (see Greenberg 1966).[2] In the second plural, the *i* of the marker *is* is realized as a glide, and it is deleted if preceded by *i*: *cantáis*, *teméis*, but *partís*.

An exception to the pattern in (2) is the use of the marker -*o* for the first person singular only in the present indicative (*canto*, *temo*, *parto*). Another exception to the marking in (2) is found in the imperative, where there is no marker in the second person singular (*¡Canta, tú!*), so that this form is identical to the third person singular of the present indicative (*canta*, *teme*, *parte*), except in the case of irregular verbs with monosyllabic imperative forms (*haz*, *ven*, *di*, …; see Section 5.6.1). The *vosotros* form of the imperative takes the suffix -*d* (*¡Cantad, vosotros!*), which is deleted before the reflexive *os* (*¡sentaos!*), except in the imperative of *ir* (*¡Idos ya!*). The imperative forms of *vos* also lack the final -*d* (*¡Cantá, vos!*). The last irregularity is found in the second person of the preterit, which takes the ending -*ste* in the *tú* form (*cantaste*) as well as in the *vosotros* form (*canta-ste-is*).

The lack of agreement markers in the first and third singular causes syncretism in some, but not all, tenses. First and third person singular forms are different only in some indicative tenses; specifically, in the present indicative, preterit, and future, as well as in the compound forms with the auxiliary *haber* shown in Table 12.2. First and third person singular are identical in all other tenses: imperfect (*cantaba*), conditional (*cantaría*), present subjunctive (*cante*), etc.

Table 12.2 Contrast between first and third singular persons.

Tense	*Contrast*	*Example*	
		1sing	*3sing*
present	-o/-a	canto	canta
	-o/-e	temo	teme
		parto	parte
future	-ré/-rá	cantaré	cantará
preterit	-é/-ó	canté	cantó
	-í/-ió	temí	temió
		partí	partió
perfect	he/ha	he cantado	ha cantado
future perfect	habré/habrá	habré cantado	habrá cantado
past anterior	hube/hubo	hube cantado	hubo cantado

3 TAM

Tense, aspect, and mood (TAM) markers are less regular than person and number markers, and present several cases of allomorphy in different conjugations and persons. We will first consider the present tenses and then the rest of the tenses, focusing on the position of the stress and specific markers.

3.1 Present tenses

The present tenses (present indicative, present subjunctive, and imperative) differ from the rest in their stress pattern. In these tenses the stress falls on the penultimate syllable, except in the second person plural, where historical evolution has produced final stress (CANTĀTIS > *canta(d)es* > *cantáis*; CANTĒTIS > *cante(d)es* > *cantéis*; CANTĀTE > *cantad*), as summarized in Table 12.3, where the stressed vowel is in bold. Note that the forms *cantáis/cantéis* could be included under the general rule of penultimate stress if we assume that the falling diphthong is a realization of an

Table 12.3 Present tenses of *cantar*.

	Pres. ind.	*Pres. subj.*	*Imper.*
1sg	c**a**nto	c**a**nte	
2sg	c**a**ntas	c**a**ntes	c**a**nta
3sg	c**a**nta	c**a**nte	
1pl	cant**a**mos	cant**e**mos	
2pl	cant**á**is	cant**é**is	cant**a**d
3pl	c**a**ntan	c**a**nten	

Table 12.4 Present subjunctive.

Verb	*Forms*
cantar	cante, cantes, cante, cantemos, cantéis, canten
temer	tema, temes, tema, temamos, temáis, teman
partir	parta, partas, parta, partamos, partáis, partan

underlying hiatus (Hualde 2005: 229); but this analysis would not be possible for the imperative or for the *voseo* forms.

In fact, in areas of *voseo*, stress also falls on the last syllable in indicative and imperative forms agreeing with *vos*: *cantás* and *cantá*, respectively, which derive from contraction of *cantáis* and *cantad* (see Section 2.1).

This stress pattern cannot, of course, be followed by irregular verbs with monosyllabic forms: *dar* (*doy*, *das*,…), *ir* (*voy*, *vas*,…), etc. The verb *estar* also has final stress in those forms that were monosyllabic in Latin and have acquired an initial epenthetic *e* (STŌ > *esto* > *estoy*; STĀS > *estás*, etc.).

Aside from these cases, there are no forms with irregular stress. In particular, unlike in the case of nouns and adjectives, there is no antepenultimate stress, which creates a contrast between nouns and verbs in cases like: *fábrica* 'factory' and *fabrica* '(s)he fabricates'; *plática* 'conversation' and *platica* '(s)he converses'; *número* 'number' and *numero* '(s)he numbers' (Harris 1969: 120; Hualde 2005: 230–231).

As for TAM marking, neither the present indicative nor the imperative display overt marking. In these verb tenses, the vowel that follows the root is the mark of first person singular in the case of *-o* (see Section 2.1) and the theme vowel for all other persons (see Section 4). The present subjunctive, on the other hand, does present TAM markers: *e* in the first conjugation and *a* in the other two conjugations, as shown in Table 12.4.

3.2 Tenses with columnar or morphological stress

3.2.1 Position of the stress Except for the present tenses, in all other tenses stress is columnar or morphological, and it falls on the same morphological constituent (Hualde 2005: 231–233). The different tenses can be classified into three groups depending on whether the stress falls on the theme vowel, on the marker of TAM or in either one or the other depending on person. In the first group, we have the imperfect indicative (II) and subjunctive (IS), as well as the future subjunctive (FS); in the second group, the future indicative (FI) and conditional (C); and, in the third, the preterit (Pt) (see Ambadiang 1993: 154–174 for different theoretical views regarding the analysis of person and number markers), as summarized in Table 12.5.

3.2.2 Imperfective indicative In the imperfect indicative, the allomorphy is due to the historical loss of Latin -B- in the second and third conjugation (TIMĒBAT > *temea* > *temía*; DORMIĒBAT > *dormía*). This did not happen in the first conjugation,

Table 12.5 TAM markers.

Verb Tense	*Person*						*Example*
	1sg	*2sg*	*3sg*	*1pl*	*2pl*	*3pl*	
II			ba				cantaba, cantabas, cantaba, cantábamos, cantabais, cantaban
			a				temía, temías, temía, temíamos, temíais, temían
IS			ra ~ se				cantara, cantaras, cantara, cantáramos, cantarais, cantaran cantase, cantases, cantase, cantásemos, cantaseis, cantasen
FS			re				cantare, cantares, cantare, cantáremos, cantareis, cantaren
FI	ré	rá		ré		rá	cantaré, cantarás, cantará, cantaremos, cantaréis, cantarán
C			ría				cantaría, cantarías, cantaría, cantaríamos, cantaríais, cantarían
Pt	é	ste	ó	Ø	ste	ro	canté, cantaste, cantó, cantamos, cantasteis, cantaron
	í	"	ió	"	"	"	temí, temiste, temió, temimos, temisteis, temieron

possibly to avoid a sequence *aa* (CANTĀBAT > *cantava* > *cantaba*). The two alternative paradigms of the imperfect subjunctive derive, respectively, from the Latin pluperfect indicative (CANTĀ(VE)RAT > *cantara*) and pluperfect subjunctive (CANTĀ(VI)-SSET > *cantase*).

3.2.3 Future indicative and conditional The future indicative and conditional derive from modal periphrases with HABEŌ, in the present indicative (CANTĀRE HABEŌ 'I have to sing' > *cantar he* > *cantaré*) and imperfect (CANTĀRE HABĒBAM 'I had to sing' > *cantar hea* > *cantar ía* > *cantaría*). Different synchronic morphological analyses have proposed that the future is formed by the infinitive followed by *haber* in the present (Harris 1969: 78–87) or followed by an inflectional suffix (Harris 1987: 83), or that it includes a specific thematic base followed by *haber* (Roca 1990) or by an inflectional suffix (Ambadiang 1993: 223). In Old Spanish, it was possible to insert clitic pronouns between the two historical elements: *hablarle has* 'le hablarás', *hablarle ía* 'le hablaría' (Company Company 2006).

3.2.4 Preterit The preterit is the most irregular tense. In one hypothesis, the preterit does not include any specific TAM markers (Harris 1987). Under another hypothesis, there are specific preterit markers, but they are irregular and are not found in all persons (Alcoba 1999: 4928; RAE 2009: 200–201).

In favor of the first hypothesis, in the first person plural the PN marker *-mos* is added directly after the theme vowel. In the second conjugation, the theme vowel of the preterit is *i*, which allows for a contrast between preterit (*temimos* 'we feared') and present (*tememos* 'we fear'). In the other two conjugations, the theme vowel is the same in both tenses, producing syncretism (*cantamos* 'we sing; we sang' *partimos* 'we leave; we left'). Under this hypothesis, the stressed vowel of the first person singular could be analyzed as a theme vowel: *e* in the first conjugation (*canté*) and *i* in the other two (*temí*, *dormí*). Notice that *i* is also the theme vowel of the imperfect (*temía*, *dormía*). As for the second person forms, the ending *-ste* can be analyzed as a person marker, which is followed by a plural marker *-is* in the *vosotros* form (*cant-a-ste-is*). The analysis of third person forms, on the other hand, is less straightforward under this hypothesis. We would need to assume that *-ó* is a person marker that appears only in this tense, and that in the third person plural, *-ro-* is either an empty morpheme or a person marker associated with *-n* in this tense.

An advantage of the second hypothesis, which is the one adopted in this chapter, is that it does not force us to postulate irregular theme vowels (such as é in *canté*), idiosyncratic person markers, or empty morphs. The drawback is that we must accept that the preterit does not have a TAM marker in one person (the first person plural) and lacks uniformity in the others. Under this analysis we can explain certain analogical developments in some dialects. On the one hand, the reinterpretation of *-ste* as a marker of TAM leads to the nonstandard addition of final *s* (*cantastes*). On the other hand, the interpretation of *é* in *canté* as a marker of TAM explains why in some dialects of Peninsular Spanish there is a tendency to generalize this vowel to the first person plural, giving rise to a nonstandard contrast between *cantemos* 'we sang' and *cantamos* 'we sing.'

3.2.5 Nonfinite forms The nonfinite forms are stressed on the theme vowel. In these forms, there are no oppositions of person, tense, or mood, but there are aspectual differences. The gerund has imperfective aspect, the participle has perfective or perfect aspect, and the infinitive has unmarked aspect. The infinitive bears the suffix *-r* (*cantar*), the gerund *-ndo* (*cantando*), and the participle *-d-*, followed by the gender and number inflectional markers of adjectives (*cantado*, *cantada*, *cantados*, *cantadas*).

4 Theme vowel and thematic base

The theme vowel has a purely morphological or morpholexical function. It allows us to classify verbs according to the type of inflectional variation that they show. It is located immediately after the root, with which it forms the thematic base. It can be either stressed or unstressed and it can be realized as a vowel or as a diphthong.

The theme vowel is not an obligatory component of verbs, and the athematic forms (i.e., forms without theme vowel) are homogeneously distributed in all three conjugations. In these forms the lack of the thematic vowel is due to the existence of another vowel (monophthong or diphthong) as PN or TAM marker in all three conjugations. Notice that this generalization excludes the imperfect, whose TAM marker is either -*ba*- (first conjugation) or -*a*- (second and third conjugations). Specifically, the following forms lack a theme vowel: (a) the first person singular of the present indicative, which has unstressed -*o* as PN marker (*canto*, *temo*, *parto*); (b) the present subjunctive, which has *e* (in the first conjugation, *cante*, *cantes*,…) and *a* (in the other two conjugations, *tema*, *temas*,…; *parta*, *partas*,…) as TAM markers, stressed or unstressed depending on the person; and (c) the first and third persons singular of the preterit, which have stressed *é, ó* in the first conjugation (*canté* y *cantó*) and *í*, *ió* in the other two conjugations (*temí*, *temió*; *partí*, *partió*) as TAM markers.

Within a paradigm, the thematic base changes according to its theme vowel and also depending on stress position, as mentioned above in Section 3. Taking these two factors into account, we can establish three different thematic bases: present (including present indicative, present subjunctive, and imperative), preterit (preterit, imperfect indicative, imperfect subjunctive, future subjunctive, gerund, and participle), and future (future indicative and conditional) (Alcoba 1999: s.75.4; RAE 2009: s.4.3).

Note that the terms 'present' and 'future' are semantically transparent as they relate to forms that have this temporal meaning. This is not so with the term 'preterit,' which includes forms that do not refer to past situations. The preterit theme is also called 'perfect' because some of these forms come from the Latin perfect theme (preterit, imperfect subjunctive, future subjunctive, participle).

In Table 12.6 we show the theme vowel of each theme as well as the position of the stress. The absence of theme vowel in certain cases follows from the general principle mentioned above.

In the present thematic base, the stress may be on the last vowel of the root or on the theme vowel (see Section 3.1). In the first and second conjugation, the same vowel is found regardless of whether or not it is stressed, but in the third conjugation we find stressed *í* and unstressed *e*, producing convergence with the second conjugation in the latter case.

Table 12.6 Variations in the thematic base.

Theme	*I conj.*	*II conj.*	*III conj.*
present	cant- -á-	tem- -é-	part- -í-
	cánt- -a-	{tém-/párt-} -e-	
preterit	cant- -á-	{tem-/part-} {-í-/-ié-}	
future	cant- -a-	tem- -e-	part- -i-

Table 12.7 Variations in the theme vowel.

Thematic base	*Tense*	*PN*	*Theme vowel*	*First conj.*	*Second conj.*	*Third conj.*
Present	Infin.		á/é/í	cantar	temer	partir
	Pres. Ind.	1pl		cantamos	tememos	partimos
		2pl		cantáis	teméis	partís
	Imper.	2pl		cantad	temed	partid
	Pres. Ind	2sg	a/e	cantas	temes	
		3sg		canta	teme	
		3pl		cantan	temen	
	Imp.	2sg		canta	teme	
Preterit	Pret.	2sg	á/í	cantaste	temiste	
		1pl		cantamos	temimos	
		2pl		cantasteis	temisteis	
	Imperf. Ind.			cantaba	temía	
	Part.			cantado	temido	
	Pret.	3pl.	á/ié	cantaron	temieron	
	Imperf. Subj.			cantara	temiera	
				cantase	temiese	
	Fut. Subj.			cantare	temiere	
Future	Fut. Ind.		a/e/i	cantará	temerá	partirá
	Cond.			cantaría	temería	partiría

In the preterit thematic base, the root is unstressed and the stress falls on the theme vowel (or in the TAM marker, in the case of athematic forms). In the first conjugation, we find *á*, and in the other two conjugations, there is alternation between í and *ié*. The future thematic base derives from the present thematic base, with de-stressing of the theme vowel since the TAM marker is stressed (see Section 3.2.3).

Taking into account the existence of variation in theme vowels, we may assume that the most basic theme vowels are those that create a contrast among all three conjugations, regardless of the position of stress: *a* in the first conjugation (*cantar*), *e* in the second (*temer*), and *i* in the third (*partir*). Theme vowels are exemplified in Table 12.7. I use the third person singular when there is no variation within a tense, and forms of *temer* when the second and third conjugation have the same theme vowel.

5 Main irregularities of verbal inflection

Irregular verbs are those whose conjugation has some difference with respect to the paradigms we have analyzed so far, exemplified with *cantar*, *temer*, and *partir*.

Traditionally, it is considered that irregularity may be of orthographic, phonological, morphological, or morphosyntactic kinds. Orthographic irregularity is not really irregularity, but, rather, graphic variation due to the regular application of orthographic rules (e.g., the use of *z* in *aplazó* and *c* in *aplacé*) because *z* is regularly used before *a*/*o*/*u* and *c* before *e*/*i* to represent the same phoneme. The morphosyntactic irregularity has to do with the existence of defective paradigms. The existential *haber*, for instance, is an impersonal verb and is conjugated only in third person singular (*hay*, *había*, *habrá*,...). On the other hand, the aspectual *soler*, has lexically an imperfective value, and, consequently, is conjugated only in imperfective tenses (*suelo*, *solía*,... but **solió*, **soldré*, etc.).

In this section, I will focus on morphological irregularities while considering also phonological ones, given the fact that the boundary between the two is not always clear, and phonological variation tends to be morphologized. Specifically, we will discuss verbs with vowel-final roots (Section 5.1), verbs with either vowel (Section 5.2) or consonant alternations in the root (Section 5.3), strong past and participles (Section 5.4), athematic futures (Section 5.5), and other more specific irregularities (Section 5.6).

5.1 Vowel-final verbal roots

Prototypically, verb roots end with a consonant which syllabifies with the vowel or diphthong of the inflectional suffix. However, there are some verbs whose root ends in a vowel (*ca-er*, *cre-er*, *envi-ar*, *inco-ar*, *actu-ar*, *hu-ir*). These verbs do not show homogeneous behavior regarding the position of the stress in the present theme (Alcoba 1999: s.75.6; Hualde 2005: 229–230; RAE 2009: 218–225), as demonstrated in Table 12.8.

In verbs whose infinite ends in *-iar*, *-uar*, the position of the stress is not predictable in the rhizotonic (i.e., with stress on the root) forms of the present theme. Since in these forms the stress is on the penultimate syllable (see Section 3.1), the differences have to do with whether the final *i*/*u* of the root is syllabified as a vowel or as a glide. In most verbs, *i*/*u* denotes a glide, so that the stress falls on the preceding syllable

Table 12.8 Present indicative of verbs with vowel-final roots.

-iar/-uar				*-uir*	*Other*
Diphthong		*Hiatus*			
anunciar	*odiar*	*enviar*	*actuar*	*huir*	*pelear*
an**u**ncio	**o**dio	env**í**o	act**ú**o	h**u**yo	pel**e**o
an**u**ncias	**o**dias	env**í**as	act**ú**as	h**u**yes	pel**e**as
an**u**ncia	**o**dia	env**í**a	act**ú**a	h**u**ye	pel**e**a
anunci**a**mos	odi**a**mos	envi**a**mos	actu**a**mos	hu**i**mos	pele**a**mos
anunci**á**is	odi**á**is	envi**á**is	actu**á**is	hu**í**s	pele**á**is
an**u**ncian	**o**dian	env**í**an	act**ú**an	h**u**yen	pel**e**an

(e.g., *anunciar: anuncia*; *averiguar: averigua*). With some verbs, however, *i*/*u* corresponds to a vowel (in hiatus with the following vowel) and receives the stress (e.g., *enviar: envía*; *actuar: actúa*). Verbs in *-uir* are different since they add a consonant *y* in forms that do not contain the theme vowel *i* (e.g., *huir: huye*, *huyamos*,…, but *huimos*, *huís*). Verb roots ending in a mid or low vowel always produce hiatus, with stress on the root-final vowel (e.g., *pelear: pelea*; *incoar: incoa*; *releer: relee*; *recaer: recae*). Exceptionally, the verbs derived from *línea* (*delinear*, *alinear*) are pronounced with a diphthong by some speakers: *alineo* (Hualde 2005: 230).

5.2 *Vowel alternations in the root*

One of the most common irregularities has to do with variation in the (last) vowel of the root. In some cases, the alternation is related to the position of the stress, but there are also other factors. In Table 12.9, I summarize the main alternations, indicating the factors that condition them. If there is alternation between two forms, it is exemplified with the first person singular of the preterit (unstressed root) and the third person singular of the present indicative (stressed root). If there is a more complex alternation, the third person singular of the preterit is included. In choosing these forms, we follow Bybee's (1985: 124–127) analysis for *dormir*. For Bybee, these are "basic" (less marked and more frequent) forms from which one can "derive" the rest of the paradigm.

As shown in Table 12.9, the alternations involve a vowel and a diphthong, two different vowels, or two vowels and a diphthong. Diphthongs are always rising and contain the vowel *e* preceded by a glide with the same backness/frontness as the vowel with which it alternates.

The first group of alternations has clear phonological conditioning: there is a diphthong when the root is stressed and a vowel when it is unstressed. The alternation between a diphthong and a mid vowel is found in a great number of verbs, whereas only a very small number of verbs have an alternation between diphthong and high vowel. In particular *i* ~ *ie* is found in *adquirir* and *inquirir* (but not in *requerir*: *requerí*, *requiere*) and *u* ~ *ue* is restricted to the single verb *jugar*

Table 12.9 Alternations in the vowel of the root.

Variation	*Example*			
	Infinitive	*Unstressed root*		*Stressed root*
e ~ ie	cerrar	c**e**rré		ci**e**rra
o ~ ue	contar	c**o**nté		c**ue**nta
i ~ ie	adquirir	adqu**i**rí		adqu**ie**re
u ~ue	jugar	j**u**gué		j**ue**ga
e ~i	pedir	p**e**dí	p**i**dió	p**i**de
e ~ i ~ ie	mentir	m**e**ntí	m**i**ntió	m**ie**nte
o ~ u ~ue	dormir	d**o**rmí	d**u**rmió	d**ue**rme

(*conjugar*, for example, is regular: *conjugué*, *conjugo*). These alternations are unpredictable starting from the infinitive (or from any other form with an unstressed root). Predictability is greater starting from the form of the stressed root. In most cases, if a verb has one of these diphthongs when the root is stressed, it has a mid vowel when unstressed. Exceptions are only some verbs derived from nouns and adjectives diphtong forms such as *adiestrar*, *arriesgar*, *amueblar*, and *frecuentar*, derived from *diestro*, *riesgo*, *mueble*, and *frecuente* (Pensado 1999: 4471–4473; Hualde et al. 2001: 153–154).

Many third conjugation verbs with *e* in the infinitive (*pedir*, *herir*, *reñir*, *vestir*, *reír*, *freír*, etc.) have a more complex alternation *e* ~ *i* ~ *ie*. In these verbs, the vowel *i* is found in a stressed syllable (3a), and, in addition, in the first and second persons plural of the present subjunctive, in the third person singular of the preterit, in all other forms of the preterit theme with theme vowel *ié*, and in the gerund (3b); all other unstressed forms have the root vowel *e* (3c).

(3) a. /í/: pido, pides, pide, piden; pida, pidas, pidan
b. /i/: pidamos, pidáis; pidió, pidieron; pidiere, pidiera, pidiese, pidiendo
c. /e/: pedir; pedimos, pedís; pedí, pediste, pedisteis; pediré, pediría, pedido

In this case as well, it has been suggested that there is phonological conditioning. We find *e* if the following syllable contains the vowel *i*, whether stressed or unstressed, and *i* is found when the following syllable contains any other nucleus, including the diphthongs *ie*, *io* (Harris 1980; Hualde et al. 2001: 124).

The third group of alternations in Table 12.9 results from the combination of the two alternations already analyzed. They are found in a set of verbs of the third conjugation. In these three-way alternations, the diphthong is found when the root is stressed (4a), the mid vowel when the following syllable contains the vowel *i* (stressed or unstressed) (4b), and the high vowel elsewhere (4c).

(4) a. /ié/: miento, mientes, miente, mienten; mienta, mientas, mientan
b. /e/: mentir; mentimos, mentís; mentí, mentiste, mentisteis; mentiré, mentiría; mentido
c. /i/: mintamos, mintáis; mintió, mintieron; mintiere, mintiera, mintiese, mintiendo

5.3 Root-final consonant alternations

Some second and third conjugation verbs add a root final velar consonant in the first person singular of the present indicative and in all forms of the present subjunctive, as shown in (5) for *salir*.

(5) a. Present indicative: sal**g**o, sales, sale, salimos, salís, salen
b. Present subjunctive: sal**g**a, sal**g**as, sal**g**a, sal**g**amos, sal**g**áis, sal**g**an

Forms with this velar consonant constitute a 'morphome' in Aronoff's (1999) terminology, that is to say, a group of forms "whiths tigh psycological links based on purely morphological factors" (O'Neill 2011: 865). The morphological relation existing in forms that constitute a morphome gives stability to the class and explains why it is analogically extended to other verbs, as happened, for instance, with *hacer* (*faço* > *fago* > *hago*).

The root-final velar consonant may be voiceless or voiced. In Peninsular varieties with a contrast between/s/and/θ/, we find/k/after an interdental fricative (6a) and/g/after all other segments (6b) (Pensado 1999: s.68.7.4; RAE 2009: s.4.11). For Latin American Spanish, this phonological rule is not valid, *lucir*/lusír/: *luzco*/lúsko/vs. *asir*/asír/: *asgo*/ásgo/.

(6) a. Consonant/k/: conocer: cono**zc**o; conducir: condu**zc**o; lucir: lu**zc**o
  b. Consonant/g/: valer: va**lg**o; salir: sa**lg**o; tener: te**ng**o; poner: po**ng**o; venir: ve**ng**o; asir: a**sg**o; caer: ca**ig**o, traer: tra**ig**o, oír: o**ig**o; decir: d**ig**o, hacer: h**ag**o

The verb *yacer* is exceptional. It admits three possibilities: two follow the patterns already mentioned (*yazco* and *yago*) and a third one does not (*yazgo*). The largest group of verbs follows the pattern in (6a). This group includes most verbs ending in *-cer* (*agradecer*, *conocer*, *padecer*, *parecer*, etc.), those ending in *-ducir* (*aducir*, *deducir*, *conducir*, etc.), and a few others such as *placer*, *lucir* and verbs derived from these. The pattern in (6b) is less common, but it includes some very frequent verbs (*hacer*, *tener*, *venir*, and *decir*).

The velar consonant is sometimes added to the regular root (*conozco*, *valgo*, *tengo*, *asgo*, …), preceded by *i* if the root ends in a vowel (*caigo* and *traigo*). In other cases, however, the velar consonant replaces a root-final fricative (*hago* and *digo*).

The use of the velar consonant is not conditioned by phonological factors, but the morphological status of this segment is also unclear. The fact that in most cases it is inserted after the regular root suggests that is an "interfix" (Malkiel 1974) or a thematic extension (Pérez Saldanya 1994) that characterizes some forms of the paradigm. However, given the fact that in other cases it replaces the final consonant of the root, one can also analyze the facts as involving allomorphy of the root.

5.4 *Strong preterits and participles*

5.4.1 Strong preterits Strong preterits are those that are stressed on the root in the first and third person singular forms, as is the case with *tener*, shown in (7):

(7) t**u**ve, tuv**i**ste, t**u**vo, tuv**i**mos, tuv**i**steis, tuv**ie**ron

The root of these preterits appears also in other forms of the preterit theme, but not in all of them. In particular, it is found in the future subjunctive (*tuviere*, *tuvieres*, …) and the imperfect subjunctive (*tuviera* or *tuviese*, *tuvieras* or *tuvieses*, . . .). These are tenses that derive from the perfect theme in Latin. They involve a 'morphome', which explains the stability of the class (Maiden 2005; O'Neill 2011).

The allomorph of the root of strong preterits (and related tenses) is always different from that of the infinitive. The difference may affect the (last) vowel of the root, the final consonant, or both. In (8) I exemplify the different types of strong preterit, grouped according to the final consonant of the preterit root, using the third person singular:

(8) a. tener: tuvo; haber: hubo; estar: estuvo; andar: anduvo
 b. decir: dijo; conducir: condujo; deducir: dedujo; traer: trajo
 c. poner: puso; querer: quiso
 d. saber: supo; caber: cupo
 e. venir: vino
 f. hacer: hizo

In spite of the fact that there is not a unitary pattern, one can make some generalizations (Bybee 1999). Notice that all strong preterits have a high vowel (*u* or *i*) in the root, except for *traer* (but cf. dialectal *trujo*), and that *u* is always used before a labial consonant (8a and d). It is harder to make generalizations regarding the root-final consonant. In any case, the most common patterns are those in (8a) and (8b).

The irregularity of these preterits involves not only the root but also some of the endings. The final unstressed *e* and *o* of the first and third person singular forms (*tuve* and *tuvo*) are irregular. In preterits with root-final *j*/x/, the use of the theme vowel *e* in the third person plural (*dijeron*, *trajeron*) instead of *ie* (*tuvieron*, *pusieron*, *vinieron*, …) is another irregularity.

5.4.2 Participles Some second and third conjugation verbs also have strong participles, with stress on the root. The irregularity of these participles has to do with the position of the stress, the lack of a theme vowel, and the presence of a consonant other than regular mark *-d-*, as illustrated in (9):

(9) Irregular participles:
 a. abrir: abier**t**o, cubrir: cubier**t**o; morir: muer**t**o
 absolver: absuel**t**o; volver: vuel**t**o
 escribir: escri**t**o; romper: ro**t**o;
 b. poner: pue**st**o; ver: vi**st**o
 c. imprimir: impre**s**o (*but also regular* imprimido)
 d. hacer: he**ch**o; decir: di**ch**o

The most common mark is *t-*, which may be preceded by *r*, *l* or a vowel (9a). Less commonly we find *-st-* (9b), *-s-* (9c) and *-ch-* (9d), after a vowel.

5.5 Athematic futures

The tenses of the future theme are highly regular. Only in a few instances do we find irregularities, which include the deletion of the theme vowel, and, sometimes, of the final consonant of the root. These irregularities are found in some second and third conjugation verbs whose root ends in a plosive (*saber*, *caber*, *haber*, *poder*), *r* (*querer*), *n* or *l* (*poner*, *tener*, *venir*, *valer*, *salir*, and others derived from these). With plosive-final roots, the root-final consonant is syllabified with the rhotic of the TAM (10a); in the case of *r*-final roots, the two taps produce a trill (10b); and, after *n* and *l*, an epenthetic *d* is inserted (10c).

(10) a. sabrá/sabría, podrá/podría
  b. querrá/querría
  c. vendrá/vendría, valdrá/valdría

The phonological context is a necessary, but not sufficient, condition for the deletion of the theme vowel: there is theme vowel deletion in *saber* but not in *deber* (*deberá/debería*), in *querer* but not in *morir* (*morirá/moriría*), in *valer* but not in *doler* (*dolerá/dolería*). The theme vowel is also missing in the future of *hacer* (*hará/haría*) and *decir* (*dirá/diría*), where the root undergoes contraction. With the exception of verbs with *l*-final roots, all other verbs with an irregular future also have an irregular preterit (*supe*, *quise*, *vine*, … but *valió*, *salió*).

5.6 Other irregularities

5.6.1 Monosyllabic roots Some of the most common verbs in Spanish are also the most irregular ones. Without attempting exhaustiveness, we will mention here some of the main irregularities of these verbs. One of the characteristic features is the existence of monosyllabic forms in the present theme, and sometimes in the preterit theme. Exemplifying with the third person singular of the present indicative, we find such forms in *dar* (*da*), *ser* (*es*), *ir* (*va*), *ver* (*ve*), and *haber* (*ha* as an auxiliary or modal, and *hay* as an existential verb). To this list we may add *poner*, *tener*, *venir*, *salir*, *decir*, *hacer*, and *yacer*, which have monosyllabic forms in the imperative singular (*pon*, *ten*, *ven*, *sal*, *di*, *haz*, and *yaz* or *yace*), and *saber*, which has *sé* in the first person singular of the present indicative.

5.6.2 Suppletive roots The verbs *ser* and *ir* have suppletive roots (i.e., forms that derive from different etyma). In the case of *ser*, suppletion is inherited from Latin and has roots with *s*-, *er*- and *fu*-. The root *s*- has several allomorphs: *so*- in several forms of the present indicative (*soy*, *somos*, *sois*, *son*); *se*- in the infinitive (*ser*), future and conditional (*será/sería*), and present subjunctive (*sea*); *si*- in the participle (*sido*), and preceded by the vowel *e* in the third person singular of the present indicative (*es*). The root *er*- is used in the imperfect indicative (*era*) and in the second person singular of the present indicative (*eres*). Finally, the root *fu*- is used in the preterit (*fui*, *fuiste*, *fue*, …), the future subjunctive (*fuere*), and the imperfect subjunctive (*fuera* or *fuese*).

The verb *ir* has forms with the root *i*-, derived from Latin ĪRE (*ir*, *iré*, *iría*, *iba*), with the root *v*-, derived from Latin VADERE (*vas*, *vaya*), and with the root *fu*-, in the preterit, future subjunctive and imperfect subjunctive that are identical to those of *ser*.

5.6.3 The -*y* ending The verbs *ser*, *estar*, *ir*, *dar* have a form in -*y* in the first person singular (*soy*, *estoy*, *voy*, *doy*). The third person singular of existential *haber* (*hay*) also ends in -*y*. This verb is defective and is only conjugated in the third person singular. It is generally assumed that this -*y* derives from the adverbial pronoun *y* 'there' in enclitic position. This explanation is reasonable for verbs that are frequently used with a locative or directional adverb (e.g., *vo* + *y* 'I go there'; *ha* + *y* 'there is'), but it is less sensible from *dar*, where one may think of analogical extension. See Lloyd (1987: 355–358) and Pensado (2000: 187–196) for a discussion of the different hypotheses on the origin of the final -*y*.

6 Conclusions

Spanish, like Latin and the other Romance languages, is an inflectional language. The conjugation or verbal inflection, in particular, has a great complexity due to the grammatical categories expressed in the verb – person and number, tense, aspect, and mood – and to the existence of different patterns of conjugation.

Person and number markers are very regular and the general pattern only has two specific exceptions: the first person singular in the present indicative, which has a -*o* (*canto*), and the second person singular in the imperative, which has no marker (*canta* or *cantá*, in areas of *voseo*). Tense, aspect, and mood (TAM) markers are less regular than person and number markers, and they present several cases of allomorphy in different conjugations and persons. In the present tenses (present indicative and subjunctive, and imperative), the stress falls on the penultimate syllable, except in the second person plural, with stress on the last syllable. In all other tenses, on the contrary, stress is columnar or morphological, and it falls on the same morphological constituent. Present tenses, moreover, are unmarked and have no specific TAM marker, unlike the rest of tenses, which have specific markers, sometimes with allomorphy depending either on the conjugation (imperfect indicative) or on the category of person and number (future indicative and preterit).

The main verbal irregularities that are found are related to changes in the root, and more specifically, to variations in the vowel or final consonant, or to the stress position (in cases of vowel-final root). Irregularities affecting inflectional endings are much more sporadic and are limited to strong preterits and participles, athematic futures, and verbs with monosyllabic forms or with suppletive roots.

ACKNOWLEDGMENTS

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NOTES

1 Compound forms in fact derive from constructions where the Latin verb *habeo* subcategorized for an absolutive participial clause with resultative value (e.g., *ego habeo epistulam scriptam* 'I have a letter written'). In Old Spanish, as in present-day Italian and French, the auxiliary could be either *haber* or *ser*, depending on the main verb (Romani 2006).

2 In a morphological opposition, an unmarked form is the basic or default form. Usually it typically occurs more often and has less semantic restrictions than a marked one. On the other hand, the marked form tends to have a specific formal marker (an affix, for instance), in contrast with the unmarked form, that may not have any specific marker.

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13 Forms of Address

BOB DE JONGE AND DORIEN NIEUWENHUIJSEN

1 Introduction to the forms of address in Modern Spanish

The formation of the pronominal system of the forms of address in Modern Spanish is characterized by a series of changes through the history of the Spanish language. The main goal of this chapter is to describe and explain the uses of the forms of address through the centuries until the present time. In this introduction, we will give an overview of the forms of address in Old Spanish and in Modern Peninsular Spanish (Section 1.1) followed by an account of the forms of address in Latin American Spanish (Section 1.2). Section 2 discusses the historical origins of the forms of address; in Sections 3 and 4, we discuss specific characteristics of the forms of address in Spain and in Latin America, respectively.

1.1 Spain: tú, usted, vosotros, ustedes

The pronominal system of the forms of address in Old Spanish is presented in Table 13.1, below.[1] Distinctions are made between forms for subject and object of a preposition (stressed forms) and oblique forms for direct and indirect object (unstressed forms). Noteworthy is the fact that different forms are used as an object of the preposition *con* 'with': *contigo* and *convusco*. These forms originated in the Latin *tecum* 'you (sg.) with' and *voscum*, 'you (pl.) with,' which through a number of (phonological) changes became *tigo* and *vusco*. Because these forms were hardly recognizable as containing the original Latin preposition *cum*, the Spanish result of it, *con*, was added again, giving *contigo* and *convusco* (Menéndez Pidal [1904] 1982: 251).

When compared with Table 13.1, the forms of address in Modern Peninsular Spanish show some notable differences, as can be seen in Table 13.2 below: *vos* has disappeared and a new polite form, *usted*, together with its plural form *ustedes*, has appeared. Although functionally a second person pronoun, formally *usted(es)*, is a

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Table 13.1 Forms of address in Old Spanish until the fourteenth century.

<table>
<tr><th></th><th colspan="3">Singular</th><th colspan="3">Plural</th></tr>
<tr><th></th><th>Subject</th><th>Object of preposition</th><th>Oblique</th><th>Subject</th><th>Object of preposition</th><th>Oblique</th></tr>
<tr><td>Second person familiar</td><td>tú</td><td>ti
contigo</td><td>te</td><td rowspan="2">vos</td><td rowspan="2">vos
convusco</td><td rowspan="2">vos</td></tr>
<tr><td>Second person formal</td><td>vos</td><td>vos
convusco</td><td>vos</td></tr>
</table>

third person pronoun and uses corresponding third person verb endings (see Section 2.3). This also explains the appearance of third person oblique *le(s)*, *lo(s)*, *la(s)*, *se*, *sí*, and *consigo* forms. The second person plural *vos* 'you (pl.)' has disappeared as a subject pronoun and object of a preposition, including the prepositional phrase *convusco*, whereas the oblique form *vos* has given way to the reduced form *os*, probably as a result of the unstressed syntactical position it has canonically. But the most important change in the second person plural is the appearance of the modifier *otros* with the subject and object of preposition pronouns, giving *vosotros*.

1.2 *Latin America:* tú, vos, usted, ustedes

Table 13.3, the forms of address in Modern Latin American Spanish, shows some clear differences with the current situation in the Peninsula. Most striking is the absolute absence of *vosotros* 'you (pl.)' in all grammatical positions, its function being taken over by *ustedes*, which in Spain is a polite form to address a group of people, whereas in Latin America it has lost its mark of politeness, being the only plural form of address for polite as well as informal relations (see also Section 3.1). Furthermore, in some variants, *vos* is still used, although not in all grammatical positions, and never as a polite form of address.

In Section 4, we will discuss the particularities of Latin American Spanish. First, however, we will enter into the historical origins and development of the forms of address.

Table 13.2 Forms of address in Modern Peninsular Spanish.

	Singular			*Plural*		
	Subject	*Object of preposition*	*Oblique*	*Subject*	*Object of preposition*	*Oblique*
Second person familiar	tú	ti contigo	te	vosotros	vosotros	os
Second person formal	usted	sí, usted consigo	le lo, la se	ustedes	sí, ustedes consigo	les los, las se

Table 13.3 Forms of address in Modern Latin American Spanish.

<table>
<tr><th></th><th colspan="3">Singular</th><th colspan="3">Plural</th></tr>
<tr><th></th><th>Subject</th><th>Object of preposition</th><th>Oblique</th><th>Subject</th><th>Object of preposition</th><th>Oblique</th></tr>
<tr><td>Second person familiar</td><td>tú
vos</td><td>ti
contigo
vos</td><td>te</td><td rowspan="2">ustedes</td><td rowspan="2">sí
ustedes
consigo</td><td rowspan="2">les
los, las
se</td></tr>
<tr><td>Second person formal</td><td>usted</td><td>sí, usted
consigo</td><td>le
lo, la
se</td></tr>
</table>

2 Origin of the forms of address

2.1 *Historical development of* vos > vosotros

In Old Spanish, the personal pronoun of the second person was *vos* 'you (sg. and pl.),' that descended directly from Latin *vōs* 'you (pl.)' (Menéndez Pidal [1904] 1982: 251) and was used for all cases as a form of address for plural and singular polite relations until the fourteenth century. As can be seen in (1)–(4), *vos* is unmarked in the sense that the forms are used in normal second person singular and plural references. If necessary, a modifier could be used to clarify the role of the referent, such as *mismos* 'selves' and *todos* 'all,' rendering *vos mismos* 'you yourselves' and *vos todos* 'you all.' This use of the pronouns with different kinds of modifiers is already apparent in the *Cantar de Mio Cid*.

(1) por siempre **vos** faré ricos, que non seades menguados. (*Cid*, I.108)
'I shall make **you** (pl.) rich forever, and you shall not be poor.'

(2) Dámos**vos** en don **a vos** treínta marcos; (*Cid*, I.196)
'We give **you** (sg.) thirty marks [**to you** (sg.)] as a gift;'

(3) Amigos, non tengades en poco a ninguno porque **vos** seades buenos caualleros de alta sangre. (*Zifar M*, p. 145)
'Friends, do not despise anyone just because **you** (pl.) are fine gentlemen of noble blood.'

(4) E porende, mios fijos, deuedes **vos** guardar de maldezir de ninguno. (*Zifar M*, p. 306)
'And therefore, my sons, you should refrain [**yourselves** (pl.)] from cursing anyone.'

In the thirteenth century, a process of change initiated in which one of these modifiers, *otros* 'others,' was introduced after the stressed pronouns *vos* and came to be used more and more regularly. The period of the thirteenth and fourteenth centuries was characterized by variation between the simple (*vos*) and compound form (*vosotros*), and by the fact that the compound form with *otros* was gaining ground at the cost of the simple form until its complete grammaticalization by the end of the fifteenth century.

The advantages of the compound form were clear: the pronoun *vos* 'you' could both refer to a singular and to a plural interlocutor (i.e., *vos* was polisemical as far as number is concerned). The compound form resolves this matter, in view of the fact that it can only refer to a plural interlocutor. Moreover, the new pronoun paradigm is more coherent in the sense that all pronouns show different forms for stressed and unstressed uses (see Tables 13.1 and 13.2), as had always been the case for all the other personal pronouns. Finally, the meaning of *otros* 'others' fits the second person very well, since a communicative setting always implies a contrast between speaker and interlocutor, the latter being 'the other(s).'

In the first known manuscript of the *Libro del cavallero Zifar*, the fourteenth-century M manuscript, the compound form only appears once and is of a clear contrastive nature (see De Jonge 1986: 133):

(5) [the king to his vassals:] …, e tajemosles las cabeças. E sobredes dos de **vos otros** al tejado de la camara con las cabeças, mostrandolas a todos, e dezit a grandes bozes: "Muertos son los traydores Rages e Joel, …" (*Zifar M*, p. 277)
'…, and let us cut their heads off. And two of **you** [*vos otros*] climb onto the roof of the house with their heads, and show them to all, and say with loud voices: "the traitors Rages and Joel are dead, …"'

Example (5) clearly shows that *otros* is not used here only to indicate plurality, for the subject is referred to by *dos de vosotros* 'two of you.' The main purpose of the use of *otros* is, then, to designate contrastively the division of tasks of the different members of the group. The growing use of the compound form implies an invasion in other contexts with diminishing contrastiveness, the reason for which the contrastive value that characterized it was gradually neutralized.

During this process of change, the corresponding form for the first person plural *nos* changed analogously to *nosotros*, while the ancient *connusco* disappeared altogether, as was the case with *convusco*. The greater advantages for the second person as compared with the first are clear: in the first place, there is the singular–plural ambiguity that does not exist for the first person plural, and, secondly, the second person is contrastive by nature because it implies a contrast with the speaker, and this is precisely what *otros* 'others' means, whereas the first person is neutral (De Jonge 1986: 133; García et al. 1990: 76–77). Therefore, as De Jonge (1986) and García et al. (1990) demonstrated, the first cases of the compound form with *otros* were observed, not surprisingly, with the second person, but the greater coherence of the paradigm, with different forms for stressed and unstressed uses undoubtedly caused the analogous change in the first person plural.

By the end of the fifteenth century, the change was completed for *vosotros*, and only very few cases of *nos* as a stressed pronoun were still observed, the great majority being *nosotros* as well. Of course, these rare uses rapidly disappeared as well, making the paradigm perfectly balanced.

Other Romance languages show similar developments, representing different stages of the process of change we have just described in Spanish. For example, French and Italian show contrastive values in their compound structures *nous autres*, *vous autres*, and *noi altri*, *voi altri*, respectively (Meyer-Lübke [1890–1906] 1974: s.75; Lenz 1925: 228; Gili Gaya 1946: 116–117). Other languages, such as Catalan and Sicilian, have completed the grammaticalization process and have *nosaltres* and *vosaltres*, *nuautri* and *vuautri*, respectively, in their pronominal paradigms (Badía Margarit 1980: 166; Privitera 1998: 22–23).

Studies of literary works, of which more manuscripts are preserved, present excellent examples of the change in progress. The didactical work *El libro de Calila e Digna* is delivered to us in two manuscripts, one from the end of the fourteenth century (manuscript A) and another one from the end of the fifteenth century (manuscript B). In example (6a), manuscript A, we find two appearances of the stressed pronoun *vos*, once as a subject and once as the object of the preposition *a* 'to'; in (6b), manuscript B, we find the compound form in both positions:

(6a) …e dixoles: "**Vos** sodes mis hermanos e mis amygos para demandar el tuerto que yo rresçibi; pues ayudadme e guysad commo aya derecho, ca bien podria acaesçer a **vos** lo que a mi acaesçio." (*Calila A*, p. 108, l. 1631–1633)

(6b) …, e dixoles: "**Vos otros** sodes mis hermanos e amigos para caluniar el tuerto que yo rreçeby; pues ayudatme e guisat commo yo aya derecho, ca puede ser que vos acaezca a **vosotros** lo que acaeçio a my." (*Calila B*, p. 108, l. 1892–1896)
'…, and said to them: "**You** [*vos*/*vos otros*] are my brothers and my friends, so you should denounce the injustice that I suffered; so help me and lead me the way I deserve, because the same thing that happened to me could very well happen to **you** [*vos*/*vosotros*]."'

The progress of the compound forms and the corresponding weakening of their contrastive value is also shown in (7a–b), where one and the same fragment is compared in two of the three existing manuscripts of the *Libro del cavallero Zifar*. We see that in the fifteenth-century manuscript, P, (7a), *nos* 'we' suffices and does not yet need *otros*. However, in S (1512), (7b), the paradigmatic advantages of the compound form caused *nosotros* to be preferred also over the simple form *nos*, thus becoming grammaticalized:

(7a) "Ay amigas sseñoras!" dixo el vno dellos, "e por que vos amanesçio mal dia por la nuestra venida? ca sabe Dios que **nos** non cuydamos fazer enojo a ninguno nin ala vuestra señora nin a **vosotras**, nin somos venidos a esta tierra por fazer enojo a ninguno; . . ." (*Zifar P*, pp. 182–183)

(7b) "Ay amigas o señoras!" dixo el vno dellos, "e por que vos amanesçio mal dia por nuestra venida? que sabe Dios que **nosotros** non pensamos fazer enojo a ninguno nin a vuestra señora nin a **vosotras**, nin somos venidos a esta tierra por fazer enojo a ninguno; . . ." (*Zifar S*, pp. 182–183)
'"Ah, dear ladies!" said one of them, "why did such a bad day come for you because of our arrival? for God knows that **we** [*nos/nosotros*] do not intend to harm any person, neither your mistress nor **you** [*vosotras*], nor did we come to this place to harm anyone; . . ."'

Moreover, in (7a), we see that *vosotros* is used to indicate a certain amount of contrast between the mistress and her maids, and that there is basically no contrast with the first person plural. This process balanced the personal pronoun paradigm in the sense that all forms now were different for subject vs. oblique uses.

2.2 *Origin and decline of* vos *as a single referent pronoun*

In Old Spanish, *tú* was used for confidential treatment among speakers, but there was also a more polite form of address, *vos* (see Table 13.1 and Section 2.1). This form was also used for plural reference, and originates from Latin, where *vōs* was used as a plural form, but began to be used as a singular form to express politeness and respect towards a singular interlocutor from the fifth century onwards, when the Roman empire had two emperors, each of whom was consequently addressed in the plural (Carricaburo 1997: 11). However, this mechanism of using a plural form in order to express courtesy is observed in many other (also non-Roman) languages (Aitchison 2001: 148–149); the basic idea is that by the use of a plural form, a speaker implicitly considers the interlocutor to be more than himself. Use of *vos* as a single reference of polite address is subsequently found in the *Cantar de Mio Cid*, where it is used to address the king and noblemen in general (Menéndez Pidal [1944] 1976: 324, see example (2)).

In French, *vous* still is in everyday use as a form of polite address, but in Spanish the value of *vos* suffered a rapid decline from the fifteenth century onwards, which affected the degree of politeness that could be indicated by it. In the fourteenth century, *vos* was still respectful, but by the end of the fifteenth century it could be used among friends. In the seventeenth century, *vos* had suffered a considerable inflation and had lost all connotation of respectfulness. In (8), we see an example of this inflated value of *vos* in the *Quijote*. In this example, an arrogant person who is boasting of his capacities and adventures is being criticized:

(8) Finalmente, con una no vista arrogancia, llamaba de ***vos*** a sus iguales y a los mismos que le conocían, y decía que su padre era su brazo, su linaje, sus obras, y que debajo de ser soldado, al mismo rey no debía nada. (*Quijote*, p. 614)
'Finally, with an arrogance never seen before, he treated his equals and-others that knew him, using ***vos*** and said that his father was his support, his pedigree, his occupation, and being a soldier, he didn't owe anything to the king himself.'

An important factor in this inflation process was the systematic appearance of *vuestra merced* 'your mercy' as a new, more respectful form from the fourteenth century onwards. In view of the fact there were now two forms of respectful address, use of the less respectful one, *vos*, could be considered by some as not respectful enough, which implied the beginning of the inflation process. By the end of the seventeenth century, use of *vos* was obsolete in Peninsular Spanish. Its role as a polite form of address had been taken over by *usted*, the grammaticalized form of *vuestra merced*.

2.3 *Historical development of* vuestra merced > usted

The first appearances of *vuestra merced* are accompanied by the definite article *la*, demonstrating that originally it was a nominal phrase functioning as a form of formal address. From the sixteenth century onwards, the use of the article is no longer observed, which may be taken as proof that the form was grammaticalized and had turned into a personal pronoun of polite address. Furthermore, its origin as a nominal phrase explains its co-occurrence with the third person singular verb inflection. Moreover, being a nominal phrase, *vuestra merced* could also take a plural form, thus creating a separate plural form of formal address, *vuestras mercedes*, something that had not existed earlier in the Spanish pronoun system (see Table 13.1).

The development of *vuestra merced* into *usted* is quite difficult to detect, for it was merely the product of phonological changes, presumably due to its increasing use. However, it is very likely that there were various, simultaneous developments, probably related to certain regions of the Peninsula. At first, there was a regional difference between the use of the (originally possessive) *vuestra* and *vuesa*; the former leading to *vuestra merced* > *vuested* > *vusted* > *usted*, which turned into the official form of formal address in Spanish; the latter has developed in the following way: *vuesa merced* > *vuesarce(d)* > *voa(r)cé*, which is very much related to the development of Modern Portuguese *você*, the general form of address in Brazil (De Jonge and Nieuwenhuijsen 2009: 1635–1651). By the beginning of the seventeenth century, the form *usted* was reaching its definitive shape and was in general use by the end of the same century.

3 Specific characteristics of the forms of address in Spain

3.1 Geographical distribution of vosotros *and* ustedes

The *vosotros* form and verbal endings, used to establish a familiar relationship with a group of persons, are typical of the language of Spain since this form of address does not exist in Latin American Spanish (see Section 4.1). However, also in Spain there are parts where *vosotros* is virtually or completely absent. This is the case on the Canary Islands, in western Andalusia, and in parts of the provinces of Cordoba, Jaen, and Granada, where, as in Latin America, it is replaced by the plural form *ustedes* with the corresponding verb endings. In these specific parts of Spain, therefore, *ustedes* is used for formal and informal contacts, being the only plural form of address. Note, however, that in popular speech in these regions, *ustedes* is combined with *vosotros* verb endings.

3.2 tú ↔ usted*: changing use of (in)formal address*

The most important general difference between *tú* 'you' (informal) and *usted* 'you' (formal) is that *tú* addresses the hearer unconventionally, without indications of respect or courtesy. The common reciprocal use of *tú* supposes that the interlocutors are known to each other, belong to the same group, or have frequent and familiar contact. If this is not the case, use of *tú* may disturb good relations between the interlocutors. Reciprocal use of *usted*, however, indicates respect or distance between interlocutors (Alonso-Cortés 1999: 4039–4041).

The reciprocal uses of *tú* and *usted* described above indicate solidarity in informal situations (*tú*) or in formal situations (*usted*). In other situations, however, where forms of address are not used symmetrically, use of *usted* indicates power of the addressed person, due to major authority, prestige, and/or higher age; use of *tú*, on the other hand, indicates that the addressed person has a lower degree of power (i.e., authority, prestige, and/or age) with respect to the speaker (Carricaburo 1997: 9).

From the 1960s onwards, a shift in the use of *usted* to *tú* has been observed. Being a process induced by younger speakers (Carricaburo 1997: 11; Calderón Campos and Medina Morales 2010: 200), it seems that the transition to democracy, and, as a consequence, greater individual liberty from 1975 onwards, has accelerated considerably the replacement of *usted* by *tú*. Solidarity, so it seems, takes over from power as being the main factor in the use of forms of address (Calderón Campos and Medina Morales 2010: 199). Nowadays, use of *tú* is common in situations where there is no obvious difference in social status and/or age; before the transition, *usted* was the norm for all interlocutors who simply had no familiar relationship.

4 Specific characteristics of the forms of address in Latin America

4.1 Generalization of ustedes

As already mentioned in Section 1.2, probably the most salient characteristic of the forms of address in Latin American Spanish is the total absence of *vosotros* in the whole area. Instead of this form, speakers use *ustedes* to address a group of persons, whether their relationship is formal or informal. *Ustedes*, therefore, has generalized as the only plural form of address in Latin American Spanish. The two examples below show the use of *ustedes* as an informal plural form of address since it is used for children (9) and friends (10).

(9) **Ustedes** son sólo niños. (Chile; *Corpus del Español*)
'**You** [*ustedes*] are only children.'

(10) En fin, amigos míos, tal vez **ustedes** coincidan en que ése es el sueño de la libertad. (Cuba; *Corpus del Español*)
'In short, my friends, perhaps **you** [*ustedes*] agree that that is the dream of freedom.'

4.2 Voseo

Whereas in the majority of Spain a symmetrical system of forms of address is used, with one form for informal treatment and one for formal treatment, in singular as well as in plural (see Table 13.2), Latin America can be divided in various parts with different systems of address, as far as the singular forms are concerned. The most striking characteristic of the Latin American singular forms is the presence of *voseo* in various varieties (i.e., the use of the personal pronoun *vos* and its corresponding verbal morphology for one interlocutor). Although the form derives from the Old Spanish deferential form of address *vos* (see Section 2.2), nowadays in Latin America it is only used to treat someone in an informal, confidential way. We will start this section by presenting the pronominal forms and verbal morphology and then have a closer look at the origin of the phenomenon and its regional distribution.

4.2.1 The morphology of *vos* The pronominal paradigm of *vos* is a hybrid one, since it is a mixture of forms inherited from the Old Spanish deferential *vos* (for subject and object of a preposition) and of forms taken from the informal *tú* (oblique and possessive). The verbal morphology also is a mixture of forms, with special *voseo*-endings in the present indicative (and therefore also in the present subjunctive) and the imperative, combined with *tuteo*-endings in all the other tenses. Moreover, the verbal morphology differs from region to region and can be roughly classified into three different types. In Table 13.4 below, we present the different

Table 13.4 Most common morphology of *voseo* verb endings (type I).

	-ar	*-er*	*-ir*
Present indicative	hablás	comés	vivís
Imperative	hablá	comé	viví

verbal forms of the so-called *voseo* type I in the two tenses mentioned above. Type I is the most extended type, since it is used in the River Plate countries Argentina, Uruguay, and Paraguay, in east Bolivia, in the Mexican state of Chiapas, and in Central America in Nicaragua and Costa Rica. In some, less frequent types, *voseo* morphology is also different in the future tense; for detailed information about the different types of morphology, see Fontanella de Weinberg (1999: 1409–1411), Moreno de Alba (1988: 173–174), Zamora Vicente (1996: 401–404), and Carricaburo (1997: 15–18).

The following examples, from an Argentinean and a Nicaraguan writer, show the use of *vos* as a subject (11) and of *voseo* forms in the present indicative ((11) and (12)) and imperative (13).

(11) ¿**Vos creés** que yo no pienso de noche? (Argentina; *Cortázar*, p. 104)
'**Do you believe** I don't think at night?'

(12) ¿Y por qué **pensás** que estoy pasando una crisis? (Nicaragua; *Belli*, p. 146)
'And why **do you think** I am going through a crisis?'

(13) **Vení** un momento. (Argentina; *Cortázar*, p. 194)
'**Come** here for a moment.'

4.2.2 Mixed *voseo* Whereas the different types of *voseo* morphology provide us with a wide range of variation in the Latin American regions where this form of address is used, the phenomenon proves to be even more complicated since there are regions in which speakers use a mixture of the pronouns and verb endings of *vos* and *tú* different from the paradigms of the authentic *voseo* as discussed in Section 4.2.1. When a speaker uses mixed *voseo*, he either uses the pronominal forms of *vos* combined with the verbal morphology of *tú*, a phenomenon that is also called 'pronominal *voseo*,' or he uses the verbal morphology of *vos* together with the pronominal forms corresponding to the second person *tú*, the so-called 'verbal *voseo*' (Carricaburo 1997: 30, 33; Fontanella de Weinberg 1999: 1405). In example (14), we find a combination of verbal *tuteo* and pronominal *voseo*; in (15) the verbal form of *voseo* is combined with the pronoun *tú*.

(14) Mixed pronominal *voseo*
No sé si te **acuerdas vos** de ella. (Bolivia; *CREA*)
'I don't know if **you remember** her.'

(15) Mixed verbal *voseo*
No, **tú** no **podés** haberte ido con ellos … (Uruguay; *CREA*)
'No, **you can't** have gone away with them …'

4.2.3 Historical origin In Old Spanish, the deferential form of address was *vos*, a form inherited from Latin. This form gradually lost its value of respect, becoming in the sixteenth century a form to treat equals or friends, and afterwards even an insult when used for equals or people with a higher position (see Section 2.2). However, alongside the confidential *vos*, the 'old' form kept being used with a deferential purpose. It was this *vos* with its various values that was brought to Latin America by the Spanish conquerors, where it kept being used to a great extent by the settlers and subsequently learned by the *criollos*, those who were born in the Latin American colonies, but whose parents were Spaniards. It was therefore the most frequent singular form of address during the colonization of the Latin American continent and the foundation of the new colonial societies, more frequent than the other informal singular form of address *tú*.

In the seventeenth century, *vos* fell into disuse in Spain since it had lost its deferential meaning, and as a singular informal form of address it had to compete with *tú*. This language change also reached Latin America, but only those regions that stood in close contact with the homeland. These regions adopted the new trend favoring the exclusive use of *tú* for informal contacts. On the other hand, regions that had been cut off from Spain and developed in relative isolation kept using *vos* as a confidential form of address, clearly disfavoring *tú*. This historical development led to the particular distribution of *voseo* in Latin America, which will be discussed in more detail in Section 4.2.4.

4.2.4 Regional distribution of *tú* and *vos* Regions with exclusive or almost exclusive use of *tú* are the Spanish Antilles: Santo Domingo, Cuba (except for a very small region on the eastern end of the island), and Puerto Rico; Mexico (except for the state of Chiapas, bordering on Guatemala); and central Peru. During the colonization, Mexico and Peru were important centers of power since there the Spaniards founded viceroyalties shortly after their arrival on the continent. Santo Domingo hosted an important university, and ships on their way to South America were almost obliged to make a stopover here. Finally, Cuba and Puerto Rico only became independent in 1898 and therefore were closely linked to Spain for a very long period. Because of these close contacts, *vos* disappeared from the scene, as in Spain, leaving *tú* as the only informal form of address (Carricaburo 1997: 20).

In all the other Latin American countries, *vos* is used, although the range of the phenomenon varies a lot. Among the regions with *voseo*, we can distinguish those where *vos* coexists with *tú* and those with exclusive use of *vos*. The most important representatives of the second group are the River Plate countries Argentina and Paraguay. Here *vos* is the normal and only informal form of address, and it is used by all social classes. In Central America, for a long time Costa Rica has been the only country where *tú* was practically nonexistent, as in the River Plate countries, but lately *tú* seems to be reappearing, probably due to instruction of the written norm (Quesada 2010: 663–664). Noteworthy is the fact that, while *tú* is invading *vos* territory in spoken language, the opposite is taking place in written language, where *vos* is expanding in *tú* contexts (Quesada 2010: 668).

In countries or regions where *vos* and *tú* are both used as informal, confidential forms of address, the former is characteristic of the spoken language and normally used by the popular classes, frequently being considered as substandard, whereas the normative *tú* is used by the higher classes, being the form learned at school. Countries with these sociolinguistic features are, for example, Guatemala, El Salvador, Venezuela, Bolivia, and Chile. In Venezuela, *voseo* is considered to be a distinctive feature of the regions of Maracaibo and Zulia (Carricaburo 1997: 41–43; Álvarez and Freites Barros 2010: 335). In Nicaragua, *voseo* won ground after the Sandinista revolution of 1979, since by using *vos* people expressed their solidarity with the new political leaders and with their fellow citizens, whereas *tú* was associated with the old dictatorial regime (Lipski 1994: 142).

Uruguay, one of the River Plate countries, is an interesting case, not because of the coexistence of *vos* and *tú*, since this is, as we have seen, a rather widespread phenomenon, but because of the fact that the two forms are used by the same speaker in different situations and with different interlocutors. *Vos* is an intimate form, whereas *tú* is confidential, but less intimate than *vos*. In Uruguay, therefore, the difference between the two forms is not related to social class and to normative or substandard use, but to the relationship one has with the interlocutor (Carricaburo 1997: 30–32; Fontanella de Weinberg 1999: 1405). On the other hand, in Chile, *voseo* is estimated to be used less than is really the case since it is identified with Argentinean Spanish (Hummel 2010: 109–110). However, it is actually growing among younger generations as a rebellious form toward older generations and authority in general (Torrejón 2010a: 423), and even seems to be expanding to other sectors of society (Torrejón 2010b: 768).

4.2.5 Attitudes towards *voseo* In countries where *vos* is the norm, being the only informal form of address, the sociolinguistic attitudes toward the phenomenon and toward speakers who use it are positive. In Argentina, for example, *vos* enjoys great prestige and is considered an important feature of the national identity. It is used in spoken language as well as written documents by all social classes. However, the generalization of *vos* for all registers and the consequent disappearance of *tú* is a relatively recent phenomenon in Argentina, since it only came into use in the second half of the twentieth century (Carricaburo 1997: 24–29; Fontanella de Weinberg 1999: 1406–1407).

In Uruguay, where, as we have seen, *vos* and *tú* coexist, but are used by the same speakers in different communicative situations, both *vos* and *tú* are prestigious forms of address. On the other hand, in countries like Guatemala or Bolivia, *vos* is considered to be a substandard form of address that reflects a lack of education and is characteristic of the lower, poor social classes. Consequently, speakers of the higher social classes feel it should be avoided and look down upon those who do use this form in their daily speech.

4.3 *Use of* usted *in different regions*

As in Spain, in most parts of Latin America the singular *usted* is a deferential form of address. However, some countries deserve special mention because of their peculiar use of *usted*. In Colombia, *usted* is not only a polite form of address to show respect, but also a form to express solidarity with and confidence in the interlocutor. In this last use, it is comparable to the use of *tú* or *vos* in other countries. In between the two opposite uses of *usted* stands *tú* as a form of address to express certain familiarity, although at the same time marking a certain distance (Lipski 1994: 213; Carricaburo 1997: 40–41). In Costa Rica, the situation is quite similar in the sense that *usted* is used both respectfully and intimately, in the last use alternating with *vos* (Quesada 2010: 663–664, 667–668; Moser 2010). In the Andean part of Venezuela, the use of *usted* is common when treating relatives, for instance, between husband and wife, parents and children, and between brothers and sisters, and is also used among friends and neighbors (Lipski 1994: 351; Carricaburo 1997: 42; Álvarez and Freites Barros 2010: 331–334; Freites Barros and Zambrano Castro 2010). Finally, in Honduras, the use of *usted* is also very common in situations marked by confidence or solidarity (Carricaburo 1997: 44).

5 Concluding remarks

In this chapter, we have discussed the forms of address in Old Spanish and Modern Peninsular and Latin American Spanish. It is hardly surprising that, compared with the Old Spanish paradigm, on which both the Peninsular and the Latin American forms of address are based, a considerable amount of changes have taken place since language is a dynamic phenomenon and speakers constantly adapt their speech to new or changing circumstances. In this particular case, the changes in Spain include the loss of the deferential form of address *vos*, as well as the creation of *usted(es)*, that took over the role of *vos* as a polite form, and the development of the originally compound form *vosotros* as an unambiguously plural pronoun to establish informal relations. In spite of the clear communicative advantages of the latter, nowadays it is virtually absent in some parts of southern Spain and the Canary Islands, where it is substituted for its polite counterpart *ustedes*, which has lost its deferential mark in the process. Also, after the return of democracy in Spain, the informal form of address *tú* has gained ground at the cost of the polite *usted*, especially in situations where differences in social status or age are not at stake.

On the other hand, the specific sociocultural Latin American setting to which the Spanish language was exported at the end of the fifteenth century also brought about certain changes in the pronominal paradigm. The plural form of informal address *vosotros* is absent in the whole Latin American territory, and subsequently *ustedes* has become the only plural form, similar to the situation in some parts of

Spain and the Canary Islands. Moreover, the pronoun *vos*, that in Spain has been lost, in Latin America has embarked on a new career, becoming in some regions the only intimate form of address, whereas in other regions it competes with the other informal form of address *tú*. Finally, in some parts of Latin America, the formal form of address *usted* has acquired a new value, since it is used as an intimate form to treat relatives and friends. In conclusion, we can say that both Spain and Latin America have exploited the forms of address they inherited from Old Spanish in their own way, making them fit their specific communicative needs.

NOTE

1 It is a fact that the formal development of the first person plural is closely related to the second plural, as will be shown in Section 2.1., but since formally speaking, the first person is not a form of address, it is not included in the figures presented in this part of the chapter.

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14 Structure of the Noun Phrase

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1 Introductory remarks

This chapter discusses several issues concerning the syntax of Spanish noun phrases. These are structures headed by a noun, its extended projections and its arguments or modifiers, if any. The strings within brackets in (1a–f) instantiate noun phrases:

(1) a. [El anillo] estaba escondido
'The ring was hidden.'
b. De allí colgaban [tres hermosos grabados de Piranesi]
from there hung [three beautiful engravings by Piranesi]
'Three beautiful engravings by Piranesi were hanging from there.'
c. No quería recordar [el bombardeo del puente]
not wanted to remember [the bombardment of the bridge]
'S/he did not want to remember the bombardment of the bridge.'
d. [Todos los comentarios] son bienvenidos
'All comments are welcome.'
e. [El que te dejé] se ha roto
'The (one) that I lent you is broken.'
f. Sólo vi a [la de Banyoles]
'(I) only saw the (one) from Banyoles.'
g. [La roja] era la más vistosa
'The red (one) was the most colorful.'

In these examples, the lexical head of the construction is a noun. It can be either overt, as in (1a–d), or it can be phonologically null in Spanish, as in (1e–g). As shown, the noun may appear with a series of phrasal complements or elements of various types (adjectives, quantifiers or determiners, among others) that may precede or follow it.

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Studies on the denotation and properties of noun phrases have a long tradition in Philosophy and Linguistics. In the generative literature, accounts on their constituency and syntactic properties have closely reflected the theoretical framework assumed at different stages of the scientific history of the field. The discussion here mainly focuses on relatively recent proposals and revolves around two intertwined themes: the difference between nouns and other predicates with respect to thematic properties, and the types of categories that have been claimed to constitute the functional architecture of noun phrases. The chapter is organized as follows: Section 2 discusses some approaches related to the argument structure of nouns; Section 3 is devoted to the possible functional projections under the Determiner Phrase (DP) domain; and Section 4 focuses on adnominal adjectives, their interpretation and the hierarchical order in which they appear.

2 The argument structure of nouns

Nonderived or absolute nouns such as *huella* 'track/(im)print,' *árbol* 'tree,' *libro* 'book,' or *novela* 'novel' are able to license a series of modifiers of various categorial types with a variety of interpretations. Consider the following examples:

(2) a. Las huellas (dactilares)
The (im)prints (dactilars)
'The fingerprints'
b. Guerras (religiosas) (fratricidas) (devastadoras)
wars (religious) (fratricidal) (devastating)
'Devastating fratricidal religious wars'
c. El árbol (de mi casa)
the tree (of my house)
'the tree of/near my house'
d. Un libro (de bolsillo)
a book of pocket
'A pocket book'
e. Las novelas (de Barcelona) (de Mendoza)
the novels (of Barcelona) (of Mendoza)
'The Barcelona novels by Mendoza'

The above modifiers, which are all optional, restrict the denotation of the nominal head in several ways. In (2a), the relational adjective[1] *dactilar* comes close to an instrumental interpretation, the expression meaning 'an imprint done with the finger(s).' The relational adjectives in (2b), *religiosas* and *fratricidas* 'religious' and 'fratricidal' are interpreted, respectively, as the cause (religion) and the participants (brothers) of the named event, the string formed with the noun and the relational adjectives being modified by the qualifying (evaluative) adjective *devastadoras* 'devastating.' In (2c), the prepositional complement introduced by the preposition *de* 'of' may be interpreted as a locative or as a possessor. In (2d), the complement restricts the denotation of the noun by classifying it as a kind of book, the same way that other modifiers of the same type could classify it according to different dimensions (*libro de ocasión* 'bargain-priced book' or *libro de segunda mano* 'second-hand book,' among others). Example (2e) contains an iconic (or representational) noun with two modifiers, also introduced by the preposition *de* 'of.' Our knowledge about contemporary Spanish literature may allow us to interpret one complement as the theme of the novels (*Barcelona*) and the other as their author (*Mendoza*), although a variety of other interpretations are available. The same sequence could mean, for example, 'the novels owned by the city of Barcelona that talk about the person Mendoza,' 'the novels that the city of Mendoza owns, which come from Barcelona,' or 'The novels written by a person named Barcelona that have the city of Mendoza as a theme,' among several other possible interpretations.

The above examples are brought to light to show that the relation between nonderived nouns and their possible modifiers can be quite varied. The semantic licensing of these modifiers is not inherent to the lexical entry of the noun because nonderived nouns are basically nonrelational syntactic objects and lack what we may properly consider a thematic grid. Nevertheless, the adnominal adjectives (relational or qualifying) or the prepositional complements modifying a noun are hierarchically organized. Adjectives, which are canonically in postnominal position in Spanish, must follow quite a strict ordering sequence, whereas the hierarchy of genitive modifiers determines their accessibility to possessive pronominalization and extraction. These facts indicate that all types of modifiers or complements occupy designated positions in the structural configuration of the noun phrase. For discussion on these issues, see Milner (1978); Giorgi and Longobardi (1991); Bosque and Picallo (1996); Picallo and Rigau (1999); Cinque (2010), to mention only a very few.

Derived nouns and their associated verbal or adjectival bases have been subject to much discussion in the generative literature. Chomsky (1970) argued that, contrary to the transformational accounts provided in previous stages of generative grammar, derivational processes of noun formation are not carried out at the transformational component, claiming that deverbal or deadjectival nominals must be listed as independent items in the lexicon and should enter the derivation categorially specified as NPs. This hypothesis, known as the *lexicalist hypothesis*, was set to account for the apparently similar syntactic behavior of all types of noun phrases (headed either by derived or by nonderived nouns) as well as for the morphological irregularities observed between a verb and its related nominal, as shown in (3). Note that non attested forms ($^{\#}$) appear to be morphologically possible:

(3) a. arrancar/arranque ($^{??}$arrancamiento/$^{\#}$arrancación)
'to uproot'/'outburst (or start)'
b. morder/mordedura/mordida ($^{\#}$mordimiento/$^{\#}$mordición)
'to bite'/'bite'/'bribe'
c. remorder/remordimiento ($^{\#}$remordedura/$^{\#}$remordición/$^{\#}$remordida)
'to fret'/'remorse'

d. masticar/masticación (#mastique/#masticamiento)
'to chew'/'chewing'
e. barnizar/barnizado (#barnizamiento/#barnización)
'to varnish'/'varnishing'

In addition to the observed morphological irregularities, derived nouns do not always inherit the argument structure of their related bases and may have idiosyncratic meanings, as shown by the verb/noun pairs in (4 a–c). Below the listed items, we exemplify the arguments or participants they can (or cannot) license in clausal structures and related nominal constructions:

(4) a. imaginar/imaginación
to imagine/imagination
(i) Juan imaginaba *(historias)
Juan imagined *(stories)
(ii) La imaginación (*de historias) de Juan
Juan's imagination (*of stories)
b. moderar/moderación
to moderate/moderation
(i) María modera *(el coloquio)
María moderates *(the colloquium)
(ii) La moderación (*del coloquio) de María
María's moderation (*of the colloquium)
c. habitar/habitación
to habitate/room
(i) Emma habitaba *(en la cima del monte)
Emma inhabited *(on the top of the mountain)
(ii) La habitación de Emma (*en la cima del monte)
Emma's room (*on the top of the mountain)

Note that the nominal expression in (4 c (ii)) is, of course, grammatical under the (here) irrelevant reading 'Emma owns a room on the top of the mountain' (where the locative is the complement of *room*), but impossible in the hypothetical intended reading 'Emma's inhabitation on the top of the mountain.' These idiosyncrasies or irregularities, together with the apparent optionality for argument realization and the existence of interpretive relations unfound in verbally headed structures (like that of possessor), suggested that nouns differ from verbs in fundamental ways and that they, unlike verbal predicates, may lack a thematic grid or argument structure.

Grimshaw (1990) claimed, however, that some of the observed differences between verbal and nominal heads are only apparent, arguing that there are several types of deverbal nominals that must be distinguished with respect to their argument structure. Some of them license grammatical arguments of the kinds that verbs have, whereas some only license semantic participants. The first type are process or complex event nouns, they denote states of affairs (not objects or entities), and inherit the thematic grid of their related verbs, taking internal arguments obligatorily; they are exemplified in (5). Note that the Agent in the Spanish (5a) is optional and surfaces as an adjunct introduced by the prepositional locution *por parte de* 'on the part of':

(5) a. [La traducción del poema (por parte de Luis)] duró muchos años
[the translation of the poem (on the part of Luis)] lasted many years
'The translation of the poem by Luis lasted many years.'
b. [La desaparición de la evidencia] tuvo lugar el año pasado
[the disappearance of the evidence] took place the last year
'The disappearance of the evidence took place last year.'

The characteristic "passive" form of transitive action nominals has been discussed in Cinque (1980) for Italian, and Picallo (1991, 1999) for Catalan and Spanish, respectively, among others.

The second type characterized by Grimshaw are the result nominals (6a, b). They name objects or entities obtained either from the application of a process or from the outcome of an event. Their semantic participants do not have to be obligatorily realized and behave, in this particular respect, like the nonderived nominals exemplified in (2) above:[2]

(6) a. Los ingenieros estudiaron minuciosamente [la rotura (de la viga)]
the engineers studied meticulously [the break (of the beam)]
'The engineers meticulously studied the breaking of the beam.'
b. [La caída (del imperio romano)] fue explicada en clase
[the fall (of the Roman Empire)] was explained in class
'The fall of the Roman Empire was explained in class.'

Optionality *versus* obligatoriness in the syntactic realization of internal arguments was assumed to be determined by the event structure of the construction which, according to Grimshaw, relates to argumental configuration and is encoded in the lexicon. The relation between event/aspect interpretation and the expression of arguments in nominal constructions has been developed since Grimshaw's original proposal. The study of aspect and its relation to the thematic grid, together with more recent transformational accounts for word formation, has been pursued within the Distributed Morphology (DM) and the Exo-skeletal (Ex-S) hypotheses. These approaches propose a radically computational approach for word formation and claim that functional elements, rather than the lexical items themselves, are what determine the syntax of the construction. These views are opposed to approaches where the lexical properties of items determine syntactic structure. For discussion on these issues see Zubizarreta (1987); van Hout (1991); Halle and Marantz (1993); Marantz (1997); van Hout and Roeper (1998); Embick and Noyer (2001), among others. For the Exo-skeletal approach see Borer (2003, 2005), and for a general overview, see Alexiadou et al. (2007).

In the DM and Ex-S systems, grammatical categories and argument structure are assumed to be the product of syntactic operations. The central hypothesis is that the grammar does not operate with syntactic objects bearing categorial labels, as argued by the proponents of the lexicalist hypothesis. Syntactic items operate as feature bundles that receive a phonetic realization at the syntactic output in DM approaches. The Ex-S approach shares DM's basic conceptual tenets, but the acategorial root (henceforth [√ROOT]) of the construction is claimed to also include phonological features. In both systems, lexical items will have one or the other category depending on the functional syntactic context in which they merge. The characteristics of the functional context determine the interpretation of the nominal and the realization of its argument structure, if any. The next section is devoted to discussing the relation between functional structure and argument realization within some DM approaches.

3 The functional structure of nominals

For ease of exposition, we can divide the structure of nominals in two major fields: a lexical level headed by a category-neutral [√ROOT] and a series of hierarchically ordered functional categories of various types. Among the latter we can distinguish three domains (which correspond to the three domains proposed in Grohmann 2000): (a) a lower functional layer immediately dominating [√ROOT] related to the expression of argument and event structure; (b) an intermediate layer related to the general morphological properties of all nouns; and (c) a higher functional set of projections (i.e., the "DP field") related to the referential and discourse oriented properties of the construction. The discussion in this chapter will not be concerned with this higher left peripheral functional set of projections (see Chapter 15). See also, among others, Abney (1986); Leonetti (1990); Giusti (1993, 1996); Longobardi (1994); Aboh (2004); Haegeman (2004); and Villalba and Bartra-Kaufmann (2010), with all references cited therein.

3.1 The lower functional layer: derivation

As mentioned, nominalizations are formed syntactically in the DM systems. Morphemes (i.e., their features) are independent entities heading nodes hierarchically organized. The different readings of complex event or result nominals are obtained from the types of functional projections dominating the category-neutral [√ROOT].

Structure (7) represents the constituency of complex event/process nominals, as suggested in Alexiadou (2001: 19). The lexical root is claimed to be dominated by the functional categories *v*P and AspectP (see Chapter 17). The functional projection $\underline{v}$ encodes eventivity and agentivity, whereas Aspect is said to be the locus of (im)perfectivity. These projections are dominated by the intermediate and the higher functional layers encoding Number and the Determiner, as in (7):

(7) [D [Num [$_{\mathrm{AspP}}$ Aspect [$_{vP}v$ [$_{\surd\mathrm{P}}$ √ROOT DP/CP]]]]]

Approaches along these lines, but with different realizations of functional structure, have been suggested by other authors; see, for example, Roeper (2005) and Harley (2009), among many others. Alexiadou et al. (2007) offer an extensive overview and discussion of different proposals.

In Alexiadou's (2001) account, result-interpreted nominals lack the AspP and the *v*P functional projections. The acategorial [√ROOT] is directly dominated by the functional projection (Number P), as shown in (8):

(8) [D [Num [$_{\surd\mathrm{P}}$ √ROOT (DP/CP)]]]

Note, however, that structure (8) is not fully able to represent Spanish nominalizations of the types exemplified in (5) or (6) because within DM each derivational morpheme affixed to a root requires a separate projection.

The following examples, showing the nominalization morphemes *-ción/-miento/-do(-da)*, can ambiguously correspond to event/process or to result nominals. The types of predicates in which these nominal constructions can be framed highlight one or the other interpretation:

(9) La destrucción de la ciudad
'The destruction of the city'

EVENT/PROCESS

a. [La destrucción de la ciudad] ocurrió en 1937
'The destruction of the city occurred in 1937.'

RESULT

b. Los periodistas describieron [la destrucción de la ciudad]
'The journalists described the destruction of the city.'

(10) El enterramiento de las víctimas del terremoto
'The burial of the victims of the earthquake'

EVENT/PROCESS

a. [El enterramiento de las víctimas del terremoto] ya tuvo lugar
'The burial of the victims of the earthquake took place already.'

RESULT

b. [El enterramiento de las víctimas del terremoto] fue descrito detalladamente
'The burial of the victims of the earthquake was described with detail.'

(11) La salida de los obreros
'the exit of the workers'

EVENT/PROCESS

a. [La salida de los obreros] era a las cinco
'The exit of the workers was at five.'

RESULT

b. Los Lumière filmaron [la salida de los obreros]
'The Lumières filmed the exit of the workers.'

The internal PP complement in both types of readings appears to have the same syntactic properties, regardless of whether it is an argument (i.e., the (a) cases) or an optional participant (i.e., the (b) cases). They behave similarly under *wh*-extraction (cf. ??**De quién fue descrito/ocurrió el enterramiento?* 'Of whom was described/occurred the burial?'), and the genitive complement can be pronominalized by a possessive pronoun in these examples (see Giorgi and Longobardi 1991 and Ticio 2006, with all references cited there, for Italian and Spanish, respectively):

(12) a. Su destrucción (cf. (9))
'its destruction'
b. Su enterramiento (cf. (10))
'their burial'
c. Su salida (cf. (11))
'their exit'

Given that derived nominals can exhibit the same nominalizing suffix, independently of their interpretation, a functional projection hosting it must be posited in either event or result nominals. In some specific cases, two different suffixes on the same base correspond to one or to the other reading as in the pair *despellejamiento* 'skinning/speaking ill' (event)/*despellejadura* 'skinning' (result) among some others. We can adopt, from Harley and Noyer (1998), the functional projection *n*P. The structure (13) could then replace Alexiadou's schemata (8) for result nominals:

(13) [D [Num [$_{nP}$–*ción/–miento/–do(–da)* [$_{\surd P}$ √ROOT (DP/CP)]]]]

Note that the exemplified nominalizations, in any reading, can be realized with (at least) three different suffixes in Spanish, which apparently differ only in the morphological spell out of the corresponding functional features in the *n*P projection, as represented in (13). Less productive suffixes are the following: */–ura/*, as in *captura* 'capture' or *envoltura* 'wrapping'; */–aje/*, as in *almacenaje* 'storing' or *ensamblaje* 'assembling'; as well as zero-derived, as in *caza* 'hunting' or *pesca* 'fishing,' among others. For event nominalizations, Fábregas (2010) suggests that the suffixes *-miento* and *-do(-da)* reflect two different structural positions for the internal argument: it either merges in the specifier position of the base projection (nominalizations in *-miento*) or in its complement position (nominalizations in *-do (-da)*). Nominalizations in *-miento* correspond to internal arguments denoting entities that do not affect the aspectual properties of the nominal. Nominalizations in *-do(-da)* select complements that have an effect on the (a)telicity of the construction, that is, whether or not the event is considered to have been completed. The suffix *-ción* can attach to bases that may select arguments of any type, according to this author. For the very limited purposes of the present discussion, we can adapt Fábregas' specific proposal and technical implementation by assuming that the relevant projection where these internal arguments merge (in their specifier or in their complement position) is [√ROOT] in configurations of the types (7) above for event nominals, as shown in (14):

(14) [D [Num [$_{\text{AspP}}$ Aspect [$_{n\text{P}}$*-ción/-miento/-do(-da)* [$_{v\text{P}}$*v* [$_{\surd\text{P}}$ (DP) [$_{\surd'}$√ROOT (DP)]]]]]]]

As said, Fábregas' account mainly focuses on event/process nominalizations, which have argument structure. His hypothesis may lead one to predict that result nominals (those that only license optional participants) should nominalize with the suffix *-ción*, which is the unmarked all-purpose option. However, we have already seen in examples (9)–(11) that this is not always so. In any case, the expression of the agent in nominals derived from transitive predicates distinguishes the event–process from the result reading. In the first case, the agent is introduced by a Prepositional Phrase, as in clausal passives:

(15) a. [La observación del cometa *por los astrónomos*] se produjo hace dos meses
'The observation of the comet by the astronomers occurred two months ago.'
b. [La descarga de mercancías *por los estibadores*] tiene lugar por la mañana
'The discharge of goods by the stevedores takes place in the morning.'
c. [El fusilamiento de los prisioneros *por el pelotón*] ocurrió en 1937
'The shooting of the prisoners by the squad occurred in 1937.'
d. [La recogida de víveres *por los voluntarios*] tiene lugar los lunes
'The collecting of foodstuffs by the volunteers takes place on Mondays.'

Alexiadou (2001: 119) suggested that the agent introduced by the preposition *por* in Spanish event–process nominals may merge in the specifier position of the category neutral [√ROOT]. However, we cannot adopt that suggestion if Fábregas' proposal is taken into consideration. Suppose, with Harley (2009: 331), that the external passive-like argument merges in the specifier of *v*P, when that argument is syntactically realized.[3] Harley notes that the *v*P projection must be present in unaccusative bases as well, such as the one exemplified in (11) above. In the case of unaccusative bases, *v* has the semantics of 'become/be' rather than the 'cause' reading of the transitive configurations exemplified in (15). We can add that *v* must also appear in nominalizations based on psychological predicates, which do not express causation. Nominalizations of the types *agitación* 'agitation,' *sorpresa* 'surprise,' *temor* 'fear,' *diversión* 'amusement,' among others, denote states and appear to be argument-taking, in the genitive in these cases:

(16) La agitación/sorpresa/temor/diversión de Juan
the agitation/surprise/fear/amusement of Juan
'Juan's agitation/surprise/fear/amusement'

The reader may have noted that DM approaches to different aspects of nominalization are not always consistent with each other. Our cursory look at a few proposals has left out a large number of details undiscussed, or even unmentioned, that need to be worked out if transformational accounts are adopted for derivational morphology. In the domain of nominal derivation, there are several facts that should be considered under an account on these lines: Spanish has about 30 productive suffixes to construct derived nominals of various types and denotations, and they show a lot of dialectal variation. Moreover, it is not always clear whether some derivational suffixes are allomorphs of the same base form or if they correspond to different suffixes. See Santiago and Bustos (1999) and all references cited there for a detailed account of the facts.

Transformational/computational treatments of nominalizations have not gone unchallenged. Baker (2003) and Don (2004) argue against the hypothesis that roots are stored in the lexicon without categorical specification. On the basis of linguistic evidence, they claim that derivation is a directional process where the category of the base root (nominal or verbal) must be lexically stored. Newmeyer (to appear) also questions these approaches on conceptual grounds with some empirical considerations based on English data. He questions as well whether nonlexicalist treatments are an improvement over Chomsky's (1970) classical lexicalist hypothesis. Let us now turn to discussing the inflectional domain of nominal constructions.

3.2 The intermediate functional layer: inflection

Spanish nouns all have number and gender. For count nouns, grammatical number is associated to the opposition "one/more than one," for mass nouns we can roughly say that plural is associated with the interpretation "types of X" or "units of X." The plural may also be associated with idiosyncratic interpretations in abstract nouns (cf. *amistad/amistades* 'frienship/group of friends,' *autoridad/autoridades* 'authority/authorities') (see Ambadiang 1999). The term *gender* refers to one of the two formal inflectional classes to which a Spanish noun may belong: either the so-called "masculine" or the "feminine." The "masculine" is the grammatically unmarked noun class and typically surfaces as the suffix /*-o*/ in Spanish (as in *hueso* 'bone-MASC,' *mantenimiento* 'maintenance-MASC' or *cuchillo* 'knife-MASC'). The "feminine" class is prototypically realized as /*-a*/ (as in *silla* 'chair-FEM,' *puerta* 'door-FEM,' *cuchara* 'spoon-FEM' or *llegada* 'arrival') (see Chapter 10).

Besides gender, all nouns, of whichever formal gender type, must be inflected for any of two possible grammatical numbers: the singular, which is phonologically null, or the plural, realized with the suffix /–(*e*)*s*/ (as in the pairs *ordenador/ordenadores* 'computer(s),' *caja/cajas* 'box(es),' *señal/señales* 'signal(s),' or *observación/observaciones* 'observation(s)'). In some cases, only singular forms are used (*grima* 'irritation' or *salud* 'health,' among others). There are also inherently plural nouns (*pluralia tantum*): *albricias* 'congratulations,' *bártulos* 'implements,' and *víveres* 'foodstuffs' are examples of some of them. Determiners, demonstratives, quantifiers, and modifying adjectives, as well as some relative forms and anaphoric pronouns agree with the gender and number features of the nominal head, shown in italics in the following examples (see also Chapters 10 and 25):

(17) a. Una *sinfonía* clásica
a-FEM,SING symphony-FEM,SING classical-FEM,SING
b. Tres *matemáticos* con los cuales hablé
three mathematicians-MASC,PLUR with the-MASC,PLUR whom-MASC,PLUR talked-1,SING
'three mathematicians with whom I talked'
c. Este *libro*, no lo he leído
this-MASC,SING book-MASC,SING, not it-ACC,MASC,SING have-1,SING read
'This book, I have not read it.'

Some authors have proposed two separate functional categories Ge(nder)P and Num(ber)P to host these inflectional features. The functional projections are hierarchically organized in the mirror-image order of the inflectional suffixes:

(18) … $[_{\text{NumP}}$ Num $[_{\text{GeP}}$ Ge $[_{\text{NP}}$ …]]]

Positing a NumP projection as a functional category has been relatively uncontroversial, given the role that number has in the interpretation of a noun phrase (for earlier proposals, see Ritter 1991 with data from Hebrew as well as Picallo 1991 and Bernstein 1993, among others, for the Romance languages). On the contrary, proposals for a separate functional category encoding gender have been challenged. In the following section, the main arguments of the controversy are discussed.

3.3 The gender controversy

In Spanish, the gender feature of a noun is morphologically opaque in many cases (cf. *tribu* 'tribe-FEM,' *mapa* 'map-MASC,' *señal* 'signal-FEM,' *animal* 'animal-MASC,' *fuelle* 'bellow-MASC,' *calle* 'street-FEM,' or *pared* 'wall-FEM'). Belonging to the feminine or to the masculine noun class is often unrelated to the referential or interpretive properties of a noun, in a similar way that the conjugation class Spanish verbs belong to is unrelated to their semantics. Common nouns denoting inanimate entities are formally either feminine or masculine (cf. *lápiz* 'pencil-MASC,' *pluma* 'feather-FEM,' *libro* 'book-MASC,' *libra* 'pound-FEM,' *cazo* 'dipper-MASC,' *olla* 'pot-FEM,' *aceite* 'oil-MASC,' *agua* 'water-FEM,' *azúcar* 'sugar-MASC,' *sal* 'salt-FEM,' *arbusto* 'bush-MASC,' *planta* 'plant-FEM,' *día* 'day-MASC' or *noche* 'night-FEM,' among thousands of others). Other nouns, traditionally known as *epicene*, may denote animate individuals but inflect for the masculine or the feminine independently of the natural sex of its referent: *jirafa* 'giraffe-FEM,' *hipopótamo* 'hipoppothamus-MASC.' *hormiga* 'ant-FEM,' *topo* 'weaver-MASC,' *ballena* 'whale-FEM,' *escorpión* 'scorpion-MASC,' or *persona* 'person-FEM' are some of them.

The formal inflectional class of a noun may also be related to some non-grammatical properties of its referent in the sense that it serves to express the natural sex of the denoted animate entity. Independent lexical entries exist in some cases to denote the natural sex of an individual, as exemplified in the pairs *hombre/mujer* 'man-MASC/woman-FEM,' *caballo/yegua* 'horse-MASC/mare-FEM,' or *carnero/oveja* 'sheep-MASC/FEM.' In other cases, the same lexical root can appear with the two typical inflections for gender, as in *niño/niña* 'boy-MASC/girl-FEM,' *perro/perra* 'dog-MASC/FEM,' or *poeta/poetisa* 'poet-MASC/FEM' (see Ambadiang 1999 and Chapter 10 for extensive discussion on the particularities of gender marking in Spanish). These few examples allow us to see that formal gender is a grammatical entity, independent from natural sex, and that formal inflectional class is idiosyncratic and quite unsystematic in common nouns. This fact has cast doubts for considering gender a functional category, prompting some researchers to argue that this feature does not project at the syntactic component because it is an inherent property of the lexical noun head and constitutes encyclopedic knowledge (see Ritter 1993 as well as Alexiadou et al. 2007 for discussion of this view with data in a number of languages). Within the framework of DM, Embick and Noyer 2001 have argued that gender should be considered a dissociated morpheme, inserted postsyntactically in the nominal configuration.

Within the Principles and Parameters framework, Picallo (1991) and Bernstein (1993) advocated for the syntactic status of gender features. They were argued to head a functional category and to trigger raising operations accounting for word order in the nominal domain, in particular the canonical postnominal position of adjectives in Romance (see, however, Section 4). Bernstein also suggested that gender is the feature that licenses noun ellipsis in languages like Spanish or Italian. The evidence adduced by Bernstein is exemplified in (19a, b), where the noun is nonovert, and gender (a bound morpheme) appears suffixed to the indefinite determiner or quantifier. Note that their basic forms are *un/algún*, as shown in the corresponding (20a, b) examples with an overt noun:

(19) a. Un*o* azul
a-MASC,SING blue
'A blue one'
b. Algun*o* de oro
some-MASC,SING of gold
'Some golden ones'

(20) a. Un cuaderno azul
a-MASC,SING notebook-MASC blue
'A blue notebook'

b. Algún collar de oro
some-MASC,SING necklace-MASC of gold
'Some gold necklace'

Formal gender serves also as a grammatical device at the interpretive component, to link pronouns to their intended linguistic antecedents when number remains invariable as in (21a, b):

(21) a. La abogada$_i$ creía que el acusado$_j$ pensaba que el fiscal$_h$ no la $_{i/*j}$ conocía
the-FEM lawyer-FEM believed that the-MASC defendant-MASC thought that the-MASC prosecutor-MASC did not *clitic*-FEM know
'The lawyer believed that the defendant thought that the prosecutor did not know her.'

b. La abogada$_i$ creía que el acusado$_j$ pensaba que el fiscal$_h$ no lo $_{*i/j}$ conocía
the-MASC lawyer-FEM believed that the-MASC defendant-MASC thought that
the-MASC prosecutor-MASC did not *clitic*-MASC know
'The lawyer believed that the defendant thought that the prosecutor did not know him.'

The examples in (22a, b) show also that when number remains invariable, only gender allows us to interpret a pronoun either as a variable, as in (22a), or as a deictic expression denoting any grammatically masculine-named entity salient in context, as in (22b). The possible bound interpretation in (22a) has been represented by coindexing; for discussion on these issues, see Chapter 26.

(22) a. Cuando un marinero tiene una barca$_i$, la$_i$ calafatea
when a sailor has a-FEM,SING boat-FEM,SING, (he) it-FEM,SING calks
'When a sailor has a boat, he calks it$_{(=\text{the boat})}$'

b. Si una mariscadora tiene una barca, lo pinta
if a-FEM,SING shellfish gatherer-FEM,SING has a-FEM,SING boat-FEM,SING, (she) it-MASC,SING paints
'If a shellfish gatherer has a boat, she paints it$_{(\neq\text{the boat})}$'

Some scholars from different theoretical traditions have pointed out that formal gender is similar to other grammatical entities known as noun classes or noun classifiers. This idea has been adopted in Picallo (2008), who claims that grammatical gender is a classifier-like device, the same kind of abstract functional element that appears in the guise of noun classes or noun classifiers in different language families. In her account, gender is a formal feature linking grammar to external systems, possibly to encode nonlinguistic processes of entity categorization. Upon observing that genderless categories (like clauses and the so-called "neuter" Spanish pronouns) are also numberless, she suggests that this abstract category feeds the expression of grammatical number in the same way that classifiers appear to feed the expression of counting or measuring devices in the nominal systems of many languages.

The functional layers dominating the NP projection may also host adnominal adjectives. We turn to discussing some of their syntactic and interpretive properties in the next section.

4 Adnominal adjectives

Adjectives canonically appear in postnominal position in the Romance languages, those of the relational types, practically without exception. Qualifying adjectives can also appear in prenominal position but, in this case, some interpretive restrictions may apply.[4] Examples (23a–c) show that the Romance canonical (postnominal) sequence of modifying adjectives is the mirror image of its canonical (prenominal) sequential order in English, as shown in examples (24a–c) (see Lamarche 1991; Bosque and Picallo 1996; Bouchard 2002; Cinque (2010) among others):

(23) a. Este gato blanco
this cat white
b. El crítico literario inteligente
the critic literary intelligent
c. Una producción automovilística japonesa impresionante
A production automobilistic Japanese impressive

(24) a. This white cat
b. The intelligent literary critic
c. An impressive Japanese car production

Relational adjectives often correspond to compounds in English, as given in (24c) for the Spanish (23c). A head parameter cannot be appealed to in order to account for the overt position of words (see Kayne 1994). Adjectives of various types appear to merge into dedicated positions in the functional structure and may affect the interpretation, or take scope, of other noun–adjective sequences, as observed in Bosque (1989: 115 ff.), Bouchard (2002: 126) and Cinque (2010), among others. If we compare (25a, b) with (26a, b), we can see that the interpretation of the string depends on the relative position of the adjectives in the sequence. In (26a), policies by government(s) towards Europe are referred to, whereas in (26b) we are talking about policies by Europe towards government(s), in both cases qualified as disastrous:

(25) a. Política europea
politics European
'European politics'
b. Política gubernamental
politics governmental
'governmental politics'

(26) a. Una política europea gubernamental desastrosa
a politics European governmental disastrous
'a disastrous governmental European politics'
b. Una política gubernamental europea desastrosa
a politics governmental European disastrous
'a disastrous European governmental politics'

The sequential pattern obtained, if we compare Spanish with English, cannot be captured by assuming successive N-raising operations in the case of Spanish because the sequence would not result in the particular mirror-image order. Cinque (2010) notes that, in addition, the N-raising approach would be unable to provide a unified analysis for the interpretive patterns found in Romance and the Germanic languages in general, which are closely related to the respective canonical postnominal or canonical prenominal position for adjectives. The following section is devoted to present an overview of Cinque's detailed proposal for modifying adjectives.

4.1 Prenominal and postnominal adjectives

As noted, relational adjectives are always postnominal in Spanish. Qualifying adjectives also are canonically in postnominal position. In this case, they can be ambiguous between an individual-level reading (a permanent, characterizing or enduring property) and a stage-level interpretation (a temporary condition). If qualifying adjectives precede the noun only the individual-level reading is usually possible:

(27) a. La mirada *hostil* de Aquiles (individual-level and stage-level interpretation)
the glance hostile of Achilles
b. La *hostil* mirada de Aquiles (individual-level interpretation)
the hostile glance of Achilles
'Achilles' hostile glance'

Demonte (1999: 146 ff) points also out that qualifying adjectives can be interpreted either as restrictive or as nonrestrictive in their canonical position. Thus, (28a) is ambiguous between these readings, whereas (28b) is only nonrestrictive. Bello (1847 § 47) had already observed that adjectives are interpreted as epithets if prenominal:

(28) a. Las togas *elegantes* de Petronio
the togas elegant of Petronius
(i) [some of Petronius' togas are elegant]
(ii) [all of Petronius' togas are elegant]
'Petronius' elegant togas'
b. Las *elegantes* togas de Petronio
the elegant togas of Petronius
[all of Petronius' togas are elegant]

Scalar adjectives denoting a pole in a range of some given property (among others, *largo/corto* 'long/short'; *ancho/estrecho* 'wide/narrow'; *alto/bajo* 'tall/short'; *grande/pequeño* 'big/small'; *viejo/joven* 'old/young'; *rápido/lento* 'quick/slow') have an ambiguous reading, either a comparative or an absolute interpretation, if the adjective is in its canonical (postnominal) position, but they have an absolute interpretation if they are prenominal (see Demonte 1999: 198 and Cinque (2010)):

(29) a. Los ríos *anchos* de la Patagonia
the rivers wide of Patagonia
(i) [Patagonia's rivers are wide water courses as compared the average wideness of rivers]
(ii) [Patagonia's rivers are wide water courses]
b. Los *anchos* ríos de la Patagonia
the wide rivers of Patagonia
[Patagonia's rivers are wide water courses]

Demonte and Cinque discuss other interpretive distinctions relative to the canonical/noncanonical position of adjectives that we can not possibly discuss here. The basic fact to note is that they are mirror-image distributed with respect to their English counterparts in the noted interpretations or restrictions. Cinque proposes a unified account for the observed patterns and readings based on the following claims: (a) adjectival modification has two different origins, either as reduced relatives or as direct modifiers; (b) adjectives merge in a strict hierarchical order in the functional projections dominating NP; and (c) the sequential orders obtained in Romance are the result of upward phrasal movements.

4.2 *Phrasal movement in NP*

Adjectives universally merge to the left of NP and are always phrasal (Bosque and Picallo 1996; Cinque (2010)). According to Cinque, the different sources of adnominal adjective modification (either direct or as reduced relatives) correspond to two different levels of adjective merging at specifiers of a series of functional projections dominating NP. The lower levels correspond to direct modification and host relational or qualifying adjectives that merge to hierarchically ordered functional projections $FP_{(x)}$. Adjectives originating from reduced relatives occupy a higher structural level, as abstractly shown in (30). The schema may correspond to the string (31) in one of its interpretations:

(30) $[_{DP} \ldots [_{FP(k)}$ Red Rel $F^0 \ldots [_{FP(i)}$ AdjP (dir mod) F^0 $[_{FP(h)}$ AdjP (dir mod) F^0 $[_{NP}]]]]]$

(31) La/una producción cestera china habilidosa
the/a production of-baskets Chinese skillful
'The/a skillful Chinese basket production'

Recall that relational adjectives appear always postnominally.[5] In the example under discussion, the adjectives *cestera* 'of-baskets' and *china* 'Chinese' occupy, respectively, the specifier positions of functional projections FP(h) and FP(i) in the abstract configuration (30). The order is obtained by phrasal raising operations, as schematically exemplified in (32) for the relevant substructure:

(32) $[_{DP}$ …$[[_{FP(i)}$ china F^0 $[[_{FP(h)}$ cestera F^0 $[_{NP}$ producción]]]]]]

a. $[_{DP}$ …$[[_{FP(i)}$ china F^0 $[[_{NP}$ producción$]_i$ $[_{FP(h)}$ cestera F^0 t_i …]]]]]

b. $[_{DP}$ …$[[[_{NP}$ producción$]_i$ $[_{FP(h)}$ cestera $F^0 t_i$ …$]]_j$ $[_{FP(i)}$ china F^0 t_j]]]

Postnominal qualifying adjectives such as *habilidosa* 'skillful' in (31) above can be ambiguous between the individual/stage level and restrictive/nonrestrictive interpretations (see discussion in Section 4.1). They can be direct modifiers or have a restrictive-relative source. In the first case, phrasal movement of the types exemplified in (32 a, b) would apply to a position below FP(k) in the schema (30), whereas if their origin is that of a restrictive relative, phrasal movement would have applied to a projection above FP(k) in the representation in (30). The two possibilities are schematically shown in (33a) and (33b), the word order remaining the same:

(33) a. $[_{DP}$ …[[producción cestera china$]_h$ $[_{FP(j)}$ habilidosa F^0 $\underline{t}_h$]]] (cf. (31))

b. $[_{DP}$ …[[producción cestera china$]_h$ $[_{FP(k)[RedRel]}$ habilidosa F^0 $\underline{t}_h$]]] (cf. (31))

Given that qualifying adjectives of a restrictive-relative source always occupy a postnominal position in Romance, prenominal qualifying adjectives can only be direct modifiers, according to Cinque. Example (34) would correspond to representation (35) after movements exemplified in (32b) have applied:

(34) La habilidosa producción cestera china

(35) $[_{DP}$ ...$[_{FP(j)}$ habilidosa F^0 [[producción cestera china] …]]]

As Cinque notes, the fact that complements of the noun (either arguments or optional participants) are stranded following the string of adjectives constitute a problem for this account. Examples (36a), corresponding to a result-interpreted nominal, and (36b), corresponding to an event/process nominal, show the order in which adjectives and PP complements should appear:

(36) a. La producción manual china de cestos (estaba almacenada en contenedores)
the production manual Chinese of baskets (was stored in containers)
'The Chinese manual production of baskets was stored in containers.'

b. La producción manual de cestos por parte de la China (se prolongó durante años)
the production manual of baskets by China (extended during years)
'The manual production of baskets by China (extended many years).'

Note that the relational adjective *manual* in the event/process nominal (36b) is not an argument of *producción*, it incorporates an instrumental modifier (see note 5).

Under an analysis where arguments/participants merge with their prepositions below the NP level (see Section 3.1), phrasal NP movement would result in ungrammaticality because adjectives would then have to ungrammatically follow the PP complements (cf. **La producción de cestos manual china* 'the production of baskets manual Chinese' or **La producción de cestos por parte de la China manual* 'the production of baskets by China manual'). The problem disappears if prepositions merge in the highest functional projections below DP (or outside DP), attracting their apparent complements and forcing the entire remnant to raise (Kayne 2000, 2002, 2004).

Summarizing, phrasal movement operations appear to be the best available account for the facts involving adjective modification. Unanswered questions are: why mirror image configurations on adnominal adjectives should occur, and why adjectives follow precisely the rigid sequential order they do. Such rigidity may be a indirect by-product of the fact that they lack *Phi* features and case, these being only in concord with the inflectional features of the noun they modify.

5 Conclusion

The hypotheses discussed in this chapter concerning several aspects of nominal syntax, and the operations currently assumed to take place under distinct theoretical approaches, leave many gaps to be filled and some asymmetries to be noted. For example, operations involving the lower functional level (broadly speaking: those functional projections related to derivational morphology, up to categorization) have been proposed to involve head raising by different authors. However, those operations argued to take place above the categorial (NP) level, roughly, those taking place in the domains of higher functional projections (inflectional morphology and the DP field) have been argued to obtain through phrasal movement or through other local syntactic relations, not necessarily involving movement. Research that takes into consideration all elements and factors concerning nominal constituency may shed some light on whether or not this operational asymmetry is possible, or theoretically desirable, and also if it leads us to a better understanding of the syntax of NPs.

NOTES

1 Relational adjectives are derived from common or proper nouns and do not denote properties, but entities that are interpreted in relation to the modified noun in various ways (see Section 4 below).

2 A third type considered by Grimshaw (1990) are nouns denoting entities related to time or time spans. *Acontecimiento* 'event' and *viaje* 'trip' are two of them. They are considered simple event nominals that, as opposed to complex event nominals, do not have to be associated with arguments.

3 It is not clear why the presence of this argument does not block possessive pronominalization of the internal argument (cf. *Su demolición por los albañiles* 'Its demolition by the masons'), see examples (12a–c). It triggers this effect in result nominals, where all PP participants are in the genitive (cf. **Su demolición de los albañiles* 'Its demolition by the masons'). This issue arises in particular if we assume Kayne's (2000, 2002, 2004) hypothesis for PPs (see Section 4).

4 As is known, a limited list of prenominal adjectives differ in interpretation from their postnominal counterparts in Spanish (i.e., *un viejo amigo* 'a long time friend' / *un amigo viejo* 'an old aged friend'; *un pobre tipo* 'a miserable fellow' / *un tipo pobre* 'a (monetarily) poor fellow'; among some other cases). A very reduced number of them can appear only in prenominal position (i.e., *la mera mención de su nombre* 'the mere mention of his name' / **la mención mera de su nombre* 'the mention mere of his name'). See Demonte (1999) for an extensive discussion on adjective types, their sequential order in the nominal structure and their interpretation in Spanish.

5 Those that appear to absorb a thematic role (as *cestera china* in (31)) can only modify result nominals, but not event/process nominals (see Bosque and Picallo 1996). This distribution among nominalization types possibly follows from the fact that only event-process nominals license internal arguments. Relational adjectives merge, however, in specifier positions of functional projections dominating NP.

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15 Indefiniteness and Specificity

MANUEL LEONETTI

1 Introduction

The syntax and the semantics of indefinite noun phrases (henceforth, INPs) have been studied extensively over the last few decades. As a consequence, the grammatical properties of indefinite determiners in Spanish are now much better known than just 50 years ago: a quick look at the contents of the recently published *Nueva gramática de la lengua española* (RAE 2009) shows how much progress has been made with respect to previous grammatical descriptions. With the aim of presenting the most significant advances in our understanding of the grammar of INPs, this chapter deals with three different topics.

Section 2 is devoted to bare nouns (i.e., those nominal expressions lacking a determiner, such as *actriz de teatro* 'theater actress' in *Ella era actriz de teatro* 'She was a theater actress'). The reasons why bare nouns (henceforth, BNs) will be discussed in this chapter are the following: on the one hand, they systematically receive indefinite-like readings in Spanish, as also occurs in the other Romance languages, and, on the other hand, they have been studied with the same basic theoretical tools that are relevant for INPs (scope, semantic types and type-shifting, information structure), and have been traditionally assimilated into the class of indefinite expressions (see Longobardi 2001 for an up-to-date version of the idea). In a few words, the grammar of Spanish BNs is intimately connected to the grammar of INPs, though the two classes of expressions show different properties.

The remaining two topics are as follows. Section 3 includes a brief overview of research on the indefinite article and other indefinite determiners. The discussion is limited to a couple of significant points, given that most of the relevant issues for indefinites are addressed in Chapter 16 of this book. Finally, Section 4 deals with specificity. After a review of different aspects of the *specific/non-specific* distinction, some syntactic phenomena are discussed where specificity has revealed itself to be a key notion for grammatical description.

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2 Nouns without determination

The problem of the grammatical behavior of nouns without determination had already been addressed in certain classical works on Spanish grammar (cf. Alonso 1967; Alarcos 1978; Lapesa 1975), but it was only in the 1990s, with the interesting exception of Sánchez de Zavala (1976), that the topic was revisited under the influence of modern approaches to syntax and semantics (see Bosque 1996; Laca 1990, 1996, 1999; and also Dobrovie-Sorin and Laca 2003 for an excellent survey of bare nouns in Romance languages). The main goal of this section is to briefly discuss the most productive contributions of recent research and to summarize what we have learnt from it.

2.1 The semantics of bare nouns

It is usually assumed that BNs, at least in languages with articles and determiners like Romance languages, are property-denoting expressions (see McNally 1995; Laca 1996, 1999; Espinal 2010; Espinal and McNally 2011). Thus, in (1a) *actriz de teatro* denotes the property of being a theater actress, and in (1b) *azúcar* denotes the property of being sugar.

(1) a. Ella era actriz de teatro.
  'She was a theater actress.'
 b. Hace falta azúcar.
  'We need sugar.'

This means that BNs are nonreferring expressions (i.e., they cannot be used to refer to singular or plural individuals). Their semantic type is the same that corresponds to predicative expressions, $<e, t>$. It is the presence of determiners that turns them into referring or quantified expressions. A number of facts relate to this basic feature (see Laca 1996, 1999 for details), as described below.

First, BNs – at least mass singulars and bare plurals – are always grammatical as nominal predicates; as shown in (2a–c), singular count nouns can be predicates when they denote roles, jobs or professional activities, but not otherwise (cf. (2d), from Bosque 1996: 58); see Zamparelli (2008) for an analysis of bare predicate nominals in Romance languages.

(2) a. Era *escritor*.
  'He was a writer.'
 b. La nombraron *reina del carnaval*.
  'They appointed her queen of the carnival.'
 c. Aspira a *director de orquesta*.
  'He aspires to become a conductor.'
 d. La mariposa es *(un) *lepidóptero*.
  'The butterfly is a lepidopter.'

Second, being nonreferential and nonquantificational expressions, BNs are "non-delimited" (i.e., unable to denote closed sets of entities). They cannot provide the delimitation or the culmination that telic contexts require, which makes the examples in (3), from RAE (2009: 1148–1149), ungrammatical (though the situation is not the same with bare singular count nouns, for instance in *Encontraron aparcamiento en cinco minutos* 'They found a parking place in five minutes'):

(3) a. *Leyó informes en dos horas. (Cf. Leyó los informes en dos horas)
'S(he) read reports in two hours.'
b. *Una vez pintados cuadros, . . . (Cf. Una vez pintados los cuadros, . . .)
'Once pictures were painted, . . .'

Third, BNs in Spanish are systematically assigned a narrow scope reading, while INPs may take variable scope (wide or narrow) with respect to operators (as already noted in Carlson 1977 for English). In (4), from Laca (1996: 253), the noun *profesores* can be understood only as being under the scope of negation, but not as denoting a number of professors that did not attend the meeting (McNally 1995; Laca 1996, 1999; Bosque 1996).

(4) A la reunión no asistieron profesores.
'The meeting was not attended by any professor.'

This is a classical argument for distinguishing BNs from INPs, as it shows that BNs do not behave like quantified nominals.

Finally, as in other Romance languages – with the interesting exception of Portuguese – BNs in Spanish can only receive existential (weak) readings: they are interpreted as denoting a nonspecified amount of individuals or stuff. Generic readings where a whole class or kind is denoted are, on the contrary, excluded, as shown in (5) (but see Longobardi 2001 for a different point of view).

(5) a. En los océanos se están extinguiendo *(los) tiburones.
'Sharks are becoming extinct in the oceans.'
b. Detestaba *(las) aglomeraciones.
'S(he) hated overcrowding.'
c. *Trajes de neopreno siempre tienen cremalleras.
(cf. Un traje de neopreno siempre tiene cremalleras)
'Wetsuits always have zippers.'

As Dobrovie-Sorin and Laca (2003: 263) rightly point out, the same principle that excludes generic readings for BNs in Spanish must be responsible for the ban against "quasi-universal" or definite interpretations of BNs. In (6) bare plurals cannot be interpreted as denoting the students of a certain university or the farmers in a certain area; however, such definite readings are available for bare plurals in the English versions.

(6) a. Se acerca el fin del semestre. *Estudiantes están agotados.
'We are reaching the end of the term. Students are exhausted.'
b. No había llovido en tres meses. *Agricultores estaban inquietos.
'It hadn't rained for three months. Farmers were worried.'

The contrast with Germanic languages, which allow both existential and generic/definite readings in BNs, has been widely discussed in recent literature (from the pioneering work of Carlson 1977 to Longobardi 2001 and Dobrovie-Sorin and Laca 2003). Carlson's original proposal for English assumes that BNs are to be analyzed as names of kinds, but the idea cannot be extended to Romance languages in view of the constraint against generic readings. This favors an alternative unified account of Romance BNs in terms of property denotation, if we accept that existential readings are the natural output of the inferential operation of existential closure on property-denoting nominals. An adequate explanation is still needed for the Romance–Germanic contrast, but such an issue falls beyond the limits of this chapter.

2.2 Semantic incorporation

If BNs denote properties in Spanish, their occurrence in argument positions calls for a mode of semantic composition that should be different from the canonical operation of saturation, as property-denoting expressions would not be able to saturate argument slots. McNally (1995) and van Geenhoven (1998) claim that a special mechanism of *semantic incorporation* is needed. The basic intuition here is that, at some level of semantic representation, BNs are fused with the predicate with which they combine to give rise to some sort of complex predicate: they are absorbed or semantically incorporated by the predicate as the restriction of its internal argument, and their existential properties are contributed by the governing predicate. This explains why semantically incorporated nominals do not have quantificational force of their own and show narrow scope with respect to operators and quantifiers. Another way to state this is to claim that the bare nominal counts as a sort of predicate modifier or qualifier. There are different proposals on how to define semantic incorporation (or pseudo-incorporation) (cf. Farkas and de Swart 2003). In any case, it is certainly useful to assume that some notion of semantic incorporation can play a prominent role in any account of BNs in Spanish.

Semantic incorporation explains how BNs combine with basic predicates and why they are neither referential nor quantificational expressions. In addition, it can be related to a widely acknowledged property of BNs, in particular bare count singulars: they can be used as complements of verbs and prepositions if they yield "appropriately classificatory" predicates, i.e., predicates denoting culturally stable, frequent, stereotypical types of events (see Bosque 1996: 43–45 and de Swart and Zwarts 2009 for the widespread correlation between BNs and stereotypical meanings). In Espinal's and McNally's (2011) terms, the resulting predicate has to denote a "characterizing property" of the subject, not necessarily prototypical or institutionalized; a property is characterizing if it is sufficiently relevant in a context to distinguish individuals that have the property from individuals that do not. Thus, the examples in (7), from Espinal and McNally (2011), obey this condition, while the examples in (8) do not contain typical characterizing properties and are odd (but notice that acceptability judgments are strongly context-dependent: the examples would be fine in a context where carrying a pencil or having some tortilla are considered as characterizing or classificatory properties):

(7) a. Lleva sombrero.
'S(he) wears a hat.'
b. Tiene apartamento.
'S(he) has an apartment.'

(8) a. ?Lleva lápiz.
'(S)he carries a pencil.'
b. ?Tiene tortilla.
'(S)he has a tortilla.'

Stereotypical interpretations, frequently giving rise to idiomatic readings, are salient also in BNs following prepositions, as in (9), and in nominal predicates, as in (10).

(9) *a casa* 'to home'
en clase 'in class'
por teléfono 'by phone'
de permiso 'on leave of absence'
con cuchara 'with a spoon'
en barrica de roble 'in oak barrels'
en prisión 'in jail' (cf. *en la prisión* 'inside the jail,' not necessarily as a prisoner)

(10) *ser modelo* 'to be a (professional) model'
ser payaso 'to be a (professional) clown' (cf. *ser un payaso*, either identificational or meaning 'to behave like a clown', from Bosque 1996: 67)

This kind of special reading arises in environments where bare singulars are integrated in some predicative expression – a preposition in (9), a verb in (10) – and occur in competition with ordinary nominals. It is reasonable to think of them as interpretive results of semantic incorporation. Cross-linguistic variation in this area is not easy to account for, since we are frequently dealing with idiom formation phenomena. Furthermore, semantic incorporation can shed light on a number of syntactic constraints on BNs, as will be shown below.

2.3 *The distribution of bare nouns*

2.3.1 Constraints on positions BNs typically exhibit a more limited distribution than full NPs. The main constraint operating on BNs in Spanish concerns preverbal subjects, and was clearly formulated in Suñer (1982: 209) as the *Naked Noun Constraint*. It states that an unmodified BN cannot be a subject in preverbal position. The constraint affects both singular common nouns and plurals, as shown in the examples in (11):

(11) a. *Gato estaba durmiendo
'Cat was sleeping.'
b. ??Gatos maullaban en el jardín.
'Cats were meowing in the garden.'

Bare plurals and mass nouns may appear as postverbal subjects under certain conditions, as in (12):

(12) a. En el jardín maullaban gatos.
'In the garden there were meowing cats.'
b. A veces sobraba comida.
'Sometimes food was left over.'

With nonstative predicates, the constraint can be circumvented: (a) when bare plurals are modified, as in (13); (b) when they are contrastively focused or topicalized, as in (14); and (c) when they are coordinated, as in (15):

(13) a. *Gatos hambrientos* maullaban en el jardín.
'Hungry cats were meowing in the garden.'
b. En ocasiones *preguntas curiosas* interrumpen la explicación.
'Sometimes curious questions interrupt the explanation.'

(14) a. GATOS maullaban en el jardín.
'It was cats that were meowing in the garden.'
b. *Gatos* sí entran, en el jardín (pero *perros* normalmente no).
'Cats enter the garden (but dogs usually do not).'

(15) Gatos y perros vagaban por el jardín.
'Cats and dogs were roaming in the garden.'

The data raise at least two basic questions. The first is why unmodified nouns cannot appear in preverbal subject position. The second is how to account in a unified way for other, similar restrictions concerning indirect objects, at least in European Spanish (**Ella prestó su coche a amigos* 'She lent her car to friends'), certain stative verbs that exclude BNs as arguments (**Me gustan naranjas* 'I like oranges'; cf. *Me gustan las naranjas*), and perfective contexts (**Nos comimos paella* 'We ate paella,' with the clitic *nos*, an 'ethical' dative, acting as a perfective marker; cf. *Comimos paella*), among others. This second question is dealt with in Section 2.3.2.

The first question addresses the motivation for the *Naked Noun Constraint*. Formal syntactic accounts usually attempted to derive the constraint from the general requirement of proper government for empty categories, assuming that BNs are preceded by an empty determiner that cannot be licensed in preverbal subject position. These accounts, however, can hardly be extended to cover other distributional restrictions and are not related to the semantic properties of BNs either. Alternative semantic proposals based on the interaction of BNs with sentential contexts look more promising. One such proposal, originally presented in Suñer (1982) and developed in Laca (1996, 1999) and Dobrovie-Sorin and Laca (2003), relies on the topic nature of preverbal subject position in so-called Null Subject Languages (for instance, Spanish and Italian, where postverbal subjects are always in focus). Assuming that BNs are property-denoting expressions, the classical *Naked Noun Constraint* results from the incompatibility of topic positions – in particular, sentence-internal topics – with nonreferential nominals. Topics usually require nominals with independently established reference, but BNs lack it as they receive their existential import from the predicate. Contrastive and Informative Foci, on the contrary, are perfectly compatible with property-denoting nominals (see Cohen and Erteschik-Shir 2002 for an approach to English bare plurals based on information structure): this explains why the examples in (11) are anomalous, while (12) and (14) are acceptable. However, yet to be explained are the questions of (a) why contrastive topics may admit bare plurals, as in (14b), and (b) why bare singulars obey much stronger constraints than bare plurals: this is a systematic asymmetry, probably due to the fact that the two classes should actually be assigned different semantic types, or even different syntactic structures, due to the crucial role of plural morphology (Farkas and de Swart 2003; Dobrovie-Sorin et al. 2005; Espinal 2010). In any case, the incompatibility of BNs with preverbal subject position follows from their incorporated status: it is well known that incorporation usually involves objects and certain internal subjects (with unaccusative verbs), but not external arguments.

Any valuable account of the constraints on preverbal subjects and indirect objects should be able to explain why the insertion of modifiers – typically restrictive modifiers – can rescue strings that are otherwise unacceptable, as shown by the following contrasts:

(16) a. *Gente no merece consideración.
'People don't merit consideration.'
b. Gente así no merece consideración.
'People like these don't merit consideration.' (RAE 2009: 1151)

(17) a. *Regaló sus muebles a amigos.
'S(he) gave her/his furniture to friends.'
b. Regaló sus muebles a amigos que los cuidaran.
'S(he) gave her/his furniture to friends that could take care of it.'

To explain why modifiers improve the acceptability of BNs, we should get a better understanding of how they contribute to determine information structure and context selection. As for the role of coordination, the grammatical status of (15) is surely due to the fact that coordination counts as a sort of quantificational mechanism.

In sum, the distributional constraints on BNs can be overridden when certain factors appear, such as plural morphology and coordination (acting as forms of quantification), restrictive modifiers, contrastive focus, and more generally the availability of a thetic interpretation for the sentence (in order to keep BNs under the projection of informative focus). At the moment, we lack a satisfactory unified account of all these factors.

2.3.2 Constraints on predicate types The second question raised above concerns certain lexical classes of predicates. Stative predicates, in particular so-called *individual-level predicates* (see Chapter 22) seem to reject BNs as subjects and objects (Laca 1990; Bosque 1996), as shown in (18):

(18) a. Me gustan *(las) naranjas.
 'I like oranges.'
 b. Adora *(la) música clásica.
 '(S)he adores classical music.'

An analysis based on semantic incorporation fits nicely with this kind of data: it is well known that individual-level predicates require independently introduced – i.e., topical–subjects (or objects, as in (18b)), thus rejecting incorporated nominals. The problem is how to obtain a precise characterization of the predicates that reject semantic incorporation. It is not entirely clear that the notion of *individual-level predicate* is the right one (see Laca and Dobrovie-Sorin 2003 and Cohen and Erteschik-Shir 2002 for some criticism). Much work is still needed on the relationship between lexical semantics and types of semantic incorporation, although research along the lines of Espinal (2010) and Espinal and McNally (2011) seems to be particularly promising.

3 Indefiniteness

3.1 The indefinite article

Most grammatical properties of indefinite determiners in Spanish are shared with the rest of the Romance languages and, generally, with all languages whose determiner system encodes the semantic distinction between definiteness and indefiniteness (in particular, those that possess an indefinite article). The core issues in a description of Spanish are the role of the indefinite article *un* and its plural form *unos*, the existence of epistemic and modal indefinites like *algún* 'some,' and the gradual conversion of a number of adjectives into indefinite determiners (*cierto* 'a certain,' *otro* 'another,' *bastante* 'enough'). Here I will briefly review some recent contributions to the field (see Chapter 16 for general considerations on indefinite quantifiers).

The indefinite article shows the widest distribution of all indefinites: it can receive specific, non-specific, cardinal and generic readings (Leonetti 1999; RAE 2009: §15), and its meaning is limited to the absence of definiteness (i.e., the absence of a uniqueness/familiarity requirement). A language can be said to have fully developed an indefinite article when an indefinite determiner historically derived from the numeral 'one' is used to express generic readings, as in (19), and occurs in predicative nominals, as in (20):

(19) *(Una) persona educada sabe disculparse.
'An educated person knows how to apologize.'

(20) Aquello fue *(un) milagro.
'That was a miracle.'

Additional evidence that Spanish *un* is no longer exclusively a numeral can be found in contexts where a numerical or purely cardinal interpretation is excluded – unless *un* is focused (examples from RAE 2009: §15.3n):

(21) a. Has tenido una buena idea.
'You've got a good idea.'
b. Tengo un terrible resfriado.
'I have a terrible cold.'

Some Romance languages have developed a partitive article from the preposition *de* 'of' (*del/della/dei/degli/delle* in Italian, *du/de la/des* in French), with an indefinite existential reading. Although there is no partitive article in Spanish, there are instances of a 'bare partitive' construction headed by *de*: as pointed out in Treviño (2010), *de* can give rise to an indefinite nominal when followed by *todo* 'all' or by a definite, as shown in (22):

(22)	a.	Había	de todo	en esa tienda.
		There was	of everything	in that store
		'There was everything in that store.'		
	b.	Te traje	del chocolate	que te gusta.
		I brought you	of the chocolate	you like
		'I brought you (some of) that chocolate you like.'		
	c.	Con las lluvias, salieron	de esos animalitos.	
		With the rains, got out	of those little animals	
		'With the rains, (some of) those little animals got out.'		

'Bare partitives' appear only in focus positions, as objects or subjects of unaccusative verbs, and obey syntactic constraints akin to those operating on BNs.

They receive only existential or weak readings, and are always assigned narrow scope.

A striking feature of the Spanish article system is the existence of a plural form of the indefinite article, *unos*. Recently the contrast between *unos* and *algunos* 'some' has been intensively studied (Gutiérrez-Rexach 2001, 2003; López-Palma 2007; Martí 2008, 2009). *Unos* exhibits a number of interesting properties: it cannot occur as a determiner in the subject of an individual-level predicate (cf. 23a)); it participates only in group predication, being excluded with distributive predicates (cf. 23b)); and it is typically non context-dependent and non discourse-linked (cf. 23c), where *unos* is not adequate to express that the sleeping persons are a subset of the set of passengers).

(23) a. En este ayuntamiento, {#unos/algunos} concejales son honestos.
'In this town council, some councilors are honest.'
b. {#Unos/Algunos}estudiantes se pusieron los pantalones.
'Some students put their trousers on.'
c. Murieron doce pasajeros. {#Unas personas/Algunas personas} estaban durmiendo en el momento del accidente.
'Twelve passengers died. Some people were sleeping when the accident took place.'

The contrast is actually less stable than one could at first sight assume, and there are contextual factors (contrast, world knowledge, insertion of modifiers) that can blur the difference with respect to *algunos*. However, the data offer a valuable testing ground for hypotheses on how contextual restrictions operate on indefinites. One of the basic intuitions about *unos* is that it seems to reject discourse connections with previously established information: this is why it is usually odd in partitive constructions (#*unos de ellos* vs. *algunos de ellos* 'some of them'), even with covert partitive readings. An adequate analysis should succeed in connecting such constraints to the fact that *unos* tends to express collective or group interpretations, blocking the access to the atoms that constitute the group.

3.2 Some Spanish indefinite determiners

Many languages exhibit so-called 'epistemic indefinites.' Epistemic indefinites are determiners that trigger modal inferences about the speaker's epistemic state (Tovena and Jayez 2006): some of them signal that the speaker is unable or unwilling to identify a referent (this is the case of Spanish *algún* 'some') (see Alonso-Ovalle and Menéndez-Benito 2010), some of them indicate that the identification is irrelevant or indifferent (as in 'free choice' indefinites like *cualquier* 'any'), and others signal that the speaker is able to identify the referent (for instance, *cierto* 'a certain'). Recent work on these items usually deals with their semantic contribution in terms of constraints they impose on their domain of quantification. According to Alonso-Ovalle and Menéndez-Benito (2010), *algún*, for instance, requires that at least two individuals in its domain be possibilities; as a consequence, it does not tolerate singleton sets. The contrast between *un* and *algún* can be derived from this constraint:

(24) a. Juan compró un libro que resultó ser el más caro de la librería.
'John bought a book that happened to be the most expensive in the bookshop.'
b. #Juan compró algún libro que resultó ser el más caro de la librería.
'John bought some book that happened to be the most expensive in the bookshop.'

With *algún*, the speaker indicates that any individual in the domain of quantification may be the one that satisfies the existential claim, which is incompatible with the uniqueness imposed by the relative clause in (24a). On the other hand, the indefinite article *un* is fine in (24b) because it does not impose specific conditions on its domain and can combine with a singleton set. The unmarked nature of *un* explains the variety of readings it can be assigned.

So-called 'specific indefinites' trigger different constraints. *Cierto* 'a certain' conveys the assumption that someone is able to identify the referent of the INP but, being indefinite, it communicates also that the hearer may not possess the information needed for such a task. As claimed in Eguren and Sánchez (2007), *cierto* is an "imprecise" identifier: it can be adequate both when the speaker is unable to fully identify a specific referent, and when (s)he is able to do it, but unwilling to be more informative, and even when the referent is known to speaker and hearer, but the speaker wants to avoid a more precise mention. A sentence like (25) could be used in all three cases. Thus, *cierto* cannot be said to encode epistemic specificity (see Section 4.1), but just some kind of identifiability.

(25) Supo que cierto colega le había mentido.
'(S)he knew that a certain colleague had lied.'

Cierto is originally an adjective, and nowadays it behaves as an indefinite determiner in some cases and still as an adjective in others (cf. *una cosa cierta* 'a true thing'). The same gradual conversion from adjective to determiner has been observed in words like *otro* 'another,' *varios* 'several,' *bastante* 'enough,' *demasiado* 'too much,' *numerosos* 'manifold,' *determinados* 'determinate,' *diferentes* 'different' (the case of *otro* has been carefully studied in Eguren and Sánchez 2003). The process has not reached the final stage in every case: some of these elements can be used as pronominals and can head partitive constructions, thus behaving like proper indefinite determiners (for instance, *otro* and *varios*), while some others cannot (*cierto* and *numerosos*; cf. **Conozco cierto* 'I know a certain', **cierta de las actrices* 'a certain of the actresses'). The study of this gradual incorporation of quantitative adjectives into the class of determiners can shed new light on what it means to be an indefinite. It is important to recall that several Spanish indefinites – in particular, cardinal numerals and 'vague' indefinites like *muchos* 'many,' *pocos* 'few,' and *demasiados* 'too many' – share a number of grammatical properties with adjectives: they make natural predicates in copulative sentences, can occur in typically adjectival positions inside NPs (as in *mis tres hermanos* 'my three brothers'), and some of them accept adjectival suffixes (*pocos* 'few' > *poquísimos* 'very few').

4 Specificity

4.1 The specific/non-specific distinction

The notion of 'specificity' was first introduced in the 1960s to account for certain cases of logical ambiguities involving INPs and intensional verbs or operators. A sentence like *John wants to marry a Norwegian girl* is usually said to show two different readings: in one of them, the speaker refers to a particular Norwegian girl whose existence is assumed and whose identity can possibly be determined; in the other one, the speaker is not making reference to a particular individual, but just to any potential referent that could satisfy the condition of being a Norwegian girl. The first reading is the specific interpretation, and the second one is a good representative of the non-specific interpretation. A classical way to deal with this ambiguity in formal semantics is to treat it as a scope ambiguity, due to the interaction of the indefinite article with the intensional verb *want*: when the indefinite, treated as a quantifier, has wide scope with respect to the intensional element, the specific reading obtains, and when the relative scope is inverted the reading is non-specific (cf. Quine 1956).

Rivero (1975) introduced the term "specificity" in the domain of Spanish linguistics, and applied the notion to both definite and indefinite NPs, thus showing that it is independent of definiteness. In the minimal pairs in (26a) and (26b), it is the mood on the verb inside the relative clause that marks the two relevant interpretations, as in other Romance languages: the indicative signals that the nominal is specific, and the subjunctive indicates a non-specific interpretation. The mood contrast had already been mentioned in Quine (1956) to distinguish *relational* (specific) and *notional* (non-specific) readings.

(26) a. Me interesa ver una guía que {tiene/tenga} mapas.
'I am interested in seeing a guide that contains maps.'
b. Me interesa ver la guía que {tiene/tenga} mapas.
'I am interested in seeing the guide that contains maps.'

Rivero noticed that the *specific/non-specific* distinction is relevant not only in modal contexts like the one in (26) but in extensional contexts too. In (27), a classical example where the INP *un médico* 'a doctor' cannot interact with the scope of any sentential operator, there is still an ambiguity.

(27) Me lo ha dicho un médico, y yo me lo creo.
'A doctor told me, and I believe it.'

In fact, in (27) the speaker may want to refer to a particular individual (on the specific construal), or just to indicate what kind of person talked to him/her (on the non-specific reading, much more natural in this context). The ambiguity seems to depend on whether it is the descriptive content of the INP that is salient, or rather it is the speaker's intention to refer to an individual that has a crucial role in the interpretation. At least two problems are raised by examples like (27), as discussed below.

First, in intensional contexts the main difference between the two readings involves existential import, as existential generalization is only guaranteed in the specific interpretation – in (26a), it is licit to infer that there is a guide that the speaker is interested in seeing, but not in (26b); in extensional contexts, on the other hand, the contrast cannot involve existence anymore because existential generalization applies in both readings, and a new characterization of the *specific/non-specific* distinction is called for, introducing the possibility that the distinction is pragmatic rather than semantic.

Second, relative scope cannot be invoked as the logical mechanism underlying the ambiguity in (27). This leaves us with two options: either we keep the original view of specificity as a scope phenomenon – and in such a case (27) must represent something different from specificity – or we include (27) under a broader notion of specificity that has to be redefined. I assume that the second option is the most interesting one. The problem is that there has been no uniform definition of specificity in recent research (see Leonetti 1999; von Heusinger 2001, 2002, 2007; Farkas 2002; Gutiérrez-Rexach 2003 for an overview of the problem).

Farkas (2002) distinguishes three different uses of the term *specificity*: epistemic, scopal, and partitive. Specificity defined as the use of a nominal by the speaker with "a referent in mind" represents the so-called "epistemic" view. According to this, a specific use of an INP is based on the speaker's knowledge or certainty about the referent. This may be true of many specific uses of indefinites, especially in extensional contexts, but it is not a necessary condition for a specific interpretation. Moreover, it is the speaker's intention to refer to a particular individual that should be decisive for specificity, instead of his/her mental state. An epistemic approach, then, is not valid as a general characterization of specificity (but see Gutiérrez-Rexach 2003: §3 for an attempt to offer a formal treatment). Scopal specificity makes reference to a hallmark of specific indefinites: their tendency to take scope over anything else in the sentence. This semantic property is actually independent from epistemic specificity, and again does not cover all cases of alleged specific use. Finally, partitive specificity (as defended in Enç 1991) appears when the referent of an INP is chosen from a contextually given set. It is not equivalent to the other two kinds and cannot work as a general characterization either, since partitive indefinites can be non-specific and have narrow scope; it is, however, quite close to epistemic specificity and definiteness, given that the three notions are finally related to familiarity.

Now the question is whether a unified notion of specificity is available that can capture the family resemblances that connect the three kinds of specific interpretation. A promising approach toward a unified definition can be found in von

Heusinger (2001, 2002, 2007). According to him, specificity indicates that an expression is referentially anchored to another argument expression in the discourse, its referent being functionally dependent on the referent of the other expression.

In any case, the varieties of specific and non-specific readings are contextual side-effects of more basic semantic properties of INPs and cannot support the hypothesis that indefinite determiners are semantically ambiguous. It seems more adequate to maintain a monosemic approach to the semantics of indefinites, together with a view of specificity as a phenomenon that belongs to the semantics–pragmatics interface. As will be clear below, the role of syntax is limited to constraining the inferential specification of the optimal reading for INPs.

4.2 *Specificity-related phenomena*

Ways of marking specificity in the grammar are usually classified in two groups: NP-internal, and NP-external or contextual (Leonetti 1999; Gutiérrez-Rexach 2003; RAE 2009). Among NP-internal devices, lexical properties of different kinds of determiners play a major role, together with adjective position, mood in relative clauses and Differential Object Marking (DOM). Adjective position and DOM will be treated in Section 4.2.1 and Section 4.2.2, respectively. Lexical properties of determiners have already been mentioned in Section 3.1 and Section 3.2. Though it is essentially correct to claim that *cierto* heads specific INPs, and that negative and free choice indefinites head non-specific INPs, it seems that (non)specificity appears as a side-effect of other basic features in the semantics of determiners (for instance, evidential or epistemic features). We will see that specificity has the status of a derived pragmatic effect in all the contexts mentioned here. As for mood in relative clauses, contrary to Rivero (1975), it is not a systematic way of differentiating specific and non-specific interpretations because relative clauses in indicative mood may occur inside non-specific INPs too (for instance, when assigned a generic reading), and subjunctive is associated with non-specific interpretations only in intensional contexts.

NP-external marks of specificity in Spanish are limited to clitic doubling in direct objects and certain aspects of word order (see Section 4.2.3 and Section 4.2.4, respectively). A general condition that has to be kept in mind in all cases is that specificity effects usually arise when formal marking is optional – and thus meaningful – but are absent when it is obligatorily triggered by some syntactic principle. The following sections discuss grammatical facts where (non)specificity certainly plays a role, but none of them can strictly be considered as devices for encoding specific or non-specific interpretations.

4.2.1 Adjective position Adjective position works as a NP-internal mark of specificity in Spanish. Picallo (1994) and Bosque (2001) (see also RAE 2009: §13.14k–n, §15.9k) have demonstrated that adjectives in prenominal position inside INPs force specific readings (in the scopal and epistemic senses), while the postnominal position is compatible with both specific and non-specific readings. In (28), the string *una interesante novela de Delibes* must be specific; the alternative order *una novela interesante de Delibes* is ambiguous.

(28) Quiero leer {una interesante novela/una novela interesante} de Delibes.
'I want to read an interesting novel by Delibes.'

Elative adjectives, like *interesantísimo* 'very interesting', produce the same effects as prenominal adjectives. Both prenominal position and elative meaning preclude a restrictive interpretation of the adjective and force an explicative or appositive one; explicative modifiers operate on referentially autonomous nominals, and this makes the specific reading the only one compatible with the presence of those modifiers. Adjective position is just a trigger for the inferential specification of an optimal interpretation of the INP, and specificity appears as the result of such inferential processes.

4.2.2 Differential object marking A second type of NP-internal device for specificity marking is Differential Object Marking. It is well known that Spanish uses the preposition *a* for distinguishing marked and unmarked direct objects; other languages resort to case morphology or object agreement. The insertion of *a* before the direct object is triggered by two basic features: animacy and specificity, the first one being the main factor (see Brugè and Brugger 1996; Torrego 1998; von Heusinger and Kaiser 2003; Leonetti 1999, 2004; Bleam 2006; López 2009; RAE 2009 for data). In (29), the object NP is animate, and the *specific/non-specific* distinction correlates with the presence or absence of *a*. Without the preposition, the speaker cannot be referring to a couple of particular nurses.

(29) Necesitan (a) dos enfermeras.
'They need two nurses.'

However, the correlation is obscured by the fact that (29), with *a*, is still ambiguous between a specific and a non-specific reading. Many marked animate objects can receive non-specific readings (for instance, modified bare nominals and negative indefinites like *nadie* 'nobody'). Thus, *a* is not a true specificity marker because its insertion is mainly determined by animacy. Moreover, in certain contexts, Spanish DOM is associated with semantic effects different from specificity, as in (30), where individuation or discourse prominence is involved:

(30) Estaba dibujando {a una niña/una niña}.
'(S)he was {portraying/drawing} a child.'

In (30), the marked object refers to a child that exists in the real world, while the unmarked object refers to a child that pertains to the world of the drawing (in this case, the absence of *a* is due to the inanimate nature of the referent). What is the semantic contribution of DOM, then? This is certainly not an easy question. Some abstract notion must underlie the interpretive facts in (29) and (30): it could be some notion tied to information structure and topicality, or to the affectedness of the object, or, rather, it could be a formal feature that makes object NPs "visible" to certain interpretive operations, as in López (2009: §5). Whatever the semantic role of DOM is, specificity – in any of its senses – seems to be just one of the possible contextual results of such an abstract notion, and it is not systematically correlated with *a*-marking.

4.2.3 Clitic doubling While adjective position and DOM count as NP-internal marks of specificity, clitic doubling is a NP-external device. Clitic doubling is an important feature in a characterization of the grammar of Spanish clitics (see Chapter 21). While doubling with dative clitics is generalized in any variety of the language and has no semantic effects on the "associate" lexical nominal, doubling with accusative clitics, although strictly limited to pronouns in (standard) European Spanish, extends to other kinds of NPs (mostly definite NPs) in American varieties, and imposes certain semantic constraints on the "associate." It has been noticed since Suñer (1988) that doubling with accusative clitics in American dialects (in particular, the Porteño or Rioplatense variety spoken in Argentina and Uruguay) requires a specific interpretation of the indefinite object. The contrast in (31), from Suñer (1988), shows that clitic doubling is possible with specific INPs, but it is ungrammatical with non-specific INPs such as the one in (31b):

(31) a. Diariamente, la escuchaba a una mujer que cantaba tangos.
'Daily, (s)he listened to a woman who sang tangos.'
b. (*La) buscaban a alguien que los ayudara.
'They were looking for somebody who could help them.'

Doubling with indefinites is actually scarcely attested even in Porteño, but it seems in any case excluded with prototypical non-specific expressions like negative and free choice quantifiers and BNs, which confirms Suñer's claim (see Gutiérrez-Rexach 2000 for a detailed analysis). Similar facts have been noticed in languages like Romanian. The constraint on specificity is absent only in American dialects of Spanish where doubling occurs without gender/number/case agreement between the clitic and the associate – varieties originated in language contact between Spanish and languages like Quechua and Aymara: in such cases, clitics have lost their original features and behave like simple object markers.

Clitic doubling and DOM share important properties: both are sensitive to animacy and definiteness/specificity factors, and both expand along the same diachronic path. This could suggest that the motivation for specificity constraints is again the same in the two phenomena. In Leonetti (2008) the hypothesis is advanced that the motivations are different. As already mentioned in Section 4.2.2, DOM does not encode a requirement for specific objects, but rather imposes some condition of discourse prominence that gives rise to specificity as a side-effect when object marking is not compulsory, and therefore significant. In clitic doubling, specificity is again a side-effect, this time triggered by the necessary feature matching between the clitic and the associate nominal, as proposed in Suñer (1988). Since clitics are definite, the matching involves definiteness: the clitic imposes upon its associate the requirement of having a uniquely identifiable referent, and moreover, a familiar and discourse-salient one – the same kind of reference that pronominal clitics are assigned. This is why doubling occurs mostly with definite expressions. When indefinites are allowed to enter the doubling construction (as in Porteño), the only way they can obey the matching condition is by means of a specific interpretation, possibly built on a link to previous contextual information. Specificity in clitic doubling, thus, seems to be related to discourse-linking and familiarity.

4.2.4 Fronting constructions Finally, fronting constructions are another NP-external grammatical device that can be relevant for specificity. One of them, in particular, counts as a marker of non-specific interpretation in Spanish. Quer (2002) first pointed out that there is a productive fronting construction in Spanish that (a) is not to be confused with Clitic Dislocation or Contrastive Focalization (see Chapter 28), and (b) accepts only certain kinds of INPs as fronted elements. He called it *Quantificational Q(uantifier)P(hrase)-Fronting*, and here I will keep using this terminology. The construction is illustrated in (32) and (33). Indefinites like *algo* 'something,' *alguien* 'someone,' *poco* 'few,' *bastante* 'enough,' and *nada* 'nothing' are perfect in QP-Fronting, while definites like *el libro* 'the book' are excluded – unless the construction is used in a context where the propositional content has been already mentioned – and even INPs like *un libro* 'a book' are odd.

(32) a. Algo debe saber.
'(S)he must know something.'
b. A alguien encontrarás.
'You will find someone.'

(33) a. #El libro habrá leído.
'(S)he must have read the book.'
b. #Un libro habrá comprado.
'(S)he must have bought a book.'

The question is why QP-Fronting should represent a 'definiteness constraint' context and why not all indefinites can be preposed. An adequate answer could be based on the fact that QP-Fronting is associated with the absence of an informational partition in the sentence: the fronted constituent is neither a topic nor a narrow focus (cf. Leonetti 2009). The crucial observation is that the set of indefinites that undergo QP-Fronting in a productive way is roughly equivalent to the set of indefinites that cannot be dislocated as topics. Moreover, they are by default non-specific. In fact, specific readings are anomalous in Fronting. An obvious way of connecting the data seems to be by resorting to information structure: it is the ban against fronted topics in the construction that constrains the kind of nominals that can be preposed, allowing only those that do not involve the individuation of particular referents (i.e., those determiners or interpretations that are incompatible with independently established reference). This explains why most indefinites are acceptable when fronted (with non-specific readings), and at the same time, why the examples in (33) are odd, given that definite and indefinite articles are perfectly able to head topical phrases. According to this, QP-Fronting is not totally equivalent to a "definiteness effect" environment: it is a construction that excludes nominals that can make good candidates for topics, and as a consequence favors the insertion of purely cardinal indefinites with non-specific interpretations.

5 Conclusion

Three main topics have been covered in this chapter. The first is the grammar of bare nouns. An overview has been presented based on well-established assumptions about Spanish BNs: in particular, their behavior as property denoting semantically incorporated expressions, the absence of generic readings, and the constraints on their distribution. The second is the variety of Spanish indefinite determiners: the role of the indefinite article *un* and its plural form *unos*, and the existence of epistemic and modal indefinites like *algún* 'some' have been identified as the most salient features of the system. Finally, the third topic is specificity. After a brief presentation of the notion, some grammatical phenomena related to specificity have been discussed: adjective position, Differential Object Marking, clitic doubling, and fronting constructions. In all of them, specificity seems to appear as a side-effect of the interaction between the semantics of indefinites and the sentential context.

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16 Quantification

JAVIER GUTIÉRREZ-REXACH

1 Reference and quantification

Natural language sentences are used by speakers to communicate facts, beliefs, queries, and information in general about entities or individuals, their properties or relations with other individuals, or generalizations about their nature as entities. Languages differ in the devices they use to express reference to individuals, or generalizations about a number of individuals. A sentence such as *Pedro is tall* consists of a subject noun phrase (*Pedro*) and a predicate (*tall*). The predicate attributes the property it describes ('being tall') to the individual the subject refers to ('Pedro'). Proper names are almost always directly referential, in that they are used to point directly to an individual. When a noun phrase is headed by a determiner (*a*, *every*, *many*, *most*, etc.), this element plays a crucial role in the predication process and may convey a generalization about individuals that goes beyond direct reference. Such noun phrases are sometimes called determiner phrases or quantifier phrases in the literature, highlighting the central functional role of the determiner (cf. Bosque and Gutiérrez-Rexach (2009: ch. 10) and Picallo (Chapter 14, above) for details on the syntactic structure of the noun phrase). For reasons of space we will mostly focus on the semantics of Spanish quantifiers in this chapter. Sánchez López (1999), Leonetti (1999, 2007), and Bosque and Gutiérrez-Rexach (2009: ch. 8) present a broader survey of several grammatical issues.

The contrast in meaning between the sentences *A student is tall* and *Every student is tall* is clearly related to the different nature of the determiner that expresses the number or quantity of individuals to which the property of being tall applies. We say that these sentences differ in quantificational force. The former sentence states that there is (at least) one student who is tall in the situation under consideration. The latter sentence states that every individual under consideration is tall. The noun phrases *a student* and *every student* are quantificational because they express generalizations about quantities of individuals.

Modern linguistics, following the insights of logicians such as Gottlob Frege and Bertrand Russell, normally assumes that the semantic contrasts between referential

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and quantificational elements, or among quantifier classes, are also related to differences in linguistic and logical representation (Gutiérrez-Rexach 2011). In other words, different noun phrases are associated with differences in content or denotation, and such differences can be formally represented via a logical statement or logical form (cf. López Palma (1999), Escandell (2004), and Szabolcsi (2010) for clear introductions to several semantic notions used in this chapter). Proper names and other referential elements are represented or translated as constants in first-order logic, to express the fixed nature of their meaning, whereas determiners are represented as quantifiers. For example, *a/some* is treated as an existential quantifier, with the following associated meaning: "There is at least one individual for which it is true that …" The determiner *every* is a universal quantifier, with the associated meaning: "It is true for all individuals that …" Quantifiers are assumed to bind a variable in the logical statement or proposition expressing the meaning of the sentence where they occur. A sentence such as *A dog was barking* would have a first-order logical form that can be paraphrased as "There is at least one individual x such that x is a dog and x is barking."

2 Constraints on determiner denotations

Representing natural language meaning through an elementary logical language has certain limitations, especially when the goal is to capture the syntactic structure of a sentence or the nuances in meaning of different quantifier classes. First-order logic cannot capture the content of the extensive variety of natural language determiners. For example, it has been shown that the English determiner *most* is not expressible in first-order logic. The sentence *Most students are happy* does not mean that most individuals under consideration are students and they are happy (first-order logical meaning). Rather, it means that most of those individuals who are students are also happy.

After the seminal contribution of the philosopher Richard Montague, generalized-quantifier theory developed during the 1980s and 1990s (Barwise and Cooper 1981; Keenan and Stavi 1986; Gutiérrez-Rexach 1998; Keenan and Westerståhl 1997; Peters and Westerståhl 2006) with the goal of improving the shortcomings of traditional logical analyses of quantification in natural language. Noun phrases in general are taken to express or describe, more technically *denote*, generalized quantifiers. The word *generalized* is intended to mean that any noun phrase expresses a quantifier, including proper names and those phrases headed by simple or complex determiners of several classes. Thus, the notion of a quantifier has to be generalized from the basic existential/universal distinction of first-order logic to a uniform pattern across noun phrases. The functional nature of the subject–predicate relation that was intuitively described in the preceding section is reversed. A predicate still expresses a property, but it is not treated as a function that applies directly to the subject (argument). Rather, the subject noun phrase is viewed as a higher-order functional expression that takes the predicate as argument, or, equivalently, as a set of properties. In general, noun phrases express generalized quantifiers (sets of properties). Consider the following sentences:

(1) a. Tres estudiantes están en clase.
'Three students are in class.'
b. Todos los estudiantes están en clase.
'All the students are in class.'

The generalized quantifiers associated with *tres estudiantes* 'three students' and *todos los estudiantes* 'all the students' are, respectively, interpreted as the set of properties that three students or all the students have. What (1a, b) assert is that being in class is one of those properties. These sentences would be respectively true or false depending on whether being in class is a property that (at least) three students or all the students have or not. As stated above, proper nouns also denote generalized quantifiers. The proper noun *Pedro* in (2) denotes the set of properties that Pedro has, and the sentence identifies being in class as one of them:

(2) Pedro está en clase.
'Pedro is in class.'

The meaning of a noun phrase such as *un estudiante* 'a student' can also be determined in a compositional fashion. The noun *estudiante* denotes a set of individuals (the set **STUDENT**) and the determiner *un* relates it to another set of individuals, namely, the one denoted by the verb phrase. Given the particular meaning of the determiner *un*, for any predicate expression *B*, a sentence of the form *un estudiante B* is true if and only if the intersection of the sets denoted by *estudiante* and *B* ($\mathbf{STUDENT} \cap \mathbf{B}$) is not empty, (i.e., if there is at least one individual who is a student and who also satisfies *B*). Other determiner meanings can be characterized in a similar fashion. In what follows we will use the term *determiner* to comprise syntactically simple determiners (*el* 'the,' *tres* 'three,' *un* 'a,' *muchos* 'many,' etc.), and those complex determiners that are not syntactically atomic (*todos los* 'all the,' *la mayoría de los* 'most (of the),' *más de tres* 'more than three,' *menos de la mitad de los* 'fewer than half of the,' etc.). Independently of the syntactic structure of these simple or complex expressions, what is of importance from a semantic point of view is that they denote determiner functions. Let **E** be a universe of individuals, and **A**, **B** subsets of **E**. Then, for any determiner **D**, **D**(**A**)(**B**) is a function mapping two arguments (**A** and **B**) to truth values (**True**/**False**). The following determiner functions, among others, can be defined over **E**:

(3) **todos**(**A**)(**B**) = **T** iff $\mathbf{A} \subseteq \mathbf{B}$
(alg)un(**A**)(**B**) = **T** iff $\mathbf{A} \cap \mathbf{B} \neq \varnothing$
ningún(**A**)(**B**) = **T** iff $\mathbf{A} \cap \mathbf{B} = \varnothing$
(más de) tres(**A**)(**B**) = **T** iff $\mathrm{Card}(\mathbf{A} \cap \mathbf{B}) > 3$
los dos(**A**)(**B**) = **T** iff $\mathbf{A} \subseteq \mathbf{B}$ & $\mathrm{Card}(\mathbf{A}) = 2$
la mayoría(**A**)(**B**) = **T** iff $\mathrm{Card}(\mathbf{A} \cap \mathbf{B}) > \mathrm{Card}\,(\mathbf{A} - \mathbf{B})$
muchos(**A**)(**B**) = **T** iff $\mathrm{Card}\,(\mathbf{A} \cap \mathbf{B}) \geq n$
etc.

Applying (3) to elucidate the meaning of a sentence is relatively straightforward. For example, **todos (ESTUDIANTE)(CONTENTO)** is true if the set of students is a subset (⊆) of the set of happy individuals; **ningún(ESTUDIANTE)(CONTENTO)** is true if the intersection (∩) of the sets denoted by *estudiante* and *contento* is empty; **la mayoría (ESTUDIANTE)(CONTENTO)** is true if the number of students who are happy is greater than the number of those students who are not happy; etc. Other Spanish determiners can be easily defined following the same strategy. In sum, a simple or complex determiner is a function with two arguments. The first argument corresponds to the noun denotation and the second argument corresponds to the content of the sentence's main predicate (verb phrase). Since noun phrases not only occur in the subject or nominative position, appropriate "case extensions" (Keenan 1989; Gutiérrez-Rexach 1998) can be defined to derive the meaning of such expressions when they are in an accusative or dative position.

Natural language determiners also satisfy a series of constraints that set them apart from their logical counterparts. For example, all determiners are conservative or "live on" their first argument (Barwise and Cooper 1981), (i.e., $\mathbf{D}(\mathbf{A})(\mathbf{B}) = \mathbf{D}(\mathbf{A})(\mathbf{A} \cap \mathbf{B})$), motivating the intuition that the determiner has a close link with its nominal restriction. From the conservativity constraint, it follows that the sentences in (4) are equivalent:

(4) a. Algunas casas son antiguas.
'Some houses are old.'
b. Algunas casas son casas antiguas.
'Some houses are old houses.'

If we substitute any other determiner for *algunas* in this sentence, the equivalence still holds: the exception being the focus determiner particle *solo* 'only.' The effect of the conservativity constraint is to make nominal quantification inherently restricted to the first argument of a determiner. In order to check whether **algunas(CASA)(ANTIGUA)** is true, we do not have to consider those entities in the relevant universe of discourse that are not houses. When we process sentence (4a), we do not first check which entities in the discourse domain are old and then determine whether any of these are houses; rather, we look at the relevant houses and determine whether some of them are old.

Determiners are also characterized by their monotonicity properties, namely the type of inferences they license: set-to-subset (decreasing) inferences, or set-to-superset (increasing) inferences. For example, sentence (5a) entails both sentence (5b) and (5c):

(5) a. Ningún senador fumaba.
'No senator smoked.'
b. Ningún senador fumaba puros.
'No senator smoked cigars.'
c. Ningún senador conservador fumaba.
'No conservative senator smoked.'

The property set denoted by *fumaba puros* 'smoked cigars' is a subset of the property denoted by *fumaba* 'smoked,' so the determiner function **ningún** licenses set-to-subset inferences for its second argument (the main predicate). The same is true for its first argument: sentence (5a) entails sentence (5c) because the property denoted by *senador conservador* is a subset of the property denoted by *senador*. Thus, we say that *ningún* is a decreasing determiner in its two arguments, the nominal restriction and the sentential predicate. The determiners *menos de n, no más de n* (for any number *n*) or *pocos* are also decreasing. The determiner *todos* is decreasing only in its first argument: it licenses set-to-subset inferences in its nominal restriction – sentence (6a) entails (6b), and set-to-superset inferences in the main predicate – (7a) entails (7b):

(6) a. Todos los estudiantes suspendieron.
'All the students failed.'
b. Todos los estudiantes mayores de veinte años suspendieron.
'All the students older than twenty failed.'

(7) a. Todos los estudiantes estaban leyendo *El Quijote*.
'All the students were reading *El Quijote*.'
b. Todos los estudiantes estaban leyendo.
'All the students were reading.'

On the other hand, the determiner function *un/algún* is an increasing determiner and has an entailment pattern that is the opposite of *ningún*; that is, it licenses set-to-superset inferences in its two arguments. We predict that sentence (8a) entails both (8b) and (8c).

(8) a. Un/algún senador conservador fumaba puros.
'A/some conservative senator smoked cigars.'
b. Un/algún senador conservador fumaba.
'A/some conservative senator was smoking.'
c. Un/algún senador fumaba puros.
'A/some senator was smoking cigars.'

Continuous determiners are those few determiners that are not monotonic (decreasing or increasing): *exactamente n* 'exactly n,' *entre n y m* 'between n and m,' etc. The monotonicity properties of determiners are of linguistic significance. Following the Ladusaw–Fauconnier generalization (Keenan 1996), it can be claimed that so-called negative words or n-words (*nadie* 'nobody,' *nunca* 'never,' *nada* 'nothing,' etc.) and negative polarity items (*ni un pelo de tonto* 'idiot at all,' *la menor idea* 'the faintest idea,' etc.) are licensed in the scope of decreasing functions. In other words, not only negation licenses such items (9a), but they also become licensed when they occur in the relevant argument of decreasing determiners (9b, 9c, and 10). N-words are also licensed when they appear in preverbal position (11), presumably because they already incorporate a negative element morphologically, and they inherently satisfy the Fauconnier–Ladusaw generalization, unlike their English counterparts. On the other hand, increasing and continuous determiners never license negative polarity items (12).

(9) a. No ha estado nunca en París.
'He has never been to Paris.'
b. Ningún senador conservador ha estado nunca en París.
'No conservative senator has ever been to Paris.'
c. Ningún senador que haya estado nunca en París está descontento.
'No conservative senator who has ever been to Paris is disappointed.'

(10) a. Todo senador que haya estado nunca en París está contento.
'Every senator who has ever been to Paris is happy.'
b. *Todo senador conservador ha estado nunca en París.
'*Every conservative senator has ever been to Paris.'

(11) a. Nunca ha estado en París.
'He has never been to Paris.'
b. *Ha estado en París nunca.
'*He has ever been to Paris.'

(12) a. *Algún senador conservador ha estado nunca en París.
'*Some conservative senator has ever been to Paris.'
b. *Algún senador que haya estado nunca en París está contento.
'Some senator who has ever been to Paris is happy.'

Recent research has uncovered that NPI-licensing is subject to additional constraints. Not all n-words and NPIs are licensed uniformly in decreasing environments. Other elements such as certain modality expressions, comparatives, certain interrogative environments, etc., may also act as licensors, etc.

3 Quantifier classes

3.1 The indefiniteness restriction

A very important classification of quantifiers emerges according to whether they can occur in an existential construction (i.e., a sentence that asserts the existence of X in Y (*Hay X en Y* … 'There are X in Y'), such as *Hay D senadores en la sala* 'There are D senators in the room'). The determiners *algunos* 'some,' *tres* 'three,' *menos de cuatro* 'fewer than four,' *varios* 'several,' *ningún* 'no,' *pocos* 'few,' and *muchos* 'many,' among others can be substituted for D and occur in this construction, whereas the occurrence of determiners such as *todos/todo* 'all/every,' *la mayoría* 'most,' *los* 'the,' *las diez* 'the ten,' *mi* 'my,' *ambos* 'both,' *este* 'this,' *cada* 'each,' and *todos menos tres* 'all but three' would make the sentence ungrammatical.

This so-called *indefiniteness restriction* partitions determiners in two groups: determiners that cannot occur in an existential construction are labeled as *strong*; and those that can are labeled as *weak*, since they only seem to express generalizations about the size of the set denoted by the noun (Milsark 1977). The indefiniteness restriction surfaces in other constructions, such as those expressing essential or inalienable possession (13), distance in time or space (14), or inherent measure (15):

(13) La silla tiene cuatro/*las patas.
'The chair has four/*the legs.'

(14) a. Hace cuatro/*los años que no te he visto.
'I have not seen you in four/*the years.'
b. Está dos/*las calles adelante.
'He is two/*the streets ahead.'

(15) a. La mesa mide cuatro/*los metros de largo.
'The table is four/*the meters wide.'
b. La piedra pesa dos/*las toneladas.
'The stone weighs two/*the tons.'

In generalized-quantifier theory, weak determiners are considered to have the property of intersectivity since they express a relation of intersection between their two arguments (Keenan 1987). For example, in order to check whether **exactamente tres**(**A**)(**B**) is true, we only need to determine whether the intersection of **A** and **B** has exactly three members. On the other hand, those determiners that cannot occur in an existential construction normally express a relation of inclusion between their two arguments, such as *todos* 'all,' *cada* 'each,' or *ambos* 'both.' They can also express proportionality, such as *la mayoría* 'most,' *un tercio* 'a third,' *la mitad* 'half,' etc.

3.2 Determiners and context sets

Definite determiners, whether simple (*los* 'the') or complex (*los diez* 'the ten'…), demonstrative determiners (*este* 'this'), possessive determiners (*mi* 'my'), and partitives (*algunos de los* 'some of the') do not occur in existential sentences either. These determiners are all inherently context-dependent and thus presuppose or do not assert existence. Semantically, they express an inclusion condition and they are relativized or restricted to a context set (Westerståhl 1985; Keenan and Westerståhl 1997; Peters and Westerståhl 2006). A possible characterization of the determiner function **los diez**, leaving initially aside the context-dependent nature of the determiner, would be as follows (Keenan 1996), for any sets of individuals **A**, **B** in **E**:

(16) $\textbf{los diez}\ (\mathbf{A})\ (\mathbf{B}) = \mathbf{T}$ iff $\mathbf{A} \subseteq \mathbf{B}\ \&\ \mathrm{Card}(\mathbf{A}) = 10$

Nevertheless, it seems that we also need to take into account that *los diez estudiantes* in (17) normally presupposes a context set or resource domain of students that is familiar in the common ground, namely the set of students under discussion (for example, the students enrolled in Spanish 101, etc.):

(17) Los diez estudiantes llegaron tarde.
'The ten students were late.'

Pragmatically, it is normally stated that definites satisfy a familiarity condition (Heim 1982). Semantically, the determination of the restriction or first argument of the determiner is mediated by intersection with a presupposed set, the context set (**C**). Using Westertåhl notation, we can write $\mathbf{D}^{\mathbf{C}}$ for the contextual restriction of a determiner function **D** to the context set **C**:

(18) $\textbf{los diez}^{\mathbf{C}}(\mathbf{A})(\mathbf{B}) = \mathbf{T}$ iff $(\mathbf{C} \cap \mathbf{A}) \subseteq \mathbf{B}$ & $\mathrm{Card}(\mathbf{C} \cap \mathbf{A}) = 10$

What (18) states is that in order to determine the truth conditions of (17) above, namely, if **los diez** (**ESTUDIANTE**)(**LLEGAR TARDE**) is true, we have to check whether the individuals in a contextually restricted set of students (**C** ∩ **STUDENT**) of cardinality 10 are late. The impact of contextual restriction on determiners is also reflected morpho-syntactically. Consider the difference between the context-independent universal singular determiner *todo* 'every' and the contextually-restricted plural *todos* 'all.' *Todo* tends to be used in generalizing or characterizing statements (19), and cannot be linked to entities already available in discourse (20):

(19) a. Todo cuerpo sumergido en un fluido experimenta un empuje vertical hacia arriba.
'Every mass in a fluid is pushed upwards.'
b. Sus argumentos carecen de toda lógica.
'His arguments lack any logic.'

(20) *Me compré la colección de cromos de futbolistas y le di a mi sobrino todo cromo de jugadores del Zaragoza.
'I bought a soccer-card collection and I gave my nephew every Zaragoza-player card.'

On the other hand, *todos los* 'all the' is normally used in statements that apply to all the members of a discourse-available set:

(21) Llegaron a la meta solo nueve corredores. Todos los keniatas estaban entre ellos.
'Only nine runners got to the finish line. All the Kenyans were among them.'

Thus, Spanish spells out as a syntactic complex (*todos* + *los*) the universal (inclusive) and contextually restricted nature of this determiner function. We can conclude that *todo(s)* is a universal determiner but lacks a contextual feature, and in its plural form it combines with a definite obligatorily: **todos estudiantes* 'lit. all students' vs. *todos los estudiantes* 'all the students,' *todos esos estudiantes* 'all those students,' etc. Singular *todo* does not combine with a definite in its generic reading (*todo estudiante* 'every student'/**todo el estudiante*). When combined with a definite determiner, it either refers to all the parts of an integrated whole (*toda la mesa* 'the whole table') or it becomes context-dependent. The quantifiers *todo el mundo* 'everybody' and *toda la clase* 'the whole class' in (22) are not used to express generalizations about the world or a class; rather, they refer to specific sets of individuals:

(22) a. Todo el mundo llegó tarde.
'Everybody was late.'
b. Se ha enfadado con toda la clase.
'He is mad at the whole class.'

Possessives and demonstratives are also contextually restricted. Possessives are restricted to a context set of objects owned by the possessor, and demonstratives are restricted to a deictic environment. They can both be considered definite determiners with a restriction established by an intrinsic relation (possession: 'the x possessed by y'; deixis: 'the x pointed at by y'). Hence, the approximate equivalence between *su libro* 'his book' and *el libro de él* lit. 'the book of he,' and *este libro* 'this book' and *el libro que está aquí* 'the book that is here.' Additionally, the relevant contextual relation is determined by certain presuppositions. For example, Spanish demonstratives have an obligatory proximity presupposition (proximal/distal): *este* and its variants in gender and number presuppose proximity to the addressee; *aquel* presupposes distance with the speaker and the addressee; and *ese* is non-specified for a proximity presupposition, so its uses are wider and include several proximity relations as well as focus (Gutiérrez-Rexach 2001, 2005).

Since they are already definite/inclusion determiners intrinsically, possessives and demonstratives cannot select for and combine with other definites or universals: **mis todos* 'my all,' **este el* 'this the,' **su este* 'his this,' etc. Selection order is important here, since the complex determiners *todos mis N*, *el N este*, or *este N suyo* are possible indeed. One may view these restrictions as the result of c-command or hierarchical relations within an expanded version of the DP architecture: different determiners would occupy different head positions with strict dominance relations between them (cf. Bosque and Gutiérrez-Rexach 2009). Alternatively, selection order may be viewed as the byproduct of semantic requirements. For example, *todos mis* is possible because the possessive relational component refines the inclusion component of the universal determiner, etc.

3.3 More determiner complexes

Splitting the definite component from the relational one (possessive/demonstrative) is allowed in Spanish, as is also the case in certain other languages under restricted conditions: *el libro este* lit. 'the book this,' *el libro mío*, lit. 'the book mine,' etc. In these instances, the relational elements with lexical content (demonstrative or possessive) occur as postnominal adjectival modifiers, and the definite remains as the determiner head of the noun phrase. It is even possible to have double demonstratives, as in *aquellos dos libros esos (de que te hablé)* 'those two books (that I talked to you about).' The second demonstrative tends to be the default *ese*, playing a focus-related role (compare with ??*esos dos libros estos* lit. 'those two books these').

Other complex universals and definites are built through the composition of a universal/definite and a weak determiner: *todas esas muchas* 'all those many,' *los cuatro* 'the four,' *mis varios* 'my several,' *esos pocos* 'those few,' etc. The definite/universal establishes a delimited set and the weak determiner acts as a counter or cardinality marker on that set. This is why the reverse combination is not possible: **dos los* 'two the,' **varios esos* 'several those,' etc. As stated above, this restriction can also be accounted for syntactically as the result of a dominance hierarchy within DP.

Partitive determiners are complex determiners in which a weak determiner and a strong (plural) determiner combine. The latter is introduced by the partitive preposition *de* and denotes the set or whole from which the relevant part originates. Given that a weak determiner does not isolate a proper whole, it is predicted that the determiner complexes in (23a) express partitive-determiner functions and those in (23b) do not:

(23) a. a Algunos de los/tres de esos/varios de mis libros se perdieron.
 'Some of the/three of those/several of my books got lost.'
 b. *Mis de algunos/*esos de cuatro/*los de pocos libros se perdieron.
 '*My of some/*those of four/*the of few books got lost.'

Since Spanish lacks a determiner head expressing proportion, proportional determiners require the presence of a lexical head (a noun) expressing the intended part or proportion: *mayoría* 'majority,' *tercio* 'third,' *mitad* 'half,' *parte* 'part,' etc. proportional-determiner functions are always syntactically complex in Spanish: *most* is expressed as *la mayoría de los* lit. 'the majority of the' or *más de la mitad de los* 'more than half of the,' etc. The complex is built like a partitive-determiner complex in that the partitive marker *de* is always required (**la mitad los estudiantes* lit. 'the half the students'). Proportional determiners can be strong (*la mayoría de los* 'most,' *la mitad de los* 'half') or weak (*dos tercios de los* 'two-thirds of the,' *tres cuartas partes de los* 'three fourths of the,' etc.), as proven by the fact that the strong proportionals cannot occur in an existential construction whereas the weak ones can. Obviously, weak or strong status is derived from the weak or strong status of the first element or head of the complex.

3.4 Pronominals, indefinites, and context

In most world languages, including Spanish, the operation of determiner pronominalization is allowed. A determiner is pronominal when it can occur as a full noun phrase without an explicit nominal restriction. The proper generalization seems to be that the context set behaves as a covert restriction in this case. For example, *todos* in (24a) is understood as 'all the members of a context set **C**':

(24) a. Todos llegaron tarde.
'All of them were late.'
b. Nunca salgo con estas.
'I never go out with those ones.'
c. Compré varios.
'I bought several (of them).'

Singular and plural indefinites are also partitioned along the contextual-dependence parameter (Gutiérrez-Rexach 2004, 2010). The determiners *algún* and *algunos* are context dependent, can be pronominalized, and the latter has an explicit partitive variant (*algunos de los*). On the other hand, *un* and *unos* tend to be context-independent, with some exceptions, for instance, when *unos* is contrasted (*unos... otros* 'some... others'). In (25a), we see that pronominal *alguno* is a genuine existential (one or more), whereas *uno* is a numeral (one), not a pronominalized existential. The dialogue in (25b) shows that *algunas* and the contrastive *unas ... otras* are used to link to previously introduced discourse referents (Shakespeare works):

(25) a. De los candidatos solo quedó alguno/uno.
'Of the candidates only some/one remained.'
b. – A: Heredé las obras completas de Shakespeare.
– B: No me gustan *unas/algunas (de ellas).
– A: A mí unas me gustan, otras no.
'– A: I inherited the complete works of Shakespeare.
– B: I do not like some of them.
– A: I like some, others I do not.'

3.5 Exceptive and floating quantifiers

Complex-determiner functions are also required to convey more sophisticated part-whole relations. A very common scenario arises when the main predicate applies to a part or subset of a group but does not apply to the other part. This type of relation is commonly expressed with complex determiners that characterize an exception. Such exceptive quantifiers (Keenan and Faltz 1985) are built by adding an exception phrase (headed by *excepto*, *salvo*, *menos* 'except') to a noun phrase, as in (26):

(26) a. Todos los estudiantes excepto Juan leyeron el libro.
'All the students except Juan read the book.'

b. Ningún soldado salvo esos dos se atrevería a levantar la voz.
'No soldier but those two would dare to raise his voice.'

Not all quantifiers can be modified by exception phrases. Only the idealizing (van Benthem 1986) determiners *todo(s)* 'all/every' and *ningún* 'no' can head an exceptive determiner function:

(27) *Tres/*los/*muchos/*la mayoría de los estudiantes excepto Juan leyeron el libro.
'*Three/*the/*many/*most students but Juan read the book.'

This restriction is due to the fact that only the quantifiers headed by *todos* and *ningún* have clear unique exception sets. For example, the quantifier *tres estudiantes* 'three students' may have several alternative exceptions depending on which three students or more making the statement true are picked, so it is not a good candidate for having an exception modifier. Consistently, an exception phrase will be valid only if it denotes such a unique and precisely delimited set, and only quantifiers allowing the identification of this set are good complements of an exception head. Proper nouns, definites, and numerals in their non-monotonic interpretation (exactly *n*) can occur as complements of the exceptive preposition:

(28) Todos los estudiantes excepto Juan/dos/los suspensos/esos vinieron.
'All the students except Juan/two/the ones who failed/those came.'

Weak determiners expressing only monotonic cardinality assessments are not allowed as complements of an exceptive preposition since they do not identify an exception set, as shown in (29a). On the other hand, exceptions that clearly identify such a set but are too big (*la mayoría*) or do not constitute exceptions (*todos, ninguno*) are not possible either (29b), because the resulting exceptive statement would be contradictory: Bigger exceptions would not be exceptions but the norm; *todos* cannot be an exception to *todos*, etc.

(29) a. Ningún/todos los estudiantes excepto *algunos/*más de tres/*muchos vinieron.
'No/all the student(s) except *some/*more than three/*many came.'
b. Ningún/todos los estudiantes excepto *la mayoría/*todos/*ninguno vinieron.
'No/all the student(s) except *most/*all/*none came.'

Certain determiners may be associated with another quantificational element with scope over the verb phrase. These associated elements are called floating quantifiers since it appears as if the two elements originated together but are split at surface syntax, and one of them has "floated" to a clause-internal position:

(30) a. Los bomberos estaban todos muy cansados.
'The firefighters were all very tired.'
b. Juan y María son los dos de Córdoba.
'Juan and María are both from Córdoba.'
c. Estos documentales tratan la mayoría de temas políticos.
'These documentaries deal most of them with political topics.'

There is a strong debate in the literature on the syntactic status of such floated elements: Whether they are really displaced or "floated" down from the clause-initial position, or it is the clause-initial determiner that has been moved upwards from an embedded position; alternatively, it can be claimed that there has not been any displacement at all and the "floated" element is just a verb-phrase modifier. Semantically, most weak determiners cannot be associated with floated quantifiers (31a). Only the indefinite *unos* and those allowing hidden partitive or strong meanings have this possibility (31b):

(31) a. *Mas de tres/*pocos bomberos estaban todos muy cansados.
'*More than three/*few firefighters were all very tired.'
b. Unos/ciertos bomberos estaban todos muy cansados.
'Some/several firefighters were all very tired.'

The noun phrases *unos/ciertos bomberos* in (31b) are used to distinguish a group of some/certain firefighters, whereas those in (31a) are mere cardinality markers. The rationale for this restriction on floating quantification is that a determiner has to be able to establish a group or set in order to associate with a floated element. The role of the floated element is to create a partition determining the extent of the main predication (i.e., to which elements the main predicate applies). *Unos* and *ciertos* are weak quantifiers that can establish or isolate a group (Gutiérrez-Rexach 2004, 2010). Consistently, weak quantifiers that only express a cardinality assessment are not allowed as floated quantifiers because they would not partition the relevant set properly:

(32) Los bomberos estaban *muchos/*algunos/*bastantes cansados.
'The firefighters were *many/*some/*enough tired.'

4 Scope, polyadicity, and plurality

4.1 *Scope interactions*

So far we have only considered sentences with one quantifier. When more than one quantifier occur in a sentence, scopal relations emerge as a form of semantic interaction between these elements. Relative scope is determined by the different order of quantifiers in the semantic representation of a clause and the associated dependence that is established. For example, sentence (33) is ambiguous because it contains two quantifiers: *todos los estudiantes* 'all the students' and *un libro* 'a book.'

(33) Todos los estudiantes leyeron un libro.
'All the students read a book.'

Under one interpretation, every student read a different book. Under the second reading, there is a unique book such that every student read it. This ambiguity is a genuine scope ambiguity. In the first reading, the scope order of the quantifiers is the one preserving the linear order of the noun phrases in the clause (i.e., the subject quantifier is more dominant or has scope over the object: *todos los estudiantes* > *un libro*). The universal quantifier headed by the determiner *todos* takes scope over the existential quantifier headed by *un* and the latter depends on the former. The second reading is an inverse scope reading in which the scopal order of the quantifiers differs from the surface linear order. In this reading, the existential quantifier takes scope over the universal quantifier: *un libro* > *todos los estudiantes*. When the object has narrow scope in (33), we are talking about several books (potentially a different one for each student); when it has wide scope, we are talking about a single book that every student under consideration read.

Although it would be logically possible, given a sentence with *n* quantifiers, to generate a number of readings corresponding to the different scopal orders that may arise, there are strong constraints on this space of possibilities in Spanish and in most natural languages to avoid unnecessary ambiguity. First, the scopal order of the quantifiers tends to be identical to their surface linear order in most sentences. Subjects normally take scope over objects, and inverse readings are made unambiguous by linear rearrangement or modification (see also below):

(34) a. Hay un libro que todos los estudiantes leyeron.
'There is a book that all the students read.'
b. Todos los estudiantes leyeron el mismo libro.
'All the students read the same book.'

In general, weak determiners (indefinites, numerals, evaluatives) do not easily allow object wide-scope readings. Consider (35):

(35) Todos los estudiantes leyeron tres libros.
'All the students read three books.'

The interpretation in which the object quantifier, for example *tres libros*, takes inverse wide scope is marginal, if available at all: 'There is a set of three books such that all the students under consideration read these books.' If a speaker wants to convey this meaning, he would normally utter (36):

(36) Todos los estudiantes leyeron los tres/los mismos tres libros.
'All the students read the three/the same three books.'

The reason why (35) is not a good expression of the intended meaning (36) is that the cardinal quantifier *tres libros* 'three books' does not normally identify a unique set, since any set with three or more books would make the sentence true. In general, for a quantifier to be able to take wide scope it has to allow the retrieval of a unique set – technically, it has to be a principal filter (Szabolcsi 1997). Thus, strong determiners (i.e., universals, definites (including possessives and demonstratives), and proportional determiners), have more straightforward object wide-scope readings, as in (37). Weak determiners usually have a wide-scope reading when they are modified by individualizers (*en concreto* 'concrete,' *específico* 'specific,' *en particular* 'in particular') or prenominal adjectives (Bosque and Gutiérrez-Rexach 2009), as shown in (38):

(37) Todos los estudiantes leyeron esos/mis libros.
'Al the students read those/my books.'

(38) Todos los estudiantes leyeron tres libros en particular/dos maravillosos libros.
'All the students read three books in particular/two wonderful books.'

4.2 The landscape of scope independent readings

Not all quantificational interactions are reducible to iterations of quantifiers in one scopal order or another. The combination of certain quantifiers is genuinely polyadic and is not reducible to the iterated application of monadic quantifiers. More explicitly, two quantifiers $\mathbf{Q_1}$ and $\mathbf{Q_2}$ occurring in a sentence *s* represent an instance of polyadic quantification when the meaning of *s* cannot be derived by applying them separately in the $\mathbf{Q_1} > \mathbf{Q_2}$ or the $\mathbf{Q_2} > \mathbf{Q_1}$ order in a traditional Fregean fashion. Rather, the quantifiers undergo absorption and form a polyadic quantifier function ($\mathbf{Q_1}$, $\mathbf{Q_2}$) that applies to the relevant sentential predicate. Every human language contains such instances that lie "beyond the Frege boundary." The most typical examples are reflexives, reciprocals and anaphoric combinations with *mismo* 'same,' *diferente* 'different,' *sendos* 'one each,' or *ambos* 'both':

(39) a. Juan y Pedro se odian (el uno al otro).
'Juan and Pedro hate each other.'
b. Los alumnos respondieron preguntas diferentes.
'The students answered different questions.'

c. Las cinco puertas tienen sendos pomos.
'The five doors have a knob each.'
d. Juan y Pedro levantaron la mesa entre ambos.
'Juan and Pedro lifted the table together.'

The main characteristic of object anaphors is that they are completely dependent on the subject quantifier. The standard observation in syntactic theory is that these elements cannot occur in the subject position because they have to be c-commanded by their antecedent (Bosque and Gutiérrez-Rexach 2009: ch. 9). Semantically, such a constraint follows from the quantificational requirements of these anaphors: They are not scope-independent and they form a polyadic complex with the subject quantifier (Q_{SUBJECT}, Q_{ANAPHOR}). For example, determining the reference of *el uno al otro* 'each other' is impossible unless we have identified the two individuals to which the reciprocity relation applies (i.e., the denotation of the antecedent expression). Apparent exceptions are sentences such the following:

(40) a. Los mismos tres críticos veían todas las películas.
'The same three critics watched all the movies.'
b. Las mismas historias se repiten.
'The same stories are always repeated.'
c. Diferentes estudiantes eligieron diferentes asignaturas.
'Different students chose different courses.'

The dependent element may be sometimes interpreted as a cataphor or prospective anaphor, as in (40a). Dependencies may also emerge with other elements in the common ground or in the preceding discourse: *mismas* in (40b) is dependent on a previously-introduced element (for example, promises by politicians during a campaign, etc.). Finally, the dependent element may just have an intensional flavor (*diferente* as *distinto* 'of a different kind or type, instantiating a different property, etc.'). These possibilities give rise to contrasting interpretations as a function of how the dependence is construed.

Sentences with crossing-coreference patterns, also called Bach–Peters sentences, are not reducible to iterations either. In (41), both quantifiers contain pronouns whose interpretation is dependent on the other quantifier in the clause, blocking the possibility that these quantifiers be applied in an iterated fashion:

(41) Los estudiantes que lo admiraban se sintieron traicionados por el profesor que los suspendió.
'The students who admired him felt betrayed by the professor who flunked them.'

In addition, resumptive quantifiers are also genuinely polyadic. Resumption takes place when there are two or more instances of the same determiner in a sentence, and it can be characterized as the absorption of the two determiners in a polyadic complex. Consider the following examples:

(42) a. Dos chicos comieron dos bocadillos.
'Two kids ate two sandwiches.'
b. Nadie conoce a nadie.
'Nobody knows anybody.'
c. Pocos detectives resolvieron pocos crímenes.
'Few detectives solved few crimes.'

In addition to the readings already discussed in which one quantifier has scope over the other, there is a clearly salient interpretation in which pair quantification takes place. For example, in (42a) we are talking only about two pairs of kids and sandwiches ($<\mathbf{k_1,s_1}>$, $<\mathbf{k_2,s_2}>$). This interpretation is genuinely polyadic, not derivable by applying *dos bocadillos* first and *dos chicos* afterwards (*dos chicos* > *dos bocadillos*) or vice versa. The resumptive interpretation of (42b) is that there is nobody in the relation denoted by *know*: no pairs $<\mathbf{x,y}>$ such that **x** knows **y**. In a similar fashion, the resumptive interpretation of (42c) is that there are few pairs <**detective, crime**> such that the detective solved the crime.

Branching or independent quantification is also polyadic. An instance of branching between two quantifiers emerges when there is a relation associating elements in their respective restriction domains. Such relation is not reducible to the one resulting from scopal ordering. Consider sentence (43):

(43) La mayoría de los chicos de tu clase y muchas de las chicas de la mía han quedado ya.
'Most boys in your class and many of the girls in mine have already gone out.'

This sentence is not equivalent to any of the two alternatives that spell out different scopal orderings:

(44) a. La mayoría de los chicos de tu clase han quedado ya con muchas de las chicas de la mía.
'Most of the boys in your class have already gone out with many of the girls in mine.'
b. Muchas de las chicas de mi clase han quedado ya con la mayoría de los chicos de la tuya.
'Many of the girls in my class have already gone out with most boys in yours.'

For example, (44a) states that it is true for most of the students in your class that they have each gone out with many girls. The most salient interpretation of (43) is stronger: there is a set containing most of the students in your class and another set with many of the girls in mine such that each member of the first set has already gone out with one of the members of the second set. Such branching interpretations are quite common, as in (45). Adding explicit distributors (*cada uno* 'each') makes them even more prominent (46, 47):

(45) Exactamente cuatro rusos y exactamente cuatro húngaros lucharon entre sí.
'Exactly five Russians and exactly four Hungarians fought.'

(46) Al menos cuatro puntos están conectados cada uno al menos con cuatro estrellas.
'At least four dots are each connected to at least four stars.'

(47) Como máximo cuatro alumnos aprobarán cada uno menos de dos asignaturas.
'At most four students will pass fewer than two courses each.'

Branching readings of transitive sentences seem to be associated in general with a maximality condition on two sets and implicit or explicit distribution over them: Every member of the first set is connected to every other member of the second set. There is also a very similar form of scope independence not requiring total connection between the sets involved, just that all the members of those sets are connected in some fashion. This type of quantificational interaction is called cumulativity. Consider (48):

(48) Tras el acuerdo, trescientas firmas españolas han comprado setecientos ordenadores chinos.
'Following the agreement, three hundred Spanish companies have purchased seven hundred computers from China.'

The most salient reading of (48) is not the subject > object one (i.e., that each one of three hundred Spanish companies bought seven hundred computers). What (48) states is that a total of three hundred firms bought a total of seven hundred computers. Some companies may have bought one computer, others two or three, etc. The same type of interaction between quantifiers emerges in the following sentence:

(49) Contaron a muchos de los niños unas veinte historias.
'They told twenty stories to many of the children.'

4.3 *Distributivity and collectivity*

The behavior of quantifiers is also conditioned by and depends on two modes of interaction that are related to predication: distributivity and collectivity (cf. Gutiérrez-Rexach 2003, vol. 3B). A predicate is distributive when it applies to each one of the members of a set individually; a predicate is collective when it is true of the whole set but not necessarily of its members. *Cada* 'each' is the most significant distributor in Spanish, and its presence as a determiner or a floating quantifier renders a predicate obligatorily distributive:

(50) a. Cada estudiante bebió un refresco.
'Each student had a drink.'
b. Los estudiantes bebieron un refresco cada uno.
'The students had a drink each.'

There are predicates and adjectives that are intrinsically collective (*congregarse* 'gather,' *reunirse* 'meet,' *numeroso* 'numerous,' etc.). These predicates block distributive subentailments to the members of a plurality. A predicate such as *llegar* 'arrive,' when applied to a plurality-denoting expression, does not block its application to the relevant members: if the students arrived and Juan is one of those students, then we can infer that Juan arrived. On the other hand, (51a) and (52a) do not respectively entail (51b) and (52b):

(51) a. Los estudiantes se congregaron.
'The students gathered.'
b. *Juan se congregó.
'*Juan gathered.'

(52) a. La multitud era numerosa.
'The crowd was dense.'
b. *Pepe es numeroso.
'*Pepe is dense.'

There are also other modifiers that can turn an expression into a collective predicate: *entre todos* 'all jointly,' *juntos* 'together,' *en conjunto* 'as a whole,' etc. For example, sentence (53) is understood as distributively applying to all the individuals in the denotation of *estudiantes*. When we add *en conjunto*, the predicate becomes collective, resulting in an anomalous sentence.

(53) Los estudiantes pesan 75 kilos (#en conjunto).
'The students weigh 75 kg (#together).'

The collectivity/distributivity distinction is critical to understand plural quantifiers since it restricts how these elements may interact. A sentence such as (54) is ambiguous between a collective, a distributive, and a cumulative interpretation:

(54) Cinco estudiantes levantaron tres mesas.
'Five students lifted three tables.'

In the cumulative reading, we only focus on the total number of students (five) and tables (three) involved but not on the liftings. On the other hand, in a collective-interpretation scenario, there are three liftings involved, in which five students lifted a table together in each one of the three cases. In the distributive reading, there would be a total of fifteen liftings since each student lifts three tables individually. As is expected, ambiguity is normally eliminated by adding the appropriate modifiers.

5 Quantification and dynamics

Definites and indefinites, in addition to the properties specified in section three above (see also Leonetti 1990, and Chapter 15, above), have a rather flexible behavior that allows them to receive additional interpretations depending on the structural environment in which they occur. This has led several authors to propose that they are not genuine quantificational elements. Rather, they receive their quantificational force from overt or covert operators in their surrounding environment.

Plural definite determiners may have generic interpretations when they combine with predicates of kind or class, as shown by the contrast in (55):

(55) a. Los holandeses son inteligentes.
'Dutch people/The Dutch are intelligent.'
b. Los holandeses llegaron tarde.
'The Dutch were late.'

Sentence (55a) is ambiguous: we may understand that we are talking about the Dutch in general or about a specific group of Dutch individuals. The generic interpretation of (55a) is that Dutch people (in general) are smart. The possibility of having a generic interpretation is related to the fact that the main predicate is a characterizing or individual-level predicate. If the same subject combines with a standard stage-level or locational predicate (55b), the generic reading disappears: the definite quantifier refers to all the members of a contextually determined group. Singular and plural definites are also used to refer to classes or kinds, and to group-level entities or collections:

(56) a. Los dinosaurios se extinguieron.
'Dinosaurs are extinct.'
b. El oro es cada vez más difícil de extraer.
'Gold is becoming increasingly difficult to extract.'
c. La familia se reunió.
'The family gathered.'

Indefinites are not always associated with the force of an existential quantifier. As Lewis (1975) and followers have claimed, sometimes they appear to vary in quantificational force. Consider the following sentence:

(57) Si un niño golpea a otro hay pelea.
'If a kid kicks another one, there is fighting.'

Sentence (57) does not state that there is a specific kid, such that if he kicks another one, fighting ensues. What (57) clearly means is that it is the case for all/most kids that when they kick another one, there is a fight. Thus, *un niño* seems to have universal/generic force, not existential. The particular force that has to be associated with the indefinite is made explicit sometimes by the presence of an adverb of quantification (*siempre* 'always,' *normalmente/a menudo* 'often,' *muchas veces* 'many times,' *a veces*, 'sometimes'):

(58) a. Siempre que un niño golpea a otro hay pelea.
b. A veces sucede que si un niño golpea a otro hay pelea.
c. Normalmente cuando un niño golpea a otro hay pelea.
'Always/usually/sometimes if a kid kicks another one, there is fighting.'

What the Lewis/Heim/Kamp (Heim 1982; Kamp and Reyle 1993) analysis of these sentences hypothesizes is that indefinites do not have a quantificational force of their own and they are not associated with a single unique quantifier in a logical-form representation. Their force comes from a quantifier that has scope over them. Typically, such element is an adverb of quantification acting as an unselective binder (i.e., giving force to all the indefinites within its scope). Thus, in (59), not only does the first indefinite receive universal force from the unselective element, but the second indefinite does so too, since we understand that any soccer player insulting any referee would get a red card from that referee.

(59) Si un futbolista insulta a un árbitro siempre recibe tarjeta roja.
'If a soccer player insults a referee, he always gets a red card.'

Definites also have dependent or weak uses. *La novia* in (60) does not necessarily refer to a unique familiar individual; it is a dependent element with several possible values (brides) associated with different weddings:

(60) Si voy a una boda, me gusta besar a la novia.
'If I go to a wedding, I like to kiss the bride.'

Finally, indefinites and definites have the property of being dynamic binders (Chierchia 1995). They can bind pronouns or other dependent expressions across sentential domains:

(61) Un/el/*todo hombre entró. *Pro* se sentó.
'A/the/*every man came in. He sat down.'

6 Questions and quantification

Qu-words and phrases differ from other determiners and pronominal expressions in that they are not restricted to declarative statements and commonly belong to other sentence types (interrogative, exclamatives). Nevertheless, *qu*-words have many similarities with regular determiners and pronouns. In general, it seems that they should be treated as existential determiners or quantifiers. First, they can occur in existential sentences:

(62) ¿Qué/cuántos hay sobre la mesa?
'What/how many is/are there on the table.'

Additionally, interrogative determiners (for example, *qué* en *qué zapato*) are conservative and intersective. This explains why the following equivalences hold, where the equivalence in (63) is due to conservativity, and in (64) to symmetry/intersectivity:

(63) a. ¿Qué zapato es más grande?
'Which shoe is bigger?'
b. ¿Qué zapato es un zapato que es más grande?
'Which shoe is a bigger shoe?'

(64) a. ¿Cuántos zapatos compró Paola?
'How many shoes did Paola buy?'
b. ¿Cuántas cosas que compró Paola son zapatos?
'How many things bought by Paola are shoes?'

Finally, like Spanish indefinites, *qu*-pronouns and determiners may be inherently restricted to context sets, and this property is morphologically encoded. We thus distinguish between the context neutral *qué* and the context-dependent *cuál*, which is obligatorily restricted to a context set:

(65) a. Hay cinco pares de zapatos. ¿Cuál/*qué quieres?
'There are five pairs of shoes. Which one/*what do you want?'
b. ¿Cuál/*qué de estos te gusta más?
'Which one/*what of these do you prefer?'

Interrogative quantifiers also participate in polyadic complexes, either with dependent elements or via resumption or branching. For example, the polyadic complex in (66a) has an anaphoric-dependent; the one in (66b) is a resumptive pair quantifier built via the absorption of two *qu*-pronouns (its answer is a pair or a list of pairs); and (66c) may be construed as an instance of branching or cumulative quantification:

(66) a. ¿Qué estudiante se odia a sí mismo?
'Which student hates himself/herself?'
b. ¿Qué estudiante leyó qué libro?
'Which student read which book?'

c. ¿Qué policías detuvieron a qué ladrones?
'Which policemen arrested which robbers?'

Most recent approaches to the semantics of questions associate the denotation of an interrogative sentence with the set of its answers (cf. Gutiérrez-Rexach 2003, vol. 5C). A sentence such as (67) may receive an individual answer (68a), a pair-list answer (68b), or a functional answer (68c):

(67) ¿Qué libro leyeron (todos) los estudiantes?
'Which books did all the/every student(s) read?'

(68) a. *El Quijote*.
b. Pepe *El Quijote*, Manuel *Ficciones* y Luisa *Rayuela*.
c. Su libro favorito.
'His favorite book.'

Such interpretations can be seen as the result of quantificational interaction between the universal and the interrogative quantifier. When the interrogative element has scope over the universal (*qué* > *todos los*) we get the individual reading; with respect to (67), we would be asking about the book that every student read. When the scopal order is *todos los* > *qué*, the resulting reading is the pair-list one in (68b), since for each student under consideration we are asking about his favorite book. Finally, (68c) can be considered a functional or intensional version of (68b), where instead of giving explicit pairings <**student, book**>, we present a way of inferring them (associate each student with his favorite book). As mentioned with respect to (66), polyadic complexes may emerge from the combination of an interrogative and a declarative quantifier. For example, (69a) has a clear cumulative reading spelled-out in the answer (69b):

(69) a. ¿Qué libros leyeron esos tres estudiantes?
'Which books did these three students read?'
b. Leyeron *El Quijote, Ficciones, Rayuela* y *Pedro Páramo*.
'They read *El Quijote, Ficciones, Rayuela*, and *Pedro Páramo*.'

The cumulative answer (69b) only associates a set of three students with the four books they read, without asserting which student read which book. Polyadic readings are subject to the same restriction as their declarative counterparts. For example, weak quantifiers rarely display cumulative or pair-list readings:

(70) a. ¿Qué libros leyeron pocos estudiantes?
'Which books did few students read?'
b. #Pedro y Juan leyeron *El Quijote* y *Ficciones*.
'#Pedro and Juan read *El Quijote* and *Ficciones*.'
c. #Pedro leyó *El Quijote* y Juan *Ficciones*.
'#Pedro read *El Quijote* and Juan *Ficciones*.'

7 Degree quantification

So far we have only dealt with quantification over individuals, which are the most important denotational domain in natural languages. It has been argued that quantification over entities in other domains is also expressible, most significantly over degrees or measures. The adverb *muy* 'very,' the degree phrase *dos metros* 'two meters,' and the *qu*-pronoun *cuánto* 'how much' all quantify over degrees (cf. Sánchez López 2006 for an overview):

(71) a. Pepe es muy alto
'Pepe is very tall.'
b. Pepe mide dos metros.
'Pepe is two meters tall.'
c. ¿Cuánto mide Pepe?
'How tall is Pepe?'

All the quantifiers involved in (71) range over degree variables *d*. In (71a), it is asserted that Pepe is tall to a degree **d** that exceeds a standard or norm; when uttering (71c) we are asking for a degree **d** such that Pepe is tall to that degree, etc. Comparatives and superlatives also involve degree quantification. In standard comparatives, one degree is compared to another one, and what is stated is that they are equal, or one is greater than the other. The degrees of height of Pedro and Luis are identical in (72a), and in (72b) Pedro is tall to a degree higher than Luis. Finally, a proper or true answer to (72c) will specify the individuals whose degree of height is less than that of Pedro:

(72) a. Pedro es tan alto como Luis.
'Pedro is as tall as Luis.'
b. Pedro es más alto que Luis.
'Pedro is taller than Luis.'
c. ¿Quiénes son menos altos que Luis?
'Who are less tall than Luis?'

Superlatives also express quantification over degrees. Absolute superlatives are not associated with an explicit comparison set, whereas relative or comparative superlatives have such an explicit association. Thus, the comparison set of absolute superlatives is normally the default comparison class of the noun denotation. In (73a), it is understood that we are referring to the highest mountain (in the world), whereas the restricted superlative in (73b) makes the comparison class explicit.

(73) a. Pedro escaló la montaña más alta.
'Pedro climbed the highest mountain.'
b. Pedro escaló la montaña más alta de Europa.
'Pedro climbed the highest mountain in Europe.'

Comparative superlatives are not uniform, and several readings emerge as a result of which element is taken to be the basis of the comparison set. In (74a), the comparison set is associated with the subject; in (74b), with the indirect object:

(74) a. Pedro (y no Juan) le dio a Luis el regalo más caro.
'Pedro (not Juan) gave Luis the most expensive present.'
b. Pedro le dio a Luis (y no a sus otros hermanos) el regalo más caro.
'Pedro gave Luis (not his other siblings) the most expensive present.'

Hence, in (74a), the relevant present is the most expensive one in comparison to the presents that other guests (Juan) gave Luis; in contrast, in (74b), the comparison set is the presents that other guests (besides Luis) received. Finally, degree quantification also extends to the realm of non-declarative sentences. Degree *qu*-words (*cuán*, *cuánto*, *cómo de* 'how') are used to express questions about degrees, and exclamative sentences almost always express emotional attitudes about the degree of a property. Sentence (75) requests information about a gradable property of events, namely time. When uttering (76), a speaker normally expresses surprise at the fact that Juan is taller than what was expected:

(75) ¿Cómo de tarde llegó?
'How late was he?'

(76) ¡Qué/cuán/cómo de alto es Juan!
'How tall Juan is!'

8 Conclusion

The study of quantification in natural language in the last three decades has uncovered a great variety of phenomena and structures with numerous connections and implications for specific languages. Although the analysis of the Spanish language has not been at the forefront until relatively recently, in this paper it has been shown that the landscape of quantificational constructions is considerably rich and varied.

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17 Structure of the Verb Phrase

JAUME MATEU

1 Introduction: VP and the lexicon–syntax interface

In this chapter, the structure of the verb phrase (VP) is analyzed. As is well known, the starting point of many works that deal with the lexicon-syntax interface is to assume that the structure of the VP is heavily determined by the lexical semantic properties of the verbal predicate. Two main approaches to the lexical semantics-syntax interface have been put forward in the literature: one based on thematic roles and one based on predicate decompositions (see Levin and Rappaport Hovav 2005 for extensive discussion of both approaches). To exemplify the first approach, a transitive verbal predicate like *romper* 'to break' is said to s (emantically)-select two *theta roles*: Causer and Theme, which are assigned to the external argument (i.e., the argument that is projected external to VP; but see Koopman and Sportiche 1991 for the so-called *VP-internal subject hypothesis*) and the direct internal argument, respectively. Note that direct internal arguments are roughly underlying objects; indirect internal arguments are other VP-internal arguments; and external arguments are underlying subjects. Although in the generative literature theta roles are often taken as mere formal diacritics contained in a *theta-grid* (e.g., *{<u>Causer</u>, Theme}*; cf. Stowell 1981), they have sometimes been mixed with their corresponding contentful thematic relationships put forward by Gruber (1965) or Fillmore (1968), among others. In some approaches to the lexicon-syntax interface, thematic roles have been replaced by aspectual roles (e.g., originator, measurer, terminus, incremental theme, i.a.); cf. Tenny's (1994: 2) Aspectual Interface Hypothesis, according to which "the universal principles of mapping between thematic structure and syntactic argument structure are governed by aspectual properties."

On the other hand, the linguistically relevant lexical semantic properties have been argued to be defined in terms of decompositions of verbal predicates into smaller event structure primitives. For example, the semantics of the verbal

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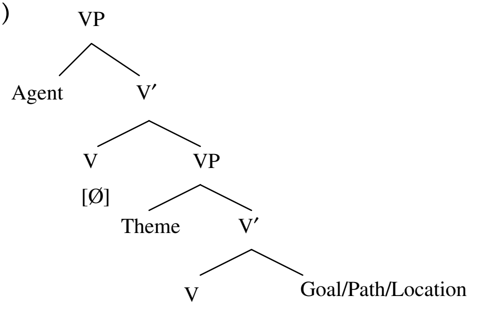
predicate *romper* 'break' can be lexically decomposed roughly as [[X ACT] CAUSE [Y BECOME <*ROTO*>]], where the angular brackets contain the constant *ROTO* 'broken'; the variable X is associated to the external/subject argument, while the variable Y is associated to the direct internal argument via so-called *linking rules* (see Levin and Rappaport Hovav 1995, 2005, among others).

The two abovementioned approaches to the lexicon-syntax interface are often referred to as "projectionist": the syntax of VP is *projected* from the lexical semantic properties of the verb. In contrast, proponents of so-called "constructivist approaches" (e.g., see Borer 2005; Marantz 1997; 2005; and Ramchand 2008, i.a.) argue that abstract syntactic constructions are provided with a structural meaning that is independent from the semantic/conceptual contribution of the verb. The notion of "mapping" from lexicon to syntax or the "linking" of arguments has no meaning in these approaches; instead, the syntax narrows down possible semantic interpretations of predicates and arguments. Cognitive constructionist approaches to argument structure (e.g., see Goldberg's 1995 Construction Grammar approach) must be distinguished from generative constructivist approaches. As pointed out by Ramchand (2008: 11), "the reason constructions have meaning is because they are systematically constructed as part of a generative system (syntactic form) that has predictable meaning correlates."

In this chapter, I concentrate on the structure of the VP and its relation to the lexicon-syntax interface. There is no general consensus with respect to the complexity of the VP: on the one hand, there are those linguists who claim that the syntax of the VP is quite simple and even flat (e.g., see Culicover and Jackendoff 2005); on the other hand, there are other authors, who, following earlier proposals by Larson (1988), argue that there is a more complex structure in the VP than meets the eye. In the next section, the so-called *VP-shell hypothesis* will be applied to Spanish; in particular, I will provide a sketch of how Hale and Keyser's (2002) proposal can be applied to this Romance language (see also Demonte 1994, 2006; Bosque and Masullo 1998; Mateu 2002; Zubizarreta and Oh 2007, i.a.).

The main differences between the two competing approaches, the semanticocentric one and the syntacticocentric one, are due to their different understanding of the locus and characterization of the syntactically relevant semantic properties. The complexity of VP in syntacticocentric approaches typically arises from the syntacticization of the linguistically relevant semantic predicate decompositions. For example, an event structure like [[X ACT] CAUSE [Y BECOME <*BROKEN*>]] turns out to be encoded in syntactic terms: [$_{VP}$ DP [$_{V}$ CAUSE] [$_{VP}$DP [$_{V}$ BECOME] *BROKEN*]] (see Hale and Keyser 1993; Baker 1997; Travis 2000; Mateu 2002; Ramchand 2008, i.a., for the claim that the semantic event structure can be equated to the layered syntactic structure of the VP). Accordingly, syntactically relevant thematic roles can be claimed to be derived from the abstract syntactic decomposition of VP: for example, according to Baker's (1997: 120–121, ex. (76)) syntactic representation in (1), Agent/Causer is the specifier of the higher VP of a Larsonian structure, Theme is the specifier of the lower VP, and Goal/Path/Location is the complement of the lower VP.

(1)



In contrast, in nonsyntacticocentric approaches, thematic structure and event structure are *not* encoded in the VP structure, but in a nonsyntactic semantic component. As a result, a simpler, surface-based VP is assumed (i.e., no VP-shell hypothesis is assumed), but a much more complex syntax-semantics interface module is posited (e.g., see Jackendoff 1990; Culicover and Jackendoff 2005). No generalized uniformity between syntax and semantics is assumed: accordingly, they reject Baker's (1988, 1997) *Uniformity of Theta Assignment Hypothesis* (UTAH), whereby identical thematic relationships between items are represented by identical structural relationships between those items at the first level of syntactic representation. For example, in Jackendovian and Bresnian (i.e., LFG-based) approaches the linking between thematic roles and grammatical structure is not argued to be carried out via the UTAH, but rather via relative ranking systems like, for example, the so-called *thematic hierarchy* (see Levin and Rappaport Hovav 2005 for a complete review).

Given their different assumptions, these two competing approaches of the lexicon-syntax interface are expected to offer different explanations of the so-called *argument structure alternations*, like the ones exemplified in (2). For example, see Demonte (1994), Levin and Rappaport Hovav (1995), Mendikoetxea (1999, 2000, ch. 23), Masullo (1999), and Schäfer (2009), i.a., for the so-called *causative alternation* in (2a–2a′); see Jackendoff (1990), Demonte (1991), Dowty (1991), Mateu (2002), Moreno Cabrera (2003), Beavers (2006), and Mayoral Hernández (2008), i.a., for the so-called *locative alternation* in (2b–2b′); see Franco (1990), González (1997), Parodi and Luján (2000), Vanhoe (2002), and Landau (2010), i.a., for the dative/accusative alternation in psychological verbs in (2c–2c′); see Demonte (1995), Romero (1997), Bleam (2003), Cuervo (2003), and Beavers and Nishida (2009), i.a., for the dative alternation in (2d–2d′) and the locative–dative alternation in (2e–2e′); see Torrego (1989), Mateu and Rigau (2002), and Mendikoetxea (2006), i.a., for the unergative/unaccusative alternation in (2f–2f′).

(2) a. Juan rompió el vaso.
Juan broke the glass.
a′. El vaso se rompió.
the glass refl. broke

b. Juan cargó heno en el carro.
John loaded hay in the cart.

b′. Juan cargó el carro {con/de} heno.
Juan loaded the cart with/of hay.

c. A Juan$_i$ le$_i$ asusta la demencia senil.
to Juan dat.cl. frightens the dementia senile

c′. Los truenos lo asustaron.
the thunders ac.cl. frightened

d. Juan envió la carta a Pepito/a la biblioteca.
Juan sent the letter to Pepito/to the library.

d′. Juan le$_i$ envió la carta {a Pepito$_i$/# a la biblioteca$_i$}
Juan dat.cl. sent the letter to Pepito/to the library.

e. Juan puso el mantel en la mesa.
Juan put the tablecloth in the table.

e′. Juan le$_i$ puso el mantel a la mesa$_i$.
Juan dat.cl. put the tablecloth to the table.

f. Los niños duermen en esta habitación.
The children sleep in this room.

f′. En esta habitación duermen niños.
In this room sleep children.

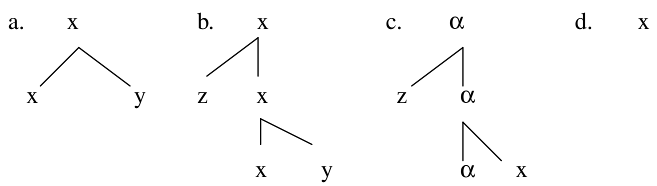
The present chapter is structured as follows: in Section 2, I apply the basics of Hale and Keyser's (2002) syntactic decomposition of VP to Spanish. Following Hale and Keyser (1993), Ramchand (2008), and others, argument structure and event structure can be claimed to be represented in a layered syntactic structure of VP. In Section 3, I analyze a case study in more detail: in particular, I exemplify some relevant contrasts between Spanish and English with respect to their Path/Result structures included in the VP. Section 4 contains some concluding remarks.

2 Argument structure and the syntactic decomposition of VP

In this section, I deal with the syntactic decomposition of the VP in Spanish by using Hale and Keyser's (2002) configurational theory of argument structure. To my view, one of the most notable insights of their program is that two apparently different questions like (1) why there are so few (syntactically relevant) thematic roles, and (2) why there are so few lexical-syntactic categories, have the very same answer (see Hale and Keyser 1993: 65–66; Mateu 2002: ch. 1). Basically, Hale and Keyser's answer is that (syntactically relevant) thematic roles are few since the number of specifier and complement positions of the abstract syntax of l(exical)–syntactic structures is also quite reduced. Interestingly, this paucity of structural positions can be shown to be related to the reduced number of lexical-syntactic categories of the abstract syntax of argument structures. Let us then see how both questions are related.

Argument structure is conceived of by Hale and Keyser as the syntactic configuration projected by a lexical item, that is, argument structure is the system of structural relations holding between heads (nuclei) and the arguments linked to them. Their main assumptions can be expressed as follows: argument structure is defined in reference to two possible relations between a head and its arguments, namely, the head-complement relation and the head-specifier relation. A given head (i.e., *x* in 3) may enter into the following structural combinations in (3): these are its argument structure properties; its syntactic behavior is determined by these properties.

(3)



According to Hale and Keyser (2002), the prototypical or unmarked morphosyntactic realizations of the *x* head in English are the following ones: V in (3a), Prep in (3b), Adj in (3c), and N in (3d). The same can be argued for Spanish (see Mateu 2002: ch. 1).

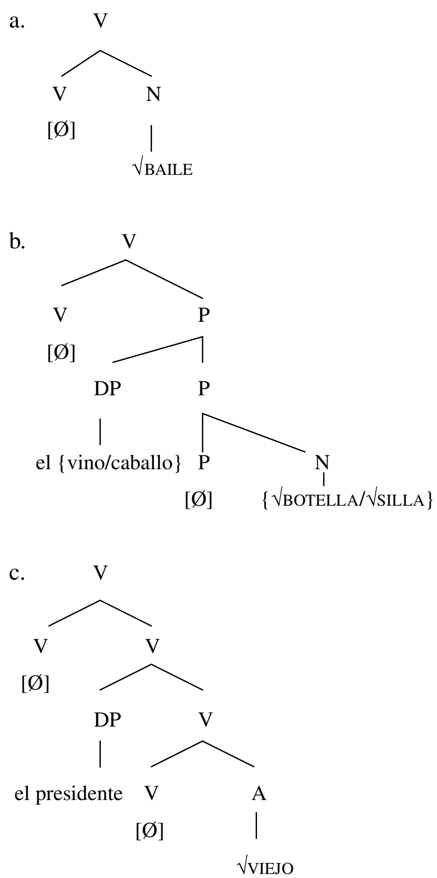
Hale and Keyser's (1993, 2002) fundamental tenet that verbs always take a complement is to be sure another insight of their work. The main empirical domain on which their hypotheses have been tested includes unergative creation verbs like *bailar* 'to dance' (see 4a), transitive location verbs like *embotellar* 'to bottle' (see 4b) or transitive locatum verbs like *ensillar* 'to saddle' (see 4c), and (anti)causative verbs like *envejecer* 'to age' (see 4d). For discussion of the syntax and semantics of these and other classes of verbs in Spanish see Demonte (1994, 2006); Bosque and Masullo (1998); Gumiel et al (1999); Masullo (1999); Mateu (2002); Fábregas (2003); Pérez Jiménez (2003); Gallego and Irurtzun (2006); Fernández Soriano and Rigau (2009), among others.

(4) a. Juan bailó.
Juan danced.
b. Juan embotelló el vino.
Juan bottled the wine.
c. Juan ensilló el caballo.
Juan saddled the horse.
d. Los malos resultados envejecieron al presidente.
The bad results aged the president.
d′. El presidente envejeció.
The president aged.

Unergative verbs are argued to be hidden transitives in the sense that they involve merging a non-relational element (typically, a Noun) with a verbal head (see 5a);

both transitive location verbs (e.g., *embotellar* 'to bottle') and transitive locatum verbs (e.g., *ensellar* 'to saddle') involve merging the structural combination in (3b) with the one in (3a): see (5b). Finally, unaccusative verbs involve the structural combination in (3c), the causative one involving two structural combinations (i.e., the one depicted in (3c) is merged with the one in (3a): see (5c)).

(5)



See Hale and Keyser (2000, 2002), for arguments distinguishing causative constructions (e.g., 5c) from transitive ones (e.g., 5b). For example, only the former enter into the causative alternation (cf. 2a–2a′ and 4d–4d′), a fact that is claimed to be related to its having a double verbal shell. See also Masullo (1999) and Mendikoetxea (1999, 2000) for different explanations of the presence vs. absence of the reflexive pronoun in the Spanish intransitive alternant.

Let us see how the abstract VP structures depicted in (5) are lexicalized into the surface verbs in (4). Applying the incorporation operation to (5a) involves copying the full phonological matrix of the nominal root *baile* 'dance' into the empty one corresponding to the verb. Applying it to (5b) involves two steps: the full phonological matrix of the noun {*botella*/*silla*} is first copied into the empty one corresponding to the preposition. Since the phonological matrix corresponding to the verb is also empty, the incorporation operation applies again from the saturated phonological matrix of the preposition to the unsaturated matrix of the verb. Finally, applying incorporation to (5c) involves two steps as well: the full phonological matrix of the adjectival root *viejo* is first copied into the empty one corresponding to the internal unaccusative verb. Since the phonological matrix corresponding to the external verb is also empty, the incorporation applies again from the saturated phonological matrix of the inner verb to the unsaturated matrix of the external verb.

Hale and Keyser (1993, 2002) argue that the external argument (i.e., the Originator/Initiator role) is truly external to argument structure configurations: for example, they typically appeal to this proposal when accounting for why unaccusative structures can be causativized (e.g., see 4d), while unergatives ones cannot (e.g., **María bailó a Juan* '*María danced Juan'; cf. *María hizo bailar a Juan* 'María made Juan dance'). Accordingly, the external argument can be claimed to occupy the specifier position of a functional projection in so-called "s(entential)-syntax"[1] (cf. Kratzer 1996; Pylkkännen 2008, among others; see also Masullo (1993); Fernández Soriano 1999, for dative/locative subjects that act as external arguments).[2] See Hale and Keyser (2002: ch. 6) on why unaccusatives of the *arrive*-type do not causativize: cf. *Juan llegó tarde* 'Juan arrived late' vs. **María llegó a Juan tarde* 'María arrived Juan late' (cf. *María hizo llegar a Juan tarde* 'María made John arrive late'). See also Zubizarreta and Oh (2007: 192; fn. 14). For s-syntactic/analytic causatives with the light verb *hacer* 'make,' see Zubizarreta (1985) and Torrego (2010), i.a.

As can be inferred from the representations in (5), Hale and Keyser (1993, 2002) argue that the derivational morphology involved in the formation of complex verbs like the denominal and de-adjectival examples in (4) is expressed in *l-syntactic* terms (see also Marantz 1997, 2005 for the explicit claim that derivational morphology is syntax). Accordingly, Hale and Keyser's proposal is that the synthetic examples in (4a), (4b), or (4d) and the analytic ones in (6a-c) have the same basic syntactic argument structure; that is, (5a), (5b), and (5c), respectively.

(6) a. Juan hizo un baile elegante y sensual.
Juan did a dance elegant and sensual.
b. Juan puso el vino en una botella nueva.
Juan put the wine in a bottle new.
c. Los malos resultados hicieron más viejo al presidente.
The bad results made more old the president.

An important tenet of Hale and Keyser's approach is that the semantics of argument structure and event structure can be claimed to be read off the syntactic structures. Four theta roles can be claimed to be read off the basic syntactic argument structures (see Mateu 2002; Harley 2005, i.a.): *Originator* is the specifier of the relevant functional projection that introduces the external argument (at s-syntax, according to Hale and Keyser); *Figure* is the specifier of the inner predicate, headed by P or Adj; *Ground* is the complement of P, and *Incremental*

Theme is the nominal complement of V (Note that Figure and Ground are semantic terms borrowed from Talmy 1985, 2000). Concerning the semantic functions associated to the eventive element (i.e., V), DO can be read off the unergative V, CAUSE can be read off the V that subcategorizes for an inner predicative complement, and CHANGE can be read off the unaccusative V. See also Marantz (2005: 5) for the claim that one does not have to posit "cause, become or be heads in the syntax … Under the strong constraints of the theoretical framework, whatever meanings are represented via syntactic heads and relations must be so constructed and represented, these meanings should always arise structurally."

One caveat on "coverage" is in order here. At this point one could wonder why Hale and Keyser (1993, 2002) discussed essentially only two verb classes; that is, those verbs that have a nominal complement (e.g., unergatives) and those ones whose members are characterized by the appearance of an internal predication and therefore an internal subject (e.g., the steadfastly transitive locative verbs and the alternating de-adjectival causative/inchoative verbs). At first sight, it seems that their theory only deals with a very small fraction of the total range of verb types. However, this apparent lack of coverage should not prevent one from realizing that, at an abstract level, most of descriptive verb types can be claimed to be reduced to those two *basic* ones: those ones that consist of V plus a nominal (*N*) complement and those ones that consist of V plus a predicative (*P* or *Adj*) complement. Recall also that causatives consist of a Larsonian V plus a verbal (V) complement: cf. (5c). The latter disjunction could in fact be eliminated assuming Mateu's (2002) and Kayne's (2009) proposal that Adjectives are not a primitive category, but are the result of incorporating a Noun (i.e., a Ground) into an adpositional marker. Similarly, assuming a Hale and Keyserian-like framework, Erteschik-Shir and Rapoport (2007: 17–18) point out that "the restricted inventory of meaning components that comprise verbal meanings includes (as in much lexical research) Manner (=means/manner/instrument) (M), State (S), Location (L) and, as far as we are able to tell, not much else" (2007: 17). Also, "Each such semantic morpheme has categorical properties … MANNERS project N, STATES project A and LOCATIONS project P . . . The restricted inventory of verbal components parallels the restricted inventory of lexical categories, restricting verb types and consequently possible interpretations, following Hale and Keyser" (2007: 18).

As an example of their sticking to the very restrictive system sketched out above, Hale and Keyser (2002: 37f) analyze agentive atelic verbs like *empujar* 'push' as involving a transitive structure like the one assigned to locative verbs (i.e., (5b)); for example, *empujar el carro* could be paraphrased as *[$_V$PROVIDE [$_P$the cart WITH a push]]* (cf. Sp. *darle un empujón* 'give it a push').[3] More controversially, Hale and Keyser (2002: 214-221) claim that stative verbs like *costar* 'cost' or *pesar* 'weigh' could be analyzed as involving the same configuration associated to copula *ser* 'be': cf. *[$_V$Este toro pesa una tonelada]* 'This bull weighs one ton' and *[$_V$Este toro es bravo]* 'This bull is brave.'

For reasons of space, in the remainder of this section, I will limit myself to pointing out some relevant properties that distinguish those verbs that have an unergative structure, which take a nominal complement, from those ones that have a predicative (PP/AP) complement. When dealing with the latter verbs, I will also discuss Hale and Keyser's (1993, 2002) famous distinction between central vs. terminal coincidence relations. Finally, I will conclude the present section by summarizing Haugen's (2009) insightful revision of the important distinction between those VPs that involve Incorporation and those VPs that involve Conflation, which will pave us the way to deal with the case study selected in Section 3: result/path structures in the VP.

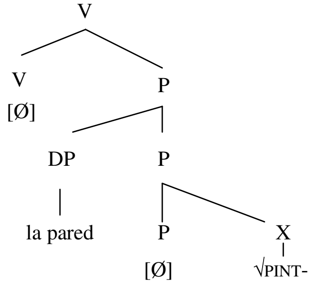
One advantage of Hale and Keyser's program is that it sheds light on the syntactic argument structure commonalities that can be found in apparently different lexical semantic classes of verbs: for example, creation verbs (e.g., 7a) and consumption verbs (e.g., 7b) can be argued to be assigned the unergative structure in (5a). (See Hale and Keyser 2002 for the claim that unergatives typically express creation or production (DO X); see also Volpe 2004, for empirical evidence that consumption verbs are unergatives.) Since these verbs can incorporate their complement, it is predicted that their object can be null. By contrast, the inner subject of change of location/state verbs cannot be easily omitted (see Hale and Keyser 1993 and subsequent work for the claim that specifiers/subjects do not incorporate) (cf. (7c–e)). Accordingly, a crucial syntactic difference is established between Incremental Theme (i.e., the complement of an unergative structure) and Figure (i.e., the subject/specifier of an inner Locative (Prep) or Stative (Adj) predication). See Zagona (ch. 18) for so-called "aspectual se" (e.g., cf. *Juan se comió la pizza* 'Juan refl. ate the pizza').

(7) a. Juan pintó (un cuadro).
Juan painted a picture.
b. Juan comió (una pizza).
Juan ate (a pizza).
c. Juan embotelló ??(el vino).
Juan bottled the wine.
d. Los malos resultados envejecieron *(al presidente).
The bad results aged the president.
e. Los fuertes vientos rompieron *(la ventana).
The strong winds broke (the window).

Let us next deal with some relevant aspectual differences among those verbs that involve a predicative complement. As pointed out above, both change of location verbs and change of state verbs involve an inner predication that is represented via a syntactic Figure-Ground organization. Although Hale and Keyser (2002: 224) point out that "in general, aspect is orthogonal to argument structure," Mateu (2002) puts forward some correlations between these two domains by using their semantic notions of central and terminal coincidence relations: for example, atelic predicates like *empujar* 'push' involve an inner predication headed by a central coincidence P, whereas telic change predicates like *romper* 'break' involve an inner predication headed by a terminal coincidence relation. According to Hale and Keyser (1993, 2002), a terminal coincidence relation involves a coincidence between one edge or terminus of the Figure's path and the Ground, while a central coincidence relation involves a coincidence between the center of the Figure and the center of the Ground. Therefore, the aspectual ambiguity of an example like the one in (8) is accounted for by positing a central P for the atelic case and a terminal P for the telic one: see (9).

(8) Juan pintó la pared {durante/en} diez minutos.
Juan painted the wall {for/in} ten minutes.

(9)



I conclude this section by pointing out an important difference between *Incorporation* processes vs. *Conflation* processes. According to Haugen (2009), there are two ways of forming denominal verbs, that is, via Incorporation or via Conflation:

(10) *Incorporation* is conceived of as head-movement (as in Baker 1988; Hale and Keyser 1993), and is instantiated through the syntactic operation of Copy, whereas *Conflation* is instantiated directly through Merge (compounding).
Haugen (2009: 260)

In Incorporation cases, the denominal verb (e.g., see 5a: *Juan bailó* 'Juan danced') is formed via *Copying* the relevant set of features of the nominal complement into the null verb. In contrast, in Conflation cases, the denominal verb (e.g., see 11a) is formed via *Compounding* a nominal root with the null verb (see 11b).

(11) a. The factory horns sirened midday and everyone broke for lunch.
(ex. from Clark and Clark 1977, *apud* Borer 2005)
b. $[_{FP}$ $[_{DP}$ the factory horns$]$ … $[_{V}$ $[_{V}\sqrt{\textit{SIREN}}$ V$]$ $[_{DP}$ midday$]]]$

I will explain below why those denominal verbs formed via Conflation are not possible in Spanish nor, more generally, in any Romance language. As we will see in Section 3, Conflation accounts for the way the so-called *Manner component* is introduced in English resultative-like constructions like the ones exemplified in (12), which are all impossible in Spanish/Romance (see Mateu 2002, 2010; Mateu and Rigau 2002; McIntyre 2004; Harley 2005; Mateu and Espinal 2007; Zubizarreta and Oh 2007, i.a.). In the next section, I will deal with some relevant l-syntactic differences between Spanish and English that have to do with Talmy's (1991, 2000) famous bipartite typology of motion events and Snyder's (2001) Compounding paramenter.

(12) a. John danced into the room.
b. John danced away.
c. John danced the puppet across the stage.
d. John danced the night away.
e. John outdanced Mary.
f. John danced his butt off.
g. John danced his feet sore.
h. John danced his way into a wonderful world.

3 Paths and results within the syntactic decomposition of VP

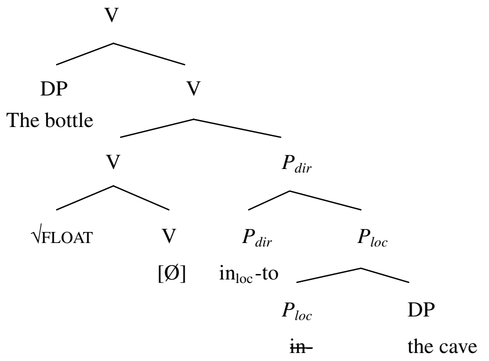
As is well-known, Talmy (1991, 2000) points out that the Germanic family belongs to the class of so-called *satellite-framed languages* (i.e., languages where path is typically expressed as a "satellite" around the verb, e.g., as a directional particle or prefix), whereas the Romance family belongs to the class of *verb-framed languages* (i.e., languages where path is typically incorporated into the verb). Consider some paradigmatic examples of his typology in (13), where Path is encoded into a preposition in the English example in (13a), but is incorporated into the verb in the Spanish one in (13b):

(13) a. The bottle floated *into* the cave.
b. La botella *entró* en la cueva flotando.
the bottle entered in the cave floating
'The bottle entered the cave floating.'

To use Talmy's (1985) terms, (13a) involves conflation of Motion with Manner, or alternatively, in Talmy's (1991) terms, (13a) involves conflation of MOVE with a SUPPORTING[EVENT]. In contrast, the corresponding counterpart of (13a) in a Romance language like Spanish (see 13b) involves a different lexicalization pattern, that is, Motion is fused with Path, the Manner component (or the so-called *Co-event*) being expressed as adjunct.

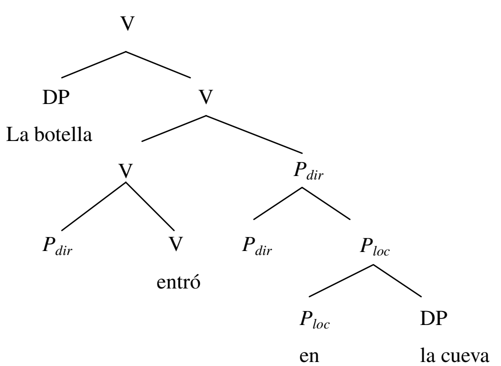
Let us show how the VP patterns exemplified in (13) can be analyzed in the present syntactic framework. The relevant syntactic analysis in (14) involves the unaccusative combination in (3c), where the inner predicate is not an Adjective but a PP (*into the cave*), which takes an inner subject (*the bottle*), thanks to the intervention of a host Verb (i.e., the morphosyntactic realization of α in 3c). This unaccusative verb can be assigned the constructional meaning of CHANGE (see Hale and Keyser 2002; Zubizarreta and Oh 2007). Concerning the double P involved in (14), P_{dir} (spelled out by *to*) corresponds to Hale and Keyser's terminal coincidence relation, while P_{loc} (spelled out by *in*) corresponds to their central coincidence relation.

(14)



The Spanish example in (13b) is analyzed in (15), where the incorporation of the complex directional P element into a null unaccusative verb gives a Path verb: *entrar* 'enter.' The adjunct *flotando* 'floating' is to be merged outside the basic argument structure. As can be inferred from the English translation of the Spanish example in (13b), a similar analysis holds for the Romance pattern of *enter the cave floating* (*versus* cf. the Germanic pattern: *float into the cave*). Accordingly, the Path lexicalization pattern is also found in (mainly, the Romance lexicon of) English; for additional discussion of other apparent counterexamples, see Talmy (1985, 2000).

(15)



The so-called *Co-event pattern* of Germanic languages is to be related to the fact that, for example, the complex P element *into* in (14) is not incorporated in the verb, this null verb being then allowed to be conflated/compounded with the root √FLOAT, which expresses Talmy's co-event. As a result, the unaccusative V(erb) in (14), whose configurational or constructional meaning is also that of CHANGE, turns out to be associated with an additional embedded meaning, that of *floating*.

According to Mateu and Rigau (2002), the Co-event pattern in the English example in (14) involves a V-V compound. Similarly, Zubizarreta and Oh (2007) provide an interesting account of the differences between Germanic and Romance by relating them to Snyder's (2001) observation that Germanic, but not Romance, has productive and compositional root compounding (N-N; e.g., *banana box*). For example, Zubizarreta and Oh (2007) argue that Romance cannot use the Compound Rule "Merge two lexical categories of the same categorical type" to compose Manner and Directed Motion in the way Germanic does. Assuming that Conflation can be redefined as a subtype of Compounding (see Haugen 2009 and Section 2, above) and that the Talmian Co-event pattern always involves Conflation, an interesting connection can be said to emerge between Talmy's (1991, 2000) typology and Snyder's (2001: 328) Compounding Parameter, initially formulated as in (16):

(16) The grammar {disallows*, allows} formation of endocentric root compounds during the syntactic derivation.
[*unmarked value]

Given (16), Snyder (2001: 329) put forward an initial correlation between productive root compounding (e.g., N-N compounds like *frog man* or *banana box*) and complex predicate formation (e.g., complex resultatives and verb–particle constructions), which cannot be maintained *stricto sensu*: for example, novel endocentric compounds can be created at will in Basque, but, contrary to Snyder's expectations, this language lacks complex resultatives (see also Beavers et al. 2010 for more counterexamples). This notwithstanding, there is a residue for the validity of Snyder's Compounding parameter iff reformulated in the following reduced sense; that is, (dis)allowance of Conflation of a root with a null light verb. This accepted, there emerges an interesting connection between Talmy's (1991, 2000) typological Co-event pattern and the Conflation/ Compounding Parameter in (17). For example, given (17), the unmarked value corresponding to Romance languages would account for the fact that Conflation cases like those ones exemplified in (11), (12), and (13a) are not possible in Spanish.

(17) The grammar {disallows*, allows} compounding of a root with a null light verb during the syntactic derivation.
[*unmarked value]

To conclude this section, let us review some qualifications and criticisms that have been put forward against Talmy's (1985, 1991, 2000) typology (e.g., see Beavers et al. 2010 for a recent review). For reasons of space, I will deal with some alleged exceptions to Talmy's claim that Romance languages are "verb-framed." For example, according to Beavers et al. (2010), *until*-markers in motion events present satellite-framed behavior since the goal is expressed via a PP (e.g., 18):

(18) La botella flotó hasta la cueva.
The bottle floated until the cave.

However, the existence of examples like the one in (18) is not problematic since, according to Mateu (2002), the syntactic notion of Path/Result that is relevant to Talmy's typology is *not* the adjunct one but the one heading the argumental Small Clause-like PP in constructions like (14). There are arguments for claiming that *until*-markers are adjuncts in the relevant cases: for example, its presence in Italian examples like the one in (19a) does not involve auxiliary BE-shift, which shows that the argument structure involved in (19a) is not the unaccusative one in (14) but the (irrelevant) unergative one plus an adjunct PP. Similarly, Aske's (1989) qualification that atelic paths like the one encoded by *hacia* 'towards' are compatible with manner of motion verbs in verb-framed languages like Spanish (see 19b) is also coherent with the fact that they are adjuncts (see the Italian example in 19c), whereby they can be argued to be irrelevant to Talmy's typology.

(19) a. Gianni {*ha*/**è*} camminato fino a casa. (Italian)
Gianni {has/is} walked until to home.
b. Juan caminó hacia/hasta el mar. (Spanish)
Juan walked towards/up-to the sea.
c. Gianni {ha/*è} camminato verso il mare. (Italian)
Gianni {has/*is}walked towards the sea.

Furthermore, Spanish examples like those ones exemplified in (20) should *not* be taken as true counterexamples to Talmy's typology, as is often claimed (e.g., see Martínez Vázquez 2001 and Beavers et al. 2010).

(20) a. … deslizándose a las habitaciones de las bailarinas…
slipping prep. the rooms of the dancers
'… slipping into the dancers' rooms …'
b. Se arrastró a su lado …
refl. crawled prep. her/his side
'S/he crawled to her/his side …'
c. … saltó a su lado.
jumped prep. his/her side
'… s/he jumped to his/her side.'
d. … volaron a Mar de Plata …
flew prep. Mar de Plata
'They flew to Mar de Plata …'
Ex. taken from Martínez Vázquez (2001)

It is a mistake to analyze the examples in (20) as instantiations of the Germanic Co-event pattern exemplified in (14): for example, one would otherwise expect the well-formedness of examples like the ones in (21), contrary to fact. Given the contrast between (20) and (21), it seems natural to conclude that what is at stake here is the contrast between *directional* manner verbs (e.g., those in 20) and *pure* (i.e., nondirectional) manner verbs (e.g., those in 21).

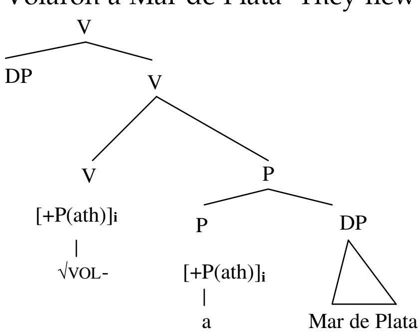
(21) a. *Juan bailó a la cocina.
Juan danced prep. the kitchen
'Juan danced to the kitchen.'
b. *Juan cojeó a la puerta.
Juan limped prep. the door
'Juan limped to the door.'

Unlike the examples in (18) and (19), those ones in (20) cannot be analyzed as unergative structures plus a prepositional adjunct since there is empirical evidence that the examples in (20) do involve an unaccusative structure. For example, the auxiliary *essere* 'be' is selected in their Italian counterparts in (22).

(22) a. Gianni {è/*ha} volato a Mar de Plata (cf. *Gianni {ha/*è} volato molto*)
Gianni {is/*has} flown prep. Mar de Plata
b. Gianni {è/*ha} saltato dalla finestra (cf. Gianni *{ha/*è} saltato molto*)
Gianni {is/*has}jumped from-the window

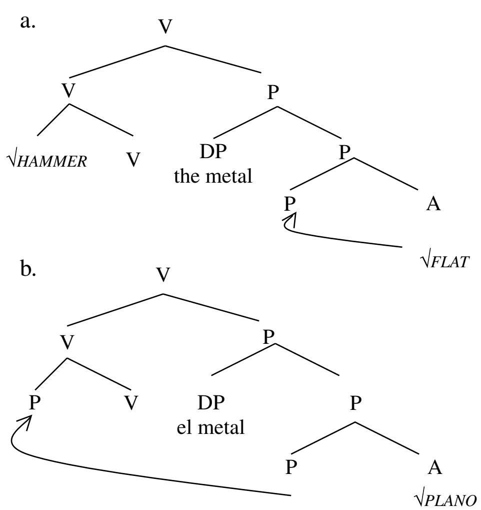
The examples in (20) and (22) do not involve the Conflation analysis exemplified in the Germanic Co-event pattern in (14), but rather the Incorporation analysis in (23). In particular, they can be analyzed as cases where the PP specifies the abstract P(ath) that has been incorporated (i.e., Copied) into the motion verb. This analysis can then be claimed to account for the abovementioned restriction that the verb cannot encode pure manner (e.g., see 21).

(23) Volaron a Mar de Plata 'They flew to Mar de Plata'



Finally, it is interesting to show that Haugen's (2009) distinction between Conflation vs. Incorporation can also be applied to adjectival resultative constructions. For example, the insertion of the nominal root √*HAMMER* in (24a) does involve Conflation/Compounding (see 25a), whereby these complex constructions are expected to be impossible in Spanish and, more generally, in Romance: see (24b). In contrast, simple resultative constructions like (24c) can be argued to involve Incorporation of the abstract P(ath) in (25b) into the light verb (see Mateu 2002, for more details).

(24) a. María hammered the metal flat.
b. *María martilleó el metal plano (*on the resultative reading)
c. María dejó el metal plano (con un martillo).
María got the metal flat with a hammer

(25)


For the claim that an abstract null P(ath) must be represented in adjectival resultative constructions, see Mateu (2005); see Demonte and Masullo (1999) for the distinction between resultative and depictive predicates. For example, *María dejó a Juan cansado* ('tired') is ambiguous: cansado can be resultative (i.e., 'María caused Juan to become tired') or depictive (i.e., 'María abandoned Juan while he was tired'). The adjective is part of the basic argument structure in the first case (as depicted in 25b), while it is an adjunct in the second case.

Furthermore, cognate resultative constructions like the one exemplified in (26a) are quite productive in Spanish (see Demonte 1991; Demonte and Masullo 1999, i.a.). As expected, (26a) would not involve Conflation but rather Incorporation: assuming Haugen's (2009) definition in (10), the latter process involves *Copy*. Finally, it has been shown that examples like (26b) do not constitute a true kind of resultative construction (see Mateu 2002; Levinson 2010, for so-called *pseudo-resultatives*).

(26) a. Juan lavó la camisa bien lavada.
Juan washed the shirt well washed
b. Juan cortó la carne fina.
Juan cut the meat thin.

All in all, most of apparent counterexamples to Talmy's typology do not seem to call his main descriptive generalization into question. Spanish (and more generally, Romance) lacks complex resultative(-like) constructions that involve

Conflation (e.g., see the examples in (12)). Despite many criticisms and qualifications (e.g., see Beavers et al. 2010), Talmy's typology wins out with an incredible force: [*pure/nondirectional* Manner verb + Result Predication] constructions that involve Conflation are predicted to be systematically absent from Spanish/Romance.

4 Concluding remarks

In this chapter, it has been claimed that the structure of VP is not flat (*pace* Culicover and Jackendoff 2005), but can be decomposed into different "layers/ shells" (see Larson 1988; Hale and Keyser 1993f.; Demonte 1994, 1995; Harley 1995, 2005; Baker 1997; Mateu 2002f.; Zubizarreta and Oh 2007; Ramchand 2008, i.a.). Following Hale and Keyser (1993f), different VP structures (e.g., unergative, unaccusative, transitive, and causative structures) can be claimed to be associated with different abstract structural meanings, which must be separated from the conceptual/encyclopedic meanings provided by the roots (see also Mateu 2002 and Ramchand 2008, i.a., for the crucial distinction between syntactically transparent structural meanings vs. syntactically nontransparent encyclopedic/ conceptual meanings). The heads of VPs are "light" (cf. CAUSE, DO, CHANGE, etc), which can be claimed to be associated with lexical/encyclopedic roots via Incorporation (i.e., the process involved in examples like (4)) or via Conflation (i.e., the process involved in examples like (12)). Spanish and, more generally, Romance can be shown to lack this second type of composition. Assuming that Conflation can be redefined à la Haugen (2009) (see Section 2) and that Talmy's (1991, 2000) Co-event pattern always involves Conflation/Compounding (see section 3), an interesting connection can be said to emerge between Talmy's (1991, 2000) bipartite typology of motion events and Snyder's (2001) Compounding Parameter.

NOTES

1 According to Hale and Keyser, the term *s-syntax* is used to refer to the syntactic structure assigned to a phrase or sentence involving both the lexical item and its arguments, and also its "extended projection" (Grimshaw 1991, 2005), and including, therefore, the full range of functional categories and projections implicated in the formation of a sentence interpretable at PF and LF.
Concerning so-called *applicatives* (see Cuervo 2003; Pylkkännen 2008, i.a.), the high ones can be claimed to appear at s-syntax (e.g., (ia), where the applicative relates an individual, *Juan*, to a static situation), while the low ones (e.g., (ib), where the applicative relates two objects: *Juan* and *beso*) could correspond to inner configurations similar to the Prepositional one involved in (5b). For example, the l-syntactic analysis of *María gave Juan a big kiss* is *(María)* [$_V$PROVIDED [$_P$*Juan* WITH *a big kiss*]].

(i) a. A Juan *le* gustan los libros.
to Juan dat.cl. like.3pl the books

b. María *le* dio a Juan un beso enorme
María dat.cl. gave to Juan a kiss big
(cf. *María gave Juan a big kiss*)

2 Hale and Keyser do not discuss the status of *adjuncts*, since those constituents are not to be encoded in basic argument structures like those ones in (5). Furthermore, for a preliminary discussion of the l-syntax of so-called "complemento preposicional/de régimen verbal" lit. "verbally governed PP complement" (cf. i), see Simoni (2005) and Gallego (2010). See also Demonte (1991: ch. 2).

(i) a. Juan insistió (en hacerlo).
Juan insisted on doing it
b. Su trabajo consiste *(en corregir pruebas).
her work consists in proof-reading

3 A different track is the one pursued by Harley (2005), who assumes the nontrivial claim that roots (nonrelational elements) can take complements (vs. cf. Mateu 2002). Accordingly, *push* verbs are, for example, analyzed by Harley (2005: 25 ex. (25)) as in (i), where the root *push* is claimed to take *the cart* as complement:

(i) Sue pushed the cart: $[_{v}\text{DO} [_{\surd P}$ push the cart]]

Interestingly, Harley's syntactic distinction between structural arguments (i.e., introduced by relational elements) and non-structural arguments (e.g., complements of root) can be said to have a nice parallel in Levin and Rappaport Hovav's (2005) semantic distinction between event structure arguments vs. mere constant participants.

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18 Tense and Aspect

KAREN ZAGONA

1 Introduction

Traditional grammars usually approach the topics of tense and aspect by describing the meanings of the tense and aspect morphemes of the language. Since languages vary in their morpheme inventories, there has been little basis in traditional grammars for considering theoretical issues: that is, questions concerning what is a possible tense, how tense and aspect interact, and why rules of interpretation vary from one context to another or from one language to another. Recent research has focused considerable attention on these general questions by investigating how temporal information is encoded in functional categories and how temporal meaning is built up compositionally in sentences. Spanish offers interesting perspectives on these issues, especially in cases where tense and aspect intersect, as in the preterite and imperfect past tenses and compound perfect tenses.

To begin with, it is useful to sketch some basic concepts and terminology. Tense and aspect encode temporal information about events. Their values are relational, in the sense that they situate an event relative to an external perspective. **Tense** characterizes the temporal location of an event (past, present, future) relative to an external time of evaluation, such as the "moment of speech" that serves as a default deictic center in main clauses: *canté* 'I sang,' *canto* 'I sing,' *cantaré* 'I will sing' are understood as preceding, coinciding with, or following the moment of speech. Spanish **compound tenses** are constructed periphrastically with auxiliary *haber* 'have' followed by a past participle (*Juan ya había comido*, 'Juan had already eaten'). Compound tenses are analyzed in some approaches as tenses, in other approaches as involving tense and aspectual relations (Reichenbach 1947; Comrie 1975, 1985; Hornstein 1990; Iatridou et al. 2001; Carrasco Gutiérrez 2008).

Aspect characterizes the boundaries of an event – its beginning and end – relative to an external temporal frame. **Perfective** aspect includes the beginning and end of the event within that frame. This is illustrated by the preterite past tense, as in:

The Handbook of Hispanic Linguistics, First Edition. Edited by José Ignacio Hualde, Antxon Olarrea, and Erin O'Rourke.

María plantó un árbol 'Mary planted a tree.' The preterite depicts the event of tree-planting from its onset through to its endpoint. The entire event – including its beginning point and endpoint – is included within the external aspectual timeframe of the sentence. **Imperfective** aspect excludes event boundaries (Smith 1997). Progressive morphology, such as *María estaba plantando un árbol* 'Mary was planting a tree,' illustrates the imperfective interpretation; it depicts only a stage of the event, without including its endpoints. Similarly, the imperfect past tense, *María plantaba un árbol* 'Mary was planting a tree/used to plant a tree,' does not include the event boundaries within the external timeframe. It can refer to one or more events, excluding some endpoints. Aspectual morphology of the clause thus specifies an external viewpoint on events, based on reference to event boundaries.

A distinction is often drawn between lexical aspect and grammatical aspect (de Miguel 1999). **Grammatical aspect** relates events to an external viewpoint, as discussed above. **Lexical aspect** categorizes events in terms of their temporal properties, such as punctual versus durative, stative versus non-stative, and telic versus atelic. **Telicity** concerns the presence or absence of a natural endpoint to an event. Events with endpoints are compatible with *in*-adverbs: *Juan limpió la casa en una hora* 'Juan cleaned the house in an hour'. Lexical aspect is sometimes analyzed as an idiosyncratic feature of individual verbs; however, it is sensitive to features of other constituents, including arguments and some adjuncts (Vendler 1967; Comrie 1976; Smith 1997; Verkuyl 1999). For example, *saber la lección* 'know the lesson' is stative, but *saber algo de repente* 'suddenly know something' is a punctual change of state. The discussion of this chapter begins with theoretical considerations below, and then reviews empirical issues for the analysis of Spanish tenses (Sections 3–5) and aspect (Section 6).

2 Overview of tense

Two different approaches have been taken to temporal reference in the formal semantics literature: the tense logic approach, which analyzes tenses as operators, and the referential approach. In the tense logic approach, a tense operator introduces a time variable whose value is supplied by semantic rules; in this approach, "times" are introduced in the semantic representation, not in the linguistic structure of the object language. The referential approach is developed in Reichenbach's theory of tense, and, independently, in work by Partee (1973), who showed that tenses behave like pronominals, in that they pick out contextually relevant times. Studies within the referential approach to tense have explored where in clause structure *times* are introduced and what grammatical and semantic features are needed to capture the temporal relations between times (see Enç 1987; Hornstein 1990; Zagona 1990, 1995; Stowell 1993; Demirdache and Uribe-Etxebarria 2000; Thompson 2005).

Concerning the structure of tenses, it is useful to take as a starting point the framework of Reichenbach (1947), which makes strong claims about the uniformity of tenses, both language-internally and universally. Reichenbach claims that all tenses are composed of three times, related by one of two semantic relations:

(1) Times:
  a. Speech-time (*S*)
  b. Reference-time (*R*)
  c. Event-time (*E*)

(2) Relations
  a. Precedence
  b. Simultaneity

The three times in (1) are related to one another by either a precedence relation (shown as '_') or a simultaneity relation (shown as a comma: ','). **Speech-time** (*S*) refers to the time at which a sentence is uttered; **Event-time** (*E*) refers to the "run-time" of an event. For example, in *Juan salió a las tres* 'Juan left at three o'clock,' *S* is the time at which the sentence is uttered, and *E* is the time of Juan's leaving – specified as 3 o'clock in this example. **Reference-time** (*R*) is a third time, claimed to be present in all tenses. Its different relationships with *E* and *S* can be seen in the simple tenses in (3) versus the compound perfect tenses in (4):

(3)	a.	*Cantó*	Preterite Past	E,R_S
	b.	*Cantaba*	Imperfect Past	E,R_S
	c.	*Canta*	Present	S,R,E
	d.	*Cantará*	Future	S_R,E

(4)	a.	*Había cantado*	Past perfect	E_R_S
	b.	*He cantado*	Present perfect	E_R,S
	c.	*Habré cantado*	Future perfect	S_R,E

In (3) and (4), the past, present, or future value of the tense is determined by the relationship between *S* and *R*. The difference between the simple and compound perfect tenses is captured in terms of how *R* and *E* are related. In the simple tenses, *R* and *E* coincide; in the compound perfect tenses, *E* precedes *R*. The past perfect thus contains two precedence relations while the simple past tense contains just one. This is supported by the ambiguity of sentences like (5), where the time adverb may refer to either *R* or *E:*

(5) Juan había almorzado a las tres.
  'Juan had eaten (lunch) at 3:00.'
  (i) 3:00 is the time when Juan ate lunch. (adverb modifies E)
  (ii) 3:00 is the time after Juan ate lunch. (adverb modifies R)

In Reichenbach's analysis, *R* is a constituent of every tense. Its presence is not detected directly in simple tenses, because it coincides with *E*. Its presence is only made obvious in contexts where it refers to a time that is distinct from *S* and *E*, such as the compound perfect tenses and the conditional, as shown by comparison between the future and conditional tenses in (6):

(6) a. Juan dice que José **cantará** a las tres. (Future *S_R,E*)
'Juan says that José will sing at 3:00.'
b. Juan dijo que José **cantaría** a las tres. (Conditional *R_E_S*)
'Juan said that José would sing at 3:00.'

In (6a), the tense structure for *cantará* is one in which *R* and *E* coincide; in (6b), the tense structure for *cantaría* is one in which the Event-time of Juan's singing is understood as in the future relative to a past reference time *R* (understood as the same as the *R,E* time of the main clause).

The claim that the three primitives – S, R, and E – occur in all tenses implies that there is no theoretical distinction between simple and compound tenses; they differ only in the semantic relationship between E and R, as noted above. The system is inherently restricted to tenses with three times because there is no mechanism of iterating times; more complex tense structures like those in (7) are impossible:

(7) a. * E_R_R_S (past past perfect)
b. * R_S_R_E (past future perfect)

The non-existence of tenses such as (7) is claimed not to be an accidental property of particular grammars but rather evidence that there are only three time primitives and no operations that introduce recursion in the system.

Reichenbach's framework gives rise to a system of ternary tense distinctions. Any pair of times *x,y* can be related in three distinct ways: *x_y*; *y_x*; *x,y*. In this approach, past, present, and future tenses are all separate, and none of the tenses is expected to form a natural class with one of the others. However, in some respects it seems that a binary distinction between past and nonpast is needed. This is suggested by the fact that the present/future distinction is not maintained morphologically in the subjunctive, and also, even in the indicative, non-past tenses vary in their present and future values, as shown by the compatibility of present tense with both present and future adverbs (*Juan enseña ahora/mañana* 'Juan teach-PRESENT now/tomorrow')

Research on tenses within the referential framework has argued that times are represented in the functional categories of clause structure (Enç 1987; Hornstein 1990; Zagona 1990, 1995; Stowell 1993; Giorgi and Pianesi 1997; Guéron and Hoekstra 1998; Demirdache and Uribe-Etxebarria 2000, 2004; Guéron 2004). For concreteness, let us take the property of reference to time as a feature, [TIME], which is distributed on certain heads in clause structure. The times *S* and *R* in the Reichenbach system may be analyzed as features of the CP and *v*P phase edges. Enç (1987) provided evidence that the evaluation time for a tense (*S* in the Reichenbach framework) is encoded in CP. Assuming Rizzi's (1997) analysis, the highest position of the left periphery is Force, and the lowest head (above TP) is the Finite phrase:

(8) CP phase: *v*P phase (preliminary):
Force – (Topic) – (Focus) – (Topic) – Finite – TP *v*P
$[\text{TIME}]_S$ $[\text{TIME}]_R$

The details of the *v*P phase will be discussed below; in the CP phase, both Force and Finite phrase have features related to tense. The Force phrase represents features for clause-typing, and for the entity responsible for it. For example, a main clause declarative has a both a [declarative] feature and a [speaker] feature, which represents the information that the speaker is responsible for (or participant in) the assertion of the sentence. The Finite phrase specifies mood; indicative mood, which is the focus of this chapter, implies that the mode of evaluation is temporal, in relation to the [TIME] feature of the Force phrase. The Finite phrase selects a TP whose features locate *v*P in relation to the external evaluation time of the clause. Thus, the time *R* is located relative to the time *S*, by virtue of features that are distributed on *v*P, Tense, Finite, and Force heads.

One consequence of encoding times in clause structure in the manner suggested above is that the semantic primitives *S*, *R*, and *E* may be replaced by a single time primitive, the feature [TIME]; the distinctions between *S*, *R*, and *E*, and variations in interpretation that each may have in certain contexts would then derive from the configuration in which the [TIME] feature appears, and from the values of other features associated with it in a given context. *S* is the value of a [TIME] feature in CP, *R* and *E* are values of [TIME] features of the verb phrase and its functional projections. Such an approach may account for variations in how [TIME] features are interpreted, such as the fact that *S* is not interpreted as the time of speech in some contexts. In main clauses, *S* is the time of the speaker's declarative speech act; in a complement clause embedded under a verb of communication (Giorgi and Pianesi 2000), *S* is the time of the main clause event. This is illustrated by (9):

(9) a. Juan dijo ayer que María había llegado el jueves.
'Juan said yesterday that Maria had arrived Thursday.'
b. [Force *Juan* [<*Juan*> *dijo* [Force *que María había llegado el jueves*]]]
$[\text{TIME}]_S$ $[\text{TIME}]_S$

In the main clause of (9), past tense is evaluated relative to Speech-time, which is the time at which the speaker engages in the speech act. However the past tense in the complement clause in (9) is not evaluated relative to Speech-time but rather to the time of the main clause event of saying. The interpretation of *S* in each clause is determined by its syntactic context: in the main clause, *S* has a pragmatically determined referent; in the complement clause, it has a syntactically determined referent in the main clause.

A second consequence of encoding times in clause structure in the way suggested here is that temporal relations hold between pairs of times in local relationships.

If *x*, *y*, and *z* are features of heads, locality considerations prohibit *x* from entering into an ordering relation with *z* if *y* is a more local time that intervenes between *x* and *z*. Thompson (2005) argues on the basis of locality that *R* is structurally higher than *E*. She proposes that *R* is encoded in an Aspect Phrase above the verb phrase, and *E* is encoded in the verb phrase itself:

(10)	Aspect	*v*	V
		[TIME](=*R*)	[TIME](=*E*)

The structural asymmetry between *R* and *E* is shown by differences in interpretation between predicate-internal and preposed adverbs. This is illustrated for Spanish by the different readings of the adverb *a las tres* 'at 3:00' in (11) versus (12):

(11) Juan había almorzado a las tres.
'Juan had eaten (lunch) at 3:00.'
(i) 3:00 is the time Juan ate lunch. (adverb modifies E)
(ii) 3:00 is the time after Juan ate lunch. (adverb modifies R)

(12) A las tres, Juan había almorzado.
'At 3:00, Juan had eaten lunch.'
(i) 3:00 is the time after Juan ate lunch. (adverb modifies R only)

The preposed adverb in (12) modifies only the Reference time. This implies that it can be interpreted only in relation to the structurally higher [TIME] feature (=R) in (10), not the lower one (=E). If the adverb is base generated in left-dislocated position, the restriction on its interpretation reflects an inability to establish connectivity inside the VP. Such a restriction is consistent with the fact observed by Lebeaux (1988), Speas (1990), that temporal adjuncts do not seem to reconstruct to (or be interpreted in) predicate-internal positions. This in turn implies that movement of a temporal adjunct from the lower position is only possible under certain restrictive conditions, such as licensing by a WH feature.

The preceding discussion implies that what are traditionally viewed as tense distinctions (past, present, future) are encoded as relations between *S* and *R* (i.e., relations between times of the CP-phase edge and the *v*P-phase edge); these two times are mediated by Finite and Tense heads, which determine the semantic relation between them. One issue that remains to be investigated is whether *R*/*E* relations are mediated by heads with similar properties. Another issue is whether the restricted number of times that occur in tenses is related to their appearance at phase edges. If so, it is interesting that three times, and not just two, are differentiated in tense systems cross-linguistically.

3 Past tense

As shown in (3) above, there are two simple past tenses: the Preterite and Imperfect past. These share a precedence relation between *S* and *R*. They differ with respect to the aspectual value of the *R*/*E* relation: only the Preterite is **perfective**, including the beginning and end of the event in the reference interval. Consider the predicate *dibujar un círculo* 'draw a circle,' a telic event – a process that has a discrete endpoint; in the Preterite sentence (13a), the endpoints are within the reference interval, which makes the continuations in (13b) and (13c) infelicitous:[1]

(13) a. Juan dibujó un círculo …
Juan draw-PRET.3sg a circle
'Juan drew a circle …'
b. # … pero no lo terminó.
but not it finish-PRET.3sg
'… but he didn't finish it.'
c. #… y todavía lo hace.
and still it do-PRES.3sg
'… and he's still doing it.'

These continuations are infelicitous because the endpoint of the event is within *R*, which implies that the endpoint occurred, but the continuations deny that the endpoint was reached. Demirdache and Uribe-Etxebarria (2000) propose that aspect is parallel to tense, with three possible relations encoded in each: precedence, subsequence, and inclusion. In the tense system, these correspond to past, future, and present tense, and in the aspect system, to perfect, prospective, and imperfective/progressive aspect. Perfective aspect, they argue, does not derive from a semantic relational feature, but from an anaphoric (co-indexing) relation between *R* and *E*. This approach correctly predicts that the temporal properties of *R* will vary with those of *E*. The examples in (13) show that when *E* has a discrete endpoint, it is within R, and the whole event is interpreted as finished. For events that have no inherent endpoint, however, it is more difficult to pinpoint that the whole event is within *R* because the event itself, and hence *R*, continues indefinitely. Thus, the durativity of a state or activity predicate does not produce the same degree of incompatibility in sequences of Preterite + continuation into the present:

(14) a. ?Juan amó a su hermano, y todavía lo ama. (stative/Preterite)
Juan love-PRET.3sg A his brother, and still does
'Juan loved his brother, and still does.'
b. Juan amaba a su hermano, y todavía lo ama. (stative/Imperfect)
Juan love-IMP.3sg A his brother, and still does
'Juan used to love his brother, and still does.'

(15) a. ??Miguel revolvió la sopa, y todavía lo hace. (activity/Preterite)
'Miguel stirred the soup, and he's still doing it.'
b. Miguel revolvía la sopa, y todavía lo hace. (activity/Imperfect)
'Miguel was stirring the soup, and he's still doing it.'

These sentences are less pragmatically odd than (13), which suggests that Preterite past does not impose a termination on an event which otherwise lacks one. At the same time, it places the event wholly within *R*, which accounts for the contrast between the (a) and (b) examples in (14)–(15). Predicates with individual-level interpretations cannot be used in the perfect without modification, as shown by examples from de Miguel (1999): *El portero del equipo {era/*fue} chileno* 'The team's goalie was Chilean' vs. *El portero del equipo fue chileno hasta que renunció a su nacionalidad para no ocupar plaza de extranjero* 'The team's goalie was Chilean until he renounced his citizenship so as not to fill a foreigner's spot.'

Summarizing to this point, the Preterite past is perfective: *R* includes the whole of *E*, but *R* does not introduce termination (delimitation or telicity) to events that otherwise lack them. The properties of the Preterite are correctly predicted by an analysis of perfectivity as a co-indexing relationship between *R* and *E*. However, this leaves unresolved the issue of how to express the difference between interpretations of Preterite and Imperfect states and processes, for example, in (14a) versus (14b). Coindexing of *R* and *E* is expected to produce an *R* interval with the properties of *E*, in this case indefinite duration. The oddness of the continuation in (14a) suggests that in the main clause, there is a time *R′* at which *E* does not hold, which is contradicted by the continuations. It is not obvious how this component of the interpretation would derive from the co-indexing approach.

The Imperfect past is compatible with both states and events. With states, the time *E* is vague in its duration because it often has no specific boundaries. With nonstatives, however, the time *E* contains subintervals with distinct attributes, such as the onset and endpoints, and internal steps in a process. Nonstatives show transparently the relationship between *R* and *E*. Nonstatives have three types of readings in the Imperfect: progressive, habitual/iterative, and intentional (Cipria 1996). The progressive refers to the internal process of an event *E*: *Juan preparaba la cena cuando lo llamó María* 'Juan was making dinner when Maria called him.' In this reading, *R* is included in *E*, and more specifically, *R* excludes the beginning and endpoint boundaries of *E.* Because the endpoint of the event is not included in *R*, progressives are compatible with continuations that deny that the endpoint was reached: *Juan preparaba la cena pero la dejó cuando llamó María* 'Juan was preparing-IMP dinner but left it when María called.' The habitual/iterative interpretation refers to a general state of affairs:

(16) a. Juan preparaba la cena los viernes.
Juan prepare-IMP the dinner on Fridays
b. Pedro revolvía la sopa demasiado.
Pedro stir-IMP the soup too much

In the habitual/iterative interpretation, an indefinite number of events has occurred; the regularity of the events, rather than their quantity, is central to the interpretation. Notice that delimited events are understood to have occurred in their entirety. In (16a), individual events of dinner preparation are telic, and are interpreted as having reached their endpoint. The interval as a whole, however, is not delimited; for example, it is compatible with *for*-phrases: *Por muchos años, Juan preparaba la cena los viernes* 'For many years, Juan prepared dinner on Fridays.' In the habitual interpretation, the inclusion relation does not hold between *R* and single events *E*. Instead, the individual events *E* comprise a collection or sequence, *E′*: $[_{E'} \ldots E_1, \ldots E_n \ldots]$, and *R* is included within *E′*.

The third characteristic reading of the imperfect is the 'intentional' reading, illustrated in (17) (Cipria 1996):

(17) Hasta ayer, íbamos a la playa de vacaciones pero hoy Pepa dijo
until yesterday go-IMP.1pl to the beach on vacation but today Pepa said
que no hay dinero para eso.
that not there is money for that
'Up until yesterday we were going to the beach on vacation but today Pepa said
that there's no money for that.'

The intentional reading of (17) does not claim that the event of going to the beach has occurred. What has occurred is a state: the intention of going to the beach. The *R*/*E* relation is again one of inclusion, although here again the relation does not hold between the basic event *E* (going to the beach) and *R*. Instead, the basic event *E* comprises a goal or 'telos' of the intentional state *E′*, and *R* is included in *E′*, but the goal – the event of going to the beach – is not included in *R*. As with the progressive interpretation, the delimiting boundary (endpoint) has not necessarily been reached. The Inclusion relation thus captures the aspectual properties of the Imperfect. The description of these readings given above suggests that there is some limited form of aspectual recursion within the verb phrase required to capture the distinct properties of *E* and *E′*, where these differ.

4 Nonpast tenses: present, future, and conditional

The nonpast tenses include the present and future tenses and the temporal interpretation of the conditional. These tenses may be grouped together based on two properties: first, they differ from past tenses in that they do not give rise to interpretations where *R* precedes *S*; second, the nonpast tenses allow both simultaneous and subsequence readings under certain conditions. For example, future tense has a subsequence interpretation in sentences like *Juan cantará mañana* 'Juan will sing tomorrow,' but has a present modal interpretation in sentences like (18) and (19):

(18) Serán las ocho
be-FUT.3sg the eight
'It must be eight o'clock.'

(19) Se pondrá mucha pimienta en esta sopa.
one put-FUT.3sg much pepper in this soup
'They must put a lot of pepper in this soup.'

The present tense is also compatible with both present and future interpretations: *Juan sale mañana/ahora mismo* 'Juan is leaving tomorrow/right now.' The conditional is ambiguous between atemporal (simultaneous) conditions and subsequence conditions. The sentence *María dijo que Juan temería al perro* 'Maria said that Juan would fear the dog' can mean either that Juan's fear of the dog could occur at the time of Maria's saying (main clause event time), under certain conditions, or that it would occur at some time subsequent to Maria's past report. The two interpretations can be disambiguated by adverbials that specify a timeframe, such as *en ese momento* 'at that moment,' *al próximo día* 'the next day.' The ambiguity between simultaneous and subsequence readings for these tenses suggests that the *S/R* relation is not specified, and is determined by factors that interact with tense rather than tense itself. This in turn leads to a constructional (syntactically-based) view of the nature of tenses, as opposed to a view in which a particular morphological form has fixed semantic features. An issue for the syntactic analysis of tense is how to represent this "class" of tenses in terms of features. A second issue is the nature of the mechanisms that determine a present or subsequence interpretation in the presence of adverbs or other constituents.

It was shown in Section 4 above that there are both perfective and imperfective Past tenses. The readings of the simple Present tense indicate that it is an imperfective tense: no perfective interpretation is possible. Consider the contrast in (20) between the perfective interpretation of the Preterite past in (20a) and the absence of such an interpretation for the Present in (20b):

(20) a. María construyó la casa el año pasado.
Maria build-PRET.3sg the house the year last
'Maria built the house last year.'
b. María construye la casa este año.
Maria build-PRES.3sg the house this year
'Maria is building the house this year.'

In (20a) the whole event of house-building is within *R*, so its endpoint is interpreted as having been reached. In (20b), the tense has an imperfective interpretation, with the event final boundary outside *R*. Other readings of the present tense are parallel to the imperfective interpretations discussed above for the Past tense: states, habitual activities, and events in progress. There is also an intentional reading, or 'scheduled future':

(21) Ese vuelo sale a las tres.
that flight leaves at the three
'That flight leaves at three o'clock.'

This reading is restricted to contexts where an animate entity is understood to plan the event in question. If no planner can be inferred, the reading is infelicitous, as in *Llueve a las tres* 'It rains at 3:00.' However, the entity engaged in the planning is not necessarily an event participant, as in (21), where the only DP argument is inanimate.

The conditional tense differs from the other tenses discussed above in that it encodes a modal feature that may or may not have a temporal value. In main clauses, the default interpretation is not principally temporal, but that of a 'logical location' for an event. In *Juan cantaría* 'Juan would sing,' the event of singing is not understood as a present or future situation, but instead is "located" relative to unspecified satisfaction conditions. A second respect in which the conditional tense differs is that, in embedded contexts, it has an interpretation of 'future-of-the-past.' This is illustrated by the contrast in (22):

(22) a. María dirá mañana que Pedro ganará el premio al próximo año.
Maria say-FUT tomorrow that Pedro win-FUT the prize the next year
'Maria will say tomorrow that Pedro will win the prize next year.'
b. María dijo ayer que Pedro ganaría el premio al próximo año.
Maria say-PRET yesterday that Pedro win-COND the prize the next year
'Maria said yesterday that Pedro would win the prize next year.'

In the embedded clause of (22a), the subsequence interpretation is expressed by future tense; in (22b), by conditional morphology. The future/conditional alternation in embedded clauses is similar to the English *will/would* alternation. In such cases, the morphological distinction between the future and conditional reflects a past/non-past distinction, albeit of a different type from that discussed above, where 'past' is equivalent to '*R* precedes *S*.' In the context illustrated in (22), it corresponds to whether the evaluation time is past or not; in both sentences, the external evaluation is understood as the time of Maria's saying (the event of the main clause): in (22a), it is a future time; in (22b), a past time.

5 Embedded clauses

The meaning of tenses in embedded clauses is affected by several contextual factors, including: (a) whether the embedded clause is a complement clause or an adjunct; (b) the tense of the matrix clause; (c) the semantic type of the matrix verb; and (d) the aspect of the embedded event. The mood of the embedded clause is also a crucial factor, as is discussed in Chapter 19; the present discussion is restricted to indicative tenses. A perennial issue for the analysis of embedded clause tense is the nature of 'sequence-of-tenses,' which show certain tense forms without the corresponding semantic value. Before illustrating this phenomenon, it is useful to look more broadly at the patterns of relationship between embedded and non-embedded tenses, reviewing the contextual factors mentioned above.

Complement and adjunct clauses differ in their relationship to the main clause. In adjuncts such as relative clauses, indicative tenses are interpreted as though they were unembedded: the external evaluation time is the time of speech, and the embedding clause does not restrict the reference of the tense. In complement clauses, however, an embedded past tense is evaluated relative to a main clause past event. The examples in (23) show the independence of relative clause tenses:

(23) a. Juan conoció al niño que llora.
Juan meet-PRET A + the child that cry-PRES
'Juan met the boy who cries/is crying.'
b. Juan conoció al niño que lloró.
Juan meet-PRET A + the child that cry-PRET
'Juan met the boy who cried.'

In (23a), the main clause is past tense while the relative clause is a present tense. The relative clause is not required to agree in a past/nonpast feature with the main clause. Furthermore, the event of crying in the relative clause is understood as simultaneous with the time of speech, exactly as is the case for a main clause present tense. The relative clause is not restricted in reference by the main clause event. Similarly in (23b), the past tense of the relative clause is independent, in the sense that the event of crying is evaluated only relative to the time of speech; it is not restricted in reference by the main clause event. Consider now the complement clauses in (24):

(24) a. *Juan oyó que el niño llora.
Juan hear-PRET that the child-M cry-PRES
'Juan heard that the child is crying.'
b. Juan oyó que el niño lloró.
Juan hear-PRET that the child-M cry-PRET
'Juan heard/understood that the child cried.'

As (24a) shows, where the intended reading is a single event of crying, a main clause past event does not allow its complement to contain a present indicative tense. If (24a) were grammatical, the interpretation would be such that the event of crying was evaluated relative to the time of speech, in other words, that it is simultaneous with the speaker's 'now'; instead, the complement is evaluated relative to the past event of the main clause. This interpretation is possible only if the embedded clause is formally a past tense. The feature 'past' in this context is associated with a past evaluation time – just as was shown above in (22b) of Section 4, for the conditional. This implies that a morphological past affix can have two different origins, according to whether *R* or *S* is situated in the past. Recall from §2 that times are encoded in both the CP and *v*P phases, repeated below:

(8) CP phase:	*v*P phase (preliminary):
Force – (Topic) – (Focus) – (Topic) – Finite – TP	*v*P
$[\text{TIME}]_S$	$[\text{TIME}]_R$

Either of these times can trigger past morphology for the clause: (i) the Time *R* in *v*P is [past] by agreement with an interpretable Past feature of Tense, or (ii) the Time *S* in the Force Phrase can be [past] by agreement with the *v*P of the matrix clause. The latter is a 'relative tense,' in the sense that the evaluation time is established relative to a linguistic antecedent rather than to the speaker's 'now.'

Several factors influence the distribution of absolute versus relative tenses: the tense of the main clause verb, the semantic type of the main clause verb, and aspectual features of the embedded clause event. The contrast in (24) above illustrates that a main clause past tense is a context in which the embedded clause tense must contain a relative tense. Formal agreement between the clauses is required in this context, with the result that the external evaluation time in the complement clause always corresponds to the main clause event time. The complement clause cannot be interpreted as occurring after the main clause event (**Juan oyó ayer que el niño lloró esta mañana* 'Juan heard/understood yesterday that the boy cried this morning'). The generalization appears to be that a past matrix verb triggers agreement with the time feature of the complement Force phrase. A nonpast matrix verb does not trigger agreement obligatorily. That is, morphological agreement is not obligatory, and where there is morphological agreement, it need not have the semantic characteristic of a relative tense:

(25) Juan entiende que el niño lloró ayer.
Juan hear/understand-PRES that the child-M cry-PRET yesterday
'Juan hears/understands that the child cried yesterday.'

(26) Juan verá mañana que Pedro ganará el premio esta noche.
Juan see-FUT tomorrow that Pedro win-FUT the prize tonight
'Juan will see tomorrow that Pedro will win the prize tonight.'

In (25), the present tense verb of the matrix clause does not trigger obligatory agreement since the complement clause contains a past tense. In (26), the occurrence of future tense in both clauses is consistent with the possibility that morphological agreement has occurred, producing a relative future tense, parallel to the relative past tense in (23b). However, the interpretation of (26) does not support such an analysis since the adverb *esta noche* imposes a reading of the event of winning the prize as occurring in the future relative to the time of speech, but not in the future relative to the main clause event. This implies that non-past agreement is not obligatory.

A further factor that affects the distribution of relative versus absolute tenses is the semantic class of the matrix verb. Verbs of belief (*pensar* 'think,' *creer* 'believe') and other verbs of mental attitude follow the pattern illustrated in (23), of obligatory agreement between the complement clause and the matrix past tense verb. Verbs of communication, however, admit present tense under past more freely (Giorgi and Pianesi 2000; Zagona 2007):

(27) María dijo que Pedro está enfermo.
Maria say-PRET that Pedro be-PRES sick
'Maria said that Pedro is sick.'

In contexts such as (27), an embedded present tense has a 'double access' interpretation: the embedded clause situation holds both at the time of the main clause event and at the time of speech.

The phenomenon of 'sequence-of-tenses' refers to embedded tenses which have a certain form, apparently due to agreement with the main clause tense, without the semantic value that is usually associated with that form. This is shown by the second reading (the 'simultaneous' reading) of (28):

(28) María creía que Juan estaba enfermo.
Maria believe-IMP that Juan be-IMP sick
(i) 'Maria believed that Juan was sick (at some time in the past).' ('shifted reading')
(ii) 'Maria believed that Juan was sick (at that time).' ('simultaneous reading')

In the 'shifted' reading, the embedded past tense has a precedence value since the time of Juan's sickness is understood as preceding the time of Maria's belief. In the 'simultaneous' reading of (28), the embedded past tense does not have a precedence value since the time of Juan's sickness is understood as simultaneous with the evaluation time of Maria's belief. With respect to the syntactic ingredients for sequence-of-tenses, the 'simultaneous' interpretation is sensitive to both tense and aspect. If the embedded clause contains perfective aspect, 'simultaneous' interpretation is impossible (*María creía que Juan estuvo enfermo* 'Maria believed that Juan was (preterite past) sick').

6 Aspect

As noted in the introduction above, lexical aspect classifies events in terms of their temporal properties, particularly such contrasts as punctual versus durative, stative versus non-stative, and telic versus atelic (Vendler 1967; Smith 1997). The contrast between telic and atelic events is not generally presented in traditional grammars alongside the other contrasts. It concerns whether or not an event has a natural endpoint, as illustrated by the contrast in (29):

(29) a. *Juan dibujó dos círculos* (en un minuto).
'Juan drew two circles (in a minute).'
b. *Juan dibujó círculos* (*en un minuto).
'Juan drew circles (*in a minute).'

The event (29a), of drawing two circles, has a natural endpoint: the point at which the two circles are completely drawn. The adverbial *en un minuto* 'in a minute' specifies how long it takes to get to that endpoint. The event of (29b), of drawing an unspecified number of circles, is atelic; it lacks a natural endpoint. Consequently, this event is not compatible with an adverbial that specifies the length of time taken to reach the endpoint.

The term *lexical aspect* suggests that the basis for the classification of events resides in lexical items, particularly verbs, and this view is adopted in some studies, based on the assumption that these properties are idiosyncratic features of verbs. Recent research has argued that lexical aspect is determined compositionally in the syntax (Borer 2005; MacDonald 2008; Ramchand 2009), on the basis of the position of a verb, and its interaction with other constituents, as illustrated by the verb *dibujar* 'draw' in (29).

Grammatical aspect situates an event relative to a temporal frame, and in so doing, reflects an external temporal *viewpoint* on the event. Grammatical aspectual morphology provides a basis for locating event intervals relative to their temporal frame. The relationship has been described along two dimensions in the literature. One is the perfective–imperfective distinction, which relates events to their temporal frames in terms of whether the boundaries are included in or excluded by the frame, as discussed in Section 3 above. A second approach focuses on ordering relations, with perfect, progressive, and prospective aspect analyzed in terms of the same ordering primitives that produce past, present, and future tenses (Demirdache and Uribe-Etxebarria 2000) (D and U-E). In (30) for example, the viewpoint interval is distinguishable from the Event-time (*R* versus *E* in the terminology of the Reichenbach framework in Section 2.

(30) a. Los niños habían comido toda la sopa.
the children have.IMP.3pl eat.PRT all the soup
'The children had eaten all the soup.'
b. Los niños estaban comiendo toda la sopa.
the children be.IMP.3pl eat.PRT all the soup
'The children were eating all the soup.'
c. Los niños comían toda la sopa.
the children eat.IMP.3PL all the soup
'The children were eating all the soup.'
'The children used to eat all the soup.'

In the Reichenbach framework, sentence (30a) refers to three times: the Speech-time, the Event-time, and a Reference-time – a time at which is the soup-eating event is already completed. The past tense locates Reference-time, a 'postevent' time, prior to Speech-time. The progressive (30b), locates only the internal part of the event, minus its beginning and end boundaries, prior to Speech-time. In the imperfect (30c), either the activity stage or a larger interval is referred to, with 'habitual' instances of the event. In each of these cases, the interval that is situated in the past is a time interval that corresponds to a temporal frame, or Reference-time, not the events themselves. This temporal frame reflects that the speaker's viewpoint on the event.

The issue of recursion has been raised in the domain of aspect as in the domain of tense. It was noted in Section 1 that only three times are available in the tense system: those described in the Reichenbach system as *S*, *R*, and *E*. The discussion of imperfective aspect above suggested that there is some limited recursion of the Event time, where *E* and an interval *E′* are distinguished (such as the intervals of habitual activity or intention). There is also limited recursion of grammatical aspectual morphology: perfect–perfect and progressive–progressive sequences are impossible, (**Juan había habido cantado ya a las tres* 'Juan had had sung already at 3:00,' but a sequence of perfect–progressive is possible. D and U-E (2000) argue that the limitation on recursion is semantic rather than structural: semantically vacuous relations are prohibited, a ban which accounts for the impossibility of iteration of identical morphemes. One residual question is the co-occurrence of imperfective verb tenses with progressive morphology (*estaban cantando* 'be-IMP.3pl singing') since both morphemes introduce inclusion relations. One issue to be explored is whether both features are interpretable in this context.

The notion of Viewpoint Aspect as parallel to Tense seems to be quite compatible with the phase-based approach to clause structure in the sense that the CP phase and the *v*P phase have (a) a time feature encoded on their periphery, as shown in (8) of Section 2, repeated below as (31), and (b) the idea may be considered as well that each phase also contains a phrase that derives temporal relations between *R* and *E*. Within the *v*P phase, the head might be TP, or Aspect phrase, as shown in (32):

(31)	CP phase:	*v*P phase (preliminary):
	Force – (Topic) – (Focus) – (Topic) – Finite – TP	*v*P
	$[\text{TIME}]_S$	$[\text{TIME}]_R$

(32)	Force …	TP	XP	TP/Aspect	*v*P
	$[\text{TIME}]_S$	[Past]	$[\text{TIME}]_R$	[Past]	$[\text{TIME}]_E$

In such an approach, the impossibility of double finite morphology could be explained by the occurrence of Finite Phrase exclusively in the CP phase. The lower TP/Aspect phrase would then have the status of a non-finite small clause. A phase-based implementation of tense/aspect relations remains to be studied in detail. One issue that calls for further study is the contrast between tense and aspect with respect to how insulated their semantic features are from the effects of features of nominals and adverbials. Main clause tenses are fixed in value, at least with respect to the past/non-past distinction, as discussed in Section 2–3. Aspect, however, is influenced by contextual features to a greater degree. In particular, a "double tense" analysis of the present perfect tense would be expected to produce a preterite-like interpretation, referring to a time *R* that is ordered after *E*. However, such an interpretation (of a post-event time) is limited to telic events. Compare telic and stative situations in (33); *ya* 'already' is intended to make salient the interpretation of *E* as preceding *R*:

(33) a. Juan ya ha llegado.
Juan already have.PRES arrive-PRT
'Juan has already arrived.'

b. ??Juan ya se ha parecido a su hermano.
Juan already REFL have.PRES resemble.PRT A his brother
'Juan has already resembled his brother.'

The telic event of arriving produces the 'prototypical' present perfect interpretation, where speech-time is included in a time R which is a postevent time – that is, *R* follows *E*. In (33b), the stative predicate *parecerse* 'resemble' does not give rise to the ordering relation. This implies that the temporal structure of the event – perhaps an implicit result state – is a necessary ingredient for licensing an ordering relation between *E* and *R*.

A further issue for the analysis of grammatical aspect is whether there are features that are specific to aspectual representation, as distinct from event-type features ('lexical aspect') and ordering (finite tense) features. Without cross-linguistic evidence for such features, the motivation for Aspect Phrase as a category in clause structure is substantially weakened. Future research may give a more precise description of where the [TIME] feature is encoded in the left periphery of *v*P, and whether the complex temporal relationships that are found within *v*P are due to a more elaborate 'fine structure,' or perhaps to a more articulated set of features for the *v*P phase heads. This characterization of the issue suggests that the alternatives lie either in the domain of hierarchical structure or functional features, although the solution may involve both phase-head features and more articulated structure.

NOTES

1 Abbreviations for tense and aspectual morphology in glosses include PRET for preterite past, IMP for imperfect past, and PRT for participles; A is 'personal *a*.'

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19 Mood: Indicative vs. Subjunctive

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1 Introduction: syntactic contexts for moods

Moods constitute a manifestation of modality. This category reflects the speaker's attitude towards propositional contents, more specifically the various forms in which statements are interpreted under the influence of semantic environments, whether hypothetical or real. Moods are verbal inflections reflecting modality. They may directly encode grammatical differences related to speech acts, as in the subjunctive form *tenga* in *¡Tenga un buen día!* 'Have a nice day!,' as opposed to the indicative *tiene* in *Tiene un buen día* 'S/he is having a nice day.' In many cases, moods are induced or triggered by various grammatical categories in restricted syntactic contexts, for example the preposition *sin* 'without' in *sin que tú lo {supieras/*sabías}* 'without you {knowing-SUBJ./knowing-IND.}.'

There exists an abundant theoretical literature on the Spanish verbal moods, written in several frameworks, as well as many traditional and descriptive studies on this topic. Overviews may be found in Lleó (1979), Manteca Alonso-Cortés (1981), Borrego et. al. (1986), Bosque (1990), Porto Dapena (1991), and Ahern (2009), among others. Extensive chapters are devoted to the description of moods in recent grammars of Spanish. See Ridruejo (1999), Pérez Saldanya (1999) and RAE-ASALE (2009: ch. 25). General presentations of the subjunctive mood in current theoretical linguistics may be found in Portner (1999), Quer (2005), and Laca (2010). Studies on Spanish moods from pragmatic, functional, or discourse-oriented perspectives include Lunn (1989, 1995), Mejías-Bikandi (1994), Maldonado (1995), Gregory (2001), Haverkate (2002), and Travis (2003).

Since I will not be able to address all the questions raised in these works, I will point out the main issues that I consider to be central, and also the controversial questions that I think are alive nowadays, focusing on those which seem to be particularly relevant for present-day theoretical concerns. Both presentation and discussion of these matters will have to be necessarily schematic. Some relevant

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grammatical issues, such as the relationship between mood and tense, will not be addressed here for reasons of space (see Chapter 18, above).

There is an almost universal agreement on the fact that Spanish has three moods: indicative, subjunctive, and imperative. With many verbs, imperative forms are preempted by others, either from the indicative (*Por favor, canta* 'Please, sing'), or the subjunctive paradigm (*Por favor, canten* 'Please, sing'), acquiring illocutionary force in both cases. The indicative is the verbal mood chosen "by default", in the sense that it may appear without a trigger in main or subordinate clauses. The imperative mood is restricted to matrix clauses, as in *Ven aquí* 'Come here.' It is incompatible with negation and with the subordinate conjunction *que* in these clauses:

(1) a. Quiero que {*ven/vengas}.
'I want you to come-IMPER./come-SUBJ.'
b. No {*ven/vengas}.
'Do not come-IMPER./come-SUBJ.'
c. ¡Que {*ven/vengas}!
'Come-IMPER./Come-SUBJ.'

All these properties are usually taken as evidence that imperative inflection is placed in the highest syntactic position available above C° (sometimes named *Force Phrase*, as in Rizzi 1997 and subsequent work), and associated with illocutionary features.

The subjunctive mood occurs in the following three syntactic structures:

(2) a. Selected by a lexical or functional head
b. In the scope of a modal operator
c. In matrix clauses.

The subjunctive is induced or triggered in types (2a) and (2b), but it lacks a trigger in (2c). The subjunctive of type (2a) corresponds to sentences such as *Lamento que llegaran tarde* 'I regret that they arrived late,' where the verbal form *llegaran* appears in a complement clause selected by the verb *lamentar* 'regret.' Nouns, adjectives, prepositions, and subordinate conjunctions may be subjunctive triggers:

(3) a. Es la hora de que el niño se *levante*.
'It is time for the child to get up.'
b. Estoy cansada de que te *quejes*.
'I am tired that you complain.'
c. Para que *estés* informada
'In order for you to be informed.'
d. Como *llamen*.
'In case they call.'

Adverbs may also be triggers in predicational contexts (Section 3). Stowell (1993), Quer (1998), and other authors apply the term *intensional* to subjunctives licensed in contexts of lexical selection. This broad use of the term *intensional* might be confusing. Since intensional predicates give rise to the non-specific interpretation of indefinites, as in *Necesito una nueva secretaria* 'I need a new secretary,' the term *intensional* excludes emotive factive verbs. These predicates do not license the nonspecific interpretation of indefinites (*Me gusta una nueva secretaria* 'I like a new secretary'), but lexically select for the subjunctive mood. Other "nonintensional" predicates which trigger the subjunctive mood include those expressing influence and causation, sometimes called *directives*, as *forzar* 'force' or *conseguir* 'get.'

Type (2b) corresponds to sentences such as *No recuerdo que Juan viviera aquí* 'I don't remember that J. lived-SUBJ. here,' where the adverb *no* is responsible for the subjunctive in *viviera* (cf. **Recuerdo que Juan viviera aquí* 'I remember that J. lived-SUBJ. here,' ungrammatical in the absence of a trigger). The subjunctive in *viviera* is induced by the operator *no* from the upper clause. Most instances of subjunctives in relative clauses correspond to type (2b) as well. For example, the trigger of the subjunctive form *sirva* in *Solo tengo un diccionario que me sirva* 'I only have one dictionary which is useful to me' is the adverb *solo* 'only,' which constitutes a triggering operator relatively similar to the adverb *no.*

The subjunctive triggered in type (2b) structures has been called *polarity subjunctive* (Stowell 1993; Quer 1998; Giannakidou 1998) since the process of mood triggering in these cases is similar to the licensing of negative polarity indefinites, as in *No recuerdo que dijera nada* 'I do not remember him saying anything.' Notice that negation does not fit in group (2a) properly (**Juan no llame* 'John does-SUBJ. not call'). A common trigger of polarity subjunctive is interrogation. This operator may induce subjunctive in complement clauses (4a), relative sentences (4b), and adverbial clauses (4c):

(4) a. ¿Recuerdas que *dijera* algo con sentido?
'Do you remember him/her saying anything meaningful?'
b. ¿Dijo algo que *tuviera* sentido?
'Did s/he say anything that made sense?'
c. ¿He de callarme porque tú me lo *digas*?
'Must I be quiet just because you tell me to do so?'

Notice that (4c) contrasts with **He de callarme porque tú me lo digas*, ungrammatical for lack of a modal trigger. Quer (1998) pointed out a number of differences between the subjunctives types in (2a) and (2b). One interesting difference is the fact that sequences of tenses are respected in type (2a), but not necessarily in (2b), as seen in (5):

(5) a. Me gusta que ella {viva/*viviera} aquí.
'I like that the she {lives-3rd PERS.INDIC.PRES./lives-3rd.SUBJ.IMPF.} here.'
b. No creo que ella {viva/viviera} aquí.
'I do not think that she {lives-3rd.INDIC.PRES./lives-3rd.SUBJ.IMPF.} here.'

Another difference between (2a) and (2b) – Quer argues – hinges on the fact that only the former structure gives rise to so-called *obviation* or *disjoint reference* configurations. This phenomenon, to which we will return in Section 6, refers to the fact that the null subject of a subordinate subjunctive clause cannot corefer with the main verb's subject. Thus, the verbal form *regrese* may display 3rd or 1st person features in (6b), with a polarity subjunctive, but has only 3rd person features in (6), with a lexical trigger. This means that the speaker can be the person who comes back in (6b), but not in (6a):

(6) a. Quiero que regrese.
'I want him/her/it to come back.'
b. No creo que regrese.
'I do not think I/he/she/it will come back.'

The distinction between (2a) and (2b) is a very productive one, but it is not free of problems. Some difficulties arise when negation lexically incorporates to a subjunctive-taking predicate. For example, sentence (6b), a standard example of subjunctive triggered by a negative operator, may be compared with (7):

(7) Dudo que regrese.
'I doubt that {I/he/she/it} will come back.'

The meaning of (7) is quite close to that of (6b), but (7) allows a nonobviative reading (namely, *dudo* [1st pers.] . . . *regrese* [1st pers.]). The verb *dudar* permits the suspension of sequences of tenses also, as in *Dudo que regresara* 'I doubt that I/he/she/it came back.' These properties are unexpected, since this is a case of lexical selection of mood by a predicate in a local context.

Another structure in which selected and polarity subjunctive get close is affixal negation, as in {*No es cierto/Es incierto*} *que las cosas sean así* 'It is {not true/untrue} that things are that way'). Finally, the relationship between (2a) and (2b) is also close in the case of so-called intensional predicates. These predicates are lexical triggers, as show in (8a), but, at the same time, they behave as modal operators, as witnessed by (8b). This similarity is not found in lexical triggers belonging to other semantic groups, as seen in (9):

(8) a. Necesito que tenga experiencia.
'I need that s/he has-SUBJ. experience.'
b. Necesito a alguien que tenga experiencia.
'I need someone who has-SUBJ. experience.'

(9) a. Le molesta que la gente fume.
'It annoys him/her that people smoke-SUBJ.'
b. *Le molesta la gente que fume.
'People who smoke-SUBJ. annoy him/her.'

C-command is required both in polarity and lexically-selected subjunctive. It is controversial whether *quizá* 'maybe' and other similar adverbs belong to (2a) or (2b). They induce subjunctive inside their sentential domain, as required in (2a), but they do not seem to be heads, and are compatible with the indicative mood. In any case, these adverbs are not able to license subjunctive from a parenthetical position. This indicates that mood triggering under c-command is a syntactic process in these cases, rather than the result of some discourse phenomenon:

(10) a. Quizá ella lo {sabe/sepa}.
'Maybe s/he knows-IND./SUBJ. it.'
b. Ella lo {sabe/*sepa}, quizá.
'She knows it, perhaps.'

Let us consider type (2c) now. Subjunctive forms substituting imperatives belong to this group, as in *Pasen ustedes* 'Come in' (affirmative imperatives with *usted*, and other DP-like subjects), *No entren* 'Do not enter' (negative orders), *¡Vivan los novios!* 'Hooray to the bride and groom!' (desiderative statements), or *¡Que empiece la fiesta!* 'Let the party begin!' (an exclamative after *que*, traditionally called *exhortative*).

Subjunctives of type (2c) are also characteristic of so-called *unconditionals* or *concessive-conditionals*. These are non-embedded disjunctive clauses expressing the meaning of two coordinated conditional clauses with opposite meanings. The unconditional sentence *Quieras o no* 'Whether you want to or not' is, thus, synonymous with *Tanto si quieres como si no (quieres)* 'No matter whether you want to or not.' They coincide with imperatives in rejecting preverbal subjects (a property that may naturally result from verb movement to C° or to a higher projection):

(11) {*Juan quiera/Quiera Juan} o no.
'Whether Juan wants to or not.'

Other variants of (2c) include the non-subordinate subjunctive imperfect with some modal verbs (*Pudiera ser así* 'It might be so') and the pluperfect in utterances such as *Hubieras venido antes* 'I wish you were here before.' The specific location of subjunctive features in subordinate clauses is controversial, but most authors take it to be a syntactic projection higher than Tense and lower than C° (arguably, Mood Phrase). See Tsoulas (1994) on this point. Kempchinsky (1990, 1998) claims that subjunctive features embedded in subordinate clause are interpreted in C° because of selectional reasons.

2 Is it possible to unify subjunctive meanings?

In a large number of analyses of Spanish subjunctive (many of them traditional or structurally-oriented), the question arises of whether or not it is possible to attribute one single meaning to all uses of the subjunctive. This issue was particularly relevant in some branches of structural linguistics in which a one-to-one correlation between grammatical morphemes and particular contents was stipulated on theoretical grounds. The point has been raised again in a number of more recent studies of Spanish moods oriented from pragmatic, cognitive, or functional perspectives.

The list of candidates to provide these unifying semantic labels is rather large: uncertainty, unreality, contingency, virtuality, nonfactuality, nonassertiveness, among others. Hummel (2001) carries out a detailed overview of the numerous grammatical analyses seeking these unifying notions in Romance, and proposes his own unificational approach. See, on the same issue, Lozano (1972, 1975), Bell (1980), De Jonge (2001), and Luquet (2004). On the history on grammatical analyses of the subjunctive mood, see the papers in Bosque (1990: pt I), and Zamorano Aguilar (2001).

There is something intuitively true in all these notions, but, at the same time, they are hard to apply in absence of a syntactic structure. For example, it has been repeatedly pointed out that the subjunctive complement of so-called emotive factive predicates refers to actual or real (rather than virtual) facts, as in *No me gusta que me grites* 'I do not like you shouting at me.' On the contrary, notions such as 'suspicion' or 'fear,' both referring to actual events, do not seem to encode a meaning difference strong enough as to expect the mood contrasts in (12):

(12) a. Sospecho que me {estas/*estés} engañando.
'I suspect you {are-IND./are-SUBJ.} cheating me.'
b. Me temo que me {estas/estés} engañando.
'I am afraid you {are-IND./are-SUBJ.} cheating me.'

In the same vein, the conditional conjunctions *si* 'if' and *en caso de (que)* 'in case' express very similar meanings, but they select opposite moods:

(13) a. Si no {llamas/*llames}.
'If you don't {call-IND./call-SUBJ.}.'
b. En caso de que no {*llamas/llames}.
'In case you don't {call-IND./call-SUBJ.}.'

As pointed out above, the subjunctive mood is able to provide a sentence its illocutionary force. In the proper modal environments, the subjunctive may also give rise to the nonspecific interpretation of indefinite NPs, as in *Busco un pantalón azul que me guste* 'I am looking for some blue pants that I might like'; that is, it permits that a NP refers to some entity that might not exist. The subjunctive also marks the scope of modal operators (Section 5), and carries other grammatical roles different enough as to reject a single, wide-ranging semantic label.

But positive straightforward answers to the old question on the unification of subjunctive values are disappointing only if the question is asked on broad semantic grounds rather than on restrictive syntactic terms. The classical idea that subjunctive is the mood of subordination is still correct. Contexts of the types (2a, b), which cover most uses of the subjunctive, have something in common: they introduce states of affairs conceived through the angle of some evaluation, possibility, necessity, emotion, intention, causation, and other nonfactual or nonvericonditional (in Giannakidou's 1998, 1999 terms). Interestingly, most of these concepts are often used to define the very notion of 'modality.' Subjunctive inflection is also the formal mark in subordinate clauses of the relevant features (to a large extent, modal) present in the lexical or functional elements selecting for it. Besides, it signals the scope taken by the operators (now "modal" in a more strict sense of the word) that license its appearance.

3 Mood and lexical selection

We have seen that mood triggers belong to several grammatical word classes. Nouns and adjectives select for mood in their sentential arguments both as predicates (as in *Es una obviedad que* ... + IND. 'It's a given that ...') or as heads introducing clausal complements (as in *Llegó el momento de que* ... + SUBJ. 'The moment arrived that ...'). This paradigm includes some adverbs. Thus, *bien* 'well' selects for subjunctive as a predicate in *No me parece bien que* ... 'It does not seem to me appropriate that ...,' but *encima* selects for indicative as a head in *Encima de que te estuvimos esperando* 'In addition to the fact that we were waiting for you.' Prepositions nonselected lexically or not integrated in a complex predicate (such as "V + P, A + P or N + P") take subjunctive also, as *para* 'for' and *sin* 'without' do.

Predicates taking indicative are often gathered in semantic classes such as those in (14):

(14) a. event: *suceder*, *ocurrir*, *acontecer* 'happen.'
b. language, communication: *decir* 'say,' *advertir* 'warn,' *anunciar* 'announce,' *prometer* 'promise.'
c. perception, judgment: *ver* 'see,' *creer* 'think, believe,' *notar* 'note,' *recordar* 'remember,' *descubrir* 'discover.'
d. acquisition, possession or loss of information: *saber* 'know,' *enterarse* 'find out,' *leer* 'read,' *olvidar* 'forget,' *indicio* 'sign.'
e. certainty, objectivity: *probar* 'prove,' *demostrar* 'show,' *seguro* 'sure,' *claro* 'clear.'

The following semantic concepts do not cover all predicates taking subjunctive, but address most of them:

(15) a. will, intention: *querer* 'want,' *aspirar (a)* 'aspire,' *decidirse (a)* 'make up one's mind,' *esforzarse (por)* 'strive,' *luchar (por)* 'fight,' *pretender* 'try,' *procurar* 'endeavour.'
b. causation, influence: *hacer* 'do,' *causar* 'cause,' *pedir* 'ask,' *recomendar* 'recommend,' *favorecer* 'favor.'

c. affection, emotion: *molestar* 'annoy,' *alegrarse (de)* 'be glad,' *lamentar* 'regret.'
d. assessment, evaluation: *convenir* 'suit, be convenient,' *bueno* 'good,' *desafortunado* 'unfortunate.'
e. possibility, necessity, eventuality: *hacer falta* 'to be necessary,' *necesitar* 'need,' *dar igual* 'be irrelevant,' *posible* 'possible.'
f. opposition, rejection: *desmentir* 'deny,' *negarse (a)* 'refuse,' *contrario (a)* 'contrary,' *renuente (a)* 'reluctant.'

Some comments are in order if we compare the groups in (14) and (15). Most factive predicates belong to (15), but *saber* 'know') is a factive verb which fits in (14d). Perception verbs are semifactive predicates, but they tend to be interpreted as factives (at least, conversationally), so that *No vi que estaba lloviendo* 'I did not see that it was raining' implies 'It was raining' (see (36) below). Most predicates in (14) are considered to be *assertive* (Terrell and Hooper 1974; Hooper 1975). Nevertheless, this notion is acknowledged to be used loosely, since to know, learn, forget, or be aware of something do not entail asserting some propositional content.

The line separating the semantic concepts in (14) and (15) is sometimes fuzzy. Take the notion of 'guilt' as an example. The noun *culpa* 'guilt,' the adjective *culpable* 'guilty' and the verb *culpar* 'blame' select subjunctive. Since 'guilt' does not have its own group in (15), one might say that it represents a subtype of the notion of 'cause' in (15b), or rather a type of emotion in (15c). Although few existing descriptions of categories such as the ones in (14) and (15) allow us to explicitly deduce these differences, paradigms of this sort may help us to characterize predicates that seem to fit in more that one group. For example, one might *a priori* think that *aconsejar* 'advise' is a verb of communication, but the fact that it takes subjunctive shows that it is interpreted as a verb of influence by Spanish grammar. It does not go with *anunciar* 'announce' in (14b), but rather with *pedir* 'ask' or *sugerir* 'suggest' in (15b). Similarly, although the nouns *sensación* 'sensation' and *temor* 'fear' apparently belong to the same group, mood selection shows that the former fits with perception predicates and the latter with affection predicates:

(16) a. Tengo la sensación de que nos {estamos/*estemos} perdiendo.
'I have the feeling that we {are-IND./are-SUBJ.} losing.'
b. Tengo el temor de que nos {*estamos/estemos} perdiendo.
'I fear that we {are-IND./are-SUBJ.} losing.'

Mood alternations with the same predicates give rise to meaning differences. The following list contains the most characteristic (the verb takes indicative if used in the sense specified at the left of ">"; in the other interpretation it takes subjunctive):

(17) a. communication > influence: Insisto en que se {comportan/comporten} correctamente.
'I insist that they {behave/must behave} properly.'

b. communication > assessment: Me reprocha que no le {hago/haga} caso.
'They reproach me for {not paying attention/to not pay attention} to him/her.'

c. thought or belief > intention or will: He pensado que {es/sea} usted el nuevo embajador.
'I've been thinking that you {are/could be} the new ambassador.'

d. perception > assessment: Entiendo que lo {has/hayas} perdonado.
'I understand {that you have forgiven him/you forgiving him}.'

e. perception > intention: Siempre veía que cada cosa {estaba/estuviera} en su sitio.
'I always {used to notice that everything was/saw to it that everything was} in its place.'

The predicate specified to the left of ">" denotes communication, perception, or thought in most cases; the meaning obtained at the right is intentional, or at least prospective. A more detailed description of these mood alternations may be found in RAE-ASALE (2009). One might think of three types of solutions for them:

(18) A. Lexical solutions
B. Syntactic solutions
C. Discursive or pragmatic solutions

In type A solutions, lexical entries are split. This is a straightforward solution to cases of homonymy or polysemy, as in *sentir*, which selects subjunctive as an affection verb 'regret': *Siento que lo perdamos* 'I regret that we are going to lose it/him,' but indicative as a perception verb 'feel, perceive': *Siento que lo perdemos* 'I feel that we are losing it/him.' However, lexical solutions do not seem to be appropriate for the groups in (17) since most alternations of this sort extend to paradigms. That is, the process illustrated in (17a) with *insistir* takes place with *decir* 'say,' *repetir* 'repeat,' *comunicar* 'communicate,' *indicar* 'point out' and other verbs. If we apply A to (17a), we certainly miss a generalization.

In option B, the meaning of the predicate does not change as a consequence of mood alternations. A strong argument for B comes from the fact (early observed by Demonte 1977; Manteca Alonso-Cortés 1981, and other authors) that sentential complements of the same verb can be coordinated even if they display different moods, as in *Me dijo que tenía* [IND.] *razón y que esperara* [SUBJ.] 'S/he told me that I was right and that I had to wait.' Ahern and Leonetti (2002) correctly point out that the choice of the subjunctive in the sentence *El director dice que la actriz sea rubia* 'The director says that the actress should be blond' does not imply that *decir* comes to mean 'demand' here. As the English gloss shows, it rather means that *sea* is interpreted as 'has to be' or 'should be.' Summing up, in option A, differences in meaning come from recategorization of matrix predicates; in option B, they result from the association of subjunctive inflection with modal verbs.

Option B seems to be better than A for groups (17a) and (17d), as the English glosses suggest:

(19) Se le ocurrió que yo {hablaba/hablara} con todo el mundo.'It occurred to him/her that I {was talking/should talk} to everybody.'

However, it is not clear that B can be applied to all groups in (17) or to similar alternations. For example, a type B solution of the following mood contrasts with the verb *explicar* 'explain' does not seem to work:

(20) a. El autor explica en la entrevista que su obra literaria {es/*sea} así.
'The author explains in the interview that his/her literary work {is-IND./is-SUBJ.} that way.'
b. La personalidad del autor explica que su obra literaria {*es/sea} así.
'The author's personality explains his/her literary work {being-IND./being-SUBJ.} that way.'

One may think of *explicar* as a verb of communication only in the presence of an animate external argument. As a verb of causal link, its embedded complement seems not to be asserted, but rather presupposed. In any case, this is a lexical difference, hence not appropriate in a type B approach, as opposed to (19) and other similar mood alternations.

Solutions of type B provide an *a priori* more principled way to account for mood alternations, since they require fewer stipulations. Nevertheless, they present some limitations still to be understood. On the other hand, paradigms such as (15) are established on semantic grounds, but they have to be lexically restricted. For example, *escribir* 'write' and *leer* 'read' are verbs of communication allowing for sentential complements, but only the former admits subjunctive, as in *Me escribió que lo esperara un día más* 'He wrote me to wait for him one more day.'

In option C, meaning changes in (17) would be the result of more general tendencies attested in discourse. For example, (17a) could be conceived as a particular case of the well-know tendency to interpret declarations as requests. For the time being, no detailed analyses of options B and C have been developed. As for A, it can only be called "a solution" properly as long as it is able to account for meaning changes that apply to whole paradigms.

4 Mood and locality

The nature of the subjunctive as a dependent mood has a number of syntactic consequences. Both mood selection by head and by an operator are locally restricted. In the first case, one might expect that mood selected by a syntactic head corresponds to its sentential complement, rather than to some other clause. That is, if X° and Y° are potential head triggers of modal inflection in V in (21):

(21) X° . . . Y° . . . V,

it is expected that Y° (rather than X°) induces these mood features (Travis 1984; Baker 1988). This is what we find in (22), where *lamentar* triggers subjunctive and *creer* takes indicative:

(22) a. Lamento que ella crea que la {hemos/*hayamos} defraudado.
'I regret that she believes that we {have-IND./have-SUBJ.} disappointed her.'
b. Creo que ella lamenta que la {*hemos/hayamos} defraudado.
'I believe that she regrets that we {have-IND./have-SUBJ.} disappointed her.'

Mood triggers in (22) are verbs. The subordinate conjunction *que* 'that' is not a mood trigger in subordinate clauses, but a transparent functional category that passes the mood selected by the matrix predicate to the subordinate clause.[1] This role strongly contrasts with that of the conjunction *si* 'whether' in indirect questions, which induces the locality effects expected in (23):

(23) a. Depende de que {*llueve/llueva}.
'It depends on whether it {rains-IND./rains-SUBJ.}'
b. Depende de si {llueve/*llueva}.
(same as (23a))

It has been observed that *creer* 'believe,' *pensar* 'think,' *suponer* 'suppose,' and other verbs of propositional attitude may be interpreted in Y° as parenthetical predicates (that is, as if they were somehow separated by commas and could be omitted or put in a background context). In these configurations, Y° is skipped by X° if the former takes indicative complement. The subjunctive mood can be induced in V whether X°, with italics in (24), is a head or an operator. On these structures, see Lleó (1979), Fukushima (1990), and RAE-ASALE (2009: § 25.8e–j):

(24) a. Es *imposible* pensar que los errores no se {hayan/han} cometido adrede.
'It is impossible to think that the mistakes {have-IND./have-SUBJ.} not been made on purpose.'
b. Ello *no* permite suponer por sí solo que ya {tengan/tienen} ganadas las elecciones.
'This alone does not allow us to assume that they {have-IND./have-SUBJ.} already won the elections.'

The syntactic structure of these sentences presents certain similarities with the one obtained when two negative triggers in successive clauses compete to license a narrow scope indefinite, as in *Nunca venía a casa sin traer {algo/nada}* 'S/he never came over without bringing {something/anything}.'

Another apparent violation of locality is discussed by Borgonovo (2003), who analyzes the unexpected choice of indicative in contexts such as the ones in the (b) sentences below:

(25) a. Para que no me {*enojo/enoje}.
'So that I do not {get-IND./get-SUBJ.} angry.'
b. Para que no creas que me {enojo/*enoje}.
'So that you don't think that I {get-IND./get-SUBJ} angry.'

(26) a. No creemos que {*es/sea} un buen candidato.
'We do not think that s/he {is-IND./is-SUBJ.} a good candidate.'
b. Si no creyéramos que {es/*sea} un buen candidato.
'If we were not to believe that s/he {is-IND./is-SUBJ.} a good candidate'

Since one expects subjunctive after *no creer que* . . ., it comes as a surprise that this mood is rejected in (25b). As Borgonovo argues, the subjunctive in *creas*, induced by the preposition *para*, disallows the subordinate clause to be interpreted as the focus of negation. In fact, the subjunctive in *creer* blocks so-called "negative raising" interpretations.

Locality in mood choices presents other problems. We saw in (8) and (9) that intensional verbs play the role of both modal operators and lexical triggers in selectional contexts. We also confirmed that other predicates which take this mood (such as *molestar* 'annoy') are rejected in polarity subjunctive contexts. Notice now that, assuming this difference, we may account for (27a, b), but not for (27c):

(27) a. Me molesta que {*dice/diga} esas cosas.
'It annoys me that s/he {says-IND./says-SUBJ.} those things}.'
b. Me molestan las cosas que {dice/*diga}.
'The things that s/he {says-IND./says-SUBJ.} annoy me.'
c. Me molesta que diga siempre lo que le {viene-IND./venga-SUBJ.} en gana.
'It annoys me that s/he says whatever s/he feels like.'

The mood alternation in (27c) raises a problem of locality, since the licensing effect of *molestar*, which pertains to group (15c) of subjunctive-taking predicates, should stop in *diga*. Contrary to what one would expect, the subjunctive *venga* is grammatical in (27c). This suggests that subjunctive inflection in *diga* provides the appropriate features required by modal operators as triggers of polarity subjunctive. Minimal pairs can be constructed in which absence of this "inflectional trigger" gives rise to an unlicensed polarity subjunctive in relative clauses:

(28) a. Insisto en que diga algo que {*tiene/tenga} sentido.
'I insist that s/he say-IND. something that {makes-IND./makes-SUBJ.} sense.'

b. Insisto en que dice algo que {tiene/*tenga} sentido.
'I insist that s/he is saying-IND. something that {makes-IND./makes-SUBJ.} sense.'

Finally, adjectival modifiers of some abstract nouns present another syntactic problem of locality in mood triggering, since nouns, rather than the adjectives modifying them, are the phrasal heads that should be lexical selectors. In (29) it is shown that these head nouns are not mood selectors:

(29) a. Era un hecho extraño que la muchacha no {*había/hubiera} telefoneado.
'It was a strange fact that the girl {had-IND./had-SUBJ.} not phoned.'
b. Era un hecho cierto que la muchacha no {había/*hubiera} telefoneado.
'It was a sure fact that the girl {had-IND./had-SUBJ.} not phoned.'

Mood alternations with *el hecho de que*... and other similar structures are often considered to be the result of informational factors. More specifically, indicative is said to be preferred if the proposition is interpreted as focus, whereas subjunctive predominates if it interpreted as thematic or topic-like information. See, on this issue, Woehr (1975), Lipski (1978), Krakusin (1992), and RAE-ASALE (2009: § 25.6). In Bosque (2001), *hecho* 'fact' is analyzed as a *light noun*. Mood selection by its adjectival modifier provides a structure similar to that in which the adjective modifying the English noun *manner* in "in an x manner" turns out to lexically restrict an external predicate (for example V, in "V *in an overwhelming manner*").

5 Mood and scope

As seen from the outset, subjunctive contexts of type (2a, b) are characteristic of, although not exclusive to, restrictive relative clauses. The operators which license the subjunctive in these clauses basically coincide with those which allow for free-choice items (Giannakidou 2001) and some other indefinites such as *alguno* 'some.' The following list is not complete, but it contains most of them:

(30) negation
a. No he encontrado aquí gente que me *guste*.
'I haven't found anybody I like here.'
interrogation
b. ¿Has visto a alguna otra persona que *reaccione* así?
'Have you ever seen any other person react like this?'
future
c. Siéntese donde *quiera*.
'Sit wherever you want.'
conditional sentences

d. Si encuentras algún libro que te *interese*.
'If you find a book that interests you.'
imperatives
e. Deme usted el dinero que *considere* justo.
'Give me the money that you consider to be fair.'
modal verbs
f. Pudo haber dicho algo que *fuera* cierto.
'S/he could have said something that was true.'
intensional predicates
g. Busco un diccionario que me *sirva*.
'I am looking for a dictionary that is of use to me.'
imperfecive tenses
h. El que *haya* escrito este artículo está loco.
'Whoever has written this article is crazy.'
the adverb *solo* 'only'
i. Solo tiene un amigo que le *haga* caso.
'S/he only has friend who pays attention to him/her.'

Licensing takes place in the complex NP headed by indefinites or other quantifiers, but definite determiners are possible if they are compatible with quantificational interpretations of the DP, as in (30e), or give rise to type (rather than token) readings, as in *Juan busca la mujer que lo haga feliz* 'J. is searching for the woman who will make him happy.'

The presence of indicative in all these contexts is generally interpreted as a sign that the proposition falls outside the scope of the modal operator. Nevertheless, the nonspecific interpretations of the subject DP is obtained in (30h) both with indicative (*ha escrito*) and subjunctive (*haya escrito*). There seems to be a general consensus nowadays in the fact that mood changes in relative clauses do not reflect Donnellan's (1966) distinction on attributive vs. referential interpretation of the NP (see Rivero 1975, 1990; Rojas 1977). A "type B solution" – recall the comments on (18B) – is not disregarded here, since the NP subject of (30h) allows for paraphrases such as 'Whoever *might* have written this article.' Quer (1998) argues that scope differences related to mood alternations in relative clauses follow from "models of evaluation" as understood in the Discourse Representation Theory framework.

As explained, the term *polarity subjunctive* attempts to reflect the parallelism between operator triggering of the subjunctive mood and licensing of negative polarity items (NPIs). In fact, negation licenses both subjunctive and NPIs in the same subordinate clauses, whether relative or not:

(31) a. No he leído un libro que {*trata/trate} de ninguno de estos temas.
'I have not read a book which {deals-IND./deals-SUBJ.} with any of these topics.'
b. No sabíamos que {*estaba/estuviera} metido en ningún lío.
'We didn't know that s/he {was-IND./was-SUBJ.} in any trouble.'

c. No me voy a enojar porque {*sacas/saques} a colación ningún asunto.
'I am not going to get mad just because you {mention-IND./mention-SUBJ.} any topic.'

C-command is necessary in both NPI licensing and subjunctive dependencies. Exceptions to the latter seem to be only apparent: lack of c-command between *busca* and *sepas* excludes (32a), whereas the higher position of the Tense (Infl) node in the sentence guarantees c-command of *guste* by the future tense in (32b):

(32) a. Quien tu {sabes/*<u>sepas</u>} <u>busca</u> este diccionario.
'Someone you {know-IND./know-SUBJ.} is looking for this dictionary.'
b. El libro que te <u>guste</u> a ti me <u>gustará</u> a mí.
'Any book you like I will like.'

A more radical difference between subjunctive dependencies and NPI licensing comes from the syntactic properties of the matrix predicates. Operator licensing of subjunctive across factive predicates is straightforward, but NPIs licensing across them is blocked, as pointed out by González Rodríguez (2003):

(33) a. No lamento que María haya dicho {algo/*nada} inapropiado.
'I do not regret that María said {something/anything} inadequate.'
b. No me extrañaría que llamaran la atención a {algún/*ningún} alumno.
'I would not be surprised if they were to scold a student.'

She argues that licensing of negative polarity items by a higher operator is not possible in temporally autonomous embedded clauses. Factives belong to the group of predicates which select those complements, whereas volitional or directive predicates do not. According to this analysis, the antithetical result shown in (34) is expected:[2]

(34) a. {*Deseo/Lamento} que nos visitara.
'I {desire/regret} that s/he visited us.'
b. No {deseo/*lamento} que nos visite nadie.
'I do not {desire/regret} that anyone visit us.'

As we saw in (8) and (9), negation and other operators may license subjunctive in complement, relative, and adverbial clauses. As a matter of fact, subjunctive marks the focus of negation in causal complements. This is shown in (35), where it opens the expectation of a contrastive sentence presenting an alternative cause (... *sino porque* ... '... but rather because ...':

(35) No aceptó el trabajo porque le {ofrecían/ofrecieran} más dinero.
'S/he didn't accept the job because they offered him/her more money.'

The subjunctive mood signals the focus of negation in *es que* clauses as well, as in *No es que me disguste* 'It is not that I dislike him/her/it.' There is much less consensus on the exact contribution of mood to other complement clauses with polarity subjunctive licensed by negation, particularly with verbs of perception and knowledge:

(36) a. Yo no recordaba que entonces me {levanté/levantara}.
'I couldn't remember that I got up then.'
b. No oí que te {estaban/estuvieran} llamando.
'I didn't hear that they were calling you.'
c. Juan no sabía que {tenía/tuviera} que firmar.
'Juan didn't know that he had to sign.'

The meaning differences obtained in (36) are clear enough (Guitart 1990; Haverkate 2002): the indicative entails the truth of the complement, hence a situation which occurred and someone did not remember, hear, or know. The variants with subjunctive lack this entailment and suspend the truth value of the complement. A controversial issue, discussed by Brugger and D'Angelo (1995), Borgonovo (2003), and other authors is whether or not these meaning differences directly follow from the lower clause's being under the scope of negation in the subjunctive variant. The question is problematic since it is not evident that in a "*no* + VP" simple structure, non-quantificational complements of V are outside the scope of negation. Quer (1998) argues for a tripartite (that is, nonbinary) analysis of structures similar to these, which, he claims, avoids this problem. In a less developed theoretical framework, Kleiman (1978) argued that subjunctive in (36) provides the external or sentential interpretation of negation (that it, 'It is not true that I remembered that ...'), whereas the indicative signals a VP internal reading of negation in these structures.

6 Mood and coreference

As pointed out in Section 1, *obviation* is a syntactic manifestation of disjoint reference. The matrix clause's subject and that of the subordinate clause cannot be coreferent in (37a, b):

(37) a. *Quiero que me quede en casa.
'I want to stay at home.'
b. Juan$_i$ desea que {*pro$_i$/pro$_j$} tenga suerte.
'John wants {himself/him-her} to be lucky.'

Predicates which give rise to obviation effects do not coincide in Romance languages. In Spanish, these predicates are typically volitional, as in (37), but the paradigm includes some expressing emotions or affective reactions, as in (38a), and causation, as in (38b):

(38) a. Pro$_i$ lamentaba que {*pro$_i$/pro$_j$} trabajase allí.
'S/he regretted that s/he worked there.'

b. Pro_i ha conseguido que {$^*pro_i/pro_j$} pierda cinco kilos haciendo ejercicio.
'S/he has managed {to lose/to get someone else to lose} five kilos doing exercise.'

The phenomenon extends to experiencer datives (sometimes called *quirky subjects*) interpreted as arguments of a few verbs related to the notions of intention and will, as in (39a). Other emotional factive verbs which select for dative arguments do not give rise to obviation effects systematically, as shown in (39b), and neither do other subjunctive-taking predicates, as in (39c, d):

(39) a. Me_i gusta que {$^*pro_i/pro_j$} cante.
'I like that s/he sings.'
b. Me_i sorprende que pro_i logre concentrarme con todo este ruido.
'It surprises me that I am able to concentrate in spite of all this noise.'
c. Pro_i espero que {pro_i/pro_j} tenga suerte en el examen.
'I hope {I/she/he get} lucky on the exam.'
d. Pro_i negó rotundamente que {pro_i/pro_j} la matara.
'S/he flatly denied that s/he killed her.'

Particularly prominent among the factors that weaken or cancel obviation effects is the presence of modal verbs, as in (40a), as well as passive sentences, as in (40b):

(40) a. Pro_i confío en que pro_i te {??ayude/pueda ayudar}.
'I am confident that {I/it-he-she} can help you.'
b. Pro_i esperaba que pro_i {*se marchase pronto/fuese elegido en la primera vuelta'}.
'S/he expected that s/he was going to {leave soon/be elected in the first round'}.

Other relevant factors have been studied in the literature on obviation, in which some remarkable differences among Romance languages (as well as Latin and Greek) are also analyzed (Costantini 2009; San Martín 2009). The main theoretical approaches on obviation can be gathered in two lines of research:

(41) A. Theories that interpret subjunctive obviation as a particular case in binding.
B. Theories that interpret subjunctive obviation as a consequence of the distribution between inflected verbs and infinitives.

A-theories hold that obviation constitutes a violation of principle B of the binding theory of Chomsky (1981). Some authors supporting them, like Picallo (1985, 1990) or Progovac (1993), take as a point of departure the defective nature of subjunctive inflection. This property implies that the syntactic environment in which the pronominal null subject of the subordinate subjunctive clause is free (that is, not bound) is extended to a higher node. From this point of view, we get an exact parallelism in the sentences in (42):

(42) a. [El jefe de [Juan]$_i$]$_j$ {*lo$_j$/lo$_i$} aprecia mucho.
'Juan's boss appreciates him very much.'
b. [El jefe de [Juan]$_i$]$_j$ desea que {pro$_i$/*pro$_j$} regrese pronto.
'Juan's boss wants {him/himself} to return soon.'

Arguing for A as well, Kempchinsky (1987, 1990) claims that the extension of the binding domain for the pronominal follows from the idea that verbs of volition select for a modal operator in C°, which provides the subjunctive morphology. The subjunctive tense moves from I° to C° at LF so that the governing category for the embedded subject will be the matrix clause rather than the subordinate clause.

Approach B is defended, with some differences, by Farkas (1992), Bouchard (1984), Schlenker (2005), and others. Bouchard (1984) argues that obviation follows from Chomsky's (1981) *Elsewhere principle*, which disallows pronominals in contexts in which anaphors are possible. A potential problem of this line of reasoning is the fact that infinitives are not in complementary distribution with inflected verbs, whether in subjunctive, as in (43a), or indicative, as in (43b):

(43) a. La animó a {que asistiera/asistir} a la reunión.
'S/he encouraged her to attend the meeting.'
b. Te prometo {que iré/ir}.
'I promise you {that I'll go/to go}.'

Other problems of B are analyzed by San Martín (2009). More recent approaches to the analysis of subjunctive obviation are discussed in Kempchinsky (2009) and Costantini (2009).

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NOTES

1 Even so, the complementizar *que* plays an important role in the syntax of subjunctive clauses. Although it is possible to delete it with indicative complements in some contexts (RAE-ASALE 2009: § 43.3g–j), its deletion is characteristic of subjunctive complements of verbs of volition, intention, and influence, as in (i):

(i) Esperamos (que) sepan ustedes disculpar este pequeño incidente.
'We hope you will excuse us for this minor incident.'

Spanish is much more restrictive than Italian as regards this structure. Complementizer deletion has been studied in great detail for Italian (see Quer 2005: § 4.4 for a review of the main proposals). In Bosque and Gutiérrez-Rexach (2009), it is suggested that in absence of the subordinate conjunction *que*, the V + SUBJUNCTIVE complex overtly moves to C° in Spanish. An argument for this comes from the rejection of preverbal subjects and preposed adverbs in this structure:

a. Nos rogaban (que) estuviéramos siempre alerta.
'They asked us to always be alert.'
b. Nos rogaban *(que) siempre estuviéramos alerta.
(same meaning)

2 I suggest that the apparent counterexamples provided by some emotive factive predicates may be analyzed as results of lexical recategorization:

(i) a. No me gusta que nos visite nadie.
'I don't like anyone visiting us.'
b. *No me sorprende que nos visite nadie.
'It doesn't surprise me that anyone is visiting us.'

Notice that *gustar* 'like' does not behave as a factive verb here, but rather as a volitional predicate. In fact, sentence (ia) does not imply 'Someone {is visiting/has visited'} us.' As opposed to *sorprender* 'surprise' and other factive verbs, *gustar* rejects *el hecho de que . . .* 'the fact that' in these structures.

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20 The Simple Sentence

HÉCTOR CAMPOS

1 Introduction

A "sentence" (Sp. *oración*) is a minimal unit of predication that relates a subject and a predicate, where the former will be typically represented as a noun phrase (NP) and the latter as a verb phrase (VP). Traditionally, sentences have been classified according to three criteria:

(1) a. according to their dependence upon or independence from other units
b. according to the nature of their predicate
c. according to the "attitude" of the speaker

Criterion (1a) distinguishes main from embedded clauses. Whether a sentence appears in a main or in a subordinate clause is relevant for the application of certain syntactic operations; however, this is too general a criterion to help us understand – and let alone classify – the variety of sentences that exist in a particular language. Criterion (1b) classifies sentences according to the nature of the verb; that is, whether it is transitive, intransitive, copular, passive, reflexive, reciprocal, or impersonal (see Gili Gaya 1961: s. 32; GLE: ch. 19). However, as noted in the NGDLE, these particular properties of verbs, while relevant for certain aspects of syntactic analysis, do not necessarily constitute "sentential types" and "do not constitute grammatical properties that will necessarily help define a sentential paradigm" (NGDLE: s. 1.13k).

Criterion (1c) is the one that proves most fruitful to classify sentences and it is the one that we will pursue in this chapter. I will be following the classification proposed by the Real Academia Española (RAE) in the NGDLE (2009) and NDGLEM (2010), which revise that of previous editions of the grammar (GLE 1931; ESB 1979; NGDLE 2009), and which incorporates terminology and concepts of modern linguistics). I will be framing the new classification proposed by the RAE on more recent theories of syntax, semantics, and pragmatics. Section 2 introduces the theoretical framework that I will be using throughout this chapter. Sections 3 and 4 propose a classification based on this framework.

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2 Classification of sentences according to the "attitude" of the speaker

Traditional grammars distinguish between the pure content of the message (the *dictum*) and the way that that message is expressed (the *modus* or "modality"). The ESB (s. 3.2.2) proposes seven different types of sentences, depending on how the *dictum* in (2a) is expressed: declarative (2b), interrogative (2c), exclamative (2d), desiderative (2e), imperative (2f), dubitative (2g), or as possibility/probability (2h) (see ESB 1979: s. 3.2.2; NGDLE: s. 1.13b –i, ch. 42).

(2)	a.	LLEGAR (Melita) ARRIVE (Melita)	(*dictum*)
	b.	Melita llegó la semana pasada. 'Melita arrived last week.'	(declarative)
	c.	¿Llegó Melita la semana pasada? 'Did Melita arrive last week?'	(interrogative)
	d.	¡Melita llegó la semana pasada! 'Melita arrived last week.'	(exclamative)
	e.	Ojalá llegue Melita pronto. 'I hope Melita will arrive soon.'	(desiderative)
	f.	Melita, ¡llega pronto! 'Melita, arrive soon!'	(imperative)
	g.	Quizás Melita llegue mañana. 'Melita may arrive tomorrow.'	(dubitative)
	h.	Melita habrá llegado ya. Melita will-have arrived already 'Melita may have arrived already.'	(probability)

The types shown in (2b –h) seem to constitute the main types of sentences in Spanish. However, the ESB (1979: s. 3.2.2) is quick to warn the reader that the classification proposed in (2) is a not "a rigorous classification in which one term excludes the others, because in the linguistic reality they can be superimposed and they create intermediate zones that may require a larger number of nuances."

The NGDLE attempts a more rigorous classification by distinguishing between "*enunciación*" and "*enunciado*," terms which, unfortunately, have no good (intuitive) translation in English. "*Enunciación*" is defined as the "verbal act that the speaker carries out with his own words and that represents, therefore, the verbal action" (NGDLE: s. 42.1b). "*Enunciado*," on the other hand, is the "linguistic structure with which the verbal act is carried out" (NGDLE: s. 42.1b,c).

Following recent developments in syntactic as well as semantic theories, the NGDLE (s. 42.1) claims that there are four basic types of "enunciación" (Sp. *modalidades de la enunciación*, *modalidades enunciativas*): interrogative, exclamative, imperative, and declarative. The NGDLE observes that exhortative as well as desiderative sentences also have enough distinctive syntactic properties in Spanish to be considered as additional "types"; as the NGDLE also observes, sometimes the boundary between imperatives and these two latter types is not very clear. To facilitate our discussion, I will assume in this work that exclamative as well as desiderative sentences constitute different types in Spanish.

I will refer to these different types of "enunciación" as "**sentential types**." It is worth noting that the notions of "*enunciado/enunciación*" allow for the classification of nonsentences. The NGDLE (s. 1.13g) observes that the expression *¡De acuerdo!* 'OK, agreed!' is equivalent to the sentence 'I agree!,' and the nominal group *¡Mi cartera!* 'My wallet!' is equivalent to the exclamative 'Someone has stolen my wallet!' For reasons of space, we restrict our study to sentences in this article. Following Chierchia and McConnell-Ginet (1990) and Portner (2004, 2005, 2009), I will also assume that each sentential type has a semantic correlate, or "**sentential force**." The relation between "sentential type" and "sentential force" is shown in (3), adapted from Portner (2009: 263). Portner only includes the first three types. Following the NGDLE, I have added exclamatives and desideratives. Notice that dubitative (2g) and probability sentences (2h) have been left out of our classification, as they also have been in the NGDLE. I return to these constructions in Section 4.

(3)	*Sentential type*	*Sentential force*
	Declarative	Asserting
	Interrogative	Asking
	Imperative	Requiring
	Exclamative	Exclamation
	Desiderative	Wishing

However, it is clear that we need yet another level to capture the complete meaning of a sentence. Consider (4):

(4) a. ¿Puedes decirme la hora?
You-can tell-me the hour
'Can you tell me the time?'
b. ¡Dime la hora, por favor!
Tell-me the hour, please
'Tell me the time, please!'

Syntactically speaking, (4a) is an interrogative, and, as such, it has the sentential force of asking. In fact, someone could (jokingly) assume that this is a real question, answer "yes" and stop there, without telling us the time! However, we generally do not mean (4a) as a question, but as a request for someone to tell us the time, similar to (4b). This suggests that there is yet another level of meaning beyond that of sentential force. Semanticists and pragmaticists refer to this level as "**illocutionary force**" (Sp. *fuerza ilocutiva/elocutiva*); that is, "the communicative action which the speaker intends to perform by getting the hearer to understand that this is the speaker's intention" (Portner 2005: 192). Following Portner (2009: 263), I will assume that "sentential force must be analyzed at the interfaces among syntax, semantics, and pragmatics, while illocutionary force is a pragmatic phenomenon having to do with the speaker's communicative intentions, analyzed in terms of speech act theory (e.g., Austin 1962; Searle 1969)."

3 Basic sentential types

According to Portner (2005: 194), "the three sentential types of declarative, interrogative, and imperative seem to be universal. There are other 'minor' clausal types, like exclamatives and optatives, in some languages. But even all of these do not match the diversity of illocutionary acts, which include promising, threatening, hinting, proposing, denying . . ." In what follows, we will concentrate on the different constructions in which these basic sentential types are realized in Spanish. Where appropriate, we will briefly discuss some of the unexpected illocutionary forces that these constructions may have.

3.1 *Declarative type*

Declarative type, with its corresponding assertive force, is the default sentential type and force. A **declarative sentence** (Sp. *oración declarativa*) like (2b) expresses the point of view of the speaker about a particular situation (whether true or not). As with all the other types discussed in this section, declarative sentences can be affirmative or negative.

3.2 *Interrogative type*

Interrogative force is usually expressed by means of a **question**, as in (5):

(5) a. ¿Vendrá Gabriela a Estados Unidos?
'Will Gabriela come to the United States?'
b. ¿Vivirá Gabriela en el este o en el oeste del país?
'Will Gabriela live in the east or the west of the country?'
c. ¿Cuándo vendrá Gabriela a los Estados Unidos?
'When will Gabriela come to the United States?'

The sentential force of interrogatives is usually that of asking. According to the NGDLE (s. 42.6a ff.), (5a) is a "**total interrogative**" (Sp. *interrogativa total*). Total interrogatives are usually answered with "yes" or "no." The example in (5b) is a "**disjunctive interrogative**" (Sp. *interrogativa disyuntiva*); the alternatives for the answer are given explicitly ("in the east or the west"). The example (5c) shows a "**partial or pronominal interrogative**" (Sp. *interrogativa parcial o pronominal*). Partial interrogatives are usually asked with interrogative words such as *cuándo* 'when,' *quién* 'who,' *qué* 'what,' etc., and they require a specific answer.

NGDLE (s. 42.6b) observes that (5a) can also be considered a disjunction since it implicitly gives the listener the choice of "will Gabriela come to the US?" or "won't Gabriela come to the US?" A reviewer has observed that (5b) can also be added a further disjunction, as in "Will Gabriela live in the east or in west, or in the north or in the south?" Thus, the main difference between (5a) and (5b) is that (5a) may be disjunctive at most. Hence linguists like Bolinger classify simple questions like (5a) as "binary," while those of (5b) as "alternative."

Total interrogatives can also be answered with other adverbs besides *sí* 'yes' and *no* 'no.' For affirmative answers, we could say *claro*, *desde luego*, *efectivamente*, *naturalmente*, *por supuesto*, *sin duda*, all meaning something close to 'sure!, of course!'. Instead of *no*, we could have *de ningún modo*, *en absoluto*, *qué va*, *quia*, *ni hablar*, etc., meaning something like 'not at all, absolutely not.' To express doubt, we could use *a lo mejor*, *probablemente*, *quizás*, *seguramente*, *tal vez*, 'perhaps, maybe' (NGDLE: s. 42.7l).

Regarding the syntactic properties of the interrogatives in (5), we can see that all three interrogatives invert the subject and the verb. Total interrogatives (5a) have a rising intonation; disjunctive interrogatives have a rising intonation on the first conjunct (*en el este?*) and then a falling intonation on the second conjunct (*en el oeste*). In partial interrogatives (5c), the interrogative word typically appears in front of the sentence and they have a falling intonation.

Explicit disjunctive questions can be interpreted as total questions; thus, (5b) can be answered (jokingly again) as *sí/no*. Intonation will mark whether the question is meant as a total or as a disjunctive interrogative. As a total interrogative, the group (*en el este o en oeste*) is pronounced as a unique melodic unit and it has a rising intonation. As a disjunctive question, it is pronounced with two melodic units, the first one with a rising and the second one with a falling intonation.

3.2.1 Interrogatives and inversion We noticed in (5) that the subject and the verb are typically inverted in interrogatives. However, the subject can also appear at the end of a sentence (6a) or in front of the verb (6b):

(6) a. ¿Vendrá a Estados Unidos Gabriela?
Will-come to States United Gabriela?
'Will Gabriela come to the United States?'
b. ¿Gabriela vendrá a Estados Unidos?
'Gabriela will come to the United States?'

With neutral intonation (6a) is very close in meaning to (5a). In (6b), on the other hand, the speaker is trying to confirm or put into doubt a previous statement and thus it is not equivalent to (6a). The example in (6b) is an echo question: we heard the statement and we are trying to reconfirm its veracity. This may be a structure of "verum focus," one where we are trying to confirm the original statement.

With compound tenses (i.e., tenses that include the auxiliary verbs *haber* 'to have' or *estar* 'to be'), the subject rarely appears between the auxiliary and the main verb in questions, unlike what is observed in English. While the subject is marginal after *estar*, it is ungrammatical after *haber*, as shown in (7). However, it is perfectly possible for the subject to appear after the main verb or the direct object:

(7) a. ¿Está (??Mema) preparando (Mema) la cena (Mema)?
'Is Mema cooking dinner now?'
b. ¿Ha (*Mema) preparado (Mema) la cena (Mema)?
'Has Mema cooked dinner?'
c. ¿Había (??Mema) preparado (Mema) la cena (Mema) (cuando llegaste)?
Had (??Mema) prepared (Mema) the dinner (Mema) (when you arrived)?

If *haber* 'to have' contains more than one syllable, then the subject between the auxiliary and the verb sounds better, as shown in example (7c). See Suñer (1987) for further details.

3.2.2 Confirmative questions Some total interrogatives can be interpreted as "**confirmative questions**" (Sp. *interrogativa confirmativa*). In these questions the speaker knows something to be a fact and is asking for confirmation. Questions like (8a) have falling intonation. These questions can also be asked with a "**confirmative question tag**" (Sp. *apéndice confirmativo/interrogativo, muletilla interrogativa*) *¿no?* 'not,' as shown in (8b):

(8) a. ¿Eres de Chile?
'Are you from Chile?'
b. Eres de Chile, ¿no?
'You are from Chile, aren't you?'

These questions can also be asked with other question tags such as *¿no es cierto/ verdad?* 'isn't it true?' or just *¿cierto/verdad?* 'true?' In many Central American countries, just *¿veá?* or *¿va?* are used. Other confirmative tags used in many dialects are *¿ves?* 'you see?,' *¿eh?*; more dialectal are tags like *¿tú sabes?*, *¿viste?* 'you know/ see?,' *¿a poco no?* 'really?' (typically used in answers), etc. There are also several lexicalized expressions: *¿ves?* 'you see?,' *¿no crees/te parece?* 'don't you think?'. More and more the confirmative tag *¿vale?* 'ok?' is gaining ground. See NGDLE (s. 42.8).

3.2.3 Disjunctive interrogatives We can add the question tag . . .*o no?* 'or not?' to a total interrogative to make it disjunctive, as in (9). The question in (9) is used when the speaker wants the hearer to clearly state his/her choice:

(9) ¿Vendrás con nosotros o no?
'Will you come with us or not?'

In (9), the question tag forms an intonation unit with the rest of the sentence. Constructions like (9) should not be confused with constructions like (10a,b):

(10) a. Vendrás con nosotros, ¿o no?
'You will come with us, or not?'
b. Vendrás con nosotros, ¿sí o no?
'You will come with us, yes or no?'

In (10) the question tags constitute a separate intonation unit, as indicated by the comma. In (10a) the speaker takes for granted the content of the declarative, but then puts it into question. The example in (10b) is a bit harsher in tone than (10a), and the speaker is asking the hearer to make up his/her mind. The questions in (9) and (10) would be considered disjunctive interrogatives.

3.2.4 Inversion in interrogatives We saw in (5c) that the subject and the verb are typically inverted in total interrogatives. The subject in partial interrogatives may also appear at the end of the sentence, as shown in (11). The subject seldom appears between the interrogative word and the verb:

(11) ¿Cuándo (*Gabriela) vendrá a Estados Unidos (Gabriela)?
When (*Gabriela) will-come to States United (Gabriela)
'When will Gabriela come to the United States?'

In Caribbean Spanish, however, a pronominal subject may appear between the question word and the verb (NGDLE: s. 42.9h):

(12) ¿Dónde tú vives?
'Where do you live?

3.2.5 Lack of inversion in interrogatives Some causal and manner interrogative words such as *por qué* 'why,' *cómo* 'how,' *a santo/cuento de qué* 'due to what,' *hasta qué punto* 'how far,' *de qué modo* 'how,' etc., allow for preverbal subjects. Rhetorical questions usually contain preverbal subjects as well (NGDLE: s. 42.9f –g):

(13) a. ¿Por qué los profesores tienen tres meses de vacaciones?
Why the teachers have three months of vacations
'Why do teachers have three-month vacations?'
b. ¿Cuándo los pueblos no han buscado la libertad?
When the peoples not have looked-for the freedom
'When have the people not looked for their freedom?'

3.2.6 Split questions "**Exploratory questions**" (Sp. *preguntas exploratorias*) are another type of interrogatives (see NGDLE: s. 42.9k, l, m). In generative grammar, these constructions have been called "**split wh-questions**" (Sp. *preguntas-wh escindidas*) (see Py 1991; Vigara Tauste 1992; for more recent analyses, see Uriagereka 1988; Lorenzo 1994; Camacho 2002; López-Cortina 2007, and references there cited). In these structures the question has been split into two parts: a part containing the wh-question word, and then, after a pause, an interrogative tag that contains the answer. The examples in (14a) and (14b) mean almost the same. In these constructions the speaker expects a confirmation from the hearer. These constructions are found mainly in colloquial speech.

(14) a. ¿A dónde vas? ¿A Oviedo?
'Where are you going? To Oviedo?'
b. ¿Vas a Oviedo, no?
'You are going to Oviedo, right?'

In colloquial Spanish, we encounter another type of "split question." I will refer to these as "**split expletive *qué* questions**" (Sp. *preguntas-wh escindidas con qué-expletivo*), as in the examples in (15):

(15) a. ¿Qué vas, al cine?
What you-go to-the cinema
lit. 'What are you going, to the cinema?'
b. ¿Qué invitaste, a José?
What you-invited, to José
lit. 'What did you invite, José?
c. ¿Qué? ¿Viene Pedro con nosotros?
What Comes Pedro with us
'So, is Pedro coming with us?'
d. ¿Vas qué, al cine?
You-go what, to-the cinema
'You are going what, to the cinema?'

These constructions are very close in meaning to those discussed in (14a). However, notice that we use *qué* 'what' instead of the specific question word that corresponds to the answer, which may include a sentence, as in (15c). Notice that the split questions in (14) and (15), even though they contain a wh-word, would be answered "yes" or "no," thus constituting instances of total, rather than partial, interrogatives. As observed in López-Cortina (2007), these constructions may also appear with *qué* 'what' *in situ*, as shown in (15d).

3.2.7 Interrogatives with an attitude Question words can be used with certain emphatic nouns to express that the speaker is upset, uncomfortable, impatient, desperate, etc. Some of these nouns are *diablo(s)/demonio(s)-diantre(s)* 'demon(s),' *narices* lit. 'noses,' *coño* 'cunt,' *carajo* 'penis,' *cojones* 'balls,' etc. (NGDLE: s. 42.12e ff; note that these words vary among the different countries):

(16) a. ¿Dónde diablos/demonios estabas?
Where demons you-were
'Where the hell were you?'
b. ¿Cómo narices dice eso?
How noses he-says that
'How on earth can he say that?'

3.2.8 Interrogatives and some of their illocutionary acts In this section we will consider different illocutionary acts that interrogative sentences may have. Syntactically speaking, all the examples in (17) are interrogatives, and as such, they have the sentential force of asking. However, their illocutionary forces are different. The illocutionary forces are shown in the parentheses following the example. We cannot include here an exhaustive list of all the possible illocutionary forces for interrogatives. However, all we need to observe here is that there is no one-to-one correspondence between sentential type (the linguistic structure we use) and illocutionary force (the communicative action we intend to perform).

(17) a. ¿Te tomarías un cafecito conmigo? (invitation)
'Would you (like to) have a coffee with me?'
b. ¿Me pasas la sal? (request)
'Can you pass me the salt?'
c. ¿Vamos al cine esta tarde? (suggestion/invitation)
'Shall we go to the movies tonight?'
d. ¿Te molesta el humo? (permission)
'Does smoke bother you?'
(i.e., Do you mind if I smoke?)

3.3 Exclamative type

Parallel to what we observed for interrogatives in Section 2.1, exclamative force has two basic syntactic realizations: "**total exclamatives**" (Sp. *exclamativas totales*) and "**partial exclamatives**" (Sp. *exclamativas parciales*) (NGDLE: s. 42.14, s. 32.2). Zanuttini and Portner (2003) argue that there is no syntactic element introducing exclamative force. In their system, exclamatives are the result of the co-occurrence of a factive and a WH operator. Thus, examples like (18a) would not qualify as exclamatives, but as declaratives. Following the RAE, I will take a more conservative approach, and assume that there is an exclamative force that allows for the realization of both types of exclamatives in (18).

Regarding their sentential and illocutionary force, they are typically used for expressing emotional reaction (surprise, annoyance, exclamation, etc.). Total exclamatives look like declarative sentences, but they differ in intonation and by the fact that we write them between exclamation marks, as shown in (18a). Partial exclamatives appear with interrogative/exclamative words, as shown in example (18b). Note that the adverb *tan* 'so' is used in exclamatives but not in declaratives.

(18) a. ¡Estoy (tan/muy/re-/requete) cansado!
'I-am (so/very/very/very) tired!'
b. ¡Qué cansado estoy!
What tired I-am
'I am so tired!'

Partial exclamatives are much more frequent in Spanish than they are in English. Partial exclamatives typically invert the subject and the object. Apart from *qué* 'what,' we also find the interrogative words *cuánto(a)(s)* 'how much/many' (19a,b), *cómo* 'how' (19b), and *cuán* 'how' (19c).

(19) a. ¡Cuántas amigas tiene Claudio!
How-many friends has Claudio
'Claudio has so many (girl) friends!'
b. ¡Cuánto/Cómo bailamos en la fiesta!
How-much/How we-danced in the party
'We danced so much at the party!'
c. ¡Cuán/Qué inteligente es Belén!
How/what intelligent is Belén
'Belén is so intelligent!'
d. ¡Qué días pasamos allí!
'What days we-had there!'

Cuánto(a)(s) 'how many' is used with nouns. *Cuánto* 'how much' is used when we want to emphasize quantity. Notice that in (19b) we can also use *cómo* 'how' instead of *cuánto* 'how much.' To form an exclamative with an adjective, we use *cuán* 'how,' as in (19c). This form, however, is archaic and is typically replaced by the interrogative *qué* 'what'.[1] *Qué* 'what' can also precede a noun, as in (19d). In this instance, we do not express the property of the noun we are trying to enhance. Thus (19d) could mean that we had some fun days, some hard days, some tough days, etc.

3.3.1 Exclamatives and expletive *que* In colloquial language, it is possible to find a (second) expletive *que* 'that' following the exclamative phrases with *qué* 'what,' as in (20):

(20) ¡Qué interesante/largo que es este libro!
What interesting/long that is this book
'This book is so interesting/long!'

Exclamative constructions with expletive *que* 'that' are found less frequently with *cuánto/cuánto(a)(s)* 'how many/how much,' but the construction is still possible nonetheless, as in ¡*Cuánto que ha nevado este invierno!* 'It has snowed so much this winter!'

3.3.2 Exclamatives, fronting and affective morphemes We typically focalize elements that we want to stress by fronting them. If we add the proper intonation to these constructions, we get an exclamative. In this case, it is impossible to use an expletive *que*. When an adjective or an adverb is fronted, it is not uncommon to use a diminutive (21a) or some affective morpheme (21b) with the fronted forms:

(21) a. ¡Rápido/Rapidito se fueron de la fiesta!
Fast/Fast-dim they-left from the party
They left the party very fast.'
b. ¡Caro/carazo pagué por el arreglo del coche!
Expensive/expensive-aug I-paid for the repair of-the car
'I paid a lot of money to repair the car!'

When a noun or a prepositional phrase is fronted, we must use expletive *que*, as in (22a). Some grammarians have argued that constructions like (22a) are (exclamative) relative clauses. But, as noted in NGDLE (s. 42.15e), when the noun appears inside a prepositional phrase, the whole prepositional phrase must be fronted to form an exclamative (22b), as opposed to what happens in the corresponding relative clause (22c); this suggests that constructions like (22a) are not noun phrases.[2]

(22) a. ¡Las tonterías que dices!
The nonsenses that you-say
'You are talking such nonsense!'
b. ¡Con los amigos que sales!
With the friends that you-go-out
'The friends you go out with!'
c. . . . los amigos con los que sales . . .
. . . the amigos with the that you-go-out . . .
'. . . the friends that you go out with . . .'
d. ¡Es increíble las tonterías que dice!
Is incredible the nonsenses that he-says
'It's incredible how many stupid things he says!'

Further evidence that constructions like (22a) are not noun phrases comes from constructions like (22d), where the main verb is used in the singular, rather than in the plural form that we would expect if (22a) were a noun phrase.

3.3.3 Exclamative determiners We observed in (19d) that the interrogative/exclamative *qué* 'what' can be used in front of nouns. Instead of *qué*, it is very common to find the "**exclamative determiners**" (Sp. *determinante exclamativo*) *vaya* or, less commonly, *menudo* (see NGDLE: s. 42.15k –l). With *menudo*, the use of expletive *que* is optional; with *vaya*, it is obligatory. *Vaya* must be fronted while *menudo* can stay *in situ*:

(23) a. ¡Vaya/menudo auto que te compraste!'
"vaya"/"menudo" car that you you-bought
'You got yourself some car!'
b. *Se compró vaya auto.
'He got himself some car!'
c. ¡Se compró menudo auto!
'He got himself some car!'

We can use *vaya que/vaya si* in front of a sentence to emphasize the quality of the adjective or adverb (sometimes implicit). We can also use the interjections *caramba* 'good heavens,' and *pucha(s)* 'jeez, wow' in front of these sentences with the same effect. The use of *que* is obligatory in these examples, as shown in example (24c):

(24) a. ¡Vaya que/vaya si eres tonto!
"Vaya" that/"vaya" if you-are foolish
'You are so foolish!'
b. ¡Vaya que/vaya si trabajas!
"Vaya" that/"vaya" if you-work
'You work so much!'
c. ¡Caramba que/Puchas que trabajas!
Wow that/Wow that you-work
'You work so hard!'

The constructions in (24) must not be confused with the construction in (25), where both *que* 'that' and *si* 'if' are used. These constructions are typically used as replies to questions and emphasize quantity:

(25) -¿Gastaste mucho dinero?
Spent-you much money?
'Did you spend a lot of money?'
-¡Vaya que si gasté (mucho) dinero!
"Vaya" that if I-spent (much) money
'I spent a lot of money!'

In some dialects the use of *vaya* is incompatible with the adverb of quantity.

3.3.4 Exclamatives with adjectives and adverbs We saw in (19c) that we typically use *cuán* 'how' and *qué* 'what' to form exclamative adjective phrases. A very common exclamative construction with adjectives and adverbs is the one found in (26a). Notice that in these constructions we always use the neuter article *lo* 'the.' The adjective agrees with the noun it modifies; there is no agreement between the article and the adjective/adverb. *Que* is obligatory in these examples:

(26) a. ¡Lo inteligentes que son esas chicas!
The intelligent that are those girls
'Those girls are so intelligent!'
b. ¡Lo bien que habla francés Maggie!
The well that speaks French Maggie
'Maggie speaks French so well!'

3.3.5 Exclamative *cómo* 'how' Adjectives and adverbs may also appear with the exclamative *cómo* 'how.' In the examples in (27), the adjective is never fronted, unlike what is observed in English and also different from what we observed for *cuán* and *qué* in (19c). *De* 'of' typically precedes the adjective and adverb in these constructions. The same constructions without *de* sound formal or archaic:[3]

(27) a. ¡Cómo son (de) inteligentes esas chicas!
How are (of) intelligent those girls
'How smart those girls are!'
b. ¡Cómo pinta (de) bonito Ximena!
How paints (of) beautiful Ximena
'How beautifully Ximena paints!'

3.3.6 Exclamatives indicating quantity We saw in (19a) that to emphasize the quantity of a noun we use the interrogative/exclamative *cuánto(a)(s)* 'how much/many.' The construction shown in (28) is equally productive in Spanish, where the noun *cantidad* 'quantity' in parentheses is typically omitted:

(28) ¡La (cantidad) de amigas que tiene Claudio!
The (quantity) of friends that has Claudio
'Claudio has so many (female) friends!'

These constructions can also appear with the exclamative *qué*:

(29) ¡Qué (cantidad) de amigas (que) tiene Claudio!
What (quantity) of friends (that) has Claudio
'Claudio has so many (girl) friends!'

3.3.7 Exclamatives and expletive *no* We may find exclamative constructions with *no* 'not' as in (30), where *no* does not add a negative value to the sentence. This is an instance of "**expletive no**" (Sp. *no expletivo*). These constructions have a somewhat literary and archaic flavor to them (see Campos 1993; Espinal 2000, NGDLE: s. 48.11n).

(30) ¡Cuántas veces no te he advertido!
How-many times not you have warned
'How many times I have warned you!'

Expletive negation can also be found with total exclamatives, although these constructions are even less common than those in (30). They usually appear in the future tense and may be preceded by *si* 'if':

(31) ¡(Si) no será tonto!
(If) not he-will-be foolish
'He is so stupid!'

3.3.8 Predicative exclamatives Finally, exclamative force can be realized without a verb, as in (32). These exclamatives are known as "**two-member exclamatives**" (Sp. *exclamativas bimembres*) or as "**predicative exclamatives**" (Sp. *exclamativas predicativas*) (NGDLE: s. 42.15i –j). We typically interpret these constructions as containing a copular verb and we can use an exclamative adjective (32a,b) or noun phrase construction with an adjective (32c) to form them:

(32) a. ¡(Qué) bonito el departamento de Guille!
(How) pretty the apartment of Guille
'How pretty Guille's apartment is!
b. ¡Difícil el examen!
Difficult the exam
'The exam was very difficult!'
c. ¡Linda idea la tuya!
Pretty idea the yours
'Nice idea (you have/had)!'

In (32), the subject is a noun phrase. It is possible for predicative exclamatives to contain a full sentence as a subject (for further analysis of exclamatives, see Gutiérrez Rexach 2001):

(33) ¡Qué bueno que puedas venir a la reunión!
What good that you-can come to the meeting
'It's great that you'll be able to make it to the meeting!'

3.4 Exhortative type

Exhortative sentences are those that urge/ask/encourage someone to do something. Following Spanish traditional grammars (GLE: s. 313, for instance), I will call "exhortative" what some grammarians call "imperative," but as we will see below, our term is more comprehensive. Since the term "imperative" refers both to a mood as well as to a modality, I assume the term "exhortative" in order to avoid confusion. The sentential force of exhortative constructions is that of ordering, requesting, or urging someone to do something. Exhortative force may also be expressed with many different constructions or forms.

Commands are a subtype of exhortatives. As we can see in (34), commands typically contain a verb in the imperative mood for some forms (34a). In standard Spanish, these forms are the forms corresponding to someone considered the same or lower in authority or power (pedagogically called the ''informal forms'') (GLE: s. 313). On the other hand, the present subjunctive is required for other commands (34b, 35). The formal forms are seen in (34b). Notice that etymologically (and morphologically) speaking, *usted(es)* 'you-formal' is a third person, deriving from *vuestra merced* 'your Mercy/Highness' and still keeps the verbal forms of the third person throughout all the tenses and moods. This pronoun is considered a second person when we consider discourse (the addressee), but a third person for morphological effects and agreement in general (i.e., it takes the same agreement patterns as the pronouns *he/she*).

The term "exhortative" (as opposed to "imperative") allows for the use of two different moods (imperative in 34a and subjunctive in 34b) without any major confusion. Invitations to carry out an action along with the speaker are also a subtype of exhortative since we are asking someone to do something (albeit along with us). I will refer to these constructions as "subjunctive invite-alongs," which in English are usually expressed as "*let's* + verb." Some grammarians have argued against treating the forms in (34c) as "imperative" as these constructions contain neither a verb in the imperative form (but neither do the forms in 33b) nor a second person subject. By treating them under the more general label of "exhortative," we avoid this problem and at the same time capture the fact that these constructions behave, syntactically speaking, similar to commands (see below). Furthermore, as I will show below, it is possible to find exhortative constructions with subjects that are not second or first person. Again, these constructions are hard to fit into an "imperative" type, but will easily fit into our "exhortative" type.

Consider the verb *venir* 'to come'; the basic affirmative forms to express command are shown in (34). With the exception of (34c), all these forms would be translated as the imperative form 'Come!' in English. I include the invite-along form for *nosotros* 'we.' The examples in (34) show the different forms depending on who the exhortative is intended for:

(34)	a.	ven (tú), you, sg. inf.	vení (vos), you, sg. inf.	venid (vosotros/as), you, pl. inf.
	b.	venga (Ud.), you, sg. form.	vengan (Uds.) you, pl. inf.	
	c.	vengamos (nosotros/as) 'let's come'		

The forms in (34a) are imperative while those of (34b,c) are subjunctive forms. To express these forms in the negative (i.e., 'don't come,' 'let's not come'), we must resort to the present subjunctive in all cases since imperative forms do not have a corresponding negative form, as shown in (35):

(35) a. no vengas (tú), no vengas (vos), no vengáis (vosotros/as)
b. no venga (Ud.), no vengan (Uds.)
c. no vengamos (nosotros/as)

We observe two general syntactic properties with these exhortative forms: (1) clitic pronouns typically follow the verb in the affirmative and precede it in the negative (*hazlo* 'do it' vs. *no lo hagas* 'don't do it'); and (2) negative forms are formed with the present subjunctive. I take these properties to be distinctive of "exhortative type."

The subject of a command is the person who receives the order, suggestion, or advice (i.e., the hearer). Proper names are never the subject of a command; rather, when proper names appear with exhortatives, they are vocatives and may appear before or after the verb. Notice that the verb takes the form of *tú* 'you-inf.' or *usted* 'you-form.,' depending on the level of formality and thus showing that *Pedro* is not the subject:

(36) a. Pedro, ¡pasa!/¡pase!
'Pedro, come in! (inf.)/come in! (form.)'
b. ¡Pasa/pase, Pedro!
'Come in, Pedro!'

The subject of a command is usually not expressed. However, when it appears, it typically follows the verb and implies a contrast or discrimination, where the listener is selected from a larger group. When the subject precedes the imperative, there is usually a contrast with another subject (37c):

(37) a. ¡Pasa!, ¡Bailemos!
'Come in!' 'Let's dance!'
b. ¡Pasa tú!, ¡Bailemos nosotros!
'You come in!' 'Let's you and I dance!'
c. Tú siéntate aquí y ustedes allí.
You-sg sit-yourself here and you-pl there
'You sit here and you (the rest) over there!'

When we use *usted/ustedes*, as in example (38a), no such contrast or discrimination is observed. In fact, this is the form used in instructions in textbooks, as shown in (38b), where no contrast or discrimination is implied. Note that in older textbooks, you may still find the abbreviation *Vd*. instead of *Ud*. This is not surprising if we remember that *usted* derives from *vuestra merced*.

(38) a. ¡Pase (Ud.)!
'Come in (you-form.)!'
b. ¡Traduzca Ud. las siguientes oraciones!

Translate you the following sentences
'Translate the following sentences!'

3.4.1 Periphrastic exhortatives To express an affirmative command, we can also use the verb *ir* 'to go' followed by a gerund (39a). The verb *ir* is used in the imperative/subjunctive form, thus qualifying this construction as a realization of exhortative type. These constructions bring into the foreground the simultaneity of the command with regard to another event. In the northwest areas of the Andes, it is common to find a gerund following the imperative form of *dar* 'to give' with commands (39b):

(39) a. ¡Ve poniendo la mesa mientras termino de cocinar!
Go setting the table while I-finish of cooking
'Set the table while I finish cooking.'
b. ¡Dame poniendo la mesa!
Give-me setting the table
'Set the table for me, will you?'

In Ecuadorian Spanish, the future seems to have grammaticalized as a command form and exhibits the same syntactic behavior as imperatives or subjunctive commands with regards to the position of the clitic pronouns:

(40) a. ¡Traerásmelo la semana que viene!
You-will-bring-me-it the week that comes
'Bring it to me next week!'
b. Lo veré (*verelo) la semana que viene.
Him I-will-see (*I-will-see-him) the week that comes
'I will see him next week.'

In (40a), where the future indicates a command, the clitic pronouns (-*melo*) follow the verb (in Standard Spanish, they would precede it). In (40b), on the other hand, where the sentence is not a command but a declarative (both in Ecuadorian as well as in Standard Spanish), the clitic must precede the future verb. I will call constructions like (40a) "future of command." Since the clitics follow the verb and the corresponding negative form is formed with the subjunctive, I suggest that these forms also are a realization of exhortative type.

3.4.2 Future tense as command or prohibition We can use a declarative sentence in the future tense to express a strong command (41a). A construction with a negative future expresses prohibition (41b).

(41) a. ¡Lo harás otra vez!
It you-will do another time
'You will do it another time!'

b. ¡No la verás más!
Not her you-will-see more
'You will not see her anymore!'

Notice that in these constructions the clitic precedes the verb and there is no subjunctive in the negative form; these constructions, therefore, do not qualify as syntactic exhortatives. I will assume that these constructions are declarative sentences with the illocutionary force of command and prohibition, respectively. Given our tripartite system (sentential type, sentential force, illocutionary force), these differences are now easy to capture.

3.4.3 Infinitives as commands or prohibitions In signs, commands usually have the form of infinitives instead of imperatives. This typically occurs with negative forms (42a), although we may also find them in affirmative forms as well, as in (42b):

(42)	a.	No fumar.	No entrar.	No acercarse.
		Not to-smoke	Not to-enter	Not to-approach-self
		'No smoking!'	'Do not enter!'	'Do not approach.'
	b.	Empezar cola aquí.	Sacarse los zapatos.	
		To-start line here	To-take-yourself the shoes	
		'Start the line here.'	'Remove your shoes!'	

Notice that the clitic form *-se* follows the infinitive in these constructions (last example in each set), thus failing the test for classifying as exhortative type (syntactically). I will thus claim that these are infinitive constructions with the illocutionary force of prohibition and command, respectively.

3.4.4 "*A* + *infinitive*" as commands In colloquial Spanish, we often find the preposition *a* 'to' followed by an infinitive to express an affirmative command, usually for a pleasant act (43a). But it can also be found in some less pleasant contexts, as in (43b).

(43)	a.	¡A comer!,	¡A bailar!
		To to-eat,	To to-dance
		'Come eat!,'	'Come dance!'
	b.	¡A callarse!	
		To to shut-up	
		'Shut up!'	

For a negative equivalent, we would resort to a present subjunctive construction. We could also resort to (44), although (44) feels more colloquial than (43). These examples are not as strong as the prohibitions in (42a).

(44) ¡Sin empujar!, ¡Sin enojarse!
Without to-push Without to-get-mad-yourself
'No pushing!' 'Don't get mad!'

Notice that in both (43) and (44) the clitic *-se* follows the infinitive. I will thus argue that these are also infinitive constructions with the illocutionary force of commands/requests.

3.4.5 Exhortatives and reflexive passive constructions Reflexive passive constructions with certain (transitive) verbs also fall under our exhortative type (NGDLE: s. 42.4e, i). Consider the examples in (45), where the verb is in the subjunctive form:

(45) a. Véase el párrafo siguiente.
Be-seen the paragraph following
'See the next paragraph.'
b. Manténganse congelados los trozos de pescado.
Be-kept frozen the pieces of fish
'Keep the pieces of fish frozen.'

These constructions differ from the constructions discussed in (34) in that the subject is not a second or first person, but rather a third person, which can be singular (45a) or plural (45b), as reflected on the verb. In spite of their third person subject, these constructions are still exhortative since the speaker (or more typically, the writer, as these constructions are typically found in written form) is urging someone to do something, as in the examples discussed previously. Observe that the clitic pronoun follows the subjunctive verb (just as in the imperatives/subjunctives discussed above). It is interesting to observe, though, that these forms are never used in negation. To express a negative instruction, a negative infinitive construction like (42a) is typically used.

3.4.6 Exhortatives or desideratives? Consider the following examples, also with a non-second person as a subject, which may or may not be expressed:

(46) a. ¡Que pase el siguiente!
That passes the next
'(Let) the next one come in!'
b. ¡Que abra alguien la puerta!
That open someone the door
'Someone open the door!'
c. ¡Que lo cuelguen!
That him they-hang
'Let them hang him!'

These constructions are also exhortative in that we are urging someone to do (or not do) something. However, syntactically they are not exhortative. They are typically formed with *que* 'that' followed by a verb in the subjunctive. As in the previous construction, the subject here is a third rather than a second person. Different from the other constructions that express exhortation, in these constructions the clitic pronoun precedes rather than follows the verb. As we will see in the next section, syntactically, these constructions resemble (syntactically speaking) desiderative constructions. I will argue that these constructions are better analyzed as desiderative in type, with an illocutionary force of order or request.

3.4.7 Spurious exhortatives Finally, there is yet another construction which can be interpreted as a command or an exhortative without it being syntactically exhortative. The examples in (47) are declarative, but they have the illocutionary force of prohibition and request, respectively.

(47)	Se prohíbe la entrada.	Se ruega silencio.
	Itself forbid the entrance	Itself begs silence
	'(The) entrance (is) forbidden.'	'Silence (is asked for)!'

We may also find constructions that look like exhortatives, as in (48):

(48)	¡Muera el dictador!	¡Pásalo bien!
	Die the dictator	Pass-it well
	'Down with the dictator!'	'Have a good time!'

In (48), we are not giving a command; we are making a wish. The examples in (48) are syntactic exhortatives in type; however, their illocutionary force is that of a wish.

3.5 Desiderative type

Desiderative sentences are those that have the illocutionary force of expressing desire and are found in two main sorts of constructions, as in (49). In the first type of construction (49a), we find the conjunction *que* 'that' followed by the verb in the present subjunctive. In the second type of construction, as in example (49b), they appear with the adverb *ojalá*, which has no equivalent adverb in English ('hopefully' is perhaps the closest), but which is very close in meaning to the verbs 'hope' and 'wish':

(49) a. ¡Que te mejores pronto!
That yourself recover soon
'(I hope) that you get better soon.'
b. ¡Ojalá te mejores pronto!
Hopefully yourself recover soon
'(I hope that) you get better soon.'

3.5.1 Desideratives without *que* Constructions like (48a) may also be found without the initial *que*. Quite common are constructions which include God, the Virgin, saints, etc., where *que* is optional (NGDLE: s. 42.4n):

(50) a. ¡En paz descanse!
In peace rest
'May he/she rest in peace.'
b. ¡Mueran los dictadores!
Die the dictators
'May all dictators die!'
c. ¡Dios te bendiga!
God you bless
'God bless you!'

The position of the clitic as well as the position of the subject in (50c) are clear indicators that these constructions are not exhortatives in spite of the similarities with these constructions.[4]

3.5.2 *Ojalá* As I mentioned above, the adverb *ojalá* is very close in meaning to the verbs 'hope' and 'wish'. It derives from Hispanic Arabic *law šá lláh*, which meant 'if God wants.' However, it has lost its original meaning completely. In fact, in certain parts of Mexico, we can hear expressions like *ojalá Dios quiera* 'God willing' (NGDLE s. 32.5o). The use of *que* 'that' still reflects is etymological verbal character; however, in modern Spanish, *que* is completely optional with *ojalá*. Depending on whether we wish to express a hope or a wish, the syntax of *ojalá* will vary. Consider first the cases where *ojalá* expresses hope:

(51) a. Ojalá (*que*) estudies mucho mañana. (pres. subj.)
'I hope you (will) study a lot tomorrow.'
b. Ojalá (*que*) estés estudiando mucho ahora. (pres. subj).
'I hope you are studying a lot now.'
c. Ojalá (*que*) hayas estudiado mucho anoche. (pres. perf. subj.)
'I hope you studied a lot last night.'

When we hope for something in the future (51a) or in the present (51b), we use the present subjunctive; when we hope for something to have happened in the past (51c), then we use the present perfect subjunctive.

Ojalá may also express a wish. Wishes in the present and in the past are necessarily counterfactual; wishes in the future may be possible (52a) or counterfactual (52b). However, in all cases, both in English and in Spanish, we need to use a verbal form in the past when we want to express a counterfactual wish.

(52) a. Estoy esperando una llamada. Ojalá suene el teléfono. (pres. subj.)

'I'm expecting a call. I wish the phone would ring.'

b. No iré a España este verano. Ojalá (que) fuera. (imp. subj.)
'I'm not going to Spain this summer. I wish I were.'

c. No estoy en España ahora. Ojalá (que) estuviera. (imp. subj.)
'I am not in Spain now. I wish I were.'

d. No fui a España el verano pasado. Ojalá (que) hubiera ido. (plup. subj)
'I didn't go to Spain last summer. I wish I had gone.'

Notice that to express counterfactual wishes in the future and in the present, we need to use the imperfect subjunctive. For (counterfactual) wishes in the past, we use the pluperfect subjunctive.

3.5.3 How *ojalá* is different from 'hope' and 'wish' As mentioned, *ojalá* is very close in meaning to the verbs 'hope' and 'wish.' When those verbs are used, the subject of the main clause and that of the embedded clauses cannot be the same, as shown in (53a); when the subject remains the same, we typically use an infinitive, as in (53b). Such a restriction is not found with *ojalá*, as shown in (53c).

(53) a. *Espero que (yo) vaya al cine el sábado.
'I-hope that (I) go to-the cinema the Saturday.'

b. Espero ir al cine el sábado.
'I-hope to-go to-the cinema the Saturday.'

c. Ojalá (que) (yo) vaya al cine el sábado.
"Ojalá" (that) (I) go to-the cinema the Saturday
'I hope to go to the cinema on Saturday.'

3.5.4 Desideratives and *así* We can also use *así* with the meaning of *ojalá* to express both hope and desire. This use is less productive and is considered literary and archaic. As we can see in (54a,b), these desires/wishes are really curses. They can also apply to the present, past, and future, just as with *ojalá*:

(54) a. ¡Así te mueras!
"así" yourself die
'I hope you die!'

b. ¡Así lo hubiera arrollado ese tren!
"así" him had run-over that train
'I wish that train had run over him!'

4 On dubitative and probability sentences

As mentioned in Section 2, all the types of sentences discussed so far (declarative, interrogative, exclamative, exhortative and desiderative) constitute "*modalidades* *de la enunciación*" (enunciative modalities). The reader may have noticed that all the different types exemplified in (2) fall within our triple classification (sentential type, sentential force, illocutionary force) with two major exceptions: dubitative (2g) and possibility/probability (2h) sentences. As observed above, in traditional grammars, these used to constitute different types of sentences, together with the five other types we have discussed. The NGDLE (s. 42.1h) proposes that besides the enunciative modalities or "*modalidades de la enunciación*" discussed above, there are also "*modalidades del enunciado*" or "*modalidades proposicionales*" ("propositional modalities"). Borrowing from modal logic and semantics, these may include the following types of propositional modalities: epistemic (knowledge), deontic (obligation), alethic (necessity, contingency, or probability), bouletic (ability, desire), teleological (goals), etc. (NGDLE: s. 42.1h; see also Portner 2009: Introduction). For reasons of space, I will only refer to dubitative and probability sentences here, which would be part of the alethic propositional modality. It must be noted here that most of these propositional modalities can be expressed through periphrastic constructions.

Traditional grammars distinguish probability/possibility from dubitative sentences (ESB: s. 3.2.5). The former express the probability/possibility through the verbal form itself (55a) or by means of an auxiliary verb (55b), while the latter express it through an adverb of possibility/probability (55c):

(55) a. Pedro estará cansado.
Pedro will-be tired
'Pedro must be tired.'
b. Pedro debe/puede/ha de estar cansado.
Pedro must/can/has of be tired.
'Pedro must be tired.'
c. Quizás/Quizá Pedro esté cansado.
'Perhaps Pedro is tired.'

In example (55c) both *quizá* and *quizás* are equally correct. Other possibility adverbs which mean 'perhaps/possibly' are: *a lo mejor*, *acaso*, *capaz (que)* (used mainly in Latin America), *igual*, *lo mismo*, *posiblemente*, and *tal vez*. *Seguramente* 'surely' denotes high probability rather than certainty, and thus should be considered part of this group.

The sentences in (55) express probability in the present. To express probability in the past, we would have the following constructions:

(56) a. Pedro estaría/habrá estado cansado.
Pedro would-be/will-have been tired
'Pedro must have been tired.'
b. Pedro debe/puede/ha de haber estado cansado.
Pedro must/can/has of have been tired
'Pedro must have been tired.'
c. Quizás Pedro haya estado/estaba cansado.

Maybe Pedro has been/was tired
'Perhaps Pedro was tired.'

Typically the future tense is used to express probability in the present while the future perfect and the conditional are used to express probability in the past. The auxiliaries *deber* 'must,' *poder* 'can,' and *haber de* 'have to' are used in the simple form of the present tense to express probability in the present; to express probability in the past, the main verb is used in the perfect form.

4.1 **Quizá(s)** *and mood sensitivity*

Adverbs that indicate probability may appear with indicative or subjunctive; however, when the adverb follows the verb, the subjunctive form is not grammatical (NGDLE: §25.14 i ff.):

(57) a. Quizás fuera/fue al cine ayer.
Perhaps he-went-subj/went-indic to-the cinema yesterday
b. *Fuera/Fue, quizás, al cine ayer.
Went-subj/Went-indic, perhaps, to-the cinema yesterday

4.2 **Quizá(s)** *and the future*

The present tense may be used to refer to an action in the future (58a). However, when we use *quizá(s)* 'perhaps,' we must use the present subjunctive if the action is to take place in the future:

(58) a. Voy al cine mañana.
I-go to-the movies tomorrow
'I'm going to the movies tomorrow.'
b. Quizás vaya/*voy al cine mañana.
Perhaps I-go-subj/I-go-indic to-the cinema tomorrow
'I may go to the cinema tomorrow.'

The use of the future tense after *quizás* (*quizás iré* 'perhaps I will go') has been showing a steady decline during the last century (NGDLE: s. 25.14m).

4.3 *Other adverbs to express probability*

The forms *a lo mejor* and *igual* 'maybe' are typically used with indicative. The former form can be found with subjunctive in Latin America, while the latter form is used exclusively with the indicative:

(59) A lo mejor/Igual voy/*vaya al cine esta noche.
Maybe I-go-ind/*I-go-subj to-the cinema this night
'I may go to the cinema tonight.'

5 Summary and conclusions

In this chapter, I have shown that sentences are best classified by a ternary system which includes *sentential type*, *sentential force*, and *illocutionary force*. Following this approach, we have re-classified the traditional types in (2) into five major sentential types: declarative, interrogative, exclamative, exhortative, and desiderative. We have seen that while there may be a one-to-one mapping between sentential type and sentential force, there is no one-to-one mapping between sentential type/force and illocutionary force. Taking sentential type and force as our starting point, we explored the different syntactic constructions that may realize these particular forces and explored some of the potential illocutionary forces that those constructions may have. In addition to the tripartite system mentioned above, we explored the possibility that there may be an additional level of *propositional modality* that is necessary to capture further types of sentences. Further work remains to be done in order to describe and analyze the types of propositional modality beyond probability/possibility and to explore whether the five sentential types and forces proposed here can be reduced even further. Finally, as mentioned in Section 2, the Spanish Royal Academy's notion of "enunciado" (utterance) does not restrict itself to sentences. The reader is referred to the NGDLE (in particular to chs 25 and 42) to see how utterances without verbs can also be classified in the system proposed by the RAE. It remains to be seen how those structures would fare in the system proposed in this chapter. As we can see, classification of sentences (and utterances) is not a boring exercise. Many issues still remain open for further study and discussion.

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NOTES

1 Questions with *cuán* are possible, although they sound even more archaic than exclamatives:

(i) ¿Cuán ocupado estarás este fin de semana?
'How busy will you be this weekend?

As observed in (19c), *qué* 'what' is typically used instead of *cuán* 'how' in modern language. *Qué* can only be used as a question meaning 'how' if *tan* 'so' precedes the adjective:

(ii) ¿Qué *(tan) inteligente es esa chica?
'How intelligent is that girl?'

In classic texts *cuál* 'which' was used as modern *cómo* 'how':

(iii) ¡Cuál gritan esos malditos! (Zorrilla, Tenorio, as cited in Gili Gaya 1961: s. 34)
Which shout those damn
'Those darned ones shout so much!'

2 It must be observed, however, that constructions like (22c) can also be used as exclamatives, as in (i). Notice the (optional) lack of agreement on the verb in (ib), thus suggesting that constructions like (ia) are ambiguous between a DP (when there of is agreement) and a CP structure (when there is no agreement):

(i) a. ¡Los amigos/amiguitos con los que sale!
The friends with the that he-goes-out
'The friends he goes out with!'

b. Me sorprende(n) los amigos con los que sale.
Me surprises the friends with the that he-goes-out
lit. 'It surprises me the friends that he goes out with.'

3 The same construction can be used as a question, but then the use of *de* of is obligatory (ia). Notice that the adjective cannot be fronted with *cómo*, as in (ib):

(i) a. ¿Cómo son de inteligentes esas chicas?
How are of intelligent those girls?
'How intelligent are those girls?'

b. *¿Cómo (de) inteligentes son esas chicas?
How (of) smart are those girls
'How smart are those girls?'

A reviewer has observed that while (ib) is ungrammatical for most speakers of Latin American Spanish, it is grammatical for speakers of Peninsular Spanish.

4 A reviewer has observed that instead of (50b), we could have (i), where the pronominal clitic *se* must follow rather than precede the verb:

(i) Muéran*se* los dictadores.
'May the dictators die.'

These constructions are not very productive in modern language, and they reflect an older stage of the language where the clitic could not appear in first position. The productive desiderative constructions are formed like those in (49). I leave this issue open for further research. For imperatives and clitic positions in imperatives, see Silva-Villar (1996), den Dikken and Blasco (2007), and Rubio Hernández (2007), among others.

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21 Clitics in Spanish

FRANCISCO ORDÓÑEZ

1 Morphology of Spanish clitics

The grammatical tradition has always distinguished two classes of pronouns in Spanish. On the one hand, there is a series of stressed pronouns, as in (1); on the other, there is set of unstressed pronouns, shown in (2):

(1) a. **Ella** habló
3rd.nom spoke
'She spoke.'

b. Me habló a **mí**
1st.dat spoke to me
'He/She spoke to me.'

(2) a. **La** vi en la plaza
3rd.acc saw in the plaza
'I saw her in the plaza.'

b. **Me** vio en la plaza
1st.acc saw in the plaza
'He/She saw me in the plaza.'

These first pronouns are traditionally called *pronombres tónicos*, or "strong pronouns"; the second, traditionally called *pronombres átonos*, are known as clitics. There are various morphological, phonological, and syntactic criteria that distinguish one paradigm from the other (see Zwicky 1977; Fernández Soriano 1993; Halpern 1998, and references cited there). The properties that distinguish them are coordination, modification, emphasis and isolation.

The Handbook of Hispanic Linguistics, First Edition. Edited by José Ignacio Hualde, Antxon Olarrea, and Erin O'Rourke.

First, clitic pronouns, unlike strong pronouns, cannot be coordinated, as shown in the examples below:

(3) *La y lo vi.
3rd.acc.fem and 3rd.acc.masc saw
'I saw him and her.'

(4) Ella y él salieron tarde.
She.nom and he.nom left late
'She and he left late.'

Second, pronominal clitics cannot be modified, unlike strong pronouns:

(5) a. *[Las dos] vi en el jardín
[3rd-fem.pl-both] saw in the garden
'I saw them both in the garden.'
b. *[Los solos] vi en el jardín
[3rd-masc.plur alone] saw in the garden
'I saw them alone in the garden.'

(6) [Ellos dos] salieron tarde
[They.nom] both left late
'They both left late.'

Also clitics cannot be emphasized or focalized, while strong pronouns can be easily emphasized:[1]

(7) *LA vi el otro día.
3rd.fem saw the other day
'I saw HER the other day.'

(8) Habla de ELLA.
Speaks about HER
'He/She speaks about her.'

Finally, only strong pronouns, not clitics, can appear in isolation as an answer to a question:

(9) ¿A quién viste?
'Who did you see?'
a. *La.
3rd.acc.fem
b. A ella.
'To her'

All these properties point to the fact that pronominal clitics and strong pronouns need different syntactic treatments. In some respects, clitics share some properties with affixes (Zwicky and Pullum 1983). For instance, affixes also cannot be coordinated, be modified, or appear in isolation:

(10) a. *No hay que **des** y **re**hacer esta madeja. (coordination)
'No need to un and re-do this reel.'
b. *Ella es pequeñimuy**ita**. (cf. ella es muy pequeña) (modification)
'She is small-very-diminutive. (She is very small)'
c. —¿Prefieres hacer la cama o deshacerla? (isolation)
'Do you prefer to make the bed or unmake it?'
—*D**es.**
'Un.'

However, unlike affixes, clitics do not adhere to rigid positions inside the word they attach to. In Spanish, clitics may appear before or after the verb, shown in (11) and (12), depending on various syntactic and morphological factors discussed below:

(11) a. **Lo** quiso comprar.
3rd.masc wanted to buy.
b. Quiso comprar**lo**
Wanted to buy-**3rd.masc**
'He/She wanted to buy it.'

(12) Cómpre**lo**
Buy-3rd.masc!
'Buy it!'

It should be noted that clitics are a category that appears in many languages, but the functions of clitics in Spanish are relatively restricted. For instance, Spanish clitics are uniquely pronominal – here I am referring to what Halpern (1998) and Zwicky (1977) call special clitics – whereas auxiliary clitics or adverbial clitics are found in Slavic languages. Furthermore, Spanish pronominal clitics correspond only to objects of the verb, with the possible exception of impersonal *se*. By contrast, subject clitics are found in many northern Italian dialects (Poletto 2000).

Another characteristic of clitics in Spanish is that they can be divided in two groups depending on their morphological composition. There are the so-called *person clitics* (Bonet 1991, 1995; Kayne 2000), which correspond to first and second person pronouns, *me nos*, *te*, and *os*, and reflexive, reciprocal, and impersonal *se*. These pronouns share a lack of expression of overt case:

(13) a. Me vio
1st.pers (DO) saw
'He saw me.'

b. Me dio un libro
1st.pers (IO) gave a book
'He gave a book to me.'

(14) a. Se vio en el espejo
reflex (DO) saw in the mirror
'He/She saw himself/herself in the mirror.'

b. Se otorgó el premio
reflex (IO) gave the prize
'He/She gave himself/herself the prize.'

The paradigm of person clitics is the following, as appears in Table 21.1. Note that the pronoun *os* for second person plural is only used in Spanish spoken in the Iberian Peninsula.

Table 21.1 So-Called person clitics.

Clitic	*1st person*	*2nd person*	*Reflexive, Reciprocal, Impersonal*
Singular	*me*	*te*	*se*
Plural	*nos*	*os*	

The other group is composed of *non-person clitics*. These clitics – which actually correspond to the 3rd person – do distinguish case in the standard dialect, as in (15) and (16). The 3rd person accusative distinguishes for gender, case, and person; dative distinguishes only for number. See Table 21.2 for summary.

(15) Lo vi.
3rd.masc.acc saw.
'I saw him.'

(16) Le di el libro
3rd.dat gave the book
'I gave him/her the book.'

Table 21.2 Non-person clitics (standard dialect system).

<table>
<tr><th>Clitic</th><th>Masculine singular</th><th>Feminine singular</th><th>Masculine plural</th><th>Feminine plural</th></tr>
<tr><td>Accusative</td><td>lo</td><td>la</td><td>los</td><td>las</td></tr>
<tr><td>Dative</td><td colspan="2">le</td><td colspan="2">les</td></tr>
</table>

The singular, accusative masculine clitic is also used for predicative complements:

(17) a. Inteligente, lo es
Intelligent, 3rd.masc.sing is
'Intelligent, he/she is.'

b. Contento, lo está.
Happy, 3rd.masc.sing is
'He is happy.'

Note that the set of non-person clitics coincides for the most part with the determiner system (Torrego 1995; Uriagereka 1995). Also like determiners and the nominal system, they contain what Harris (1991) calls word markers, including: *-o* (generally associated with masculine gender); *-a*, (generally associated with feminine gender); *-e*, (default marker associated with either gender); and the plural marker -s. These are shown below:

(18) L-o-s niñ-o-s
The-masc-plural boy-masc-plural

(19) L-a-s niñ-a-s
The-fem-plural girl-fem-plural

(20) L-a-s noch-e-s
The-fem-plural night-e-plural

Harris (1996) proposes that non-person clitics in Spanish should be analyzed in the same way, and they therefore must have the same composition with number and gender:

(21) L-o-s, L-a-s, L-e-s

Although there is consistency in all dialects in the form, dialects differ in how third person clitics represent the various arguments of the verb. For instance, the clitic *le* in (22), which in standard Spanish corresponds to dative, is also used for animate accusative clitics in many dialects. Traditional grammarians have called this phenomenon *leísmo*, and it is very common all over Spain and in many parts of Latin America (see Súñer and Yépez 1988; Fernández-Ordóñez 1999; Romero and Ormazábal 2006).

(22) A Juan le vi ayer
P-John 3rd.dat.sing saw yesterday.
'John, I saw him yesterday.'

This effect appears linked to the fact that Spanish also has differential object marking for specific animate objects with the preposition *a*. Thus, the preposition used for differential object marking in (23) is the same preposition as indirect object arguments in (24):

(23)	Vi	**a**	**Juan.**
	saw	P	Juan (DO)
	'I saw Juan.'		

(24)	Regalé	el libro	a	Juan .
	gave	the book	P	Juan (IO)
	'I gave the book to Juan.'			

The *leísta* system is thereby marking a difference in animacy, which is already marked in lexical animate objects in all dialects. One way to interpret this phenomenon is to assume that direct object animates get dativized in this dialect (see Ormazábal and Romero 2006 for alternative views on whether this is a process of dativization). However, dialects differ in the extent of *leísmo.* In a *pure leísta dialect*, all animate direct objects are marked as *le* and all inanimate direct objects as *lo* or *la*, as shown in Table 21.3 and (25) below. However, the most common system is the one system in which *le* is only used for masculine animate. For an extensive treatment of the phenomenon of *leísmo*, see Fernández-Ordóñez (1999) and Ormazába and Romero (2007). The *leísta* dialect spoken in Quito (Ecuador) does not seem sensitive to gender. However the *leísta* dialects spoken in Catalonia and Basque country are generally restricted to masculine singular or plural.

(25) A Juan le vi ayer.
P-Juan le saw yesterday
'I saw Juan yesterday.'

There are also *pure gender systems* in which the case distinction has been lost for non-person clitics, and the form is selected by gender and the count/non-count distinction instead; see Table 21.4. This is mostly the dialect spoken in the Santander

Table 21.3 Pure Leísta system.

	Inanimate masculine	*Inanimate feminine*	*Animates*
Accusative	*lo*	*la*	*le*

Table 21.4 Pure gender system, neutralizing datives and accusatives.

	Count referents		*Non-count referents*
Referent	*Masculine*	*Feminine*	*Both genders*
Singular	*le*	*la*	*lo*
Plural	*les*	*las*	

region in Spain (see Fernández-Ordóñez 1999). Note that there are also some intermediate systems.

To conclude, the fact that we find all this variability in the third person, but not in the first or second, confirms the division between third person and first and second person in the terms proposed by Harris (1995) and Bonet (1995).

2 Clitics: their source

The diagnostics reviewed in the previous section show that the behavior of clitics is not that of a full-fledged phrase. They are morphological heads that need to be part of a phonological prosodic unit. Therefore, it is natural to assume that clitics are heads that attach to another head in which they can form a prosodic unit, and therefore have a different distribution from corresponding XPs. For most cases, the source of the clitics in Spanish is transparent: either DO or IO, as shown above. However, there are interesting features that differentiate clitics from their non clitic counterparts. For instance, DO clitics must necessarily have a specific referent, as shown by the fact that they cannot co-occur with an indefinite interpreted as non-specific (see Roca 1992; Sportiche 1996):

(26)	*Un reloj,	lo	compré	ayer.
	A watch	3rd.masc.sg	bought	yesterday
	'A watch I bought yesterday.'			

(27)	*Dinero, no	lo	tengo.
	Money, not	3rd.masc.sg	have
	'Money I do not have.'		

Clitics usually correspond to arguments of verbs. However, dative clitics might correspond to arguments of an adjective:

(28)	Es	fiel	a Juan.
	Is	faithful	to Juan
	'He/She is faithful to Juan.'		

(29)	Le	es	fiel.
	3rd.dat	is	faithful
	'He/She is faithful to him/her.'		

Spanish, contrary to most other Romance languages, lacks partitive and locative clitics. Thus, when a partitive or locative argument is dislocated, no clitic appears with the verb:

(30) A Barcelona, no voy a ir.
'To Barcelona, I am not going.'

(31) Manzanas, no venden en ese supermercado.
Apples, not sell in that supermarket
'They do not sell apples in the supermarket.'

In some cases the clitics might lack an argumental source. This is what it is found in the clitic of interest or ethical datives. These dative clitics indicate that the action might benefit or not a third party in many cases:

(32) Este niño no me come nada.
This boy not 1st.sing eat anything
'This boy won't eat anything for me.'

(33) Por favor, no le bebas la leche.
Please, not 3rd.dat drink the milk
'Please, don't drink the milk that's for him/her.'

That there is no argument source for the clitic can be shown because they are always optional. Moreover, unlike typical arguments of the verb, they cannot be questioned or focalized:

(34) ?*Fue a mí a quien no me comió nada este niño
(from Gutiérrez-Ordóñez 1999)
it was to me to who not 1p ate anything this boy
'This boy did not eat anything for me.'

(35) ?*Fue a él a quien no le bebió la leche.
It was to him to whom not 3rd.dat drank the milk
'He/She did not drink the milk for him.'

3 Proclisis and enclisis

Clitics must precede tensed verbs in Spanish (a process called proclisis), and follow non-finite verbs and imperatives (called enclisis). These different distributional patterns have been analyzed as evidence that clitics originate in the argument position, as in (36a), and because of their phonological and morphological deficiency, have to move to form a prosodic and syntactic unit, as in (36b–39):

(36) a. Vi lo_i
b. Lo_i vi t_i
3rd.masc.acc saw
'I saw it.'

(37) Antes de verlo
before P see.INF-3rd.masc.sing
'Before seeing it'

(38) Viéndolo
see.Gerund-3rd.masc.sing
'Seeing it'

(39) Velo!
see-3rd.masc.sing
'See it!'

The main point of contention is where the clitic moves to and what explains the difference between proclisis in tensed verbs and enclisis with infinitives, gerunds, and imperatives. For proclisis, Kayne (1975) assumed that clitics adjoined directly to the verb. No adverbial or negation may intervene between tensed verb and the clitic. See examples in (40)–(42). For a different state of affairs on what is called interpolation in Medieval Spanish, see Rivero (1986).

(40)	a.	*Lo	no	vio.
		3rd	not	saw
	b.	No	lo	vio.
		not	3rd	saw
		'He/She did not see it.'		

(41)	a.	*Lo	nunca	vio
		3rd	never	saw
	b.	Nunca	lo	vio
		Never	3rd	saw
		'I never saw it.'		

(42)	a.	*Mira	bien	lo
		see	well	3^{rd}
	b.	Míralo	bien	
		see-3rd	well	
		'See it carefully.'		

However, when auxiliaries are present, the clitic must necessarily move to the left of the auxiliary:

(43)	*(Lo) había	(*Lo)	visto	(*Lo)	
	3rd had	3rd	seen	3rd	
	'I had seen it.'				

Since auxiliaries are taken to be in a tense projection, this movement leads to the natural conclusion that clitics are attracted to tense. The fact that negation and adverbs cannot intervene is a consequence of the fact that negation and adverbs must be in a higher projection than tense in Spanish. However, other elements may in fact intervene between verb or auxiliary and clitic. For instance, colloquial Rioplatense Spanish uses reiterative *re-* morpheme to indicate epistemic modality or intensification according to Kornfeld and Kuguel (2006). This particle appears before auxiliaries, thus it is not a regular verbal lexical prefix. The interesting fact is that clitics appear before the *re-* marker, as we show in (44). Another particle that might intervene between tense and clitics is the evidential particle *dizque* used in Mexican, shown in (45), and other dialects of Latin American Spanish (Treviño 2008).

(44) Y me **re** habían estafado.[3] (Rioplatense Spanish)
and 1st. part had had fooled
'They had fooled me again and again.'

(45) Me lo **dizque** arregló (Mexican Spanish)
1st 3rd.acc Evidential repaired
'Apparently he/she said that he had fixed it for me.'

The evidence from these two varieties of Spanish shows that clitics in preverbal position are sitting in a higher projection than the verb-final position:

(46) CL … (dizque/Re) . . .V

On the other hand, in the enclitic order there is no possibility of intervention of any particle between the verb and the clitic. This signals that the relation between clitic and verb in enclisis is more head-like that the one of clitic and verb in proclisis. This is expressed in the spelling system in which proclisis is written separately and enclisis is not. Another difference between Spanish proclisis and enclisis comes from their different behavior with respect to what looks like coordination of verbs with one clitic. While coordination of CL + V AND V is permitted in (47), no such coordination is allowed in the case V + CL AND V in (48) (Matos 2000):

(47) Lo compró y vendió en una sola operación (from Soriano 1993)
3rd bought and sold in one operation.
'He/She bought and sold it in one operation.'

(48) *Cómpralo y vende en una sola operación
buy-3rd and sell in one operation
'Buy it and sell in one operation.'

One natural way to understand such asymmetries is to assume that the clitic has been extracted across the board in (49) and moved to a higher inflectional projection above the verbs in proclisis. Crucially, the clitic does not form a lexical unit with the verb in proclisis. No such configuration is permitted in this enclitic case, most likely due to the fact that the verb and clitic form a lexical head unit (Kayne 1994; see also Benincà and Cinque 1993).

(49) a. CL_1 [t_1 V and t_1 V]

Finally, enclitics may enter into morphophonological interactions with the verb, which is unattested for proclisis. For instance, the plural marker for plural imperatives *-n* surfaces after the clitics in many varieties studied in Halle and Harris (2005) and Kayne (forthcoming). The 3rd person agreement might be reduplicated before or after the clitic or not:

(50) a. vendan le n
sell-3pp 3rd.dat 3pp
b. venda le n
sell 3rd.dat 3pp
'Sell them to him/her.'

All these differences between proclisis and enclisis have led linguists to the insight that the two processes must be characterized differently in syntactic terms. Any hypothesis that treats enclisis and proclisis as pure right-versus-left adjunction falls short of accounting for them. Instead, in the spirit initiated by Pollock (1989) on verb movement, the most likely characterization of the differences between proclisis and enclisis is verb movement (Kayne 1991; Rivero 1994; Raposo and Uriagereka 2005). From this perspective, enclisis involves movement of the verb to the clitic site, whereas in proclisis the verb fails to reach the clitic site.

4 Clitics and movement

So far it has been shown that clitics must move to an inflectional projection that verbs reach in enclisis but not in proclisis. However, clitics themselves must also have moved to this projection. The reasons for assuming movement for clitics are due to the complementary distribution with overt arguments in the following examples:

(51) a. Vi el libro.
saw the book
'I saw the book.'
b. Lo vi.
3rd saw
'I saw it.'
c. *Lo vi el libro.
3rd saw the book.
'I saw it the book.'

Further evidence for the idea of movement for clitics can also explain certain parallelisms with other movement processes like NP movement. Clitics, like NP subjects, can license floating quantifiers:

(52) a.

[Los estudiantes]	salieron	[todos]	antes.
The student-plur	left	all-plur	before.

'The students all left before.'

b.

[Los]	vi	[a todos]	antes.
3p.pl	saw	all-plur	before.

'I saw them all before.'

In both examples there is agreement in number and gender between the NP subject and the clitic with a floating quantifier in postverbal position. According to Sportiche (1988), this relation must be very local at an initial point of the derivation to explain the agreement in gender and number. In both examples, the floating quantifier is left stranded behind after movement of the NP subject and movement of the clitic:[4]

(53) a.

[$_{SV}$ [[todos]	[los estudiantes]]	salieron]	antes de la clase.
all	the students	left	before the class

b.

[$_{ST}$ [los estudiantes]$_i$	salieron [$_{SV}$ [[todos] t_i]]	antes de la clase.
the student-masc-plur	left all-masc-plur	before the class.

(54) a.

*vi [[a todos]	[Los]]	antes de la clase.
saw P-all-masc-plur	3rd-masc-plur	before the class

b.

[Los]$_i$	vi [[a todos t_i]]	antes de la clase.
3rd-masc-plur	saw P-all-masc-plur	before the class.

This movement of the clitic can go beyond the verb that subcategorizes it. This phenomenon, called *clitic climbing*, can be clearly shown in the following example:

(55)

[Lo$_i$]	tengo que	empezar	a poder	entender	t_i
3rd.sing	must	to start	to be able	to understand	

'I must be able to start to understand it.'

The clitic can also appear in any intermediate or final position:

(56)

Tengo que	empezar[lo]	a poder[lo]	entender[lo]
must	to start[3rd.sing]	to be able[3rd.sing]	to understand[3rd.sing]

'I must be able to start to understand it.'

These different possibilities are also found in sequences involving causative and perception verbs; intermediate positions are always available:

(57) a.

Me	[lo]	hizo empezar[lo]	a reparar[lo]
1st.sing	[3rd.sing]	made to start[3rd.sing]	to repair[3rd.sing]

'He made me start to repair it.'

b.

Me	[la]	vio	acabar[la]	de diseñar[la]
1st.sing	[3rd.sing]	saw	to finish[3rd.sing]	to design[3rd.sing]

'He saw me finish designing it.'

The different distributional possibilities can be understood as cases in which clitics move to intermediate tense projections. There are, however, important restrictions on this movement in Spanish. The intervention of wh-elements blocks the movement of clitics, as shown in the following contrasts noticed by Luján (1980), Rizzi (1982), and Kayne (1989):

(58) No me sé callar.
Not 1st know to be quiet
'I do not know how to be quiet myself.'

(59) *No me sé **cómo** callar.
Not 1st know how to be quiet.
'I do not know how to be quiet myself.'

This restriction is parallel to the one found with extraction of wh-elements from embedded interrogatives (see also Chapter 25, below). It only occurs when the wh-element in Comp has the +wh feature. There is no blocking effect in sentences with *que* or any other preposition with infinitive, as in (60). Note that Rizzi (1982) and Kayne (1989) show that these prepositions occupy the Comp position. Negation between the two modal verbs also blocks clitic climbing, as appears in (61). For some exceptional examples, see Treviño (1993).

(60) a. [Lo] tiene que leer[lo]
[3rd] have to read[3rd]
'He/She has to read it.'

b. [Lo] empieza a decir[lo]
[3rd] starts to say[3rd]
'He/She starts to say it.'

(61) a. *Lo intentó **no** hacer.
3rd attempted not to do
'He/She tried not to do it.'

b. *La quiso **no** leer.
3rd wanted not to read
'He/She wanted not to read it.'

The reason for the ungrammaticality of these sentences is the blocking effect of negation or overt wh-elements. Kayne (1989) defends the idea that clitic movement involves head movement of the clitic with the tense head. The head unit moves beyond its CP in order to reach the higher tense. Thus the movement of the complex T + V proceeds through the embedded CP:

(62) Lo $+ T_i$ quiso $[_{CP}\ t_i\ [_{IP}$ comprar $t_i\ ]]$

According to this analysis, head movement of the complex CL + T is blocked by the interference of an overt wh-element in the COMP area or negation, as in (63) and (64). Note that in case a wh is in the Spec CP area, we have to assume that the head of C is filled with an empty + *wh* head which blocks the clitic.

(63) *No lo$_i$ + sé [$_{SComp}$ si [$_{SFlex}$ comprar t$_i$]]
not 3rd know whether to buy

(64) *Lo$_i$ + quiere [$_{SComp}$ [$_{NegP}$ no [$_{SV}$ comprar t$_i$]]]
3rd wanted not to buy

The adjunction of the clitic to a tense projection is a precondition for the later climbing of the clitic above its CP clause. Only a language with null subjects such as Spanish, Italian, Catalan, or European Portuguese permits clitic climbing of the type mentioned above (Kayne 1989). In order to capture this correlation, T in Spanish should be characterized as strong and allows movement of its clitic above its CP:

(65) No lo + T$_1$ quiere [SComp [hacer t$_1$ [SV]]
Not 3rd + T wants to do

Most of the verbs that admit clitic climbing are subject control verbs.[5] According to Kayne (1989), this is due to the fact that the tense of the infinitive and the tense of the matrix tenses share the same subject reference. Thus, most object control verbs do not admit it:

(66) a. *Me los$_i$ sugirieron leer t$_i$ pronto.
1st 3rd suggested to read soon
'They suggested to me to read them soon.'
b. *Me los$_i$ suplicaron dejar t$_i$ en mi maleta.
1st 3rd supplicated to leave in my bag
'They begged me to leave them in my bag.'

Another kind of restriction on clitic climbing can be rooted in the character of the tense dependency or lack thereof between the embedded and the infinitive verb of the matrix. Luján (1980) proposes a restriction according to which verbs independently allow the indicative in the embedded clause to bar clitic climbing (For criticisms of this proposal, see Suñer 1980). It follows that factive verbs and raising verbs, which permit indicative in their embedded clauses, block the possibility of clitic climbing:[6]

(67) *Lo$_i$ lamentó leer t$_i$.
3rd lamented to read
'He/She regretted to read.'

(68) *Los médicos me$_i$ lo$_j$ parecen prohibir t$_i$ t$_j$.
The doctors 1st 3rd seem to prohibit
'Doctors seem to prohibit it to me.'

Finally, no clitic climbing is permitted either from an embedded indicative or subjunctive clause, shown in (69) and (70). The fact that they head an independent tense with independent nominative specification blocks movement of the clitic.

(69) *Tú me$_i$ quieres que vea t$_i$
you 1st want that see
'You want to see me.'

(70) *Ellos me$_i$ dijeron que vieron t$_i$
They 1st said that saw
'They said that they saw me.'

In sum, in Spanish, clitic climbing is allowed for embedded infinitives of subject control verbs with a tense dependency specification. The precondition for that is that the language must be pro-drop, which, of course, is the case of Spanish. So far, all these factors indicate clearly that the characterization of tense of the embedded infinitive is fundamental to understand clitic climbing.

The hypothesis presented above assumes the idea that all these control verbs have CP and IP and are therefore biclausal. Nevertheless, some linguists have proposed otherwise, that is, that the sentences that permit clitic climbing are monoclausal (Rizzi 1982; Zubizarreta 1982; Cinque 2006, and references cited in Wurmbrand 2001).[7] For example, Cinque 2006 proposes that modal verbs are part of functional projections of the sentence. He argues that the different modal verb would be inserted in the different functional projections in (71), but lexical verbs would be part of the VP.

(71) Frequentative Modal > Volitive Modal > Celerative Modal > Terminative Modal > Continuative Modal > . . .

For the monoclausal proposal, the ability of the clitic to appear in front of the modal verb is explained by the fact that clitics usually appear before auxiliary verbs. Each functional projection would need to be associated with a clitic position. From this perspective, there are many functional projections that permit clitics to pass through, purely optionally. The fact that clitic climbing is not allowed in factive verbs, raising verbs, and object control verbs is due to the fact that, on the one hand, these verbs are not part of the functional spine of the clause, and, on the other, clitics cannot move beyond the functional spine of the clause in which they originate; see Cinque (2006) for details. Some questions remain for this approach or any restructuring approach, like, for instance, the correlation between pro-drop, clitic climbing, and the blocking effects of negation. The question for Cinque (2006) and the restructuring analysis is why this correlation exists.

Finally, not only do modal and aspectual verbs allow clitic climbing, but causative and perception verbs also do. All these constructions share the property of permitting the subject of the embedded infinitival clause to appear overtly. That subject is always preceded by the Prepositional Differential Object Marker *a*:

(72) a. Pedro [lo] hizo leer[lo] a Juan
Pedro [3rd] made read[3rd] to Juan
'Pedro made Juan read it.'

b. Pedro [lo] vio leer[lo] a Juan
Pedro [3rd] saw read[3rd] to Juan
'Pedro saw Juan read it.'

In most dialects of Spanish, the subject might appear between the main verb and the infinitive:

(73) a. Pedro hizo a Juan leer[lo]
Pedro made to Juan read[3rd]
'Pedro made Juan read it.'

b. Pedro vio a Juan leer[lo]
Pedro saw Juan read[3rd]
'Pedro saw Juan read it.'

When the subject causee interferes between causative and infinitive, clitic climbing is blocked, as in (74). Kayne (1975) and Sportiche (1988) point a similar process out for French as well. They attribute this blocking effect to the specified subject condition.

(74) a. *Pedro [lo] hizo a Juan leer.
Pedro 3rd made to Juan read
'Pedro made Juan read.'

b. *Pedro [lo] vio a Juan leer.
Pedro 3rd saw to Juan read
'Pedro saw Juan read.'

Interestingly, this blocking of clitic climbing is also attested for main subjects, not only causees. Thus for instance, Spanish permits inversion of the subject in a wide range of cases. Subjects are permitted to appear before causative or perception verbs and infinitives, as appears in (75). I do not discuss here the discourse conditions that allow subjects in Spanish to appear in these specific positions; see Ordóñez (2000) y Zubizarreta (1998) for detailed studies of the matter.

(75) a. Ayer me vio Juan beberla antes.
Yesterday, 1st saw Juan drink-3rd before
'Yesterday, Juan saw me drinking it.'

b.	Ayer	me	hizo	Juan	beberla	antes.	
	yesterday,	1st	made	Juan	drink-3rd	before	
	'Yesterday, Juan made me drink it.'						

However, clitic climbing is barred when the subject interferes, as in (76). Also Suñer (1980) and Luján (1980) note that certain object control verbs permit clitic climbing (77a). However, clitic climbing is blocked when the argument object controller interferes (Kayne 1989; Cinque 2006), shown in (77b). Note that there is obligatory doubling of the object because it is pronominal (see section below on doubling).

(76)	a.	*Ayer	me	la	vio	Juan	beber	antes.
		Yesterday,	1st	3rd	saw	Juan	drink	before
		'Yesterday Juan saw me drinking it.'						
	b.	*Ayer	me	la	hizo	Juan	beber	antes
		Yesterday,	1st	3rd	made	Juan	drink	before
		'Yesterday Juan made me drink it.'						

(77)	a.	Nos	lo	permitieron	leer	a nosotros.
		1st	3rd	permitted	to read	to us
	b.	*Nos	lo	permitieron	a nosotros	leer.
		1st	3rd	permitted	to us	to read
		'They permitted us to read it'				

Also the insertion of adverbials might block clitic climbing, as pointed out by Luján (1980):

(78)	a.	Deseaba	mucho	verla.	
		desired	very much	to see-3rd	
	b.	*La	deseaba	mucho	ver.
		3rd	desired	very much	to see
	c.	La	deseaba	ver	mucho.
		3rd	desired	to see	very much
		'He/She wanted very much to see her.'			

There have been different proposals to account for these restrictions on clitic climbing. Under the restructuring analysis, the impossibility of the examples above can be taken as evidence that the rule that creates a restructuring gets blocked by the insertion of overt subjects or adverbial material. In Kayne's or Cinque's analysis, the blocking effects of the overt material must create a problem with the configurations that inserts adverbials, causes, or overt postverbal subjects in these sentences. Recall that all the sentences are perfectly grammatical when the interfering element appears after the infinitive. Ordóñez (forthcoming) proposes that the reason for the lack of clitic climbing must be found in the analyses of these constructions in terms of remnant movement. In this view, infinitives end up after the intervening elements due to movement, as in (79a). Then there is movement of the intervening subject causee or adverbial, as in (79b), and finally, there is remnant movement of the finite clause, as in (79c):

(79) a. [[$_{\text{SComp}}$han hecho [$_{\text{SComp}}$ Juan [$_{\text{SV}}$comerla]]]] Movement of infinitive SV ⇒
have made Juan eat-3rd
b. [$_{\text{SV}}$comerla]$_j$ [[$_{\text{SComp}}$han hecho [$_{\text{SComp}}$ Juant$_j$]]] Mov. of causee Subj, adverb ⇒
eat-3rd have made Juan
c. [a Juan] [$_{\text{SV}}$comerla]$_j$ [[$_{\text{SComp}}$han hecho [$_{\text{SComp}}$t$_j$]]] Remnant mov.⇒
to Juan eat-3rd have made
d. [[$_{\text{SComp}}$han hecho [$_{\text{SComp}}$t$_j$]] [a Juan] [$_{\text{SV}}$comerla]$_j$]
have made to Juan eat-3rd
'They have made Juan eat it.'

When the clitic remains in the infinitival clause as above, the sentence yields a grammatical result. However, clitic climbing is impossible in that configuration since there is no c-command relationship between a clitic before the finite verb and the clitic in the infinitive:

(80) *[[$_{\text{SComp}}$ Nos_ la$_i$ hizo [$_{\text{SComp}}$_t$_j$]] [a Juan] [$_{\text{SV}}$ comer t$_i$]$_j$]
1st 3rd$_i$ made to Juan eat t$_i$.

Thus, from this perspective, clitic climbing is blocked as a consequence of the fact that remnant movement dissolves the required c-command relationship.

5 Clitic doubling

The evidence presented so far shows that there are good reasons to believe that the best way to characterize the distribution of clitics in Spanish involves movement to a projection close to the verb. This simple movement analysis leads to the expectation that there should be complementarity in distribution between clitic and source, as shown above in (51). However, Spanish fails to show complementarity between the source complement and the clitic in many cases. This phenomenon, dubbed *clitic doubling*, is obligatory when the source is a strong pronominal. Thus, all overt pronominal arguments, DO's or IO's, require the corresponding clitic.

(81) a. *(Las$_i$) vi a ellas$_i$.
3rd.plur saw P them
'I saw them.'
b. *(les$_i$) di el libro a ellas$_i$.
3rd.plur gave the book to them
'I gave the book to them.'

Clitic doubling, however, is optional with lexical IOs in all dialects of Spanish, except for the IO of psych-verbs or verbs indicating inalienable possession, for which clitic doubling is obligatory:

(82) a. (Le$_i$) regalé las flores a María$_i$.
3rd.dat gave the flowers to María
'I gave the flowers to María.'

b. Estas flores *(le) gustan a María
These flowers 3rd.dat like to María
'Maria likes these flowers.'

c. *(Le$_i$) saqué el diente [al caballo]$_i$
3rd.dat. took the tooth to the horse.
'I took the horse tooth out.'

One general property of clitic doubling is that it is only permitted with the differential object marker *a*. This has been called Kayne's generalization:

(83) Lo vi *(a) él.
3rd saw to him
'I saw him.'

The relationship between the clitic and argument also has to be very local. Thus, no doubling relationship can be established outside the clause level (from Suñer 1988):

(84) La profesora que le$_i$ dio una F [a Paco]$_i$ la semana pasada se rompió la pierna esquiando.
The professor who 3rd$_i$ gave an F [to Paco]$_i$ last week broke her leg skiing
'The professor who gave an F to Paco last week broke her leg skiing.'

(85) *La profesora que le dio una F la semana pasada se rompió la pierna esquiando [a Paco]$_i$
The professor that 3rd$_i$ gave an F last week broke her leg skiing [to Paco]$_i$
'The professor that gave an F to Paco last week broke her leg skiing.'

Dialects might differ with respect to the different possibilities of doubling a DO. Thus, most dialects do not allow clitic doubling with a DO, unless they are pronominal or the quantifier type we discussed before. However, Rioplatense and other dialects of Spanish mostly in the Andean area do permit doubling with DO optionally. Only animates or specific DP's might be permitted to double as DO in this dialects. So no DO bare negative quantifiers or not specifics permit doubling (86b,c,d) (see Suñer 1988 for details):

(86) a. (La) vi a [Mafalda$_i$]
3rd saw to Mafalda
'I saw Mafalda.'

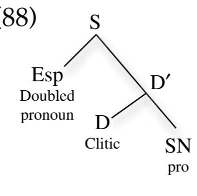
b.	No (*lo$_i$)	vi	[a nadie$_i$]
	not 3rd	saw	to nobody
	'I didn't see anybody.'		
c.	A quién$_i$	(*lo$_i$)	viste?
	who	3rd	saw
	'Who did you see?'		
d.	(*La$_i$)	busco	[a la chica que tenga dinero]$_i$.
	3rd	looking for	the girl that has-subjunctive money
	'I am looking for a girl that might have money.'		

This characteristic has lead Suñer (1988) to propose a matching principle for clitic doubling in Spanish. Assuming that DO clitics are inherently specific, the impossibility of the clitic in (86) (b), (c), and (d) would be due to the fact that there is no matching in features of specificity between clitic and the double.[8] The co-appearance of clitic and source, the matching requirements, and the locality between clitic and doubled element have led Rivas (1977), Strozer (1976), Jaeggli (1982), and Suñer (1988) to consider the relation between clitic and doubled element to be base generated agreement. According to Sportiche (1996), the agreement approach and the movement approach are not mutually exclusive if we consider a more articulated theory in which agreement and doubling get related by movement at some point of the syntactic derivation. As proposed above, despite the fact that floating quantifiers and clitics appear separate, they could have formed a single constituent at a starting point of the derivation, which then becomes separated by the movement of the clitic:

(87)	a.	vi [[a todos]	[Los]]	antes de la clase.
		saw P- all-masc-plur	CL-masc-plur before the class	
	b.	[Los]$_i$	vi [[a todos t$_i$]]	antes de la clase.
		CL-masc-plur	saw P-all-masc-plur	before the class
		'I saw all of them before class.'		

Uriagereka (1995) and Torrego (1995) propose this approach for clitics in Spanish. Their hypothesis has its roots in Postal's (1969) analysis of pronouns as determiners. In this view, clitics are determiners, and they form a big DP constituent with the doubled element in Spanish. The head clitic moves and leaves the rest stranded behind, as in (88). Note that Torrego assumes this approach for 3rd person accusative clitics only. When no doubling is attested, the structure must leave an empty category behind obligatorily.

(88)



In sum, lack of complementarity, matching effects, and movement can all be accounted for with a theory that assumes a complex DP structure for clitics. The head of this complex head has to move to an inflectional projection in the structure, leaving the rest of the DP structure stranded. Note that nothing in this theory implies that the big DP vacated of its head clitic should be left in its base position. Big DP moving to the specifier of a past participle must be responsible for the agreement in past participles in French and Italian (see Sportiche 1998; Belletti 1990).

6 Clitic combinations

When more than two clitics are together, they must move as a unit to the same projection. Spanish does not permit split clitics combinations:

(89)	a.	*Juan	me	quiere	poder	regalarla.
			1st	want to be able		to give-3rd
	b.	*Juan	la	quiere	poder	regalarme.
		Juan	3rd	wants	to be able	to give-1st
	c.	Juan	me	la	quiere	poder regalar
		Juan	1st	3rd	wants	to be able to give
		'Juan wants to be able to give it to me.'				

Perlmutter (1972) presented the following filter or template that captures all the combinatorial possibilities of Spanish clitics. This filter contains restrictions on ordering by person, by case, and by the presence of the pronoun *se*, and it generates the orders of clitics in (90):

(90) se- II - I -III (DAT)-III (ACC)

(91) se me/se nos/se te/se os/se le/se lo/me lo/te lo/te me/te le/me le

Two aspects of clitic combinations remain challenging for syntacticians and morphologists. The first involves a change in form that clitics undergo when they appear in combination, as in object clitics in the third person in Spanish. Since both dative and accusative clitics may appear in isolation, as shown in (92) and (93), it is to be expected that the combination of the two forms would be available. However, the output is ungrammatical, as in (94). Instead, we find that otherwise reflexive, reciprocal clitic *se* appears instead of the clitic *le*. This was labeled by Perlmutter as the *spurious* se *rule*, as in (95).

(92)	Juan	**le**	compró un libro.
	Juan	3rd.dat	bought a book
	'Juan bought him/her a book.'		

(93) Juan **lo** compró.
Juan 3rd.acc bought
'Juan bought it.'

(94) *Juan **le** **lo** compró.
Juan 3rd.dat 3rd.acc bought
'Juan bought him/her it.'

(95) Juan **se** **lo** compró.
Juan refl 3rd bought
'Juan bought him/her it.'

Moreover, *se* is not a pure phonological transformation of *le*. One important argument against such phonological explanation is that it does not present inflection for the plural, just like the reflexive/reciprocal pronouns:

(96) *A estos chicos, el libro se-s lo di.
to these boys the book SE-plur 3rd gave
'I gave the book to these boys.'

(97) *Los niños se-s vieron en el espejo.
The boys SE-plur saw in the mirror
'The boys saw themselves in the mirror.'

The spurious *se* rule poses important questions as to what is blocking the appearance of the regular dative clitic. Laenzlinger (1998) proposes that the problem might be explained by a restriction on having two morphologically-specified case 3rd person (accusative and dative) clitics together. Another line of thought proposed recently by Cardinaletti (forthcoming) is that NO morphologically complex bi-morphemic clitic *L-E*, which contains a word marker in the sense of Harris (1991), can adjoin to an already complex clitic like *L-O* which also contains a word marker. Cardinaletti (forthcoming) argues that *se* is a simple mono-morphemic clitic with the *e* as a pure epenthetic vowel. Bonet (1995) and Harris (1996) present an alternative in terms of distributive morphology. They claim that spurious *se* is just the product of a morphological rule that deletes the case features of third person clitic *le* so that it becomes a pure argument with no case or number specifications.

This unexpected *se* is at the root of another important phenomenon discussed by Bonet (1995) and Harris (1996): the so-called *parasitic plurals* in Mexican and Uruguayan Spanish. Since *se* is incapable of carrying the plural marker, the marker of plurality of the dative argument appears instead in the accusative clitic, even when the accusative clitic is referring to a singular DP:[10]

(98) El libro$_j$, a ellos$_i$, quién se$_i$ lo$_j$-s prestó?
the book.sing to them.plur who SE 3rd.plur gave
'The book, who gave it to them?'

Parasitic plurals can be explained if the plural marker is attached to the constituent formed by the two clitics, instead of attaching to the accusative clitic itself (Harris 1996):

(99) [SE LO]-S
Se 3rd-plur

Another well-known puzzle is that the combinatorial possibilities of arguments exceeds the combinatorial possibilities of clitics. Perlmutter (1971) observed that the combination of a 3rd person dative with a 1st or 2nd person clitic referring to the DO is impossible, shown in (100). However, when the dative clitic is left out as a strong pronoun, as in (101), the output becomes grammatical:

(100)	*Ellos	me	le	presentaron.
	they	1st	3rd.dat	introduced
	'They introduced me to him.'			

(101)	Ellos	me	presentaron	a él.
	They	1st	recommended/presented	to him.
	'They recommended/presented me to him.'			

The same restriction is found with causative constructions, in which the source of the dative is different from the source of the 1st and 2nd person clitic:

(102)	a.	Me	hizo	hablarle	por teléfono.
		1st	made	to speak-3rd.dat	by phone
	b.	*Me le	hizo	hablar	por teléfono.
		1st	3rd.dat	made to speak	by phone
		'He made me speak to him on the phone.'			

Bonet (1991) attributes this restriction to a morphological filter, the so-called *me-lui* constraint (also called the *person case restriction*), according to which the first and second persons are incompatible with a third person clitic. There have been attempts to reduce this morphological restriction to syntactic mechanisms by Anagnostopoulou (2003), Ormazábal and Romero (2007), and Adger and Harbour (2007). According to these authors, the person-case restriction can be explained by the feature checking mechanisms in the Minimalist program (Chomsky 1995). These authors assume – although with differences in the details – that the reason for the ungrammaticality of the above combinations is due to the fact that 1st and 2nd person clitics compete for the same feature as the 3rd dative clitic. For instance, Anagnostopoulou (2003) assumes that the feature that dative and 1st and 2nd person clitics share is the feature *person* but the accusative 3rd person lacks this features. A functional projection related to the verb must check this person feature. Checking succeeds when only one of the clitics contains this person feature, as we present in (103). Thus, when the 3rd person, which contains the person clitics, combines with clitics of a person clitic, the derivation is rendered ungrammatical, as in (104):

(103)	Me [~~Person~~]	LO [~~Number~~]	F [~~Person~~, ~~Number~~] presentaron
	1st	3rd.acc	introduced

(104)	*Me	LE [~~Number~~ Person]	F [~~Person~~, ~~Number~~] presentaron
	1st	3rd	presented.

Another surprising restriction in the combinatorial possibilities of clitics is found in causative and perception verb constructions. Climbing of the clitic of the infinitive above the causative or perception verb is forbidden when the clitic has animate reference as shown in the following contrasts (Luján 1980):

(105)	a.	Me	hicieron	educarla.[+ animate]	
		1st	made	educate[3rd]	
	b.	*Me la.[+ animate]		hicieron	educar.
		1st 3rd		made	educate
		'He made me educate her.'			

(106)	a.	Me	hicieron	leerlo.[-animado].	
		1st	made	read[3rd]	
	b.	Me	lo	hicieron	leer.
		1st	3rd	made	read
		'They made me read it.'			

A final interesting point of variation in Spanish is presented by the possibility in many non-standard dialects to prepose the first and second person clitics to the reflexive *se*, as in (106) (Ordóñez 2002):[11]

(107)	a.	**Se**	**me**	escapa.
		se	1st	escape-3s
	b.	**Me**	**se**	escapa.
		me	se (REFL)	escape-3s
		1st	SE	losing it
		'It's getting away from me.'		

However, none of these varieties permit the non standard combination in post-verbal position with infinitives, as shown in (108):

(108) a. Puede escapár**seme**.
it can escape-se-1st

b. *Puede escapár**mese**.
it can escape-1st-se
'I could lose it.'/'It could get away from me.'

The facts above show that certain clitic combinations are sensitive to their syntactic distribution. This is an important challenge since it implies that the way adjunction works for enclisis (see above) is determining the possibilities of clitic combinations with infinitives.

7 Conclusion

Clitics remain an important topic of discussion since they interface with three important modules of grammar: phonology, morphology, and syntax. Clitics are dependent phonologically on a word with more prosodic structure. Their morphological makeup is not always simple, and the differences between different dialects provide interesting clues on how we want to characterize the different clitic categories in Spanish. There are many diagnostics for syntactic movement, but this movement is nevertheless highly restricted. Finally, clitic doubling in Spanish, far from being a problem for movement theories, suggests that the initial structure for clitics should be considered more complex than previously thought. Clitic combinations are probably the topic in which syntax and morphology interact in such a way that there is no perfect mapping between arguments and the clitics that represent them. The combinations with clitics are a proper subset of the possible combinations of arguments.

NOTES

1 The dialect of Río de la Plata in Argentina and Uruguay apparently stress what we call clitics in this chapter. This very peculiar pattern occurs only when clitics appear in final position in imperatives and infinitives. See Moyna (1999), Huidobro (2003), for some treatments of this phenomenon.

2 Spanish does not permit clitics with past participles in absolutive constructions contrary to Italian. See Belletti (1990) Kayne (1991):

(i) *vístolo
see-3rd

(ii) *vistome
see-1st

3 This is not the prefix RE which in all dialects of Spanish attaches to the lexical verb. This can never appear before the auxiliary:

(i)	Juan	había	rehecho	el nudo.
	'Juan	had	retied	the knot.'
(ii)	*Juan	re	había hecho	el nudo.
	'Juan	re	had tied	the knot.'

4 Contrary to examples with subjects and floating quantifiers, clitics cannot move with the quantifier.

5 Suñer (1980) and Luján (1980) point out that object control verbs like *permitir* 'allow' or *prohibir* 'prohibit' can admit clitic climbing as well. However, Kayne (1989) and Cinque

(2006) propose that these cases should be taken as examples of hidden causative verbs, which permit clitic climbing more generally.

(i) Me los permitieron leer
1st 3rd permitted to read
'They permitted me to read it.'

6 In the case of factive verbs, one might think of an alternative in which the CP is embedded in a complex NP *the fact that*, which would be silent. This complex NP would block climbing of the clitic:

(i) *Lo lamento [$_{CP}$**el hecho de** [$_{IP}$ hacer t]]
3rd regret the fact of doing it
'I regret the fact of doing it'

7 A variant of the monoclausal analysis is proposed by Rizzi (1982) and Zubizarreta (1982). According to Rizzi (1982) monoclausality with modal verbs derives from a biclausal structure which enters into a process of restructuring. Restructuring consists of erasing CP dividing the two clauses. Modal verbs and infinitives end up forming a head unit. Observe that the fact that this head unit in many cases would have to include a prepositional element we mentioned before:

(i) Lo [acaba de hacer t]
3rd finishes of to do
(ii) Lo [empieza a leer]
3rd starts to read
'She starts to read it'

8 There are some apparent cases of mismatch of clitics in person. These examples probably involve an appositive structure with a silent pronoun as in (1):

(i) a. Nos vieron a los estudiantes
1st saw to the students
b. Nos vieron [a nosotros] los estudiantes.
$1st_i$ saw [to us]$_i$ the students.
'They saw us the students'

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22 *Ser* and *Estar*: The Individual/Stage-level Distinction and Aspectual Predication

JOSÉ CAMACHO

1 Introduction

The distribution of copular verbs in Spanish is a widely researched topic in Spanish linguistics: from traditional grammarians to contemporary linguists, many have attempted to capture a very complex and challenging pattern. Spanish's two main copular verbs, *ser* and *estar*, overlap in many contexts. As a first approximation *ser* combines with predicates that denote a permanent characterization of the subject, whereas *estar* + predicate denotes non-permanent traits, as seen in (1). In the first example, *being pleasant* is perceived as a character trait of Alejandro's, whereas in the second one, it is seen as a temporary situation that happens to be currently true.

(1) a. Alejandro es agradable.
'Alejandro is.IL pleasant.'
b. Alejandro está agradable.
Alejandro is.SL pleasant
'Alejandro is being pleasant (today).'

One influential line of thought on the difference between *ser* and *estar* assumes that this contrast embodies the distinction between stage-level (SL) and individual-level (IL) predicates (cf. Carlson 1977; Kratzer 1995; Diesing 1988, 1992; Escandell and Leonetti 2002; Asociación de Academias de la Lengua Española 2009, henceforth AALE). Stage-level properties hold for specific stages or situations in which

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the subject is involved, whereas individual-level predicates denote properties that hold for the subject absolutely. According to this dichotomy, speakers conceptualize non-linguistic properties like 'being intelligent/Chinese' as IL and 'being available/in my house' as a stage. While these are conceptual distinctions, the SL/IL partition has clear syntactic and semantic correlates.

As AALE (2009: 2812) points out, the SL/IL distinction belongs to the category of aktionsart or lexical aspect, but other researchers have proposed different explanations for the distribution of *ser/estar*. Among them, several have focused on slightly different conceptions of aspect, such as whether the predicate indicates change of state or not (cf. Fernández Leborans 1995), or whether it denotes perfectivity (Luján 1981; Roby 2009). Specifically, Luján (1981) defines a perfective predicate as one that holds over a delimited, bounded period of time, whereas an imperfective predicate holds of "a stretch of time with no beginning or end assumed and extending a number of delimited time periods" (Luján 1981: 177). Roby connects this notion to viewpoint aspect (the one expressed by the imperfect/preterit distinction). Others have suggested that the relevant distinction is pragmatic, not lexical or syntactic (cf. Maienborn 2005).

In this chapter I will review the distribution of *ser/estar* and how it fits within the broad family of aspectual analyses proposed for it. The chapter is organized as follows. In Section 2, I introduce the basic distribution of *ser* and *estar*. In Section 3, I sketch the conceptual SL/IL distinction and its linguistic correlates, as well as the challenges it faces. In Sections 4 and 5, I propose a slightly different conceptualization of the SL/IL distinction and contrast it with the observations from Sections 2 and 3. Finally, Section 6 discusses the notion of coercion, by which predicates shift their meaning depending on the context.

2 The distribution of *ser* and *estar*

2.1 Non-overlapping contexts

Ser and *estar* appear in several contexts (cf. Fernández Leborans 1999; AALE 2009: ss. 37.7–37.9), and in most of them, they cannot be interchanged, specifically in the following contexts.

(a) S*er*, but not *estar* can be used as an identificational predicate; that is, contexts where the speaker selects an alternative from among a number of contextually determined possibilities, as in (2).

(2) La presidenta es/*está ella.
'The president is.IL/*is.SL she.'

(b) As auxiliaries, *ser* appears in passive constructions (cf. (3)), *estar* as an aspectual auxiliary (cf. (4)): *estar* + gerund denotes durativity, *estar* + *a punto de* + infinitive indicates prospective aspect. In their auxiliary use, then, the two verbs cannot be interchanged. Furthermore, aspectual auxiliary *estar* cannot appear with *estar* as a main verb (cf. (5a)).

(3) El tiburón fue/*estuvo visto en la playa por los bañistas.
'The shark was.IL/*was.SL seen at the beach by the bathers.'

(4) a. Domitila está/*es cantando.
'Domitila is.SL/*is.IL singing.'
b. Domitila está/*es a punto de dormirse.
'Domitila is.SL/*is.IL about to fall asleep.'

(5) a. *Blanca está estando cansada.
'Blanca is.SL being.SL tired.'
b. Chuck está siendo sarcástico.
'Chuck is.SL being.IL sarcastic.'
(example from http://www.youtube.com/watch?v=J933-H2BU9g)

(c) Only *ser* can appear with a DP predicate (cf. (6)). When preceded by *de*, the DP becomes grammatical with *estar* but not with *ser* (cf. (7)). By using (7), a speaker presents the presidency as a role or a function ('acting as …'), whereas (6) simply asserts that Obama is president.

(6) Obama es/*está (el) presidente desde el 2009.
'Obama is.IL/*is.SL (the) president since 2009.'

(7) Obama está/*es de presidente desde el 2009.
'Obama is.SL/*is.IL of president since 2009.'

2.2 *Overlapping contexts*

2.2.1 Adjectival predicates Many predicates appear with both *ser* and *estar* (cf. (8)). Predicates with *ser* characterize the subject in absolute terms: in (8a), 'being cold' is an absolute characteristic of the North Pole, whereas predicates used with *estar* denote transient properties, such as *cold* in (8b).

(8) a. El polo norte es frío.
'The North Pole is.IL cold.'
b. La carne está fría.
'The meat is.SL cold.'

Some adjectives appear only with *estar*, as seen in (9). Notice that some of these do not necessarily imply temporary meaning (like *arruinado* 'ruined,' *muerto* 'dead,' *lleno* 'full,' *contento* 'happy,' *ausente* 'absent,' cf. Luján 1981: 172). Conceptually, they do not seem to denote transient properties either, although, as we will see, they tend to have the distribution of stage-level predicates. These adjectives cannot usually be coerced into use with *ser*:

(9) a. El millonario está/*es arruinado.
'The millionaire is.SL/*is.IL ruined.'
b. La ajedrecista está/*es cansada.
'The chess-player is.SL/*is.IL tired.'

Other adjectives appear mostly with *ser* (cf. (10a). Among them, some can be coerced into uses with *estar* (cf. (10b); Luján 1981: 172; Escandell and Leonetti 2002). Other adjectives can never appear with *estar* (cf. (11) and Fernández Leborans 1995: 265). As Fernández Leborans points out, those adjectives that denote properties inherent to a genus or species are usually not coercible into use with *estar*.

(10) a. La ajedrecista es/*está inteligente.
'The chess-player is.IL/*is.SL intelligent.'
b. Aunque ese ajedrecista no suele mostrar mucha inteligencia en sus jugadas, hoy estuvo bastante inteligente.
'Although this chess-player usually doesn't show much intelligence in his moves, today, he was fairly intelligent.'

(11) Todos los seres vivos son mortales, pero los que están en peligro de extinción son/*están muy mortales.
'All living beings are mortal, but those in danger of extinction are.IL/*are.SL very mortal.'

2.2.2 Prepositional phrase predicates The absolute/transient distinction can also apply to PP predicates. Most of them appear only with *ser*, while a few appear only with *estar*, as seen in (12)–(13) (the first three examples are from Fernández Leborans 1999: 2368).

(12) a. El vestido es/*está a rayas, con cuello y sin mangas.
'The dress is.IL/*is.SL to stripes (striped), with neck and without sleeves.'
b. El anillo es/*está de oro.
'The ring is.IL/*is.SL (made) of gold.'
c. Juan es/*está de Madrid.
'Juan is.IL/*is.SL from Madrid.'

(13) María está/*es con una amiga.
'Maria is.SL/*is.IL with a friend.'

With locative prepositions, the distribution depends on several variables. First, *ser* is used with path-denoting PP predicates (cf. Zagona 2010, ex. 10):

(14) Este regalo es/*está para José. (Path → *ser*)
'This gift is.IL/*IS.SL for Jose.'

Second, location prepositions appear with *estar* when the subject refers to a movable entity (animate or inanimate cf. (15a,b)), as observed by Luján (1981: 187). If the subject refers to a motionless inanimate individual, the copula can be either *ser* or *estar* (cf. (15c)), depending on dialect and speaker. In this example, copula choice does not strictly correlate with a transient/absolute distinctions.

(15) a. Los libros están/*son en el estante.
'The books are.SL/*are.IL on the shelf.'
b. Mi hermano está/*es en Buenos Aires.
'My brother is.SL/*is.IL in Buenos Aires.'
c. El baño está/es a la entrada de la casa.
'The bathroom is.SL/is.IL at the entrance of the house.'

On the other hand, if the subject is interpreted as an event, *ser* is required:

(16) La fiesta es/#está en la discoteca.
'The party is.IL/#is.SL at the disco.'

3 Stage-level vs. individual-level predicates

The notion of a transient/absolute distinction corresponds to the SL/IL partition. Indeed, *ser* and *estar* have frequently been considered primary candidates to lexicalize the SL and IL distinction, and several proposals have explicitly assumed this to be the case, but as far as I know, the only explicit development of that proposal and its consequences is Arche (2006) (cf. also Mejías-Bikandi 1993; Fernández Leborans 1999; Escandell and Leonetti 2002; Gulmiel 2008; AALE 2009; Roby 2009 for summary and discussion; and Marín (2010)). As I mentioned earlier, the conceptual SL/IL distinction has been correlated with several linguistic phenomena, which I will summarize in this section. In the next one, I will explore how *ser*/*estar* pattern with respect to those linguistic properties.

3.1 Properties of stage-level and individual-level predicates

Jaeger (2002) offers the following summary of the linguistic properties for SL/IL predicates (cf. also Carlson 1977; Stump 1985; Diesing 1992; Kratzer 1988/1995; McNally 1994; Fernald 2000; Arche 2006, among others).

3.1.1.1 Perception reports and small clause predicates. Perception verbs (*see*, *hear*, etc.) only take SL predicates (typically AP and VP):

(17) a. María vio a Oscar sentado/en su casa/listo para el colegio. (SL)
'Maria saw Oscar sitting/in his/her house/ready for school.'
b. *María vio a Pedro inteligente/alemán. (IL)
'Maria saw Pedro intelligent/German.'

Small clause predicates of verbs like *considerar* 'consider' only take IL predicates (AALE 2009: 2813).

3.1.1.2 Restrictions on modifying adverbials. Only SL predicates can be modified by temporal, frequency or locative adverbials (Kratzer 1988/1995):

(18) a. Luisa habló inglés hoy/varias veces. (SL)
'Luisa spoke English today/several times.'
b. *Luisa supo inglés hoy/varias veces. (IL)
'Luisa knew English today/several times.'

(19) a. Marta habla francés en Paris.
'Marta speaks French in Paris.'
b. ??Marta sabe francés en Paris.
'Marta knows French in Paris.'

In Kratzer's (1988/1995) and Diesing's (1992) influential analyses, this contrast follows from an additional event argument that SL predicates have, which anchors them spatiotemporally and allows for modification by adjuncts such as 'in Paris.'

3.1.1.3 Appearance as the restrictor of a when *conditional.* In the examples in (20), the first clause expresses a temporal condition. That predicate (the so-called restrictor) can only be SL as seen in (20a)'s SL 'speak French' vs. b's IL 'know French' (cf. Kratzer 1988/1995; Fernández Leborans 1999: 2438–2439, her examples):

(20) a. {Siempre que/Cuando} María habla francés, lo habla muy bien.
'Whenever Maria speaks in French, she speaks very well.'
b. *{Siempre que/Cuando} María sabe francés, lo habla muy bien.
'Whenever Maria knows French, she speaks it very well.'

3.1.1.4 Depictive adjuncts. IL predicates cannot appear as depictive secondary predicates (Rapoport 1991; Demonte 1991: 168; McNally 1994; Demonte and Masullo 1999: 2473–2474, examples from this work):

(21) a. Juan manejó nervioso.
'Juan drove nervous.'

b. *Juan manejó mortal.
'Juan drove mortal.'

3.1.1.5 Lifetime effects. Tense modifies different elements depending on the SL/IL nature of the predicate (Kratzer 1995: 155–157). In (22a) with the IL predicate, tense applies to the DP subject, so we can draw the implicature that Doris is dead, whereas in (22b), tense applies to the predicate 'having black gloves.'

(22) a. Doris tenía manos largas.
'Doris had long hands.'
b. Doris tenía guantes negros.
'Doris had black gloves.'

3.1.1.6 Generic vs. existential interpretation. IL predicates tend to be interpreted generically in certain contexts, SL predictates existentially.

(23) a. Firemen are altruistic. (IL, generic)
b. Firemen are available. (SL, existential)

3.2 SL/IL properties and ser/estar

In this section, I apply the tests reviewed in the preceding section for the distinction between stage and individual-level predicates to *ser* and *estar* (cf. also Maienborn 2005, Arche 2006 and Marín 2010).

3.2.1.1 Perception reports. Only predicates that appear with *estar* are also acceptable complements of perception verbs (cf. (17) above vs. (24) below). However, *estar* itself cannot appear as complement to a perception verb (cf. (25) and Maienborn 2005: 166).

(24) a. Oscar está sentado/en su casa/listo para el colegio. (SL)
'Oscar is.IL sitting/in his/her house/ready for the school.'
b. Pedro es inteligente/alemán. (IL)
'Pedro is.IL intelligent/German.'
c. *Pedro está inteligente/alemán.
'Pedro is.SL intelligent/German.'

(25) *María vio a Oscar estar sentado/en su casa/listo para el colegio.
'Maria saw Oscar be.SL sitting/in his/her house/ready for the school.'

3.2.1.2 Restrictions on modifying adverbials. Both *ser* and *estar* predicates are possible with frequency adverbs. Interestingly, the interpretation one gets in (26b) is that Luisa used to act in an altruistic way several times a day. In other words, it seems like the predicate is coerced into an SL interpretation, but without requiring a switch to the copular verb *estar*. Notice, by the way, that such coercion into SL meaning is not readily available with a regular IL predicate, such as in (18b) above.

(26) a. En esa época, Luisa estaba disponible varias veces al día.
'During that period, Luisa was.SL available several times a day.'
b. En esa época, Luisa era altruista varias veces al día.
'During that period, Luisa was.IL alutristic several times a day.'

Regarding locative adverbs, *estar* is the default option, as observed in Section 2.2.2 above and in (27). With eventive subjects, locative predicates are possible, as already noted.

(27) a. La camisa está sobre la silla.
'The shirt is.SL on the chair.'
b. *La camisa es sobre la silla.
'The shirt is.IL on the chair.'

Maienborn (2005: 163) observes that if one adds an adjective to *estar*, the full copula + predicate cannot be modified by location PP adjuncts, as seen in (28), from Maienborn (2005: 163). From this observation, she draws the conclusion that PP locatives do not modify SL predicates, contrary to expectations. However, this example does not show that *estar* does not select for SL predicates, only that the resulting constituent *estar* + predicate cannot be modified by a locative.[4]

(28) a. *La camisa está mojada sobre la silla.
'The shirt is.SL wet on the chair.'
b. *El champán está tibio en la sala.
'The champagne is.SL warm in the living-room.'

3.2.1.3 Appearance as the restrictor of a when *conditional.* Typically, only *estar* can appear in the restriction of the temporal conditional (cf. (29)), although Schmitt (1992: 414) has observed examples like (30) in Brazilian Portuguese where *ser* can appear in the antecedent of a *when* conditional. In this case, one gets the impression that *ser grosera/cruel/amable* is coerced into an SL reading; that is, (30) is interpreted as 'María acts rudely, cruelly, nicely,' not as an absolute property of María. Once again, no change of copula is required.

(29) a. {Siempre que/Cuando} María está alegre, todo le sale bien.'Whenever Maria is.SL in-a-good-mood, everything turns out well for her.'
b. *{Siempre que/Cuando} María es alegre, todo le sale bien.
'Whenever Maria is.IL in-a-good-mood, everything turns out well for her.'

(30) {Siempre que/Cuando} María es grosera/cruel/amable, es bastante grosera/cruel/amable.
'Whenever Maria is rude/cruel/nice, she is really rude/cruel/nice.'

3.2.1.4 Depictive adjuncts. Only adjectives compatible with *estar* can appear as depictive adjuncts:[5]

(31) Juan llegó nervioso/*mortal.
'Juan arrived nervous/*mortal.'

(32) a. Juan está/#es nervioso.
'Juan is.SL/#is.IL nervous.'
b. Juan es/*está mortal.
'Juan is.IL/*is.SL mortal.'

3.2.1.5 Lifetime effects. These effects are observed with *ser* (cf. Arche 2006, ch. 6): (33a) implies that Doris is dead, (33b) does not.

(33) a. Doris era de Bogotá.
'Doris was.IL from Bogota.'
b. Doris estaba en Bogotá.
'Doris was.SL in Bogotá.'

3.2.1.6 Generic vs. existential interpretation. *Ser* predicates tend to be interpreted generically, *estar* predicates are not, with the exception of participial predicates of *estar*, which can be generic (cf. AALE 2009: 2813–2814).

(34) a. Los perros son peligrosos. (IL, generic)
'Dogs are.IL dangerous.'
b. Los perros están furiosos. (SL, non-generic)
'Dogs are.SL furious.'

(35) Los perros están prohibidos en este edificio. (SL, generic)
'Dogs are.SL forbidden in this building.'

As this brief survey of SL/IL linguistic properties shows, *estar* tends to pattern with SL predicates characteristics, whereas *ser* tends to pattern with IL, although in some cases an SL interpretation can be coerced while maintaining *ser*.

The SL/IL characterization of *ser/estar* runs into a few objections. As mentioned earlier, several adjectives that cannot be characterized as conceptually SL appear with *estar* obligatorily (*arruinado* 'ruined,' *muerto* 'dead,' *lleno* 'full,' *contento* 'happy,' *ausente* 'absent'). In terms of the tests just described, these adjectives have a mixed pattern. For example, they behave like IL predicates because they cannot appear in the antecedent of a *whenever* clause (cf. (36a) or with frequency adverbs (cf. (36b). On the other hand, they can appear as depictive adjuncts, like other *estar* predicates do (cf. (37)).[6]

(36) a. *Siempre que Napoleón estaba muerto/vivo/loco, todo le salía mal.
'Whenever Napoleon was dead/alive/crazy, everything would turn out wrong for him.'

b. *Napoleón a veces estaba muerto/vivo/loco.
'Napoleon was sometimes dead/alive/crazy.'

(37) a. El tiburón apareció muerto/vivo en la playa.
'The shark appeared dead/alive on the beach.'
b. El secuestrado fue liberado vivo/loco.
'The kidnapped person was freed alive/crazy.'

A second type of challenges for the SL/IL characterization of *ser/estar* was illustrated in (16) above: locative PPs can appear with both *ser* and *estar*, but when the subject is eventive, *estar* cannot be used. Notice that the nature of the subject does not affect the SL/IL interpretation of the predicate when it is not a locative PP (cf. (38)), suggesting that something in the structure of *estar* + locative PPs accounts for this pattern.

(38) a. La fiesta fue/estuvo divertida.
'The party was.IL/was.SL fun.'
b. La pelota es/está grande.
'The ball is.IL/is.SL big.'

A related issue concerns the use of geographical locative predicates. In (39) (from Roby 2009: 16), it is hard to argue that the location of México or San Sebastián is somehow a stage in the existence of those cities, although *estar* is possible (and preferred by some speakers).

(39) a. México está al sur de los Estados Unidos.
'Mexico is.SL south of the United States.'
b. San Sebastián está al este de Santander.
'San Sebastian is.SL to the East of Santander.'

A fourth challenge for the *estar*~SL connection stems from the so-called evidential uses of *estar* (Roby 2009: 17). If the speaker has direct sensory evidence, s/he can use *estar* in cases that do not denote a stage of the subject. Thus, example (40) from Roby (2009: 17) can be uttered in a context where the ham is excellent in an absolute sense, not as a stage in its life cycle. Many have observed that in these cases there is an implicit comparison with the speaker's expected quality of ham in general: 'as far as ham goes, this one is excellent,'

(40) Este jamón serrano está fenomenal.
'This Serrano ham is.SL phenomenal.'

Finally, Luján (1981) observes that *ser* predicates can be coerced into uses with *estar* much more easily than the converse (cf. Luján 1981; Querido 1976), although as Fernández Leborans (1995) points out, a few *ser* predicates cannot be coerced into *estar*: *ser/*estar mortal* 'be.IL/*be.SL mortal.' Luján's observation is consistent with generalizations made in the literature on semantic coercion (cf. Fernald 1999).

(41) a. Este perro es inteligente. (*intelligent*, a canonically *ser* predicate)
'This dog is.IL intelligent.'
b. Este perro no está muy inteligente hoy.
'This dog is.SL not (being) very intelligent today.'

(42) a. Este cine está siempre vacío. (*vacío*, a canonically *estar* predicate)
'This movie-theater is.SL always empty.'
b. *Este cine es siempre vacío.
'This movie-theater is.IL always empty.'

To summarize, the observations made so far are the following:

(43) a. *Ser/estar* predicates track the SL/IL distinction, with a few mismatches.
a1. Certain *estar* predicates have mixed syntactic patterns as SL/IL predicates. Unexpected outcomes usually involve *when* conditionals and locatives.
a2. Evidential *estar* does not involve a strict SL reading.
a3. Semantic coercion (i.e., changes from IL to SL interpretation) need not involve changes in copula.
a4. *Ser* predicates can be coerced into SL readings, but *estar* predicate coercion is much more restricted (ex. (26)).
b. Predicates that appear with *estar* coincide with predicates that appear as secondary predicates (ex. (21)).
c. *Estar* predicates can also appear as complements of perception verbs
d. (ex. (17a,b).
e. *Estar* itself cannot appear as a complement of a perception verb (ex. (17c)).
f. *Estar* is an aspectual auxiliary (ex. (4)) but cannot be the main verb in a progressive (*estar* + gerund) construction (ex. (50).
g. Locative predicates diverge from the expected pattern in systematic ways (exs. (15), (16), (28)).
h. NPs/DPs can appear with *ser*, not with *estar* (ex. (6)).

4 The formalization of aspect with *ser/estar*

The preceding description suggests that the SL/IL distinction approximates but does not perfectly match the distribution of *ser* and *estar*. In this section, I propose a formalization of the SL/IL conceptual distinction that builds on Luján's (1981) and Zagona's (2010) proposals.

4.1 *The feature content of* estar *and its predicate*

Luján proposes a calculus between the copula and the predicate whereby the copula selects for the appropriate feature on the predicate. In her terms, *ser* selects for a [-Perfective] adjective, *estar* for a [+Perfective] feature. A [+Perfective] predicate holds for "a delimited period of time whose beginning and end are both known or assumed or at least one of them is" (Luján 1981: 176). On the other hand, [-Perfective] holds for an unbounded period of time. Luján's conception of perfectivity is closer to lexical aspect than to situation aspect. Zoby's (2009) conception, on the other hand, relates the [±Perfective] feature of *ser/estar* with the aspect of the imperfect and preterit tense. However, as Luján points out, *ser* and *estar* show viewpoint aspect (i.e., imperfect/preterite) in addition to their selectional restrictions. In other words, aspect as manifested on the preterit and imperfect does not show the same kind of asymmetries that it does when it appears on the combination of copula and predicate.

Fernández Leborans (1995) and Zagona (2010) also argue that the crucial property driving the *ser/estar* distinction is related to lexical or situational aspect. For Fernández Leborans (1995: 271), *estar* selects for the transition to an ending state, following Pustejovsky's (1990) aspectual typology. However, it is clear that in some instances, *estar* does not select for transitions, as in the so-called evidential cases in (40) above, where the predicate simply suggests a comparison with other ideal or expected situations.

Zagona formalizes aspectual distinctions indirectly: she suggests that *estar* selects for a categorial feature, an uninterpretable prepositional feature, [*u*P], which must be checked by a complement (cf. also Gallego and Uriagereka 2009 for a similar analysis). The complement has two properties: in addition to being prepositional, it cannot contain a certain lexical–aspectual content, which Zagona characterizes as a Path (roughly comparable to the transition just described).

4.1.1 Beginning and end boundaries I agree with Luján, Fernández Leborans, and Zagona that the notion of event boundary is relevant for the distribution of *estar*. However, rather than invoking a transition (which calls for a preceding event), I will suggest that *estar* selects for the beginning boundary of a state. In other words, when a speaker says *Juan está alegre* 'Juan is.SL happy,' *estar* selects for the inception of the state of being happy. I will formalize this feature as [INCH] (inchoative). Notice that this conception of *estar*'s selectional properties is consistent with Luján's (1981: 185) observation that *estar* predicates are compatible with the periphrastic construction *quedar(se)* 'remain, become' + adjective, which does not necessarily imply a transition:

(44)		Adjective	*Quedar* + A
	a.	*soltero* 'single'	quedar(se) soltero
	b.	*maltrecho* 'battered'	quedar maltrecho
	c.	*listo* 'ready'	quedar listo
	d.	*intacto* 'intact'	quedar intacto

How does the notion of [INCH] relate to the notion of stage-level? Conceptually, a spatiotemporal SL entails a beginning boundary, but a beginning boundary does not necessarily imply a stage. For example, in the so-called evidential uses of *estar* ((40) above, *este jamón está fenomenal* 'this ham is excellent'), there is no explicit or implicit notion that the predicate holds as a stage of the subject. However, if what matters is the beginning of a situation, one can draw the implicature that in other similar situations, ham (in general, not this one in particular) may be less good (cf. Maienborn 2005).

As shown above, the distribution of *ser/estar* and its predicates requires a mechanism for the verbs to interact with the predicate, so I will assume that the predicate itself contains an aspectual feature that matches *estar*'s [INCH] requirements. By contrast, *ser* predicates do not match *ser* in any respect. Certain predicates that appear with both copulas (like *alegre* 'happy') will be assumed to be ambiguous between a lexical entry with an aspectual feature to match *estar*'s and an entry without it. For other predicates, I will assume that their lexical setting is unique: it either contains the aspectual feature or it lacks it. Hence, they will appear with either *estar* or *ser* by default. When forced by some linguistic item or contextual queue, their default meaning may be semantically coerced into a different interpretation. As we will see, semantic coercion will be seen as a process that repairs the anomalous semantic matching between two constituents.

Formally, I draw from a number of recent proposals on how features from two distinct categories are matched (cf. Chomsky 1995, 2000, 2001; Pesetsky and Torrego 2001, 2006, 2007; Baker 2010; Camacho 2010). I define agreement as an operation by which a functional category A with a feature α ($A_{[\alpha]}$) identifies a category B with the same feature α ($B_{[\alpha]}$), and the value of the features becomes shared (as in Pesetsky and Torrego).[7] $A_{[\alpha]}$ (the probe), is defined as a functional category, and in this I depart from Chomsky's and Pesetsky's and Torrego's assumptions.

Features can be valued or unvalued (i.e., have some kind of content), and interpretable or uninterpretable (i.e., relevant or irrelevant for the conceptual interface). In (45a), pro is unvalued for person and number but interpretable;[8] in (45b), the adjective is valued for number and gender, but uninterpretable. When two categories agree, they share the same feature-value as in Pesetsky and Torrego (2007), thus when pro agrees with a functional projection that has number and person as in (46b), the features of the valued category will be shared with those of the unvalued category, as in (46c).

(45)	a.	pro$_{[i\text{NUM}[\,]/i\text{PERS}[\ ]]}$:	Interpretable, unvalued number and person
	b.	Sospechosa$_{[u\text{NUM}[\text{SG}]/u\text{GEN}[\text{FEM}]]}$:	Uninterpretable, valued number and gender
		'Suspicious'	

(46) a. pro$_{[i\textsc{num}[\ ]/i\textsc{pers}[\ ]]}$ llegaron$_{[u\textsc{num}[\textsc{pl}]/u\textsc{pers}[3]]}$
'They arrived'

b. pro$_{[i\textsc{num}[\ ]/i\textsc{pers}[\ ]]}$ llegaron$_{[u\textsc{num}[\textsc{pl}]/u\textsc{pers}[3]]}$

c. pro$_{[i\textsc{num}[\textsc{pl}]/i\textsc{pers}[3]]}$ llegaron$_{[u\textsc{num}[\textsc{pl}]/u\textsc{pers}[3]]}$

Only uninterpretable features that have undergone agreement and valued can be deleted. Uninterpretable features or features that remain unvalued cannot be processed at the interface level, and lead to a non-convergent derivation or undergo a repair strategy, coercion.

The specific proposal for *ser/estar* is represented in (47)–(49). *Estar* heads an aspectual projection but *ser* does not (cf. (47a) vs. (47b). Additionally, *estar* has an uninterpretable, valued aspectual feature. An adjective like *alegre* 'happy' has two alternative lexical entries, one aspectually marked (but unvalued), the other one not marked for aspect, as in (48). When merged with *estar*, both categories agree and the values of *estar*'s feature are copied onto those of the predicate (cf. (49a)). By contrast, the unmarked version of *alegre* can merge with *ser*, as in (49b).

(47) a. [$_{\text{AspP}}$ estar$_{[u\textsc{asp}[\textsc{inch}]]}$]]
b. [$_{\text{CopP}}$ ser]

(48) a. alegre$_{[i\textsc{asp}[\ ]}$
b. alegre

(49) a. [$_{\text{AspP}}$ estar$_{[u\textsc{asp}[\textsc{inch}]]}$ [$_{\text{AP}}$ alegre$_{[i\textsc{asp}[\textsc{inch}]]}$]]
b. [$_{\text{CopP}}$ ser alegre]

Should the aspectual entry of *alegre* attempt to merge with *ser*, its unvalued feature would remain unvalued. If the entry in (48b) merges with *estar*, *estar*'s aspectual, uninterpretable feature will not be deleted.

Recall that certain predicates only appear with *ser* (i.e., *mortal* 'mortal'), and others only with *estar* (*muerto* 'dead'). Following the logic above, these adjectives only have one lexical entry: *mortal* lacks an aspectual feature and *muerto* always has it. When *mortal* appears with *estar*, the verb's uninterpretable features are not deleted, as in (50). When *muerto* appears with *ser*, its aspectual feature will remain unvalued (cf. (51)).

(50) *[$_{\text{AspP}}$ estar$_{[u\textsc{asp}[\textsc{inch}]]}$ [$_{\text{AP}}$ mortal]]

(51) *[$_{\text{CopP}}$ ser [$_{\text{AP}}$ muerto$_{[i\textsc{asp}[]]}$]]

We now turn to the other cases described in previous sections.

4.2 *Verbal passive versus adjectival predication*

As suggested earlier, both *ser* and *estar* appear with participals (cf. (3) above). Following Zagona's (2010) suggestion (derived from Carrasco et al. 2006), I will assume that true passives with *ser* take a verbal complement (a verbal participle), whereas predicates with *estar* appear with an adjectival participial (cf. (52), from Zagona (2010 ex. 39)). Additionally, if the adjective has a beginning boundary, but the verbal participle has a more complex structure (in particular, a change of state feature), use of *estar* with the participle will leave *estar*'s [*u*ASP [INCH]] undeleted.

(52) a. *ser* + participle: b. *estar* + participle:

```
onset      telos                    onset    telos
-----o-----------o------>           -----o--------o------------>
 |///////////////|                              |///////////|
    Includes Event boundaries        Excludes Event endpoint
    (Non-homogenous event)           (Homogenous situation)
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4.3 *Secondary predicates and complements of perception verbs*

Secondary predicates and complements of perception verbs match *estar*-predicates, as suggested. I assume that in each case, the bounded nature of the predicate is selected. In the case of a perception verb, I assume that its complement is a small clause selected by the functional projection associated with accusative case (so-called little v). This projection carries the relevant aspectual feature inherited from the adjective, as in (53b). If the predicate lacks the aspectual feature, then no agreement will take place and the unvalued feature will remain as in (53c).

(53) a. Greta vio a Miguel contento/*inteligente.
'Greta saw Miguel happy/*intelligent.'
b. vio v$_{[u\text{ASP}[\text{INCH}]]}$ [a Miguel [$_{\text{AP}}$ contento$_{[i\text{ASP}[\text{INCH}]]}$]]
c. *vio v$_{[u\text{ASP}[\text{INCH}]]}$ [a Miguel [$_{\text{AP}}$ inteligente]]

In certain dialects, *ser* is possible in the context of perception verb complements (cf. n. 3): *nunca lo vio ser figura* 'she never saw him being a figure.' Note that this use still retains a delimited interpretation, and can be subsumed under the notion of coercion, see below.

With respect to secondary predicates, I assume that they are also selected by some aspectually marked head which has the same feature as that of *estar*, as in (54). Only aspectually marked predicates will delete that feature, as in (54b,c).

(54) a. Greta llegó contenta/*inteligente.
'Greta arrived happy/*intelligent.'
b. llegó $[_{\text{Asp}}$ $\text{Asp}_{[\cancel{u\text{ASP}[\text{INCH}}]]}$ $[_{\text{AP}[i\text{ASP}[\text{INCH}]]}$ contenta]]
c. *llegó $[_{\text{Asp}}$ $\text{Asp}_{[uP\text{-INCH}]}$ $[_{\text{AP}}$ inteligente]]

Notice that overt *estar* is possible in (54) (*Greta llegó estando contenta* 'Greta arrived (while) being happy'). In this case, I assume *estar* overtly heads the Asp projection. By contrast, overt *estar* is not possible in (53) (cf. **Greta vio a Miguel estando contento*). It seems that the relevant difference has to do with whether the secondary predicate is subject or object oriented, the latter case being ungrammatical. One possible explanation suggests that in the object-oriented case, *estar* is structurally too low and falls within the temporal scope of the main verb. Since *estar* has an independent tense, this leads to an interpretive clash.

4.4 **Estar** *as progressive auxiliary*

One of *estar*'s vexing paradoxes is its role as a progressive auxiliary. Under most accounts, progressive tenses are unbounded (see, for example, Smith 1997: 84), but the above analysis assumes that *estar* agrees with a bounded predicate (cf. Luján 1981; Zoby 2009). Zagona (2010) suggests that gerunds denote a different type of aspect than *estar* predicates. As verbs, gerunds can have situation aspect, and their gerund morphology denotes an additional aspectual layer (viewpoint aspect, cf. Smith 1997; Arche 2006; Zagona 2010).

Smith (1997: 74) argues that progressive tenses cannot be closed. If one compares her examples in (55), in the first one is anomalous because the *after* clause requires a closed, bounded interpretation, but the progressive cannot force that reading. The perfective, on the other hand, can.

(55) a. ?*Herbert was hiding the loot after the telephone rang.
b. Herbert hid the loot after the telephone rang.

In Spanish, however, a closed reading is possible in this context (cf. (56)), which suggests that we need at least two layers of aspect: the one provided by the gerund, and the one provided by the auxiliary's tense. Furthermore, if the main auxiliary is absent, the gerund is still compatible with a bounded interpretation, as seen in (57). In fact, Smith suggests that "by a default inference, one can conclude that the initial point of the event has occurred. The inference follows from the fact that part of the event is visible" (1997: 63). Examples (56) and (57) suggest that the initial point is more than an inference.

(56) Enrique estuvo escondiendo el botín después de que el teléfono sonó.
'Enrique was.SL.PERF hiding the loot after the telephone rang.'

(57) Escondiendo el botín después de que sonó el teléfono, Enrique miró por la ventana.
'Hiding the loot after the phone rang, Enrique looked out the window.'

Let us assume, then, that *estar* merges with a different aspectual projection, one that contains a "full event structure, including both the external argument and a higher temporal head that instantiates the onset of the event" (Zagona 2010: 15). Further evidence to this effect comes from another test for boundedness (cf. Smith 1997: 64). Use of the progressive in the first clause of (58a) yields a contradiction, because the second clause denies the beginning point asserted by the first clause. By contrast, (58b), with prospective aspect, is not contradictory.

(58) a. #Adrián estaba comiendo, pero no había empezado/empezó a comer.
'Adrian was.SL eating, but hadn't begun/didn't begin to eat.'
b. Adrián estaba a punto de comer, pero no había empezado/empezó a comer.
'Adrián was.SL about to eat, but hadn't begun/didn't begin to eat.'

These considerations lead me to propose that a progressive aspect projection is selected by *estar* that involves a beginning boundary, just like the other predicates selected by *estar* (cf. Fernández Leborans 1995), as in (59).

(59) [AspP$_{[u\text{ASP[INCH]}]}$ estar [Asp$_{\text{prog}}$P Asp$_{[i\text{ASP[INCH]}]}$ comiendo]]

As mentioned earlier, *estar* is ungrammatical as the main verb in progressive constructions (cf. (60a)). The representation in (60b) shows why: the highest aspectual feature remains unchecked. Since *ser* is aspectually transparent, it will not interfere with deletion of the higher feature, and it yields a grammatical representation.

(60) a. Marta está siendo/*estando alegre.
'Marta is.SL being.IL/*being.SL happy.'
b. *V$_{[u\text{F-INCH}]}$ . . . [$_{\text{AspP}}$ estando$_{[\cancel{u\text{F-INCH}}]}$ [$_{\text{AP}}$ A$_{[i\text{F-INCH}]}$]]

Note that if the higher progressive auxiliary is absent, the gerund by itself is grammatical, as seen in (cf. (61)), as predicted by the analysis.

(61) Estando yo sentado en el parque, vi un pájaro.
'(Being.SL) sitting in the park, I saw a bird.'

4.5 *Estar + *DP*

Two alternative explanations come to mind for the restriction against DP predicates with *estar* (cf. (6) above). In the first one, DPs lack the relevant aspectual feature (cf. Camacho 1994). Alternatively, *estar* selects for a predicative complement, and DPs are by default referential. This explanation extends Chung and McCloskey's (1987: 179, 180 n. 4) account for the observation that Irish must insert a particle *ina* before an NP in a tenseless copular construction (cf. (62)). For them, *ina* marks the NP predicatively, rather than referentially (cf. also Baker 2003, ch. 2).

(62) agus é *('na) dhíodóir
and him PRED lawyer
'while he was a lawyer'

If this is correct, then *de*, which renders DPs grammatical (cf. (7) above) would be a predicational head. However, note that *de* also introduces a stage-like reading (cf. comments to (7) above).

5 Locative PP predicates

In the preceding section, I proposed that *estar* selects for the inception of a state/event, whereas *ser* is unmarked for the aspectual properties of its complement. In this section, I extend this proposal to locative PP predicates. In general, the default copula with locative PPs is *estar*, as seen in (15) above, repeated below. However, locative PPs systematically diverge from other *estar*-predicates in that the interpretation of the subject can influence copula choice, in particular, that an eventive subject blocks *estar* with a locative PP (cf. (16) above, repeated below).

(15) Los libros están/*son en el estante.
'The books are.SL/*are.IL on the shelf.'

(16) La fiesta es/#está en la discoteca.
'The party is.IL/#is.SL at the disco.'

Since locative PPs appear with *estar* and as secondary predicates and complements of perception verbs, I assume the same analysis that has already been proposed: their lexically specified $[i\text{ASP}_{[\,]}]$ feature must be valued by *estar*. I further argue that this is their only lexical entry, since they do not typically alternate with *ser*.

In the case of subjects with an eventive interpretation (cf. (16)), I propose that they block agreement between the verb and the PP: the eventive subject is aspectually marked, but with the wrong feature, so it cannot agree with *estar*. At the same time, the fact that it is aspectually specified means that it can block further agreement with the PP, as in (63).

(63) $[_{\text{AspP}}$ estar$_{[u\text{ASP[INCH]]}}$ DP$_{[i\text{ASP[DURAT]]}}$ $[_{\text{PP}}$ P$_{[i\text{ASP[\ \ ]}}$ la discoteca]]

However, since eventive subjects do not affect non-locative predicates (cf. (64)) the structures of locative and non-locative predicates must differ. Following evidence presented in AALE (2009: 2815–2816), I propose that locative PPs are adjoined. Specifically, AALE notes that locative PPs cannot cliticize (unlike other predicates, cf. (65)), and they can be deleted, whereas other predicates cannot (cf. (66)).

(64) La fiesta es/está divertida.
'The party is.IL/is.SL fun.'

(65) a. El perro está contento$_i$ pero el gato no lo$_i$ está.
'The dog is happy but the cat isn't.'
b. ??El perro está [en la casa]$_i$ pero el gato no lo$_i$ está.
'The dog is in the house but the cat isn't.'

(66) a. ¿Está Pedro Ø/en su casa?
'Is Pedro (there)/at his house?'
b. ¿Está Pedro cansado/*Ø?
'Is Pedro tired/Ø?'

If the relevant difference between locative PPs and other predicates is that locative PPs are adjuncts, their structure would be the one in (67). As suggested, the eventive subject blocks agreement between the verb and the PP. If the eventive DP is absent, agreement proceeds as expected. When an adjective is a predicate, it is a complement, hence the subject DP and the predicate are in the same agreement domain as *estar* (cf. (68)), they are equidistant, and the subject does not block agreement.

(67) $[_{\text{AspP}}$ estar$_{[u\text{ASP[INCH]]}}$ DP$_{[i\text{ASP[DURAT]]}}$ $[_{\text{AspP}}$ $[_{\text{PP}}$ P$_{[i\text{ASP[\ \ ]]}}$ DP]]] (locative PP)

(68) $[_{\text{AspP}}$ estar$_{[u\text{ASP[INCH]]}}$ [DP$_{[i\text{ASP[DURAT]]}}$ $[_{\text{XP}}$ X$_{[i\text{ASP[\ \ ]]}}$]]] (non-locative predicates)

6 Coercion[9]

The analysis developed so far treats *estar* as a functional head that agrees in aspect with its complement, explaining why *estar* correlates with a certain (bounded) interpretation. *Ser*, on the other hand, is unmarked for aspect. As mentioned earlier, certain predicates are ambiguous (they appear with *ser* or *estar*), others are not. The typology of predicates in (69) proposes a tripartite distinction between adjectives: (a) those that are ambiguous because they have two lexical entries; (b) those that are lexically unambiguous but that can be contextually coerced into the opposite meaning; and (c) lexically unambiguous adjectives that cannot be coerced.

(69)

Predicate	*Status*	*Lexical entry*	*Example*
(a) *alegre* 'happy'	Lex. ambiguous	(a) $\text{alegre}_{[\text{iASP}[\ ]]}$	*La niña es/está alegre.*
		(b) alegre	'The girl is.IL/is.SL happy.'
(b) *simpático* 'charming'	Lex. unambiguous – coercible	(a) simpático	*Es simpático, pero hoy no está simpático.* 'S/he is.IL charming, but today s/he is.SL not charming.'
(c) *mortal* 'mortal,' ausente 'absent'	Lex. unambiguous – not coercible	(a) mortal	**Este mes la guerra ha estado mortal.* 'This month, the war has been.SL mortal.'
		(b) $\text{ausente}_{[\text{iASP}[\ ]]}$	**Pedro siempre es ausente: es su carácter.* 'Pedro is.IL always absent: it's his character.'

Semantic coercion (cf. Fernald 1999; Escandell and Leonetti 2002) has been conceived as a process that repairs the anomalous semantic result of combining two constituents. However, as Escandell and Leonetti note, the process must be syntactically constrained. Within the current proposal, coercion can be seen as reparation of an anomalous agreement operation that leaves an unvalued or an uninterpretable feature after the syntactic computation.

Consider *está simpático* 'is.SL charming.' The adjective lacks an aspectual feature, so *estar*'s uninterpretable feature will remain undeleted. Coercion will repair this anomaly by providing a boundary that will match the uninterpretable feature. This feature can come from an adjunct, or even from the context. Needless to say, this strategy is more costly because it must repair the anomalous outcome of the regular computation.

In the opposite situation, the predicate has an unvalued, interpretable feature at the output of the derivation. As noted above, this situation is very infrequent, although it seems to be what happens in examples like (26b) above, where the adverbial 'several times a day' appears with *ser*. This is also the case in Schmitt's examples in (30) above, where *ser* appears within the scope of a *when* conditional. Recall that in those cases, the SL interpretation was possible even though *estar* did not appear. Rather, coercion affected the predicate with the aspectually unmarked copula *ser*. The generalization seems to be that it is easier to coerce a construction where there is a lingering uninterpretable feature or an unmarked head than it is to force a change on a lexically marked predicate or copula.

7 Summary

In this chapter, I have summarized the distribution of *ser* and *estar* along with the different proposals in the literature based on the aspectual properties of the copulas and/or the predicates. I have concluded that *estar* agrees with its complements in an aspectual, inchoative feature, which encodes the beginning point of an event, whereas *ser* lacks any aspectual properties. Certain predicates are lexically ambiguous, whereas others are lexically marked. For a subset of the latter, it is possible to coerce their meaning into the opposite setting, and this meaning coercion has been argued to involve either uninterpretable or unmarked feature settings.

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NOTES

1 I will gloss *ser* as be.IL and *estar* as be.SL, as a mnemonic device for 'individual-level' and 'stage-level,' which will be more precisely defined below.

2 However, whereas (15c) is acceptable with *ser* for some speakers, ??*la puerta es a la entrada de la casa* 'the door is.IL at the entrance of the house' seems to suggest that doors aren't usually located at the entrance and this one happens to be. In effect, use of *ser* forces a puzzling SL-like reading.

3 Maienborn (2005: 166) claims that both *ser* and *estar* are impossible as complements to perception verbs. As Schmitt (1992) has argued, *ser* complements are attested in Brazilian Portuguese, and, I might add, also in several varieties of Spanish. The following example is a comment about soccer player Messi, taken from http://valechumbar.com/9189/como-ver-barcelona-vs-arsenal/ (accessed April 29, 2010), and I have found several others from Mexico, Southern Spain, Chile, and other Spanish-speaking regions. However, for some speakers of Castilian Spanish, this example is not acceptable. (a) Todo el equipo fue fundamental pero en las bravas nunca lo vi ser figura como en partidos como el Almeria . . . "All the team was essential, but in the difficult situations, I never saw him be(ing) a star like in the games against Almeria . . ."

4 Thanks to an anonymous reviewer for clarification on this point.
5 *Nervioso* 'nervous' can appear with *ser*, but in such case, its meaning becomes individual-level.
6 As Roger Schwarzchild (p.c.) points out, the same is true of 'dead' in English: it patterns like an SL predicate, for example in allowing an existential reading:
(a) Birds are dead in that field.
(b) There are dead birds in that field.
7 Like Baker (2010) and Camacho (2010), I assume that the search for a matching category can be up the tree or down the tree, subject to locality constraints such as a strong phase.
8 *i*/*u*F stands for 'interpretable/uninterpretable feature F,' and F[] for 'unvalued feature F.'
9 Thanks to an anonymous reviewer for suggesting development of this section.

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23 Passives and *se* Constructions

AMAYA MENDIKOETXEA

1 Introduction: a classification of *se* constructions

The clitic *se* in Spanish, like its Romance equivalents, is crucially involved in constructions representing core grammatical phenomena such as coreferentiality, impersonality, voice, causation, telicity, and so on. It is, therefore, not surprising that the properties of *se* constructions, of which we offer a representative (though not exhaustive) sample in (1), are amongst the most researched topics in Spanish linguistics.

(1) a. *Se* observa(n) cambios en la economía. *impersonal/passive*
se observe-SG(-PL) changes in the economy
'One can observe changes in the economy.'

b. Las casas prefabricadas *se* construyen fácilmente. *middle*
the houses prefabricated-PL *se* build-PL easily
'Prefabricated houses are easily built.'

c. Los hermanos *se* miraron. *reciprocal*
the brothers *se* looked-3PL
'The brothers looked at each other.'

d. Los niños *se* lavaron. *reflexive*
the children *se* washed-3PL
'The children washed.'

e. Ana *se* desmayó. *pseudo-reflexive*
Ana *se* fainted-3SG
'Ana fainted.'

f. El cristal *se* rompió. *unaccusative/ inchoative*
the glass *se* broke-3SG
'The glass broke.'

The Handbook of Hispanic Linguistics, First Edition. Edited by José Ignacio Hualde, Antxon Olarrea, and Erin O'Rourke.

g. Juan *se* comió las manzanas. *aspectual/telic*
Juan *se* ate-3SG the apples
'Juan ate (up) the apples.'

h. Juan *se* fue. *aspectual*
Juan *se* went-3SG
'Juan went (away).'

A descriptive account of *se* constructions can be found, among others, in Molina Redondo (1974) (see Mendikoetxea 1999a; Sánchez López 2002; and, especially, Dobrovie-Sorin 2006 for a review of Romance *se* in the theoretical literature). Although I will deal with most of the constructions in (1) individually, for expository purposes, I will distinguish between constructions with "arbitrary" *se* (1a,b) and those with "anaphoric" *se* (1c–1f). In the latter, but not in the former, *se* alternates with 1p (first-person) and 2p (second-person) clitics like *me*, *te*, etc. Constructions with "aspectual" *se* (1g,h) can be considered as a subtype of "anaphoric" *se* constructions, but will not be dealt with here, as they show distinct properties; for example, while the constructions in (1a–f) show some degree of "reduced transitivity," this analysis cannot easily be extended to (1g,h) (see e.g., Sanz and Laka 2002; De Miguel and Fernández Lagunilla 2000). I will use SE/SI to refer to this clitic in Romance in general, as opposed to its realization in particular languages: Spanish *se*, Italian *si*, etc.

The chapter is organized as follows. In Section 2, I examine the status of *se* and its place in the clitic paradigm. Analyses of the syntax and semantics of "arbitrary" *se* constructions are given in Section 3. In Section 4, I turn to the properties of 'anaphoric' *se* constructions. Section 5 contains some concluding remarks.

2 The status of *se*

The variety of constructions that *se* appears in makes it difficult to extract the morpho-syntactic properties of *se* from the syntactic and semantic characteristics of these structures. Indeed, within the context of Romance cliticization, SE/SI has been considered an "oddity." Furthermore, there is little consensus on crucial questions such as its grammatical category and whether a unified account can be given for all its occurrences in structures like (1).

2.1 *What kind of element is SE/SI?*

A distinction commonly found is that between the 'pronominal' uses of *se*, in constructions where it alternates with 1p and 2p clitics, and *se* as some sort of non-alternating INFL-related element, a morphological marker of 'passivization' or 'impersonalization' in constructions like (1a,b) and of 'unaccusativization' or 'decausativization' in (1f) (Burzio 1981, 1986; Otero 1986; Zubizarreta 1987). The introduction of functional heads in the grammar in the late 1980s and early 1990s allowed for more precise analyses of the categorial nature of SE/SI. For Spanish, Mendikoetxea (1992) considers *se* to be the realization of the features of an AGR (element) head, under an analysis which considers object clitics to be affix-like elements (see, among others, Fernández Soriano 1989; Franco 1993; Sportiche 1996, 1999). De Miguel (1992) regards *se* as the realization of the features of an ASP(ect) head in an analysis designed to explain some aspectual restrictions found in Spanish impersonal constructions (see also Masullo1999 for an analysis in which *se* is an external argument in ASP and Kempchinsky 2004 for some instances of *se*). Along similar lines, Sanz and Laka (2002) analyze *se* as the head of an Event Phrase in constructions like (1g), with *se* as a telicity marker. The observation that SE/SI is the only truly reflexive clitic has led to recent analyses of this element as an anaphor, as argued by Dobrovie-Sorin (1998) for Romanian *se*, Rivero (2000, 2002) for Romance and Slavic SE/SI, and Teomiro (2010) for Spanish *se*.

2.2 *The feature content of SE/SI and the clitic paradigm*

The characterization of SE/SI as deficient or featureless (e.g., in Burzio 1991) is based on the hypothesis that SE/SI is not a 3p clitic, hence its occurrence in arbitrary constructions, lacking specific 3p (third-person) reference ((1a,b). As opposed to *me*, *te*, etc. and their Romance equivalents in constructions like (2), (3a) can only be interpreted as reflexive, and (3b), with a 1p subject, is simply uninterpretable (see also Chapter 26, below) because *se* cannot have independent reference.[1]

(2) (Ellos) *me/ te/ nos/ os* miran.
(they) CL-1SG/CL-2SG/CL-1PL/CL-2PL look-3PL
'They look at me/you/us/you (people).'

(3) a. Juan *se* miró.
Juan *se* looked-3SG
'Juan looked at himself.' (e.g., in the mirror)
b. *(yo) *se* miró
(I) *se* looked-1SG

Other facts appear to indicate that *se* lacks syntactic number and gender features. First, in reflexive constructions with 3p subjects like in (1c) above, *se* remains invariable, regardless of the number features of the subject. Secondly, predicative elements in *se* constructions like (4) show "default" (masculine singular) agreement:

(4) *Se* viene content*o*.
se come-SG happy-MASC-SG
'One arrives happy.'

The implicit human referent in Spanish, as in Romance in general, can be made explicitly feminine; for example, when the adjective describes a property typical of women like *embarazada* 'pregnant' in (5), while *se* remains invariable (see also Rivero 2002). This suggests that gender (and number) features on the adjective in

(5) are pragmatically determined (see Egerland 2003) and that *se* is indeed featureless:

(5) Cuando *se* está embarazad*a*, hay que comer bien.
When *se* is pregnant-FEM-SG, has that to-eat well
'When one (= a woman) is pregnant, one has to eat well.'

It is also possible to find masculine plural predicative elements in constructions like (4) and (5) in Spanish (see Martín Zorraquino 1979: 168; Otero 1986: fn 35), but they are slightly less acceptable than feminine predicative elements. A featureless element would certainly constitute an 'oddity' in the clitic paradigm. Moreover, under current theoretical assumptions (after Chomsky 1995), there is no such thing as a "featureless" grammatical element. Kayne (2000: ch. 8) argues that SE/SI is to be considered among 'person' clitics in Romance. Mendikoetxea (2008a) distinguishes between: (i) D(eterminer)-CL(itic)s (*lo(s)*, *la(s)*, and *le(s)* in Spanish), which share properties with D and nominal morphology and, like definite articles, with which they are historically related, head DPs; and (ii) AGR(eement)-CL(itic)s like *me*, *te*, *nos*, *os*, and *se*, which share properties with affixes and verbal morphology (see Kayne 2000: ch. 8 and Chapter 21, above). As is well known, Romance D-CLs, like definite articles, are historically related to the Latin demonstrative *ille*, *illa*, *illud*, as opposed to 1p and 2p CLs (as well as SE/SI) which already existed in Latin as pronouns. As an AGR-CL, SE/SI appears in reflexive constructions, from which D-CLs are banned (3a). Unlike 1p and 2p CLs, however, it lacks independent reference (3b) and can appear in 'arbitrary' constructions (1a,b), where it is the lexicalization of a 0-person feature (in the spirit of Kayne 1993) in CL_1, a functional projection in the temporal (T) domain, as in (6) (following Manzini and Savoia 2001):

(6) **C** [CL_1 SE/SI … [**T** … [**V**]]]

Note that Kayne's (1993) 0-person is distinct from Benveniste's non-person, which refers to 3p pronouns. It is a kind of defective person feature which is "less strong than a 'positively numbered' person" (Kayne 1993: 16). It is a person morpheme that lacks specification for first or second person. The defective nature of the person feature of *se* is responsible in Mendikoetxea (1992) for the fact that the AGR heads this element is the realization of cannot assign/check Case. For Manzini and Savoia (2001), *si* is the lexicalization of the feature 'quantifier' because of its denotational properties as a free variable (Manzini 1986).

3 The syntax and semantics of arbitrary *se* constructions: passives, impersonals, and middles

Though there is earlier theoretical work on these constructions (see references in Mendikoetxea 1999a; Sánchez López 2002), the most influential analyses belong to the 1980s and early 1990s, as cited below. Work within the Government and

Binding (GB) framework (e.g., Chomsky 1981, 1986) focused mostly on the morphosyntactic properties of *se* constructions and analyzed *se* (and its Romance equivalents) as either a pronominal/argumental element with Case and theta-role (e.g., Cinque 1988 for argumental *si*) or as an affixal/nonargumental element, with properties similar to those of passive morphology (among others, Grimshaw 1982; Burzio 1986; Otero 1986; Zubizarreta 1987; Cinque 1988 for non-argumental *si*). In the 1990s, efforts were made to integrate *se* within the clitic paradigm and offer an analysis of *se* constructions in terms of functional projections (Mendikoetxea 1992; De Miguel 1992). Recent analyses have focused on the anaphoric status of *se* (Dobrovie-Sorin 1998; Rivero 2000; Teomiro 2010) and/or the agreement/checking operations triggered by the features of *se* and other elements in the constructions in (1) (see e.g., Mendikoetxea 2008a).

Constructions like those in (1a,b) are interpreted as having arbitrary subjects, like *one* and *people,* in English or as agentless passives. Under (1a), we have two structures, depending on whether there is verbal agreement with the internal argument or whether the verb shows default 3SG agreement. The agreeing construction, repeated as (7a), is often referred to as "passive" (also "middle-passive" or "impersonal-passive"), while the non-agreeing constructions, repeated as (7b), is referred to as "impersonal"; this is also the term used for constructions like those in (8), with intransitive verbs:

(7) a. *Se* observa*n* cambios en la economía. *passive*
se observe-PL changes in the economy
b. *Se* observa cambios en la economía. *impersonal*
se observe-SG changes in the economy
'One can observe changes in the economy/Changes in the economy can be observed.'

(8) a. *Se* bebe mucho en las fiestas. *unergative*
se drink-SG a-lot in the parties
'One drinks a lot at parties.'
b. Con estos atascos *se* llega siempre tarde. *unaccusative*
with these traffic-jams *se* arrive-SG always late
'With these traffic-jams, one is always late.'

The existence of the two constructions in (7) figured prominently in early theoretical analysis of *se* constructions (among others, Otero 1972, 1973, 1976; Contreras 1973; Knowles 1974; Westfal 1980), but there is no consensus regarding the nonagreeing construction. It is often disallowed in Castilian Spanish (north and central Peninsular varieties), but there are syntactic contexts in which nonagreement is favored even by speakers who rule out (7b) as part of their grammar (see Mendikoetxea 1999a).

'Middle' *se* constructions are regarded as syntactically identical to constructions with "passive" *se*. Semantically, middles have generic interpretation and ascribe a property to the subject, while passives refer to events (see Mendikoetxea

1998; Dobrovie-Sorin 2006: 122). Thus, in (1b), repeated as (9) below, a property is being predicated of prefabricated houses independently of the event of building them:

(9) Las casas prefabricadas se construyen fácilmente.
the houses prefabricated-PL*se* build-PL easily
'Prefabricated houses are easily built.'

Much of the literature devoted to SE/SI has focused on the constructions in (7)–(9). For expository purposes, we refer to the constructions in (7)–(8) as 'Clitic Impersonal Constructions' (CL-ICs) and we keep the term "middle" for the construction in (9).

3.1 Cl-ICs: "impersonal" and "passive" se

3.1.1 Syntactic analyses Analyses of CL-ICs within the GB framework (Chomsky 1981, 1986) have often focused on the differences and similarities between passive and impersonal SE/SI (e.g., for Italian *si*: Belletti 1982; Manzini 1986; Hyams 1986; Burzio 1986; Cinque 1988; for Spanish *se*: Zubizarreta 1982; Otero 1986; Campos 1989; De Miguel 1992; Mendikoetxea 1992). Belletti (1982) regards impersonal *si* in Italian as the realization of the 'pronominal' features of INFL in Null Subject languages (Rizzi 1982), where it is assigned Case and theta-role in association with an empty category in subject position ((10a) below). Passive *si*, however, is a passivization marker: it 'absorbs' the external theta-role, blocking ACC(usative) Case assignment to the internal argument, which must be assigned NOM(inative) and triggers agreement ((10b) below) (see Jaeggli 1986a for the standard GB analysis of passives).

(10) a. $[_{NP}\ e^{i}]\ [_{INFL}\ \text{SE/SI}^{i}{}_{\text{NOM}}]\ [_{VP}\ \text{V (NP)}]$ *impersonal* SE/SI
b. $[_{NP}\ e^{i}]\ [_{INFL}\ \text{SE/SI ACC}]\ [_{VP}\ \text{V NP}^{i}]$ *passive* SE/SI

Analyses of Spanish *se*, however, have often rejected the association of *se* with Case (and theta-role) (but see Masullo 1992), based on the assumption that this element lacks argumental features. Otero (1986) claims that impersonal *se* is a marker in INFL whose role is to absorb the [+Def] feature of this element in finite contexts, while passive *se* is regarded as a true passivizer. De Miguel (1992) distinguishes between two functionally different *se* in CL-ICs: (1) arbitrary *se*, in transitive and unergative contexts with specific time reference; and (2) generic *se* in all other contexts (see Section 3.1.2, below). The former provides the sentence with the interpretation of an action performed by an unspecified subject; the latter makes the event generic. Both head the functional category ASP(ect). Mendikoetxea (1992) captures the Case differences between impersonal and passive *se* constructions observed by Belletti (1982), while at the same time providing a unified analysis of the clitic *se*. She regards *se* as the realization of a 0-person feature in AGR S(ubject) (impersonal) and AGR O(bject) (passive) (see Section 2.2, above). The defective content of *se*, as a 0-person clitic, is responsible for the absence of NOM in (8) and ACC in (7a), where the internal argument is associated with NOM. In this analysis, *se* is not an argument and is not associated with either theta or Case features, but it is the realization of the features of AGR, responsible for Case assignment/checking. In particular, when *se* realizes the person feature of AGR, AGR is inactive for Case assignment/checking.

Like Belletti's analysis, this analysis is designed to capture differences between Romance languages. Thus, in Spanish a (roughly) specific, animate internal argument in transitive contexts is introduced by the preposition *a*, often analyzed as a Case marker. Thus, in constructions like (11a), the presence of *a* clearly indicates that ACC is available for the internal argument so that *se* is the realization of the person feature in AGR S, as represented in (11b), with PRO in subject position due to the unavailability of NOM:

(11) a. *Se* vio a los niños.
se saw-SG to the children
'One saw the children.'/'The children were seen.'
b. $[_{\text{DP}}$ PRO$]$ $[_{\text{AGRs}}$ *se*$_{[\text{-NOM}]}]$ $[_{\text{AGRo}[+\text{ACC}]}$ $[_{\text{VP}}$ vio a los niños$]]$

Belletti's account of the absence of impersonal *se* in French can be reinterpreted in terms of the features of AGR S: while Spanish and Italian can have pronominal AGR S (i.e., they can have SE/SI as the realization of the person feature in AGR S), French, as a non Null Subject language, cannot. Thus, French can have structures like (12), with passive *se*, but not structures like (13), with impersonal *se* (examples and glosses from Cinque 1988: 567):

(12) Les pommes *se* mangent. *French*
the apples *se* eat-PL

(13) a. *(*Il) *se* mange souvent les gateaux ici. *French*
'It *se* often eats cakes here.'
b. *(*Il) *se* dort beaucoup ici.
'It *se* sleeps much here.'

Also, Romanian, despite being a Null Subject language, can have *se* as the realization of AGR O, but not as the realization of AGR S, a reinterpretation of Dobrovie-Sorin's (1992) analysis that Romanian has ACC *se* (in transitive and unergative contexts) but no NOM *se* (in unaccusative contexts).

An often parallel debate concerns the argumental status of SE/SI. Cinque (1988) postulates the existence of two types of *si* in Italian: (i) [+arg] *si*, associated with <Spec, IP> and assigned NOM and the external theta-role, in transitive and unergative contexts, and (ii) [-arg] *si*, in contexts lacking an external theta-role (with unaccusatives, copula, and passives). This element, which is associated with NOM, but no theta-role, is characterized as "a syntactic marker for unspecified (generic) *person*" (Cinque 1988: 530), whose function is to supplement

Agr with the features needed to "identify" (in the sense of Rizzi 1986a) the content of the element **pro** in subject position as unspecified or generic. This system is devised to account for the behavior exhibited by constructions with *si* in non-finite contexts. In particular, Cinque (1988) observes that while sentences with *si* are uniformly excluded from untensed **control** structures, where no nominative Case is available for the clitic, in contexts in which such nominative Case is available (e.g., **raising** structures) there is an asymmetry involving transitive and unergative Vs, on the one hand, and unaccusative Vs in general, on the other hand. This is illustrated in (14) (examples taken from Cinque 1988: 524–525), though the asymmetry is found across Romance in general (see Mendikoetxea 1992, ch. 5, for discussion on this matter and an analysis which does not require the distinction proposed by Cinque 1988 in order to account for the facts):

(14) a. (Trans) Sembra non esserSI ancora scoperto il vero colpevole.
'It seems one not to have yet discovered the true culprit.'
b. (Unerg) Sembra non esserSI lavorato a sufficenza.
'It seems one not to have worked sufficiently.'
c. (Unacc) *Sembra esserSI arrivati troppo tardi.
'It seems one to have arrived too late.'
d. (Cop) *Sembra non esserSI benvenuti qui.
'It seems one not to be welcome here.'
e. (Pass) *Sembra non esserSI stati invitati da nessuno.
'It seems one not to have been invited by anyone.'

The theoretical and empirical problems of this proposal have been discussed in great detail in the literature (e.g., Dobrovie-Sorin 1998). Recent analyses consider SE/SI to be either an argument (Raposo and Uriagereka 1995: 4.2 for European Portuguese and Teomiro (2010) for Spanish) or a nonargumental clitic (Mendikoetxea 2008a; as represented in (6) above). Having a clitic in argument position raises well-known problems regarding cliticization and highlights the oddity of SE/SI, the only "subject" clitic in Spanish, if this analysis is on the right track. Otherwise, if the clitic is in a nonargument position (e.g., in a functional projection), the argument position in CL-ICs must be occupied by a null pronominal of some kind. Most analyses postulate some kind of non-referential or generic *pro* in the subject position of CL-ICs (see Suñer 1990). Mendikoetxea (2008a, b) refers to this element as **G**(eneric)-**pro**, akin to Holmberg's (2005) G-pronouns, which is only present in the structure when T lacks a referential person feature (e.g., when it has a 0-person feature).

3.1.2 Semantic analyses While the syntax of Romance CL-ICs has been researched widely, there is very little in the literature concerning their semantics. Semantic analyses of these constructions have to account for the following descriptive facts: (a) CL-ICs are interpreted as having unspecified human subjects (see for Spanish, Molina Redondo 1974; Suñer 1990; Jaeggli 1986b: 55, among others); (b) CL-ICs may have inclusive and exclusive readings (see for Spanish, Morales 1997); and (c) CL-ICS have variable quantificational force: they may express generic/universal or episodic/existential readings (see for Spanish, De Miguel 1992; Mendikoetxea 2002, 2008a).

Descriptive facts (a) and (b) belong to the syntax–discourse interface. Regarding (a), the problem is how to explain the human reference of CL-ICs given that they do not contain a noun or property expression that univocally entails such reference. Chierchia (1995) stipulates that the range of the variable x_{arb} in the semantic representation of CL-ICs is restricted to humans (but see Aranovich 2002, against the use of such indexes). It has also been argued that impersonal pronouns like SE/SI, German *man*, and French *on* are inherently specified for the feature [+human] (Egerland 2003). A different view is that the human reference of these structures results from a pragmatic convention: it follows from the fact that it is assumed that non-humans are not legitimate discourse participants (Kański 1992). If that assumption is suspended, human implication may also be suspended (or rather extended to non-humans), as in (15b), which is acceptable when a dog is a participant in the relevant speech act:

(15) a. #*Se* ladra mucho.
 se bark-SG a lot
 b. ¡No *se* ladra a los invitados!
 not *se* bark-SG at the guests
 'One does not bark at guests!'

As for (b), the "human" referent in CL-ICs may include or exclude the speaker, as opposed to other impersonal pronouns which show either inclusive readings (*uno* 'one' and non-referential *tú* 'you') or exclusive readings (3p plural impersonal sentences). Montrul and Slabakova (2000) find an association between the two readings of *se* and perfective/imperfective aspect (see also D'Alessandro and Alexiadou 2002).[2]

Aspect also seems to play a role in the quantificational readings of CL-ICs ((c) above). De Miguel (1992), following Cinque's (1988) influential analysis, establishes an association between, on the one hand, generic time reference and universal interpretation and, on the other, specific time reference and existential interpretation in Spanish CL-ICs. According to this, we would expect the implicit subject of (16a) to have a universal interpretation (similar to 'people,' 'everybody'), while the implicit subject of (16b) should have an existential interpretation, compatible with the existence of a single individual satisfying the description (examples from Mendikoetxea 2002: 264):[3]

(16) a. En estas reuniones siempre *se* habla de lo mismo
 in these meetings always *se* speak-SG of the same
 b. Ayer *se* habló de política en la Facultad
 yesterday *se* spoke-SG about politics in the Faculty

Mendikoetxea (2002, 2008a), however, has shown that there is no such association between aspect and quantificational force in CL-ICs. The null pronominal in subject position in (16), like indefinite NPs in Diesing (1990), introduces a variable that has to be bound by a quantifier. Under this analysis, a universal quantifier may either bind the variable introduced by the null pronominal in (16a) ('Everybody talks about the same in those meetings') or a situational variable so the variable introduced by the null pronominal is interpreted existentially ('Always in those meetings there are people who talk about the same'). Similarly, (16b) can have an existential reading ('Yesterday there were people who talked about politics in the faculty') or a universal reading, by which something is predicated of all the individuals in a certain context ('Everybody in the faculty talked about politics yesterday'). Aranovich (2002) offers an analysis of the semantics of CL-ICs in terms of second-order predication, akin to Kański (1992). Sanz (2010) provides a novel and interesting account of the semantics of impersonal, passive, middle and unaccusative constructions with *se*, by establishing a parallelism between the verbs in these constructions and bare NPs. The verb with *se* predicates of events as groups or as sets and some kind of plural event quantifier is involved: they denote 'bare plural events.'

3.2 *Middles as a special type of CL-ICs*

A prototypical Romance 'middle' like that in (1b), repeated as (9) above, consists of the following elements: (a) the clitic SE/SI; (b) a preverbal DP, which is the internal argument of the verb; (c) a transitive verb which agrees (in number) with the preverbal element; and (d) a manner or adverbial expression, like *fácilmente* 'easily' in (9). Semantically, middles attribute properties to entities; they are non-eventive (*stative*) statements and involve the modality factor of ability or possibility: (9) expresses that it is an inherent property of prefabricated houses that they are easy to build. In contrast, the CL-ICs we have dealt with in Section 3.1 are *eventive*: (17) informs about a particular event that took place last year, the DP triggering verbal agreement is often postverbal, and no manner adverbs are required:

(17) El año pasado *se* construyó un puente sobre el rio Guadalix.
the year past *se* built-3SG a bridge over the river Guadalix
'Last year a bridge was built over the river Guadalix.'

Middles should be distinguished from CL-ICs like (17), with a preverbal subject: (18) does not express an inherent property of bridges (that they are quick to build in this country); rather, it expresses a generic statement (bridges are generally built quickly in this country).

(18) (En este país) los puentes *se* construyen con mucha rapidez.
(in this country) the bridges *se* build-PL with great speed

This is important, as once we separate middles from CL-ICs, Romance middle constructions are more similar to English and German middle constructions than commonly assumed (see Fagan 1992; Roberts 1987; González Romero 2002).

Not much attention has been devoted to Spanish middles as defined here, which they have often been studied together with other constructions under the term 'middle.' We are using 'middle' in a restricted sense. In the Romance literature, 'middle' or 'middle voice' is often used more widely to include (a) all the constructions that we are referring to in this chapter or (b) all constructions lacking an explicit human subject (i.e., not reflexives) (see Mendikoetxea (1999a) for extensive discussion on the use of the term 'middle' and the existence of a middle voice in Spanish; a descriptive account of the properties of middles can be found in Mendikoetxea (1998, 1999a) and García Negroni (2002)). Regarding the syntax of these constructions, one of the most interesting aspects, in so far as it distinguishes middles from other CL-ICs, is the preverbal nominal. Much of the theoretical discussion on middles in Romance and Germanic languages has centered on whether the preverbal nominal in middles is generated in that position (no movement analysis) or whether it moves to that position (movement analysis), and whether middle formation is a lexical or syntactic process (see, among others, Zubizarreta 1987; Stroik 1992; Hoekstra and Roberts 1993; Ackema and Schoorlemmer 1995). Mendikoetxea (1998, 1999a) argues that this element is base-generated in topic position, on the basis of evidence from Clitic Left Dislocation structures. As a topic, it occupies the initial preverbal position, unless there is a focalized element, in which case it occurs postverbally, as in (19):

(19) ¡BIEN *se* lavan las camisas blancas! (¡Y NO MAL!)
WELL se wash-PL the shirts white-PL (AND NOT BADLY!)

As for the semantics of middles, it is commonly accepted that they do not express generic events (i.e., they do not involve event quantification) and their implicit subject is obligatorily interpreted as generic or universal. More recently, Sanz (2010) has argued that middles are expressions of kinds of events: a plurality of potential events (see Section 3.1.2, above); their meaning is interpreted intensionally and they do not establish a predication between a verb and its subject. The relationship between the bare plural verb and its arguments is established through second-order and third-order predication, which accounts for the obligatory presence of adjuncts and other elements in the structure.

4 The syntax and semantics of anaphoric "*se*" constructions

We now turn our attention to the structures in (1c–f), repeated as (20) below, which we refer to as clitic-reflexive structures (CL-REFLs):

(20) a. Los hermanos *se* miraron. *reciprocal*
the brothers *se* looked-3PL
'The brothers looked at each other.'

b. Los niños *se* lavaron. *reflexive*
the children *se* washed-3PL
'The children washed (themselves).'

c. Ana *se* desmayó. *pseudo-reflexive*
Ana se fainted-3SG
'Ana fainted.'

d. El cristal *se* rompió. *unaccusative/inchoative*
the glass *se* broke-3SG
'The glass broke.'

A distinction is established between 'extrinsically' reflexive structures (20a,b), which express 'true' reflexive/reciprocal meanings and have agentive subjects, and 'intrinsically' reflexive structures like (20c,d), which have non-agentive subjects (see Otero 1999; Sánchez López 2002). In what follows, I deal first with (20a–c) (Section 4.1) and then with unaccusative structures in (20d) (Section 4.2), which, though putatively a subtype of intrinsic reflexivization, have properties that distinguish them from all the other structures in (20).

4.1 *Extrinsic and intrinsic reflexive constructions with* se *in Spanish*

4.1.1 Extrinsic reflexivization Spanish has three ways of marking a predicate as extrinsically reflexive (see Chapter 26, below): (a) with a SELF-anaphor in terms of Reinhart and Reuland's (1993) theory of reflexivization (R&R henceforth) when the anaphoric element is an object of the verb (21a); (b) with a simplex SE-anaphor in R&R terms, when the anaphoric object is prepositional (21b); and (c) by using the clitic *se*, as in (21c) (and (20a,b) above) in CL-REFLs:

(21) a. María se critica a *sí misma*.
'Maria criticizes *herself*.'

b. Juan solo piensa en *sí*.
'Juan only thinks about *himself*.'

c. Los niños *se* lavaron.
the children *se* washed-3PL
'The children washed (themselves).'

Under the hypothesis that *se* in (21a) is just an instance of clitic doubling (see Torrego 1995), the emerging pattern is not very different from that found in other Romance languages like Italian, where the anaphor *se stesso* and the clitic *si* are regarded as two different reflexivization strategies with the same predicate as shown in (22):

(22) a. Gianni ama *se stesso*. [equivalent to (21a)]
Gianni love-3SG himself

b. Gianni *si* ama [equivalent to (21c)]
Gianni *si* love-3SG

Within the Principles and Parameters approach, two competing analyses can be found for CL-REFLs: (1) the **pronominal** approach, in which CL-REFLs are biargumental (transitive) structures with an external argument (*los niños* in (21c)) and an internal argument (SE/SI or the empty category associated with it); and (2) the **suppression** or **reduction** approach, in which CL-REFLs are monoargumental (unaccusative) structures with an internal argument (*los niños* in (21c)), and no external argument. The two analyses differ crucially in the role attributed to the element SE/SI: (a) as a syntactic anaphor bound by the DP subject (among others, Rizzi 1986b; Dobrovie-Sorin 1988; Fontana and Moore 1992), or (b) as a marker for unaccusativity or as an intrasitivizing affix, which operates in the lexicon either to absorb the external theta-role and/or to reduce a verb's Case feature (among others, Bouchard 1984; Marantz 1984; Grimshaw 1990; Kayne 1991).

These two approaches reflect the set of apparently contradictory properties exhibited by CL-REFLs, which seem to behave as transitive or as unaccusative structures in relation to different processes. Arguments for unaccusativity include, among others, Italian auxiliary selection and participle agreement (Burzio 1986), French causatives (Grimshaw 1982, following Kayne 1975), nominalized infinitives in Italian (Zucchi 1993) and Spanish (Mendikoetxea 1997), participial constructions and arbitrary subjects in Catalan (Alsina 1996) and Spanish (Mendikoetxea 1997), and missing causes in causative constructions in Spanish (Mendikoetxea 1997). Evidence for the transitive-like properties of CL-REFLs is found in the ungrammaticality of partitive clitics in Italian (Burzio 1986) and Catalan (Alsina 1996), the fact that the subject cannot be realized as a bare NP in Spanish (Mendikoetxea 1997), subject-oriented secondary predicates in Catalan (Alsina 1996) and, crucially, reflexive constructions with dative clitics in Romance languages like Spanish, Portuguese, and Italian, as illustrated in (23) for Spanish (see Dobrovie-Sorin 2006: 2.2; Labelle 2008):

(23) Juan *se* ha comprado un coche.
Juan *se* have-3SG bought a car
'Juan has bought himself a car.'

On the basis of this evidence some authors have adopted a 'mixed' approach (e.g., Burzio 1986). Note that the problem does not arise in frameworks in which two arguments can be mapped onto a single grammatical function or can bear two thematic roles, as in the version of LFG adopted in Alsina (1996) and the Multi-attachment Hypothesis of Relational Grammar (Rosen 1988) (see Aranovich (2000) for a semantic account of the dual properties of reflexives in terms of Dowty's (1991) proto-role theory). Similarly, Mendikoetxea (1997) concludes that CL-REFLs have two apparently contradictory characteristics: (i) they have derived subjects, that is, the grammatical subject of reflexives is an underlying object, as claimed by the suppression approach; and (ii) they have an external argument, as claimed by the pronominal approach. These properties can be accounted for with an analysis in which *se* is generated as the head of an AGR O(bject) head, making ACC unavailable, and PRO is the external argument, in the spirit of Kayne (1991). The internal argument moves to the subject position (either overtly or covertly), where it controls PRO, as represented in (24) for (21c) (Mendikoetxea 1997: 96) (cf. the impersonal construction in (11b), with *se* in AGR S):

(24) $[_{\text{AGRsP}}$ los niños$^{i}]$ … $[[_{\text{AGRoP}}$ $[_{\text{AGRo}}$ *se*$]$ $[_{vP}$ PRO^{i} $[_{\text{VP}}$ $[_{\text{V}'}$ lavaron $[_{\text{DP}}$ t$]]]]]]$

'Externalization' of this DP is obligatory; hence, though a derived subject, it shows properties of an external argument; for example, it cannot be realized as a bare plural like the subject in the unergative structure in (25b) and unlike the unaccusative subject in (25a), as shown in (26):

(25) a. Vienen $[_{\text{NP}}$ mujeres$]$.
come-3PL women
'Women are coming.'

b. *Duermen $[_{\text{NP}}$ mujeres$]$.
sleep-3PL women
'Women are sleeping.'

(26) a. **Se* visten $[_{\text{NP}}$ niños$]$.
se dress-3PL children

b. **Se* abrazaron $[_{\text{NP}}$ hombres$]$.
se hugged-3PL men

More recently, Rodríguez Ramalle (2007) has argued that reflexive constructions are syntactically transitive, but semantically unaccusative, as reflexivization is an LF operation (following Reinhart and Siloni 2004). Current analyses of Romance CL-REFLs, however, deny the unaccusative status of these constructions. Reinhart (1997) and Reinhart and Siloni (2004, 2005) analyze these structures as unergative, resulting from a reduction operation which suppresses the internal argument of a verb (see also Labelle 2008 for French and Otero 1999; Teomiro 2010 for Spanish).

It is now commonly assumed that *se* does not turn a predicate into a reflexive predicate. With *se* not regarded as an anaphor, reflexive interpretation is the result of the relation established between *se* and the subject DP (Torrego 1995) or the result of 'clause-internal' control in structures like (24). For Reinhart (1997), SE/SI is simply a marker of lexical reduction. Labelle (2008), however, considers French *se* directly responsible for reflexive interpretation, though not an anaphor: it introduces a (near-) reflexive function *f* ranging over entities sufficiently close to the external argument to stand proxy to it. As mentioned in Section 2.1 above, there are also proposals that stress the anaphoric status of *se* (see e.g., Rivero 2000, 2002; Dobrovie-Sorin 2006; Teomiro 2010).

4.1.2 Intrinsic reflexivization With intrinsically reflexive verbs like *desmayarse* 'faint' (also referred to as 'pronominal' or 'pseudo-reflexive' in traditional grammar) in (20c), repeated as (27a) below, *se* has commonly been regarded as a marker of unaccusativity, as in unaccusative constructions like (20d), repeated as

(28a) below.[4] The only difference is that unaccusatives have transitive counterparts (28b) (see Section 4.2 below), as opposed to intrinsic reflexives (27b). See Labelle (1992) for French and Maldonado (1999: 113–123) for Spanish on the relation between the verbs in (27) and (28), and Sánchez López (2002: 77) for discussion and references on the characterization of these sentences as 'middle' in traditional grammar.

(27) a. Ana *se* desmayó.
Ana *se* fainted-3SG
'Ana fainted.'
b. *El calor/La noticia/Su madre desmayó a Ana.
the heat/the news/her mother fainted to Ana

(28) a. El cristal *se* rompió.
the glass *se* broke-3SG
'The glass broke.'
b. El calor/La piedra/Juan rompió el cristal.
'The heat/the stone/Juan broke the glass.'

By analogy with (28), intrinsically reflexive predicates have been analyzed as 'conceptually transitive': derived from a transitive lexical entry, in the same way as unaccusative *romperse* in (28a) is derived from transitive *romper* in (28b) (see Labelle 1992), with SE/SI as responsible of, or as a marker of, the suppression of the external-theta role.[5] This is essentially the reduction analysis in Reinhart (1997) and Chierchia (2004), partially adopted by Otero (1999: 1470–1471) for Spanish, who argues that there are two subclasses of intrinsically reflexive verbs: unaccusative and unergative, depending on which of the two arguments of a transitive verb is reduced. Along similar lines, Teomiro (2010) claims that intrinsically reflexive verbs are unergative: they are derived from a lexical operation of reflexivization by which the internal θ-cluster is reduced and bundled with the external θ-cluster and the verb is no longer able to assign accusative Case (as in Reinhart and Siloni 2004, 2005). The unergative–unaccusative controversy simply highlights the difficulty of identifying unaccusativity in Spanish (see Sánchez López 2002 for discussion on this matter regarding intrinsic reflexives). Alternatively, Masullo (1999) following Levin and Rappapport Hovav's (1995) distinction between external causation and internal causation (see Section 4.2 below), argues that these are dyadic internally-caused verbs, which require obligatorily the presence of *se* in Spanish. A similar analysis is given for unaccusative *se* by Mendikoetxea (2000) in the next section. Note that in Massullo (1992), intrinsically reflexive verbs are analyzed as 'antipassive.'

4.2 *Unaccusative "se"*

Unaccusative (or ergative) constructions with SE/SI have been studied mostly within the context of the causative alternation in (28), found in all Romance languages. The occurrence of *se* in constructions like (28a) has commonly been associated with the absence of the external argument and the unavailability of ACC (under Burzio's generalization in Burzio 1981, 1986) (as represented in (29)), as opposed to the transitive structure in (28b):

(29) [$_{\text{IP}}$ __ [$_{\text{CLP}}$ *se* [$_{\text{VP}}$ [$_{\text{V}}$ rompió [$_{\text{DP}}$ el cristal]]]]]

$-\theta$	$+\theta$
[+CASE]	[−CASE]

The analysis in (29) is in line with Perlmutter's (1978) initial formulation of the *Unaccusative Hypothesis* and fits in with influential proposals which establish predictable relations between thematic roles and syntactic positions (Perlmutter and Postal 1984; Baker 1985). It associates structures like (28a) with (periphrastic) passives and passive *se* constructions (in Section 3.1 above), with which they share certain morphological and syntactic properties associated with the lack of an external argument (e.g., *ne*-cliticization, participle agreement, etc. in Italian). The difference between them is that while the 'absorbed' external argument is absent in unaccusative structures like (28a), this argument is 'active' in periphrastic passives and passive *se* constructions. The contrast, which is often illustrated by the (in)compatibility of constructions like (27a) with agentive adverbs and purpose clauses, has been accounted for by analyses in which absorption of the external argument is a syntactic process in passives (Jaeggli 1986a; Baker et al. 1991), but a lexical process in unaccusatives (among others, Zubizarreta 1987; Mendikoetxea 1992, 1999b; but cf. Masullo 1992, for whom unaccusativization in Spanish is a syntactic process).

Because unaccusativization is regarded as a lexical process and the lexicon is the place of idiosyncrasies, standard GB approaches do not offer principled explanations for (a) which transitive verbs can have unaccusative counterparts and (b) which unaccusative verbs do *not* have transitive counterparts (i.e., are not derived from transitive entries). The lexical nature of the process is also responsible for its lack of productivity when compared to passive formation. Current analyses have explored these issues within the context of the lexicon–syntax interface. One of Levin's and Rappaport Hovav's (1995) achievements within this line of research is the realization that unaccusative verbs are not semantically homogeneous but belong to different (sub) classes which share grammatical properties. Thus, all verbs in (30) are unaccusative but they differ crucially in their semantic and syntactic properties:

(30) a. El cristal *se* rompió.
'The glass broke.'
b. El rosal floreció.
'The rose tree blossomed.'
c. Juan vino.
'Juan came.'

The verbs in (30a,b) express change of state, but while (30a) denotes 'external' causation' (like *secar(se)* 'dry,' *agrietar(se)* 'crack,' *quemar(se)* 'burn,' etc.), (30b) denotes 'internal' causation (like *palidecer* 'go white,' *adelgazar* 'get thin,' *crecer* 'grow,' etc.): "some property inherent to the argument of the verb is 'responsible' for bringing about the eventuality" (Levin and Rapapport Hovav 1995: 92). As for (30c), it is a verb expressing existence and appearance (like *existir* 'exist', *aparecer* 'appear,' etc.).

Only verbs like those in (30a) can enter the causative alternation (i.e., can have transitive counterparts) and occur with *se* in Spanish (see Masullo 1992, 1999; Mendikoetxea 1999b, 2000). This is so because unaccusativization is basically 'decausativization.' According to Levin and Rappaport Hovav (1995), unaccusative predicates like (30a) are derived from causative lexical semantic representations (LSR) via a process of existential binding in the lexicon which prevents the projection of the external argument in the argument structure, as shown in (31). Neither internally caused verbs, nor unaccusative verbs of existence and appearance, which are all monadic, are derived from transitive counterparts because their lexical semantic representations are not like that in (31).

(31) Unaccusative *break*

LSR:	[*x* DO-SOMETHING]CAUSE [*y* BECOME *BROKEN*]
lexical binding	0
Argument structure	<y>

Though based on different assumptions regarding the nature of lexical representations, this approach is not radically different from the standard GB approach in that passives and unaccusatives are derived via a similar process, which for unaccusative takes place in the mapping between the lexical–semantic representation and argument structure in (31) and for passives in the mapping between argument structure and the syntax. In both, *se* can only be regarded as a marker of the operation of this process either in the lexicon or the syntax, which cannot account in a principled way for the presence of 'reflexive' morphology in these constructions in Romance as well as in many other languages (see e.g., Kemmer 1988). Under this view, the anticausative variant is derived from the transitive/causative variant. In recent years, the idea that there is a direct derivational relationship between the two variants of the causative alternation has been questioned. Instead, it has been argued that both variants are derived from one source (e.g., a category-neutral verbal root in Alexiadou et al. (2006)) (see Schäfer (2008: ch. 4) for a review of different 'common base' proposals).

An alternative analysis is that which puts together unaccusatives and CL-REFLs as discussed in Section 4.1.1. Both are derived, according to Reinhart (1997) and Chierchia (2004), by a reduction rule, as opposed to passives, which are the result of a saturation rule. For Reinhart (1997), unaccusatives are distinguished from true reflexives in that the latter involve the reduction of the internal argument. For Chierchia (2004), the nature of this distinction is semantic; that is, unaccusatives involve *stative,* rather than *dynamic*, reflexivization: a property of the glass causes it to break, so *el cristal* in (28a) is both (stative) cause and theme. Mendikoetxea (2000) combines the idea that unaccusatives are derived from verbs expressing external causation with the idea that unaccusatives are basically like reflexives. The syntactic properties of these constructions are like those of true reflexives in (24). These are basically dyadic, bieventive constructions whose external argument is realized as PRO, and which involve clause-internal control, but express stative causation. Thus, they do not involve any (reduction) lexical operation, but the presence of PRO as the external argument explains why they share properties with both passives and CL-REFLs. The presence of *se* in the structure is explained in terms of the checking/agreement operations it enters into with other (functional) elements in the clause. The term 'unaccusative' can be applied to these verbs only in so far as the internal argument is the syntactic subject of the construction.

The idea that 'unaccusative' structures with *se* are basically dyadic is also present in Masullo (1999) (and Mateu 2002), for whom the clitic *se* incorporates into an internal aspectual head (where it absorbs ACC) (see also Kempchinsky 2004) and is the expression of the external argument denoting either internal causation or inanimate external causation. Externally caused change of state 'unaccusative' verbs differ from internally caused change of state verbs like *florecer* 'blossom' (30b) and verbs of existence and appearance like *venir* 'come' (30c), which lack *se*. Absence of *se* is related to absence of (external) causation, and this for Masullo (1999) and Mendikoetxea (2000) means simple monoeventive structures. These are true unaccusatives, which exclude the projection of *se* and/or the external argument that *se* is related to.[6]

5 Concluding remarks

It is clear from what has been said in this chapter that most endeavors to characterize the distinct properties of SE/SI constructions involve postulating the existence of different types of SE/SI. However, the issue of whether a unified analysis is possible for all its uses is a recurrent topic in the Romance literature. Postulating one SE/SI would simplify the lexicon, and, thus, the acquisition process. According to Dobrovie-Sorin (2006), a unified analysis of SE/SI in passive, middle, and reflexive contexts is possible under an analysis of this element as anaphoric (see also Rivero 2000 and Teomiro 2010). For other authors, however, a unified analysis involves denying SE/SI an anaphoric status. Raposo and Uriagereka (1994) uniformly analyze Portuguese *se* as an argument: external in arbitrary constructions and internal in reflexives. Mendikoetxea (1997) argues that Spanish *se* is always an AGR-CL (like *me, te,* etc.) and that the different interpretations of *se* constructions are due to the presence of PRO as the external argument, which may be non-controlled (arbitrary) or controlled by the internal argument (reflexive). For Kayne (1988), the unifying property of all uses of SE/SI is that this element absorbs the external theta-role. Manzini (1986) regards Italian *si* as a 'free variable,' which is coindexed with the subject NP.

Though this chapter has concentrated mostly on Spanish, the facts discussed are found in all Romance languages, with differences attributed to the syntactic properties that distinguish between them, perhaps at the microparametric level (see Kayne 2000). Moreover, the use of the same morphology to express reflexive, impersonal, middle, and anticausative meanings is found in other languages of the world, one of the clearest example being Slavic languages (see Rivero 2000, 2002). This uniformity is doubtlessly interesting from a theoretical perspective, but so is the cross-linguistic and intralinguistic variation one finds in the way languages express impersonal and reflexive meaning. Holmberg (2005, 2010) associates the different devices employed by languages to express impersonal meanings with the N(ull) S(ubject) parameter: Strictly non-NS languages, like French and English, make use of overt generic pronouns like *on* and *one*, respectively; partially NS languages, like Finnish, which allows null subjects for 1p and 2p but not for 3p subjects, have null generic pronouns; and consistent NS languages, like Italian and Spanish, make use of overt morphology: SE/SI (among other strategies: generic 'you,' overt quantifiers, etc.) (see also Egerland 2003). This proposal is explored in Mendikoetxea (2008b), where it is argued that the availability of different devices to express genericity or impersonality depends ultimately on the nature of the EPP feature on T.

When we look at the devices employed to express reflexive meanings, the situation is quite similar to that outlined for impersonals. What this situation indicates is that SE/SI is not responsible for the interpretation of these constructions as reflexive, anticausative, or middle, but the surface manifestation of some syntactic process, placing variation possibly at the level of PF, a possibility currently explored in Teomiro (2010). This hypothesis may also account for differences observed at the microparametric level when one looks at verbs with and without the clitic in the different Romance languages in areas of indeterminacy such as inherently reflexive predicates and internally-caused verbs of change of state.

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NOTES

1 The sentence in (3b) could be interpreted as support for the existence of a 3p feature in *se*, in conflict with the 1p subject. Here, it simply illustrates that *se* cannot have independent

reference. Note that in some Slavic languages, the equivalent to the structure in (3b) is possible with a reflexive meaning (= 'I look at myself'): the reflexive clitic remains invariable regardless of the person features of the subject. This phenomenon is also found in North Italian dialects, as shown in (i) (from Kayne 2000: 148). In Spanish, however, 1p and 2p subjects trigger the occurrence of 1p and 2p clitics.

(i) Nun *se* lavom
we *se* wash-1pl
'We wash (ourselves).'

2 As has been shown throughout, CL-ICSs refer to an undefined group of individuals participating in the event. This group may or may not include the speaker. The 'inclusive' reading is particularly prominent in some contexts in Italian, as discussed in detail by D'Alessandro (2004, ch. 5), from whom the following examples are taken. In (ia) the subject is a generic group of humans, which may or may not include the speaker, while in (ib) it is a group which obligatorily includes the speaker (see also Cinque 1988: 3.4).

(i) a. In quell ristorante *si* mangiava bene
in that restaurant si ate-IMPF well
'People ate well in that restaurant.'
b. Ieri *si* è arrivati tardi.
yesterday si is arrived late
'Yesterday we arrived late.'

In Mendikoetxea (2008, n. 17 and s. 6.4), it is argued, following Kayne (2009), that constructions like (ib) involve a silent 1pl pronoun in subject position, but see D'Alessandro (2004, ch. 5) for a different account, based on the (un)boundedness of the event.

3 See Mendikoetxea (2002) on why the existential interpretation is not available in unaccusative and passive contexts, as shown by the ungrammaticality of (i), based on Kański's (1992) distinction between minimal and nonminimal predicates (see also De Miguel 1992, for a different approach).

(i) *Ayer *se* estuvo cansado.
yesterday *se* was tired

4 Two subclasses are distinguished depending on whether the verb has a prepositional object as *arrepentirse* 'regret' in (i) (also *jactarse* 'boast,' *quejarse* 'complain,' *esforzarse* 'make an effort') or not, like *desmayarse* 'faint' in (27a) (also *acatarrarse* 'catch a cold,' *enfadarse* 'get angry,' *rebelarse* 'rebel,' etc.)

(i) Pedro *se* arrepintió de su reacción
Pedro *se* regretted-3SG of his reaction
'Pedro regretted his own reaction.'

5 This analysis often involves the characterization of *se* as a clitic which 'absorbs' ACC. Evidence for this is found in the existence of pairs like (i), with verbs like *lamentar(se)* 'regret,' *olvidar(se)* 'forget,' *incautar(se)* 'apprehend,' *aprovechar(se)* 'take advantege,' etc., which alternate between a transitive variant and an intrinsically reflexive variant, in which the internal argument is obligatorily introduced by the preposition *de* (see Rigau 1990 for Catalan), signaling the unavailability of ACC:

(i) a. Pepe lamentó [$_{DP}$ su decisión]
Pepe regretted-3SG his decision
b. Pepe *se* lamentó [$_{PP}$ de su decisión]
Pepe *se* regretted-3SG of his decision
'Pepe regretted his decision.'

6 There are also apparent cases of the causative alternation without *se*, which have often been taken as lexical idiosyncrasies (see, for instance, Hernanz and Brucart 1997)

(i) a. Juan hirvió la leche.
'John boiled the milk.'
b. La leche hirvió.
'The milk boiled.'

Alternatively, these examples can be analyzed as instances of the causativization of a monadic internally caused verb (see the arguments for this analysis in Mendikoetxea 1999b), which allows us to maintain our account of *se* as an element occurring in the 'unaccusative' variant of the causative alternation.

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24 Coordination and Subordination

RICARDO ETXEPARE

1 Coordination and subordination: basic properties of two clause-linking strategies

The junction of two finite clauses may involve a certain number of restrictions which make one of the joined clauses a less autonomous syntactic object. As a starting point, let us take a simple declarative clause:

(1) No me ha gustado
neg cl has liked
'I didn't like it'

Example (1) can stand by itself, and it is an assertion that contributes a new proposition to the common ground. An assertion requires a context of utterance, characterized by a set of indexical parameters, the so-called Kaplanian parameters. Those indexical parameters include at least a speaker (s), an addressee (h), the utterance time (UT), and an (actual) world-index (w_0). Ascribing the saying represented in (1) to someone else, as in (2), requires a certain degree of manipulation. The clitic pronoun in the embedded clause is not directly anchored to the speaker (s) indexical parameter, as in (1), but to the subject of the verb *say* (for reported discourse in Spanish, see Maldonado 1991, 1999).

(2) Juan dice que no le ha gustado
Juan says that neg cl has liked
'Juan says that he did not like it'

This asymmetry regarding the speaker parameter can be reproduced with other indexical parameters. In what is called the Double Access Reading (Abusch 1997), the temporal location of an embedded present tense depends on both the utterance

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time and the time of the matrix verb. (For a general discussion of this and other matters related to embedded Tense construal in Spanish, see Carrasco 2000, and also Zagona 2004). It may also happen that the finite dependent is evaluated not with respect to the actual world, but with respect to a set of worlds different from it. In (3), from Kempchinsky (1986), the set of worlds that the embedded proposition denotes is the set of worlds compatible with the subject's beliefs. In that case, the embedded proposition shows up in the subjunctive mood, and can be contrasted with an independent assertion, anchored to the actual world:

(3) No pensaba que el bar estuviera cerrado; es más, estaba abierto
Neg I-thought that the bar was-subj closed; in fact, it was open
'I did not think that the bar was closed; in fact, it was open'

A general way of describing the kind of asymmetries illustrated in (1)–(3) is to say that finite dependents introduced by *que* involve relativized deixis (Anderson and Keenan 1985). This type of dependency seems to be associated with a more impoverished syntactic structure. The set of functional projections that can be syntactically activated in dependent clauses is a subset of those operative in main clauses. Contrastive focalization, for instance, is odd in dependents of epistemic verbs:

(4) a. UN LIBRO ha comprado Pedro, no un disco
a book has bought Pedro not a CD
'What Pedro bought is a BOOK, not a CD'
b. ??Juan cree que UN LIBRO ha comprado Pedro (, no un disco)
Juan believes that a book has bought Pedro, not a CD

There is a noticeable asymmetry between the contrastive focalization of objects, as in (4a,b), which involves overt displacement of the focus, and contrastive focalization of subjects, which does not (5a,b):

(5) a. PEDRO ha comprado un libro, no Juan
'Pedro has bought a book, not Juan'
b. Juan cree que PEDRO (no María) ha comprado un libro
'Juan believes that Pedro (not Maria) has bought a book'

The fact that the contrast depends on the actual displacement of the focus operator clearly shows that the reason for the degraded status of (4b) is syntactic, not semantic.

Dependent clauses are also sensitive to the lexical properties of the matrix verb. Volitional predicates like *querer* 'to want' require subjunctive complements:

(6) Pedro quiere que vayas/*vas a casa
Pedro wants that you-go-subj/indic prep home
'Pedro wants you to go home'

This is a fundamentally asymmetric relation: no 'reverse mood selection' is attested. Lexical selection can also involve the form of the complementizer. Interrogative verbs select for *si*, instead of *que*:

(7) Juan pregunta si/*que vendrás
Juan asks whether/that you-come-fut
'Juan asks whether you will come'

Also, a subclass of verbs in Spanish, basically verbs of volition and semifactives, seem to be able to select for a null complementizer (Torrego 1983):

(8) Recuerdo Ø bebimos un vino excelente
I-remember we-drank a wine excellent
'I remember we drank an excellent wine'

Finally, dependent *que*-clauses can behave as canonical arguments of the predicates that select them (cf. 2), and as arguments of lexical prepositions, as in (9):

(9) Piensa [$_{\text{PrepP}}$ en [$_{\text{CP}}$ que va a morir]]
he-thinks in that he-will prep die
'He thinks about the fact that he is going to die'

The following properties can be seen to delimit a certain type of relation that two clauses may entertain together. In this relation: (a) one of the clauses is unlike an independent clause in that at least a subset of its indexical parameters are anchored to the matrix clause, not the speech context; (b) it is structurally impoverished, lacking functional structure present in main clauses; (c) it is selected by an argument taking element of the other clause; (d) it is introduced by dedicated functional structure that marks its dependent status on another clause; (e) it shows special paradigmatic forms as a result of lexical selection; and (f) it can be substituted by other grammatical formatives that are known to occupy argument positions in a clause. Let us informally call this relation 'subordination.'

Some of the properties that characterize subordination extend to adverbial clauses. Consider in this regard (10) (Rigau 1995; García 2000):

(10) Al llegar yo, María se fue
Pre + D arrive-inf I, Maria cl left
'When I arrived, Maria left'

If we concentrate on the relation between the preposition *a* and the following clause, we see that some of the properties of subordination also apply in this case. First, the following clause (*el llegar yo*) can not be an independent clause. As such, it is structurally impoverished. Compared to independent infinitives, for instance, the adverbial structure does not support a preverbal subject, as in (11) (note that this is subject to dialectal variation: see Pöll 2007):

(11) a. Al (*yo) llegar tarde
Prep + D I arrive-inf late
b. ¡Yo llegar tarde!
I arrive-inf late
'Me arrive late!'

Finally, we can say that the temporal clause is introduced by a prepositional head that selects a particular paradigmatic form, in this case an infinitival. The preposition *a* may thus be seen as an equivalent of the formative *que* in the domain of finite dependents. There is one important way, however, in which an adverbial clause like (11a) cannot be equated with what we informally called subordinate clauses, and this is precisely with respect to its relation with the matrix clause. There is no selection relation between any element of the matrix clause and the temporal clause, and no particular syntactic restriction arising from the lexical nature of the matrix predicate. This is understandable if adverbial clauses modify the matrix clause. Adverbial clauses impose different constraints on the temporal preposition and its first argument, on the one hand, and the prepositional phrase and the matrix clause on the other. A head-complement or first merge (Chomsky 2000) relation relates the preposition and its nominal argument in the former case:

(12) … $[_{\text{PrepP}}$ a $[_{\text{DP}}$ -l $[_{\text{InfP}}$ llegar yo]]

For the second type of relation, the traditional notion of adjunction or pair-merge seems more appropriate:

(13) $[_{\text{TP}}$ $[_{\text{PrepP}}$ a $[_{\text{DP}}$ el $[_{\text{InfP}}$ llegar yo]]] $[_{\text{TP}}$ Maria se fue]]

Coordinative conjunctions such as *y* 'and' and *o* 'or,' stand in a particular relation to the lexical head with which they combine: the conjoined terms must each be capable to satisfy the selectional restrictions of that head, thus giving the impression that the conjunction itself is transparent to outside selection. That is, in something like (14), each of the conjuncts (a participle, a prepositional, and an adjectival phrase) can (and in fact, must) separately satisfy the selectional restrictions of the locative copula *estar*:

(14) Pedro está arruinado, sin esperanza y harto de todo
Pedro is ruined without hope and fed up of everything
'Pedro is ruined, without hope and fed up with everything'

At the same time, the conjunction itself does not seem to impose any particular selectional restriction on the kind of elements it can conjoin, as far as syntactic category is concerned. An illustrative sample is provided in (15):

(15) a. Son Juan y María [DP and DP]
They-are Juan and María

b. Trabajo en casa y en la oficina [PP and PP]
I-work at home and at the office
c. Estoy muy cansado y bastante aburrido [AdjP and AdjP]
I am very tired and quite bored
d. Dice que viene y que le esperes [CP and CP]
he-says that he-comes and that cl you wait

It thus seems that coordination does not obey some of the fundamental properties that we used to characterize subordination.

Coordinative conjunctions, on the other hand, seem to impose constraints of a more global type. Thus, a parallelism constraint applies to the different conjuncts for something like semantic likeness. Under this view, (16a) would be ungrammatical, despite the well formedness of (16b,c), because the conjuncts receive a different theta-role (Camacho 2003: 5):

(16) a. *Juan y el cuchillo cortaron el pan
'Juan and the knife cut the bread'
b. Juan cortó el pan
'Juan cut the bread'
c. El cuchillo cortó el pan
'The knife cut the bread'

This semantic likeness constraint is apparent also in the interpretation of temporal phrases. Temporal/aspectual adverbs with scope over both conjuncts require temporal/aspectual parallelism (Camacho 2003: 13):

(17) a. Este jugador siempre [[$_{TP}$ se cae] y [$_{TP}$ pierde la pelota]]
this player always cl falls and loses the ball
'This player always falls and loses the ball'
b. *Este jugador siempre [[$_{TP}$ se cae] y [$_{TP}$ va a perder la pelota]]
this player always cl falls and is-going to lose the ball

Assuming that *siempre* marks the edge of the TP, (17a,b) show that TPs cannot be conjoined unless their temporal features are identical. The symmetric properties of the conjuncts and the apparent invisibility of conjunction heads to selection seem to set apart conjunctive phrases from the domain of subordination as stated here. We will come back to the apparent exceptional status of coordination in Section 3.[1]

2 Subordination and mood

In embedded clauses, unlike in main ones, paradigmatic mood alternations are restricted to the subjunctive and the indicative. The received view is that indicative mood is the unmarked mood in Spanish. Subjunctive mood requires special licensing.

2.1 Intensional and polarity subjunctives

If we focus on the relation that the subjunctive Mood entertains with its licensing elements, we can make a distinction between two different types: one based on lexical selection relations, and another based on the sensitivity of subjunctive mood in the domain of certain syntactic operators. Quer (1998), following Stowell (1994), calls those two different types of subjunctive Intensional and Polarity subjunctives. The two types of subjunctive can be described according to the following contrasting properties.

2.1.1 Sequence of tense Intensional subjunctives show constraints on possible sequences of Tenses (see Picallo 1985; Suner and Padilla 1987). For instance, Spanish does not admit sequences of the form PRESENT [PAST], as shown in the contrasts below:

(18) a. Deseo que hayas acabado la tesis entonces
I-wish that you-have-subjPres finished the dissertation then
'I wish you have finished the dissertation then'
b. *Deseo que hubieras acabado la tesis entonces
I-wish that you-had-subjPast finished the dissertation then
'I wish you had finished the dissertation then'

Polarity subjunctives are licensed by a syntactic operator like negation or question. One such case is illustrated in (19). As seen below, in this case the subjunctive dependent does not show any temporal restriction:

(19) a. No recuerdo que trabaje [PRESENT[PRESENT]]
neg I-recall that she-works-subjPres
'I don't recall that she is working'
b. No recuerdo que trabajara [PRESENT[PAST]]
neg I-recall that she-works-subjPast
'I don't recall that she was working'

2.1.2 Alternation with indicative Intensional subjunctives do not alternate with indicative. Subjunctives licensed by syntactic operators, such as yes–no interrogative features, alternate with the indicative mood:

(20) Quiero que vengas/*vienes
I-want that you-come-subj/ind
'I want you to come'

(21) a. ¿Recuerdas que trabajara?
You-remember that he worked-subj
'Do you remember if he worked?'

b. ¿Recuerdas que trabajaba?
You-remember that he-worked
'Do you remember that he worked?'

2.1.3 Locality Subjunctives licensed by a lexical predicate do not extend to further embedded CPs. They are licensed only in the clause immediately embedded by the higher predicate (22). Polarity subjunctives on the other hand, can appear in consecutively embedded complements, independently of the choice of the selecting predicate (23).

(22) Quiero que crean que nos gusta/*guste
I-want that they believe-subj that cl like-ind/subj
'I want them to believe that we like it'

(23) No creo que piense que nos guste
neg I-believe that he-thinks-subj that cl like-subj
'I don't believe that he thinks that we like it'

The subjunctive in the second embedded clause is not licensed if the first one is not marked subjunctive too:

(24) No creo que piensa que nos gusta/*guste
neg I-believe-ind that he-thinks-ind that cl like-ind/subj
'I don't believe that he thinks that we like it'

2.1.4 Obviation The last criterion distinguishing the two subjunctive types has to do with disjoint reference effects between the embedded and the matrix subject. Intensional subjunctives tend to display obviation effects (25a). Those effects do not arise in the case of polarity subjunctives (25b).[2]

(25) a. *pro_i quiero que pro_i la invite
I-want that cl I-invite-her
'I want to invite her'
b. pro_i no creo que pro_i la invite
neg I-believe that cl I-invite
'I don't think I will invite her'

2.2 *Semantic factors in mood selection*

2.2.1 Indicative and subjunctive The prototypical selectors of Intensional Subjunctive fall into one of the following predicate classes (Quer 1998: 43): (1) directives, such as *ordenar* 'to order,' *pedir* 'to ask for something'; (2) modals, such as *puede que* 'maybe'; (3) semimodals, like *necesitar* 'to need'; and (4) volitionals, like *querer* 'to want' or *preferir* 'to prefer.' The determining factor in the relation between the lexical head and the subjunctive clause is semantic selection. As noted by Bosque (1990: 19), the subjunctive complement can be selected by heads of different categories, as shown in (26a,b). What those different categories have in common is a semantic feature encoding volition:

(26) a. Deseando que vuelva
wishing that she-comes-back
'Wishing that she comes back'
b. El deseo/Deseoso de que vuelva
The desire/desirous of that she-comes-back
'The desire that she comes back/Desirous that she-comes back'

Much of the cross-linguistic work in romance subjunctives has as its base the idea that the truth of the complements of verbs like *querer* 'to want' and *creer* 'to believe' are evaluated in a different way. The former are *intensionally anchored*, in Farkas' terms (1992), whereas the latter are *extensionally anchored*. Thus, the proposition expressed by the complement clause (27a) must be true in the world that models reality according to Maria, while in (27b) it must be true in the set of worlds that Maria takes to be the future alternatives of her actual world (Quer 1998: 45):

(27) a. María cree que los turistas de su grupo son pesados
Maria believes that the tourists of her group are-ind boring
'Maria believes that the tourists in her group are boring'
b. María quiere que los turistas de su grupo sean menos pesados
Maria wants that the tourists of her group are-subj less boring
'Maria wants the tourists in her group to be less boring'

As illustrated in (27), the divide between intensional and extensional anchoring correlates with the major division between subjunctive taking and indicative taking predicates. Extensionally anchored complements surface in the indicative mood, whereas intensionally anchored complements appear in the subjunctive mood.

2.2.2 Causative and factive predicates Causative predicates require subjunctive complements (Quer 1998: 47):

(28) Juan hizo que volviera/*volví
Juan made that I-came-subj/ind
'Juan made me come back'

Subjunctive complements of causative predicates show all the properties of complements of intensional predicates, yet the predicates are not intensional (they do not involve a modal base). Most of them, furthermore, are implicative: the truth of their complement is presupposed on the part of the speaker.

Factive-emotive predicates constitute another problematic case. The propositional complement of these predicates is interpreted factively, that is, in the evaluation model of the speaker, not the subject. This class involves verbs such as *lamentar* 'to regret' and psych verbs like *agradar* 'to please' or *preocupar* 'to worry.' Quer notes that the psychological verbs in this class are only factive under an episodic predicate (29a), not otherwise (29b):

(29) a. Me agradó que me preguntaran, #pero no lo hicieron
Cl pleased that cl they-ask-past-subj but they neg cl they-do-past-ind
'It pleased me that they asked questions, #but they didn't do it'
b. Me agrada que me pregunten, pero no lo hicieron
cl pleases that cl they-ask-pres-subj but neg cl do-past-ind
'It pleases me that they ask, but they didn't do it'

Quer observes that in the nonepisodic cases, factive-emotive predicates can occur with a conditional complement, so-called nonlogical *if* (see Quer 2002):

(30) Me preocupa si los estudiantes no hacen preguntas
Cl worries if the students neg they-make questions
'It worries me if the students do not ask questions'

Quer (1998: 93–103) tries to assimilate the nonfactive members of the class to causative predicates. Causation is linked to counterfactuality: to say that x causes y is to say that if x had not happened, y would have not taken place (Lewis 1973). The presence of the conditional *if*-clause signals the presence of an implicit modal operator. The occurrence of the subjunctive in the clausal argument of psych predicates would then follow from the implicit modality that is tied to the causative component in those verbs.

2.2.3 Double mood selection A different problematic case for the semantic determination of intensional subjunctives is the existence of verbs that allow for double mood selection. These are typically verbs of communication:

(31) a. Dice que te vas
He-says that cl you-leave-ind
'He/she says that you are leaving'
b. Dice que te vayas
he-says that cl you-leave-subj
'He/she says that you must leave'

In (31b), the verb *decir* takes a subjunctive complement. In that case, the predicate is interpreted as directive. The subjunctive dependent in (31b) displays all the properties of intensional subjunctives. In cases such as (31b), the presence of subjunctive mood does not automatically follow from the lexical properties of the verb. Kempchinsky (1986, 2009) develops an analysis that associates subjunctive mood with the presence of interpretable modal features in the left periphery of the embedded clause. Concretely, she argues that intensional subjunctives are characterized by the presence of a quasi-imperative operator in Comp. The relation between the imperative mood and subjunctive in Spanish is manifest in several contexts, as in 3rd person subject imperatives (32):

(32) Que vengan
That they-come-subj
'Let them come'

2.2.4 Polarity licensing operators and the subjunctive/indicative alternation

Unlike intensional subjunctives, polarity subjunctives alternate with the indicative mood. The alternation has semantic effects in the anchoring of the embedded proposition. Consider the following minimal pair (adapted from Quer 2001: 91):

(33) a. El jurado no cree que sea inocente
the jury neg believes that he-is-subj innocent
b. El jurado no cree que es inocente
the jury neg believes that he-is-ind innocent
'The jury does not believe that he/she is innocent'

Whereas in (33a) the embedded proposition is evaluated in the belief world of the jury, in (33b) it is evaluated in the model of the speaker. In other words, the proposition denoted by the complement is presupposed. A similar semantic effect can be found in relative clauses under intensional predicates (from Rivero 1971; see also Gutiérrez-Rexach 2003):

(34) a. Luisa espera a un reportero que la entrevistará
Luisa awaits a journalist that cl interview-fut
b. Luisa espera a un reportero que la entreviste
Luisa awaits a journalist that cl interview-subj
'Luisa awaits a journalist that will interview her'

Example (34a) favors a specific interpretation of the indefinite *un reportero*. Specific indefinites require accommodation of the common ground to entail the existence of an individual with those characteristics (see Condoravdi 1994).

3 Infinitive dependents

3.1 Temporal properties and selection

Infinitivals, together with gerunds and participles, lack person and tense morphology. Unlike gerunds and participles, which possess an unambiguous aspectual value, the temporal contribution of infinitivals seems to depend on the nature of the element that selects them. The status of infinitivals as temporally indeterminate entities partly explains their selectional properties. They combine easily with predicates that determine a clear temporal orientation for the embedded clause. This is the case with directive or volitional predicates that temporally orient their dependent to the future. With predicates like *decir* 'say,' which do not constrain the temporal value of their dependent, their presence is excluded (Hernanz 1999):

(35) *Juan dijo ir a la fiesta
Juan said go-inf to the party

However, this is not the case if the infinitival itself is able to specify its temporal relation vis-à-vis the matrix event:

(36) Juan dijo haber ido a la fiesta
Juan said have-inf gone to the party
'John said that he went to the party'

3.2 Control infinitivals

Infinitives may show different degrees of clausal integration with the matrix clause. We can take this to mean that they project a different number of functional projections (Wurmbrandt 2001). Infinitivals may show overt elements related to CP, as complementizers and wh-pronouns:

(37) a. No sabemos si ir
neg we-know if go-inf
'We don't know whether to go'
b. No nos dijeron dónde ir
neg cl they-tell where go-inf
'They did not tell us where to go'

The subject is implicit in the infinitival clause when the latter is selected by a matrix predicate:[3]

(38) Juan quiere [ir (*Pedro)]
Juan wants go-inf Pedro
'Juan wants Pedro to go'

Example (38) exemplifies what Landau (2001) calls Exhaustive Control, where the denotation of the implicit infinitival subject is coextensive with the denotation of the controller. It is a matter of debate whether in this case the embedded infinitival projects a subject or not, and if so, whether it should be analyzed as being syntactically independent from the controller in the matrix clause (see Hornstein 2001).

Easier to establish is the implicit subject in so-called Partial Control cases (Landau 2001), where the denotation of the implicit subject cannot be fully recovered from the antecedent:

(39) Sugiero [_ ir juntos al partido]
I-suggest go-inf together to the match
'I suggest going together to the match'

In (39), the adverbial *juntos* requires a plural antecedent, which can only be provided by the implicit subject in the infinitival clause. Infinitival complements that are optionally introduced by a preposition show an interesting contrast in this regard (40). Only the prepositional infinitival allows partial control, suggesting that partial control cases involve a more complex syntactic structure. The two structures also present a subtle difference in meaning. The bare one, unlike the prepositional one, implies the subject's will to accomplish the action expressed by the embedded clause.

(40) a. *Pienso ir juntos al partido
I-think go-inf together to the match
'I am thinking of going (*together) to the match'
b. Pienso en ir juntos al partido
I-think in go-inf together to the match
'I am thinking about going together to the match'

3.3 *Raising constructions*

Spanish has a handful of subject-raising verbs, one of which is *parecer* 'seem':

(41) (*Juan) parece que ha venido María
Juan seems that has come Maria
'*Juan/it seems that Maria has arrived'

Given that *parecer* does not license a subject in (41), the overt subject in (42) must have raised from the embedded nonfinite complement:

(42) Juan parece [_ estar contento]
Juan seems be-inf happy
'Juan seems to be in a happy state'

Parecer must be distinguished in this regard from *parecer* plus a dative argument. In this case, the meaning of *parecer* resembles that of a psychological predicate, and controls an implicit argument in the embedded clause (see Torrego 1996b; Ausín 2001; Gallego 2010 for discussion). Note that *parecer* plus dative, unlike bare *parecer*, does not license a metereological predicate in its infinitival complement. Those predicates possess impersonal subjects that cannot be controlled by a higher argument:

(43) (*Me) parecía [_ nevar mucho]
Cl seemed snow-inf much
'It seemed (to me) to snow a lot'

Spanish also has instances of raising to object, out of complements of verbs of perception (44a) and causative verbs (44b).

(44) a. Los niños nunca han visto reír a María
the children never have seen laugh prep Maria
'The children never saw Maria laugh'
b. Hicimos bailar a Julia
we-made dance-inf prep Julia
'We made Julia dance'

Conceptually, the dative arguments in (44a,b) represent the agents of both dancing and laughing. Syntactically, however, they behave as objects of the matrix verb. They are pronominalized by an accusative clitic:

(45) La he visto reír/La hice bailar
cl have-I seen laugh-inf/cl I-made dance-inf
'I saw her laugh/I made her dance'

The embedded origin of the object is clear for perception verbs, which may include an impersonal argument that cannot be the direct object of *ver* (Hernanz 1999):

(46) Los niños nunca han visto [_ nevar]
the children never have seen snow-inf

A contrast observed by Treviño (1994: 74–75) between *hacer* causatives and control causatives makes the same point for causatives:

(47) a. El presidente se obligó a comer pescado
the president refl forced to eat fish
'The president forced himself to eat fish'
b. *El presidente se hizo comer pescado
the president refl made eat fish
'The president made himself to eat fish'

Whereas *obligar* in (47a) is a transitive verb allowing the object to adopt a reflexive form under identity of reference with the subject, the causative *hacer* does not allow it. The contrast suggests that the matrix subject and the dative marked argument in the causative construction are not coarguments, and therefore cannot give rise to a reflexive configuration. In other words, the dative DP in *hacer*-causatives is an argument of the embedded infinitival. For causatives, we refer the reader to Bordelois (1974), Treviño (1994), and Torrego (1998, 2010), and references therein; for perception verbs, see De Tullio (1998); pseudorelatives of the sort *Vi a María que cantaba* are studied by Campos (1994) and Rafel (2000).

4 The status of the finite complementizer

4.1 *Nominal properties of* que-*clauses*

The syntactic distribution of *que*-clauses suggests that dependents headed by *que* possess a nominal feature. Three types of evidence converge on this hypothesis: first, there is the fact that *que*-clauses seem to occupy ordinary subject and object positions (see Section 1). Second, *que*-clauses can be immediately preceded by the definite determiner (48):[4]

(48) El que no hayas llamado me resulta inconcebible
Det that neg has-subj called cl results inconceivable
'I find inconceivable that you did not call'

Finally, *que*-clauses combine with a small subset of the available prepositions. The prepositions that can precede a *que*-clause are *a* 'to,' *de* 'of,' *en* 'in,' *con* 'with,' and *por* 'for,' which alternates with *para* (final preposition) with some verbs (Demonte 1994; Delbecque and Lamiroy 1999; Barra-Jover 2002). Demonte (1994) suggests (focusing mainly on *a*, *de*, *en*, and *con*) that the prepositions have an aspectual value, typically involving nondelimited aspect. Take, for instance, (49):

(49) Pienso (en) que no vendrá
I-think in that neg she-come-fut
'I (frequently) think that she will not come'

Whereas in the absence of a preposition, the predicate *pensar* only has a stative interpretation, the addition of the preposition results in a gerundive or frequentative reading.

The nominal status of *que*-clauses also surfaces in complex nominal constructions. Since nouns are not Case assigning categories, the presence of a preposition *de* is necessary to license the CP (Plann 1986), which must therefore be an entity requiring case:

(50) La idea *(de) que no volveremos me deprime
the idea of that neg we-come-back-fut cl depresses
'The idea that we will not come back saddens me'

Against the idea that *que*-clauses possess a nominal feature stands the fact that *que* clauses do not trigger agreement in either gender, number, or person (see Picallo 2002 for discussion):

(51) Que la tierra es redonda y que gira alrededor del sol ha(*n) sido probado (*s)
That the earth is round and that it-moves around the sun have been proved

4.2 *Queísmo and dequeísmo*

A common dialectal variant of embedded *que*-clauses involves the addition of the preposition *de*. This is the so-called *dequeísmo* (see Gómez Torrego 1999):

(52) Pienso de que los conozco poco
I-think prep that cl I-know little
'I think I know them little'

Demonte and Soriano (2005) show that *de* in that case is not a genuine preposition. One clear piece of evidence in this regard is the fact that the *deque* clause can be resumed by a clitic pronoun, which can never pronominalize a PP:

(53) [Lo_i primero que le diré] es [de que no tengamos $miedo]_i$
the first that cl I-tell-fut is prep that neg we-have-subj fear
'The first thing I'll tell her is not to fear anything'

Deque sequences can also occupy a subject position, which otherwise does not admit PPs:

(54) De que todas esas niñas se vayan a enfermar es mucho más dificil
prep that all those girls cl go-subj to fall-sick is much more difficult
'It is much more difficult that all those girls end up being sick'

The presence of *de* in those cases does not signal the existence of an underlying null noun, as in complex noun phrase constructions of 'the fact that' type. For instance, *deque* structures cannot occur as dependents of factive verbs, a natural context for such a null noun to be licensed. Also, they do not admit a preceding determiner, as one could conceivably expect in the case of an elided noun. Demonte and Soriano take *de* to spell out features related to epistemic modality (see also Schwenter 1999).

A phenomenon opposite to *dequeísmo* is the absence of a preposition in contexts where the CP would seem to require one, so called *queísmo*. *Queísmo*, as the term indicates, is strictly associated with finite *que*-complements. The preposition cannot be absent before DPs or infinitivals. *Queísmo* is very frequent with pronominal verbs like *acordarse* 'to remember':

(55) Me acuerdo (de) que teníamos seis años
cl remember prep that we-had six years
'I remember we were six years old'

The distribution of *queísmo* and *dequeísmo* differs in some important regards. *Dequeísmo* only affects the occurrence of the preposition *de*. *Queísmo* results from the absence of a wider set of prepositions, such as *a*, *con*, *por*, *en*, *de*. This set of prepositions is identical to the one involved in the case marking of CPs. This may be related to the fact that the majority of the contexts where *queísmo* arises can be reasonably analyzed as allowing appositional clauses (see Leonetti 1993).

4.3 *Predicative and modifying uses of* que-*clauses*

Que-clauses may occur in contexts other than typical argumental ones (Etxepare 2008):

(56) a. Ricardo hablaba que no callaba
Ricardo talked that neg he shut up
'Ricardo talked in such a way that he wouldn't shut up'
b. Juan llegó a la meta que no se tenía en pie
Juan got into the line that neg cl hold on
'Juan got into the line in such a state that he could not hold on'

The CPs in (56) modify different parts of the underlying event configuration: with unaccusatives, they modify a resultant state (56b); with unergatives, they must modify the process part (56a). CPs can also function as attributive modifiers (Demonte and Masullo 1999):

(57) Tu padre está que salta de contento
your father is that jumps of joy
'Your father is in such a state that he is jumping for joy'

Alvarez (1999), summarizing the received view on attributive constructions, suggests that they are consecutive clauses from which an antecedent term of degree has been omitted:

(58) Juan llegó a la meta (tan cansado) que …
Juan got into the line so tired that

This solution clearly overgenerates. As noted by Alarcos (1994), for instance, bare *que*-clauses cannot combine with the copula *ser* 'be.' Compare in this regard (59a,b):

(59) a. *Pedro es que llama la atención
Pedro is that he-calls the attention
'(intended meaning) Pedro is such that he draws other people's attention'
b. Pedro está que llama la atención
Pedro is-loc that he-calls the attention
'Pedro is in such a state that he draws other people's attention'

The full consecutive clause on the other hand, is possible with *ser*:

(60) Pedro es tan guapo que llama la atención
Pedro is so handsome that he-calls the attention
'Pedro is so handsome that he draws everyone's attention'

The distinction between *ser* and *estar* in Spanish has been defined in aspectual terms (Schmitt 1996; Arche 2006): only *estar* contributes an internal temporal dimension. This suggests that bare CPs, unlike consecutive clauses, directly modify elements of the aspectual configuration of the clause. Further indirect evidence in this direction is provided by patterns of grammaticalization across different Spanish varieties. Thus, in Peruvian Spanish bare CPs have taken over some of the functions of the gerund (see Arrizabalaga 2010):

(61) Está que llueve
It-is that it-rains
'It is raining'

4.4 *Speech act dependents*

Verbs of communication like *decir* 'say' can take dependents that have interrogative force (Plann 1982; Uriagereka 1988; Brucart 1993; Suñer 1993):

(62) Juan dice que qué quieres
Juan says that who is-coming
'Juan says: what do you want?'

Plann (1982) notes that only those verbs of communication which can take a direct question quote can receive a true interrogative interpretation in their embedded clause. This correlation can be extended to sentence moods other than the interrogative. Thus, one may have true embedded exclamatives (63), and as shown by Rivero (1994), one can also have true commands (64):

(63) Juan exclamó que qué hermoso era el paisaje
Juan exclaimed that how beautiful was the view
'Juan exclaimed: how beautiful is the view!'

(64) El jefe ordenó que a trabajar
the boss ordered that prep work-inf
'The boss ordered: work'

Lahiri (2002) proposes that Spanish has a special type of complementizer *que*$_{quotative}$ whose function is to take an expression of the type of questions, commands, exclamatives, and declaratives, and yield utterances of that value. The embedded complements are not to be confused with independent speech acts, though: the indexical parameters of the embedded clause are anchored to the matrix clause, not the speech context.

4.5 *Nonselected* que-*clauses*

A particular feature of Spanish is that *que* can occur in main clauses, as a marker of reported speech. Consider the following contrast (Etxepare 2010):

(65) a. Oye, el Barça ha ganado la Champions
hey the Barça has won the Champions
b. Oye, que el Barça ha ganado la Champions
hey that the Barça has won the Champions

Example (65a) is an ordinary assertion while example (65b) contributes the additional meaning that someone else, who is not the speaker, has said (65b).

Other instances of nonselected *que* seem to introduce adverbial clauses, typically with a causal function. These represent so-called conjunctive *que* (Alarcos 1994), and are always sentence-final:

(66) Me voy, que hace frío
cl I-leave that it-is cold
'I'm leaving since it is cold'

The *que*-clause can provide the grounds for the speech act, in cases like (67a,b):

(67) a. ¿Quién ha venido, que he oído voces?
Who has come that I-have heard voices
'Who arrived? (I tell you) because I heard voices'
b. ¡Eh, que me estás pisando!
Hey that cl you-are stepping on
'Hey, you stepped on me!'

In (67a), the grounds of the question are provided by the *que*-clause. In (67b), the *que*-clause gives the reason why the speaker drew the addressee's attention by uttering *hey*.

5 Coordination

The symmetric properties of coordination, and in particular, of copulative coordination, have traditionally inspired syntactic formalizations that set apart conjunction phrases from other ordinary phrases. Chomsky (1981) introduced the following phrase structure rule for conjunction:

(68) XP $\rightarrow$ XP Conj XP

The rule in (68) directly captures the intuition that a conjunction of DPs behaves as a DP, and that a conjunction of PPs behaves as a PP. This basically makes conjunction structures a multiheaded configuration, with the conjunct a spell out of the juxtaposition of two terms (see Progovac 1998).

But a simple paratactic approach to conjunction will not do for many cases. Consider for instance (69). Bare nouns cannot occupy subject positions in Spanish (69a), but conjoined ones can (69b), showing that the conjunction must be present at the level of the derivation where thematic relations hold (from Bosque 1996).

(69) a. *Hijo permaneció poco tiempo de visita
son stayed little time visiting
b. Madre e hijo permanecieron poco tiempo de visita
mother and son stayed little time visiting

The underlying presence of the conjunction also surfaces in its ability to define local domains for purely syntactic relations. Coordinative conjunctions block case assignment by an external head, as shown below (Goodall 1987: 47–48):

(70) a. Para mí b. Para [él y yo] c. Para [mí y él]
for me for him and me
d. *Para [yo y él] e. *Para [él y mí]

The preposition *para* 'for' assigns oblique case in Spanish. When we conjoin two pronouns under *para*, the second conjunct, which follows the copulative head, cannot surface in the oblique form, but must do so in the unmarked, nominative form. The asymmetry between the first and the second conjunct suggests that the conjoined terms are not parallel. In its ability to block case assignment, *y* behaves as a syntactic head. The head status of the copulative conjunction is supported by typological evidence: Stassen (2001) shows that postposed copulative conjunctions of the type *X Y-and* tend to correlate strongly with OV order, whereas VO order correlates with preposed coordination of the type *X and Y*. In other words, conjunctions seem to follow the general ordering trend of other heads in the language.

It would seem that the conjunct head has a closer relation with the second conjunct than with the first one. This observation is supported by those cases in which the conjunction phrase lacks a first conjunct, as in extraposition configurations or in discourse oriented conjunction:

(71) a. Vino Juan ayer, y María y Pedro también
came Juan yesterday, and Maria and Pedro too
b. ¿Y a dónde vas?
And prep where you-go
'And where are you going?'

The structural representation of (copulative) coordination must thus include a partial representation where the conjunction and its second term form a phrasal unit (Munn 1993; Johannessen 1998):

(72) $[_{\text{Conjunction Phrase}}$ y [XP]]

The structure of (72) conforms to the general picture emerging since the late 1980s and the 1990s that takes coordinate structures to instantiate ordinary asymmetric structures (see Munn 1993; Johannessen 1998; Camacho 2003).

5.1 *Asymmetries in (copulative) coordination*

5.1.1 Agreement and clausal coordination Conjoined subjects give rise to the following agreement asymmetries in pre- and postverbal positions:

(73) a. Vino/vinieron un director y su ayudante
came-sing/pl a boss and his aid
b. Un director y su ayudante vinieron/*vino el lunes
a director and his aide came-pl/sing Monday
'A director and his aide came last Monday'

Example (73a), with singular agreement, is reminiscent of first conjunct agreement. First conjunct agreement is a well-attested phenomenon in many languages of the world (Johannessen 1998). A common analysis involves agreement with the structurally closer conjunct, in an asymmetric representation of coordination where the first conjunct occupies the specifier position of a coordination phrase (van Koppen 2005):

(74) Aux …$[_{\text{Conjunction Phrase}}$ XP [and [YP]]]
_____________________/

As noted by Vicente (2010), however, Spanish does not behave like other languages that display genuine first conjunct agreement. In such languages, a first conjunct agreement effect can be observed in cases where a plural reading of the subject is enforced (e.g., by modification of the subject by *together*). Take for instance Arabic (from Vicente 2010):

(75) 3a?a-t Hind-un wa ¿amr-un ma?an
came-3s Hind-nom and Amr-nom together
'Hind and Amir came together'

Spanish, on the other hand, invariably requires full conjunct agreement in these environments:

(76) Vinieron/*vino un director y su ayudante juntos
came-pl/sing a director and his aid together

A more plausible approach involves conjunction reduction (clausal coordination plus ellipsis in the second conjunct). In that case, agreement is between each of the DPs and its verb, and it is necessarily singular:

(77) Vino un director y [~~vino~~ su ayudante]

If (77) underlies the singular agreement in (73a), as Vicente (2010) notes, we may also have an explanation of why singular agreement is impossible in (75b): the Backward Anaphora Constraint (Langacker 1969) prohibits backward ellipsis within coordinate structures.

5.1.2 Asymmetric coordination It has been observed that in many cases, coordination does not seem to involve symmetric relations. Consider for instance cases such as (78) (from Camacho 1999: 2643; see also Culicover and Jackendoff 1997):

(78) Haz el mínimo gesto y verás
do the slightest gesture and you'll see

In (78), the first conjunct clearly has a conditional meaning. Syntactically too, it behaves as if it were a conditional (79a). Imperatives do not license polarity items such as *el mínimo NP* (79b).

(79) a. Si haces el mínimo gesto, verás
if you make the slightest gesture, you'll see
b. *Haz el mínimo gesto
do the slightest gesture

To the extent that conditional phrases behave as syntactic adjuncts, we may want to admit the following configuration as a possible representation for coordinate phrases, where the first conjunct is adjoined to the Conjunction Phrase:

(80) $[_{\text{Conjunction Phrase}}$ [XP] $[_{\text{Conjunction Phrase}}$ y YP]]

5.1.3 Feature resolution Other agreement asymmetries arising in conjunction do not arguably have a syntactic basis. This is possibly the case of feature resolution involving person agreement features. Conjunctions of third person and first or second person always trigger first or second person agreement (Camacho 2003: 82–88), regardless of the order of the pronouns:

(81) a. Juan y yo hemos/*han ido juntos
Juan and I have-we/have-they gone together
'Juan and I went together'
b. Juan y tú habeis/*han ido juntos
Juan and you have-you/have-they gone together
'You and Juan went together'

The auxiliary shows plural number agreement, corresponding to the sum of the conjuncts, and first or second person agreement, corresponding to one of the conjuncts. First person has prominence over second:

(82) Tú y yo hemos ido juntos
you and I have-we gone together
'You and I went together'

5.2 *Plurality, distributivity, and conjunction*

Conjunctive and disjunctive coordination give rise to both distributive and collective interpretations of the sum denoted by the conjunction phrase. Spanish simple disjunctions linked by *o* 'or' do not make a difference between exclusive and inclusive interpretations. However, as observed by Jiménez Juliá (1986), *o* can appear more than once. In that case, the interpretation is obligatorily exclusive:

(83) a. Juan o Marta traerán un regalo, y posiblemente ambos
Juan or Marta bring-fut a present, and possibly both of them
'Juan or Marta will bring a present, and possibly they both will'
b. O Juan o Marta traerán un regalo, # y posiblemente ambos
Either Juan or Marta bring-fut a present, and possibly both of them

The iteration of the conjunction head also forces a distributive reading in the case of *y* 'and' (from Camacho 1999: 2669):

(84) a. Pedro, Ana y Juana pelaron, lavaron y cortaron las manzanas
Pedro, Ana and Juana peeled, cleaned and cut the apples
b. Pedro, Ana y Juana, pelaron y lavaron y cortaron las manzanas
Pedro, Ana and Juana peeled and cleaned and cut the apples

In the case of (84a), a cumulative reading in which, say, Juana peeled, cleaned, and cut the apples, and Pedro and Ana only cut them, is possible. The iterated conjunct in (84b) on the other hand, forces a reading in which each of the agents participated in each of the tasks.

Negative coordination can be simple (*ni*) or iterated (*ni* … *ni*) (see Bosque 1994). The latter is distributive, as shown by the fact that it is incompatible with collective predicates (Camacho 1999: 2681):

(85) *Ni mi hermano ni mi sobrina se encontraron en el metro
Neither my brother nor my niece refl met in the metro
'Neither my brother nor my niece met in the metro'

The simple conjunction is impossible in contexts enforcing obligatory distributivity, as the floating pronoun structures studied by Sánchez López (1995):

(86) Los niños no cantan *(ni) ellos boleros ni ellas tangos
the children neg sing neither them-masc boleros nor them-fem tangos

At the opposite pole, obligatorily collective readings arise when the conjunction phrase is preceded by a preposition like *entre* 'between, among' (Rigau 1990):

(87) Entre tú y yo terminaremos la tarea
Between you and me will-finish the task
'We will finish the task between you and me'

As noted by Camacho (1999: 2668), the collective reading must be of the cumulative type. *Entre* … *y* phrases cannot combine with events which require equal participation of the participants:

(88) *Entre Esti y Livia se encontraron a las cuatro
between Esti and Livia refl found et four

Many varieties of Iberian and South American Spanish also have comitative conjunctions, which force a collective reading (Camacho 2000: 366):

(89) Con Juan vamos al cine
With Juan we-go to-the movies
'Juan and I we go to the movies'

Comitative conjunction is also bad with noncumulative collective readings, such as (90):

(90) *Con Juan nos encontramos a las cuatro
With Juan refl we-met at four
'Juan and I met at four'

5.3 *Adversative coordination*

Consider the following case of adversative coordination:

(91) Juan es joven, pero experto
Juan is young, but knowledgeable

In principle, a sentence like (91) admits two different possible analyses: one involving clausal coordination and ellipsis (signaled by strikethrough in (92a)), and another one involving subclausal coordination (92b):

(92) a. [$_{\text{Coordination Phrase}}$ [$_{\text{IP}}$ Juan es joven] pero [$_{\text{IP}}$ ~~Juan es~~ experto]]
b. Juan es [$_{\text{Coordination Phrase}}$ [$_{\text{AdjP}}$ joven] pero [$_{\text{AdjP}}$ experto]]

Some cases of adversative coordination only admit the first analysis since there is no subclausal term that may be coordinated:

(93) a. El joven apareció, pero muerto
the young appeared, but dead
'The young man appeared, but appeared dead'
b. [$_{\text{Coordination Phrase}}$ [$_{\text{IP}}$ El joven apareció] pero [$_{\text{IP}}$ ~~el joven apareció~~ muerto]]

According to Vicente (2010) adversative coordination can be divided in at least two different types, that he calls *corrective* and *counterexpectational* (see Bosque 1980; Horn 1989; Lakoff 1968, for the latter). Corrective adversative coordination results in the denial of the first conjunct, and it shows the special contrastive nexus *sino*:

(94) Juan no vino tarde, sino pronto
Juan neg came late, but early
'Juan didn't come late, but early'

Example (96) illustrates counterexpectational adversatives, which do not involve the denial of the first conjunct, but imply that the second conjunct is a somewhat unexpected continuation. Vicente (2010) claims that, unlike counterexpectational adversatives, corrective ones necessarily correspond to conjunction reduction. In corrective coordinations, a first conjunct agreement effect necessarily arises:

(95) No se presentó/*presentaron un pianista, sino tres trombonistas
not SE showed-up.sing/pl a pianist, but three trombone players
'A pianist did not show up, but three trombonists did'

This is unlike counterexpectational *pero*, which readily allows plural agreement with both conjuncts:

(96) Van a participar en la operación un único neurocirujano pero tres cardiólogos
go-3pl to participate in the operation a single neurosurgeon, but three cardiologists

NOTES

1 The special character of coordinate structures has also been underlined by the existence of grammatical constraints that seem to make specific reference to the parallel status of the conjuncts. One such constraint is Ross's Coordinate Structure Constraint, which requires Across the Board (ATB) extraction of wh-words. Another one is gapping, a form of ellipsis, which seems to apply to the second conjunct in copulative coordination. See Chapter 27 on ellipsis and Chapter 25 on wh-movement.

2 As stressed by Kempchinsky (1990, 2009), obviation affects the matrix and the embedded subjects. Obviation does not arise between the embedded subject and an object in the matrix clause (Kempchinsky 2009: 1791):
(i) Animé a Elisa$_i$ a [que pro$_i$ estudiara en el extranjero]
I encouraged Elisa that she studied-subj abroad
'I encouraged Elisa to study abroad'

3 I exclude floating pronouns of the sort discussed by Piera (1987), as well as floating quantifiers (see Torrego 1996a). Some adverbial infinitival clauses license overt subjects (see Fernández Lagunilla 1987).

4 For arguments that the structure does not involve an underlying null nominal, of the sort of *el hecho de que* 'the fact that,' see Leonetti (1993) and Picallo (2002). *El* is restricted to subject clauses and to complements of factive verbs.

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25 Wh-movement: Interrogatives, Exclamatives, and Relatives

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1 Introduction

Movement is a metaphor for a syntactic configuration in which a syntactic object surfaces in a position where it is not ultimately interpreted. There are a number of movement-related constructions that have generated a large base of descriptive and theoretical research on language in general, and Spanish in particular: left and right dislocation (Contreras 1976; Rivero 1980; Villalba 2000; López 2003), scrambling phenomena (Ordóñez 1998), focus movement (Zubizarreta 1998), etc. In this chapter I will focus on one type – Wh-movement – and provide a descriptive overview capturing the lexical, syntactic, and semantic variation across the syntax of Spanish. I will also review theoretical approaches addressing previous and ongoing research in the area.

The study of Wh-movement has primarily focused on Interrogatives (1) in the theoretical literature, where Spanish has offered key comparative linguistic data.

(1) ¿Qué tienes?
 'What do you have?'

However, in this chapter I will expand our exploration to include a discussion of Wh-movement in its broader sense, including Exclamatives (2) and Relative Clauses (3), in order to provide a more comprehensive look at the continuities and discontinuities that characterize Wh-movement in a range of syntactic contexts.

(2) ¡Qué inteligente es Pedro!
 'How intelligente Pedro is!'

The Handbook of Hispanic Linguistics, First Edition. Edited by José Ignacio Hualde, Antxon Olarrea, and Erin O'Rourke.

(3) Esos libros que me compraste ayer, los voy a vender.
'Those books that you bought me yesterday, I'm going to sell them.'

To begin this survey and in order to highlight Wh-movement and related syntactic phenomena, it will be important to reign in the empirical focus. There are various types of interrogatives, exclamatives, and relatives, each with their similarities and differences. As a class, interrogatives (4) function to query some unknown information of some type, exclamatives (5) provide an emotive evaluation of some presupposed information, and relatives relay truth-value information (6).

(4) a. ¿Tienes las entradas para el concierto? *Yes/No question*
'Do you have the tickets for the concert?'
b. ¿Qué tienes? *Wh-interrogative*
'What do you have?'

(5) a. ¡La de gente que vino! *Nominative exclamative*
'The (amount) of people that came!'
b. ¡Qué inteligente es Pedro! *Wh-exclamative*
'How intelligent Pedro is!'

(6) a. Esos libros, que no me gustan, los voy a vender. *Appositive Relative*
'Those books, which I don't like, I'm going to sell them.'
b. Esos libros que me compraste ayer, los voy a vender. *Restrictive Relative*
'Those books that you bought me yesterday, I'm going to sell them.'

However, each of these classes varies internally in syntactic and semantic terms not relevant to Wh-movement. Therefore, the current discussion will only concern Wh-interrogatives (4b), Wh-exclamatives (5b), and Restrictive Relative Clauses (6b), given they share three particular lexical, syntactic, and semantic properties characteristic of Wh-movement: (1) the use of a common set of Wh-words; (2) the obligatory 'fronting' of these Wh-words to a clause-initial position; and (3) a strict relationship between the fronted Wh-word and its interpreted, or base, position.

As the data in (7–9) show, the set of Wh-words employed overlaps between each of these structures, but is not completely shared.

Wh-interrogatives

(7) a. ¿Qué libro tienes en la mano?
'What book do you have in your hand?'
b. ¿A quiénes viste en la fiesta?
'Who did you see at the party?'
c. ¿Cuántos libros has comprado?
'How many books have you bought?'

Wh-exclamatives

(8) a. ¡Qué cosas dice tu hermano!
'What things your brother says!'
b. ¡Cómo son de exagerados!
'How dramatic you (all) are!'
c. ¡Cuántos libros has comprado!
'You've bought quite a number of books!'

Restrictive relatives

(9) a. Esos libros que compré ayer no valen para nada.
'Those books (that) I bought yesterday are worthless.'
b. La mujer a quien le diste el sobre
'The woman who you gave the letter to'
c. El cuchillo con el cual cortamos el pastel
'The knife with which we cut the cake'

Interrogatives demonstrate the widest variety of Wh-words and include: *qué* 'what,' *cuál(es)* 'which,' *cuánto/a(s)* 'how much/many' *cómo* 'how,' *cuándo* 'when,' *quién(es)* 'who,' *dónde* 'where,' *por qué* 'why,' and *por qué* 'for what'; relatives follow with the second largest set: *que* 'that'/'which,' cual(es) 'which,' *cuanto/a(s)* 'how many,' *como* 'how,' *donde* 'where,' *cuando* 'when,' *quien(es)* 'who,' *porque* 'because,' and *cuyo* 'whose'; and exclamatives appear with the most restricted set: *qué* 'what'/'how), *cuánto* 'how much,' *cuánto/a(s)* 'how much/many,' *cómo* 'how.' Wh-words, named such given the common wh-string associated with the comparable English set of interrogative pronouns 'what,' 'where,' 'why,' and 'who' are inextricably linked to the lexical function they perform and to the semantic import and pragmatic force of the utterance in which they appear, and thus are restricted accordingly.

Fronting is also a fundamental feature of Wh-movement, as is illustrated in (7–9). As opposed to Wh-interrogatives, Wh-exclamatives (10a) and relatives (10b) disallow the Wh-phrase from remaining in its base position.

(10) a. *¡Dice tu hermano qué cosas!
'Your brother says what things!'
b. *El cuchillo cortamos el pastel con el cual
'The knife we cut the cake with which'

Simple interrogatives allow a Wh-pronoun to appear in-situ, but its interpretation is semantically and pragmatically marked (11) as distinct from the fronted version (12).

(11) a. ¿Tienes qué? *Echo question*
'You have what?'

b. # Las entradas para el concierto.
'The concert tickets.'

(12) a. ¿Qué tienes? *Wh-interrogative*
'What do you have?'
b. Las entradas para el concierto.
'The concert tickets.'

Finally, the Wh-pronoun and its base position, often referred to as a 'trace' in Generative accounts, share a strict coreference relationship in which the Wh-pronoun acts as an operator binding the trace position, or variable.

(13) a. ¿Qué libro$_i$ tienes $_{ti}$ en la mano?
b. ¡Cuántos libros$_i$ has comprado $_{ti}$!
c. El cuchillo con el cual$_i$ cortamos el pastel $_{ti}$.

At this point, it can be seen how the metaphor 'movement' has come to be used to describe this particular structural configuration. As seen in (14), a Wh-word cannot co-occur with an overt element in its interpreted position, suggesting that the Wh-pronoun is not a new syntactic object that has been inserted into the construction, but rather it is the pronominal form of the trace which has 'moved' from its base position to a clause-initial position.

(14) a. ¿Qué libro$_i$ tienes (*el libro azul)$_i$ en la mano?
'What book do you have (*the blue book) in your hand?'
b. ¡Cuántos libros$_i$ has comprado (*cuatro libros)$_i$!
'You've bought quite a number of books (*four books)!'
c. El cuchillo con el cual$_i$ cortamos el pastel (*con el cuchillo)$_i$.
'The knife with which we cut the cake (*with the knife)'

In the following sections, I provide a more thorough review of the descriptive characteristics of Wh-movement that interrogatives (Section 2), exclamatives (Section 3), and relatives (Section 4) share, as well as key aspects where they diverge. These data will serve to complement (Section 5), where I turn to more nuanced data leveraged in the formal and applied literature to provide theoretical accounts for Wh-movement.

2 Interrogatives

Wh-interrogatives are also known as constituent questions. This title aptly makes reference to the fact that Wh-words in Wh-interrogatives can query information from a range of constituent types (NP, VP, AP) and functions (Subject, Object, and Adverbial and Adjectival modifiers), giving rise to a host of interrogative pronouns.

(15) a. ¿Quién$_{\text{subject NP}}$ quiere ir al parque hoy?
'Who wants to go to the park today?'
b. ¿Qué$_{\text{object VP}}$ quiere Juan hoy?
'What does John want today?'
c. ¿A dónde$_{\text{object NP}}$ quiere ir Juan hoy?
'Where does John want to go today?'
d. ¿Cuándo$_{\text{modifier AP}}$ quiere ir al parque Juan?
'When does John want to go to the park?'

As noted earlier, Wh-movement involves a dependency relationship between an overt Wh-pronoun and a nonovert antecedent in base position. In Spanish, if the antecedent is selected by a verb which requires a prepositional marker or selected directly by a preposition, that prepositional element must appear as part of the fronted Wh-phrase.

(16) a. ¿[*(Con) quién] quieres ir?
'Who do you want to go with?'
b. ¿[*(A) cuál] libro te refieres?
'Which book are you referring to?'

Some Wh-pronouns have the capability to absorb the prepositional marker given the lexical–semantics of the Wh-word, but only when the Wh-pronoun receives Oblique case (i.e., is not part of the verbal subcategory).

(17) a. ¿Dónde comiste el bocadillo?
'Where did you eat the sandwich?'
b. Comí el bocadillo [$_{\text{oblique}}$ en mi cuarto].
'I ate the sandwich in my room.'

(18) a. ¿*(A) dónde irás después de escribir el capítulo?
'Where will you go after writing the chapter?'
b. Iré [a Disneylandia].
'I will go to Disneyland.'

Although there is a high level of freedom in word ordering in declaratives (see Chapter 28), in Wh-interrogatives it is typically observed that fronting of the Wh-phrase triggers a reordering of overt subjects and the verb, known as subject-verb inversion, or inversion, such that the verb precedes the subject (Rivero 1980; Torrego 1984; Contreras 1989; Goodall 1993; Baković 1998).

(19) a. ¿Qué come María en la mañana?
'What does Mary eat in the morning?'
b. *¿Qué María come en la mañana?

Notice that inversion is not a direct property of interrogatives as preverbal subjects are allowed in yes/no questions (20a) and echo questions (20b), where no fronting occurs.

(20) a. ¿María come manzanas en la mañana?
'Does Mary eat apples in the morning?'
b. ¿María come qué en la mañana?
'Mary eats what in the morning?'

Not only are subjects barred from appearing in the domain between the Wh-phrase and the verb, so are objects (21a) and many adverbial expressions (21b).

(21) a. *¿Qué a Juan le dio María?
'What to John did Mary give?'
b. *¿Qué a tiempo le dio María a Juan?
'What on time did Mary give John?'

However, these elements are free to appear in absolute clause-initial position before Wh-phrase as Topic elements (Contreras 1976, Rivero 1978).

(22) a. María, ¿qué le dio a Juan a tiempo?
'Mary, what did (she) give John on time?'
b. A Juan, ¿qué le dio María?
'To John, what did Mary give (him)?'
c. A tiempo, ¿qué le dio María a Juan?
'On time, what did Mary give John?'

Indirect questions in subordinate clauses also follow the matrix inversion pattern, a particular property of Spanish not found in many languages where subject–verb inversion is active in matrix clauses (Emonds 1976). Embedded interrogatives must be lexically selected by verbs such as *preguntar* 'to ask,' *saber* 'to know/find out,' *decir* 'to say/tell,' etc.

(23) a. Quiero saber qué tiene de malo ese bar.
'I want to know what's so bad about that bar.'
b. *Quiero saber qué ese bar tiene de malo.

Despite the robust expression of subject–verb inversion in matrix and subordinate clauses, it is not obligatory in all Wh-movement in interrogatives. There are three main cases in which subjects intervene between Wh-words and the verb. The first concerns the thematic role of the Wh-pronoun. Wh-pronouns that are not selected by the verb; that is, adjuncts such as *por qué* 'why,' *cuándo* 'when,' *cómo* 'how,' and *en qué medida* 'in what way' allow subjects to intervene between the Wh-phrase and the verb (Torrego 1984; Suñer 1994; Goodall 1993).

(24) a. *¿A quién Juan odia?
'Who does John hate?'
b. ¿Por qué Juan odia a Luis?
'Why does John hate Luis?'

To highlight this contrast, when the Wh-word *por qué* 'for what' is an argument of the verb, it triggers obligatory inversion (Contreras 1989).

(25) a. *¿Por qué Juan votó?
'For what did John vote?'
b. Juan votó por paz.
'John voted for peace.'

The second case in which subject-verb inversion is not obligatory concerns the 'complexity' of the Wh-phrase. Complex Wh-phrases which make more specified discourse reference also tend to allow preverbal subjects (Goodall 2004; Ordóñez and Olarrea 2006)

(26) a. *¿A quién María conoció en Paris?
'Who did Mary meet in Paris?'
b. ¿A cuál de estas chicas María conoció en Paris?
'Which of these girls did Mary meet in Paris?'

The final case deals with dialect variation and does not apply to all varieties in Spanish. The fact that varieties of Caribbean Spanish allow preverbal subjects in Wh-interrogatives even in cases where the Wh-word is an argument and is not complex, as in (27), has attracted much attention in the descriptive and theoretical literature (Lipski 1977; Toribio 2000).

(27) ¿Qué tú sabes?
'What do you know?'

It appears, however, a matter of some debate to what extent subjects of all types are allowed preverbally in Wh-interrogatives in Caribbean dialects as a whole. The literature suggests that some particular Caribbean varieties only allow pronominal elements, as in (27) (Ordóñez and Olarrea 2006), where others allow full determiner phrases (DP), as in (28) (Suñer 1994).

(28) ¿Qué Juan sabe?
'What does John know?'

Extraction from subordinate clauses is also a key aspect of Wh-movement in interrogatives. As we have seen in this section, Spanish Wh- dependency relationships in interrogatives can be established in matrix and in subordinate contexts. It is also the case that Wh- dependencies can be established between matrix and subordinate clauses as well.

(29) ¿Qué$_i$ quiere Juan [que le compre $_{ti}$]?
'What does John want he/she to buy him/her?'

Notice that a long-distance dependence can apply in multiple embedded contexts, yet subject–verb inversion only applies to the matrix subject and verb (Contreras 1989).

(30) ¿A quién$_i$ dice María [que Juan no quiere [que su hijo no conozca $_{ti}$]]?
'Who does Mary say that John doesn't want that his/her son doesn't meet?'

However, there are structural configurations that restrict Wh-movement in Spanish, as for a number of languages shown in work in the late 1960s and early 1970s (Chomsky 1973). Often referred to as Syntactic Islands (Ross 1967) in the theoretical literature, Spanish shows sensitivity to many (31) but not all of these Islands (32) (Perlmutter 1971; Suñer 1991).

(31) a. *¿A quién$_i$ habló José con Irma [después de ver $_{ti}$]? *Adjunct Island*
'Who did Joseph speak with Irma after seeing?'
b. *¿Cuántos$_i$ compró María [$_{ti}$ libros]? *Left branch Condition*
'How many did Mary buy books?'
c. *¿Qué$_i$ tocas [el piano y $_{ti}$]? *Coordinate Constraint*
'What do you play the piano and?'
d. *¿Qué$_i$ se pregunta Juan dónde María fue a comprar $_{ti}$? *Wh- Island*
'What does John wonder where Mary went to buy?'
e. *¿Qué$_i$ defendió Juan [la propuesta de que se venda $_{ti}$]? *Complex NP Constraint*
'What did John defend the proposal that be sold?'
f. *¿De qué$_i$ sabe Juan que [una botella $_{ti}$] se cayó de la mesa? *Subject Constraint*
'What does John know that a bottle of fell off the table?'

(32) a. ¿A quién$_i$ se pregunta Juan [si María quiere $_{ti}$]? *Whether Islands*
'Who does John wonder whether Mary loves?'
b. ¿Quién$_i$ cree Juan [que $_{ti}$ votó por Clinton]? *That-trace Effect*
'Who does John believe that voted for Clinton?

3 Exclamatives

Turning to Wh-exclamatives, we see that they demonstrate many grammatical characteristics similar to those of Wh-interrogatives. Most obviously, Wh-exclamatives share a set of Wh-words with Wh-interrogatives, albeit a smaller set. As in the case of Wh-interrogatives, these Wh-words provide evidence that they too undergo movement and are associated with an antecedent trace in base position.

(33) a. ¡Qué$_i$ cosas dice $_{ti}$ tu hermano !
'What things your brother says!'
b. ¡Cuánto$_i$ te quiero $_{ti}$!
'I love you so much!'
c. ¡Cuántos libros$_i$ hay que leer $_{ti}$!
'What a bunch of books there is to read!'

One notable difference in the Wh-word inventory is the fact that the Wh-pronoun *qué* 'what' is more productive in Wh-exclamatives than in Wh-interrogatives (Alonso-Cortés 1999).

(34) a. ¡Qué increíble$_{ADJ}$!
'How incredible!'
b. ¡Qué bien$_{ADV}$ que lo viste!
'How great that you saw it!'
c. ¡Qué en forma$_{PP}$ estás!
'How in shape you are!'
d. ¡Qué de gasolina$_{PP}$ come ese carro!
'This car uses so much gasoline!'

The more productive behavior of *qué* and general restriction on the Wh-word inventory in exclamatives, however, is semantically conditioned: Wh-exclamatives convey an emotive, evaluative response that exceeds expectation to a presupposed proposition (Gutiérrez-Rexach 1996, 2008). Wh-exclamatives only occur with elements compatible with a degree or scalar interpretation (35a) and not with non-degree based elements (35b) or categorical elements (35c).

(35) a. ¡Qué extraordinariamente feo es Pedro!
'How extraordinarily ugly Pedro is!'
b. *¡Qué prácticamente feo es Pedro!
'How practically ugly Pedro is!'
c. *¡Qué soltero está Pedro!
'How single (not married) Pedro is!'

The modifier *tan* 'so' is optional in exclamatives, but obligatory in degree or scalar Wh-interrogatives. Given that exclamatives by their nature provide an evaluative reading, optionality is motivated (36a). Yet interrogatives do not inherently inquire into the evaluative status of a proposition, and therefore require overt marking, as seen in (36b).

(36) a. ¡Qué (tan) inteligente es tu amigo!
'How intelligent your friend is!'
b. ¿Qué *(tan) inteligente es tu amigo?
'How intelligent is your friend?'

In contrast with Wh-interrogatives which request unknown information, Wh-exclamatives refer to presupposed information. This is highlighted by the observation that Wh-exclamatives are selected by factive predicates in embedded contexts (37a) (Elliott 1974; Grimshaw 1979) (37), that negation is standardly blocked in exclamatives due to the conflict of denying a fact that one supposes to be true (38), and that exclamatives are sensitive to specificity; the Wh-pronoun must be related to a specified, presupposed antecedent (39).

(37) a. Me parece horroroso qué torpes son los políticos. (exclamative reading)
'I think it's horrible how clumsy politicians are.'
b. Me pregunto qué torpes son los políticos. (interrogative reading)
'I wonder how clumsy politicians are.'

(38) a. ¡Qué barbaridades cometería alguien así!
'What atrocities would someone like that commit!'
b. *¡Qué barbaridades no cometería nadie así!
'What atrocities wouldn't anyone like that commit!'
(Villalba 2004)

(39) a. ¡Cuánto cuesta el vino!
'The wine is so expensive!'
b. *¡Cuánto cuesta un vino!
'A wine is so expensive!'
(Villalba 2008)

Exclamatives also show similar fronting and inversion patterns to interrogatives. Indeed, the surface word order of exclamatives can be indistinguishable from Wh-interrogatives (40) (Bosque 1984). Furthermore, word order patterns are limited in similar ways in for Wh-movement in Wh-exclamatives as was described for Wh-interrogatives (41–42) (i.e., subject–verb inversion is active in matrix and embedded clauses).

(40) a. ¿Cuántos idiomas hablas?
'How many languages do you speak?'
b. ¡Cuántos idiomas hablas!
'You speak quite a number of languages!'

(41) a. ¡Qué inteligente es tu amigo!
'How intelligent you friend is!'
b. *¡Qué inteligente tu amigo es!

(42) a. ¡Me olvido qué inteligente es tu amigo!
'I forget how intelligente you friend is!'
b. *¡Me olvido qué inteligente tu amigo es!

Dislocated elements can also appear in clause-initial position before Wh-pronouns, as in Wh-interrogatives (43). And, much like Wh-interrogatives, subject–verb inversion is optional with Wh-adverbial expressions such as *cómo* 'how' (44), and where complex Wh-phrases appear with preverbal subjects (45).

(43) a. María, ¡qué alta (que) es!
'Mary, how tall she (Mary) is!'
b. María, ¿qué quiere?
'Mary, what does she (Mary) want?'

(44) a. ¡Mira cómo reluce el cuchillo!
'Look how the knife shines!'
b. ¡Mira cómo el cuchillo reluce!

(45) ¡Qué libros más difíciles Juan nos asignó leer!
'What difficult books John assigned us to read!'

One salient difference between Wh-exclamatives and Wh-interrogatives is the fact that exclamative Wh-phrases are strongly clause-bound and cannot be extracted outside of the originating clause where an antecedent trace is found (Villalba 2008).

(46) a. ¡Qué forrado estás!
'How loaded you are!'
b. *¡Qué forrado dice Juan que estás!
'How loaded John says you are!'

(47) a. ¿Cuántos libros dice Juan que tiene la biblioteca?
'How many books does John say that the library has?'
b. *¡Cuántos libros dice Juan que tiene la biblioteca!
'There are so many books that John says that the library has!'

Another particular feature of Wh-exclamatives concerns optional elements in matrix clauses. Exclamatives allow relativization (48a) and the elision of copular verbs (49a) (*ser* 'to be,' *estar* 'to be,' *hay* 'there is/are,' and *parecer* 'to seem') in matrix clauses but not in indirect exclamatives, as in (48b) and (49b) (Alonso-Cortés 1999).

(48) a. ¡Cuánto (que) te quiere!
'He/she loves you so much!'
b. Me impresiona cuánto (*que) te quiere.
'It impresses me how much she loves you!'

(49) a. ¡Qué difícil (es) la jardinería!
'How difficult gardening is!'
b. Me asombra qué difícil *(es) la jardinería.
'It amazes me how difficult gardening is.'

4 Relatives

A relative clause is a subordinate clause headed by a pronoun, adjective, or adverbial relative element. Relatives are included in the present discussion as they show a similar Wh-word inventory, fronting of Wh-words to the head of the clause (subordinate in this case), and an operator/variable relationship between a Wh-pronoun and a trace position, as illustrated in (50).

(50) a. El hombre [que$_i$ te debe $_{ti}$ dinero] está aquí.
'The man that owes you is here.'
b. Ese es el libro [del cual$_i$ te hablé $_{ti}$].
'This is the book that I talked to you about.'
c. Los miembros del comité [con quienes$_i$ tienes que hablar $_{ti}$] se fueron.
'The members of the committee with whom you have to speak left.'

A key difference between the Wh-movement in Wh-interrogatives and Wh-exclamatives is that there are three elements to be coindexed: an antecedent, a relative pronoun, and a trace in the antecedent's base thematic position. Furthermore, antecedents in relatives are often overt (51a) given the declarative, descriptive nature the grammatical construction plays, but can also be nonovert (51b) (Plann 1980).

(51) a. No saben la hora$_{\text{antecedent } i}$ [$_{RC}$ cuando$_i$ van a partir $_{ti}$.]
'They don't know the hour (time) when they are going to embark.'
b. Juan es $_{\text{non-overt antecedent } i}$ [$_{RC}$ el que$_i$ repara las televisiones.]
'John is the one who repairs televisions.'

When the relative pronoun is the object of a preposition (oblique), in many cases the definite article is combined with the pronoun. This article agrees in number and gender with the antecedent.

(52) Es la persona a la que le mandé la carta de recomendación.
'He/she is the person to whom I sent the letter of recommendation.'

(53) Es el candidato por el que voté en las elecciones pasadas.
'He is the candidate for whom I voted in the past elections.'

Word ordering also behaves distinctly in relatives. On the one hand, the subject position is more flexible than in Wh-interrogatives and Wh-exclamatives in that relatives allow preverbal (54a) and postverbal (54b) subjects much in the same way as in simple declarative sentences.

(54) a. Me compré el libro que María quería.
'I bought the book that Mary wanted.'
b. Me compré el libro que quería María.
'I bought the book that Mary wanted.'

On the other hand, relatives show a more strict linear ordering. Antecedents must precede the relative pronoun in left-to-right linear order.

(55) a. La persona a la cual le di el regalo no ha llegado todavía.
'The person to which I gave the gift hasn't arrived yet.'
b. *A la cual le di el regalo la persona no ha llegado todavía.
'To which I gave the gift the person hasn't arrived yet.'

However, antecedents need not be adjacent, as is clear if we consider a preposition and/or article interveners; but, more interestingly, full constituents can intervene in Heavy NP shift contexts (Larson 1988).

(56) a. Le entregué una [lista$_{\text{antecedent}}$ [que$_{\text{RC}}$ contenía los nombres de todos los profesores]] a María.
'I provided a list that contained all the professors' names to Mary.'
b. Le entregué una [lista$_{\text{antecedent]}}$ a María [que$_{\text{RC}}$ contenía los nombres de todos los profesores.]]
'I provided Mary a list that contained all the professors' names.'

As in other Wh-movement constructions, Topical elements can appear. Note that in contrast to Topicalization in Wh-interrogatives and Wh-exclamatives, the preposed topical phrase appears after the relative pronoun (57a), and is ungrammatical before the relative pronoun (57b) (Arregi 1998).

(57) a. La habitación en la que, a la hora del asesinato, estaba Juan ya no se usa.
'The room in which at the time of the assassination John was, is no longer used.'
b. *La habitación a la hora del asesinato, en la que estaba Juan ya no se usa.
'The room at the time of the assassination, in which John was, is no longer used.'

Long-distance dependencies are also allowed in relative clauses, as in Wh-interrogatives. The antecedent can be extracted out of the clause where it is thematically selected, as in (58a). However, as the distance between antecedent and the trace increases, interpretability decreases (58b).

(58) a. Ojalá que me regale la pulsera$_i$ [que sabe que me gusta$_{ti}$].

'I hope (that) he/she gives me the bracelet (that) he/she knows (that) I like.'

b. Ojalá que me regale la pulsera$_i$ [que creo [que dijeron [que sabe [que me gusta $_{ti}$]]]].
'I hope (that) he/she gives me the bracelet that I think they said that he/she knows that I like.'

5 Accounts

Turning to the theoretical discussions concerning Wh-movement, there are three main questions driving research: (1) where do Wh-words appear in the clause structure?; (2) what formal properties do matrix and embedded complementizer phrases share?; and (3) what is the nature of the relationship between Wh-operators and antecedent trace positions across clause boundaries?

5.1 Landing site of Wh-phrases and inversion patterns

Many accounts for the position of Wh-words in Spanish Wh-movement have focused on Wh-interrogatives. Early accounts responded to analyses proposed for English, and other so-called V2 languages, in which subject–verb (auxiliary) inversion is active.

(59) a. Who is John?
b. *Who John is?

The Wh-criterion of Rizzi (1996) capitalizes on the apparent adjacency requirement between the verb and Wh-phrases to account for the restriction of intervening syntactic objects in matrix clauses. Rizzi (1996) proposes that Wh-words raise to the Specifier position of the [+ wh]-marked Complementizer Phrase (CP) ([Spec, CP]) and the verb moves to the head of this phrase (C′) to license Wh-movement.

However, a straight-forward analysis of this type for Spanish is complicated by two pieces of evidence. First, the Wh-criterion only targets matrix clauses capturing the matrix/embedded asymmetry in languages like English (60). Yet Spanish displays obligatory inversion patterns in matrix and embedded clauses (61).

(60) a. Mary wonders who John is.
b. *Mary wonders who is John.

(61) a. *¿María se pregunta quién Juan es?
b. ¿María se pregunta quién es Juan?
'Mary wonders who John is?'

Second, Spanish allows intervening elements between the Wh-phrase and the verb depending on the nature of the Wh-phrase. These particular cases include non-thematic Wh-phrases (62a) and complex Wh-phrases (62b).

(62) a. ¿Cómo (que) Juan no quiere ir al parque?
'Why Juan doesn't want to go to the park?'
b. ¿Cuál de estos libros Juan devolvió a la biblioteca?
'Which of these books did John return to the library?

Furthermore, in Caribbean varieties of Spanish, non-inversion is allowed much more robustly (63), as intervening elements may appear regardless of the status of the Wh-phrase.

(63) ¿Qué tú sabes?
'What do you know?'

Following the assumption that Wh-phrases surface in [Spec, CP], accounts for Spanish inversion patterns take two primary forms, on the one hand focusing on the nature of the Wh-phrase and/or verb's relationship to the Wh-phrase, and on the other hand emphasizing the nature of the intervener phrase. Considering the Wh-phrase, Contreras (1989) argues that when the Wh-phrase is nonthematic, it serves as a sentential operator and does not bind a variable in base position. As a base-generated Wh-phrase, syntactic elements are allowed to surface preverbally. Suñer (1994), on the other hand, suggests that the (non)inversion in (non)thematic Wh-phrases is primarily based on the verb. Her approach includes a refinement of the Wh-criterion in which two processes are delimited: one that holds for all Wh-phrases and a second that is required for thematic arguments in which the verb requires strict locality with the Wh-operator.

A key advantage to Suñer's approach is that it can be used to leverage a principled account for non-inversion patterns in Caribbean Spanish. Suñer proposes that strict locality restrictions on Wh-phrases are language-specific and are the cross-linguistically marked case. From this angle, she suggests that standard varieties of Spanish (and perhaps English) display the more marked Wh-movement condition and Caribbean Spanish only applies the more general condition.

Despite the apparent gains from this insight, two issues remain: one concerns the empirical evidence from Caribbean Spanish. In Suñer's approach, cases of non-inversion include all subject types, pronominal (64a) and full DPs (64b and 64c).

(64) a. ¿Quién tú eres?
'Who you are?'
b. ¿Qué Juan dijo de eso?
'What John did say about that?'
c. Yo no sé qué la muchacha quería.
'I don't know what the girl wanted.'

This appears to be the case in Puerto Rican Spanish, the focus of her investigation – yet reports suggest other Caribbean varieties only allow interveners of a particular subset of subject types (Lipski 1977; Baković 1998; Ordóñez and Olarrea 2001). And two, a dual-component system based on thematic licensing does not account for cases of noninversion in complex Wh-phrases.

Addressing the first issue, we return to the second approach to Wh- non-inversion patterns: a focus on the intervener phrase. Ordóñez and Olarrea (2006) provide evidence that in Dominican Spanish, preverbal subjects in Wh-interrogatives are generally limited to particular pronominal elements *tú* 'you,' *usted* 'you' (formal), *ustedes* 'you' (plural), *él* 'he,' and *ella* 'she,' with the second person singular *tú* 'you' as the most common. Their account points to a tripartite pronominal system made up by tonic pronouns, 'weak' pronouns, and clitics. In Dominican Spanish, subject pronouns have become 'weak' pronouns which, according to their analysis, are contained within the Inflectional Phrase (IP) and therefore are not true structural interveners.

(65) ¿[$_{CP}$ Qué [$_{IP}$ tú quieres]]?
'What do you want?'

In order to account for the broader variation of intervener subject types in Caribbean Spanish, Ordóñez and Olarrea conjecture that the 'weak' pronominal system may be extending from pronouns to DPs in some speakers [and therefore some varieties].

In sum, the approaches discussed to this point underline the difficulties involved in proposing a unified syntactic account for Spanish Wh-movement. However, recent applied research has made claims that non-inversion patterns for adjunct Wh-phrases, complex Wh-phrases, and Caribbean dialects should be attributed to differential costs on working memory, and not be considered fundamentally syntactic. Goodall (2004), building on well-known evidence that a syntactic object can be well-formed but perceived as unacceptable (66) (Chomsky and Miller 1963; Bever 1970), argues that preverbal subjects in Wh-interrogatives are syntactically licit, but are perceived as ungrammatical due to processing mechanisms.

(66) The woman$_i$ the man$_j$ the host$_k$ knew$_k$ brought$_j$ left$_i$ early.

In this framework, acceptability hinges on the degree to which the relationship between the 'filler' (Wh-phrase) and the 'gap' (trace) can be recovered. Two dynamics lead to graded performance: (1) the structural distance between the filler and gap; and (2) the referential status of Wh-phrase and/or intervener DPs (Gibson 1998; Frazier and Clifton 2002). Through Experimental Syntax procedures (Cowart 1997; Sprouse 2007), Goodall provides data from non-Caribbean Spanish speakers showing expected acceptability contrasts for (non)thematic Wh-phrases (67) and contrasts in the referential status of Wh-phrases (68), as well as degraded acceptability based on the nature of the intervener (pronominal/non-pronominal) (69).

(67) a. ¿Por qué Miguel trabaja tanto? 4.8/5
'Why does Michael work so much?
b. ¿Qué Juan leyó en la biblioteca? 2.1/5
'What did John read in the library?'
N=26

(68) a. ¿Cuáles de esos libros Ana leyó? 3.9/5
'Which of those books did Ana read?'
b. ¿Qué Ana leyó? 2.2/5
'What did Ana read?'
N=26

(69) a. ¿Qué tú leíste en la biblioteca? 2.2/5
'What did you read in the library?'
b. ¿Qué el niño leyó en la biblioteca? 1.9/5
'What did the boy read in the library?'
N=23

In this light, Spanish inter-dialectal variation in inversion patterns can be seen as a matter of degree (processing-based), not category (syntactic-based). Yet Goodall (2011) suggests that inversion patterns in English are syntactic. In a Satiation study (Synder 2000; Francom 2009) contrasting Spanish versus English inversion, data reveal that under repeated exposure, mean acceptability ratings increase for Spanish noninversion sentences over the course of the experiment but not for English noninversion, pointing to a categorical source of inversion patterns in English.

5.2 The nature of CP in matrix and embedded clauses

A second line of inquiry on Wh-movement deals with the properties that matrix and embedded CP share. Early analyses of the structure of CP made the assumption that that there was a single, basic complementizer phrase type for interrogatives, exclamatives, and relatives.

(70) a. [$_{CP}$ [$_{spec}$ Cómo [$_{C'}$]] te fue?
'How did it go?'
b. ![$_{CP}$ [$_{spec}$ Qué grande [$_{C'}$]] está tu niño!
'Your son is getting so big!'
c. Esa es la película [$_{CP}$ [$_{spec}$ [$_{C'}$ que]] quería ver.
'That is the film that I wanted to see.'

However, there are problematic cases that challenge this assumption in which a simple CP does not appear to be adequate, given that multiple Wh-phrases appear within the same finite clause.

(71) a. Me pregunto (que) quién vendrá esta noche.
'I wonder who will come tonight.'
b. ¡Cuántos libros (que) tiene!
'He has so many books!'
c. Dice Mamá que a tu hermana (que) no la dejes salir.
'Mother says that your sister, don't let her go outside.'

Rizzi (1997) argues that the CP is a multi-faceted syntactic layer which incorporates phrasal projections into the computational system dedicated to discourse mechanisms, such as Topic and Focus, which were once understood to be 'periphery' features (Chomksy 2004; Demonte and Fernández-Soriano 2009).

(72) CP Layer
$[_{\text{CP layer}}$ Force Phrase > Topic Phrase > Focus Phrase > Finite Phrase] > Tense Phrase > …

A fleshed-out CP provides projections for multiple Wh-phrases within the same clause, addressing the issues posed by doubly-filled complementizers (73a), relativized Wh-exclamatives (73b), and dislocation in subordinate clauses (73c) by taking advantage of the host of discourse projections contained within the CP layer.

(73) a. Me pregunto $[_{\text{CP}}$ $[_{\text{ForceP}}$ (que) $_{\dots}$ $[_{\text{FocusP}}$ quién $_{\dots}$]]] vendrá esta noche $_{\dots}$
b. ¡$[_{\text{CP}}$ $[_{\text{FocusP}}$ Cuántos libros $[_{\text{FiniteP}}$ (que) $_{\dots}$]]] tiene!
c. Dice Mamá $[_{\text{CP}}$ $[_{\text{ForceP}}$ que $[_{\text{TopicP}}$ a tu hermana $[_{\text{FiniteP}}$ (que) $_{\dots}$]]]] no la dejes salir.

However, there are questions still to be addressed. First, a unified syntactic account for the Wh-phrase in Wh-exclamatives and Wh-interrogatives is questionable given that relativization is possible in Wh-exclamatives (74) but not in Wh-interrogatives, and Wh-exclamatives allow Wh-phrases in both cardinal number and quantifier readings (75), while Wh-interrogatives only allow cardinal number readings (76) (Bosque 1984).

(74) a. ¡Cuántas historias (que) tienes!
'You have a lot of stories!'
b. ¿Cuántas historias (*que) tienes?
'How many stories do you have?'

(75) a. ¡Cuántos libros más leerías si tuvieras tiempo! *Cardinal*
'You could read so many books if you had the time!'
b. ¡Cuántos más libros leerías si tuvieras tiempo! *Quantifier*
'You could read so many more books if you had the time!'

(76) a. ¿Cuántos libros más leerías si tuvieras tiempo? *Cardinal*

'How many more books would you read
if you had the time?'

b. *¿Cuántos más libros leerías si tuvieras tiempo? *Quantifier*
'How many books more would you read
if you had the time?'

Second, whereas embedded clauses can take an optional *que* 'that' in (73c), relative pronouns are not optional.

(77) Esta no es la lección con la *(que) quieres comenzar el semestre.
'This is not the lesson with which you want to start the semester.'

Against the proposal that [article + *que*] and [article + *cual*] are stylistic equivalents (Rivero 1982), Brucart (1992) suggests that *que* is always a subordinating clause marker, and not a 'true' relative. Building on this proposal, Arregi (1998) argues that relative operators (*cual*, *quien*) appear overtly only to recover reference and that *que* appears as a Last Resort, in terms of Chomsky (1991), to mark subordination. Given the distinction between overt and non-overt wh-operators, a single projection (FiniteP) for subordination and relative pronouns may not be adequate.

5.3 Wh-phrase extraction across clause boundaries

Wh-movement shows that the Wh-operator can bind a variable that is selected in a multiply-embedded clause in Wh-interrogatives (78a) and relatives (78b) but not in Wh-exclamatives (78c).

(78) a. ¿Qué$_i$ dice María [que Juan sabe [que Inés comió $_{ti}$]]?
'What does Mary say that John knows that Agnes ate?

b. Espero que me de el chocolate [que dijeron [que sabe [que me gusta $_{ti}$]]].
'I hope that he/she gives me the chocolate that they said that he/she knows that I like.'

c. *¡Qué inteligente$_i$ dice Juan [que estás $_{ti}$]!
'How intelligent John says you are!'

Although it does not appear to be the case that there are restrictions on the absolute distance between operator and variable in Wh-exclamatives and relatives, there appear to be limitations on the type of structural configurations in which this relationship can hold.

(79) *¿Qué$_i$ tocas el piano y $_{ti}$?
'What do you play the piano and?'

As was the case for formal accounts of inversion, much of the early theoretical work on Wh-extraction restrictions was based on English. Ross's (1967) influential survey of English Wh-movement identified a set of configurations, referred to as Islands, that disallow extraction from certain modifier (80), noun phrase (81), and clausal (82) types.

(80) a. *Who$_i$ did John talk with Mary [after seeing $_{ti}$]? *Adjunct Island*
b. *How many$_i$ did John buy [$_{ti}$ books]? *Left Branch Condition*

(81) a. *Who$_i$ does Mary believe [the claim that John likes $_{ti}$]? *Complex NP Constraint*
b. *What$_i$ does John know that [a bottle of $_{ti}$]fell on the floor? *Subject Island*

(82) a. *Who$_i$ does John wonder [whether Mary likes $_{ti}$]? *Whether Island*
b. *Who$_i$ does Mary think [that $_{ti}$ likes John]? *Comp-trace Effect*

There are a number of formal accounts for these restrictions (Subjacency (Chomsky 1977); Parasitic Gaps (Chomsky 1982); Relativized Minimality (Rizzi 1990), etc.) that span various frameworks (Chomsky 1973, 1981, 1995). Equivalent structural limitations appear to apply in Spanish as well for many of these Islands, suggesting common grammatical underpinnings explaining their ungrammaticality.

(83) a. *¿A quién$_i$ habló Juan con María [después de ver$_{ti}$]? *Adjunct Island*
b. *¿Cuántos$_i$ compró Juan [$_{ti}$ libros]? *Left Branch Condition*

(84) a. *¿Quién$_i$ cree María [la propuesta de que Juan quiera $_{ti}$]? *Complex NP Constraint*
b. *¿Qué$_i$ sabe Juan que [una botella de $_{ti}$] se cayó al suelo? *Subject Island*

One key area where English and Spanish data diverge concerns the nature of embedded complementizer phrases. Islands for Wh-movement in English, Whether Islands (85), and Comp-trace violations (86) are grammatical in Spanish.

(85) ¿Qué libro no sabías si Juan había comprado ya?
'Which book didn't you know if John had bought yet?'

(86) a. ¿Who did John say (*that) saw Mary?
b. ¿Quién dijo Juan *(que) vio a María?

These data, in conjunction with contrasts with English inversion in embedded clauses and the fact that verbally selected complementizers are not optional in Spanish, lead to the conclusion that embedded CPs in the two languages are not syntactic equivalents (Torrego 1983, 1984).

Other important differences between Spanish and English regarding Wh-movement restrictions also may be found in evidence from applied investigations. Recent investigation has suggested that the acceptability contrasts in Complex

NP (87) (Sag et al. 2007) and Subject Condition (88) (Kluender 2004) structures reflect processing difficulty, not syntactic restriction.

(87) a. *What does Mary believe the claim that John likes?
b. ?Which restaurant does Mary believe claims that Mary likes?

(88) a. *Who$_i$ did Mark say a fight with $_{ti}$ started a national scandal?
b. ?Who$_i$ did Mark say fighting with $_{ti}$ started a national scandal?

In general, little work has been done to investigate possible processing sources for Islands in Spanish. But in a Satiation study investigating anomalous structures in Spanish and English, Goodall (2011) observes corroboration for English processing effects in CNPC and Subject Islands, but not for equivalent Spanish structures. Although no explanation for the Subject Island contrast is given, it is noted that a key difference between English and Spanish Complex NPs is found in the extraction site of Wh-phrase: in Spanish the Wh-phrase is extracted out of a PP (89a) and in English an NP (89b) – suggesting these may not be comparable structural configurations.

(89) a. *¿Quién$_i$ cree María [la propuesta de que Juan quiera $_{ti}$]?
b. *What$_i$ does Mary believe [the claim that John likes $_{ti}$]?

6 Concluding remarks

In this chapter, it has been shown that Wh-movement displays a series of very similar effects in three semantically and pragmatically diverse constructions: interrogatives, exclamatives, and relatives. There are also a number of aspects in which Wh-movement is not uniform across these three structures. Given this descriptive variation, I have addressed some of the major themes that have characterized past research and continue to shape ongoing investigation. Although much of the theoretical research on Wh-movement has been based on Wh-interrogatives, recent integration of formal and applied research has opened fresh avenues for interdisciplinary investigation and has encouraged more comprehensive study of Wh-movement in exclamatives and relatives.

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26 Binding: Deixis, Anaphors, Pronominals

LUIS EGUREN

1 Introduction

This chapter offers an overview of the state of the art on the topics of binding and deixis in Spanish, centering on a number of constructions and phenomena that are characteristic of this language. In all cases, relevant facts will be presented, main accounts of those facts will be reviewed, and open issues will be outlined. The chapter is organized as follows. Section 2 is concerned with deixis, focusing on demonstratives, a class of Spanish indexicals with idiosyncratic properties. Section 3 briefly introduces Binding Theory, and aims to help the general reader follow the discussion in the rest of the chapter. Sections 4 and 5 are devoted to the description and analysis of some distinctive features of Spanish 'anaphors' and 'pronominals,' two terms that are to be understood here as in the generative tradition. Finally, Section 6 addresses the issue of the complementary distribution of these two types of nominal expressions.

2 Deixis

2.1 Demonstrative determiners and pronouns

Spanish has a three-term system of demonstrative determiners and pronouns: *este-ese-aquel* 'this-that-that yonder' (with their corresponding variants in gender and number). It also instantiates a series of neuter pronouns (*esto-eso-aquello*) for abstract deixis and anaphora. With respect to spatial deixis, this tripartite system is commonly characterized in terms of positional reference to the participants of the utterance, so that *este* indicates proximity of the demonstrated object with respect to the speaker, *ese* denotes proximity with respect to the addressee, and *aquel* points to an object which is distant with respect to both the speaker and the addressee (see the references in Eguren 1999: fn. 29). This traditional view has been challenged in

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recent work by Gutiérrez-Rexach (2002, 2005), who claims that the tripartite system reduces to a basic opposition between a proximal term (*este*) and a distal one (*aquel*), with *ese* playing a neutral role in between or acting as a substitute for either of the two other demonstratives. Gutiérrez-Rexach brings up the following facts (amongst others) in support of this insight.

First, in parallel to the demonstrative series, Spanish has a ternary locative adverbial system (*aquí-ahí-allí* 'here-there-there yonder'), in which the deictic adverbs *aquí* and *allí* are also opposed by a proximal/distant distinction, and *ahí* plays the same role as *ese* in the nominal domain. Demonstratives can combine with locative deictic adverbs:

(1) a. Este de {aquí/ahí/*allí}.
 b. Ese de {aquí/ahí/allí}.
 c. Aquel de {*aquí/ahí/allí}.

As Gutiérrez-Rexach points out, the pattern in (1) shows that there has to be a matching in the value of the [+/-proximal] feature since *este* cannot co-occur with *allí* and *aquel* cannot co-occur with *aquí*. *Este* and *aquel* can both combine, however, with the neutral adverb *ahí*. As for demonstrative *ese*, its neutral role is made explicit by the fact that it can combine both with the proximal adverb *aquí* and the distal one *allí*, as well as with the neutral *ahí*.

Another piece of evidence for the neutral role of *ese* can be found in its inability to participate in a particular type of discourse deixis based on the proximal/non proximal distinction:

(2) Juan no dijo lo mismo que Pedro. Yo le creo a {este/*ese}, no a {aquel/*ese}.
 'John didn't say the same as Peter. I believe the latter, not the former.'

In the example in (2), *este* identifies the linearly closest of two potential antecedents, and *aquel* demonstrates the linearly most distant one. *Ese*, however, cannot refer to any of the two antecedent NPs.

In his study of the use of Spanish demonstratives in oral discourse, Gutiérrez-Rexach (2005) finally finds out that only 30% of the occurrences of *ese* could be unambiguously associated with objects proximal to the addressee. In most cases, *ese* worked as a neutral term with respect to spatial positioning. For example, in one of the recorded sessions, a student, pointing at a car passing by (which was not closer to the addressee than to the speaker), stated the following: *No me gusta ese coche* 'I don't like that car.' And in another dialogue, a student uttered, pointing at a glass of beer that was closer to him than to other participants: *Esa cerveza es la más cara* 'That beer is the most expensive one.'

All these facts lead Gutiérrez-Rexach to conclude that, when referring to nonabstract objects, the ternary opposition in the Spanish demonstrative system has been recategorized as a bipartite system. The new system is built on the proximal/distant distinction, in which *este* ([+proximal]) opposes to *aquel* ([-proximal]), and *ese* is unspecified with respect to the [+/-proximal] feature. This picture is certainly consistent with the attested simplification of the Spanish demonstrative system in anaphoric uses and in deixis *am phantasma* (Eguren 1999: fn. 31; see also on this topic RAE 2009: 17.2m–o).

Following the same line of research, Gutiérrez-Rexach and Zulaica (in press) carry out a corpus-based study of the combinatorial properties of Spanish neuter demonstratives with tenses and events. The data in their study shows that [-proximal] *aquello* is used to anaphorically refer to events in the past, whereas *esto* and *eso* freely co-occur with both present and past tenses. In the light of these results, these authors hypothesize a different reduction for the system of neuter demonstrative pronouns when used by speakers to refer to events in discourse: the tripartite system is again recategorized as a basic binary distinction based on the proximity parameter; but, in this case, *esto* and *eso* are unspecified with respect to a proximity presupposition, while *aquello* still retains a marked [+distal] feature.

The reviewed work certainly sheds new light on the characterization of the Spanish demonstrative system. On the basis of this novel research, more corpus-based studies are needed to check the tripartite–bipartite reduction in other uses of both demonstrative determiners and pronouns, focusing on how this reduction might take place in each case, and also paying attention to dialectal variation (see RAE 2009: 17.2n).

2.2 *Postnominal demonstratives*

Demonstratives precede the noun in Spanish, but they can also appear postnominally when the nominal expression is introduced by the definite article:

(3) a. Este libro.
this book
b. El libro este.
the book this

As underlined in Roca (1996), NPs with a postnominal demonstrative basically behave like NPs with a prenominal one. Both types of nominal expressions disallow, for instance, a generic reading (4a), and are excluded in a superlative construction (4b):

(4) a. {La ballena/#Esta ballena/#La ballena esta} está en peligro de extinción.
'The whale is in danger of extinction.'
b. {La mujer/*Esta mujer/*La mujer esta} más guapa del mundo.
'The most beautiful woman in the world.'

This shows that the properties of the whole NP are determined by the demonstrative also in the case of postnominal demonstratives. To account for this fact, both Brugè (1996) and Roca (1996) develop a unified analysis for the two different word orders, making use of the technical apparatus of generative grammar, including DP projections headed by determiners. Brugè claims, in particular, that the demonstrative is always generated as an XP in a low position inside the extended nominal projection (5a), and that, at PF, it is realized either in this position (5b), or in the prenominal position (5c). In the first option, the demonstrative moves to [Spec, DP] at LF, whereas the second option is due to movement of the demonstrative before Spell Out. In both cases, the demonstrative raises to [Spec, DP] to check the referential feature of D^0 (as can be seen, she also assumes that the noun head moves to a higher functional head).

(5) a. [$_{\text{DP}}$ [$_{\text{D}^{\text{o}}}$ [… [$_{\text{FP}}$ este [$_{\text{F}^{\text{o}}}$] [$_{\text{NP}}$ [$_{\text{N}^{\text{o}}}$ libro]]]]]]
 b. [$_{\text{DP}}$ [$_{\text{D}^{\text{o}}}$ el [$_{\text{XP}}$ [$_{\text{X}^{\text{o}}}$ libro$_j$] […[$_{\text{FP}}$ este [$_{\text{F}^{\text{o}}}$ t_j] [$_{\text{NP}}$ [$_{\text{N}^{\text{o}}}$ t_j]]]]]]]
 c. [$_{\text{SpecDP}}$ este$_i$ [$_{\text{D}^{\text{o}}}$ [$_{\text{XP}}$ [$_{\text{X}^{\text{o}}}$ libro$_j$] [… [$_{\text{FP}}$ t_i [$_{\text{F}^{\text{o}}}$ t_j] [$_{\text{NP}}$ [$_{\text{N}^{\text{o}}}$ t_j]]]]]]]

As Roca (1996) points out, a main shortcoming in Brugè's proposal is that it cannot capture the fact that the postnominal demonstrative can either precede or follow a noun modifier in Spanish (for the details of this criticism, see Roca's dissertation; a full description of the ordering of postnominal demonstratives and noun modifiers in Spanish can be found in Roca 1996 and in RAE 2009: 17.5h–l):

(6) El libro (ese) {viejo/de latín} (ese).
 the book (that) old/of latin (that)

Roca tries to solve this problem with an alternative analysis, in which the demonstrative heads a DemP that is the complement of D^0 (7a). In his view, in the case of postnominal demonstratives, D^0 is occupied by the definite article, and the noun moves to [Spec, DemP] (7b). In this construction, the article works as an expletive, forming a chain with the demonstrative, which raises to D^0 at LF. In DPs with a prenominal demonstrative, the head D^0 is empty, and the demonstrative moves to D^0 (7c).

(7) a. [$_{\text{DP}}$ [$_{\text{D}^{\text{o}}}$ [$_{\text{DemP}}$ [$_{\text{Dem}^{\text{o}}}$ este [… [$_{\text{NP}}$ [$_{\text{N}^{\text{o}}}$ libro]]]]]]]
 b. [$_{\text{DP}}$ [$_{\text{D}^{\text{o}}}$ el [$_{\text{DemP}}$ [$_{\text{NPi}}$ libro] [$_{\text{Dem}^{\text{o}}}$ este [… [$_{\text{NP}}$ t_i]]]]]]
 c. [$_{\text{DP}}$ [$_{\text{D}^{\text{o}}}$ este$_j$ [$_{\text{DemP}}$ [$_{\text{NPi}}$ libro] [$_{\text{Dem}^{\text{o}}}$ t_j [… [$_{\text{NP}}$ t_i]]]]]]

Under this analysis, Roca accounts for the free ordering of the postnominal demonstrative and a noun modifier as follows. He considers, following Kayne (1994), that the noun and its complements form a clause-like (CP) projection selected by the determiner (8a). The order in which the postnominal demonstrative precedes the noun modifier is then obtained by raising only the noun to [Spec, DemP] (8b). The reversed order results from moving the whole CP to the same position (8c).

(8) a. [$_{\text{DP}}$ [$_{\text{DemP}}$ [$_{\text{Dem}^{\text{o}}}$ ese [$_{\text{CP}}$ [$_{\text{NPi}}$ libro] [$_{\text{C}'}$ [$_{\text{C}}$ de [$_{\text{IF}}$ t_i [$_{\text{I}'}$ I [$_{\text{NP}}$ latín]]]]]]]]]
 b. [$_{\text{DP}}$ [$_{\text{D}^{\text{o}}}$ el [$_{\text{DemP}}$ [$_{\text{NPi}}$ libro] [$_{\text{Dem}^{\text{o}}}$ ese [$_{\text{CP}}$ [$_{\text{NP}}$ t_i] [$_{\text{C}'}$ [$_{\text{C}}$ de [$_{\text{IF}}$ t_i [$_{\text{I}'}$ I [$_{\text{NP}}$ latín]]]]]]]]]]
 c. [$_{\text{DP}}$ [$_{\text{D}^{\text{o}}}$ el [$_{\text{DemP}}$ [$_{\text{CPi}}$ libro de latín] [$_{\text{Dem}^{\text{o}}}$ ese [$_{\text{CP}}$ t_i]]]]]

As the discussion above shows, both Brugè's and Roca's proposals (tacitly) treat the prenominal and the postnominal linear orderings of demonstratives as equivalent alternatives. Bernstein (2001) claims, instead, that there are interpretive differences between pre- and postnominal demonstratives. In her opinion, the prenominal position yields a neutral interpretation, whereas the postnominal DP-final position gives rise to a focus interpretation of demonstratives, matching the pattern found in the Romance clause. In recent work, Roca (2009) also holds that there is a difference in the interpretation of prenominal and postnominal demonstratives in Spanish, but characterizes this difference in terms of their deictic or anaphoric readings. He notes that the demonstrative can be used deictically or anaphorically in both positions, but remarks that, whereas a DP with a prenominal demonstrative is fine in a sentence initiating a discourse, a postnominal demonstrative sounds odd in the same position. This suggests, Roca says, that the anaphoric (topic-related) reading is the preferred one for postnominal demonstratives. This idea is corroborated by the fact that, unlike prenominal demonstratives (9a), postnominal demonstratives cannot be contrastively focused (9b) (unless in an explicit deictic context, like the one induced by the presence of a locative reinforcer (9c)):

(9) a. Me llevaré este libro, no aquel.
(I will take) this book, not that
b. ??Me llevaré el libro este, no el libro aquel.
(I will take) the book this, not the book that
c. Me llevaré el libro este de aquí, no aquel (de allí).
(I will take) the book this of here, not that (of there)

A detailed corpus study is also required in this case in order now to clarify the informative import of both prenominal and postnominal demonstratives. A more refined structural account of the similarities and differences of the two constructions could then be given, which ought to take into consideration the existing analyses of nominal expressions containing the definite article and a demonstrative in other languages (see Alexiadou et al. 2007: 1.4).

3 Binding theory

As is well known, Binding Theory is a subtheory within the Principles and Parameters framework (Chomsky 1981, 1986; Chomsky and Lasnik 1993) that determines the conditions that govern the intrasentential distribution and referential dependencies of different types of nominal expressions. Chomsky (1981) formulates the principles of Binding Theory as in (10):

(10) Principle A: an anaphor is bound in its governing category.
Principle B: a pronominal is free in its governing category.
Principle C: an R-expression is free everywhere.

(11) Binding:
α binds β iff α and β are coindexed and α c-commands β.

(12) C-command:
α c-commands β iff α and β do not dominate each other, and the first branching node dominating α also dominates β.

(13) Governing category:
γ is the governing category for α iff γ is the minimal category containing α, a governor of α, and a SUBJECT to α (where SUBJECT refers to either a subject NP or AGR(eement)).

The classical Binding Theory in (10) has been thoroughly revised and reinterpreted in the last three decades. These changes include the redefinition of 'binding domain' (Chomsky 1986), the reformulation of binding relations in non strictly configurational terms (Reinhart and Reuland 1993; Williams 1994), the substitution of the binding conditions for discourse-based/pragmatic principles (Kuno 1987; Levinson 2000), and the minimalist reduction of principles A and B to core syntactic operations (Hicks 2009, and the references therein). Leaving all these revisions and reinterpretations aside, in the discussions to follow I will make use of the version of Binding Theory in (10), as it suffices for the purposes and scope of this chapter.

The paradigm in (14) illustrates the kind of empirical facts covered by principles A and B of Binding Theory (in these examples, coindexation indicates coreference):

(14) a. John$_i$ likes {himself$_i$/him$_{*i/j}$}.
b. John$_i$ said that Mary likes {himself$_{*i}$/him$_{i/j}$}.

In (14a), the reflexive 'anaphor' *himself* must corefer with *John* since *John* binds the anaphor in its governing category (i.e., the clause, which contains the anaphor, its governor: the verb, and an NP subject, *John*). The 'pronominal' *him*, however, has to be free (not bound) in the same binding domain. It cannot accordingly corefer with *John*, and has disjoint reference. In (14b), the binding domain is now the subordinate clause. In this case, the anaphor cannot be referentially dependent on the matrix subject (*John*), as this NP lies outside its governing category, but the pronominal is free in this domain, and nothing prevents it from coreferring with the subject of the main clause.

Within Binding Theory, an 'anaphor' is therefore defined as an item that is bound in a specific local domain, whereas a 'pronominal' is an element that is free in the same domain. In this technical sense, the class of Spanish anaphors is made up by reflexive pronouns (*se*, *sí*, *sí mismo* 'himself,' etc.) and the reciprocal pronominal phrase *(el) uno P (el) otro* 'each other,' amongst other expressions, whereas the category of pronominals includes nonreflexive personal pronouns (*él/ella* 'he/she,' *lo/la* 'him/her,' etc.) and the null category *pro* (see Chapter 27). The rest of this chapter summarizes some distinctive properties of both anaphors and pronominals in Spanish.

4 Anaphors

4.1 Sí (mismo), (el) uno P (el) otro

The example in (15) shows that the Spanish reflexive pronoun *sí (mismo)* and the reciprocal phrase *(el) uno P (el) otro* are anaphors that obey principle A of Binding Theory:

(15) Los padres$_j$ esperan que sus hijos$_i$ confíen {en sí$_{i/*j}$ mismos/el uno en el otro$_{i/*j}$}.
'Parents hope that their children will trust {themselves/each other}.'

Spanish has both simple reflexive anaphors (*mí*, *ti*, *sí*, etc.) and complex ones (*mí/ti/sí mismo*), which are formed by adding the adjective *mismo* to a reflexive pronoun. Various proposals have been made on the meaning and role of this adjective. López-Díaz (1999) holds that it is a reflexive anaphor with an identificative meaning. The optionality of *mismo* in some reflexive constructions (see below) suggests, however, that this modifier is not related to reflexivity. Otero (1999) considers, as an alternative, that the adjective in Spanish complex anaphors corresponds to the emphatic *mismo* that follows definite NPs, proper nouns, or personal pronouns (*{el profesor/Juan/él} mismo* '{the teacher/John/he} himself'), and takes this element to be an intensifier that yields a focalized contrastive interpretation of the nominal expression it modifies. A plausible analysis of the internal structure of Spanish complex anaphors might go as follows: the reflexive is generated in N^0 and moves to D^0, whereas *mismo* occupies a postnominal position inside the NP. As depicted in (16), this would also be the structure of DPs with a non-reflexive personal pronoun and emphatic *mismo*:

(16) $[_{DP}\ [_{D'}$ sí/él$_i$ $[_{NP}\ t_i\ [_{AP}$ mismo$]]]$

The structure and the interpretation of complex reflexive anaphors in Spanish are therefore fairly well known. The *sí/sí mismo* alternation remains, however, an open issue that still calls for an explanation: *mismo* must often be present (17a), but it can also be optional in some cases (17b) (see RAE 2009: 16.4q–s).

(17) a. Se perjudica a sí *(mismo).
'He harms himself.'
b. Se siente muy seguro de sí (mismo).
'He is feeling self-confident.'

As for the reciprocal anaphor *(el) uno P (el) otro*, the combinatorial and formal properties of this phrase have been precisely characterized by Bosque (1985). Amongst other clarifying insights,[1] Bosque reveals that reciprocals have a more restricted distribution than reflexives in Spanish: the reciprocal phrase cannot be the complement of a noun (18a), and cannot have an indirect object as its antecedent either (18b); for some exceptions to (18a), see RAE (2009: 16.5o).

(18) a. Las declaraciones de los políticos {sobre sí mismos/*los unos sobre los otros}.
'The politicians' declarations on {themselves/each other}.'
b. Les hablé {de sí mismos/*el uno del otro}.
'I talked to them about {themselves/each other}.'

He furthermore demonstrates that the Spanish reciprocal (that always contains a preposition, in contrast to other languages) is a prepositional phrase.[2] This explains, for instance, why it can be coordinated with another PP (19a), or alternates with a PP selected by the predicate (19b):

(19) a. No sé si hablaron de ti o el uno del otro.
'I don't know if they talked about you or about each other.'
b. Hablaban {de fútbol/el uno del otro}.
'They talked about {soccer/each other}.'

Bosque (1985) also deals with the morphological properties of the reciprocal phrase, and points out that *(el) uno* and *(el) otro* match in gender (20a,b,c), number (20d), and (partially) also in definiteness (20e):

(20) a. Juan y Pedro hablan el uno del otro.
b. Ana y María hablan la una de la otra.
c. Juan y María hablan {el uno del otro/*el uno de la otra}.
'X and Y talk about each other.'
d. Hablan {el uno del otro/los unos de los otros/*el uno de los otros}.
e. Hablan {el uno del otro/uno de otro/uno del otro/*el uno de otro}.
'They talk about each other.'

Note that when the coordinated nouns in the antecedent differ in gender (20c), both members of the reciprocal phrase take the unmarked masculine form. Nevertheless, some examples are attested (RAE 2009: 16.5i) in which the gender distinction in the antecedent is maintained in the reciprocal phrase.

As regards the semantics of the reciprocal phrase, it could finally be argued, following the proposal in Heim, Lasnik and May (1991) for English reciprocals, that *(el) uno* is a distributive anaphor that adjoins to the group-denoting antecedent at LF, whereas *(el) otro* acts as a reciprocator, establishing the bidirectionality relation that holds in the reciprocal construction. This analysis would capture the three main semantic components in this construction: anaphoricity, distributivity, and reciprocality.

4.2 The se-*construction*

Both reflexive and reciprocal anaphors in Spanish force the verbal clitic *se* (or its variants) to be present when they function as a direct or an indirect object (21a,b). Since it has no gender, number, or case specifications, this anaphoric clitic is an impoverished item in terms of features in contrast with more complex anaphors. As shown in (22), the reflexive or reciprocal clitic can also appear without an explicit object:

(21) a. Ana y María *(se) critican {a sí mismas/una a otra}.
'Ann and Mary criticize {themselves/each other}.'
b. Ana y María *(se) regalaron un libro {a sí mismas/una a otra}.
'Ann and Mary gave {themselves/each other} a book as a present.'

(22) a. Ana y María se critican.
'Ann and Mary criticize {themselves/each other}.'
b. Ana y María se regalaron un libro.
'Ann and Mary gave {themselves/each other} a book as a present.'

Torrego (1995) gives a comprehensive account of the Spanish *se*-construction in (21) and (22), focusing on reflexives. This linguist adheres to the view (see also Bosque 1985; Otero 1999) that the presence of the clitic in sentences like (21a,b) is an instance of the obligatory pronominal redundancy (or 'clitic-doubling') that obtains with tonic pronouns in object positions in Spanish (cf. *María *(la) critica a ella* 'Mary criticizes her'), and attributes this fact to the specifics of Case with (both reflexive and non-reflexive) personal pronouns in marked accusative (or dative) case. In her opinion, the obligatory presence of the clitic follows from a morphological requirement of pronouns in Spanish. She assumes, in particular, that the interpretable person feature of Spanish tonic pronouns bears a Case feature that has to be checked off, and holds that entering into agreement with the clitic eliminates this feature.

Torrego also offers an analysis for the structure of Spanish doubled reflexives (*se-a sí mismo*), in which the tonic reflexive *(a sí mismo)* is the argument, and the clitic *se* functions as a pleonastic pronoun. In the structure she has in mind, the doubling clitic is first placed in the specifier of the argumental reflexive phrase, and then raises to the verb due to its affixal nature ($[se_i + V\ [\ a\ [_{DP} t_i [\ sí\ mismo]]]]$). Torrego finally characterizes the reflexive clitic that stands alone in cases like the ones in (22) as an operator that binds a null pronoun.

A different widespread approach to the reflexive (and reciprocal) *se*-construction in Spanish (and other Romance languages) takes the clitic to be a lexical operator that reduces the argument structure of the verb, and treats reflexive/reciprocal constructions as unaccusative. A discussion of the unaccusativity hypothesis for reflexives can be found in Chapter 23, above, which addresses all types of *se*-constructions in Spanish; in this chapter, different analyses of the clitic *se* are compared, and the properties of so-called 'pronominal' verbs are also examined.

4.3 Other anaphoric expressions

There are some other anaphoric expressions in Spanish besides reflexive and reciprocal pronouns. As first observed by Guéron (1983) for French, the determiner introducing NPs denoting inalienable possession also behaves like an anaphor in Spanish (see Section 3):

(23) a. María_i levantó la_i mano.
Mary raised the hand
'Mary raised her hand.'
b. La_{*i} mano fue levantada por María_i.
the hand was raised by Mary
c. Ana_j espera que María_i levante $\text{la}_{i/*j}$ mano.
Ann hopes that Mary raises the hand

Bosque (1992) shows, on his part, that the Spanish distributive quantifier *sendos* 'one each' has the same distribution than reflexives and reciprocals, and therefore concludes that this determiner belongs to the syntactic class of anaphors as well:

(24) a. Juan y Ana han escrito sendos libros.
John and Ann have written one each books
'John and Ann have written one book each.'
b. *Sendos libros han sido escritos por Juan y Ana.
one each books have been written by John and Ann
'One book each has been written by John and Ann.'
c. Sus amigos_j dijeron que Juan y Ana_i escribieron $\text{sendos}_{i/*j}$ libros.
their friends said that John and Ann wrote one each books
'Their friends said that John and Ann wrote one book each.'

Together with the reciprocal phrase *(el) uno P (el) otro*, another two expressions trigger a reciprocal reading of verbal predicates in Spanish: the prepositional anaphor *entre sí* 'amongst themselves' and the adverbial adjunct *mutuamente* 'mutually.' It has been pointed out that these two markers of reciprocity replace *(el) uno P (el) otro* when the reciprocal functions as a direct or an indirect object (25), but are excluded when it is a selected or an adjunct PP (26) (cf. Bosque 1985; Otero 1999):

(25) a. Se apoyan {el uno al otro/entre sí/mutuamente}.
'They support each other.'
b. Se hicieron un regalo {el uno al otro/entre sí/mutuamente}.
'They made a present for each other.'

(26) a. Dependen {el uno del otro/*entre sí/*mutuamente}.
'They rely on each other.'
b. Trabajan {el uno para el otro/*entre sí/*mutuamente}.
'They work for each other.'

The combinatorial properties of *entre sí* and *mutuamente* should be analyzed in detail though, given that these expressions, amongst other facts, do not always alternate with a reciprocal in object position (27a), and may seemingly be used instead of some selected PPs (27b):

(27) a. Se vieron {el uno al otro/*entre sí}.
'They saw each other.'
b. Hablaron {el uno con el otro/entre sí}.
'They talked to each other.'

5 Pronominals

5.1 *Obviation in subjunctive clauses*

One of the more intricate (and more studied) topics on Spanish pronominals is the obligatory disjoint reference interpretation of the subject of some subjunctive clauses with respect to the matrix subject. This 'obviation effect' (which is also observed in other Romance languages) is illustrated in (28):

(28) a. María$_i$ desea que pro$_{*i/j}$ venga.
Mary wishes that pro comes.SUBJ
b. Juan$_i$ le ordenó que pro$_{*i/j}$ viniera.
John him ordered that pro comes.SUBJ

Obviation in subjunctive clauses may be, *prima facie*, a counterexample for principle B of Binding Theory since the empty pronominal in the examples in (28) is not bound in its governing category, and could corefer with the higher subject (see Section 3).

As a first approximation to this phenomenon, it could be argued that disjoint reference obtains in subjunctive clauses because the coreference reading is expressed by infinitival subordination (cf. *María$_i$desea PRO$_i$venir* 'Mary wishes to come'). Raposo (1985), Suñer (1986), Kempchinsky (1990), San Martin (2007), and Kempchinsky (2009) offer, however, both theoretical and empirical evidence showing that this blocking-type analysis of obviation is not adequate.

The standard analysis of obviation in subjunctive clauses is the so-called 'domain extension approach.' Under this approach, the binding domain of the subordinate pronominal subject is extended into the matrix clause, where it cannot be bound by the matrix subject, and can only have a disjoint reference interpretation.

The idea of domain extension has been implemented in various ways. The initial proposals within the extended domain approach took Tense to be the determining factor in obviation (Picallo 1985; Raposo 1985). On the basis of *consecutio temporum* data, both Picallo and Raposo assumed, in particular, that subjunctive clauses are not specified for tense, and further hypothesized that binding domains are defined with respect to tense (and not to AGR; see Section 3). It follows from these assumptions that the binding domain for the pronominal subject in subjunctive clauses is the matrix clause since this is the minimal domain containing tense. As stressed by Suñer (1986), Padilla-Rivera (1990), and Kempchinsky (1986, 1990), the main objection to these tense-based analyses of obviation is that the characterization of subjunctive clauses as tenseless is misleading since tense markers in subjunctive clauses often express a semantic tense which is different from the semantic tense in the matrix clause.

A different approach to binding domain extension is developed by Kempchinsky (1986, 1990). Kempchinsky emphasizes that obviation only obtains with verbs of volition or influence, like the ones in (28), and points out that the semantic distinctive feature of subjunctive clauses selected by volitional and influence verbs is their lack of truth value, a property they share with simple imperatives. She then proposes that these subjunctive clauses are embedded imperatives that have a subjunctive/imperative operator in COMP, which must be identified at LF, and further argues that the subjunctive morphology on the subordinate verb identifies this operator. The subjunctive INFL thereby moves to C^0 at LF, and, as a result, the binding domain of the subjunctive subject is extended into the superordinate clause. This analysis allows her to offer an explanation for both obviation in subjunctive clauses with derived nominals (29a), and for the fact that the disjoint reference effect does not arise with other classes of verbs, like factives (29b), and does not take place either when the subjunctive on the subordinate verb is induced by negation (29c) (vs. **María cree que venga*): in Kempchinsky's view, only in the case of derived nominals, as in the corresponding sentences with volitional verbs (28a), the subjunctive clause is interpreted as an embedded imperative, and contains a subjunctive/imperative operator in COMP.

(29) a. Su$_i$ deseo de que pro$_{*i/j}$ venga.
her wish of that pro comes.SUBJ
b. Juan$_i$ lamenta que pro$_i$ viniera.
John regrets that pro comes.SUBJ
c. María$_i$ no cree que pro$_i$ venga.
María not believes that pro comes.SUBJ

Note that the different behavior of volitional and factive predicates in particular with respect to obviation could also be explained within the tense-based analysis by claiming, as in González Rodríguez (2003), that the matrix verb determines the temporal (in)dependence of the subordinate subjunctive clause.

San Martin (2007) has recently re-examined the topic of obviation, under the domain extension approach, from a cross-linguistic and minimalist perspective. In her view, the property that explains cross-linguistic variation regarding obviation is the (non)existence of subjunctivity markers in the left periphery of embedded clauses, so that languages with subjunctive complementizers and/or modal particles allow for free reference, whereas languages lacking such markers show obviation effects. San Martin formally captures this insight through a minimalist reformulation of binding domains that include phase heads, and argues that domain extension only takes place in languages that do not distinguish between indicative and subjunctive mood in the left periphery of embedded clauses.

There are, however, some problems with the binding domain extension approach to obviation. As shown in (30), both a subcategorized pronoun in the complement clause to a volitional verb, and the subject pronoun in the most embedded clause in cases of double embedding, can corefer with the matrix subject:

(30) a. Juan$_i$ desea que María lo$_i$ bese.
John wishes that Mary him kisses.SUBJ
b. Juan$_i$ quiere que Ana$_j$ desee que pro$_{i/*j}$ venga.
John wants that Ann wishes that pro comes.SUBJ

These two problems could be circumvented by saying that what now counts in defining the binding domain for the pronoun is the presence of an NP subject (see Section 3); that is, the lexical subject in the complement clause (*María*) in (30a), and the intermediate subject (*Ana*) in (30b) (cf. Raposo 1985).

The main empirical challenge to the domain extension approach is, anyhow, the lack of obviation between the subjunctive subject and an object in the matrix clause:

(31) Juan le$_i$ ordenó que pro$_i$ viniera.
John him ordered that pro comes.SUBJ

Both Picallo (1985) and Kempchinsky (1990) suggest that coreference with matrix objects is allowed because the object does not c-command the subjunctive clause, which has been extraposed and VP-adjoined. In a recent paper, Kempchinsky (2009) holds, however, on both theoretical and empirical grounds, that the matrix object does c-command the embedded subject in examples like (31). And she also mentions the fact, already noted in the literature, that passive subjects (32a), modal verbs (32b), and focused overt subjects (32c) in the subjunctive clause may override the disjoint reference effect:

(32) a. Ana$_i$ espera que pro$_i$ sea elegida.
Ann hopes that pro is.SUBJ elected
b. Ana$_i$ espera que pro$_i$ pueda acompañaros.
Ann hopes that pro is able.SUBJ to accompany you
c. Ana$_i$ espera que ELLA$_i$ apruebe el examen.
Ann hopes that SHE passes.SUBJ the exam

Kempchinsky (2009) therefore concludes that a purely structurally-based binding theory approach fails to capture the full range of facts. With this idea in mind, she re-analyzes subjunctive obviation with volitional and influence predicates, and now suggests that obviation is due to the effect of a quasi-imperative subjunctive operator in the left periphery of the embedded clause, which instructs the semantic component to interpret the pronominal subject as 'anyone other than the matrix subject'.

To summarize in Kempchinsky's words, it could well be deduced from the discussion above that the accurate characterization of the syntactic and semantic properties of subjunctive clauses (including obviation) has proven to be "tantalizingly challenging," and "remains an incomplete task."

5.2 *Overt and null pronouns*

As shown in (33), both overt and null pronouns in Spanish comply with condition B of Binding Theory:

(33) a. *Juan$_i$ no lo conoce {a él$_i$/pro$_i$}.
'John doesn't know him.'
b. Juan$_i$ dice que {él$_i$/pro$_i$} no lo conoce.
'John says that he doesn't know him.'

This being certainly the case, it has also been claimed that there are some differences between overt and null pronouns as regards their binding properties. Luján (1987, 1999) first observes that an overt pronoun in fronted adverbial clauses, such as the one in (34), is naturally interpreted by most speakers, unlike the covert form, as disjoint in reference with respect to the noun in the matrix clause:

(34) Cuando {él/pro} trabaja, Juan no bebe.
'When he works, John does not drink.'

Nevertheless, as Luján herself acknowledges, the overt pronoun can perfectly have a coreferential reading if these clauses are included in an appropriate context in which the pronoun plays a contrastive role, identifying its referent in contraposition to other individuals:

(35) Sus empleados beben en el trabajo, pero cuando él$_i$ trabaja, Juan$_i$ no bebe.
'His employees drink at work, but when he works, John does not drink.'

A second difference between overt and null pronouns was pointed out by Montalbetti (1984). In (36a), the empty pronominal in the subordinate clause can have three different interpretations: it can be a free pronoun, it may corefer with the matrix subject, and it can also function as a bound variable, giving rise to a distributed reading ('each of the students believes that he is intelligent'). The overt pronoun in (36b) can have both the free pronoun and the coreferential readings, but in Montalbetti's view, it cannot be interpreted as a bound variable.

(36) a. Muchos estudiantes creen que pro son inteligentes.
b. Muchos estudiantes creen que ellos son inteligentes.
'Many students believe that they are intelligent.'

Montalbetti holds that a similar pattern obtains in (37). In this case, the overt and empty pronouns cannot have the coreferential reading since *nadie* 'nobody' is a non-referential quantifier. The example in (37a) is ambiguous between a free and a bound reading, but again (37b) can only be interpreted as containing a free pronoun (Picallo 1994 observes similar facts in the domain of Spanish possessives: *Nadie busca a su hijo* 'Nobody looks for his son' vs. *Nadie busca al hijo de él* 'Nobody looks for the son of his').

(37) a. Nadie cree que pro es inteligente.
b. Nadie cree que él es inteligente.
'Nobody believes that he is intelligent.'

The overt pronoun can be interpreted, however, as a bound pronoun when it occurs as a complement of a preposition, a syntactic position where empty pronouns cannot appear:

(38) a. Muchos estudiantes quieren que María se case con ellos.
'Many students want Mary to marry them.'
b. Nadie quiere que María hable de él.
'Nobody wants Mary to talk about him.'

To account for the difference between overt and null pronouns with respect to the bound variable reading, Montalbetti resorts to an LF constraint stating that overt pronouns cannot be bound when the alternation overt/empty is available.

Montalbetti's judgments on the data above are not, however, precise enough. As in the case of obviation in adverbial clauses, an overt pronoun can be bound by a quantifier in a context in which a comparison is established and the pronoun has a contrastive interpretation (cf. Luján 1999; Bosque and Gutiérrez-Rexach 2009):

(39) Todo estudiante piensa que él es inteligente y los demás no.
'Every student thinks that he is intelligent, but the others are not.'

These two differences between overt and null pronouns in Spanish are therefore only apparent: when the overt pronoun is used in a contrastive setting, both the obviation effect in adverbial clauses and the restriction on the bound variable reading dissolve. What the aforementioned facts really show is that, in those positions in which pronouns can be omitted, the overt form functions as a distinctive or focused term, which requires a context that justifies its use.

6 The problem of complementary distribution

Principles A and B of Binding Theory predict that anaphors and pronominals must have a complementary distribution; that is, a coreferential pronoun cannot appear where an anaphor occurs, and vice versa (see Section 3):

(40) a. Juan$_i$ {se$_i$/lo$_{*i}$} odia.
'John hates {himself/him}.'
b. Pedro$_i$ dice que Juan {se$_{*i}$/lo$_i$} odia.
'Peter says that John hates {himself/him}.'

As has been noted, there are, however, some problematic cases, both in Spanish and in other languages, in which this prediction does not seem to be borne out, such as binding of pronominals in prepositional phrases or the anaphor-like behavior of so-called 'empathic' and 'distributive' pronouns.

6.1 Binding in PPs

One well-known central case that, apparently, questions the claim that anaphors and pronominals can never occupy the same position is the possibility of a coreferential interpretation with the subject of a simple sentence for both anaphors and pronominals contained in a selected prepositional phrase:

(41) a. María$_i$ siempre habla de sí$_i$ misma.
b. María$_i$ siempre habla de ella$_{i/j}$.
'Mary always talks of {herself/her}.'

Different explanations of the pattern in (41) can be found in the literature on binding in Spanish, which ultimately maintain the basic insight that anaphors and pronominals are in complementary distribution. Some of these accounts will be briefly reviewed next.

Demonte (1989) suggests that morphological pronouns are syntactic anaphors in this context, and offers in support of her idea the fact that only the reflexive interpretation obtains in (41b) when the emphatic adjective *mismo* (see Section 4.1) is added to the pronoun (cf. *María$_i$ siempre habla de ella misma$_{i/*j}$*). A second proposal is offered by de Jong (1996), who holds that the direct object marker *a* impedes binding of pronouns whereas bound pronouns that are objects of any other preposition are permitted, and offers a Case-based analysis for this contrast. De Jong's description of the data is not, however, adequate. As illustrated in (45) below, some "fixed" prepositions take a bound pronoun as a complement (*María habla de ella* 'Mary talks of her'), while others do not (*#María influyó en ella* 'Mary influenced her'). Moreover, as shown by Torrego (1995), a coreferential non-reflexive pronoun can follow the accusative marker *a* in some cases (*María se critica a ella* 'lit. Mary se-criticizes to she'), yielding a more "objective" interpretation than the reflexive form.

Another account of coreferential pronouns in PPs in Spanish is discussed in Bosque and Gutiérrez-Rexach (2009), which takes the non-reflexive pronoun in (41b) to be a true pronominal, and allows the syntax to provide two different analyses for a sequence like *hablar de*. This expression could be analyzed, on the one hand, as a complex predicate. In this case, the binding domain is the whole sentence, and the reflexive (*sí misma*) is bound in its governing category, as required. On the other hand, the preposition *de*, like other prepositions, may function as an independent predicate, creating its own binding domain. The relevant governing category would now be the PP headed by the preposition, and the pronominal (*ella*), being free in this domain, can be referentially dependent upon the sentential subject. Bosque and Gutiérrez-Rexach remark that a syntactic analysis along these lines for the coreferential reading of the nonreflexive pronoun in (41b) is independently motivated, as it has also been applied to those cases in which a pronominal within an object NP (42), or a nonselected adverbial phrase (43), corefers with the subject in a simple sentence (see Raposo 1985; Demonte 1989, and the references therein):

(42) a. Los niños vieron fotos de ellos.
'The boys saw pictures of them.'
b. Los niños$_i$ vieron [PRO$_{arb}$ fotos de ellos$_i$]$_{GC}$

(43) a. María vio una serpiente cerca de ella.
'Mary saw a snake near her.'
b. María$_i$ vio una serpiente$_j$ [PRO$_j$ cerca de ella$_i$]$_{GC}$

To end up, Otero (1999) explains the coexistence of an anaphor and a coreferential pronoun in examples like the ones in (41a,b) in a different way. He claims that these two sentences do not have the same meaning. As represented in (44), the reflexive pronoun must be interpreted as a bound variable (the second x in the formula in (44a)), whereas the non-reflexive pronoun gives rise to a coreferential reading without binding (hence the occurrence of *María*, and not the variable x, in the formula in (44b)):

(44) a. María siempre habla de sí misma.
'For María = x, x always talks of x'.
b. María siempre habla de ella.
'For María = x, x always talks of María.'

The construction in (44b) would therefore involve some sort of personality splitting (i.e., María as a talker is seen as dissociated from María as the theme of talking). Otero also points out that the use of this construction implies the attribution of a mental state to an animate entity, a property that characterizes the phenomenon known as 'logophoricity' (in general terms, as he reminds us, a logophoric pronoun takes as its antecedent an individual with a semantic role including the property of 'mental state', whose "words, thoughts, feelings or general state of consciousness" are described in the discourse). Otero then concludes that, in cases like the one in (44b), the pronoun functions as an unbound anaphor with a logophoric interpretation.

The empirical challenge for any account of the phenomenon under consideration is the observed fact that the coreferential reading of a nonreflexive pronoun included in a selected PP easily obtains with some verbs, but is highly anomalous with other verbs:

(45) a. Pedro$_i$ solo piensa en él$_i$.
Peter only thinks in he
b. Pedro$_i$ duda de él$_i$.
Peter doubts of he
c. #Pedro$_i$ abusó de él$_i$.
Peter abused of he
d. #Pedro$_i$ influyó en el$_i$.
Peter influenced in he

Otero's proposal might help us explain the contrasts in (45) since logophoric pronouns, as said above, are generally arguments of predicates of mental experience or communication. There is, however, considerable variation amongst native speakers in their judgments on the acceptability of coreferential pronouns in complement PPs, which makes it difficult to even describe the facts properly. An in-depth study of all the relevant data bearing on this issue should, accordingly, still be carried out.

6.2 *Emphatic and distributive pronouns*

So-called 'emphatic pronouns' are another case of a nonreflexive pronoun coreferring with the subject of a simple sentence in Spanish, which at first sight goes against the generalization that anaphors and pronominals have a complementary distribution. As defined in Piera (1987), emphatic pronouns, like *ella* in (46), are nominative forms that appear in a non-argumental position, together with a nominal expression in the canonical subject position on which they are referentially dependent:

(46) Julia$_i$ telefoneó ella$_i$.
Julia phoned she
'Julia herself phoned.'

Piera analyzes these forms as VP-adjuncts, and claims that they are syntactic anaphors that are bound in their governing category. Building on Piera's insights, Sánchez López (1996) takes a step forward in the characterization of emphatic pronouns, and holds that their occurrence is constrained by the aspectual properties of the predicate. She shows, in particular, that only predicates involving two subevents (a process plus a state) can combine with an emphatic pronoun (see the contrast in (47)), and proposes that this anaphor behaves as a non-argumental VP modifier, which is licensed by the existence of a final state in the subeventive structure of the predicate.

(47) a. María escribió ella la novela.
Mary wrote she the novel
'Mary herself wrote the novel.'

b. *Juan sabe él inglés.
John knows he English

To conclude, in the construction with distributive pronouns in a coordinated phrase illustrated in (48), the non-reflexive pronouns also behave syntactically as anaphors:

(48) Sus amigos$_{i+j}$ son él$_i$ de Madrid y ella$_j$ de Sevilla.
their friends are he from Madrid and she from Seville
'As for their friends, he is from Madrid and she is from Seville.'

Sánchez López (1995) attributes the fact that the pronouns in (48) do not satisfy principle B of Binding Theory to the presence of a covert distributive operator modifying the coordinated phrase whose quantificational nature allows these pronominal forms to occupy a bound position. As this linguist suggests, distributive operators would therefore function in this respect like floating universal quantifiers, which also license a non-reflexive pronoun as a bound anaphor:

(49) Sus amigos vendrán todos ellos a la fiesta.
their friends will.come all they to the party
'All of their friends will come the party.'

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NOTES

1 In addition to the issues reviewed in the text, Bosque (1985) compares the properties of the reciprocal construction with those of symmetrical predicates, linear structures, and distributive sentences.
2 Belletti (1982) claims that Italian reciprocals have a small clause-like internal structure, in which *l'uno* is the configurational subject of the subcategorized PP: *[$_{PP}$l'uno [$_{PP}$P l'altro]]*. For a different analysis of the internal structure of Spanish reciprocals see Quintana (2001).

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27 Empty Categories and Ellipsis

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1 Introduction: the limits of ellipsis

A rather striking and cross-linguistically prominent property of natural language is the presence of phonetically unrealized elements in a sentence necessary for its correct interpretation. This general phenomenon falls under the heading of ellipsis. There is much variety with respect to the distribution and interpretive conditions of the *gap* left behind by ellipsis. Nevertheless, three fundamental questions crucial for understanding a gap are: (1) is it internally atomic or internally complex?; (2) what syntactic conditions allow for its presence?; and (3) how is its semantic information recovered? In Section 2, we outline the range of elliptical constructions, and offer some brief remarks regarding their construction specific properties. In Section 3, we address the three fundamental questions just introduced.

2 Elliptical constructions

2.1 *Empty (pro)nominal categories*

As a starting point, we discuss cases of empty pronominal and nominal categories. Consider first the sentence in (1), which contains the null pronominal subject characterizing null subject languages (NSL):

(1) *pro* vino.
    pro come-PST.3sg

There is an unpronounced definite pronoun in subject position, represented as *pro*, an empty category first proposed by Chomsky (1982) (see also Jaeggli 1982). Consider another type of unpronounced subject in (2), represented by PRO:

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(2) Juan intentó PRO salir.
Juan try-PST.3sg PRO leave

Although both null elements are subjects, they show significant differences. As illustrated in (1) and (2), respectively, *pro* appears in finite contexts, while PRO is limited to nonfinite contexts. Additionally, whereas PRO is arguably found universally, *pro* is not; it is typically associated with "rich" verbal inflection (cf. Jaeggli and Safir 1989 for a different approach) or "strong agreement" (Chomsky 1982).

With respect to PRO, Chomsky (1981) argues that its properties and distribution follow from the binding conditions under the assumption that PRO is both [+pronominal] and [+anaphoric]. This entails that it must be both bound and free within its governing domain. Since these contradictory conditions cannot simultaneously be met, PRO must be ungoverned. Nonfinite Infl is not a governor, thus its position as the subject of a nonfinite clause. Observe in (2) that the controller (that is, the antecedent) of PRO must be the matrix subject *Juan*. This illustrates a case of *obligatory control*, which contrasts with cases of *non-obligatory control*, illustrated in (3):

(3) Juan piensa que PRO jugar a la lotería es malo.
Juan think-PRES.3sg that PRO play the lottery be-PRES.3sg bad

The subject of *jugar* need not be controlled by *Juan*.[1] In this case, an arbitrary interpretation of PRO results, such that playing the lottery is bad for *anyone*. PRO can also appear in adjunct clauses:

(4) PRO caminando despistado, Juan se cayó.
PRO strolling absentminded Juan SE fall-PST.3sg

In minimalist literature, obligatory control has been reanalyzed as movement of the controller from the position occupied by PRO (Hornstein 1999, 2000). This eliminates the need to posit the existence of PRO as a separate syntactic element since PRO is just a copy of the controller itself.

Returning to *pro*, it does not appear to be a single homogeneous element. We can differentiate between expletive subjects of unaccusative verbs in (5a), unspecified subjects in (5b), and null subjects of weather verbs in (5c):

(5) a. *pro* parece que Juan está enfermo (Jaeggli and Safir 1989)
pro seem-PRES.3s that Juan be-PRES.3sg ill
b. *pro* llaman a la puerta
pro knock-PRES.3pl at the door
c. *pro* llueve intensamente
pro rain-PRES.3sg intensely

There are also several ways in which *pro* differs from overt definite pronouns in Spanish (Fernández Soriano 1988, 1999; Luján 1999). One interesting difference regards the subject of embedded verbs. As observed by Montalbetti (1984), only when the embedded subject is *pro* can the matrix subject be coreferential; otherwise, non-coreference is obligatory, a situation known as obviation, as illustrated in (6a).[2] Note, however, that when the overt pronoun receives contrastive stress, the obviation effect is not present, as illustrated in (6b).

(6) a. Muchos chicos$_i$ dijeron que *pro*$_{\{i,j\}}$/ellos$_{\{*i,j\}}$ vendrían.
Many boys say-PST.3pl. that *pro*/they come-COND.3pl
b. Muchos chicos$_{\{i,k\}}$ dijeron que ELLOS$_{\{i,j\}}$ vendrían.

Studies on *pro* led to the proposal of an influential parameter, the *null subject* (or *pro drop*) *parameter* (Rizzi 1982, 1986). Several properties were argued to cluster around this parameter, lending support to the Principles and Parameter theory. In more recent literature, however, doubts have been raised regarding the co-occurrence of all of these properties (see Perlmutter 1971; Jaeggli and Safir 1989; Rizzi 1999; Boeckx (to appear) for relevant discussion). Moreover, there are languages without rich agreement, yet show rampant null subjects (and objects), such as Chinese. In these so-called "topic drop" languages, both subject and object pronouns can be dropped under the appropriate discourse conditions.

In addition to the empty pronominal elements *pro* and PRO, there are cases of ellipsis in Spanish in which only the head of the NP is elided, illustrated in (7). Two approaches have been offered that account for the gap in these constructions. The most traditional is syntactic in which a phonetically realized determiner licenses the gap, illustrated in (7a). The other is semantic and relies on the obligatory contrastive nature of the remnant. Eguren (2010) puts forth cases like (7b) as support for this latter approach:

(7) a. El ~~café~~ de Colombia es el mejor café que conozco.
the ~~coffee~~ from Columbia be-PRES.3sg the best coffee that know-PRES.1sg
b. Aquí crecen árboles grandes y ahí crecen ~~árboles~~ pequeños.
here grow-PRES.3pl trees big-pl and there grow-PRES.3pl ~~trees~~ small-pl

Eguren (2009, 2010) also argues that the empty nominal head is not the result of a PF-deletion operation. Ticio (2005) and Saab (2009) argue in favor of the deletion analysis. In general, the interpretation of the nominal gap is obtained by means of an antecedent in the discourse. However, there are also cases of nonanaphoric interpretation, as in *Los del gobierno nunca reconocen sus errores* 'People in government never admit their faults.' In such cases, the empty noun has to be interpreted as including the feature [+ human]. On the functioning of non anaphoric empty nouns, cf. Panagiotidis (2003).

If the DP is headed by the definite article, the complement of the elided N is obligatory, possibly due to its clitic nature, which requires obligatory cliticization in the domain of the DP. Note the contrast between (7a) and (8). Finally, only prepositional phrases headed by *de* can accompany the determiner. Contrast the use of *de* and *a* in (9).

(8) *El ~~café de Colombia~~ es el mejor café que conozco.
(9) *El tren {*a/de} Sevilla y el ~~tren~~ {*a/de} Barcelona han salido con retraso.
the train{*to/of} Seville and the ~~train~~ {*to/of} Barcelona have-PRES.3pl left with delay

Other cases of N-ellipsis involve *alguno* or *uno*, as in *algún profesor* > *alguno* 'some teacher > someone' and *un profesor* > *uno* 'a teacher > one,' respectively. The change in morphological form depends on the presence of the noun head; this suggests that these pairs may be two categorially different lexical items, a stance representative of traditional grammars. However, this categorial duplication raises doubts, since it should affect all classes of determiners and quantifiers. See Bosque (1989) for some arguments against the general categorial proliferation approach.

2.2 *Verbal ellipsis*

The unifying trait of verbal ellipsis is its lack of an overt verb, although other VP elements can also be phonetically unexpressed. The relevant constructions include gapping (Section 2.2.1), TP-ellipsis (Section 2.2.2), null complement anaphora (Section 2.2.3), and sluicing (Section 2.2.4), among others.

2.2.1 Gapping Characteristically, gapping occurs in coordinated structures. The verb in the second conjunct is elided when its meaning is recoverable from the preceding verb, as in (10). Some remnant must remain (contrast 10 and 11a), whether an argument or an adjunct.

(10) Álex toca el violín y Marta ~~toca~~ el piano.
Alex play-PRES.3sg the violin and Marta ~~play~~ the piano

(11) a. *Álex toca el violín y Marta ~~toca el violín~~.
b. Álex toca el violín desde pequeño y
Alex play-PRES.3sg the violin since little-masc.sg and
Marta ~~toca el violín~~ desde hace sólo dos años.
Marta ~~play the violin~~ since been only two years

Remnants of gapped sentences include contrastive topics, appearing before the gap, and contrastive focus (the VP-remnant), to its right. Although normally the subject, the contrastive topic can also be a topic at the left edge of the sentence: *A Antonia la convocó Luis y a María, ~~la convocó~~ Julia* (Lit.: 'To Antonia CLacc.3fem.sg call-PST.3sg Luis and to Maria ~~CLacc.3fem.sg call~~ Julia').

The gap can be interpreted by a transitive relation in which in a series of three verbs, the first provides the content for the second two: *Álex toca el violín, Marta ~~toca~~ el piano y Pascual ~~toca~~ el saxo* 'Alex plays the violin, Marta the piano and Pascual the sax.' Moreover, the antecedent of gapping must fulfill a strict requirement of parallelism; it has to be the first overt predicate of the most immediately preceding coordinate conjunct. Subordination fails this strong parallelism requirement.

The gap can contain material that is discontinuously arranged in the antecedent. So, for instance, it is not possible to interpret María in (12) as playing the piano in a place or day different from the ones contained in the first conjunct:

(12) Luis toca el violín el viernes en aquel bar y
Luis play-PRES.3sg the violin the Friday in that bar and
María ~~toca~~ el piano ~~el viernes en aquel bar~~.
Maria ~~play~~ the piano ~~the Friday in that bar~~

Given the discontinuous nature of the gap in (12), it appears not to form a unitary constituent, a unique property not found in other types of verbal ellipsis. Nevertheless, some authors – for instance, Brucart (1987) and Coppock (2001) – propose that the remnant right-adjoins higher in the structure, allowing for the target of deletion to form a unitary constituent.[2]

Nevertheless, some authors question gapping's status as ellipsis proper (see, for instance, Vicente 2010), in part due to the important role of information structure and discourse and in part due to the lack of strict structural requirements. In this vein, gapping has been proposed, originally by Goodall (1987) and more recently by Johnson (2009), to result from across-the-board (ATB) movement applied to coordinate structures. On this story, no deletion is necessary, only internal merge operations are required. The most controversial point of this coordination approach is that subject raising out of the first conjunct violates the Coordinate Structure Constraint (CSC) (Ross 1967) and the ATB rule application (Williams 1978) since its counterpart in the second conjunct is not involved in any movement operation. By adhering to the sideward movement theory of Nunes (2001), however, the CSC constraint can be circumvented. For an analysis of gapping in these terms, see Agbayani and Zoerner (2004).

2.2.2 Tp-ellipsis Unlike English, Spanish lacks VP-ellipsis. Spanish, however, does have TP-ellipsis, illustrated in (13):[3]

(13) Juan no invitó a María a la fiesta, pero
Juan not invite-PST.3sg to María to the party, but
Álex sí ~~invitó a María a la fiesta~~.
Álex yes ~~invite PST.3sg to María to the party~~

The second conjunct does not contain any tense affix, only the affirmative polarity adverb *sí*. Thus, the elliptical site in Spanish corresponds not to VP, but to the next higher projection: TP. Assuming that Laka's (1990) ΣP, a projection that houses negative and emphatic affirmative polarity items, is the functional head licensing TP-ellipsis, the structure of TP-ellipsis would be: [$_{\Sigma P}$*sí/no/también/tampoco* [$_{TP}$ ~~TP~~ …]]. Tense here is included in the gap. Therefore, the tense of the elliptical clause cannot differ from the tense of the antecedent clause. One of the four polarity items in Spanish is required to license TP-ellipsis. Suñer (1995) and Laka (1990) place *sí* and *no* in Σ^{o} and *también* and *tampoco* in Spec, Σ^{o}. These polarity elements can be preceded by modal, aspectual, or quantificational adverbs. Additionally, it appears that the overt constituent that appears in the first position of the elliptical clause in Spanish is a topic (López 1999), as the following data suggest:

(14) A María la han aprobado, pero
To María CLacc.3sg.fem have-PRES.3pl passed, but
a Luisa no ~~la han aprobado~~.
to Luisa not ~~CLacc.3sg.fem have-PRES.3pl passed~~

After López (1999), Depiante (2004) and Saab (2010) also point out the close relation that exists between TP-ellipsis and (clitic) left-dislocation.

Contrasting with gapping, observe that TP-ellipsis is available in turn-taking by different speakers. Thus, (15) can be a reply to *tengo prisa*:

(15) Yo también ~~tengo prisa~~.
I also ~~have-PRES.1sg hurry~~

Note also that in TP-ellipsis, a cataphoric relation can hold between the antecedent and the gap, but only when the elliptical clause and the one containing the antecedent belong to the same utterance: *Quizás usted no ~~se equivoca~~, pero la mayoría nos equivocamos* (Jorge Volpi, *En busca de Klingsor*) 'Maybe you don't, but most of us make mistakes.' Finally, as Brucart (1987) first observes, TP-ellipsis is compatible with subordination contexts:

(16) Yo creo que María tiene razón, pero Luis no.
I think-PRES.1sg that Maria have-PRES.3sg reason, but Luis not

Since either the antecedent or the elliptical clause can be embedded, (16) is ambiguous: the counterpart of *Luis* can be *yo* or *María*, illustrated in (17):

(17) a. [Yo creo que María tiene razón, pero Luis no
I think-PRES.1sg that Maria have-PRES.3sg reason, but Luis not
~~cree que María tenga razón~~]
~~believe-PRES.3sg that Maria have-PRES.3sg reason~~
b. Yo creo que [María tiene razón, pero Luis no
I think-PRES.1sg that Maria have-PRES.3sg reason, but Luis not
~~tiene razón~~]
~~have-PRES.3sg reason~~

Nonetheless, the interaction between TP-ellipsis and subordination is complex. As noted by López (1999), TP-ellipsis in Spanish includes parallelistic and contrastive requirements with respect to the antecedent. On the one hand, the relation with the antecedent cannot be hypotactic, but paratactic (Gallego 2009), precluding the matrix predicate as an antecedent. On the other hand, there are also parallelistic constraints related to information structure. Consider the example in (18):

(18) Luis tiene razón, pero creo que María no ~~tiene razón~~.
Luis have-PRES.3sg reason, but think-PRES.1sg that Maria not ~~have-PRES.3sg reason~~

In (18), the elliptical site is in a complement clause, but the relation with the antecedent is paratactic, which satisfies the parallelistic and contrastive requirements.

Bosque (1984) points out that only assertive predicates, which do not presuppose the truth of the subordinate clause, can select a clause with TP-ellipsis. There are, however, factive predicates that can take a subordinate clause with TP-ellipsis, as *reconocer* 'admit' or *saber* 'know.' Only emotive factive predicates (as *lamentar* 'regret'), which include an evaluation of the situation denoted by the complement and select subjunctive mood in the subordinate, exclude TP-ellipsis.

2.2.3 Sluicing Sluicing refers to a phenomenon in which a wh-phrase introducing an indirect question appears in isolation, as in (19):

(19) Alguien me habló, pero
Somebody CLdat.1sg speak-PST.3sg, but
no recuerdo quién ~~me habló~~.
not remember-PRES.1sg who ~~CLdat.1sg speak-PST.3sg~~

Note that some authors include *Focus Sluicing* as a variant of this pattern. In these constructions the remnant is not a wh-word, but a focused NP: *Alguien me habló, creo que tu madre* 'Somebody told me I think that it was your mother' (cf. van Craenenbroek and Lipták 2009).

The unpronounced portion in (19) is often analyzed as an entire CP whose TP has been elided (originating with Ross 1969, see also Merchant 2001; cf. van Riemsdijk 1978 for a different account), illustrated in (20):

(20) Alguien me habló, pero no recuerdo [$_{CP}$ quién [$_{TP}$ ~~me habló~~]]

One important argument in favor of the full CP account is that only predicates that select for indirect question CP complements allow sluicing (see Merchant 2001 and references therein for several other arguments for a full CP analysis of sluiced wh-phrases from several languages), as illustrated in (21), taken from Brucart (1999):

(21) Se entrevistó con alguien, pero no {dijo/*aceptó} con quién.
SE meet-PST.3sg with somebody, but not {say/*accept}-PST.3sg with whom

Sluices need to be licensed by an antecedent in the clause (although there are cases of discourse antecedents as well; see Brucart 1999). One leading idea, coming from studies on VP ellipsis (see for instance Fiengo and May 1994), is that there is a structural isomorphism between the elided TP and its antecedent. However, this is easily shown not to hold:

(22) Admite haberle conocido, pero no recuerda
admit-PRES.3sg have-CLacc.3sg known, but not remember-PRES.3sg
cuándo [~~le conoció~~]/[*~~haberle conocido~~]
when [~~CLacc.3sg know-PST.3sg~~]/[*~~have-CLacc.3sg known~~]

These facts suggest that there are semantic conditions on the licensing of a sluice (see Merchant 2001).

Although there is no structural isomorphism with the antecedent, the elided TP is often taken to have fundamentally the same internal structure of non-elided TPs. In this respect, the lack of pronunciation is just the result of a PF-phenomenon. However, under this assumption, the contrasts in (23), noted by Ross (1969), raise questions:

(23) a. Es cierto que ha alegado una prueba concluyente,
be-PRES.3sg true that have-PRES.3sg provided a proof conclusive,
pero no pienso decir cuál.
but not think-PRES.1sg say which
b. *No pienso decir qué prueba concluyente es cierto
*not think-PRES.1sg say which proof conclusive be-PRES.3sg true
que ha alegado.
that have-PRES.3sg provided

If it is the case that the internal structure of the elliptical site has the same syntax as overt TPs, then how is it that sluicing can avoid the island violation in (23b)? The complex NP is just one example where sluicing avoids island violations. See Ross (1969), Brucart (1999) and Chung et al. (1995) for others. One possibility is explored in Merchant (2001), who proposes that some islands are PF-phenomena, and thus, given that the elliptical site is the result of PF-deletion, there is no island violation at all.[4]

2.2.4 Null-complement anaphora In constructions with null-complement anaphora, the infinitival or sentential complement of a restricted class of predicates is covertly realized, as in (24):

(24) Intentó entrar en el edificio, pero no pudo ~~entrar en el edificio~~.
try-PST.3sg enter in the building but not can-PST.3sg ~~enter in the building~~

Verbs allowing null-complement anaphora (henceforth, NCA) constitute a miscellaneous class: modals (*poder* 'can,' *querer* 'want,' *deber* 'must,' *saber* 'be able'), aspectuals (*empezar a/comenzar a* 'start,' *acabar de/terminar de* 'finish,' *dejar de* 'leave,' *soler* 'do usually'), causatives (*dejar* 'let,' *autorizar a* 'authorize,' *ayudar a* 'help,' *enseñar a* 'teach,' *informar de* 'report,' *invitar a* 'invite,' *obligar a* 'force,' *persuadir de* 'persuade,' *disuadir de* 'dissuade'), and verbs that – in terms of Bosque (1984) – denote "attitudes, predispositions and purposes" of the subject (*aceptar* 'accept,' *aprender a* 'learn,' *dudar de* 'doubt,' *rehusar* 'refuse,' *renunciar a* 'give up'). The latter class also includes pronominal verbs that take prepositional clausal complements (*acordarse de* 'remember,' *decidirse a* 'make up one's mind.' *enfadarse por* 'get angry,' *extrañarse de/por* 'be surprised,' *negarse a* 'refuse,' *olvidarse de* 'forget,' *oponerse a* 'be opposed,' *quejarse de* 'complain,' among others).

Many NCA verbs take a prepositional clausal complement, which precludes the use of the accusative clitic *lo*. This accords with Brucart's (1999) observation that, in general, predicates that accept NCA do not allow clitic *lo* to represent their clausal complement. For nonprepositional NCA verbs, the incompatibility with *lo* can be explained semantically. Modal and aspectual verbs tend to reject accusative clitics (**Lo* {*puede/suele/quiere*} vs. {*Puede/suele/quiere*} *hacerlo*). The only potential exception is *saber*, although only on its nonmodal use: *Sabía que pasaría* > *Lo sabía* '(S)he knew that it would happen > (S)he knew it.' When referring to some capacity of the subject, *lo* is only available with the verbal proform: *Intentó resolver el problema, pero no* {(**lo*) *supo/supo hacerlo*} '(S)he intended to solve the problem, but (s)he didn't know to do it,' a behavior also found with causative verbs like *dejar*.

Consider *aceptar* and *rehusar*. Neither take prepositional clausal complements, yet both accept NCA and *lo*. When these verbs allow NCA, however, they refer to performative acts of accepting or refusing an offer ('saying yes or no'). In contrast, with *lo*, they denote mental activities related to making a decision ('accepting/ refusing something').

Hankamer and Sag (1976) characterize NCA as deep anaphora, in which the gaps are atomic, lacking internal structure. Depiante (2001) reaches the same conclusion, based on the impossibility of extracting material from the elliptical clause – previously observed by Zubizarreta (1982) – illustrated in (25) by the unavailability of clitic climbing:

(25) *Juan las puede ver y María también *las* puede.
Juan CLacc.3pl.fem. can-PRES.3sg see and Maria also CLacc.3pl.fem can-PRES.3sg

However, as noted by Saab (2009), extraction from a NCA is allowed in examples like *La chica* [*a la que quisiste entrevistar pero no pudiste* Ø] *está hablando por TV* 'The girl you tried to interview but couldn't is speaking on TV,' where the relative pronoun is the direct object of *entrevistar* and the covert predicate.

A fact related to the atomic nature of the gaps in NCA, is the lack of epistemic readings of modals with NCA, originally noted by Sáez del Álamo (1987). In (26), the subjects only receive a root interpretation in which they have the ability, or permission, to translate the article:

(26) Juan puede traducir ese artículo y Pedro también puede.
Juan can-PRES.3sg translate this paper and Pedro also can-PRES.3sg

In contrast, *Juan puede traducir ese artículo* has the root and epistemic interpretation ('It is possible that Juan translate that article'). Analyzing only epistemic modals as raising verbs, the asymmetry falls out since raising the subject from the second conjunct in (26) is unavailable in NCA.

2.2.5 Other kinds of verbal ellipsis In this section we merely point out some other cases of verbal ellipsis. The first is *Coordination Reduction* (CR), exemplified in (27):

(27) Juan le regaló a María un libro y
Juan CLdat.3sg gave-PST.3sg to Maria a book and
~~le regaló~~ a Pedro un atlas.
~~CLdat.3sg gave-PST.3sg~~ to Pedro an atlas

Like gapping, CR always occurs in coordinated structures. However, unlike gapping, the elliptical site appears immediately after the coordinate conjunction.[5]

It appears that the overt material in the second conjunct in (27) forms a single constituent, entailing that there is no phrasal coordination without ellipsis here, although treatments of CR without ellipsis employing a three-dimensional approach to coordination can be found. Another alternative employs sideward movement (see Nunes and Uriagereka 2000; Nunes 2001; and Fernández-Salgueiro 2008, for Spanish data).

The second is *Comparative ellipsis*, which occurs in comparative clauses taking a complement headed by *que* or *como*:

(28) Juan tiene tantos hermanos como ~~hermanos tiene~~ María.
Juan have-PRES.3sg as-many brothers as ~~brothers have-PRES.3sg~~ Maria

In (28) the material covertly realized is lexically identical to its correlate. Only the overt elements are different and, consequently, contrastive. This is reminiscent of other parallelistic ellipses, although note that the correlate of the quantified phrase is preposed (cf. *Juan tiene tantos hermanos como hermanas tiene María* 'Juan has as many brothers as Maria has sisters'). For a detailed analysis, see Reglero (2007); Brucart (2003). In comparatives of inequality, the complement can also be introduced by *de* (*Juan tiene más hermanos de los que tiene María*). In such cases, the subordinate clause is a quantitative relative and a verb must be overtly realized (**Juan tiene más hermanos de los que María*). Nevertheless, in a completive clause selected by the verb of the quantificational relative, the repeated predicate can be elided: *Juan tiene más hermanos de los que decía que tenía*' Juan has more brothers than he said he had.'

Last, there is *Stripping*. Stripping consists of an affirmative or negative polarity operator preceding an argument of the covert verb:

(29) Se lo dijo a Pedro, pero
CLdat.3sg CLacc.3sg.masc tell-PST.3sg to Pedro, but
no ~~se lo dijo~~ a María.
not ~~CLdat.3sg CLacc.3sg.masc tell-PST.3sg~~ to Maria

The example in (29) could be a stylistic version of TP-ellipsis, but Depiante (2000, 2004) has shown that both differ in a number of ways. Stripping resembles gapping, and differs from TP-ellipsis, in two important aspects. First, it is only available in coordination. *A María no* can be a valid answer to a question like *¿A quién se lo ha dicho Pedro?* 'Who has Peter told it to?,' but, in the same context, it would be less natural to utter *No a María*. This argument, nevertheless, should be qualified. As Vicente (2006) notes, negative answers are possible in dialogue contexts: RENATA: [...] *El número trece trae mala suerte.*/MALVINA: *No a toda la gente* (Emilio Carballido, *Las cartas de Mozart*) ' – The number 13 brings bad luck. / – Not to everybody.' Second, stripping cannot occur in a subordinate structure. For an analysis of these constructions relying on information structure principles, see Vicente (2006) and Depiante and Vicente (2009).

2.3 *Fragments*

Sentence fragments constitute the most radical version of ellipsis since they do not present the customary binary structure – subject–predicate or topic–comment – that characterizes clausal projections. There are many variants, but the most common are answer fragments:

(30) Q: –¿Qué hará el rector? A: –Avisar a la decana.
what do-FUT.3sg the rector call to the dean-fem

Structurally, any phrasal projection below TP can constitute a fragment, including a sentential answer to (30), which reproduces tense information: *Avisará a la decana*. A pseudo-cleft construction makes the different grammatical statuses of these two answers apparent since only subsentential constituents can function as a focus:

(31) Lo que hará el rector {es/será} avisar/*avisará a la decana.[6]
It that do-FUT.3sg the rector {be-PRES.3sg/be-FUT.3sg} call/call- FUT.3sg to the dean-fem

Nevertheless, see Barton (1990), Ginzburg and Sag (2000), and Valmala (2007) for a nonellipsis account of fragments. Those that deny their elliptical status tend to avoid the term 'fragment', preferring to call them nonsentential constituents.

In spite of their apparent nonsentential nature, fragments have propositional content. When argumental, they receive theta-roles, other grammatical indices, as well as accusative preposition *a*:

(32) Q: –¿A quién avisará el rector? A: –A la decana.
to whom call- FUT.3sg the rector to the dean-fem

The accusative preposition is obligatory even though the interrogative element in the question also contains it, and thus the fragment behaves as if it occurs within a sentential context. In fact, Morgan (1973) and Hankamer (1979) propose a sentential approach in which fragments result from in-situ deletion of all the constituents that are void of phonetic content. Thus, (32) would be: [$_{TP}$ ~~el rector avisará~~ a la decana].

One important problem for such a view is that the covertly realized elements need not form a unitary constituent, which clashes with the standard view that syntactic processes are constituent-governed. To circumvent this problem, Merchant (2004) proposes what is presently the standard sentential analysis for fragments. Establishing a parallelism with sluicing and relying on the fact that fragments have focal content in the discourse, Merchant analyzes fragments as sentential constituents that undergo leftward movement to the specifier of a Focus Projection (FP), located immediately above CP. Moreover, a feature E in C^{o} induces the covert realization of all material contained in the complement of this head. The derivational mechanism that Merchant (2004) proposes for sluicing is similar, but not identical, to the one advocated for fragments. In sluicing, the wh-word is placed in the specifier of CP. This difference can account for their different sensitivities to islands. Although, the sensitivity of fragments to island effects is not uncontroversial for some authors.

Not all fragments are linked to a discourse antecedent, however; deictically dependent fragments also exist whose interpretation is based on the extra-linguistic situation, as discourse-initial fragments illustrate. To accommodate these cases, Merchant (2004) proposes the existence of covertly realized proforms present in the structure that can be pragmatically recovered when they represent an entity or action perceptually salient in the speech situation. Although this handles fragments without introducing special syntactic processes, it is not void of empirical or theoretical problems, as noted in Valmala (2007). Perhaps the most remarkable is that the movement analysis Merchant proposes is equivalent to focus fronting, an operation usually restricted to contrastive focus and not available in contexts of information focus. Information focus merely identifies the value of a variable among a list of potential candidates. In contrast, contrastive focus presents the value of a variable as opposed to a concrete value previously present in the discourse context. The answer to a wh-interrogative is interpreted as information focus. Fragments can have contrastive import, although they do not require it. Finally, it is interesting to note that fragments seem more related to pseudo-clefts than to contrastive focus constructions: *Lo que hará el rector es avisar a la decana* 'What the rector will do is call the dean' constitutes a perfect paraphrase of (31), a construction available to all fragments.

3 The nature, licensing, and interpretation of the gap

Elliptical constructions vary widely. Nevertheless, each construction raises the same general questions about the nature, licensing, and interpretation of the gaps resulting from the process of ellipsis. In this section, we briefly discuss the issues relevant to the internal nature of the gap (in Section 3.1), its licensing conditions (in Section 3.2), and the recovery of the semantic information contained in it (in Section 3.3).

3.1 The nature of the gap: deep and surface anaphora

Elliptical elements are phonetically void and interpretively contentful. Theoretically, there are different ways to analyze them. One is to not represent them syntactically; their interpretation would be obtained via semantic rules of a contextual nature (Ginzburg and Cooper 2004). Although a possibility, it is commonly held that there are null categories in the syntax, thus we do not pursue the semantic route here, nor its consequences for the grammar.

There are two different ways of employing null categories in the syntax. The first option – *the empty category analysis* – assumes that the covert unit is atomic, lacking any syntactically relevant internal structure, but is visible at the conceptual–intentional (C–I) interface. Its interpretation is a function of its antecedent via a process identical to that applied to pronouns (Lobeck 1995) or a semantic rule that copies the antecedent's information at LF (Fiengo and May 1994). The second option – *the deletion analysis* – assumes that the elliptical material is present in the syntax with its full lexical content, and a deletion mechanism activates its non-pronunciation at the articulatory–perceptual (A–P) interface (Merchant 2001, Johnson 2001). Within generative approaches, both analyses have been extensively argued for and evaluated. Therefore, the current approach to elliptical phenomena tends to be non-unitary, a situation that is reflected in the very title of this chapter: whereas argument ellipsis is accounted for with pronominal empty categories like *pro* and PRO, verbal ellipsis is frequently analyzed as the merger of overt material into the syntactic derivation and its non-pronunciation activated at the A–P interface.

A twofold approach to ellipsis phenomena was first advocated by Hankamer and Sag (1976), a seminal work that classifies all anaphoric constructions (both overt and covert) into two different groups: deep anaphora and surface anaphora. The distinction arises from several diagnostics reflected in (33):

(33)

	Deep anaphora	*Surface anaphora*
a. Is it a syntactically atomic entity?	Yes	No
b. Can it be deictically controlled?	Yes	No
c. Are missing antecedents allowed?	No	Yes
d. Does it require syntactic parallelism?	No	Yes
e. Is extraction from inside the anaphoric site possible?	No	Yes

Deep anaphors are syntactically atomic, and their interpretation is obtained from semantic rules establishing an anaphoric or deictic relation with other entities present in the discourse or in the situational context. Personal pronouns are paradigmatic of deep anaphora. Hankamer and Sag (1976) also include certain verbal and nominal proforms, as well as one elliptical construction: NCA. In contrast, surface anaphora is syntactically complex, possessing fully-fledged structure, although opaque due to its covert realization. For reasons of space, we cannot analyze all the diagnostics in (33), but see Saab (2008), for a critical summary of the arguments relevant to the deep/surface anaphora distinction, and Depiante (2000), for a detailed study supporting it. In this section, we will briefly comment on (33b,c).

Deixis can pragmatically orient the interpretation of deep anaphora, contrasting it with surface anaphora, which only permits an anaphoric interpretation. This is illustrated in (34), where # stands for pragmatic inadequacy:

(34) a. [Speaker is trying to open door]
– No puedo.
not can-PRES.1sg
b. [Speaker is looking for lost document]
– #No sé dónde.
not know-PRES.1sg where

Some linguists have argued that pragmatic control is not impossible in gapping, especially in highly conventionalized situations. So, for instance, an utterance like *Yo una caña* (Lit.: 'I a small beer') could be an order addressed to a waiter. Nevertheless, in general, deictic resolution is far more difficult in surface than in deep anaphora constructions.

Now consider (35), which illustrates missing antecedents (Grinder and Postal 1971):

(35) Luis no ha estado en ningún país asiático, pero María sí
Luis not have-PRES.3sg been in any country Asian but Maria yes
~~ha estado en un país asiático~~ y *pro* le encantó.
have-PRES.3sg been in a country asian and *pro* CLdat.3sg love-PST.3sg

The antecedent of *pro* cannot be *ningún país asiático*, but only the covert argument contained in the elided TP. If that TP were atomic, anaphorically referring to one of its subcomponents would be mysterious.

With respect to this test, TP-ellipsis and NCA go in opposite directions, as shown by (36), from Saab (2008: 54):

(36) a. Pablo no pudo encontrar ningún libro, pero yo sí
Pablo not can-PST.3sg find any book but I yes
{??pude/~~pude~~} ~~encontrar un libro~~ y me gustó leerlo.
{can-PST.3sg} ~~find a book~~ and CLdat.1sg like-PST.3sg read-CLacc.3sg

Deep anaphora, like NCA, does not accept missing antecedents, while surface anaphora, like TP-ellipsis, does.

3.2 *The licensing of the gap*

In general, licensing ellipsis requires some local structural recoverability conditions to detect the gap. These conditions follow directly from X-bar theory principles. So, for instance, the occurrence of a complement or an adjunct inside a VP implies the presence of the corresponding verb since heads are structurally obligatory, contrasting with complements and specifiers. This holds for gapping, as noted in Section 2.2.1; the presence of a VP remnant is crucial for its licensing. Likewise, TP-ellipsis is licensed by the presence of a polarity adverb, as discussed Section 2.2.2.

Another strategy of licensing ellipsis derives from the close relation between certain functional categories and their related lexical projections. So, for instance, determiners and quantifiers take NPs, polarity heads take TPs, and modal and aspectual verbs take VPs. These functional elements can license ellipsis:

(37) a. Le gustaban los grabados y
cldat.3sg like-impf.3pl the engravings and
compró *dos* ~~grabados~~ de artistas italianos.
buy-pst.3sg two ~~engravings~~ of artists Italian-pl
b. Luis aceptó la propuesta, pero María *no* ~~aceptó la propuesta~~.
Luis accept-pst.3sg the proposal but María not ~~accept-pst.3sg the proposal~~

Finally, a third mechanism of local recoverability relates to the presence of the gap's phi-features in the sentence, as in null subject constructions. Only the subject's tense and person features are expressed by the tensed verbal form, not the object's. Thus, only the subject can be elided:

(38) *María solicitó el puesto y yo entrevisté *pro*
*María apply- pst.3sg the position and I inerview-pst.1sg *pro*

Additionally, as implemented in various forms, clitic pronouns affixed to the verb, with the corresponding phi-features, can license an empty pronominal internal argument (Jaeggli 1982), illustrated in (39):

(39) María solicitó el puesto y yo la entrevisté *pro*
Maria apply-pst.3sg the position and I clacc.3sg.fem interview-pst.1sg *pro*

3.3 *The interpretation of the gap*

The lack of phonetic content inhibits elliptical constituents from functioning as focus or contrastive elements in the information structure of the sentence (Merchant 2001). Therefore, a question like *¿Quién ha hecho eso?* 'Who has done this?' can be answered with *He sido yo* 'It was I' or just with *Yo*, but not with a null subject (**He sido*), as it is the informative focus of the sentence in this case. The same holds when its function is a contrastive topic.

A gap's semantic content must be accessible via the discourse or situational context, as in (40), where the elliptical element is interpreted via its antecedent in the first conjunct:

(40) Juan trajo un pastel y María ~~trajo~~ chocolate
Juan bring-PST.3sg a cake and Maria bring-PST.3sg chocolate

In (40), the pronominal features of the covert subject are recovered from the finite verb, which carries number and person information, as well as from the discourse context itself, which must make available the gender information not provided by the verb:

(41) ~~Ella~~ salió de la habitación sigilosamente.
~~She~~ leave-PST.3sg from the room stealthily

There also exist covert elements that are interpreted cataphorically, such that the element that fixes the interpretation of the gap appears after it:

(42) Si puedes ~~traerlo~~, tráelo.
if can-PRES.2sg ~~bring-CLacc.3sg.masc~~ bring-IMPTV.2sg CLacc.3sg.masc

In all the previous examples, the interpretation of the elliptical element is carried out via the discourse. In other cases, the process is deictic: there is some element in the situational context in which the sentence is produced that fixes the content of a phonetically empty element:

(43) ~~Tú~~ no preguntes. ¡Y ~~tú~~ ven ~~aquí~~!
~~You~~ not ask-PRES.SUBJ.2sg and ~~you~~ come-IMPTV ~~here~~
~~Yo~~ voy ~~allí~~.
~~I~~ go-PRES.1sg ~~there~~
(Demetrio Aguilera Malta, *Una pelota, un sueño y diez centavos*).

In the dialogue in (43), the null subjects denote participants in the speech act (hearer and speaker, respectively). On the other hand, the complement that determines the goal of motion for the verbs *venir* and *ir* is also interpreted deictically: *venir* expresses motion that converges with the position of speaker and *ir* motion that diverges with respect to this same reference point. When the referent of the elliptical category cannot be recovered anaphorically or deictically, the empty pronominal receives a non-specific interpretation, illustrated in (44). This also holds for impersonal and passive *se* sentences in Spanish:

(44) *pro* llaman a la puerta
pro knock-PRES.3pl at the door

Finally, when the elliptical element does not have any interpretive import, as in (45), its presence is due to strictly structural principles:

(45) *pro* nevó intensamente
pro snow-PST.3sg intensely

3.3.1 Partial identity in ellipsis In some cases, the elliptical constituent does not contain the same grammatical features as its antecedent. There is only partial identity, as in (46):

(46) a. Julián es sincero, pero María no ~~es sincera~~.
Julian be-PRES.3sg honest but María not ~~be-PRES.3sg honest~~
b. Andrea ha estado en México una vez y
Andrea have-PRES.3sg been in Mexico one time and
yo ~~he estado en México~~ dos ~~veces~~
I ~~have-PRES.3sg been in Mexico~~ two ~~times~~

In (46a), the antecedent and the elided constituent have different gender features. In (46b), the verbal forms have different values for person, and the number features of the QPs are distinct. Regardless, ellipsis is possible. Nevertheless, mismatch of these same grammatical features is not always acceptable under ellipsis, as the contrasts in (47) show:

(47) a. *El gato de Alicia y la ~~gata~~ de Luisa son de la misma raza.
b. El gato de Alicia y los ~~gatos~~ de Luisa son de la misma raza.
The cat of Alicia and the-{*fem.sg/masc.pl} of Luisa be-PRES.3pl of the same breed

When a noun is elided, its number can differ from its antecedent's, as in (47b); in contrast, gender must be the same (47a). Note that when the antecedent is an adjective, both phi-features can diverge: *Julián es sincero, pero tus amigas no ~~son sinceras~~*, literally Julián be-PRES.3sg honest-masc.sg, but your friends not ~~be-PRES.3sg honest-fem.pl~~.

This asymmetry is reminiscent of the opposition between interpretable and uninterpretable features proposed in Chomsky (1995), although not identical to it. On the one hand, uninterpretable features, licensed by an agreement relation with an interpretable feature, do not have to be licensed for ellipsis to take place. In contrast, interpretable features show a mixed pattern. As (47b) shows, strict number identity with the antecedent is not obligatory for a noun to be elided, although this does appear to be the case for gender. A possible approach to this contrast is to elaborate on the different nature of gender and number in nouns. As has been remarked in the literature (Ritter 1991), whereas number is a purely syntactic category, gender is part of the lexical information of nouns as an inherent property, possibly as a word-marker (Bernstein 1993). The difference between both categories with respect to partial identity in ellipsis, then, would be rooted in the fact that gender is a non-interpretable feature, as proposed in Alexiadou et al. (2007: 239). A slightly different approach is taken in Saab (2008), who provides an analysis of partial identity in ellipsis using the framework of Distributed Morphology

(Embick and Noyer 2001): the difference comes from the fact that gender features are included in the elliptical NP, whereas number information is contained in a functional projection placed above the lexical layer of the noun. Therefore, gender features are obligatorily included in the elliptical site, as opposed to number.

Consider tense features with respect to partial identity. As (48) shows, the tense information of an elliptical verb, in contrast to person features, cannot differ from its antecedent's:

(48) a. *Antonio viajó ayer y yo ~~viajaré/viajo~~ mañana.
Antonio travel-PST.3sg yesterday and I ~~travel-FUT/PRES.1sg~~ tomorrow
b. Antonio viaja hoy y yo ~~viajo~~ mañana.
Antonio travel-PST.3sg today and I ~~travel-PRES.1sg~~ tomorrow

This contrast derives directly from the interpretability of tense features, which must be strictly identical in ellipsis, and the uninterpretability of person features, which are attached via agreement with the subject, and need not be identical. The grammaticality of (48b), despite the different temporal references of the time adverbs, indicates that the constraints governing identity in ellipsis are purely formal since present tense admits a prospective or futurate interpretation, thus allowing the presence of both temporal adverbs and complying with strict identity under ellipsis.

Nevertheless, in comparative clauses it is possible to delete a verb with a tense differing from the antecedent's:

(49) Ahora tiene más experiencia que ~~experiencia tenía~~ hace un año.
now have-PRES.3sg more experience than ~~experience have-PRES.3sg~~ been one year

Not all linguists regard comparative clauses as elliptical (cf. Sáez del Álamo 1999, for instance), but the standard view since Bresnan (1973) is that they include covert elements (see Lechner 2004; Brucart 2003 for an updated discussion). If the latter view is endorsed, something additional is needed to explain the contrast between (49) and (48a), most likely two factors: (1) gapping occurs in coordination, whereas comparative deletion is related to a subordinate structure; and (2) the corresponding interpretations also differ from one another: a propositional event in gapping and a quantified magnitude in comparative ellipsis. Note indeed that the clause introduced by *que* in (49) is interpreted as 'the degree of experience that (s)he had a year ago.'

3.3.2 Nonidentity in ellipsis The most extreme instances of nonidentical antecedents are examples in which the covert element includes a lexical unit that is different from the one functioning as its counterpart in the antecedent, as in (50):

(50) Luis se presentó a sí mismo al ver que el presidente de la sesión
Luis SE introduce-PST.3sg to him self to see that the president of the session

se olvidaba {~~de presentarse a sí mismo~~/~~de presentar a Luis~~}.
SE forget-IMPF.3sg {~~of introduce to him self~~/~~of introduce to Luis~~}

Example (50) is ambiguous, as the reconstruction of the corresponding elliptical sites displays: they admit a *strict identity reading*, which reproduces the lexical material of the antecedent, and a *sloppy identity reading*, where the reconstructed material is different from the antecedent. These facts are referred to as *Vehicle Change Effects* by Fiengo and May (1994), who propose an LF coindexing mechanism of reconstruction that allows the elliptical site to include a pronominal token different from the corresponding counterpart in the antecedent if certain syntactic identity conditions are met. Accordingly, variables or R-expressions can be reconstructed with a pronominal correlate. Consequently, the sloppy reading of (50) is possible because there is a similar relation between the anaphor *sí mismo* and the pronominal clitic *lo*: *Luis se presentó a sí mismo al ver que el presidente de la sesión se olvidaba de presentarlo*, where the enclitic form *lo* is coreferential with *Luis*. See Merchant (2001) for an alternative proposal based on the contrast between focus and given information.

4 Conclusion

The process of ellipsis results in gaps in a sentence, gaps that are syntactically present, contribute to interpretation, yet have no phonetic realization. This chapter has reviewed a variety of constructions that fall under the heading of ellipsis: empty nominal and pronominal categories, verbal ellipsis (including gapping, TP-ellipsis, sluicing, and null-complement anaphora), and fragments. Although in many ways a heterogeneous set of constructions, we have seen that each raises important questions regarding the nature, licensing and interpretation of the gaps that unite them all under the heading of ellipsis. Providing answers to these important questions raised by this phenomenon of natural language promises to offer a deeper understanding of the mechanisms of language that allow us to interpret the variety of empty elements that surface in syntax. The present overview is a first step toward such an understanding.

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NOTES

1 However, it can be, as illustrated by *Juan piensa que jugar a la lotería es malo para su bolsillo* 'John thinks that playing the lottery is bad for his wallet.'
2 Our anonymous reviewer suggests a related approach in which the remnant moves leftward to the specifier of some functional projection above VP, the gap being its complement. Oehrle (1987) and Johnson (1994) propose that the category of the second conjunct in gapping is vP, not TP. See also López and Winkler (2003).
3 See Zagona (1988) and Lobeck (1999), who investigate the lack of VP-ellipsis in Spanish and the lack of TP ellipsis in English.
4 The avoidance of island violations by Sluicing is only possible when the counterpart of the wh-word is an explicit indefinite NP. See Chung and al. (1995) and Saab (2010) on this point.
5 It would be possible to argue that (27) is an instance of gapping where the first constituent is topicalized, or, alternatively, by proposing that there is a null empty pronominal subject (*pro*) in the second coordinated conjunct. However, these analyses are inadequate for languages that license structures like in (27) but are not pro-drop languages or have more constrained topicalization procedures, such as English.
6 In colloquial Spanish, if a complementizer precedes the focus constituent, (31) becomes grammatical, as spoken at an interview in Venezuela: *Mi mamá lo que hizo fue que le puso adhesivo ahí* 'What my mother did was that she put plasters there.' As a subordinator, the complementizer allows the clausal CP to function as focus.

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28 Word Order and Information Structure

ANTXON OLARREA

1 Introduction: free word order in Spanish

Spanish allows for a fairly unrestricted ordering of constituents in simple and embedded declaratives. Thus, a transitive verb like *comprar* 'to buy,' realizing all its arguments, an (optionally overt) subject pronoun *él* 'he,' and a direct object *el periódico* 'the newspaper,' and with a temporal adjunct *todos los días* 'every day,' can appear in a sentence in, at least, the following constituent orders:

(1) a. (Él) compraba el periódico todos los días
 (He) used-to-buy the newspaper every day
 b. (Él) compraba todos los días el periódico
 c. El periódico, (él) lo compraba todos los días
 The newspaper, (he) cl-it used-to-buy everyday
 d. Él, el periódico lo compraba todos los días
 e. El periódico lo compraba (él) todos los días
 f. El periódico es lo que (él) compraba todos los días
 The newspaper is what (he) used-to-buy everyday
 g. El periódico es lo que compraba todos los días (él)
 h. Compraba (él) el periódico todos los días
 i. Todos los días compraba (él) el periódico.

All the sentences in (1) have the same propositional content: it would be difficult to imagine a situation in which any one of them is true but the rest are not. The example (1a) has been considered in the grammatical tradition the basic, unmarked S(ubject) V(erb) O(bject) order in Spanish. Examples (1b–i), on the other hand, show derived word orders. These syntactic varieties of the same sentence differ from each other not only in the order of their constituents, but also in their discourse informational content and in their prosodic properties: in (1) specific constituents

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are interpreted as either background or new information, or as belonging to a restricted set of options. They are emphasized with respect to other sentential constituents or may show different intonation patterns to express syntactic and informational prominence. Even though all of them are well-formed sentences that satisfy all the syntactic, morphological, semantic and phonological principles of Spanish grammar, not all of them can be used interchangeably in a given discourse context.

The purpose of this chapter is to offer a brief overview of the research related to the interaction between syntax, informational content, and prosodic phonology in Spanish that accounts for the differences exemplified above, and to outline open issues and venues for future research. Due to space limitations the discussion will be restricted entirely to simple declarative sentences. The chapter is organized as follows: Section 2 describes the properties of the neutral SVO order in Spanish and defines the notions topic and focus as associated with the concept of Information Structure; Section 3 is dedicated to the constructions that codify topic and focus in Spanish from a strictly descriptive point of view; Section 4 summarizes the different accounts on topicalizing and focalized constructions. Finally, Section 5 offers some open research questions and concluding remarks.

2 The SVO order and information structure

2.1 *Focus and topic in SVO*

The traditional way to posit a neutral or unmarked word order in a language is to establish the compatibility of a sentence with an informational or out-of-the-blue question, such as *what happens/happened?* In Spanish, the unmarked SVO order in *Juan compró el periódico* 'Juan bought the newspaper' is a felicitous response to such a question, a sentence in which the entire utterance is new or asserted. This neutral order also presents a characteristic neutral prosodic pattern in Spanish in which the Nuclear Stress or intonation peak of the sentence falls in the rightmost stressed syllable, in this case the stressed syllable in *periódico* (see Zubizarreta 1994 and Chapters 7 and 8, above).

Juan compró el periódico, with the same neutral intonation described above, can also be a felicitous answer to questions that establish a different discourse context (e.g. *What did Juan buy?*). Now only the DP object *el periódico* in the answer provides new information, while *Juan compró* is given or previously known information. In this new discourse context, the SVO order also exemplifies the well-known tendency for languages to order given, old, presupposed, or background information before new or asserted information. The part of an utterance that is new, asserted, or not-presupposed information is referred to as the **focus** of the sentence (Jackendoff 1972), whereas the rest of the utterance, which represents what is already known by the speakers, is the background or **presupposition**. In a question like *What happened?*, the presupposition is that there is an *x, x is an unspecified event, such that x happened*. In *What did Juan buy?*, on the other hand, the presupposition is that there is an *x, x is an unspecified object, such that Juan bought x*. In both cases, the wh-element in the question stands for a variable and the corresponding answer, the entire sentence *Juan compraba el periódico* in the first case, and the DO *el periódico* in the second, provide a value for that variable. Focus is thus understood as the constituent that provides a resolution for a variable established previously in a presupposition structure.

Different types of foci and many different definitions for the concept have been proposed in the literature, based on semantic, pragmatic, and syntactic distinctions (see Jackendoff 1972; Rochemont 1986; Vallduví 1992; Lambrecht 1994; Kiss 1998, among others). For expositional purposes, such distinctions will be initially ignored here, and focus will be treated in this section as a unified phenomenon based on the fact that most types of foci satisfy the wh-test commonly assumed to be the diagnostic to identifying the focal part of an utterance (Rooth 1992; Szendröi 2005): the segment that provides the answer to a wh-question is the focus.

This wh-test can be used to show that, with a neutral intonation pattern, different constituents in an SVO sentence in Spanish may be the focus, as shown in the examples in the following examples where the focus appears in [F] brackets, following Jackendoff's 1972 notation:

(2) Juan compró [$_F$el pe**rió**dico]
(Context: What did John buy?/¿Qué compró Juan?)

(3) Juan [$_F$compró el pe**rió**dico]
(Context: What did John do?/¿Qué hizo Juan?)

(4) [$_F$ Juan compró el pe**rió**dico]
(Context: What happened?/¿Qué sucedió?)

Examples (2–4) show that the SVO order in Spanish, in sentences with neutral intonation, is compatible with information structure interpretations in which the focus is either the DO in (2), the Verb Phrase in (3), or the entire sentence in (4). In all these cases, the focal segment contains the primary stress of the sentence, the salient syllable of the rightmost constituent, marked above **in bold**. The non-bracketed segments constitute the presupposition, as defined above.

Nevertheless, these are not all the possible focus readings of the SVO order in Spanish, as shown in (5, 6):

(5) [$_F$**Juan**] compró el periódico
(Context: Who bought the newspaper/¿Quién compró el periódico?)

(6) Juan [$_F$ com**pró**] el periódico
(Context: What did J. do with the newspaper?/¿Qué hizo Juan con el periódico?)

Examples (5) and (6) are sentences in which the main accent has been shifted from final position, and in them an emphatic or contrastive stress rather than

N(uclear) S(tress) is assigned to the focal constituent, to the subject *Juan* in (5) and to the verb *compró* in (6). This exemplifies the relationship between focus, stress, and the most embedded position in the sentence: on the one hand, under unmarked or neutral intonation, the highest syntactic node marked as focus must dominate the constituent that contains the Nuclear Stress or final stressed element; on the other, focused elements that are not final are not compatible with neutral intonation. This correspondence between focus and Nuclear Stress is illustrated in Bosque and Gutiérrez-Rexach (2009: 681) in the following way:

(7) a. [$_F$ … **xx** …]
b. * [$_F$ …] … **xx** …

Only (7a), in which the constituent marked as focal contains the NS is possible, as exemplified in (8):

(8) a. ¿Quién compró el periódico?
b. *[$_F$ Juan] compró el pe**rió**dico

The question in (8) establishes the VP as presupposed and the subject is therefore the focal segment in the answer. The fact that the [F] constituent is unstressed leads to ungrammaticality. Thus, there is clearly a strong correlation between stress and focus in Spanish: focus is always prosodically prominent.

This dichotomy focus/presupposition is one of the many used to describe the contribution of sentential components to the flow of information within the discourse, along with theme/rheme, topic/comment, focus/background, presupposition/assertion, etc. (see Erteschik-Shir 2007 and Casielles-Suárez 2004 for recent and meticulous overviews of these terms.) In addition to the criterion that separates the information content of a sentence in focus/presupposition, a different criterion to establish this information partition is to consider what the sentence is about (Reinhart 1981). Following the *aboutness* criterion, sentences are partitioned between the **topic** and the **comment**, as first proposed in Hockett (1958). **Topic** is the expression which names what the sentence is about, and, like focus, it is also defined with respect to a particular discourse context. For example, given the context in (9), the answer will be a sentence in which the topic is *Juan*, as shown in (10):

(9) ¿Qué le pasó a Juan?
'What happened to Juan?'

(10) [$_{topic}$Juan] se hizo torero
'Juan became a bullfighter'

Sentences with SVO order that are discourse-initial, or that are answers to out-of the-blue questions such as *What happened?*, like our initial example *Juan compró* *el periódico*, have thus no topic, unless we consider their topic the particular time or place about which it is asserted (Gundell 1974). These constructions are called **thetic** or all-focus sentences. When there is a previous discourse in which speaker and hearer share certain presuppositions, a topic may be newly introduced if preceded in Spanish by what Contreras (1976) calls "topicalizing expressions" such as *en cuanto a*, 'as for,' *por lo que afecta a*, 'regarding,' *hablando de* 'speaking of':

(11) a. En cuanto a Juan, (él) se hizo torero
b. Por lo que afecta a Juan, (él) compraba el periódico todos los días

In (11), the constituents *(En cuanto a)* and (*Por lo que afecta a*) *Juan* are followed by a juncture/pause, and are unambiguously interpreted as topics (Zagona 2002).

In an excellent attempt to summarize the different definitions of the term topic in the literature, Casielles (2004: 22) points out that the topic of a sentence is usually taken to be "what the speaker takes as a point of departure of the sentence, that which is not part of the focus, placed towards the beginning of a sentence, often a subject, often active or salient in the discourse, and often expressed by pronominal or unaccented lexical phrases." It is clear from the necessary vagueness of the definitions that topic is a rather difficult notion to define, and that there is no agreement on the syntactic or discourse requirement for topics. Nevertheless, the following generalizations can be taken as uncontroversial: (1) the same phrase cannot be at the same time topic and focus of a sentence; (2) there is a tendency for a topic to be associated with certain positions, particularly to appear in sentence-initial position (and thus often to be identified with the subject); and (3), the order topic-focus is the unmarked one in declarative sentences under neutral intonation.

The clear parallelism between the topic/comment and focus/presupposition dichotomies is such that in some cases the topic of a sentence has been identified with the nonfocal portion, or, similarly, the focus has been with the comment. The question arises as to whether these two dichotomies are equivalent so that a sentence can be exhaustively partitioned in topic/focus. Nevertheless, although not complimentary or able to exhaustively partition a sentence (see Vallduví 1992 for an influential trinary partition of the information structure of a sentence in Catalan, and Casielles 2004 for a similar analysis in Spanish), topic and focus are widely considered information structure primitives that interact in such a way that "all types of topic and focus can be derived from them without the need of further primitives" (Erteschik-Shir 2007: 27).

2.2 *Non-SVO unmarked orders*

Preverbal subjects do not always constitute the unmarked order in Spanish. Leaving aside the order in interrogatives and exclamatives in Spanish (chs. 15 and 20 and references therein), in certain Spanish declarative constructions the subject does not precede the verb but follows it (Fernández Soriano 1993, 2001). The subject appears in postverbal position in unmarked sentences in Spanish in the following circumstances:

(12) A. when it is a bare NP
B. in certain infinitival and imperative constructions
C. with unaccusatives like *llegar* 'to arrive' or *crecer* 'to grow'
D. with SE passives and in middle voice constructions
E. with verbs like *faltar* 'to lack/to be missing,' *suceder* or *ocurrir* 'to happen'
F. with psychological verbs like *gustar* 'to like,' *asustar* 'to frighten,' *preocupar* 'to worry'

The examples in (13) illustrate the variation in acceptability of these constructions under prosodically unmarked and informational neutral contexts:

(13) a. Llegaron niños — a'. #Niños llegaron
'Kids arrived'
b. Al salir el sol — b'. *Al sol salir
To-the sun to-rise ('As the sun raises')
c. Que lo diga Pedro — c'. #Que Pedro lo diga
That Pedro says it ('(I command) for Pedro to say it')
d. Se convocaron los premios Goya — d'. #Los premios Goya se convocaron
'The Goya awards were announced'
e. Ocurrieron demasiadas catástrofes — e'. #Demasiadas catástrofes ocurrieron
'Too many catastrophes occurred'
f. Me gustan las películas de Billy Wilder — f'. #Las películas de Billy Wilder me gustan.
'I like Billy Wilder's movies'

The sentences in (13) with preverbal subjects are not felicitous responses to out-of-the-blue questions and cannot appear in discourse-initial contexts. With the exception of (13b), ungrammatical in all possible informational contexts in Standard Spanish (but see Ordóñez and Olarrea 2002, 2006 for Caribbean Spanish), the rest are acceptable only when the preverbal subject is focalized and receives contrastive stress. In this sense, they are diametrically opposed to final subject sentences in Spanish like the ones in (14):

(14) a. Compraba el periódico todos los días [$_{F}$**Juan**]
b. Se hizo torero [$_{F}$**Juan**]

In (14), variations of the SVO unmarked order discussed previously, the postverbal subject *Juan* is focal, following a well-established tendency in other Romance languages like Italian in which the postverbal subject is always the focus of the sentence (see, for example, Frascarelli 1999; Dubrig 2003). Thus, predicates that appear in unmarked SVO orders in Spanish allow for postverbal subjects when these are focused. On the other hand, constructions like the ones in (13), with postverbal subjects as the unmarked order, allow for preverbal subjects when they are focal. In addition, these examples show that not all preverbal subjects are topics in Spanish.

The constructions in (12–13) have received wide attention in the literature, mostly from a syntactic point of view (see, among others, Mendikoetxea 1999a, 1999b, 2008 and Chapter 23, above, for passives with *se*, middle, and unaccusative constructions; Parodi and Luján 2000 for psych verbs; Bonilla 1997 for an introduction to infinitival constructions, and Chapter 17, above, for further references). Out of all of them, only (13a) and (13e), with bare NP subjects in mandatory postverbal position, have been studied in Spanish in terms of information structure. The syntactic description for these constructions was initially formulated in Suñer (1982) as the *Naked Noun Constraint*, a constraint that prevents unmodified bare NPs to appear in preverbal position. This constraint has been usually derived from the proposal that Bare Nouns are DPs with an empty determiner that cannot be properly governed by Inflection in preverbal subject position, a violation of the condition on proper government of empty categories (Contreras 1986; Bosque 1996; Espinal 2010). This ban on nonfocused bare NPs in preverbal subject positions in Spanish has also been approached from a perspective that combines semantic and information structure principles (Laca 1996, 1999; Casielles 1996, 2004; Dobrovie-Sorin and Laca 2003). Under this approach, the constraint is explained by the fact that bare NPs lack independently established reference while topics are incompatible with nonreferential expressions. Foci, on the other hand, are compatible with nominals that denote properties. This explains the acceptability of (15), where a nonreferential bare NP with emphatic intonation, marked here in small caps, can appear preverbally:

(15) NIÑOS llegaron después de la fiesta.
'It was kids that arrived after the party.'

I will return to the analysis and interpretation of this type of sentence in Section 3.2.

Summarizing this second section, linear order of syntactic constituents is determined in part by what is contextually known or presupposed and by what is not. In Spanish, like in many other languages, prosody interacts with word order to determine information structure: in unmarked word orders, the highest syntactic node marked as focus must dominate the constituent that contains the Nuclear Stress. SVO orders are usually interpreted either as all-focus sentences or, in nonfocal environments, as sentences in which the subject is the topic. Nevertheless, there are in Spanish several constructions, as in (13), in which the postverbal subject is also the unmarked order. In these cases, preverbal subjects, when grammatical, are not topics but always focal.

3 Topic and focus structures: a descriptive overview

This section is dedicated to the constructions that codify topic and focus in Spanish from a strictly descriptive point of view. First, the structures in which a topic appears in a peripheral sentence position are discussed. Next, those constructions in which a focal element appears in the left periphery are examined.

3.1 *Topicalizing constructions*

Following Cinque's (1990) proposal for Italian, the Romance linguistics literature has distinguished three types of constructions in which a topic appears in a sentential peripheral position, to the left, as in (16a,b), or to the right (16c). I will refer to these constructions as Hanging Topic Left Dislocation (HTLD), Clitic Left Dislocation (CLLD), and Clitic Right Dislocation (CLRD) respectively, in accord with López (2009) terminology[1]:

(16) a. Las rosas, me encantan esas flores (HTLD)
The roses, to-me are pleasing those flowers
'Roses, I love those flowers'
b. Las flores las compré ayer (CLLD)
'The flowers them I bought yesterday'
c. El bibliotecario los encontró ayer, esos libros
'The librarian found them yesterday, those books' (CLRD)

These constructions are characterized by the presence of a phrase in a peripheral position, connected with the clause by some anaphoric element referred to as the resumptive element (the DP *esas flores* in (16a), the clitic *las* in (16b), and the DP *esos libros* in (16c). We can differentiate them on syntactic considerations (Hernanz and Brucart 1987; Olarrea 1996; Zubizarreta 1999; Benincà and Poletto 2004; Alexiadou 2005, and others). In the following section, I will discuss first the differences between HTLDs and CLLDs from a strictly descriptive point of view; I will then concentrate on what minimally separates CLLDs and CLRDs.

3.1.1 Clitic left dislocations and hanging topics: similarities and differences

Hanging Topic Left Dislocations and Clitic Left Dislocation differ in Spanish with respect to the following seven parameters:

(17) A. their phrasal type
B. the properties of the resumptive element
C. the type of constituent that can precede them
D. the possibility of embedding
E. the "connectivity" between the displaced and the resumptive elements
F. their recursive properties
G. their sensitivity to syntactic islands

3.1.1.1 In the HTLD construction, the left dislocated element can only be a DP (18), while in CLLD any phrasal type can be dislocated (19):

(18) a. Juan, no me acuerdo de él (HTLD)
'John, I don't remember (of) him'

b. *De Juan, no me acuerdo de él (HTLD)
Of John, I don't remember (of) him

(19) a. A Juan lo vimos en la fiesta (CLLD)
'Juan, we saw him at the party'

b. Rápido sí que lo es Usain Bolt (CLLD)
'Fast is indeed what is Usain Bolt'

c. Que María haya podido decir eso no puedo creerlo (CLLD)
'That Mary could have said that I can't believe'

In (19a) a PP is dislocated, while in (19b) the dislocated element is an AdjP and a CP in (19c).

3.1.1.2 The resumptive element in HTLD constructions can be a phrase (an epithet) or a pronoun, either a clitic or a tonic pronoun (20), while in CLLD the coreferential element has to be an empty pronominal licensed by agreement or a clitic, never a tonic pronoun or a phrase (21) (but see Suñer 2006 for CLLDs constructed with epithets in certain dialects of Spanish):

(20) a. John Coltrane, este saxofonista me encanta. (HTLD-phrase)
'John Coltrane, that saxophone player I love'

b. Miles Davis, él sí que me fascina (HTLD-tonic pron.)
'Miles Davis, he is indeed fascinating to me'

c. En cuanto a Buddy Guy, hace años que no lo veo en concierto (HTLDcl.)
'As for Buddy Guy, it's been years since I (don't) see him in concert'

(21) a. En Juan no es posible confiar (CLLD)
In John not is possible to trust
'It is impossible to trust John'

b. *En Juan no es posible confiar en él (*CLLD-tonic pr.)
In John not is possible to trust in him

c. A María no la vi nunca tan enfadada (CLLD)
To Mary not her I saw never so irritated
'Mary, I have never seen her so irritated'

d. *A María nunca vi a esa chica tan enfadada (*CLLD-epithet)
To Mary never I saw that girl so irritated

3.1.1.3 The HTLD constituent can be preceded by topicalizing expressions such as *en cuanto a* 'as for,' *por lo que afecta a* 'regarding,' *hablando de* 'speaking of' (22a). CLLD constructions are ruled out when preceded by expressions of this type (22b):

(22) a. En cuanto a Antxon, él nunca va a terminar su capítulo (HTLD)
'As for Antxon, he will never finish his chapter'
b. *Te he dicho que en cuanto a Juan lo vi ayer (CLLD)
'I have told you that regarding John, I saw him yesterday'

The fact that not all topics can be preceded by a topicalizing expression shows that this cannot be used as a test for topic-hood, against Reinhart (1981).

3.1.1.4 HTLDs cannot be embedded. They have to appear in absolute first position (23) while CLLDs can be freely embedded and thus can appear as complements to verbs that subcategorize for a CP, like *pensar* 'to think' or *sorprender* 'to find surprising' (24):

(23) a. *Todos dicen que John Coltrane, ese saxofonista es el mejor (HTLD)
'Everybody says that John Coltrane, that sax player is the best'
b. *No me sorprende que Miles Davis, él sí que supo desafiar a sus críticos
'It does not surprise me that Miles Davis, he indeed knew how to challenge his critics'

(24) a. Todos piensan que de Juan no deberíamos hablar (CLLD)
everybody thinks that of John we should not talk
'Everybody thinks that we shouldn't talk about John'
b. No me sorprende que de María nadie se haya quejado
'It is not surprising that about Mary nobody has complained'

3.1.1.5 There is obligatory identity of Case and subcategorization between the dislocated element and the resumptive pronoun in CLLDs. Cinque (1987) refers to this obligatory identity as "connectivity." But this is not the case in HTLD constructions:

(25) a. Nosotros, nadie nos ha visto (HTLD)
we, nobody us has seen
'Nobody has seen us'
b. Juan, estaba pensando en él en este momento.
'John, (I) was thinking about him right this moment'

(26) a. A nosotros no nos han dicho nada (CLLD)
to us not us have said nothing
'Nobody told us anything'
b. *Nosotros no nos han dicho nada

(27) a. En Juan estaba pensando en este momento
'In Juan (I) was thinking right this moment'
b. *Juan estaba pensando en$_{ec}$ este momento

In (25) the dislocated element shares neither the Case (accusative in (25a)) nor the subcategorization (the verb *pensar* 'to think' selects for preposition *en*) of the resumptive element within the sentence. In the CLLD constructions in (26), the dislocated element must bear an accusative marker (the preposition *a*) in order for the constructions to be correct. In (27), the need for identity of subcategorization is shown.

Furthermore, the lack of connectivity in HTLD structures can be shown by the possibility of having a left dislocated element that does not agree in gender and number with the coreferential element. This is ruled out in CLLD constructions:

(28) a. El ordenador, yo odio esas máquinas infernales
'The computer, I hate those evil machines'
b. *El ordenador las odio.
the computer them I hate

Cinque (1990) also claims that it is impossible to dislocate a pronoun bound by a quantifier when there is no connectivity. As predicted, HTLDs cannot dislocate a bound pronoun (29):

(29) a. *Su$_i$ madre, cada chico$_i$ le regalará flores a ella
his mother, each child will give her flowers
b. A su$_i$ madre cada chico$_i$ le regalará flores
'To his mother each child will give her flowers'

3.1.1.6 While these two constructions under analysis here, CLLDs and HTLDs, present a whole set of distinctive syntactic properties, they also share others. In both constructions more than one constituent can be dislocated, as shown in (30):

(30) a. En cuanto al dictador y al pueblo, éste repudia a aquél (Contreras 1978)
'As for the dictator and the people, the former hates the latter'
b. Ese libro a Pedro no se lo dio nadie.
that book to Peter did not give it to him nobody
'Nobody gave Peter that book'

But notice that the HTLD construction requires the presence of a conjoined phrase. If the dislocated elements are not conjoined, only one dislocated phrase is allowed:

(31) a. *Juan, el libro, él no lo ha comprado
'John, the book, he hasn't bought it'
b. *El libro, Juan, este no ha comprado aquél
the book, John, this one not has bought that one

Villalba (2000) separates this *as for construction* from HTLDs and CLLDs based in the fact that there is no need for a resumptive element in these sentences:

(32) En cuanto a la rueda pinchada, Juan explicó que había clavos en la carretera
'As for the flat tire, John explained that there were nails on the road'
(adapted from Villalba 2000: 105)

Crucially, when both types of dislocation are present, the HTLD constituent must precede the CLLD (33):

(33) a. En cuanto a Juan, esa carta se la escribió Pedro
as for John, that letter to-him it wrote Peter
'As for John, Peter wrote him that letter'
b. *Esa carta en lo que se refiere a Juan, se la escribió Pedro
this letter as for John, to-him it wrote Peter

3.1.1.7 While HTLDs are insensitive to strong and weak islands, CLLDs are insensitive only to weak islands:

(34) a. En cuanto a ese trabajo, no puedo aceptar la idea de que ya lo han conseguido
'As for that job, I can't accept the idea that they have already gotten it'
b. Hablando de 'Freaks', un amigo que ha visto esa película me ha dicho que es magnífica
'As for "Freaks," a friend who has seen that movie has told me that it is great'
c. Por lo que se refiere a ese libro, te tomas un par de días de descanso y seguro que lo acabas
'As for that book, you take a couple of days off and it is obvious that you will finish it'

These examples of HTLDs represent coreference with an element within a Complex Noun Phrase, a Relative Clause, and a Coordinate Island, respectively.

On the other hand, both HTLD and CLLD structures can be constructed with a coreferential element inside a Wh-island (a weak island), as shown in (35):

(35) a. Dinero, me pregunto [quién tiene]. (HTLD)
b. A esos espías no sé [cómo se puede saber [quién los traicionó]]. (CLLD)

It has also been proposed that, in addition to their syntactic properties, there are prosodic factors that also differentiate these constructions. For some authors, there is a sharp intonation break between the dislocated phrase in HTLD and the rest of the sentence, while such pause is not present in CLLD. Although the left dislocated phrase in CLLD is in some cases separated from the rest of the sentence by a comma, it is normally assumed that the intonation break between the two is much weaker than in HTLD (see Face 2001, 2002; Beckman et al. 2002, and Chapter 8, above, for description of prosodic patterns in Spanish dislocated sentences and focal constructions).

3.1.2 Clitic right dislocations Whereas dislocations to the left of the sentence have received considerable attention, CLRDs has been a construction neglected in modern syntactic studies, probably due to the oral character of the construction or due to its inclusion in the class of sentential afterthoughts (see Villalba 2000 for insightful comments). Right dislocation is a construction in which a clitic/agreement co-occurs with a phrase in the right periphery of the sentence, as illustrated in (36):

(36) Yo no las traje, las llaves
'I did not bring them, the keys'

The CLRD constituent forms in all cases its own intonational phrase, separated from the rest of the sentence with a sharp break. In this sense, it patterns with HTLDs. Most studies, though, relate CLRDs to CLLDs and not to HTLDs. Note that, unlike the latter, CLRDs cannot be preceded by a topicalizing expression (**Yo no las traje, en cuanto a las llaves* 'I did not bring them, as far as the keys'). Also, CRLDs can be constructed with any constituent type that can be CLLDed, a DP as in (37a), or a AdjP (37a), a CP (37b) or a PP(37c) – the examples are parallel to the CLLDs in (19):

(37) a. Sí que lo es, muy rápido
'Yes it is, very fast'
b. Ya te lo he dicho, que te calles
'I have already you (it), that you be quiet'
c. Lo vimos en la fiesta, a Juan
'We saw him at the party, John'

Apart from the fact that CLRD cannot be preceded by a topicalizing expression, and that they have a sharp intonational break marked orthographically by a comma, these constructions share all the characteristics of CLLDs: they are iterative (38a); they occur both in root and embedded contexts (38b); the presence of a resumptive clitic is mandatory (38c); and they are freely ordered when more than one is present (38d):

(38) a. Las venden muy caras, las cervezas, en Barcelona
'They sell them very expensive, the beers, in Barcelona'

b. Parece que María lo vio ayer, a Juan
it seems that Mary him saw yesterday to John
'It seems that Mary saw him yesterday, John'
c. *María vio ayer, a Juan.
d. Lo vio, a Juan, ayer, María.

Zubizarreta (1998: 198, n. 57) proposes that CLRD is just CLLD followed by leftward movement of the rest of the sentence across the left-dislocated constituent. This approach helps her explain the fact that right dislocated elements occupy a peripheral position in the sentence, unlike the similar Double Clitic Constructions *Yo no las traje las llaves hoy* 'I did not bring them, the keys, today' in Spanish (see Chapter 21, above). Villalba (2000) and López (2001), on the other hand, derive CLLD from CLRD.

3.2 Clitic Left Dislocation vs. Focus Fronting

HTLDs may or may not be constructed with a clitic, while CLLD require the obligatory presence of one to license the gap. In this respect, these two constructions can be differentiated from the constructions we will refer to as Focus Fronting:

(39) ESTAS FLORES quiere María
these flowers wants Mary

Now the leftmost phrase constitutes the melodic peak of the sentence (even though the peak is syllabic, represented here by an entire constituent in small caps following the syntactic tradition), triggers Subject–Verb inversion (40a) and cannot license a resumptive pronoun (40b):

(40) a. *ESTAS FLORES María quiere (FF)
b. ESTAS FLORES (*las) quiere María (CLLD)

It is important to point out the strict correlation between the emphasis and the obligatory inversion in (40), unless the element in focus is the subject and it receives contrastive emphasis and interpretation (Section 4.3).

There are then several characteristics that distinguish this construction from CLLDs, besides the obligatory absence of a clitic and the inversion of the subject. In FF, only one constituent is fronted (notice that in this respect, FFs pattern with HTLDs):

(41) a. *LA CARTA A JUAN escribió Pedro
the letter to John wrote Peter
b. *A JUAN LA CARTA escribió Pedro
to John the letter wrote Peter

The preposed focus element in these examples is interpreted as a quantifier, as argued in Hernanz and Brucart (1987) and Cinque (1990):

(42) LAS ACELGAS detesta María (Hernanz and Brucart 1987: 88)
(The) chards hates Mary
'Chards, Mary hates them'
For all x, x=chards, [$_{TP}$María hates x]

The quantificational nature of fronted foci and the fact that they trigger subject–verb inversion has led many authors to point out their crucial similarities with Wh-phrases (Cinque 1990; Zubizarreta 1994,1999; Zagona 2000; Belletti 2001; Bosque and Gutiérrez-Rexach 2009). Unlike the other left-peripheral constructions describe above, and like Wh-phrases, FFs do obey strong islands:

(43) a. *LAS ACELGAS no sé quién detesta (Wh- Island)
'It is chard I don't know who hates'
b. *LAS ACELGAS conozco a la mujer que detesta (Complex NP Island)
'It is chard I know the woman who hates'

Similarly, FFs cannot be fronted in a question (44a) but can be preceded by a dislocated topic (44b) (examples from Zagona 2002):

(44) a. *¿Cuándo LAS MANZANAS compraron?
'When was it the apples that they bought?'
b. En octubre, POCAS MANZANAS compraron
'In October, few apples they bought'

4 Topicalizing and focus structures: formal accounts

4.1 Topicalizing constructions: movement or base-generation?

Before Rizzi (1997), it was widely assumed that HTLLDs occupy a CP-adjunct position while CLLDs are TP adjuncts, the resumptive element contained within the TP in both cases:

(45) [$_{CP}$ [$_{CP}$ HTLD] … C [$_{TP}$ [$_{TP}$ CLLD] … T [VP]]]

Several of the properties of HTLDs described in Section 3.1.1 follow from this simple analysis: HTLDs occur only in root contexts (23) and precede CLLDs when both are present in a sentence (33). Since there is no direct grammatical link between the HTLD and the rest of the sentence, other facts can also be accounted for: the coreferential element can be overt (20), and there is no *connectivity* between this element and the dislocated phrase (25, 26). Furthermore, by positing that the HTLD element is base-generated in the left-peripheral position and not the result of movement, and is in that sense different from Wh-constructions (Rivero 1980; Hernanz and Brucart 1987 and subsequent work), we can explain their insensitivity to syntactic islands (34).

While HTLDs are always analyzed as base-generated adjuncts, whether the dislocated phrase in CLLD is the result of movement or not remains controversial due to the fact that CLLDs do not behave like Wh-constructions in all respects: CLLD constructions, unlike interrogatives, do not license parasitic gaps (46) or induce Weak Crossover (WCO) effects (47) (from Suñer 2006):

(46) A una candidata el jefe la descartó sin entrevistar *(la)
A one candidate, the boss her-Acc rejected her without interviewing *(her)
'One candidate, the boss rejected her without interviewing her'

(47) A Teresa$_k$ su$_k$ madre la$_k$ quiere mucho
A Teresa-Acc her mother her-Acc loves a lot
'Teresa, her mother loves her a lot'

Examples (46) and (47) show that the resumptive clitic does not seem to have the properties of a variable since neither the parasitic gap is allowed nor WCO effects are induced.

In addition, CLLDs do not trigger mandatory subject inversion, unlike Wh-constructions in Spanish (41) and can easily skip over Wh-islands (35). Based on these properties, Hernanz and Brucart (1987), Contreras (1991), Anagnostopoulou (1994), and Zagona (2002), among others, claim that CLLDs are not the result of movement but rather generated/merged in a left peripheral sentential adjunct position.

On the other hand, the fact that CLLDs are sensitive to strong syntactic islands such as Complex NPs, Adjuncts or Sentential subjects, as shown in (48), favors a movement analysis (see Cinque 1977; Dobrovie-Sorin 1990; Kayne 1994; Villalba 2000; López 2003):

(48) a. ??A Juan Pedro conoció al médico que lo operó (Complex NP Island)
Juan-Acc, Pedro met the doctor who him-Acc operated
b. ??A Juan el médico estornudó mientras lo operaba (Adjunct Island)
Juan-Acc, the doctor sneezed while him-Acc operated
c. *A Juan, creíamos que [que el médico lo operara tendría consecuencias imprevisibles] (Sentential Subject Island)
John-Acc, we-believed that [that the doctor him-Acc operated-Subj would have unpredictable consequences]

If the CLLD constituents are base-generated, the data in (48) is paradoxical: islands constrain movement, so the relationship between the CLLD-ed constituent and the resumptive pronoun should not be blocked by any type of syntactic configuration that restricts movement. Cinque (1990) proposes a solution: syntactic chains can be created by movement or by base-generation. In the latter case, the chain between the dislocated element and the resumptive pronoun is a "binding chain." Binding chains, of which CLLD constructions are an example, require for the head phrase to be in a nonargument position, have referential properties, and receive a referential theta-role. While strong islands constrain the binding relationship between the head and the tail of any chain, only the chains created by movement are constrained by weak (wh- or factive) islands. This explains not only why CLLDs are insensitive to weak islands, but could also explain reconstruction effects in CLLDs (see Zubizarreta 1998 and Suñer 2006 for discussion). In a similar way, Rizzi (1997) explains the absence of WCO effects in CLLDs by assuming that WCO is a distinctive characteristic of nonargumental relations involving genuine quantification. We can distinguish, following Lasnik and Stowell (1991), between nonargumental dependencies involving a quantifier that binds a variable and those that involve nonquantificational binding. Wh-movement and Focus Fronting are examples of true quantificational binding, while CLLDs are proposed to be examples of the type of nonquantificational binding that does not induce WCO effects.

To summarize this section, HLTDs are uniformly analyzed as involving base-generation of the dislocated phrase, while there is no total agreement in the literature with respect to whether CLLDs are the result of movement or not. The main problem for base-generation analyses of CLLD is that they have to appeal to special mechanisms in order to derive the connectivity property of CLLD and their sensitivity to strong islands, generally by assuming that the dislocated phrase forms a chain that binds a nonvariable within the clause. Movement analyses, on the other hand, have to account for the insensitivity of CLLDs to weak islands, the absence of WCO effects, and the ability to license parasitic gaps, and, in Spanish, the lack of subject–verb inversion that otherwise characterizes movement constructions.

4.2 *The left periphery*

Since Rizzi (1997), the term *left periphery* refers to a complex functional field that expands the area external to the clause, the CP, and whose cartography, subject of numerous studies, is much richer and more articulated than it has been traditionally assumed. In the initial proposal, the complementizer domain is divided into specific functional projections whose main goal is to capture the informational articulations of a sentence (topic/comment; focus/presupposition). This is shown in the schema in (49):

(49) [ForceP [TopicP [FocusP [TopicP* [FinP]]]]]

Processes of Topicalization and Focalization are analyzed as involving movement of a phrase to a specific dedicated position in the left periphery. Topic is, under this approach, a functional head that carries a [+ Topic] feature and that needs to attract a compatible phrase in a local relationship in order to satisfy a Spec-Head criterion analogous to the Wh-Criterion and the Neg-Criterion (see Chapter 25, above). The highest complementizer-like head is Force and the lowermost is Finiteness (Chomsky and Lasnik 1993). Two different Topic projections and a single Focus projection are located between the two. Since processes of Topicalization and Focalization are analyzed as involving movement of a phrase to a specific dedicated position in the left periphery, the interpretation of a peripheral constituent as Topic or as Focus is an automatic consequence of it filling the specifier of one of the appropriate heads in this configuration. An agreement relation is established between the head of a phrase and the constituent filling its specifier position in order to satisfy a Spec-Head criterion analogous to the Wh-Criterion and the Neg-Criterion (Chapter 25). A Topic head, carrying a [+ topic] feature, and the phrase in its Spec will share the topic feature/interpretation; an identical relation will account for the focus interpretation of a phrase in the specifier of FocusP.

In (49), the highest Topic phrase is able to host HTLDs in its specifier position, as a result not of movement but of base-generation/merge. The lowest Topic P is recursive, which accounts for the fact that there can be more than one CLLDed phrase (30b). The relative order of topicalized and focus constructions with respect to each other can also be accounted for once it is assumed that Wh-phrase and focus constructions compete for the [Spec, FocusP] position (at least in main clauses, see Rizzi 2001). This explains why left-peripheral foci cannot be fronted in a question (44a) but can be preceded by a dislocated topic (44b). It also accounts for the fact that HTLDs can precede a Wh-phrase (50):

(50) a. [$_{\text{TopicP}}$ En cuanto a Juan, [$_{\text{FocusP}}$ ¿qué quiere comer hoy?]]
as for John, what wants to eat today?
'As for John, what does he feel like eating today?'
b. [$_{\text{FocusP}}$ *¿ [$_{\text{TopicP}}$En cuanto a Juan qué le pasa?]]
'As for John, what is the matter with him?'

This cartographical approach to the left-periphery explains in part the relative word order of elements in the left periphery of the sentence. In order to account for the examples of focus *in situ* and nonperipheral foci, these strictly syntactic movement analyses of focus have been extended so that the full CP structure of the left periphery is repeated immediately above the VP (see Belleti 2001; Butler 2003). These approaches are based on the stipulation that the Topic head, unlike other functional heads, is able to license more than one projection (Benincà 2001 offers a detailed description of the left periphery in Italian that eliminates a recursive TopicP, and Beas 2007 and López-Cortina 2007 do so for Spanish). Furthermore, they propose a proliferation of functional heads whose explanatory power has been widely questioned, while the need to posit a principled way of limiting the number of functional categories for a particular language has been defended (see Erteschik-Shir 2006 for a treatment of topic not as a functional head but as a syntactic feature assigned to lexical items that percolates to their maximal projections). Finally, the need to stipulate the syntactic features [±topic] or [±focus] as theoretical primitives has been also questioned (see López 2009 for a stimulating proposal in which the crucial syntactic features that account for information structure are not [+focus] and [+topic] but [±anaphor] and [±contrast]).

4.3 Focus constructions and syntactic movement

Focus constructions show all the properties of operator-variable constructions that are absent in CLLDs: they not only force subject inversion in Spanish (40a) and are sensitive to weak and strong islands (43), but also license parasitic gaps (51a) and show WCO effects (51b):

(51) a. LAS ACELGAS comió María sin cocer (pg)
'Chard Mary ate without cooking'
b. *A JUAN$_i$ detesta su$_i$ madre
John-Acc hates his mother-Nom
'It is John that his mother hates'

These characteristics have led to positing identical movement analyses for focalized constructions and interrogative sentences. Focus Fronting, just like Wh-movement, is triggered by the Focus Criterion: the focus/wh-element has to appear in a specifier–head relationship with its licensor, a Focus functional head (49). In general, analyses that are centered on the wide similarities between Wh-movement and Focus Fronting regard focus as a purely syntactic property. In order to account for the well-established generalization that states that the focus constituent always bears main prosodic prominence (Section 2.1), these analyses must assume that main stress gets assigned to the constituent that is marked as focus in the syntax. Thus, stress contour and intonation are determined by syntactic configurations and the prosodic characteristics of focus are secondary. This is what Ladd (1996) called the *accent-to-focus view*.

There is, nevertheless, the possibility of taking just the opposite perspective, the *focus-to-accent view*: the syntactic constituent that bears the accent is the one that receives focus interpretation, rather than the accent being the one that matches the constituent carrying a focus feature. The idea that the focus of an utterance is determined by the prosodic make-up of the utterance led to the formulation of analyses that take focus movement to be prosodically driven, thus diminishing the parallelism between focus and Wh-movement (see Szendrői 2005 for a detailed discussion). This line of research was pioneered for Spanish in Zubizarreta's (1994, 1998) influential work, and is discussed in the next section.

4.3.1 Focus and prosodic movement: zubizarreta (1998) Zubizarreta, expanding on Cinque (1993), claims that the key correspondence between focus and prosody is derived form the fact that the syntactic constituent marked as [+Focus] must contain the rhythmically most prominent word in that phrase, also known as the nuclear stress of the sentence. Rhythmic prominence is derived from syntactic structure since, in Romance languages, a N(uclear) S (tress) R(ule) assigns the main stress of the sentence to the rightmost syllable that can receive stress and is contained in the most deeply embedded constituent.

Zubizarreta distinguishes two types of nuclear stress: neutral and contrastive stress. The former is the one that identifies neutral focus. Recall from Section1 that, in Spanish, neutral stress in SVO orders (*Juan compraba el pe**rió**dico*) is assigned to the rightmost accented syllable and is compatible with three different focal interpretations: the entire sentence, the VP *compraba el periódico*, or just the object *el periódico* can be the focus. In either of them, the constituent interpreted as focus is an example of neutral focus and contains the most embedded stressed syllable. Thus, a neutral focus constituent must be in a position where it can receive the nuclear stress by the NSR, that is, must appear in sentence final position in Spanish (all examples form Zubizarreta 1999):

(52) a. El gato(S) se comió (V) un ratón (O)
'The cat ate a mouse'
b. Se comió un ratón [$_F$ el gato](S)

While (52a) is ambiguous (the entire sentence, the VP or the object can be neutral focus), (52b) is not: it can only be the answer to the question *Who ate a mouse*?, which requires narrow focus on the subject, but not to the question *What happened?* Zubizarreta claims also that (52b) is neither right-dislocated nor receives a contrastive interpretation, as shown by the fact that it can be a felicitous answer to a question–answer pair such as *Who ate what?* in which both the object and the subject are interpreted as focus (*Se comió un ratón el gato y un bistec el perro/* Se comió un ratón el gato y no el perro* 'The cat ate a mouse and the dog a steak/*The cat ate a mouse and the dog did not).

The neutral nuclear stress in (52b) must fall on the most embedded constituent *el**ga**to*. A way to derive the VOS order in (52b) is to reorder the constituents by means of a process that aligns sentential stress and information focus. This syntactic movement that is motivated by the alignment requirement of the NSR is called p (rosodic) movement. Since Zubizarreta assumes, with Kayne (1994), that all movement is leftward, VOS is obtained either from movement of the VO constituent from an SVO order, or from movement of the O from a VSO order. In either case, it is assumed that movement is local, that the constituent that moves is deaccented, and that it adjoins to the VP (but see also Domínguez 2004, who allows for rightward movement of a constituent in order to receive nuclear stress, and Ordóñez 1997, who offers a strictly syntactic account of focal interpretation of sentence final subjects in Spanish that involves two movements: subject movement to [Spec, FocP] and remnant movement of the remaining TP above the Focus constituent.)

It is also possible to use a SVO order as a felicitous answer to the question *Who ate a mouse?* In this case, the subject receives emphatic stress, marked in small caps:

(53) EL GATO se comió un ratón

Unlike the nuclear stress type that can force p(movement) and is ruled by the NSR, this emphatic stress is ruled by the E(mphatic) S(tress) R(ule): it can only be applied to constituents *in situ*, and allows for no movement of the focal constituent. Focus revealed by emphatic stress is also subject to a different semantic interpretation, that of **contrastive focus**. A contrastive interpretation negates the value of a variable (in our example *it is not the case that x, x=the dog ate the mouse*) and introduces an alternative value for such a variable (*it was the case that x, x=the cat ate a mouse*). Any *in situ* constituent in a SVO sentence can be interpreted as contrastive if it receives emphatic stress:

(54) a. EL GATO se comió un ratón (y no el perro/*and not the dog*)
b. El gato SE COMIÓ un ratón (y no lo mató/*and did not kill it*)
c. El gato SE COMIÓ UN RATÓN (y no se bebió un plato de leche/*and did not drink a bowl of milk*)
d. El gato se comió UN RATÓN (y no un trozo de queso/*and not a piece of cheese)*

The prosodic requirements of contrastive focus are different from those of neutral focus. Unlike the NRS, the ESR only requires for the word that receives emphatic stress to be dominated by all the constituents marked with the feature [+F]. This eliminates the need to align the most embedded constituent with the rhythmically most prominent element of the sentence.

Since Focus Fronting places a focus phrase in the left periphery of the sentence, out of alignment with the NSR, Zubizarreta (1998: 103) assumes that the initial phrase receives emphatic stress and is always interpreted as contrastive (*LAS ACELGAS detesta María (y no las patatas)* '(The) chard Mary detests (and not the potatoes)'). She proposes that the peripheral position for emphasis, topic and focus is [Spec, TP], but no prosodic account of movement for Focus Fronting is offered (see Domínguez 2004 for discussion).

The NRS also explains some common properties of several possible word orders in Spanish: VOS, VO-Prep-S, VS-Prep-O, etc., exemplified in (55):

(55) a. Compraba el periódico Juan
b. Compraba el periódico todos los días Juan
c. Compraba Juan todos los días el periódico
d. Todos los días compraba Juan el periódico

In all of them, the last constituent must carry the nuclear (unmarked) stress and is then interpreted as the focus of the sentence. Interestingly, in Spanish, unlike in other Romance languages like Italian or French, the VSO order in (55d) is possible and requires no associated emphatic stress or intonation. Ordóñez (1997) offers a purely syntactic account of this phenomenon, motivated by the sole need to check syntactic features and is in no way related to prosody.

5 Concluding remarks

Several important topics regarding the information structure of Spanish declarative sentences have been left out in this chapter due to space constraints. Three of them should be mentioned here: first, cleft sentences, like the ones illustrated in (1f–g) carry a focus interpretation. Relevant information about these constructions can be found in Fernández-Leborans (2001) and Moreno-Cabrera (1999). Second, the analysis and interpretation of fronted quantifiers that occupy a left-peripheral position in Spanish presents problems that are insightfully discussed in Arregi (2003), Leonetti (2009), and Leonetti and Escandell-Vidal (2009). Lastly, focalizing adverbs are crucial in determining the scope properties of focalized constructions. A detailed description of these adverbs in Spanish can be found in RAE-ASALE 2009 (ss. 40.5–40.10).

Studies about information structure, and especially about the relationship between focus and intonation, have a strong import on the overall organization of grammatical models. Prevalent formal models (Chomsky 1995, for example) postulate no interaction between the phonological component and semantic interpretation, a difficult position to maintain if algorithms based on stress can predict the syntactic position of focus constituents. Szendrői (2001), Erteschik-Shir (2007), and López Cortina (2007) offer stimulating discussions about models of grammar in which phonology is either present or absent from syntactic and semantic levels of representation.

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NOTES

1 The first obstacle when trying to analyze these constructions is terminological, and it has been so since the initial ground-breaking studies in Spanish: Rivero (1980), following Chomsky (1972), uses the term *Topicalization* for all the constructions that I have referred to as *Left Dislocation*, even though she argues convincingly that these constructions are not

identical to their English counterparts. She reserves that term *Left Dislocation* for our *CLLDs*. Contreras (1991) assumes that all of them are LDs, and does not distinguish between *Focus Fronting* and the other two structures. Hernanz and Brucart (1987) use the term "*tematización*" instead of *Left Dislocation* (and oppose it to "*topicalization*") when referring to both types, HTLDs and CLLDs. The construction we refer to here as *Focus Fronting* has been also called *Focus Movement* (Zagona 2002), *Focus Preposing* (Casielles 2004), "*rematización*" (Hernanz and Brucart 1987) or, confusingly, *Topicalization* (Chomsky 1972; Rivero 1980).

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29 Speech Acts

VICTORIA ESCANDELL-VIDAL

1 Introduction

The idea that using language is a form of action has revealed a major source of inspiration for modern thinking. Linguistic utterances are not merely a way to truly describe states of affairs, but can be used to perform various activities, such as request, apologize and promise, among many others:

(1) ¿Me prestas un boli?
'Can I borrow a pen?'

(2) No sabes cuánto lo siento.
'I am really sorry.'

(3) Allí estaré.
'I'll be there.'

Similarly, there are actions, such as betting, declaring and condemning, which cannot be carried out unless certain words are uttered:

(4) a. Te apuesto 10 €.
'I bet 10 €.'
b. Declaro inaugurado este congreso.
'I hereby declare this conference open.'

Some utterances have the power to count as actions:

(5) a. Te prometo que estaré allí.
'I promise to be there.'
b. Te invito a cenar.
'I invite you for dinner.'

The Handbook of Hispanic Linguistics, First Edition. Edited by José Ignacio Hualde, Antxon Olarrea, and Erin O'Rourke.

The examples in (5) do not describe a fact, but rather perform an action: just by saying *I promise*, I am actually making a promise. These are *performative* utterances. In all these cases, language is a means for creating new states of affairs and introducing changes in the world, including not only the activities and the internal states of the participants, but also the relations among them. Utterances are, thus, *speech acts*, that is, intentional actions brought about through the interplay of the language faculty with other cognitive systems and the social and situational environment.

To account for them, a multidisciplinary perspective is required that can encompass insights from different areas, including philosophy of language, grammar, human cognition, and social interaction: this is the realm of Pragmatics, a discipline developed over the last 50 years. Of course, the relationship between language and action had not gone unnoticed in earlier works, but only in the last decades have researchers gained full awareness of the difference between what an expression means – the field of Semantics-, and what it can be used for – the concern of Pragmatics. This has resulted in the conscious adoption of theoretical frameworks making specific claims about how to relate, and distinguish between, the contribution of linguistic items and that of contextual and situational factors. In this respect, current Spanish linguistics has followed the mainstream Anglo-American tradition, with the adaptation of the ideas in Austin (1962), Searle (1969), Grice (1975), Sperber and Wilson (1986) and Brown and Levinson (1987) (see Escandell-Vidal 1996a for an overview). Though original and innovative theoretical proposals are rather scarce – Sánchez de Zavala (1997) is possibly the most salient exception – significant contributions have been made in the form of very detailed accounts of the major kinds of speech acts and their conditions of adequacy.

The rest of the paper is organized as follows. The basic notions will be presented in Section 2. Sections 3–5 will discuss the main trends and findings on speech acts in Spanish linguistics, with particular attention to the relations between sentence type and illocutionary force. Section 6 is devoted to presenting some conclusions and suggesting directions for future research.

2 Basic notions

Uttering any linguistic expression involves various kinds of acts (Austin 1962). The production of a string of sounds linked to a conventional meaning is by itself a form of action – the act *of* saying something; it is called *locutionary act*. The act performed *in* saying something – informing, asking, commanding – is called an *illocutionary act* and the speaker's intention is referred to as its *illocutionary force*. Finally, the effects obtained in the addressee *by* saying something – effects like convincing, amusing and saddening – are *perlocutionary acts*. Though these can be identified in any utterance, the literature on speech acts has generally focused on the illocutionary side.

Languages provide specific structures to encode different illocutions in a rather conventional way; thus, declarative sentences seem to specialize as statements, interrogatives as questions, and imperatives as commands:

(6) Juan duerme.
'John is sleeping.'

(7) ¿Juan duerme?
'Does John sleep?

(8) (Juan,) duerme.
'(John,) sleep!'

In each sentence two main constituents can be distinguished: the *propositional content* (roughly, the predication expressed: *Juan + dormir*) and the *mood indicator* (the linguistic elements that determine the sentence type, that is, the syntactic, morphologic or prosodic features that make it possible to distinguish a declarative, an interrogative and an imperative from each other). Thus, the examples in (6)–(8) share the same propositional content but differ in mood, so the illocutionary acts conventionally associated with each of these sentences vary accordingly. Mood is considered an *illocutionary force indicating device* (IFID), a linguistic constituent that delimits the intention with which an utterance is made (Searle and Vanderveken 1985).

It should be noted, however, that there is no one-to-one correspondence between sentence type and illocutionary force. On the one hand, the same sentence can be used to serve different illocutionary points on different occasions. The example in (6) can be uttered to inform, but also to show disapproval or hint at the need to change, with the interpretation shown in (9):

(9) He shouldn't sleep so much > He's irresponsible/lazy/... > He should take things more seriously.

The interrogative in (7) can be a question, but could also have the force of an emphatic assertion, as in (10):

(10) John never seems to sleep! > He's workaholic!

The example in (8) does not necessarily convey a command: it can be a request, a suggestion, a piece of advice, a desire, even an ironic challenge ('Don't you dare sleep!'), depending on the circumstances. A speech act is described as *direct* when its illocutionary force corresponds with its sentence type; when this is not the case and other factors (contextual, situational or social information) have to be taken into account, the speech act is *indirect*.

On the other hand, it is obvious that there are far more illocutionary intentions than sentence types: in Spanish there are no specific mood indicators for complaints, greetings or offers, so these actions must be carried out by indirect means. In (1), for example, an interrogative was used to make a request, not to pose a genuine question. Interestingly, for many languages asking whether one can do a specific action counts as a rather conventional way to request permission from the addressee to do it.

This brings into play further conditions on the success of a speech act, namely, appropriate internal states or external requirements for the participants. For instance, apologies require the expression of regret to be sincere; and for a wedding formula to take legal effect, it has to be uttered by a person with the right to marry. Furthermore, like actions, utterances have social consequences as well: promises and oaths create a commitment for the speaker, bound by her/his own utterance; if someone has been declared guilty, s/he can even serve a life sentence as a result! Both requirements and consequences are predictable to a certain extent; in this sense, speech acts seem to follow some sort of general guidelines whose precise nature has to be determined and explained.

Taking into account both linguistic and nonlinguistic conventions, Searle (1975) has suggested a classification of speech acts into five main categories, depending on the speaker's attitude and intended results: *Assertive*: the speaker presents the propositional content as a representation of a state of affairs in the real world (reporting, announcing, answering, etc.); *Commissive*: the speaker commits her/himself to doing some future action (promising, swearing, guaranteeing, etc.); *Directive*: the speaker attempts to get the hearer to take a particular course of action (ordering, requesting, advising, forbidding, etc.); *Expressive*: the speaker manifests her/his attitude towards the propositional content (apologizing, thanking, greeting, etc.); and *Declarations*: the speaker brings about the state of affairs represented in the proposition by successfully performing the speech act (christening, resigning, etc.).

This does not mean that all indirect acts are necessarily supported by some pre-existing convention. For example, asserting that someone is doing something is not a conventional way to criticize them; yet this is a possible interpretation for (6), as shown in (9). The intended illocutionary force can be arrived at as the result of an inference based on world knowledge and certain assumptions about the individual and the situation. General processing devices and abilities involved in interpreting utterances must have a role as well in any account of linguistic actions. Thus, there is much more to interpreting speech acts than decoding mood indicators. Determining the illocutionary force of an utterance is, ultimately, a matter of identifying the speaker's intention, a process that involves linguistic knowledge, but also awareness of social conventions and inferential elaborations of world knowledge.

Research on speech acts has attempted to account for the complex relations among all these factors and three main trends can be identified. Studies oriented towards explaining the language faculty privilege the grammatical markers indicating the kind of speech act being performed. When attention is placed on the social side, cultural conventions determining the choice of linguistic expressions come to the foreground. Finally, when the focus falls on cognition, the goal is to search for the general principles linking language and action. These three trends are well represented in Spanish scholarship, as will be shown in the next sections.

3 Sentence type and illocutionary force

The approach deriving from speech act theory has favored a change in the way sentence type, sentence meaning and illocutionary force are dealt with in current research. This change has come about particularly in the last 25 years and can be perceived in various respects. To begin with, it is commonly accepted that a speech act is a unit of communication, not of grammar; therefore, it has to be primarily defined by its communicative intention, not by its linguistic features. An apology, for instance, can be performed either by a nonsentential constituent, a sentence or a series of sentences and fragments forming a whole text, as shown in the following examples:

(11) a. ¡Perdón!
'Sorry!'
b. Le presento mis más sinceras disculpas.
'I offer you my humblest apologies.'
c. Perdón. Ya sé que debería haber llegado hace media hora . . . El despertador, que no sonó. . . Lo siento muchísimo, de verdad. No volverá a ocurrir.
'Pardon me. I know I should have arrived half an hour ago . . . the alarm clock-it didn't go off . . . I'm ever so sorry, really. It won't happen again.'

3.1 Together, but different

This move has fostered the independence of syntactic and pragmatic notions. In traditional works, grammatical and pragmatic considerations were mixed together, so each sentence type was usually described not by its grammatical features, but by its typical illocutionary force: interrogatives were characterized as questions, imperatives as commands, etc.; no distinction was made between what a sentence means and what it can be used for in discourse. (Among the exceptions, it is worth mentioning Bello's 1847 grammar; see Haverkate 1982 for a discussion.)

Current grammatical analyses, on the contrary, offer a clear-cut distinction between linguistic features on the one hand, and illocutionary uses on the other. As a consequence, in recent works, speech act distinctions are integrated with, though kept different from, grammatical notions. The *NGDLE* (Bosque and Demonte 1999) – the largest comprehensive grammar of Spanish to date – is the best exponent of this change: the chapter on the syntax of constructions involving *wh*-movement (interrogatives, relatives and exclamatives) is different from, but related to, those examining the semantic side and the pragmatic import of interrogatives (*NGDLE* ch. 61 =Escandell-Vidal 1999) and exclamatives (*NGDLE* ch. 62 =Alonso-Cortés 1999); similarly, the grammatical properties of imperative mood and the interpretation of imperative utterances are presented in different sections (*NGDLE* ch. 60.2 =Garrido-Medina 1999). The same can be observed in the recent *NGRAE* (Real Academia Española 2009), a grammar with both a descriptive and a normative purpose, endorsed by all the academies of Spanish-speaking countries: *NGRAE* follows in this respect the line opened in *NGDLE*, so as not to mix grammatical explanation with pragmatic aspects, while paying attention to both. The present work is a further example of this trend.

3.2 Imperatives

Imperatives can be used in discourse with a variety of directive illocutionary forces, ranging from orders to pleas and from requests and suggestions to wishes (Garrido-Medina 1999: § 60.2.2; Haverkate 2002: § 3.1.2; *NGRAE*: § 42.5). The relation between speaker and addressee (in terms of social distance and power) and the kind of text are two crucial dimensions that determine the actual illocutionary force of an imperative utterance:

(12) a. ¡Rompan filas!
'Break ranks!'
b. ¡Perdona a tu pueblo, Señor!
'Forgive your people, oh Lord!'

The example in (12a) is understood as an order due to the institutional setting in which is presumably uttered (namely, the army), whereas (12b) is a prayer in a religious context.

Requests issued in imperative mood are far more common in Spanish than in English. In fact, they are the usual way to convey most directive illocutions, particularly in European Spanish. In service encounters, where the rights and obligations of the participants are conventionally established, a bare imperative can be an adequate way to request:

(13) Ponnos dos cañas y un vino. [to the bartender]
'Serve us two beers and a (glass of) wine.'

When the speaker feels that s/he should smooth the request, s/he can add *por favor* 'please':

(14) Cierra la ventana, por favor.
'Close the window, please.'

Bare imperatives are also used in offers, invitations, advice and suggestions, among others:

(15) a. Tómate otro bombón.
'Have another chocolate!'
b. Toma asiento.
'Take a seat!'
c. Ten cuidado.
'Be careful!'
d. Echale más sal.
'Put more salt on it!'

As the above examples show, there is a correlation between illocutionary force and some aspects of propositional content. The identification of an utterance as a command or as an offer depends finally on who is to benefit from the action: commands are carried out to the speaker's advantage, whereas offers – a class of commissive acts – benefit the hearer. Similarly, control over the action is the crucial factor for distinguishing directive from expressive illocutions. Thus, the very same expression will be conceived as a directive, if understood as a subject-controlled activity, and as a desire, otherwise:

(16) ¡Ponte bien!
'Sit up straight!'/'Get well!'

Imperatives can also occur in so-called 'conditional constructions':

(17) Da un paso más y eres hombre muerto.
'Take one more step and you are a dead man.' (Cf. If you go one step further, you will be a dead man. > Don't you dare move one step further or you will be a dead man.)

The examples examined so far reveal that a single sentence type can give rise to a wide range of different illocutions: identifying the speaker's intended meaning is not merely a matter of decoding; other factors, such as the kind of event, the social relations among the participants, the consequences of the action, and the extra-linguistic situation play a major role as well.

Some uses of infinitives have been related with imperative mood in the sense that they can issue commands. In fact, infinitives are an alternative form to give directions to multiple and generic addressees, particularly when negative (where subjunctive is the suppletive form):

(18) a. ¡Correr! (Cf. ¡Corred!)
'Run!'
b. No fumar (Cf. No fuméis; No fumen)
'Do not smoke'

Infinitives also appear introduced by the preposition *a* 'to', in the so-called 'directive infinitive construction' (Rivero 1994):

(19) a. ¡A correr!
'Let's run!'
b. ¡Todo dios a correr!
'Let's run, everyone!'

These require an identifiable addressee and are specialized for commands, being thus incompatible with mitigating devices used in requests, such as *por favor* (as pointed out by a reviewer). Another infinitive construction, the perfect infinitive, is used to give directions referred to the past (Bosque 1980):

(20) ¡Haber venido antes!
'You should've come earlier!' (Lit.: Have come earlier!)

Though these have been labeled as 'retrospective imperatives,' one should bear in mind that these are imperatives neither from a formal point of view nor from an illocutionary perspective: they have a counterfactual import, aiming not at amending a past course of events, but at expressing a criticism.

In Spanish, as in other Romance languages, the present tense can be used with a directive force:

(21) Mañana vienes a la misma hora.
'Come tomorrow at the same hour.' (Lit. Tomorrow you come at the same hour.)

Interestingly, this use is restricted to expressing commands. Requests and suggestions cannot be performed in this way:

(22) Ahora te tomas otro bombón. [a description only; unacceptable as a suggestion]
'Now you take another chocolate.'

3.3 Interrogatives

Interrogatives illustrate the split between sentence type and illocutionary force in an even more perspicuous way. Though they are usually associated with questions – and even confused with these – obtaining new information about some unknown entity or event is just one of the possible interpretations. In fact, the communicative intentions that can be pursued by means of an interrogative almost exhaust the whole range of illocutions (Fernández Ramírez 1959, Escandell-Vidal 1988, 1999; Haverkate 1998; *NGRAE* § 42.6–11). Consider (23):

(23) a. ¿Estuvo Colón alguna vez en China?
'Did Columbus ever go to China?'

b. ¿Cuándo fue Colón a China?
'When did Columbus go to China?'

They can be uttered as genuine questions (i.e., the speaker does not know the answer), as exam questions (i.e., the speaker knows the answer and wants to check whether the addressee knows it as well), and even as emphatic assertions (with the interpretation 'Columbus never went to China'; cf. infra).

Interrogatives can be used as an indirect way to make requests, offers and suggestions, to ask for permission, and to express criticism, as shown in (24). Note that in Spanish no modal auxiliaries are needed for this purpose, while in English modal elements are required to obtain the same effect:

(24) a. ¿Me pasas la sal?
'Can you pass the salt?' (Lit.: Do you pass me the salt?)
b. ¿Has probado la tarta de manzana?
'Have you tried the apple pie?' > Have a piece of apple pie.
c. ¿Le pones pimienta?
'Are you putting some pepper in? (Lit.: Do you put pepper in?) > You should put some pepper in.
d. ¿Abro la ventana?
'Shall I open the window?' (Lit.: Do I open the window?)
e. ¿Te vas sin abrigo?
'Are you going out without your coat?' (Lit.: Do you go out without coat?) > You shouldn't go out without it.

Requesting and asking for directions or permission by means of a bare interrogative is rather conventional, though the propositional content and the knowledge of the situation help identify the intended interpretation. As in the case of imperatives, adding *por favor* 'please' to an interrogative guides the interpretation towards a request.

Spanish offers different intonation patterns for interrogatives, each specialized in a particular range of interpretations. In this respect, it is worth mentioning the pioneering work of Fernández Ramírez (1959). Though his analyses are not related to the Anglo-American speech act literature, he has provided very detailed descriptions of the various acts performed by interrogatives (cf. Escandell-Vidal 1998, 2002).

Separate consideration should be given to the cases in which negation occurs in an interrogative environment – an issue extensively discussed in the literature (Escandell-Vidal 1988: ch. 7, 1999: § 61.3.3; Gutiérrez-Rexach 1997; *NGRAE* § 42.10). This could seem surprising, particularly in the case of *yes–no* interrogatives, since these bear precisely on polarity (affirmative vs. negative); yet, negative interrogatives are quite common and give rise to complex interpretive effects. Consider (25):

(25) ¿No estás de acuerdo?
a. Don't you agree/You agree, don't you?
b. (Is it the case that) you don't agree?

In the (a) interpretation the speaker favors the positive answer (i.e., that the addressee agrees) and the negation can be detached as a question-tag in Spanish too (cf. *Estás de acuerdo, ¿no?* 'You agree, don't you?'): this corresponds to a confirmation-seeking question. In the (b) interpretation, on the contrary, it is a negative predication that is being expressed: the speaker has just gathered from the situation that the addressee does not agree, and s/he puts forward this idea to explain a somewhat unexpected behavior. In this reading, the sentence can be introduced by discourse markers such as *entonces* 'then', or *de manera que* 'so', overtly indicating that a conclusion is being drawn (cf. *Entonces, ¿no estás de acuerdo?*, 'Then/so you don't agree.'). Thus, the occurrence of a negation indicates that the speaker is not neutral with respect to the expected reply. The ambiguity between the two interpretations arises in the written form only: in spoken discourse, the two readings are clearly distinguished by intonation patterns (Fernández Ramírez 1959; Escandell-Vidal 1998; *NGRAE*: § 42.10h).

Research on interrogatives has paid particular attention to two further classes: rhetorical and echo interrogatives. Rhetorical interrogatives (sometimes referred to as *rhetorical questions*) are not requests for information nor do they expect an answer; on the contrary, they have the force of an assertion, with a typical polarity-reversal effect (Escandell-Vidal 1984, 1999: § 61.5.2; Gutiérrez-Rexach 1998; *NGRAE* § 42.12):

(26) a. ¿Podemos olvidar a los que luchan por la justicia?
'Can we forget those who fight for justice?' > We cannot forget those who fight for justice.
b. ¿No te dije que vinieras pronto?
'Didn't I tell you to come early?' > I did tell you to come early.

A similar effect is obtained in *wh*-questions:

(27) a. ¿Quién dijo que era fácil?
'Who said this was going to be easy?' > Nobody said this was going to be easy.
b. ¿Qué no ha hecho por ti?
'What hasn't s/he done for you?' > S/he has done everything for you.

As for their pragmatic import, rhetorical interrogatives are assertive acts, usually intended as reminders: the speaker poses a fictive question for which both the addressee and her/himself should know the answer. In this sense, they can be considered as emphatic forms of assertion, reinforcing a piece of information already present in the common ground. It is no surprise, then, that they can be used as answers to other questions:

(28) —¿Puedo confiar en ti?
—¿Y por qué no habías de confiar?
'—Can I trust you?'
'—Why shouldn't you trust (me)?' > There is no reason for which you shouldn't trust me.

As the above examples show, there can be nothing in the form of the utterances that forces a rhetoric interpretation; in fact, all of them could be understood as genuine questions in the adequate circumstances. This suggests again that illocutionary force is not merely a matter of decoding, so other situational and contextual factors should be taken into account as well.

However, there are some structures that can only obtain a rhetorical interpretation. The most classical one involves negative polarity items (NPIs) (cf. Gutiérrez-Rexach 1997): interrogatives containing NPIs such as *mover un dedo* 'lift a finger,' *pegar ojo* 'sleep at all', *dar un duro* 'give a damn' are intended as rhetorical:

(29) a. ¿Ha movido {alguna vez/nunca} un dedo por ti?
'Has s/he ever lifted a finger for you?' > S/he has never lifted a finger for you.
b. ¿Cuándo ha movido un dedo por ti?
'When has s/he lifted a finger for you?' > S/he has never lifted a finger for you.

Similarly, the use of modal forms calls for rhetorical interpretations. Compare (27a), with the indicative *era* 'it was' in the embedded clause, which is neutral, with (30), where the subjunctive *fuera* 'it were' demands a rhetorical interpretation.

(30) ¿Quién dijo que fuera fácil?
'Who said this was going to be easy?'

Adding *acaso* 'by any chance' to a *yes–no* interrogative forces a rhetorical reading as well.

As for echo interrogatives (also referred to as *echo-questions*), they have been described in the literature as reproducing actual (or attributed) utterances of any type, with a wide variety of illocutionary purposes (Dumitrescu 1991, 1992, 1993; Reyes 1994; Escandell-Vidal 1999: § 61.5.1, 2002). They can be expressive acts, conveying the speaker's surprise or rejection with respect to the previous turn:

(31) —¿Qué te has comprado?
—¿Que qué me he comprado? Pero si no tengo ni un duro . . .
—'What did you buy?'
—'What did I buy? (Lit. That what did I buy?) I don't even have a cent!'

(32) —Juan no ha venido.
—¡Que Juan no ha venido! ¡Es increíble!
—'John didn't come!'
—'John didn't come? (Lit. That John didn't come?) That's incredible!'

Echoes can also be requests for clarification, if the speaker needs more information in order to understand a vague expression or has not properly understood some previous words. The dialogue in (33) illustrates the two situations in which each echo is pronounced with a different intonation pattern:

(33) —¿Lo leíste?
—¿Si leí qué? [falling intonation]
—¿Leíste lo de los cercopitecos?
—¿Que si leí lo de los QUÉ? [rising intonation]
—Los cercopitecos, esos monos que hacen diferentes señales de alarma
—'Did you read it?'
—'Did I read what?'
—'Did you read (the article) about the cercopitheci?'
—'Did I read about WHAT?'
—'The cercopitheci, those monkeys that make different alarm calls . . .'

As the examples show, echo interrogatives are not a uniform class from an illocutionary perspective, nor are they homogeneous from the point of view of grammar: they are syntactic environments with specific licensing conditions. Though some advances have been made, their structure is far from being well explained.

A different interrogative structure that can be worth mentioning is the 'root infinitive construction' (Escandell-Vidal 1987; Etxepare and Grohmann 2003):

(34) a. ¿Asustarte tú? No puedo creerlo . . .
'You frightened? I can't believe it'
b. ¿Yo leer novelas rosas? Qué dices . . .
'I read romance novels? What are you talking about?'

These are always used as replies rejecting the content of a previous utterance or of a thought attributed to a different speaker. This explains their limited distribution in discourse and their special intonational properties (Escandell-Vidal 1998, 2002).

3.4 Exclamatives

It is common to label as *exclamations* both sentences of a certain grammatical type and utterances that convey admiration, joy, surprise, anger, disgust, etc., regardless of their syntactic features: this is, then, a further case of the need to differentiate grammatical notions from pragmatic ones (Alonso-Cortés 1999; Villalba 2003, 2008; Munaro 2006; Castroviejo-Miró 2007, 2008; Gutiérrez-Rexach 2008). Consider the examples in (35):

(35) a. ¡Qué alto es Gasol!
'How tall Gasol is!'

b. ¡Lo alto que es Gasol!
'How tall Gasol is!'

By uttering any of these sentences, the speaker seems to be doing several things at the same time: s/he is highlighting the degree to which Gasol is tall; in addition, s/he is taking for granted that Gasol is tall (i.e., presenting Gasol's tallness as an uncontroversial part of the common ground); and, finally, s/he is conveying an emotive attitude towards Gasol's tallness – an attitude that can be described as unexpectedness (Villalba 2008). Exclamatives are thus specialized for expressive illocutions; however, the particular kind of attitude conveyed is not encoded, but has to be inferentially gathered from different aspects of the situation. There is agreement among the researchers about the exclamative status of the two examples in (35): they both contain a *wh*-element (either a fronted *wh*-constituent, or a DP with a sentential interpretation containing a degree relative), are factive (i.e., present their content as a fact), and are assigned an extreme degree interpretation.

On the other hand, any sentence type (declarative, interrogative or imperative) can be turned into an exclamation (i.e., a speech act with an expressive force, regardless of its syntactic structure) by means of specific prosodic contours:

(36) a. ¡Han llegado los primos!
'The cousins have arrived!'
b. ¡Qué dirá usted que me he encontrado?
'Guess what I've found! (Lit.: What will you say that I've found!)
c. ¡Déjame en paz!
'Leave me alone!'

These constructions express the speaker's attitude, but they lack the two defining properties of *wh*-exclamatives: they do not take for granted their propositional content and they do not emphasize the high degree of a property.

Some other constructions seem to be halfway in between, as illustrated by the examples in (37) (cf. Hernanz 1999, 2001, 2007; Villalba 2003, 2008; Munaro 2006; Castroviejo-Miró 2008; González-Rodríguez 2008):

(37) a. ¡Es tan mono . . .!
'He's so cute . . .!'
b. ¡Es de mono . . .!
'He's so cute . . .!'
c. ¡Sí que es alto!
'Boy, is he tall! (lit.: Yes that is tall!)'
d. Bien te lo decía yo . . .
'I warned you . . .! (lit.: Well I told you)'
e. ¡Será tonto!
'Is he stupid or what! (lit.: Will he be silly!)'
f. ¡Fantástica, la película!
'Wonderful, that film!'

Even if they lack a canonical *wh*-syntax or a clearly marked prosodic pattern, these examples have specific grammatical properties that exclude nonexpressive interpretations.

3.5 Cartographic approaches

Grammatical studies with a formal orientation (particularly in the Chomkyan tradition) systematically neglected semantic and pragmatic considerations for methodological reasons – a stance fully consistent with the basic assumptions of the model. However, in the last decades, since Rizzi's (1997) cartographic hypothesis of the sentence left-periphery, a whole range of functional projections (ForcePhrase, TopicPhrase, FocusPhrase) have brought the correlation between sentence type and direct illocution back into the grammar. The projection of ForcePhrase, which is reminiscent of the performative hypothesis in Katz and Postal (1964), is primarily intended as a syntactic device accounting for clause types; nevertheless, it can be a useful tool to account for the interface between syntax and discourse. The claim is that some grammatical features contribute to putting further constraints in the way the sentence can be interpreted and used in discourse.

Some examples can be offered to illustrate this perspective. Recent research has emphasized that several particles usually seen as "accessories" are indeed a means for guiding the hearer towards the identification of the speaker's intention. This is the case of some occurrences of *que* (complementizer 'that') in root structures, like the following (from Hernanz and Rigau 2006):

(38) a. Evidentemente, María está enfadada.
'Obviously, Mary is angry.'
b. ¡Evidentemente que María está enfadada!
'Of course, Mary should be angry!'

The contrast between (38a,b) involves grammatical positions and has clear consequences in terms of illocutionary import: (38a) can be discourse-initial assertion; (38b), on the contrary, is adequate only as a reaction to a previous utterance.

The occurrence of *que* in root environments can also trigger a quotative interpretation, adding a new speech event (Rivero 1994; Extepare 2008; Demonte and Fernández-Soriano 2009). The examples in (39) from Etxepare (2008) illustrate this. In (39a), the information is presented as reproducing a previous assertion by a different speaker (i.e., as reporting); in (39b), a directive speech act is added:

(39) a. Oye, que el Barcelona ha ganado la Champions.
'Listen, [they say] that Barcelona has won the Champions league.'
b. Si viene mi madre, que el tabaco es tuyo.
'If my mother comes, [tell her] that the tobacco is yours.'

Research on exclamatives evolved similarly. Much recent work is devoted to the formalization of expressive illocutionary force by means of operators occupying dedicated syntactic positions (Hernanz 1999, 2001, 2007; Gutiérrez-Rexach 2008; Batllori and Hernanz 2009).

A more finely articulated organization of the functional projections at the left periphery can thus explain many aspects of the correlation between syntactic properties and discourse potential. The move to bring illocutionary force back into the syntactic representation, however, is not free from difficulties; discussion on this issue is, however, beyond the limits of this chapter.

4 Illocutionary force and politeness

While grammarians tried to establish sentence type and illocutionary force as two different, though related, domains of inquiry, other researchers worked on the social side of linguistic actions: speaking takes place in a social environment, so the social requirements and consequences of language usage become a matter of interest. This approach has produced a huge amount of descriptions that would deserve an independent chapter. In this section I can only present a very brief overview. Two main lines can be identified in the literature on Spanish: research on the way in which social categories and discourse situations determine linguistic choices, and research on how politeness strategies vary across cultures.

4.1 Politeness strategies and the structure of speech acts

Leech (1983) and Brown and Levinson (1987) have been the major source of inspiration for politeness studies in Spanish linguistics. Leech develops Grice's conversational principles by suggesting new maxims to account for social interaction. Central to Brown and Levinson's approach is Goffman's notion of *face*: the public image that any individual wants to maintain. The need for preserving social relations represents a reason for resorting to indirect ways, especially with potentially threatening illocutions (such as requests); apparently impolite expressions can be nevertheless acceptable if used among close friends. Politeness is thus understood as a means for managing social interaction. In this way, a very attractive connection between speech acts research and social issues was established that opened a fruitful line of research on how social categories (particularly, distance and power) determine linguistic usage.

Soon after the publication of those two works, Haverkate (1987) was among the very first to discuss politeness strategies in Spanish, combining Searle's, Leech's, and Brown's and Levinson's insights. The relations between speaker and addressee and the situation are crucial to determining the degree of politeness of each illocutionary type. A scale of cost–benefit to the hearer is a necessary tool to assess the social impact of a given utterance: thus, whereas expressives and commissives are inherently polite, directives are not, so politeness strategies have to be adopted if one wants to mitigate the negative impact of an utterance. Performing a potentially impolite illocution (i.e., an act that can "threaten" the addressee's face), such as requesting, triggers the (unconscious) search for linguistic strategies to reduce the imposition. Politeness has thus been considered as the main reason for using indirect speech acts in an iconic way: the more indirect the means of expression, the more polite. Much work has been devoted to analyzing politeness strategies in Spanish (Haverkate 1994; Escandell-Vidal 1995; Hernández-Flores 2002), also focusing on different illocutions: assertions (Haverkate 1991), expressives and commissives (Haverkate 1993), directives (Mulder 1993, 1998), and suggestions (Koike 1998), among others.

4.2 *Speech acts and politeness across cultures*

The works referenced in the previous section are devoted to answering the question of how different kinds of illocutions are performed in Spanish. In more recent years, research about politeness strategies has shifted towards a different purpose, namely, accounting for conversational styles (i.e., the set of communicative routines of a given culture). Each cultural group has its own conventions and strategies, which can be different from those of other cultures; this has brought to the foreground the search for underlying social values.

This move has opened the field of Pragmatics to a research methodology adopted from the social sciences, mostly with large corpora and statistical analysis. The main source of inspiration was the *Cross-Cultural Speech Act Realization Project* (CCSARP), first implemented in Blum-Kulka et al. (1989), where Argentinean Spanish was compared to Australian English, Canadian French, and Hebrew with respect to requests and apologies. This has produced works with two orientations: cross-linguistic studies, where Spanish is compared to other languages; and studies on intralinguistic variation, where the comparison is focused on geographic and sociopragmatic aspects.

Research on cross-cultural politeness has experienced exponential growth in the last two decades, due in part to the development of other related fields, such as second language teaching and intercultural communication. Much has been done both on interlinguistic comparison (Fant 1989; le Pair 1996; Márquez-Reiter 2000; Hickey and Steward 2005; Pütz and Neff 2008) and on intralinguistic variation (Placencia and Bravo 2002; Bravo 2003; Murillo-Medrano 2005; Briz et al. 2007; Placencia and García 2007; Rodríguez Alfano 2009).

Despite the huge amount of work, the picture on Spanish conversational style is far from being neat: the authors agree on the tendency to positive politeness strategies and to affiliation, whereas Anglo-Saxon culture seems to favor negative politeness strategies and privacy. However, this impressionistic view disappears in a kaleidoscope when large language samples are used and when language-internal variation is analyzed. The result is that identifying culture-specific values seems more and more out of reach. In addition, postmodern approaches to politeness (Watts 2003) do not favor the search for generalizations either: they lead away from speech act classifications and from the view of culture as a monolithic construct.

5 Cognition and inferential processes

The third main trend in the approach to speech acts in the recent literature on Spanish follows the ground-breaking work by Grice (1975), postulating the existence of an internal logic, founded on principles of rationality, which governs conversation; in this way, research on cognitive processes entered the area of speech acts. If utterance interpretation is a matter of recognizing the speaker's intended meaning, there must be some principles explaining how the illocutionary force of an utterance is successfully identified in absence of other conventions – of a linguistic nature, or otherwise. The key in this approach is the notion of *implicature* – i.e., the extra piece of meaning that a speaker conveys without explicitly saying; implicatures depend crucially on the context and are the result of inference.

Other authors have developed Grice's ideas further. Among them, the most influential model in Spanish pragmatics is Sperber's and Wilson's (1986) relevance theory (see Escandell-Vidal 1996; Yus 2003). Relevance theory aims at accounting for human communication in terms of the design features of our mind. Communication is a matter of expressing and recognizing intentions; if we are able to do this, it is because we are endowed with some dedicated cognitive subsystems that specialize in the interpretation of evidence provided. A pragmatic theory is ultimately a theory of the interaction between the language faculty and other cognitive modules.

Though *speech acts* as such do not play any role in inferential models, illocutionary labels are sometimes used for convenience. Illocutionary forces are not directly encoded in the form of the utterance; rather, they are inferred as the result of combining the linguistic and nonlinguistic information available. What sentence types encode, according to this approach, is an abstract, underspecified semantic representation, common to all the uses of a sentence; the integration of linguistic and nonlinguistic information is a crucial step to flesh out the schematic semantic representation and identify the speaker's intended meaning. Imperatives, for example, encode that the event is both potential and desirable; imperative mood, however, does not specify either for whom the event is desirable or how the event is supposed to come into existence. From this perspective, sentence meaning is separated from illocutionary force in a precise way, and a characterization of the semantic contribution of sentence type seems within reach.

A general discussion on illocutionary force as such from a relevance-theoretic perspective is rather scarce (Figueras 2000; Rosales-Sequeiros 2002; Jary 2007). However, the influence of this model in the way researchers approach the relations between formal indicators and interpretation owes a lot to Sperber's and Wilson's insights and can be found, to some extent, in most of the works mentioned in Section 3. Some attempts to integrate social and cognitive approaches have also been made (Escandell-Vidal 1996b, 2004).

This section on cognitive models could not end without a reference, though necessarily very brief, to Sánchez de Zavala: probably the most original thinker in Spanish pragmatics. In his posthumous 1997 book, he set the foundations for an alternative way of doing pragmatics with a more psychological stance. From current mainstream pragmatics, he criticized two assumptions: first, the idea that language is basically a means for communication; and second, the idea that pragmatic explanation can be cast in terms of rationality. Instead, he advocated for an approach where language has dimensions that go far beyond communication and where emotion also plays a major role. Unfortunately, he was not able to fully develop his ideas: should he have been able to do so, this chapter would have ended in a different way.

6 Conclusion and directions for future research

The aim of the previous sections has been to provide a roadmap for research on speech acts in Spanish linguistics in three main areas: grammar, society, and cognition. After this review, some conclusions emerge. The evidence for distinguishing between sentence type and illocutionary force is overwhelming. This has produced several welcome results in grammatical theory. On the one hand, it has made grammatical explanation both more powerful and more economic: grammar can resort to a well-defined set of primitive notions, cast exclusively in grammatical terms (word order, verbal mood, prosodic patterning, etc.), thus abstracting from the various uses an utterance can have depending on the circumstances. On the other hand, considerations on the conventional illocutionary import of sentence types have become a necessary part of linguistic description, thus providing a more complete perspective on language knowledge and use. A further consequence has to do with semantics: it has been shown that grammatical features do not encode illocutionary forces in a direct way; rather, they convey a more abstract, underspecified meaning, along the lines suggested in Sperber and Wilson (1986) (cf. *NGRAE* § 42.2k), which has to be inferentially completed with other nonlinguistic assumptions during the process of utterance interpretation. In this way, a semantic account of sentence types that is autonomous with respect to illocutionary force can be formulated.

Much has been done in this field, but there are still many aspects that require further research. The classic inventory of speech acts is not without problems. It is not always easy to ascribe an utterance to a single class. As shown in the examples in (37), the border between assertive and expressive acts is fuzzy, so further factors have to be brought into consideration to define non-overlapping categories. This is due to the fact that speech act classes are not primitives. Advances in syntactic and semantic theory and new formal models on discourse representation can contribute to reconsidering speech act classification by taking into account other parameters, such as the contribution to updating common ground.

One area that has been usually excluded from formal analyses is intonation. Attention has been focused on the written form, thus disregarding a crucial component not only of utterances, but also of sentences. Research is moving towards the necessary integration of intonation in a model of grammar. Many researchers talk about particular marked, expressive, emphatic prosodic patterns, without explaining exactly what they mean and without providing an explicit characterization of the acoustic and phonological properties of such patterns. These are crucial properties for the identification of illocutionary force. Much work will be needed as well to establish the line between phonological and expressive aspects of intonation.

The focus on the written form has also favored the analysis of speech act realization in rather formal registers. There are, however, a large amount of structures – most belonging to an informal register – with interesting grammatical and pragmatic properties that have not received sufficient attention. Only in the last years are we beginning to disentangle the various forms in which assertions are modulated: emphasis, intensification, emotion, irony or humor have linguistic correlates that deserve a more detailed account. Similarly, there is still much to discover about the best treatment of so-called hidden exclamatives and concealed questions. Recent developments in syntactic and pragmatic theory offer very sophisticated tools to deal with all these issues in the near future.

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30 Discourse Syntax

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1 Introduction

In this chapter we present a perspective that places language use in discourse at the forefront of syntax. The central postulate of USAGE-BASED theory is that the basis for grammatical knowledge is speakers' linguistic experience – the frequency and contexts of use of forms (Bybee 2010) – in contradistinction to the view that language use and knowledge about language (or "performance" and "competence" cf. Chomsky 1965) are independent. These two perspectives on syntax have been broadly characterized as FUNCTIONAL (according to which grammatical structures conventionalize out of discourse patterns) and FORMAL (according to which explanations of linguistic structure are based on grammatical form, independent of function) (cf. Croft (1995) vs. Newmeyer (1998)).

The discourse-based perspective on syntax sees the observation of spontaneous language use, rather than reflection about language use, as the appropriate way to obtain data for linguistic analysis. Conversation receives particular primacy in this area, as the form of language that we spend most of our time using, that which is acquired first and without any formal instruction, and that which provides the most systematic data for the analysis of grammar as it is produced in real time (Labov 1984; Chafe 1994; Ono and Thompson 1995; Briz 1998; Du Bois 2003; Schegloff 2007). Community-based research has further highlighted the need to discover patterns in sociolinguistically-sampled speech corpora rather than collections of individual data (e.g., Labov 1984).

Functionalist analysis has given rise to a questioning of some basic formalist assumptions, such as the discreteness of categories, the notion that syntax, the lexicon, and phonology are separate components, and that grammar consists of generalizations made about abstract features. From a functionalist perspective, the following claims are made: linguistic categories (e.g., word classes, transitivity, subordination, etc.) are continua, with a prototype or exemplar structure; grammatical constructions combine syntactic, semantic, and phonological information;

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and grammatical generalizations derive "from the repetition of many local events" in discourse (Bybee 200: 714) (see Hopper and Thompson 1980, 1984; Du Bois 1987; Hopper 1998; Croft and Cruse 2004; Goldberg 2006; Langacker 2008).

Here we present an overview of work in Hispanic Linguistics that approaches syntactic patterning from the perspective of discourse function, with a particular focus on community-based studies of spoken corpora. We begin Section 2 with information flow considerations in the realization and distribution of NPs. Next, in Section 3, we discuss transitivity as a scalar phenomenon. Nonreferential uses of lexical Noun Phrases in discourse are reviewed in Section 4. In Section 5, we consider constructions as the basic unit of grammar, including both particular and general expressions. Finally, in Section 6, we present variable first-person singular subject expression as a case study of discourse effects in grammar.

2 Information flow

One of the richest areas of study within Hispanic Linguistics from the functionalist approach has been that of Noun Phrase realization and syntactic role. Such work has shown that the form of the subject (unexpressed/null, pronoun or full lexical NP), the syntactic role an NP takes (subject, direct object), and the positioning of NPs in relation to the verb are driven by several issues. These include the referents' relationship with the preceding discourse (whether they are opening a new sequence or continuing an old one), their topicality and, especially, information flow properties, as explained below (e.g., Bentivoglio 1983; Bolinger 1991; Myhill 1992; Ocampo 1992, 1993; Silva-Corvalán 1994; Morris 1998; López-Meirama 2006; García Salido 2008).

A wide body of research on Spanish and other languages has demonstrated that the information flow properties of the referent affect the way it is linguistically expressed. Specifically, a referent that is being introduced to the discourse for the first time is more likely to be expressed with "lexically heavy" material (full lexical NPs) than a referent that has already been talked about, which tends to be realized with less material (pronouns, unexpressed mentions) (e.g., Prince 1981; Givón 1983; Chafe 1994; Bentivoglio and Sedano 2007). This is illustrated in the following example, where *Don Julio Restrepo* is introduced into the discourse with a proper name, then referred to by a demonstrative pronoun, then a personal pronoun, followed by an unexpressed mention (represented by Ø) (see Dumont 2006: 290). (Notice also a similar pattern for the mentions of *mi tía Juanita*.)

(1) C: Recuerdas los nombres de otros o otras curanderas,
o curanderos.
V: <u>Don Julio Restrepo</u> curaba de brujería.
<u>Ése</u> era de Santa Cruz.
<u>Ése</u> curaba de brujería una vez.
a mi tía Juanita le hicieron mal,
y <u>él</u> vino y <u>Ø</u> la curó y Ø sanó.
de brujería. [NMCOSS: 4][1]

C: 'Do you remember the names of other (female) healers,
or (male) healers.'
V: 'Mr Julio Restrepo cured (people) from witchcraft.
He was from Santa Cruz.
He cured of witchraft once.
they put a spell on my Aunt Juanita,
and he came and (he) cured her and (she) got better.
of witchcraft.'

It is not just syntactic form, but also syntactic role that interacts with discourse function. Specifically, NPs introducing referents into the discourse for the first time tend not to occur as subjects of transitive clauses, and instead speakers prefer the roles of subjects of intransitive clauses or objects of transitive clauses. This pattern has been dubbed PREFERRED ARGUMENT STRUCTURE, and its attestation in a wide range of typologically diverse languages questions the notion of "subject" and "object" as categories even for traditionally nominative/accusative languages (see Du Bois 2003 for a review).

In Spanish, the tendency has been extensively tested, and found to hold (Bentivoglio 1993, Bentivoglio and Ashby 2003). In an analysis of the distribution of full NPs, pronouns, verbal clitics, and unexpressed mentions extracted from sociolinguistic interviews of speakers in Caracas, Venezuela, Ashby and Bentivoglio (1993: 65) found that while full NPs occurred much less frequently than pronominal and unexpressed NPs overall (representing just 28% of the total number of NPs in the data, 591/2,121), their distribution was skewed along the lines predicted by Preferred Argument Structure. Full NPs occurred proportionately less as subjects in transitive clauses (6% vs. 94% pronominals and unexpressed mentions), followed by subjects in intransitive clauses (22%), and only for objects in transitive clauses did they reach a majority (60%). A similar skewing was found for information status: approximately one-third (200/591) of all full NPs expressed "brand new information" (that is, following Prince (1981), those referents that had not occurred in the preceding discourse nor were they identifiable through a frame or schema), and this was distributed differently across the syntactic roles: 42% of full NP direct objects expressed brand new information compared with 26% of full NP subjects of intransitive clauses and just 6% of full NP subjects of transitive clauses (1993: 71). This pattern can be seen in example (2) below (cf. Dumont 2006: 288). The referents of the underlined full NPs are all introduced into the discourse for the first time, *el marido* 'my husband' and *un muchito*[2] 'a little boy' as subjects of intransitive clauses, and *su grandma* 'his grandmother' as the direct object of the verb *tener* 'have.' Note that while all three referents are introduced into the discourse for the first time, *un muchito* is unidentifiable, or brand new, while *el marido* can be considered identifiable through a familial relationship with the speaker, and *su grandma* identifiable through a familial relationship with *un muchito*, who was introduced in the previous clause (cf. Thompson 1997). This is reflected in their morphosyntax, with the speaker expressing the identifiable referents in the form of definite NPs in their canonical positions (preverbal for subjects and postverbal for objects), and the unidentifiable referent as an indefinite NP in postverbal noncanonical subject position.

(2) Ahí iba yo sola.
porque el marido en veces estaba trabajando.
y en ve- --
luego,
me acompañaba un muchito.
que tenía su grandma allá en Parki View. [NMCOSS: 8]
'I would go there alone,
because my husband was sometimes working.
and some- --
later,
a little boy would accompany me.
who had his grandmother in Park View.'

Thus, here we see patterning in grammatical structures (NP realization, distribution across syntactic roles, and word order) with a parallel patterning in discourse function (information status of the referents), an illustration of how grammar is shaped by language use, or the way in which "grammars code best what speakers do most" (Du Bois 2003: 49).

3 Transitivity

The distribution of new information in the clause challenges not only the categories of "subject" and "object", but also the traditional distinction between "transitive" and "intransitive" verb classes. Ashby and Bentivoglio (1993), for example, observe distinct patterning with different kinds of verbs within each class. In particular, they find that the copulas *ser* and *estar* pattern more similarly to transitive verbs in terms of their discourse pragmatics in that they disfavor new information more than do other intransitive verbs (though less than transitive verbs as a class) (1993: 71). Furthermore, they find that Spanish speakers use the single argument of *haber* and the direct object of *tener* (as in (2) above) in existential-presentational constructions to introduce new information significantly more than the other roles favored for the introduction of new information, subjects of other intransitive clauses, and objects of transitive clauses (1997: 22). We find in operation, then, a local strategy that functions at the level of constructions with specific verbs, regardless of the class they pertain to, thus giving cause to question the discreteness of transitivity (see also Section 5).

Hopper and Thompson's (1980) work presents cross-linguistic evidence to challenge this widely accepted dichotomy, proposing that transitivity is not based on the mere presence of a direct object, but is manifested to different degrees depending on the "effectiveness" of the action, as reflected in components (or parameters) such as aspectual features of the verb, agentivity and volitionality of the subject, and affectedness and individuation of the object. Hopper and Thompson (1980) hypothesize that these components of scalar transitivity covary such that features indicative of higher transitivity tend to co-occur in the clause. Testing this hypothesis in Spanish, Vázquez Rozas (2006: 101–108) argues that verbs like *atraer* 'attract,' *fascinar* 'fascinate,' and *interesar* 'interest' (which may appear either with a direct object or as *gustar* 'please'-type verbs with an indirect object experiencer) tend to be used in the higher transitivity direct object construction when the object is physically affected, the subject agentive, and the situation has an inherent endpoint (i.e., is telic), but in the lower transitivity indirect object construction with aspectually stative situations, an inanimate subject, and an object that is affected psychologically. In the following pair of examples, the direct object experiencer construction in (3) involves an activity of the stimulus participant (the growls of the lion), while the indirect object experiencer construction in (4) involves a stative property of the stimulus (the leather binding of the book).[3]

(3) Los rugidos del león atrajeron al cazador,
quien [. . .] lo metió en una gran jaula.
(Vázquez Rozas 2006: 104)
'The growls of the lion attracted the hunter,
who [. . .] put it in a big cage.'

(4) El libro estaba encuadernado en piel [. . .], pero a ella no le atraía.
(Vázquez Rozas 2006: 104)
'The book was bound in leather [. . .], but it didn't attract her.'

A further challenge that presents itself to the traditional understanding of transitivity concerns how to deal with different kinds of "objects". "Objects" with low-content verbs, often in conventionalized combinations (cf. Ashby and Bentivoglio 1993: 67–68; Chafe 1994: 111–113), may actually form unitary predicates, or "V-O compounds", for example, *have fun, make no sense, get sleep* (Thompson and Hopper 2001: 37). Examples from Spanish are given in (5), (6) (Torres Cacoullos and Aaron 2003: 321–322), and (7) (Burgos 2007: 29). Note that in (7) the predicate is jointly produced by the two interlocutors, evidence that this is a conventionalized schema which they share (cf. Lerner 2002).

(5) hago escultura. [NMCOSS: 102]
'I do sculpture.'

(6) Y le puse complaint a ese chota. [NMCOSS: 219]
'and I filed a complaint against that cop'

(7) P: no me gusta mucho trabajar,
porque le sacan --
A: Ay,
la paciencia. [CCS: ESP]

P: 'I don't like much to work,
because they try --'
A: 'Oh,
your patience.'

Sentential objects may also be of issue for transitivity. Rather than viewing the material following *creo* 'I think' in (8) below as a subordinate clause (specifically an object complement), Thompson (2002) argues that apparent biclausal combinations often behave as single clauses which include a speaker-stance discourse formula. That is, certain apparent complement-taking predicates such as *yo creo* and *yo (no) sé* (see Section 6 below) may serve as frames of speaker stance toward the content of their putative subordinate clause, which actually carries out "the work that the utterance is doing" (Thompson 2002: 155). Evidence comes from their distributions as well as pragmatics; for example, *(yo) creo* 'I think' shows the same kind of variability in grammatical and prosodic position as epistemic/evidential adverbial expressions such as *a lo mejor* 'probably' by being variable in position (occurring preceding in (8), following in (9), or even within, the material it marks). It is also often prosodically independent, i.e., occurring in an Intonation Unit of its own, as in (9) (cf. Travis 2006). (Following the Du Bois et al. (1993) transcription method, Intonation Units are represented on distinct lines in the transcription; see Ono and Thompson 1995; Sánchez-Ayala 2001; Shenk 2006, on the use of the Intonation Unit as a meaningful unit for syntactic analysis.)

(8) Yo,
Creo que estoy --
casi que seguro que puede ser eso. [CCS: breakfast]
'I,
think that I'm --
almost sure that it might be that.'

(9) Él es de --
Friburgo,
creo. [CCS: Tumaco]
'He is from --
Freiburg,
I think.'

In sum, this body of research has shown that transitivity is usefully viewed as a continuum which is subject to local effects of frequently used discourse patterns. These considerations reinforce our understanding of the discourse functions of transitivity. Namely, speakers use higher transitivity for foregrounded material, especially in a narrative (Hopper and Thompson 1980: 280–294), and lower transitivity as a reflection of subjective "talk about how things are from our perspective", especially in conversation (Thompson and Hopper 2001: 53).

4 Referentiality in discourse

Another illustration of grammatical patterns driven by discourse function is found in the area of referentiality. Discussions of referentiality often conflate specificity and discourse referentiality, which are correlated but not coterminous. On the one hand, specificity has to do with the way an NP refers to entities. Specific (particular) NPs refer to unique people or things that are not considered to be interchangeable (10), while nonspecific (general) NPs indicate any member of a class of entities (11) (cf. Ashby and Bentivoglio 1993: 69–70). A third type is generic NPs, which refer to an entire class of entities (12) (Torres Cacoullos and Aaron 2003: 308). In (New Mexican) Spanish this functional distinction is grammatically encoded: nonspecific NPs are more likely than specific NPs to appear bare (without a determiner) but, unlike in English, generic NPs line up with specific NPs in disfavoring bare forms.

(10) Y luego una vez que se me % --
.. quemó el generador de <u>la troca mía</u>, [NMCOSS: 311]
'and then once when --
the generator of <u>my truck</u> burnt out,'

(11) Lo que me ayudó a mí,
que yo arrié quince años <u>troca</u>. [NMCOSS: 214]
'what helped me,
Is that I drove <u>trucks</u> for fifteen years.'

(12) Pero el nombre que la gente usaba aquí pa' los --
pa' <u>los buzzards</u>,
... I CAN'T REMEMBER. [NMCOSS 217]
'But the name that people used here for --
for <u>buzzards</u>,
I can't remember.'

Discourse referentiality, on the other hand, has to do with tracking participants. Discourse referential (Tracking) NPs are used "to speak about an object as an object, with continuous identity" in the discourse (Du Bois 1980: 208). In Hopper and Thompson's terms, Tracking NPs are used for "manipulable" discourse participants, people, or things about which something is said (1984: 711). In contrast, non-Tracking NPs serve to classify (as predicate nominals) or to orient (as adverbials, giving the location or time of an entity or situation) (Thompson 1997: 69). For example, in (13) the unexpressed subject is a discourse-referential NP but the predicate nominal *train master* is nonreferential, serving to characterize the subject. The proper noun *Corpus Christi* in the *en* locative prepositional phrase is also used nonreferentially, serving to orient the predication rather than to introduce the city in order to talk about it.

(13) es train master en Corpus Christi. [NMCOSS: 190]
'He's a train master in Corpus Christi.'

A third kind of non-Tracking NP concerns lexical NPs used in the "V-O compounds" discussed in Section 3. The underlined NPs in examples in (5), (6), (7), and also (11) are not used referentially as arguments (objects) but rather as "Predicating NPs [which] function as part of naming a type of event, activity, or situation" (Thompson 1997: 71–72): 'sculpting,' 'accusing,' 'trying (one's) patience,' and 'truck driving.'

Applying this discourse-based understanding of referentiality to English-origin nouns that are not established loanwords in New Mexican Spanish, Torres Cacoullos and Aaron (2003) are able to account for a slightly higher rate of bare (determinerless) forms in English-origin nouns than in native Spanish ones (36%, 98/270 vs. 30%, 413/1,386). These scholars show that such nouns are English in origin only, following Spanish grammatical patterns for predicate nominals in a classifying function, especially ones designating occupations, as in (13), and for Predicating NPs in combinations with low content verbs such as *hacer* 'do' and *poner* 'put,' seen in (5) and (6) above. The higher rate of bare forms is thus due not to lack of grammatical integration but to the use by bilinguals of English-origin nouns as non-Tracking NPs.

In sum, studies have profited from distinguishing discourse referentiality and specificity in characterizing NPs. Specificity in particular has been found to affect the form of objects (e.g., García and Putte 1989; Schwenter 2006; Reig Alamillo 2009; Balasch (2010). This is evidence, once again, of the role of discourse considerations in linguistic structures.

5 Constructions and prefabs

CONSTRUCTIONS have been viewed as the units of grammar on the basis of observations about the idiomaticity of language (Fillmore et al. 1988), the interaction of syntax and lexicon (Langacker 2008), cross-linguistic variation (Croft 2001), language acquisition and learning (Tomasello 2003; Goldberg 2006), and grammaticalization (Bybee and Torres Cacoullos 2009). Constructions are conventionalized form-function pairings, ranging from fixed expressions (including lexical items, idioms and prefabs: see below) to productive morphosyntactic structures that are at least partially schematic (for an overview, see Croft and Cruse 2004: 225–290; Goldberg 2006: 3–17). For example, the idiom *subirse a la cabeza* 'get intoxicated' (lit. 'go to your head') would be considered a construction, as would the more schematic *subir* 'go up' STAIRS (e.g., *escaleras*, *peldaños*) and *subirse* 'climb' TREE (e.g., *higuera*, *mezquite*). Torres Cacoullos and Schwenter (2008) present evidence for constructional status in terms of the differential patterning with (middle) *se*: in their data *subir* was overwhelmingly not marked with *se* for stairs (3/66), but was *se*-marked for trees (9/9) (cf. Maldonado 1999; Aaron 2004; Clements 2006 for other functionalist accounts of *se*).

Social and pragmatic use is included as part of the meaning of a construction. Consider, for example, the variation between *estar* VERB-*ndo* and *andar* VERB-*ndo* as expressions of progressive and other imperfective aspects in Mexican varieties, as illustrated in (14) and (15).

(14) estás hablando de una forma de vida (Lope Blanch 1971: 261)
'you are talking about a lifestyle'

(15) ando buscando unas tijeras (Lope Blanch 1971: 415)
'I am looking for some scissors'

Torres Cacoullos (2001) shows that this variation does not reflect aspectual differences, as is often assumed, but collocational routines: *estar hablando* 'speaking' is the conventional way to express 'be speaking', and *andar buscando* the conventional way to 'be looking for something.' Auxiliary *estar* is generally more likely to co-occur with verbs of speech (*platicando* 'chatting'), as well as verbs denoting perceptible bodily activities (*llorando* 'crying') and mental activities (*pensando* 'thinking'), whereas auxiliary *andar* is more likely with verbs denoting motion such as *dando la vuelta* 'going around' and physical activities, particularly outdoor ones, such as *trabajando* 'working' (*en las pizcas* 'in the harvest,' *en el campo* 'in the country'). In tracing the evolution (grammaticalization) of the two constructions, Torres Cacoullos (2001) suggests that this distribution is the residue of source-construction meanings, 'be located (stationary)' for *estar* and 'go around' for *andar*. Moreover, the *andar* VERB-*ndo* construction has social meaning in that its relative frequency is higher in popular than educated speech. This social meaning is related to the lexical associations observed, since rural activities in large spaces are more compatible with 'going around'. Thus, the educated / popular social stratification of *estar/andar* VERB-*ndo* may derive from particular instances of use, developing from an indoor–outdoor and urban–rural difference.

Collocational patterns, or lexical associations, are also shown in the combinations of adjectives with verbs of becoming (change of state) such as *quedarse*, *ponerse* and *volverse*, all meaning something similar to English 'become.' Studies attempting to establish generalizations about abstract features (such as the duration of the state or the degree of involvement of the subject) have failed to predict which adjectives go with which verbs. Bybee and Eddington (2006) argue that speakers' choices of verb–adjective combinations are shaped by more local exemplar-based categorization. For example, they propose several categories of adjectives that occur with *quedarse* 'become (lit. remain + reflexive).' Importantly, these categories are based on clusters of adjectives which are semantically related to particular lexical types that occur frequently with this same verb in oral and written corpora. For instance, with high frequency *quedarse solo* 'end up alone' are grouped *soltera* 'single,' *aislado* 'isolated,' and other semantically related adjectives, whereas *parado* 'stopped' and *tieso* 'stiff,' for example, are grouped with high frequency *quedarse inmóvil* 'immobile.' In acceptability judgments, Bybee and Eddington found that high frequency verb–adjective combinations were ranked as more acceptable than those with low corpus frequency. Furthermore, of the low-frequency items, those that were semantically related to a high-frequency phrase (e.g., *quedarse parado* 'be stopped,' related to *quedarse inmóvil*) were rated as more acceptable than those that had no such semantic relation (e.g., *quedarse orgullosísimo* 'be proud'). Thus, the most frequent *quedarse* ADJECTIVE collocations form the basis for different groups of adjectives through analogical comparison. These results suggest that highly frequent central members of a category attract like members, such that category formation is based on local exemplars.

Usage frequencies affect the formation of constructions, as frequent items may become automatized as single units (Bybee 2003) (cf. Company Company 2006). For example, the collocation *yo no sé* 'I don't know' (cf. Travis 2006) need not be produced by speakers through the application of grammatical rules to the individual items that make it up, but rather may be produced (and stored) as a conventionalized single unit. Evidence is found in its entrenchment or relative fixedness in form (e.g., the limited variability in subject expression, with a strong tendency for *yo* to be expressed), and its autonomy from other uses of the verb *saber* as seen in its semantics (e.g., *yo no sé* does not necessarily refer to lack of knowledge, but is also used as a general mitigating device). Constructions that occur with high (token and relative) frequency with particular lexical types (such as *andar buscando* 'be looking for,' *quedarse solo* 'end up alone,' and *yo no sé* 'I don't know') are known as PREFABS or "word sequences that are conventionalized, but predictable in other ways" (Bybee 2006: 713). Thus, in usage-based theory, both idioms with an unpredictable meaning (such as *subirse a la cabeza* 'get intoxicated') and prefabs may be stored in memory as units.

Should prefabs be relegated to a lexical component that is viewed as separate from a syntactic component as in the formalist perspective, or should prefabs be considered part of grammar? From a usage-based functionalist perspective, the lexically particular and the grammatically general interact, existing as points along a continuum. Diachronic evidence for the interaction of the particular and the general can be seen in the evolution of the *estar* VERB-*ndo* Progressive construction. Bybee and Torres Cacoullos (2009) find that the prefab *estar hablando* 'be talking' takes a lead in the semantic-structural changes observed in the grammaticalization of this construction, and may serve as the center of a verbs-of-speech subclass of *estar* VERB-*ndo*, attracting more lexical types and thereby contributing to the productivity of the general Progressive construction. Wilson (2009) shows similarly that *quedarse* 'become' prefabs have persisted over the centuries, with gradient categories of adjectives clustering around them.

The role of prefabs in language variation and change belies a lexicon–syntax dichotomy, as does gradualness in constituency change. Complex prepositions provide evidence that changes in constituency, or cohesion among units, are gradual. For example, concessive *a pesar de* 'in spite of' developed from a meaning of sorrow > opposition > contradiction, increasing in relative frequency from 2% (4/199) of all occurrences of *pesar* 'sorrow, regret' in Old Spanish to 96% (167/174) in twentieth-century data (Torres Cacoullos 2006). As the erstwhile independent item *pesar* was absorbed into the new unit *a pesar de*, it underwent decategorialization (Hopper 1991: 22), shedding its nominal trappings, as measured by the loss of plural marking and a drop in the use of determiners and adjectival modification. Rather than the abrupt reanalysis illustrated in (16), (where X is at first an NP and over time also an infinitive or a finite clause), gradualness in loss of analyzability is shown by changing distribution patterns.

(16)	from	[*a pesar*] + [*de* X]	to	[*a pesar de*] + [X]
		'[in spite] + [of X]'		'[in spite of] + [X]'

For example, in coordinated adnominal NPs, repetition of *de* 'of' for each adnominal NP, as in (17), suggests the relative independence of *de* from *a* + *pesar*; the alternative configuration, a single *a pesar de* with scope over two NPs, as in (18), has increased in relative frequency over time.

(17) algo de atrevido y varonil en todo el ademán, <u>a pesar de</u>l recogimiento <u>y de</u> la mansedumbre clericales
(nineteenth century, Juan Valera, *Pepita Jiménez*: 316)
'something bold and virile in his whole look, <u>in spite of</u> the withdrawal <u>and of</u> the tameness of the cleric'

(18) olía a lavanda y espliego, pero por debajo del perfume olía como yo, la fisiología nos igualaba <u>a pesar de</u> los potingues <u>y</u> las abluciones diarias
(twentieth century, Juan Manuel de Prada, *La tempestad*: 135)
'she smelled of lavendar, but underneath the perfume she smelled like me, our physiologies making us equals <u>in spite of</u> the concoctions <u>and</u> the daily ablutions'

Thus, in a usage-based view, constituent structure crystallizes from the frequent co-occurrence of items (Bybee 2010). In this section we have seen the gradience of constituent structure (e.g., *a pesar de*), the contributions of prefabs to grammaticalization and their centrality in synchronic categories (*estar hablando*, *quedarse solo*), and the social meaning of grammatical expressions (*andar* VERB-*ndo)*. All are compatible with usage-based constructions–on a lexicon–syntax continuum–as units of grammar.

6 Exemplifying discourse syntax: variable first-person singular subject expression

The discovery of grammatical patterning in discourse has been the preoccupation of cumulative research following the Variationist method over the last four decades. Variationist research has examined the form–function asymmetry widespread in speech, that is, the variation among different forms serving generally similar discourse functions. This variation has been shown to be structured, as observed in the LINGUISTIC CONDITIONING of variant forms, that is, probabilistic statements about linguistic contexts that differ significantly in the relative frequencies of the forms (Labov 1969; Sankoff 1988). In the Variationist method, hypotheses about constraints on usage are operationalized based on contextual elements with which the variants co-occur and are then tested as factors in multivariate analysis. This has been profitably applied to the study of variable subject expression in Spanish (for reviews see Silva-Corvalán 2001: 154–169), revealing a range of lexical and discourse conditioning factors of the nature discussed above. Note that while distinct genres such as conversation and narrative have been found to differ in the token frequencies of subject expression, the linguistic conditioning has been shown to be similar (Travis 2007) (cf. Poplack and Tagliamonte 2001, for comparable findings in relation to other linguistic variables in African American speech).

Our own research on New Mexican Spanish (Travis 2007; Torres Cacoullos and Travis 2010, 2011) has identified very similar effects as studies for first-person singular expression in other varieties of Spanish. Table 30.1 shows the results of a multivariate (Variable-rule) analysis (Sankoff et al. 2005) of the contribution of environmental factors selected as significant to the choice of expressed *yo* in a corpus of sociolinguistic interviews of New Mexican Spanish. In the first column, the numbers represent the probability (or factor weight) that each factor contributes to the occurrence of the variant: the closer to 0, the less likely that *yo* will be

Table 30.1 Factors contributing to the choice of expressed *yo* in New Mexican Spanish speakers (N = 22).

	N = 1,833; Input: .28 (Overall rate: 32%)		
	Probability	*% yo*	*N*
Previous realization			
Expressed	.68	45%	968
Not expressed	.39	20%	610
Semantic class of verb			
Psychological	.70	51%	290
Other	.46	28%	1543
Subject continuity (switch reference)			
Different subject	.58	39%	986
Same subject	.41	23%	847
Ambiguity of verb morphology (person)			
Ambiguous	.61	37%	503
Not ambiguous	.46	29%	1330
Factors not selected as significant			
Clause type			
Position in turn			

expressed and the closer to 1, the more likely that it will be. The second column shows the percentages of expressed *yo* and the third column the total number of tokens in the environment defined by each factor. The factor groups which jointly account for the largest amount of variation in a statistically significant way are previous realization, semantic class of verb, subject continuity, and ambiguity of verb morphology (shown in bold in Table 30.1).

We begin our discussion with a very strong lexical effect observed, namely that implicated in the semantic class of verb. These results indicate that *yo* is most favored by psychological (.70) as opposed to other semantic classes of verbs (.46), a result also reported in numerous previous studies (e.g., Enríquez 1984: 240; Bentivoglio 1987: 60; Silva-Corvalán 1994: 162; Travis 2007: 115). However, close analysis of this class reveals that one half of the tokens occurring with psychological verbs (49%, 143/290) occur in just two particular discourse formulas or constructions, namely *yo creo* 'I think' and *yo (no) sé* 'I (don't) know,' both of which show much higher rates of expression than the overall average of 32% (*yo creo* as high as 87%, 46/53, and *yo (no) sé* at 52%, 47/90). The behavior of these two highly frequent constructions seems to be pulling up the rate of expression for the class as a whole, suggesting that the effect may be one of particular constructions rather than of a general class of psychological verbs. This is not unique to cognitive verbs; García-Miguel (2005) makes a similar point regarding the high frequency of verbs of sight (in particular *ver* 'see') within the class of perception verbs and the different constructions in which they occur in Spanish written and spoken data.

Consider, next, what appears to be a morphosyntactic effect, that is, the favoring of *yo* expression by verb forms which are morphologically ambiguous in person marking (.61) (for example in first- and third-person singular Imperfect forms such as *yo tenía cuidado de ellos* 'I would take care of them' in (19) below). It has been proposed that subjects are expressed with morphologically ambiguous verbs in order to resolve the ambiguity (e.g., Hochberg 1986; García Salido 2008), but it has also been found that true ambiguity is rare in natural discourse, as it is typically resolved by context (e.g., Bentivoglio 1987: 45; Silva-Corvalán 1994: 154). In relation to this controversy, it is interesting to note that this morphosytnactic feature is affected by lexical patterning: in our data, psychological verbs are rarely ambiguous (9%, 25/290, compared with 31%, 478/1,543 of other verbs), and we find no ambiguity effect for psychological verbs, with the rate of *yo* expression being nearly identical in ambiguous and unambiguous contexts (48%, 12/25 and 51%, 136/265, respectively). As an alternative to the notion of resolving ambiguity, Silva-Corvalán (2001: 161–163) proposes that it is the discourse function of the different tense, moods and aspects that motivates their use with expressed or unexpressed subjects, a hypothesis supported by the present data: Preterit (perfective) forms, which have a foregrounding function, show a significantly lower *yo* rate (24%, 114/478) than Imperfect, Conditional, and Subjunctive forms, which have a backgrounding function (37%, 187/499).

We now move on to clear discourse effects, beginning with that of subject continuity. The subject continuity, or switch reference, effect follows the predicted pattern noted above (Section 2): a subject is less likely to be expressed when it is coreferential with the immediately preceding subject, and is more likely to be expressed when it is not. This can be seen in (19), where *yo* is expressed when there is a switch in reference with respect to the subject of the preceding clause (*mi mamá*, *ellos*), but is left unexpressed in the last clause where it is coreferential with the preceding subject.

(19) y mi mamá se alistaba pa' el trabajo.
y yo tenía cuidado de ellos,
ellos se iban a la escuela y yo me quedaba alzando la casa y,
y,
(Ø) me quedaba en la casa con los niños chiquitos. [NMCOSS: 117]
'and my mom would get ready for work.
and I would take care of them,
they would go to school and I would stay maintaining the house and,
and,
(I) would stay at home with the little kids.'

Finally, the results for previous realization exhibit another discourse effect. What we observe here is that *yo* is more likely to be expressed when the preceding coreferential subject is also expressed (factor weight .68) than when it is unexpressed (.39), a constraint which has been observed for subject expression in several dialects of Spanish (Cameron 1994; Flores-Ferrán 2002; Travis 2007). This can be seen in example (20) below, where we have strings of expressed subjects in (a–c) and strings of unexpressed subjects in (d–f).

(20) a. O: yo les planchaba porque en ese tiempo no había AUTOMATIC WASHERS, okay.
A: Hm.
@@ <@ No @>.
O: so=,
b. Yo tenía una WRINGER TYPE MACHINE,
c. so yo me les vinía por la ropa ONCE A WEEK,
OR TWICE A WEEK y=,
d. (Ø) la llevaba pa' mi casa,
e. y (Ø) se las lavaba,
f. (H) (Ø) se las planchaba y (Ø) se las traiba. [NMCOSS 117]
a. O: 'I would iron for them because at that time there were no AUTOMATIC WASHERS, okay.
A: 'Hm.
@@ <@ No @>.'
O: 'so,
b. I had a WRINGER TYPE MACHINE,
c. So I would come for the clothes ONCE A WEEK,
OR TWICE A WEEK and,
d. (I) would take it home,

e. and (I) would wash it for them,
f. (H) (I) would iron it for them and (I) would take it to them.'

This finding is consonant with a robust constraint uncovered in sociolinguistic and psycholinguistic research known as structural priming (or parallel structure or perseveration) (cf. Labov 1994: 547–568 for a review). Structural priming refers specifically to "the unintentional and pragmatically unmotivated tendency to repeat the general syntactic pattern of an utterance" (Bock and Griffin 2000: 177). This effect illustrates that each clause is not constructed independently, but is shaped by what precedes in the discourse. As Travis (2007) concludes, the finding that priming has an effect in spontaneous language use has profound implications for our view of grammar, as it indicates that the grammar of discourse is developed on-line and in real time as discourse is constructed.

In sum, like the other grammatical features discussed above, variable subject expression in Spanish is not determined by abstract syntactic rules. Rather, it is shaped by a confluence of lexical and discourse factors. We stress that the effect of such factors can only be observed through empirical analysis of spontaneous discourse data.

7 Conclusion

In this chapter we have presented a sample of studies of discourse, in which grammatical structure is manifested in, and seen as deriving from, patterns of use. We have reviewed empirical studies on information flow, transitivity, discourse referentiality and constructions, all of which support a discourse basis for syntax. We look forward to the continued growth of this line of research in Hispanic linguistics, both in terms of deeper analyses of problems in Hispanic linguistics under the lens of discourse function and further contributions of empirical studies of Spanish-speaking speech communities to cross-linguistic insights about usage-based grammar.

APPENDIX

Transcription conventions (Du Bois et al. 1993)

LETTER:	speaker label	.	final intonation contour
Carriage return:	new Intonation Unit	,	continuing intonation contour
Capital initial restart		--	truncated intonation contour
@	one syllable of laughter	-	truncated word
<@ @>	words spoken while laughing	=	lengthened syllable
		%	glottal stop

NOTES

1 This information gives the corpus from which the examples are drawn, and the number or name of the specific transcript. Those marked as NMCOSS come from the New Mexico Colorado Spanish Survey (cf. Bills and Vigil 2008); those marked CCS come from the Corpus of Colombian Spanish (cf. Travis 2005); others come from the respectively cited literature. Speech produced in English is in small caps in both the original and the translation.
2 *Muchito* is a variation of the form *muchachito*.
3 Thanks to Bill Croft for pinpointing the semantic difference between (3) and (4).

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31 Historical Morphosyntax and Grammaticalization

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1 Introduction: the scope of morphosyntactic change

Interaction of language levels is the usual way in which grammatical changes arise. Most grammatical changes impact the structure of morphology, syntax, and semantics simultaneously, and many language changes are triggered by phonological, morphological, syntactic, and semantic–pragmatic motivations. Multicausality is the standard manifestation of change in the history of any language. The borderline between morphology and syntax is weakened, or even eroded, in language change; consequently, there are no clear-cut distinctions between the two language levels, so that it is better to work with the concepts of *morphosyntactic* change and *morphosyntax*. Because of that, in diachrony there is neither autonomous syntax nor autonomous morphology, it being very difficult to treat morphology or syntax as independent enough areas of grammar. If the language under analysis is mostly of the inflectional type, as Spanish is, the difficulty of dividing syntax from morphology increases.

Language intersection is the logical effect of the fact that a linguistic sign has two faces: form and meaning. There is no morphosyntactic change without change in meaning, and semantic change generally involves changes in distribution and selection of forms. In consequence, different forms entail different meanings.

Morphosyntactic change is concerned with changes in distribution and meaning of constructions, words, clitics, and morphemes (when these result from a process of grammaticalization), changes in sentences, and sometimes, changes in discourse patterns. Morphosyntactic change also has to do with interactions of linguistic levels, both downwards and upwards in the language level.

Historical Spanish morphosyntax gives support to multicausality and constant level interaction as main features of grammaticalization. The rise of the synthetic future *cantaré* 'I will sing' < Latin *cantare habeo* 'I have to sing' constitutes a paradigmatic example. The change was motivated by phonological, morphophonemic,

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morphological, and syntactic causes and also by a new way of conceptualizing time, all of them acting concurrently (Company Company 2006a and references cited there in).

1.1 Phonological causes

Three mergers and a change in stress contributed to the loss of Latin future tense. The first was the merger of /b/and/w-v/ > /b/, which gave rise to an uncomfortable homophony between simple preterite and future tenses in the 1st conjugation. Simple past *cantauit* '(s)he sang' – *cantavimus* 'we sang,' etc. began to sound the same as future *cantabit* '(s)he will sing' – *cantabimus* 'we will sing,' etc. The second merger was the loss of vowel length, *ĭ* > *ē* in final syllable, which eroded the distinction between present and future in many verbs, requiring the context to make the temporal distinction: pres. *agĭt* '(s)he leads' = fut. *agēt* '(s)he will lead.' The third merger was the general loss of vowel length, provoking displacement of stress in Spanish almost systematically towards the penultimate syllable: *légere* > *legére* 'to read', eliminating the 3rd conjugation, which was absorbed into the 2nd or the 4th conjugation.

1.2 Morphological causes

Lack of regular paradigms is a well-known motivation for language change. As to the first morphological cause, the Latin future morphology was irregular interparadigmatically: the 1st and 2nd conjugations took *-bo*: *cantabo* 'I will sing,' *monebo* 'I will advise,' while the 3rd and 4th took *-am*: *legam* 'I will read,' *audiam* 'I will listen (to).' As to the second cause, both the 3rd and 4th conjugations, but not the 1st or 2nd, had intraparadigmatic apophony, *a* > *e*: *legam* 'I will read,' *audiam* 'I will listen (to),' but *leges* 'you will read,' *audies* 'you will listen (to),' etc. As to the third cause, Latin had a homonymic clash between future tenses and present subjunctive: the first person of the 3rd and 4th conjugations was identical to the present subjunctive: *legam* 'I will read' ~ 'I might read,' *audiam* 'I will listen (to)' ~ 'I might listen (to)'; in other words, Latin had no distinction between future and subjunctive, since both tenses have the common feature of 'nonfactual event.'

1.3 Syntagmatic causes

In Indoeuropean there was a strong constraint keeping clitics from occupying the absolute first position in the syntagmatic chain. This morphophonemic restriction gave rise to two types of future coexisting in Old Spanish: the synthetic future, *cantaré(lo)*, and the analytic future, *cantar-lo-hé* 'I will sing.' The distinction was syntactically and semantically exploited, each one having a different distribution and meaning.

1.4 Semantic–pragmatic causes

Christianity brought about a fundamental change in the conception of time. In Classical Latin the future comes to the individual: egocentric time, in such a way that the human being is static on the time axis and cannot change future events. In the Christian conception of time, the individual is moving toward the future and can change future events with good actions: egodeictic time. This deep change in the conception of time gave rise to many modal, obligative, and epistemic periphrastic constructions in late Latin. One of them, *cantare habeo*, coalesced into a single word in Western Romance languages, VP > V, forming the Modern Spanish simple future tense *cantaré, cantarás*, etc.[1] A resegmentation of morphological boundaries took place also: *cantar-é* > *canta-ré*.

The change was both a conservative and an innovative transformation. The change was conservative because it did not actually create a new category in Spanish; the category 'future' already existed, rather it was preserved. At the same time, the change was innovative because the meaning of that category changed deeply after a new conception of future time arose. As in the case of the history of the Spanish future tense, many morphosyntactic changes in this language show that language level interaction is the normal path along which diachronic changes operate.

This chapter has two aims: first, to describe some major morphosyntactic changes in the history of Spanish within the framework of Grammaticalization, and second, to characterize Grammaticalization. The chapter has four sections besides this introduction. Section 2 presents the various definitions of grammaticalization. Section 3 focuses on the relationship between synchronic and diachronic variation, the way in which the innovative form emerges and advances over time, the role of context and frequency in grammaticalization, the grammatical consequences of change, and the pathways that arise from grammaticalization. Section 4 is devoted to briefly characterize the mechanism of grammaticalization: reanalysis. Brief conclusions close the chapter in Section 5.

2 Grammaticalization

2.1 *Traditional definition*

Grammaticalization is a process by which a lexical form or construction, in specific pragmatic and morphosyntactic contexts, assumes a grammatical function or by which an already-grammatical form or construction, in specific contexts, acquires an even more grammatical one (Kuryłowicz 1965; Lehmann [1982] 1995; Heine et al. 1991; Hopper and Traugott [1993] 2003; Company Company 2003). That is, lexical or less grammaticalized items and constructions are pressed into service for the expression of more grammatical functions.

2.2 *Complementary definition*

A nonstandard, but traditional enough, definition of grammaticalization consists of fixing discourse strategies. The linguistic phenomena that operate at the text or discourse level at any given stage of language to achieve special expressive effects become, over time, conventional grammatical structures lacking any pragmatic conditioning. In this second sense, which is complementary to the aforementioned one, grammaticalization is the conventionalization of tendencies or routines which have emerged from the discourse.

The first definition is well exemplified by the grammaticalization of the Old Spanish adverb *ý* < Lat. *ībi* 'there' into a bound form *-y* of the present indicative existential verb *hay* < *haber* 'there is/are' (Hernández-Díaz 2007). In Old Spanish, the conservative-etymological existential verb was *ha* < Lat. *habet*, lacking any affix (1). Examples in (2) show the typical context whereby the grammaticalization of *ha-y* started: in the sentence there is a remarkable locative redundance because, besides the locative adverb *ý*, there are other locative complements: *en la montaña do yo moro* 'on the mountain where I live,' encoding a scene where the direct object (DO), *un lago muy grande* 'a very big lake,' is located. The pronoun *y* functions as a full anaphor which retrieves the locative complement previously mentioned. At the beginning of the grammaticalization, the DO always was a singular count, well-known and salient entity, positively valued in the discourse fragment, and very frequently the DO is a location itself, as (2) shows. That is, at the beginning of the change, the shifting form always appears in contexts that are closely related to the etymological meaning of the form undergoing the change, location in this case. Example (3) shows both the etymological-conservative *ha* and the innovative-grammaticalized *ay* coexisting in the same sentence. The example enlightens the typical contexts which favor each existential verb form at the beginning of the grammaticalization. The conservative *ha* usually appears with a generic nonspecific DO, *peçes e agua* 'fishes and water,' while the innovative form *ay* subcategorizes a concrete, definite specific DO, *un galápago mi amigo* 'a turtle who is my friend'. In (4) the free, lexical, stressed adverb *ý* may appear in any position in the sentence, post- and preverbally (4a) and (4b), may appear with tenses other than the present, and may co-occur with verbs other than *haber* (4b).

(1) Ca en las cosas en que tan gran mal **ha**, que se non pueden cobrar si se fazen (14th, *Lucanor*, 55)
'because in those things that are greatly evil, that cannot be avenged if they are embarked upon'

(2) *En la montaña do yo moro* **ay** *un lago muy grande* (14th, *Cisne*, 116)
'on the mountain where I live there is a big lake'

(3) E yo se de *un lugar apartado e muy viçioso* do **ha** *peçes e agua*, e **ay** *un galapago mi amigo* (13th, *Calila*, 156)
'I know a nice, faraway place where there are fish and water, and there is a turtle who is my friend'

(4) a. seyendo la tierra de suso sana e entera, que nunqua **ouiera ý** poblança alguna (13th, GEII, 435.30b)
'being the land healthy and entire by itself, never there was any settlement'

b. E quando la uio, marauillos ella e todos los que **ý estauan** (13th, *GEII*, 435.28)
'and when she did see her, she and everybody who was there was delighted'

As grammaticalization proceeded in the fifteenth and sixteenth centuries, the locative *y* became an unstressed, obligatory form, and it was fused with the verb, becoming a bound form that can only appear in present indicative. The etymological locative meaning of *y* 'there' has been opaqued to the point that another locative adverb, *allá* 'there,' with a very similar meaning to the etymological *ý*, appears in the sentence (5a); it means that the locative meaning of *y* is completely weakened or eroded. The locative complement which co-occurs with the existential *hay* becomes more abstract, *entre los naturales* 'among the natives' (5b), although concrete locative complements may also co-occur with existential verbs at any stage of Spanish. The example in (6) shows the final stages of the grammaticalization: the present indicative of the existential verb *haber* has a fixed form, *hay*, and the sentence may lack a locative complement.

(5) a. porque *alla* **ay** mucha abundancia y aca falta (16th, *DLNE*, 1.3)
'because there is great abundance and here dearth'
b. El mesmo desasosiego **ay** *entre los naturales* (16th, *DLNE*, 13.200)
'there is the same unrest among the natives'

(6) **Hay** tiempo para todo (21st)
'there is time for everything'

Summarizing, the change was that *hay* succeeded in ousting *ha* and that *y* was eliminated as an adverb of the Spanish language. The free adverb has become a morphological form integrated into a single verbal base; the adverb was reinterpreted as a bound form, and the full verb was also reinterpreted as a bound form: a root. It is an instance of grammaticalization of a construction: verb + adverb (Traugott 2003, 2008). At the same time, the stress was completely removed from *ý*, and nowadays -*y* is a unstressed morpheme, with no lexical meaning at all.

The examples in (1)–(6) show various characteristic traits of morphosyntactic change. First, as grammaticalization progresses, the innovative form is gradually released from the original contexts. Second, there is permanent language level interaction in language change: lexicon > morphology, syntax > morphology, free form > bound form, etc. Third, the fact that both *y* and *ha* had a very light phonetic weight, a simple vowel/i/and a simple vowel/a/, no doubt favored the grammaticalization, giving support to language level interaction again: phonology, syntax, and morphology are tightly linked.

Many morphosyntactic changes in Spanish fit the first definition of grammaticalization, as the following examples in (a)–(f) show: (a) Free full verb > auxiliary verb: Lat. *habeo scriptas litteras* 'I have letters and they are written' > Sp. *he escrito las cartas* 'I have written the letters'; (b) Adjective + noun *mente* > root + affix:

manner adverbs in *-mente*: Lat. *deuota mente* 'with a pious mind' 'with a pious intention' > Sp. *devotamente* 'devoutly'; (c) Locative preposition *a* > object marker: *voy **a México*** 'I go to Mexico' > *dedicó el libro **a sus amigos*** '(s)he dedicated the book to his/her friends' ~ *envió **a su hija** de vacaciones* '(s)he sent his daughter on vacations'; (d) Directive verb with purposive clause > temporal future marker: ***voy** a la iglesia **a rezar*** 'I am going to the church to pray' > ***voy a pensarlo*** 'I am going to think about it'; (e) Full sentence > simple or compound word: Medieval Sp. *haga en él **cual** castigo **quiera*** 'give him any punishment' > *haga en él **cual quiera** castigo* > Sp. *haga **cualquier(a)** castigo* 'do whatever punishment'; and (f) Deictic anaphoric free pronoun > clitic definite-article: Lat. *ille-illa*: *Medea **illa*** 'Medea, that one which was named previously' > Sp. *el-la*: ***las** mujeres están incorporadas al mundo laboral* 'women are part of the work force.'

The changes attested to in (1)–(6) above conform to the second definition of grammaticalization also. At the beginning of the process the use of the adverb *ý* in existential sentences is pragmatically motivated: it establishes a sufficiently abstract locative setting where the existence of relevant, concrete, specific, well-known entities is predicated. As grammaticalization takes place, the use of *y* is released from appearing obligatorily in locative contexts and with salient entities, appearing with any kind of existing entity: concrete or abstract, specific or nonspecific, known or unknown. Once released from locative contexts, *y* generalizes, becoming a conventional, obligatory, grammatical form lacking any pragmatic-textual conditioning. In Modern Spanish, *-y* only has the grammatical meaning 'morpheme of the present indicative of the 3rd person of *haber*.'

There are several morphosyntactic changes from the history of Spanish which fit the second definition of grammaticalization, as shown in (a)–(d): (a) Temporal subordination *mientras* 'while' > conditional subordination *mientras* 'until' 'as long as': ***mientras** María lee, Juan escucha música* 'while Mary reads, John listens to music' > ***mientras** no te tomes la sopa, no sales a jugar* 'you can't go out to play until you finish your soup'; (b) Deontic-external modality > epistemic-internal modality: ***puedes** escribir* 'you can write' (physical or intellectual capacity) > ***puedes** escribir* 'you may write' (permission); (c) Full movement verb > intensive deverbal discourse-marker: ***anda** más rápido* 'walk faster' > *¡**ándale**!, eso es lo que quería decir* 'yes, indeed, that's what I meant'; (d) Full noun > discourse-marker: *tanta pobreza da **lástima*** 'such poverty is pitiful' > *¡**lástima**!, no te sacaste el premio* 'a pity!, you didn't win the prize,' etc.

Two traits arise from this second type of grammaticalization. One trait is that the form or construction itself does not usually change its external form, but there are new contexts for the selection of the form or construction. The extension to new contexts and the accumulation of new meanings are actually the grammaticalization. Another trait is that the form does not necessarily acquire obligatory use.

Most grammaticalizations match both the first and the second definition. Initially the form or construction is pragmatically or discoursively motivated, and it is only used in contexts with semantic affinity to the meaning of the form or construction undergoing the change. Gradually the form generalizes, acquiring independence from its etymological meaning and able to appear in many more contexts. It gains grammatical meaning, with no dependence of any pragmatic-textual conditioning. All grammaticalizations involve semantic and syntactic conventionalization: new meaning and new distribution.

Some theoretical conclusions may be derived from the foregoing data and its analysis, as described in (a)–(e) below.

(a) Grammaticalization, in any of its definitions, presupposes that there are pre-existing linguistic forms and that, therefore, in morphosyntactic change there is no creation *ex novo*; rather previously-existing lexical discourse and grammatical material is recruited or renewed in some way. On the other hand, there is no absolute loss in morphosyntactic change, because there is always a syntactic way of expressing a given semantic content.

(b) It is possible to distinguish two complementary basic usage types of grammaticalization: a broad concept which basically means 'becoming part of grammar' and a narrow concept which means 'acquiring greater grammaticality of linguistic items.' Under the first concept, grammaticalization is a traditional term in historical linguistics. Under the second one, it is a new and enriched approach to historical linguistics, which considers syntagmatic and paradigmatic impact on language change simultaneously, with no sharp distinction between synchrony and diachrony.

(c) The standard definition of grammaticalization is that it is essentially unidirectional or asymmetrical in direction. It may be characterized as a down in the pathway or cline of grammaticalization, with a consequent demotion in the level of language at which a form or construction operates. The form or construction usually begins in the lexicon and/or in discourse and ends in the syntax or the morphology. Spanish examples (1)–(6) above show an orthodox and well-known example of unidirectionality in the history of Spanish: the form is decategorized, lexical free adverb > bound form, moving from syntax to morphology.[2]

(d) Grammaticalization is both evolution and structure preservation (Bybee 2009). New distributions, contexts, and meanings are added to the old ones, which may persist over centuries along with the innovative construction. The conservative form or construction does not necessarily disappear from the grammar. For instance, old existential *ha* is preserved in present-day Spanish, but only in legal discourse, in a formulaic, fixed expression: *no ha lugar a su pregunta* 'there is no place for your question.'

(e) Finally, grammaticalization may be defined as a macrochange, because it affects phonetic substance, function, distribution, meaning, and use frequency, all at the same time. Phonetic substance, generally, but not as a rule, is eroded: *ibi* > *y*. Function is changed via recategorization or reinterpretation of forms: full verb > auxiliary verb, preposition > case-marker, etc. Distribution is also modified when the form acquires a new function. The meaning changes from concrete > abstract, inasmuch as grammatical meaning is more abstract than lexical meaning. In addition, the form or construction undergoing grammaticalization increases its use frequency because the form generalizes to new contexts and many times becomes obligatory.

3 Emergence and advance of the innovative form in grammaticalization

Synchronic variation is a prerequisite for morphosyntactic change to take place in the history of a language. In turn, diachronic variation usually leaves traces in any synchronic stage of language, creating permanent synchronic variation. The traditional separation between synchrony and diachrony is weakened, even eroded, in grammaticalization. As a consequence, language, and also grammar, is simultaneously high stable and a never-ending imperceptible transformation. Both stability and change are defining attributes of language. Speakers basically look for efficiency and communicative success, adapting constantly and unnoticeably to new social, cultural, economic, and technical situations. From this viewpoint, a new conception of morphosyntactic-semantic change has come onto the linguistic scene in the last 20 years: change is no longer a breakdown that decomposes the system, but a series of creative micro-innovations that allow language to keep working as a successful communicative tool.

3.1 Emergence of the innovative form

For new forms to arise and grammaticalization to take place, it is necessary for some kind of synchronic variation to exist in the language, making it possible to express the "same" referential meaning in at least two ways. If no choice between two linguistic options exists, there will be little or no grammaticalization. The choice is always between two alternatives: the conservative, etymological form, construction, or meaning and the innovative one. Options may be characterized into what might be called *explicit and implicit alternants*. Both end in the same results: the innovative form, construction, or meaning emerges, competing with the conservative one, and, many times, overtaking it.

3.1.1 Explicit linguistic alternants In this case, there are two explicit forms – 'form' must be understood in a wide sense – encoding the "same" communicative situation. For example, one speaker may choose between two alternants in different communicative situations or two speakers use different forms to code "similar" referential situations. In Spanish, conservative and innovative forms may contrast in four ways.

(i) *Two different lexical or grammatical forms*: article vs. possessive heading the NP: ***la*** *casa de Juan* ~ ***su*** *casa de Juan* 'John's house'; IO pronoun with number agreement and without it: ***les*** *dije a los niños* ~ ***le Ø*** *dije a los niños* 'I said it to the children.'

(ii) *Two different constructions*: synthetic future tense vs. analytic future tense: ***cantaré*** ~ ***cantar-lo-hé*** 'I will sing (it).'

(iii) *Two different contexts*: adverb expressing temporal sequence vs. adverb expressing the speaker's attitude: *primero se corta la cebolla, después las papas y* ***finalmente*** *se echa todo a freír* 'first, onions are cut, later potatoes, and finally all ingredients are fried' ~ ***finalmente****, el examen no estuvo tan difícil* 'the test was not so difficult, after all'. In this third kind of contrast, the form or construction remains invariable, but there are suprasegmental changes.

(iv) *Absence vs. presence of a form*: presence of the definite article vs. its absence: *poner* ***los*** *pies en polvorosa* ~ *poner* ***Ø pies*** *en polvorosa* 'to beat it'; prepositionless DO vs. prepositional DO: *mató* ***Ø el caballo****, ya estaba muy enfermo* ~ *mató* ***al caballo****, ya estaba muy enfermo* '(s)he put the horse down, it was very sick.' Absence vs. presence may be considered an explicit alternative also, because they form a minimal pair, inasmuch as absence is a zero sign.

3.1.2 Implicit linguistic alternants Synchronic variation may be manifested in the way of nonexplicit alternants. The form or construction in this case acquires an innovative meaning from the pragmatic context, whether linguistic or extralinguistic, when the speaker charges that form or construction with new semantic nuances that are not explicitly coded in the form, but are inferred from the context only. Signs are pragmatically enriched via this contextual charging, the new meanings usually being added to the pre-existing ones. The addition of the new meaning impacts the distribution of the form, and also, at the end, may impact its morphology, in such a way that a synchronic variation emerges, being equivalent to (i) and (iii) above.

This second emergence of synchronic variation is completely inferential in nature because there is nothing explicit in the form to suggest the innovative meaning. It is worth noting that grammaticalization, in general, is an inferential process. That is, the speakers charge the forms with new meanings inferred from the context, and, in consequence, new forms are grammaticalized once they are conventionalized without needing the original motivating pragmatic context.

American Spanish provides an interesting example of such a contextual inferencing. The adjective *sendos* 'one thing for each one of them' 'apiece' comes from Lat. *sĭngŭlos*, giving **senlos* in late Latin. The late Lat. **senlos* had a distributive meaning, and it was obligatorily plural in form because two or more entities were required: *les dio* ***sendos*** *golpes* '(s)he gave each of them a blow.' In American Spanish *sendos* has been evolved into a degree quantifier with an intensive meaning of 'big,' 'enormous'; it no longer needs two or more entities, and, in consequence, it may appear in singular: *sendos* > *senda botella* 'a very big bottle.' In American Spanish, this second meaning is completely conventionalized, being much more frequent than the etymological one. Mexican Missionary theatre, sixteenth–seventeenth centuries, gives us evidence of how the change took place. In Christmas dramas, known as *pastorelas*, the three Wise Men give the Holy Child *sendos regalos, oro, incienso y mirra* 'one of them gives gold, the other one incense and the other gives myrrh,' that is, each one of the three kings gives Jesus one present; the adjective was initially used in the etymological distributive meaning. But from the pragmatic context, the dramatic scene itself, the hearer infers that there were not only three presents, one from each Wise Man, but that must be 'royal,' 'excellent,' 'big,' 'enormous,' coming, as they do, from kings. The innovative meaning, in adequate pragmatic situations, was gradually added to *sendos*, and this modifier now has two meanings: distributive + intensive. The intensive meaning has completely displaced the distributive one in usual speech of most American Spanish dialects, the distributive meaning being a very learned form.

3.2 *Advance of the innovative form*

Morphosyntactic change is a gradual and very slow phenomenon. Graduality and slowness mean that there are internal phases between two language stages, preserving communication within the speech community and between different generations. Graduality is also a symptom that most morphosyntactic changes have internal motivation. Grammaticalization gradually spreads through both linguistic and social contexts.

Table 31.1 (Company Company 2003) schematically shows the progress of the innovative form. The letter C represents both the conservative form and the contexts where this form usually appears. The letter I represents the innovative form and its akin contexts. The table intends to reflect the fact that grammaticalization begins shyly, i, later advances to more and more contexts, i, **i**, **i**, farther from the etymological meaning and distribution of the innovative form, and finally it conventionalizes, I, evolving into the standard form or construction to express certain area of grammar. At the same time, C gradually loses contexts of usage: C > c > (c).

The creation and progress of the indefinite of generalization *cualquier(a)* 'any,' instantiates the model in Table 31.1. This indefinite appeared originally in Medieval Spanish in very specific contexts backing the etymological meanings of the two forms composing the new word: a relative pronoun–adjective, *cual* < Lat. *quale* 'which,' 'that,' and the Latin full verb *quaero*, which originally meant 'to look for,' that is, looking for or making a choice between two or more things or persons.[3] The original context explicitly coded a choice between two or more entities located in the context (7a). Gradually, the indefinite advanced to contexts released from the selection meaning, evolving into an indifference meaning 'whatever' (7b). Later, the indefinite acquired a generic nonspecific meaning, a true indefinite of generalization (7c), and finally, it has acquired a valorative meaning, positive in a negative polarity setting (7d), or negative when referring to human beings 'nobody' (7e), 'a whore' (7f) (Company Company and Pozas 2009). The five meanings are attested to in present-day Spanish.

(7) a. para sanar el que fuere mordido de la *culebra*, o de *alacran*, o d'*escorpion* o de **otra qualquier ralea veninosa** … far esta atriaca (15th, *Tratado de cirugía*, 192)
'to cure anyone who was bitten by a snake, a scorpion or any other poisonous species … make up this prescription'

Table 31.1 Advance of the innovative form in grammaticalization.

C	**c**	c	**c**	c	c	(c)
i	i	i	**i**	**i**	I	**I**

b. serán muy despreciables a los ojos de **cualquiera hombre imparcial** cuanto nos digan y repitan (18th, *Cartas marruecas*, 9.37)
'what they are saying and repeating to us would be despictable in the eyes of any impartial man'

c. y para ello a todas horas y en **qualquier tiempo** pueden entrar los fieles executores en las tiendas y tabernas (18th, NLBL, *Genaro García*, G349.72)
'to that effect at any given time, the loyal enforcers of the law enter stores and taverns'
Eso **cualquiera** lo hace (21st)
'anyone can do that!'

d. Estás bebiendo un excelente vino, no es **un vino cualquiera** (21st)
'you're drinking an excellent wine; it's not just any old wine'

e. Y no es ningún catedrático, ni bachiller, ni nada de eso, sino **un cualquiera** (19th, *Sí de las niñas*, 214)
'and he's not a professor or a teacher or anything like that; he's just a nobody'

f. Pobre, es **una cualquiera**, la veo siempre por la noche en la banqueta (21st)
'she's a loose woman; I always see her on the sidewalk at night'

3.3 *Grammaticalization and markedness*

The emergence and advance of new forms and constructions in grammaticalization are inherently related to markedness. Grammaticalization does not affect a category homogeneously, but rather advances progressively from very specific or marked contexts to unmarked ones. That is, grammaticalization is a process evolving to unmarkedness. The marked context is the original and very specific context where the innovative form or construction arises at the beginning of grammaticalization: *ha* + *ý* + locative complement; the unmarked contexts are all other contexts where the conservative form appears: *ha* +/− complement, locative or not. The new form or construction will progressively move forward to more and more contexts, generalizing and becoming the unmarked form or construction to code a certain area of the grammar.

Markedness may be understood as a binary relationship between two opposites which are related in terms of their privileges of occurrence; one of them, the marked member, is assigned to specific distributions, and the other, the unmarked one, is assigned elsewhere. The unmarked member of the opposition has fewer distributional restrictions, is more flexible, and may appear in a greater number of contexts. Frequency is an important factor in determining the (un)marked status of a form or construction: the range of application of the unmarked member is always higher than the range of application of the marked member. Changes in markedness always motivate changes in frequency.

The gradualness of morphosyntactic change is proof that change never affects all the members of a category and all the possible contexts at the same time. If this were the case, communication would break down. The progress of grammaticalization is a strong proof that the internal structure of categories is not homogeneous, but rather is essentially asymmetrical, comprised of items that show different grammatical behaviors: certain members can be considered more genuine, prototypical, or basic representatives of the category than others. Another consequence of asymmetry and gradualness is that the arrangement of the members is also hierarchical: the central or typical ones are unmarked for the properties of the category, inasmuch as they display all the defining morphosyntactic and semantic properties of that category, whereas the entities placed at the edges of the category are very marked as to those properties, and do not undergo the full range of processes that generally apply to central entities.

A tight connection holds between the focal zone of the category and diachronic stability, on the one hand, and between category margins and diachronic instability, on the other hand. Items instancing the prototype are usually diachronically more stable than nonprototypical ones. Grammaticalization affects the margins of the category first; later, it affects less marginal entities, and finally, but not necessarily, it will affect the items that instance the prototype. In general, those entities with a low degree of categoriality, placed at the edges of the category and which exhibit properties belonging to two or more categories, are prone to grammaticalize first (Company Company 2002). Entities at the category margins are vulnerable to linguistic change because they can have a double, and many times doubtful, categorial interpretation, a fact which creates permanent potential structural ambiguity. This ambiguity may instigate change.

3.4 Hierarchy in the advance of the innovative form

Grammaticalization follows a hierarchy from more > less favorable contexts. The grammaticalization of the Latin locative directive preposition *ad* 'to' into the Spanish object case-marker *a* 'direct object' illustrates how grammaticalization follows a hierarchy. The innovative *a*-marking began with human DOs, lexically close to indirect objects (IO), it later affected other DOs, and now is slowly invading the prototypical inanimate DO. The change was an extension of meaning in which the original sense of 'direction towards an entity or a place,' that is, a locative function, is extended to mark an entity which is in some way reached by the action of the verb, the IO, recipient or goal, and this marking in turn is extended to mark an entity which is affected by the verbal action, the DO, patient or theme. The result of this grammaticalization was the creation of an object case-marker in Spanish, via the reinterpretation of the locative preposition *a* as an object case-marker. Both meanings, locative and case-marking, are essentially the same: 'a type of direction,' concrete in the first case, abstract in the second case. Both have coexisted for centuries in Spanish. To sum up: LOC > IO > DO.

The new *a*-marking was extended according to an individuation hierarchy working together with an animacy and definiteness hierarchy (Laca 2006). Inanimates, which may be considered the prototype of DO, were, and in some way still are, reluctant to take the innovative *a*-marking. The progression of *a*-marking is

Table 31.2 Advance of grammaticalization of DO case-marker.

	13th (%)	*14th* (%)	*15th* (%)	*16th* (%)	*20th* (%)
Personal pron.	100	100	99	99	100
Proper noun	99	99	96	88	100
Humans	42	35	35	50	57
Animates	3	3	6	7	—
Inanimates	1	0	3	8	17

shown in a simplified way in Table 31.2 (locative DOs are left aside because they offer particular problems with regards to *a*-marking).

Table 31.2 shows that frequency of use is the cue of change in progress. There are two strong breakdowns as regards the frequency of *a*-marking: one between very individuated human items (the first two lines) and the rest of the DOs, and the other one between human and nonhuman DOs. The percentages indicate that the new DO case-marker initially affected highly individuated nouns. Personal pronouns and proper nouns, that is, DOs sharing semantic features with the other object, the IO, systematically take the preposition *a* from very early times. *A*-marking gradually advanced to human common nouns: 37% in Old Spanish > 50% in the sixteenth century > 57% in the twentieth century. Animate beings show a minimum of *a*-marking, with a slight increase in the fifteenth and sixteenth centuries; they are not attested in the twentieth century corpus. Inanimate entities, the last line in the table, do not in general accept the innovative *a*-marking in Medieval Spanish. In the fifteenth century, *a*-marking on inanimates barely comes onto the scene, 3%; it increases to 8% in the sixteenth century, although it is still not significant in the whole DO structure, and in the twentieth century it shows a notable increase to 17%. In general, the progress of *a*-marking in Spanish conforms to the following hierarchies (Laca 2006): (1) personal pronoun > proper noun > common noun; (2) individuated noun > nonindividuated; (3) human > animate > inanimate; (4) definite > indefinite; (5) specific nonspecific; (6) concrete > abstract.

This grammaticalization led to a complex and well-known synchronic variation, that of differential object-marking. Present-day Spanish has two devices for marking a DO: a prepositionless-noun phrase (NP), the conservative coding, or an NP headed by the preposition *a*. As a general rule, concrete, inanimate DOs usually lack prepositional marking (8a), whereas the entities placed far from the prototype, such as abstract nouns and animate nonhuman beings, can take *a*-marking or not (8b,c). However, the entities found at the edges of the category, lexically close to the IO, that is, personal pronouns, proper nouns and individuated human nouns, obligatorily take the prepositional case-marker (8d). As regards inanimate DOs, *a*-marking is much more frequent with abstract nouns (9a), but *a*-marking with singular concrete nouns is not infrequent (9b), at least in American Spanish.

Degrees of transitivity, verb aspectual class, presence/absence of the agent and degrees of agentivity are also important factors in allowing *a*-marking on an inanimate DO.

(8) a. Comió **Ø peras verdes** y le hicieron daño/***a peras verdes**
'(s)he ate unripe pears and (s)he got sick'
b. Los medios de producción han rebasado ***a*** **los programas gubernamentales**/**Ø** los programas gubernamentales
'the production resources surpassed all government programs'
c. Mató ***a*****l caballo**, estaba muy enfermo/**Ø el caballo**
'(s)he put the horse down, it was very sick'
d. Miraba siempre ***a*** **Juan** de reojo/***Ø Juan**
'(s)he always looked at John out of the corner of her eye'
Deja ***a*** **la pobre niña** en paz/***Ø la pobre niña**
'leave the poor girl alone'

(9) a. Después de conocer mucho ***a*** **la vida**, ya no me interesa tanto el teatro
'After becoming acquainted with life, I am not as interested in the theater'
b. Para que no nos peleemos, puse **a la silla** en medio
'To keep us from fighting, I put the chair in the middle'

3.5 *The role of context: inferential process, metaphor, and metonymy*

Forms, obviously, do not change in isolation, but when they are actually used in real syntagmatic contexts, in actual discourse, that is, in specific speech acts. There are two units of change, simultaneously working on two complementary levels: (1) a superior level, that constitutes a macro-unit, consisting of the context, which embodies the change and is the *locus* of it; and (2) an inferior level, a micro-unit, made up of forms and constructions that change when speakers use them within the macro-unit: scenario + actors. In consequence, grammaticalization is always a context-dependent process and has more to do with constructions (Traugott 2003, 2008) than with bare words, because these are always embodied in specific morphosyntactic structures.

Language obviously does not change by itself, but rather speakers change language; hearers are even more fundamental agents of change because many morphosyntactic changes are a consequence of creative misinterpretations on the part of hearers. A speaker's message is not completely explicit. It is full of implicatures and presuppositions and is anchored in pragmatic situations, both linguistic and extralinguistic. Because of that, the hearer needs to make an "effort" to decode the message adequately. One consequence of this "effort" is that creative misinterpretations emerge.

As I said before, the process of change is always inferential. The speaker places new semantic nuances onto forms or constructions that by themselves do not have the new meaning. The new semantic nuance may be textually or extratextually anchored. The first case is exemplified by the innovative existential *hay*, the second case by the intensive quantifier *sendo*.

The idea that context-discourse is a generator of change, and thus of grammar, has a natural theoretical consequence: a main part of grammar emerges from usage, habituation and repetition; it may be defined as the crystalization of use (Hopper 1998).

There is some controversy about which cognitive tool facilitates the inferential, semantic, and functional process in grammaticalization: metaphor or metonymy. Two early and seminal works on grammaticalization adopt different positions: Heine et al. (1991) consider that grammaticalization is basically metaphoric, while Hopper and Traugott ([1993] 2003) view it as basically metonymic. Recently, positions have been fluctuating: a defense of metaphor may be seen in various papers appearing in Barcelona (2000); Barcelona (2000) himself argues that it is difficult to establish a sharp line between metaphor and metonymy, instead suggesting a mixed model of transfer from one cognitive space to another one. A vindication of metonymy appeared recently in many papers collected in Panther et al. (2009). Espinosa (2006) contains a comprehensive characterization of metaphor and metonymy, with metaphor as the basic cognitive tool in semantic change. Pérez-Saldanya (2003) considers that if the form preserves its etymological meaning projecting it onto a more abstract predication, the change is a metaphor; metonymy takes place when the form weakens its original meaning in a stronger way.

My position, based on Spanish historical morphosyntactic evidence, is this: the fundamental fact to bear in mind is that change is a very creative, associative cognitive process whereby speakers usually code abstract areas of knowledge in terms of concrete areas. Metaphor might be used as a cover term for this creative association, but both metaphor and metonymy may be attested to in the history of any language. Basically, if the association has either textual contiguity, or a whole-part relationship, or a cognitive textual extension, the change is metonymic. For example, the extension of the preposition *a* from locative > object case-marker is metonymic. If there is not clear contiguity or the association is somewhat abrupt, then the change is metaphorical. For instance, the grammaticalization of *volver* + infinitive: *volver a casa* 'to go back home' > *volver a trabajar* 'to go back to work': locative movement > aspectual movement (Melis 2006) is a metaphor.

3.6 Consequences of grammaticalization

When traditional grammaticalization begins and progresses, certain semantic and formal changes take place. The main consequences attested in the history of Spanish are summed up in the following (i)–(xiv). In some cases there is an equal sign which means that the consequence is a direct result from the preceding case.

(i) Weakening or loss of lexical-referential meaning = increase of more abstract grammatical meaning = decategorization/recategorization.

(ii) Extension across contexts = generalization and frequently obligatorification of the sign = increase of use = less diatopic variation.

(iii) Lessening of autonomy = weakening or loss of morphosyntactic freedom.

(iv) Freeing of contextual restraints = growth in frequency.

(v) Reduction of scope = intrapositional predication or intra-word scope.

(vi) Grammatical integration = paradigmatization: the new manner adverb in *-mente* paradigmatizes with other adverbs, for instance, adverbs of uncertainty: *probablemente* ~ *quizá* 'probably,' 'maybe.'

(vii) Frequently, but not necessarily, univerbation: two words = one word.

(viii) Frequently, but not necessarily, erosion and loss of phonological weight.

(ix) Layering: a functional domain, over time, may accumulate more than one construction to express that domain: 'future' may be coded by the synthetic form *cantaré* 'I will sing (it)' and by the periphrasis *voy a cantar* 'I'm going to sing.'

(x) Divergence: the same etymon splits into different analysis: in late Medieval Spanish *haber* is both a full possessive verb, a defective existential verb, a perfective auxiliary verb, and a modal auxiliary verb.

(xi) Persistence of syntactic–semantic etymology: the original meaning, quite weakened, usually persists when grammaticalization progresses, and, somewhat paradoxically, that etymological meaning facilitates the advancement to new contexts.

(xii) More polysemy: the preposition *a* has now many more meanings than in Latin.

(xiii) Renewal of already extant categories: the category of 'future' already existed in Latin, but was renewed in Romance.

(xiv) Lexicalization, understood in two senses: on the one hand, the lexicon as well as dictionaries are enlarged because the form or construction, having new functions and meanings, needs more specifications into the lexicon, and the lexicographic entry must be enlarged. On the other hand, there is lexicalization because of loss of transparency, or opacity, between the two faces of the sign or between the sign and its contexts of use, and the new reinterpreted sign must be specified into the lexicon.[4]

3.7 Clines of grammaticalization

Morphosyntactic change in grammaticalization generates a grammatical and temporal channel, known also as a cline or pathway. The path constitutes a diachronic continuum, and the progress along it is gradual. Clines get narrow because the possible initial forms that may constitute the source of change are various, at least potentially, but only one of them actually enters into grammaticalization in a given language. In consequence, the path is narrowed and must be read from right to left: if the target is known, it is possible to know the source, but from the possible sources there is no way to predict which one might grammaticalize. For example, at the beginning of the grammaticalization of definite articles in Romance languages, any of the Latin demonstratives could have been the source of the change, but only two demonstratives entered into grammaticalization: *ille* in most Romance languages > Sp. *el*; Fr. *le*; It. *il*, etc., *ipse* in Sardinian: *su*, *sa*, and in some Catalan dialects > *so*, *sa*.

Some well-known grammaticalization clines are: Discourse > syntax > morphology > morphophonology (> loss); Lexical or content word > grammatical word > clitic > inflectional affix (> Ø); Full verb > auxiliary > clitic > inflectional morpheme; Individual inference, invited by context > generalized invited inference > conventionalized meaning; Etymological context > bridging context > switch context > conventionalization.

4 Mechanism(s) of grammaticalization

A traditional viewpoint is that the major mechanism of grammaticalization is reanalysis, and that there is no grammaticalization without reanalysis. However, the analysis of this mechanism has been left till the end of this chapter because there are several issues that are highly controversial, such as: (a) what the mutual relationship between grammaticalization and reanalysis is; (b) whether grammaticalization and reanalysis are (almost) synonymous; (c) what the relative chronology between grammaticalization and reanalysis is: whether reanalysis precedes or follows grammaticalization; if it follows, some preparatory contexts of use are required, a kind of previous phase to which reanalysis applies later; (d) whether all changes in grammaticalization need to involve reanalysis, that is, whether there is grammaticalization without reanalysis; and (e) whether both reanalysis and grammaticalization behave, or not, in a similar manner as to gradualness. I will not enter into these problems (Detges and Waltereit 2002; Eckardt 2006).

Reanalysis is defined as a change in the structure of an expression or class of expressions which do not necessarily involve an immediate modification of their phonological form (Langacker 1977: 57). Reanalysis is the reinterpretation of a form or construction with a consequent refunctionalization, or recategorization, of it. Reanalysis undoubtedly makes both the creativity of language change – better, the creativity of hearers–speakers using the language – and the inherent link between form and meaning in language change more evident. The immediate cause of reanalysis is ambiguity and the need to recover transparency or isomorphism between form and meaning, and the ultimate cause is the creative inferential process by which speakers, in a veiled way, charge the forms with new semantic nuances.

Langacker (1977) distinguishes two types of reanalysis: resegmentation and reformulation. The first one is more superficial in that it is more apparent, implying a readjustment of the grammatical boundaries of a form or construction. It embodies three kinds of resegmentations: boundary loss, boundary creation, and boundary shift. Reformulation is deeper, the phonological form remains the same, but a refunctionalization is produced. Both reanalyses may work together (Heine and Kuteva 2007: ch. 2). Also, multiple resegmentations and/or reformulations may be undergone within a grammaticalization process.

Examples of resegmentation are numerous in spontaneous speech and in child language acquisition. The history of Spanish also provides constant evidence of resegmentation: the future tense was resegmented by a boundary shift: *cantar-é* > *canta-ré*; the univerbation of the relative pronoun, *cual* and the verb *querer* to *cualquiera* constitutes a boundary loss, etc. Reformulation also provides numerous examples, many of which were exemplified above.

5 Concluding remarks

Grammaticalization is a theoretical framework that has greatly enriched historical linguistics, feeding it from different fields, such as typology, pragmatics, psycholinguistics, cognition, etc., eroding the distinction between synchrony and diachrony, and generating an extended definition of Grammar. It has shed light on new theoretical insights while preserving the data-oriented tradition of historical linguistics. Within this theoretical framework, I have analyzed the scope of morphosyntactic change and how it operates. I have also examined some morphosyntactic changes of Spanish, relating them to the major topics embraced by grammaticalization.

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NOTES

1 There are a few words in Spanish meaning 'future' time, which preserve the Latin conception of future as moving towards the speaker. These items are coded with the verb *venir* 'to come': *lo venidero* 'what will come,' *el porvenir* 'the future,' *devenir* 'to evolve into.' There are also some lexicalized NP with egocentric time meaning: *la semana que viene* 'the next week.'

2 Spanish, and other languages, shows evidence that grammaticalization may be multidirectional: discourse > grammar, grammar > grammar, grammar > discourse. The most common is unidirectional from up to down. Directionality has generated a lot of

theoretical literature in the last ten years (cf. Company Company 2008 for a review of (uni) directionality and evidence of multidirectionality in Spanish).

3 There is some controversy about the origin of these indefinites. For some scholars, they are a calque of the Latin composite indefinites *quivis* 'anyone,' *quilibet*, *uterlibet*, *ibilibet* 'whatever.' For other scholars, they are genuine Romance formations coming from relative sentences: *haga en él* ***qual*** *castigo* ***quiera***.

4 The subtype of grammaticalization known as degrammaticalization, and also as antigrammaticalization, postgrammaticalization, etc., offers special problems as to the consequences. Usually, degrammaticalization widens the predicational scope of forms or constructions undergoing the change, acquires autonomy, goes up the cline: morphology > syntax, syntax > discourse, shows reduction or loss of the syntactic properties, many times reaching complete syntactic isolation (Heine 2003; Company Company 2006b).

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32 First Language Acquisition of Spanish Sounds and Prosody

CONXITA LLEÓ

1 Introduction

Utterances produced by children acquiring their first language (L1) are nontarget-like. From a phonetic perspective child utterances are simplifications or reductions of the adult ones. What is the reason for such simplifications? This question underlies most of research on early phonology, and has given way to various theoretical models, some of them favoring explanations based on lack of motor control, others based on an immature perception, while still others favor more abstract explanations related to incomplete phonological representations. Around this central question of the field there are further questions, like: (a) what are the basic units of acquisition: words, syllables, segments, or features? In other words, what is the nature of child phonological representations?; (b) is the chronological order of acquisition the same in all languages?; (c) what is the weight of input frequency and of innate knowledge in relation to the outcome of the acquisition of sounds? Or put in another way: do children learn by imitation or because they are preprogrammed to learn language?; (d) is the process of acquisition continuous or discontinuous?; (e) does grammar emerge given certain conditions or is grammar on place from the beginning?

These and related questions have led research in the field of L1 phonetic acquisition for the last four decades. They have not yet been answered in a conclusive manner, but through the years, we have gained knowledge in several areas of this research. For instance, we now know that babbling results from a combination of unmarked sounds and the most frequent sounds produced around the baby. Thus, babbling has many sounds in common, but crucially differs depending on the language being acquired. Moreover, we know that babies are able to perceptually discriminate sounds from very early on, even if they are not

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part of the target system. However, when sounds are learned as belonging to a lexical item, the phonological representation requires some time before it can be stored. We also know that the phonotactics of the target language plays a crucial role in child language development.

At first, research focused on the acquisition of English, but it was gradually extended to the acquisition of other languages, like Spanish, which will be the topic of this chapter. After this brief introduction, in Section 2 we will succinctly discuss various models of acquisition, especially phonological acquisition. Section 3 contains the core of the chapter, namely it discusses existing research on the acquisition of the phonological component of Spanish, its sounds, and its prosody. The chapter ends with some thoughts on the acquisition of Spanish in a bilingual context (Section 3.4), concluding remarks and perspectives for future research (Section 4).

2 Modeling acquisition

2.1 *Nativist and empiricist approaches*

Studies on the acquisition of Spanish align themselves along certain theoretical tendencies and models. With the advent of Chomsky's Generative Grammar and its concomitant interest on language acquisition, an open debate on the explanatory adequacy of theories of language and their acquisition was fostered. This debate gave way to two extreme positions, which can be characterized as nativist, on the one hand, and empiricist, on the other, with various nuances between these two ends. Chomsky's view assumes innateness and deductiveness, based on some endowment we are born with as human beings, which allows us to organize linguistic knowledge in a short time. Input is thus organized by innate principles of language, which speed up the process of acquisition. This position is generally contrasted with an inductive view, exclusively based on input, without innate linguistic support for organizing it. In this latter view, language is learned like cognitive processes are, without any specific language acquisition endowment.

In the area of phonology, Chomsky and Halle (1968) found ideas that were akin to theirs in Jakobson's (1941) work, *Kindersprache, Aphasie und allgemeine Lautgesetze*. Their common view focuses on the abstract representation of sounds and especially on the underlying representation of the sounds of words in our brains. The child acquiring the phonological component of her language must develop such abstract phonological representations of words.

2.2 *Rules, principles and parameters, and constraints*

In the area of phonology, various approaches have been applied to the acquisition of L1, depending on the theoretical premises advocated by the researcher. We will consider three different points of view or models, namely classical Generative Phonology, based on specific rules, Principles and Parameters, distinguishing universal principles vs. language-dependent parameters, and Optimality Theory, based on constraints.

In the 1970s and 1980s, phonologists working in Generative Phonology looked for rules to account for child production data. Rules were supposed to explain the different phenomena appearing in child data in all languages. For instance, Macken (1979: 20) proposes that "phonological rules [...] are formalizations of the strategies that a particular child has adopted to represent words and classes of phonetically similar words." Even if defined in this manner, rules are a mere description of the facts, another way of formulating generalizations found in the data, and do not have any psycholinguistic function in a model of acquisition, as they are supposed to be learned by the very young child, to be unlearned later on.

The Model of Principles and Parameters represented a clear advance in the field of phonological acquisition, as it made spurious rules unnecessary. The phonological theories on which the model is based are the constructs of Nonlinear Phonology. Child production is seen as the development of constituents from unmarked to more marked, and the association of certain segments or features to their corresponding positions in the prosodic skeleton. Development results from the setting of the various parameters: As soon as a certain parameter has been set, the child can begin producing segments attached to those emerging positions. The model has been best studied in the area of the syllable, and in some other areas of prosodic development (see Section 3.3, below).

At the beginning of the 1990s the new model of Optimality Theory, based on surface constraints, made its appearance, and in the years to come it attracted phonologists working in language acquisition. One of the main reasons for its appeal in the area of acquisition was the notion of markedness, on which the model was based. The proposal that in the grammars of very young children markedness constraints are highly ranked soon appeared very appealing to many researchers in the field. In the Spanish acquisition literature, the model has been applied to various areas: syllable, stress, prosodic word, etc. We will see some of its results in relation to various phenomena, especially in Section 3.3.

2.3 *Creative and imitative behaviors*

Depending on the approach to acquisition advocated, input frequency and innate knowledge are weighted differently as factors leading to the outcome of phonetic and phonological acquisition. Children simplify the production of their target language, but there are great differences between the incomplete phonological components of the children exposed to different languages and between their courses of acquisition. In this respect, a theory that is gaining support is that based on imitation. This theory, as formulated by Meltzoff and colleagues, is not limited to a behavioristic view of language, but assumes that "infant imitation is mediated by a stored representation" (Meltzoff 2002: 24). Kuhl and Meltzoff (1996) did a study on infant vocal imitation between 3 and 5 months of age. Infants listened to one of three vowels, /a/, /i/, or/u/for 15 minutes. Results showed that infants produced significantly more vowels of the type they had been exposed to, in spite of the very short time of exposure.

Moreover, several studies with very young children at the babbling stage have demonstrated that children babble differently depending on the language they are exposed to, in spite of producing many similar sounds (Oller and Eilers 1982; Boysson-Bardies et al. 1984; Boysson-Bardies and Vihman 1991). Lleó et al. (1994) and Lleó et al. (1996) compared babbling in Spanish and German, and concluded that Spanish babies produced more continuants than German babies, and also more labials, whereas German babies produced more plosives and more coronals.

2.4 The relationship between perception and production

Whereas the beginnings of phonological acquisition research had been done on production, in the 1990s intensive work was carried out on perception. Jusczyk and colleagues defined the basic questions of that research program (see, e.g., Jusczyk 1997): Do children at the initial state have a bias to perceiving trochees rather than iambs? Do children perceive sounds better in certain prosodic positions; that is, better in onset than in coda position? Do they hear lexical items and tend to ignore functional words? Do infants acquire language under the influence of the phonotactic patterns of their target language?

In Spanish, research on perception is related to bilingualism (Bosch and Sebastián-Gallés 2001, 2003; Sebastián-Gallés and Bosch 2003). Questions posed in this context refer to the perception of a large set of phonological categories by a bilingual speaker whose other language has a smaller set, as in the case of Catalan in relation to Spanish. Catalan has two degrees of opening for mid vowels (e.g., front/ɛ/and /e/), whereas Spanish has one single degree of opening,/e/. Perception tests on the pair/ɛ/–/e/showed that 4-month-old bilingual babies are able to discriminate the two sounds independently of which language is dominant in the environment, whereas 8-month-old babies as well as adults tend to ignore the category only present in Catalan if their linguistic background is Spanish dominant.

2.5 Continuity vs. discontinuity

Attempts at showing a discontinuous course of development, for example, Jakobson's (1941) view of babbling as occurring independently of language, have been put into question by empirical research. Children seem to abide by the Prosodic Hierarchy, using syllables, feet, words, and phrases much as adults do (see Section 3.3, below). Very soon the minimal units applied to representing words are features, organized much the same way as in adult languages. There certainly are differences, but these are of a gradual rather than categorical nature (e.g., the prosodic word plays a greater role in child than in adult language). In fact, it has often been shown that the various stages that the phonological component of the child goes through coincide with some natural language (Carreira 1991). All the evidence available seems to lead to the child grammar as being made out of the same categories as the adult grammar. For instance, markedness constraints, which have been shown to be outranking in child grammar, are also present in adult grammar. The difference is rather one of degree, because in adult grammars they are outranked by (some) faithfulness constraints. However, in spite of all the evidence in favor of continuity, sounds are not organized from the start; that is, grammar gradually emerges after the child has had some experience producing words of her target language (Lleó 2007; Fikkert and Levelt 2008).

3 The acquisition of Spanish sounds

3.1 *Early studies on the acquisition of Spanish*

Spanish began to attract some research on acquisition in the 1960s and 1970s. Alarcos (1968) described the development involved in the acquisition of the phonology of L1 as compared with diachronic sound change. This study is of a very general nature, with many ideas going back to Jakobson (1941). It draws on data from a French–Spanish bilingual baby to offer a few examples of phonological processes and feature acquisition. According to Alarcos (1968), the system of phonemes is supposed to develop from about 9 months up to 3 years. The author speculates that the reason for the nontarget-like child production is not a motor one, but rather a miss-match between the acoustic sensations and the motoric ones. He claims (Alarcos 1968: 343f) that the chronological order of acquisition is about the same in all languages, although hardly any evidence is offered to support such a strong claim (as we will see below, this is not tenable in such a strong form).

A journal called *Infancia y Aprendizaje* (childhood and learning), dealing largely with language acquisition, began to be published in the year 1977 in Madrid: articles having to do with phonology appeared only about every two years, and were always related to the learning of orthography. The first descriptive articles dealing with the acquisition of Spanish phonology are Montes (1970, 1971), which focus on the productions of four Colombian children from the beginning of word production (see Section 3.2.2).

3.2 *The acquisition of segments*

Given that features are organized into segments in different ways, depending on the language, the question emerges about how and in what order children acquire the various phonemes of their language. However, some caution is necessary, as it is important to note that it is difficult to compare the order of acquisition of certain sounds and units from language to language; the reason for this is that different authors use different criteria to decide when a certain unit has been acquired and use different procedures to measure development. For instance, Anderson and Smith (1987: 61) require "correct adult target" production "at least 75% of the time" to consider a sound acquired, whereas Núñez-Cedeño (2007) only requires 50%

accuracy. There are very few proposals to lead to possible comparisons; but see some precise quantification procedures in Ingram (2002).

3.2.1 Acquisition of vowels Spanish vowels, comprised of five cardinal units organized in a triangle, do not pose much of a problem to young children. Hernández Pina (1984), based on data from one child, reports that vowel acquisition begins with /a/, which is followed by the rest of vowels in the following order: /e/, /i/, /o/ and/u/, and is accomplished at 20 months. Goldstein and Cintrón (2001) confirm what has been noticed in other studies, namely that children acquiring Spanish show very few errors in vowel production, much less than those acquiring English or German (cf. Kehoe 2002). In this latter study two monolingual Spanish children growing up in Madrid are observed: at age 1;6 they produce about 80% target-like vowels.

3.2.2 Acquisition of consonants According to Alarcos (1968), the acquisition of consonantal features in Spanish begins with bilabial stops, nasality and voicing being acquired immediately thereafter. Fricatives come in later, preceded by the lateral /l/, whereas /r/ is still substituted by a nasal or a lateral. Montes (1970, 1971) reports a great preference for the labial place of articulation, which he attributes to a universal tendency. The last sounds to be acquired are /ɾ/ and /r/. He also notices several processes, such as assimilations, cluster reductions, substitutions, and metathesis. Hernández Pina (1984) dealt with the development of consonants, as well, beginning at 12 months, although criteria for considering a sound as acquired are not made explicit. Consonants begin with [w], [β] and [b], followed by /t/ and /p/. That is, approximants begin to be produced before stops. Nasals begin with/m/(at 13 months), followed by/n/, appearing at about the same time as /s/. The affricate /tʃ/ appears at about 21 months, /ɲ/ is consolidated at about 27 months. The last phoneme to be acquired is /r/, which is not yet produced at 3 years of age.

López Valero et al. (1989) assess the productions of three children between 2;0 and 2;6, and notice that the last sounds to be acquired are /x/, /f/, /r/, and /θ/, as these are not yet pronounced at 2;0, but do not identify a chronological order between them. Consonant clusters appear late. Much of the work done on the acquisition of L1 phonology in Spanish-speaking countries revolves around processes (or rules) of deletion, substitution, or assimilation (see, e.g., following articles in Pérez Pereira 1996: Chillemi and Martínez de Urquiza; Albalá et al.; Domínguez and Maldonado; Fernández Viader et al.).

A series of studies on Spanish segments were carried out by Macken at Stanford, as part of Charles Ferguson's Child Phonology Project. Macken (1978) concentrates on the acquisition of consonants in Spanish by one Mexican child (J) living in Redwood City, California. Interesting in this study for phonological acquisition in general is the fact that the author finds a negative correlation between the segmental and the syllabic complexity, as if complexity were only developing in one of the areas. As far as the timing for the acquisition of certain sounds in Spanish, J started out at about 1;9 with voiceless stops/p t k/, nasals/m n/, glides [w j], and the laryngeal [h], which was first used preceding words without consonantal onset or following a vowel at the end of a word; that is, corresponding to devoicing of vowels initially or finally. At 1;9 J deletes final nasals, which begin to be produced at 1;11, but only in monosyllabic words, not within disyllables. Fricatives, and specifically the affricate/tʃ/begin to be produced at 1;9; /f s x/ are correctly produced at about 1;10 in the onset. It is important to notice that besides Spanish, J was exposed to English. This might have played more of a role than assumed by Macken (see Section 3.4 below).

Macken (1979) is mainly interested in finding the basic unit of acquisition: the word? the phoneme? distinctive features? She posits "word patterns" as the most basic units at the initial stage, which later proceed to phonemes. These are acquired in certain sequences, which are presented in an in-depth study of Si, a girl from the same project reported in Macken (1978). Si's phonological system at 1;7 comprised the labial and dental voiceless stops/p t/, the nasals/m n/, and the glides [w j]. Velar/k/is produced at 1;9, affricate/tʃ/at 1;10, liquid/l/and palatal nasal/ɲ/at 1;11, fricative/s/at 2;0, and fricative/f/at 2;1.

Anderson and Smith (1987) investigated spontaneous productions by six 2-year-old monolingual Puerto Rican children (ages from 2;4 to 2;10). This study observes that, if all syllable positions are combined, "nasals, glides and stops were the most accurately produced sounds," whereas "the least accurately produced sounds were the laterals, fricatives and vibrants" (1987: 66). Goldstein and Cintrón (2001) also analyzed the sound production by three 2-year-olds (ages: 1;10, 2;4, and 2;5) acquiring Puerto Rican Spanish. Results partly confirmed the findings of previous studies on Spanish, as the consonants first acquired are stops, nasals, fricatives and the affricate. The lateral/l/was never substituted, whereas the flap/ɾ/was often substituted by the voiced interdental fricative [ð] or the lateral [l]; the trill/r/was generally substituted by [l] and also by [h]. The latter is explained by the similarity with [ʀ] and [x], which are the most normal variants of the trill in Puerto Rican pronunciation.

Acquisition of VOT for stop consonants has also been a topic of research. Languages like English and German distinguish long lag and aspirated (voiceless stops) from short lag (voiced stops), whereas Spanish bases its voice distinction on short lag (voiceless stops) and lead or prevoicing (voiced stops). Macken and Barton (1980a, 1980b) found that whereas English-speaking young children produce the voice contrast at the age of 2;6, children acquiring Spanish do not distinguish voiceless and voiced stops by means of VOT, as all stop consonants tend to be produced with short lag. It appears that children acquiring Spanish distinguish voiced (with continuants or spirants as allophones) from voiceless stops (without such allophones), and thus produce continuants from very early on. This seems to be in contradiction with some of the results sketched on the order of segment acquisition in this section, but agrees with for example, Hernández Pina (1984). There has also been some research on the acquisition of VOT in Spanish compared with English or German, especially by an English–Spanish bilingual (Deuchar and Clark 1996) and by three German–Spanish bilinguals (Kehoe et al. 2004).

In sum, it appears that monolingual Spanish children tend to produce labial approximants very soon, followed by voiceless stops and fricatives. Voiced stops with their prevoicing are acquired relatively late. The liquid/l/is produced soon, but the vibrants/ɾ/and especially the trill/r/are acquired much later.

3.3 Acquisition of prosody

Although the initial work on phonological acquisition in the 1970s and 1980s mainly focused on the acquisition of segments, the 1990s saw prosody emerge as the most flourishing field in the realm of acquisition. It had been known for some time that syntactic and prosodic constituents often did not match, and the Prosodic Hierarchy proposed by Nespor and Vogel (1986) provided a tool to better understand the prosodic side of the duality. Much work has been done on the role of the Prosodic Hierarchy in the acquisition of English, and of other Germanic and Romance languages. Among the prosodic constituents, most studies have focused on the acquisition of the syllable, the foot (i.e., stress), and the prosodic word. Lately, other prosodic topics like intonation and rhythm have received attention, too.

3.3.1 Acquisition of the syllable Spanish has relatively simple syllables. Onsets may contain two consonants (a stop or the fricative/f/with the liquid /l/or the vibrant/ɾ/), and rhymes may contain a coda, comprised of a glide or a single consonant, mainly a coronal (typical codas in Spanish are the consonants /s n l ɾ/ and the glide [j]). The nucleus can also branch, as raising diphthongs are quite common in Spanish. Carreira (1991) was the first study trying to bring data on early phonological acquisition into line with metrical and parametric theory of the syllable in Spanish. The occurrence of CV (and V) syllables before CVC ones confirms the relevance of the Subset Principle in early acquisition, as CV is less inclusive than CVC, because the latter implies the existence of the former, but the reverse does not hold. One of the aims of the study was to show that acquisition is continuous in the sense that all stages in acquisition correspond to existing natural languages.

Syllable development is relatively slow in Spanish: The child Si in Macken (1979) deletes final codas until 2;1. Complex onsets are still reduced by J at the age of 2;6 (Macken 1978). Lleó (2003) studied coda production by two monolingual Spanish children growing up in Madrid (José and María), from the beginning of word production until 2;2 and 2;3, respectively. Both children produce a very small percentage of codas, and tend to substitute glides for certain consonants; both children prefer codas of medial stressed syllables. A third Spanish child (Miguel) growing up in Madrid under the same monolingual conditions was studied in Lleó et al. (2003): Miguel already produced 70% of codas at 1;7, which is rather early for a Spanish child. Another child, Seihla, whose coda production is analyzed in Núñez-Cedeño (2007), begins to produce medial and final codas simultaneously, at about 2;0.

Kehoe and Lleó (2003), following a study by Levelt and Van de Vijver (1998), observe the order of syllable structure development by the three monolingual Spanish children, José, María, and Miguel. The order of acquisition is CV and V before CVC and VC, that is, all three Spanish children acquired onsetless syllables before syllables with codas. Falling diphthongs were also produced before word-final codas, often having a glide as a substitute for a target consonantal coda. The presence of glides instead of consonants in coda position seems to be a typical property of the Spanish children's productions, as such substitutions are not found as frequently in the productions of children acquiring German, for example.

Clusters generally undergo a process of reduction. For tautosyllabic clusters, many studies have found the so-called "sonority pattern," by which onset clusters are reduced in favor of the least sonorous segment. Barlow (2003) observes the acquisition of Spanish by a girl aged 2;8, who reduces most of the clusters comprised of obstruent plus liquid to the obstruent. Heterosyllabic clusters tend to be produced target-like by this girl. However, Macken (1979) had observed that Si, at an earlier age, also reduced heterosyllabic clusters: clusters comprised of nasal plus voiceless stop were reduced to the stop, and those comprised of nasal plus voiced stop were reduced to the nasal. De Zuluaga (1979), in a study of the acquisition of Colombian Spanish by one young child, found that, in general, tautosyllabic cluster reduction favored the least sonorous segment, but there were many exceptions to this general solution. Lleó and Prinz (1996), in a study based on four monolingual Spanish children, found a certain tendency to favor the more sonorous segment, both in tautosyllabic as well as heterosyllabic cluster reduction. However, the preference for the more sonorous segment was prevalent only in the productions of two children. The preference for the less sonorous segment is confirmed for Seihla in Núñez-Cedeño (2008), who also reports the presence of a vowel between the two consonants of the cluster at a stage immediately preceding target-like production.

In their study of three 2-year-old children acquiring Puerto Rican Spanish, Goldstein and Cintrón (2001) also found that CV syllables were acquired first, immediately followed by V, CVC and VC. Onset clusters were often reduced to the obstruent. Goldstein and Iglesias (1996), in a study with twenty four 3-year-old and thirty 4-year-old Spanish-speaking children in Puerto Rico, found that the only phonological process occurring relatively often at 3 years of age was cluster reduction.

In sum, although the basic syllable CV is the preferred one, Spanish children produce syllables without onsets very soon. Codas develop slowly in most cases. Onset clusters appear to be reduced to the obstruent, although some Spanish children prefer the sonorant.

3.3.2 Acquisition of segments in different prosodic positions After having focused on the development of segments and syllables separately, researchers began to observe the mutual relationships between these two types of units. Anderson's and Smith's (1987: 63) study differentiated between onset and coda position: in onset position, "stops were about twice as frequent as both fricatives and nasals," whereas in coda position "there was little difference in frequency of occurrence among the stops, fricatives, and nasals." Lleó (1996) reported on different styles for the acquisition of Spanish: one child (José) developed multisyllabic words very soon, but his inventory of consonants was very reduced, for a while producing just /p/ as the sole consonant of a trisyllabic word (e.g., [papapa] for *zapato* 'shoe'). The other child (Miguel) went through a stage of truncation of long words (e.g., from trisyllabic to disyllabic), but his inventory of consonants was much larger. Here, again, a trade-off between segments and prosodic positions seems to take place, as in Macken (1978). See Bosch (2004), as well, who discusses in-depth profiles for the acquisition of Spanish at ages 3, 4, 5, and 6, based on syllabic positions, in order to diagnose possible deviations from normal phonological development.

3.3.3 Acquisition of the prosodic word Languages like Spanish or Italian have relatively long words. Trisyllabic words, especially, are rather numerous in the early stages of the acquisition of Spanish, as words like *pelota* 'doll,' *zapato* 'shoe,' *abuela* 'grandma,' *trompeta* 'trumpet,' and *muñeca* 'doll,' are among the early words of young children's vocabularies. Metrically, such words comprise a trochee preceded by an unstressed, unfooted syllable (they are so-called amphibrachs). Moreover, in early Spanish vocabularies, monosyllabic words are very scarce. The situation is very different for children acquiring English or German, with a great number of monosyllables at the early stages (Demuth 1996; Lleó and Demuth 1999). Lleó (2006) described the order of appearance of prosodic words in Spanish on the basis of production data by three monolingual Spanish children (José, María, and Miguel). All three children began producing disyllables stressed on the initial syllable at 1;2 or 1;3, followed both by trisyllables with stress on the second syllable and by disyllables stressed on the final syllable. Monosyllables are first produced a few months later, in all three cases at 1;7. Target-like production of words like *mariposa* 'butterfly,' comprised of two syllabic trochees begins later, at 1;10 in the case of José and María, and at 2;2 in the case of Miguel. Truncation of unfooted syllables, which is a long-lasting process in the acquisition of languages like English or German, hardly plays any role in the acquisition of Spanish (Lleó 2002, Lleó and Demuth 1999).[1]

Macken (1979: 17) had reported that "the restriction of words to one or two syllables is probably universal during early acquisition," and that the "weak-syllable deletion rule" and the "initial syllable deletion," which had been found in the data of children acquiring English, are also normal for children acquiring Spanish (Macken 1979: 18). However, there is some evidence that truncation of unfooted syllables is a feature that separates the acquisition of Romance languages like Spanish or Italian from the acquisition of Germanic languages like German or English. The fact that these children, Si in Macken (1979) as well as J in Macken (1978), tend to reduce disyllables to monosyllables may show the influence of English, which, although not actively present in the recording sessions, was present in the daily life of these children. Lleó (2002) describes truncation of pretonic syllables by means of alignment, and reaches the conclusion that the constraint requiring that the prosodic word be aligned with a foot at the left edge, is demoted much sooner in the grammar of Spanish children than in the grammar of German children. Goldstein and Cintrón (2001) confirmed that the children in their study produced longer words than children acquiring English. Whereas the latter produce a great number of monosyllables, children exposed to Spanish, as those in their study, had a vast majority of disyllables and a few trisyllables.

3.3.4 Acquisition of stress Hochberg (1988b) sets out to study the acquisition of stress in Spanish in order to elucidate the question of whether stress is acquired by means of rules or on a word-by-word basis. Production data from 50 Mexican-American preschoolers (ages 3 and 5) show that children find irregular words harder to produce than regular ones and tend to adapt the former to Spanish regular stress (see Harris 1983). Such results show that children develop rules that assign stress rather than storing it lexically (but see Eddington 2000 for an alternative interpretation of Hochberg's results). In fact, even the 3-year-olds seem to already master the stress rules of Spanish, which leads the author to the conclusion that "the suprasegmental domain is among the first that children master" (Hochberg 1988b: 701). She further found that words with penultimate stress were easier if they ended in a vowel, whereas among words with final stress those ending in a consonant were preferred. All proparoxytones turned out to be difficult, independently of syllable structure.

The main focus of Hochberg (1988a) lies on the initial state in stress learning, in finding out whether there is a bias towards trochees, as proposed by authors like Allen and Hawkins (1980), or whether the start is neutral. The article studies stress acquisition (beginning at 1;5–1;6) by four Mexican-American children living in Redwood City, California. Their spontaneous productions had been recorded weekly during ten months, and they had been complemented with data based on imitation. Results point to a neutral or nonbiased initial state, as children produced paroxytones as well as oxytones correctly, whereas in the case of a trochaic bias, paroxytones should have been preferred. Lleó and Arias (2006) have found a certain tendency to substitute the trochaic word pattern for the iambic one, which does not happen at the initial stage, but later, at about 1;7–1;9, when children have already acquired paroxytonic stress as the most normal stress pattern in Spanish.

3.3.5 Acquisition of intonation Many phonologists doing research on L1 acquisition believe that prosody is acquired before segments, and intonation is a good starting point to validate this claim. There has not been much work done on the acquisition of Spanish intonation. Lleó et al. (2004) set out to determine whether a 3-year-old monolingual Spanish child, Miguel, had already acquired the $L > H^*$ (delayed peak) pitch accent typical of Spanish prenuclear phrases. Miguel's broad-focus utterances comprised of two phrases showed a clear difference in the pitch accents of the two phrases, the prenuclear being produced with L>H* in most cases, and the nuclear one with H*L. Lleó and Rakow (2006b) showed that at age 2;0, out of three Spanish monolingual children, only one (María) was able to produce the delayed peak of prenuclear pitch accent. It can thus be concluded that Spanish children acquire the prenuclear pitch accent between 2;0 and 3;0 years of age.

In Lleó and Rakow (2011), the tone contours of *yes–no* questions by two monolingual Spanish children (José and Miguel), aged 3;0, were analyzed. Spanish tone contours of yes–no questions are characterized by the following tone properties, when compared with declaratives (Face 2004): Interrogatives have a higher initial F0 peak than declaratives; they show a medial F0 fall; the final stressed syllable has the lowest F0, which then abruptly raises from there to the end of the utterance. The two monolingual Spanish children did indeed produce the tonal contours of interrogatives in target-like fashion at about age 3;0, but at 2;0, alignment and scaling were not yet target-like.

3.3.6 Acquisition of rhythm Spanish, together with most Romance languages, is considered to be a syllable-timed language, whereas English and German, together with other Germanic languages, are considered to be stress-timed. Taking advantage of this difference and by means of applying the Pairwise Variability Index (PVI) measurements of Grabe and Low (2002), Kehoe and Lleó (2005) and Lleó et al. (2007) studied the productions of three monolingual Spanish children and three monolingual German children at about age 3;0. PVIs measure the variability of vocalic (PVI-V) and intervocalic (PVI-C) intervals, and according to these measurements, variability should be low in syllable-timed languages and high in stress-timed languages. As expected, the authors found that both vocalic and intervocalic PVIs concentrate in the low values of the scale for the Spanish children, but reach higher values for the German children. Bunta and Ingram (2007) have investigated rhythm in the productions of 10 Spanish monolinguals, together with 10 English monolinguals and 10 English–Spanish bilinguals, all of them aged 4;0 to 5;2. Results for the monolingual children show that both the older (aged 4;6 to 5;2) and younger children (aged approximately 4;0 to 4;5) have different PVIs depending on the language, the values for Spanish being lower than those for English.

3.4 L1 acquisition of the phonetics and phonology of Spanish as the weak language

Most studies discussed in this chapter refer to monolingual Spanish children. In our present way of life, characterized by mobility and migration, multilingualism has become the norm rather than the exception. The literature on the acquisition of Spanish phonology reports many cases of acquisition of Spanish together with another language, often English or German. Both the acquisition of segmental categories and the acquisition of prosody in such bilingual studies have shown much interaction between the two phonological components of the bilingual child

(Lleó 2008: 369ff). Bilingual children acquiring Spanish have often been observed in countries like the United States or Germany, where the language of the environment is not the one spoken at home. This may lead to a dominance of the other language, which thus exerts some influence on Spanish. In such cases, interaction between the two systems of the bilingual child may take very subtle forms, and it is to be found in areas that are hardly perceptible to the human ear. Studies of Spanish based on children who grow up in the United States (or in Germany) may not be equivalent to monolingual studies. In Section 3.3 we have referred to children acquiring Spanish in the United States, who at first preferred monosyllables and disyllables (according to Macken 1979: 17), whereas monolingual Spanish children clearly favor disyllables over monosyllables (Lleó and Demuth 1999). We have also questioned the universality of Macken's weak-syllable deletion rule in Section 3.3.3 because truncation is prevalent in English or German, but hardly active in monolingual Spanish. These and other phenomena found in data stemming from an area in which the ambient language is not the same as the language the child is acquiring should be carefully scrutinized, as influence of the ambient language may be affecting the results more than previously assumed.

In the bilingual context, both languages may develop a competence that is indistinguishable from the monolingual one. However, the language not supported by the larger social environment may be acquired under the influence of the ambient language. In bilingual research, the question is often posed about the critical or so-called sensitive period: from what age on is a language acquired as a second language? In the area of phonology, it seems that even in the case of exposure to two languages from birth, the "other" language, the one not supported by the environment outside of the family, may develop under the influence of the language spoken in the larger social context, manifesting transfer phenomena at a very early age.

4 Conclusions and future perspectives

Research on L1 acquisition of Spanish sounds has produced many studies. At first, some authors wanted to test Jakobson's universalist proposals, but later on phonologists have tried to find the "truths" of Spanish acquisition. Children very soon learn the categories of their language and also the phonetic language-specific substance that they involve. It has become clear that the frequency of a certain phenomenon in the input can lead to tendencies that differ from those found in the acquisition of other languages. Studies in this realm are not old, and it is to be expected that the many questions that could not yet be answered during these ca. forty years will be further explored. Among such open questions are the relationship of perception and production in the first years of acquisition, the form of underlying representations and their development, as well as the weight of the input frequency of certain categories on the order of acquisition of segments and prosodic constituents.

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NOTES

1 Pretonic syllables of iambic-shaped words like *camión* 'truck' and of amphibrachs like *pelota* 'ball' are the targets of such truncations (underlined in the examples).

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33 Spanish as a Second Language and Teaching Methodologies

CRISTINA SANZ

1 Introduction

In 2006, 14 million people around the world were studying Spanish as a second language, a number surpassed only by the number of English learners. In the United States alone, there are 6 million students of Spanish. In Europe, Spanish has 3.5 million students scattered throughout 38 countries. In Ivory Coast, on the African continent, 74% of all high schoolers chose Spanish as their required foreign language, and in China, the number of students registered in Spanish classes has grown 160%. Since 2006, Brazil has instituted Spanish as a high school requirement, adding 11 million students and bringing the total number to 25 million learners of Spanish as a foreign language worldwide (Instituto Cervantes 2006).

Calculations place the wealth generated by activities associated with the teaching of Spanish at 15% of Spain's GDP (Delgado et al. 2007), of which one-third comes from courses. Unfortunately, we do not have numbers for Latin American countries such as Mexico, Argentina, Chile, and Costa Rica that also attract growing numbers of students through study abroad programs. The future of the teaching of Spanish is as promising as it is challenging: not enough qualified teachers, small presence of Spanish on the Internet, and the need to adapt teaching approaches to optimize language learning in different contexts. For example, US Latinos should not be taught Spanish as if it were a foreign language.

This chapter is divided into three sections. The first section provides a condensed history of Spanish language teaching, and is followed by a description of current teaching methods in Section 2. Section 3 presents an overview of the pedagogical research that only recently has begun to motivate teaching practices. The chapter concludes with a brief overview of current and future issues in the teaching of Spanish, including technology, study abroad, assessment, and teacher education.

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2 History of the teaching of Spanish

Compared with other European languages, the history of the teaching of Spanish is exceptional in two ways. On the one hand, Spanish was the first European language to be taught as a foreign language en masse in 1492, with the campaign to teach Spanish to the indigenous peoples of the Americas. This led to the development of the first materials, manuals, bilingual lexicons, and grammars. Much of the success was due to the official intervention of the monarchy and the church. On the other hand, and despite examples from other colonizers who established a network of language schools, like the French Alliance Française, the German Goethe Institute and the British Council, only recently has the Spanish government discovered in the Spanish language its economic and political value, and the potential for cultural influence. This in part explains the lack of progress on approaches to the teaching of foreign languages, including Spanish, in Spain.

According to Aquilino Sánchez Pérez (1992), in Europe the history of Spanish teaching started in the sixteenth century in the Low Countries, an important publishing center, at the time part of Phillip II's empire. Under the crown's influence, the language of Castile extended to what are today's Germany, France, and Italy. The materials published did not show much of a sensibility towards the learning of foreign languages (i.e., were not developed with the foreign language learner in mind). This was in part because Castilians were not interested in learning foreign languages, and because they assumed the crown subjects living beyond the Pyrenees had no option but to learn Spanish (1992: 79–80). Even then, there was a tension between learning foreign languages for communication – learning to *speak* the language for commerce and diplomacy – and grammar books geared toward philological study. Sanchez's *Historia de la enseñanza del español* is an excellent history of *pedagogical grammars* of Spanish from the sixteenth to the nineteenth centuries; it was not until the nineteenth century that pedagogues/teachers such as Berlitz developed and published *methods*. At that point, the tension between the "traditional" and the "natural" methods became apparent and the discussion reached the public, a discussion that occupied the nineteenth and a good part of the twentieth centuries, and that, some will argue, is very much alive today.

In Europe and the United States, changes in language pedagogy took place after the Second World War, when high school education became compulsory and curricula included a foreign language requirement. This new situation meant a substantial change in the context in which foreign languages were taught: away from a reduced number of highly motivated, usually highly educated members of the upper classes, to pre-teens of uneven abilities completing a requirement. However, even then, the approach to foreign language teaching remained practically unchanged and mirrored the way Latin, the other "foreign language" in the curriculum, was taught. The focus was on reading and writing, grammar, and literary texts.

In the United States, the solution to this clash between reality and method came with the introduction of the Army Method. Based on behaviorist psychology and structural linguistics, the Army Method sees language development as habit formation, emphasizing oral production, dialogue memorization, and repetition.

The Army Method introduced a shift from the teacher to the materials as the centerpiece of the method, and encouraged scientifically developed materials (Sánchez Pérez 1992: 387). It also introduced technology, sometimes to replace the teacher. The advent of Chomskyan linguistics and the lackluster results produced by the implementation of the Army Method in high schools led to confusing and frustrating times for language teaching practitioners in the United States. In Europe, the notional–functional method (Van Eck 1976) followed. Its major contribution was a switch in focus to what to teach: a selection of structures grouped around abstract notions and more concrete functions based on the needs of the adult language learner, thus opening the door to the Communicative Method.

In Spain, the teaching of foreign languages had not seen any changes. French was taught in levels 6 and up, but high school was not required. Also, the presence of foreigners and travel outside Spain was rare during Franco's isolationist dictatorship, so the emphasis was on writing skills and the literary canon. In this context, Martín Alonso and Francisco de Borja Moll wrote the first "gramáticas para extranjeros" in 1949 and 1954, respectively. These were pedagogical grammars, not an attempt to introduce new methods (Sánchez Pérez 1992). The situation ended around the 1980s, at which time English had replaced French as the foreign language of choice in schools. Teachers had more opportunities to travel to the United Kingdom and Ireland to be trained in new approaches and improve their language proficiency. Students were, for the first time, overcrowding high schools. And, in the now strong middle class, Spain's accession to the European Union triggered a fever to obtain a college degree and to learn English: "academias de lenguas" mushroomed, finding many parents eager to pay extra money to have their children learn English effectively in small classes taught by – in the best of cases – trained native speakers.

It is at that time also that Spanish grew in importance as a language of business, and Spain became, slowly but surely, an important destination for study abroad. The wave reached Latin America soon after (especially after 2002, when the euro/dollar exchange became unfavorable to US students). Around that time, a handful of Spanish universities started offering courses in foreign language teaching. The Universitat de Barcelona and ESADE developed Masters courses in the Teaching of Foreign languages, and a select cadre of language teachers, including teachers of Spanish as a foreign or second language, were trained in the Communicative Approach. Finally, in 1990, the Instituto Cervantes opened its doors and established a network of schools of Spanish as a foreign language populated by instructors who benefited from on-site teacher education workshops. In this way, it positioned itself at the forefront of advances in language pedagogy and teacher education.

3 State-of-the-art in foreign/second language pedagogy

This section presents and overview of three widely implemented approaches: Task-based Instruction; Processing Instruction; and Content-based Instruction.

3.1 Task-based instruction

All over the world, the so-called Communicative Method (Langs 1988; Wilkins 1975) has been the method of choice for over 30 years now. The label "communicative" has become a must for any teaching practice or textbook. Its success and its failure are both due to its flexibility. Unlike previous methods, the Communicative Method has not produced a set of scientifically developed dialogues, lists of notions or functions, or structures. Instead, it has provided a set of simple and commonsensical guidelines that are attractive to all involved: students, teachers, parents, and administrators and that fits learners of all ages and needs. Equal emphasis is placed on the materials and the teacher. In fact, the main emphasis is placed on the learner, now an active participant. A communicative classroom is learner-centered: what is taught is decided based on learners' needs. These needs will call for differential focus on the four abilities and the choice of topics, genres and registers that will themselves determine the choice of context. The emphasis is on communicating content rather than on form, which is a vehicle, not an end to itself. Communication is defined as the exchange of information with a minimum of two interlocutors, so pair and group interactions are commonplace under this approach. Interactions have a clear purpose and end: learners interact to "get something done," like completing a map, agreeing on a decision, preparing a report, or presenting a report.

The label communicative became so common as to be practically meaningless, especially because it was often used to qualify materials and procedures that were not at all communicative. Many will argue that a drill is a drill whether items are disconnected or whether they are related content-wise, and that 50 minutes of these "communicative" drills do not make for a communicative lesson plan. So, while the term communicative is still widely used, Task-based Language Instruction (Long and Crookes 1993), *la enseñanza por tareas*, is the official name for the most widely implemented teaching approach in Spain and in the courses offered at the Instituto Cervantes. Task-based Instruction does not place a premium on the method, technique, dialogues, grammars or materials, but on the task. Tasks integrate all four skills, and their organization, sequencing, and development are at the heart of the teaching plan. Tasks are determined by students' goals. Continuous assessment is a natural part of task-based instruction: have students completed the task? And, in so doing, have they reached its pedagogical goal?

3.2 Processing instruction

The teaching profession did not wait long before some began to question the constant, exclusive focus on content and called for a greater focus on form. Development of fluency did not seem to be so much of a problem any more, but what about accuracy? Is there no role for explicit references to form, to grammar in the classroom? Michael Long proposed a distinction between *Focus on Form* and *Focus on Forms* (Doughty and Williams 1998), the latter being the equivalent of what previous treatments of grammar had been. In contrast, the context for Focus on

Form is the communicative classroom, with Focus on Form episodes that do not last long, and are rarely pre-emptive: the teacher moves the student's attention to focus it on the language in reaction to the ineffective way in which students are carrying out the task, or in response to a student's question. Focus on Form can be delivered in many ways using a number of techniques, such as recast (an implicit way of providing feedback), enhanced input (using all caps, underlining, or intonation), or structured input in Processing Instruction (Van Patten 2005).

Processing Instruction (PI) is a Focus on Form technique compatible with a communicative syllabus. Informed by insights from input processing, processing instruction is, like text enhancement, input-based. In PI, learners are: (1) given information about a linguistic form (for example, object clitics in Spanish); (2) given instructions on how to go about processing that form, based on input processing principles (e.g., in Spanish, it is necessary to process clitic and verb morphology because the clitic is in preverbal position and the sentence does not follow English SVO word order); (3) pushed to process the form presented through structured input–input that has been manipulated so that learners have to rely on the form to decide on the meaning (i.e., practice is task-essential). Not every structure is amenable to PI, and not all forms deserve a Focus on Form episode. For example, adjective–noun order is acquired without the need for an explicit move towards form, while research shows that PI optimizes the development of clitics, *ser/estar* contrast, aspect (preterit vs. imperfect), mood (subjunctive vs. indicative) and gender agreement. These forms have been the focus of most research in Spanish SLA and specifically in pedagogically oriented research (see Koike and Klee 2003) with results showing a positive response to the question of a possible role for explicit grammar. However, researchers are still debating how, when, for whom, and on what forms to focus.

3.3 Content-based instruction

All these years the teaching profession has been discussing how to teach language, but what about 'content'? What is the role of content in the Spanish classroom? Typically, content has been divided into two levels: culture with small "c," and Culture, capitalized. For grades K-12, except for students preparing for the advanced placement (AP) Spanish literature exam, content is mostly limited to culture with a small c: food, schedules, climate, celebrations, and folklore. That is usually also the case for the first two years of language instruction at the college level. Therefore, there is a clear frontier between courses with small c and courses *on* capital C, as students move on to literary readings produced by Spanish and Latin American writers in the third year.

An alternative method departing from the traditional curriculum is the integration of foreign language and subjects in schools across Europe, implementing the underlying principles of bilingual education. Content and Language Integrated Learning (CLIL), also known as Content-Based Language Teaching (CBLT) (Byrnes 2005), is a generic term to describe a variety of curricular designs in which a second language (a foreign, regional, or minority language and/or another official state language) is used to teach certain subjects in the curriculum other than the language lessons themselves. CLIL has gained a high level of support in all of Europe in response to the EU's language policy, by which every citizen is to be able to speak at least two languages of the Union in addition to his native tongue. CLIL schools are different from immersion schools where all instruction is carried out in the foreign/ second language. In CLIL, instruction is accompanied by normal – in part expanded – foreign language instruction, especially if instruction in the subject – usually but not necessarily in the areas of the humanities and social sciences – requires it. CLIL instruction is a truly integrated form of language and subject instruction.

In the context of American higher education, the Modern Language Association's 2007 report identified the need to move beyond an instrumentalist vs. constitutive view of language. The first view sees language learning as a skill to use for communicating thought and information. For the latter, language is understood as an essential element of a human being's thought processes, perceptions, and self-expressions. The report points to the split in the standard configuration of university foreign language curricula, with a sequence of little c courses followed by a number of core courses primarily focused on canonical literature. The report traces that split to the labor and class division within foreign language departments where the language curriculum is taught by graduate students and part-time faculty, while tenure-track literature professors are responsible for content courses. The report proposes two solutions: to do away with the canon and add nonliterary reading and audiovisual materials to the curriculum; and to organize modern language departments in ways that are more balanced and less hierarchical. It has been three years since the report was produced. But change in institutions of higher education and their units is slow. One possible solution is to apply CLIL principles to higher education (Martínez and Sanz 2008) as a truly integrated form of language and content instruction, where language instruction takes place as the subject or task requires it. Finally, because CLIL is not limited to traditional foreign languages but also includes minority languages, it would work well for K-12 schools and colleges with high percentages of Latino students.

4 Pedagogical research: theoretical underpinnings and key issues

4.1 Theoretical background

This introduction contextualizes broadly the research on the acquisition and teaching of Spanish within the general trends in second language acquisition research. Lack of space prevents a full listing. Instead, Table 33.1 summarizes the results from a search of three main databases with counts for articles in four main areas: the historical and social context of the teaching of Spanish; general background and Sociocultural Theory (SCT); research on three key pedagogical issues, namely input, pedagogical conditions, and practice; and articles on approaches

Table 33.1 Search results in four databases for nine years of research on the teaching of Spanish as a second/foreign language.

Database	*2001–2009*			
	History, social context	*Background, SCT*	*Input, conditions, practice*	*Approaches, techniques*
Linguistics and language behavior abstracts	17	40	16	36
Psychological abstracts	0	4	13	23
MLA	5	2	10	15
TOTAL	22	46	39	74

and techniques in the teaching of Spanish. Although unevenly spread across databases, the area that shows the most amount of articles is approaches and techniques, while the area with the least amount of scholarly attention is, not surprisingly, focused on the history and social contexts of second language teaching of Spanish.

A number of models and theories have generated research that varies on the relative weight they have placed on internal and external variables and their interaction. The 1950s were dominated by a combination of structuralist linguistics and behaviorist psychology (Skinner 1957), that conceived of all learning, including L2 learning, as the result of repeated response to stimuli leading to habit formation. Learning was externally driven. Correction was necessary to prevent or correct the effects of negative transfer from the first language and to rid the learner of bad habits. The Army Method's "scientifically developed" materials resulted from descriptions and contrastive analyses completed at the time.

In the 1960s, Chomsky's (1969) publication of *Aspects of the Theory of Syntax*, together with the cognitive revolution in psychology, signaled the beginning of a period that emphasized internal factors with almost total disregard for those external to the language learner. Different from previous positions, errors were seen as a window into the learner's *interlanguage*, and a natural part of it. Krashen's Monitor Hypothesis (1985), of great influence at the time and in following years, attempted to explain the natural order of acquisition identified in morpheme studies, which was found to be constant in adults and children, in both the classroom and in naturalistic contexts, and in ESL learners of different backgrounds. Krashen interpreted these error patterns as evidence of an internal syllabus, unalterable by neither individual differences nor contextual conditions, and triggered by comprehensible input containing language beyond the learners' current level ("i + 1"). Control of affective conditions was necessary to avoid filtering out the input. Because different elicitation techniques produced different orders of acquisition, Krashen proposed a difference between *acquired* knowledge, true competence ready for use in comprehension and production, and *learned* knowledge, conscious knowledge that could only be used for self-correction by the Monitor. While exposure to comprehensible i + 1 leads to the former, exposure to explicit input (i.e., grammatical explanation and correction) only results in learned knowledge. The reaction – both positive and negative – to the Monitor Theory was important and resulted in an explosion of publications delving into the nature of the issues it raised. For the first time and since then, terms such as input, interaction, and explicit vs. implicit learning have been the centerpiece of pedagogical research.

The review started in the 1950s with a conceptualization of learning as habit formation driven by stimuli and the learner as passive recipient. We moved through the 1970s to an idealized native speaker and L2 learner driven by internal parameters impervious to any external conditions. In the last two decades, the field has become aware that SLA, like all human accomplishments, is a complex phenomenon that cannot be explained by reductionist approaches. In this context, the research on Spanish SLA started late – not until the 1980s – with production that is yet not comparable to that of ESL, but showing a notable rate of growth. Currently, the field is dominated by processing (e.g., Sanz 2005) and somewhat by formal approaches to the acquisition of nonprimary languages (e.g., Montrul 2008). However, some scholars, a number of whom work with L2 Spanish data within what is called Sociocultural Theory (SCT) (for a review and research timeline, see Lantolf and Beckett 2009), are calling for a more flexible perspective that incorporates the social context and that is more qualitative in its approach (e.g., Byrnes 2005). The next two sections are devoted to SCT and to key issues in processing-oriented SLA research, respectively.

4.2 Sociocultural theory and Spanish as an L2

The tenets of Vygotskian psychology have had a profound influence in several areas of pedagogy and human development over the last century, including SLA, most notably in the principles of Sociocultural Theory (henceforth, SCT) with a steady increase in SLA research within this area. For Lev Vygotsky, learning and development take place through interaction with the social environment. One important premise is that this interaction is always mediated by artifacts, of which language is considered the most powerful. This mediation aids the child in the acquisition of new skills and complex mental processes. The internalization of these higher-order skills proceeds from an external, mediated, and peer-assisted stage to an internal and more self-regulated stage.

Spanish as an L2 has been a particularly fertile ground for SCT research. Much of the early research on L2 Spanish focused on learning within the *zone of proximal development* (ZPD), which is defined as the distance between the child's *actual* and *potential* levels of development, or, in other words, what the child can do independently and what he or she can do only with the guidance of an adult or more capable peer (Vygotsky 1978). Language learning takes place in the ZPD. In his 16-week study examining the development of mood and aspect in Spanish among university students of Spanish, Negueruela (2003) argues that a collective ZPD can be successfully applied in the classroom context if the instructor creates mediated conditions for learning, promotes verbalization and internalization of linguistic concepts, and pays attention to students' individual differences and learning outcomes.

Internalization of linguistic concepts results in the development of *inner speech*, which appears when assisted learning becomes more independent and self-regulated. According to Vygotsky, inner speech can be indirectly observed through *private speech*, its previous stage, during which concepts and rules are directed to oneself. In Antoìn and DiCamilla (1999), for instance, beginning L2 Spanish learners were recorded in a laboratory setting while collaborating on a writing task. Participants were found to use private speech in the L1 English as a means to scaffold their thoughts and overcome more cognitively demanding aspects of writing in Spanish, with advanced Spanish learners increasingly shifting to the L2 to accomplish similar goals (DiCamilla and Antón 2004). Research into private speech has also shown that students can be actively involved in the Spanish classroom even when they do not overtly manifest social interaction (Lantolf and Yáñez 2003).

In recent years, SCT research has refined and expanded the scope of artifacts that seem to aid mediation. One such development is the so-called Schema for Orienting Basis of Action (SCOBA) (Lantolf 2008), a meditational artifact that presents knowledge imagistically rather than verbally, such as clay models for prepositions (Serrano-López and Poehner 2008) or graphs for the preterit vs. imperfect distinction (Negueruela 2003). In SCT, explicit grammatical rules tend to be dismissed since they are usually too broad in scope and do not capture the abstract meaning of the linguistic process at hand. The use of SCOBAs, therefore, is supposed to remedy this situation by giving the learner an opportunity to transform this rule into an imagistic realization (Lantolf 2008). Language play (Lantolf 1997) and gestures in L2 Spanish learning are other popular themes in SCT research (Negueruela et al. 2004).

SCT presents itself to language practitioners as a more attractive alternative to mainstream SLA because it has made *praxis* a central tenet and classroom research its preferred approach, with studies that look at classroom-based interventions to facilitate the development of forms known to be problematic for L2 learners. Nevertheless, cognitive research in SLA has been and continues to be the dominant paradigm, with classroom and laboratory studies on key issues such as the role of pedagogical conditions, individual variables, and the interaction among them in explaining second language development, with clear implications for classroom and computer language instruction.

4.3 Key issues in processing-oriented pedagogical SLA research

This section summarizes research on the role of input, pedagogical interventions (e.g., feedback practice), and individual differences in SLA in general and in Spanish SLA in particular.

The fundamental role of input. This focus on input makes sense: there is absolute agreement that input is a requirement for acquisition to take place. Disagreement ensues on what is required, how much is sufficient, and what kind of input is relevant for language acquisition. For studies focusing on Spanish, see the following: Leow (2009) on input enhancement; Gass and Torres (2005) on interaction; and Long et al. (1998) on implicit feedback.

Heavily influenced by earlier L1 research on Caretaker Speech (the language that adults use when addressing children), the goal at first was to understand the mechanisms that made input comprehensible (e.g., simplification) and that led to acquisition (e.g., frequency, saliency; see Chaudron 1988 for an excellent review of this early literature). Early research provided descriptions of the L2 (mostly ESL) learners are exposed to: Teacher Talk (the language that teachers use when addressing their students), and Foreigner Talk (the language native speakers use when addressing nonnative speakers). Michael Long's Interaction Hypothesis (1981), closely related to Task-based Instruction, proposed the research agenda for the next two decades. As part of its attention to the didactic nature of input, this strand of literature also generated a sizable number of empirical studies on the effects of models, task complexity, task characteristics, and correction on L2 acquisition (Mackey and Abbuhl 2005).

The issue remains, however, whether it is possible for adults to reach native-like accuracy and fluency without exposure to grammar explanation and overt, repeated error correction (i.e., through classroom interaction or immersion in a natural setting). Does providing adult learners with metalinguistic information (i.e., information about how the language works) help them by (a) accelerating the process, (b) enabling it to progress farther than it would in a naturalistic environment, or (c) both? These are pertinent questions to both L2 researchers and language practitioners alike. The language teaching profession expects applied linguists to provide guidelines for the most effective pedagogical approaches. Information is needed regarding what grammar to explain, when to explain it, whether to correct errors or not, how to correct errors, and how much and what kind of practice to provide second language learners. But more research is needed on whether provision of explicit input facilitates the acquisition of an L2 (Sanz and Morgan-Short 2005).

For 30 years, in an attempt to find answers to the questions posed above, SLA researchers have conducted empirical studies on *pedagogical interventions*, the second topic in this section. Pedagogical interventions can be preemptive or reactive. The typical structural syllabus is an example of the first (i.e., it presents grammar rules before it provides learners with practice on the forms to be learned). Reactive pedagogical interventions are often used in task-based approaches to L2 teaching, whereby instructors react to their students' errors either during or after task completion by providing a mini-lesson on the problematic form, which has developed through the students' performance of the task at hand.

Pedagogical interventions result when one or more of the following are combined with practice: explicit rule presentation, manipulated input, and feedback. Each of these three variables can be placed along a continuum from more explicit to more implicit. That is, the more metalinguistic a learning condition is, the more explicit it is, whereas the more 'naturalistic' a learning condition is, the more implicit it is considered to be.

Both DeKeyser (2003) and Norris and Ortega (2000) conclude that explicit types of instruction are more effective than implicit types of instruction. However, these conclusions must be read with caution, due to the following limitations: (1) tests are usually biased towards explicit knowledge; (2) treatments are short, which puts the implicit groups at a disadvantage; (3) long-term effects of explicit conditions disappear after a few days; and (4) participants in the implicit conditions do learn and retain. Sometimes, other elements of the procedure, such as the requirement to verbalize while completing the task, tilts the balance even more in favor of the explicit group (Sanz et al. 2009).

In fact, some studies indicate that grammar presentation before or during practice is not necessary for acquisition of some target forms (see Sanz and Morgan-Short 2004 for a summary). The common factor in such studies is that learners are provided with *task-essential practice*: it seems that with certain tasks, the practice itself, and not the explicit condition under which the practice is carried out, may lead to acquisition. Furthermore, since Pica (1983), other researchers have identified negative effects of exposure to grammatical rules, in that, at least in the short term, learners' production indicates overuse and over-generalization as two sources of errors. More research is needed in order to evaluate these possible long-term effects.

So, do pedagogical interventions help, hinder, or do not make any difference at all in language development? Researchers have proceeded in many directions making generalizations and pattern identification across studies difficult. Also, researchers have been looking at the product but not the processes involved in language development under different conditions. That is why some researchers are turning to cognitive neuroscience and brain imaging techniques in search for answers (Morgan-Short et al. 2010). This research suggests that, although explicit instruction speeds up development in the early stages of acquisition, learners exposed to enough implicit practice retain more and their processing is neuro-cognitively comparable to that of native speakers.

Another focus of processing-oriented practice has been *Practice.* SLA research has typically focused on explicit rule presentation, manipulated input, and feedback, rather than the actual practice that contextualizes them. However, practice – "task" or "activity" in more pedagogical terms – is a key variable. Input-based practice is when students are asked to listen to or read a sentence and choose the picture described. Output-based activities are always a form of practice in that they require learners to produce orally or in writing. Practice can be more or less *explicit* depending on the purpose for using the language – to extract meaning or to manipulate form. For example, reading for comprehension is more *implicit* than a grammatical fill-in-the-blanks exercise. Finally, practice can be *task-essential* or not. Task-essentialness is defined by Loschky and Bley-Vroman (1993: 132) as "the most extreme demand a task can place on a structure … the task cannot be successfully performed unless the structure is used." Current research is looking at how practice interacts with other pedagogical components (rule explanation and feedback), and how practice as a construct needs to be further dissected to reveal its practice characteristics (input vs. output-based, explicitness, and task-essentialness) and their role in language development. Examples of recent empirical research on the effects of practice in L2 Spanish are Farley (2001), Morgan-Short and Bowden (2006), Toth (2006), and Leeser (2008).

Finally, *the role of individual differences in explaining the effects of instruction* has also received the attention of SLA research, as there is general agreement that individual differences (IDs) seem to have a greater effect on the acquisition of an L2 than the L1. In fact, IDs are considered responsible for the most clear difference between L1 and L2 acquisition, namely, that achieving native-like proficiency in a second language (L2) seems to be the exception rather than the norm. Surprisingly though, SLA research has focused most of its energy on identifying universals to the detriment of individual variables, in part due to the influence of Chomskyan approaches to language. Also, from a methodological standpoint, ID research is difficult to conduct. As in other areas, constructs are not precise enough for clear operationalization, and the designs are often correlational: relationships between the ID and outcome are established, but a cause-effect or even directionality cannot be identified. The question remains: how much is universal and how much is individual? Much of the focus has been on the effects of IDs on outcomes, but how do individual differences affect processes? Also, the nature of the specific IDs and the degree to which they affect specific aspects of the acquisition of the L2 – phonology vs. vocabulary, for example – are still debated in the literature (Doughty and Long 2003; Bowden et al. 2005; Ellis 2008). The list of IDs is long and continues to grow, to include, like this section, age, gender, aptitude, and prior experience motivation, but also anxiety (Young 1999), risk-taking, empathy, inhibition, tolerance of ambiguity, and cognitive style (which includes field independence/dependence, category width, reflectivity/impulsivity, aural/visual learning style, and analytic/gestalt learning style), among others.

A widely and unquestioningly accepted assumption is that achieving native-like competence in an L2 requires learning in childhood. Assumptions and anecdotes formed the basis of much early writing on *age* and second language acquisition and stand in contrast with results from more recent empirical research and from reinterpretations of previous data too. In scientific terms, these assumptions would point to a critical period which necessarily includes an onset and an offset. Outside of it, native-like success in language learning is impossible. A weaker version, a *sensitive period*, proposes a time during which the organism is especially receptive to learning, but outside of which successful learning is not precluded. A third position (Birdsong 1999) maintains that age effects are not described by a period per se, but rather by a linear decline in performance that persists throughout the lifespan, which moves the discussion from critical age effects to aging (Lenet et al. 2011).

Gender differences in L2 learning have attracted little attention. Current research indicates that there is indeed a processing difference between males and females, when processing both the native language and the L2. As is the case with age affects, these differences in processing seem related to verbal memory and the influence of estrogen upon it (see Bowden, Sanz and Stafford 2005).

Although the interest in motivation and its role in SLA has seen a recent revival, one individual difference that has attracted significant attention from those who take an information-processing perspective is *aptitude*, a largely stable trait and the individual difference most predictive of L2 learning. The most commonly used test of L2 aptitude is the Modern Language Aptitude Test (MLAT), developed when behaviorist learning theory was prevalent. Current processing approaches to SLA, however, underscore the role of working memory as the place where input is held, attended to, and processed for subsequent representation in the developing system. Working memory would seem to be a better predictor of L2 success and with more explanatory power than the MLAT, which results from combining several components.

Finally, the study of *multilingual acquisition* is an ideal area for those interested in the role of IDs in language learning since no two bilinguals are the same. For example, bilinguals differ in age of L2 acquisition, context and frequency of use of both languages, and degree of bilingualism. Laboratory as well as classroom research conducted in the Basque Country by Cenoz (1994) and by Sanz (2000) in Catalonia has shown a positive relationship between level of proficiency in the minority and majority languages and acquisition of subsequent languages, in their case English, with a number of factors posited to explain it. Some evidence exists that it is heightened awareness at the level of noticing (i.e., an advantage to focus on key elements of the language during input processing, that gives experienced language learners the edge). Given the number of Spanish bilinguals around the world and the need to motivate retention of the minority language among them – Spanish, in the case of US Latinos – identifying the socioeducational and individual circumstances under which life with two languages has a positive effect on cognition is of utmost importance. This is the goal of this strand of literature.

A tighter relationship between research and pedagogical applications is also needed, especially in the areas of assessment, study abroad, and computer-assisted instruction (CALL). Only now is the Diploma de Español como Lengua Extranjera (DELE), which applies the guidelines put together by the Council of Europe, beginning to incorporate linguistic varieties other than Castilian Spanish, which naturally limits the tests' popularity. Another problem is the nature of the test. Unlike the Test of English as a Foreign Language (TOEFL), required to enter a US college, results from the DELE are not expressed as a score in a range, but as levels achieved, which does not allow for use in program assessment, very much needed to respond to calls for accountability in higher education. Also, CALL has many advantages over traditional instruction (Blake and Delforge 2007), such as making second language instruction available to those who cannot study in a foreign country or a language classroom. At the same time, CALL is replacing instructors in large college foreign language departments. Not all CALL applications are equally valuable to all learners, so research is needed to identify the advantages in CALL to optimize language development first, and then help budgets. A final challenge is the study of the conditions that foster language development and their interaction with individual differences in stay/study abroad contexts (Collentine and Freed 2006), an educational experience for millions of US and European students. More research is needed that includes measures of motivation, attitudes, working memory/executive control, and language aptitude. Finally, also needed is research on the relationship between development of self-identity and second language development.

5 Conclusion

Without a doubt, in the last 30 years the teaching of Spanish has grown in quality and in number of students and teachers. But challenges lie ahead. Up-to-date methodology is slowly but surely reaching the classrooms due in part to advances in teacher education. However, the growing number of students meets a shortage of teachers, especially well-prepared teachers, and this need is especially acute in countries like Brazil. New audiovisual materials are now added to traditional materials by presses large and small, but the presence of Spanish on the Internet is low compared with the number of speakers. Another challenge is the special social, political, and educational contexts of Spanish teaching in the United States, where Spanish is learned both as a foreign language – like in Germany, or China – but also as a second language by millions of minority Spanish speakers. Finally, research needs to better inform pedagogical applications and decisions, especially in the areas of assessment, computer-assisted instruction (CALL), and study abroad.

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34 The L2 Acquisition of Spanish Phonetics and Phonology

MIQUEL SIMONET

1 Introduction

Perhaps somewhat surprisingly, the acquisition of the phonetics and phonology of Spanish as an L2 has received relatively little attention, much less than other aspects of the grammar (see Montrul, Chapter 35, below). Apparently, applied linguists working on the L2 learning of Spanish have been discouraged by the recurrent view that phonology is "too difficult" for L2 learners and perhaps less important for effective communication than other aspects of the language. For instance, Pennington and Richards (1986: 207, cited in Elliot 1997: 95) claim that "pronunciation, traditionally viewed as a component of linguistic rather than communicative competence or as an aspect of accuracy rather than conversational fluency, has come to be regarded as of limited importance." As a consequence, relatively little is known about the stages of the L2 acquisition of Spanish sounds, the difficulties that may be caused by certain Spanish sound patterns (and not others) to learners of different linguistic backgrounds, the potential factors (social, instructional, cognitive, anatomical, etc.) that may affect the learning of some sounds (and perhaps not others), etc. Much remains to be investigated. In particular, there is a need for studies that use sophisticated experimental methods and tools available to researchers in other fields since these tools can be more sensitive in revealing the relative relevance of different factors. Finally, systematic comparisons should be performed with language learners differing in native language and linguistic experiences. A need is evident even at the basic level of data acquisition, theorization, and modeling.

This is not to say, of course, that no work has been carried out on the topic that concerns us here. In fact, the study of the L2 acquisition of Spanish sounds has a long tradition. For instance, working within the tenets of Lado's Contrastive Analysis Hypothesis (CAH) (Lado 1957), early research was carried out which attempted to predict the difficulties native English learners of Spanish would face.

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This was done by systematically comparing the phonetics and phonology of English and Spanish (Lado 1956; Bowen and Stockwell 1960; Cárdenas 1960; Stockwell and Bowen 1965). The main hypothesis was that the "deviations in pronunciation" seen in L2 learners are mainly (or solely) due to the specific characteristics of their L1. This early view (Trubetzkoy 1939), which is based on the anecdotal observation that learners with different L1's have non-native accents with drastically different characteristics, has been incorporated in modified form in (and largely confirmed by) modern, fully-developed theoretical models, such as Best's Perceptual Assimilation Model (PAM) (Best 1995) and Flege's Speech Learning Model (Flege 1995). Finally, some recent work has focused on investigating the potential effects of formal instruction or context of learning on the acquisition of pronunciation patterns (e.g., Elliot 1997; Díaz-Campos 2004), thus concentrating on progress or development. The present chapter reviews a few recent experimental studies on the learning of Spanish vowels and consonants. Prior to discussing laboratory data, we proceed to introduce some general aspects of second language phonetic learning.

2 Second-language speech learning and bilingual phonetics

Learning a second language (L2) is a challenging enterprise. One of the aspects of the second language that presents a greater challenge for the learner is the development of speech patterns that resemble those possessed by native speakers of the target L2. Many highly fluent, proficient adult bilinguals display the following: retention of a noticeable non-native accent in their L2 (as judged by monolingual speakers of the L2) even after having had extensive experience with the language (e.g., Oyama 1976; Flege et al. 1995a; Flege et al. 1999a); production of L2 speech sounds that differ from those produced by native speakers of the target L2 (e.g., Caramazza et al. 1973; Bohn and Flege 1992; Flege et al. 1992, 1995b, 1999b; Munro et al. 1995; MacKay et al. 2001; Piske et al. 2002); and demonstration of some difficulties perceiving and processing L2-specific sounds and sound contrasts (e.g., Caramazza et al. 1973; Pallier et al. 1997; Sebastian-Gallés and Soto-Faraco 1999; Bosch et al. 2000; MacKay et al. 2001; Flege and MacKay 2004; Hojen and Flege 2006). Numerous factors have been found to affect the ability of L2 learners to acquire the speech skills of native speakers of the L2. In a review, Piske et al. (2001) discuss the following factors: age of L2 learning, length of residence in an L2-speaking environment (length of experience with the L2), gender, length of formal instruction, motivation, individual language-learning aptitude, and amount of L1 vs. L2 use. The single factor that has received the most attention in the experimental phonetic literature is chronological age at the onset of L2 learning.

It is by now a proven fact that age of learning of the L2 is a significant, robust predictor of perceived foreign accent (e.g., Oyama 1976; Flege et al. 1995a), it affects the acoustic characteristics of sounds produced by L2 speakers (e.g., Caramazza et al. 1973; Guion 2003), and it also affects the perceptual abilities of L2 learners (e.g., Caramazza et al. 1973; Pallier et al. 1997; Sebastián-Gallés and Soto-Faraco 1999). However, it is unclear what the cause of this might be (Flege 1987a; Long 1990; Flege et al. 1995a). Lenneberg (1967) proposed that the finding was due to the existence of a critical period for language learning (i.e., the Critical Period Hypothesis or CPH). According to this explanation, the complete mastery of a language is no longer possible if the onset of learning occurs after the end of some period in life during which human beings retain a full language-learning capacity. The existence of a critical period tends to be attributed to an age-dependent loss of neural plasticity or to maturational changes in the structure and functioning of the brain that occur during puberty (e.g., Lenneberg 1967; Patkowski 1980, 1990; Scovel 1988; DeKeyser 2000).

An alternative hypothesis to the CPH that has received recent attention is that age effects on L2 speech performance are due to differences in the type of interactions that take place between the L1 and the L2 sounds (e.g., Flege 1995, 2002, 2007; see also Pallier et al. 2003). According to this proposal, individuals who have had a longer experience with their L1 at the time of onset of L2 learning have greater difficulties in learning the speech skills of their L2 because the likelihood that L1 sounds interfere in the learning process of specific L2 sounds is greater. It is known that children become attuned to their native language very early in life (Werker and Tees 1984; Kuhl et al. 1992; Kuhl et al. 2008). On the one hand, very young infants are able to correctly perceive the differences among virtually all sound contrasts (Werker and Tees 1984; Kuhl et al. 2006; see Kuhl et al. 2008). On the other hand, Kuhl et al. (2008) discuss some evidence suggesting that, by the end of the first year of life, the abilities of individuals to discriminate among non-native sound categories diminishes and, importantly, their ability to correctly identify native phonetic units increases. Most surprisingly, children who retain robust perception of non-native phonemic contrasts by the end of their first year of life have been found to show slower subsequent advancement in their native language (Kuhl et al. 2008 and references therein). Therefore, it is not inconceivable that the development of robust native sound categories will affect the development of L2 sound representations and that L1 interference on the L2 will depend on the stage of development of L1 sound categories. Consequently, recent research has pursued the idea that differences between native and non-native speakers are due to "perceptual interference" of one system on the other (e.g., Iverson et al. 2003, 2008).

Two recent theoretical models have offered operationalizations of the interactions that may develop between L1 and L2 sounds: that is, Flege's Speech Learning Model (SLM) (e.g., Flege 1995) and Best's Perceptual Assimilation Model (PAM) (e.g., Best 1995). While Best's theory was designed to explain cross-language speech perception in general (that is, perception of non-native sounds by naive listeners), Flege's theory was designed to model the *development* of non-native speech skills by L2 learners. In other words, while both models are useful to predict the "problems" non-native listeners may experience, only Flege's model includes an account of how these "difficulties" might be overcome by experienced learners. Both models make explicit and testable predictions and are backed up by decades of experimental research. Flege's SLM assumes that the development of speech skills remains intact across the life span of individuals and that L1 and L2 sound categories reside in a common phonetic space. Therefore, the model assumes that interactions will arise between L1 and L2 sound categories. On the one hand, if there is little acoustic-perceptual distance between an L1 and an L2 sound, learners are likely to "equate" the two sounds (i.e., perceive an L2 sound through the grid of the L1) and thus postulate a single sound category for the two sounds. Although bilinguals might eventually be able to overcome equivalence classification provided sufficient and appropriate input, this phenomenon may be responsible for blocking the development of L2-specific sounds in the case of many individuals or for delaying it in the case of others. On the other hand, the development of "new," L2-specific sounds is more likely to occur when there are larger acoustic-perceptual differences between the target L2 sound and the closest L1 sound (Flege and Hillenbrand 1984; Flege 1987b). In the event that an L2 learner develops a separate category for an L2-specific sound, the SLM contemplates the possibility that the L1 and the L2 sound categories might deflect away from each other in order to increase contrastiveness between the two sounds (see also Guion 2003).

The development of explicit models of L2 speech learning that look into L1–L2 interactions make it possible to meaningfully address research questions that seem to be basic for our understanding of the L2 acquisition of phonetics and phonology. For instance, since L2 learners may indeed make progress in acquiring the sounds of their L2, how is it that they develop this knowledge? May we explain the stages and processes L2 learners experience in their acquisition of new sounds and articulatory patterns? Are these stages and processes stable across individuals learning the same language? Are there similarities in the experiences of learners trying to acquire different languages? Are there principled reasons why some sounds appear to be "harder" than others in general (e.g., Colantoni and Steele 2008)? How may we address and understand individual differences in L2 speech learning? How may we better train learners to acquire the speech skills of their L2? What constitutes sufficient and appropriate input? Most of these questions have yet to receive a satisfying answer. We now proceed to discuss some experimental Spanish data available to us. We will focus on segmental phenomena.

3 Spanish vowels

Recent research has investigated the production and perception of Spanish vowels by native speakers of several languages, including Dutch (Escudero and Boersma 2002; Iruela 2003), English (Morrison 2003; García de las Bayonas 2004; Cobb 2009; Menke and Face 2010), Japanese (Carranza 2008), Ecuadorian and Peruvian Quechua (Guion 2003; O'Rourke 2010), K'iche' Maya (Baird 2010) and Mandarin Chinese (Chen 2007), among others. However, the study of the L2 perception of Spanish vowels has attracted very little attention. This is hardly a surprise if we take into consideration that Spanish is a language with a relatively small vowel inventory. Recent evidence shows that learners whose L1 has a larger vowel inventory than their L2 tend to be faster or more successful in developing native-like L2 vowel perception skills than learners whose L1 has a smaller vowel inventory than their L2 (e.g., Iverson and Evans 2007, 2009). It is expected that L2 learners of Spanish whose native language has a larger vowel inventory will assimilate the five Spanish vowels to five different L1 vowel categories. Therefore, even though it is likely that the Spanish vowel categories of L2 learners will not mirror those of native speakers (at least not in production), it is expected that the learners will experience no difficulty in discriminating the five vowels of Spanish from each other. For instance, Spanish /i e a o u/ may be assimilated to the English vowels in *sea*, *day* or *debt*, *black*, *mow*, and *moon*, respectively. García de las Bayonas (2004, cited in Menke and Face 2010) asked native English-speaking L2 learners of Spanish to identify Spanish vowels. Identification accuracy was very high, between 94% and 96%.

A language with a relatively small vowel inventory may allow for a wider acoustic distribution of each of these vowels than one with a larger vowel inventory. If that occurs, it is possible that the acoustic space occupied by a single L2 vowel overlaps with those of two or more L1 vowels. How does an L2 learner, solely on the basis of positive evidence, learn that there is no phonemic contrast between variants (allophones) of an L2 sound that perceptually correspond to two L1 phonemes? Escudero and Boersma (2002) referred to this theoretical possibility with the term "multiple category assimilation," following the terminology of Best's PAM. In the case of multiple-category assimilation, a further source of error could arise if one of the two (or more) L1 vowels linked to a single L2 vowel partially overlaps with a second L2 vowel (Morrison 2003). In the most likely case, however, if a vowel category results from the merger of a single L1 and a single L2 vowel with different acoustic-perceptual distributions (i.e., with a partial overlap between the two or with one being a subset of the other), the production of the L2 learner will deviate from the native speaker norm and his perceptual boundaries may also be dislocated.

Even though significant difficulties in Spanish-vowel discrimination by speakers of languages with a larger vowel inventory are not expected (i.e., "significant" in the sense that they are unlikely to affect lexicalization or lexical access) this does not mean that perceptual differences between natives and non-natives do not arise. Morrison (2003) showed that native English speakers assimilated Spanish /a/ to the English vowel in *hat* at a rate of 70%, to the vowel in *hut* 14% of the time, to the vowel in *hot* 5% of the time and to the vowel in *het* at a rate of 6%. Also, both Spanish /o/ and /e/ were assimilated to English /e/ and /o/ 81% of the time, and 19% of the time to other vowels. On the other hand, Spanish /i/ was assimilated to the English vowel in *heat* at a rate of 100% and Spanish /u/ was assimilated to the English vowel in *hood* 91% of the time. That is, even though English speakers' discrimination of Spanish vowel contrasts was very good, it was not at ceiling. Additionally, Escudero and Boersma (2002) showed that Dutch-speaking learners of Spanish made significantly more errors when identifying Spanish /i/ and /e/ than when identifying Spanish /o/ and /u/. Further tests by these authors suggested that this was due to the patterns of assimilation of Spanish vowels with native Dutch vowels. Dutch has three front vowels that virtually occupy the acoustic-perceptual space of the two Spanish front vowels (three-to-two correspondence) while it has two back vowels that virtually occupy the space of the two Spanish back vowels (two-to-two correspondence). Escudero and Boersma's interpretation is that the existence of more contrasts in the L1 than the L2 of these learners in, approximately, the same acoustic-perceptual region affects the formation of perceptual category boundaries. These two papers show that knowledge of the acoustic and perceptual (phonetic) characteristics of both the L1 and the L2, in addition to the system of categorical contrasts (phonology), is fundamental in order to make robust hypotheses about the perceptual development of L2 sounds. This suggests that purely symbolic (phonological) learning models should be abandoned in favor substance-based (phonetic) models.

We now turn to vowel production. Guion (2003) carried out a thorough acoustic study of the Spanish and Quichua (Ecuadorian Quechua) vowels produced by four groups of Quichua–Spanish bilinguals residing in Otavalo. These four bilingual groups differed in their self-reported age of onset of learning (AOL) of Spanish: (1) simultaneous bilinguals (AOL = 0); (2) early bilinguals (AOL = 5–7); (3) mid bilinguals (AOL = 9–13); and (4) late bilinguals (AOL = 15–25). Quichua has a system of three vowels/i a u/. Spanish/a/is lower (i.e., has a higher F1) than Quichua/a/. Quichua/i/falls somewhere in between the acoustic (F1–F2) distribution of Spanish/i/and/e/, with more overlap with Spanish/i/than/e/. Quichua/u/is acoustically located somewhere between the acoustic distribution of Spanish/u/and/o/.

In addition to contrasting the different speaker groups to investigate potential AOL effects, Guion (2003) examined the potential L1–L2 interactions within each speaker's vowel systems. Regarding the production of front vowels, the results revealed that bilinguals could be classified into three groups: (1) speakers who had only one phonetic category for their Quichua/i/and both their Spanish/i/and/e/ (one-vowel group); (2) speakers who assimilated their Quichua/i/to their Spanish/ i/but maintained a separate category for their Spanish/e/(two-vowel group); and (3) speakers who developed new categories for both their Spanish/i/and/e/and thus maintained a separate category for their Quichua/i/(three-vowel group). "Success" in new-category formation was indeed directly affected by AOL, with most simultaneous bilinguals being able to maintain three separate front vowel categories and most late bilinguals presenting a system with a single front vowel. Regarding the back vowels, also three groups were found: (1) speakers who used fundamentally the same vowel for their Quichua/u/and their Spanish/u/and/ o/(one-vowel group); (2) speakers who assimilated their Spanish/o/to their Quichua/u/but kept a separate phonetic category for their Spanish/u/(separate Spanish/u/); and (3) speakers who assimilated their Spanish/u/with their Quichua/u/but maintained a difference with their Spanish/o/(separate Spanish/u/). Interestingly, none of the twenty bilinguals was able to develop a three back-vowel system. Finally, Guion (2003) found that the bilinguals who had been able to develop new categories for either their Spanish front *or* back vowels also presented two separate statistical distributions for their Quichua/a/ and their Spanish/a/.

In summary, Guion (2003) infers that late learners tend to use Quichua vowel categories in their Spanish, unlike early and simultaneous bilinguals, who seem to have been able to develop L2 categories that resemble those used by Spanish-speaking monolinguals. Guion (2003) is a fundamental study in that it provides a full acoustic characterization of both the L1 and L2 vowel systems of a large controlled group of bilinguals and late L2 learners. This robust analysis, with its many comparisons, was able to reveal differences across speakers as well as the nature of phonetic–phonological systems in bilinguals. In that sense, this article stands as a capital study of the L2 production of Spanish vowels.

While Quichua-speaking learners of Spanish need to develop a system with *more* vowels than the one they possess in their L1, the situation is the opposite for native speakers of languages such as English, which has more vowel categories than Spanish. We have already considered that these two situations pose different kinds of perceptual learning difficulties for L2 learners. Thus, while Quichua natives will have to learn new contrasts and thus develop new categories for Spanish, English natives will have to learn the appropriate acoustic (F1–F2) boundaries of Spanish vowels although they may not have to develop a completely new phonetic category for most of them. Menke and Face (2010) and Cobb (2009) (see also Cobb and Simonet 2010) examine the acoustic characteristics (F1–F2) of Spanish vowels as produced by native English speakers differing in length of experience with Spanish and, thus, level of proficiency in the language. Perhaps the most important conclusion of both Menke and Face (2010) and Cobb (2009) is that, while native English-speaking learners differ greatly in their production of Spanish vowels from native Spanish-speaking controls at the initial state, they tend to improve with increased experience with the language. Interestingly, the English-speaking learners display different patterns of development or "progress" as a function of the different Spanish vowel categories.

Although there are some interesting similarities between the Menke and Face (2010) and Cobb (2009) studies, there are also some differences. Both studies confirmed that Spanish/a/,/u/and/e/present some initial problems to native English learners. While beginners'/u/tokens are maximally different from those of native controls (apparently more so than any other vowel), advanced learners do not differ, for the most part, from native speakers in this respect. On the other hand, the differences between these two studies are somewhat puzzling. While Menke and Face (2010) found significant differences between learners and natives in their production of /i/and/o/, Cobb (2009) failed to find these. The/a/tokens produced by all three groups of learners in Menke and Face (2010) differed from the native controls, suggesting that the pronunciation of/a/is affected even at a very advanced stage. However, Cobb (2009) did not find significant differences between advanced learners' and natives'/a/, while they were found between beginners and advanced learners. Finally, while Cobb (2009) did not find evidence for pronunciation "improvement" in the case of/e/with even advanced learners differing from native controls, Menke and Face (2010) did. Therefore, the pattern of /a/ and /e/improvement was reversed between the two studies. It is unclear why these differences arose, which calls for more research on this specific topic. One aspect to keep in mind is that the two studies introduce different types of noise (or error) in their data. On the one hand, while Cobb (2009) uses only five speakers per group, Menke and Face (2001) use as many as twenty. On the other hand, while Cobb (2009) carries out a careful normalization of formant values (see Cobb and Simonet 2010 for a third normalization procedure), Menke and Face (2010) pool data from many speakers without prior normalization.

A second aspect investigated in both Menke and Face (2010) and Cobb (2009) is unstressed vowel production. Menke and Face (2010) compared tokens of each of the five stressed vowels with their corresponding counterparts in unstressed position. The analysis was limited to establishing how many (and which) of the vowels would show an effect of lexical stress. Stress effects were found for the native controls in the F2 of/e/and/u/, but not of/i a o/. In the beginners' productions, effects were found for F2 in/e/and/u/and for both F1 and F2 in /a/, but not in/i/and/o/. Significant effects were revealed for the intermediate learners'/a/and/i/in both F1 and F2 and in the F2 of/e/and/u/, but not in/o/. Finally, the advanced learners displayed stress effects for all of the vowels (i.e., F1 and F2 in/a/, F1 in/o/and F2 in/i e u/). This finding is unexpected. Greater stress effects were found for the advanced learners than for the beginners with the intermediate learners located in the middle. Paradoxically, while the stressed vowels of native English speakers tend to get "better" with improvement in proficiency, their unstressed vowels tend to get worse.

Cobb (2009) investigated this topic from a different perspective. She calculated the centroid F1 and F2 for the entire vowel system of each individual participant and then measured the Euclidean distance between all the vowels, stressed and unstressed, produced by a given participant and their own centroid. The findings were also unexpected. Native controls were found to centralize more than the two learner groups and the advanced learners were found to centralize more than the beginners. The paradox revealed by Menke's and Face's (2010) findings is not evident in Cobb's (2009) data; that is, learners' productions of *both* stressed and unstressed vowels "improve" (i.e., approach native speakers' productions) with an improvement in the overall proficiency level. An important difference is that Cobb (2009) found robust effects of stress in the natives' vowel productions while Menke and Face (2010) did not.

Cobb and Simonet (2010) reanalyzed Cobb's data by investigating the effects of stress on both the F1 and F2 of all vowels. Effects of stress were found for all groups and all vowels, but these effects differed as a function of both factors. It was found that the three groups centralized their/i a u/vowels in unstressed positions and that this effect was of a similar size across groups. However, there were significant differences in the mid vowels. For/e/, native speakers were found to modify F2 (centralize), while beginners were found to modify only F1 (raise). For/o/, natives were also found to modify F2 (centralize) while the two groups of learners were found not to modify the acoustic characteristics of their unstressed mid-back vowel as a function of stress.

To summarize, acquiring language-specific patterns of conditioned phonetic variation is apparently "difficult." While English-speaking learners may make rapid progress in their pronunciation of stressed vowels in optimal positions, acquiring vowel-specific patterns of stress reduction takes more time. Perhaps it is now the appropriate time to address the study of the acquisition of L2-specific conditioned phonetic variation and move beyond the acquisition of L2-specific sound contrasts in salient prosodic positions. Importantly, the different conclusions reached by the three studies suggest that findings largely depend on the analytical procedures or tools we decide to use to examine our data. Future studies would benefit from more advanced and reliable acoustic and statistical analyses and a more sensitive placement of subjects into groups.

4 Spanish consonants

A series of studies have investigated the production of Spanish consonants by nonnative speakers. We will focus on a number of recent articles that have investigated the production and perception of the following Spanish sounds: (1) voiceless and voiced stops produced by early Spanish–English bilinguals and late English-speaking L2 learners of Spanish (e.g., Zampini 1998; Magloire and Green 1999; Zampini and Green 2001); (2) intervocalic voiced stops/approximants produced by late English-speaking L2 learners of Spanish (e.g., Zampini 1994; González-Bueno 1995; Díaz-Campos 2004; Face and Menke 2009); (3) the intervocalic rhotic contrast between the tap and the trill produced by English-speaking adult learners of Spanish (e.g., Major 1986; Reeder 1998; Waltmunson 2005; Face 2006; Rose 2010); and (4) alveolar laterals produced by early Catalan–Spanish bilinguals (Simonet 2010a, 2010b).

Perhaps the feature of the sound system of Spanish that has received more attention from the L2 speech learning literature is the production and perceptual discrimination of the contrast between voiceless and voiced stops (e.g.,/p/–/b/; see Zampini and Green 2001 for a review). Spanish/p/differs from/b/(when realized as a stop) in at least two phonetic features: (1) voice onset time (VOT) pattern and (2) length of stop closure. Voice onset time (VOT) pattern refers to differences in the relative synchronization of the release of the articulators after closure and the onset of voicing (e.g., Lisker and Abramson 1964). Spanish/p/has a short-lag positive VOT (i.e., the onset of voicing occurs slightly after the time of release of the articulators), while Spanish/b/has a negative VOT (i.e., the onset of voicing occurs during closure, that is, before the release of the articulators). Additionally, Spanish/b/(when realized as a stop) presents a significantly shorter closure than Spanish/p/(Martínez-Celdrán 1993). English/p/differs from/b/in that the latter has a short-lag positive VOT while the former has a long-lag positive VOT. There are no differences in the length of the stop closure in word-initial position between these two sounds (e.g., Lisker and Abramson 1964; Zampini 1998; Zampini and Green 2001).

Zampini (1998) (see Zampini and Green 2001) studied the production and perception of the English and Spanish/p/–/b/contrast by adult L2 learners of Spanish. These subjects were examined four times, three in Spanish (week 3, week 6, and week 15 of an academic semester) and one in English. The subjects were found to produce Spanish/p/with a much shorter VOT than English/p/ and a slightly longer VOT than English/b/. Spanish/b/was produced with a short-lag positive VOT. Importantly, the VOT of Spanish/b/did not differ from that of English/b/. In sum, these subjects had developed a new category for Spanish/p/but, apparently, not for Spanish/b/. Spanish/p/and/b/were found to be significantly different in their VOTs but this difference was of a much smaller magnitude than that between English/p/and/b/. The results for length of stop closure revealed that learners produced Spanish/p/with a much longer closure than Spanish/b/while English/p/and/b/did not differ in this acoustic feature. These results suggested that the L2 learners had been able to develop new categories for *both* Spanish/p/and/b/and were successfully marking the contrast in their production; however, Spanish/p/resembled that of Spanish natives more than Spanish/b/did. Apparently, thus, Spanish/b/is an especially "difficult" sound for native English speakers (i.e., although they are able to develop ways of producing the contrast, it is not the case that they do so in a native-like manner).

Zampini (1998) (see Zampini and Green 2001) also examined the auditory discrimination of the Spanish/p/–/b/contrast of these L2 learners. This was done by investigating the/p/–/b/VOT perceptual boundary; that is, the VOT value that triggers a higher percentage of selection of one category than the other in a perceptual identification experiment in which a continuum of/p/–/b/tokens is created by modifying VOT values. In addition to the L2 learners of Spanish tested at three different times, Zampini (1998) recruited English monolinguals and early Spanish–English bilinguals. The results showed that the L2 learners did not differ from the English monolinguals in the first session while they did in the second and third sessions. The learners did not differ from the early bilinguals in the second and third sessions. In other words, evidence of development (i.e., a reorganization of the perceptual abilities of late L2 learners) was found. The significance and completeness of Zampini's (1998) study is twofold: (1) it investigates the production and perception, and the relationship between the two, of both English and Spanish contrasts as produced by late L2 learners of Spanish, early Spanish–English bilinguals and English-speaking monolinguals; and (2) it "provides longitudinal data, not often seen in the literature" (Zampini and Green 2001: 38). For these reasons, Zampini's research stands as a very significant step towards our understanding of the L2 acquisition of Spanish stop consonants.

Spanish/b d g/are realized as approximants or spirants in most contexts, including the intervocalic position. At least four recent papers have investigated the production of these sounds by native English-speaking L2 learners of Spanish (see above). Since English/b d g/are not often realized as approximants, there is an initial tendency for English-to-Spanish transfer. Zampini (1994) found that a group of late English-speaking L2 learners produced many more stops than spirants in the contexts where native Spanish speakers almost categorically produced spirantized variants. Another difference between the two groups of speakers was that learners produced some/b/tokens as labiodental fricatives instead of bilabials. Zampini attributes this difference to the interference of Spanish orthography since the/b/ tokens that received a labiodental realization seemed to be those written with *v* in the standard orthography (see Face and Menke 2009 for further tests on the effects of orthography). Another interesting result of Zampini (1994) is that the learners produced less spirantized variants (i.e., more "errors") of/d/than of/b/or/g/. Zampini attributed this finding to the phonemic status of the English voiced interdental fricative, which is phonetically very similar to the spirantized realization of Spanish/d/. Thus, apparently, learners were less likely to produce approximant/d/than/b/or/g/ because they have a native phonemic contrast between two sounds that are allophonic in Spanish. Therefore, they might have been less ready to accept the interdental approximant as an allophonic variant of/d/. Similar results were found in later research. Thus, even though González-Bueno (1995) found an overall higher percentage of spirant realizations in the speech of late L2 learners than Zampini (1994) and also Díaz-Campos (2004), one finding was very similar:/d/was the phoneme that showed the lower percentage of spirant productions.

Face and Menke (2009) investigated the productions of a large number of English-speaking L2 learners as a function of their L2 proficiency (i.e., advanced, intermediate, beginners). This is the only paper on L2/b d g/production to include advanced learners. The results showed cross-sectional evidence of "progress" in production since advanced learners produced more approximants than intermediate learners and intermediate learners produced more approximants than beginners. The rates of approximant realizations were much higher across groups than those in Zampini (1994) and Díaz-Campos (2004) but similar to those in González-Bueno (1995). However, one finding was identical: learners produced less spirant realizations of/d/than of/b/or/g/. The finding that the development of spirant realizations of/d/is somewhat more "difficult" than that of/b/and/g/ for native English L2 learners of Spanish seems therefore to be a robust one. In light of these findings, Zampini's (1994) original hypothesis deserves to be submitted to further empirical scrutiny.

The remainder of this section focuses on the L2 acquisition of liquids (rhotics, laterals and stop + liquid clusters) for which mostly production data are available. Face (2006) studied the production of the Spanish intervocalic rhotic contrast as produced by English-speaking late L2 learners of Spanish (see also Major 1986; Reeder 1998; Waltmunson 2005; Rose 2010). Forty-one native English speakers who were learning Spanish in a formal setting were recruited and divided in two groups (i.e., beginners and intermediate learners). Face quantified the type of "errors" the learners made and classified these in two groups: (1) transfer errors, in which learners produce an English-like approximant instead of a Spanish-like tap or trill; and (2) developmental errors, in which learners produce a sound other than the expected Spanish-like tap, the Spanish-like trill or the English-like approximant. First, an important finding was that English-speaking L2 learners of Spanish are generally able to acquire a native-like pronunciation of the Spanish tap at a relatively early stage in their development. Face attributes this to the fact that there exists a very similar phonetic category in English. Second, regarding the production of the trill, beginners produced native-like trills in only 5% of the instances and intermediate learners produced them in only 26.6% of the instances. Interestingly, the beginners produced as many English-like approximants as other sounds, including taps. On the other hand, the intermediate learners produced many more taps than English-like approximants. Notice that the overall number of Spanish-like (i.e., "correct") taps is much higher overall than that of trills. Thus, cross-sectional evidence of "progress" was evident in the acquisition of both Spanish rhotics. While the beginners produced large numbers of English-like approximants for both the Spanish tap and the trill, the intermediate learners were mostly "successful" in avoiding English-like approximants in their Spanish production. However, the intermediate learners were not at all successful in acquiring the Spanish rhotic *contrast* since they seemed to overgeneralize the tap in their production of both Spanish rhotic categories.

Simonet (2010a, 2010b) investigated the production of alveolar laterals in two groups of early Spanish–Catalan bilinguals residing on Majorca. Catalan laterals are known to be velarized while Spanish ones are known to be nonvelarized. Ten Catalan-dominant bilinguals were found to produce Spanish laterals with a lower F2 (more velarized) than ten Spanish-dominant speakers. Simonet (2010b) also examined the within-speaker patterns of interaction between L1 and L2 categories and found that the Catalan-dominant males and females, as well as the Spanish-dominant males, but crucially not the Spanish-dominant females, produced Catalan and Spanish laterals with different acoustic distributions. In other words, while most bilinguals produced two different phonetic categories in their two languages, some did not. Simonet attributes this finding to the effect of the social indexicality of the Catalan-like velarized /l/, which seems to be a sociophonetic stereotype in Mallorca. Simonet's main finding was that social indexical constraints may greatly affect bilingual speech performance, perhaps as much as cognitive factors do.

Colantoni and Steele (2006) examined the production of Spanish stop + liquid clusters such as /pr/, /br/, /pl/, /bl/, etc. Ten native English-speaking L2 learners of Spanish were recruited and classified in two groups (i.e., intermediate and advanced learners). The authors tested very specific hypotheses regarding the stages of development of native-like pronunciation of the different acoustic characteristics of the Spanish stop + liquid clusters. Learners differed from native Spanish speakers in most of the features of the target sound sequences, but especially in their production of the rhotics and in their use of epenthetic vowels. Finally, an interesting hypothesis that received support in these data was that L2 learners were more successful in acquiring L2 features that possessed a smaller degree of variability in native speech while sounds or features that were highly variable in native productions were "difficult" to learn. Future research on the L2 acquisition of these or other sound categories will do well to investigate the magnitude of variability of the target sounds in the speech of native speakers prior to analyzing L2 productions.

5 Conclusions

The present chapter has reviewed some of the recent experimental work on the L2 acquisition of Spanish phonetics and phonology. We have focused on papers that studied the development of native-like production and perception of Spanish segments in early and late bilinguals or L2 learners. Regarding Spanish vowels, we have discussed research on (1) the production of these segments by native Quichua-speaking simultaneous, early and late bilinguals (e.g., Guion 2003), and by native English-speaking late L2 learners (e.g., Cobb 2009; Menke and Face 2010), and also on (2) the perception of vowels by native speakers of English (Morrison 2003; García de las Bayonas 2004) and Dutch (Escudero and Boersma 2002). Regarding Spanish consonants, we have discussed research on (1) the production and perception of voiced and voiceless stops by native English-speaking late learners and by Spanish–English bilinguals (see Zampini and Green 2001), (2) the variable production of spirantized /b d g/ by late English-speaking learners (e.g., Zampini 1994; Face and Menke 2009), and (3) the production of liquids and stop + liquid sequences in late English-speaking learners (e.g., Face 2006; Colantoni and Steele 2009) and Catalan–Spanish early bilinguals (Simonet 2010a, 2010b). The acquisition of many more Spanish sound categories should be investigated. Different native languages should also be examined. It is to be expected that surprising findings will be revealed by an improvement in both the depth and the breadth of our research questions, an adoption of fully developed theoretical models such as Flege's SLM and Best's PAM, and a careful use of sophisticated laboratory techniques.

Even though we have decided to focus our attention on the L2 acquisition of Spanish segments, this is not to say that prosodic features have not been investigated. In fact, a number of authors have examined the production of some Spanish prosodic features by both early and late L2 learners, for example, native Mandarin-speaking late L2 learners (Chen 2007), early Basque–Spanish bilinguals (Elordieta and Calleja 2005), simultaneous and early Quechua–Spanish bilinguals (O'Rourke 2005, 2008), and early Catalan–Spanish bilinguals (Simonet 2011). However, while the study of the L2 acquisition of segments presents the advantage of being able to rely on very robust theoretical models, such as Flege's SLM (e.g., Flege 1995), which allow for the operationalization of specific hypotheses, research on the acquisition of prosodic features is still in need of a fully-fledged, well-grounded model. For instance, while there is no *a priori* reason to assume that the acquisition of prosodic categories will be different from that of segmental categories, it is actually still unclear how to identify what the conceptual equivalent of a segmental phonetic category might be in prosody (see e.g., Flege 1995 for the concept of "phonetic category"). Research on all these fronts is necessary for our understanding of the acquisition of L2 Spanish sounds and of the factors that affect bilingual speech production and perception.

Robust theoretical models are now available that make specific, testable predictions for the behavior of different types of bilinguals and L2 learners. These theories rely on complete cognitive models. Bridges have been built between findings in L2 speech learning research and findings in fields such as L1 acquisition, sociophonetics, speech perception, and psycholinguistics. Sensitive and reliable methods have been developed in order to be able to obtain a better understanding of speech performance in bilinguals and L2 learners. Researchers working on the L2 acquisition of Spanish phonetics and phonology and/or on speech performance in bilinguals who have Spanish as one of their languages (either as a dominant or a nondominant language) will do well to ground their work on such theories and models and to use the reliable methods that have been developed recently.

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35 Theoretical Perspectives on the L2 Acquisition of Spanish

SILVINA MONTRUL

1 Introduction

The study of language acquisition seeks to explain how the individual constructs an abstract mental grammar through exposure to input. For example, how does the infant starting with apparently no grammar eventually develops native-speaker competence and use of the language? How does an adult who already knows a language learn a second language? The nature–nurture debate has figured prominently in theoretical accounts of this mystery, although few will deny that some sort of special human capacity *together with* exposure to the target language bring about full native language attainment. Theories, however, tend to prioritize one factor over the other: whereas the theory of Universal Grammar (Pinker 1989; Chomsky 1981, 1995; Crain and Lillo-Martin 1999′ Snyder 2007) emphasizes the role of the genetic endowment in systematic language growth (nature), empiricist approaches (Bates 1979; Bybee and Hopper 2001; Tomasello 2003; Müller Gathercole and Hoff 2009) stress instead how language emerges piecemeal from interaction with the environment (nurture). This chapter presents the learning problem of second language acquisition (Section 2) and the basic tenets of these two major theoretical positions (Section 3). I discuss empirical findings from the acquisition of Spanish morphosyntax and lexical semantics that speak to this theoretical divide (Section 4). I show that because the theoretical assumptions of each position are different, the questions they ask, and the empirical methods they use are also very different (Section 5). As a result, each position fares better than the other at explaining different facets of the second language process and outcome.

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2 The problem of second language acquisition

Both monolingual children learning their first language (L1) and adult second language (L2) learners must construct a grammar of the target language on the basis of input. A fundamental difference between children acquiring one or two first languages and adult second language learners is that L2 learners already possess a mature linguistic system acquired early in life. Moreover, adult L2 learners are already cognitively mature. Another crucial difference between L1 acquisition and adult L2 acquisition is that full acquisition of the target grammar is not universal or guaranteed in the L2 (Schachter 1990). Although native-like attainment in many linguistic domains is possible (Birdsong 1992; Montrul and Slabakova 2003), it is not equally likely in all linguistic areas. Unlike L1 grammars, interlanguage grammars are prone to fossilization or developmental arrest, such that localized errors may persist at a later steady state (Lardiere 2007). Very often, no amount of evidence from the input, explicit instruction, or correction is sufficient to override non-native patterns.

Much research on second language acquisition is concerned with a characterization of L2 learners' linguistic competence and how it develops by exposure to explicit (instructed) and naturalistic input. L2 acquisition is characterized by significant transfer from the L1, especially at initial stages of development. In addition, L2 learners go through systematic developmental sequences and make errors like those produced by child learners. The challenge for theories of L2 acquisition is to explain the L2 learning process from initial state to ultimate attainment, differential outcomes between L1 child and L2 adult acquisition, and similarities between the two learning situations. Although nativist and emergentist approaches seek to explain the problem of L1 acquisition, they have also been extended to L2 acquisition.

3 Nativism and empiricism

3.1 Basic tenets of nativism

The theory of Universal Grammar is representative of linguistic nativism. Universal Grammar (UG) (Chomsky 1981, 1986, 1995) is a theory of the human biological endowment for language. It assumes that linguistic behavior is unique to the human species: complex aspects of linguistic knowledge must be encoded in the genes. Innate knowledge of language involves knowledge of abstract rules (what is possible) as well as constraints (what is not possible). This knowledge is represented symbolically in the mind of human beings. Innate linguistic principles emerge early in life, are universal, and appear without decisive evidence from the environment.

UG is like a Language Acquisition Device, a mental construct that mediates between the primary linguistic data or input and actual linguistic behaviors (understanding and producing novel utterances by combining a finite set of elements). UG contains the set of abstract rules and constraints that limit the hypothesis space of variation in human language. UG also defines the range of variation observed across languages (parameters, in Chomsky 1981) and characterizes the notion of possible human language. According to this theory, children are born with abstract grammatical knowledge (grammatical categories such as noun, verb, determiner, tense, agreement, etc., and knowledge of principles for combinations of these elements), which is triggered by exposure to input. The child selects the appropriate elements from UG (parameters, principles, functional categories, features, etc.) on the basis of the linguistic environment. Language acquisition is seen as a logical problem: how can the child acquire such a complex grammatical capacity when the evidence from the environment is limited and insufficient to bring about such knowledge? The nativist solution has been to attribute innate linguistic knowledge of language to the child.

3.2 Basic tenets of empiricism

Emergentism, an example of the empiricist approach, seeks to explain how the interaction between genes and the environment produces language (Elman et al. 1999; Ellis 2002, 2003; O'Grady 2008). Innateness is also part of the emergentist position, but what is considered innate is independent of language. If language consists of pairing multiple sounds and meaning onto a low-dimensional channel constrained by the limits of the human processing system space, at least some aspects of the speech perceptual system and of the conceptual system must be innate. Many of the perceptual and categorizational abilities of human infants are present in other animals (Juczyck 1997). Similarly, a great deal of conceptual structure belongs to the realm of thought (Jackendoff 2002), and cannot be part of UG.

In contrast to linguistic nativism as expresses in the Universal Grammar approach, emergentism emphasizes the environment for language learning. Grammar is learned by general learning mechanisms and linguistic complexity is an emergent property of lexical learning. The input is seen as more informative and revealing of linguistic structure than the nativists make it seem. A central claim is that the child initially learns language by exposure to frequent patterns (defined at the phonological, morphological, semantic and syntactic levels) and statistical learning. Statistical learning comes from computer science. It means that a learner can take advantage of examples (data) from how language is used to capture its underlying probability distributions and automatically learn to recognize complex patterns (from stress, prosody, syllable, and lexical patterns). The learner then generalizes from the given examples, so as to be able to produce a useful output in new cases. Lacking grammatical knowledge *a priori*, the child initially learns holophrases by rote, which can be either words or made-up chunks (e.g. *Where is X?*, *It's a*, *wanna X*, *X gone*). These phrases are not analyzed at first. But once the child has learned a critical mass of such words and chunks (a substantial amount), the child is said to construct a grammar. Hence, abstract grammatical knowledge *emerges* from language use, the interaction of input and innate aspects of cognition.

Under this usage-based perspective, the child is portrayed as doing more work than in the UG approach. The linguistic representations of young children are not as abstract as nativists claim them to be, at least initially. Rather, children use very specific linguistic expressions in very limited ways and do not generalize to other instances, as nativists contend. There is no productivity and no rule-governed behavior at the beginning.

3.3 Second language (L2) acquisition

These positions have been formulated primarily to explain language acquisition by children, but the nativist perspective has generated an important body of research in other learning situations (simultaneous bilingualism, child and adult L2 acquisition, incomplete acquisition, and impaired acquisition) and in languages other than English, more so than the emergentist perspective. With respect to the acquisition of Spanish, the nativism–emergentism debate in child language has mainly focused on the very early period, around 1–3 years of age, and has addressed the emergence of inflectional morphology, including determiners and nominal morphology, verbal paradigms, clitics, and negation (López Ornat 1994, 1997; Mueller Gathercole et al. 1999, 2002; Bel 2001; Lleó 1998, 2001; Torrens 2002; Grinstead 2000; Pérez-Leroux 2011). The issues have centered on evidence of a grammatical system, inductive versus deductive learning, and productivity.

The nature–nurture debate has been less emphasized in L2 acquisition, maybe because other factors come into play in this situation. For example, even if we accept that children start the language acquisition process with no grammar, it is hard to say the same for L2 learners, who already constructed a grammar. Both the nativists and emergenstists can successfully account for the fact that the native language plays a significant role in L2 learning. On MacWhinney's Competition Model (1987, 1992), L2 learners rely, especially initially, on the transfer of phonological, syntactic, and lexical cues from the L1 to the L2. Learners must discover which cues map directly onto L2 from L1, which ones will map only partially, and which ones will not apply. In Schwartz's and Sprouse's (1996) Full Transfer/Full Access Hypothesis, the entire L1 system (abstract rules and parameter settings) is the initial state of L2 acquisition. L2 learners analyze L2 input through the linguistic structure of the L1. However, once the L1 structure can no longer accommodate the L2 input, learners restructure their system, incorporating elements from UG (Full Access). Thus, according to this model, it is possible to overcome L1 influence and reach native-like competence in a second language. The point of debate between the two approaches is whether L2 learners show evidence of being constrained by specific linguistic mechanisms in addition to their L1, or whether they work their way into the L2 grammar by staying close to the input, using their L1 grammar, and applying other general learning mechanisms.

Nativists have looked at the L2 learning process since the initial state and up to ultimate attainment in near-native speakers. Emergentists so far have mostly focused on early stages of development. Due to their interest in language use, emergentists have also focused predominantly on oral production data, while nativists have used other experimental tasks, like grammaticality and truth–value judgment tasks, to investigate structures and interpretations that are not obvious from the input and are unlikely to occur very often in production. Another important difference is that due to the universal character of the theory, nativists often include cross-linguistic comparisons of different L2 learners' native languages or target languages. The few existing L2 studies conducted within the emergentist framework tend to focus on only one language or language group.

There exists a large body of research on nativist perspectives on L2 acquisition in general (e.g., White 1989, 2003), and on the L2 acquisition of Spanish in particular pioneered by Liceras (1985) (see Montrul 2004 and Liceras 2010 for overviews). Unfortunately, the potential contribution of emergentism to L2 acquisition (N. Ellis 1996, 2002; McWhinney 2002) has not been explored much in Spanish. Except for Pérez-Leroux and Glass (1999), who actually contrast the predictions of the two paradigms in their study of pragmatic constraints on null subjects, I know of no other L2 acquisition studies that take this approach.

The next section exemplifies how the L2 acquisition problem is addressed by each perspective. To be able to compare the relative merits of the two approaches, I have selected examples of generative and emergentist studies that have investigated the same grammatical phenomenon in L2 acquisition. These include: object clitics, null objects, and transitivity alternations.

4 Empirical evidence

4.1 Object clitics

Spanish has a rich system of pronominal clitics. Although many languages have clitics, clitics in these languages differ in their syntactic distribution with respect to the verb, the distribution of clitics vs. null objects, and the availability of clitic-doubling constructions. Under the generative approach, Spanish object clitics are analyzed as affix-like elements that head their own functional categories (Uriegereka 1995). They are part of the functional architecture of the sentence, interacting with verbs, negation, and other elements in the syntax. English lacks clitics and does not select these categories from the inventory of properties available through UG. For the type of functional approached assumed by the emergentist perspective, clitics are constructions that evolved from agreement markers diachronically, and link form and meaning in discourse (Belloro 2007). Clitics are analyzed on the basis of the semantic and pragmatic functions they perform and are not necessarily portrayed as part of the syntactic architecture.

The L2 acquisition of Spanish clitics has been a prominent testing ground for debates on the nature of the initial state in L2 acquisition, the role of the native language, and the availability of UG (i.e., parameter resetting or acquisition of functional categories not instantiated in the L1) from initial to endstate (Montrul

2004), and as part of the interlanguage system during development (Liceras 1985). For example, Montrul (1998) investigated the acquisition of the functional category AgrIOP (indirect object agreement phrase) by English- and French-speaking learners of Spanish. The study assumed a parameter resetting model, according to which a number of seemingly unrelated structures are actually related at a very abstract level and learned simultaneously once the learner has acquired the crucial structure that triggers the acquisition of the system in a deductive way. This is called clustering. The question was whether dative clitics were the particular triggers for the projection of the functional category AgrIOP, and whether L2 learners were aware of other syntactic consequences associated with AgrIOP Spanish. The study also investigated the role of L1 influence in early interlanguage development.

The experiment was based on the following cross-linguistic differences. When structural dative case was lost in Middle English, a number of structures emerged, like Exceptional Case Marking or ECM as in (1), preposition stranding as in (2), prepositional passives as in (3), double object constructions as in (4), and indirect passives as in (5).

(1) Mary believes John to be a good friend.

(2) What is this book about?

(3) This bed was slept in.

(4) John gave Mary a present.

(5) Mary was given a present.

In Spanish, where there is morphological dative case as instantiated in clitics, all the translations of the English sentences are ungrammatical, as examples (6)–(10) show.

(6) *María cree Juan ser un buen amigo.

(7) *Qué es el libro sobre?

(8) *Esta cama fue dormida en.

(9) *Juan dio María un regalo.

(10) *María fue dada un regalo.

Assuming the Full Transfer/Full Access Hypothesis (Schwartz and Sprouse 1996), Montrul hypothesized that French-speaking learners, whose L1 has dative clitics and AgrIOP, should have no problem acquiring all the properties related to AgrIOP in Spanish: they should reject all the equivalents of the English sentences in Spanish (examples (6)–(10)). Once the English-speaking learners acquire that dative clitics are the overt manifestation of AgrIOP in Spanish, they should also realize that the counterparts of the English sentences are ungrammatical in Spanish, as the parameter resetting model would predict.

The experiment involved English-speaking and French-speaking learners of Spanish at the low intermediate and intermediate level of instruction, as well as a control group of Spanish native speakers. The main tasks were a written production task and a written grammaticality judgment task, the latter consisting of grammatical and ungrammatical sentences with dative clitics, indirect objects, and *gustar*-type verbs. The learners in the two groups performed at around 85% accuracy in the elicited production task on structures with clitics, suggesting that they had learned dative clitics in Spanish, the assumed trigger. The grammaticality judgment task investigated the acquisition of sentences with dative clitics not produced in the oral task and the knowledge that all the structures in (6)–(10) were ungrammatical in Spanish. The results showed no differences between the Spanish native speakers and the French-speaking learners, but differences were found between the Spanish native speakers and the English-speaking learners. The higher accuracy of the French-speaking over the English-speaking learners suggested an advantage for the French group due to L1 transfer of the AgrIOP projection from French.

With the ungrammatical sentences in (6)–(10), the English learners performed poorly: they incorrectly accepted indirect passives, preposition stranding, double objects, and ECM constructions in Spanish. If the dative clitics are the trigger for the acquisition of the dative case system of Spanish, the results also indicated that the preemption process (or unlearning) is gradual for the English-speaking learners, casting doubt on the nativist view that much of language acquisition is a matter of deductive learning and switching parameters automatically. In other words, there was no evidence of clustering for the English-speaking group even when they appeared to use dative clitics correctly in the oral production task. Although in another study Bruhn de Garavito (2000) showed that Spanish dative constructions are acquired at a native-like level by English speakers, Montrul's study of less proficient learners shows these learners are very much constrained by their L1 at this level, consistent with the Full Transfer stage of the Full Transfer/Full Access Hypothesis. Because Montrul's study did not include more advanced groups, it does not explain what happens in the transition from the intermediate to near-native levels for English speakers learning Spanish.

Coming from the usage-based perspective, Zyzik (2006a) investigated the early emergence of dative clitics in beginner to advanced English-speaking learners of Spanish (12 beginner, 12 intermediate, 12 high intermediate, and 14 advanced in addition to 12 native speakers). Assuming that early grammatical knowledge is guided by semantic and pragmatic principles rather than syntax, Zyzick wanted to find out whether L2 learners of Spanish create prototype categories to learn complex structures, even though these categories might not coincide with the system of native speakers. The four groups of learners completed four oral production tasks: a picture book narration; two video narrations; and a structured interview. Zyzik focused on an analysis of the dative clitic (*le/les*) as encoding the semantic roles of recipient, source, experiencer, beneficiary, possessor, patient (masc./fem.), and others in a variety of constructions. Of a total of 309 dative clitics elicited, the beginner learners produced 2% (5), the intermediate learners 22% (68), the high intermediate learners 30% (93), and the advanced 46% (143). Zyzik claims that the few clitics produced by the beginner learners were unanalyzed chunks (*le gusta*, *me gusta*). Even when these structures are introduced very early and highly stressed in language classrooms, the L2 learners failed to use the correct agreement morphology on the verb.

The semantic role analysis revealed that for the three groups that produced clitics (intermediate and beyond), the greater percentage of clitics occurred with recipients (e.g., *servirle un café*, *sacarle el collar*). The second most frequent use of dative clitics was experiencer in psych verb constructions for the intermediate learners, while it was patient (masc. and fem.) for the high intermediate and advanced groups. Uses of dative clitics in inalienable possession constructions (*le seca la cabeza*) and as beneficiaries were almost nonexistent in the data. Zyzik found overextensions of dative clitics to accusative contexts with increasing proficiency (*le tapa con una manta para que ella pueda dormir*), regardless of the animacy or gender of the patient. According to Zyzik, learners overgeneralized dative to accusative context guided by animacy. After all, most dative objects tend to be animate, and very few extensions of *le/les* to inanimate direct objects were found in the data. Zyzik proposes that L2 learners build a prototype semantic category based on frequency patterns in the input. According to what Zyzick reports from corpus studies, 90% of dative objects are animate. She claims that frequency effects in the input such as these are bound to affect the learners' developing grammars by leading them to assume that animate direct objects should be marked with the dative. Zyzik questions how UG-derived knowledge would contribute to learners' ability to make case distinctions.

Contrary to what Zyzick portrayed, UG-derived knowledge does account for the acquisition of case distinctions. As agreement and referentiality markers, accusative and dative clitics are assumed to project their own functional projections. Although not about clitics, a recent study by Montrul and Bowles (2010) shows that intermediate L2 learners distinguish between animate direct objects and indirect objects (both marked by the preposition a, which is a dative case marker with indirect objects but an accusative case marker with direct objects), suggesting that they do in fact have knowledge of the syntactic properties of accusative and dative case, even when there may be some shared semantic characteristics (e.g., animacy) between indirect objects and animate direct objects.

4.2 Null objects

In certain syntactic environments, Spanish allows object drop, but this is hardly taught in Spanish language classrooms.

(11) a. ¿Compraste pan?
'Did you buy bread?'
b. Sí, compré.
*'Yes, I bought.'

(12) a. ¿Compraste el diario?
'Did you buy the newspaper?'
b. *Sí, compré.
*'Yes, I bought.'

Fujino and Sano (2002) proposed that L1 acquiring Spanish-speaking children go through a developmental grammatical stage during which they produce null objects in contexts like (12b). They reported up to 50% object omission between the ages of 1;7 and 1;9 in the two children. But when the children began to use object clitics, null objects dramatically decreased. On their nativist interpretation of these patterns, children have grammatical knowledge of clitics from the beginning, but fail to spell them out. Usage-based accounts, by contrast, claim that object omissions such as these are due to processing factors because the child does not yet have enough processing resources to plan and produce sentences with two arguments.

Zyzik (2008) investigated Fujino's and Sano's model for L1 acquisition in L2 learners of Spanish. Participants and methods were the same as in Zyzik (2006a). This study added a written grammaticality judgment task with grammatical and ungrammatical sentences with null objects, questions, clitic doubling with strong pronouns, and clitic placement. The oral data were analyzed for frequency and distribution of different types of objects: lexical NP, DO clitic, null object, anaphoric se, strong pronoun, and phrasal complement. The percentage distribution of lexical NPs ranged from 91.6% in beginners to 72% in advanced. As for the distributions of clitics and null objects, these are summarized in Table 35.1. Zyzik discusses all the uses of null objects, noting that the vast majority referred to inanimate objects, as in different varieties of Spanish where specific null objects are licensed. Interestingly, and crucial for the hypothesis tested, Zyzik does not discuss in any detail the relationship between clitics and null objects, summarized in Table 35.1.

Table 35.1 shows that at the earliest stages, beginner L2 learners hardly produce clitics but they produce null objects, as has been found in L1 Spanish. However, the rate of these constructions in the L2 data is much lower than reported in the L1 data. Starting at the intermediate level, as the use of clitics increased, the use of

Table 35.1 Percentage production of direct object clitics and null objects in English-speaking learners of Spanish (adapted from Zyzik 2008).

	Beginner	*Intermediate*	*High intermediate*	*Advanced*
DO clitic	0.6% (5)	8.8% (83)	18% (175)	22.5% (297)
Null object	3.7% (31)	4.5 (29)	3% (29)	1.2% (16)
Total objects	846	934	973	1317

ungrammatical null objects remained relatively stable. By the advanced level, there is now 22% production of clitics, and null objects are very infrequent (1.2%). Although the frequencies are much lower than in Fujino and Sano's L1 acquisition data, there seems to be a trade-off between null objects and clitics in L2 acquisition as well.

The grammaticality judgments of written sentences confirmed that the L2 learners accepted ungrammatical null objects, especially at the lowest levels of proficiency. The correlation between the oral production data and the judgment data on null objects was significant. Zyzik claims that the low incidence of null objects in speech makes it impossible to claim that there is a distinguishable stage of null objects in L2 acquisition. The fact that there was variability by verbs and individual variability by subjects (including some self-corrections) led her to conclude that null objects in L2 grammars are not a grammatical stage but a simple performance phenomenon. Interestingly, Zyzick also interpreted poor performance on the ungrammatical items in the judgment task as mere performance errors, as claimed for children. This claim is odd if L2 learners are adults who already have the mature cognitive capacity to be able to process and produce more than one argument per sentence.

Viewed from a nativist standpoint, the fact that L2 learners perform consistently in production and grammaticality judgments suggests that this is a legitimate grammatical stage, during which the English speakers briefly entertain the grammar of another language, maybe European Portuguese, where specific direct objects can be null. Although this is not a pattern that can be explained by the learner's L1, it certainly falls within the range of options available from Universal Grammar. This seems to be the same option of which Spanish children avail themselves. Furthermore, a study by Bruhn de Garavito and Guijarro Fuentes (2002) with intermediate and advanced English-speaking and European Portuguese-speaking learners of Spanish found that the English-speaking learners knew the syntactic constraints on null objects (subjacency) in Spanish, assessed via a written grammaticality judgment task. For Bruhn de Garavito and Guijarro Fuentes, their results showed that L2 learners were capable of resetting parameters and acquire the syntactic constraints on null objects, even when the syntactic structures related to the null object system they tested are not very frequent in the input.

4.3 *Transitivity alternations*

Transitivity alternations are another topic that has been addressed from both nativist and emergentist perspectives in L2 acquisition. In Spanish and other languages, there is a class of transitive verbs that alternate in transitivity, appearing in transitive and intransitive configurations, as in (13a,b).

(13) a. El ladrón rompió la ventana.
'The thief broke the window.'
b. La ventana se rompió.
'The window broke.'

The transitive version in (13a) expresses a causative situation; the intransitive version in (13b) focuses on the change of state, and the verb is unaccusative. This is called the causative–inchoative alternation, and it only occurs with a specific set of verbs. Other transitive verbs, which also express some change of state (*pintar* 'pintar,' *cortar* 'cut,' *escribir* 'write'), do not alternate in transitivity. Other intransitive verbs (*desaparecer* 'disappear,' *salir* 'leave,' *escapar* 'escape') do not alternate in transitivity.

Languages vary as to the morphological realizations of the causative–inchoative alternation. In Spanish, the reflexive clitic *se* is obligatory in the intransitive version. How do children and L2 learners come to know that some verbs alternate in transitivity but others do not, even when they seem to be related in meaning? Children initially distinguish between transitive and intransitive verbs, but also later at some point start to make overgeneralization errors with argument structure and with reflexive morphology (Montrul 2004). The same learnability problems that transitivity alternations present for L1 learners apply to L2 learners: namely, which verbs alternate and what the reflexive clitic entails. In addition, L1 influence plays a role. A language like Turkish has overt morphology in the causative or the inchoative form, as in Spanish, depending on the verb. In the Turkish verb *batmak* 'sink,' the causative suffix -*DIr* or any of its allomorphs attaches to the verb root to form the causative form. With Turkish verbs like *kırmak* 'break' the anticausative morpheme -*Il* is like the Spanish reflexive clitic, attaching to the intransitive form. This alternation is not morphologically marked in English.

Montrul (1999) investigated how English and Turkish-speaking L2 learners learn that superficially transitive and intransitive verbs manifest different syntactic behavior, and how the clitic *se* affects the argument structure and aspectual properties of different intransitive verbs in Spanish, as in verbs that participate in the causative–inchoative alternation. If UG constrains the hypotheses that L2 learners can entertain about lexicosyntactic knowledge, L2 learners should in principle be able to identify common syntactic aspects of related verbs because this knowledge is also available from their native language. However, if L2 learners do not initially differentiate between intranstive verbs like *fall*, whose sole argument is an object (unaccusative verbs), from intransitive verbs like *run*, whose sole argument is a subject (unergative verbs), or have trouble classifying particular verbs, it is also possible that they might overgeneralize the transitivity alternation to verbs that do not alternate. Montrul also expected the acquisition of the morphological marking of argument structure alternations with overt–nonovert affixes to be affected by L1 influence. English speakers were expected to have more difficulty with the reflexive morphology of alternating verbs than the Turkish speakers, and to overgeneralize the reflexive clitic *se* to other intransitive verbs, especially to unaccusatives.

Native speakers of Spanish, Turkish-speaking intermediate learners of Spanish, and English-speaking intermediate learners of Spanish completed an acceptability judgment task with pictures and were asked to judge the grammatical and semantic appropriateness of written sentences. Some pictures represented actions or activities with transitive verbs and others with intransitive verbs. The clitic *se* was manipulated in the intransitive sentences and the verb *hacer* 'make' in the transitive sentences.

Overall, the L2 learners distinguished between semantically-defined verb classes: they knew that alternating verbs alternated in transitivity, but that other intransitive verbs did not. However, learners were less accurate than native speakers at rejecting transitivity errors with all the nonalternating classes, including transitives, suggesting that like L1 learners, they also overgeneralize the alternation. As in L1 acquisition, the L2 learners tended to apply a transitive template to those verbs for which acquisition of grammatically relevant aspects of meaning was in some sense incomplete. A transitive template applied to nonalternating verbs would explain why child and adult learners occasionally overgeneralized the causative/inchoative alternation to other verbs. Cabrera and Zubizarreta (2002) expanded Montrul's original work by controlling for proficiency and investigating individual subjects' performance. They found that the learners who overgeneralized with intransitives appeared to use the same syntactic strategy (a transitive template) proposed by Montrul. What Cabrera and Zubizarreta were able to tell is that this strategy does not work for all learners, depending on their proficiency.

Montrul (1999) also found that errors due to L1 influence were very obvious with the reflexive morphology of alternating verbs. The Turkish group was more accurate than the English groups at accepting -*se* with the intransitive forms. Results of alternating verbs without *se* showed that the overwhelming majority of English speakers rejected forms with the clitic, while the Turkish speakers did not. There was also overgeneralization of *se* to other intransitive sentences (with unaccusatives, unergatives, and nonalternating verbs), but again the Turkish speakers oversupplied this morpheme more often than the English native speakers.

Zyzik (2006b) contends that the variability by verbs and individuals documented by the Montrul and the Cabrera and Zubizarreta studies challenges the conclusion that L2 learners know verb classes. Therefore, Zyzik (2006b) examined whether L2 learners initially learn these verbs, and the intransitive version with *se*, on a verb-by-verb basis, following the basic tenets of emergentism and piecemeal learning. The acquisition of the clitic *se* represents a challenging form-meaning mapping task for L2 learners, given its multiple functions. Zyzik hypothesized that if L2 learners learn verbs like *maquillarse* 'apply make up to oneself,' *afeitarse* 'shave oneself,' and *bañarse* 'give oneself a bath' first as chunks, they will think that *se* is part of the verb and incorrectly extend *se* to transitive verbs that do not require *se* like **Miguel se baña el perro* 'Miguel gives the dog a bath.' Zyzik also predicted significant variability by verb with-*se* overgeneralization or omission in required contexts.

English-speaking L2 learners (same as in Zyzik 2006a, 2008) were shown a picture of a man on a scale and were told: here *El hombre se pesa* 'The man weighs himself.' When they were asked to describe a picture of a man putting apples on a scale, the expected response was *El hombre pesa las manzanas* 'The man weighs the apples.' The task included four reflexive–transitive verb pairs and four causative–inchoative verb pairs. In half of the sentences, the intransitive response was modeled and the transitive response elicited; in the other half, the transitive response was modeled and the intransitive elicited.

Overall, the L2 learners made a significant percentage of transitivity errors: 64% (beginners), 44% (intermediate), 40% (high intermediate), 26% (advanced). The directionality analysis revealed that the L2 learners performed better on the intransitive-to-transitive condition than on the transitive-to-intransitive condition. The results also showed highly variable performance on individual verbs. Zyzik concluded that, after one year of study, beginner learners do not have productive knowledge of *se* as a reflexive or inchoative marker. Those first uses are unanalyzed chunks. Only at the advanced level do learners show productive use and knowledge of these alternations, although errors of *se* omission persisted. Zyzik also agrees with Montrul and with Cabrera and Zubizarreta that these errors stem from L1 influence due to existing zero-derived counterparts in English.

Thus, although these studies started from different theoretical assumptions and their explanations for the results may differ, they all found that errors with intransitive verbs are more common than errors with transitive verbs, as in L1 acquisition. This error cannot come from input frequencies but from a generalized pattern about verb meaning. All these studies agree that English-speaking learners' difficulty with *se* in Spanish stem from L1 influence.

5 Discussion

In this chapter, I have introduced how the problem of second language acquisition is conceived by the nativist (UG) and the empiricist (emergentist) perspectives, and I have presented a sample of empirical studies on the L2 acquisition of morpshosyntax and lexical semantics. Evaluating which approach provides a better account of any particular phenomenon is not straightforward, for the following reasons. Nativism and empiricism start out from very different assumptions about the nature and development of linguistic knowledge in children and adults. While nativists assume that children and adults already start with innately-specified complex grammatical principles triggered by the input, empiricists believe that grammatical knowledge is not prewired, but emerges from experience with the input. Given their different assumptions, researchers working within these approaches typically pose different research questions. Different research questions require attention to different grammatical phenomena, which in turn calls for different methodologies and data analyses. Therefore, we tend to reach different conclusions about the matter at hand.

When it comes to explaining and accounting for the process and outcome of adult L2 acquisition, both linguistic nativism and emergentist approaches can account for the role of the native language or L1 transfer. Because nativists take both the notion of linguistic system and cross-linguistic comparisons as fundamental aspects of their theory, the issue of L1 influence and notions of parameter resetting take center stage. The difference between the Competition Model and the Full Transfer/Full Access Hypothesis is that in the Competition Model, what transfers are overt surface cues (word order, case markers, etc.), whereas for the Full Transfer/Full Access Hypothesis, what transfers are mainly abstract properties of the L1 system (features, functional categories, parameter settings), not always evident from the surface structure.

Another issue in L2 acquisition studies is to explain the L2 language learning process from initial state to ultimate attainment, and theories of transfer fare well to explain the initial state. But how does development occur? For emergentists, initial L2 learning relies on chunking and rote memorization, with no evidence of productivity. Guided by semantic and pragmatic notions, the learner creates prototype categories from the input data and eventually constructs a grammar, as Zyzick's study on dative clitics shows.

Nativists see properties of language and specific constructions as part of a complex system: acquiring clitics or null objects implies much more than producing them in appropriate semantic or pragmatic contexts. It also implies knowledge of the syntactic environments where the overt or nonovert expression of objects is licensed or not. By contrast, the studies representing the emergentist perspective discussed in this chapter tend to focus on specific structures. When other grammatical phenomena are included, such as the sentences in the grammaticality judgment task testing other aspects of object clitics in Zyzik (2008), these are not linked by a specific linguistic analysis. Rather, language is presented as a collection of individual structures or constructions with no relationship with each other because the most important factor driving acquisition is the input frequency of the particular constructions. Although frequency effects are important, it is not clear that they drive the sequence and outcome of language learning very faithfully. Pérez-Leroux and Glass (1999) found that, although null subjects in Spanish are more frequent in the input in topic/focus structures than in other very infrequent syntactic contexts, they had not been acquired earlier or better than null subjects in the less frequent contexts.

Because nativists pose hypotheses for the initial stage, intermediate stages, and ultimate attainment, studies of L2 acquisition include learners beyond the initial state of acquisition, when the complexity of grammatical knowledge can be assessed. However, nativists studies have been unsuccessful in providing evidence for deductive learning, or the idea that learning does not take place step-by-step or incrementally: once a parameter is reset, a number of seemingly unrelated constructions are acquired all of a sudden. Thus, the specific steps in the transition from initial to intermediate, advanced, and near-native grammars has not been properly accounted for in generative approaches. By staying closer to the data, emergentists have been more successful in this realm.

One crucial aspect of L2 acquisition that is accounted for by the nativist approach and hard to reconcile with the basic tenets of emergentism is the concept of interlanguage, the system that L2 learners build in their learning process. Although many errors that L2 learners make can be traced back to either imposing the structure of the native language or L1 from cues on L2 input, many other nontarget features cannot be due to L1 influence or frequency effects. For example, when English-speaking learners of Spanish go through a null object stage for definite objects, this cannot be due to English or to the frequency of definite null objects in Spanish because they are very rare. But specific null objects are common in other languages, like European Portuguese. Similarly, Perpiñán (2010) recently uncovered that, in forming prepositional relative clauses, Arabic and English-speaking learners of Spanish produce and incorrectly accept in grammaticality judgments sentences without prepositions (Arabic L1: **Esta chica es la secretaria que los compañeros de oficina la piensan* 'This girl is the secretary that the office mates think,' English L1: **Esta es la enfermera que la chica se depende* 'This is the nurse that the girl depends'). This same phenomenon of Null Prep has been already been reported in learners of English from different language backgrounds by Klein (1993) (**The clerk who Jill had complained was fired*) and in child English (Hilldebrand 1987). If prepositionless relative clauses are not licensed in English or Spanish, why do learners from different language backgrounds learning different languages produce structures such as these that are not part of their L1 or the input? Where are these systematic patterns coming from? These type of data are very hard to reconcile with the emergentist perspective. Under the generative approach, this phenomenon emerges from the operation of specific grammatical mechanisms available from UG. Interlanguages are constrained by UG like any other natural language.

6 Conclusion

As we have seen, nativists and emergentists reach different conclusions about the nature of language and the language learning process because they focus on different aspects of language. The data discussed in this chapter suggest that there seems to be a very early period in both L1 and L2 acquisition when the learner is learning the lexical items of the target language on an item-by-item basis and for which evidence of deductive learning is hard to come by. However, this stage is quite short-lived as there is also evidence that the learner does not learn these items in isolation, but as part of a system. Once the system emerges, it is hard to see how everything learners know can be inducted from input frequencies. There is also evidence that not everything the learner says is a direct reflection of the input: both L1 and L2 learners make some similar errors (e.g., transitivity alternations, null objects), and, while many errors in L2 acquisition can be traced to the L1 grammar, other errors that are common in L1 and L2 acquisition and across the acquisition of different languages suggest that some sort of universal, internal mechanism is also driving the process. Hence, the truth is somewhere in the middle. Although the tension between the two perspectives might be difficult to alleviate for philosophical reasons, as they stand, emergentists can account for the earliest periods of acquisition, while nativists appear to be more accurate on the period when basic grammatical knowledge is in place but still restructuring toward ultimate attainment.

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36 Spanish as a Heritage Language

MARÍA M. CARREIRA

1 Introduction

This chapter presents an overview of the field of the teaching of Spanish to bilingual Latinos in the United States. Building on previous overviews (Roca 1992, 1997; Kreeft Peyton et al., 2001; Potowski 2007; Kreeft Peyton 2008), it describes critical landmarks in the development of the field, summarizes ongoing research and advancements in teaching, and identifies outstanding issues. Since its inception, this field has gone by a variety of names (e.g., Spanish for native speakers, Spanish for bilinguals, Spanish as a heritage language, etc.). The students at the center of this field have also gone by different names (native speakers, quasi-native speakers, home-background speakers, Hispanic bilinguals, and most recently, heritage language learners). Following current practices, I will use the term "Spanish as a heritage language" (SHL) to refer to the field and "Spanish heritage language learners" (SHL learners) to refer to the students. It should be noted that though widely adopted, the use of this terminology is not without its critics. For example, Corson (1999), Wiley (2001, 2005), and García (2005) have argued that "heritage language" may not be the most effective term, because it evokes images of the past and tradition rather than of contemporary reality.

2 From the Limited Normative Approach to the Comprehensive Approach

SHL first gained official recognition from the profession in 1972, when the American Association of Teachers of Spanish and Portuguese (AATSP) issued the following statement:

> whenever in the United States there are pupils or students for whom Spanish is the native tongue, at whatever level from kindergarten to the baccalaureate,

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> there be established in the schools and colleges special sections for developing literacy in Spanish and using it to reinforce or complement other areas of the curriculum, with correspondingly specialized materials, methods, and teachers (cited in Roca 1997: 38).

The actual teaching of SHL, however, well predates such recognition, with instruction stretching as far back as the 1930s (Valdés-Fallis 1978a). With a prescriptive orientation, early approaches to teaching SHL have little in common with current ones, which are oriented towards expanding learners' command of registers and addressing their socioaffective needs. Following a "Limited Normative Approach," early textbooks and other materials made widespread use of labels such as 'arcaísmos,' 'barbarismos,' and 'anglicismos' to call students' attention to their use of nonstandard linguistic features and eradicate them from their speech. The Limited Normative Approach came under attack in the 1970s and 1980s for failing to develop general proficiency in speaking, reading, and writing and for contributing to students' linguistic insecurities (Valdés-Fallis 1978a; Faltis 1990)

Spurred by the growing presence of bilingual Latinos in Spanish classes and concerned about the shortage of teachers with training and experience in SHL teaching, in 1978 the National Endowment for the Humanities sponsored a summer institute which brought together SHL instructors with the principal objective of improving curricula. Under the direction of Guadalupe Valdés Fallis and Richard Teschner, the participants produced syllabi for teaching SHL to different student populations in different learning contexts. The model curricula articulated a rationale for SHL courses, set forth goals and objectives for instruction, proposed teaching methodologies and strategies, evaluated pedagogical materials, and addressed issues of assessment and articulation. Widely influential at the time, these syllabi remain remarkably relevant (Valdés-Fallis and Teschner 1978b).

The year 1978 also saw the publication of the first textbook that followed a Comprehensive Approach to teaching SHL, *Español escrito: curso para hispanohablantes* by Guadalupe Valdés Fallis and Richard Teschner. The Comprehensive Approach rejected the intense preoccupation of the Limited Normative Approach with extirpating nonstandard linguistic features and focused instead on giving students "the opportunity to grow in their mother tongue, the opportunity to use it in meaningful communication and creative expression" (Valdés et al. 1981: 14). Numerous SHL textbooks followed suit, among these, Mejías and Garza-Swan (1981), Burunat and Starcevic (1983), Miguélez and Sandoval (1987), and Marqués (1986). Appearing in the wake of the Civil Rights movements of the 1960s and 1970s, these textbooks explored issues of access, inclusion, and social justice, and sought to help Latino students explore their identity vis-à-vis local communities of Spanish speakers (Leeman and Martínez 2007).

A seminal volume in 1981, edited by Guadalupe Valdés, Anthony Lozano, and Rodolfo García-Moya, mapped out the field's research and pedagogical priorities, including teacher preparation, student placement, the language of instruction, and the design of syllabi, curricula, and programs. Also in this volume, Valdés made the case for modeling SHL teaching after language arts teaching in English. In keeping with this perspective, she called for a curriculum focused on developing students' listening, speaking, reading, and writing skills, and their ability to use Spanish effectively for vocational and practical purposes in their everyday lives. Building on this proposal, Potowski and Carreira (2004) recently recommended bringing the SHL curriculum into closer alignment with the language arts standards of the National Council of Teachers of English. They argued that this approach not only leads to a better SHL curriculum, but also helps address general academic deficiencies in Latino students which inhibit their development in Spanish and prevent them from advancing in school.

Valdés (1981) also advocated that SHL teaching should have as one of its goals to contribute to the maintenance of Spanish in the United States. Research indicates that the Spanish of US Latinos declines with each generation in the United States, creating a bilingual continuum (Silva-Corvalán 1994). Typically, the foreign born retain strong skills in Spanish, while second- and especially third-generation speakers show evidence of incomplete acquisition and loss of linguistic structures. Beyond the third generation, few Latinos retain a functional command of Spanish (Fishman 1991; Veltman 2000; Silva-Corvalán 2003). Notwithstanding this reality and Valdes' advocacy, the question of how to address language maintenance through SHL instruction remains largely unexplored by researchers and practitioners (Valdés 1995, 2006). Furthermore, according to a recent survey, most SHL programs do not see their core mission as involving the maintenance of US Spanish. Rather, they see it as involving the teaching of grammar and literature (Valdés et al. 2006).

3 The 1990s and early 2000s

Between 1990 and 2000, the US Latino population increased by 62%, from 22.4 million to 35.3 million (by comparison, the total US population grew by 13.2% during this time) (United States Census 2000). With more Latinos attending school and studying Spanish, SHL specialists revisited longstanding questions and practices, identified new ones, and advanced the production of a new generation of pedagogical materials. Four edited collections were particularly influential in this regard: Merino et al. (1993), Colombi and Alarcón (1997), AATSP (2000), and Roca and Colombi (2003).

Contributors to Merino et al. (1993) proposed that SHL pedagogies and curricula should be inclusive of students' home language and culture(s) and offered activities and curricula consistent with this perspective. They also advocate using Vygotskian principles to develop learning contexts in which students are active participants in the learning process and in which learning emerges from the interaction of learners, educators, and community members (see also Faltis 1990; Rodríguez-Pino 1994).

Valdés (1995, 1997) set forth four goals for SHL instruction: (1) Spanish-language maintenance; (2) the acquisition of a prestige variety of Spanish; (3) the transfer of literacy skills from English to Spanish; and (4) the expansion of the bilingual range. At the same time, she laid out different priorities for different student populations. For example, for students who are well schooled in English, she proposed that instruction should aim to transfer literacy skills from English to Spanish. On the other hand, for the newly arrived who attended school in a Spanish-speaking country, the emphasis should be on the continued development of age-appropriate language competencies. With learner diversity emerging as a particular challenge of SHL teaching, much effort went into cataloguing student characteristics that were believed to be pedagogically significant, such as generational status in the United States, amount and type of exposure to Spanish, and literacy skills in Spanish and English (Valdés 1995, 1997, 2001; Walqui 1997; Carreira 2003). Also, assessment and placement emerged as critical to managing learner diversity by facilitating the formation of maximally homogeneous classes (Otheguy and Toro 2000). Recently, however, Carreira (2007) has argued that placement alone cannot solve the problem of learner diversity. To fully address this problem, there must be a paradigmatic shift in language teaching from a one-size-fits-all approach to a learner-centered, differentiated approach.

Exploring the issue of which variant(s) to teach, Torreblanca (1997) prioritized maximizing communication and singled out Mexican Spanish as being the most consistent with this goal. Porras (1997) endorsed a bidialectal approach whereby students were freely allowed to use their local variants in the classroom while also being taught an educated, standard variant of Spanish. Villa (1996) noted that many jobs in the American service industry require a mastery of local variants of Spanish, and that such variants are also the most appropriate for students whose goal in studying Spanish is to communicate with family members and friends. In addition, Villa (2002) argued that classroom practices that single out US Spanish variants as inferior or less adequate than monolingual varieties are pedagogically counterproductive because they foment insecurities among learners. He observed that US Latinos face a "double jeopardy" for their use of Spanish, criticized, on the one hand, by those who would eradicate Spanish from the United States, and, on the other hand, by some Spanish speakers, who deem US Latinos to be imperfect speakers of this language.

This line of research brought attention to the importance of attending to students' affective needs. Elucidating how the rebuke of those who want to eradicate Spanish erodes the linguistic self-esteem of US Latinos and contributes to their abandonment of Spanish, MacGregor-Mendoza (2000) documented the historical use of punitive practices in the Southwest against students who used Spanish in school. She found that many students who were subjected to these practices developed language shyness and became reluctant to speak Spanish. Against this backdrop, she argued that the passage of California's proposition 227 and Arizona's Proposition 203, which eliminated bilingual education in these states, could have detrimental effects on the psychological and academic development of Latino children because it represented a societal rejection of the Spanish language and Latino cultures.

The critical stance of some Spanish speakers has also been argued to have an adverse impact on Latino students. For example, a study of a university Spanish department conducted by Valdés et al. (2003) revealed that faculty members held highly negative views of US Spanish and that "US Latinos occupied the lowest levels of the power structure, and, through numerous interactions with faculty and fellow students, were encouraged to accept the dominant definitions of appropriate and correct language" (2003: 23). Similarly, Riegelhaupt's and Carrasco's (2000) study of a bilingual Chicana student-teacher in Mexico documented how her host family's negative attitudes about US Spanish and Latinos undermined her linguistic confidence and compromised her development in Spanish.

In light of their findings, Riegelhaupt and Carrasco advocated raising SHL learners' awareness of "the limitations of their home dialects when in more formal and/or new contexts … (and allowing) them the opportunity to make their own decisions as to the necessity or desirability of acquiring new dialects and registers" (2000: 407). However, Carreira (2000) cautioned that raising awareness of the value of the standard language without teaching students about the linguistic validity of their home dialect could exacerbate their linguistic insecurities and increase their ambivalence towards Spanish. To validate home dialects and inoculate students against harsh criticism, she proposed incorporating basic principles of linguistic science, dialectology, and historical linguistics in the SHL curriculum. In the same vein, Martínez (2003) proposed exploring the functions, distribution, and evaluation of dialects, and raising student awareness of language, power, and social inclusion (see also Fairclough 2005).

As these discussions unfolded, resources for SHL teaching multiplied. Several organizations played a leading role in this regard. In 1994, the American Council on the Teaching of Foreign Languages (ACTFL) established a special interest group on Spanish for Native Speakers to facilitate the sharing of information among members. ACTFL also collaborated with Hunter College High School and Hunter College on what is arguably the most important professional development project in the field of heritage language teaching. Funded by a grant from Funds for the Improvement of Post Secondary Education (FIPSE) and led by John Webb and Jamie Draper, this three-year project brought together public school teachers in SHL and Haitian creole to work on teacher preparation and curriculum development. An edited book (Webb and Miller 2000) emerged from this project, which was distributed to all registered participants in the 2000 ACTFL conference. More recently, Kim Potowski's *Fundamentos de la enseñanza del español a hispanohablantes en los EE.UU.* (2005) is the first and only book written expressly for teachers of SHL.

In 1999, the National Foreign Language Center (NFLC) and the AATSP launched Project REACH (Recursos para el aprendizaje y la enseñanza del español (http://www.nflc.org/reach, site discontinued)), offering teaching modules, an annotated bibliography of SHL textbooks for K-16, language exercises for students, and links to reference materials and sites of interest to teachers, students, and community members. In 2000, the AATSP published *Spanish for native speakers*, a guide to establishing and developing SHL programs at the secondary and post-secondary levels. Also this year, the Center for Applied Linguistics (CAL) and the University of California, Los Angeles (UCLA) conducted the third NEH Summer Institute for Teachers of Spanish for Native speakers (see Kreeft Pyeton 2008 for a summary of the work accomplished). Over the years, CAL has also produced a wide number of ERIC Digests and MiniBibs in the area of SHL teaching (see SNS CAL Guide 2010).

An unprecedented number of SHL textbooks arrived on the market in the 1990s and early 2000s. In contrast to earlier books, which focused on connecting students to local communities of Spanish speakers, this new generation of books tended to construct Spanish as a commodity for economic competitiveness in the global market. Leeman and Martínez (2007) attribute this change in focus to the rise of globalization and the expansion of the US Latino market, which raised awareness of the economic potential of US Latinos among educators, the US government, and the private sector. Responding to this reality, this new generation of SHL textbooks focused on teaching standard Spanish so as to prepare students to compete as professionals in the global market. While Leeman and Martínez (2007) and Train (2003) saw this approach as privileging foreign, monolingual variants of Spanish and devaluing bilingual variants spoken by Latinos in the United States, other researchers maintained that it could help reverse language shift by raising awareness of the economic value of US Spanish (see Roca and Lynch 2000; Villa 2000; Carreira 2002).

Notwithstanding the growth of the US Latino population and market, the wide availability of pedagogical materials, and significant advancements in the field of SHL teaching, access to SHL courses remained surprisingly limited during the 1990s. According to a 2001 national survey by the NFLC and the AATSP, most bilingual Latinos studying Spanish at the college level did so in L2 classes. Of the 146 institutions that participated in this survey, only 26 (17.8%) reported having an SHL program. The study concluded: "absolute numbers of Spanish speakers in many institutions are too low to make a separate series of courses economically viable" (Ingold et al. 2002: 328). Currently, CAL and the National Heritage Language Resource Center at UCLA are collaborating on a database of K-16 heritage language programs in the US. In the meantime, the question of how to teach SHL learners in L2 classes remains largely unexplored.

4 A place within the field of heritage languages

The new millennium ushered in important advancements in the teaching of other heritage languages. As was the case for Spanish, interest in this area correlated with the growing presence of HL learners in language classrooms and the increased awareness on the part of educators, the US government, and the private sector of the economic potential of HL speakers in a globalized market (Bretch and Ingold 2002). The NFLC and CAL launched the Heritage Language Initiative to build "an educational system that is responsive to heritage communities and national language needs and capable of producing a broad cadre of citizens able to function professionally in both English and another language" (Bretch and Ingold 2002: 23). An important piece of this initiative was the First National Conference on Heritage Languages (Long Beach, CA, 1999), which produced a pioneering volume edited by Joy Peyton, Donald Ranard, and Scott McGinnis, *Heritage languages in America: preserving a national resource* (2001).

The *Heritage Language Journal*, an online journal dedicated to heritage language learning and teaching, was established in 2002 by the Center for World Languages at UCLA and the UC Consortium for Language Learning and Teaching. These two organizations also established the first Foreign Language Center dedicated to heritage languages. Housed at UCLA, the National Heritage Language Resource Center (NHLRC) aims to develop "effective pedagogical approaches to teaching heritage language learners, first by creating a research base and then by pursuing curriculum design, materials development, and teacher education" (NHLRC 2010). In keeping with these goals, the NHLRC conducted a national survey of 1700 HL learners from twenty-two languages, including Spanish. Some of the findings for Spanish are discussed below in (a)–(d) (a complete report can be found at NHLRC 2010; see also Carreira and Kagan, 2011); in the next section, we delve deeper into the significance of the first three sets of findings in the context of discussing background factors that correlate with grammatical competency:

(a) *Age of onset of bilingualism*. A large majority of SHL learners (78.4%) spoke only Spanish before the age of 5. Surprisingly, this is true although the overwhelming majority are US-born (75.3%) or arrived in this country before the age of 5 (13.8%). Being exposed to English during early childhood correlates with having grammatical gaps in the HL. Conversely, late onset of bilingualism correlates with having a fully formed and stable grammatical system (Montrul 2008). The survey's findings, therefore, bode well for SHL learners' grammatical system.

(b) *Input*. SHL learners receive a lot of exposure to Spanish. The overwhelming majority speaks Spanish at home: 46.3% speak only Spanish and 47.6% speak both English and Spanish. Nearly all watch TV in Spanish (95%), listen to Spanish-language music (93.7%), and listen to radio in Spanish (87.3%) on a regular basis. Also, a sizable majority (62%) visits their country of origin every year or visited this country three to five times during their childhood.

These high levels of exposure may explain SHL learners' high proficiency ratings. Most rate their aural skills in the range of advanced to native-like (99.2% for listening and 66% for speaking). Conversely, very few rate their aural skills in the low range (0.5% for listening and 4.6% for speaking).

(c) *Literacy and schooling*. SHL learners are relatively unschooled in Spanish: 45% of respondents did not study Spanish in elementary school, either in the US or abroad. Despite this, they give themselves high literacy ratings: 95% rate their reading skills in the intermediate to native-like range and 86.5% rate their writing skills in this range. This is likely due to a combination of two factors: first, that Spanish, being highly phonetic, is relatively easy to read and write; and second, that literacy is taught in the home environment, as evidenced by the fact that Spanish speakers are more likely than other survey respondents to have learned to read in their HL before English (50% vs. 36.8%).

(d) *Motivation*. A significant majority of learners (71.1%) cite career goals as their main reason for studying Spanish. Other top reasons are to communicate better with family and friends in the United States and to learn about their cultural and linguistic roots. These findings relate back to an earlier discussion about instructional priorities in SHL, namely, whether SHL teaching should focus on connecting students to their community of residence through their home language or whether it should promote a standard variant with professional currency in the global economy. These results indicate that attention should be played to both goals.

The near absence of low proficiency Spanish-speakers from the survey suggests they do not enroll in SHL classes and raises the question of what happens to these students in Spanish departments. As Wiley (2001) explains, the answer to this question is closely tied to the meaning that practitioners attach to the term "HL learner":

> The labels and definitions that we apply to heritage language learners are important, because they help to shape the status of the learners and the languages they are learning. Deciding on what types of learners should be included under the heritage language label raises a number of issues related to identity and inclusion and exclusion … Some learners, with a desire to establish a connection with a past language, might not be speakers of that language yet. (2001: 35)

While in some languages HL learners might not speak their language, in Spanish they do, to varying degrees. The degree of proficiency required of students that enroll in SHL classes is set fairly high, as evidenced by the findings of the NHLRC survey, by placement tests for SHL learners (see Otheguy and Toro 2000; Fairclough et al. forthcoming), and by SHL textbooks, which assume that students will have basic reading and writing skills. It is also evident in the treatment of two categories of students: (1) "transitional bilinguals" and (2) culturally Latino students who have little or no facility in Spanish.

Lipski (1993) uses the term "transitional bilingual" in reference to low-proficiency bilinguals who make errors that are more typical of L2 learners than of SHL learners. He argues that because SHL textbooks address the errors of SHL learners and are written for students with communicative facility in Spanish, they are not appropriate for transitional bilinguals. On the other hand, L2 books, which do target such mistakes, are not appropriate either, because they do not tap into transitional bilinguals' relatively well-developed receptive skills and their knowledge of the colloquial registers. Lipski concludes that with no materials or classes for them, transitional bilinguals' needs went unmet. To date, this situation remains largely unchanged. Beaudrie (2009) found that most universities and colleges with HL courses do not have courses for receptive bilinguals, and Reynolds et al. (2009) found that receptive bilinguals often are mistakenly placed in beginning or intermediate level L2 classes.

Carreira (2004) examines another underserved category of learners: those who have strong attachments to Latino culture and self-identify as Latinos, but who have no facility in Spanish. While placing these students in L2 courses may be appropriate from a linguistic standpoint, it is otherwise problematic because L2 classes are not set up to respond to the affective needs of these learners and build on their cultural knowledge. Furthermore, because L2 classes are attended primarily by Anglophone students, some Latino students may reject them out of the sense that being assigned to them constitutes a de facto rejection of their Latino identity by Spanish instructors and departments. To confront these problems, Carreira proposes that Spanish departments infuse their L2 courses with a "heritage focus," that is, that they adopt a cultural curriculum that focuses on the US Latino experience and taps into the cultural expertise of Latino students, regardless of whether or not they speak Spanish.

5 Recent developments in the field of SHL

This final section of the chapter discusses three important recent developments in the field of SHL: (1) research on HL grammatical systems; (2) research on the effects of instruction on HL learners; and (3) research on sociolinguistic factors that bear on Spanish language use and maintenance by US Latinos.

5.1 Research on HL learner's grammatical competence

For linguists, research on the grammatical competence of HL learners is fundamental to understanding the structure of bilingual grammars and the factors governing the acquisition and attrition of minority languages. For practitioners, it is on the critical path to developing a theory of heritage language teaching that will undergird the design of materials, methodologies, and placement tools. In keeping with the thrust of this article, the present discussion will focus on the implications of this research for SHL teaching. For a discussion of its theoretical implications, see Chapters 36 and 38, below. As previously discussed, a long-standing practice of SHL teaching has been to categorize learners by background factors. Recent research singles out four factors as being particularly important in this regard: (1) the age at which HL learners acquire English; (2) the order in which they acquired Spanish and English; (3) the language(s) they speak at home; and (4) the amount of schooling and other input received in the HL.

Age has long been understood to be a pivotal factor in L2 acquisition. In rough terms, there is a negative correlation between age of acquisition and proficiency in the L2: the older the age of the learner, the lower the level of proficiency (Montrul 2008). Age also plays a critical role in HL acquisition. The earlier a child comes in contact with the dominant language (in this case, English) and starts to use this language more than the HL, the more compromised his knowledge of the HL will be (Montrul 2008). It is believed that this happens because contact with the dominant language diminishes access to input in the HL at a time when the HL's grammatical system is being formed. This time is known as the "critical period" for language acquisition. If input is diminished before the closing of this period (somewhere between the ages of 8 and 10), the child's HL grammar will not develop to a mature state and it will be susceptible to attrition (Montrul 2008). On the other hand, if a child receives sufficient input during this critical period, his grammatical system will reach full maturity and be protected from attrition. Illustrating this, Montrul (2002) found that Spanish-speaking children who learned English before the age of 7 evidenced signs of incomplete acquisition and attrition of tense/aspect distinction in Spanish. On the other hand, those who learned English between the ages of 8 and 12 performed largely like monolingual speakers of Spanish with respect to this distinction.

The order or sequence of acquisition of a bilingual's two languages, that is, whether they are learned simultaneously or sequentially, is also grammatically consequential. Research indicates that simultaneous bilinguals, that is, individuals who are exposed to two languages from birth or before the age of 3, have more problems with core elements of the L1 grammar than sequential bilingual children, that is, individuals who learn their second language during childhood after acquiring the structural foundations of their first language some time after the age of 4 or 5 (Silva-Corvalán 2003; Montrul 2008). It is believed that the grammatical superiority of sequential bilinguals may stem from the fact that they receive more input to the L1 in the home environment than simultaneous bilinguals. This allows them to form more complete grammatical systems.

Demonstrating the importance of input in the home environment, Muller Gathercole (2002) and Silva-Corvalán (2003) found that children who spoke only Spanish at home had a better command of particular aspects of the grammar (e.g., gender, aspect, and mood) than those who spoke both Spanish and English. The latter, in turn, had better command of these phenomena than those who spoke only English at home. Muller and Gathercole (2002) also found that children in two-way bilingual programs had a better command of gender than those in English-immersion programs, which indicates that bilingual programs are also providers of critical input for grammatical development (see Montrul and Potowski (2007) for similar findings). Other input-related factors that have been shown to correlate with a better control of grammatical phenomena include having two or more years of schooling abroad, speaking the L1 outside the home, and traveling abroad (Fairclough 2005; Montrul 2008).

Much like background factors, lexical knowledge has been shown to correlate with grammatical competency. For example, Polinsky (2006) found a strong correlation between speakers' control of particular grammatical phenomena and their knowledge of a basic word list of about two hundred items. Emerging research on Spanish indicates that a much higher number of words may be needed to assess grammatical competency with SHL learners. Fairclough et al. (forthcoming) discuss the use of lexical proficiency scores in the context of presenting a placement test for SHL learners that employs a variety of measures. Their lexical recognition task, which used the 5,000 most frequently used words in Spanish, proved useful for separating receptive bilinguals from other bilinguals, but it was not sufficient to distinguish between more advanced learners. In all, this research validates the practice of using background factors as indicators of proficiency and underscores the significance of four factors in particular. Because these factors correlate with competency in specific areas of the grammar (notably, gender, tense/aspect, and mood), they can inform placement decisions, syllabus design, and curriculum development.

5.2 *Research on the effects of instruction*

As our discussion of the 2001 survey by the NFLC and AATSP indicated, for departments that have relatively few SHL learners, the question of whether a separate SHL track is justified is an important one. Recent research indicates that it is not only justified, but that without it, SHL learners may be disadvantaged relative to L2 learners. Montrul et al. (2008) found that L2 and SHL both have gaps in their knowledge of gender and make similar types of mistakes. However, they differ with regard to where they make their mistakes: for L2 learners, it is in the oral domain; for HL learners, it is in the written domain. The authors conclude that form-focused instruction is warranted for both types of learners, but for L2 learners the focus should be on increasing accuracy in the oral domain, while for HL learners it should be on the written domain. Bowles (2011) found that task-based paired interactions between L2 and HL learners work best when they address the particular needs of each type of learner and tap into their complimentary strengths. Accordingly, a task with both a written and oral component will benefit HL learners by developing their literacy skills and it will benefit L2 learners by developing their oral fluency.

Looking at the effects of instruction on the development of the past subjunctive, Potowski et al. (2009) found that both types of learners benefit from form-focused instruction, as evidenced by improvements in interpretation and production tasks. However, L2 learners benefited more in general and only they showed improvements in grammaticality judgments. The authors suggest that a possible explanation for this may be that "L2 instructional methods, although potentially beneficial for linguistic development, may not be the most effective type of instruction for heritage learners" (2009: 564).

Fairclough (2005) also found that L2 learners benefit more from form-focused instruction than HL learners. Even more central to the question of whether it is necessary to separate SHL and L2s, this study calls attention to the fact that SHL learning is not just about filling grammatical gaps, it is also about mastering a second dialect (i.e., the standard dialect). Fairclough shows that learning a second dialect is a lengthy and complex process. To explain its inherent difficulty, she invokes the concept of "language distance," which states that when two systems are very similar, it is more difficult to keep them apart and the tendency is to merge them. While form-focused instruction appears to be an effective approach for addressing grammatical gaps, Fairclough finds that it is not as effective at teaching a second dialect. For that purpose, she recommends an approach that involves validating students' dialect first and then identifying relevant differences between the learner's dialect and the target dialect.

Overall, research on the effects of instructions is supportive of the idea that L2s and SHL learners benefit from different types of instruction. This is true, even when the two populations make the same types of mistakes and are matched for proficiency. Furthermore, this research raises a red flag that placing SHL learners in L2 classes disadvantages them because the instructional methods and approaches of such classes are better suited to L2 learners.

5.3 Sociolinguistic research on language use and maintenance by US Latinos

Recent sociolinguistic research yields important insights on longstanding issues regarding the focus of SHL instruction and curricula. Ana Celia Zentella's (2005) edited book, *Building on strength: language and literacy in Latino families and communities*, examines how Latino children are socialized to speak, read, and write Spanish by their caregivers. The book shows that for US Latinos, the Spanish language is inextricably connected to familism, *cariño* 'affection,' mores, and *respeto* 'respect.' Specifically, Latino caregivers use Spanish to maintain a moral order in the family, instill cultural and religious values in children, and demand respect outside the home. Given the central importance of these issues for Latinos who speak Spanish, they should figure prominently in SHL teaching and curricula. Another key finding of the book is that Latinos' language socialization practices vary significantly depending on their personal and group migration history, their connection to transnational networks, local attitudes toward Spanish, etc. The practical significance of this is that SHL teaching and curricula must be anchored in the particular linguistic experiences of students in any given class.

Focusing on the experiences of Mexican Americans, Glenn Martínez's (2006) *Mexican Americans and language: del dicho al hecho*, examines the complex interplay of factors underlying language maintenance and shift, notably, dominant language ideologies, government policies, the socioeconomic status and ethnic vitality of the immigrant group, and institutional support for the immigrant groups' language. Though Spanish language maintenance has been a concern of SHL research for a long time, it has not played a role in instruction. As previously noted, this is due in part to the lack of clarity about how to address language maintenance through instruction. Martínez's findings suggest that a critical step in this direction is to raise student awareness of the variety of factors involved in language maintenance and shift. One way to do this is through projects that compare and contrast the experiences of different communities of US Latinos, in present as well as past times.

Kim Potowski's (2007) *Language and identity in a dual immersion school* sheds light on another longstanding concern of SHL teaching: addressing students' affective needs vis-à-vis Spanish. This study found that children were more likely to use Spanish when it represented a positive identity investment, that is, when it furthered an image of themselves that they liked, such as being a good student, being knowledgeable, etc. Potowski also found that children frequently switched to English to connect with their peers, express feelings, and to facilitate communication. Potowski concludes that heritage speaker teaching be inclusive in their conceptualizations of the identities that students seek to portray and enact through their expertise, affiliations, and inheritance of Spanish.

6 Closing remarks

In her 1997 overview of the field of SHL, Ana Roca asked:

> (1) What sort of progress have we made? For example, how far have we come as a profession in shaping coherent theoretical paradigms on which to base the special teaching approaches that we have continuously called for and occasionally demanded? (2) In the last twenty-some years, how much have we influenced professional organizations such as the MLA, the AATSP, and ACTFL on Spanish for native speakers instruction and other heritage language instruction? … We lack basic information and surveys. We need to ask how much progress we have really achieved or not achieved in assessing the bilingual student's language proficiency for placement purposes … What's the point of placement testing if there are no courses developed to place students in? That is, if there is no native-speaker curriculum to speak of, where do we place the Hispanic bilingual students of such wide-ranging linguistic levels? How do we work with classes containing students of different levels in colleges that cannot afford a separate track? (Roca 1997: 39–40)

Judging by these criteria, it evident that much progress has been made in the recent past. Research on HL grammatical systems is bringing us closer to the development of placement tools, theories, and teaching paradigms that are responsive to the needs of HL learners. Not only are many professional organizations engaged in advancing SHL teaching, but there is now a National Foreign Language Center, the NHLRC, dedicated to this purpose which has carried out a national survey of HL learners. Sociolinguistic research is also yielding valuable insights into longstanding concerns and identifying new areas of concern.

Still, these important advancements remain to be explicitly connected to what happens in the classroom. That is, they need to inform the design of methodologies, curricula, materials, and teacher preparation programs. Much work also remains to be done concerning what to do when separate tracks are not available, as well as concerning the treatment of transitional bilinguals and students who have a cultural connection to Spanish but who do not speak it.

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37 Acquisition of Spanish in Bilingual Contexts

CARMEN SILVA-CORVALÁN

1 What is bilingual first language acquisition?

Two major patterns in bilingual language acquisition have been identified in studies of bilingualism: simultaneous bilingualism and sequential bilingualism, but no agreement exists with respect to the age at which bilingual development would be considered to be sequential. In simultaneous bilingualism, the child acquires two languages at the same time from birth, or, as some researchers propose, before 3 years of age. Here, we use the term Bilingual First Language Acquisition (BFLA, or "2L1") to refer to situations where the child's exposure to two languages begins before 6 months of age, that is, before the child produces the first identifiable word. This means that the question of the time of first exposure to the languages is eliminated in BFLA but not in sequential bilingualism, which may be further differentiated, depending on when acquisition of a second language begins, into: (a) successive bilingualism, when the child's exposure to a second language starts after the sixth month of age; and (b) bilingual second language acquisition (BSLA), a form of early bilingualism that happens when a child has one established language before learning a second language, whether in preschool or later (the age of three usually separates successive bilingualism and BSLA). This chapter focuses on the grammatical aspects of BFLA, that is, on the acquisition of two languages from birth; no detailed attention is given to social and affective aspects of language development.

The term bilingualism refers to the use of two languages in everyday life. It has been estimated that half of the world's population, if not more, is bilingual or multilingual (i.e., they use three or more languages) (Tucker 1998). Bilingualism is indeed present in all social classes and in all age groups, in every country where Spanish is also the official or co-official language. A few examples of languages that share their communicative space with Spanish should suffice: Mayan languages in Guatemala and parts of Mexico, Guarani in Paraguay, Quechua (mainly) in Peru,

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Bolivia, and Ecuador, Welsh in Patagonia (Argentina), Mapuche in parts of Chile and Argentina, Galician in Spain. Yet despite this situation of widespread bilingualism, which would appear to constitute a typical BFLA learning context, practically no research has been done to document this process outside Spain and the United States. Nor do we know what the proportion of BFLA children is in these bi- and multilingual societies.

The first in-depth longitudinal study of a bilingual child whose languages were Spanish and English was reported by Fantini (1974, 1985). This linguist recorded the Spanish language development of his son, from birth to age ten. Before him, two well-known studies had been published, one by Ronjat (1913) of his son's development in French and German to the age of 4;10 (the conventional notation to indicate age is years;months.days), and the other by Leopold (1939–1949), who recorded the bilingual development of his two daughters in German and English. But until the 1980s, research on child bilingualism was scant, to a large extent limited, in the United States, to psychometric studies (i.e., measuring intelligence, aptitude, or memory) based on school tests obtained from bilingual children. Since then, there has been an enormous growth of interest in the process of BFLA, perhaps made possible by advances in theoretical models of language and language acquisition (Hamers and Blanc 1983), coupled with the realization that child bilingualism is indeed prevalent throughout the world.

Indeed, a crucial motivation for the study of BFLA is the fact that the simultaneous acquisition of two languages may allow us to see more clearly what principles are at work in the process of language acquisition. The bilingual children to be studied are in fact the ideal subjects for cross-linguistic research because such factors as personality, cognitive development and social environment are under control and not confounded as they might be in studies of monolinguals. Thus, scholars have been attempting to answer many important questions, some specific to BFLA, but most relevant as well to general theories of acquisition. As Genesee (2000: 167) states, "The capacity of the human mind to represent two or more linguistic systems, sometimes with radically different structural properties … [e.g. Spanish and Quechua, Spanish and Basque, *this author's examples*], in functionally compatible ways has implications for our conceptualization of the neurocognitive architecture that makes this possible."

2 Research questions in BFLA

Studies of BFLA have been done within the framework of the various theories proposed for monolingual acquisition, in particular, *nativist* and *constructivist* approaches. Briefly, nativist theories affirm that children are genetically endowed with a Universal Grammar, a set of linguistic principles common to all languages, which, together with language specific parameters and sufficient input, guide acquisition. By contrast, constructivist theories state that what is innate are cognitive abilities and general learning mechanisms that make possible the learning of language from the language input which the child receives in situated instances of communication. Some of the principles defended by constructivists are that learning is gradual, contextualized and *piecemeal*, that abstract structures emerge from specific constructions once a 'critical mass' of these constructions is achieved, and that the sequence of acquisition of a grammatical feature is determined by its complexity and frequency in the target language.

Studies of bilingual acquisition are relatively recent, despite their undeniable theoretical and practical value. From a practical perspective, it is necessary to put an end to some myths about bilingual acquisition, such as that it fosters language confusion, or that it causes cognitive and language problems. Quite to the contrary, numerous studies have demonstrated that child bilingualism offers cognitive, linguistic, and obvious social advantages (see Bialystok 2001; De Houwer 2009).

From a theoretical perspective, the high frequency of child bilingualism requires that a theory of language acquisition seriously consider the acquisition of two languages from birth or from the first three years of life. Thus, the fundamental goal (though long term) that research on 2L1 has set up is the development of models that may account for how bilinguals acquire two linguistic systems at a time, and how these two systems are represented in the mind of the bilingual. Some researchers have observed that the principles put forward by constructivists, for instance, are valid as well to account for the acquisition of 2L1 (e.g., Gathercole 2007). But in addition, 2L1 researchers have proposed models and questions specific to bilingual and multilingual acquisition, such as: *the differentiated development* model (DDM) vs. *the unitary language system* model (ULS), and the model of *autonomy* vs. the *interdependence* of the systems of the bilingual.

The DDM defends that from the earliest appearance of phonology, morphology and syntax, forms are used in a language-specific manner. When a child is regularly exposed to two languages from birth and is acquiring a fairly balanced level of proficiency in both, each language system develops in a self-sufficient and independent manner (e.g., Meisel 1989; Deuchar and Quay 2000). On the other hand, the ULS, defended by Volterra and Taeschner (1978), affirms that although children are exposed to distinct sets of linguistic input, they go through an initial stage in which their two or more languages are represented in a unitary or fused system until they reach a stage of differentiation (see also Murrell 1966; Schnitzer and Krasinsky 1996).

The model of *interdependence* assumes differentiated development but with some degree (to be empirically determined) of interaction between the two languages. If the language systems develop autonomously, the patterns of acquisition and linguistic representation would match those of monolinguals. If there is interdependence of systems, one of the languages may guide development and may have some influence on the development of the other language by delaying or accelerating acquisition, or by causing the production of nontarget constructions not attested in monolingual development. Interdependence would be reflected in delayed acquisition of a grammatical phenomenon present in one of the languages (call it "A") but not in the other ("B") if, compared with monolingual acquisition, the language feature is acquired at a later age in language "B" (e.g., delay in the acquisition of obligatory subjects in English by Spanish–English bilinguals).

The opposite would be acceleration of acquisition when a grammatical feature is marked similarly in both languages (e.g., plural marking in Spanish and English). Interdependence may also be reflected in the transfer of a linguistic feature from one to the other language (e.g., the use of stranded prepositions in the Spanish of Spanish–English bilinguals (*¿Qué abres la lata* <u>*con*</u>*?* 'What do you open the can <u>with</u>?').

If both languages are activated in bilingual contexts, transfer phenomena are to be expected. Müller and Hulk (2001), for instance, have shown that cross-linguistic influence is manifested when the structures in question involve the interface between syntax and pragmatics. They study the omission of object pronouns by bilinguals who speak a Germanic and a Romance language, and argue that the bilinguals are able to separate the two languages from early on, but that the languages have an influence on each other. The effect of the influence, not dependent on language dominance, is that object omission persists in the Romance languages of the bilinguals at a higher rate and longer than in the speech of monolinguals. This difference would be due to delay in the acquisition of the discourse constraints on object drop in the Romance languages in contact with a less restrictive object-drop Germanic language. Silva-Corvalán's (forthcoming) study of subject expression and placement in the speech of Spanish–English bilinguals supports Müller's and Hulk's findings about differentiated development, but she goes further in proposing that language dominance and proficiency play a central role in determining the direction and extent of the influence across the languages of a bilingual.

In light of these theoretical considerations, studies of BFLA have sought answers to such questions as:

a. How are the two systems organized at various stages in the attainment of bilingualism? Are the systems differentiated from birth or is there a stage of undifferentiated representation in the early stages of language development? If the systems are interdependent, how is the influence manifested within and across language subcomponents?
b. To what extent are the mechanisms of monolingual and bilingual acquisition similar or different?
c. What is the effect of the nature of the linguistic input (i.e., linguistic feature frequency and complexity) on the development of each of the bilingual's language systems?
d. What is the effect of the amount of exposure to and use of the languages on the development of each of the bilingual's language systems?
e. What is the minimum amount of exposure required for the successful acquisition of two languages, and for the maintenance of the two languages beyond early childhood?
f. What is the effect of the social factors that may underlie different forms of bilingual behavior?
g. What is the effect of bilingualism on mental processes of language recognition and production?

3 Contextual factors in the development of child bilingualism

Bilingual infants and children are ideal subjects for language acquisition research, but not everything about them is perfect. Indeed, research has found much interindividual variation, comparable to that exhibited by monolingual children. This is especially problematic in cases where only one language is the majority language or enjoys the status of being official. The outcomes of case studies are not easily generalizable in these situations because the child's acquisition of each language depends upon his sociolinguistic history in each language, and these histories can be quite different. Some children grow up speaking two languages fluently, while others attain a reduced form of one of the languages, or do not acquire productive proficiency at all.

A number of contextual factors play a crucial role in the development of bilingualism. Among them are: the age at which the child is exposed to the two languages (not relevant in BFLA), the number of speakers of each language and their social status, the presence of other languages in the child's environment, the frequency with which the languages are spoken at home and in the community, family and community attitudes toward each of the languages and toward bilingualism. The language environment in which acquisition takes place to a large extent determines the input to the child and the outcome. BFLA children may grow up in a bilingual family environment located in a community that makes regular use of both, or only one, or neither language. The family may be one where (a) both parents are bilingual and address the child in both languages at home and in public, or one language at home and the other in public; or (b) only one parent is bilingual and the child receives input in two languages, one from each parent (the "one parent-one language" approach). There are in fact many different configurations of the bilingual environment which give rise to different types of bilingualism.

Depending on these ecological factors, individual speakers will attain different levels of proficiency in their two languages. The bilingual community, in fact, is characterized by a continuum of bilingualism which ranges from full proficiency in both languages (*balanced bilingualism*) to reduced proficiency in, normally, the minority language (*unbalanced bilingualism*). The continuum may be identified in practically every bilingual community involving Spanish and another language, for example, in Catalan and Spanish in the Balearic Islands, in Guarani and Spanish in Paraguay, in Otomi and Spanish in Mexico, in Basque and Spanish in the Basque Country. This unbalanced proficiency also characterizes Spanish–English bilinguals in the United States, even in heavily Hispanic communities (Zentella 1997). For example, while by age 5;6 monolingual Spanish children have acquired the Present Perfect and the Pluperfect Indicative and Subjunctive tenses, most bilingual children in the United States have not. In English, by contrast, the same children have acquired the Present Perfect and Pluperfect Indicative, much like monolingual English speakers (Silva-Corvalán 2003a).

Differences in contextual factors result in the possibility to develop higher or lower levels of bilingual proficiency along the continuum. This raises the question of how much exposure children need in order to gain a productive command of a construction. Some have argued that a *critical mass* of input data has to be accumulated for a child to generalize beyond stored instances; this has also been suggested for bilinguals (e.g., several contributions in Oller and Eilers 2002; Silva-Corvalán 2003a). But the question of what constitutes "a critical mass" in BFLA is virtually unexplored. Social factors such as the prestige of the languages and political attitudes, which determine government and educational policies, also have an important impact on the extension and degree of bilingual development at the individual and societal level. Political changes in Spain, for example, have resulted in an increased number of bilingual children who can maintain the minority language until adulthood (Siguán 2008).

4 Bilingual children's language development: from words to sentences to continuous discourse

BFLA children do not differ from monolingual children with respect to stages of language development: they move on from babbling, to single words, to word combinations, to sentences, and to fluent conversations. Babbling occurs during the first year of life and lays the foundation for the development of speech. Although there is interindividual variation, it is generally the case that by the end of the first year BFLA children will understand words and simple sentences in their two languages. It has been observed that bilingual children start producing language later than monolinguals, yet this is not necessarily the case; it depends in great measure on the amount of input the child receives from his caretakers. Nico, for example, a Spanish–English developing child, produced his first words toward the end of his first year of life: *papa* (for *papá* 'dad'), *mama* (for *mamá* 'mom'), *abú* (for *la luz* 'the light'), *aga* (for *agua* 'water') in Spanish; *ap*^{h} 'apple,' *hot* in English.

Babbling disappears as the children start to say more and more identifiable words in both languages. The question that arises is whether these words are represented as one or two independent systems. Deuchar and Quay's (2000) study of a Spanish–English bilingual (Manuela), and Montanari's (2010) study of a Spanish–Tagalog–English trilingual (Kathryn) have shown incontestably that BFLA children develop two (or three) separate lexicons. Their conclusion is based mainly on the existence of translation equivalents (including a few three equivalents in the case of the trilingual) identified from the very beginning of word production in the data from the children studied, for example, *bye–tatai*, *more–más*, reported by Deuchar and Quay (2000: 59) for Manuela; *papa–daddy* (Spanish/Tagalog–English), *susi–llave* (Tagalog–Spanish), reported by Montanari (2006) for Kathryn. One of the problems faced by these researchers is that many words could belong to either of the languages of the child. An utterance like [ten] for Spanish *tren* and English *train* when the child is not yet producing the subtle phonetic differences that differentiate a Spanish from an English /t/ could not be assigned to a language only on the basis of linguistic considerations. Words that mean and sound similarly across languages, usually cognates, are more common when the languages are closely related at the lexical level, for example, Spanish, Catalan, Galician, and to a lesser extent English. The issue of differentiation should be easier to examine with pairs of languages genetically unrelated, such as Spanish and Quechua, Nahuatl, or Basque.

Similarly to monolinguals, when bilingual children start to produce what adults interpret as words, they use these utterances to convey broad meanings, equivalent to one or more phrases or sentences in adult speech. For instance, Nico produces *on* (addressing a Spanish- or an English-speaking interlocutor) when he describes that a light is on, or wants a light to be turned on, when he wants to listen to music, and when he wants a faucet to be turned on. These single words with multiple meanings are called *hollophrases* and also characterize monolingual acquisition.

Around the age of 18 months, monolingual children may have about fifty words. These children obviously receive language input in just one language and produce words in only one language. BFLA children share their waking time between two (or more) languages, and, expectedly, also share their productive vocabulary between these languages. It has been observed that once a monolingual child reaches fifty words, there occurs a "vocabulary spurt" (O'Grady 2005: 7), and the child may since start adding up to ten words per day. Deuchar and Quay (2000: 56) report that Manuela had reached 330 words at age 1;9.30 (more words than the average monolingual at the same age); 172 of these are English words and 129 are Spanish words, 25 are ambiguous between English and Spanish, and 4 "are utterances with neither an English nor a Spanish adult source word" (2000: 56).

Around the second half of the second year, monolinguals start producing word combinations, and around the age of two or so "there is rapid growth in the ability to produce a wide variety of complex constructions" (O'Grady 2005: 80). The same applies to BFLA children, though just as in the case of monolinguals, production of words and the rate of growth of vocabulary development may vary greatly from child to child. Deuchar and Quay (2000: 69) record Manuela's first two-word combinations at age 1;6.25; Silva-Corvalán records the beginning of these combinations at 1;4.12 and 1;7.24 for the two bilinguals she studies; Montanari (2006: 81) records the first two-word combination for Kathryn at age 1;5.08. But this trilingual child produces most combinations after 1;9; the few verbs produced in Tagalog and English between this age and 2;1 lacked inflection, and only two verbs are recorded in Spanish, both in an inflected form (*tiene* '(it) has' and *prendo* 'I turn on').

BFLA children produce many mixed two-word utterances before age 2;0 (and later on many sentences that incorporate words from both languages, i.e., code-switching). Examples are '*más* paper' and 'more *galleta*' (Deuchar and Quay 2000: 74), 'hot *sol*' (Montanari 2006: 82), 'pushing *niño*', '*más* ice,' and 'close *puerta*' (from Spanish–English bilingual children). This type of mixing was considered at first to be evidence of undifferentiated syntactic development (Volterra and Taeschner 1978; Redlinger and Park 1980), but more recent studies have shown that mixing results from vocabulary gaps and the acceptance of mixing in the child environment rather than from a unified syntax (Lanza 1997; Deuchar and Quay 2000; Montanari 2009).

Indeed, a number of studies have shown that the surface orders of utterances produced by bilingual children with a Mean Length of Utterance (MLU) of 1.75 and above reflect the structural properties specific to each input language. For instance, Ezeizabarrena (1991, cited in Montanari 2006: 73) examines the acquisition of negative constructions in Basque–Spanish bilingual children and "finds that while the clausal negator is always and exclusively placed preverbally in Spanish, Basque combinations display both the NEG + (NON-) FINITE VERB and the NON-FINITE VERB + NEG order in accordance with the 'grammatical operations' (i.e. right-headedness) specific to Basque" (Montanari 2006: 74).

Much earlier, in their detailed description of the acquisition of interrogatives, adverbs, and adjectives by Spanish–English bilingual children, Padilla and Lindholm (1976: 141) had concluded that these children "learn each of their languages separately" and "that little if any interference was apparent" in the speech of the nineteen bilinguals they studied. The conclusion about the absence of interference needs to be evaluated in light of the authors' methodological constraint to include only children who heard and used approximately equal amounts of Spanish and English, that is, in a situation of balanced bilingualism that may not be very common.

Be that as it may, the trilingual child studied by Montanari (2009) gives evidence that, even when growing up with three languages, a child is able to differentiate the three codes and produce different word orders in her word combinations following the patterns specific to her three languages (Tagalog, Spanish, and English). Thus, Montanari (2009: 516) concludes that "the fact that trilinguals, and not only bilinguals, can combine the words of their languages differently from the start demonstrates that trilingual exposure does not slow down the process of differentiation and that learning three languages is not necessarily more 'costly' than learning two."

By age 2;0 BFLA children typically start producing complex constructions, making it possible to study a variety of syntactic structures and a wealth of code-switched utterances. Their development follows the same or a similar path as that of monolinguals, and also at the same rate, at least in the bilingual's dominant language (De Houwer 2005). Combinations of two or three clauses are normally produced in the fourth year of life or earlier: for example, *Cuando el papi frenó se me cayó la silla para el otro lado* 'When daddy stopped the car my seat fell to the other side.' (Nico, 2;10.26); *Si lo sueltas, no puede vivir* [a bird] 'If you let it go it won't live' (Bren, 3;4.14).

In their fifth or sixth year of life (i.e., after their fourth or fifth birthdays), BFLA children move on to re-telling stories that have been read to them, or to narrating anecdotes about themselves or others. How soon and how often they perform these communicative activities depends to a large extent on the quality and quantity of input they receive in each language. There seems to be a close correlation between amount of input and caretakers' attention to the child's bilingual learning experience and success in achieving proficiency in two languages (De Houwer 2009). By age 5;6, a child who has received about 30% input in Spanish and 70% in English is able to tell the story in a picture book in Spanish, with only some lexical items inserted in English to fill in vocabulary gaps (e.g., *Había un hoyo. Y dició. Y el niño dis, dijo, 'Sapo, Sapo, ¿estás aquí?' Pero no, solamente era un* groundhog. 'There was a hole. And (he) said. And the boy said, "Frog, frog, are you here?" But no, it was only a groundhog.' Likewise, in a study of a Spanish–English bilingual child whose bilingual development is encouraged by the parents, Alvarez (2003) shows no developmental lag in either of the two languages in data obtained from story telling from ages 6;11 to 10;11.

The prevalence of code-switching in the bilingual's language depends on both its occurrence in the adult input, and adult attitudes to it (Lanza 1997). When adults code-switch themselves or accept the child's language mixing by showing that mixed utterances are understood and that the use of another language is permissible, then children will feel freer to code-switch and thus develop a colloquial communicative style that incorporates switching. Otherwise, children may refrain from saying something or find roundabout ways of expression, as the child studied by Alvarez (2003: 234) does when he refers to a *colmena* 'beehive' as *el sitio donde viven las abejas* 'the place where the bees live.'

When a bilingual attains unbalanced levels of proficiency, the dominant language may show no negative effects, but reduced language resources in the language that has received less input and fewer possibilities of use by the child appear to have a negative effect on narration. Silva-Corvalán (2003b), for example, studies the narrative skills of a 5-year-old bilingual who has attained lower proficiency in Spanish than in English. This author shows that two narrative components that are considered to reflect the level of cognitive maturity: the structure and the evaluation of a story are well developed in the dominant language, but less developed in the weaker one. Obviously, this outcome has implications for education inasmuch as testing bilingual children in their weaker language in school may run the risk of viewing them as cognitively deficient, a risk that would be avoided if both languages were taken into account in any evaluation of cognitive and linguistic levels of development.

5 Research methods in BFLA

Research on bilingual acquisition is broad in scope and encompasses many aspects of language development, including preverbal speech perception (see Bosch and Sebastián-Gallés 2010), emerging phonology (see Montanari 2010), bilingual language disorders (see Goldstein 2004), lexical development, (morpho)syntactic development, and communicative aspects of bilingual development. These various facets of bilingual development have been studied using a diversity of methods for the collection of language and language environment data. These methods may be grouped into two categories: collection of spontaneous data and collection of experimental data. Obviously, preverbal speech perception may only be studied experimentally; all other areas of research have used the two methods, assigning more or less weight to one or the other.

Spontaneous data from bilinguals are collected in naturalistic environments where the children may be observed and audio- and/or video-recorded while engaging in naturally occurring interaction. But even these data are sometimes interspersed with passages in which children respond to questions or stimuli presented by the researcher, which could be considered experimentally obtained data. By contrast, the collection of strictly experimental data is done in an experimental setting, frequently the researcher's lab, following a predetermined procedure that will ensure that the children will produce the type of language data needed by the researcher. Performance in these highly-controlled elicited tasks differs from performance in spontaneous tasks (Restrepo and Gutiérrez-Clellen 2004). Furthermore, the design and application of experimental protocols for the study of bilingual development face considerable difficulties (Genesee 2006; Myers-Scotton 2006), and the outcomes are "lower in ecological validity than are naturalistic studies" (Fernald 2006). Regardless of these drawbacks, experimental methods need to continue to be designed and applied since they could teach us much about language production and comprehension.

In addition to spontaneous and experimental data collection, there is a third method that has been used in numerous studies of early lexical and grammatical development. The instrument used is the MacArthur–Bates Communicative Development Inventory (CDI 1992), which consists of a series of questionnaires used to ask parents to report on the words and combinations of words used by their child, on the use of communicative gestures, imitation of adult actions, etc. Also included is a form where parents are asked to provide nonlinguistic information (e.g., the number of hours a week each member of the family spends talking with the child and in what language, the parents' occupation, ethnic background).

The questionnaires were designed in the United States to assess a wide range of English language milestones that are appropriate for typically developing children under the age of 3 years. They have since been translated and adapted to other languages, including Spanish, Galician, and Basque. Barreña, Ezeizabarrena and García (2008), for instance, report on a study of children of 16 to 30 months of age based on data collected using the Basque version of the MacArthur–Bates CDI. This adapted questionnaire is written in Basque, so the authors use it only to obtain information about the children's use of this language. Pérez Pereira (2008) uses the Galician version of the CDI to evaluate Galician–Spanish bilingual children's communicative and linguistic development. The children were assessed only in Galician, however, and were compared with monolinguals. The results showed that the bilingual children were not below their monolingual counterparts in comprehension or production of language, and that they seemed to have more highly developed grammatical development than the monolingual children of a similar age (16–30 months). Similar results were obtained by Pearson, Fernández and Oller (1993) by applying the CDI to a small group of children in both English and Spanish. They found that the bilingual children did not show lower results than the monolingual ones when the total vocabulary produced in the two languages was considered.

The studies conducted with the CDI have included fairly large groups of children. Barreña, Ezeizabarrena and García (2008) and Pérez Pereira (2008) included close to 1,000 children each. But most studies of BFLA children's language production to date have been based on data from case studies. Indeed, another important methodological difference in bilingualism research concerns the number of children studied, which ranges from the case study of one child (Silva-Corvalán and Montanari 2008) or two children (Liceras et al. 2008), to studies of large groups of children.

Case studies of one or two children, usually done by a parent or grandparent, have also normally been based on dense longitudinal data collected over several months (e.g., in the early stages of development, Deuchar and Quay 2000; Krasinski 2005) or spanning several years (e.g., Fantini's 1974, 1985 pioneer descriptive study of his son; Silva-Corvalán's (forthcoming) study of her grandsons). This type of research is conducted within the backdrop of other case studies (which may or may not involve Spanish) and thus allow comparisons and generalizations. In-depth longitudinal case studies of bilingual acquisition are more likely to reveal the variable performance of individuals, as well as important unforeseen findings and new understanding of the processes involved in bilingual development, including how a particular learner's language behavior changes over time in similar situations in the *process* of acquisition. Finally, the language environment is crucial in determining different types of bilingualism and has been carefully considered and described in the research on bilingual language development (see Li and Moyer 2008 for further information on methods).

6 Morphosyntactic development: some case studies

One of the more studied grammatical aspects for the examination of hypotheses about the interaction of syntax and discourse in monolingual Spanish and in L1 and L2 acquisition has been the expression and placement in the sentence of grammatical subjects (Montrul 2004; Otheguy et al. 2007; Ortiz López to appear, among many others). By contrast, fewer studies have examined this aspect in BFLA, and these few have focused on Spanish–English or Catalan–English bilinguals (in comparison with monolinguals in each language), who are studied from the time of first verb productions to varying ages (Juan-Garau and Pérez-Vidal 2000; Paradis and Navarro 2003; Silva-Corvalán and Sánchez 2007; Liceras et al. 2008). English differs from Spanish and Catalan with respect to these grammatical features: (1) English is non-null-subject with Subject–Verb–Object (SVO) order; and (2) Spanish and Catalan are flexible pro-drop SVO languages; when expressed, subjects may be placed after the verb. The expression of subjects and the order of major arguments in Spanish and Catalan are mainly constrained by discourse-pragmatic factors; they are syntacticized in English. The linguistic features in (1) and (2), then, have provided a nice test case to examine the view that a language is vulnerable to influence from a contact language where these phenomena do not involve an interface, but not vice versa. Here, we focus on Spanish–English BFLA.

Scholars agree that BFLA children realize at a very early age that English requires overt subjects and Spanish does not, thus providing clear evidence that the model of the input languages guides the development of autonomous syntactic systems in BFLA. There is some disagreement, on the other hand, about the possibility of cross-linguistic influence and about the degree of the child's knowledge of the discourse-pragmatic rules for subject use in Spanish. Paradis and Navarro (2003) found that the frequency of subject expression in a Spanish–English bilingual (aged 1;9 to 2;6) was higher than in two Spanish monolinguals, but conclude that this increased frequency could be attributed to the input the child received from caretakers as much as to internal cross-linguistic influence in the child's language processing. They also add that by 2;6 the child's grasp of the discourse-pragmatic functions of subject realization in Spanish was lagging behind in comparison with monolinguals at the same age. In this regard, Silva-Corvalán and Sánchez (2007) reach a different conclusion since their study of a BFLA child to age 2;11 indicates that this child does not violate the discourse-pragmatic rules of Spanish: he expresses subjects, as monolinguals do, when they are focal (contrastive or new), and does not express them when they are coreferential (i.e., when the subject of a verb has the same referent as that of the preceding verb). Liceras et al.'s (2008) study of two Spanish–English twins (aged 2;04–4;11) does not examine the discourse properties of subjects, but agrees with earlier studies that bilinguals differentiate the two language systems from early on.

In her longitudinal study of two BFLA siblings (Nico and Bren to age 5;11), Silva–Corvalán (forthcoming) provides clear evidence that the child who has received a more reduced input in Spanish (Bren) fails to omit subject pronouns quite frequently in contexts that do not require them. Beyond age 2;11, both children steadily increase their production of overt subject pronouns in Spanish, reaching frequencies that are much higher than those of monolingual children and of the adults they interact with in Spanish. Bren maintains a much higher frequency of overt subjects than his brother, a result interpreted to indicate that language dominance plays a role in determining the degree of possible influence from a contact language. This interpretation is supported by the outcome of Liceras et al.'s (2008) study of the twins. These BFLA children, who are growing up in Spain and are most likely Spanish dominant, do not use a higher proportion of overt subjects than monolinguals, thus highlighting that cross-linguistic influence in this area of the grammar of Spanish is sensitive to dominance relations between languages.

The position of subject pronouns in Spanish is by far preverbal in all varieties of this language. Other subject types may occupy the pre- or postverbal position depending to a large extent on the information weight of the subject relative to the rest of the sentence constituents. The quantification of subject placement in written data gives the following results for percentage of preverbal subjects by type of verb: unaccusative, 66.2%; unergative, 87.8%; copulative, 84.6%; transitive, 88.5% (Mayoral Hernández 2008: 138). This is similar to what Silva-Corvalán (forthcoming) obtains in spoken adult data and in the data from the two BFLA siblings, which suggests that the placement of nominal subjects in pre- or postverbal position does not appear to be sensitive to dominance levels. For example, both children appropriately place the subject in postverbal position with verbs of the *gustar* 'like' type (*Me duele el pie* 'My foot hurts'), and when the subject conveys new information (*Me empujó Christian* 'Christian pushed me').

The morphological marking of subjects and other arguments has been examined in Basque–Spanish BFLA children. In Basque, a split ergative case-marking system indicates verb–argument relations, and in Spanish these relations conform to a nominative case system. A study of Basque–Spanish BFLA children aged 2;01 to 3;04 (Austin 2007) indicates that there is some cross-linguistic influence in argument marking, reflected in a higher percentage of omission of ergative case-marking in Basque compared to the rate of omission in monolingual Basque children's speech. Omission of the ergative is reinforced by the nominative pattern of Spanish, in which no morphological difference is made between transitive and intransitive subjects. The importance of language dominance is highlighted by a comparison of Austin's study with Almgren's and Barreña's (2001), whose study of a well-balanced Basque–Spanish bilingual shows no differences with a monolingual Basque with respect to the marking of ergativity. The same applies to a comparison with monolingual Spanish children: following the Spanish standard, the bilingual produces unmarked transitive and intransitive subject arguments.

The Basque–Spanish bilingual child also obeys the word order rules of each of his two languages, placing subjects in both pre- and postverbal position in Spanish, and only preverbally in Basque. The direct objects in Spanish are placed after the verb as expected; Almgren and Barreña (2001) find only one SOV sentence and one OV sentence which conform to Basque word order and are not correct in Spanish, but cite similar examples produced by Spanish monolinguals. It is clear, then, that a BFLA child in balanced contact with both languages develops the two grammar systems independently from the moment morphological and syntactic elements first appear and without any interference effects.

The possibility of cross-linguistic interaction affecting subordinate sentences has also been examined in Basque–Spanish bilingual data. Subordinate conjunctions are placed at the beginning of a subordinate sentence in Spanish, while in Basque they are placed to the right of the sentence, mainly suffixed to the verb. Causal conjunctions are an exception: they may be placed at the beginning of the sentence, as in Spanish. In this situation in which two alternative structures are allowed by the grammar of one of the languages, it is expected that the language with multiple options (Basque) will be affected by the language with only one (Spanish). This expectation is supported by Barreña's (2001) study, who reports that one of two bilinguals he examines, Mikel, produces at first (to age 2;9) only the subordinate structure in Basque that is parallel with the Spanish one. No errors characteristic only of the bilinguals' data are reported, so the only effect that Spanish appears to have for some bilinguals is that of delaying somewhat the acquisition of the suffixed subordinators.

The expression of possession in Galician and Spanish has lent itself to the examination of Slobin's (1973) hypothesis about the effect of form transparency in child language development. Spanish offers different forms to indicate possession depending on whether the possessive is an adjective (*tu libro* 'your book'), or a pronoun (*es tuyo* 'it's yours'). Galician has the same form functioning as an adjective or a pronoun, and this form is marked for gender, thus making it more complex and less transparent than the corresponding Spanish form (Pérez Pereira et al. 1996). The results of Pérez Pereira et al.'s study of Galician–Spanish bilinguals support the proposal that multifunctional forms (i.e., nontransparent) are learned later. The Galician–Spanish bilinguals, and even a Galician monolingual, learn possessives in Galician later than in Spanish.

BFLA children who receive unequal amounts of input in their languages evidence delay in the acquisition of verb morphology in the weaker language. Spanish–English bilinguals in the United States, dominant in English, have not acquired any subjunctive nor compound tenses in Spanish by age 5;6, while in English, their stronger language, all compound tenses have been acquired by this age. By contrast, monolingual Spanish learners have typically acquired compound and subjunctive forms by 5;6 (Silva-Corvalán 2003a).

The effect of bilingualism has also been examined in the process of acquisition of *ser* and *estar* 'to be' in Spanish and English. Krasinski (2005) and Silva-Corvalán and Montanari (2008) agree in their observation that the Spanish–English bilinguals they study to ages 3;0 and 4;0 (Zevio and Nico, respectively), exhibit early uses of the copulas that are uncharacteristic of monolingual children's forms. Compared to monolinguals, the bilinguals evidence a slightly longer period of use of what appears to be one "shared" copula form in both languages ([ə, əs, e, es]), that is, they pass through an early stage during which the Spanish and English copulas are not phonetically differentiated in every possible context. Otherwise, the trajectory followed by the bilinguals is not very different from that of monolinguals at similar stages in development (cf. Hernández Pina 1990; Sera 1992): both groups of children make errors of omission of the copula or of the wrong selection of copula with predicate adjectives.

Even if short, the period of delay the bilinguals undergo bears out the claim that bilingualism causes delay in acquisition. Yet some researchers have shown that the bilingual child learns some syntactic structures faster than a monolingual (subject realization, for instance). The outcome of Silva-Corvalán's and Montanari's (2008) study suggests that delay may affect those areas of the grammar at the interface of semantics and pragmatics when the languages do not share the same semantic and pragmatic constraints. Thus, different subcomponents of language are affected differently. In conclusion, we note that bilinguals' morphosyntactic development proceeds in the same manner as that of monolinguals for the dominant language, albeit with some slight delay in the weaker language of unbalanced bilinguals.

7 Further research

BFLA is indeed a barely explored territory, wide open to every possible area of research. Indeed, much research is needed that will focus on phonological and morphosyntactic development in a variety of language pairs (e.g., Spanish–Quechua, Spanish–Mayan languages, Spanish–Nahuatl) and in a variety of sociolinguistic situations. Longitudinal studies that go beyond age 3;0 could be carried out to test if the paucity of cross-linguistic effects identified in early acquisition might disappear or persist in balanced bilingualism. Likewise, as one of the languages might become stronger, it is necessary to examine if and in what ways uneven input might affect the weaker language. The quality and quantity of input required for the development of functionally adequate bilingualism remains as well to be ascertained. Furthermore, studies based on naturalistic data need to be complemented with experimental tasks that test children's interpretation of language and also their use of language, and that aim at explaining how comprehension is related to production.

More specifically, to answer the question whether English motivates a higher proportion of subject expression in Spanish, an issue that has important implications for language change, it would be necessary to study bilingualism involving Spanish and another prodrop language (e.g., Spanish–Portuguese, Spanish–Italian). If rates of expression are lower than in the case of Spanish–English bilingualism, then we might conclude that a non prodrop language affects this area of the grammar of Spanish. If rates are just as high, then bilingualism *per se*, that is, the demands posed by having to learn and use two languages, might be a more valid explanation for the increased proportion of overt subjects.

It is crucial at the present state of (lack of) knowledge about childhood bilingualism in Spanish-"other language" to examine comprehensive language samples from a few children thoroughly, preferably from early on to the end of kindergarten and under diverse environmental conditions. The outcomes of such studies could serve as a valuable baseline for generating further hypotheses and evaluating methodologies to be used in larger studies, as well as for suggesting valid generalizations about the process of bilingual language acquisition. Finally, as Genesee (2000: 168) has aptly stated, "Research on children acquiring two languages simultaneously can make a unique contribution to our understanding of the human language faculty and by extension the human mind because it permits us to examine the limits of the mind's capacity to acquire and use language."

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38 Reading Words and Sentences in Spanish

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1 Introduction

The aim of this chapter is to sketch an overview of the research that has been carried out on reading in Spanish at the word and sentence level. The study of Spanish psycholinguistics has played an important role in the converging evidence supporting current theories of processing at the word and sentence level. Moreover, it has also challenged the enduring bias toward models of language processing based on data collected exclusively from the English language. This has also contributed to the larger goal of moving towards a comprehensive theory of language processing that is built on data that are not language specific, while taking into consideration specific language features that can modulate processing. The main subtopics addressed and the main findings from Spanish psycholinguistic research will be described. The first half of the chapter will be devoted to the recognition of printed words, while the second part will describe the research on sentence processing.

Efficient reading is based on the correct recognition and processing of individual printed words, which constitute the primary building blocks of language processing, and on the assembly of the words into phrases and sentences. Research on visual word recognition in Spanish can help to characterize the universal and the language-specific mechanisms of word recognition. In this chapter, we will refer to some of the unique features of Spanish and describe how they have an impact on the core processes of word processing. These features are: alphabetical orthography; quasi-univocal grapheme-to-phoneme mappings; clear syllabic structure; rich morphology; and close coexistence with other languages. In Section 2, we will present some of the basic findings from each of these research topics that have provided important insights into how printed words are perceived, encoded, and processed.

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Understanding a text involves more than just recognizing words and retrieving word meanings. Reading is a complex process that involves a set of independent but interconnected subprocesses. Words are assembled into sentences not in a linear fashion, but through hierarchical grammatical dependencies between constituents such as subjects, verbs, predicates, pronouns, and antecedents. Thus, reading requires an analysis of constituent structure, that is, an analysis of the relative ordering of words in the sentence and of the grammatical roles played by these words. In order to figure out "who did what to whom" we need to be able to identify the *who*, the *whom*, and the *what* in the internal structure of the sentence: that is, the argument structure of the sentence. Languages signal grammatical relations between constituents mainly through agreement, dependent marking (case-marking), and fixed word order. In Spanish, agreement plays a major role, and research on sentence comprehension in Spanish has investigated agreement processes between different elements. Another major focus of research has been syntactic ambiguity resolution. In Section 3, we will describe the main findings in these topics and their contribution to the universal aim of understanding how sentences are processed.

2 Word reading

2.1 *Alphabetical orthography*

A vast number of languages in the world use alphabetical orthographies (i.e., words are represented by concatenating letters). It is widely accepted that words in alphabetic languages are processed via their constituent letters. Thus, accessing the semantic knowledge related to a printed word (its meaning and related concepts) is preceded by the correct recognition of the individual letters that constitute that given string. Nonetheless, how letters are identified and precisely ordered to form a meaningful string is still under debate (see Grainger 2008 and Grainger; Rey and Dufau 2008 for a review). The visual word recognition system has to distinguish between strings that are highly similar in terms of visual overlap (*cat* and *rat*) and of intervening letter identities and their positions (*dog* and *god*), but at the same time, it has to overlook other differences between allographs or graphemic variations of the same word (*horse*, *HORSE*, *HoRsE*, and *hOrSe*). However, it is not totally clear yet how each of these processes is efficiently attained.

In recent decades, Spanish psycholinguists and cognitive neuroscientists have devoted effort to framing and defining the processes involved in visual word recognition. There is now general agreement that letter identity assignment (i.e., what are the letters of the string?) and letter position assignment (i.e., what is the relative and absolute order of those letters?) are the two basic processes. Even though some models of visual word recognition do not explicitly distinguish between letter identity and letter position assignment processes (e.g., Grainger and Jacobs 1996; Coltheart Rastle et al. 2001,), there is evidence showing that these two parts of letter processing are indeed different (e.g., Perea and Lupker 2004).

A fairly large number of psycholinguistic studies in which Spanish has been the test language have highlighted the apparent inefficiency of the visual word recognition system to codify the precise position that a letter has within a string. A considerable part of the evidence concerning letter ordering has been obtained from studies dealing with the transposed-letter similarity effect. According to this effect, the human word processor can correctly process words that include jumbled letters, showing that, to certain extent, "*it deosn't mttaer in waht oredr the ltteers in a wrod are*" (see Perea et al. 2008 for a review). Many of these studies have shown that a nonword like *CHOLOCATE* (which includes a letter transposition) strongly activates the real word *CHOCOLATE*, as compared with other nonword like *CHOTONATE* (which includes two letter replacements).

Evidence has been obtained from a number of different paradigms and techniques, ranging from purely behavioral tasks measuring participants' performance in terms of reaction times and/or accuracy rates (e.g., Perea and Carreiras 2006a, 2008; Acha and Perea 2008), to experiments recording participants' eye movements or electrophysiological correlates (e.g., Carreiras et al. 2007). This evidence obtained in Spanish refutes traditional models of letter coding that proposed that each letter is tagged into a concrete and precise position (e.g., Rumelhart and McClelland 1982), and as a direct consequence of these studies and parallel research carried out in other languages, several computational models of letter position coding have recently been developed which capture the imprecise way in which humans assign the constituent letters to a given location within a string (e.g., Davis, in press; Gómez et al. 2008; Grainger and van Heuven 2003; Whitney 2001). However, a number of questions remain unsolved, particularly as regards the special status of some word-internal letters. At the moment, there is lively debate regarding how specific letter positions that represent a morphemic boundary are computed (e.g., the two letters that separate the two constituent lexemes of a compound word, such as "*NS*" in *ONSET*, or the two letters that separate a root morpheme and an affix, such as "*KE*" in *WALKER*; Perea and Carreiras 2006b; Duñabeitia et al. 2007a).

With regard to the way in which the visual word recognition system extracts and processes information concerning letter identities, psycholinguistic studies conducted in Spanish have also fruitfully contributed to a growing body of evidence suggesting that readers accommodate their way of perceiving and processing an input to the specific situations. Two key pieces of evidence in this respect are the studies showing that: (1) consonantal information plays a critical role in lexical access, while vowel information is not so crucial; and that (2) under appropriate circumstances, readers process letter-like characters as letters by increasing their tolerance to distortion in the basic features that determines what a letter is. On the one hand, several recent studies have shown that in order to efficiently process words, readers first extract the basic information from a written string based on the consonants (and not the vowels) that are included in the printed material (e.g., Carreiras et al. 2009; Carreiras, Gillon-Dowens et al. 2009). These studies carried out in Spanish, together with others carried out in other languages (e.g., New et al. 2008), show that lexical access is mainly mediated by the correct identification of the consonantal pattern or skeleton of a word, thus posing constraints on word recognition models that do not explicitly establish *a priori* processing differences between consonants and vowels. On the other hand, an increasing body of evidence suggests that readers are able to accommodate their perceptual knowledge to inputs that are not entirely made of letters when they are required to perform tasks based on lexical access. For instance, a handful of studies show that strings like *M4T3R14L* or *MΔT€R!ΔL* are read as the word *MATERIAL* when they are flashed at participants for some milliseconds (Carreiras et al. 2007; Perea et al. 2008; Molinaro et al. 2010), or when this sort of stimuli are embedded within correctly spelled sentences (Duñabeitia et al. 2009). Importantly, recent related data have shown that this only happens when the to-be-recognized object is a letter string and not a number string (Perea, Duñabeitia et al. 2009). Hence, these results have contributed to a better understanding of how bottom-up and top-down interactions between the different perceptual levels of letter processing take place at initial stages of accessing a word, highlighting the tolerance of the human visual word recognition system to deviations from standard or canonical letter representations.

2.2 Grapheme-to-phoneme mappings

Alphabetic languages greatly differ in terms of the degree of their orthographic transparency or consistency (i.e., the systematicity of the mapping between letter and phoneme sequences). Languages in which an individual letter consistently maps onto a single phoneme are characterized as *orthographically shallow* or *transparent*. Spanish (together with some other languages such as Italian, Greek, and Finnish) is considered to be one of the most orthographically transparent languages, with almost one-to-one letter-to-phoneme mapping. Given the considerable level of orthographic transparency of Spanish, speakers of this language have been found to reach a ceiling level of reading expertise faster than readers of more opaque languages (Seymour et al. 2003; see also Defior et al. 2002). This characteristic of Spanish has been also connected to the performance of Spanish dyslexic children, highlighting potential differences between dyslexic readers of transparent languages and dyslexic readers of opaque languages (Davies et al. 2007). It has been generally proposed that readers of languages with very transparent orthographies rely mainly on the activation of phonological codes and that they use preferably the nondirect grapheme-to-phoneme conversion processing route rather than the direct lexical route in recognizing words (e.g., Ziegler and Goswami 2006).

Investigating phonological vs. orthographic processes in Spanish is a real challenge because of the very consistent letter-to-phoneme mappings and the scarce number of inconsistencies. Pollatsek, Perea, and Carreiras (2005) used the context-dependent phonemic mappings of the letter *c* in Spanish (the associated phoneme varies depending on the annexed vowel) to test automatic activation of the phonological code during the earliest stages of visual word recognition. The authors compared the lexical decision performance of Spanish readers on target words such as *CANAL* preceded by nonword primes that had exactly the same extent of orthographic overlap. Critically, these nonword primes included the same letter *c*, which in some cases was pronounced the same as in the target (i.e.,/k/, e.g., conal), while in others it took its alternative pronunciation (i.e.,/θ/, e.g., cinal). Their results showed a larger facilitation when the targets were briefly preceded by nonwords with the same sound for the letter *c*, indicating the early involvement of phonology in visual word recognition (see also Carreiras, Perea et al. 2009 for electrophysiological evidence about the time course of orthographic and phonological processing). Hence, despite the high consistency in grapheme–phoneme mappings in Spanish, the few inconsistencies that are found in the language have also been useful for researchers aiming at uncovering the involvement of phonological information at the earliest stages of word processing and the time course of this involvement.

2.3 *Syllabic processing*

The clear syllabic boundaries of Spanish words have been repeatedly used to explore whether the syllable is a processing unit in visual word recognition. As compared to other languages with less clear syllabic structures (e.g., English), Spanish is a key language to uncover the importance of the syllable in language comprehension. For almost two decades, Spanish psycholinguists have highlighted the relevance of the syllable and have consistently shown that readers take into account syllabic information while accessing the mental lexicon. As explained below, data obtained from studies testing Spanish syllables constitute the strongest evidence in favor of the syllable as a processing unit in the domain of visual word recognition (see Duñabeitia, Cholin et al. 2010 for a descriptive summary).

Probably the two most important findings related to syllabic processing in visual word recognition are the syllable congruency effect (Carreiras and Perea 2002) and the inhibitory effect of first syllable positional frequency (Carreiras et al. 1993). The syllable congruency effect establishes that a word is processed easier (faster and more accurately) when preceded by a string containing the same syllable, as compared to when it is preceded by a string in which the initial syllable does not coincide. An example is the Spanish word PASIVO (*passive*), which is syllabified as PA.SI.VO. Spanish readers recognize this more effortlessly when it is preceded by a string like PA**** than when it is preceded by a string like PAS****, despite the greater orthographic overlap in the latter condition. This finding corroborates the claim that the initial syllable is a crucial processing unit in visual word recognition, over and above initial graphemes (see also Álvarez et al. 2004; Carreiras et al. 2008; Carreiras et al. 2005).

The other landmark effect in the syllabic processing literature is the inhibitory effect of the frequency of the first syllable. This effect, first reported by Carreiras et al. (1993), shows that Spanish words starting with a high-frequency syllable yield longer recognition and reading latencies as compared to words starting with a low-frequency syllable. The syllable frequency inhibitory effect highlights the importance of the initial syllable as a basic access unit in visual word processing since a word containing a high-frequency initial syllable activates many other words sharing that syllable in that given position (i.e., syllabic neighbors), which demands more recognition time for that given input. Many other studies in Spanish have subsequently replicated this effect with behavioral, electrophysiological, and neuroimaging measures (Perea and Carreiras 1998; Álvarez et al. 2001; Barber et al. 2004; Carreiras et al. 2006; Conrad et al. 2009).

Curiously, while most words in many languages are polysyllabic, research in the field of visual word recognition has focused on reading monosyllabic words, and, in fact, current computational models deal exclusively with the processing of monosyllabic words (Grainger and Jacobs 1996; Ziegler et al. 1998; Perry and Coltheart 2000; Coltheart et al. 2001; but see Ans et al. 1998 for an exception). Research on polysyllabic words in Spanish suggests that the processing of polysyllabic words is mediated by syllabic structure, with syllables being functional units of the reading process. Moreover, a new model of bisyllabic visual word recognition, fully representing the principles of interactive activation between the three representation layers of letter, syllable, and word units, has proved successful in correctly parsing all Spanish bisyllabic words with different types of syllabic structure (Conrad et al. 2010).

2.4 *Morphological processing*

Recent decades have been especially fertile in psycholinguistic research on the morphological decomposition of polymorphemic words, looking at inflectional and derivational morphology. Mainly (but not exclusively) two opposing views have been proposed in order to account for how readers process polymorphemic words: the prelexical and the postlexical account. The prelexical account proposes that affixed words are initially decomposed into their corresponding morphemes by automatically stripping off the affixes from the root morphemes before lexically searching for the stems of those words (Taft 1979; Rastle et al. 2004). Alternatively, supporters of the postlexical account of morphological decomposition propose that the decomposition process takes place only when whole-word access is achieved, that is, postlexically (e.g., Grainger et al. 1991; Giraudo and Grainger 2001, 2003). As we will briefly mention in the following paragraphs, a good number of studies conducted in Spanish have provided important insights about morphological decomposition processes, helping to tip the scales in favor of the prelexical account.

Due to the rich morphological properties of Spanish, a robust (and increasing) part of the psycholinguistic literature on this topic stems from experiments carried out in this language. Empirical evidence favoring early (prelexical) automatic decomposition of polymorphemic words has been obtained in several Spanish studies testing inflectional morphology (Barber et al. 2002; Domínguez et al. 2002; Sánchez-Casas et al. 2003), as well as derivational morphology (Duñabeitia et al. 2007a, 2008) and compound word creation (Duñabeitia et al. 2007b; Duñabeitia, Marín et al. 2009; Vergara-Martínez et al. 2009). An illustrative example is the study conducted by Duñabeitia, Perea, and Carreiras (2008), who, in a series of masked priming lexical decision experiments, tested target words that could be briefly preceded by a morphologically or orthographically related string, or by an unrelated prime. Critically, under masked priming conditions, participants are unaware of the existence of the prime words, but their influence can still be observed in target processing. Participants were presented with words like *CERTAMEN* or *IGUALDAD*, that were preceded either by related masked prime words like *VOLUMEN* or *BREVEDAD* (only orthographically or morpho-orthographically related, respectively), or by unrelated masked prime words like *TOPACIO* or *PLUMAJE*. Duñabeitia et al. found a clear dissociation between morphological and orthographic effects under these conditions, showing that masked suffix priming effects (*BREVEDAD–IGUALDAD*) are greater than purely form-based priming effects (*VOLUMEN–CERTAMEN*). Hence, these data help to distinguish purely form-based relationships from morphological relationships, and support a prelexical view of morphological decomposition in which polymorphemic words are initially decomposed into their constituent morphemes (leading to suffix priming effects that are similar in nature to the constituent priming effects observed in Spanish compound words; see Duñabeitia, Laka et al. 2009).

2.5 Close coexistence of Spanish with other languages

Spanish is a widely-spoken language that coexists in many places with other languages. Considering the vast number of native speakers of Spanish, and that approximately 150 million speakers in the world have direct contact with Spanish as a second or third language, it is not surprising that this language provides an excellent opportunity to test different hypotheses related to how the human word recognition system faces the difficult task of dealing with two (or more) interacting lexicons (e.g., Dimitropoulou et al., in press a). Furthermore, the nature of Spain as a multilingual state with locally co-official languages such as Basque, Catalan, or Galician, makes Spanish an appropriate language in which to study automatic processes of word translation in the bilingual mind. It clearly goes beyond the scope of this chapter to examine bilingual word processing (we refer the reader to Chapter 40, below), but it is important to mention that several studies have shown evidence favoring the automatic activation of translation equivalents with high orthophonological overlap during reading (i.e., cognate words; e.g., Sánchez-Casas et al. 1992; Davis et al 2010; Duñabeitia et al. 2010). More importantly, some studies have shown that coactivation of translation equivalents also extends to word pairs that do not share orthographic or phonological units (i.e., noncognate translations; e.g., Basnight-Brown and Altarriba 2007, see Duñabeitia et al. 2010, for a comprehensive summary). Moreover, recent data from Greek–Spanish bilinguals suggests that multilingual speakers automatically activate the representation of a word in a nontarget language when presented with its noncognate translation equivalent, even at the lowest levels of proficiency in the non-native language (see Dimitropoulou et al., in press b). Hence, as seen, the close coexistence of Spanish and other neighboring languages has promoted extensive research on bilingual visual word processing, which will undoubtedly increase in forthcoming years.

3 Sentence comprehension

Two major theoretical perspectives in psycholinguistics have been proposed to describe the processing routines involved in sentence comprehension. *Syntax-first models* (Ferreira and Clifton 1986; Frazier 1987) assume that the interpretation of a sentence is obtained through independent stages of analysis (modules), is serially ordered (grammatical analysis precedes semantic interpretation), and informationally encapsulated (grammatical analysis is not influenced by meaning). Thus, initial stages of sentence comprehension would be pursued on the basis of syntactic information only because no other knowledge would be available to the cognitive system. A critical assumption is that the language processor has a limited amount of resources (due to limited short-term memory capacity) to perform the operations required to construct the mental representation of a sentence. Thus, to reduce memory load during comprehension, the parser conducts a single analysis of the sentence, the "simplest" and more economical in terms of computational costs. Principles such as Minimal Attachment (Frazier and Fodor 1978), Late Closure (Frazier 1987), and Minimal Chain (De Vincenzi 1991) predict what is the preferred (i.e., less complex) interpretation in cases of ambiguity. If an initial analysis proves to be incorrect, a reanalysis of the sentence structure is required.

The nature of this second step has been widely discussed (Fodor and Ferreira 1998): all the proposals assume that, at this later stage of analysis, the language processor can rely on syntactic, semantic, and pragmatic information to grasp the intended meaning of a sentence. Many authors (Frazier 1987; Friederici et al. 2002; De Vincenzi et al. 2003) hypothesize a similar process to occur also for the repair of grammatical violations such as, for example, morphosyntactic violations. When the syntactic information does not allow the construction of a well-formed structure, the system modifies the so-far-built representation to obtain a coherent interpretation of the utterance. These "repair" processes are thus assumed to be similar to "reanalysis," which involves recovery from an ambiguous syntactic analysis when the sentence allows an alternative grammatical interpretation.

Over the last twenty years or so, there has been a vast amount of behavioral research on attachment preferences in different languages to test empirically the predictions of the mechanisms of "minimal attachment" and "late closure." Interestingly, Cuetos and Mitchell (1988) challenged the universality of the "late closure" mechanism in a seminal paper on Spanish sentences containing a relative clause (RC) preceded by a complex NP with two possible hosts such as:

(1) Alguien disparó contra el criado de la actriz que estaba en el balcón con su marido.
'Someone shot the male servant of the actress who was in the balcony with her husband.'

Instead of attaching the RC (who was on the balcony) to the N2 of the complex NP (N1 of N2) (the actress) as predicted by the Late Closure strategy, Spanish readers preferred to take the N1 (the male servant) as the host of the RC. Further on-line data has indicated that while English readers show either no preference (Carreiras and Clifton 1993) or low attachment preference (Carreiras and Clifton 1999), Spanish readers show high attachment (Carreiras and Clifton 1993, 1999). These results, together with others from many different languages, obliged a modification of the "garden path" "syntax-first model" resulting in the "Construal Hypothesis". In this version (Frazier and Clifton 1996), the universality of attachment strategies was retained but not the ubiquity. The Construal Hypothesis assumes that universal strategies remain valid in specific syntactic domains. Recent theoretical approaches share this distinction between primary and nonprimary relations (Chomsky 1995, 2000, 2001). Elements that constitute primary relations are incorporated into the sentence by means of a formal operation called "Merge" that operates following category- and semantic-selectional criteria (Chomsky 1995). In contrast, constituents forming nonprimary relations are incorporated into the sentence by using a different operation that does not create a new syntactic object, but simply modifies (or expands) one of the old ones, as happens with relative clauses.

Multiple-constraints models constitute a very different theoretical approach (e.g., MacDonald et al. 1994; Trueswell et al. 1994). These models propose an alternative approach to sentence processing according to which information is not hierarchically processed by distinct modules. On the contrary, as soon as each type of information is available, it is interactively integrated with the already available information. The critical differences between these models and the modular approaches are the following: (1) syntactic and semantic information interact from initial stages of processing; (2) there are no serially ordered modules dealing with qualitatively distinct types of information – information is processed in parallel; and (3) the cognitive system does not pursue only one single possible interpretation of the sentence, but (when facing an ambiguity) multiple interpretations are kept available and their level of activation re-ranked based on the amount of information supporting each alternative. Thus, there is no need to postulate a separate stage of reanalysis if an initial analysis is revealed to be incorrect. These two theoretical perspectives, syntax-first models and multiple-constraints models, have also influenced the research on sentence processing in Spanish, not only on syntactic ambiguity, as described previously with relative clauses, but also in other domains (e.g., processing of anaphors, empty categories, agreement, etc.). Below we will describe the main findings in the research carried out in Spanish in some of these domains.

3.1 Anaphoric relationships

The term anaphor usually describes any expression coreferring with an antecedent. In recent decades, anaphor resolution has become a crucial topic for understanding the mechanisms and representations involved in language comprehension. For instance, in *The woman crossed the street. She was very careful*, the personal pronoun *She* unambiguously corefers with the only possible antecedent, *The woman*, that fills the same subject position in the sentence. In this case, the perfect alignment of grammatical, semantic, and structural information facilitates the computation of the coreferential relationship necessary to comprehend the sentence introduced by the personal pronoun. However, there are cases in which only a particular piece of information is available during pronoun resolution.

One important question that has been investigated in Spanish is the contribution of formal cues, such as grammatical gender, to pronoun resolution. We have shown that not only semantic but also purely grammatical gender agreement speeds up pronoun resolution in Spanish. In Carreiras, Garnham, and Oakhill (1993) (see also Garnham et al. 1995), a sentence containing a pronoun was preceded by two antecedents that could either have the same gender (e.g., both feminine, if the pronoun was feminine) or two different genders (i.e., masculine and feminine). The reading times of pronominal sentences were faster when they were preceded by antecedents of different gender (gender cue condition) rather than by antecedents of the same gender (no gender cue condition). The reading time advantage of the gender cue condition occurred with antecedents characterized by natural gender, as in (2), and by arbitrary grammatical gender, as in (3):

(2) Ricardo/Alicia arrestó a Pablo porque lo descubrió robando un coche.
Richard (MASC)/Alice (FEM) arrested Paul(MASC) because (OMITTED SUBJECT) him (MASC-ACC) found stealing a car.

(3) El camión/La grúa remolcó al autobús porque lo inmovilizó la nieve.
The truck (MASC)/breakdown truck (FEM) towed the bus (MASC) because (OMITTED SUBJ) it (MASC-ACC) immobilized the snow.

These results suggest that pronoun assignment can be solved using only grammatical gender cues (Carreiras et al. 1993; Garnham, et al. 1995).

In addition, we investigated the contribution of high-level semantic information to anaphor resolution. Conceptual information is another important factor that modulates pronoun resolution. Some studies on conceptual anaphora (e.g., Gernsbacher 1991; Carreiras and Gernsbacher 1992) have shown that collective nouns that are grammatically singular (e.g., jazz band) were mentally represented as plural concepts. In fact, the reading times were faster when jazz band was followed by a plural than by a singular pronoun. Further evidence on the role of conceptual knowledge in pronoun interpretation has come from gender stereotypes. Carreiras et al. (1996) showed that the information relating to stereotypical gender was incorporated into the mental representation of a character introduced by a role name and used for pronominal assignment. In particular, a pronoun was read faster when it agreed with the stereotypical gender associated with the role name than when it disagreed (e.g., he vs. she for carpenter; she vs. he for secretary). The studies on conceptual anaphora, focus, and stereotype information suggest a direct and early impact of high-level semantic information in addition to formal cues such as gender.

The time course of the two main factors that have been shown to affect pronoun resolution – high-level and low-level lexical–morphosyntactic cues – has been the focus of debate. According to the proposed two-stage model, lower and higher levels of information intervene at different stages in anaphor resolution (Garrod and Sanford 1990, 1994; Garrod and Terras 2000). In an initial bonding stage, low-level automatic processes that rely on morphosyntactic features, such as gender, establish a link between an anaphor and a potential antecedent. Later this link is tested and solved at the integration stage. Here, context makes an important contribution. However, according to constraint–satisfaction models of language comprehension (e.g., MacDonald et al. 1994; Trueswell et al. 1994) morphosyntactic, semantic, and discourse information simultaneously interact from the first moments of sentence processing. Work carried out in another Romance language, Italian, with epicenes (Cacciari et al. 1997; Cacciari et al. in press) seems to show that, in line with the two-stage model, when morphosyntactic cues are present, they will enter first in the model, with high-level semantic effects requiring more time to influence anaphor resolution. Further research will shed light on the time course of formal and semantic cues on pronoun comprehension.

3.2 *The empty category PRO: processing what cannot be seen*

Another arena to test theories of sentence processing has been the resolution of PRO in Spanish. An Empty Category PRO is an element that lacks phonological realization, standing at the subject position of infinitive constructions. This element must establish a relationship with an antecedent in order to acquire its meaning. Thus, it is an attractive structure to test predictions from different models of syntactic processing. Empty category constructions are common in Spanish, a null subject language, although psycholinguists have paid little attention to them (e.g., Demestre et al. 1999; Alonso-Ovalle et al. 2002; Betancort et al. 2004; Betancort et al. 2006). Studies by Betancort et al. (2006) and Demestre et al. (1999) have investigated PRO in subordinate infinitive context sentences. In an ERP study, Demestre et al. (1999) tested how fast readers activated the legal antecedent of a PRO in sentences like (4) and (5):

(4) Pedro/*María quiere [PRO] ser rico.
'Peter/*Mary wants [PRO] to be rich (MASC).'

(5) Pedro/María ha aconsejado a María/*Pedro [PRO] ser educada con los trabajadores.
'Peter/Mary has advised Mary/*Peter [PRO] to be polite (FEM) with the workers.'

In the mismatched conditions, the authors found an early negativity distributed over frontal and central scalp regions following the mismatched target ("rich/polite") presentation. They concluded that this pattern of electrophysiological response reflected the rapid detection of gender agreement violations in infinitival clauses, which implies that the parser had established the coreference relation between the null subject (PRO) and its antecedent.

Betancort et al. (2006) investigated the processing and time course of the empty category PRO. Eye movements were recorded while participants read sentences in which a matrix clause was followed by a subordinate infinitival clause, so that the subject or the object of the main clause could act as controller of PRO, and therefore as implicit grammatical subject of the infinitive. In Experiment 1, verb control information was manipulated: The matrix clause contained either subject-control verbs like *prometer* ('promise') or object-control verbs like *exigir* ('demand'), (Mismatch conditions are preceded with an asterisk), as in shown in (6)–(9):

(6) María$_i$ prometió a Pedro$_j$ [PRO$_i$] ser bastante cauta con los comentarios.
Mary$_i$ promised Peter$_j$ [PRO$_i$] to be quite careful (FEM) with her comments.

(7) María$_i$ prometió a Pedro$_j$ [PRO$_i$] ser bastante cauto con los comentarios.
*Mary$_i$ promised Peter$_j$ [PRO$_i$] to be quite careful (MASC) with her comments.

(8) María$_i$ exigió a Pedro$_j$ [PRO$_j$] ser bastante cauto con los comentarios.
Mary$_i$ demanded that Peter$_j$ [PRO$_j$] be quite careful (MASC) with his comments.

(9) María$_i$ exigió a Pedro$_j$ [PRO$_j$] ser bastante cauta con los comentarios.
*Mary$_i$ demanded that Peter$_j$ [PRO$_j$] be quite careful (FEM) with his comments.

In Experiment 2, the preposition that headed adverbial subordinate clauses was manipulated: two different kinds of infinitival adverbial clauses were used, expressing purpose (preposition *para*) and reason (preposition *por*), and in which control information is primarily induced by the prepositions (*para* tends to trigger subject-control; *por* object-control). Experiment 1 showed that readers make immediate use of verb control information to recover the antecedent of the empty category PRO in Spanish obligatory control constructions. The data obtained in Experiment 2 suggest that during the processing of the empty category PRO in purpose vs. reason adverbial subordinate infinitival clauses the control information induced by the prepositions *por* vs. *para* is not initially used as a constraint to guide the selection of the nominal antecedent of PRO. In addition, both experiments showed that PRO antecedent selection is a very fast process and that, together with verb control information, recency plays an important role. Further eyetracking experiments (a) on the recovery of unpronounced or unwritten material with the grammatical particle '*se*' which licenses different sorts of gaps (Meseguer et al. 2009), and (b) on the processing of subject vs. object relative clauses (Betancort et al 2009) have shown that formal cues and structural preferences, but sometimes also conceptual cues, play important early roles, with different time courses, in sentence processing.

3.3 Neurobiological data

Recently, the debate about the relationship between syntax and semantics, as well as the comparison between single-analysis mechanisms and multiple-constraint models, has been enriched by considerations stemming from the analysis of neurobiological data derived from *ERP* studies. Within the "information processing" paradigm, event-related potentials (ERPs) are a suitable technique for the study of language, due to their excellent temporal resolution that permits measurement of voltage variations occurring within time intervals of tens of milliseconds. The electroencephalography (EEG) can be recorded from a number of electrodes placed on the scalp while the participant performs a cognitive task, for instance, reading a number of grammatically correct and ungrammatical sentences. While the cognitive processes of interest are time-locked to the stimuli, "noise" is assumed to be randomly distributed (in terms of polarity, latency and amplitude) in the EEG trace. Therefore, averaging on individual trials will tend to reduce noise while increasing the signal, revealing Event Related Potential (ERP) components. ERPs obtained after averaging on trials belonging to different experimental conditions, for example, grammatically correct vs. ungrammatical sentences, can be overlayed and inspected for differential effects. Assuming that the conditions differ only with respect to the experimental manipulation, the differences between waveforms can be attributed to the cognitive processes of interest, such as the detection of the grammatical violation.

On the basis of information concerning the time course and the scalp distribution of the observed differential effects, inferences about the neural implementation of the processes of interest can be drawn. Given their excellent temporal resolution compared with other imaging techniques such as PET and fMRI, ERPs allow reliable inferences with respect to the relative time course of cognitive processes. In particular, their high temporal resolution makes them suitable for testing psycholinguistic models of language comprehension. Three main components of ERP have been reported for sentence processing difficulties, as described in Sections 3.3.1–3.3.3 below.

3.3.1 ***N400*** This first component is a negative deflection starting at 250 ms and peaking around 400 ms after the onset of a lexically anomalous word compared to the preceding context. As a consequence of its functional characteristics, this ERP component has been called N400 (Kutas and Hillyard 1984). It is the classical correlate of semantic violations, in particular of selective restriction violation: when a word is not consistent with the preceding context, the N400 effect is triggered.

Interestingly, N400 amplitude correlates with the *cloze-probability* of the critical word. The cloze-probability of a word corresponds to the percentage of times a word is selected as completion for a particular sentence fragment. The lower this index, the larger the amplitude of the N400: this component is thus sensitive to the level of preactivation of a particular item induced by the preceding context (Barber and Kutas 2007; Molinaro, Conrad et al. 2010).

3.3.2 LAN In many reports, an anterior negativity within a window ranging from 150–500 ms, known as the Left Anterior Negativity or LAN (Neville et al. 1991; Friederici 1995; Barber and Carreiras 2005) has been observed in correlation to syntactic violations. The typical distribution of this component is left frontal (Osterhout and Mobley 1995; Coulson et al. 1998; Molinaro et al. 2008), hence its name; although some studies have reported a bilateral or left central distribution (Hagoort 2003; Silva-Pereyra and Carreiras 2007). A LAN with a latency of 150 ms has been observed with phrase structure violations (*earlyLAN*: Neville et al. 1991; Friederici et al. 1996) and a component with a latency of 300 ms has been reported for morphosyntactic agreement violations, as in the sentences below from Barber and Carreiras (2005; see also similar effect in Osterhout and Mobley 1995; Coulson et al. 1998).

(10) Agreement: El piano estaba viejo y desafinado.
the (MASC-SING) piano (MASC-SING) was old and off-key.

(11) Gender disagreement: La piano estaba viejo y desafinado.
the (FEM-SING) piano (MASC-SING) was old and off-key.

(12) Number disagreement: Los piano estaba viejo y desafinado.
the (MASC-PL) piano (MASC-SING) was old and off-key.

LAN effects have been attributed to syntax-specific processes, although it has been argued that these effects may reflect nonspecific working memory processes (King and Kutas 1995; Coulson et al. 1998).

3.3.3 *P*600 Syntactic anomalies also elicit a positive deflection starting 500 ms after the onset of the incorrect item, peaking around 600 ms and continuing for 500 ms at least. Because of its polarity (positive) and latency (600 ms following the critical word), this ERP component is called P600 (Osterhout and Holcomb 1992; Hagoort et al. 1993). The P600 has shown to be sensitive to: (1) words that are *ungrammatical* continuations of the preceding sentence fragment; (2) words that are *nonpreferred* but grammatical continuations; and (3) words that are *complex* continuations.

Although the precise cognitive and neural events underlying this effect are not known, some authors interpret the P600 as an index of generic syntactic integration difficulty (Osterhout et al. 2004). Other authors give instead a more strict interpretation of the P600 as an ERP correlate of later stages of parsing: failure of parse (Hagoort and Brown 2000; Carreiras et al. 2004), reanalysis and violation resolution (Friederici 1995, 2002; Kaan and Swaab 2003). Interestingly, Kaan and Swaab (2003) distinguish two subcomponents: P600s with a posterior distribution that would index syntactic integration difficulties, and frontally distributed P600s that would signal an increased complexity at the discourse level. The two subcomponents of the P600 were clearly seen in the experiment carried out in Spanish by Carreiras et al. (2004) using sentences with a relative clause preceded by a complex NP (N1 of N2):

(13) Juan felicitó a la cocinera del alcalde que fue premiada.
'John congratulated the cook (FEM) of the mayor who was honored (FEM).'

(14) Juan felicitó al cocinero de la alcaldesa que fue premiada.
'John congratulated the cook (MASC) of the mayoress who was honored (FEM).'

In this type of structure, the sentence is ambiguous at the relative pronoun until the perception of the disambiguating participle. Given the preference for a high-attachment strategy reported in Spanish by Carreiras and Clifton (1993, 1999), we could expect processing difficulties for "cocinero (MASC) de la alcaldesa (FEM) que fue premiada (FEM)" as compared to "cocinera (FEM) del alcalde (MASC) que fue premiada (FEM)." This is exactly what Carreiras and colleagues (2004) found in an ERP experiment at the position of the disambiguated participle. Interestingly, there was no sign of a LAN component, but only a P600 component. The P600 effect was widely distributed in the 500–700 ms window, including frontal areas, while the distribution was mainly posterior in the 700–1000 ms window. Recent findings extend the functional interpretation of this component beyond purely grammatical anomalies. P600s have, in fact, been elicited also by thematic and orthographic violations within a sentence context (Kim and Osterhout 2005; Leone et al. under review).

3.4 Semantics and syntax in the early stages of processing

Semantic and syntactic violations elicit effects in the same time window (i.e., between 300 and 500 ms). The difference between the LAN and the N400 is mainly identified in the topography of the effects. While the LAN has a left-frontal distribution (Barber and Carreiras 2005; Silva-Pereyra and Carreiras 2007), the N400 shows the maximum of its effect on the right-posterior areas of the scalp (Barber and Kutas 2007; Molinaro, Conrad et al. 2010). From a neurophysiological perspective, topographical differences between two ERP components in the same time window imply two different neural populations subserving different types of processing (Luck 2005). A series of studies in Spanish has focused on the interaction between syntax and semantics, trying to evaluate the differential impact of these two types of violations on ERPs in the 300–500 ms time window.

Event-related potential (ERPs) studies have found effects of agreement violation mainly in two components: the Left Anterior Negativity (LAN) and the P600 (e.g., Kutas and Hilliard 1983; Münte and Heinze 1994; Osterhout and Mobley 1995; Gunter et al. 2000; Deutsch and Bentin 2001; Barber and Carreiras 2003, 2005; Barber et al. 2004; Silva-Pereyra and Carreiras 2007). In particular, Barber and Carreiras

(2005) studied the agreement of gender and number mechanisms in Spanish between word pairs presented in isolation or embedded in sentences. Disagreement in word pairs formed by a noun and an adjective (e.g., *faro–alto* 'lighthouse–high') produced an N400-type effect, while word pairs formed by a determiner and a noun (e.g., *el–piano* 'the-piano') showed an additional left anterior negativity effect (LAN). Agreement violations with the same words inserted in sentences (e.g., *El piano estaba viejo y desafinado* 'the (MASC-SING) piano (MASC-SING) was old (MASC-SING) and off-key') resulted in a pattern of LAN-P600.

However, other studies seem to show that the LAN-P600 pattern for morphosyntactic violations can be modulated by the nature of information involved. For instance, in a recent study Mancini et al. Carreiras (submitted a) presented subject–verb mismatches to Spanish speakers. The mismatch could either be due to an unexpected number value in verb position (number mismatch) or to an unexpected person value (person mismatch). Those two types of mismatch are supposed to be processed as pure syntactic violations. However, while the number agreement violation elicited a LAN effect in the 300–500 ms time window, the person mismatch elicited a negativity that was widely distributed, being evident both in the anterior and in the posterior areas of the scalp. According to the authors, in the person mismatch, formal information guides the interpretation of the syntactic anomaly, which in order to be understood requires access to high-level representations, and for this reason access to semantic representations.

Another interesting piece of information that does not completely fit with the exclusive use of syntactic information at initial stages of processing comes from another experiment carried out by Mancini et al. (submitted b) with a very particular structure of Spanish. We presented Spanish speakers with morphophonological mismatches that are, however, syntactically acceptable:

(15) a. Los periodistas *3rd.plu* trabajamos *1st.plur* mucho.
b. Los periodistas *3rd.plu* trabajaron *3rd.plur* mucho.
'The journalists work hard.'

These structures are defined Unagreement patterns. *Los periodistas trabajamos* presents an apparent person mismatch in the values expressed on the subject (third person plural) and verb (first person plural, but the verb is syntactically acceptable also with second person). These structures are licit in Spanish, and they are interpreted assigning the value of the subject to the lexical subject (in English it would sound like: *We journalists work hard*). We found a reduction in the amplitude of the earlyP600 compared to a control syntactic condition that did not show any inflectional mismatch. Unagreement patterns require the interpretation of the Person value in the verb position without integrating it with the morphological properties expressed by the noun in subject position. Somehow, the system has to inhibit the integration of the verb feature with those expressed on the previous noun, while this integration process has to be pursued for the control condition. This suggests that the earlyP600 interval represents integration processes at work also during normal full agreement computations.

More specifically, this earlyP600 interval would correlate with a stage of processing at which, after an initial identification of the structurally-related constituent (sensitive to morphophonological cues, Wagers et al. 2009), the target element is integrated within a structurally-organized sentence representation. This sentence-level representation would not only depend on syntactic information (such as the matching of ϕ-features), but also on semantic- and discourse-level information shown to increase the earlier phase of the P600 (see Kuperberg 2007; Bornkessel-Schlesewsky and Schlesewsky 2008).

In sum, most of the data from different domains of sentence processing in Spanish seem to suggest that formal operations are at work at earlier stages of processing, but it is also true that there is some data showing that sometimes nonsyntactic representations are accessed during these initial stages. Future work will shed light on questions of under what circumstances and when this information is used during the time course of sentence processing. Moreover, it is important to mention that studies in sentence processing using Spanish as a second language have been also carried out, showing that the computation of formal cues is different in second language learners (e.g., Gillon-Dowens et al. 2010). However, a review of the literature on sentence processing in Spanish with bilingual and second language learners is beyond the scope of this chapter.

4 Concluding remarks

There is a clear and enduring bias in psycholinguistics to build models of language processing based on data collected from English alone. The research carried out in Spanish has contributed at least in two ways to the broader goal of formulating comprehensive theories of language processing which are not tied to a particular language. On the one hand, it has sometimes provided converging evidence to support existing theories of processing at the word and sentence level. On the other hand, it has raised and highlighted specific questions that had not been previously considered, thus advancing the field.

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39 Language Impairments

JOSÉ MANUEL IGOA

1 Introduction

The reader may wonder what the point is of including a chapter on language impairments in a text on Hispanic linguistics. The coverage of such a topic in this book might suggest that there are particular disturbances or specific symptoms therein associated with the Spanish language in contrast to those found in other languages. If such were the case, we should ask what the distinguishing characteristics of language breakdown are in Spanish-speaking populations, and more importantly, what reasons could be given to account for these idiosyncrasies. The aim of this chapter is to disclose the most relevant findings drawn from research on language impairments in Spanish speaking populations in order to emphasize the most salient features of such impairments that can be traced down to specific characteristics of the Spanish grammar and its various components: namely, phonology, morphology, and syntax. This, of course, does not imply that the patterns of language breakdown are entirely dependent on the features of particular grammars. It is widely assumed that the general design of the neurocognitive architecture of the language faculty is the same for every language and every speaker. However, as we will see, the ways in which language deficits surface in the form of symptoms differ from one language to another as a function of the options that each grammar makes available to its users in its various components.

A disturbance of the linguistic competencies in a given language may come about from a number of different sources. Accordingly, there is an ample variety of linguistic pathologies and associated symptoms. These may be classified according to different criteria. One criterion is the specificity of the impairment, which yields two broad categories of language deficits: selective impairments that are found to affect only (or predominantly) the patient's linguistic abilities, with no noticeable damage to other cognitive skills; and language impairments that are associated to cognitive disabilities in nonlinguistic domains. The former may, in turn, be divided into (1) deficits that arise from a genetic impairment, which usually yield an atypical profile of linguistic abilities from early childhood, as that found in the so-called

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"specific language impairment" (SLI) or genetic dysphasia, as well as in other cases of genetic malformations, such as Williams syndrome, in which spatial intelligence is severely impaired while language performance shows a mixed pattern of spared and damaged abilities (Karmiloff-Smith 1998); and (2) acquired language impairments in adults resulting from focal brain damage (e.g., aphasia and dyslexia), usually located in the left (language-dominant) hemisphere (Rapp 2001).

As for language deficits that develop as signs of a broader cognitive impairment, these are usually defined in terms of the more general condition in which they are embedded. The most common and widely studied language deficits in this regard are those associated with schizophrenia and other psychotic disorders, those linked to degenerative conditions like dementia or Parkinson's disease, and those that arise within severe developmental disorders, such as autism or Down's syndrome, among others. The usual, but by no means the only, picture in such cases is a major conceptual disability, a key cognitive deficit that goes beyond linguistic capacities and arrests or obliterates other cognitive abilities or resources, such as (1) the ability to control or inhibit certain behaviors or beliefs (which appears to be impaired in schizophrenia or dementia; see Reed et al. 2002, Voss and Bullock 2004), (2) the capacity to reason about others' beliefs and mental states (usually deficient in mild and moderate forms of autism; see Baron-Cohen et al. 1985), or (3) the ability to store and retrieve information from semantic memory (one of the central deficits found in dementia; see Hodges et al. 1992).

Another way of classifying language impairments that is implicit in the above taxonomy consists of distinguishing between disorders that arise during the course of the development of language and disorders that emerge in adulthood. Either of them may be specific to language or may have a broader scope. However, most developmental disorders of language are presumed to have a genetic basis, whereas adult language impairments may originate from genetic causes or be acquired through an external agent (i.e., brain damage). Finally, one may categorize language disturbances in a more fine-grained fashion, by looking at the tasks that are impaired and the resulting behavioral manifestations, regardless of the pathological condition to which they belong (Coltheart 2001). According to this symptom-based criterion, a patient might be impaired in his or her ability to recognize, repeat, or produce phonological segments, strings of phonological segments, morphemes, or words, to prove understanding of the meaning of written or spoken words, phrases, or sentences by matching them with drawings, to produce lexical items from pictures, to parse strings of syntactic constituents in order to derive their meaning, or to understand or generate inferences when listening to or reading texts and discourses, among other possible deficits.

In this chapter, I will follow this latter approach, and will focus on those deficits that selectively impair basic linguistic processes, drawing mainly on data from studies of adult monolingual aphasic Spanish-speaking patients who suffer different forms of language loss in the spoken modality. However, I will also make some remarks on the reading difficulties experienced by patients with various forms of acquired dyslexia, as well as reading problems shown by aphasic adult bilinguals having Spanish as one of their languages. In a later section, I will provide some information about the patterns of deficits revealed by Spanish-speaking children with specific language impairment.

Researchers in the field of neuropsychology, the discipline that explores the patterns of breakdown of different cognitive faculties and processes and their neural underpinnings, often emphasize the importance of providing accurate and detailed descriptions of the impaired and preserved abilities of patients in order to devise more effective tools to assess and rehabilitate the patients' performance. However, knowledge of the patterns of associations and dissociations of symptoms observed in different forms of language pathologies also affords researchers helpful clues to understand the structure and functioning of the healthy human language processing apparatus. In this particular aspect, the study of language impairments has also become a valuable testing ground for linguistic and psycholinguistic theories and hypotheses about language knowledge and processing.

2 Spoken language impairments in Spanish

Most neuropsychological studies carried out with Spanish-speaking aphasics have yielded evidence for a modular structure of the basic processes involved in language use. In this regard, they line up with studies of patients with other languages. In addition, a number of studies have been conducted with Spanish patients with the aim of testing hypotheses concerning the mechanisms underlying a given linguistic deficit or the most likely variables (linguistic or otherwise) that account for its symptoms. The picture, however, is far from uniform, for it is quite common that different patients with the same type of impairment exhibit dissimilar deficits if one goes beyond the general diagnostic label and takes a closer look at the symptoms. I will start this review by examining the scant data on the phonological deficits in language production found in various types of aphasic disturbances. Phonological deficits usually differ from one patient to another, both in severity and kind, as a function of the aphasic condition involved and its associated symptoms. Next, I will examine a few studies concerned with deficits in word comprehension and production, concentrating on the ensuing patterns of associations and dissociations of symptoms. Finally, I will focus on the patterns of deficits found in *agrammatic* patients, both in comprehension and production at the sentence level, as shown in a more extensive body of work. *Agrammatism* is a cover term for language impairments affecting grammatical morphology and syntax. Research carried out in this area is particularly valuable for linguists, given the peculiar characteristics of the Spanish grammar in terms of its richer grammatical morphology and greater flexibility in word order when compared with English.

2.1 *Phonology*

Phonological errors in word and sentence production in Spanish monolingual patients are relatively common in most *aphasic disorders*. These errors, usually referred to as "literal" or "phonemic" *paraphasias*, consist of substitutions, deletions, additions, or exchanges of phonemes. Phonemic paraphasias are more common cross-linguistically in so-called "motor aphasias," such as agrammatic Broca's aphasia, conduction aphasia, and anomic aphasia, than in "receptive aphasias" like Wernicke's or jargon aphasia. Broca's aphasia is one of the classic syndromes found in the literature on acquired language disorders. Its major symptom is agrammatism, a deficit in morphosyntactic abilities that results in the production of effortful speech with simple sentences and omission of some functor words and inflections. Conduction aphasia is characterized by difficulties in repetition, with unimpaired spontaneous language comprehension and production in most cases. As for anomic aphasia, its most distinctive feature is the difficulty in word finding, which brings about a nonfluent speech style, but with spared comprehension abilities. Wernicke's aphasia, another classic syndrome reported in the literature, consists of a severe disturbance of language comprehension and the production of a jargon-like discourse with many lexical errors (i.e., paraphasias).

As for phonemic paraphasias in Spanish-speaking patients, substitutions appear to be the most common mistake, and exchanges the least, in all kinds of aphasic disorders in Spanish (Ardila et al. 1989). This kind of errors should be distinguished from "verbal" paraphasias or semantic substitutions. These appear most commonly in Wernicke's aphasia, a very severe language disturbance characterized by a lack of control in the retrieval of linguistic units, and a substantial breakdown of both comprehension and production abilities.

The distribution of phonemic paraphasias in Spanish patients is heterogeneous across different types of deficit and various kinds of tasks (e.g., oral description of pictures, object naming, and repetition). Thus, omissions of phonemes tend to be more frequent in agrammatic aphasia, whereas substitutions are more characteristic of Wernicke's and conduction aphasics. In addition, performance in word and sentence repetition appears to be significantly damaged in agrammatic, conduction, and Wernicke's aphasics, but fairly spared in anomic patients, who conversely experience more difficulties in picture description and word naming relative to the rest of their linguistic abilities, due to their word-finding problems (Ardila 2001). This heterogeneous pattern in the distribution of phonological errors is not peculiar to Spanish, and conforms to the specific underlying deficit that is distinctive of each aphasic syndrome. It should be stressed at this point that the phonological deficits described in this section are arguably grounded in more central underlying impairments. For instance, it has been claimed that agrammatic as well as conduction patients have trouble integrating linguistic information in working memory, and are therefore prone to having difficulties in repetition tasks, especially those involving sentences of relative structural complexity. In contrast, the central (and sometimes the only) deficit of anomic aphasics is a word finding difficulty, which accounts for their poor performance in naming in the face of a comparatively better preserved repetition ability.

An interesting peculiarity of phonological deficits found in Spanish-speaking patients is that phonological errors more frequently involve consonants than vowels (Ardila 2001). This observation squares with the results of the analysis of phonological errors (slips of the tongue) of healthy Spanish speakers (García-Albea et al. 1989). The greater vulnerability of consonants relative to vowels has been reported in different forms of aphasia, both nonfluent (i.e., agrammatism), and fluent (i.e., Wernicke), and in both production and perception deficits (Ferreres et al. 2003). The latter finding lends support to the hypothesis that the advantage of vowels over consonants is not a matter of relative articulatory difficulty, but rather stems from the fact that vowels are more perceptually salient than consonants in Spanish, and that syllables in this language are assembled around the vocalic nucleus. Moreover, given the comparative simple Spanish vocalic system (with only five categories), vowels are more frequent than consonants, which makes them less vulnerable to failure (Ferreres 1990). Of course, this is not just a matter of frequency yielding more opportunities for failure in absolute terms. The relevance of frequency in this regard is that it leads to increased practice, which results in more robust phonemic categories. The "frequency effect" is a widespread phenomenon across all language processing levels, with more frequent units (be they segments, words, or morphemes) being more resistant to error than less frequent ones on a regular basis.

Phonological deficits are also claimed to be the underlying cause of some forms of both acquired and developmental reading impairments (identified under the labels of *alexia* and *dyslexia*, respectively). This is the case especially in languages like Spanish that have a shallow orthography, that is, a mostly regular print-to-sound correspondence, and thus allow for or facilitate a grapheme-to-phoneme conversion process during reading. However, I will postpone the discussion of this matter until a later section of this chapter.

2.2 *Lexicon*

Difficulties involving the recognition and production of spoken words are prevalent in most, if not all, varieties of language disturbances. Impairments of lexical processing, which come under the general label *anomia*, may occur at different levels in the array of processes that range from the recognition of speech sounds from acoustic cues, the mapping of phonemes to word forms stored in the input lexicon and the activation of word meanings, to the selection of word forms from lexical meanings, the activation of phonological codes in the output lexicon, and the mapping of an ordered sequence of segments onto articulatory gestures. The most commonly studied problems along this chain of events are the different varieties of *word deafness* (i.e., problems of word recognition), impairments involving the semantic processing of words, and naming difficulties. However, the label "anomia" is usually restricted to word naming problems. The problems in lexical processing just described affect the "open class" vocabulary, that is, words belonging to the categories of nouns, verbs, and adjectives, as well as nonproductive derivational morphemes, as opposed to the closed class set of words and morphemes, which includes functor words and grammatical morphemes.

A review of the studies of lexical impairments in Spanish yields a very similar picture to that found in other languages in terms of the major associations and dissociations observed among symptoms. The most common dissociation (in fact, one of the best documented "double dissociations"[1] found in the literature on aphasia) is that shown by two different kinds of anomia, one involving a semantic impairment with no phonological alteration, and the other consisting of a phonological deficit with a spared semantic system (Martín 1999). Another dissociation reported in previous case studies of anomic patients in other languages is that between "word meaning deafness," a severe impairment in the auditory comprehension of language, and a preserved ability to carry out semantic judgments with nonlinguistic (visual) material (e.g., picture matching and categorization). This dissociation has recently been reported in Spanish as well (Martín et al. 2006), and allegedly demonstrates a functional (and likely anatomical) distinction between conceptual–semantic processing performed in a nonlinguistic fashion and the processes and cerebral mechanisms that provide access to conceptual information from linguistic input.

Besides supplying cross-linguistic evidence about the recurrence of certain patterns of associations and dissociations of symptoms in aphasic disturbances, research on acquired language impairments in Spanish also provides valuable descriptions of the patterns of errors found in different pathological conditions. In this respect, Spanish need not follow the same pattern as other languages. As regards lexical disturbances in Spanish, it is worth reporting a different distribution of types of lexical errors (paraphasias) across aphasic conditions. An interesting contrast is found between fluent and nonfluent aphasias in terms of the relative distribution of literal (phonological) and semantic paraphasias. Both types of errors are reported to occur to a similar extent in fluent aphasias (i.e., Wernicke) in Spanish, whereas literal paraphasias outnumber semantic paraphasias in nonfluent aphasias (agrammatism) in a 5-to-1 ratio (see Ardila 2001). Furthermore, it is interesting to note that literal paraphasias are virtually absent in phonological anomia, a disconnection between conceptual representations (or word meanings) and phonological codes of words. Apparently, when word search fails, patients resort to the strategy of using circumlocutions (i.e., definitions or descriptions of the concept or object whose name they are seeking) and occasionally incur semantic paraphasias. As for the similar proportion of both kinds of lexical errors in fluent aphasias, it is the result of the severe breakdown of both semantic and phonological representations that is at the root of this impairment, which consequently disturbs word comprehension as much as word production.

A final point that deserves some attention is the use of neuropsychological data to test hypotheses concerning the variables that govern the processes under scrutiny (lexical selection, in this case), and that are impaired in aphasic disturbances. There is an old debate in Psycholinguistics and Neuropsychology of language about the role of word frequency, as opposed to other lexical variables, in lexical access and recognition. Frequency of use has been shown to affect word recognition and production in a number of tasks, including word and picture naming. What is under discussion is the locus of the frequency effect (the processing level at which it operates) and the source of the effect itself. Some authors claim that it is not frequency *per se* that affects word processing, but other confounding factors, such as familiarity or age of acquisition (AoA), that are highly correlated with word frequency. The study of aphasic patients who experience difficulties in naming pictures has provided an opportunity to tease apart the effect of frequency, AoA, imageability, and other lexical variables in the difficulties faced by anomic patients in naming tasks. Recent results from studies of Spanish patients seem to show that AoA is a stronger determinant of lexical selection, with words learned later in life being more difficult to access than those acquired earlier, and thus more prone to failure in picture naming (Cuetos et al. 2005).

2.3 *Morphosyntax*

The study of the disorders of morphosyntactic processing in Spanish-speaking aphasics has a relative short history, beginning in the mid-1990s. The picture that emerges from these early studies is not very consistent in terms of the patterns of deficits found across patients. Broadly speaking, there is a general trend towards a selective deficit in agrammatic patients affecting some aspects of grammatical morphology (omission or substitution of some functor words and grammatical morphemes) and sentential syntax (difficulty in processing noncanonical constituent order). These problems are displayed both in comprehension and production tasks. I will sketch a brief review of the main findings of this research, and then I will address a handful of studies that have tested hypotheses about the major causes of the loss of grammatical abilities in agrammatic patients who are native speakers of Spanish.

The main two difficulties shown by Spanish-speaking agrammatic patients are the omission of articles in spoken sentences and the inability to understand and produce sentences with noncanonical SVO constituent order (Benedet et al. 1998, Ostrosky-Solís et al. 1999, Ardila 2001). Other reported symptoms are the frequent omission of clitics and a failure to correctly interpret topicalized active sentences of the form VSO (Reznick et al. 1995; Benedet et al. 1998). In contrast, Spanish agrammatics exhibit a good performance in subject–verb agreement, as well as a better than chance ability to use a range of grammatical markers and functor words, especially those that have a high degree of "cue-validity" (Bates and MacWhinney 1989). Cue-validity can be defined as the degree of consistency in meaning and reliability in use that a given morpheme exhibits in the language. In Spanish, for instance, the "-s" plural marker has a high degree of cue-validity since it is mostly consistent (bears the same meaning) and highly reliable (it is regularly used to express plurality as a morpheme). Likewise, some prepositions like "a" (*to*) are also fairly high in cue-validity, since they are consistently used to mark animate verb-objects (e.g., 'He visto *a* Juan,' 'I've seen (to) John'). Different studies on agrammatism in Spanish have shown a preserved use of the preposition "a" in object position (particularly in canonical active sentences), although the patients' performance in this particular issue is still significantly poorer than that of neurologically healthy control subjects. On the other hand, performance drops when grammatical markers that have a low degree of cue-validity (e.g., prepositions like "por"; translated as *by* when introducing postverbal passive agents, but having many other uses as well, such as locative and causative) are embedded in noncanonical structures, such as passives. Recall that passive constructions are not only noncanonical in Spanish, like cleft structures, but also very seldom used. A more recent study focusing on prepositions (Reyes 2007) has shown that agrammatic performance is linearly related to grammatical markers' degree of semantic transparency and consistency in use, with patients achieving the best results with locative prepositions, followed by temporal, case-marking, and phrasal prepositions, which are lexically dependent, at the poorest end of the continuum.

There are other features of agrammatism in Spanish that are worth mentioning. These include: overuse of strong pronouns in subject position (as known, Spanish is a pro-drop language in which the omission of subject pronouns is the default case), the avoidance of complex and less frequently used verb forms, and the consequent over-reliance on simpler forms like the infinitive, the present and the preterite (Reznick et al. 1995; Centeno and Obler 2001). The relevance of this latter symptom will become evident later.

Over and above the description of agrammatic symptoms provided by the above mentioned studies, researchers have also been concerned with the underlying deficits in agrammatism in an attempt to make sense of the pattern of impaired and preserved grammatical abilities shown by patients. A hotly debated issue in the aphasia literature is whether the agrammatic impairment is linguistic in nature, that is, whether it is due to a disruption of morphological or syntactic processes *per se*, or should be better explained by a shortage or limitation in the working memory capacity of the affected subjects. To my knowledge, there is one study that has addressed this issue with Spanish-speaking aphasics (Miera and Cuetos 1998). By comparing two groups of patients (agrammatic and anomic) with a matched control group of healthy subjects in sentence-picture matching tasks and several other working memory tasks, it was found that agrammatic patients showed a significant difficulty in processing certain syntactic structures when compared to the other two groups. On the other hand, the three groups of participants did not differ in their working memory abilities. These results lead us to conclude that the agrammatic deficit is probably not caused by a problem of memory capacity, but is rooted in a structural deficit.

Another issue under discussion is the role played in grammatical deficits by the frequency of use of morphemes, with less frequent units being more vulnerable to damage than more frequent ones, in comparison with the regular or irregular character of the forms, that is, their construction by rule or their retrieval from memory as ready-made entities. According to some studies, Spanish-speaking agrammatic aphasics reveal a reduced use of verb forms, practically restricted in spontaneous conversation to the present tense, and including the preterite in more controlled tasks like repetition or elicitation. This result suggests that high-frequency verb forms are more resistant to damage because they have become rutinized and overlearned (Centeno and Obler 2001; Centeno and Smith Cairns 2010).

On the other hand, the contrast of two highly frequent forms that differ in regularity provides the opportunity to test the role of this second factor in aphasics' performance and to disentangle its effects from those brought about by sheer frequency. This requirement is met by the Spanish verbs 'ser' and 'estar,'[2] two highly frequent copular verbs that differ in that the former is highly irregular, whereas the latter is regular. A sentence completion study administered to two agrammatic aphasics, where the subjects had to fill in the correct form of either of the two Spanish copular verbs, yielded a better performance for 'estar' than for 'ser,' with more substitution errors for the latter verb (O'Connor et al. 2007). Moreover, in accordance with results from previous studies, subjects showed a tendency to overuse the present tense and to substitute singulars for plurals in tensed forms. These results align with those found in a previous study with Spanish–Catalan bilinguals, who showed a better performance in a broad range of regular verbs (as opposed to irregulars) in a morphological transformation task (i.e., switching the verb from one tense to another in a sentence completion task) (De Diego et al. 2004).

The relative preservation of regular morphology in agrammatism does not match traditional descriptions of this linguistic disorder, which emphasized a selective damage of a hypothetical syntactic module caused by a loss of the closed-class vocabulary. Moreover, it poses a challenge to extant models of sentence processing based on a sharp distinction between lexical–associative processes, based on retrieval operations, and rule-based combinatorial processes of grammar, like the declarative–procedural model (Pinker and Ullman 2002). At the least, these data should compel researchers to take a closer look at the structure of the syntactic component of language and distinguish between different subcomponents that may be selectively vulnerable to damage in aphasic disorders. Studies on languages with a rich morphology, like Spanish, will surely provide valuable insights and data to tackle this complex issue.

The cross-linguistic analysis of agrammatic deficits has also proved to be relevant in the assessment of hypotheses about the linguistic mechanisms involved in sentence derivation. In particular, the observation that agrammatic speakers perform at chance levels when processing sentences that contain traces resulting from movement operations (e.g., dislocated or cleft sentences, passive sentences) has led some authors of the generative grammar persuasion to postulate an underlying deficit that disrupts the process that assigns thematic roles to structural positions in the phrase marker. This deficit may be caused by an inability to identify the traces of moved constituents (the Trace Deletion Hypothesis) (Grodzinsky 1995), by a difficulty in mapping thematic roles to syntactic positions (Linebarger et al. 1983), or by a more specific problem in assigning thematic roles to traces where there are "double dependencies" between traces and their antecedents (the so-called Double-Dependency Hypothesis) (Mauner et al. 1993). Double dependencies arise when there are two referential dependencies in the sentence, as it occurs in full passives, where the external argument (the patient) is linked to a null element licensed by the passive morphology of the verb (e.g., A boy$_i$ was chased t$_i$) while the agent role has to be linked to the past participle inflection (e.g., 'was chased$_j$ by a girl$_j$').

The study of languages that admit a range of different word orders offers an advantageous chance to test these hypotheses. Spanish is a case in point since it admits a number of possibilities in this regard: from canonical active sentences to cleft actives with clitic doubling (e.g., 'A la jirafa la mujer *la* está empujando' lit. 'To the giraffe the woman *her* is pushing'; 'It is the giraffe that the woman is pushing'), to passives, and even scrambled passives (e.g., 'Por la mujer la jirafa está siendo empujada' lit. 'By the woman the giraffe is being pushed'). So far, the results appear to favor hierarchical theories that predict special difficulties when processing sentences with double dependencies when compared to other structures with traces (Beretta et al. 2001). However, more research is surely needed to settle this issue.

In connection with this, I would like to underscore an important contribution to linguistic theory drawn from the studies of agrammatic speakers of Spanish. The dissociation between tense inflection, which is usually disrupted in Spanish agrammatic patients, and subject–verb agreement, which is spared (note that the complementary dissociation has never been reported), shows that agrammatism does not involve a complete loss of syntax, as it had been claimed over the years in the literature on aphasic disorders. At the same time, it provides psycholinguistic evidence supporting the distinction between tense and agreement as two separate functional categories in the syntactic tree (Friedmann 2001).

3 Written language impairments in Spanish

Part of our understanding of normal reading and writing comes from the study of reading and writing disorders that appear either as a consequence of brain damage in adults (acquired *alexia* and *agraphia*) or as a result of difficulties in the acquisition of literacy skills in children. In this section, I will review some relevant findings drawn from neuropsychological studies of patients with reading disorders in Spanish. Studies on acquired writing impairments in Spanish are rather scarce, and generally show parallel symptoms to those found in reading deficits. Contemporary models of the processes subserving the extraction of meaning from printed texts are largely based on the study of patterns of associations and dissociations of symptoms in reading-disabled people (Coltheart et al. 2001). These models agree on the fact that, at least in languages that have alphabetical writing systems, readers rely on two alternative pathways when recognizing written words, hence the name of *dual-route model* to characterize this sort of models. One route is known as the *lexical–semantic* pathway. This route is available for reading known words by accessing the semantic system directly from the input orthographic lexicon, but it also includes a bypass route (the *direct lexical* pathway) that enables the reader to decode texts without accessing the meaning of words, thus sidestepping the semantic system. The other route is the *phonological (nonlexical)* pathway, which is used to read nonwords as well as uncommon or foreign words. This latter route requires the translation of graphemes to phonemes by means of rules and the subsequent activation of phonological codes, whereas in the lexical pathway words are recognized solely on the basis of orthographic representations.

The study of dyslexic patients has shown that either pathway (or parts thereof) can become impaired as a consequence of brain damage, leading to double dissociations among patients. Thus, damage to the phonological pathway results in an inability to read nonwords as well as long and uncommon words, yielding a reading disturbance known as *phonological dyslexia*. Conversely, a disruption of the lexical pathway causes failure to read irregularly spelled words (such as 'yacht' or 'pint'), which are often regularized (e.g., /yatched/,/pɪnt/), leaving intact the capacity to read nonwords. This difficulty is referred to as *surface dyslexia*. A third kind of reading disorder is *deep dyslexia*, a more severe condition that affects both the phonological route and the access to the semantic system, and which results in semantic substitutions (e.g., reading 'wallet' for 'money', or 'ship' for 'captain') and incapacity to read nonwords.

However, the availability and the actual use of the reading pathways appear to vary from one language to another, and even among readers of the same language, depending on their expertise. Thus, languages may be roughly classified in two broad categories according to the systematicity of letter-to-sound correspondences: opaque languages, like English or French, where irregular spellings (also called "exception" words) are quite usual, and transparent languages, like Spanish, Italian, or Turkish, whose orthographies convey the pronunciation of words in a systematic way. Nonalphabetic or logographic languages, like Chinese and the Japanese kanji script where there is no rule-based correspondence between script and sound, are an extreme case of orthographic opacity. Spanish, on the other hand, has orthographic rules that reliably allow readers to retrieve the correct pronunciation of words and nonwords, though it contains ambiguous letters ('c,' 'g') whose pronunciation depends on the context (i.e., the following vowel) and letter pairs (like 'b–v,' or 's–z') that are pronounced the same or differently depending on the dialect, which gives rise to some confusion. A further source of difficulty in this language is the system of rules that establish the use of graphic stress.

The interesting point to make in this regard is that the transparent nature of the Spanish orthographic system, and the reliability of grapheme-to-phoneme correspondences, would appear to render unnecessary the lexical pathway, and would lead us to expect a regular use of nonlexical strategies in reading. If this were the case, there would be no trace of surface dyslexia in this language. However, as the research literature on acquired reading and writing disorders clearly shows, Spanish is no different from orthographically opaque languages in this particular issue. A number of studies on dyslexic patients report cases of phonological, surface, and deep dyslexia in Spanish-speaking patients (Ardila 1991; Davies and Cuetos 2005; Ferreres et al. 2005; Iribarren 2007; Weekes 2007; for a comprehensive review of earlier studies of acquired and developmental dyslexia in Spanish, see Valle-Arroyo 1996).

Reports of phonological dyslexia in Spanish exhibit a typical pattern of breakdown in the phonological route, with severe difficulties in nonword reading and a close to normal performance in word reading (Cuetos et al. 1996; Iribarren et al. 1999). These patients read short words better than long ones, show a similar ability to read regular and irregular words, and usually make some morphological and visual errors (substituting inflections and changing letters). These cases suggest that reading through the lexical pathway is an option in Spanish.

Cases of surface dyslexia have also been reported in Spanish, despite its transparent orthography. Given the *a priori* difficulty to detect a failure in the lexical route in Spanish, different assessment strategies have been developed to uncover a possible damage in this reading pathway. One consists of using printed words without graphical stress that depart from the most common stress pattern in Spanish, which falls on the penultimate syllable; words like 'corazón' (*heart*) or 'aquí' (*here*), bearing stress on the last syllable, might be mispronounced as /ko-ˈɾa-θon/or/ˈa-ki/, with penultimate stress, when presented in capitals ('CORAZON,' 'AQUI'). Another strategy that has been used in Spanish adaptations of neuropsychological tests consists of using abbreviations (e.g., 'Avda' for 'avenida' (*avenue*) or 'Ud' for 'usted', the 2nd person singular pronoun used in formal treatment) in recognition or reading aloud tasks. These lexical items are a good test of the use of the lexical route in reading, due to their irregular pronunciation. Thus, they are likely to yield errors when their regular pronunciation is activated. A third way to reveal a disruption of the lexical route is the presence of errors when reading heterographic homophones, that is, words that are spelled differently but share the same pronunciation (e.g., 'plain' and 'plane,' 'I' and 'eye'). Homophones become indistinguishable when they are read through the phonological pathway, due to their identical pronunciation. Despite the fact that they are very scant in Spanish, virtually restricted to minimal pairs having ambiguous letters (e.g., 'vaca–baca' *–cow–rack*) or involving the silent 'h' (e.g., 'hola–ola' – *hello–wave*), they are regularly used in neuropsychological tests to reveal possible difficulties with the lexical route.

A few reports have been published of cases of surface dyslexia in Spanish. Iribarren et al. (1996) reported the case of a patient who made stress assignment mistakes when reading. More recently, Ferreres et al. (2005) have described the case of another surface dyslexic who showed a relatively good, albeit slower than normal, performance with words and nonwords in recognition and reading aloud tasks, but made a substantial amount of errors and confusions when reading homophones and pseudohomophones, the latter being nonwords that are pronounced as real words (e.g., 'votella' or 'relijión,' misspellings of the Spanish words 'botella' (*bottle*) and 'religión' (*religion*)). Arguably, these patients have the lexical pathway for reading damaged. This failure causes them to rely on the phonological pathway, which results in the slower performance and the high rate of confusions observed in reading tasks.

Finally, there are also a few neuropsychological studies reporting cases of deep dyslexia in Spanish (Ruiz et al. 1992; Ferreres and Miravalles 1995; Iribarren 2007; Davies and Cuetos 2005). In agreement with reports of cases from other languages, Spanish-speaking deep dyslexics also show the two typical features of this disturbance, namely, the presence of semantic paralexias (word-meaning substitutions) and a severe deficit in nonword reading. These symptoms are also accompanied by naming difficulties (anomia), as reflected in the overuse of circumlocutions (e.g., 'otro por mí' (*someone in my place*) when reading 'representación' (*representation*)), and visual and morphological errors.

By way of a closing remark, the study of acquired reading impairments in Spanish has served to underscore the adequacy of dual-route models. That is, insofar as the patterns of impairment and preservation of reading skills that issue from this research reveal the availability of both lexical and phonological pathways for readers of an orthographically transparent language. Thus, in this particular research area, the specific features of the Spanish orthography do not deliver any language-dependent outcomes.

4 Language impairments in bilingual Spanish Speakers

The concern for linguistic impairments in bilingual populations of speakers of Spanish, preferentially as a native language, is manifold. On the one hand, there is a theoretical interest in finding out whether knowledge of more than one language has any bearing on the pattern of deficits found and the course of recovery experienced by bilingual speakers, as compared to monolinguals, when they suffer some form of linguistic impairment (Paradis 2001). On the other hand, bilingual language impairments (and bilingual processing at large) offer the opportunity to test hypotheses drawn from models of language processing that apply to normal and impaired performance. Such hypotheses address issues like controlling the activation of languages and shifting between them when required, the type of links and relations that hold between the bilingual's lexicons, or the specificity of linguistic disorders in bilingual aphasics, among others. Another point of concern for researchers and clinicians in this field are the teachings that can be garnered from research on language pathology in bilinguals for the purpose of designing more efficient intervention procedures to help patients recover from their illness. I will briefly address each of these questions.

There is an ample variety of patterns of deficit and recovery in bilingual aphasics, depending on a number of factors, such as the typological distance between the languages, the age of acquisition and degree of proficiency of the speaker in each one, his or her usage preferences, and even the patient's premorbid resources and abilities (e.g., literacy). Generally speaking, language loss may occur in a parallel fashion in both languages (the most frequent outcome), or may selectively affect one of them, usually the lesser used or the last one to be acquired. The patterns of spontaneous recovery are more varied, though the most usual result is once more the parallel recovery of skills in both languages, and to a much lesser extent, the differential (i.e., partial) or selective (i.e., total) recovery of one of them at the expense of the other. There are other more exotic patterns of recovery, which account for a few cases of bilingual aphasia, such as the successive retrieval of languages, or the "antagonistic" recovery, in which gains in one language occur at the expense of losses in the other (Fabbro 2001).

Case reports of aphasia in Spanish-speaking bilinguals show the same categories of disturbances and similar frequency distributions as in the general bilingual aphasic population (see Paradis 2001, and references therein, for a general review). In particular, when the patient's languages are closely related typologically (such as Galician, Catalan, and Spanish), the patterns of impairment and recovery are usually proportional to the degree of premorbid proficiency in each language. Nevertheless, some cases have been reported of language-selective impairment of particular abilities or tasks, such as a specific naming deficit occurring in one language while the same ability is spared in the other. These selective deficits usually affect the most recently acquired language (Méndez et al. 2004), but they may also injure the less dominant regardless of the order of acquisition (Gómez-Tortosa et al. 1995; García-Caballero et al. 2006; Ardila 2008). In either case, such reports have been sometimes taken as evidence for an anatomical (as well as functional) segregation of languages in the brain (but see Paradis 1996 for a critical appraisal of this view).

A handful of recent studies carried out with Spanish–Catalan bilingual patients showing naming difficulties has revealed an interesting dissociation between verbs and nouns. In one study (Hernández et al. 2007), a patient with Alzheimer's disease showed greater difficulties with nouns than with verbs, while another study of the same research team (Hernández et al. 2008) reported the case of another patient with nonfluent progressive aphasia (a neurodegenerative disorder) showing the opposite pattern. Interestingly, both patients were highly proficient in their two languages, and had very similar difficulties in both. These results may be taken as evidence of a similar organization of lexical knowledge in highly proficient bilinguals, especially if they speak closely related languages. Also drawing on evidence from Spanish–Catalan balanced bilinguals, it has been shown that a semantic deficit (e.g., in patients with dementia of the Alzheimer type) affects to the same extent the transition from one language to the other in either direction (i.e., forward and backward translation) (Hernández et al. 2010). This result supports the view that even in languages that have a high degree of lexical overlap (i.e., with many cognate words), translation is performed through conceptual links.

Another issue that has roused the interest of researchers in bilingualism is the practice of mixing or switching languages during conversation. This practice is quite standard in many bilingual communities, and has been studied from different points of view: as a conventional practice subject to social norms and customs, as a linguistic activity that conforms to grammatical constraints and rules, and as a cognitive phenomenon that requires control mechanisms to switch languages at appropriate moments and thus may expend attention and memory resources. Language mixing appears to be more common in bilingual aphasics than in neurologically healthy bilinguals (Muñoz et al. 1999). More importantly, in some cases it stands out as an aberrant behavior, insofar as it is not under conscious control and often results in a violation of linguistic constraints that are usually obeyed by normal speakers. However, some authors view language mixing as a strategy used by people with language impairments to overcome their linguistic deficits (Ansaldo and Marcotte 2007). In this regard, the specific characteristics of language mixing in individual patients may provide clues to understanding the type and severity of their linguistic impairment, and may be a useful tool in the rehabilitation of bilingual aphasia.

5 Developmental disorders of language in Spanish: The Case of Specific Language Impairment

Specific Language Impairment (SLI) is a language learning disability that occurs without major neurological, cognitive, or motor and sensory disturbances. The adjective "specific" is perhaps misapplied to SLI as a whole since this label encompasses a heterogeneous collection of language delays and deficits that arise in the course of acquisition of the first language. The most (maybe the only) selective impairment within SLI is the morphosyntactic variety, which consists of a deficit in grammatical morphemes, pronominal clitics, and inflections. This yields morpheme omissions and agreement errors at the output, and causes related difficulties in language understanding.

The data supplied by studies of SLI with Spanish-speaking children exhibit a somewhat varied pattern. The main symptoms reported are the omission of articles in noun phrases (especially definite articles like 'el' or 'la'), gender-agreement errors within the NP, omission of direct object clitics ('lo' or 'la'), and errors in verb inflections, especially involving number (i.e., substituting singular for plural forms) (Bosch and Serra 1997; Bedore and Leonard 2001; Anderson 2007). This general pattern of deficits is consistent with the findings reported in other Romance languages, such as Italian or French. However, these symptoms are not uniformly present across studies of SLI in Spanish, and seem to be dependent on other modulating factors. For instance, according to some reports, gender substitution errors are more frequent in Spanish-speaking children that present with a phonological deficit associated to the grammatical impairment. This observation supports the claim that SLI is a secondary disorder that originates in a more basic phonological disturbance affecting the perception of unstressed morphemes (see Anderson 2007). An alternative to this processing account is the view that SLI is grounded in a linguistic deficit which has been named *feature blindness* and consists of a failure to process formal features of grammar that do not have a transparent form-meaning correspondence, that is, that are not semantically motivated; such is the case of agreement features and verb inflections. Needless to say, cross-linguistic research provides a privileged means to examine these issues, especially if the languages studied have the kinds of processing components that are targets of a potential disturbance.

The SLI label includes other varieties of language disorders, such as the semantic–pragmatic deficit, where basic phonological and grammatical skills are preserved yet there is a failure located at the discourse level and the child's communicative abilities. Children with a semantic–pragmatic deficit experience difficulties in integrating meaning in connected discourse, adjusting to their partners in conversation, and understanding their communicative intentions. In fact, this variety of SLI is similar in many respects to the impairments that are found in other cognitive disturbances like the autism spectrum disorder and Williams syndrome.

6 Concluding remarks

This chapter has addressed the issue of linguistic impairments in Spanish-speaking populations. Of all the language disorders described in the literature, I have focused on those that selectively disturb one or more components of the user's linguistic competence. These impairments may arise from genetic malformations, brain focal damage, or degenerative neurological diseases. A key assumption that guides research on this topic is that a close examination of the patterns of disturbed and preserved linguistic and cognitive abilities of patients (i.e., associations and dissociations) provides valuable information about the structure and operation of the computational systems that subserve language behavior. Cross-linguistic data are extremely useful to settle this issue because they set constraints on the possible variation of language impairments, and at the same time they allow us to explore language processes in more detail.

Most of the evidence supplied by Spanish data has contributed to confirm hypotheses and theories originally grounded on data from other languages. As an example, recall the evidence provided by the study of acquired dyslexia in Spanish supporting the existence of alternative pathways in reading: a lexical route whereby written words are recognized as a whole from "undecomposed" orthographic representations, and a phonological route in which words are identified (and nonwords read) by activating phonological representations. However, part of the evidence reviewed in this chapter has been used to shed light on controversial processing issues. For instance, the study of Spanish-speaking agrammatic aphasics suggests that syntactic disorders are more wide-ranging than it was previously believed, and that certain conceptual distinctions made in linguistic theory (e.g., between tense and agreement functional categories) may turn out to be relevant in terms of processing.

Finally, a conclusion that can be drawn from the preceding review of facts is that the scientific study of language impairments, like any other scientific endeavor, ought to be a joint enterprise of different disciplines. Language pathology could not even begin to be understood without the contribution of linguistic theory, on the one hand, and psycholinguistic processing models, on the other. This chapter has attempted to argue for the complementary contribution of the study of language pathology to linguistics and psycholinguistics.

NOTES

1 Double dissociations are complementary patterns of symptoms observed in different patients, with one patient showing a damaged process or component of language (say, the ability to understand written words) while keeping intact a different one (e.g., understanding spoken words), and another patient showing the opposite pattern of impaired and spared abilities. Double dissociations are commonly understood as evidence of a modular organization of linguistic processes, each supported by a different neural structure.

2 Both mean 'to be,' but whereas 'ser' expresses an existential meaning denoting a permanent trait or condition ('Juan es español' = 'John is Spanish'), 'estar' conveys a temporary meaning that depicts a state or transitory condition ('Juan está contento' = 'John is happy').

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40 Lexical Access in Spanish as a First and Second Language

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1 Introduction

Bilingualism plays an increasingly important role in contemporary society. Since the twentieth century, globalization has helped to open linguistic frontiers and increase the contact between different languages and cultures. As a result, knowing several foreign languages is often an absolute necessity. For many educational institutions it has become important to adopt appropriate language-teaching programs, and more and more parents have to make decisions concerning the linguistic education of their children (how many languages can and should be learned simultaneously in infancy, when the learning of foreign languages should start, etc.). Crucially, resolving such issues of both social and personal significance hinges on a thorough knowledge of what, linguistically and cognitively, being a bilingual involves, thus highlighting the need for sustained research into the different facets of bilingualism.

The issues largely defining bilingual cognitive research revolve around the coexistence of two languages in the linguistic system of individuals. More specifically, researchers have wondered how the two languages interact with each other during language use, what the consequences of such interaction are for linguistic performance, and how bilinguals manage to control their language use so as to avoid interference from a nonintended language. Albeit to date our understanding of these issues remains relatively poor, some significant advances have been made in recent years. An important role in this research has been played by populations speaking Spanish as a first or second language, given the presence of large

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Spanish-speaking bilingual communities (United States, Spain, Mexico, etc.) and the fact that Spanish is a popular foreign language to learn. In this chapter, we review some recent work on bilingualism carried out with speakers of Spanish, focusing on the part of bilingualism research concerned with one of the key stages of speech production: lexical access.

This chapter is structured as follows. In Section 2, we outline some basic characteristics of lexical access in monolinguals and bilinguals. In Section 3, we describe how Spanish as a first and second language is affected by the other language spoken by bilinguals, citing some recent evidence speaking to this issue. In Section 4, we turn to the issue of how Spanish-speaking bilinguals control their lexicalization in a target language. In Section 5, we dwell on the consequences of learning Spanish in an immersion context. In Section 6, we summarize the main conclusions reached by the work we have reviewed. Throughout, we focus on studies belonging to the experimental psycholinguistics literature in order to add another perspective to the Hispanic studies described in this handbook.

2 Lexical access in speech production: representations, processes and variables

One of the central issues in speech production research is how speakers access and select from the lexicon those words that match their communicative intentions. In everyday language use, these processes are carried out with great agility: the average speech rate is estimated at two or three words per second, and the number of words stored in the lexicon of an adult literate monolingual is at least around 30,000 (note that for bilinguals this number is larger when the two languages are considered together). But what cognitive machinery allows speakers to achieve such remarkable efficiency?

There is a relative agreement that lexical access entails at least three different levels of representation (Dell 1986; Levelt 1989; Caramazza 1997; Levelt et al. 1999). First, a speaker has to select the concepts corresponding to entities in the preverbal message (e.g., "dog" or "a furry domestic animal"). Second, s/he has to access those lexical items from the mental lexicon which correspond to the chosen concepts, along with their grammatical properties (e.g., DOG, *noun*, *neuter*). Third, s/he has to retrieve the phonological units composing each lexical item (e.g.,/d//ɑ//g/).

It is furthermore generally assumed that the process of lexicalization follows a spreading activation principle (Collins and Loftus 1975) that operates within and between levels of representation. For instance, in the course of naming the word *dog*, not only is the concept "dog" activated, but also semantically related or associated concepts (e.g., "cat," "bark"). All activated concepts further spread part of their activation to their corresponding lexical items (CAT, BARK), although the lexical items corresponding to unintended concepts would be activated less than the one of the intended concept (DOG). The selection mechanism is sensitive to the level of activation of lexical items, so that the node which reaches the highest activation level is selected for production.

Lexical selection in bilinguals, however, posits additional demands on speakers. On the one hand, it is usually assumed that the two languages of a bilingual share a common conceptual store, at least unequivocally so for concrete objects (Potter et al. 1984; Green 1986, 1998; De Bot 1992; Kroll and Stewart 1994; Poulisse and Bongaerts 1994; Finkbeiner et al. 2002; Li and Gleitman 2002; but see Van Hell and De Groot 1998). However, given the arbitrariness of language and language-specific word formation rules, lexical items must be organized into two separate stores for bilinguals' two languages. In other words, for each concept (e.g., "dog"), a Spanish–English bilingual will have two lexical items associated with it: DOG and PERRO [Sp. dog]. Thus, in view of the spreading activation principle assumed to operate in monolingual lexical access, an important question emerges: does the conceptual system activate in parallel bilinguals' two languages when they wish to restrict lexicalization to one language? That is, when a bilingual decides to speak one of his/her languages (a process assumed to happen at the conceptual level, similarly to register choice: La Heij 2005), does the other language also receive some activation?

The most economical solution might be to restrict the activation flow to the response language (as activation of the nonintended language would potentially produce interference). Still, most models postulate that the nonresponse language also receives activation from the conceptual system; that is, that activation flow is language nonspecific (e.g., Potter et al. 1984; Green 1986, 1998; De Bot 1992; Kroll and Stewart 1994; Poulisse and Bongaerts 1994; Poulisse 1997; Costa et al. 1999; for a review, see Costa 2005; Costa et al. 2006). Furthermore, it also seems that the activation of the nonresponse language is not limited to the lexical level but spreads to the phonological level. Some evidence for such a conclusion comes from tasks involving cognate words. Cognates are translation equivalents in two languages which have a very similar phonological and orthographic make-up (e.g., *guitar* – *guitarra* for English and Spanish). Importantly, cognate words enjoy a robust processing benefit over noncognates (e.g., *dog* – *perro*). For example, in a picture-naming task consisting in naming black-on-white line drawings presented one by one on a computer screen, Catalan–Spanish bilinguals were faster in naming pictures with cognate names than the ones with noncognate names (Costa et al. 2000). Such facilitation was interpreted as suggesting that the phonological make-up of nontarget language words also receives activation. This is because, in the process of naming a picture with a cognate name, not only the target word (e.g., *guitar*) but also its translation (e.g., *guitarra*) would get activated at the lexical level. Then these words would both spread activation to their respective phonological segments, and, given the large phonological overlap, the segments corresponding to the target word *guitar* would receive additional activation.

In sum, evidence suggests that both languages of a bilingual get activated up to the phonological level in the course of speech production (although see Kroll et al. 2006 for a slightly different view). That is, in our example of a Spanish–English bilingual naming the word *dog*, the shared concept "dog" will send a proportion of its activation to the lexical representations DOG and PERRO (and also to e.g., CAT and GATO [Sp. cat], among others), which in turn will activate the phonological segments /d/, /ɑ/, /g/, /p/, /e/, /r/, /o/, /k/, /æ/, /t/, etc. This parallel activation may imply that the nonresponse language of a bilingual would influence production in the target language. Indeed, there is some recent evidence that this is the case, which we turn to next, focusing on those studies conducted with bilingual speakers of Spanish.

3 Lexical access in speech production in Spanish in bilingual contexts

Although being able to speak two languages has obvious advantages, this ability appears to come at a cost: bilinguals seem to incur certain linguistic disadvantages in both of their languages when compared to monolinguals. While in everyday life such disadvantages seem negligible and do not in any way detract from the communicative benefits of being a bilingual, in scientific terms it is important to describe them and pinpoint their origin. So far, research has shown that bilingual children have a reduced vocabulary size in both of their languages relative to monolinguals (Oller et al. 2007; Bialystok et al. 2009), a condition which may not be overcome even in adulthood (e.g., Portocarrero et al. 2007; but see Bialystok et al. 2008; Luo et al. 2010, for a different view on this issue). Furthermore, bilingual adults perform worse than monolinguals in various speech production tasks, some of which we describe below (e.g., Rosselli et al. 2000; Gollan and Silverberg 2001; Gollan et al. 2002; Rosselli and Ardila 2002; Gollan and Acenas 2004; Portocarrero et al. 2007; Ivanova and Costa 2008; Bialystok et al. 2008).

3.1 Effects of a second language on the production of L1-Spanish

Somewhat strikingly, bilingualism seems to result in a disadvantage in lexical processing even in the first-acquired language of bilinguals, reflected in the worse performance of bilinguals in tasks requiring access to the lexicon. More concretely, there is evidence that the lexical processing in Spanish spoken as an L1 by bilingual individuals is less efficient than that of monolingual speakers of Spanish. Such evidence comes from a variety of studies employing vocabulary tests, verbal fluency, tip-of-the-tongue state (TOT) elicitation, and picture naming. For example, when asked to name pictures from the Boston Naming Test (BNT), Spanish–English bilinguals were able to retrieve fewer words than English monolinguals in both of their languages (Kohnert et al. 1998; Roberts et al. 2002; Gollan et al. 2007). Similarly, when asked to name pictures corresponding to unfamiliar and infrequent objects, Spanish–English bilinguals compared to monolinguals experienced more TOT states (such states are characterized by a feeling that one knows the word, and sometimes an ability to retrieve grammatical gender and partial phonological information but not the object name in its entirety) (Gollan and Acenas 2004; Gollan et al. 2005; Gollan and Brown 2006).

The performance of Spanish–English bilinguals has further been examined by means of the verbal fluency task. In this task, participants are asked to generate, normally for one minute, as many exemplars as possible of a given semantic category (e.g., animals) or phonetic category (e.g., words starting with the letter F). Both modalities rely on vocabulary knowledge and executive control, but differ in the degree to which each one of these processes is involved. While the semantic fluency task is considered to rely more on vocabulary knowledge, the phonetic fluency task relies more on executive control. With this task, Gollan et al. (2002) found that college-aged English monolinguals produced more exemplars than Spanish–English bilinguals when they were instructed to use both English only and whichever language (English or Spanish) came to mind. Rosselli and Ardila (2002) obtained a similar disadvantage with healthy older adults. Apart from replicating the bilingual disadvantage in verbal fluency, Sandoval et al. (2010) set forth to distinguish between different explanations thereof. These authors observed fewer correct exemplars, slower first response times, and proportionally delayed retrieval for bilinguals relative to monolinguals, and in the nondominant relative to the dominant language of bilinguals. Interestingly, bilinguals also produced significantly more lower-frequency responses and more cognates than monolinguals. Sandoval et al. interpreted their findings as suggesting that cross-language interference had an important role in explaining the bilingual disadvantage regarding verbal fluency, as opposed to other explanations attributing the disadvantage to the reduced frequency of use of each of the bilinguals' languages compared with that of monolinguals (see below) or bilinguals' smaller vocabulary size.

The bilingual disadvantage effect in lexical access has also been observed in picture naming. Gollan et al. (2005) compared a Spanish–English bilingual group and an English monolingual group, and found that the bilinguals were reliably slower and less accurate when naming pictures in their dominant language (English). Importantly, however, when these two groups were tested on a picture classification task, they did not show any differences in performance, implying that the bilingual disadvantage does not originate at the semantic level.

Furthermore, Gollan et al. (2008) tested young and older Spanish–English bilinguals and English monolinguals in a picture-naming task conducted in both English and Spanish for the bilinguals. These authors specifically compared the bilingual disadvantage effect with the effects of lexical frequency. They found that the young bilinguals naming in English (their dominant language) were slower than monolinguals, but this bilingual naming cost was larger for low- than for high-frequency names. For older adults, however, while such a cost was also present, it was of a similar size for high- and low-frequency words. Gollan et al. interpreted these findings in relation to the origin of the bilingual disadvantage. In particular, they argued in favor of an account placing the origin of the bilingual disadvantage in frequency effects in speech production, and against an explanation of the disadvantage in terms of the interference caused by the nontarget language (see below for more on the explanations of the bilingual disadvantage effect).

One important point to keep in mind regarding the findings cited above concerns the language history of the Spanish–English bilinguals tested in the respective studies. More specifically, all of them were switched-dominance bilinguals: individuals whose L2, acquired around 3.5 to 4 years of age, has become their more dominant language. Since age of acquisition (AoA) is known to have robust effects on naming performance (e.g., Morrison and Ellis 1995, 2000; Gerhand and Barry 1999; Izura and Ellis 2004), it is important to establish to what extent the cost incurred by Spanish spoken as an L1 (and, more generally, by any L1) holds for populations for whom it is also the dominant language.

Such evidence was recently provided by Ivanova and Costa (2008). These authors conducted a picture naming task with three groups of speakers: Spanish monolinguals, Spanish–Catalan bilinguals (L1 Spanish), and Catalan–Spanish bilinguals (L2 Spanish; for the discussion of the results of this latter group, see the next section). Importantly, Spanish was the dominant language for the Spanish–Catalan group. Still, Ivanova and Costa also observed a naming disadvantage for this group relative to the Spanish monolinguals. In addition, similarly to Gollan et al. (2008), they found that the bilingual disadvantage was greater for low- than for high-frequency words.

While Ivanova and Costa's participants were tested on bare naming only (production of the referent noun alone: e.g., *car*), worse performance by bilinguals has also been observed in entire noun phrase production (e.g., *el coche rojo* ['the red car']), with the same population of Spanish–Catalan bilinguals (Sadat et al. 2011). These authors detected a bilingual disadvantage in both naming latencies and articulatory durations in L1. Interestingly, although the magnitude of the bilingual disadvantage in naming latencies was similar for noun phrase production and bare naming, there was a larger difference between bilinguals and monolinguals for the former task in articulatory durations. Such a pattern of results might indicate that, in order to achieve efficient performance, bilinguals do not plan the entirety of a multi-word utterance prior to articulation, but rather initiate the production process after retrieval of only one part of the utterance and continue with retrieval in the course of production.

3.2 Explanations for the bilingual disadvantage

So far, we have reviewed a number of experimental findings from various tasks and different populations, pointing to a bilingual disadvantage when speaking a first and nondominant, and even a first and dominant language, relative to monolinguals. But what might be the origin of such a disadvantage? As advanced in the previous section, a ready explanation for it is provided by the fact that bilinguals activate both of their languages in the course of speech production (see Costa 2005), that is, that they suffer from cross-language interference (e.g., Green 1998). As a matter of fact, however, it is not clear that dual language activation produces interference under all circumstances. Importantly, Costa and collaborators (e.g., Costa and Caramazza 1999; Costa et al. 1999) found that target pictures, presented together with the written translation of the picture name, actually facilitated performance. Converging work by Gollan and collaborators showed that bilinguals performed better with target words which they found easy to translate into their other language than with target words which they found difficult to translate (target words being of the same difficulty for monolinguals: Gollan and Acenas 2004; Gollan et al. 2005). In view of such data, it seems somewhat unlikely that cross-language interference, especially coming from a weaker L2, is large enough to account for the observed disadvantage. Still, such a possibility at present cannot be categorically excluded.

Another explanation for the bilingual disadvantage, known as the "weaker links hypothesis," postulates that the disadvantage is not due to the coactivation of the two languages, but is essentially reducible to a frequency effect (Mägiste 1979; Ransdell and Fischler 1987; Gollan and Silverberg 2001; Gollan et al. 2002; Gollan and Acenas 2004; Gollan et al. 2008). The logic behind this hypothesis is that bilinguals use each of their languages less often than monolinguals do. As a result, the lexical representations in each language of a bilingual would be retrieved less frequently than the corresponding ones of a monolingual, and, hence, the links between semantics and phonology will be weaker in the case of bilingual speakers. Note that the "weaker links hypothesis" predicts that lexical frequency effects would be most marked for the weakest language of bilinguals under the assumption that reducing the frequency of high-frequency words (as for the high-frequency words of bilinguals' dominant language) would still render them of high frequency, while reducing the frequency of low-frequency words (such as the low-frequency words of bilinguals' nondominant language) would render them of extremely low frequency. However, while such a prediction was supported in the study of Gollan et al. (2008), it was not by that of Ivanova and Costa (2008) (see below), and thus more work is needed in support of the validity of this hypothesis.

3.3 *Effects of a first language on the production of L2-Spanish*

We now turn to the effects in lexical processing of an L1 on the production of Spanish spoken as an L2. Given that the studies speaking to this issue were conducted in Spain and did not employ completely balanced bilinguals, L2 Spanish was the nondominant language of the bilinguals tested. Thus, even though these bilinguals were highly-proficient in both of their languages, there was a lexical processing cost to their L2 as compared to their L1 (see also the previous section).

For example, in picture naming, Ivanova and Costa (2008) showed that Catalan–Spanish bilinguals speaking in their L2 (the third group tested in this study, in addition to the Spanish monolinguals and Spanish–Catalan bilinguals) performed the task more slowly than both other groups (see also Sadat et al. 2011, for further evidence). These bilinguals, however, did not show a greater frequency effect than the bilinguals naming in their L1. This latter aspect of Ivanova and Costa's findings is at odds with the result obtained by Gollan et al. (2008) that the bilinguals naming in their nondominant language (L1 Spanish) showed a larger frequency effect than the bilinguals naming in their dominant language (L2 English). To explain the discrepancy of their results with those of Gollan et al. (2008), the authors tentatively proposed a contribution of AoA to the naming latencies observed in the two studies, since AoA largely correlates with frequency. In particular, while the AoA was the same for both the Spanish monolinguals and the bilinguals tested in L1 in their study, it was higher for the L1 bilinguals tested in Gollan et al.'s study relative to the monolinguals, and might therefore have added to what these latter authors reported as frequency effects.

With regard to the explanation of the disadvantage of the L2 Spanish of highly proficient bilingual speakers, there are many factors which conspire to produce the observed effects. In particular, this language is spoken less frequently (cf. the "weaker links hypothesis" put forward for the L1 disadvantage of bilinguals), is acquired later, and is of lower proficiency (although note that these effects may be highly correlated in practice). Thus, it is also more likely that the L2 is subject to interference from the more dominant L1 (see also the section on bilingual language control).

4 Language control in bilingual contexts (Spanish as L1 and L2): evidence from language-switching tasks

We now turn to the issue of how bilinguals control lexicalization in each of their languages so as to prevent interference from the currently nontarget language. In fact, bilingual speech presents an interesting paradox. On the one hand, there is evidence of parallel activation of both languages (e.g., Hermans et al. 1998; Costa et al. 2000; Colomé 2001). On the other hand, bilinguals rarely commit cross-linguistic involuntary intrusions (Poulisse and Bongaerts 1994; Poulisse 1997). Thus, these speakers must possess a powerful control mechanism, allowing them to avoid interference from the language they are not speaking in a given communicative situation.

Two different approaches have tried to explain how this control mechanism functions by making different assumptions about lexical access. More specifically, many authors agree that the lexical representations within a single language compete for selection, that is, that the lexical selection mechanism considers for production all activated lexical items (e.g., Roelofs 1992; Roelofs 1993; Caramazza and Costa 2000; but see Mahon et al. 2007). However, selection by competition is not universally embraced for lexical representations between languages. Still, one class of accounts considers lexical selection to be nonspecific to the target language (Green 1986, 1998; De Bot 1992; Poulisse and Bongaerts 1994; Hermans et al. 1998; Lee and Williams 2001), so that activated lexical items from both languages compete for selection. For instance, in the course of producing the word *dog* by a Spanish–English bilingual, the lexical node PERRO will be a strong competitor, and the activation level of this lexical node will be higher than the activation level of semantically-related lexical items (e.g., CAT) because they share a common semantic representation. As a solution to the ensuing massive competition between languages, the Inhibitory Control Model (IC) proposed by Green (1998) assumes that production in a target language is achieved by an inhibitory mechanism. Such a mechanism ensures selection of the lexical node in the intended language by suppressing the activation of the lexical items in the nonintended language.

Alternatively, some authors (Costa and Caramazza 1999; Costa et al. 1999) have argued that the two languages of a bilingual do not compete for selection. That is, lexical selection is language-specific in the sense that the lexical selection mechanism only considers for selection the lexical nodes in the intended language. In this model, while the lexical nodes DOG and PERRO would be highly activated to an equal extent by the intention of a bilingual to say the word *dog*, the activation of PERRO will be, so to say, ignored by the selection mechanism.

Work conducted with Spanish–English and Spanish–Catalan bilinguals by Costa and collaborators have formed a substantial part of the growing body of research aimed at delineating the language control mechanisms used by bilingual speakers. This work is based on a version of the language-switching experimental paradigm (Meuter and Allport 1999), in which bilinguals are asked to switch between naming pictures in their two languages. The language in which a given picture has to be named is indicated by the color of the picture (if blue, name in language A; if red, name in language B). In this design, there are pictures in which the response language is the same as the one used for the immediately preceding picture (nonswitch trials), and there are pictures in which the response language is different (switch trials). The magnitude of the cost incurred by switching from naming in one language to naming in the other is calculated by subtracting the naming latencies for nonswitch trials from those for switch trials.

In a replication of the seminal findings of Meuter and Allport (1999), Costa and Santesteban (2004: Experiments 1 and 2) tested low-proficiency Spanish–Catalan and Korean–Spanish bilinguals switching between their dominant (L1) and nondominant language (L2). Interestingly, switching to the dominant L1 took more time than switching to the nondominant L2 (i.e., asymmetrical switching costs). Such a result, counterintuitive as it may be, fits well with the predictions of the IC model. This is because inhibition in this model is reactive and proportional to the level of activation of the lexical items of the language to be suppressed, predicting that the time and effort needed to overcome this inhibition is proportional to the suppression previously applied to a given language. In this sense, inhibition is proportional to the relative dominance of the two languages of a bilingual: the greater the dominance of a language, the greater the inhibition that should be applied to prevent it from interfering when naming in the nonintended language. Following this logic, the asymmetry of the switching costs as found by Meuter and Allport (1999) and Costa and Santesteban (2004) should decrease with an increase in proficiency of the two languages. And this is precisely what has been reported for high-proficiency Spanish–Catalan, Spanish–Basque and Spanish–English bilinguals by Costa and collaborators (Costa and Santesteban 2004; Costa et al. 2006).

Very surprisingly, however, such symmetrical switching costs were also observed when high-proficiency Spanish–Catalan bilinguals switched between a dominant L1 (Spanish) or a nondominant but proficient L2 (Catalan) and a rather weak L3

(English). This result would not be predicted by the IC Model since the two languages used in these experiments differed considerably with regard to their proficiency. This led Costa and Santesteban (2004) to suggest that perhaps high-proficiency bilinguals develop different lexical selection mechanisms from low-proficiency bilinguals, such that, early in second language learning, inhibition would be operative to control the two languages (Green 1998), but with an increase in second language proficiency bilinguals would revert to a language-specific selection mechanism (Costa and Caramazza 1999; Costa et al. 1999; but see Costa et al. 2006, for a modification of this hypothesis in accordance with subsequent results).

In sum, work with bilingual speakers of Spanish has helped inform models of bilingual language control which aim to answer the question of how bilinguals prevent massive interference from the nonresponse language in a given communicative situation. To reconcile various results obtained with the language-switching paradigm used to shed light on this issue, it has been proposed that inhibitory control mechanisms (Green 1998) operate to avoid interference with low-proficiency bilinguals, while both inhibitory control and language-specific selection (Costa and Caramazza 1999; Costa et al. 1999) function to this aim with high-proficiency bilinguals.

5 Learning Spanish in an immersion context

Learning an L2 is nowadays a very common activity around the world. However, it is by no means easy: learners have to acquire new sounds, new words, new grammatical rules, new prosodic patterns, and new pragmatic rules, and, moreover, develop efficient control mechanisms to prevent interference from their dominant L1 (especially during the first stages of L2 learning). In popular belief, an efficient way to overcome such challenges is direct language immersion in a country where the language-to-be-learned is spoken as an L1.[1] Beyond layman convictions, however, research with Spanish learned as an L2 in an immersion context suggests that, while such a context enhances (relative to a classroom context) the chances to master some aspects of the L2 such as vocabulary and overall oral fluency, it does not confer any extra benefit on the acquisition of phonology (Díaz-Campos 2004) or new grammatical features (e.g., Collentine and Freed 2004, showed that students in a classroom context improved more than the immersed learners after a semester of study; for general immersion effects on L2 learning see Freed 1995, 1998).

Since our focus here is lexical access in speech production, in what follows, we will concentrate on the positive effects of an immersion context for verbal fluency and word retrieval in L2 Spanish, and then point out some effects of immersion on word retrieval in the L1. To begin with, some authors (DeKeyser 1991; Freed 1995) have shown that English speakers immersed in Spain made greater improvement in their narrative abilities (speech recording analysis in the Oral Proficiency Interview) than English speakers attending classes in their own country. Furthermore, it was also found that the words used to convey ideas were semantically denser for the immersed group, that is, language samples included more nouns, attributive adjectives, and prepositions, among other features (see also Collentine and Freed 2004). More recently, Linck, Kroll, and Sunderman (2009) demonstrated that a brief linguistic immersion also favors growth in vocabulary size in comparison to a classroom context. In a semantic fluency task, immersed learners produced more words than classroom learners when the task was performed in Spanish (L2).

An interesting observation, however, arose in Linck et al.'s (2009) study when immersed and classroom learners performed the verbal fluency task in English (their L1): immersed learners produced fewer category exemplars in English than classroom learners. The authors argued that the disadvantage in retrieving words in the L1 was a consequence of the inhibition applied to L1 lexical representations in order to successfully speak in the L2 during immersion. More specifically, especially at the beginning of an immersion period, L1 lexical representations would be highly activated and strongly compete for selection when production in L2 is desired. In order to avoid interference, an inhibitory mechanism (Green 1998) is required to attenuate cross-language competition with the aim of ensuring L2 production. Furthermore, in the course of the immersion period, L2 representations would be increasingly activated due to continued exposure and use. As a consequence of such increased activation of L2 and increased inhibition of L1, access to L2 representations will become easier and the recovery of L1 representations will become harder (Green 1998; see also Levy et al. 2007, for suggestive supporting evidence). Still, note that, in a follow-up experiment, Linck et al. observed that the efficiency of retrieval of L1 words is recuperated as soon as immersed learners return home: the number of exemplars produced in L1 in the fluency task was higher in the return period, compared with those produced during the immersion period. Importantly, however, the number of L2 Spanish exemplars did not decrease after returning, suggesting that controlling (suppressing) L1 activity during immersion seems to be an efficient way to learn an L2, although at the expense of momentarily losing efficiency in L1 performance.

Furthermore, in some recent work, Baus, Costa and Carreiras (submitted) showed that such changes in the activation of both languages during the time of immersion affect not only lexical processing but also phonological processing. The authors compared the picture-naming performance of a group of German speakers enrolled in an Erasmus program (*European Region Action Scheme for the Mobility of University Students*) in Spain at the beginning and at the end of the immersion period. In order to explore how phonological activation of the nonresponse language influenced production in German (L1) and Spanish (L2), the authors manipulated the cognate status of the picture names. As pointed out above, cognate words tend to be produced faster than noncognate words, especially when speaking an L2. Accordingly, Baus et al. (in preparation) showed a cognate facilitation effect, that is, when naming in Spanish, German speakers were faster and more accurate when the word to be produced was phonologically similar between German and Spanish. Furthermore, the effect was present at both testing periods, suggesting that, even though the L2 becomes more activated during the time of immersion, L2 words continue to benefit from the extra activation received from the phonological segments shared with L1. Results on L1 naming (German) revealed two interesting effects across the time of immersion: first, as expected, at the beginning of the immersion period there were no differences between cognate and noncognate words. Importantly, after four months of immersion, German speakers were overall slower when producing words in L1 (German), and a cognate effect in L1 emerged; that is, cognate L1 words were named faster than noncognate L1 words. These results reveal how rapidly immersion modulates the activation of the two languages and, hence, the selection processes at the lexical and phonological levels of representation.

Altogether, the reviewed studies on the effects of immersion in L2 and L1 lexical access suggest that the linguistic system of individuals rapidly attunes to the context of learning an L2 by regulating the activity of the two languages. Relative to a classroom context, immersion enhances production abilities. This is achieved by offering the learner a natural way to increase the activity of the L2 by weakening at the same time the L1 activity.

6 Conclusions

Research with bilingual speakers of Spanish has given significant insights into bilingual speech production. On the one hand, such work has shown a bilingual disadvantage in lexical access relative to monolingual speakers, both when speaking a nondominant and a dominant L1 and a nondominant but highly proficient L2. On the other hand, it has suggested that bilinguals possess more efficient language control mechanisms than monolingual speakers learning an L2 (referred to as low-proficiency bilinguals in the literature). In fact, it seems that such powerful mechanisms of language control extend to nonlinguistic domains: some fascinating results in recent years have suggested that bilinguals have advantages in executive control and nonlinguistic task switching compared to monolinguals. Such advantages hold throughout the life span of bilingual individuals, aiding young adults in conflict resolution and monitoring (e.g., Bialystok et al. 2004; Bialystok et al. 2005; Costa et al. 2008; Costa et al. 2009), helping preverbal bilingual children to engage executive control processes to keep apart the representations of the two languages (Kovács and Mehler 2009), and acting as a cognitive reserve against dementia later in life (Bialystok et al. 2007). Thus, it seems that being bilingual is a positive attribute after all. Sociolinguistically, we should be aware of these experimental contributions to develop new educational policies in the future. In scientific terms, bilingualism is associated with a lexical disadvantage, but it is compensated by more efficient mechanisms of language and executive control.

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NOTE

1 An example of the impact of such beliefs is the increasing number of American students deciding to enroll in study-abroad programs every year, Spain being the third most popular destination (2009 Open Doors Report on the International Education Exchange, http://www.iie.org).

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